

**TABLE 9-17 Depreciation Class Lives and MACRS Recovery Periods**

| Asset class | Description of asset                                                 | Class life, yr | MACRS recovery period, yr |
|-------------|----------------------------------------------------------------------|----------------|---------------------------|
| 00.12       | Information systems                                                  | 6              | 5                         |
| 00.4        | Industrial steam and electric generation and/or distribution systems | 22             | 15                        |
| 13.3        | Petroleum refining                                                   | 16             | 10                        |
| 20.3        | Manufacture of vegetable oils and vegetable oil products             | 18             | 10                        |
| 20.5        | Manufacture of food and beverages                                    | 4              | 3                         |
| 22.4        | Manufacture of textile yarns                                         | 8              | 5                         |
| 22.5        | Manufacture of nonwoven fabrics                                      | 10             | 7                         |
| 26.1        | Manufacture of pulp and paper                                        | 13             | 7                         |
| 28.0        | Manufacture of chemicals and allied products                         | 9.5            | 5                         |
| 30.1        | Manufacture of rubber products                                       | 14             | 7                         |
| 30.2        | Manufacture of finished plastic products                             | 11             | 7                         |
| 32.1        | Manufacture of glass products                                        | 14             | 7                         |
| 32.2        | Manufacture of cement                                                | 20             | 15                        |
| 32.3        | Manufacture of other stone and clay products                         | 15             | 7                         |
| 33.2        | Manufacture of primary nonferrous metals                             | 14             | 7                         |
| 32.4        | Manufacture of primary steel mill products                           | 15             | 7                         |
| 49.223      | Substitute natural gas-coal gasification                             | 18             | 10                        |
| 49.25       | Liquefied natural gas plant                                          | 22             | 15                        |

SOURCE: "How to Depreciate Property," Publication 946, Internal Revenue Service, U.S. Department of Treasury, Washington, 1999.

method with a switch to straight-line in the later years is used. The IRS adopted a half-year convention for both depreciation methods. Under this convention, a property placed in service is considered to be only one-half year irrespective of when during the year the property was placed in service. Table 9-17 is a listing of the class lives, and Table 9-18 contains factors with the half-year convention for both the MACRS and straight-line methods.

Depreciation is entered as an indirect expense on the manufacturing expense sheet based upon the straight-line method. However, when one is determining the after-tax cash flow, straight-line depreciation is removed from the manufacturing expense and the MACRS depreciation is entered. This is illustrated under the section on cash flow.

There are certain terms that apply to depreciation:

- *Depreciation reserve* is the accumulated depreciation at a specific time.
- *Book value* is the original investment minus the accumulated depreciation.
- *Service life* is the time period during which an asset is in service and is economically feasible.
- *Salvage value* is the net amount of money obtained from the sale of a used property over and above any charges involved in the removal and sale of the property.
- *Scrap value* implies that the asset has no further useful life and is sold for the amount of scrap material in it.
- *Economic life* is the most likely period of successful operation before a need arises for subsequent investment in additional equipment as the result of product or process obsolescence or equipment due to wear and tear.

## DEPLETION

Depletion is concerned with the diminution of natural resources. Generally depletion does not enter into process economic studies. Rules for determining the amount of depletion are found in the IRS Publication 535.

## AMORTIZATION

Amortization is the ratable deduction for the cost of an intangible property over its useful life, perhaps a 15-year life, via straight-line calculations. An example of an intangible property is a franchise, patent, trademark, etc. Two IRS publications, Form 4562 and Publication 535 (1999), established the regulations regarding amortization.

## TAXES

Corporations pay an income tax based upon gross earnings, as shown in Table 9-19. Most major corporations pay the federal tax rate of 34 percent on their annual gross earnings. In addition, some states have a stepwise corporate income tax rate. State income tax is deductible as an expense item before the calculation of the federal tax. If  $T_s$  is the incremental tax rate and  $T_f$  is the incremental federal tax, both expressed as decimals, then the combined incremental rate  $T_c$  is

**TABLE 9-18 Depreciation Rates for Straight-Line and MACRS Methods**

| Straight-line<br>half-year convention |        |        |       |      |       |      | MACRS <sup>a</sup><br>half-year convention |        |        |        |        |       |        |
|---------------------------------------|--------|--------|-------|------|-------|------|--------------------------------------------|--------|--------|--------|--------|-------|--------|
| Year                                  | 3      | 5      | 7     | 10   | 15    | 20   | Year                                       | 3      | 5      | 7      | 10     | 15    | 20     |
| 1                                     | 16.67% | 10.00% | 7.14% | 5.0% | 3.33% | 2.5% | 1                                          | 33.33% | 20.00% | 14.29% | 10.00% | 5.00% | 3.750% |
| 2                                     | 33.33  | 20.00  | 14.29 | 10.0 | 6.67  | 5.0  | 2                                          | 44.45  | 32.00  | 24.49  | 18.00  | 9.50  | 7.219  |
| 3                                     | 33.33  | 20.00  | 14.29 | 10.0 | 6.67  | 5.0  | 3                                          | 14.81  | 19.20  | 17.49  | 14.40  | 8.55  | 6.677  |
| 4                                     | 16.67  | 20.00  | 14.28 | 10.0 | 6.67  | 5.0  | 4                                          | 7.41   | 11.52  | 12.49  | 11.52  | 7.70  | 6.177  |
| 5                                     |        | 20.00  | 14.29 | 10.0 | 6.67  | 5.0  | 5                                          |        | 11.52  | 8.93   | 9.22   | 6.93  | 5.713  |
| 6                                     |        | 10.00  | 14.28 | 10.0 | 6.67  | 5.0  | 6                                          |        | 5.76   | 8.92   | 7.37   | 6.23  | 5.285  |
| 7                                     |        |        | 14.29 | 10.0 | 6.67  | 5.0  | 7                                          |        |        | 8.93   | 6.55   | 5.90  | 4.888  |
| 8                                     |        |        | 7.14  | 10.0 | 6.66  | 5.0  | 8                                          |        |        | 4.46   | 6.55   | 5.90  | 4.522  |
| 9                                     |        |        |       | 10.0 | 6.67  | 5.0  | 9                                          |        |        |        | 6.56   | 5.91  | 4.462  |
| 10                                    |        |        |       | 10.0 | 6.66  | 5.0  | 10                                         |        |        |        | 6.55   | 5.90  | 4.461  |
| 11                                    |        |        |       | 5.0  | 6.67  | 5.0  | 11                                         |        |        |        | 3.28   | 5.91  | 4.462  |
| 12                                    |        |        |       |      | 6.66  | 5.0  | 12                                         |        |        |        |        | 5.90  | 4.461  |
| 13                                    |        |        |       |      | 6.67  | 5.0  | 13                                         |        |        |        |        | 5.91  | 4.462  |
| 14                                    |        |        |       |      | 6.66  | 5.0  | 14                                         |        |        |        |        | 5.90  | 4.461  |
| 15                                    |        |        |       |      | 6.67  | 5.0  | 15                                         |        |        |        |        | 5.91  | 4.462  |
| 16                                    |        |        |       |      | 3.33  | 5.0  | 16                                         |        |        |        |        | 2.95  | 4.461  |
| 17                                    |        |        |       |      |       | 5.0  | 17                                         |        |        |        |        |       | 4.462  |
| 18                                    |        |        |       |      |       | 5.0  | 18                                         |        |        |        |        |       | 4.461  |
| 19                                    |        |        |       |      |       | 5.0  | 19                                         |        |        |        |        |       | 4.462  |
| 20                                    |        |        |       |      |       | 5.0  | 20                                         |        |        |        |        |       | 4.461  |
| 21                                    |        |        |       |      |       | 2.5  | 21                                         |        |        |        |        |       | 2.231  |

<sup>a</sup>General depreciation system. Declining-balance switching to straight-line. Recovery periods 3, 5, 7, 10, 15, and 20 years.

SOURCE: "How to Depreciate Property," Publication 946, Internal Revenue Service, U.S. Department of Treasury, Washington, 1999.

TABLE 9-19 Corporate Federal Income Tax Rates

| Taxable income                                                                                                         | Tax rate, % |
|------------------------------------------------------------------------------------------------------------------------|-------------|
| Annual gross earnings less than \$50,000                                                                               | 15          |
| Annual gross earnings greater than \$50,000 but not over \$75,000                                                      | 25          |
| Annual gross earnings greater than \$75,000 plus 5% of gross earnings over \$100,000 or \$11,750, whichever is greater | 34          |
| Corporations with gross earnings of at least \$335,000 pay a flat rate of 34%                                          |             |

SOURCE: U.S. Corporate Income Tax Return, Form 1120, Internal Revenue Service, U. S. Department of Treasury, Washington, 1999.

$$T_c = T_s + (1 - T_s)T_f \quad (9-12)$$

If the federal rate is 34 percent and the state rate is 7 percent, then the combined rate is

$$T_c = 0.07 + (1 - 0.07)(0.34) = 0.39$$

Therefore, the combined tax rate is 39 percent.

### TIME VALUE OF MONEY

In business, money is either borrowed or loaned. If money is loaned, there is the risk that it may not be repaid. From the lender's standpoint, the funds could have been invested somewhere else and made a profit; therefore, the interest charged for the loan is compensation for the foregone profit. The borrower may look upon this interest as the cost of renting money. The amount of interest charged depends on the scarcity of money, the size of the loan, the length of the loan period, the risk that the lender feels that the loan may not be repaid, and the prevailing economic conditions. Engineers involved in the presentation and/or the evaluation of an investment of funds in a venture, therefore, need to understand the time value of money and how it is applied in the evaluation of projects.

The amount of the loan is called the *principal*  $P$ . The longer the time for which the money is loaned, the greater the total amount of interest paid. The *future amount of the money*  $F$  is greater than the *principal or present worth*  $P$ . The relationship between  $F$  and  $P$  depends upon the type of interest used. Table 9-20 is a summary of the nomenclature used in time value of money calculations.

**Simple Interest** The relationship between  $F$  and  $P$  is  $F = P(1 + in)$ . The interest is charged on the original loan and not on the unpaid balance (Couper and Rader, 1986). The interest is paid at the end of each time interval. Although the simple interest concept still exists, it is seldom used in business.

**Discrete Compound Interest** In financial transactions, loans or deposits are made using compound interest. The interest is not withdrawn but is added to the principal for that time period. In the next time period, the interest is calculated upon the principal plus the interest from the preceding time period. This process illustrates compound interest. In equation format,

$$\text{Year 1: } P + Pi = P(1 + i) = F_1$$

TABLE 9-20 Interest Nomenclature

| Symbol | Definition                                                    |
|--------|---------------------------------------------------------------|
| $F$    | Future sum<br>Future value<br>Future worth<br>Future amount   |
| $P$    | Principal<br>Present worth<br>Present value<br>Present amount |
| $A$    | End of period payment in a uniform series                     |

$$\text{Year 2: } P + Pi(1 + i) = P(1 + i)^2 = F_2 \quad (9-13)$$

$$\text{Year } n: P(1 + i)^n = F$$

An interest rate quoted on an annual basis is called *nominal interest*. However, interest may be payable on a semiannual, quarterly, monthly, or daily basis. To determine the amount compounded, the following equation applies:

$$F = P \left( 1 + \frac{i}{m} \right)^{mn} \quad (9-14)$$

where  $m$  = number of interest periods per year  
 $n$  = number of years  
 $i$  = nominal interest

Interest calculated for a given time period is known as *discrete compound interest*, with *discrete* referring to a discrete time period. Table 9-21 contains 5 and 6 percent discrete interest factors.

Examples of the use of discrete factors for various applications are found in Table 9-22, assuming that the present time is when the first funds are expended.

**Continuous Compound Interest** In some companies, namely, petroleum, petrochemical, and chemical companies, money transactions occur hourly or daily, or essentially continuously. The receipts from sales and services are invested immediately upon receipt. The interest on this cash flow is continuously compounded. To use continuous compounding when evaluating projects or investments, one assumes that cash flows continuously.

In continuous compounding, the year is divided into an infinite number of periods. Mathematically, the limit of the interest term is

$$\lim_{n \rightarrow \infty} \left( 1 + \frac{r}{m} \right)^{mn} = e^r \quad (9-15)$$

where  $n$  = number of years  
 $m$  = number of interest periods per year  
 $r$  = nominal interest rate  
 $e$  = base for naperian logarithms

The numerical difference between discrete compound interest and continuous compound interest is small, but when large sums of money are involved, the difference may be significant. Table 9-23 is an abbreviated continuous interest table, assuming that time zero is when start-up occurs. A summary of the equations for discrete compound and continuous compound interest is found in Table 9-24.

**Compounding-Discounting** When money is moved forward in time from the present to a future time, the process is called *compounding*. The effect of compounding is that the total amount of money increases with time due to interest. *Discounting* is the reverse process, i.e., a sum of money moved backward in time. Figure 9-8 is a

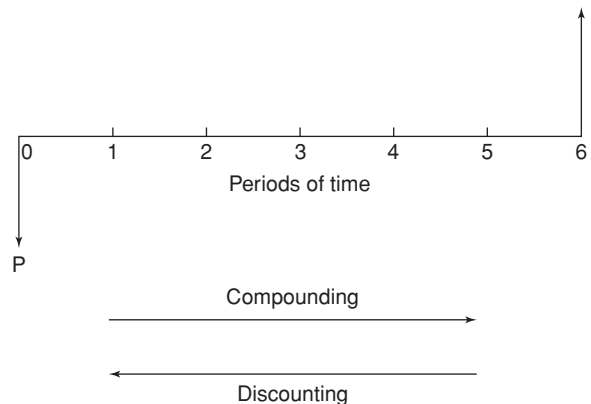


FIG. 9-8 Compounding-discounting diagram.

TABLE 9-21 Discrete Compound Interest Factors\*

| Single payment               |                                    |                                              | Uniform annual series                            |                                                         |                                                  |                                               | Single payment                     |                                              | Uniform annual series                            |                                                         |                                                  |                                               |     |  |
|------------------------------|------------------------------------|----------------------------------------------|--------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|-----------------------------------------------|------------------------------------|----------------------------------------------|--------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|-----------------------------------------------|-----|--|
|                              | Compound-amount factor             | Present-worth factor                         | Sinking-fund factor                              | Capital-recovery factor                                 | Compound-amount factor                           | Present-worth factor                          | Compound-amount factor             | Present-worth factor                         | Sinking-fund factor                              | Capital-recovery factor                                 | Compound-amount factor                           | Present-worth factor                          |     |  |
|                              | Given $P$ to find $F$<br>$(1+i)^n$ | Given $F$ to find $P$<br>$\frac{1}{(1+i)^n}$ | Given $F$ to find $A$<br>$\frac{i}{(1+i)^n - 1}$ | Given $P$ to find $A$<br>$\frac{i(1+i)^n}{(1+i)^n - 1}$ | Given $A$ to find $F$<br>$\frac{(1+i)^n - 1}{i}$ | Given $A$ to find $P$<br>$\frac{1}{i(1+i)^n}$ | Given $P$ to find $F$<br>$(1+i)^n$ | Given $F$ to find $P$<br>$\frac{1}{(1+i)^n}$ | Given $F$ to find $A$<br>$\frac{i}{(1+i)^n - 1}$ | Given $P$ to find $A$<br>$\frac{i(1+i)^n}{(1+i)^n - 1}$ | Given $A$ to find $F$<br>$\frac{(1+i)^n - 1}{i}$ | Given $A$ to find $P$<br>$\frac{1}{i(1+i)^n}$ | $n$ |  |
| 5% Compound Interest Factors |                                    |                                              |                                                  |                                                         |                                                  |                                               | 6% Compound Interest Factors       |                                              |                                                  |                                                         |                                                  |                                               |     |  |
| 1                            | 1.050                              | 0.9524                                       | 1.00000                                          | 1.05000                                                 | 1.000                                            | 0.952                                         | 1.060                              | 0.9434                                       | 1.00000                                          | 1.06000                                                 | 1.000                                            | 0.943                                         | 1   |  |
| 2                            | 1.103                              | .9070                                        | 0.48780                                          | 0.53780                                                 | 2.050                                            | 1.859                                         | 1.124                              | .8900                                        | 0.48544                                          | 0.54544                                                 | 2.060                                            | 1.833                                         | 2   |  |
| 3                            | 1.158                              | .8638                                        | .31721                                           | .36721                                                  | 3.153                                            | 2.723                                         | 1.191                              | .8396                                        | .31411                                           | .37411                                                  | 3.184                                            | 2.673                                         | 3   |  |
| 4                            | 1.216                              | .8227                                        | .23201                                           | .28201                                                  | 4.310                                            | 3.546                                         | 1.262                              | .7921                                        | .22859                                           | .28859                                                  | 4.375                                            | 3.465                                         | 4   |  |
| 5                            | 1.276                              | .7835                                        | .18097                                           | .23097                                                  | 5.526                                            | 4.329                                         | 1.338                              | .7473                                        | .17740                                           | .23740                                                  | 5.637                                            | 4.212                                         | 5   |  |
| 6                            | 1.340                              | .7462                                        | .14702                                           | .19702                                                  | 6.802                                            | 5.076                                         | 1.419                              | .7050                                        | .14336                                           | .20336                                                  | 6.975                                            | 4.917                                         | 6   |  |
| 7                            | 1.407                              | .7107                                        | .12282                                           | .17282                                                  | 8.142                                            | 5.786                                         | 1.504                              | .6651                                        | .11914                                           | .17914                                                  | 8.394                                            | 5.582                                         | 7   |  |
| 8                            | 1.477                              | .6768                                        | .10472                                           | .15472                                                  | 9.549                                            | 6.463                                         | 1.594                              | .6274                                        | .10104                                           | .16104                                                  | 9.897                                            | 6.210                                         | 8   |  |
| 9                            | 1.551                              | .6446                                        | .09069                                           | .14069                                                  | 11.027                                           | 7.108                                         | 1.689                              | .5919                                        | .08702                                           | .14702                                                  | 11.491                                           | 6.802                                         | 9   |  |
| 10                           | 1.629                              | .6139                                        | .07940                                           | .12950                                                  | 12.578                                           | 7.722                                         | 1.791                              | .5584                                        | .07587                                           | .13587                                                  | 13.181                                           | 7.360                                         | 10  |  |
| 11                           | 1.710                              | .5847                                        | .07039                                           | .12039                                                  | 14.207                                           | 8.306                                         | 1.898                              | .5268                                        | .06679                                           | .12679                                                  | 14.972                                           | 7.887                                         | 11  |  |
| 12                           | 1.796                              | .5568                                        | .06283                                           | .11283                                                  | 15.917                                           | 8.863                                         | 2.012                              | .4970                                        | .05928                                           | .11928                                                  | 16.870                                           | 8.384                                         | 12  |  |
| 13                           | 1.886                              | .5303                                        | .05646                                           | .10646                                                  | 17.713                                           | 9.394                                         | 2.133                              | .4688                                        | .05296                                           | .11296                                                  | 18.882                                           | 8.853                                         | 13  |  |
| 14                           | 1.980                              | .5051                                        | .05102                                           | .10102                                                  | 19.599                                           | 9.899                                         | 2.261                              | .4423                                        | .04758                                           | .10758                                                  | 21.015                                           | 9.295                                         | 14  |  |
| 15                           | 2.079                              | .4810                                        | .04634                                           | .09634                                                  | 21.579                                           | 10.380                                        | 2.397                              | .4173                                        | .04296                                           | .10296                                                  | 23.276                                           | 9.712                                         | 15  |  |
| 16                           | 2.183                              | .4581                                        | .04227                                           | .09227                                                  | 23.657                                           | 10.838                                        | 2.540                              | .3936                                        | .03895                                           | .09895                                                  | 25.673                                           | 10.106                                        | 16  |  |
| 17                           | 2.292                              | .4363                                        | .03870                                           | .08870                                                  | 25.840                                           | 11.274                                        | 2.693                              | .3714                                        | .03544                                           | .09544                                                  | 28.213                                           | 10.477                                        | 17  |  |
| 18                           | 2.407                              | .4155                                        | .03555                                           | .08555                                                  | 28.132                                           | 11.690                                        | 2.854                              | .3503                                        | .03236                                           | .09236                                                  | 30.906                                           | 10.828                                        | 18  |  |
| 19                           | 2.527                              | .3957                                        | .03275                                           | .08275                                                  | 30.539                                           | 12.085                                        | 3.026                              | .3305                                        | .02962                                           | .08962                                                  | 33.760                                           | 11.158                                        | 19  |  |
| 20                           | 2.653                              | .3769                                        | .03024                                           | .08024                                                  | 33.066                                           | 12.462                                        | 3.207                              | .3118                                        | .02718                                           | .08718                                                  | 36.786                                           | 11.470                                        | 20  |  |
| 21                           | 2.786                              | .3589                                        | .02800                                           | .07800                                                  | 35.719                                           | 12.821                                        | 3.400                              | .2942                                        | .02500                                           | .08500                                                  | 39.993                                           | 11.764                                        | 21  |  |
| 22                           | 2.925                              | .3418                                        | .02597                                           | .07597                                                  | 38.505                                           | 13.163                                        | 3.604                              | .2775                                        | .02305                                           | .08305                                                  | 43.392                                           | 12.042                                        | 22  |  |
| 23                           | 3.072                              | .3256                                        | .02414                                           | .07414                                                  | 41.430                                           | 13.489                                        | 3.820                              | .2618                                        | .02128                                           | .08128                                                  | 46.996                                           | 12.303                                        | 23  |  |
| 24                           | 3.225                              | .3101                                        | .02247                                           | .07247                                                  | 44.502                                           | 13.799                                        | 4.049                              | .2470                                        | .01968                                           | .07968                                                  | 50.816                                           | 12.550                                        | 24  |  |
| 25                           | 3.386                              | .2953                                        | .02095                                           | .07095                                                  | 47.727                                           | 14.094                                        | 4.292                              | .2330                                        | .01823                                           | .07823                                                  | 54.865                                           | 12.783                                        | 25  |  |
| 26                           | 3.556                              | .2812                                        | .01956                                           | .06956                                                  | 51.113                                           | 14.375                                        | 4.549                              | .2198                                        | .01690                                           | .07690                                                  | 59.156                                           | 13.003                                        | 26  |  |
| 27                           | 3.733                              | .2678                                        | .01829                                           | .06829                                                  | 54.669                                           | 14.643                                        | 4.822                              | .2074                                        | .01570                                           | .07570                                                  | 63.706                                           | 13.211                                        | 27  |  |
| 28                           | 3.920                              | .2551                                        | .01712                                           | .06712                                                  | 58.403                                           | 14.898                                        | 5.112                              | .1956                                        | .01459                                           | .07459                                                  | 68.528                                           | 13.406                                        | 28  |  |
| 29                           | 4.116                              | .2429                                        | .01605                                           | .06605                                                  | 62.323                                           | 15.141                                        | 5.418                              | .1846                                        | .01358                                           | .07358                                                  | 73.640                                           | 13.591                                        | 29  |  |
| 30                           | 4.322                              | .2314                                        | .01505                                           | .06505                                                  | 66.439                                           | 15.372                                        | 5.743                              | .1741                                        | .01265                                           | .07265                                                  | 79.058                                           | 13.765                                        | 30  |  |
| 31                           | 4.538                              | .2204                                        | .01413                                           | .06413                                                  | 70.761                                           | 15.593                                        | 6.088                              | .1643                                        | .01179                                           | .07179                                                  | 84.802                                           | 13.929                                        | 31  |  |
| 32                           | 4.765                              | .2099                                        | .01328                                           | .06328                                                  | 75.299                                           | 15.803                                        | 6.453                              | .1550                                        | .01100                                           | .07100                                                  | 90.890                                           | 14.084                                        | 32  |  |
| 33                           | 5.003                              | .1999                                        | .01249                                           | .06249                                                  | 80.064                                           | 16.003                                        | 6.841                              | .1462                                        | .01027                                           | .07027                                                  | 97.343                                           | 14.230                                        | 33  |  |
| 34                           | 5.253                              | .1904                                        | .01176                                           | .06176                                                  | 85.067                                           | 16.193                                        | 7.251                              | .1379                                        | .00960                                           | .06960                                                  | 104.184                                          | 14.368                                        | 34  |  |
| 35                           | 5.516                              | .1813                                        | .01107                                           | .06107                                                  | 90.320                                           | 16.374                                        | 7.686                              | .1301                                        | .00897                                           | .06897                                                  | 111.435                                          | 14.498                                        | 35  |  |
| 40                           | 7.040                              | .1420                                        | .00828                                           | .05828                                                  | 120.800                                          | 17.159                                        | 10.286                             | .0972                                        | .00646                                           | .06646                                                  | 154.762                                          | 15.046                                        | 40  |  |
| 45                           | 8.985                              | .1113                                        | .00626                                           | .05626                                                  | 159.700                                          | 17.774                                        | 13.765                             | .0727                                        | .00470                                           | .06470                                                  | 212.744                                          | 15.456                                        | 45  |  |
| 50                           | 11.467                             | .0872                                        | .00478                                           | .05478                                                  | 209.348                                          | 18.256                                        | 18.420                             | .0543                                        | .00344                                           | .06344                                                  | 290.336                                          | 15.762                                        | 50  |  |
| 55                           | 14.636                             | .0683                                        | .00367                                           | .05367                                                  | 272.713                                          | 18.633                                        | 24.650                             | .0406                                        | .00254                                           | .06254                                                  | 394.172                                          | 15.991                                        | 55  |  |
| 60                           | 18.679                             | .0535                                        | .00283                                           | .05283                                                  | 353.584                                          | 18.929                                        | 32.988                             | .0303                                        | .00188                                           | .06188                                                  | 533.128                                          | 16.161                                        | 60  |  |
| 65                           | 23.840                             | .0419                                        | .00219                                           | .05219                                                  | 456.798                                          | 19.161                                        | 44.145                             | .0227                                        | .00139                                           | .06139                                                  | 719.083                                          | 16.289                                        | 65  |  |
| 70                           | 30.426                             | .0329                                        | .00170                                           | .05170                                                  | 588.529                                          | 19.343                                        | 59.076                             | .0169                                        | .00103                                           | .06103                                                  | 967.932                                          | 16.385                                        | 70  |  |
| 75                           | 38.833                             | .0258                                        | .00132                                           | .05132                                                  | 756.654                                          | 19.485                                        | 79.057                             | .0126                                        | .00077                                           | .06077                                                  | 1,300.949                                        | 16.456                                        | 75  |  |
| 80                           | 49.561                             | .0202                                        | .00103                                           | .05103                                                  | 971.229                                          | 19.596                                        | 105.796                            | .0095                                        | .00057                                           | .06057                                                  | 1,746.600                                        | 16.509                                        | 80  |  |
| 85                           | 63.254                             | .0158                                        | .00080                                           | .05080                                                  | 1,245.087                                        | 19.684                                        | 141.579                            | .0071                                        | .00043                                           | .06043                                                  | 2,342.982                                        | 16.549                                        | 85  |  |
| 90                           | 80.730                             | .0124                                        | .00063                                           | .05063                                                  | 1,594.607                                        | 19.752                                        | 189.465                            | .0053                                        | .00032                                           | .06032                                                  | 3,141.075                                        | 16.579                                        | 90  |  |
| 95                           | 103.035                            | .0097                                        | .00049                                           | .05049                                                  | 2,040.694                                        | 19.806                                        | 253.546                            | .0039                                        | .00024                                           | .06024                                                  | 4,209.104                                        | 16.601                                        | 95  |  |
| 100                          | 131.501                            | .0076                                        | .00038                                           | .05038                                                  | 2,610.025                                        | 19.848                                        | 339.302                            | .0029                                        | .00018                                           | .06018                                                  | 5,638.368                                        | 16.618                                        | 100 |  |

\*Factors presented for two interest rates only. By using the appropriate formulas, values for other interest rates may be calculated.

**TABLE 9-22 Examples of the Use of Compound Interest Table**

*Given:* \$2500 is invested now at 5 percent.

*Required:* Accumulated value in 10 years (i.e., the amount of a given principal).

*Solution:*

$$F = P(1 + i)^n = \$2500 \times 1.05^{10}$$

$$\text{Compound-amount factor} = (1 + i)^n = 1.05^{10} = 1.629$$

$$F = \$2500 \times 1.629 = \$4062.50$$

*Given:* \$19,500 will be required in 5 years to replace equipment now in use.

*Required:* With interest available at 3 percent, what sum must be deposited in the bank at present to provide the required capital (i.e., the principal which will amount to a given sum)?

*Solution:*

$$P = F \frac{1}{(1 + i)^n} = \$19,500 \frac{1}{1.03^5}$$

$$\text{Present-worth factor} = 1/(1 + i)^n = 1/1.03^5 = 0.8626$$

$$P = \$19,500 \times 0.8626 = \$16,821$$

*Given:* \$50,000 will be required in 10 years to purchase equipment.

*Required:* With interest available at 4 percent, what sum must be deposited each year to provide the required capital (i.e., the annuity which will amount to a given fund)?

*Solution:*

$$A = F \frac{i}{(1 + i)^n - 1} = \$50,000 \frac{0.04}{1.04^{10} - 1}$$

$$\text{Sinking-fund factor} = \frac{i}{(1 + i)^n - 1} = \frac{0.04}{1.04^{10} - 1} = 0.08329$$

$$A = \$50,000 \times 0.08329 = \$4,164$$

*Given:* \$20,000 is invested at 10 percent interest.

*Required:* Annual sum that can be withdrawn over a 20-year period (i.e., the annuity provided by a given capital).

*Solution:*

$$A = P \frac{i(1 + i)^n}{(1 + i)^n - 1} = \$20,000 \frac{0.10 \times 1.10^{20}}{1.10^{20} - 1}$$

$$\text{Capital-recovery factor} = \frac{i(1 + i)^n}{(1 + i)^n - 1} = \frac{0.10 \times 1.10^{20}}{1.10^{20} - 1} = 0.11746$$

$$A = \$20,000 \times 0.11746 = \$2349.20$$

*Given:* \$500 is invested each year at 8 percent interest.

*Required:* Accumulated value in 15 years (i.e., amount of an annuity).

*Solution:*

$$F = A \frac{(1 + i)^n - 1}{i} = \$500 \frac{1.08^{15} - 1}{0.08}$$

$$\text{Compound-amount factor} = \frac{(1 + i)^n - 1}{i} = \frac{1.08^{15} - 1}{0.08} = 27.152$$

$$F = \$500 \times 27.152 = \$13,576$$

*Given:* \$8000 is required annually for 25 years.

*Required:* Sum that must be deposited now at 6 percent interest.

*Solution:*

$$P = A \frac{(1 + i)^n - 1}{i(1 + i)^n} = \$8000 \frac{1.06^{25} - 1}{0.06 \times 1.06^{25}}$$

$$\text{Present-worth factor} = \frac{(1 + i)^n - 1}{i(1 + i)^n} = \frac{1.06^{25} - 1}{0.06 \times 1.06^{25}} = 12.783$$

$$P = \$8000 \times 12.783 = \$102,264$$

sketch of this process. The time periods are years, and the interest is normally on an annual basis using end-of-year money flows. The longer the time before money is received, the less it is worth at present.

**Effective Interest Rates** When an interest rate is quoted, it is *nominal* interest that is stated. These quotes are on an annual basis, however, when compounding occurs that is not the actual or effective interest. According to government regulations, an effective

rate APY must be stated also. The effective interest is calculated by

$$i_{\text{eff}} = \left(1 + \frac{i}{m}\right)^{(m)(1)} - 1 \quad (9-16)$$

The time period for calculating the effective interest rate is 1 year.



**TABLE 9-23 Condensed Continuous Interest Table\***

Factors for determining zero-time values for cash flows which occur at other than zero time.

| <b>Compounding of Cash Flows Which Occur:</b>   |                       | 1%    | 5%    | 10%   | 15%   | 20%   | 25%   | 30%   | 35%   | 40%   | 50%   | 60%   | 70%   | 80%    | 90%    | 100%   |
|-------------------------------------------------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|
| <b>A. In an Instant</b>                         |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| $\frac{1}{2}$                                   | year before           | 1.005 | 1.025 | 1.051 | 1.078 | 1.105 | 1.133 | 1.162 | 1.191 | 1.221 | 1.284 | 1.350 | 1.419 | 1.492  | 1.568  | 1.649  |
| 1                                               | " "                   | 1.010 | 1.051 | 1.105 | 1.162 | 1.221 | 1.284 | 1.350 | 1.419 | 1.492 | 1.649 | 1.822 | 2.014 | 2.226  | 2.460  | 2.718  |
| $1\frac{1}{2}$                                  | " "                   | 1.015 | 1.078 | 1.162 | 1.252 | 1.350 | 1.455 | 1.568 | 1.690 | 1.822 | 2.117 | 2.460 | 2.858 | 3.320  | 3.857  | 4.482  |
| 2                                               | " "                   | 1.020 | 1.105 | 1.221 | 1.350 | 1.492 | 1.649 | 1.822 | 2.014 | 2.226 | 2.718 | 3.320 | 4.055 | 4.953  | 6.050  | 7.389  |
| 3                                               | " "                   | 1.030 | 1.162 | 1.350 | 1.568 | 1.822 | 2.117 | 2.460 | 2.858 | 3.320 | 4.482 | 6.050 | 8.166 | 11.023 | 14.880 | 20.086 |
| <b>B. Uniformly until Zero Time</b>             |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| From $\frac{1}{2}$                              | year before to 0 time | 1.002 | 1.013 | 1.025 | 1.038 | 1.052 | 1.065 | 1.079 | 1.093 | 1.107 | 1.136 | 1.166 | 1.197 | 1.230  | 1.263  | 1.297  |
| " 1                                             | " " " " " "           | 1.005 | 1.025 | 1.052 | 1.079 | 1.107 | 1.136 | 1.166 | 1.197 | 1.230 | 1.297 | 1.370 | 1.448 | 1.532  | 1.622  | 1.718  |
| " $1\frac{1}{2}$                                | " " " " " "           | 1.008 | 1.038 | 1.079 | 1.121 | 1.166 | 1.213 | 1.263 | 1.315 | 1.370 | 1.489 | 1.622 | 1.769 | 1.933  | 2.117  | 2.321  |
| " 2                                             | " " " " " "           | 1.010 | 1.052 | 1.107 | 1.166 | 1.230 | 1.297 | 1.370 | 1.448 | 1.532 | 1.718 | 1.933 | 2.182 | 2.471  | 2.805  | 3.194  |
| " 3                                             | " " " " " "           | 1.015 | 1.079 | 1.166 | 1.263 | 1.370 | 1.489 | 1.622 | 1.769 | 1.933 | 2.321 | 2.805 | 3.412 | 4.176  | 5.141  | 6.362  |
| <b>Discounting of Cash Flows Which Occur:</b>   |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| <b>C. In an Instant</b>                         |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| 1 year later                                    |                       | .990  | .951  | .905  | .861  | .819  | .779  | .741  | .705  | .670  | .606  | .549  | .497  | .449   | .407   | .360   |
| 2 " "                                           |                       | .980  | .905  | .819  | .741  | .670  | .606  | .549  | .497  | .449  | .368  | .301  | .247  | .202   | .165   | .135   |
| 3 " "                                           |                       | .970  | .861  | .741  | .638  | .549  | .472  | .407  | .350  | .301  | .223  | .165  | .122  | .091   | .067   | .050   |
| 4 " "                                           |                       | .961  | .819  | .670  | .549  | .449  | .368  | .301  | .247  | .202  | .135  | .091  | .061  | .041   | .027   | .018   |
| 5 " "                                           |                       | .951  | .779  | .606  | .472  | .368  | .286  | .223  | .174  | .135  | .082  | .050  | .030  | .018   | .011   | .007   |
| 10 years later                                  |                       | .905  | .606  | .368  | .223  | .135  | .082  | .050  | .030  | .018  | .007  | .002  | .001  | —      | —      | —      |
| 15 " "                                          |                       | .861  | .472  | .223  | .105  | .050  | .024  | .011  | .005  | .002  | .001  | —     | —     | —      | —      | —      |
| 20 " "                                          |                       | .819  | .368  | .135  | .050  | .018  | .007  | .002  | .001  | —     | —     | —     | —     | —      | —      | —      |
| 25 " "                                          |                       | .779  | .286  | .082  | .024  | .007  | .002  | .001  | —     | —     | —     | —     | —     | —      | —      | —      |
| <b>D. Uniformly over Individual Years</b>       |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| 1st year                                        |                       | .995  | .975  | .952  | .929  | .906  | .885  | .864  | .844  | .824  | .787  | .752  | .719  | .688   | .659   | .632   |
| 2nd " "                                         |                       | .985  | .928  | .861  | .799  | .742  | .689  | .640  | .595  | .552  | .477  | .413  | .357  | .309   | .268   | .232   |
| 3rd " "                                         |                       | .975  | .883  | .779  | .688  | .608  | .537  | .474  | .419  | .370  | .290  | .226  | .177  | .139   | .109   | .086   |
| 4th " "                                         |                       | .966  | .840  | .705  | .592  | .497  | .418  | .351  | .295  | .248  | .176  | .124  | .088  | .062   | .044   | .032   |
| 5th " "                                         |                       | .956  | .799  | .638  | .510  | .407  | .326  | .260  | .208  | .166  | .106  | .068  | .044  | .028   | .018   | .012   |
| 6th year                                        |                       | .946  | .760  | .577  | .439  | .333  | .254  | .193  | .147  | .112  | .065  | .037  | .022  | .013   | .007   | .004   |
| 7th " "                                         |                       | .937  | .723  | .522  | .378  | .273  | .197  | .143  | .103  | .075  | .039  | .020  | .011  | .006   | .003   | .002   |
| 8th " "                                         |                       | .928  | .687  | .473  | .325  | .224  | .154  | .108  | .073  | .050  | .024  | .011  | .005  | .002   | .001   | .001   |
| 9th " "                                         |                       | .918  | .654  | .428  | .280  | .183  | .120  | .078  | .051  | .034  | .014  | .006  | .003  | .001   | —      | —      |
| 10th " "                                        |                       | .909  | .622  | .387  | .241  | .150  | .093  | .058  | .036  | .022  | .009  | .003  | .001  | —      | —      | —      |
| <b>E. Uniformly over 5-Year Periods</b>         |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| 1st 5 years                                     |                       | .975  | .885  | .787  | .704  | .632  | .571  | .518  | .472  | .432  | .367  | .317  | .277  | .245   | .220   | .199   |
| 6th through 10th year                           |                       | .928  | .689  | .477  | .332  | .232  | .164  | .116  | .082  | .058  | .030  | .016  | .008  | .004   | .002   | .001   |
| 11th through 15th year                          |                       | .883  | .537  | .290  | .157  | .086  | .047  | .026  | .014  | .008  | .002  | .001  | —     | —      | —      | —      |
| 16th through 20th year                          |                       | .840  | .418  | .176  | .074  | .032  | .013  | .006  | .002  | .001  | —     | —     | —     | —      | —      | —      |
| 21st through 25th year                          |                       | .799  | .326  | .106  | .035  | .012  | .004  | .001  | —     | —     | —     | —     | —     | —      | —      | —      |
| <b>F. Declining to Nothing at Constant Rate</b> |                       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |
| 1st 5 years                                     |                       | .983  | .922  | .852  | .791  | .736  | .687  | .643  | .603  | .568  | .506  | .456  | .413  | .377   | .347   | .320   |
| " 10 " "                                        |                       | .968  | .852  | .736  | .643  | .568  | .506  | .456  | .413  | .377  | .320  | .278  | .245  | .219   | .198   | .180   |
| " 15 " "                                        |                       | .952  | .791  | .643  | .536  | .456  | .394  | .347  | .309  | .278  | .231  | .198  | .172  | .153   | .137   | .124   |
| " 20 " "                                        |                       | .936  | .736  | .568  | .456  | .377  | .320  | .278  | .245  | .219  | .180  | .153  | .133  | .117   | .105   | .095   |
| " 25 " "                                        |                       | .922  | .687  | .506  | .394  | .320  | .269  | .231  | .203  | .180  | .147  | .124  | .108  | .095   | .085   | .077   |

\*From tables compiled by J. C. Gregory, The Atlantic Refining Co.

TABLE 9-24 Summary of Discrete and Compound Interest Equations

| Factor           | Find | Given | Discrete compounding                                    |                | Continuous compounding                                    |                       |
|------------------|------|-------|---------------------------------------------------------|----------------|-----------------------------------------------------------|-----------------------|
| Single payment   |      |       |                                                         |                |                                                           |                       |
| Compound amount  | $F$  | $P$   | $F = P(1 + i)^n$                                        | $P(F/P, i, n)$ | $F = P(e^{rn})$                                           | $P(F/P, r, n)^\infty$ |
| Present worth    | $P$  | $F$   | $P = F \left[ \frac{1}{(1 + i)^n} \right]$              | $F(P/F, i, n)$ | $P = F(e^{-rn})$                                          | $F(P/F, r, n)^\infty$ |
| Uniform series   |      |       |                                                         |                |                                                           |                       |
| Compound amount  | $F$  | $A$   | $F = A \left[ \frac{(1 + i)^n - 1}{i} \right]$          | $A(F/A, i, n)$ | $F = A \left( \frac{e^{rn} - 1}{e^r - 1} \right)$         | $F(F/A, r, n)^\infty$ |
| Sinking fund     | $A$  | $F$   | $A = F \left[ \frac{i}{(1 + i)^n - 1} \right]$          | $F(A/F, i, n)$ | $A = F \left( \frac{e^r - 1}{e^{rn} - 1} \right)$         | $F(A/F, r, n)^\infty$ |
| Present worth    | $P$  | $A$   | $P = A \left[ \frac{(1 + i)^n - 1}{i(1 + i)^n} \right]$ | $A(P/A, i, n)$ | $P = A \left[ \frac{e^{rn} - 1}{e^r(e^{rn} - 1)} \right]$ | $A(P/A, r, n)^\infty$ |
| Capital recovery | $A$  | $P$   | $A = P \left[ \frac{i(1 + i)^n}{(1 + i)^n - 1} \right]$ | $P(A/P, i, n)$ | $A = P \left[ \frac{e^{rn}(e^r - 1)}{e^{rn} - 1} \right]$ | $A(P/A, r, n)^\infty$ |

**Example 6: Effective Interest Rate** A person is quoted an 8.33 percent nominal interest rate on a 4-year loan compounded monthly. Determine the effective interest rate.

**Solution:**

$$i_{\text{eff}} = \left( \frac{1 + i}{m} \right)^{(m)(1)} = \left( \frac{1 + 0.833}{12} \right)^{(12)(1)} - 1 = (1.00694)^{12} - 1$$

$$= 1.0865 - 1 = 0.0865$$

The effective interest rate is 8.65 percent.

## CASH FLOW

Cash flow is the amount of funds available to a company to meet current operating expenses. Cash flow may be expressed on a before- or after-tax basis. *After-tax cash flow* is defined as the net profit (income) after taxes plus depreciation. It is an integral part of the net present worth (NPW) and discounted cash flow profitability calculations.

The cash flow diagram, also referred to as a cash flow model (Fig. 9-9), shows the relationship between revenue, cash operating expenses, depreciation, and profit. This diagram is similar in many respects to a process flow diagram, but it is in dollars. Revenue is generated from the

sale of a product manufactured in “operations.” Working capital is replenished from sales and may be considered to be in dynamic equilibrium with operations. Leaving the operations box is a stream, “cash operating expenses.” It includes all the cash expenses incurred in the operation but does not include the noncash item depreciation. Since depreciation is an allowance, it is reported on the operating expense sheet, in accordance with the tax laws, as an operating expense item. (See the section “Operating Expense Estimation.”) Depreciation is an internal expense, and this allowance is retained within the company. If the cash operating expenses are subtracted from the revenue, the result is the operating income. If depreciation is subtracted from the operating income, the net profit before taxes results. Federal income taxes are then deducted from the net profit before taxes, giving the net profit after taxes. When depreciation and net profit after taxes are summed, the result is the after-tax cash flow. The terminology in Fig. 9-9 is consistent with that found in most company income statements in company annual reports.

An equation can be developed for cash flow as follows:

$$CF = (R - C - D)(1 - t) + D \quad (9-17)$$

where  $R$  = revenue  
 $C$  = cash operating expenses  
 $D$  = depreciation  
 $t$  = tax rate  
 $CF$  = after-tax cash flow

Equation (9-17) can be rearranged algebraically to yield Eq. (9-18)

$$CF = t(D) + (1 - t)(\Delta R) - (1 - t)(\Delta C) \quad (9-18)$$

The term  $t \times D$  is only the result of an algebraic manipulation, and no interpretation should be assumed. This term  $t \times D$  is the contribution to cash flow from depreciation, and  $(1 - t) \times R$  and  $(1 - t) \times C$  are the contributions to cash flow from revenues and cash operating expenses, respectively. Example 7 is a sample calculation of the after-tax cash flow and the tabulated results.

**Example 7: After-Tax Cash Flow** The revenue from the manufacture of a product in the first year of operation is \$9.0 million, and the cash operating expenses are \$4.5 million. Depreciation on the invested capital is \$1.7 million. If the federal income tax rate is 35 percent, calculate the after-tax cash flow.

**Solution:** The resulting after-tax cash flow is \$3.52 million. See Fig. 9-10.

**Cumulative Cash Position Table** To organize cash flow calculations, it is suggested that a cumulative cash position table be prepared by using an electronic spreadsheet. For this discussion,

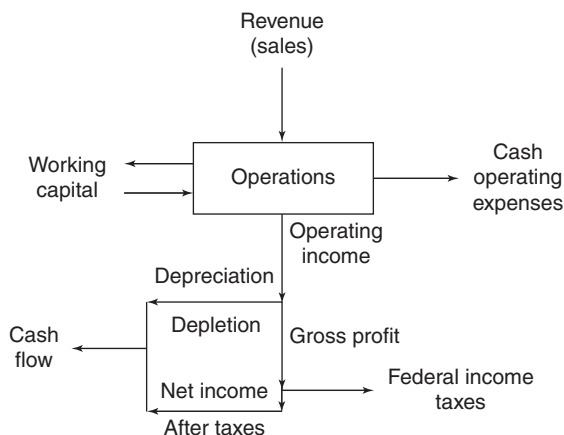
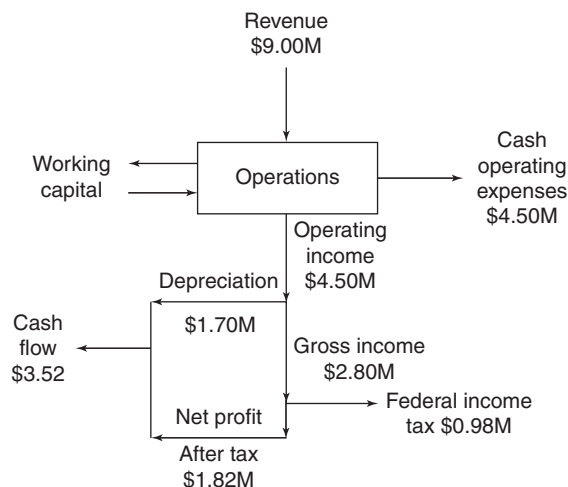


FIG. 9-9 Cash flow model.



**FIG. 9-10** Cash flow model for Example 7. M = million.

time zero is assumed to be at project start-up. Expenditures for land and equipment occurred prior to time zero and represent negative cash flows. At time zero, working capital is charged to the project as a negative cash flow. Start-up expenses are charged in the first year, and positive cash flow from the sale of product as net income after taxes plus depreciation begins, reducing the negative cash position. This process continues until the project is terminated. At that time, adjustments are made to recover land and working capital. An example of a cumulative cash position table is Table 9-25.

When equipment is added for plant expansions to an existing facility, it may be more convenient to use time zero when the first expenditures occur. The selection of either time base is satisfactory for economic analysis as long as consistency is maintained.

**Example 8: Cumulative Cash Position Table (Time Zero at Start-up)** A specialty chemical company is considering the manufacture of an additive for use in the plastics industry. The following is a list of production, sales, and cash operating expenses.

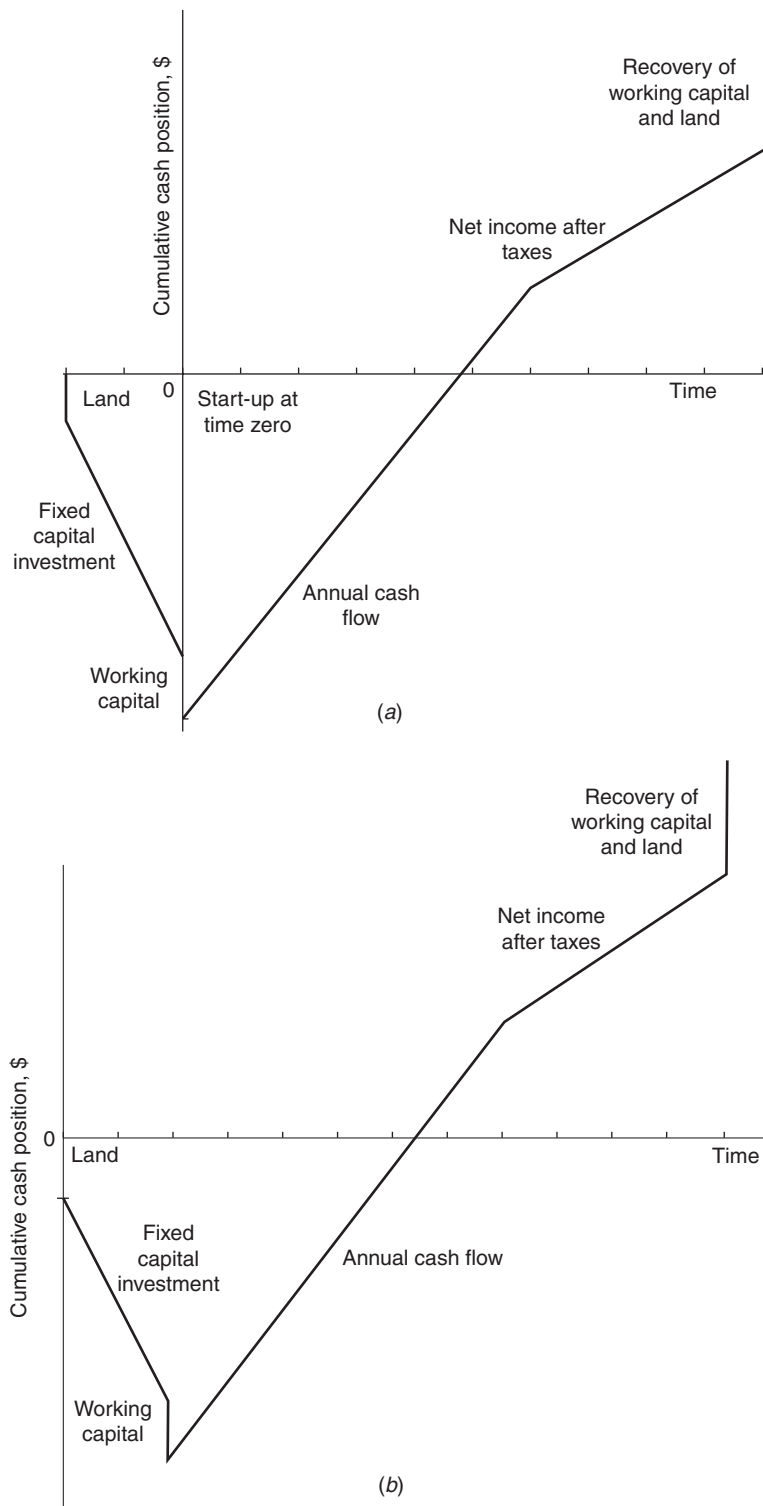
| Year | Production,<br>Mlb | Sales,<br>\$1,000 | Cash operating expenses,<br>\$1,000 |
|------|--------------------|-------------------|-------------------------------------|
| 1    | 40                 | 20,000            | 10,320                              |
| 2    | 42                 | 21,000            | 10,800                              |
| 3    | 45                 | 23,400            | 11,520                              |
| 4    | 48                 | 24,960            | 12,240                              |
| 5    | 50                 | 27,500            | 13,470                              |
| 6    | 50                 | 28,000            | 13,970                              |
| 7    | 47                 | 23,500            | 13,175                              |
| 8    | 45                 | 21,600            | 12,645                              |
| 9    | 40                 | 18,800            | 11,320                              |
| 10   | 35                 | 15,750            | 9,995                               |

Land for the project is available at \$300,000. The fixed capital investment was estimated to be \$12,000,000. A working capital of \$1,800,000 is needed initially for the venture. Start-up expenses based upon past experience are estimated to be \$750,000. The project qualifies under IRS guidelines as a 5-year class life investment. The company uses MACRS depreciation with the half-year convention. At the conclusion of the project, the land and working capital are returned to management. Develop a cash flow analysis for this project, using a cumulative cash position table (Table 9-25).

**Cumulative Cash Position Plot** A pictorial representation of the cumulative cash flows as a function of time is the cumulative cash position plot. All expenditures for capital as well as revenue from sales are plotted as a function of time. Figure 9-11 is such an idealized plot showing time zero at start-up in part *a* and time zero when the first funds are expended in part *b*. It should be understood that the plots have been idealized for illustration purposes. Expenditures are usually stepwise, and accumulated cash flow from sales is seldom a straightline but more likely a curve with respect to time.

**Time Zero at Start-up** Prior to time zero, expenditures are made for land, fixed capital investment, and working capital. It is assumed that land had been purchased by the company at some time in the past, and a parcel is allocated for the project under consideration. Land is allocated instantaneously to the project sometime prior to the purchase of equipment and construction of the plant. The fixed capital investment is purchased and installed over a period of time prior to start-up. For the purpose of this presentation, it is assumed that it occurs uniformly over a period of time. Both land and fixed capital investment are compounded to time zero by using the appropriate compound interest factors. At time zero, working capital is charged to the project. Start-up expenses are entered in the first year of operation after start-up. After time zero, start-up occurs and then manufacturing begins and income is generated, so cash flow begins to accumulate if the process is sound. At the end of the project life, land and working capital are recovered instantaneously.

**TABLE 9-25 Cash Flow Analysis for Example 8**[illegible]



**FIG. 9-11** Typical cumulative cash position plot. (a) Time zero is start-up. (b) Time zero occurs when first funds are spent.

## PROFITABILITY

In the free enterprise system, companies are in business to make a profit. Management has the responsibility of investing in ventures that are financially attractive by increasing earnings, providing attractive rates of return, and increasing value added. Every viable business has limitations on the capital available for investment; therefore, it will invest in the most economically attractive ventures. The objectives and goals of a company are developed by management. Corporate objectives may include one or several of the following: maximize return on investment, maximize return on stockholders' equity, maximize aggregate earnings, maximize common stock prices, increase market share, increase economic value, increase earnings per share of stock, and increase market value added. These objectives are the ones most frequently listed by executives.

To determine the worthiness of a venture, quantitative and qualitative measures of profitability are considered.

### QUANTITATIVE MEASURES OF PROFITABILITY

When a company invests in a venture, the investment must earn more than the cost of capital for it to be worthwhile. A profitability estimate is an attempt to quantify the desirability of taking a risk in a venture.

The *minimum acceptable rate of return (MARR)* for a venture depends on a number of factors such as interest rate, cost of capital, availability of capital, degree of risk, economic project life, and other competing projects. Management will adjust the MARR depending on any of the above factors to screen out the more attractive ventures. When a company invests in a venture, the investment must earn more than the cost of capital and should be able to pay dividends.

Although there have been many quantitative measures suggested through the years, some did not take into account the time value of money. In today's economy, the following measures are the ones most companies use:

- Payout period (POP) plus interest
- Net present worth (NPW)
- Discounted cash flow (DCFRR)

**Payout Period Plus Interest** Payout period (POP) is the time that will be required to recover the depreciable fixed capital investment from the accrued after-tax cash flow of a project with no interest considerations. In equation format

$$\text{Payout period} = \frac{\text{depreciable fixed capital investment}}{\text{after-tax cash flow}} \quad (9-19)$$

This model does not take into account the time value of money, and no consideration is given to cash flows that occur in a project's later years after the depreciable investment has been recovered. A variation on this method includes interest, called payout period plus interest (POP + I); and the net effect is to increase the payout period. This variation accounts for the time value of money.

Payout period plus interest (POP + I) =

$$\left( \frac{\text{depreciable fixed capital investment}}{\text{after-tax cash flow}} \right)_i \quad (9-20)$$

Neither of these methods makes provision for including land and working capital, and no consideration is given to cash flows that occur in a project's later years after the depreciable fixed investment has been recovered for projects that earn most of their profit in the early years.

**Net Present Worth** In the net present worth method, an arbitrary time frame is selected as the basis of calculation. This method is the measure many companies use, as it reflects properly the time value of money and its effect on profitability. In equation form

Net present worth (NPW) = present worth of all cash inflows - present worth of all investment items (9-21)

When the NPW is calculated according to Eq. (9-21), if the result is positive, the venture will earn more than the interest (discount) rate used; conversely, if the NPW is negative, the venture earns less than that rate.

**Discounted Cash Flow** In the discounted cash flow method, all the yearly after-tax cash flows are discounted or compounded to time zero depending upon the choice of time zero. The following equation is used to solve for the interest rate  $i$ , which is the discounted cash flow rate of return (DCFRR).

$$\text{DCFRR} = \sum_{t=0}^n (\text{after-tax cash flows}) = 0 \quad (9-22)$$

Equation (9-22) may be solved graphically or analytically by an iterative trial-and-error procedure for the value of  $i$ , which is the discounted cash flow rate of return. It has also been known as the *profitability index*. For a project to be profitable, the interest rate must exceed the cost of capital.

The effect of interest on the cash position of a project is shown in Fig. 9-12. As interest increases, the time to recover the capital expenditures is increased.

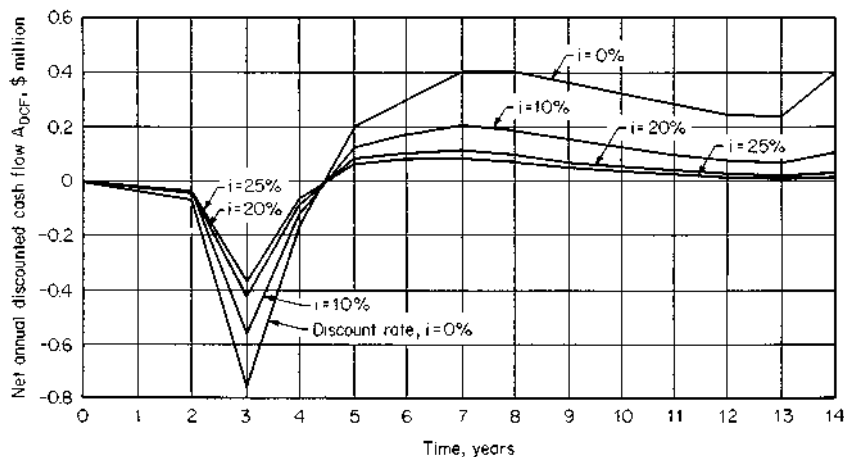


FIG. 9-12 Effect of interest rate on cash flow (time zero occurs when first funds are expended).

TABLE 9-26 Profitability Analysis for Example 9

| Time                                | Cash flow | 20% interest<br>factors | Present worth<br>20% interest | 30% interest<br>factors | Present worth<br>30% interest | 40% interest<br>factors | Present worth<br>40% interest |
|-------------------------------------|-----------|-------------------------|-------------------------------|-------------------------|-------------------------------|-------------------------|-------------------------------|
| -2                                  | -300      | 1.492                   | -448                          | 1.822                   | -547                          | 2.226                   | -668                          |
| -2 to 0                             | -12,000   | 1.230                   | -14,760                       | 1.370                   | -16,440                       | 1.532                   | -18,384                       |
| 0                                   | -2,250    | 1.000                   | -2,250                        | 1.000                   | -2,250                        | 1.000                   | -2,250                        |
| 1                                   | 6,040     | 0.906                   | 5,472                         | 0.864                   | 5,219                         | 0.824                   | 4,977                         |
| 2                                   | 7,974     | 0.742                   | 5,917                         | 0.640                   | 5,103                         | 0.552                   | 4,402                         |
| 3                                   | 8,527     | 0.608                   | 5,184                         | 0.474                   | 4,042                         | 0.370                   | 3,155                         |
| 4                                   | 8,754     | 0.497                   | 4,351                         | 0.351                   | 3,073                         | 0.248                   | 2,171                         |
| 5                                   | 9,602     | 0.407                   | 3,908                         | 0.260                   | 2,497                         | 0.166                   | 1,594                         |
| 6                                   | 9,361     | 0.333                   | 3,117                         | 0.193                   | 1,807                         | 0.112                   | 1,048                         |
| 7                                   | 6,711     | 0.273                   | 1,832                         | 0.143                   | 960                           | 0.075                   | 503                           |
| 8                                   | 5,821     | 0.224                   | 1,304                         | 0.106                   | 617                           | 0.050                   | 291                           |
| 9                                   | 4,862     | 0.183                   | 890                           | 0.078                   | 379                           | 0.034                   | 165                           |
| 10                                  | 3,471     | 0.150                   | 561                           | 0.058                   | 217                           | 0.022                   | 82                            |
| End 10                              | 2,100     | 0.135                   | 284                           | 0.050                   | 105                           | 0.018                   | 284                           |
| Net present worth                   |           |                         | 15,362                        |                         | 4,782                         |                         | -2,360                        |
| Discounted cash flow rate of return |           |                         | 33.90%                        |                         |                               |                         |                               |

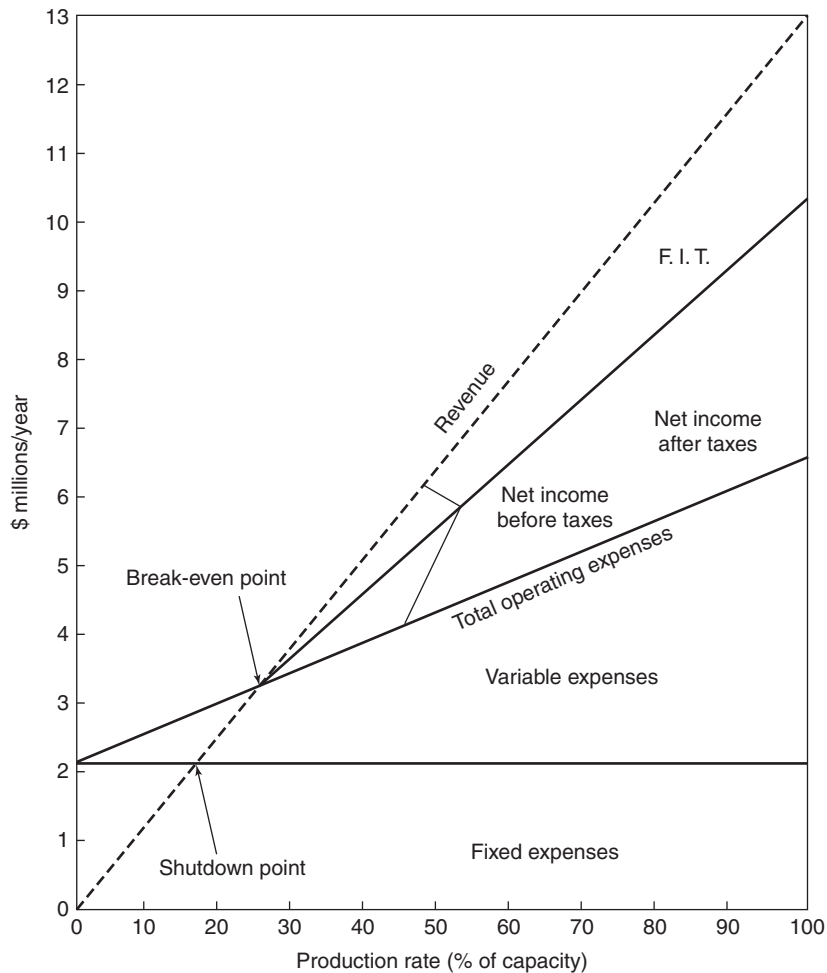


FIG. 9-13 Break-even plot.

In the chemical business, operating net profit and cash flow are received on a nearly continuous basis, therefore, there is justification for using the condensed continuous interest tables, such as Table 9-23, in discounted cash flow calculations.

**Example 9: Profitability Calculations** Example 8 data are used to demonstrate these calculations. Calculate the following:

- Payout period (POP)
- Payout period with interest (POP + I)
- NPW at a 30 percent interest rate
- DCF rate of return

**Solution:**

- From Table 9-26, the second column is the cash flow by years with no interest. The payout period occurs where the cumulative cash flow is equal to the fixed capital investment, \$12,000,000 or 1.7 years.
- In Table 9-26, the payout period at 30 percent interest occurs at 2.4 years.
- The results of the present worth calculations for 20, 30, and 40 percent interest rates are tabulated. At 30 percent interest, the net present worth is \$4,782,000, and since it is a positive figure, this means the project will earn more than 30 percent interest.
- Discounted cash flow rate of return is determined by interpolating in Table 9-26. At 30 percent interest the net present worth is positive, and at 40 percent interest it is negative. By definition, the DCFROR occurs when the summation of the net present worth equals zero. This occurs at an interest of 33.9 percent.

## QUALITATIVE MEASURES

In addition to quantitative measures, there are certain qualitative measures or intangible factors that may affect the ultimate investment decision. Those most frequently mentioned by management are employee morale, employee safety, environmental constraints, legal constraints, product liability, corporate image, and management goals. Attempts have been made to quantify these intangibles by using an

ordinal or a ranking system, but most have had little or no success. Couper (2003) discussed in greater detail the effect of qualitative measures on the decision-making process.

## SENSITIVITY ANALYSIS

Whenever an economic study is prepared, the marketing, capital investment, and operating expense data used are estimates, and therefore a degree of uncertainty exists. Questions arise such as, What if the capital investment is 15 percent greater than the value reported? A sensitivity analysis is used to determine the effect of percentage changes in pertinent variables on the profitability of the project. Such an analysis indicates which variables are most susceptible to change and need further study.

**Break-Even Analysis** Break-even analysis is a simple form of sensitivity analysis and is a useful concept that can be of value to managers. *Break-even* refers to the point in an operation where income just equals expenses. Figure 9-13 is a pictorial example of the results of a break-even analysis, showing that the break-even point is at 26 percent of production capacity. Management wants to do better than just break even; therefore, such plots can be used as a profit planning tool, for product pricing, production operating level, incremental equipment costs, etc. Another significant point is the *shutdown point* where revenue just equals the fixed expenses. Therefore, if a proposed operation can't make fixed expenses, it should be shut down.

**Strauss Plot** R. Strauss (*Chem. Eng.*, pp. 112–116, Mar. 25, 1968) developed a sensitivity plot, in Fig. 9-14, in which the ordinate is a measure of profitability and the abscissa is the change in a variable greater than (or less than) the value used in the base case. Where the abscissa crosses the ordinate is the result of the base case of NPW, return, annual worth, etc. The slope of a line on this "spider" plot is the degree of change in profitability resulting from a change in a

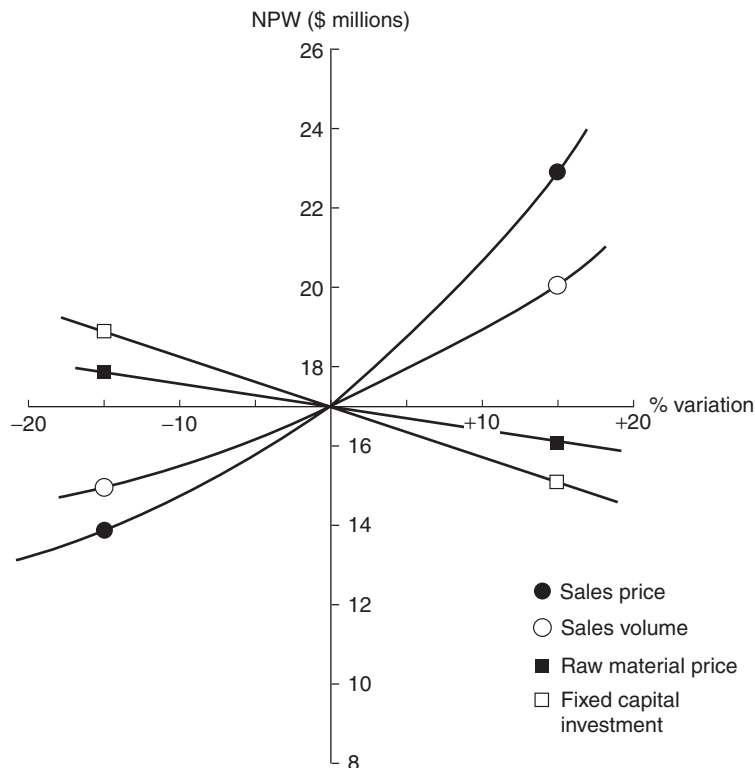


FIG. 9-14 Strauss plot.

variable, selling price, sales volume, investment, etc. The length of the line represents the sensitivity of the variable and its degree of uncertainty. Positive-slope lines are income-related, and negative-slope lines are expense-related. A spreadsheet is useful in developing data for this “what if” plot since numerous scenarios must be prepared to develop the plot.

**Tornado Plot** Another graphical sensitivity analysis is the “tornado” plot. Its name is derived from the shape of the resulting envelope. As in other methods, a base case is solved first, usually expressing the profitability as the net present worth. In Fig. 9-15, the NPW is a vertical line, and variations in each selected variable above and below the base case are solved and plotted. In this figure, the variables of selling price, sales volume, operating expenses, raw material expenses, share of the market, and investment are plotted. It is apparent that the selling price and sales volume are the critical factors affecting the profitability. A commercial computer program known as @RISK® developed by the Palisade Corporation, Newfield, N.Y., may be used to prepare a tornado plot.

**Relative Sensitivity Plot** Another type of analysis developed by J. C. Agarwal and I. V. Klumpar (*Chem. Eng.*, pp. 66–72, Sept. 29, 1975) is the relative sensitivity plot. The variables studied are related to those in the base case, and the resulting plot is the relative profitability.

Although sensitivity analyses are easy to prepare and they yield useful information for management, there is a serious disadvantage. Only one variable at a time can be studied. Frequently, there are synergistic effects among variables; e.g., in marketing, the variables such as sales volume, selling price, and market share may have a synergistic effect, and that effect cannot be taken into account. Other inter-related variables such as fixed capital investment, maintenance, and other investment-based items also cannot be represented properly. These disadvantages lead to another management tool—uncertainty analysis.

## UNCERTAINTY ANALYSIS

This analysis allows the user to account for variable interaction that is another level of sophistication. Two terms need clarification—

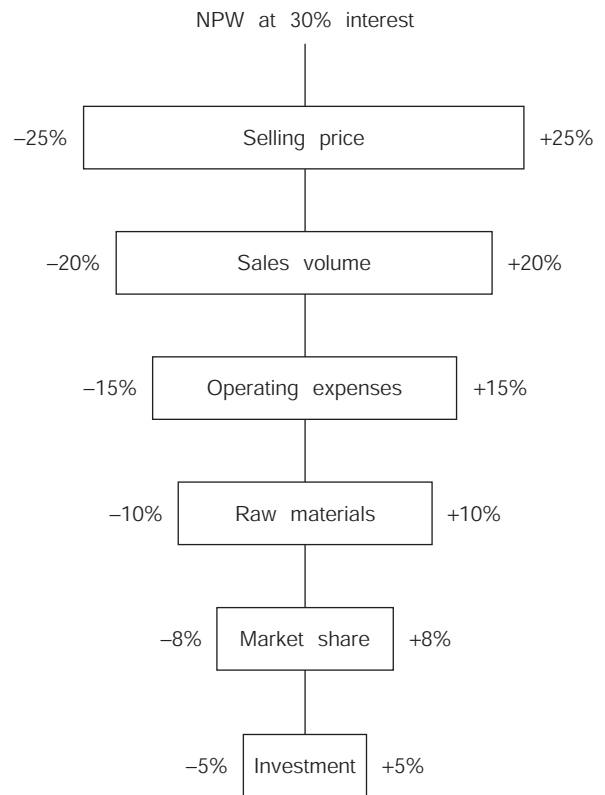


FIG. 9-15 Typical tornado plot. (Source: Adapted from Couper, 2003.)

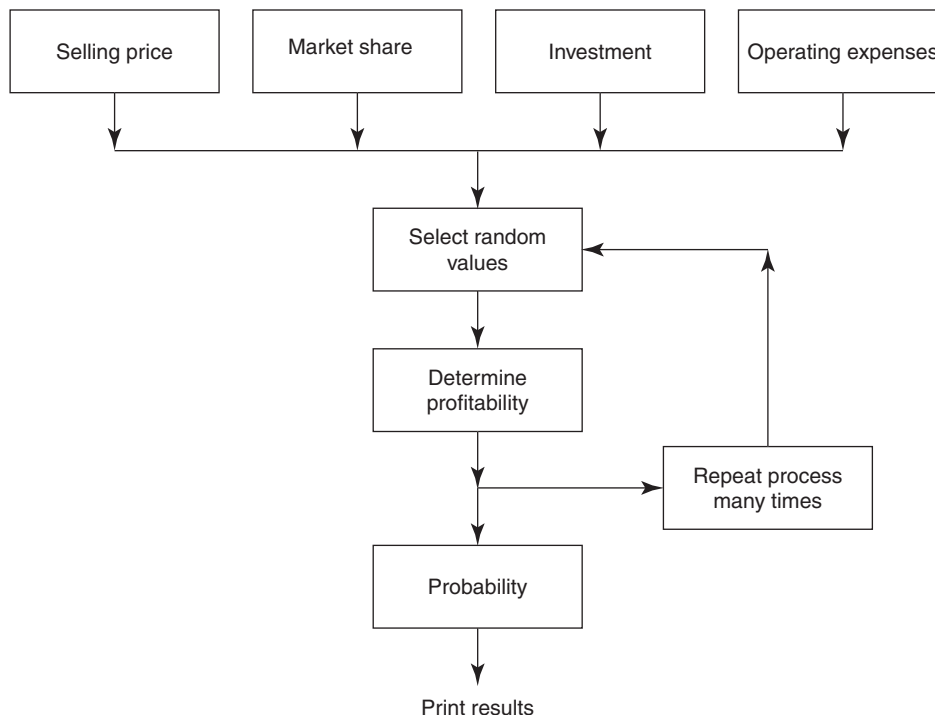


FIG. 9-16 Schematic diagram of Monte Carlo simulation.



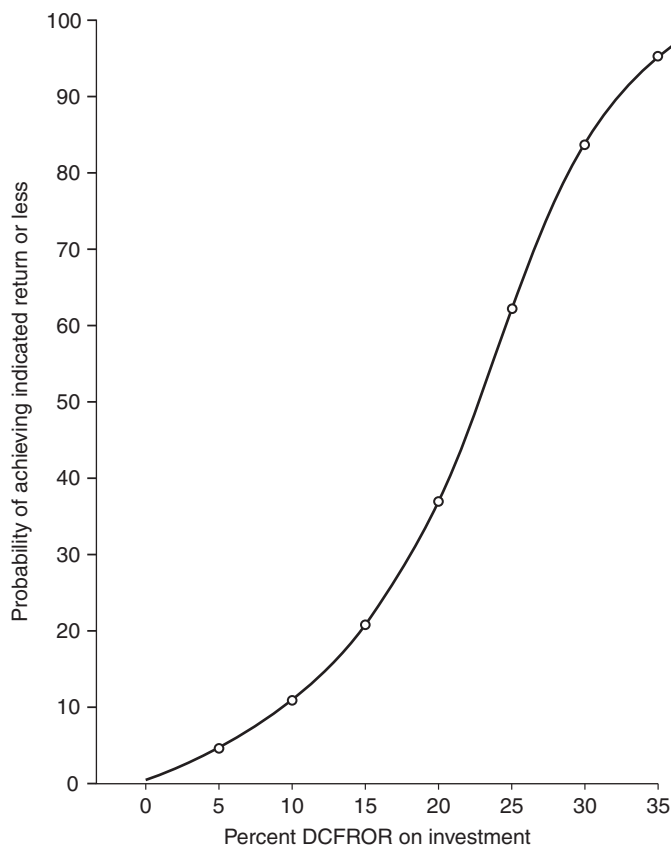


FIG. 9-17 Probability curve for Monte Carlo simulation.

*uncertainty and risk.* Uncertainty is exactly what the word means—not certain. Risk, however, implies that the probability of achieving a specific outcome is known within certain confidence limits.

Since sensitivity analysis has the shortcoming of being able to inspect only one variable at a time, the next step is to use probability risk analysis, generally referred to as the Monte Carlo technique. R. C. Ross (*Chem. Eng.*, pp. 149–155, Sept. 20, 1971), P. Macalusa (*BYTE*, pp. 179–192, March 1984), and D. B. Hertz (*Harvard Bus. Rev.*, pp. 96–108, Jan-Feb 1968) have written classic articles on the use of the Monte Carlo technique in uncertainty analysis. These articles incorporate subjective probabilities and assumptions of the distribution of errors into the analysis. Each variable is represented by a probability distribution model. Figure 9-16 is a pictorial representation of the steps in the Monte Carlo simulation. The first step is to gather enough data to develop a reasonable probability model. Not all variables follow the normal distribution curve, but perhaps sales volume and sales-related variables do. Studies have shown that capital investment estimates are best represented by a beta distribution. Next the task is to select random values from the various models by using a random number generator and from these data calculate a profitability measure such as NPW or rate of return. The procedure is repeated a number of times to generate a plot of the probability of achieving a given profitability versus profitability. Figure 9-17 is a typical plot. Once the analysis has been performed, the next task is to interpret the results. Management must understand what the results mean and the reliability of the results. Experience can be gained only by performing uncertainty analyses, not just one or two attempts, to develop confidence in

the process. The stakes may be high enough to spend time and learn the method. Software companies such as @RISK or SAS permit the user to develop probability models and perform the Monte Carlo analysis. The results may be plotted as the probability of achieving at least a given return or of achieving less than the desired profitability.

## FEASIBILITY ANALYSIS

A feasibility analysis is prepared for the purpose of determining that a proposed investment meets the minimum requirements established

TABLE 9-27 Checklist of Required Information for a Feasibility Analysis

|                                |
|--------------------------------|
| Fixed capital investment       |
| Working capital requirements   |
| Total capital investment       |
| Total manufacturing expense    |
| Packaging and in-plant expense |
| Total product expense          |
| General overhead expense       |
| Total operating expense        |
| Marketing data                 |
| Cash flow analysis             |
| Project profitability          |
| Sensitivity analysis           |
| Uncertainty analysis           |

**TABLE 9-28 Marketing Data Template**

|                                                                 |        |                |        |                |        |                |
|-----------------------------------------------------------------|--------|----------------|--------|----------------|--------|----------------|
| Project title:                                                  |        |                |        |                |        |                |
| Basis: Sales and marketing data are not inflated (20__ dollars) |        |                |        |                |        |                |
|                                                                 | Amount | 20__<br>%Total | Amount | 20__<br>%Total | Amount | 20__<br>%Total |
| Total market                                                    |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Average realistic price, \$/unit                                |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |
| Estimated product sales (with AR)                               |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Average realistic price, \$/unit                                |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |
| Current product sales (without AR)                              |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Average realistic price, \$/unit                                |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |
| Incremental product sales                                       |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Average realistic price, \$/unit                                |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |
| Current product sales displaced by improved product sales       |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |
| Total improved product sales                                    |        |                |        |                |        |                |
| Units                                                           |        |                |        |                |        |                |
| Value, \$                                                       |        |                |        |                |        |                |

NOTE: Table extends to the right to accommodate the number of project years.

AR = appropriation request.

**TABLE 9-29 Cash Flow Analysis Template**

|                          | Cash flow summary |      |             |
|--------------------------|-------------------|------|-------------|
|                          | 200X              | 200Y | 200Z, etc.* |
| Investment               |                   |      |             |
| Land                     |                   |      |             |
| Fixed capital investment |                   |      |             |
| Offsite capital          |                   |      |             |
| Allocated capital        |                   |      |             |
| Working capital          |                   |      |             |
| Start-up expenses        |                   |      |             |
| Interest                 |                   |      |             |
| Catalysts and chemicals  |                   |      |             |
| Licenses, patents, etc.  |                   |      |             |
| Total capital investment |                   |      |             |
| Income statement         |                   |      |             |
| Income                   |                   |      |             |
| Expenses                 |                   |      |             |
| Cash operating expenses  |                   |      |             |
| Depreciation             |                   |      |             |
| Total operating expenses |                   |      |             |
| Operating income         |                   |      |             |
| Net income before taxes  |                   |      |             |
| Federal income taxes     |                   |      |             |
| Net income after taxes   |                   |      |             |
| Cash flow                |                   |      |             |
| Capital recovery         |                   |      |             |
| Cumulative cash flow     |                   |      |             |

\*Table may be extended to the right to accommodate the number of years of the project.

by management. It should be in sufficient detail to provide management with the facts required to make an investment decision. All the basic information has been discussed in considerable detail in the earlier parts of Sec. 9.

The minimum information required should include, but not be limited to, that in Table 9-27. Forms and spreadsheets are the most succinct method to present the information. The forms should state clearly the fund amounts and the date that each estimate was performed. The forms may be developed so that data for other scenarios may be reported by extending the tables to the right of the page. It is suggested that blank lines be included for any additional information. Finally the engineer preparing the feasibility analysis should make recommendations based upon management's guidelines.

The development of the information required for Table 9-27 was discussed previously in Sec. 9 with the exception of marketing information. An important document for a feasibility analysis is the marketing data so that the latest income projections can be included for management's consideration. As a minimum, the tabulation of sales volume, sales prices, and market share both domestically and globally should be included. Table 9-28 shows a sample of such marketing information.

Other templates may be prepared for total capital investment, working capital, total product expense, general overhead expense, and cash flow. Table 9-29 may be used to organize cash flow data by showing investment, operating expenses, cash flow, and cumulative cash flow.

## OTHER ECONOMIC TOPICS

### COMPARISON OF ALTERNATIVE INVESTMENTS

Engineers are often confronted with making choices between alternative equipment, designs, procedures, plans or methods. The courses of action require different amounts of capital and different operating expenses. Some basic concepts must be considered before attempting

to use mathematical methods for a solution. It is necessary to clearly define the alternatives and their merits. Flow of money takes the form of expenditures or income. Savings from operations are considered as income or as a reduction in operating expenses. Income taxes and inflation as well as a reasonable return on the investment must be included. Money spent is negative and money earned or saved is positive.

Expenditures are of two kinds; instantaneous like land, working capital and capital recovery or uniformly continuous for plant investment, operating expenses, etc. A methodology involving after-tax cash flow is developed to reduce all the above to a manageable format.

In an earlier part of this section, after-tax cash flow was defined as

$$CF = (R - C - D)(1 - t) + D \quad (9-17)$$

where CF = after-tax cash flow

D = depreciation

t = tax rate

S = sales or revenue

C = cash operating expenses (COE)

For the situation in which each case will produce the same revenue or the same benefit,  $R$  will equal 0. Rearranging Eq. (9-17) algebraically yields

$$CF = (t)(D) + (1 - t)(S - C) \quad (9-17a)$$

$$\text{or} \quad CF = (t)(D) + (1 - t)(-C) \quad (9-17b)$$

$$CF = (t)(D) - (1 - t)(C) \quad (9-17c)$$

This expression is applied to each alternative. [Note: As mentioned under cash flow, the first term in the above equations,  $(t)(D)$ , is the result of an algebraic rearrangement, and no other significance should be assumed.]

Several methods are available for determining the choice among alternatives:

Net present worth

Rate of return

Capitalized cost

Cash flow

Uniform annual cost

Humphreys in Jelen and Black, *Cost and Optimization Engineering* (1991), has shown that each of these methods would result in the same decision, but the numerical results will differ.

**Net Present Worth Method** The NPW method allows the conversion of all money flows to be discounted to the present time. Appropriate interest factors are applied depending on how and when the cash flow enters a venture. They may be instantaneous, as in the purchase of capital equipment, or uniform, as in operating expenses. The alternative with the more positive NPW is the one to be preferred. In some instances, the alternatives may have different lives so the cost analysis must be for the least common multiple number of years. For example, if alternative A has a 2-year life and alternative B has a 3-year life, then 6 years is the least common multiple. The rate of return, capitalized cost, cash flow, and uniform annual cost methods avoid this complication.

Rate of return and capitalized cost methods are discussed at length in Humphreys (1991).

**Cash Flow Method** Cash flows for each case are determined, and the case that generates the greater cash flow is the preferred one.

**Uniform Annual Cost (UAC) Method** In the uniform annual cost method, the cost is determined over the entire estimated project life. The least common multiple does not have to be calculated, as in the NPW method. This is the advantage of the UAC method; however, the result obtained by this method is more meaningful than the results obtained by other methods.

The UAC method begins with a calculation for each alternative. If discrete interest is used, the annual cost  $C$  is found by multiplying the present worth  $P$  by the appropriate discrete interest factor, found in Table 9-21, for the number of years  $n$  and the interest rate  $i$ . If continuous interest is preferred, the UAC equation is

$$UAC = \frac{NPW}{(\text{years of life})(\text{continuous interest factor})} \quad (9-23)$$

The continuous interest factor may be found from continuous interest equations or from the continuous interest table, Table 9-30. In this table time zero is the present, and all cash flows are discounted back to the present. Note that there are three sections to this table,

depending on the cash flow: uniform, instantaneous, or declining uniformly to zero. One enters the table with the argument  $R \times T$ , where  $R$  is the interest rate expressed as a whole number and  $T$  is the time in years to obtain a factor. This factor is then used to calculate the present worth of the cash flow item. All cash flows are summed algebraically, giving the net present worth which is substituted in Eq. (9-23). This procedure is followed for both alternatives, and the alternative that yields the more positive UAC (or the least negative) value is the preferred alternative. In Eq. (9-23) the "factor" is always the uniform factor that annualizes all the various cash flows.

This method of comparing alternatives is demonstrated in Example 10.

**Example 10: Choice among Alternatives** Two filters are considered for installation in a process to remove solids from a liquid discharge stream to meet environmental requirements. The equipment is to be depreciated over a 7-year period by the straight-line method. The income tax rate is 35 percent, and 15 percent continuous interest is to be used. Assume that the service life is 7 years and there is no capital recovery. Data for the two systems are as follows:

| System                    | B        | C        |
|---------------------------|----------|----------|
| Fixed investment          | \$18,000 | \$30,000 |
| Annual operating expenses | 14,200   | 4,800    |

Which alternative is preferred?

*Solution:*

System B:

| Year | Item                                             | Cash flow, \$         | Factor | PW, \$  |
|------|--------------------------------------------------|-----------------------|--------|---------|
| 0    | Investment                                       | -18,000               | 1.0    | -18,000 |
| 0-7  | Contribution to cash flow from depreciation      | (0.35)(18,000)        | 0.6191 | +3,900  |
| 0-7  | Contribution to cash flow from operating expense | (1 - 0.35)(7)(14,200) | 0.6191 | -40,000 |
|      |                                                  |                       | NPW B  | -54,100 |

$$UAC_B = \frac{NPW}{(\text{years of life})(\text{uniform factor})} = \frac{-\$54,100}{(7)(0.6191)} = -\$12,484$$

System C:

| Year | Item                                             | Cash flow, \$         | Factor | PW, \$  |
|------|--------------------------------------------------|-----------------------|--------|---------|
| 0    | Investment                                       | -30,000               | 1.0    | -30,000 |
| 0-7  | Contribution to cash flow from depreciation      | (0.35)(30,000)        | 0.6191 | +6,500  |
| 0-10 | Contribution to cash flow from operating expense | (1 - 0.35)(10)(4,800) | 0.6191 | -19,316 |
|      |                                                  |                       | NPW C  | -42,816 |

$$UAC_C = \frac{NPW}{(\text{years of life})(\text{uniform factor})} = \frac{-\$42,816}{(10)(0.6191)} = -\$6,916$$

Alternative C is preferred because it has the more positive UAC.

## REPLACEMENT ANALYSIS

During the lifetime of a physical asset, continuation of its use may make it a candidate for replacement. In this type of analysis, a replacement is intended to supplant a similar item performing the same service without plant or equipment expansion. In a chemical plant, *replacement* usually refers to a small part of the processing equipment such as a heat exchanger, filter, or compressor. If the replacement is required due to "physical" deterioration, there is no question of whether to replace the item, but the entire plant may be shut down if it is not replaced. The problem then becomes whether the equipment

TABLE 9-30 Factors for Continuous Discounting

| $R \times T$  | 0      | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
|---------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Uniform       |        |       |       |       |       |       |       |       |       |       |
| 0             | 1.0000 | .9950 | .9901 | .9851 | .9803 | .9754 | .9706 | .9658 | .9610 | .9563 |
| 10            | .9516  | .9470 | .9423 | .9377 | .9332 | .9286 | .9241 | .9196 | .9152 | .9107 |
| 20            | .9063  | .9020 | .8976 | .8933 | .8891 | .8848 | .8806 | .8764 | .8722 | .8681 |
| 30            | .8639  | .8598 | .8558 | .8517 | .8477 | .8437 | .8398 | .8359 | .8319 | .8281 |
| 40            | .8242  | .8204 | .8166 | .8128 | .8090 | .8053 | .8016 | .7979 | .7942 | .7906 |
| 50            | .7869  | .7833 | .7798 | .7762 | .7727 | .7692 | .7657 | .7622 | .7588 | .7554 |
| 60            | .7520  | .7486 | .7453 | .7419 | .7386 | .7353 | .7320 | .7288 | .7256 | .7224 |
| 70            | .7192  | .7160 | .7128 | .7097 | .7066 | .7035 | .7004 | .6974 | .6944 | .6913 |
| 80            | .6883  | .6854 | .6824 | .6795 | .6765 | .6736 | .6707 | .6679 | .6650 | .6622 |
| 90            | .6594  | .6566 | .6538 | .6510 | .6483 | .6455 | .6428 | .6401 | .6374 | .6348 |
| 100           | .6321  | .6295 | .6269 | .6243 | .6217 | .6191 | .6166 | .6140 | .6115 | .6090 |
| 110           | .6065  | .6040 | .6015 | .5991 | .5967 | .5942 | .5918 | .5894 | .5871 | .5847 |
| 120           | .5823  | .5800 | .5777 | .5754 | .5731 | .5708 | .5685 | .5663 | .5640 | .5618 |
| 130           | .5596  | .5574 | .5552 | .5530 | .5509 | .5487 | .5466 | .5444 | .5423 | .5402 |
| 140           | .5381  | .5361 | .5340 | .5320 | .5299 | .5279 | .5259 | .5239 | .5219 | .5199 |
| 150           | .5179  | .5160 | .5140 | .5121 | .5101 | .5082 | .5063 | .5044 | .5025 | .5007 |
| 160           | .4988  | .4970 | .4951 | .4933 | .4915 | .4897 | .4879 | .4861 | .4843 | .4825 |
| 170           | .4808  | .4790 | .4773 | .4756 | .4738 | .4721 | .4704 | .4687 | .4671 | .4654 |
| 180           | .4637  | .4621 | .4604 | .4588 | .4572 | .4555 | .4539 | .4523 | .4507 | .4492 |
| 190           | .4476  | .4460 | .4445 | .4429 | .4414 | .4399 | .4383 | .4368 | .4353 | .4338 |
| 200           | .4323  | .4309 | .4294 | .4279 | .4265 | .4250 | .4236 | .4221 | .4207 | .4193 |
| 210           | .4179  | .4165 | .4151 | .4137 | .4123 | .4109 | .4096 | .4082 | .4069 | .4055 |
| 220           | .4042  | .4029 | .4015 | .4002 | .3989 | .3976 | .3963 | .3950 | .3937 | .3925 |
| 230           | .3912  | .3899 | .3887 | .3874 | .3862 | .3849 | .3837 | .3825 | .3813 | .3801 |
| 240           | .3789  | .3777 | .3765 | .3753 | .3741 | .3729 | .3718 | .3706 | .3695 | .3683 |
| 250           | .3672  | .3660 | .3649 | .3638 | .3627 | .3615 | .3604 | .3593 | .3582 | .3571 |
| 260           | .3560  | .3550 | .3539 | .3528 | .3518 | .3507 | .3496 | .3486 | .3476 | .3465 |
| 270           | .3455  | .3445 | .3434 | .3424 | .3414 | .3404 | .3394 | .3384 | .3374 | .3364 |
| 280           | .3354  | .3344 | .3335 | .3325 | .3315 | .3306 | .3296 | .3287 | .3277 | .3268 |
| 290           | .3259  | .3249 | .3240 | .3231 | .3222 | .3212 | .3203 | .3194 | .3185 | .3176 |
| 300           | .3167  | .3158 | .3150 | .3141 | .3132 | .3123 | .3115 | .3106 | .3098 | .3089 |
| 310           | .3080  | .3072 | .3064 | .3055 | .3047 | .3039 | .3030 | .3022 | .3014 | .3006 |
| 320           | .2998  | .2990 | .2982 | .2974 | .2966 | .2958 | .2950 | .2942 | .2934 | .2926 |
| 330           | .2919  | .2911 | .2903 | .2896 | .2888 | .2880 | .2873 | .2865 | .2858 | .2850 |
| 340           | .2843  | .2836 | .2828 | .2821 | .2814 | .2807 | .2799 | .2792 | .2785 | .2778 |
| 350           | .2771  | .2764 | .2757 | .2750 | .2743 | .2736 | .2729 | .2722 | .2715 | .2709 |
| 360           | .2702  | .2695 | .2688 | .2682 | .2675 | .2669 | .2662 | .2655 | .2649 | .2642 |
| 370           | .2636  | .2629 | .2623 | .2617 | .2610 | .2604 | .2598 | .2591 | .2585 | .2579 |
| 380           | .2573  | .2567 | .2560 | .2554 | .2548 | .2542 | .2536 | .2530 | .2524 | .2518 |
| 390           | .2512  | .2506 | .2500 | .2495 | .2489 | .2483 | .2477 | .2471 | .2466 | .2460 |
| 400           | .2454  | .2449 | .2443 | .2437 | .2432 | .2426 | .2421 | .2415 | .2410 | .2404 |
| Instantaneous |        |       |       |       |       |       |       |       |       |       |
| 0             | 1.0000 | .9900 | .9802 | .9704 | .9608 | .9512 | .9418 | .9324 | .9231 | .9139 |
| 10            | .9048  | .8958 | .8869 | .8781 | .8694 | .8607 | .8521 | .8437 | .8353 | .8270 |
| 20            | .8187  | .8106 | .8025 | .7945 | .7866 | .7788 | .7711 | .7634 | .7558 | .7483 |
| 30            | .7408  | .7334 | .7261 | .7189 | .7118 | .7047 | .6977 | .6907 | .6839 | .6771 |
| 40            | .6703  | .6637 | .6570 | .6505 | .6440 | .6376 | .6313 | .6250 | .6188 | .6126 |
| 50            | .6065  | .6005 | .5945 | .5886 | .5827 | .5769 | .5712 | .5655 | .5599 | .5543 |
| 60            | .5488  | .5434 | .5379 | .5326 | .5273 | .5220 | .5169 | .5117 | .5066 | .5016 |
| 70            | .4966  | .4916 | .4868 | .4819 | .4771 | .4724 | .4677 | .4630 | .4584 | .4538 |
| 80            | .4493  | .4449 | .4404 | .4360 | .4317 | .4274 | .4232 | .4190 | .4148 | .4107 |
| 90            | .4066  | .4025 | .3985 | .3946 | .3906 | .3867 | .3829 | .3791 | .3753 | .3716 |
| 100           | .3679  | .3642 | .3606 | .3570 | .3535 | .3499 | .3465 | .3430 | .3396 | .3362 |
| 110           | .3329  | .3296 | .3263 | .3230 | .3198 | .3166 | .3135 | .3104 | .3073 | .3042 |
| 120           | .3012  | .2982 | .2952 | .2923 | .2894 | .2865 | .2837 | .2808 | .2780 | .2753 |
| 130           | .2725  | .2698 | .2671 | .2645 | .2618 | .2592 | .2567 | .2541 | .2516 | .2491 |
| 140           | .2466  | .2441 | .2417 | .2393 | .2369 | .2346 | .2322 | .2299 | .2276 | .2254 |
| 150           | .2231  | .2209 | .2187 | .2165 | .2144 | .2122 | .2101 | .2080 | .2060 | .2039 |
| 160           | .2019  | .1999 | .1979 | .1959 | .1940 | .1920 | .1901 | .1882 | .1864 | .1845 |
| 170           | .1827  | .1809 | .1791 | .1773 | .1755 | .1738 | .1720 | .1703 | .1686 | .1670 |
| 180           | .1653  | .1637 | .1620 | .1604 | .1588 | .1572 | .1557 | .1541 | .1526 | .1511 |
| 190           | .1496  | .1481 | .1466 | .1451 | .1437 | .1423 | .1409 | .1395 | .1381 | .1367 |
| 200           | .1353  | .1340 | .1327 | .1313 | .1300 | .1287 | .1275 | .1262 | .1249 | .1237 |
| 210           | .1225  | .1212 | .1200 | .1188 | .1177 | .1165 | .1153 | .1142 | .1130 | .1119 |
| 220           | .1108  | .1097 | .1086 | .1075 | .1065 | .1054 | .1044 | .1033 | .1023 | .1013 |
| 230           | .1003  | .0993 | .0983 | .0973 | .0963 | .0954 | .0944 | .0935 | .0926 | .0916 |
| 240           | .0907  | .0898 | .0889 | .0880 | .0872 | .0863 | .0854 | .0846 | .0837 | .0829 |
| 250           | .0821  | .0813 | .0805 | .0797 | .0789 | .0781 | .0773 | .0765 | .0758 | .0750 |
| 260           | .0743  | .0735 | .0728 | .0721 | .0714 | .0707 | .0699 | .0693 | .0686 | .0679 |
| 270           | .0672  | .0665 | .0659 | .0652 | .0646 | .0639 | .0633 | .0627 | .0620 | .0614 |
| 280           | .0608  | .0602 | .0596 | .0590 | .0584 | .0578 | .0573 | .0567 | .0561 | .0556 |
| 290           | .0550  | .0545 | .0539 | .0534 | .0529 | .0523 | .0518 | .0513 | .0508 | .0503 |
| 300           | .0498  | .0493 | .0488 | .0483 | .0478 | .0474 | .0469 | .0464 | .0460 | .0455 |
| 310           | .0450  | .0446 | .0442 | .0437 | .0433 | .0429 | .0424 | .0420 | .0416 | .0412 |

TABLE 9-30 Factors for Continuous Discounting (Concluded)

| $R \times T$             | 0      | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
|--------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Instantaneous            |        |       |       |       |       |       |       |       |       |       |
| 320                      | .0408  | .0404 | .0400 | .0396 | .0392 | .0388 | .0384 | .0380 | .0376 | .0373 |
| 330                      | .0369  | .0365 | .0362 | .0358 | .0354 | .0351 | .0347 | .0344 | .0340 | .0337 |
| 340                      | .0334  | .0330 | .0327 | .0324 | .0321 | .0317 | .0314 | .0311 | .0308 | .0305 |
| 350                      | .0302  | .0299 | .0296 | .0293 | .0290 | .0287 | .0284 | .0282 | .0279 | .0276 |
| 360                      | .0273  | .0271 | .0268 | .0265 | .0263 | .0260 | .0257 | .0255 | .0252 | .0250 |
| 370                      | .0247  | .0245 | .0242 | .0240 | .0238 | .0235 | .0233 | .0231 | .0228 | .0226 |
| 380                      | .0224  | .0221 | .0219 | .0217 | .0215 | .0213 | .0211 | .0209 | .0207 | .0204 |
| 390                      | .0202  | .0200 | .0198 | .0196 | .0194 | .0193 | .0191 | .0189 | .0187 | .0185 |
| 400                      | .0183  | .0181 | .0180 | .0178 | .0176 | .0174 | .0172 | .0171 | .0169 | .0167 |
| Declining Uniformly to 0 |        |       |       |       |       |       |       |       |       |       |
| 0                        | 1.0000 | .9968 | .9934 | .9902 | .9867 | .9836 | .9803 | .9771 | .9739 | .9707 |
| 10                       | .9675  | .9643 | .9612 | .9580 | .9549 | .9518 | .9487 | .9457 | .9426 | .9396 |
| 20                       | .9365  | .9335 | .9305 | .9275 | .9246 | .9216 | .9187 | .9158 | .9129 | .9100 |
| 30                       | .9071  | .9042 | .9013 | .8985 | .8957 | .8929 | .8901 | .8873 | .8845 | .8817 |
| 40                       | .8790  | .8763 | .8735 | .8708 | .8681 | .8655 | .8628 | .8601 | .8575 | .8549 |
| 50                       | .8522  | .8496 | .8470 | .8445 | .8419 | .8393 | .8368 | .8343 | .8317 | .8292 |
| 60                       | .8267  | .8242 | .8218 | .8193 | .8169 | .8144 | .8120 | .8096 | .8072 | .8048 |
| 70                       | .8024  | .8000 | .7977 | .7953 | .7930 | .7906 | .7883 | .7860 | .7837 | .7814 |
| 80                       | .7792  | .7769 | .7746 | .7724 | .7702 | .7679 | .7657 | .7635 | .7613 | .7591 |
| 90                       | .7570  | .7548 | .7526 | .7505 | .7484 | .7462 | .7441 | .7420 | .7399 | .7378 |
| 100                      | .7358  | .7337 | .7316 | .7296 | .7275 | .7255 | .7235 | .7215 | .7195 | .7175 |
| 110                      | .7155  | .7135 | .7115 | .7096 | .7076 | .7057 | .7038 | .7018 | .6999 | .6980 |
| 120                      | .6961  | .6942 | .6923 | .6905 | .6886 | .6867 | .6849 | .6830 | .6812 | .6794 |
| 130                      | .6776  | .6757 | .6739 | .6721 | .6704 | .6686 | .6668 | .6650 | .6633 | .6615 |
| 140                      | .6598  | .6581 | .6563 | .6546 | .6529 | .6512 | .6495 | .6478 | .6461 | .6445 |
| 150                      | .6428  | .6411 | .6395 | .6378 | .6362 | .6345 | .6329 | .6313 | .6297 | .6281 |
| 160                      | .6265  | .6249 | .6233 | .6217 | .6202 | .6186 | .6170 | .6155 | .6139 | .6124 |
| 170                      | .6109  | .6093 | .6078 | .6063 | .6048 | .6033 | .6018 | .6003 | .5988 | .5973 |
| 180                      | .5959  | .5944 | .5929 | .5915 | .5900 | .5886 | .5872 | .5857 | .5843 | .5829 |
| 190                      | .5815  | .5801 | .5787 | .5773 | .5759 | .5745 | .5731 | .5718 | .5704 | .5690 |
| 200                      | .5677  | .5663 | .5650 | .5636 | .5623 | .5610 | .5596 | .5583 | .5570 | .5557 |
| 210                      | .5544  | .5531 | .5518 | .5505 | .5492 | .5480 | .5467 | .5454 | .5442 | .5429 |
| 220                      | .5417  | .5404 | .5392 | .5379 | .5367 | .5355 | .5342 | .5330 | .5318 | .5306 |
| 230                      | .5294  | .5282 | .5270 | .5258 | .5246 | .5234 | .5223 | .5211 | .5199 | .5188 |
| 240                      | .5176  | .5165 | .5153 | .5142 | .5130 | .5119 | .5108 | .5096 | .5085 | .5074 |
| 250                      | .5063  | .5052 | .5041 | .5029 | .5018 | .5008 | .4997 | .4986 | .4975 | .4964 |
| 260                      | .4953  | .4943 | .4932 | .4922 | .4911 | .4900 | .4890 | .4879 | .4869 | .4859 |
| 270                      | .4848  | .4838 | .4828 | .4818 | .4807 | .4797 | .4787 | .4777 | .4767 | .4757 |
| 280                      | .4747  | .4737 | .4727 | .4717 | .4707 | .4698 | .4688 | .4678 | .4669 | .4659 |
| 290                      | .4649  | .4640 | .4630 | .4621 | .4611 | .4602 | .4592 | .4583 | .4574 | .4564 |
| 300                      | .4555  | .4546 | .4537 | .4527 | .4518 | .4509 | .4500 | .4491 | .4482 | .4473 |
| 310                      | .4464  | .4455 | .4446 | .4438 | .4429 | .4420 | .4411 | .4402 | .4394 | .4385 |
| 320                      | .4376  | .4368 | .4359 | .4351 | .4342 | .4334 | .4325 | .4317 | .4308 | .4300 |
| 330                      | .4292  | .4283 | .4275 | .4267 | .4259 | .4251 | .4242 | .4234 | .4226 | .4218 |
| 340                      | .4210  | .4202 | .4194 | .4186 | .4178 | .4170 | .4162 | .4154 | .4147 | .4139 |
| 350                      | .4131  | .4123 | .4115 | .4108 | .4100 | .4092 | .4085 | .4077 | .4070 | .4062 |
| 360                      | .4055  | .4047 | .4040 | .4032 | .4025 | .4017 | .4010 | .4003 | .3995 | .3988 |
| 370                      | .3981  | .3973 | .3966 | .3959 | .3952 | .3945 | .3937 | .3930 | .3923 | .3916 |
| 380                      | .3909  | .3902 | .3895 | .3888 | .3881 | .3874 | .3867 | .3860 | .3854 | .3847 |
| 390                      | .3840  | .3833 | .3826 | .3820 | .3813 | .3806 | .3799 | .3793 | .3786 | .3779 |
| 400                      | .3773  | .3766 | .3760 | .3753 | .3747 | .3740 | .3734 | .3727 | .3721 | .3714 |

SOURCE: Adapted and abridged from Couper, 2003.

should be replaced like for like or whether an alternative should be chosen that may be different in cost and/or efficiency. If the replacement is due to technical obsolescence, the timing of the replacement may be important, especially if a plant expansion may be imminent in the near future. Whatever the situation, the replaced item should not present a bottleneck to the processing. The engineer should understand replacement theory to determine if alternative equipment is adequate for the job but with different costs and timing.

Certain terminology has been developed to identify the equipment under consideration. The item in place is called the *defender*, and the candidate for replacement is called the *challenger*. This terminology and methodology was reported by E. L. Grant and W. G. Ireson in *Engineering Economy*, Wiley, New York, 1950. To apply this method, there are certain rules. The value of the defender asset is a sunk cost and is irrelevant except insofar as it affects cash flow from depreciation for the rest of its life and a tax credit for the book loss if it is

replaced sooner than its depreciation life. A capital cost for the defender is the net capital recovery forgone and the tax credit from the book loss of the defender asset that was not realized. The UAC method will be used and will be computed for each case, using the time period most favorable to each. For the defender it is 1 year and for the challenger it is the full economic life. The UAC for the challenger is handled in the same manner as in the comparison of alternatives. The method is demonstrated in Example 11.

**Example 11: Replacement Analysis** A 3-year-old reciprocating compressor is being considered for replacement. Its original cost was \$150,000, and it was being depreciated over a 7-year period by the straight-line method. If it is replaced now, the net proceeds from its sale are \$50,000, and it is believed that 1 year from now they will be \$35,000. A new centrifugal compressor can be installed for \$160,000, which would save the company \$2000 per year in operating expenses for the 10-year life. At the end of the 10th year, its net proceeds are estimated to be zero. The 7-year depreciation applies also to the centrifugal

compressor. A 35 percent tax rate may be assumed. The company requires a 15 percent after-tax return on an investment of this type. Should the present compressor be replaced now?

**Solution:** The UAC method will be used as a basis for comparison. It is assumed that all money flows are continuous, and continuous interest will be used.

Defender case: The basis for this unit will be 1 year. If it is not replaced now, the rules listed above indicate that there is an equivalent of a capital cost for two benefits forgone (given up). They are

1. Net proceeds now at 3 years of \$50,000
2. Tax credit for the loss not realized

Thus net loss forgone = book value at the end of 3 years minus net capital recovery, or

$$NLF_3 = BV_3 - NCR_3$$

where  $NLF_3$  = net loss forgone at end of 3 years  
 $BV_3$  = book value at end of 3 years  
 $NCR_3$  = net capital recovery at end of 3 years

$$NLF_3 = \$150,000 \left( 1 - \frac{1}{7} \right) - \$50,000 = \$35,714$$

$$\text{Depreciation for 4th year} = \$150,000 \left( \frac{1}{7} \right) = \$21,429$$

$$NLF_4 = BV_4 - NCR_4 = \$150,000 \left( 1 - \frac{4}{7} \right) - \$35,000 = \$29,286$$

| Year              | Item                                      | Cash flow, \$       | Factor              | PW, \$  |
|-------------------|-------------------------------------------|---------------------|---------------------|---------|
| At 0 <sup>e</sup> | Tax credit for net loss forgone           | (0.35)(-35,714)     | 1.0                 | -12,500 |
| At 0 <sup>e</sup> | Net cash recovery forgone                 | -50,000             | 1.0                 | -50,000 |
| 0-1               | Contribution to CF from depreciation      | (0.35)(-21,429)     | 0.9286 <sup>b</sup> | 6,965   |
| 0-1               | Contribution to CF from operating expense | (1 - 0.35)(-15,000) | 0.9286 <sup>b</sup> | -9,054  |
| End 1             | Tax credit for net loss                   | (0.35)(29,286)      | 0.8607 <sup>c</sup> | 8,822   |
| End 1             | Net cash                                  | 35,000              | 0.8607 <sup>c</sup> | 30,125  |
|                   |                                           |                     | NPW                 | -25,642 |

<sup>a</sup>For the defender case, 0 year is the end of the third year.

<sup>b</sup>From Table 9-30, uniform section, for the argument  $R \times T = 15 \times 1 = 15$ .

<sup>c</sup>From Table 9-30, instantaneous section, for the argument  $R \times T = 15 \times 1 = 15$ .

$$UAC = \frac{NPW}{(\text{years of life})(\text{uniform factor})} = \frac{-\$25,642}{(1)(0.9286)} = -\$27,614$$

Challenger case:

| Year | Item                                 | Cash flow, \$           | Factor              | PW, \$   |
|------|--------------------------------------|-------------------------|---------------------|----------|
| 0    | First cost                           | -160,000                | 1.0                 | -160,000 |
| 0-7  | Contribution to CF from depreciation | (0.35)(160,000)         | 0.6191 <sup>d</sup> | 56,000   |
| 0-10 | Contribution                         | (1 - 0.35)(10)(-13,000) | 0.5179 <sup>e</sup> | -43,763  |
|      |                                      |                         | NPW                 | -147,763 |

<sup>d</sup>From Table 9-30, uniform section,  $R \times T = 15 \times 7 = 105$ .

<sup>e</sup>From Table 9-30, uniform section,  $R \times T = 15 \times 10 = 150$ .

$$UAC = \frac{NPW}{(\text{years of life})(\text{uniform factor})} = \frac{-\$147,763}{(10)(0.5179)} = -\$28,531$$

The UAC for the defender case is less negative (more positive) than that for the challenger case; therefore, the defender should not be replaced now. But there will be a time in the near future when the defender should be replaced, as maintenance and deterioration will increase.

## OPPORTUNITY COST

Opportunity cost refers to the cost or value that is forgone or given up because a proposed investment is undertaken, often used as a base case. Perhaps the term should be *lost opportunity*. For example, the profit from production in obsolete facilities is an opportunity cost of

replacing them with more efficient ones. In cost analysis on investments, an incremental approach is often used, and if it is applied properly, the correct cost analysis will result.

## ECONOMIC BALANCE

An engineering cost analysis can be used to find either a minimum cost or a maximum profit for a venture. This analysis is called an *economic balance* since it involves the balancing of economic factors to determine optimum design or optimum operating conditions. Such an analysis involves *engineering tradeoffs*. It may be more beneficial to invest more capital to reduce operating expenses or, conversely, incur more operating expenses without the addition of costly capital. An economic balance, then, is a study of all costs, expenses, revenues, and savings that pertain to an operation or perhaps an equipment item size. In this presentation, certain terminology is used; e.g., the term *cost* refers to a one-time purchase of capital equipment. A recurring expense is called an *operating expense*, such as utilities, labor, and maintenance. All costs and operating expenses are related to an arbitrarily designated *controllable* variable such as heat exchanger area, thickness of insulation, or number of units.

There are certain practical considerations that must be recognized in attempting to find the best or optimum condition. Occasionally, a solution may lead to a result for which industrial equipment is not available in the optimum size; therefore, engineering judgment must be exercised. A smaller-diameter pipe might lead to a higher pumping expense but lower pipe costs while a larger-diameter pipe would result in lower pumping expense but higher capital cost. The engineer then encounters an engineering tradeoff that must be resolved.

The essential elements of an economic balance are

- Fixed and variable expenses
- An allowance for depreciation
- An acceptable return on the investment

A total expense equation is

$$TE = FE + VE \quad (9-24)$$

where TE = total expenses

FE = fixed expenses

VE = variable expenses

The guidelines for solving a single-variable economic balance consist of the following steps:

1. Determine all expenses that vary as the controllable variable changes and that need to be considered in the balance.
2. Determine whether any operating limitations exist, such as pressure drop in columns where flooding occurs or limiting heads for pipelines for gravity flow.
3. Mathematically express the expenses as a function of the variables related to the equipment; otherwise use variables that define the operation such as temperature, pressure, and concentration. The final expression should include all pertinent expenses, eliminating those that are not significant. Frequently only one variable is used.
4. Ascertain if the optimum size must be one of a number of discrete sizes commercially available or whether it can be any size.
5. Solve the total expense equation either analytically or graphically.

The solution to Eq. (9-24) may be found analytically or graphically. In the analytical method, the fixed and variable expenses are related in equation format to the controllable variable. Example 12 is an engineering balance example of this method. Equation (9-24) can be differentiated with respect to the controllable variable and that result set equal to zero to find the optimum condition. A graphical method involves determining the fixed and variable expenses for a range of equipment sizes. A plot of TE, FE, and VE as a function of the controllable sizes yields a plot identifying the optimum. Figure 9-18 is a graphical solution to the TE equation. Should the optimum result fall between two commercial sizes, then the engineer must exercise judgment in the tradeoff.

**Example 12: Optimum Number of Evaporator Effects** Determine the number of evaporator effects for the minimum total annual operating

## 9-40 PROCESS ECONOMICS

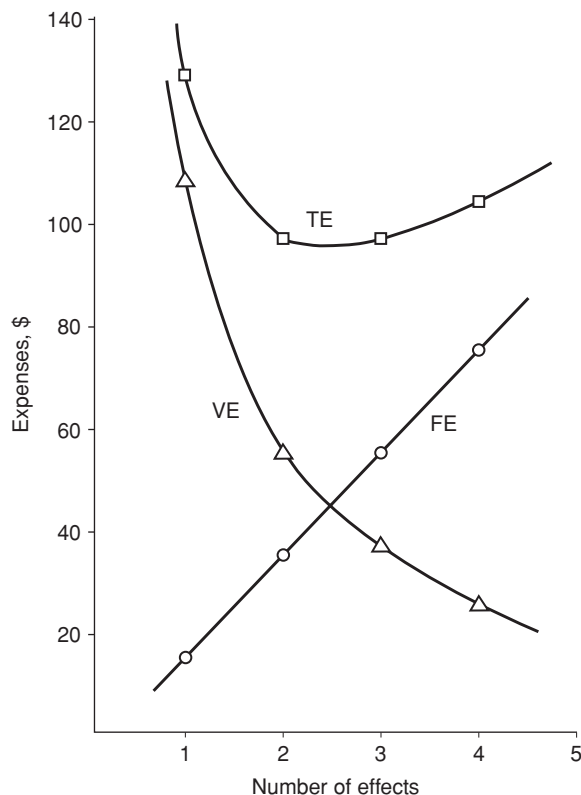


FIG. 9-18 Optimum number of evaporator effects in Example 12.

expense of a small evaporator system to concentrate a colloidal suspension. Steam costs \$3.00 per million Btu (MBtu), and each pound of steam will evaporate 0.8N lb water, with N being the number of effects. The total capital cost of each effect is \$40,000 and has an estimated life of 10 years. The annual maintenance expense is 10 percent of the capital cost. Labor and other expenses not mentioned may be considered to be independent of the number of effects. The system will operate 300 days/yr with 100,000,000 Btu/day evaporator duty. Depreciation on the equipment is by 7-year straight-line method, and the tax rate is 35 percent. Annual net profit after taxes on the investment must be 15 percent for the installed equipment.

Basis: 1 year of operation, duty  $10^8$  Btu/day. Let N equal the number of effects.

Solution:

Analytical method:

Annual fixed expenses = FE

$$FE = \$40,000N \left( 0.143 + 0.100 + \frac{0.15}{1 - 0.35} \right) = \$18,960N$$

dep.    maint.    profit (return)

Annual variable expenses VE in this example are essentially the steam expenses, and all other variable expenses do not enter into the equation. Therefore

$$VE = 10^8 \text{ Btu/day} \times 300 \text{ day/yr} \times \$3.00/10^6 \text{ Btu} \times 1/(0.8N) = \frac{\$112,500}{N}$$

$$TE = FE + VE = \$18,960N + \frac{\$112,500}{N}$$

Take the derivative of TE with respect to N and set the result equal to zero.

$$\frac{d(TE)}{dN} = 18,960 - \frac{112,500}{N^2} = 0 \quad N = 2.44 \text{ effects}$$

Therefore 3 effects are required.

Graphical method: Assume the number of effects and calculate the fixed, variable, and total expenses for effect.

| N | 18,960N  | 112,500/N | TE        |
|---|----------|-----------|-----------|
| 1 | \$18,960 | \$112,500 | \$131,460 |
| 2 | 37,920   | 56,250    | 94,170    |
| 3 | 56,880   | 37,500    | 94,380    |
| 4 | 75,840   | 28,125    | 103,965   |

The minimum TE occurs between 2 and 3 effects, but 3 effects are recommended to evaporate the colloidal solution. Figure 9-18 is a plot of the graphical solution.

In another example involving the reclaiming of a product using an evaporator and a dryer in series, the product is pumped from the discharge of the evaporator to the dryer. See Fig. 9-19. An economic analysis indicated that a 55 percent slurry is optimum, but perhaps such a slurry is too thick to be pumped. Therefore, engineering judgment must be exercised. The slurry was pumped not at 55 percent but at a lesser concentration of 50 percent, although it was not optimum.

When more than one controllable variable affects the economic balance, the solution approach is essentially the same as that for the single-variable case, but determining the optimum is tedious.

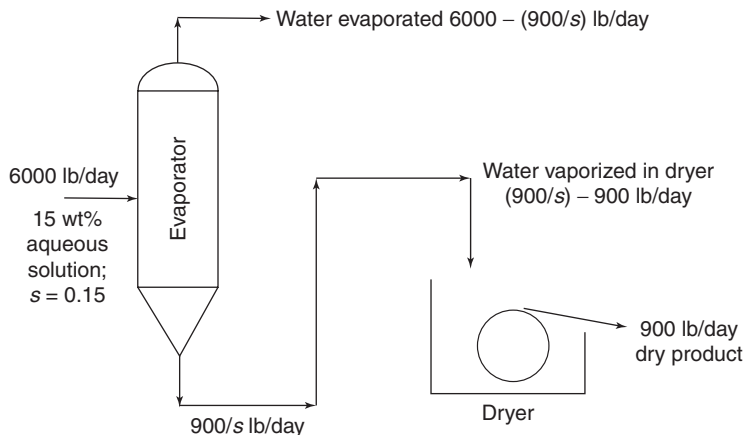


FIG. 9-19 Evaporator-dryer system. (Source: Adapted from Couper, 2003.)

## INTERACTIVE SYSTEMS

If the TE does not pass through a minimum or maximum, but continues to decline or to increase with the number of equipment items or equipment size, the next step is to look at the flow sheet for equipment upstream or downstream from the selected item. It may be necessary to group two or more items and treat them as one in the analysis. Such a system is said to be *interactive*, since more than one item affects the optimum results. An example of such an interactive system is the

removal of nitrogen from helium in a natural gas stream. Carbon adsorption is a method for removing nitrogen, but compressors are also required since this is a high-pressure process. If one attempts to find the optimum operating pressure, optimizing on compressor pressure will not result in an optimum condition; and conversely, optimizing on the size of the carbon bed will not yield an optimum. This is an example of an interactive system. Therefore, to find the optimum pressure, both the size of the carbon bed and the compressor pressure must be considered together.

## CAPITAL PROJECT EXECUTION AND ANALYSIS

The demands made on business organizations with the arrival of global free trade have made sophisticated management of capital projects, for the purposes of minimizing capital costs and maximizing project profitability, a necessity. Two elements of such sophisticated project management practice are *front-end loading (FEL)* and *value-improving practices (VIPs)*. These two management practices are, and must be, closely integrated activities, as seen in Fig. 9-20. As shown, they are performed during the early stages of a project life cycle, when they can be, and are, effective at influencing a project's profitability. However, they have very different characteristics, as are detailed in the following sections. Properly performed together, FEL and VIPs maximize project profitability by ensuring that all matters that influence project profitability are considered in the most productive manner and at the most optimal time.

### FRONT-END LOADING

**GENERAL REFERENCES:** Porter, James B., E. I. DuPont de Nemours and Company, DuPont's Role in Capital Projects, Proceedings of Government/Industry

Forum: The Owner's Role in Project Management and Pre-project Planning, 2002. Smith, C. C., Improved Project Definition Insures Value-added Performance—Part 1, *Hydrocarbon Processing*, August 2000, pp. 95–99. KBR Front-end Loading Program, data compiled from selected large projects from 1993 through 2003, <http://www.halliburton.com/kbr/index.jsp>. Merrow, E. W., Independent Project Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000. Merrow, E. W., Independent Project Analysis, Inc., 30th Annual Engineering & Construction Contracting Conference, September 1998. HPI Impact, *Hydrocarbon Processing*, August 2002, p. 23, data obtained from Independent Project Analysis, Inc. PDRI: Project Definition Rating Index—Industrial Projects, p. 5, Construction Industry Institute (CII), University of Texas at Austin, <http://construction-institute.org>, July 1996. Merrow, E. W., Independent Project Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000.

**Introduction** Front-end loading (FEL) is the process by which a company develops a detailed definition of the scope of a capital project that meets corporate business objectives. The term *front-end loading* was first coined by the DuPont company in 1987 and has been used throughout the chemical, refining, and oil and gas industries ever since (Porter, James B., E. I. DuPont de Nemours and

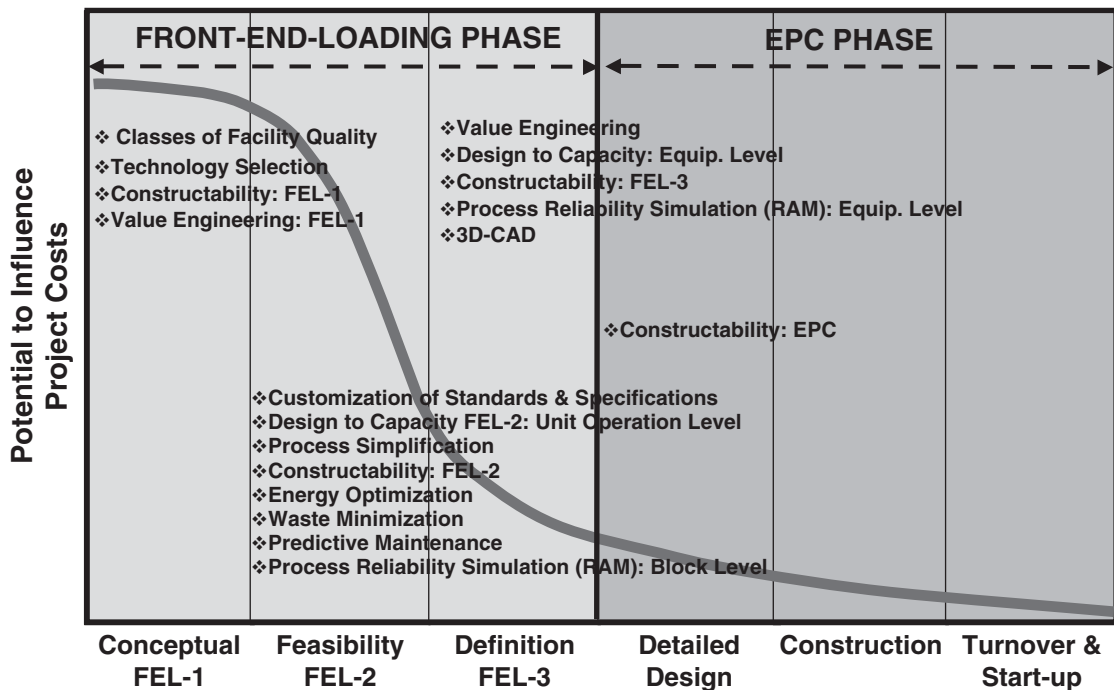


FIG. 9-20 Front-end loading and the implementation of value-improving practices.



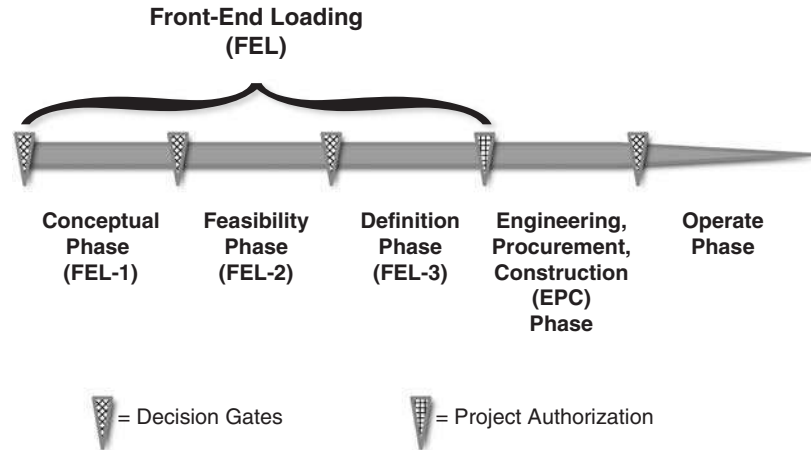


FIG. 9-21 FEL in the capital project life cycle.

Company). The product of the FEL process is a design-basis package of customized information used to support the production of detailed engineering design documents. Completion of the FEL design-basis package typically coincides with project AFE (Authorization for Expenditure) or project authorization. Project authorization is that point in the project life cycle where the owner organization commits the majority of the project's capital investment and contracts.

FEL starts when an idea for a project is first conceived by a research and development group, project engineering group, plant group, or business unit. FEL activity continues until the project is authorized. After initial conception of an idea, organized interaction is required among the various project stakeholders to assemble the project design-basis package for subsequent authorization.

Within the FEL phases, decision points are formally established by the operating company authorizing the initiation of a capital project development effort. These formal decision gates allow for continuity across the enterprise for authorization of additional funding for the next phase of engineering and project definition. Figure 9-21 illustrates the typical decision gates or stage gates for capital projects.

When the level of project definition is sufficient to support a definitive cost estimate for both the entire project and its projected rate of return, major project funding is authorized for expenditure. This is the conclusion of the FEL process and any significant involvement of the process design engineer. Not until the conclusion of the engineering, procurement, and construction (EPC) phase does the process engineer again become involved. At this time, commissioning and start-up become the focus where the validation occurs for all that was done in the FEL phases many months earlier.

Differing terminology used by companies, engineers, and project management teams is often a point of confusion. Most people seem to think they know what all the terms mean. This is never the case. Confirmation of which terminology will be used by all involved in the project is a must. Nearly every operating company and engineering contractor use differing terminology. FEL terminology is often misunderstood and further confused by differing references to which FEL phase the project is actually in. Figure 9-22 provides some idea of the differing terminology for each project phase used by only a few major oil and gas and chemical companies today. These terms change periodically, so diligence in confirming such terminology is a key task for the process engineers to finish, before beginning their work.

The influence of changes on capital projects is considerably affected by when those changes occur. The earlier a change is considered and incorporated into the project scope, the greater its potential influence on the project's profitability and the greater the ease of

incorporating the change. This means that late changes (e.g., in the EPC phase) are far more expensive to implement and are considered very undesirable. Late changes which are potentially advantageous are often not implemented because the cost to implement the change exceeds the benefits of doing so. Conversely, the cost to implement a change at the earlier phases of the project is far lower than making the same change after detailed engineering is underway.

Figure 9-23 shows how quickly this influence curve changes as the typical project progresses (Smith, C. C., *Improved Project Definition Insures Value-added Performance—Part 1*). This is why proactively seeking changes during FEL is far more advantageous to profitability than is allowing those needed changes to be “discovered” during later project phases. This also means that potentially beneficial changes (value improvements) must be sought during FEL, or else they stand a good chance of not being cost-effective to implement during the EPC phase. This is also why seeking operations, maintenance, and construction experience during FEL offers significant profitability advantages over practices which bring such experience onto the project team following FEL.

**Characteristics of FEL** Front-end loading is a specialized and adaptable work process. This work process translates financial and marketing opportunities to a technical reality in the form of a capital project. It is particularly important that the project be defined in sufficient detail by the engineering deliverables, which are generated by the FEL work process, prior to the point where major funds are authorized. In this manner, overall project risks are identified and sufficiently mitigated to have project funding approved. To achieve this important level of definition, critical decisions must be made and adhered to. In addition, the FEL project team should proactively seek value improvement alternatives and challenge the project premises, scope, and design until such time as implementation of those alternatives loses their profitability and/or technical advantage. By doing so, such value improvements will not develop into costly corrections, which surface later, during the EPC phase.

**Goals and Objectives of FEL** The FEL work process must enable nearly constant consideration of changes as the work progresses. FEL phases must consider the long-term implications of every aspect of the design. Predictability of equipment and process system life cycle costs must always be balanced with operations and maintenance preferences, as well as the need for the project to maintain its profitability or ROI (return on investment). Additional important goals and objectives of FEL projects are as follows:

- Develop a well-defined and acceptably profitable project.
- Define the primary technical and financial drivers for capital project investment.

# ← FRONT-END LOADING →

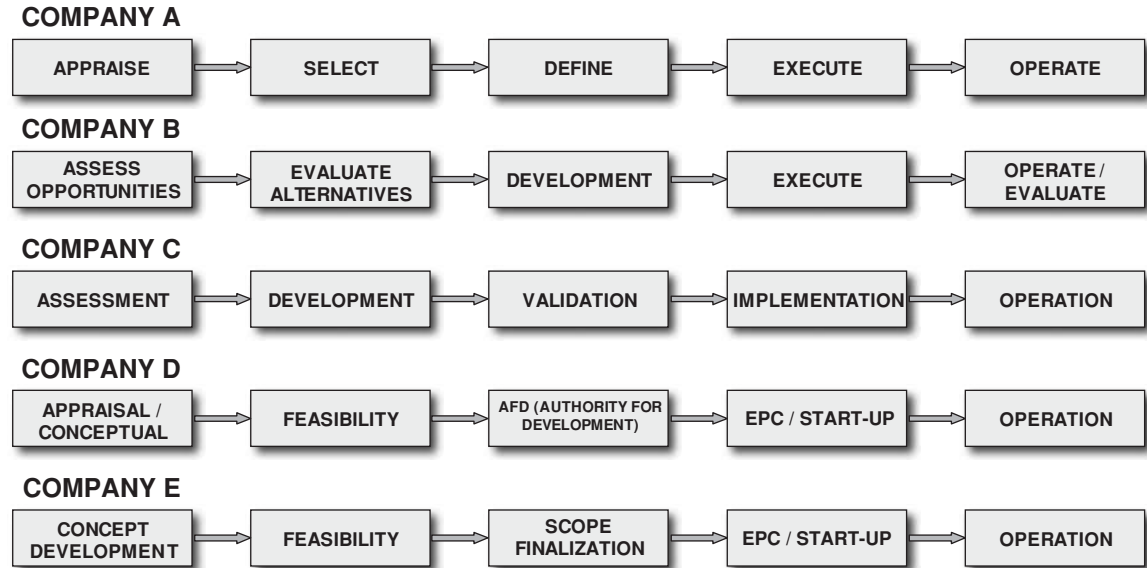


FIG. 9-22 Project life cycle terminology.

- Challenge baseline premises, and purposely seek out and evaluate alternatives and opportunities.
- Minimize changes during the EPC, turnover, and start-up phases.
- Reduce project schedule and capital cost.
- Reduce the business and project execution risk.
- Balance project technical, financial, and operational profitability drivers.

**Comparison of FEL Projects with EPC Projects** FEL projects are very different from EPC projects. Engineers and project managers having significant experience only with projects in the EPC phase often are unfamiliar with the significant differences between the philosophies and challenges of the FEL phase and the EPC phase of projects. One of the most important (but most subtle) aspects of FEL is the demand during FEL for more highly experienced staff and

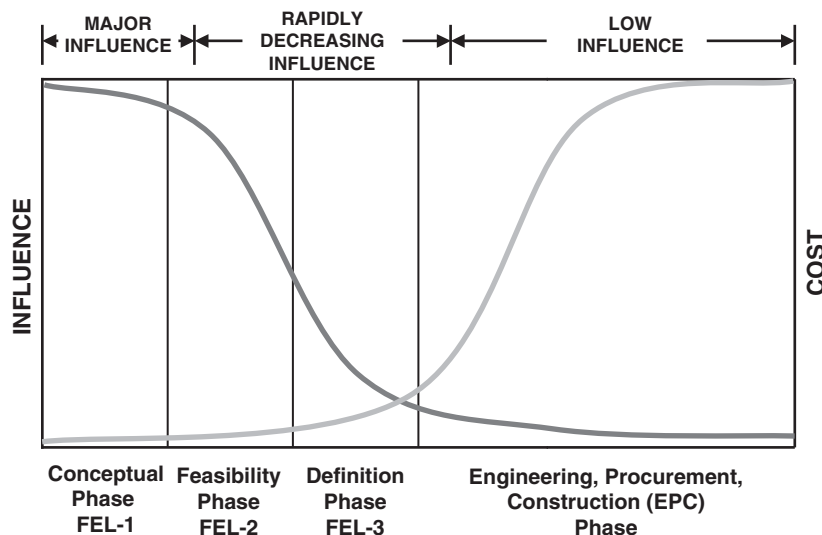


FIG. 9-23 Project life cycle cost-influence curve.

|                                    | FEL                    | EPC                      |
|------------------------------------|------------------------|--------------------------|
| <b>Project State</b>               | Undefined              | Defined                  |
| <b>Changes</b>                     | Actively Seeks Changes | Actively Resists Changes |
| <b>Impact of Change</b>            | Low                    | High                     |
| <b>Opportunity for Change</b>      | High                   | Low                      |
| <b>Contract Type</b>               | Typically Reimbursable | Typically Lump Sum       |
| <b>Value Improvement Potential</b> | High                   | Low                      |
| <b>Client Participation</b>        | Encouraged             | Discouraged              |
| <b>Philosophy</b>                  | Information-Driven     | Deliverable-Driven       |

FIG. 9-24 FEL projects versus EPC projects.

more sophisticated analysis tools, as compared to EPC projects which have achieved a well-defined project prior to authorization. This is so because of the need in FEL to create, analyze, and implement improvements to what many might consider a “good” design.

In spite of its relatively short duration, FEL proactively seeks to implement the best possible design. The nearly constantly changing environment requires people of many different disciplines and functions to work together to communicate effectively. A well-integrated team always seems to perform best during FEL, if FEL has well-established, informal, and personal interfaces between project groups and organizations. The following describes how the FEL phase is distinguished from the EPC phase:

- FEL proactively seeks data, resources, support, and decision making.
- Projects in the FEL phase place a higher level of importance on close and effective owner-contractor management interfaces.
- FEL demands continuous realignment of client desires and requirements with contractor needs.
- FEL requires greater development of personal relationships that result in respect and trust.
- FEL demands significantly higher frequency of feedback of owner satisfaction.
- FEL emphasizes elimination of low- or zero-value scope.
- FEL improves the capital productivity of projects by using best-available technology.
- FEL focuses on overall project profitability rather than on only cost, schedule, and workhours.
- FEL focuses almost entirely on the owner's business needs.

Figure 9-24 lists further differences between FEL and EPC projects. Understanding these many differences is very important to the process engineer, in that awareness of them, and the driving forces behind them, will prepare the chemical engineer for the challenging and rewarding environment of FEL projects.

**Parameters of FEL Phases** Important aspects of each phase of FEL are cost estimate accuracy, cumulative engineering hours spent, and the contingency assigned to the cost estimate. Figure 9-25 lists the typical parameters encountered industrywide (KBR Front-end Loading Program, data compiled from selected large projects from 1993 through 2003). For the capital cost estimate, each operating company may request a slightly different accuracy, which is often project-specific. What is important is the level of engineering required to support such estimating accuracy. This determination is the responsibility of both the owner and the engineering contractor. Agreement on this is critical prior to initiating project work.

The engineering hours spent during each phase of FEL vary widely between small and large projects. This is also true for those projects where new or emerging technology is being applied or where higher

throughput capacities are being applied than previously commercially demonstrated. Projects such as these may require additional engineering to achieve the desired estimate accuracy and project contingency.

**FEL Project Performance Characteristics** Overall project performance can be enhanced by ensuring that the following characteristics are emphasized during the FEL phases.

- Methodical business and project execution planning is necessary.
- Effective integration of workforce between owner and contractor staff is important.
- Projects with an integrated management team (owner and engineering contractor) have the lowest number of design changes at any project stage.
- Engineering contractor should be brought into project in early FEL phases.
- Clear roles must exist for project team members that relate to the expertise of both owner and contractor staff.
- Effective personal communication is required between owner and contractor organizations and their project team representatives, ensuring extensive site and manufacturing input.
- Schedule and cost goals are set by integrated business and technical project team composed of owner and contractor representatives.

Figure 9-26 illustrates the benefit of good FEL performance on project costs (Morrow, E. W., Independent Project Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000). Figure 9-27 illustrates the benefit of good FEL performance on critical path schedule (Morrow, E. W., Independent

|                                           | FEL-1<br>(CONCEPTUAL) | FEL-2<br>(FEASIBILITY) | FEL-3<br>(DEFINITION) |
|-------------------------------------------|-----------------------|------------------------|-----------------------|
| <b>Cost Estimate Accuracy</b>             | ±40%                  | ±25%                   | ±10%                  |
| <b>Cumulative Engineering Hours Spent</b> | 1–5%                  | 5–15%                  | 15–30%                |
| <b>Contingency</b>                        | 15–20%                | 10–15%                 | 8–12%                 |

FIG. 9-25 Parameters of FEL phases.

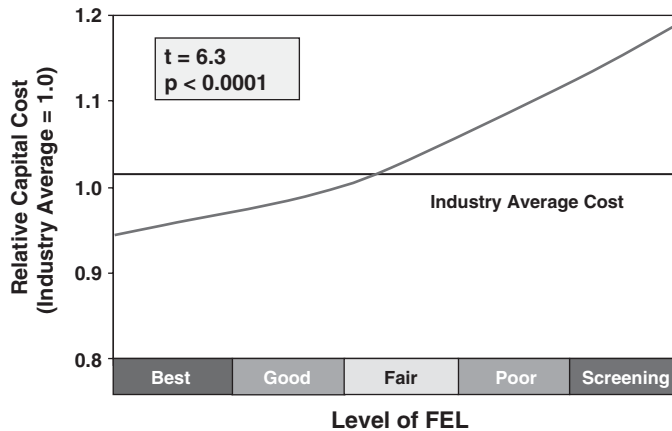


FIG. 9-26 FEL drives better cost performance.

Project Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000). IPA statistics indicate that significant project financial and schedule benefits can be realized by implementing a thorough FEL effort prior to the EPC phase.

Figure 9-28 presents the benefits of having an integrated project team during FEL on the overall project performance. This performance impacts overall project costs as well as schedule and operability. An integrated project team produces fewer late changes. This means lower capital costs, better and more predictable schedules, and a slightly better operability, as compared statistically to similar projects lacking an integrated management team (Morrow, E. W., Independent Project Analysis, Inc., 30th Annual Engineering & Construction Contracting Conference, September 1998).

In addition, project data indicate that a well-integrated FEL team can produce significantly better project performance in terms of lower capital investment, as compared to projects where FEL teams were not properly integrated. This illustrates the benefits for each engineering team member working closely together with each other team member, to produce the most profitable project results.

Although project teams, once integrated and functioning with clear roles and responsibilities, perform better, this edge can be quickly lost

if key members of that team are changed. The impacts of changes of project managers to a well-integrated FEL team are shown in Fig. 9-29 (HPI Impact, *Hydrocarbon Processing*, p. 23, August 2002, data obtained from Independent Project Analysis, Inc.).

**Investment in FEL for Best Project Performance** The cost and schedule required to optimally complete the FEL phase of a project are always under pressure and must be justified. This is especially true for “fast-track” projects where the time pressures can be significant. The Construction Industry Institute (CII) has shown that higher levels of preproject planning (i.e., front-end loading) effort can result in significant cost and schedule savings, as seen in Fig. 9-30 (PDRI: Project Definition Rating Index—Industrial Projects, Construction Industry Institute, University of Texas at Austin, July 1996). The process engineer produces the best project performance, when he or she strives, with the entire integrated FEL project team, to define the overall project (not just the process design) as well as possible, prior to AFE.

The level of definition of a project during the FEL phases has a direct influence on the project’s ultimate outcome in terms of the number and impacts of changes in the EPC phase. This level of FEL performance translates to fewer major changes in engineering, construction, and during start-up (Morrow, E. W., Independent Project

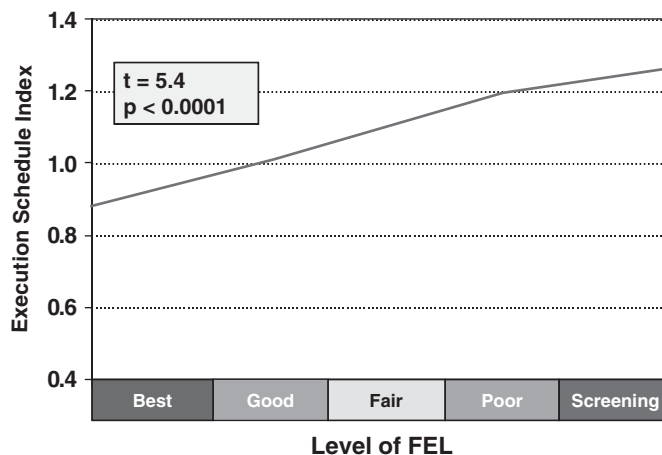


FIG. 9-27 Good FEL speeds execution time.

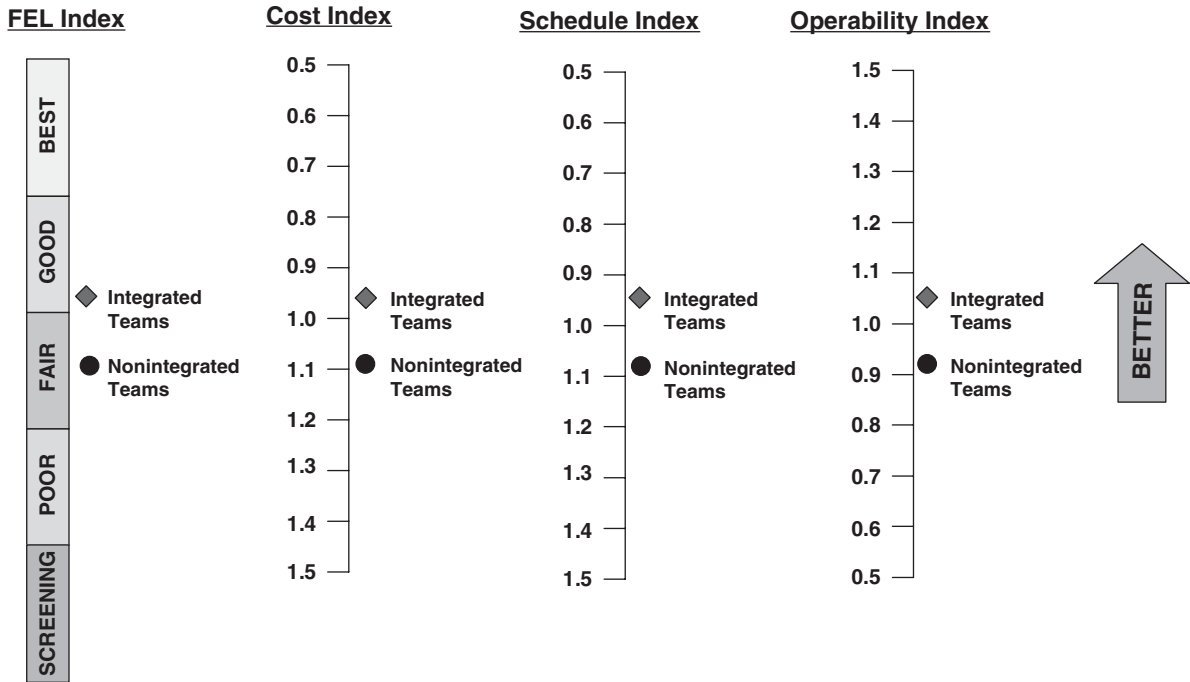


FIG. 9-28 Integrated teams result in better FEL and better overall performance.

Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000). These conclusions are depicted by Fig. 9-31. A major late change is defined by IPA's data to mean changes made after the start of detailed engineering and involving impacts greater than either 0.5 percent of the total project capital investment or 1 month in critical path project schedule.

These graphs illustrate why better project performance is produced through proactively seeking profit-improving changes as early

as possible. One of the reasons for this observation is that operation, maintenance, and construction expertise is incorporated into the project at the very beginning—during FEL.

This means that the process design engineer should be working closely with these real-world experts as they design processes and their support systems. This also means that to improve overall project performance, achieving the best practical or highest level of definition during FEL is critical. Finally, this high level of definition results in a

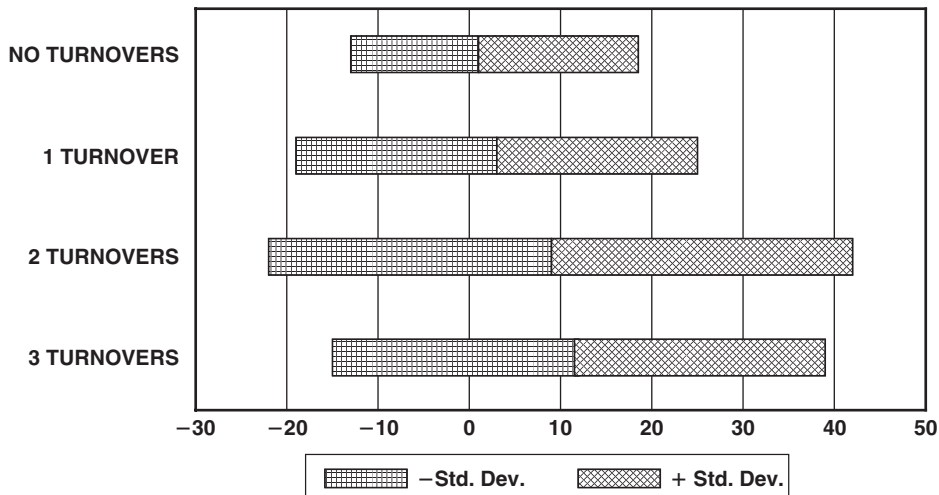


FIG. 9-29 Effects of project manager turnover on cost growth.

| LEVEL OF FEL EFFORT | COST | SCHEDULE |
|---------------------|------|----------|
| <b>High</b>         | -4%  | -13%     |
| <b>Medium</b>       | -2%  | +8%      |
| <b>Low</b>          | +16% | +26%     |
| <b>Difference</b>   | 20%  | 39%      |

FIG. 9-30 Project performance versus level of FEL effort.

reduced number of changes during the EPC phase. These observations should be the critical goals of all project teams.

The size of capital project also has an influence on FEL outcome based on IPA statistics. IPA's data indicate that small projects benefit more from better project definition prior to the EPC phase than do major projects. The data also indicate that small projects typically have more late changes than do larger projects.

Figure 9-32 illustrates the effect of large projects versus small projects in terms of the impact of late changes (Morrow, E. W., Independent Project Analysis, Inc., 32d Annual Engineering & Construction Contracting Conference, Sept. 28–29, 2000). Figure 9-32 also illustrates that the level of FEL performance directly impacts the number of, and the consequences of, late changes made in projects of any size.

**Typical FEL Deliverables** Every process engineer assigned to a project should be acutely aware of which deliverables or end products are required by those who must commission their work. This should be very well understood by all parties prior to starting the work. Further, the splits of work (who will do which aspect of the work) must be well understood. Today, it is very common to have multiple operating companies form a joint venture to authorize major projects. It is also common for multiple engineering contractors to form joint ventures to execute the engineering for the FEL phases of the project.

**Typical Conceptual Phase (FEL-1) Deliverables** These are listed below. Each project will customize these deliverables to suit the particular needs of the project. There is no such thing as a "standard" FEL. Therefore, the process engineer must understand what the details are for each deliverable needed, what the *minimum* level is for the engineering required to meet those requirements, and in which formats that information and data will be needed.

- Strategic business assessment
- Key technology selected and risk identified
- Market assessment for feed, products, and capacity
- Potential sites identified and under evaluation
- Cost estimate ( $\pm 40$  percent)
- Preliminary project milestone schedule
- Block flow diagrams completed
- Process cases identified
- Critical long-lead equipment identified
- Value-improving practices reports

**Typical Feasibility Phase (FEL-2) Deliverables** These are listed below. In this phase, emphasis is on determining the best technical and economic flow scheme, as well as the support systems required to provide the necessary annual production rate at the sales quality required. The focus for the process engineer should be on confirming the number and type of process and technology studies needed, as well as the number of alternate cases required to be evaluated and/or simulated.

- Strategic business assessment
- Project schedule level 1
- Cost estimate ( $\pm 25$  percent)
- Overall project execution strategy
- Contracting and purchasing strategies
- Permitting and regulatory compliance plan
- Soil survey and report
- Project alternatives analysis
- Process flow diagrams for selected option(s)
- Preliminary utility flow diagrams and balances
- Preliminary equipment list and equipment load sheets
- Materials of construction
- Process hazards analysis report
- Value-improving practices reports

**Typical Definition Phase (FEL-3) Deliverables** These are listed below. In this phase, emphasis is typically on optimizing the best flow scheme and support systems combination. This optimum includes consideration of the plot plan and equipment arrangements for the entire facility. Process optimization cannot be done in isolation. Significant and continuous interaction with operations, maintenance, and construction experts always produces the best results. The emphasis in this phase is on achieving the best practical level of project definition and a good-quality project estimate of  $\pm 10$  percent. This level of project definition and cost estimate quality is normally required in order to present to management a candidate project which has the right combination of overall risk and projected economic performance, and thereby secure an AFE.

- Strategic business assessment
- Detailed EPC phase project execution plan
- Detailed EPC phase project master schedule

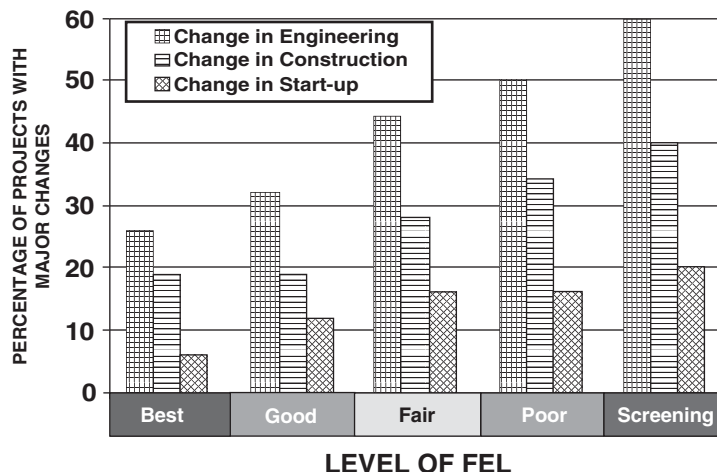


FIG. 9-31 Good FEL drives late changes down.

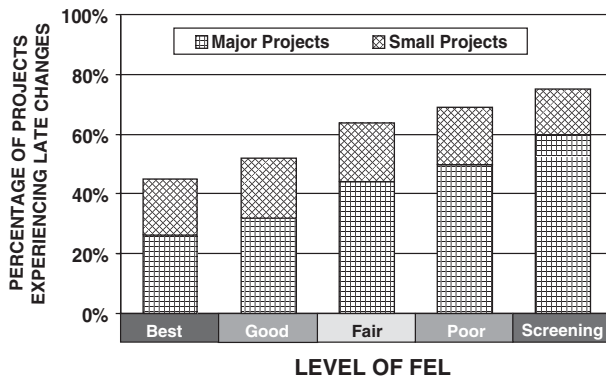


FIG. 9-32 Rate of late major changes is higher for small projects.

- Completed environment permit submittal
- Project plan/project execution plan
  - a. Cash flow plan for EPC phase
  - b. Training, commissioning, and start-up plans
  - c. Contracting plans
  - d. Materials management plan
  - e. Safety process and quality management plan
- Cost estimate ( $\pm 10$  percent)
- Finalized utility flow diagrams and balances
- P&ID's—issue IPL (issue for plant layout)
- Plot plans and critical equipment layouts
- Equipment list and equipment data sheets
- Single-line electrical diagrams
- Control system summary and control room layout
- Materials of construction
- VIPs reports
- Hazard and operability studies (HAZOP) report

### VALUE-IMPROVING PRACTICES

**GENERAL REFERENCES:** Independent Project Analysis, Inc. (IPA), <http://www.ipaglobal.com>. The Construction Industry Institute (CII), University of Texas at Austin, <http://construction-institute.org>. Lavingia, Jr., N. J., Improve Profitability Through Effective Project Management and TCM, 36th Annual Engineering & Construction Contracting Conference, Sept. 4, 2003. KBR Value Improving Practices Program, 1995 through 2005, <http://www.halliburton.com/kbr/index.jsp>. PDRI: Project Definition Rating Index—Industrial Projects, Construction Industry Institute (CII), University of Texas at Austin, <http://construction-institute.org>, July 1996. Society of American Value Engineers International (SAVE), <http://www.value-eng.org>. KBR experience.

**Introduction** Value-improving practices (VIPs) are formal structured practices applied to capital projects to improve profitability (or “value”) above that which is attained through the application of proven good engineering and project management practices. VIPs are formal analyses of project characteristics and features performed by small multidisciplinary teams at identified optimum times during the engineering design and development of capital projects.

Application of VIPs to capital projects has been statistically proved to significantly improve project profitability according to Independent Project Analysis, Inc. (IPA) and the Construction Industry Institute (CII). IPA data presented in Fig. 9-33 have been gathered from many capital projects since 1987. These data indicate that about 2.5 percent reduction in the relative capital cost can be expected for high-performing projects due to implementation of good front-end loading, work processes. The best-performing projects are often referred to as “Best Practical” or “Best in Class” projects and represent the upper 20 percent of projects.

However, when FEL improvement is combined with rigorous application of VIPs, the project performance improves to about 10 percent in reduction of relative capital cost (Lavingia, Jr., N. J., Improve Profitability Through Effective Project Management and TCM). Experience of at least one major engineering contractor indicates that about a 20 percent capital cost improvement can be expected through judicious use of their modified VIPs (KBR Value Improving Practices Program, 1995 through 2005). These improved results come about from continual adaptation and improvement of the VIPs themselves to maintain their relevance and ability to improve projects above what the project teams can accomplish by themselves.

The VIPs that have been statistically verified by IPA benchmarking of capital projects are listed below. Each has a different purpose and focus, but all produce project profitability improvements that the project team cannot achieve on its own.

- Classes of facility quality
- Technology selection
- Process simplification
- Constructability
- Customization of standards and specifications
- Energy optimization
- Predictive maintenance
- Waste minimization
- Process reliability simulation
- Value engineering
- Design to capacity
- 3D-CAD

Selection of the most applicable VIPs to be performed during a specific project is the focus of the VIP planning session, which should be held just following project kickoff. Figure 9-34 presents the optimal times during a large project for consideration of VIPs. The duration for FEL has been assumed to be 12 months. Every project will have a

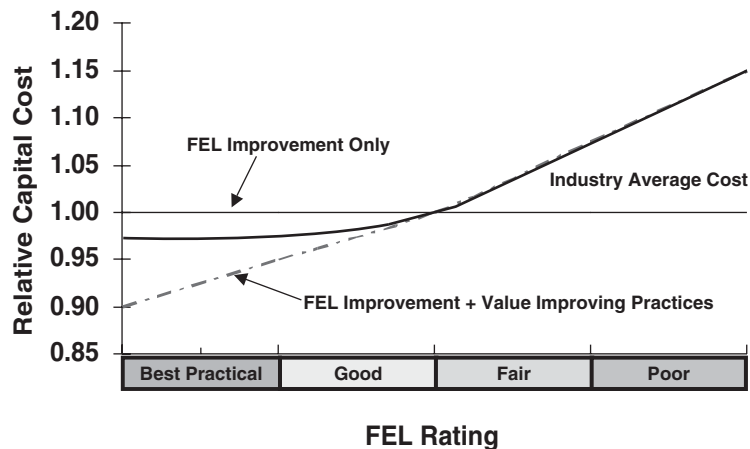


FIG. 9-33 FEL and VIPs drive lower capital investment.







unique duration for FEL with each phase dictated by the owner organization. The best times to conduct the VIP workshops should be considered at the project outset. Key anchor points for VIPs are the first appearance of the process flow diagrams (PFDs). The second anchor point for VIPs is the issue of preliminary piping and instrumentation diagrams (P&IDs).

When capital projects are benchmarked by third-party organizations such as IPA or through CII's Project Definition Rating Index (PDRI: Project Definition Rating Index—Industrial Projects, Construction Industry Institute), the implementation of applicable VIPs to the project is a part of their analysis. All the VIPs, when properly implemented, focus on producing better project definition and resultant economic improvements. The level of project definition achieved during FEL phases of the project is the focus of such benchmarking efforts.

VIPs as a group of practices are often described by their characteristics:

- Out-of-the-ordinary practices are used to improve cost, schedule, and/or reliability of capital projects.
  - They are used primarily during FEL project phases.
  - Formal and documented practices involve repeatable work processes.
  - All involve a formal facilitated workshop to confirm the value gained to the project and to formally approve VIP team recommendations.
  - All involve stated explicit support from the owner's corporate executive team.
  - VIPs must be performed by a trained experienced VIP facilitator—someone who is not a member of the project team
  - VIPs are characterized by statistical links between the use of the practice and better project performance which are demonstrated, systematic, repeatable, and proven correlations
- VIPs are also further clarified by what they are not, as below.
- Just "good engineering"
  - Simple brainstorming or strategy sessions
  - Business as usual
  - A special look at some aspect of the project
  - Cost reduction or scope reduction exercises
  - PFD or P&ID reviews
  - Safety reviews
  - Audits
  - Project readiness reviews

#### VIP Descriptions

**Classes of Facility Quality VIP** The class of facility quality VIP determines the appropriate classes of facility quality that would produce the highest value or profitability in terms of

- Capital investment (CAPEX)
- Planned facility life
- Expandability
- Level of automation
- Equipment selection,
- Operating expense (OPEX)
- Environmental controls
- Capacity
- Technology

This VIP individually confirms the best overall design philosophy for the project team, for each of the parameters listed above. Here, the designer first learns how aggressive the owner organization wants the facility design and operation to be in terms of overall risk. For example, if the plant is to have the lowest possible OPEX, then the designer will incorporate greater levels of automation, instrumentation, and robustness of mechanical design in the overall facility.

The results of this VIP are used by the project management team to update its project execution plan for each FEL phase. The class of facility quality VIP provides the best results when conducted prior to executing any other VIP effort in the conceptual phase (FEL-1) of the project.

**Technology Selection VIP** The technology selection VIP is the application of evaluation criteria aligned with the project's business objectives to identify manufacturing and processing technology that may be superior to that currently used. The goal is to ensure that the technology suite finally selected is the most competitive available. This requires a systematic search, both inside and outside the operating company's organization, to identify emerging technology alternatives. This formal facilitated process is also meant to ensure due diligence for all parties involved and that all emerging and near-commercial alternative

technologies for accomplishing a particular processing function are objectively considered. This VIP is most commonly applied at the unit operation level, although it has also been successfully applied down to the major equipment level (KBR Value Improving Practices Program, 2000 through 2005). This VIP is particularly effective for combating the NIH syndrome ("not-invented-here").

The goals of this VIP are to document which technology evaluation criteria are applicable and then to conduct a formal technology screening and evaluation assessment. The result is a prioritized listing of technology options for each selected application for the project. The preferred time to execute this VIP is the midpoint in the conceptual phase (FEL-1).

**Process Simplification VIP** The process simplification VIP uses the value methodology and is a formal, rigorous process to search for opportunities to eliminate or combine process and utility system steps or equipment, ultimately resulting in the reduction of investment and operating costs. The focus is the reduction of installed costs and critical path schedule while balancing these value improvements with expected facility operability, flexibility, and overall life cycle costs.

The process simplification VIP does far more than just evaluate and simplify processing steps. This very productive VIP ensures that low- or zero-value functions or equipment included in the project scope are challenged by experienced world-class experts and eliminated, if possible. This VIP tries to systematically differentiate "wants" from "needs" and remove the "wants." It can be especially effective for providing a neutral professional environment for identifying and challenging "sacred cows" and then removing them. Removal of these low- or zero-value functions yields significant profitability improvements to the overall project. Process simplification results in

- Reduced capital costs (CAPEX)
- Improved critical path schedule
- Reduced process inventory
- Increased yields
- Reduced operating and maintenance costs (OPEX)
- Increased productivity
- Incremental capacity gains
- Reduced utility and support systems requirements
- Reduced waste generation

Process simplification is executed in a formal workshop with a trained experienced facilitator. This VIP should always include key participants from each of the project owner's organizations, the engineering contractor organization, key third-party technology licensors, and equipment or systems vendors, where possible. One or more "cold eyes" reviewers or subject matter experts, who have extensive experience, should be included to provide an objective and unbiased perspective.

This VIP also provides a means for integrating overall plantwide systems. The process simplification VIP is typically performed during the feasibility phase (FEL-2) after the preliminary PFDs and heat and material balances become available. However, for very large and complex projects, considerable value has been gained by also performing this VIP at the midpoint or later in the conceptual phase (FEL-1).

**Constructability VIP** The constructability VIP is the systematic implementation of the latest engineering, procurement and construction concepts and lessons learned, which are consistent with the facility's operations and maintenance requirements. The goal is to enhance construction safety, scope, cost, schedule, and quality.

Since the constructability VIP has seen widespread implementation in industry for capital projects over the last 20 years, in order for this VIP to remain consistent with the definition of a VIP (i.e., above what project teams can do on their own), at least one large engineering and construction company has enhanced this VIP to include a formal facilitated workshop that seeks profitability improvements above those already identified by the project team in the course of its normal work (KBR Value Improving Practices Program, 2001 through 2005). Both work processes described below are mutually additive, flexible, and compatible.

The traditional constructability work process includes the following characteristics:

- Starts at the FEL-1 phase and continues through facility start-up
- Is an ongoing structured program

- Optimizes the combined use of operations, maintenance, engineering, procurement, key vendors, and construction knowledge and experience
  - Enhances the achievement of project objectives
  - Has construction experts working with the engineering and procurement process that results in construction safety, cost, schedule, and quality savings
  - Uses on-project and off-project expertise
- The enhanced constructability VIP adds the following to the traditional approach:
- Includes a formal facilitated workshop
  - Is held in every engineering phase of the project with a focus on the pertinent aspects of that phase
  - Identifies value improvements and their benefits above those already being considered by the traditional constructability work process
  - Focuses on the systematic implementation of the latest engineering, procurement, construction concepts, and lessons learned
  - Involves a detail review of planning, design, procurement, fabrication, and installation functions to achieve the best overall project safety performance, lowest CAPEX, and the shortest reasonable schedule
  - Applies operations and maintenance requirements and expertise
  - Includes considerations for operability and maintainability
  - Enhances construction safety, scope, cost, schedule, and quality

A formal constructability VIP workshop conducted in the conceptual phase (FEL-1) should focus on the overall project construction strategies regarding site layout, construction and turnaround laydown areas, access to the site for large equipment and modules, modularization, sequencing of heavy lifts, limitations regarding procurement, limitations regarding fabrication and transport, area labor limitations, and coordination with any existing or nearby structures or facilities.

A formal constructability VIP workshop conducted in the feasibility phase (FEL-2) should focus on more specific topics of layout optimization, using a preliminary plot plan and equipment layout for the project. Considerations should include optimum site layout in terms of construction laydown areas; optimum equipment arrangement to reduce piping and steel for structures and piperack; specific sizes and weights for modules; which components will be included in each module; crane locations for heavy lifts; equipment requiring early purchase to allow project schedule to be achieved; further analysis of limitations regarding procurement, fabrication, and area labor availability; and precommissioning, commissioning, and start-up considerations.

A formal constructability VIP workshop conducted in the definition phase (FEL-3) focuses on even greater detail for what was discussed above. In the detailed engineering stage (EPC phase), considerable detail will be reviewed to evaluate how the project can best construct what will be needed. Here, significant application of detailed lessons learned is reviewed and considered.

Constructability VIP workshops should be formal facilitated workshops drawing on personnel from operations, maintenance, and construction in addition to project and owner organization representation.

**Customization of Standards and Specifications VIP** The customization of standards and specifications VIP is a direct and systematic method to improve project value by selecting the most appropriate codes, standards, and specifications for the project. The goal is to make helpful changes to meet the actual project requirements, ensuring that the codes, standards, and specifications selected do not exceed those required for the project, and maximizing the use of specifications from equipment vendors to obtain the best overall value. This VIP is beyond typical good engineering practices and should not be confused with ongoing systematic improvements in corporate standards and specifications, or with required identification of applicable procurement specifications to be used for the project. This formal VIP takes a combination of project owner and engineering contractor corporate specifications and aggressively seeks profitability improvements consistent with the project's goals and limitations. This VIP maximizes the procurement of off-the-shelf equipment over equipment customized for the project.

Industry experience indicates that project-specific "Fit for Purpose" standards and specifications on the average cost less than the general application of traditional standards. This VIP is best performed early in the feasibility phase (FEL-2), and should include project team members involved from both the project owner and

engineering contractor, as well as appropriate suppliers of major packaged subsystems, modularized equipment, etc.

**Energy Optimization VIP** The energy optimization VIP is the systematic process for the evaluation of the thermal efficiency of a process (or multiple subunits within a larger process or facility). The goal is to improve the economic utilization of energy. This optimization starts by using the "pinch" technology branch of process energy integration (energy pinch) to identify better process energy exchange options.

Energy pinch (usually just called *pinch*) is a methodology for the conceptual design of process heating, utility, and power systems. Pinch allows the maximization of energy utilization within a process, while minimizing the use of plant utilities. Such minimization is achieved by reusing energy, via process stream-to-process stream heat exchange. A pinch analysis is performed by analyzing the trade-off between the energy which can be recovered and the additional capital costs which must be added to do so. It includes the project's design and thermodynamic constraints (performance targets) available during this preliminary design phase.

The benefits of pinch technology include lower operating costs, occasionally reduced capital cost, improved operability/flexibility, increased throughput, and site-specific process optimization and reduced emissions. Pinch technology can be applied for both grass-roots and retrofit applications. Typical applications include process heat integration as well as sitewide heat and power integration. However, this methodology is profitably applied to the optimization of high-value complex mass flow problems, such as refinery hydrogen network optimization (hydrogen pinch) and wastewater minimization (water pinch).

Once the minimum theoretical energy requirements and applicable process options have been determined, a formal facilitated workshop follows to modify the process or facility to bring the design closer to the thermodynamic optimum within project economic constraints.

The energy optimization VIP is most beneficial for processes where energy and related capital expense are a relatively large fraction of the total operating cost. The benefits result in reduced energy requirements and environmental emissions in balance with project economics. This VIP should be implemented in the feasibility phase (FEL-2) when preliminary PFDs and heat and material balances are available.

**Predictive Maintenance VIP** The predictive maintenance VIP is the proactive use of sensors and associated controls to monitor the machinery mechanical "health," using both current state and historical trends, to optimize effective planning of all shutdowns and maintenance, thereby detecting equipment abnormalities and diagnosing potential problems before they cause permanent equipment damage. Examples include real-time corrosion monitoring and equipment vibration monitoring. This additional instrumentation is generally economically justified in the case of critical equipment items and key operations. Predictive maintenance reduces maintenance costs, improves the confidence of extending time between turnarounds, improves reliability, and provides a more predictable maintenance schedule for key process equipment. It also minimizes the amount of remaining equipment life that is lost through using only preventive maintenance practices. Preventive maintenance is an older practice which is limited to periodic inspections and repairs to avoid unplanned equipment breakdowns.

For the predictive maintenance VIP to be effective, maintenance personnel from the project owner's organization must be involved in determining key predictive maintenance requirements. Suppliers of critical equipment items (i.e., compressors) are also important participants in this process.

The predictive maintenance VIP is considered by some operating companies and engineering contractors to have become standard practice. For those where it is not already standard practice, this VIP should be initiated in the feasibility phase (FEL-2) and concluded with a formal facilitated workshop and report of recommendations to the project management team.

**Waste Minimization VIP** The waste minimization VIP involves a formal process stream-by-stream analysis to identify ways to eliminate or reduce the generation of wastes or nonuseful streams within the chemical process itself. For those streams not eliminated or converted to salable by-products, it provides the method for managing the resulting wastes. This VIP incorporates environmental requirements into the facility design and combines life cycle environmental

benefits and positive economic returns through energy reductions, reduced end-of-pipe treatment requirements, and improved raw material yields. The waste reduction hierarchy is to

- Eliminate or minimize the generation of waste through source reduction
- Recycle by use, reuse, or reclamation those potential waste materials that cannot be eliminated or minimized
- Treat all waste that is nevertheless generated to reduce volume, toxicity, or mobility prior to storage or disposal

This VIP is considered by some engineering contractors to have become standard practice. For those where it is not standard practice, the waste minimization VIP should be executed in a formal workshop with an experienced facilitator with project owner and engineering contractor representatives always involved. A "cold eyes" reviewer with extensive experience should also be included to add a nonbiased perspective. The waste minimization VIP should be implemented at the feasibility phase (FEL-2) when preliminary PFDs and heat and material balances are available.

**Process Reliability Simulation VIP** The process reliability simulation VIP is the use of reliability, availability, and maintainability (RAM) computer simulation modeling of the process and the mechanical reliability of the facility. A principal goal is to optimize the engineering design in terms of life cycle cost, thereby maximizing the project's potential profitability. The objective is to determine the optimum relationships between maximum production rates and design and operational factors. Process reliability simulation is also applied for safety purposes, since it considers the consequences of specific equipment failures and failure modes.

This VIP is typically led by an engineer experienced in plant operations and the use of the RAM simulation modeling software. The VIP should also directly involve the project owner since that organization would most often supply the historical operating and maintenance information required for the development of the simulation model. This process provides the project team with a more effective means of assessing, early in the design, the cost/benefit impact of changes in design, identification of bottlenecks in the system, simulation of key operating scenarios, determination of equipment-sparing needs, training and maintenance requirements of a facility.

The process reliability simulation VIP should be initiated in the feasibility phase (FEL-2) to produce a block-level RAM model. Based on the results of that model, a more detailed equipment-level RAM model should be developed starting in the definition phase (FEL-3).

**Value Engineering VIP** The value engineering VIP is a flexible, organized, multidisciplinary team effort directed at analyzing the functions, issues, and essential characteristics of a project, process, technology, or system. The goal is to satisfy those functions, issues, and essential characteristics at the lowest life cycle cost. The value engineering VIP rigorously examines what is needed to meet the business objectives of a project and the elimination of non-value-adding investment. An open-minded attitude by participants is required to effectively remove unneeded scope and in doing so reduce the installed costs of the project. This VIP tries to systematically differentiate "wants" from "needs" and remove the "wants." Tests for non-income-producing investments include redundancy, overdesign, manufacturing add-ons, upgraded materials of construction, and customized design versus vendor standards.

The value engineering VIP also ensures that low- or zero-value functions or equipment included in the project scope are challenged to be the highest value possible for the project. Removal of these low- or zero-value functions from the project scope, if possible, will most likely yield significant profitability improvements to the overall project. These can encompass the following:

- Misalignment of unit or system capacity or operations capability with respect to the overall facility
- Overly conservative assumptions of the basic design data
- Overly conservative interpretation of how the facilities will be used during peak, seasonal, or upset conditions
- Preinvestment included in the project scope that may not be value added
- Overdesign of equipment or systems to provide uneconomic added flexibility

The value engineering VIP is executed in a formal workshop with a trained experienced technical workshop facilitator. Both the project

owner and the engineering contractor are always involved. Third-party licensors and equipment/system vendors should be included where applicable. One (or more) cold-eyes reviewer with extensive experience should also be included to provide an unbiased perspective.

This VIP leverages the growing accumulation of more detailed project knowledge to test the value of earlier, more generalized scope assumptions. It also tests the presumed added value of different stakeholder requirements, which have influenced the evolution of the project scope. This highly adaptable VIP results in reduced capital costs (CAPEX), improved critical path schedule, reduced process inventory, increased yields, reduced operating and maintenance costs (OPEX), increased productivity, incremental capacity gains, reduced utility and support systems requirements, and reduced waste generation.

The value engineering VIP should be conducted in the definition phase (FEL-3) when the first issue of P&IDs is available.

**Design to Capacity VIP** The design to capacity VIP systematically evaluates the maximum capacity of major equipment, ancillary piping, valving, instrumentation, and associated engineering calculations and guidelines. The goal is to improve life cycle costs (profitability) by eliminating preinvestment and overdesign. This VIP requires the systematic and formal evaluation of the maximum capacity of each piece of equipment instead of the traditional practice of designing with an extra safety factor or margin to allow for additional catch-up capacity or some future production increase. The goal is also to eliminate overdesign in both calculations and engineering guidelines. This VIP is conducted as a facilitated workshop with both project owner and engineering contractor representation.

This VIP reduces capital investment by confirming minimum required capacities and flexibility necessary only to meet current project business objectives. The workshop drills down to each specific system and subsystem and finally scrutinizes the design of each equipment item. This workshop is often combined with the value engineering VIP, which overlaps significantly. The design to capacity VIP should be conducted in the definition phase (FEL-3) when the first issue of P&IDs is available.

**3D-CAD VIP** The 3D-CAD VIP is the creation of a detailed three-dimensional (3D) computer model depicting the proposed process and associated equipment along with the optimized plant layout and specific equipment arrangements and orientations. The 3D model can then be used to generate computerized interference checks of bulk material configurations and equipment and extraction of error-free fabrication drawings and material quantities. The goals of this VIP are to reduce engineering and construction rework, improve operability and maintainability, and confirm the incorporation into the design of advantageous human factors (a.k.a. ergonomics) focused on ease of operation and maintenance.

A number of industry-accepted state-of-the-art 3D computer-aided design (CAD) systems have been used for this purpose. The specification-driven 3D-CAD system allows a computer model to be built to allow extraction of drawings from the model for fabrication and erection. The extracted drawings are enhanced in their accuracy by the computer interference detection system which greatly reduces field rework.

The principal benefit of utilizing 3D-CAD is the ability to produce an electronic model that accurately resembles the completed facility. This enables project teams, clients, and constructors to review and agree on the plant design before construction starts. The model can then be used to generate interference checks of bulk material configurations and equipment, as well as the extraction of error-free fabrication drawings and material quantities. The system also utilizes its design review capabilities to confirm proposed designs and obtain approvals from key project stakeholders and owners.

This VIP is considered by most major engineering contractors to have become standard practice. The 3D-CAD VIP model development should be initiated in the feasibility phase (FEL-2) after the plot plan has been finalized and the first issue of P&IDs is available.

**VIP Planning and Implementation** Each VIP has a unique character, and it should be performed at a certain time and in a certain way to produce the best results for the project. Part of the power of VIPs is that they can be used to improve the overall economics of the project without the need for inordinate additional time or expense. Ironically, the return on investment (ROI) for the cost of implementing each VIP is usually much greater than that ROI for the overall proposed project. For one engineering contractor, *the typical ROI for*

implementing VIPs is at least one order of magnitude higher (KBR Value Improving Practices Program, 2000 through 2005).

It is important to reiterate that the benefits of VIPs cannot be realized by just executing "good engineering." The application and implementation of VIPs to any project must have the explicit commitment of the owner's corporation executives. VIP execution must be deliberately and carefully planned in the initial phase of the project. For all projects, this VIP planning meeting should take place immediately following project kickoff. The project management team and the selected VIP facilitator should

- Confirm which VIPs should be applied to the project and when
- Incorporate the planned VIPs into the project scope of work and schedule
- Determine the required workshop resources and best combination of engineering, operations, maintenance, construction, and other expertise for each selected VIP workshop team

**VIPs That Apply the Value Methodology** Nearly all VIPs are conducted only once in a project at a "sweet spot" where maximum benefit is found. For example, the process simplification VIP is anchored at the first appearance of the preliminary PFDs, while the value engineering VIP and the design to capacity VIP are anchored at the first issue of the P&IDs. Both of these apply the value methodology [Society of American Value Engineers International (SAVE)] that has produced excellent results in industry for more than 55 years. The typical approach and steps for these three unique VIPs are presented below.

**Preparation and Planning** Before the VIP is begun, the goals, objectives, and scheduled time for the formal workshop must be agreed upon by the integrated project management team. The workshop facilitator must ensure that all the information required for the workshop is available and that the workshop team members have been fully briefed on the VIP's objectives, methodology, and expectations.

**The Formal Workshop** The formal workshop is always structured to make maximum use of the multi-disciplinary team's time and effort. Such workshops typically require no less than 2 days and as many as 5 days depending on the size and complexity of the project. The required workshop length should be determined by the VIP facilitator. A typical process simplification VIP, value engineering VIP, and design to capacity VIP workshop includes the following phases of a typical "job plan" that are supported by the Society of American Value Engineers International.

**The information phase** In this phase, team members review important background materials and confirm their understanding of the basis for the design of the project, the constraints, and the sensitivity of the relevant capital and operating costs. Here, incorporating important unresolved project issues into the workshop produces more meaningful financial and technical results. Discussion of the issues' validity and basis are determined during the first day of the workshop. These issues very often become some of the best brainstorming targets for cost and schedule reduction ideas.

A very specific and structured methodology is used which is known as the *Function Analysis System Technique (FAST)*. This function analysis diagramming illustrates the logical or functional relationships and dependencies between different process systems and project activities. These diagrams are then reviewed and critiqued together with the associated costs of selected groups of process functions or project functions. The function analysis can be performed at this stage, but often time can be saved by preparing a draft of these FAST diagrams prior to the workshop with a small group of the workshop team members.

**The speculation phase** Once the pertinent information and issues have been reviewed and the important functions of each process and project step identified, the team is encouraged to speculate on alternative methods to perform each function and to solve each major project issue. Brainstorming sessions within a creative environment encourage the team to strive for new and innovative ideas.

**The conceptual phase** The team then reviews the ideas against relevant project criteria such as potential impact on long-term project economics, impact on operations and maintenance costs, effect on the capital cost for the project, validity to the project scope of work, technical risks associated with implementation of the new concept, impact on project schedule, and cost required to implement the improvement. Each study has specific criteria against which proposed alterna-

tives are judged. The ideas are weighted, sorted, grouped, linked, and ranked so that the best of the technically viable ideas are efficiently identified for further detailed study.

**The feasibility phase** The ideas with the most merit are developed into preliminary two-page written proposals with potential benefits approximated as part of the workshop. Performing this important activity following the formal workshop has been shown to often result in significant loss of potential for implementation. The VIP team expands the ideas ranked highest to obtain additional technical and economic insights and information to support the idea. The proposals are then presented internally, to the assembled VIP workshop team, and discussed to determine whether the ideas retain sufficient technical and economic merit to be recommended by the VIP team to a separate steering committee or project management team.

Experience indicates that having the VIP team perform this stage within the formal workshop produces the best results.

**The presentation phase** The VIP team formally presents the profitability recommendations consistent with the objectives and constraints of the workshop and their implementation plans to the steering committee or the project management team. The steering committee then approves those recommendations that pass muster and authorizes the project team to begin the implementation effort. Often, this approval is conditional on early validation by subject matter experts within the project owner's organization, but not present within the workshop. This external feasibility check is meant to provide support to the project team for any additional resources and schedule time needed to fully incorporate the improvements into the project scope of work.

**Report and follow-up** After completing the intensive VIP workshop, the workshop facilitator completes the written VIP final report for the project record. During this time, the project management team assigns each approved recommendation to a member of the project team, estimates the engineering time and resources required to incorporate the improvement into the project scope of work, and communicates the results of the VIP within the integrated project team. This follow-up action plan creates a very positive and cost-conscious attitude within the project team that leads to further improvements in project value.

**Sources of Expertise** VIP workshops should be planned and led by a trained experienced facilitator who has significant experience in effectively conducting such VIP workshops. Technical expertise for VIP workshops should be a combination of senior project team members and subject matter experts from the operating company's organization, the engineering contractor's organization, licensed technology providers, and any key fabrication or installation subcontractors to be used. Figure 9-35 illustrates the best balance of expertise for VIP workshops (KBR Value Improving Practices Program, 1995 through 2005).

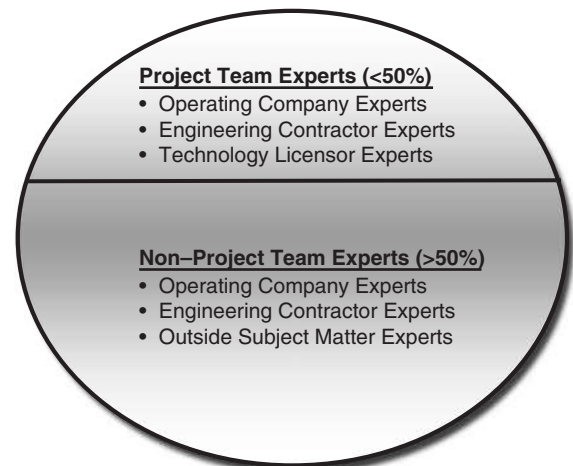


FIG. 9-35 The ideal VIP team makeup.

## GLOSSARY

**Accounts payable** The value of purchased goods and services that are being used but have not been paid.

**Accounts receivable** Credit extended to customers, usually on a 30-day basis. Cash is set aside to take care of the probability that some customers may not pay their bills.

**Accrual basis** The accounting method that recognizes revenues and disbursement of funds by receipt of bills or orders and not by cash flow, distinguished from *cash basis*.

**Administrative expense** An overhead expense due to the general direction of a company beyond the plant level. It includes administrative and office salaries, rent, auditing, accounting, legal, central purchasing and engineering, etc., expenses.

**Allocation of expenses** A procedure whereby overhead expenses and other indirect charges are assigned back to processing units or to products on what is expected to be an equitable basis. All allocations are somewhat arbitrary.

**Amortization** Often used interchangeably with *depreciation*, but there is a slight difference depending on whether the life of an asset is known. If the period of time is known to be usually more than a year, this annual expense is *amortization*; however, if the life is estimated, then it is *depreciation*.

**Annual net sales** Pounds of product sold times the net selling price. *Net* means that any allowances have been subtracted from the gross selling price.

**Annual report** Management's report to the stockholders and other interested parties at the end of a year of operation showing the status of the company, its activities, funds, income, profits, expenses, and other information.

**Appurtenances** The auxiliaries to either process or nonprocess equipment: piping, electrical, insulation, instrumentation, etc.

**Assets** The list of money on hand, marketable securities, monies due, investments, plants, properties, intellectual property, inventory, etc., at cost or market value, whichever is smaller. The assets are what a company (or person) owns.

**Balance sheet** This is an accounting, historical tabulation of assets, liabilities, and stockholders' equity for a company. The assets must equal the liabilities plus the stockholders' equity.

**Battery limit** A geographic boundary defining the coverage of a specific project. Usually it takes in the manufacturing area of a proposed plant, including all process equipment but excluding provision for storage, site preparation, utilities, administrative buildings, or auxiliary facilities.

**Bonds** When one purchases a bond, the company (or person) acquires an interest in debt and becomes a creditor of the company. The purchaser receives the right to receive regular interest payments and the subsequent repayment of the principal.

**Book value** Current investment value on the company books as the original installed cost less depreciation accruals.

**Book value of common stock** Net worth of a firm divided by the number of shares of common stock issued at the time of a report.

**Break-even chart** An economic production chart depicting total revenue and total expenses as functions of operation of a processing facility.

**Break-even point** The percentage of capacity at which income equals all fixed and variable expenses at that level of operation.

**By-product** A product made as a consequence of the production of a main product. The by-product may have a market value or a value as a raw material.

**Capacity** The estimated maximum level of production on a sustained basis.

**Capital ratio** Ratio of capital investment to sales dollars; the reciprocal of capital turnover.

**Capital recovery** The process by which original investment in a project is recovered over its life.

**Capital turnover** The ratio of sales dollars to capital investment; the reciprocal of capital ratio.

**Cash** Money that is on hand to pay for operating expenses, e.g., wages, salaries, raw materials, supplies, etc., to maintain a liquid financial position.

**Cash basis** The accounting basis whereby revenue and expense are recorded when cash is received and paid, distinguished from accrual basis.

**Cash flow** Net income after taxes plus depreciation (and depletion) flowing into the company treasury.

**Code of accounts** A system in which items of expense or fixed capital such as equipment and material are identified with numerical figures to facilitate accounting and cost control.

**Common stock** Money paid into a corporation for the purchase of shares of common stock that becomes the permanent capital of the firm. Common stockholders have the right to transfer ownership and may sell the stock to individuals or firms. Common stockholders have the right to vote at annual meetings on company business or may do so by proxy.

**Compound interest** The interest charges under the condition that interest is charged on previous interest plus principal.

**Contingencies** An allowance for unforeseeable elements of cost in fixed investment estimates that previous experience has shown to exist.

**Continuous compounding** A mathematical procedure for evaluating compound interest based upon continuous interest function rather than discrete interest periods.

**Conversion expense** The expense of converting raw materials to finished product.

**Corporation** In 1819, defined by Chief Justice Marshall of the Supreme Court as "an artificial being, invisible, intangible and existing only in contemplation of law." It exists by the grace of a state, and the laws of a state govern the procedure for its formation.

**Cost of capital** The cost of borrowing money from all sources, namely, loans, bonds, and preferred and common stock. It is expressed as an interest rate.

**Cost center** For accounting purposes, a grouping of equipment and facilities comprising a product manufacturing system.

**Cost of sales** The sum of the fixed and variable (direct and indirect) expenses for manufacturing a product and delivering it to a customer.

**Decision or decision making** A program of action undertaken as a result of (1) an established policy or (2) an analysis of variables that can be altered to influence a final result.

**Depletion** A provision in the tax regulations that allows a business to charge as current expense a noncash expense representing the portion of limited natural resources consumed in the conduct of business.

**Depreciation** A reasonable allowance by the Internal Revenue Service for the exhaustion, wear and tear, and normal obsolescence of equipment used in a trade or business. The property must have a useful life of more than 1 year. Depreciation is a noncash expense deductible from income for tax purposes.

**Design to cost** A management technique to achieve system designs that meet cost parameters. Cost as a design parameter is considered on a continuous basis as part of a system's development and production processes.

**Direct expense** An expense directly associated with the production of a product such as utilities, labor, and maintenance.

**Direct labor expense** The expense of labor involved in the manufacture of a product or in the production of a service.

**Direct material expense** The expense associated with materials consumed in the manufacture of a product or the production of a service.

**Distribution expense** Expense including advertising, preparation of samples, travel, entertainment, freight, warehousing, etc., to distribute a sample or product.

**Dollar volume** Dollar worth of a product manufactured per unit of time.

**Earnings** The difference between income and operating expenses.

**Economic life** The period of commercial use of a product or facility. It may be limited by obsolescence, physical life of equipment, or changing economic conditions.

**Economic value added** The period dollar profit above the cost of capital. It is a means to measure an organization's value and a way to determine how management's decisions contribute to the value of a company.

**Effective interest** The true value of interest computed by equations for the compound interest rate for a period of 1 year.

**Equity** The owner's actual capital held by a company for its operations.

**Escalation** A provision in actual or estimated cost for an increase in equipment cost, material, labor, expenses, etc., over those specified in an original estimate or contract due to inflation.

**External funds** Capital obtained by selling stocks or bonds or by borrowing.

**FEL (front-end loading)** The process by which a company develops a detailed definition of the scope of a capital project that meets corporate business objectives.

**FIFO (first in, first out)** The valuation of raw material and supplies inventory, meaning first into the company or process is the first used or out.

**Financial expense** The charges for use of borrowed funds.

**Fixed assets** The real or material facilities that represent part of the capital in an economic venture.

**Fixed capital** Item including the equipment and buildings.

**Fixed expense** An expense that is independent of the rate of output, e.g., depreciation and plant indirect expenses.

**Fringe benefits** Employee welfare benefits; expenses of employment over and above compensation for actual time worked, such as holidays, vacations, sick leave, insurance.

**Full cost accounting** Method of pricing goods and services to reflect their true costs, including production, use recycling, and disposal.

**Future worth** The expected value of capital in the future according to some predetermined method of computation.

**Goods manufactured, cost of** Total expense (direct and indirect expenses) including overhead charges.

**Goods-in-process inventory** The holdup of product in a partially finished state.

**Goods sold, cost of** The total of all expenses before income taxes that is deducted from income (revenue).

**Grass-roots plant** A complete plant erected on new site including land, site preparation, battery-limits facilities, and auxiliary facilities.

**Gross domestic product** An indicator of a country's economic activity. It is the sum of all goods and services produced by a nation within its borders.

**Gross margin (profit)** Total revenue minus cost of goods manufactured.

**Gross national product** An economic indicator of a country's economic activity. It is the sum of all the goods and services produced by a nation both within and outside its borders.

**Income** Profit before income taxes or gross income from sales before deduction of expenses.

**Income statement** The statement of earnings of a firm as approximated by accounting practices, usually covering a 1-year period.

**Income tax** The tax imposed on corporate profits by the federal and/or state governments.

**Indirect expenses** Part of the manufacturing expense of a product not directly related to the amount of product manufactured, e.g., depreciation, local taxes, and insurance.

**Internal funds** Capital available from depreciation and accumulated retained earnings.

**Inventory** The quantity of raw materials and/or supplies held in a process or in storage.

**Last in, first out (LIFO)** The valuation of raw materials and supplies, meaning the last material into a process or storage is the first used or out.

**Leverage** The influence of debt on the earning rate of a company.

**Liabilities** An accounting term for capital owed by a company.

**Life cycle cost** Cost of development, acquisition, support, and disposal of a system over its full life.

**Manufacturing expense** The sum of the raw material, labor, utilities, maintenance, depreciation, local taxes, etc., expenses. It is the sum of the direct and indirect (fixed and variable) manufacturing expenses.

**Marginal cost** The incremental cost of making one additional unit without additional investment in facilities.

**Market capitalization** The product of the number of shares of common stock outstanding and the share price.

**Market value added** A certain future economic value added for a company. It is the present value of the future economic value (EVA) generated by a company. It is a measure of how much value a firm has created.

**Minimum acceptable rate of return (MARR)** The level of return on investment, at or above the cost of capital, chosen as acceptable for discounting or cutoff purposes.

**Net sales price** Gross sales price minus freight adjustments.

**Net worth** The sum of the stockholders' investment plus surplus, or total assets minus total liabilities.

**Nominal interest** The number applied loosely to describe the annual interest rate.

**Obsolescence** The occurrence of decreasing value of physical assets due to technological changes rather than physical deterioration.

**Operating expense** The sum of the manufacturing expense for a product and the general, administrative, and selling expenses.

**Operating margin** The gross margin minus the general, administrative, and selling expenses.

**Opportunity cost** The estimate of values that are forgone by undertaking one alternative instead of another one.

**Payout time (payback period)** The time to recover the fixed capital investment from profit plus depreciation. It is usually after taxes but not always.

**Preferred stock** Stock having claims that it commands over common stock, with the preference related to dividends. The holders of such stock receive dividends before any distribution is made to common stockholders. Preferred stockholders usually do not have voting rights as common stockholders do.

**Present worth** The value at some datum time (present time) of expenditures, costs, profits, etc., according to a predetermined method of computation. It is the current value of cash flow obtained by discounting.

**Production rate** The amount of product manufactured in a given time period.

**Profitability** A term generally applied in a broad sense to the economic feasibility of a proposed venture or an ongoing operation. It is generally considered to be related to return on investment.

**Rate of return on investment** The efficiency ratio relating profit or cash flow to investment.

**Replacement** A new facility that takes the place of an older facility with no increase in capacity.

**Revenue** The net sales received from the sale of a product or a service to a customer.

**Sales, administration, research, and engineering expenses (SARE)** Overhead expenses incurred as a result of maintaining sales offices and administrative offices and the expense of maintaining research and engineering departments. This item is usually expressed as a percentage of annual net sales.

**Sales volume** The amount of sales expressed in pounds, gallons, tons, cubic feet, etc., per unit of time.

**Salvage value** The value that can be realized from equipment or other facilities when taken out of service and sold.

**Selling expense** Salaries and commissions paid to sales personnel.

**Simple interest** The interest charges in any time period that is only charged on the principal.

**Sinking fund** An accounting procedure computed according to a specified procedure to provide capital to replace an asset.

**Surplus** The excess of earnings over expenses that is not distributed to stockholders.

**Tax credit** The amount available to a firm as part of its annual return because of deductible expenses for tax purposes. Examples have been research and development expenses, energy tax credit, etc.

**Taxes** In a manufacturing cost statement, usually property taxes. In an income statement, usually federal and state income taxes.

**Time value of money** The expected interest rate that capital should or would earn. Money has value with respect to time.

**Total operating investment** The fixed capital investment, backup capital, auxiliary capital, utilities and services capital, and working capital.

**Utilities and services capital** Electrical substations, plant sewers, water distribution facilities, and occasionally the steam plant.

**Value added** The difference between the raw material expense and the selling price of that product.

**Value-improving practices (VIPs)** Formal structured practices applied to capital projects to improve profitability ("or value") above

that which is attained through the application of proven good engineering and project management practices.

**Variable expense** Any expense that varies directly with production output.

**Working capital** In the accounting sense, the current assets minus the current liabilities. It consists of the total amount of money invested in raw materials, supplies, goods in process, product inventories, accounts receivable, and cash minus those liabilities due within 1 year.



SECTION 10

Transport and  
Storage of  
Fluids

# PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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# Transport and Storage of Fluids

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## 10-4 TRANSPORT AND STORAGE OF FLUIDS

### Nomenclature and Units

In this listing, symbols used in the section are defined in a general way and appropriate SI and U.S. customary units are given. Specific definitions, as denoted by subscripts, are stated at the place of application in the section. Some specialized symbols used in the section are defined only at the place of application.

| Symbol     | Definition                                                                                 | SI units                       | U.S. customary units               | Symbol     | Definition                                                                                       | SI units              | U.S. customary units                     |
|------------|--------------------------------------------------------------------------------------------|--------------------------------|------------------------------------|------------|--------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------|
| $A$        | Area                                                                                       | $m^2$                          | $ft^2$                             | $K$        | Fluid bulk modulus of elasticity                                                                 | $N/m^2$               | $lbf/ft^2$                               |
| $A$        | Factor for determining minimum value of $R_1$                                              |                                |                                    | $K_1$      | Constant in empirical flexibility equation                                                       |                       |                                          |
| $A_\infty$ | Free-stream speed of sound                                                                 |                                |                                    | $k$        | Ratio of specific heats                                                                          | Dimensionless         | Dimensionless                            |
| $a$        | Area                                                                                       | $m^2$                          | $ft^2$                             | $k$        | Flexibility factor                                                                               |                       |                                          |
| $a$        | Duct or channel width                                                                      | $m$                            | $ft$                               | $k$        | Adiabatic exponent $c_p/c_v$                                                                     |                       |                                          |
| $a$        | Coefficient, general                                                                       |                                |                                    | $L$        | Length                                                                                           | $m$                   | $ft$                                     |
| $B$        | Height                                                                                     | $m$                            | $ft$                               | $L$        | Developed length of piping between anchors                                                       | $m$                   | $ft$                                     |
| $b$        | Duct or channel height                                                                     | $m$                            | $ft$                               | $L$        | Dish radius                                                                                      | $m$                   | $in$                                     |
| $b$        | Coefficient, general                                                                       |                                |                                    | $M$        | Molecular weight                                                                                 | $kg/mol$              | $lb/mol$                                 |
| $C$        | Coefficient, general                                                                       |                                |                                    | $M_b, m_i$ | In-plane bending moment                                                                          | $N\cdot mm$           | $in\cdot lbf$                            |
| $C$        | Conductance                                                                                | $m^3/s$                        | $ft^3/s$                           | $M_o$      | Out-plane bending moment                                                                         | $N\cdot mm$           | $in\cdot lbf$                            |
| $C$        | Sum of mechanical allowances (thread or groove depth) plus corrosion or erosion allowances | $mm$                           | $in$                               | $M_t$      | Torsional moment                                                                                 | $N\cdot mm$           | $in\cdot lbf$                            |
| $C$        | Cold-spring factor                                                                         |                                |                                    | $M_\infty$ | Free stream Mach number                                                                          |                       |                                          |
| $C$        | Constant                                                                                   |                                |                                    | $m$        | Mass                                                                                             | $kg$                  | $lb$                                     |
| $C_a$      | Capillary number                                                                           | Dimensionless                  | Dimensionless                      | $m$        | Thickness                                                                                        | $m$                   | $ft$                                     |
| $C_1$      | Estimated self-spring or relaxation factor                                                 |                                |                                    | $N$        | Number of data points or items                                                                   | Dimensionless         | Dimensionless                            |
| $c_p$      | Constant-pressure specific heat                                                            | $J/(kg\cdot K)$                | $Btu/(lb\cdot ^\circ R)$           | $N$        | Frictional resistance                                                                            | Dimensionless         | Dimensionless                            |
| $c_v$      | Constant-volume specific heat                                                              | $J/(kg\cdot K)$                | $Btu/(lb\cdot ^\circ R)$           | $N$        | Equivalent full temperature cycles                                                               |                       |                                          |
| $D$        | Diameter                                                                                   | $m$                            | $ft$                               | $N_s$      | Strouhal number                                                                                  | Dimensionless         | Dimensionless                            |
| $D, D_o$   | Outside diameter of pipe                                                                   | $mm$                           | $in$                               | $N_{De}$   | Dean number                                                                                      | Dimensionless         | Dimensionless                            |
| $d$        | Diameter                                                                                   | $m$                            | $ft$                               | $N_{Fr}$   | Froude number                                                                                    | Dimensionless         | Dimensionless                            |
| $E$        | Modulus of elasticity                                                                      | $N/m^2$                        | $lbf/ft^2$                         | $N_{Re}$   | Reynolds number                                                                                  | Dimensionless         | Dimensionless                            |
| $E$        | Quality factor                                                                             |                                |                                    | $N_{We}$   | Weber number                                                                                     | Dimensionless         | Dimensionless                            |
| $E_a$      | As-installed Young's modulus                                                               | $MPa$                          | $kip/in^2$ (ksi)                   | $NPSH$     | Net positive suction head                                                                        | $m$                   | $ft$                                     |
| $E_c$      | Casting quality factor                                                                     |                                |                                    | $n$        | Polytropic exponent                                                                              |                       |                                          |
| $E_j$      | Joint quality factor                                                                       |                                |                                    | $n$        | Pulsation frequency                                                                              | $Hz$                  | $1/s$                                    |
| $E_m$      | Minimum value of Young's modulus                                                           | $MPa$                          | $kip/in^2$ (ksi)                   | $n$        | Constant, general                                                                                |                       |                                          |
| $F$        | Force                                                                                      | $N$                            | $lbf$                              | $n$        | Number of items                                                                                  | Dimensionless         | Dimensionless                            |
| $F$        | Friction loss                                                                              | $(N\cdot m)/kg$                | $(ft\cdot lbf)/lb$                 | $P$        | Design gauge pressure                                                                            | $kPa$                 | $lbf/in^2$                               |
| $F$        | Correction factor                                                                          | Dimensionless                  | Dimensionless                      | $P_{ad}$   | Adiabatic power                                                                                  | $kW$                  | $hp$                                     |
| $f$        | Frequency                                                                                  | $Hz$                           | $1/s$                              | $p$        | Pressure                                                                                         | $Pa$                  | $lbf/ft^2$                               |
| $f$        | Friction factor                                                                            | Dimensionless                  | Dimensionless                      | $p$        | Power                                                                                            | $kW$                  | $hp$                                     |
| $f$        | Stress-range reduction factor                                                              |                                |                                    | $Q$        | Heat                                                                                             | $J$                   | $Btu$                                    |
| $G$        | Mass velocity                                                                              | $kg/(s\cdot m^2)$              | $lb/(s\cdot ft^2)$                 | $Q$        | Volume                                                                                           | $m^3$                 | $ft^3$                                   |
| $g$        | Local acceleration due to gravity                                                          | $m/s^2$                        | $ft/s^2$                           | $Q$        | Volume rate of flow (liquids)                                                                    | $m^3/h$               | $gal/min$                                |
| $g_c$      | Dimensional constant                                                                       | $1.0 (kg\cdot m)/(N\cdot s^2)$ | $32.2 (lb\cdot ft)/(lbf\cdot s^2)$ | $Q$        | Volume rate of flow (gases)                                                                      | $m^3/h$               | $ft^3/min$ (cfm)                         |
| $H$        | Depth of liquid                                                                            | $m$                            | $ft$                               | $q$        | Volume flow rate                                                                                 | $m^3/s$               | $ft^3/s$                                 |
| $H, h$     | Head of fluid, height                                                                      | $m$                            | $ft$                               | $R$        | Gas constant                                                                                     | $8314 J/(K\cdot mol)$ | $1545 (ft\cdot lbf)/(mol\cdot ^\circ R)$ |
| $H_{ad}$   | Adiabatic head                                                                             | $N\cdot m/kg$                  | $lbf\cdot ft/lbm$                  | $R$        | Radius                                                                                           | $m$                   | $ft$                                     |
| $h$        | Flexibility characteristic                                                                 |                                |                                    | $R$        | Electrical resistance                                                                            | $\Omega$              | $\Omega$                                 |
| $h$        | Height of truncated cone; depth of head                                                    | $m$                            | $in$                               | $R$        | Head reading                                                                                     | $m$                   | $ft$                                     |
| $i$        | Specific enthalpy                                                                          | $J/kg$                         | $Btu/lb$                           | $R$        | Range of reaction forces or moments in flexibility analysis                                      | $N$ or $N\cdot mm$    | $lbf$ or $in\cdot lbf$                   |
| $i$        | Stress-intensification factor                                                              |                                |                                    | $R$        | Cylinder radius                                                                                  | $m$                   | $ft$                                     |
| $i_i$      | In-plane stress-intensification factor                                                     |                                |                                    | $R$        | Universal gas constant                                                                           | $J/(kg\cdot K)$       | $(ft\cdot lbf)/(lbm\cdot ^\circ R)$      |
| $i_o$      | Out-plane stress-intensification factor                                                    |                                |                                    | $R_a$      | Estimated instantaneous reaction force or moment at installation temperature                     | $N$ or $N\cdot mm$    | $lbf$ or $in\cdot lbf$                   |
| $I$        | Electric current                                                                           | $A$                            | $A$                                | $R_m$      | Estimated instantaneous maximum reaction force or moment at maximum or minimum metal temperature | $N$ or $N\cdot mm$    | $lbf$ or $in\cdot lbf$                   |
| $J$        | Mechanical equivalent of heat                                                              | $1.0 (N\cdot m)/J$             | $778 (ft\cdot lbf)/Btu$            | $R_1$      | Effective radius of miter bend                                                                   | $mm$                  | $in$                                     |
| $K$        | Index, constant or flow parameter                                                          |                                |                                    |            |                                                                                                  |                       |                                          |

**Nomenclature and Units (Concluded)**

| Symbol      | Definition                                                                          | SI units                       | U.S. customary units             | Symbol                  | Definition                                      | SI units           | U.S. customary units |
|-------------|-------------------------------------------------------------------------------------|--------------------------------|----------------------------------|-------------------------|-------------------------------------------------|--------------------|----------------------|
| $r$         | Radius                                                                              | m                              | ft                               | $v$                     | Specific volume                                 | m <sup>3</sup> /kg | ft <sup>3</sup> /lb  |
| $r$         | Pressure ratio                                                                      | Dimensionless                  | Dimensionless                    | $W$                     | Work                                            | N·m                | lb·ft                |
| $r_c$       | Critical pressure ratio                                                             |                                |                                  | $W$                     | Weight                                          | kg                 | lb                   |
| $r_k$       | Knuckle radius                                                                      | m                              | in                               | $w$                     | Weight flow rate                                | kg/s               | lb/s                 |
| $r_2$       | Mean radius of pipe using nominal wall thickness $\bar{T}$                          | mm                             | in                               | $x$                     | Weight fraction                                 | Dimensionless      | Dimensionless        |
| $S$         | Specific surface area                                                               | m <sup>2</sup> /m <sup>3</sup> | ft <sup>2</sup> /ft <sup>3</sup> | $x$                     | Distance or length                              | m                  | ft                   |
| $S$         | Fluid head loss                                                                     | Dimensionless                  | Dimensionless                    |                         | Value of expression $[(p_2/p_1)^{(k-1/k)} - 1]$ |                    |                      |
| $S$         | Specific energy loss                                                                | m/s <sup>2</sup>               | lb/ft                            | $Y$                     | Expansion factor                                | Dimensionless      | Dimensionless        |
| $S$         | Speed                                                                               | m <sup>3</sup> /s              | ft <sup>3</sup> /s               | $y$                     | Distance or length                              | m                  | ft                   |
| $S$         | Basic allowable stress for metals, excluding factor $E$ , or bolt design stress     | MPa                            | kip/in <sup>2</sup> (ksi)        | $y$                     | Resultant of total displacement strains         | mm                 | in                   |
| $S_A$       | Allowable stress range for displacement stress                                      | MPa                            | kip/in <sup>2</sup> (ksi)        | $Z$                     | Section modulus of pipe                         | mm <sup>3</sup>    | in <sup>3</sup>      |
| $S_E$       | Computed displacement-stress range                                                  | MPa                            | kip/in <sup>2</sup> (ksi)        | $Z$                     | Vertical distance                               | m                  | ft                   |
| $S_L$       | Sum of longitudinal stresses                                                        | MPa                            | kip/in <sup>2</sup> (ksi)        | $Z_e$                   | Effective section modulus for branch            | mm <sup>3</sup>    | in <sup>3</sup>      |
| $S_T$       | Allowable stress at test temperature                                                | MPa                            | kip/in <sup>2</sup> (ksi)        | $Z$                     | Gas-compressibility factor                      | Dimensionless      | Dimensionless        |
| $S_b$       | Resultant bending stress                                                            | MPa                            | kip/in <sup>2</sup> (ksi)        | $z$                     | Vertical distance                               | m                  | ft                   |
| $S_c$       | Basic allowable stress at minimum metal temperature expected                        | MPa                            | kip/in <sup>2</sup> (ksi)        | Greek symbols           |                                                 |                    |                      |
| $S_h$       | Basic allowable stress at maximum metal temperature expected                        | MPa                            | kip/in <sup>2</sup> (ksi)        | $\alpha$                | Viscous-resistance coefficient                  | 1/m <sup>2</sup>   | 1/ft <sup>2</sup>    |
| $S_t$       | Torsional stress                                                                    | MPa                            | kip/in <sup>2</sup> (ksi)        | $\alpha$                | Angle                                           | °                  | °                    |
| $s$         | Specific gravity                                                                    |                                |                                  | $\sigma$                | Half-included angle                             | °                  | °                    |
| $s$         | Specific entropy                                                                    | J/(kg·K)                       | Btu/(lb·°R)                      | $\alpha, \beta, \theta$ | Angles                                          | °                  | °                    |
| $T$         | Temperature                                                                         | K (°C)                         | °R (°F)                          | $\beta$                 | Inertial-resistance coefficient                 | 1/m                | 1/ft                 |
| $T_s$       | Effective branch-wall thickness                                                     | mm                             | in                               | $\beta$                 | Ratio of diameters                              | Dimensionless      | Dimensionless        |
| $\bar{T}$   | Nominal wall thickness of pipe                                                      | mm                             | in                               | $\Gamma$                | Liquid loading                                  | kg/(s·m)           | lb/(s·ft)            |
| $\bar{T}_b$ | Nominal branch-pipe wall thickness                                                  | mm                             | in                               | $\Gamma$                | Pulsation intensity                             | Dimensionless      | Dimensionless        |
| $\bar{T}_h$ | Nominal header-pipe wall thickness                                                  | mm                             | in                               | $\delta$                | Thickness                                       | m                  | ft                   |
| $t$         | Head or shell radius                                                                | mm                             | in                               | $\varepsilon$           | Wall roughness                                  | m                  | ft                   |
| $t$         | Pressure design thickness                                                           | mm                             | in                               | $\varepsilon$           | Voidage—fractional free volume                  | Dimensionless      | Dimensionless        |
| $t$         | Time                                                                                | s                              | s                                | $\eta$                  | Viscosity, nonnewtonian fluids                  | Pa·s               | lb/(ft·s)            |
| $t_m$       | Minimum required thickness, including mechanical, corrosion, and erosion allowances | mm                             | in                               | $\eta_{ad}$             | Adiabatic efficiency                            |                    |                      |
| $t_r$       | Pad or saddle thickness                                                             | mm                             | in                               | $\eta_p$                | Polytropic efficiency                           |                    |                      |
| $U$         | Straight-line distance between anchors                                              | m                              | ft                               | $\theta$                | Angle                                           | °                  | °                    |
| $u$         | Specific internal energy                                                            | J/kg                           | Btu/lb                           | $\lambda$               | Molecular mean free-path length                 | m                  | ft                   |
| $u$         | Velocity                                                                            | m/s                            | ft/s                             | $\mu$                   | Viscosity                                       | Pa·s               | lb/(ft·s)            |
| $V$         | Velocity                                                                            | m/s                            | ft/s                             | $\nu$                   | Kinematic viscosity                             | m <sup>2</sup> /s  | ft <sup>2</sup> /s   |
| $V$         | Volume                                                                              | m <sup>3</sup>                 | ft <sup>3</sup>                  | $\rho$                  | Density                                         | kg/m <sup>3</sup>  | lb/ft <sup>3</sup>   |
|             |                                                                                     |                                |                                  | $\sigma$                | Surface tension                                 | N/m                | lb/ft                |
|             |                                                                                     |                                |                                  | $\sigma_c$              | Cavitation number                               | Dimensionless      | Dimensionless        |
|             |                                                                                     |                                |                                  | $\tau$                  | Shear stress                                    | N/m <sup>2</sup>   | lb/ft <sup>2</sup>   |
|             |                                                                                     |                                |                                  | $\phi$                  | Shape factor                                    | Dimensionless      | Dimensionless        |
|             |                                                                                     |                                |                                  | $\phi$                  | Angle                                           | °                  | °                    |
|             |                                                                                     |                                |                                  | $\phi$                  | Flow coefficient                                |                    |                      |
|             |                                                                                     |                                |                                  | $\psi$                  | Pressure coefficient                            |                    |                      |
|             |                                                                                     |                                |                                  | $\psi$                  | Sphericity                                      | Dimensionless      | Dimensionless        |

## MEASUREMENT OF FLOW

**GENERAL REFERENCES:** ASME, *Performance Test Code on Compressors and Exhausters*, PTC 10-1997, American Society of Mechanical Engineers (ASME), New York, 1997. Norman A. Anderson, *Instrumentation for Process Measurement and Control*, 3d ed., CRC Press, Boca Raton, Fla., 1997. Roger C. Baker, *Flow Measurement Handbook: Industrial Designs, Operating Principles, Performance, and Applications*, Cambridge University Press, Cambridge, United Kingdom, 2000. Roger C. Baker, *An Introductory Guide to Flow Measurement*, ASME, New York, 2003. Howard S. Bean, ed., *Fluid Meters—Their Theory and Application—Report of the ASME Research Committee on Fluid Meters*, 6th ed., ASME, New York, 1971. Douglas M. Considine, Editor-in-Chief, *Process/Industrial Instruments and Controls Handbook*, 4th ed., McGraw-Hill, New York, 1993. Bela G. Liptak, Editor-in-Chief, *Process Measurement and Analysis*, 4th ed., CRC Press, Boca Raton, Fla., 2003. Richard W. Miller, *Flow Measurement Engineering Handbook*, 3d ed., McGraw-Hill, New York, 1996. Ower and Pankhurst, *The Measurement of Air Flow*, Pergamon, Oxford, United Kingdom, 1966. Brian Price et al., *Engineering Data Book*, 12th ed., Gas Processors Suppliers Association, Tulsa, Okla., 2004. David W. Spitzer, *Flow Measurement*, 2d ed., Instrument Society of America, Research Triangle Park, N.C., 2001. David W. Spitzer, *Industrial Flow Measurement*, 3d ed., Instrument Society of America, Research Triangle Park, N.C., 2005.

### INTRODUCTION

The flow rate of fluids is a critical variable in most chemical engineering applications, ranging from flows in the process industries to environmental flows and to flows within the human body. *Flow* is defined as mass flow or volume flow per unit of time at specified temperature and pressure conditions for a given fluid. This subsection deals with the techniques of measuring pressure, temperature, velocities, and flow rates of flowing fluids. For more detailed discussion of these variables, consult Sec. 8. Section 8 introduces methods of measuring flow rate, temperature, and pressure. This subsection builds on the coverage in Sec. 8 with emphasis on measurement of the flow of fluids.

### PROPERTIES AND BEHAVIOR OF FLUIDS

Transportation and the storage of fluids (gases and liquids) involves the understanding of the properties and behavior of fluids. The study of fluid dynamics is the study of fluids and their motion in a force field.

Flows can be classified into two major categories: (a) incompressible and (b) compressible flow. Most liquids fall into the incompressible-flow category, while most gases are compressible in nature. A perfect fluid can be defined as a fluid that is nonviscous and nonconducting. Fluid flow, compressible or incompressible, can be classified by the ratio of the inertial forces to the viscous forces. This ratio is represented by the Reynolds number ( $N_{Re}$ ). At a low Reynolds number, the flow is considered to be laminar, and at high Reynolds numbers, the

flow is considered to be turbulent. The limiting types of flow are the inertialess flow, sometimes called Stokes flow, and the inviscid flow that occurs at an infinitely large Reynolds number. Reynolds numbers (dimensionless) for flow in a pipe is given as:

$$N_{Re} = \frac{\rho V D}{\mu} \quad (10-1)$$

where  $\rho$  is the density of the fluid,  $V$  the velocity,  $D$  the diameter, and  $\mu$  the viscosity of the fluid. In fluid motion where the frictional forces interact with the inertia forces, it is important to consider the ratio of the viscosity  $\mu$  to the density  $\rho$ . This ratio is known as the kinematic viscosity ( $\nu$ ). Tables 10-1 and 10-2 give the kinematic viscosity for several fluids. A flow is considered to be *adiabatic* when there is no transfer of heat between the fluid and its surroundings. An isentropic flow is one in which the entropy of each fluid element remains constant.

To fully understand the mechanics of flow, the following definitions explain the behavior of various types of fluids in both their static and flowing states.

A perfect fluid is a nonviscous, nonconducting fluid. An example of this type of fluid would be a fluid that has a very small viscosity and conductivity and is at a high Reynolds number. An ideal gas is one that obeys the equation of state:

$$\frac{P}{\rho} = RT \quad (10-2)$$

where  $P$  = pressure,  $\rho$  = density,  $R$  is the gas constant per unit mass, and  $T$  = temperature.

A flowing fluid is acted upon by many forces that result in changes in pressure, temperature, stress, and strain. A fluid is said to be isotropic when the relations between the components of stress and those of the rate of strain are the same in all directions. The fluid is said to be Newtonian when this relationship is linear. These pressures and temperatures must be fully understood so that the entire flow picture can be described.

The *static pressure* in a fluid has the same value in all directions and can be considered as a scalar point function. It is the pressure of a flowing fluid. It is normal to the surface on which it acts and at any given point has the same magnitude irrespective of the orientation of the surface. The static pressure arises because of the random motion in the fluid of the molecules that make up the fluid. In a diffuser or nozzle, there is an increase or decrease in the static pressure due to the change in velocity of the moving fluid.

**Total pressure** is the pressure that would occur if the fluid were brought to rest in a reversible adiabatic process. Many texts and engineers use the words *total* and *stagnation* to describe the flow characteristics interchangeably. To be accurate, the stagnation pressure

**TABLE 10-1 Density, Viscosity, and Kinematic Viscosity of Water and Air in Terms of Temperature**

| Temperature |      | Water                                                      |                                                              |                                                                       | Air at a pressure of 760 mm Hg (14.696 lbf/in <sup>2</sup> ) |                                                              |                                                                       |
|-------------|------|------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
|             |      | Density $\rho$<br>(lbf sec <sup>2</sup> /ft <sup>4</sup> ) | Viscosity<br>$\mu \times 10^6$<br>(lbf sec/ft <sup>2</sup> ) | Kinematic<br>viscosity<br>$\nu \times 10^6$<br>(ft <sup>2</sup> /sec) | Density $\rho$<br>(lbf sec <sup>2</sup> /ft <sup>4</sup> )   | Viscosity<br>$\mu \times 10^6$<br>(lbf sec/ft <sup>2</sup> ) | Kinematic<br>viscosity<br>$\nu \times 10^6$<br>(ft <sup>2</sup> /sec) |
| (°C)        | (°F) |                                                            |                                                              |                                                                       |                                                              |                                                              |                                                                       |
| -20         | -4   | —                                                          | —                                                            | —                                                                     | 0.00270                                                      | 0.326                                                        | 122                                                                   |
| -10         | 14   | —                                                          | —                                                            | —                                                                     | 0.00261                                                      | 0.338                                                        | 130                                                                   |
| 0           | 32   | 1.939                                                      | 37.5                                                         | 19.4                                                                  | 0.00251                                                      | 0.350                                                        | 140                                                                   |
| 10          | 50   | 1.939                                                      | 27.2                                                         | 14.0                                                                  | 0.00242                                                      | 0.362                                                        | 150                                                                   |
| 20          | 68   | 1.935                                                      | 21.1                                                         | 10.9                                                                  | 0.00234                                                      | 0.375                                                        | 160                                                                   |
| 40          | 104  | 1.924                                                      | 13.68                                                        | 7.11                                                                  | 0.00217                                                      | 0.399                                                        | 183                                                                   |
| 60          | 140  | 1.907                                                      | 9.89                                                         | 5.19                                                                  | 0.00205                                                      | 0.424                                                        | 207                                                                   |
| 80          | 176  | 1.886                                                      | 7.45                                                         | 3.96                                                                  | 0.00192                                                      | 0.449                                                        | 234                                                                   |
| 100         | 212  | 1.861                                                      | 5.92                                                         | 3.19                                                                  | 0.00183                                                      | 0.477                                                        | 264                                                                   |

Conversion factors: 1 kp sec<sup>2</sup>/m<sup>4</sup> = 0.01903 lbf sec<sup>2</sup>/ft<sup>4</sup> (= slug/ft<sup>3</sup>)  
 1 lbf sec<sup>2</sup>/ft<sup>4</sup> = 32.1719 lb/ft<sup>3</sup> (lb = lb mass; lbf = lb force)  
 1 kp sec<sup>2</sup>/m<sup>4</sup> = 9.80665 kg/m<sup>3</sup> (kg = kg mass; kp = kg force)  
 1 kg/m<sup>3</sup> = 16.02 lb/ft<sup>3</sup>

TABLE 10-2 Kinematic Viscosity

| Liquid          | Temperature        |                    | $\nu \times 10^6 \text{ (ft}^2/\text{s)}$ |
|-----------------|--------------------|--------------------|-------------------------------------------|
|                 | $^{\circ}\text{C}$ | $^{\circ}\text{F}$ |                                           |
| Glycerine       | 20                 | 68                 | 7319                                      |
| Mercury         | 0                  | 32                 | 1.35                                      |
| Mercury         | 100                | 212                | 0.980                                     |
| Lubricating oil | 20                 | 68                 | 4306                                      |
| Lubricating oil | 40                 | 104                | 1076                                      |
| Lubricating oil | 60                 | 140                | 323                                       |

is the pressure that would occur if the fluid were brought to rest adiabatically or diabatically.

Total pressure will only change in a fluid if shaft work or work of extraneous forces are introduced. Therefore, total pressure would increase in the impeller of a compressor or pump; it would remain constant in the diffuser. Similarly, total pressure would decrease in the turbine impeller but would remain constant in the nozzles.

Static temperature is the temperature of the flowing fluid. Like static pressure, it arises because of the random motion of the fluid molecules. Static temperature is in most practical installations impossible to measure since it can be measured only by a thermometer or thermocouple at rest relative to the flowing fluid that is moving with the fluid. Static temperature will increase in a diffuser and decrease in a nozzle.

Total temperature is the temperature that would occur when the fluid is brought to rest in a reversible adiabatic manner. Just like its counterpart *total pressure*, *total* and *stagnation temperatures* are used interchangeably by many test engineers.

Dynamic temperature and pressure are the difference between the total and static conditions.

$$P_d = P_T - P_s \quad (10-3)$$

$$T_d = T_T - T_s \quad (10-4)$$

where subscript *d* refers to dynamic, *T* to total, and *s* to static.

Another helpful formula is:

$$P_k = \frac{1}{2} \rho V^2 \quad (10-5)$$

For incompressible fluids,  $P_k = P_d$ .

## TOTAL TEMPERATURE

For most points requiring temperature monitoring, either thermocouples or resistive thermal detectors (RTDs) can be used. Each type of temperature transducer has its own advantages and disadvantages, and both should be considered when temperature is to be measured. Since there is considerable confusion in this area, a short discussion of the two types of transducers is necessary.

**Thermocouples** The various types of thermocouples provide transducers suitable for measuring temperatures from  $-330$  to  $5000^{\circ}\text{F}$  ( $-201$  to  $2760^{\circ}\text{C}$ ). Thermocouples function by producing a voltage proportional to the temperature differences between two junctions of dissimilar metals. By measuring this voltage, the temperature difference can be determined. It is assumed that the temperature is known at one of the junctions; therefore, the temperature at the other junction can be determined. Since the thermocouples produce a voltage, no external power supply is required to the test junction; however, for accurate measurement, a reference junction is required. For a temperature monitoring system, reference junctions must be placed at each thermocouple or similar thermocouple wire installed from the thermocouple to the monitor where there is a reference junction. Properly designed thermocouple systems can be accurate to approximately  $\pm 2^{\circ}\text{F}$  ( $\pm 1^{\circ}\text{C}$ ).

**Resistive Thermal Detectors (RTDs)** RTDs determine temperature by measuring the change in resistance of an element due to temperature. Platinum is generally utilized in RTDs because it remains mechanically and electrically stable, resists contaminations, and can be highly refined. The useful range of platinum RTDs is

$-454$ – $1832^{\circ}\text{F}$  ( $-270$ – $1000^{\circ}\text{C}$ ). Since the temperature is determined by the resistance in the element, any type of electrical conductor can be utilized to connect the RTD to the indicator; however, an electrical current must be provided to the RTD. A properly designed temperature monitoring system utilizing RTDs can be accurate  $\pm 0.02^{\circ}\text{F}$  ( $\pm 0.01^{\circ}\text{C}$ ).

## STATIC TEMPERATURE

Since this temperature requires the thermometer or thermocouple to be at rest relative to the flowing fluid, it is impractical to measure. It can be, however, calculated from the measurement of total temperature and total and static pressure.

$$T_s = \frac{T_o}{\left(\frac{P_o}{P_s}\right)^{(k-1)/k}} \quad (10-6)$$

## DRY- AND WET-BULB TEMPERATURES

The moisture content or humidity of air has an important effect on the properties of the gaseous mixture. Steam in air at any relative humidity less than 100 percent must exist in a superheated condition. The saturation temperature corresponding to the actual partial pressure of the steam in air is called the dew point. This term arose from the fact that when air at less than 100 percent relative humidity is cooled to the temperature at which it becomes saturated, the air has reached the minimum temperature to which it can be cooled without precipitation of the moisture (dew). Dew point can also be defined as that temperature at which the weight of steam associated with a certain weight of dry air is adequate to saturate that weight of air.

The dry-bulb temperature of air is the temperature that is indicated by an ordinary thermometer. When an air temperature is stated without any modifying term, it is always taken to be the dry-bulb temperature. In contrast to dry-bulb, or air, temperature, the term *wet-bulb temperature of the air*, or simply *wet-bulb temperature*, is employed. When a thermometer, with its bulb covered by a wick wetted with water, is moved through air unsaturated with water vapor, the water evaporates in proportion to the capacity of the air to absorb the evaporated moisture, and the temperature indicated by the thermometer drops below the dry-bulb, or air, temperature. The equilibrium temperature finally reached by the thermometer is known as the wet-bulb temperature. The purpose in measuring both the dry-bulb and wet-bulb temperature of the air is to find the exact humidity characteristics of the air from the readings obtained, either by calculation or by use of a psychrometric chart. Instruments for measuring wet-bulb and dry-bulb temperatures are known as psychrometers. A sling psychrometer consists of two thermometers mounted side by side on a holder, with provision for whirling the whole device through the air. The dry-bulb thermometer is bare, and the wet bulb is covered by a wick which is kept wetted with clean water. After being whirled a sufficient amount of time, the wet-bulb thermometer reaches its equilibrium point, and both the wet-bulb and dry-bulb thermometers are then quickly read. Rapid relative movement of the air past the wet-bulb thermometer is necessary to get dependable readings.

For other methods of measuring the moisture content of gases, see Sec. 8.

## PRESSURE MEASUREMENTS

*Pressure* is defined as the force per unit area. Pressure devices measure with respect to the ambient atmospheric pressure: The absolute pressure  $P_a$  is the pressure of the fluid (gauge pressure) plus the atmospheric pressure.

Process pressure-measuring devices may be divided into three groups:

1. Those that are based on the height of a liquid column (manometers)
2. Those that are based on the measurement of the distortion of an elastic pressure chamber (mechanical pressure gauges such as Bourdon-tube gauges and diaphragm gauges)



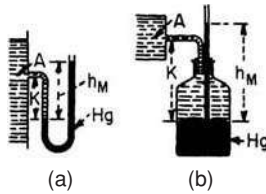


FIG. 10-1 Open manometers.

3. Electric sensing devices (strain gauges, piezoresistive transducers, and piezoelectric transducers)

This subsection contains an expanded discussion of manometric methods. See Sec. 8 for other methods.

**Liquid-Column Manometers** The **height**, or **head**,  $p_h = \rho h g/g_c$  to which a fluid rises in an open vertical tube attached to an apparatus containing a liquid is a direct measure of the pressure at the point of attachment and is frequently used to show the level of liquids in tanks and vessels. This same principle can be applied with U-tube gauges (Fig. 10-1a) and equivalent devices (such as that shown in Fig. 10-1b) to measure pressure in terms of the head of a fluid other than the one under test. Most of these gauges may be used either as **open** or as **differential manometers**. The manometric fluid that constitutes the measured liquid column of these gauges may be any liquid immiscible with the fluid under pressure. For high pressures and large pressure differences, the gauge liquid is a high-density liquid, generally mercury; for low pressures and small pressure differences, a low-density liquid (e.g., alcohol, water, or carbon tetrachloride) is used.

The **open U tube** (Fig. 10-1a) and the **open gauge** (Fig. 10-1b) each show a reading  $h_M$  (ft) of manometric fluid. If the interface of the manometric fluid and the fluid of which the pressure is wanted is  $K$  m (ft) below the point of attachment,  $A$ ,  $\rho_A$  is the density of the latter fluid at  $A$ , and  $\rho_M$  is that of the manometric fluid, then gauge pressure  $p_A$  (lbf/ft<sup>2</sup>) at  $A$  is

$$p_A = (h_M \rho_M - K \rho_A)(g/g_c) \quad (10-7)^*$$

where  $g$  = local acceleration due to gravity and  $g_c$  = dimensional constant. The head  $H_A$  at  $A$  as meters (feet) of the fluid at that point is

$$h_A = h_M(\rho_M/\rho_A) - K \quad (10-8)^*$$

When a gas pressure is measured, unless it is very high,  $\rho_A$  is so much smaller than  $\rho_M$  that the terms involving  $K$  in these formulas are negligible.

The **differential U tube** (Fig. 10-2) shows the pressure difference between taps  $A$  and  $B$  to be

$$p_A - p_B = [h_M(\rho_M - \rho_A) + K_A \rho_A - K_B \rho_B](g/g_c) \quad (10-9)^*$$

where  $h_M$  is the difference in height of the manometric fluid in the U tube;  $K_A$  and  $K_B$  are the vertical distances of the upper surface of the

\*The line leading from the pressure tap to the gauge is assumed to be filled with fluid of the same density as that in the apparatus at the location of the pressure tap; if this is not the case,  $\rho_A$  is the density of the fluid actually filling the gauge line, and the value given for  $h_A$  must be multiplied by  $\rho_A/\rho$  where  $\rho$  is the density of the fluid whose head is being measured.

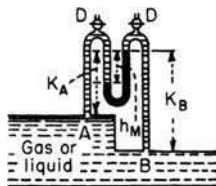


FIG. 10-2 Differential U tube.

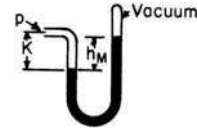


FIG. 10-3 Closed U tube.

manometric fluid above  $A$  and  $B$ , respectively;  $\rho_A$  and  $\rho_B$  are the densities of the fluids at  $A$  and  $B$ , respectively; and  $\rho_M$  is the density of the manometric fluid. If either pressure tap is above the higher level of manometric fluid, the corresponding  $K$  is taken to be negative. Valve  $D$ , which is kept closed when the gauge is in use, is used to vent off gas which may accumulate at these high points.

The **inverted differential U tube**, in which the manometric fluid may be a gas or a light liquid, can be used to measure liquid pressure differentials, especially for the flow of slurries where solids tend to settle out.

**Closed U tubes** (Fig. 10-3) using mercury as the manometric fluid serve to measure directly the absolute pressure  $p$  of a fluid, provided that the space between the closed end and the mercury is substantially a perfect vacuum.

The **mercury barometer** (Fig. 10-4) indicates directly the absolute pressure of the atmosphere in terms of height of the mercury column. Normal (standard) barometric pressure is 101.325 kPa by definition. Equivalents of this pressure in other units are 760 mm mercury (at 0°C), 29.921 inHg (at 0°C), 14.696 lbf/in<sup>2</sup>, and 1 atm. For cases in which barometer readings, when expressed by the height of a mercury column, must be corrected to standard temperature (usually 0°C), appropriate temperature correction factors are given in ASME PTC, op. cit., pp. 23–26; and Weast, *Handbook of Chemistry and Physics*, 62d ed., Chemical Rubber, Cleveland, 1984, pp. E36–E37.

**Tube Size for Manometers** To avoid capillary error, tube diameter should be sufficiently large and the manometric fluids of such densities that the effect of capillarity is negligible in comparison with the gauge reading. The effect of capillarity is practically negligible for tubes with inside diameters 12.7 mm (1/2 in) or larger (see ASME PTC, op. cit., p. 15). Small diameters are generally permissible for U tubes because the capillary displacement in one leg tends to cancel that in the other.

The capillary rise in a small vertical open tube of circular cross section dipping into a pool of liquid is given by

$$h = \frac{4\sigma g_c \cos \theta}{gD(\rho_1 - \rho_2)} \quad (10-10)$$

Here  $\sigma$  = surface tension,  $D$  = inside diameter,  $\rho_1$  and  $\rho_2$  are the densities of the liquid and gas (or light liquid) respectively,  $g$  = local acceleration due to gravity,  $g_c$  = dimensional constant, and  $\theta$  is the contact angle subtended by the heavier fluid. For most organic liquids and water, the contact angle  $\theta$  is zero against glass, provided the glass is wet with a film of the liquid; for mercury against glass,  $\theta = 140^\circ$  (*International Critical Tables*, vol. IV, McGraw-Hill, New York, 1928, pp. 434–435). For further discussion of capillarity, see Schwartz, *Ind. Eng. Chem.*, 61(1), 10–21 (1969).

**Multiplying Gauges** To attain the requisite precision in measurement of small pressure differences by liquid-column manometers,

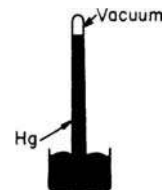


FIG. 10-4 Mercury barometer.

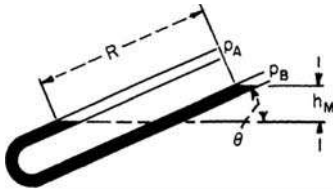


FIG. 10-5 Inclined U tube.

means must often be devised to magnify the readings. Of the schemes that follow, the second and third may give tenfold multiplication; the fourth, as much as thirtyfold. In general, the greater the multiplication, the more elaborate must be the precautions in the use of the gauge if the gain in precision is not to be illusory.

1. *Change of manometric fluid.* In open manometers, choose a fluid of lower density. In differential manometers, choose a fluid such that the difference between its density and that of the fluid being measured is as small as possible.

2. *Inclined U tube (Fig. 10-5).* If the reading  $R$  m (ft) is taken as shown and  $R_0$  m (ft) is the zero reading, by making the substitution  $h_M = (R - R_0) \sin \theta$ , the formulas of preceding paragraphs give  $(p_A - p_B)$  when the corresponding upright U tube is replaced by one inclined. For precise work, the gauge should be calibrated because of possible variations in tube diameter and slope.

3. *The draft gauge (Fig. 10-6).* Commonly used for low gas heads, this gauge has for one leg of the U a reservoir of much larger bore than the tubing that forms the inclined leg. Hence variations of level in the inclined tube produce little change in level in the reservoir. Although  $h_M$  may be readily computed in terms of reading  $R$  and the dimensions of the tube, calibration of the gauge is preferable; often the changes of level in the reservoir are not negligible, and also variations in tube diameter may introduce serious error into the computation. Commercial gauges are often provided with a scale giving  $h_M$  directly in height of water column, provided a particular liquid (often not water) fills the tube; failure to appreciate that the scale is incorrect unless the gauge is filled with the specified liquid is a frequent source of error. If the scale reads correctly when the density of the gauge liquid is  $\rho_0$ , then the reading must be multiplied by  $\rho/\rho_0$  if the density of the fluid actually in use is  $\rho$ .

4. *Two-fluid U tube (Fig. 10-7).* This is a highly sensitive device for measuring small gas heads. Let  $A$  be the cross-sectional area of each of the reservoirs and  $a$  that of the tube forming the U; let  $\rho_1$  be the density of the lighter fluid and  $\rho_2$  that of the heavier fluid; and if  $R$  is the reading and  $R_0$  its value with zero pressure difference, then the pressure difference is

$$p_A - p_B = (R - R_0) \left( \rho_2 - \rho_1 + \frac{a}{A} \rho_1 \right) \frac{g}{g_c} \quad (10-11)$$

where  $g$  = local acceleration due to gravity and  $g_c$  = dimensional constant.

When  $A/a$  is sufficiently large, the term  $(a/A) \rho_1$  in Eq. (10-11) becomes negligible in comparison with the difference  $(\rho_2 - \rho_1)$ . However, this term should not be omitted without due consideration. In applying Eq. (10-11), the densities of the gauge liquids may not be taken from tables without the possibility of introducing serious error,

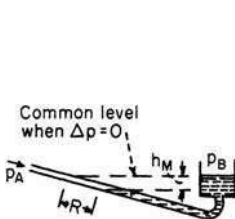


FIG. 10-6 Draft gauge.

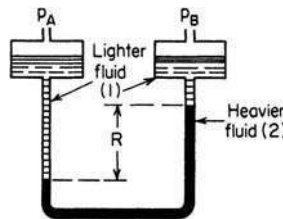


FIG. 10-7 Two-fluid U tube.

for each liquid may dissolve appreciable quantities of the other. Before the gauge is filled, the liquids should be shaken together, and the actual densities of the two layers should be measured for the temperature at which the gauge is to be used. When high magnification is being sought, the U tube may have to be enclosed in a constant-temperature bath so that  $(\rho_2 - \rho_1)$  may be accurately known. In general, if highest accuracy is desired, the gauge should be calibrated.

Several **micromanometers**, based on the liquid-column principle and possessing extreme precision and sensitivity, have been developed for measuring minute gas-pressure differences and for calibrating low-range gauges. Some of these micromanometers are available commercially. These micromanometers are free from errors due to capillarity and, aside from checking the micrometer scale, require no calibration.

**Mechanical Pressure Gauges** The **Bourdon-tube gauge** indicates pressure by the amount of flexion under internal pressure of an oval tube bent in an arc of a circle and closed at one end. These gauges are commercially available for all pressures below atmospheric and for pressures up to 700 MPa (about 100,000 lbf/in<sup>2</sup>) above atmospheric. Details on Bourdon-type gauges are given by Harland [*Mach. Des.*, 40(22), 69-74 (Sept. 19, 1968)].

A **diaphragm gauge** depends for its indication on the deflection of a diaphragm, usually metallic, when subjected to a difference of pressure between the two faces. These gauges are available for the same general purposes as Bourdon gauges but are not usually employed for high pressures. The aneroid barometer is a type of diaphragm gauge.

Small **pressure transducers with flush-mounted diaphragms** are commercially available for the measurement of either steady or fluctuating pressures up to 100 MPa (about 15,000 lbf/in<sup>2</sup>). The metallic diaphragms are as small as 4.8 mm ( $3/16$  in) in diameter. The transducer is mounted on the apparatus containing the fluid whose pressure is to be measured so that the diaphragm is flush with the inner surface of the apparatus. Deflection of the diaphragm is measured by unbonded strain gauges and recorded electrically.

With nonnewtonian fluids the pressure measured at the wall with non-flush-mounted pressure gauges may be in error (see subsection "Static Pressure").

Bourdon and diaphragm gauges that show both pressure and vacuum indications on the same dial are called **compound gauges**.

**Conditions of Use** Bourdon tubes should not be exposed to temperatures over about 65°C (about 150°F) unless the tubes are specifically designed for such operation. When the pressure of a hotter fluid is to be measured, some type of liquid seal should be used to keep the hot fluid from the tube. In using either a Bourdon or a diaphragm gauge to measure gas pressure, if the gauge is below the pressure tap of the apparatus so that liquid can collect in the lead, the gauge reading will be too high by an amount equal to the hydrostatic head of the accumulated liquid.

For measuring pressures of corrosive fluids, slurries, and similar process fluids which may foul Bourdon tubes, a **chemical gauge**, consisting of a Bourdon gauge equipped with an appropriate flexible diaphragm to seal off the process fluid, may be used. The combined volume of the tube and the connection between the diaphragm and the tube is filled with an inert liquid. These gauges are available commercially.

Further details on pressure-measuring devices are found in Sec. 8.

**Calibration of Gauges** Simple **liquid-column manometers** do not require calibration if they are so constructed as to minimize errors due to capillarity (see subsection "Liquid-Column Manometers"). If the scales used to measure the readings have been checked against a standard, the accuracy of the gauges depends solely upon the precision of determining the position of the liquid surfaces. Hence liquid-column manometers are primary standards used to calibrate other gauges.

For **high pressures** and, with commercial mechanical gauges, even for quite moderate pressures, a deadweight gauge (see ASME PTC, op. cit., pp. 36-41) is commonly used as the primary standard because it is safer and more convenient than use of manometers. When manometers are used as high-pressure standards, an extremely high mercury column may be avoided by connecting a number of the usual U tubes in series. Multiplying gauges are standardized by comparing

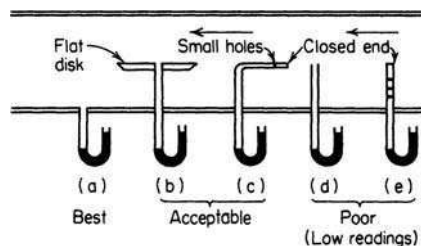


FIG. 10-8 Measurement of static pressure.

them with a micromanometer. Procedure in the calibration of a gauge consists merely of connecting it, in parallel with a standard gauge, to a reservoir wherein constant pressure may be maintained. Readings of the unknown gauge are then made for various reservoir pressures as determined by the standard.

Calibration of **high-vacuum gauges** is described by Sellenger [*Vacuum*, 18(12), 645–650 (1968)].

### STATIC PRESSURE

**Local Static Pressure** In a moving fluid, the local static pressure is equal to the pressure on a surface which moves with the fluid or to the normal pressure (for newtonian fluids) on a stationary surface which parallels the flow. The pressure on such a surface is measured by making a small hole perpendicular to the surface and connecting the opening to a pressure-sensing element (Fig. 10-8a). The hole is known as a piezometer opening or pressure tap.

Measurement of local static pressure is frequently difficult or impractical. If the channel is so small that introduction of any solid object disturbs the flow pattern and increases the velocity, there will be a reduction and redistribution of the static pressure. If the flow is in straight parallel lines, aside from the fluctuations of normal turbulence, the flat disk (Fig. 10-8b) and the bent tube (Fig. 10-8c) give satisfactory results when properly aligned with the stream. Slight misalignments can cause serious errors. Diameter of the disk should be 20 times its thickness and 40 times the static opening; the face must be flat and smooth, with the knife edges made by beveling the underside. The piezometer tube, such as that in Fig. 10-8c, should have openings with size and spacing as specified for a pitot-static tube (Fig. 10-12).

Readings given by open straight tubes (Fig. 10-8d and 10-8e) are too low due to flow separation. Readings of closed tubes oriented perpendicularly to the axis of the stream and provided with side openings (Fig. 10-8e) may be low by as much as two velocity heads.

**Average Static Pressure** In most cases, the object of a static-pressure measurement is to obtain a suitable average value for substitution in Bernoulli's theorem or in an equivalent flow formula. This can be done simply only when the flow is in straight lines parallel to the confining walls, such as in straight ducts at sufficient distance downstream from bends (2 diameters) or other disturbances. For such streams, the sum of static head and gravitational potential head is the same at all points in a cross section taken perpendicularly to the axis of flow. Thus the exact location of a piezometer opening about the periphery of such a cross section is immaterial provided its elevation is known. However, in stating the static pressure, the custom is to give the value at the elevation corresponding to the centerline of the stream.

With flow in curved passages or with swirling flow, determination of a true average static pressure is, in general, impractical. In metering, straightening vanes are often placed upstream of the pressure tap to eliminate swirl. Figure 10-9 shows various flow equalizers and straighteners.

**Specifications for Piezometer Taps** The size of a static opening should be small compared with the diameter of the pipe and yet large compared with the scale of surface irregularities. For reliable results, it is essential that (1) the surface in which the hole is made be substantially smooth and parallel to the flow for some distance on either side of the

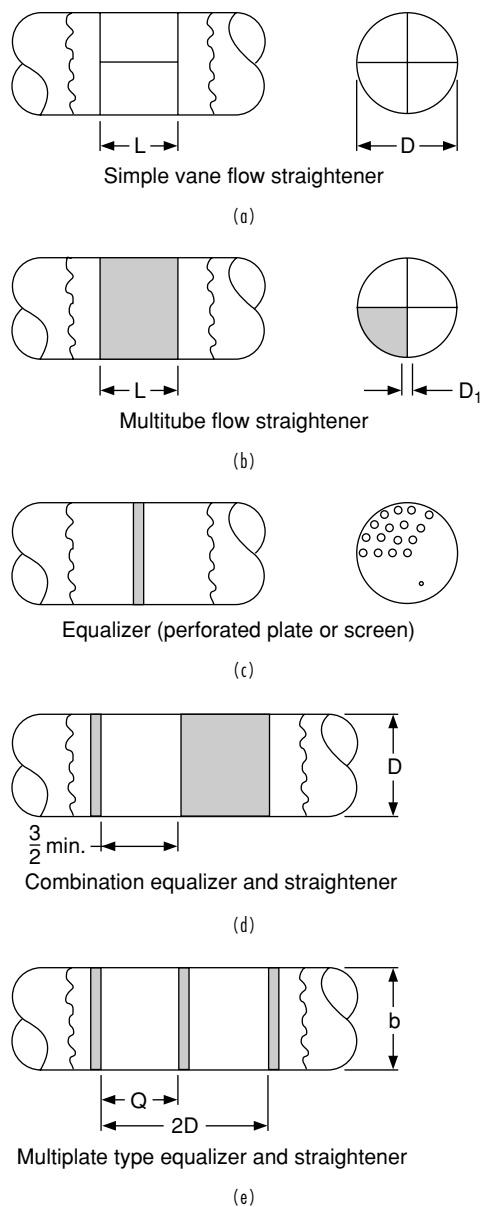


FIG. 10-9 Flow equalizers and straighteners [*Power Test Code 10, Compressors and Exhausters*, Amer. Soc. of Mechanical Engineers, 1997].

opening, and (2) the opening be flush with the surface and possess no “burr” or other irregularity around its edge. Rounding of the edge is often employed to ensure absence of a burr. Pressure readings will be high if the tap is inclined upstream, is rounded excessively on the upstream side, has a burr on the downstream side, or has an excessive countersink or recess. Pressure readings will be low if the tap is inclined downstream, is rounded excessively on the downstream side, has a burr on the upstream side, or protrudes into the flow stream. Errors resulting from these faults can be large.

Recommendations for **pressure-tap dimensions** are summarized in Table 10-3. Data from several references were used in arriving at these composite values. The length of a pressure-tap opening prior to any enlargement in the tap channel should be at least two tap diameters, preferably three or more.

TABLE 10-3 Pressure-Tap Holes

| Nominal inside pipe diameter, in | Maximum diameter of pressure tap, mm (in) | Radius of hole-edge rounding, mm (in) |
|----------------------------------|-------------------------------------------|---------------------------------------|
| 1                                | 3.18 (1/8)                                | <0.40 (1/64)                          |
| 2                                | 6.35 (1/4)                                | 0.40 (1/64)                           |
| 3                                | 9.53 (3/8)                                | 0.40–0.79 (1/64–1/32)                 |
| 4                                | 12.7 (1/2)                                | 0.79 (1/32)                           |
| 8                                | 12.7 (1/2)                                | 0.79–1.59 (1/32–1/16)                 |
| 16                               | 19.1 (3/4)                                | 0.79–1.59 (1/32–1/16)                 |

A **piezometer ring** is a toroidal manifold into which are connected several sidewall static taps located around the perimeter of a common cross section. Its intent is to give an average pressure if differences in pressure other than those due to static head exist around the perimeter. However, there is generally no assurance that a true average is provided thereby. The principal advantage of the ring is that use of several holes in place of a single hole reduces the possibility of completely plugging the static openings.

For information on prediction of static-hole error, see Shaw, *J. Fluid Mech.*, **7**, 550–564 (1960); Livesey, Jackson, and Southern, *Aircr. Eng.*, **34**, 43–47 (February 1962).

For nonnewtonian fluids, pressure readings with taps may also be low because of fluid-elasticity effects. This error can be largely eliminated by using flush-mounted diaphragms.

For information on the pressure-hole error for nonnewtonian fluids, see Han and Kim, *Trans. Soc. Rheol.*, **17**, 151–174 (1973); Novotny and Eckert, *Trans. Soc. Rheol.*, **17**, 227–241 (1973); and Higashitani and Lodge, *Trans. Soc. Rheol.*, **19**, 307–336 (1975).

## VELOCITY MEASUREMENTS

Measurement of flow can be based on the measurement of velocity in ducts or pipes by using devices such as pitot tubes and hot wire anemometers. The local velocity is measured at various sections of a conduit and then averaged for the area under consideration.

$$\frac{w}{\rho} = A \times V = Q \quad (10-12)$$

where  $w$  = mass flow rate, lb<sub>m</sub>/s, kg/s  
 $\rho$  = density, lb<sub>m</sub>/ft<sup>3</sup>, kg/m<sup>3</sup>  
 $A$  = area, ft<sup>2</sup>, m<sup>2</sup>  
 $V$  = velocity, ft/s, m/s  
 $Q$  = volumetric flow rate, ft<sup>3</sup>/s, m<sup>3</sup>/s

Equation (10-12) shows that the fluid density directly affects the relationship between mass flow rate and both velocity and volumetric flow rate. Liquid temperature affects liquid density and hence volumetric flow rate at a constant mass flow rate. Liquid density is relatively insensitive to pressure. Both temperature and pressure affect gas density and thus volumetric flow rate.

**Variables Affecting Measurement** Flow measurement methods may sense local fluid velocity, volumetric flow rate, total or cumulative volumetric flow (the integral of volumetric flow rate with respect to elapsed time), mass flow rate, and total mass flow.

**Velocity Profile Effects** Many variables can influence the accuracy of specific flow measurement methods. For example, the velocity profile in a closed conduit affects many types of flow-measuring devices. The velocity of a fluid varies from zero at the wall and at other stationary solid objects in the flow channel to a maximum at a distance from the wall. In the entry region of a conduit, the velocity field may approach plug flow and a constant velocity across the conduit, dropping to zero only at the wall. As a newtonian fluid progresses down a pipe, a velocity profile develops that is parabolic for laminar flow [Eq. (6-41)] and that approaches plug flow for highly turbulent flow. Once a steady flow profile has developed, the flow is said to be fully developed; the length of conduit necessary to achieve fully devel-

oped flow is called the entrance region. For long cylindrical, horizontal pipe ( $L < 40D$ , where  $D$  is the inside diameter of the pipe and  $L$  is the upstream length of pipe), the velocity profile becomes fully developed. Velocity profiles in flowing fluids are discussed in greater detail in Sec. 6 (p. 6-11).

For steady-state, isothermal, single-phase, uniform, fully developed newtonian flow in straight pipes, the velocity is greatest at the center of the channel and symmetric about the axis of the pipe. Of those flowmeters that are dependent on the velocity profile, they are usually calibrated for this type of flow. Thus any disturbances in flow conditions can affect flowmeter readings.

Upstream and downstream disturbances in the flow field are caused by valves, elbows, and other types of fittings. Two upstream elbows in two perpendicular planes will impart swirl in the fluid downstream. Swirl, similar to atypical velocity profiles, can lead to erroneous flow measurements. Although the effect is not as great as in upstream flow disturbances, downstream flow disturbances can also lead to erroneous flow measurements.

**Other Flow Disturbances** Other examples of deviations from fully developed, single-phase newtonian flow include nonnewtonian flow, pulsating flow, cavitation, multiphase flow, boundary layer flows, and nonisothermal flows. See Sec. 6.

**Pitot Tubes** The combination of pitot tubes in conjunction with sidewall static taps measures local or point velocities by measuring the difference between the total pressure and the static pressure. The pitot tube shown in Fig. 10-10 consists of an impact tube whose opening faces directly into the stream to measure impact pressure, plus one or more sidewall taps to measure local static pressure.

Dynamic pressure may be measured by use of a pitot tube that is a simple impact tube. These tubes measure the pressure at a point where the velocity of the fluid is brought to zero. Pitot tubes must be parallel to the flow. The pitot tube is sensitive to yaw or angle attack. In general angles of attack over 10° should be avoided. In cases where the flow direction is unknown, it is recommended to use a Kiel probe. Figure 10-11 shows a Kiel probe. This probe will read accurately to an angle of about 22° with the flow.

The combined pitot-static tube shown in Fig. 10-12 consists of a jacketed impact tube with one or more rows of holes, 0.51 to 1.02 mm (0.02 to 0.04 in) in diameter, in the jacket to measure the static pressure. Velocity  $V_0$  m/s (ft/s) at the point where the tip is located is given by

$$V_0 = C \sqrt{2g_c \Delta h} = C \sqrt{2g_c (P_T - P_s)/\rho_0} \quad (10-13)$$

where  $C$  = coefficient, dimensionless;  $g_c$  = dimensional constant;  $\Delta h$  = dynamic pressure ( $\Delta h/g_c$ ), expressed in (N·m)/kg[(ft·lbf)/lb or ft of fluid flowing];  $\Delta h_s$  = differential height of static liquid column corresponding to  $\Delta h$ ;  $g$  = local acceleration due to gravity;  $g_c$  = dimensional constant;  $p_i$  = impact pressure;  $p_0$  = local static pressure; and  $\rho_0$  = fluid density measured at pressure  $p_0$  and the local temperature. With gases at velocities above 60 m/s (about 200 ft/s), compressibility becomes important, and the following equation should be used:

$$V_0 = C \sqrt{\frac{2g_c k}{k-1} \left( \frac{p_0}{\rho_0} \right) \left[ \left( \frac{p_i}{p_0} \right)^{(k-1)/k} - 1 \right]} \quad (10-14)$$

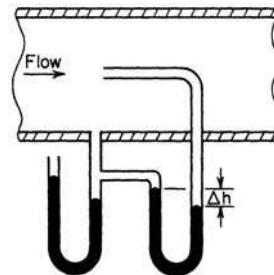


FIG. 10-10 Pitot tube with sidewall static tap.

## 10-12 TRANSPORT AND STORAGE OF FLUIDS

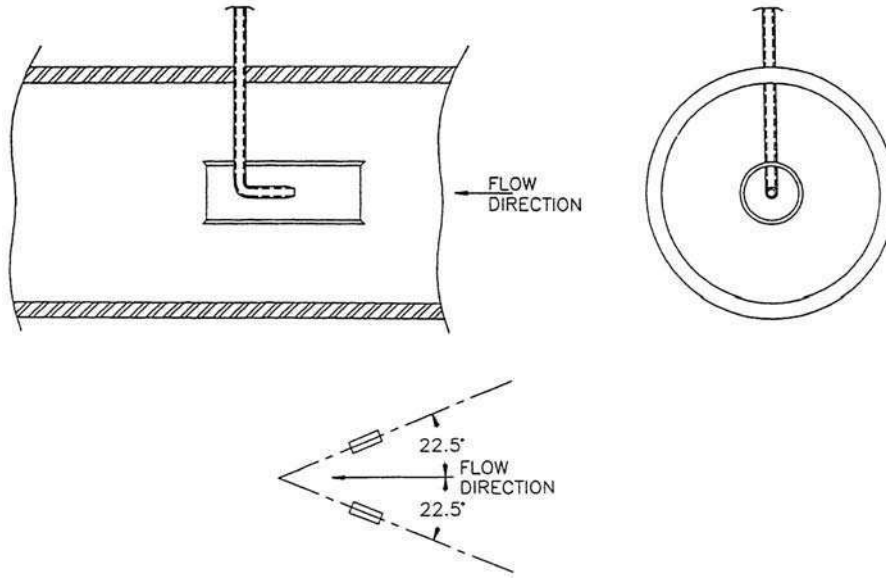


FIG. 10-11 Kiel probe. Accurate measurements can be made at angles up to 22.5° with the flow stream.

where  $k$  is the ratio of specific heat at constant pressure to that at constant volume. (See ASME Research Committee on Fluid Meters Report, op. cit., p. 105.) Coefficient  $C$  is usually close to 1.00 ( $\pm 0.01$ ) for simple pitot tubes (Fig. 10-10) and generally ranges between 0.98 and 1.00 for pitot-static tubes (Fig. 10-12).

There are certain limitations on the range of usefulness of pitot tubes. With gases, the differential is very small at low velocities; e.g., at 4.6 m/s (15.1 ft/s) the differential is only about 1.30 mm (0.051 in) of water (20°C) for air at 1 atm (20°C), which represents a lower limit for 1 percent error even when one uses a micromanometer with a precision of 0.0254 mm (0.001 in) of water. Equation does not apply for Mach numbers greater than 0.7 because of the interference of shock waves. For supersonic flow, local Mach numbers can be calculated from a knowledge of the dynamic and true static pressures. The free stream Mach number ( $M_\infty$ ) is defined as the ratio of the speed of the stream ( $V_\infty$ ) to the speed of sound in the free stream:

$$A_\infty = \sqrt{\left(\frac{\partial P}{\partial \rho}\right)_{s=c}} \quad (10-15)$$

$$M_\infty = \frac{V_\infty}{\sqrt{\left(\frac{\partial P}{\partial \rho}\right)_{s=c}}} \quad (10-16)$$

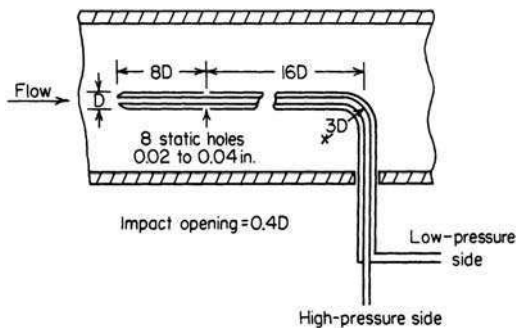


FIG. 10-12 Pitot-static tube.

where  $S$  is the entropy. For isentropic flow, this relationship and pressure can be written as:

$$M_\infty = \frac{V_\infty}{\sqrt{kRT_s}} \quad (10-17)$$

The relationships between total and static temperature and pressure are given by the following relationship:

$$\frac{T_T}{T_s} = 1 + \frac{k-1}{2} M^2 \quad (10-18)$$

$$\frac{P_T}{P_s} = \left(1 + \frac{k-1}{2} M^2\right)^{(k-1)/k} \quad (10-19)$$

With **liquids** at low velocities, the effect of the Reynolds number upon the coefficient is important. The coefficients are appreciably less than unity for Reynolds numbers less than 500 for pitot tubes and for Reynolds numbers less than 2300 for pitot-static tubes [see Folsom, *Trans. Am. Soc. Mech. Eng.*, **78**, 1447–1460 (1956)]. Reynolds numbers here are based on the probe outside diameter. Operation at low Reynolds numbers requires prior calibration of the probe.

The pitot-static tube is also sensitive to **yaw** or **angle of attack** than is the simple pitot tube because of the sensitivity of the static taps to orientation. The error involved is strongly dependent upon the exact probe dimensions. In general, angles greater than 10° should be avoided if the velocity error is to be 1 percent or less.

**Disturbances** upstream of the probe can cause large errors, in part because of the turbulence generated and its effect on the static-pressure measurement. A calming section of at least 50 pipe diameters is desirable. If this is not possible, the use of straightening vanes or a honeycomb is advisable.

The effect of **pulsating flow** on pitot-tube accuracy is treated by Ower et al., op. cit., pp. 310–312. For sinusoidal velocity fluctuations, the ratio of indicated velocity to actual mean velocity is given by the factor  $\sqrt{1 + \lambda^2/2}$ , where  $\lambda$  is the velocity excursion as a fraction of the mean velocity. Thus, the indicated velocity would be about 6 percent high for velocity fluctuations of  $\pm 50$  percent, and pulsations greater than  $\pm 20$  percent should be damped to avoid errors greater than 1 percent. The error increases as the frequency of flow oscillations approaches the natural frequency of the pitot tube and the density of

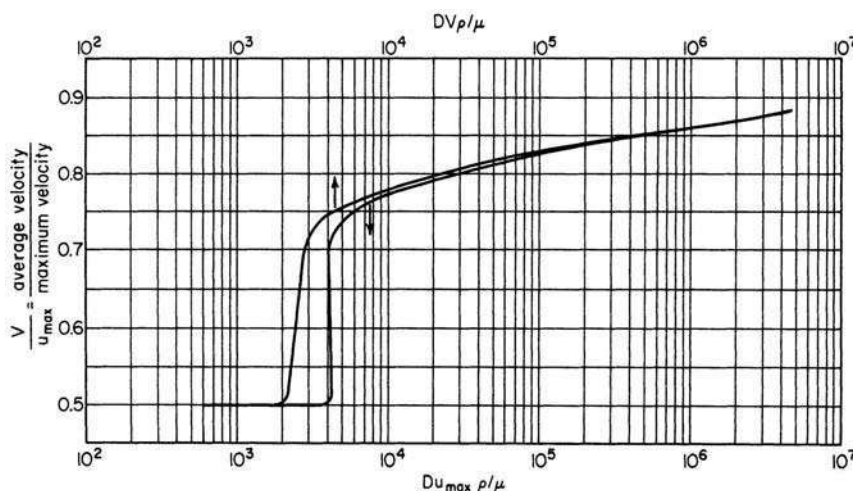


FIG. 10-13 Velocity ratio versus Reynolds number for smooth circular pipes. [Based on data from Rothfus, Archer, Klimas, and Sikchi, *Am. Inst. Chem. Eng. J.*, 3, 208 (1957).]

the measuring fluid approaches the density of the process fluid [see Horlock and Daneshyar, *J. Mech. Eng. Sci.*, **15**, 144–152 (1973)].

Pressures substantially lower than true impact pressures are obtained with pitot tubes in turbulent flow of dilute polymer solutions [see Halliwell and Lewkowicz, *Phys. Fluids*, **18**, 1617–1625 (1975)].

**Special Tubes** A variety of special forms of the pitot tube have been evolved. Folsom (loc. cit.) gives a description of many of these special types together with a comprehensive bibliography. Included are the impact tube for **boundary-layer** measurements and **shielded total-pressure tubes**. The latter are insensitive to angle of attack up to 40°.

Chue [*Prog. Aerosp. Sci.*, **16**, 147–223 (1975)] reviews the use of the pitot tube and allied pressure probes for impact pressure, static pressure, dynamic pressure, flow direction and local velocity, skin friction, and flow measurements.

A reversed pitot tube, also known as a **pitometer**, has one pressure opening facing upstream and the other facing downstream. Coefficient  $C$  for this type is on the order of 0.85. This gives about a 40 percent increase in pressure differential as compared with standard pitot tubes and is an advantage at low velocities. There are commercially available very compact types of pitometers which require relatively small openings for their insertion into a duct.

The **pitot-venturi** flow element is capable of developing a pressure differential 5 to 10 times that of a standard pitot tube. This is accomplished by employing a pair of concentric venturi elements in place of the pitot probe. The low-pressure tap is connected to the throat of the inner venturi, which in turn discharges into the throat of the outer venturi. For a discussion of performance and application of this flow element, see Stoll, *Trans. Am. Soc. Mech. Eng.*, **73**, 963–969 (1951).

**Traversing for Mean Velocity** Mean velocity in a duct can be obtained by dividing the cross section into a number of equal areas, finding the local velocity at a representative point in each, and averaging the results. In the case of **rectangular passages**, the cross section is usually divided into small squares or rectangles and the velocity is found at the center of each. In circular pipes, the cross section is divided into several equal annular areas as shown in Fig. 10-13. Readings of velocity are made at the intersections of a diameter and the set of circles which bisect the annuli and the central circle.

For an  $N$ -point traverse on a circular cross section, make readings on each side of the cross section at

$$100 \times \sqrt{(2n-1)/N} \text{ percent} \quad (n = 1, 2, 3 \text{ to } N/2)$$

of the pipe radius from the center. Traversing several diameters spaced at equal angles about the pipe is required if the velocity distribution is unsymmetrical.

With a normal velocity distribution in a circular pipe, a 10-point traverse theoretically gives a mean velocity 0.3 percent high; a 20-point traverse, 0.1 percent high.

For normal velocity distribution in straight circular pipes at locations preceded by runs of at least 50 diameters without pipe fittings or other obstructions, the graph in Fig. 10-13 shows the ratio of mean velocity  $V$  to velocity at the center  $u_{\max}$  plotted against the Reynolds number, where  $D$  = inside pipe diameter,  $\rho$  = fluid density, and  $\mu$  = fluid viscosity, all in consistent units. Mean velocity is readily determined from this graph and a pitot reading at the center of the pipe if the quantity  $Du_{\max}\rho/\mu$  is less than 2000 or greater than 5000. The method is unreliable at intermediate values of the Reynolds number.

Methods for determining mean flow rate from probe measurements under nonideal conditions are described by Mandersloot, Hicks, and Langejan [*Chem. Eng. (London)*, no. 232, CE370–CE380 (1969)].

The **hot-wire anemometer** consists essentially of an electrically heated fine wire (generally platinum) exposed to the gas stream whose velocity is being measured. An increase in fluid velocity, other things being equal, increases the rate of heat flow from the wire to the gas, thereby tending to cool the wire and alter its electrical resistance. In a constant-current anemometer, gas velocity is determined by measuring the resulting wire resistance; in the constant-resistance type, gas velocity is determined from the current required to maintain the wire temperature, and thus the resistance, constant. The difference in the two types is primarily in the electric circuits and instruments employed.

The hot-wire anemometer can, with suitable calibration, accurately measure velocities from about 0.15 m/s (0.5 ft/s) to supersonic velocities and detect velocity fluctuations with frequencies up to 200,000 Hz. Fairly rugged, inexpensive units can be built for the measurement of mean velocities in the range of 0.15 to 30 m/s (about 0.5 to 100 ft/s). More elaborate, compensated units are commercially available for use in unsteady flow and turbulence measurements. In calibrating a hot-wire anemometer, it is preferable to use the same gas, temperature, and pressure as will be encountered in the intended application. In this case the quantity  $I^2 R_w / \Delta t$  can be plotted against  $\sqrt{V}$ , where  $I$  = hot-wire current,  $R_w$  = hot-wire resistance,  $\Delta t$  = difference between the wire temperature and the gas bulk temperature, and  $V$  = mean local velocity. A procedure is given by Wasan and Baid [*Am. Inst. Chem. Eng. J.*, **17**, 729–731 (1971)] for use when it is impractical to calibrate with the same gas composition or conditions of temperature and pressure. Andrews, Bradley, and Hundy [*Int. J. Heat Mass Transfer*, **15**, 1765–1786 (1972)] give a calibration correlation for measurement

of small gas velocities. The hot-wire anemometer is treated in considerable detail in Dean, *op. cit.*, chap. VI; in Ladenburg et al., *op. cit.*, art. F-2; by Grant and Kronauer, *Symposium on Measurement in Unsteady Flow*, American Society of Mechanical Engineers, New York, 1962, pp. 44–53; ASME Research Committee on Fluid Meters Report, *op. cit.*, pp. 105–107; and by Compte-Bellot, *Ann. Rev. Fluid Mech.*, **8**, pp. 209–231 (1976).

The hot-wire anemometer can be modified for liquid measurements, although difficulties are encountered because of bubbles and dirt adhering to the wire. See Stevens, Borden, and Strausser, David Taylor Model Basin Rep. 953, December 1956; Middlebrook and Piret, *Ind. Eng. Chem.*, **42**, 1511–1513 (1950); and Piret et al., *Ind. Eng. Chem.*, **39**, 1098–1103 (1947).

The **hot-film anemometer** has been developed for applications in which use of the hot-wire anemometer presents problems. It consists of a platinum-film sensing element deposited on a glass substrate. Various geometries can be used. The most common involves a wedge with a 30° included angle at the end of a tapered rod. The wedge is commonly 1 mm (0.039 in) long and 0.2 mm (0.0079 in) wide on each face. Compared with the hot wire, it is less susceptible to fouling by bubbles or dirt when used in liquids, has greater mechanical strength when used with gases at high velocities and high temperatures, and can give a higher signal-to-noise ratio. For additional information see Ling and Hubbard, *J. Aeronaut. Sci.*, **23**, 890–891 (1956); and Ling, *J. Basic Eng.*, **82**, 629–634 (1960).

The **heated-thermocouple anemometer** measures gas velocity from the cooling effect of the gas stream flowing across the hot junctions of a thermopile supplied with constant electrical power input. Alternate junctions are maintained at ambient temperature, thus compensating for the effect of ambient temperature. For details see Bunker, *Proc. Instrum. Soc. Am.*, **9**, pap. 54-43-2 (1954).

A glass-coated bead **thermistor anemometer** can be used for the measurement of low fluid velocities, down to 0.001 m/s (0.003 ft/s) in air and 0.0002 m/s (0.0007 ft/s) in water [see Murphy and Sparks, *Ind. Eng. Chem. Fundam.*, **7**, 642–645 (1968)].

The **laser-Doppler anemometer** measures local fluid velocity from the change in frequency of radiation, between a stationary source and a receiver, due to scattering by particles along the wave path. A laser is commonly used as the source of incident illumination. The measurements are essentially independent of local temperature and pressure. This technique can be used in many different flow systems with transparent fluids containing particles whose velocity is actually measured. For a brief review of the laser-Doppler technique see Goldstein, *Appl. Mech. Rev.*, **27**, 753–760 (1974). For additional details see Durst, Melling, and Whitelaw, *Principles and Practice of Laser-Doppler Anemometry*, Academic, New York, 1976.

## FLOWMETERS

In the process industries, flow measurement devices are the largest market in the process instrumentation field. Two web sites for process equipment and instrumentation, [www.globalspec.com](http://www.globalspec.com), and [www.thomasnet.com](http://www.thomasnet.com), both list more than 800 companies that offer flow measurement products. There are more than one hundred types of flowmeters commercially available. The aforementioned web sites not only facilitate selection and specification of commercial flowmeters, but also provide electronic access to manufacturers' technical literature.

Devices that measure flow can be categorized in two areas as follows:

1. All types of measuring devices in which the material passes without being divided into isolated quantities. Movement of the material is usually sensed by a primary measuring element which activates a secondary device. The flow rate is then inferred from the response of the secondary device by means of known physical laws or from empirical relationships.

2. A positive-displacement meter, which applies to a device in which the flow is divided into isolated measured volumes. The number of fillings of these known volumes are measured with respect to time.

The most common application of flow measurement in process plants is flow in pipes, ducts, and tubing. Table 10-4 lists widely used

flowmeters for these closed conduits as well as the two major classes of open-channel flowmeters. Table 10-4 also lists many other types of flowmeters that are discussed later in this subsection.

This subsection summarizes selection and installation of flowmeters, including the measurement of pressure and velocities of fluids when the flow measurement technique requires it.

## INDUSTRY GUIDELINES AND STANDARDS

Because flow measurement is important, many engineering societies and trade organizations have developed flow-related guidelines, standards, and other publications (Table 10-5). The reader should consult the appropriate standards when specifying, installing, and calibrating flow measurement systems.

There are also numerous articles in scholarly journals, trade magazines, and manufacturers' literature related to flow measurement.

Different types of flowmeters differ markedly in their degrees of sensitivity to flow disturbances. In the most extreme cases, obtaining highly accurate flow measurements with certain types of flowmeters may require 60D upstream straight pipe and 20D downstream. Valves can be particularly problematic because their effects on a flowmeter vary with valve position. Numerous types of flow straighteners or conditioners, as shown in Fig. 10-9, can significantly reduce the required run of straight pipe upstream of a given flowmeter.

## CLASSIFICATION OF FLOWMETERS

Table 10-4 lists the major classes of flowmeters, along with common examples of each. Brief descriptions are provided in this subsection, followed by more details in subsequent subsections.

**Differential Pressure Meters** Differential pressure meters or head meters measure the change in pressure across a special flow element. The differential pressure increases with increasing flow rate. The pitot tubes described previously work on this principle. Other examples include orifices [see also Eqs. (6-111) and (8-102)], and Fig. 10-14], nozzles (Fig. 10-19), targets, venturis (see also Sec. 8 and Fig. 10-17), and elbow meters. Averaging pitot tubes produce a pressure differential that is based on multiple measuring points across the flow path.

Differential pressure meters are widely used. Temperature, pressure, and density affect gas density and readings of differential pressure meters. For that reason, many commercial flowmeters that are based on measurement of differential pressure often have integral temperature and absolute pressure measurements in addition to differential pressure. They also frequently have automatic temperature and pressure compensation.

**Velocity Meters** Velocity meters measure fluid velocity. Examples include electromagnetic, propeller, turbine, ultrasonic Doppler, ultrasonic transit time, and vortex meters. Section 8 describes the principles of operation of electromagnetic, turbine, ultrasonic, and vortex flowmeters.

**Mass Meters** Mass flowmeters measure the rate of mass flow through a conduit. Examples include Coriolis flowmeters and thermal mass flowmeters. Coriolis flowmeters can measure fluid density simultaneously with mass flow rate. This permits calculation of volumetric flow rate as well. Section 8 includes brief descriptions of Coriolis and thermal mass flowmeters.

**Volumetric Meters** Volumetric meters (also called positive-displacement flowmeters) are devices that mechanically divide a fluid stream into discrete, known volumes and count the number of volumes that pass through the device. See Spitzer (2005, *op. cit.*).

**Variable-Area Meters** Variable-area meters, which are also called rotameters, offer popular and inexpensive flow measurement devices. These meters employ a float inside a tube that has an internal cross-sectional area that increases with distance upward in the flow path through the tube. As the flow rate increases, the float rises in the tube to provide a larger area for the flowing fluid to pass.

**Open-Channel Flow Measurement** Open-channel flow measurements are usually based on measurement of liquid level in a flow channel constructed of a specified geometry. The two most common flow channels used are weirs and flumes. See Spitzer (2005, *op. cit.*).

TABLE 10-4 Comparison of Flowmeter Technologies

| Flowmeter technology                        | Accuracy*<br>(+/-) | Turndown          | Fluids†     | Pipe sizes, ‡<br>in | Maximum pressure, ‡<br>psig | Temperature range, ‡<br>°F | Pipe run | Relative pressure loss |
|---------------------------------------------|--------------------|-------------------|-------------|---------------------|-----------------------------|----------------------------|----------|------------------------|
| Differential Pressure Meters                |                    |                   |             |                     |                             |                            |          |                        |
| Pitot                                       |                    | 8:1               | L, G        | 1 to 96             | 2.76E-4 to 200              | -200 to 750                |          | L                      |
| Averaging pitot                             | 1% R               | 8:1               | L, G, S     | 0.25 to 72          | 8800                        | -20 to 2370                |          | L                      |
| Orifice                                     |                    |                   |             |                     |                             |                            | Long     |                        |
| Square-edged                                | 0.5 to 1.5% R      | 4:1               | L, G, S     | 0.5 to 40           | 8800                        | -4 to 2300                 |          | M                      |
| Eccentric                                   | 2% R               | 4:1               | L, G, S     | 0.5 to 40           | 8800                        | -4 to 2300                 |          | M                      |
| Segmental                                   | 2% R               | 4:1               | L, G, S, SL | 0.5 to 40           | 8800                        | -4 to 2300                 |          | M                      |
| Orifice and multi-variable flow transmitter | 0.5 to 1% R        | 10:1              | L, G, S     | > 0.5               | 4000                        | 1000                       |          | M                      |
| Venturi                                     | 0.5 to 1.5% R      | 10:1              | L, G, S, SL | 1 to 120            | 8800                        | -4 to 2300                 |          | L                      |
| Flow nozzle                                 | 0.5 to 2% R        | 8:1               | L, G, S     | 2 to 80             | >1000                       | <1000                      |          | M                      |
| V cone                                      | 0.5% F             | 10:1              | L, G, S, SL | 0.25 to 120         | 6000                        | To 400                     |          | M                      |
| Wedge                                       | 0.5 to 5% R        | 10:1              | L, G, S, SL | 0.5 to 24           | >600                        | ‡                          |          | M                      |
| Velocity Meters                             |                    |                   |             |                     |                             |                            |          |                        |
| Correlation                                 | 0.5% R             | 10:1              | L, G, SL    | 1 to 60             | Piping limits               | To 600                     | Long     | L                      |
| Electromagnetic                             | 0.2 to 2% R        | 10:1              | L           | 0.15 to 60          | 5000                        | -40 to 350                 | Short    | L                      |
| Propeller                                   | 2% R               | 15:1              | L           | 2 to 12             | 230                         | 0 to 300                   |          | M                      |
| Turbine                                     | 0.15 to 1% R       | 10:1              | L, G        | 0.5 to 30           | 6000                        | -450 to 600                | Short    | M                      |
| Ultrasonic Doppler                          | 1 to 30% R         | 50:1              | L, G, SL    | 0.5 to 200          | 6000                        | -40 to 250                 | Long     | L                      |
| Ultrasonic transit time                     | 0.5 to 5% R        | Down to zero flow | L, G        | 1 to 540            | 6000                        | -40 to 650                 | Long     | L                      |
| Vortex                                      | 0.5 to 2% R        | 20:1              | L, G, S     | 0.5 to 16           | 1500                        | -330 to 800                | Short    | M                      |
| Mass Meters                                 |                    |                   |             |                     |                             |                            |          |                        |
| Coriolis                                    | 0.1 to 0.3% R      | 10:1 to 80:1      | L, G        | 0.06 to 12          | 5700                        | -400 to 800                | None     | L, M                   |
| Thermal (for gases)                         | 1% F               | 50:1              | G           | 0.125 to 8          | 4500                        | 32 to 572                  | Short    | L                      |
| Thermal (for liquids)                       | 0.5% F             | 50:1              | L           | 0.06 to 0.25        | 4500                        | 40 to 165                  | Short    | L                      |
| Volumetric                                  | 0.15 to 2% R       | 10:1              | L           | 0.25 to 16          | 2000                        | -40 to 600                 | None     | M to H                 |
| Variable area                               | 1 to 5% F          | 10:1              | L, G        | 0.125 to 6          | 6000                        | <1000                      | None     | M                      |
| Open-Channel Flowmeters                     |                    |                   |             |                     |                             |                            |          |                        |
| Weirs                                       | 2 to 5% R          | 25:1              | L           | Wide range          | NA                          | NA                         |          | L                      |
| Flumes                                      | 3 to 10% R         | 40:1              | L           | Wide range          | NA                          | NA                         |          | L                      |

\*F = full scale, R = rate. †L = liquid, G = gas, S = steam, SL = slurry.

‡Dependent on the material selection and application. Readers should consult manufacturers for current capabilities.

Adapted from J. Pomroy, *Chemical Engineering*, pp. 94–102, May 1996; J. W. Dolenc, *Chemical Engineering Progress*, pp. 22–32, Jan. 1996; R. C. Baker, *Introductory Guide to Flow Measurement*, American Society of Mechanical Engineers, New York, 2003; R. W. Miller, *Flow Measurement Engineering Handbook*, 3d ed., McGraw-Hill, New York, 1996; D. W. Spitzer, *Industrial Flow Measurement*, 3d ed., The Instrumentation, Systems, and Automation Society, Research Triangle Park, N.C., 2005; and manufacturers' literature at [www.globalspec.com](http://www.globalspec.com).

## DIFFERENTIAL PRESSURE FLOWMETERS

**General Principles** If a constriction is placed in a closed channel carrying a stream of fluid, there will be an increase in velocity, and hence an increase in kinetic energy, at the point of constriction. From an energy balance, as given by Bernoulli's theorem [see Sec. 6, subsection "Energy Balance," Eq. (6-16)], there must be a corresponding reduction in pressure. Rate of discharge from the constriction can be calculated by knowing this pressure reduction, the area available for flow at the constriction, the density of the fluid, and the coefficient of discharge  $C$ . The last-named is defined as the ratio of actual flow to the theoretical flow and makes allowance for stream contraction and frictional effects. The metering characteristics of commonly used differential pressure meters are reviewed and grouped by Halimi [*J. Fluids Eng.*, **95**, 127–141 (1973)].

The term **static head** generally denotes the pressure in a fluid due to the head of fluid above the point in question. Its magnitude is given by the application of Newton's law (force = mass  $\times$  acceleration). In the case of **liquids** (constant density), the static head  $p_h$  Pa (lbf/ft<sup>2</sup>) is given by

$$p_h = h\rho g/g_c \quad (10-20)$$

where  $h$  = head of liquid above the point, m (ft);  $\rho$  = liquid density;  $g$  = local acceleration due to gravity; and  $g_c$  = dimensional constant.

The head developed in a compressor or pump is the energy force per unit mass. In the measuring systems it is often misnamed as (ft) while the units are really ft-lb/lbm or kilojoules.

For a compressor or turbine, it is represented by the following relationship:

$$E = U_1 V_{\theta 1} - U_2 V_{\theta 2} \quad (10-21)$$



**TABLE 10-5 Guidelines, Standards, and Other Publications Related to Flow Measurement**

| Technical society                                                                   | Number of guidelines and standards* |
|-------------------------------------------------------------------------------------|-------------------------------------|
| American Gas Association (AGA)                                                      | 2                                   |
| American Petroleum Institute (API)                                                  | 11                                  |
| American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE) | 5                                   |
| American Society of Mechanical Engineers (ASME)                                     | 18                                  |
| ASTM International (ASTM)                                                           | 17                                  |
| British Standards Institution (BSI)                                                 | 100                                 |
| Deutsches Institut für Normung E. V. (DIN)                                          | 48                                  |
| International Electrotechnical Commission (IEC)                                     | 6                                   |
| Instrumentation, Systems, and Automation Society (ISA)                              | 3                                   |
| International Organization for Standardization (ISO)                                | 212                                 |
| SAE International (SAE)                                                             | 6                                   |

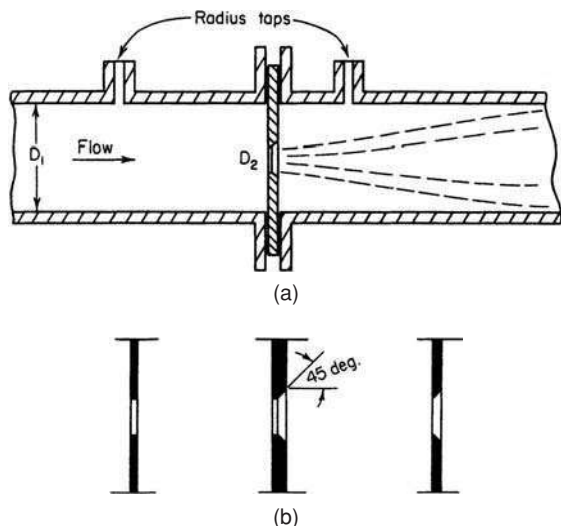
\*Number of documents identified by searching for flow measurement on <http://global.ihl.com>, the web site of a clearinghouse of industry guidelines, codes, and standards.

where  $U$  is the blade speed and  $V_\theta$  is the tangential velocity component of absolute velocity. This equation is known as the Euler equation.

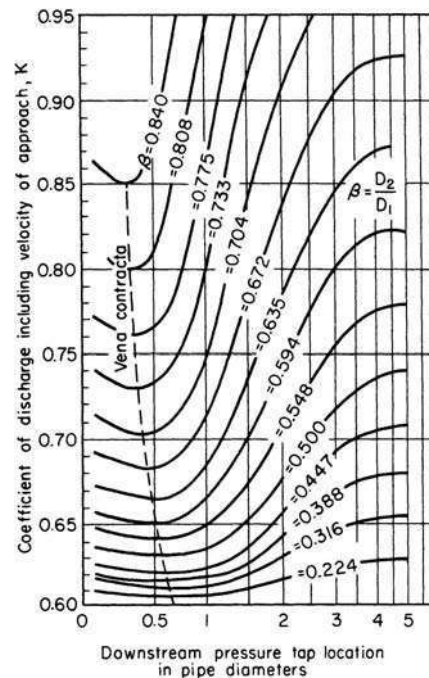
**Orifice Meters** A square-edged or sharp-edged orifice, as shown in Fig. 10-14, is a clean-cut square-edged hole with straight walls perpendicular to the flat upstream face of a thin plate placed crosswise of the channel. The stream issuing from such an orifice attains its minimum cross section (vena contracta) at a distance downstream of the orifice which varies with the ratio  $\beta$  of orifice to pipe diameter (see Fig. 10-15).

For a centered circular orifice in a pipe, the pressure differential is customarily measured between one of the following pressure-tap pairs. Except in the case of flange taps, all measurements of distance from the orifice are made from the upstream face of the plate.

1. **Corner taps.** Static holes drilled one in the upstream and one in the downstream flange, with the openings as close as possible to the orifice plate.
2. **Radius taps.** Static holes located one pipe diameter upstream and one-half pipe diameter downstream from the plate.
3. **Pipe taps.** Static holes located  $2\frac{1}{2}$  pipe diameters upstream and eight pipe diameters downstream from the plate.



**FIG. 10-14** Square-edged or sharp-edged orifices. The plate at the orifice opening must not be thicker than one-thirtieth of the pipe diameter, one-eighth of the orifice diameter, or one-fourth of the distance from the pipe wall to the edge of the opening. (a) Pipe-line orifice. (b) Types of plates.



**FIG. 10-15** Coefficient of discharge for square-edged circular orifices for  $N_{Re} > 30,000$  with the upstream tap located between one and two pipe diameters from the orifice plate. [Spitzglass, Trans. Am. Soc. Mech. Eng., 44, 919 (1922).]

4. **Flange taps.** Static holes located 25.4 mm (1 in) upstream and 25.4 mm (1 in) downstream from the plate.

5. **Vena-contracta taps.** The upstream static hole is one-half to two pipe diameters from the plate. The downstream tap is located at the position of minimum pressure (see Fig. 10-15).

Radius taps are best from a practical standpoint; the downstream pressure tap is located at about the mean position of the vena contracta, and the upstream tap is sufficiently far upstream to be unaffected by distortion of the flow in the immediate vicinity of the orifice (in practice, the upstream tap can be as much as two pipe diameters from the plate without affecting the results). Vena-contracta taps give the largest differential head for a given rate of flow but are inconvenient if the orifice size is changed from time to time. Corner taps offer the sometimes great advantage that the pressure taps can be built into the plate carrying the orifice. Thus the entire apparatus can be quickly inserted in a pipe line at any convenient flanged joint without having to drill holes in the pipe. Flange taps are similarly convenient, since by merely replacing standard flanges with special orifice flanges, suitable pressure taps are made available. Pipe taps give the lowest differential pressure, the value obtained being close to the permanent pressure loss.

The practical working equation for weight rate of discharge, adopted by the ASME Research Committee on Fluid Meters for use with either gases or liquids, is

$$w = q_1 p_1 = C Y A_2 \sqrt{\frac{2g_c(p_1 - p_2)\rho_1}{1 - \beta^4}} \\ = K Y A_2 \sqrt{2g_c(p_1 - p_2)\rho_1} \quad (10-22)$$

where  $A_2$  = cross-sectional area of throat;  $C$  = coefficient of discharge, dimensionless;  $g_c$  = dimensional constant;  $K = C/\sqrt{1 - \beta^4}$ , dimensionless;  $p_1, p_2$  = pressure at upstream and downstream static pressure taps respectively;  $q_1$  = volumetric rate of discharge measured at upstream pressure and temperature;  $w$  = weight rate of discharge;  $Y$  = expansion factor, dimensionless;  $\beta$  = ratio of throat diameter to

pipe diameter, dimensionless; and  $\rho_1$  = density at upstream pressure and temperature.

For the case of subsonic flow of a gas ( $r_c < r < 1.0$ ), the expansion factor  $Y$  for orifices is approximated by

$$Y = 1 - [(1 - r)/k](0.41 + 0.35\beta^4) \quad (10-23)$$

where  $r$  = ratio of downstream to upstream static pressure ( $p_2/p_1$ ),  $k$  = ratio of specific heats ( $c_p/c_v$ ), and  $\beta$  = diameter ratio. (See also Fig. 10-18.) Values of  $Y$  for supercritical flow of a gas ( $r < r_c$ ) through orifices are given by Benedict [J. Basic Eng., **93**, 121–137 (1971)]. For the case of **liquids**, expansion factor  $Y$  is unity, and Eq. (10-27) should be used, since it allows for any difference in elevation between the upstream and downstream taps.

**Coefficient of discharge  $C$**  for a given orifice type is a function of the Reynolds number  $N_{Re}$  (based on orifice diameter and velocity) and diameter ratio  $\beta$ . At Reynolds numbers greater than about 30,000, the coefficients are substantially constant. For square-edged or sharp-edged concentric circular orifices, the value will fall between 0.595 and 0.620 for vena-contracta or radius taps for  $\beta$  up to 0.8 and for flange taps for  $\beta$  up to 0.5. Figure 10-15 gives the coefficient of discharge  $K$ , including the velocity-of-approach factor ( $1/\sqrt{1 - \beta^4}$ ), as a function of  $\beta$  and the location of the downstream tap. Precise values of  $K$  are given in ASME PTC, op. cit., pp. 20–39, for flange taps, radius taps, vena-contracta taps, and corner taps. Precise values of  $C$  are given in the ASME Research Committee on Fluid Meters Report, op. cit., pp. 202–207, for the first three types of taps.

The discharge coefficient of sharp-edged orifices was shown by Benedict, Wyler, and Brandt [J. Eng. Power, **97**, 576–582 (1975)] to increase with edge roundness. Typical as-purchased orifice plates may exhibit deviations on the order of 1 to 2 percent from ASME values of the discharge coefficient.

In the transition region ( $N_{Re}$  between 50 and 30,000), the coefficients are generally higher than the above values. Although calibration is generally advisable in this region, the curves given in Fig. 10-16 for corner and vena-contracta taps can be used as a guide. In the laminar-flow region ( $N_{Re} < 50$ ), the coefficient  $C$  is proportional to  $\sqrt{N_{Re}}$ . For  $1 < N_{Re} < 100$ , Johansen [Proc. R. Soc. (London), **A121**, 231–245 (1930)] presents discharge-coefficient data for sharp-edged orifices with corner taps. For  $N_{Re} < 1$ , Miller and Nemecek [ASME Paper 58-A-106 (1958)] present correlations giving coefficients for sharp-edged orifices and short-pipe orifices ( $L/D$  from 2 to 10). For short-pipe orifices ( $L/D$  from 1 to 4), Dickerson and Rice [J. Basic Eng., **91**, 546–548 (1969)] give coefficients for the intermediate range ( $27 < N_{Re} < 7000$ ). See also subsection "Contraction and Entrance Losses."

**Permanent pressure loss** across a concentric circular orifice with radius or vena-contracta taps can be approximated for turbulent flow by

$$(p_1 - p_4)/(p_1 - p_2) = 1 - \beta^2 \quad (10-24)$$

where  $p_1$ ,  $p_2$  = upstream and downstream pressure-tap readings respectively,  $p_4$  = fully recovered pressure (four to eight pipe diameters downstream of the orifice), and  $\beta$  = diameter ratio. See ASME PTC, op. cit., Fig. 5.

See Benedict, J. Fluids Eng., **99**, 245–248 (1977), for a general equation for pressure loss for orifices installed in pipes or with plenum inlets. Orifices show higher loss than nozzles or venturis. Permanent pressure loss for laminar flow depends on the Reynolds number in addition to  $\beta$ . See Alvi, Sridharan, and Lakshmana Rao, loc. cit., for details.

For the case of **critical flow** through a square- or sharp-edged concentric circular orifice (where  $r \leq r_c$ , as discussed earlier in this subsection), use Eqs. (10-31), (10-32), and (10-33) as given for critical-flow nozzles. However, unlike nozzles, the flow through a sharp-edged orifice continues to increase as the downstream pressure drops below that

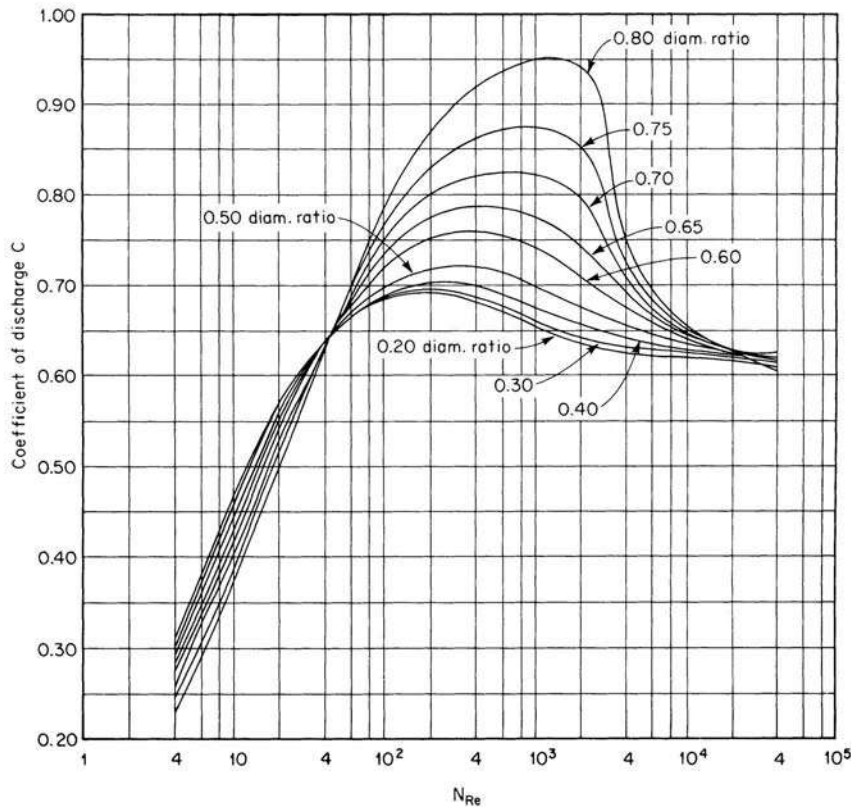


FIG. 10-16 Coefficient of discharge for square-edged circular orifices with corner taps. [Tute and Sprenkle, Instruments, **6**, 201 (1933).]

corresponding to the critical pressure ratio  $r_c$ . This is due to an increase in the cross section of the vena contracta as the downstream pressure is reduced, giving a corresponding increase in the coefficient of discharge. At  $r = r_c$ ,  $C$  is about 0.75, while at  $r \approx 0$ ,  $C$  has increased to about 0.84. See Grace and Lapple, loc. cit.; and Benedict, *J. Basic Eng.*, **93**, 99–120 (1971).

Measurements by Harris and Magnall [*Trans. Inst. Chem. Eng. (London)*, **50**, 61–68 (1972)] with a venturi ( $\beta = 0.62$ ) and orifices with radius taps ( $\beta = 0.60 - 0.75$ ) indicate that the discharge coefficient for **nonnewtonian fluids**, in the range  $N_{Re}$  (generalized Reynolds number) 3500 to 100,000, is approximately the same as for newtonian fluids at the same Reynolds number.

**Quadrant-edge orifices** have holes with rounded edges on the upstream side of the plate. The quadrant-edge radius is equal to the thickness of the plate at the orifice location. The advantages claimed for this type versus the square- or sharp-edged orifice are constant-discharge coefficients extending to lower Reynolds numbers and less possibility of significant changes in coefficient because of erosion or other damage to the inlet shape.

Values of discharge coefficient  $C$  and Reynolds numbers limit for constant  $C$  are presented in Table 10-6, based on Ramamoorthy and Seetharamiah [*J. Basic Eng.*, **88**, 9–13 (1966)] and Bogema and Monkneyer [*J. Basic Eng.*, **82**, 729–734 (1960)]. At Reynolds numbers above those listed for the upper limits, the coefficients rise abruptly. As Reynolds numbers decrease below those listed for the lower limits, the coefficients pass through a hump and then drop off. According to Bogema, Spring, and Ramamoorthy [*J. Basic Eng.*, **84**, 415–418 (1962)], the hump can be eliminated by placing a fine-mesh screen about three pipe diameters upstream of the orifice. This reduces the lower  $N_{Re}$  limit to about 500.

Permanent pressure loss across quadrant-edge orifices for turbulent flow is somewhat lower than given by Eq. (10-24). See Alvi, Sridharan, and Lakshmana Rao, loc. cit., for values of discharge coefficient and permanent pressure loss in laminar flow.

**Slotted orifices** offer significant advantages over a standard square-edged orifice with an identical open area for homogeneous gases or liquids [G. L. Morrison and K. R. Hall, *Hydrocarbon Processing* **79**, 12, 65–72 (2000)]. The slotted orifice flowmeter only requires compact header configurations with very short upstream pipe lengths and maintains accuracy in the range of 0.25 percent with no flow conditioner. Permanent head loss is less than or equal to that of a standard orifice that has the same  $\beta$  ratio. Discharge coefficients for the slotted orifice are much less sensitive to swirl or to axial velocity profiles. A slotted orifice plate can be a “drop in” replacement for a standard orifice plate.

**Segmental and eccentric orifices** are frequently used for gas metering when there is a possibility that entrained liquids or solids would otherwise accumulate in front of a concentric circular orifice. This can be avoided if the opening is placed on the lower side of the pipe. For liquid flow with entrained gas, the opening is placed on the upper side. The pressure taps should be located on the opposite side of the pipe from the opening.

Coefficient  $C$  for a square-edged eccentric circular orifice (with opening tangent to pipe wall) varies from about 0.61 to 0.63 for  $\beta$ 's from 0.3 to 0.5, respectively, and pipe Reynolds numbers  $> 10,000$  for either vena-

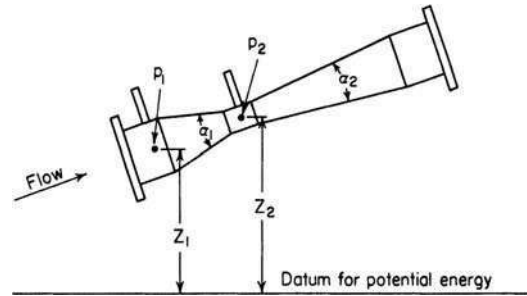


FIG. 10-17 Herschel-type venturi tube.

contracta or flange taps (where  $\beta$  = diameter ratio). For square-edged segmental orifices, the coefficient  $C$  falls generally between 0.63 and 0.64 for  $0.3 \leq \beta \leq 0.5$  and pipe Reynolds numbers  $> 10,000$ , for vena-contracta or flange taps, where  $\beta$  = diameter ratio for an equivalent circular orifice =  $\sqrt{\alpha}$  ( $\alpha$  = ratio of orifice to pipe cross-sectional areas). Values of expansion factor  $Y$  are slightly higher than for concentric circular orifices, and the location of the vena contracta is moved farther downstream as compared with concentric circular orifices. For further details, see ASME Research Committee on Fluid Meters Report, op. cit., pp. 210–213.

For permanent pressure loss with segmental and eccentric orifices with laminar pipe flow see Lakshmana Rao and Sridharan, *Proc. Am. Soc. Civ. Eng., J. Hydraul. Div.*, **98** (HY 11), 2015–2034 (1972).

**Annular orifices** can also be used to advantage for gas metering when there is a possibility of entrained liquids or solids and for liquid metering with entrained gas present in small concentrations. Coefficient  $K$  was found by Bell and Bergelin [*Trans. Am. Soc. Mech. Eng.*, **79**, 593–601 (1957)] to range from about 0.63 to 0.67 for annulus Reynolds numbers in the range of 100 to 20,000 respectively for values of  $2L/(D - d)$  less than 1 where  $L$  = thickness of orifice at outer edge,  $D$  = inside pipe diameter, and  $d$  = diameter of orifice disk. The annulus Reynolds number is defined as

$$N_{Re} = (D - d)(G/\mu) \quad (10-25)$$

where  $G$  = mass velocity  $\rho V$  through orifice opening and  $\mu$  = fluid viscosity. The above coefficients were determined for  $\beta$ 's ( $= d/D$ ) in the range of 0.95 to 0.996 and with pressure taps located 19 mm ( $3/4$  in) upstream of the disk and 230 mm (9 in) downstream in a 5.25-in-diameter pipe.

**Venturi Meters** The standard Herschel-type venturi meter consists of a short length of straight tubing connected at either end to the pipe line by conical sections (see Fig. 10-17). Recommended proportions (ASME PTC, op. cit., p. 17) are entrance cone angle  $\alpha_1 = 21 \pm 2^\circ$ , exit cone angle  $\alpha_2 = 5$  to  $15^\circ$ , throat length = one throat diameter, and upstream tap located 0.25 to 0.5 pipe diameter upstream of the entrance cone. The straight and conical sections should be joined by smooth curved surfaces for best results. **Rate of discharge** of either gases or liquids through a venturi meter is given by Eq. (10-22).

For the flow of **gases**, expansion factor  $Y$ , which allows for the change in gas density as it expands adiabatically from  $p_1$  to  $p_2$ , is given by

$$Y = \sqrt{r^{2/k} \left( \frac{k}{k-1} \right) \left( \frac{1-r^{(k-1)/k}}{1-r} \right) \left( \frac{1-\beta^4}{1-\beta^4 r^{2/k}} \right)} \quad (10-26)$$

for venturi meters and flow nozzles, where  $r = p_2/p_1$  and  $k$  = specific heat ratio  $c_p/c_v$ . Values of  $Y$  computed from Eq. (10-26) are given in Fig. 10-18 as a function of  $r$ ,  $k$ , and  $\beta$ .

For the flow of **liquids**, expansion factor  $Y$  is unity. The change in potential energy in the case of an inclined or vertical venturi meter must be allowed for. Equation (10-22) is accordingly modified to give

$$w = q_1 \rho = C A_2 \sqrt{\frac{[2g_c(p_1 - p_2) + 2g\rho(Z_1 - Z_2)]\rho}{1 - \beta^4}} \quad (10-27)$$

TABLE 10-6 Discharge Coefficients for Quadrant-Edge Orifices

| $\beta$ | $C \dagger$ | $K \dagger$ | Limiting $N_{Re}^*$ for constant coefficient† |         |
|---------|-------------|-------------|-----------------------------------------------|---------|
|         |             |             | Lower                                         | Upper   |
| 0.225   | 0.770       | 0.771       | 5,000                                         | 60,000  |
| 0.400   | 0.780       | 0.790       | 5,000                                         | 150,000 |
| 0.500   | 0.824       | 0.851       | 4,000                                         | 200,000 |
| 0.600   | 0.856       | 0.918       | 3,000                                         | 120,000 |
| 0.630   | 0.885       | 0.964       | 3,000                                         | 105,000 |

\*Based on pipe diameter and velocity.

†For a precision of about  $\pm 0.5$  percent.

‡Can be used with corner taps, flange taps, or radius taps.

where  $g$  = local acceleration due to gravity and  $Z_1, Z_2$  = vertical heights above an arbitrary datum plane corresponding to the centerline pressure-reading locations for  $p_1$  and  $p_2$  respectively.

Value of the **discharge coefficient**  $C$  for a **Herschel-type venturi meter** depends upon the Reynolds number and to a minor extent upon the size of the venturi, increasing with diameter. A plot of  $C$  versus pipe Reynolds number is given in *ASME PTC*, op. cit., p. 19. A value of 0.984 can be used for pipe Reynolds numbers larger than 200,000.

**Permanent pressure loss** for a Herschel-type venturi tube depends upon diameter ratio  $\beta$  and discharge cone angle  $\alpha_2$ . It ranges from 10 to 15 percent of the pressure differential ( $p_1 - p_2$ ) for small angles (5 to 7°) and from 10 to 30 percent for large angles (15°), with the larger losses occurring at low values of  $\beta$  (see *ASME PTC*, op. cit., p. 12). See Benedict, *J. Fluids Eng.*, **99**, 245–248 (1977), for a general equation for pressure loss for venturis installed in pipes or with plenum inlets.

For flow measurement of **steam and water mixtures** with a Herschel-type venturi in 2½-in- and 3-in-diameter pipes, see Collins and Gacesa, *J. Basic Eng.*, **93**, 11–21 (1971).

A variety of **short-tube venturi meters** are available commercially. They require less space for installation and are generally (although not always) characterized by a greater pressure loss than the corresponding Herschel-type venturi meter. Discharge coefficients vary widely for different types, and individual calibration is recommended if the manufacturer's calibration is not available. Results of tests on the Dall flow tube are given by Miner [*Trans. Am. Soc. Mech. Eng.*, **78**, 475–479 (1956)] and Dowdell [*Instrum. Control Syst.*, **33**, 1006–1009 (1960)]; and on the Gentile flow tube (also called Beth flow tube or Foster flow tube) by Hooper [*Trans. Am. Soc. Mech. Eng.*, **72**, 1099–1110 (1950)].

The use of a **multiventuri system** (in which an inner venturi discharges into the throat of an outer venturi) to increase both the differential pressure for a given flow rate and the signal-to-loss ratio is described by Klomp and Sovran [*J. Basic Eng.*, **94**, 39–45 (1972)].

**Flow Nozzles** A simple form of flow nozzle is shown in Fig. 10-19. It consists essentially of a short cylinder with a flared approach section. The approach cross section is preferably elliptical in shape but may be conical. Recommended contours for long-radius flow nozzles are given in *ASME PTC*, op. cit., p. 13. In general, the length of the straight portion of the throat is about one-half throat diameter, the upstream pressure tap is located about one pipe diameter from the nozzle inlet face, and the downstream pressure tap about one-half pipe diameter from the inlet face. For subsonic flow, the pressures at points 2 and 3 will be practically identical. If a conical inlet is preferred, the inlet and throat geometry specified for a Herschel-type venturi meter can be used, omitting the expansion section.

**Rate of discharge** through a flow nozzle for subcritical flow can be determined by the equations given for venturi meters, Eq. (10-22) for gases and Eq. (10-27) for liquids. The expansion factor  $Y$  for nozzles is the same as that for venturi meters [Eq. (10-26), Fig. 10-18]. The value of the discharge coefficient  $C$  depends primarily upon the pipe Reynolds number and to a lesser extent upon the diameter ratio  $\beta$ . Curves of recommended coefficients for long-radius flow nozzles with pressure taps located one pipe diameter upstream and one-half pipe diameter downstream of the inlet face of the nozzle are given in *ASME PTC*, op. cit., p. 15. In general, coefficients range from 0.95 at a pipe Reynolds number of 10,000 to 0.99 at 1,000,000.

The performance characteristics of pipe-wall-tap nozzles (Fig. 10-19) and throat-tap nozzles are reviewed by Wyler and Benedict [*J. Eng. Power*, **97**, 569–575 (1975)].

**Permanent pressure loss** across a subsonic flow nozzle is approximated by

$$p_1 - p_4 = \frac{1 - \beta^2}{1 + \beta^2} (p_1 - p_2) \quad (10-28)$$

where  $p_1, p_2, p_4$  = static pressures measured at the locations shown in Fig. 10-19; and  $\beta$  = ratio of nozzle throat diameter to pipe diameter, dimensionless. Equation (10-28) is based on a momentum balance assuming constant fluid density (see Lapple et al., *Fluid and Particle Mechanics*, University of Delaware, Newark, 1951, p. 13).

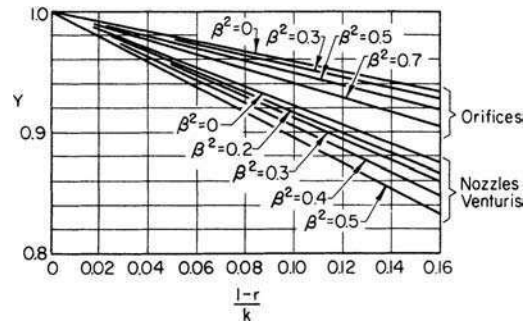


FIG. 10-18 Values of expansion factor  $Y$  for orifices, nozzles, and venturis.

See Benedict, loc. cit., for a general equation for pressure loss for nozzles installed in pipes or with plenum inlets. Nozzles show higher loss than venturis. Permanent pressure loss for laminar flow depends on the Reynolds number in addition to  $\beta$ . For details, see Alvi, Sridharan, and Lakshmana Rao, *J. Fluids Eng.*, **100**, 299–307 (1978).

**Critical Flow Nozzle** For a given set of upstream conditions, the rate of discharge of a gas from a nozzle will increase for a decrease in the absolute pressure ratio  $p_2/p_1$  until the linear velocity in the throat reaches that of sound in the gas at that location. The value of  $p_2/p_1$  for which the acoustic velocity is just attained is called the critical pressure ratio  $r_c$ . The actual pressure in the throat will not fall below  $p_1 r_c$  even if a much lower pressure exists downstream.

The **critical pressure ratio**  $r_c$  can be obtained from the following theoretical equation, which assumes a perfect gas and a frictionless nozzle:

$$r_c^{(1-k)/k} + \left( \frac{k-1}{2} \right) \beta^2 r_c^{2/k} = \frac{k+1}{2} \quad (10-29)$$

This reduces, for  $\beta \leq 0.2$ , to

$$r_c = \left( \frac{2}{k+1} \right)^{k/(k-1)} \quad (10-30)$$

where  $k$  = ratio of specific heats  $c_p/c_v$  and  $\beta$  = diameter ratio. A table of values of  $r_c$  as a function of  $k$  and  $\beta$  is given in the ASME Research Committee on Fluid Meters Report, op. cit., p. 68. For small values of  $\beta$ ,  $r_c = 0.487$  for  $k = 1.667$ , 0.528 for  $k = 1.40$ , 0.546 for  $k = 1.30$ , and 0.574 for  $k = 1.15$ .

Under **critical flow conditions**, only the upstream conditions  $p_1$ ,  $v_1$ , and  $T_1$  need be known to determine flow rate, which, for  $\beta \leq 0.2$ , is given by

$$w_{\max} = CA_2 \sqrt{g_k \left( \frac{p_1}{v_1} \right) \left( \frac{2}{k+1} \right)^{(k+1)/(k-1)}} \quad (10-31)$$

For a **perfect gas**, this corresponds to

$$w_{\max} = CA_2 p_1 \sqrt{g_k \left( \frac{M}{RT_1} \right) \left( \frac{2}{k+1} \right)^{(k+1)/(k-1)}} \quad (10-32)$$

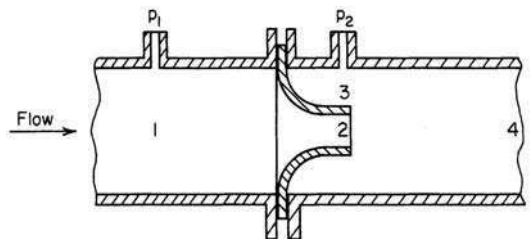


FIG. 10-19 Flow-nozzle assembly.

For air, Eq. (10-31) reduces to

$$w_{\max} = C_1 C A_2 p_1 / \sqrt{T_1} \quad (10-33)$$

where  $A_2$  = cross-sectional area of throat;  $C$  = coefficient of discharge, dimensionless;  $g_c$  = dimensional constant;  $k$  = ratio of specific heats,  $c_p/c_v$ ;  $M$  = molecular weight;  $p_1$  = pressure on upstream side of nozzle;  $R$  = gas constant;  $T_1$  = absolute temperature on upstream side of nozzle;  $v_1$  = specific volume on upstream side of nozzle;  $C_1$  = dimensional constant, 0.0405 SI units (0.533 U.S. customary units); and  $w_{\max}$  = maximum-weight flow rate.

Discharge coefficients for critical flow nozzles are, in general, the same as those for subsonic nozzles. See Grace and Lapple, *Trans. Am. Soc. Mech. Eng.*, **73**, 639–647 (1951); and Szanislo, *J. Eng. Power*, **97**, 521–526 (1975). Arntberg, Britton, and Seidl [*J. Fluids Eng.*, **96**, 111–123 (1974)] present discharge-coefficient correlations for circular-arc venturi meters at critical flow. For the calculation of the flow of natural gas through nozzles under critical-flow conditions, see Johnson, *J. Basic Eng.*, **92**, 580–589 (1970).

**Elbow Meters** A pipe elbow can be used as a flowmeter for liquids if the differential centrifugal head generated between the inner and outer radii of the bend is measured by means of pressure taps located midway around the bend. Equation (10-27) can be used, except that the pressure-difference term ( $p_1 - p_2$ ) is now taken to be the differential centrifugal pressure and  $\beta$  is taken as zero if one assumes no change in cross section between the pipe and the bend. The discharge coefficient should preferably be determined by calibration, but as a guide it can be estimated within  $\pm 6$  percent for circular pipe for Reynolds numbers greater than  $10^5$  from  $C = 0.98 \sqrt{R_c/2D}$ , where  $R_c$  = radius of curvature of the centerline and  $D$  = inside pipe diameter in consistent units. See Murdock, Foltz, and Gregory, *J. Basic Eng.*, **86**, 498–506 (1964); or the ASME Research Committee on Fluid Meters Report, op. cit., pp. 75–77.

**Accuracy** Square-edged orifices and venturi tubes have been so extensively studied and standardized that reproducibilities within 1 to 2 percent can be expected between standard meters when new and clean. This is therefore the order of reliability to be had, if one assumes (1) accurate measurement of meter differential, (2) selection of the coefficient of discharge from recommended published literature, (3) accurate knowledge of fluid density, (4) accurate measurement of critical meter dimensions, (5) smooth upstream face of orifice, and (6) proper location of the meter with respect to other flow-disturbing elements in the system. Care must also be taken to avoid even slight corrosion or fouling during use.

Presence of **swirling flow** or an **abnormal velocity distribution** upstream of the metering element can cause serious metering error unless calibration in place is employed or sufficient straight pipe is inserted between the meter and the source of disturbance. Table 10-7 gives the minimum lengths of straight pipe required to avoid appreciable error due to the presence of certain fittings and valves either upstream or downstream of an orifice or nozzle. These values were extracted from plots presented by Sprenkle [*Trans. Am. Soc. Mech. Eng.*, **67**, 345–360 (1945)]. Table 10-7 also shows the reduction in spacing made possible by the use of straightening vanes between the fittings and the meter. Entirely adequate straightening vanes can be provided by fitting a bundle of thin-wall tubes within the pipe. The center-to-center distance between tubes should not exceed one-fourth of the pipe diameter, and the bundle length should be at least 8 times this distance.

The distances specified in Table 10-7 will be conservative if applied to venturi meters. For specific information on requirements for venturi meters, see a discussion by Pardoe appended to Sprenkle (op. cit.). Extensive data on the effect of installation on the coefficients of venturi meters are given elsewhere by Pardoe [*Trans. Am. Soc. Mech. Eng.*, **65**, 337–349 (1943)].

In the presence of **flow pulsations**, the indications of head meters such as orifices, nozzles, and venturis will often be undependable for several reasons. First, the measured pressure differential will tend to be high, since the pressure differential is proportional to the square of flow rate for a head meter, and the square root of the mean differential pressure is always greater than the mean of the square roots of the differential pressures. Second, there is a phase shift as the wave passes through

**TABLE 10-7 Locations of Orifices and Nozzles Relative to Pipe Fittings**

| Type of fitting upstream                                 | $\frac{D_2}{D_1}$ | Distance, upstream fitting to orifice |                          | Distance, vanes to orifice | Distance, nearest downstream fitting from orifice |
|----------------------------------------------------------|-------------------|---------------------------------------|--------------------------|----------------------------|---------------------------------------------------|
|                                                          |                   | Without straightening vanes           | With straightening vanes |                            |                                                   |
| Single 90° ell, tee, or cross used as ell                | 0.2               | 6                                     |                          |                            | 2                                                 |
|                                                          | 0.4               | 6                                     |                          |                            |                                                   |
|                                                          | 0.6               | 9                                     | 9                        |                            |                                                   |
|                                                          | 0.8               | 20                                    | 12                       | 8                          | 4                                                 |
| 2 short-radius 90° ells in form of S                     | 0.2               | 7                                     |                          |                            | 2                                                 |
|                                                          | 0.4               | 8                                     | 8                        |                            |                                                   |
|                                                          | 0.6               | 13                                    | 10                       | 6                          |                                                   |
|                                                          | 0.8               | 25                                    | 15                       | 11                         | 4                                                 |
| 2 long- or short-radius 90° ells in perpendicular planes | 0.2               | 15                                    | 9                        | 5                          | 2                                                 |
|                                                          | 0.4               | 18                                    | 10                       | 6                          |                                                   |
|                                                          | 0.6               | 25                                    | 11                       | 7                          |                                                   |
|                                                          | 0.8               | 40                                    | 13                       | 9                          | 4                                                 |
| Contraction or enlargement                               | 0.2               | 8                                     | Vanes have no advantage  |                            | 2                                                 |
|                                                          | 0.4               | 9                                     |                          |                            |                                                   |
|                                                          | 0.6               | 10                                    |                          |                            |                                                   |
|                                                          | 0.8               | 15                                    |                          |                            | 4                                                 |
| Globe valve or stop check                                | 0.2               | 9                                     | 9                        | 5                          | 2                                                 |
|                                                          | 0.4               | 10                                    | 10                       | 6                          |                                                   |
|                                                          | 0.6               | 13                                    | 10                       | 6                          |                                                   |
|                                                          | 0.8               | 21                                    | 13                       | 9                          | 4                                                 |
| Gate valve, wide open, or plug cocks                     | 0.2               | 6                                     | Same as globe valve      |                            | 2                                                 |
|                                                          | 0.4               | 6                                     |                          |                            |                                                   |
|                                                          | 0.6               | 8                                     |                          |                            |                                                   |
|                                                          | 0.8               | 14                                    |                          |                            | 4                                                 |

the metering restriction which can affect the differential. Third, pulsations can be set up in the manometer leads themselves. Frequency of the pulsation also plays a part. At low frequencies, the meter reading can generally faithfully follow the flow pulsations, but at high frequencies it cannot. This is due to inertia of the fluid in the manometer leads or of the manometric fluid, whereupon the meter would give a reading intermediate between the maximum and minimum flows but having no readily predictable relation to the mean flow. Pressure transducers with flush-mounted diaphragms can be used together with high-speed recording equipment to provide accurate records of the pressure profiles at the upstream and downstream pressure taps, which can then be analyzed and translated into a mean flow rate.

The rather general practice of producing a steady differential reading by placing restrictions in the manometer leads can result in a reading which, under a fixed set of conditions, may be useful in control of an operation but which has no readily predictable relation to the actual average flow. If calibration is employed to compensate for the presence of pulsations, complete reproduction of operating conditions, including source of pulsations and waveform, is necessary to ensure reasonable accuracy.

According to Head [*Trans. Am. Soc. Mech. Eng.*, **78**, 1471–1479 (1956)], a pulsation-intensity limit of  $\Gamma = 0.1$  is recommended as a practical pulsation threshold below which the performance of all types of flowmeters will differ negligibly from steady-flow performance (an error of less than 1 percent in flow due to pulsation).  $\Gamma$  is the peak-to-trough flow variation expressed as a fraction of the average flow rate. According to the ASME Research Committee on Fluid Meters Report (op. cit., pp. 34–35), the fractional metering error  $E$  for **liquid flow** through a head meter is given by

$$(1 + E)^2 = 1 + \Gamma^2/8 \quad (10-34)$$

When the pulsation amplitude is such as to result in a greater-than-permissible metering error, consideration should be given to installation of a pulsation damper between the source of pulsations and the

flowmeter. References to methods of pulsation-damper design are given in the subsection "Unsteady-State Behavior."

Pulsations are most likely to be encountered in discharge lines from reciprocating pumps or compressors and in lines supplying steam to reciprocating machinery. For **gas flow**, a combination involving a surge chamber and a constriction in the line can be used to damp out the pulsations to an acceptable level. The surge chamber is generally located as close to the pulsation source as possible, with the constriction between the surge chamber and the metering element. This arrangement can be used for either a suction or a discharge line. For such an arrangement, the metering error has been found to be a function of the Hodgson number  $N_H$ , which is defined as

$$N_H = Qn \Delta p_s / q p_s \quad (10-35)$$

where  $Q$  = volume of surge chamber and pipe between metering element and pulsation source;  $n$  = pulsation frequency;  $\Delta p_s$  = permanent pressure drop between metering element and surge chamber;  $q$  = average volume flow rate, based on gas density in the surge chamber; and  $p_s$  = pressure in surge chamber.

Herning and Schmid [*Z. Ver. Dtsch. Ing.*, **82**, 1107–1114 (1938)] presented charts for a simplex double-acting compressor for the prediction of metering error as a function of the Hodgson number and  $s$ , the ratio of piston discharge time to total time per stroke. Table 10-8a gives the minimum Hodgson numbers required to reduce the metering error to 1 percent as given by the charts (for specific heat ratios between 1.28 and 1.37). Schmid [*Z. Ver. Dtsch. Ing.*, **84**, 596–598 (1940)] presented similar charts for a duplex double-acting compressor and a triplex double-acting compressor for a specific heat ratio of 1.37. Table 10-8b gives the minimum Hodgson numbers corresponding to a 1 percent metering error for these cases. The value of  $Q \Delta p_s$  can be calculated from the appropriate Hodgson number, and appropriate values of  $Q$  and  $\Delta p_s$  selected so as to satisfy this minimum requirement.

## VELOCITY METERS

**Anemometers** An anemometer may be any instrument for measurement of gas velocity, e.g., a pitot tube, but usually the term refers to one of the following types.

The vane **anemometer** is a delicate revolution counter with jeweled bearings, actuated by a small windmill, usually 75 to 100 mm (about 3 to 4 in) in diameter, constructed of flat or slightly curved radially disposed vanes. Gas velocity is determined by using a stopwatch to find the time interval required to pass a given number of meters (feet) of gas as indicated by the counter. The velocity so obtained is inversely proportional to gas density. If the original calibration was carried out in a gas of density  $\rho_0$  and the density of the gas stream being metered is  $\rho_1$ , the true gas velocity can be found as follows: From the calibration curve for the instrument, find  $V_{t,0}$  corresponding to the quantity  $V_m \sqrt{\rho_1/\rho_0}$ , where  $V_m$  = measured velocity. Then the actual velocity  $V_{t,1}$  is equal to  $V_{t,0} \sqrt{\rho_0/\rho_1}$ . In general, when working with air, the effects of

atmospheric-density changes can be neglected for all velocities above 1.5 m/s (about 5 ft/s). In all cases, care must be taken to hold the anemometer well away from one's body or from any object not normally present in the stream.

Vane anemometers can be used for gas-velocity measurements in the range of 0.3 to 45 m/s (about 1 to 150 ft/s), although a given instrument generally has about a twentyfold velocity range. Bearing friction has to be minimized in instruments designed for accuracy at the low end of the range, while ample rotor and vane rigidity must be provided for measurements at the higher velocities. Vane anemometers are sensitive to shock and cannot be used in corrosive atmospheres. Therefore, accuracy is questionable unless a recent calibration has been made and the history of the instrument subsequent to calibration is known. For additional information, see Ower et al., op. cit., chap. VIII.

**Turbine Flowmeters** They consist of a straight flow tube containing a turbine which is free to rotate on a shaft supported by one or more bearings and located on the centerline of the tube. Means are provided for magnetic detection of the rotational speed, which is proportional to the volumetric flow rate. Its use is generally restricted to clean, noncorrosive fluids. Additional information on construction, operation, range, and accuracy can be obtained from Baker, pp. 215–252, 2000; Miller, op. cit.; and Spitzer, pp. 303–317, 2005.

The **current meter** is generally used for measuring velocities in open channels such as rivers and irrigation channels. There are two types, the cup meter and the propeller meter. The former is more widely used. It consists of six conical cups mounted on a vertical axis pivoted at the ends and free to rotate between the rigid arms of a U-shaped clevis to which a vaned tailpiece is attached. The wheel rotates because of the difference in drag for the two sides of the cup, and a signal proportional to the revolutions of the wheel is generated. The velocity is determined from the count over a period of time. The current meter is generally useful in the range of 0.15 to 4.5 m/s (about 0.5 to 15 ft/s) with an accuracy of  $\pm 2$  percent. For additional information see Creager and Justin, *Hydroelectric Handbook*, 2d ed., Wiley, New York, 1950, pp. 42–46.

Other important classes of velocity meters include electromagnetic flowmeters and ultrasonic flowmeters. Both are described in Sec. 8.

## MASS FLOWMETERS

**General Principles** There are two main types of mass flowmeters: (1) the so-called true mass flowmeter, which responds directly to mass flow rate, and (2) the inferential mass flowmeter, which commonly measures volume flow rate and fluid density separately. A variety of types of true mass flowmeters have been developed, including the following: (a) the Magnus-effect mass flowmeter, (b) the axial-flow, transverse-momentum mass flowmeter, (c) the radial-flow, transverse-momentum mass flowmeter, (d) the gyroscopic transverse-momentum mass flowmeter, and (e) the thermal mass flowmeter. Type *b* is the basis for several commercial mass flowmeters, one version of which is briefly described here.

**Axial-Flow Transverse-Momentum Mass Flowmeter** This type is also referred to as an angular-momentum mass flowmeter. One embodiment of its principle involves the use of axial flow through a driven impeller and a turbine in series. The impeller imparts angular momentum to the fluid, which in turn causes a torque to be imparted to the turbine, which is restrained from rotating by a spring. The torque, which can be measured, is proportional to the rotational speed of the impeller and the mass flow rate.

**Inferential Mass Flowmeter** There are several types in this category, including the following:

1. **Head meters with density compensation.** Head meters such as orifices, venturis, or nozzles can be used with one of a variety of densitometers [e.g., based on (a) buoyant force on a float, (b) hydraulic coupling, (c) voltage output from a piezoelectric crystal, or (d) radiation absorption]. The signal from the head meter, which is proportional to  $\rho V^2$  (where  $\rho$  = fluid density and  $V$  = fluid velocity), is multiplied by  $\rho$  given by the densitometer. The square root of the product is proportional to the mass flow rate.

**TABLE 10-8a Minimum Hodgson Numbers**  
Simplex double-acting compressor

| $s$   | $N_H$ | $s$   | $N_H$ |
|-------|-------|-------|-------|
| 0.167 | 1.31  | 0.667 | 0.60  |
| 0.333 | 1.00  | 0.833 | 0.43  |
| 0.50  | 0.80  | 1.00  | 0.34  |

**TABLE 10-8b Minimum Hodgson Numbers**

| Duplex double-acting compressor |       | Triplex double-acting compressor |       |
|---------------------------------|-------|----------------------------------|-------|
| $s$                             | $N_H$ | $s$                              | $N_H$ |
| 0.167                           | 1.00  | 0.167                            | 0.85  |
| 0.333                           | 0.70  | 0.333                            | 0.30  |
| 0.50                            | 0.30  | 0.50                             | 0.15  |
| 0.667                           | 0.10  | 0.667                            | 0.06  |
| 0.833                           | 0.05  | 0.833                            | 0.00  |
| 1.00                            | 0.00  | 1.00                             | 0.00  |

2. *Head meters with velocity compensation.* The signal from the head meter, which is proportional to  $\rho V^2$ , is divided by the signal from a velocity meter to give a signal proportional to the mass flow rate.

3. *Velocity meters with density compensation.* The signal from the velocity meter (e.g., turbine meter, electromagnetic meter, or sonic velocity meter) is multiplied by the signal from a densitometer to give a signal proportional to the mass flow rate.

**Coriolis Mass Flowmeter** This type, described in Sec. 8, offers simultaneous direct measurement of both mass flow rate and fluid density. The Coriolis flowmeter is insensitive to upstream and downstream flow disturbances, but its performance is adversely affected by the presence of even a few percent of a gas when measuring a liquid flow.

## VARIABLE-AREA METERS

**General Principles** The underlying principle of an ideal area meter is the same as that of a head meter of the orifice type (see subsection "Orifice Meters"). The stream to be measured is throttled by a constriction, but instead of observing the variation with flow of the differential head across an orifice of fixed size, the constriction of an area meter is so arranged that its size is varied to accommodate the flow while the differential head is held constant.

A simple example of an area meter is a gate valve of the rising-stem type provided with static-pressure taps before and after the gate and a means for measuring the stem position. In most common types of area meters, the variation of the opening is automatically brought about by the motion of a weighted piston or float supported by the fluid. Two different cylinder- and piston-type area meters are described in the ASME Research Committee on Fluid Meters Report, op. cit., pp. 82–83.

**Rotameters** The rotameter, an example of which is shown in Fig. 10-20, has become one of the most popular flowmeters in the chemical-process industries. It consists essentially of a plummet, or "float," which is free to move up or down in a vertical, slightly tapered tube having its small end down. The fluid enters the lower end of the tube and causes the float to rise until the annular area between the float and the wall of the tube is such that the pressure drop across this constriction is just sufficient to support the float. Typically, the tapered tube is of glass and carries etched upon it a nearly linear scale on which the position of the float may be visually noted as an indication of the flow.

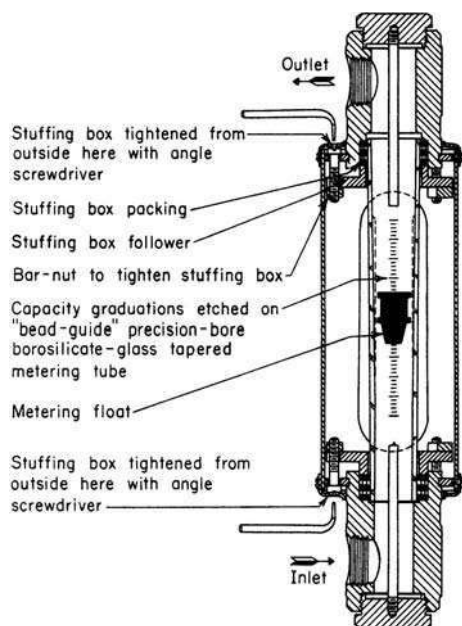


FIG. 10-20 Rotameter.

Interchangeable precision-bore glass tubes and metal metering tubes are available. Rotameters have proved satisfactory both for gases and for liquids at high and at low pressures. A single instrument can readily cover a tenfold range of flow, and by providing floats of different densities a two-hundredfold range is practicable. Rotameters are available with pneumatic, electric, and electronic transmitters for actuating remote recorders, integrators, and automatic flow controllers (see Considine, op. cit., pp. 4-35–4-36, and Sec. 8 of this *Handbook*).

Rotameters require no straight runs of pipe before or after the point of installation. Pressure losses are substantially constant over the whole flow range. In experimental work, for greatest precision, a rotameter should be calibrated with the fluid which is to be metered. However, most modern rotameters are precision-made so that their performance closely corresponds to a master calibration plot for the type in question. Such a plot is supplied with the meter upon purchase.

According to Head [*Trans. Am. Soc. Mech. Eng.*, **76**, 851–862 (1954)], flow rate through a rotameter can be obtained from

$$w = q\rho = KD_f \sqrt{\frac{W_f(\rho_f - \rho)\rho}{\rho_f}} \quad (10-36)$$

and 
$$K = \phi \left[ \frac{D_t}{D_f} \cdot \frac{\mu}{\sqrt{W_f(\rho_f - \rho)\rho}} \right] \quad (10-37)$$

where  $w$  = weight flow rate;  $q$  = volume flow rate;  $\rho$  = fluid density;  $K$  = flow parameter,  $m^{1/2}/s$  ( $ft^{1/2}/s$ );  $D_f$  = float diameter at constriction;  $W_f$  = float weight;  $\rho_f$  = float density;  $D_t$  = tube diameter at point of constriction; and  $\mu$  = fluid viscosity. The appropriate value of  $K$  is obtained from a composite correlation of  $K$  versus the parameters shown in Eq. (10-37) corresponding to the float shape being used. The relation of  $D_t$  to the rotameter reading is also required for the tube taper and size being used.

The ratio of flow rates for two different fluids  $A$  and  $B$  at the same rotameter reading is given by

$$\frac{w_A}{w_B} = \frac{K_A}{K_B} \sqrt{\frac{(\rho_f - \rho_A)\rho_A}{(\rho_f - \rho_B)\rho_B}} \quad (10-38)$$

A measure of self-compensation, with respect to weight rate of flow, for fluid-density changes can be introduced through the use of a float with a density twice that of the fluid being metered, in which case an increase of 10 percent in  $\rho$  will produce a decrease of only 0.5 percent in  $w$  for the same reading. The extent of immunity to changes in fluid viscosity depends upon the shape of the float.

According to Baird and Cheema [*Can. J. Chem. Eng.*, **47**, 226–232 (1969)], the presence of square-wave pulsations can cause a rotameter to overread by as much as 100 percent. The higher the pulsation frequency, the less the float oscillation, although the error can still be appreciable even when the frequency is high enough so that the float is virtually stationary. Use of a damping chamber between the pulsation source and the rotameter will reduce the error.

Additional information on rotameter theory is presented by Fischer [*Chem. Eng.*, **59**(6), 180–184 (1952)], Coleman [*Trans. Inst. Chem. Eng.*, **34**, 339–350 (1956)], and McCabe, Smith, and Harriott (*Unit Operations of Chemical Engineering*, 4th ed., McGraw-Hill, New York, 1985, pp. 202–205).

## TWO-PHASE SYSTEMS

It is generally preferable to meter each of the individual components of a two-phase mixture separately prior to mixing, since it is difficult to meter such mixtures accurately. Problems arise because of fluctuations in composition with time and variations in composition over the cross section of the channel. Information on metering of such mixtures can be obtained from the following sources.



**Gas-Solid Mixtures** Carlson, Frazier, and Engdahl [*Trans. Am. Soc. Mech. Eng.*, **70**, 65–79 (1948)] describe the use of a **flow nozzle** and a **square-edged orifice** in series for the measurement of both the gas rate and the solids rate in the flow of a finely divided solid-in-gas mixture. The nozzle differential is sensitive to the flow of both phases, whereas the orifice differential is not influenced by the solids flow.

Farbar [*Trans. Am. Soc. Mech. Eng.*, **75**, 943–951 (1953)] describes how a **venturi meter** can be used to measure solids flow rate in a gas-solids mixture when the gas rate is held constant. Separate calibration curves (solids flow versus differential) are required for each gas rate of interest.

Cheng, Tung, and Soo [*J. Eng. Power*, **92**, 135–149 (1970)] describe the use of an **electrostatic probe** for measurement of solids flow in a gas-solids mixture.

Goldberg and Boothroyd [*Br. Chem. Eng.*, **14**, 1705–1708 (1969)] describe several types of solids-in-gas flowmeters and give an extensive bibliography.

**Gas-Liquid Mixtures** An empirical equation was developed by Murdock [*J. Basic Eng.*, **84**, 419–433 (1962)] for the measurement of gas-liquid mixtures using **sharp-edged orifice** plates with either radius, flange, or pipe taps.

An equation for use with **venturi meters** was given by Chisholm [*Br. Chem. Eng.*, **12**, 454–457 (1967)]. A procedure for determining steam quality via pressure-drop measurement with upflow through either venturi meters or sharp-edged orifice plates was given by Collins and Gacesa [*J. Basic Eng.*, **93**, 11–21 (1971)].

**Liquid-Solid Mixtures** Liptak [*Chem. Eng.*, **74**(4), 151–158 (1967)] discusses a variety of techniques that can be used for the measurement of solids-in-liquid suspensions or slurries. These include metering pumps, weigh tanks, magnetic flowmeter, ultrasonic flowmeter, gyroscope flowmeter, etc.

Shirato, Gotoh, Osasa, and Usami [*J. Chem. Eng. Japan*, **1**, 164–167 (January 1968)] present a method for determining the mass flow rate of suspended solids in a liquid stream wherein the liquid velocity is measured by an electromagnetic flowmeter and the flow of solids is calculated from the pressure drops across each of two vertical sections of pipe of different diameter through which the suspension flows in series.

## FLOWMETER SELECTION

Web sites for process equipment and instrumentation, such as [www.globalspec.com](http://www.globalspec.com) and [www.thomasnet.com](http://www.thomasnet.com), are valuable tools when selecting a flowmeter. These search engines can scan the flowmeters manufactured by more than 800 companies for specific products that meet the user's specifications. Table 10-4 was based in part on information from these web sites. Note that the accuracies claimed are achieved only under ideal conditions when the flowmeters are properly installed and calibrated for the application.

The purpose of this subsection is to summarize the preferred applications as well as the advantages and disadvantages of some of the common flowmeter technologies.

Table 10-9 divides flowmeters into four classes. Flowmeters in class I depend on wetted moving parts that can wear, plug, or break. The

potential for catastrophic failure is a disadvantage. However, in clean fluids, class I flowmeters have often proved reliable and stable when properly installed, calibrated, and maintained.

Class II flowmeters have no wetted moving parts to break and are thus not subject to catastrophic failure. However, the flow surfaces such as orifice plates may wear, eventually biasing flow measurements. Other disadvantages of some flowmeters in this class include high pressure drop and susceptibility to plugging. Very dirty and abrasive fluids should be avoided.

Because class III flowmeters have neither moving parts nor obstructions to flow, they are suitable for dirty and abrasive fluids provided that appropriate materials of construction are available.

Class IV flowmeters have sensors mounted external to the pipe, and would thus seem to be ideal, but problems of accuracy and sensitivity have been encountered in early devices. These comparatively new technologies are under development, and these problems may be overcome in the future.

Section 8 outlines the following criteria for selection of measurement devices: measurement span, performance, reliability, materials of construction, prior use, potential for releasing process materials to the environment, electrical classification, physical access, invasive or noninvasive, and life-cycle cost.

Spitzer, op. cit., 2005, cites four intended end uses of the flowmeter: rate indication, control, totalization, and alarm. Thus high accuracy may be important for rate indication, while control may just need good repeatability. Volumetric flow or mass flow indication is another choice.

Baker, op. cit., 2003, identifies the type of fluid (liquid or gas, slurry, multiphase), special fluid constraints (clean or dirty, hygienic, corrosive, abrasive, high flammability, low lubricity, fluids causing scaling). He lists the following flowmeter constraints: accuracy or measurement uncertainty, diameter range, temperature range, pressure range, viscosity range, flow range, pressure loss caused by the flowmeter, sensitivity to installation, sensitivity to pipework supports, sensitivity to pulsation, whether the flowmeter has a clear bore, whether a clamp-on version is available, response time, and ambient conditions. Finally, Baker identifies these environmental considerations: ambient temperature, humidity, exposure to weather, level of electromagnetic radiation, vibration, tamperproof for domestic use, and classification of area requiring explosionproof, intrinsic safety, etc.

Note that the accuracies cited in Table 10-4 can be achieved by those flowmeters only under ideal conditions of application, installation, and calibration. This subsection has only given an introduction to issues to consider in the choice of a flowmeter for a given application. See Baker, op. cit., 2003; Miller, op. cit., 1996; and Spitzer, op. cit., 2005, for further guidance and to obtain application-specific data from flowmeter vendors.

## WEIRS

Liquid flow in an open channel may be metered by means of a weir, which consists of a dam over which, or through a notch in which, the liquid flows. The terms “rectangular weir,” “triangular weir,” etc., generally refer to the shape of the notch in a notched weir. All weirs considered here have flat upstream faces that are perpendicular to the bed and walls of the channel.

**Sharp-edged weirs** have edges like those of square or sharp-edged orifices (see subsection “Orifice Meters”). Notched weirs are ordinarily sharp-edged. Weirs not in the sharp-edged class are, for the most part, those described as **broad-crested weirs**.

The head  $h_0$  on a weir is the liquid-level height above the crest or base of the notch. The head must be measured sufficiently far upstream to avoid the drop in level occasioned by the overflow which begins at a distance about  $2h_0$  upstream from the weir. Surface-level measurements should be made a distance of  $3h_0$  or more upstream, preferably by using a stilling box equipped with a high-precision level gauge, e.g., a hook gauge or float gauge.

With sharp-edged weirs, the sheet of discharging liquid, called the “nappe,” contracts as it leaves the opening and free discharge occurs. Rounding the upstream edge will reduce the contraction and increase the flow rate for a given head. A clinging nappe may result

**TABLE 10-9 Flowmeter Classes**

| Class I:<br>Flowmeters with wetted<br>moving parts | Class II:<br>Flowmeters with no wetted<br>moving parts               |
|----------------------------------------------------|----------------------------------------------------------------------|
| Positive displacement<br>Turbine<br>Variable-area  | Differential pressure<br>Vortex<br>Target<br>Thermal                 |
| Class III:<br>Obstructionless flowmeters           | Class IV:<br>Flowmeters with sensors mounted<br>external to the pipe |
| Coriolis mass<br>Electromagnetic<br>Ultrasonic     | Clamp-on ultrasonic<br>Correlation                                   |

Adapted from Spitzer, op. cit., 2005.



## 10-24 TRANSPORT AND STORAGE OF FLUIDS



FIG. 10-21 Rectangular weir.

if the head is very small, if the edge is well rounded, or if air cannot flow in beneath the nappe. This, in turn, results in an increase in the discharge rate for a given head as compared with that for a free nappe. For further information on the effect of the nappe, see Gibson, *Hydraulics and Its Applications*, 5th ed., Constable, London, 1952; and Chow, *Open-Channel Hydraulics*, McGraw-Hill, New York, 1959.

Flow through a **rectangular weir** (Fig. 10-21) is given by

$$q = 0.415(L - 0.2h_0)h_0^{1.5} \sqrt{2g} \quad (10-39)$$

where  $q$  = volume flow rate,  $L$  = crest length,  $h_0$  = weir head, and  $g$  = local acceleration due to gravity. This is known as the modified Francis formula for a rectangular sharp-edged weir with two end corrections; it applies when the velocity-of-approach correction is small. The Francis formula agrees with experiments within 3 percent if (1)  $L$  is greater than  $2h_0$ , (2) velocity of approach is 0.6 m/s (2 ft/s) or less, (3) height of crest above bottom of channel is at least  $3h_0$ , and (4)  $h_0$  is not less than 0.09 m (0.3 ft).

**Narrow rectangular notches** ( $h_0 > L$ ) have been found to give about 93 percent of the discharge predicted by the Francis formula. Thus

$$q = 0.386Lh_0^{1.5} \sqrt{2g} \quad (10-40)$$

In this case, no end corrections are applied even though the formula applies only for sharp-edged weirs. See Schoder and Dawson, *Hydraulics*, McGraw-Hill, New York, 1934, p. 175, for further details.

The **triangular-notch weir** has the advantage that a single notch can accommodate a wide range of flow rates, although this in turn reduces its accuracy. The discharge for sharp- or square-edged weirs is given by

$$q = (0.31h_0^{3.5} \sqrt{2g}) / \tan \phi \quad (10-41)$$

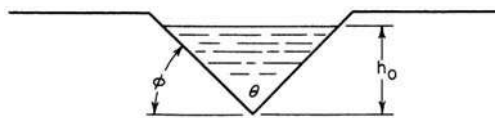


FIG. 10-22 Triangular weir.

See Eq. (10-39) for nomenclature. Angle  $\phi$  is illustrated in Fig. 10-22. Equations (10-39), (10-40), and (10-41) are applicable only to the flow of water. However, for the case of triangular-notch weirs Lenz [*Trans. Am. Soc. Civ. Eng.*, **108**, 759–802 (1943)] has presented correlations predicting the effect of viscosity over the range of 0.001 to 0.15 Pa·s (1 to 150 cP) and surface tension over the range of 0.03 to 0.07 N/m (30 to 70 dyn/cm). His equation predicts about an 8 percent increase in flow for a liquid of 0.1-Pa·s (100-cP) viscosity compared with water at 0.001 Pa·s (1 cP) and about a 1 percent increase for a liquid with one-half of the surface tension of water. For fluids of moderate viscosity, Ranga Raju and Asawa [*Proc. Am. Soc. Civ. Eng., J. Hydraul. Div.*, **103** (HY 10), 1227–1231 (1977)] find that the effect of viscosity and surface tension on the discharge flow rate for rectangular and triangular-notch ( $\phi = 45^\circ$ ) weirs can be neglected when

$$(N_{Re})^{0.2}(N_{We})^{0.6} > 900 \quad (10-42)$$

where  $N_{Re}$  (Reynolds number) =  $\sqrt{gh_0^3}/\nu$ ,  $g$  = local acceleration due to gravity,  $h_0$  = weir head,  $\nu$  = kinematic viscosity;  $N_{We}$  (Weber number) =  $\rho gh_0^3/g_c \sigma$ ,  $\rho$  = density,  $g_c$  = dimensional constant, and  $\sigma$  = surface tension.

For the flow of high-viscosity liquids over rectangular weirs, see Slocum, *Can. J. Chem. Eng.*, **42**, 196–200 (1964). His correlation is based on data for liquids with viscosities in the range of 2.5 to 500 Pa·s (25 to 5000 cP), in which range the discharge decreases markedly for a given head as viscosity is increased.

Information on other types of weirs can be obtained from Addison, op. cit.; Gibson, *Hydraulics and Its Applications*, 5th ed., Constable, London, 1952; Henderson, *Open Channel Flow*, Macmillan, New York, 1966; Linford, *Flow Measurement and Meters*, Spon, London, 1949; Lakshmana Rao, "Theory of Weirs," in *Advances in Hydrosience*, vol. 10, Academic, New York, 1975; and Merritt, *Standard Handbook for Civil Engineers*, 2d ed., McGraw-Hill, New York, 1976.

## PUMPS AND COMPRESSORS

**GENERAL REFERENCES:** Meherwan P. Boyce, P.E., *Centrifugal Compressors: A Basic Guide*, Pennwell Books, Tulsa, Okla., 2002; Royce N. Brown, *Compressors: Selection and Sizing*, 3d ed., Gulf Professional Publishing, Houston, Tex., 2005; James Corley, "The Vibration Analysis of Pumps: A Tutorial," Fourth International Pump Symposium, Texas A & M University, Houston, Tex., May 1987; John W. Dufor and William E. Nelson, *Centrifugal Pump Sourcebook*, McGraw-Hill, New York, 1992; *Engineering Data Book*, 12th ed., vol. I, Secs. 12 and 13, Gas Processors Suppliers Association, Tulsa, Okla., 2004; Paul N. Garay, P.E., *Pump Application Desk Book*, Fairmont Press, 1993; *Process Pumps*, IIT Fluid Technology Corporation, 1992; Igor J. Karassik et al., *Pump Handbook*, 3d ed., McGraw-Hill, New York, 2001; Val S. Lobanoff and Robert R. Ross, *Centrifugal Pumps: Design and Application*, 2d ed., Gulf Professional Publishing, Houston, Tex., 1992; A. J. Stephanoff, *Centrifugal and Axial Flow Pumps: Theory, Design, and Application*, 2d ed., Krieger Publishing, Melbourne, Fla., 1992.

### INTRODUCTION

The following subsections deal with pumps and compressors. A pump or compressor is a physical contrivance that is used to deliver fluids from one location to another through conduits. The term *pump* is used when the fluid is a liquid, while the term *compressor* is used when the fluid is a gas. The basic requirements to define the applica-

tion are suction and delivery pressures, pressure loss in transmission, and flow rate. Special requirements may exist in food, pharmaceutical, nuclear, and other industries that impose material selection requirements of the pump.

The primary means of transfer of energy to the fluid that causes flow are gravity, displacement, centrifugal force, electromagnetic force, transfer of momentum, mechanical impulse, and a combination of these energy-transfer mechanisms. Displacement and centrifugal force are the most common energy-transfer mechanisms in use.

Pumps and compressors are designed per technical specifications and standards developed over years of operating and maintenance experience. Table 10-10 lists some of these standards for pumps and compressors and for related equipment such as lubrication systems and gearboxes which, if not properly specified, could lead to many operational and maintenance problems with the pumps and compressors. These standards specify design, construction, maintenance, and testing details such as terminology, material selection, shop inspection and tests, drawings, clearances, construction procedures, and so on.

There are four (4) major types of pumps: (1) positive displacement, (2) dynamic (kinetic), (3) lift, and (4) electromagnetic. Piston pumps are positive displacement pumps. The most common centrifugal

**TABLE 10-10 Standards Governing Pumps and Compressors**

|                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASME Standards, American Society of Mechanical Engineers, New York                                                                                          |
| B73.1-2001, <i>Specification for Horizontal End Suction Centrifugal Pumps for Chemical Process</i>                                                          |
| B73.2-2003, <i>Specification for Vertical In-Line Centrifugal Pumps for Chemical Process</i>                                                                |
| PTC 10, 1997 <i>Test Code on Compressors and Exhausters</i>                                                                                                 |
| PTC 11, 1984 <i>Fans</i>                                                                                                                                    |
| B19.3-1991, <i>Safety Standard for Compressors for Process Industries</i>                                                                                   |
| API Standards, American Petroleum Institute, Washington                                                                                                     |
| API Standard 610, <i>Centrifugal Pumps for Petroleum, Petrochemical, and Natural Gas Industries</i> , Adoption of ISO 13709, October 2004                   |
| API Standard 613, <i>Special Purpose Gear Units for Petroleum, Chemical and Gas Industry Services</i> , February 2003                                       |
| API Standard 614, <i>Lubrication, Shaft-Sealing, and Control-Oil Systems and Auxiliaries for Petroleum, Chemical and Gas Industry Services</i> , April 1999 |
| API Standard 616, <i>Gas Turbines for the Petroleum, Chemical, and Gas Industry Services</i> , August 1998                                                  |
| API Standard 617, <i>Axial and Centrifugal Compressors and Expanders—Compressors for Petroleum, Chemical, and Gas Industry Services</i> , June 2003         |
| API Standard 618, <i>Reciprocating Compressors for Petroleum, Chemical, and Gas Industry Services</i> , June 1995                                           |
| API Standard 619, <i>Rotary-Type Positive Displacement Compressors for Petroleum, Petrochemical, and Natural Gas Industries</i> , December 2004             |
| API Standard 670, <i>Machinery Protection Systems</i> , November 2003                                                                                       |
| API Standard 671, <i>Special Purpose Couplings for Petroleum, Chemical, and Gas Industry Services</i> , October 1998                                        |
| API Standard 672, <i>Packaged, Integrally Geared, Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services</i> , March 2004           |
| API Standard 673, <i>Centrifugal Fans for Petroleum, Chemical, and Gas Industry Services</i> , October 2002                                                 |
| API Standard 674, <i>Positive Displacement Pumps—Reciprocating</i> , June 1995                                                                              |
| API Standard 675, <i>Positive Displacement Pumps—Controlled Volume</i> , March 2000                                                                         |
| API Standard 677, <i>General Purpose Gear Units for Petroleum, Chemical, and Gas Industry Services</i> , April 2006                                         |
| API Standard 680, <i>Packaged Reciprocating Plant and Instrument Air Compressors for General Refinery Services</i> , October 1987                           |
| API Standard 681, <i>Liquid Ring Vacuum Pumps and Compressors for Petroleum, Chemical, and Gas Industry Services</i> , June 2002                            |
| API Standard 682, <i>Pumps—Shaft Sealing Systems for Centrifugal and Rotary Pumps</i> , September 2004                                                      |
| API Standard 685, <i>Sealless Centrifugal Pumps for Petroleum, Heavy Duty Chemical, and Gas Industry Services</i> , October 2000                            |
| Hydraulic Institute, Parsippany, N.J. ( <a href="http://www.pumps.org">www.pumps.org</a> )                                                                  |
| ANSI/HI Pump Standards, 2005 (covers centrifugal, vertical, rotary, and reciprocating pumps)                                                                |
| National Fire Protection Association, Quincy, Mass. ( <a href="http://www.nfpa.org">www.nfpa.org</a> )                                                      |
| Standards for pumps used in fire protection systems                                                                                                         |

pumps are of dynamic type; ancient bucket-type pumps are lift pumps; and electromagnetic pumps use electromagnetic force and are common in modern reactors. Canned pumps are also becoming popular in the petrochemical industry because of the drive to minimize fugitive emissions. Figure 10-23 shows pump classification:

## TERMINOLOGY

**Displacement** Discharge of a fluid from a vessel by partially or completely displacing its internal volume with a second fluid or by mechanical means is the principle upon which a great many fluid-transport devices operate. Included in this group are reciprocating-piston and diaphragm machines, rotary-vane and gear types, fluid piston compressors, acid eggs, and air lifts.

The large variety of displacement-type fluid-transport devices makes it difficult to list characteristics common to each. However, for most types it is correct to state that (1) they are adaptable to high-pressure operation, (2) the flow rate through the pump is variable (auxiliary damping systems may be employed to reduce the magnitude of pressure pulsation and flow variation), (3) mechanical considerations limit maximum throughputs, and (4) the devices are capable of efficient performance at extremely low-volume throughput rates.

**Centrifugal Force** Centrifugal force is applied by means of the centrifugal pump or compressor. Though the physical appearance of the many types of centrifugal pumps and compressors varies greatly, the basic function of each is the same, i.e., to produce kinetic energy by the action of centrifugal force and then to convert this energy into pressure by efficiently reducing the velocity of the flowing fluid.

In general, centrifugal fluid-transport devices have these characteristics: (1) discharge is relatively free of pulsation; (2) mechanical design lends itself to high throughputs, capacity limitations are rarely a problem; (3) the devices are capable of efficient performance over a wide range of pressures and capacities even at constant-speed operation; (4) discharge pressure is a function of fluid density; and (5) these are relatively small high-speed devices and less costly.

A device which combines the use of centrifugal force with mechanical impulse to produce an increase in pressure is the axial-flow compressor or pump. In this device the fluid travels roughly parallel to the shaft through a series of alternately rotating and stationary radial blades having airfoil cross sections. The fluid is accelerated in the axial direction by mechanical impulses from the rotating blades; concurrently, a positive-pressure gradient in the radial direction is established in each stage by centrifugal force. The net pressure rise per stage results from both effects.

**Electromagnetic Force** When the fluid is an electrical conductor, as is the case with molten metals, it is possible to impress an electromagnetic field around the fluid conduit in such a way that a driving force that will cause flow is created. Such pumps have been developed for the handling of heat-transfer liquids, especially for nuclear reactors.

**Transfer of Momentum** Deceleration of one fluid (motivating fluid) in order to transfer its momentum to a second fluid (pumped fluid) is a principle commonly used in the handling of corrosive materials, in pumping from inaccessible depths, or for evacuation. Jets and eductors are in this category.

Absence of moving parts and simplicity of construction have frequently justified the use of jets and eductors. However, they are relatively inefficient devices. When air or steam is the motivating fluid, operating costs may be several times the cost of alternative types of fluid-transport equipment. In addition, environmental considerations in today's chemical plants often inhibit their use.

**Mechanical Impulse** The principle of mechanical impulse when applied to fluids is usually combined with one of the other means of imparting motion. As mentioned earlier, this is the case in axial-flow compressors and pumps. The turbine or regenerative-type pump is another device which functions partially by mechanical impulse.

**Measurement of Performance** The amount of useful work that any fluid-transport device performs is the product of (1) the mass rate of fluid flow through it and (2) the total pressure differential measured immediately before and after the device, usually expressed in the height of column of fluid equivalent under adiabatic conditions. The first of these quantities is normally referred to as **capacity**, and the second is known as **head**.

**Capacity** This quantity is expressed in the following units. In SI units capacity is expressed in cubic meters per hour (m<sup>3</sup>/h) for both liquids and gases. In U.S. customary units it is expressed in U.S. gallons per minute (gal/min) for liquids and in cubic feet per minute (ft<sup>3</sup>/min) for gases. Since all these are volume units, the density or specific gravity must be used for conversion to mass rate of flow. When gases are being handled, capacity must be related to a pressure and a temperature, usually the conditions prevailing at the machine inlet. It is important to note that all heads and other terms in the following equations are expressed in height of column of liquid.

**Total Dynamic Head** The total dynamic head  $H$  of a pump is the total discharge head  $h_d$  minus the total suction head  $h_s$ .

**Total Suction Head** This is the reading  $h_{gs}$  of a gauge at the suction flange of a pump (corrected to the pump centerline\*), plus the barometer reading and the velocity head  $h_{vs}$  at the point of gauge attachment:

$$h_s = h_{gs} + \text{atm} + h_{vs} \quad (10-43)$$

If the gauge pressure at the suction flange is less than atmospheric, requiring use of a vacuum gauge, this reading is used for  $h_{gs}$  in Eq. (10-43) with a negative sign.

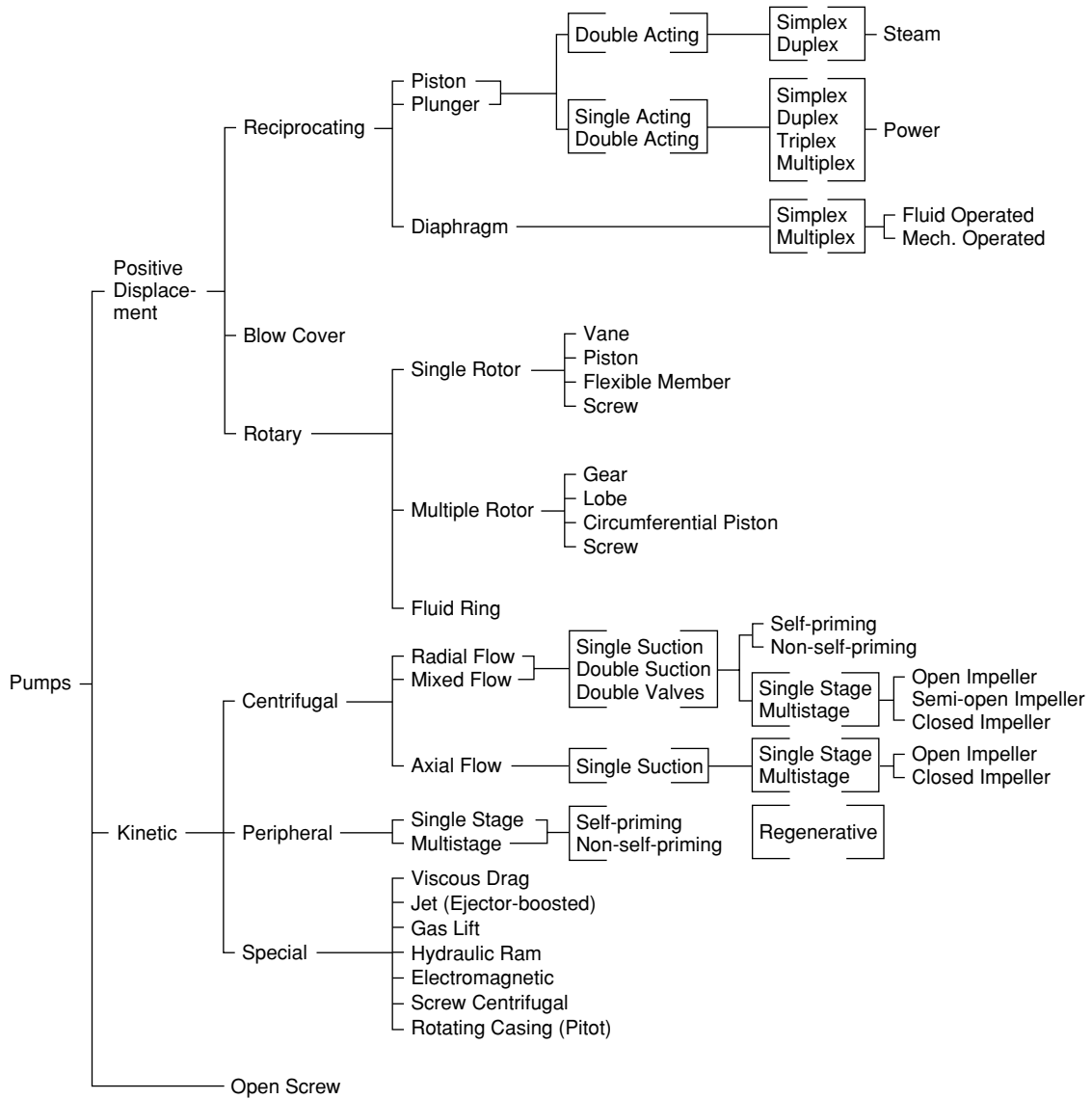


FIG. 10-23 Classification of pumps. (Courtesy of Hydraulic Institute.)

Before installation it is possible to estimate the total suction head as follows:

$$h_s = h_{ss} - h_{fs} \quad (10-44)$$

where  $h_{ss}$  = static suction head and  $h_{fs}$  = suction friction head.

**Static Suction Head** The static suction head  $h_{ss}$  is the vertical distance measured from the free surface of the liquid source to the pump centerline plus the absolute pressure at the liquid surface.

**Total Discharge Head** The total discharge head  $h_d$  is the reading  $h_{gd}$  of a gauge at the discharge flange of a pump (corrected to the pump centerline\*), plus the barometer reading and the velocity head  $h_{vd}$  at the point of gauge attachment:

$$h_d = h_{gd} + \text{atm} + h_{vd} \quad (10-45)$$

Again, if the discharge gauge pressure is below atmospheric, the vacuum-gauge reading is used for  $h_{gd}$  in Eq. (10-45) with a negative sign.

Before installation it is possible to estimate the total discharge head from the static discharge head  $h_{sd}$  and the discharge friction head  $h_{fd}$  as follows:

$$h_d = h_{sd} + h_{fd} \quad (10-46)$$

**Static Discharge Head** The static discharge head  $h_{sd}$  is the vertical distance measured from the free surface of the liquid in the receiver to the pump centerline,\* plus the absolute pressure at the liquid surface. **Total static head**  $h_{ts}$  is the difference between discharge and suction static heads.

\*On vertical pumps, the correction should be made to the eye of the suction impeller.

**Velocity** Since most liquids are practically incompressible, the relation between the quantity flowing past a given point in a given time and the velocity of flow is expressed as follows:

$$Q = Av \quad (10-47)$$

This relationship in SI units is as follows:

$$v \text{ (for circular conduits)} = 3.54 Q/d^2 \quad (10-48)$$

where  $v$  = average velocity of flow, m/s;  $Q$  = quantity of flow, m<sup>3</sup>/h; and  $d$  = inside diameter of conduit, cm.

This same relationship in U.S. customary units is

$$v \text{ (for circular conduits)} = 0.409 Q/d^2 \quad (10-49)$$

where  $v$  = average velocity of flow, ft/s;  $Q$  = quantity of flow, gal/min; and  $d$  = inside diameter of conduit, in.

**Velocity Head** This is the vertical distance by which a body must fall to acquire the velocity  $v$ .

$$h_v = v^2/2g \quad (10-50)$$

**Viscosity** (See Sec. 6 for further information.) In flowing liquids the existence of internal friction or the internal resistance to relative motion of the fluid particles must be considered. This resistance is called viscosity. The viscosity of liquids usually decreases with rising temperature. Viscous liquids tend to increase the power required by a pump, to reduce pump efficiency, head, and capacity, and to increase friction in pipe lines.

**Friction Head** This is the pressure required to overcome the resistance to flow in pipe and fittings. It is dealt with in detail in Sec. 6.

**Work Performed in Pumping** To cause liquid to flow, work must be expended. A pump may raise the liquid to a higher elevation, force it into a vessel at higher pressure, provide the head to overcome pipe friction, or perform any combination of these. Regardless of the service required of a pump, all energy imparted to the liquid in performing this service must be accounted for; consistent units for all quantities must be employed in arriving at the work or power performed.

When arriving at the performance of a pump, it is customary to calculate its **power output**, which is the product of (1) the total dynamic head and (2) the mass of liquid pumped in a given time. In SI units power is expressed in kilowatts; horsepower is the conventional unit used in the United States.

In SI units,

$$kW = HQ\rho/3.670 \times 10^5 \quad (10-51)$$

where  $kW$  is the pump power output, kW;  $H$  = total dynamic head, N·m/kg (column of liquid);  $Q$  = capacity, m<sup>3</sup>/h; and  $\rho$  = liquid density, kg/m<sup>3</sup>.

When the total dynamic head  $H$  is expressed in pascals, then

$$kW = HQ/3.599 \times 10^6 \quad (10-52)$$

In U.S. customary units,

$$hp = HQs/3.960 \times 10^3 \quad (10-53)$$

where  $hp$  is the pump-power output, hp;  $H$  = total dynamic head, lbf·ft/lbm (column of liquid);  $Q$  = capacity, U.S. gal/min; and  $s$  = liquid specific gravity.

When the total dynamic head  $H$  is expressed in pounds-force per square inch, then

$$hp = HQ/1.714 \times 10^3 \quad (10-54)$$

The **power input** to a pump is greater than the **power output** because of internal losses resulting from friction, leakage, etc. The efficiency of a pump is therefore defined as

$$\text{Pump efficiency} = (\text{power output})/(\text{power input}) \quad (10-55)$$

## PUMP SELECTION

When selecting pumps for any service, it is necessary to know the liquid to be handled, the total dynamic head, the suction and discharge heads, and, in most cases, the temperature, viscosity, vapor pressure, and specific gravity. In the chemical industry, the task of pump selection is frequently further complicated by the presence of

solids in the liquid and liquid corrosion characteristics requiring special materials of construction. Solids may accelerate erosion and corrosion, have a tendency to agglomerate, or require delicate handling to prevent undesirable degradation.

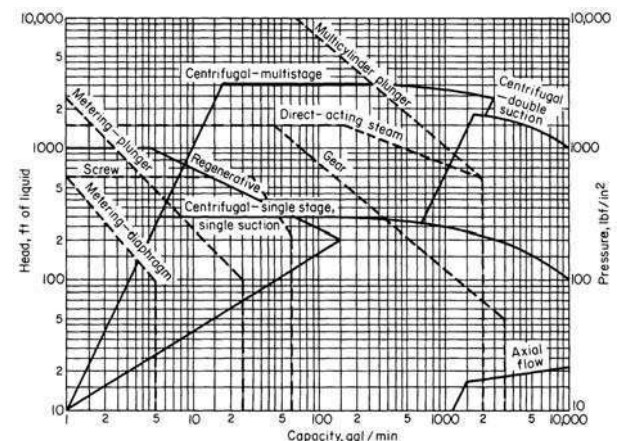
**Range of Operation** Because of the wide variety of pump types and the number of factors which determine the selection of any one type for a specific installation, the designer must first eliminate all but those types of reasonable possibility. Since range of operation is always an important consideration, Fig. 10-24 should be of assistance. The boundaries shown for each pump type are at best approximate. In most cases, following Fig. 10-24 will select the pump that is best suited for a given application. Low-capacity pumps with high discharge head requirements are best served by positive-displacement pumps. Reciprocating pumps and rotary pumps such as gear and roots rotor-type pumps are examples of positive-displacement pumps. Displacement pumps provide high heads at low capacities which are beyond the capability of centrifugal pumps. Displacement pumps achieve high pressure with low velocities and are thus suited for high-viscosity service and slurry.

The centrifugal pump operates over a very wide range of flows and pressures. For low heads but high flows the axial pump is best suited. Both the centrifugal and axial flow pumps impart energy to the fluid by the rotational speed of the impeller and the velocity it imparts to the fluid.

## NET POSITIVE SUCTION HEAD

Net positive suction head available (NPSH)<sub>A</sub> is the difference between the total absolute suction pressure at the pump suction nozzle when the pump is running and the vapor pressure at the flowing liquid temperature. All pumps require the system to provide adequate (NPSH)<sub>A</sub>. In a positive-displacement pump the (NPSH)<sub>A</sub> should be large enough to open the suction valve, to overcome the friction losses within the pump liquid end, and to overcome the liquid acceleration head.

**Suction Limitations of a Pump** Whenever the pressure in a liquid drops below the vapor pressure corresponding to its temperature, the liquid will vaporize. When this happens within an operating pump, the vapor bubbles will be carried along to a point of higher pressure, where they suddenly collapse. This phenomenon is known as **cavitation**. Cavitation in a pump should be avoided, as it is accompanied by metal removal, vibration, reduced flow, loss in efficiency, and noise. When the absolute suction pressure is low, cavitation may occur in the pump inlet and damage result in the pump suction and on the impeller vanes near the inlet edges. To avoid this phenomenon, it is necessary to maintain a **required net positive suction head**



**FIG. 10-24** Pump coverage chart based on normal ranges of operation of commercially available types. Solid lines: use left ordinate, head scale. Broken lines: use right ordinate, pressure scale. To convert gallons per minute to cubic meters per hour, multiply by 0.2271; to convert feet to meters, multiply by 0.3048; and to convert pounds-force per square inch to kilopascals, multiply by 6.895.

$(\text{NPSH})_R$ , which is the equivalent total head of liquid at the pump centerline less the vapor pressure  $p$ . Each pump manufacturer publishes curves relating  $(\text{NPSH})_R$  to capacity and speed for each pump.

When a pump installation is being designed, the **available net positive suction head**  $(\text{NPSH})_A$  must be equal to or greater than the  $(\text{NPSH})_R$  for the desired capacity. The  $(\text{NPSH})_A$  can be calculated as follows:

$$(\text{NPSH})_A = h_{ss} - h_{fs} - p \quad (10-56)$$

If  $(\text{NPSH})_A$  is to be checked on an existing installation, it can be determined as follows:

$$(\text{NPSH})_A = \text{atm} + h_{gs} - p + h_{es} \quad (10-57)$$

Practically, the NPSH required for operation without cavitation and vibration in the pump is somewhat greater than the theoretical. The actual  $(\text{NPSH})_R$  depends on the characteristics of the liquid, the total head, the pump speed, the capacity, and impeller design. Any suction condition which reduces  $(\text{NPSH})_A$  below that required to prevent cavitation at the desired capacity will produce an unsatisfactory installation and can lead to mechanical difficulty.

The following two equations usually provide an adequate design margin between  $(\text{NPSH})_A$  and  $(\text{NPSH})_R$ :

$$(\text{NPSH})_A = (\text{NPSH})_R + 5 \text{ ft} \quad (10-58)$$

$$(\text{NPSH})_A = 1.35(\text{NPSH})_R \quad (10-59)$$

Use the larger value of  $(\text{NPSH})_A$  calculated with Eqs. (10-58) and (10-59).

**NPSH Requirements for Other Liquids** NPSH values depend on the fluid being pumped. Since water is considered a standard fluid for pumping, various correction methods have been developed to evaluate NPSH when pumping other fluids. The most recent of these corrective methods has been developed by the Hydraulic Institute and is shown in Fig. 10-25.

The chart shown in Fig. 10-25 is for pure liquids. Extrapolation of data beyond the ranges indicated in the graph may not produce accurate results. Figure 10-25 shows the variation of vapor pressure and NPSH reductions for various hydrocarbons and hot water as a function of temperature. Certain rules apply while using this chart. When using the chart for hot water, if the NPSH reduction is greater than one-half of the NPSH required for cold water, deduct one-half of cold water NPSH to obtain the corrected NPSH required. On the other

hand, if the value read on the chart is less than one-half of cold water NPSH, deduct this chart value from the cold water NPSH to obtain the corrected NPSH.

**Example 1: NPSH Calculation** Suppose a selected pump requires a minimum NPSH of 16 ft (4.9 m) when pumping cold water. What will be the NPSH limitation to pump propane at 55°F (12.8°C) with a vapor pressure of 100 psi? Using the chart in Fig. 10-25, NPSH reduction for propane gives 9.5 ft (2.9 m). This is greater than one-half of cold water NPSH of 16 ft (4.9 m). The corrected NPSH is therefore 8 ft (2.2 m) or one-half of cold water NPSH.

## PUMP SPECIFICATIONS

Pump specifications depend upon numerous factors but mostly on application. Typically, the following factors should be considered while preparing a specification.

1. Application, scope, and type
2. Service conditions
3. Operating conditions
4. Construction application-specific details and special considerations
  - a. Casing and connections
  - b. Impeller details
  - c. Shaft
  - d. Stuffing box details—lubrications, sealing, etc.
  - e. Bearing frame and bearings
  - f. Baseplate and couplings
  - g. Materials
  - h. Special operating conditions and miscellaneous items

Table 10-11 is based on the API and ASME codes and illustrates a typical specification for centrifugal pumps.

## POSITIVE-DISPLACEMENT PUMPS

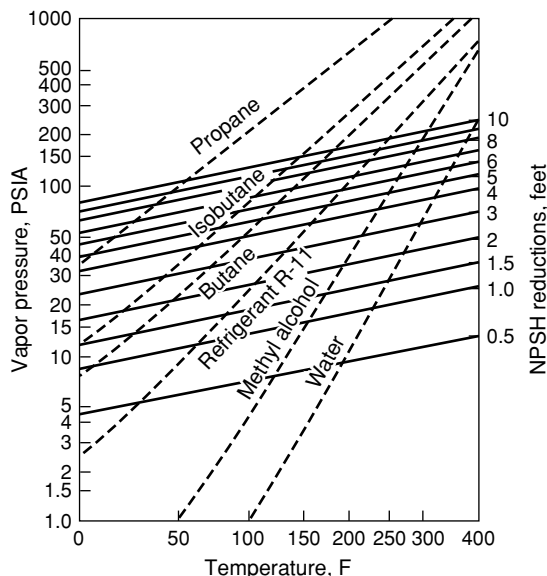
Positive-displacement pumps and those that approach positive displacement will ideally produce whatever head is impressed upon them by the system restrictions to flow. The maximum head attainable is determined by the power available in the drive (slippage neglected) and the strength of the pump parts. A pressure relief valve on the discharge side should be set to open at a safe pressure for the casing and the internal components of the pump such as piston rods, cylinders, crankshafts, and other components which would be pressurized. In the case of a rotary pump, the total dynamic head developed is uniquely determined for any given flow by the speed at which it rotates.

In general, overall efficiencies of positive-displacement pumps are higher than those of centrifugal equipment because internal losses are minimized. On the other hand, the flexibility of each piece of equipment in handling a wide range of capacities is somewhat limited.

Positive-displacement pumps may be of either the **reciprocating** or the **rotary** type. In all positive-displacement pumps, a cavity or cavities are alternately filled and emptied of the pumped fluid by the action of the pump.

**Reciprocating Pumps** There are three classes of reciprocating pumps: **piston pumps**, **plunger pumps**, and **diaphragm pumps**. Basically, the action of the liquid-transferring parts of these pumps is the same, a cylindrical piston, plunger, or bucket or a round diaphragm being caused to pass or flex back and forth in a chamber. The device is equipped with valves for the inlet and discharge of the liquid being pumped, and the operation of these valves is related in a definite manner to the motions of the piston. In all modern-design reciprocating pumps, the suction and discharge valves are operated by pressure difference. That is, when the pump is on its suction stroke and the pump cavity is increasing in volume, the pressure is lowered within the pump cavity, permitting the higher suction pressure to open the suction valve and allowing liquid to flow into the pump. At the same time, the higher discharge-line pressure holds the discharge valve closed. Likewise on the discharge stroke, as the pump cavity is decreasing in volume, the higher pressure developed in the pump cavity holds the suction valve closed and opens the discharge valve to expel liquid from the pump into the discharge line.

The *overall efficiency* of these pumps varies from about 50 percent for the small pumps to about 90 percent or more for the larger sizes.



**FIG. 10-25** NPSH reductions for pumps handling hydrocarbon liquids and high-temperature water. This chart has been constructed from test data obtained using the liquids shown (*Hydraulic Institute Standards*).

**TABLE 10-11 Typical Pump Specification**

| Specifi-<br>cation | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Specifi-<br>cation | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0                | Scope:<br>This specification covers horizontal, end suction, vertically split, single-stage centrifugal pumps with top centerline discharge and "back pullout" feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    | Suitable space shall be provided in the standard and oversized stuffing box for supplying a (throttle bushing) (dilution control bushing) with single seals. Throttle bushings and dilution control bushings shall be made of (glass-filled Teflon) (a suitable metal material).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.0                | Service Conditions:<br>Pump shall be designed to operate satisfactorily with a reasonable service life when operated either intermittently or continuously in typical process applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    | 4.7.2.1 <i>Lubrication—Stuffing Box with Mechanical Seals.</i> Suitable tapped connections shall be provided to effectively lubricate, cool, flush, quench, etc., as required by the application or recommendations of the mechanical seal manufacturer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3.0                | Operating Conditions:<br>Capacity _____ U.S. gallons per minute<br>Head ( _____ ft total head) ( _____ psig). Speed _____ rpm<br>Suction Pressure ( _____ ft head) (positive) (lift) ( _____ psig)<br>Liquid to be handled _____<br>Specific gravity _____ Viscosity ( _____ )<br>Temperature of liquid at inlet _____ °F<br>Solids content _____ % _____ Max. size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4.8                | Bearing Frame and Bearings:<br>4.8.1 <i>Bearing Frame.</i> Frames shall be equipped with axial radiating fins extending the length of the frame to aid in heat dissipation. Frame shall be provided with ductile iron outboard bearing housing. Both ends of the frame shall be provided with lip-type oil seals and labyrinth-type deflectors of metallic reinforced synthetic rubber to prevent the entrance of contaminants.<br>4.8.2 <i>Bearings.</i> Pump bearings shall be heavy-duty, antifriction ball-type on both ends. The single row inboard bearing, nearest the impeller, shall be free to float within the frame and shall carry only radial load. The double row outboard bearing (F4-G1 and F4-I1) or duplex angular contact bearing (F4-H1), coupling end, shall be locked in place to carry radial and axial thrust loads. Bearings shall be designed for a minimum life of 20,000 hours in any normal pump operating range. |
| 4.0                | Pump Construction:<br>4.1 <i>Casing.</i> Casing shall be vertically split with self-venting top centerline discharge, with an integral foot located directly under the casing for added support. All casings shall be of the "back pullout" design with suction and discharge nozzles cast integrally. Casings shall be provided with bosses in suction and discharge nozzles, and in bottom of casing for gauge taps and drain tap. (Threaded taps with plugs shall be provided for these features.)<br>4.2 <i>Casing Connections.</i> Connections shall be A.N.S.I. flat-faced flanges. [Cast iron (125) (250) psig rated] [Duron metal, steel, alloy steel (150) (300) psig rated]<br>4.3 <i>Casing Joint Gasket.</i> A confined-type nonasbestos gasket suitable for corrosive service shall be provided at the casing joint.<br>4.4 <i>Impeller.</i> Fully-open impeller with front edge having contoured vanes curving into the suction for minimum NPSH requirements and maximum efficiency shall be provided. A hex head shall be cast in the eye of the impeller to facilitate removal, and eliminate need for special impeller removing tool. All impellers shall have radial "pump-out" vanes on the back side to reduce stuffing box pressure and aid in eliminating collection of solids at stuffing box throat. Impellers shall be balanced within A.N.S.I. guidelines to ISO tolerances.<br>4.4.1 <i>Impeller Clearance Adjustment.</i> All pumps shall have provisions for adjustment of axial clearance between the leading edge of the impeller and casing. This adjustment shall be made by a precision microdial adjustment at the outboard bearing housing, which moves the impeller forward toward the suction wall of the casing.<br>4.5 <i>Shafts.</i> Shafts shall be suitable for hook-type sleeve. Shaft material shall be (SAE 1045 steel on Duron and 316 stainless steel pumps) or (AISI 316 stainless steel on CD-4MCu pumps and #20 stainless steel pumps). Shaft deflection shall not exceed .005 at the vertical centerline of the impeller.<br>4.6 <i>Shaft Sleeve.</i> Renewable hook-type shaft sleeve that extends through the stuffing box and gland shall be provided. Shaft sleeve shall be (316 stainless steel), (#20 stainless steel) or (XH-800 Ni-chrome-boron coated 316 stainless steel with coated surface hardness of approximately 800 Brinnell).<br>4.7 <i>Stuffing Box.</i> Stuffing box shall be suitable for packing, single (inside or outside) or double-inside mechanical seal without modifications. Stuffing box shall be accurately centered by machined rabbit fits on case and frame adapter.<br>4.7.1 <i>Packed Stuffing Box.</i> The standard packed stuffing box shall consist of five rings of graphitized nonasbestos packing; a stainless steel packing base ring in the bottom of the box to prevent extrusion of the packing past the throat; a Teflon seal cage, and a two-piece 316 stainless steel packing gland to ensure even pressure on the packing. Ample space shall be provided for repacking the stuffing box.<br>4.7.1.1 <i>Lubrication-Packed Stuffing Box.</i> A tapped hole shall be provided in the stuffing box directly over the seal cage for lubrication and cooling of the packing. Lubrication liquid shall be supplied (from an external source) (through a by-pass line from the pump discharge nozzle).<br>4.7.2 <i>Stuffing Box with Mechanical Seal.</i> Mechanical seal shall be of the (single inside) (single outside) (double inside) (cartridge) type and (balanced) (unbalanced).<br>Stuffing box is to be (standard) (oversize) (oversize tapered). | 4.9                | <i>Bearing Lubrication.</i> Ball bearings shall be oil-mist—lubricated by means of a slinger. The oil slinger shall be mounted on the shaft between the bearings to provide equal lubrication to both bearings. Bulls-eye oil-sight glasses shall be provided on both sides of the frame to provide a positive means of checking the proper oil level from either side of the pump. A tapped and plugged hole shall also be provided in both sides of the frame to mount bottle-type constant-level oilers where desired. A tapped and plugged hole shall be provided on both sides for optional straight-through oil cooling device.                                                                                                                                                                                                                                                                                                           |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5.0                | Baseplate and Coupling:<br>5.1 <i>Baseplate.</i> Baseplates shall be rigid and suitable for mounting pump and motor. Baseplates shall be of channel steel construction.<br>5.2 <i>Coupling.</i> Coupling shall be flexible-spacer type. Coupling shall have at least three-and-one-half-inch spacer length for ease of rotating element removal. Both coupling hubs shall be provided with flats 180° apart to facilitate removal of impeller. Coupling shall not require lubrication.*                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6.0                | Mechanical Modifications Required for High Temperature:<br>6.1 <i>Modifications Required, Temperature Range 250–350°F.</i> Pumps for operation in this range shall be provided with a water-jacketed stuffing box.<br>6.2 <i>Modifications Required, Temperature Range 351–550°F (Maximum).</i> Pumps for operation in this range shall be provided with a water-jacketed stuffing box and a water-cooled bearing frame.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 7.0                | Materials:<br>Pump materials shall be selected to suit the particular service requirements.<br>7.1 <i>Cast Iron—316 SS Fitted.</i> 15" only; pump shall have cast iron casing and stuffing box cover. 316 SS metal impeller; shaft shall be 1045 steel with 316 SS sleeve.<br>7.2 <i>All Duron Metal.</i> All pump materials shall be Duron metal. Shaft shall be 1045 steel, with 316 SS sleeve. 316 SS metal impeller optional.<br>7.3 <i>All AISI 316 Stainless Steel.</i> All pump materials shall be AISI 316 stainless steel. Shaft should be 1045 steel, with 316 SS sleeve.<br>7.4 <i>All #20 Stainless Steel.</i> All pump materials shall be #20 SS stainless steel. Shaft shall be 316 SS, with #20 SS sleeve.<br>7.5 <i>All CD-4MCu.</i> All pump materials shall be CD-4MCu. Shaft shall be 316 SS, with #20 SS sleeve.                                                                                                            |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 8.0                | Miscellaneous:<br>8.1 <i>Nameplates.</i> All nameplates and other data plates shall be stainless steel, suitably secured to the pump.<br>8.2 <i>Hardware.</i> All machine bolts, stud nuts, and capscrews shall be of the hex-head type.<br>8.3 <i>Rotation.</i> Pump shall have clockwise rotation viewed from its driven end.<br>8.4 <i>Parts Numbering.</i> Parts shall be completely identified with a numerical system (no alphabetical letters) to facilitate parts inventory control and stocking. Each part shall be properly identified by a separate number, and those parts that are identical shall have the same number to effect minimum spare parts inventory.                                                                                                                                                                                                                                                                   |

\*Omit if not applicable.

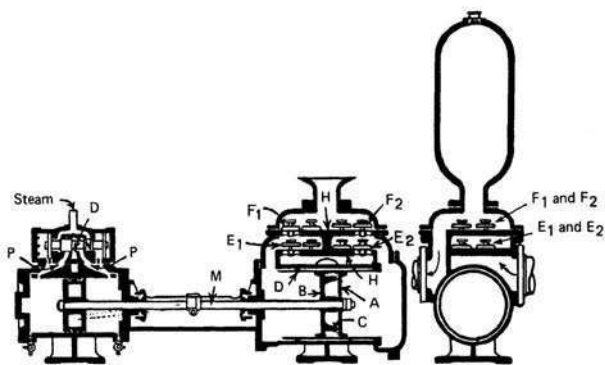


FIG. 10-26 Double-acting steam-driven reciprocating pump.

As shown in Fig. 10-26, reciprocating pumps, except when used for metering service, are frequently provided on the discharge side with gas-charged chambers, the purpose of which is to limit pressure pulsation and to provide a more uniform flow in the discharge line. In many installations, surge chambers are required on the suction side as well. Piping layouts should be studied to determine the most effective size and location. If surge chambers are used, provision should be made to keep the chamber charged with gas. A surge chamber filled with liquid is of no value. A liquid-level gauge is desirable to permit checking the amount of gas in the chamber.

Reciprocating pumps may be of **single-cylinder** or **multicylinder** design. Multicylinder pumps have all cylinders in parallel for increased capacity. Piston-type pumps may be single-acting or double-acting; i.e., pumping may be accomplished from one or both ends of the piston. Plunger pumps are always single-acting. The tabulation in Table 10-12 provides data on the flow variation of reciprocating pumps of various designs.

**Piston Pumps** There are two ordinary types of piston pumps, simplex double-acting pumps and duplex double-acting pumps.

**Simplex Double-Acting Pumps** These pumps may be direct-acting (i.e., direct-connected to a steam cylinder) or power-driven (through a crank and flywheel from the crosshead of a steam engine). Figure 10-26 is a direct-acting pump, designed for use at pressures up to 0.690 MPa (100 lbf/in<sup>2</sup>). In this figure, the piston consists of disks A and B, with packing rings C between them. A bronze liner for the water cylinder is shown at D. Suction valves are E<sub>1</sub> and E<sub>2</sub>. Discharge valves are F<sub>1</sub> and F<sub>2</sub>.

**Duplex Double-Acting Pumps** These pumps differ primarily from those of the simplex type in having two cylinders whose operation is coordinated. They may be direct-acting, steam-driven, or power-driven with crank and flywheel.

A duplex outside-end-packed **plunger pump** with pot valves, of the type used with hydraulic presses and for similar service, is shown in Fig. 10-27. In this drawing, plunger A is direct-connected to rod B, while plunger C is operated from the rod by means of yoke D and tie rods.

Plunger pumps differ from piston pumps in that they have one or more constant-diameter plungers reciprocating through packing glands and displacing liquid from cylinders in which there is consider-

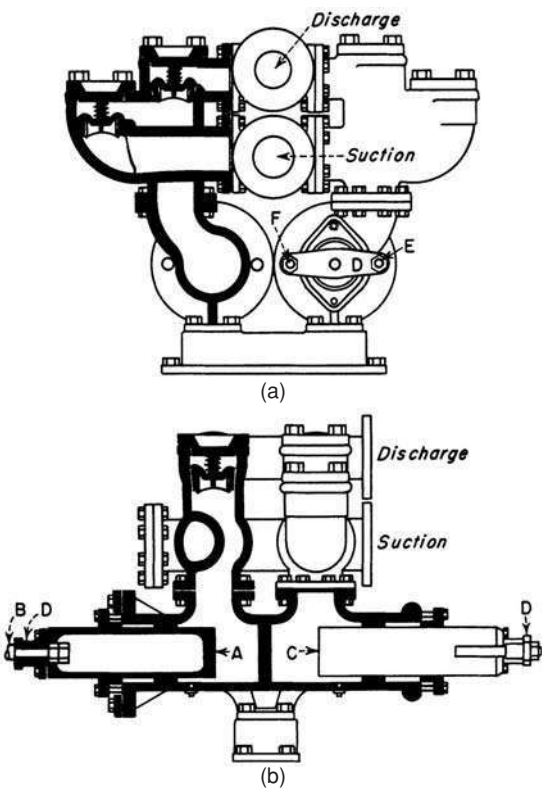


FIG. 10-27 Duplex single-acting plunger pump.

able radial clearance. They are always single-acting, in the sense that only one end of the plunger is used in pumping the liquid.

Plunger pumps are available with one, two, three, four, five, or even more cylinders. Simplex and duplex units are often built in a horizontal design. Those with three or more cylinders are usually of vertical design. The driver may be an electric motor, a steam or gas engine, or a steam turbine. This is the common type of **power pump**. An example, arranged for belt drive, is shown in Fig. 10-28 from which the action may be readily traced.

Occasionally plunger pumps are constructed with opposed cylinders and plungers connected by yokes and tie rods; this arrangement, in effect, constitutes a double-acting unit.

Simplex plunger pumps mounted singly or in gangs with a common drive are quite commonly used as **metering** or **proportioning pumps** (Fig. 10-29). Frequently a variable-speed drive or a stroke-adjusting mechanism is provided to vary the flow as desired. These pumps are designed to measure or control the flow of liquid within a deviation of  $\pm 2$  percent with capacities up to 11.35 m<sup>3</sup>/h (50 gal/min) and pressures as high as 68.9 MPa (10,000 lbf/in<sup>2</sup>).

**Diaphragm Pumps** These pumps perform similarly to piston and plunger pumps, but the reciprocating driving member is a flexible diaphragm fabricated of metal, rubber, or plastic. The chief advantage of this arrangement is the elimination of all packing and seals exposed to the liquid being pumped. This, of course, is an important asset for equipment required to handle hazardous or toxic liquids.

A common type of low-capacity diaphragm pump designed for metering service employs a plunger working in oil to actuate a metallic or plastic diaphragm. Built for pressures in excess of 6.895 MPa (1000 lbf/in<sup>2</sup>) with flow rates up to about 1.135 m<sup>3</sup>/h (5 gal/min) per cylinder, such pumps possess all the characteristics of plunger-type metering pumps with the added advantage that the pumping head can be mounted in a remote (even a submerged) location entirely separate from the drive.

TABLE 10-12 Flow Variation of Reciprocating Pumps

| Number of cylinders | Single- or double-acting | Flow variation per stroke from mean, percent |
|---------------------|--------------------------|----------------------------------------------|
| Single              | Single                   | +220 to -100                                 |
| Single              | Double                   | +60 to -100                                  |
| Duplex              | Single                   | +24.1 to -100                                |
| Duplex              | Double                   | +6.1 to -21.5                                |
| Triplex             | Single and double        | +1.8 to -16.9                                |
| Quintuplex          | Single                   | +1.8 to -5.2                                 |

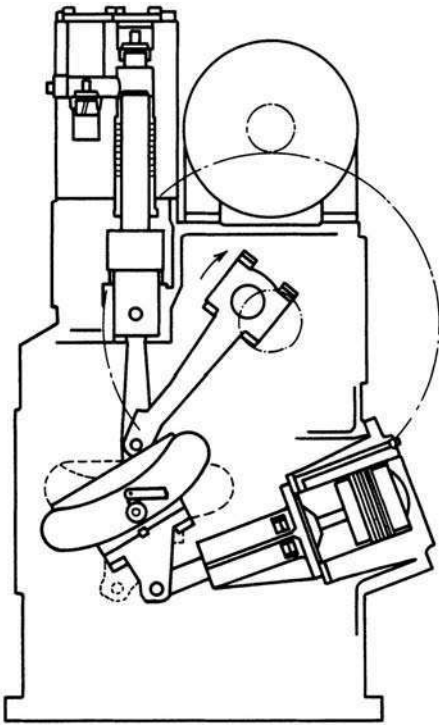


FIG. 10-28 Adrich-Groff variable-stroke power pump. (Courtesy of Ingersoll-Rand.)

Figure 10-30 shows a high-capacity 22.7-m<sup>3</sup>/h (100-gal/min) pump with actuation provided by a mechanical linkage.

**Pneumatically Actuated Diaphragm Pumps** (Fig. 10-31) These pumps require no power source other than plant compressed air. They must have a flooded suction, and the pressure is, of course, limited to the available air pressure. Because of their slow speed and large valves, they are well suited to the gentle handling of liquids for which degradation of suspended solids should be avoided.

A major consideration in the application of diaphragm pumps is the realization that diaphragm failure will probably occur eventually. The consequences of such failure should be realistically appraised before selection, and maintenance procedures should be established accordingly.

**Rotary Pumps** In rotary pumps the liquid is displaced by rotation of one or more members within a stationary housing. Because internal clearances, although minute, are a necessity in all but a few special types, capacity decreases somewhat with increasing pump

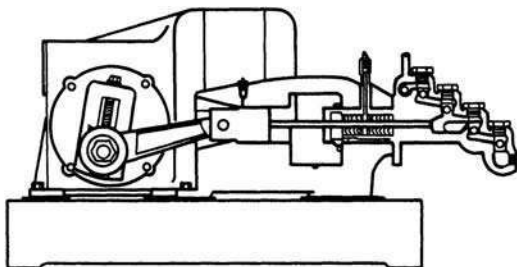


FIG. 10-29 Plunger-type metering pump. (Courtesy of Milton Roy Co.)

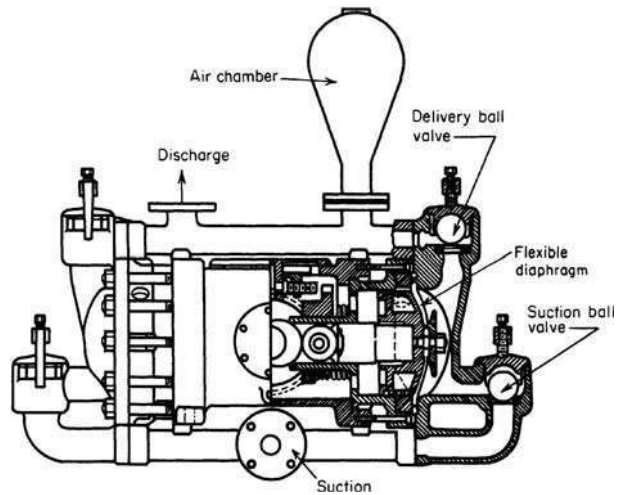


FIG. 10-30 Mechanically actuated diaphragm pump.

differential pressure. Therefore, these pumps are not truly positive-displacement pumps. However, for many other reasons they are considered as such.

The selection of materials of construction for rotary pumps is critical. The materials must be corrosion-resistant, compatible when one part is running against another, and capable of some abrasion resistance.

**Gear Pumps** When two or more impellers are used in a rotary-pump casing, the impellers will take the form of toothed-gear wheels as in Fig. 10-32, of helical gears, or of lobed cams. In each case, these impellers rotate with extremely small clearance between them and between the surfaces of the impellers and the casing. In Fig. 10-32, the two toothed impellers rotate as indicated by the arrows; the suction connection is at the bottom. The pumped liquid flows into the spaces between the impeller teeth as these cavities pass the suction opening. The liquid is then carried around the casing to the discharge opening, where it is forced out of the impeller teeth mesh. The arrows indicate this flow of liquid.

Rotary pumps are available in two general classes, interior-bearing and exterior-bearing. The **interior-bearing type** is used for handling

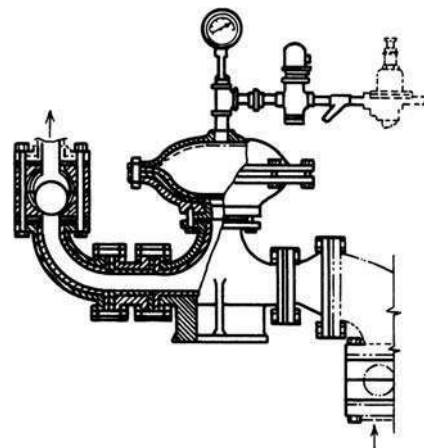


FIG. 10-31 Pneumatically actuated diaphragm pump for slurry service. (Courtesy of Dorr-Oliver Inc.)



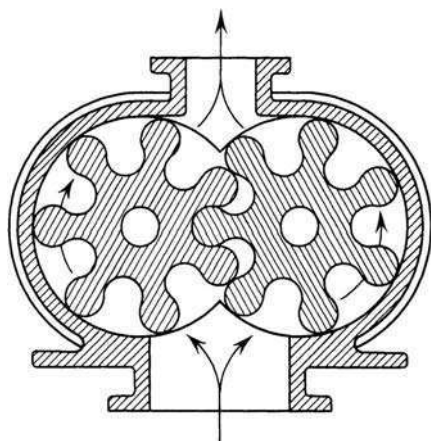


FIG. 10-32 Positive-displacement gear-type rotary pump.

liquids of a lubricating nature, and the **exterior-bearing type** is used with nonlubricating liquids. The interior-bearing pump is lubricated by the liquid being pumped, and the exterior-bearing type is oil-lubricated.

The use of spur gears in gear pumps will produce in the discharge pulsations having a frequency equivalent to the number of teeth on both gears multiplied by the speed of rotation. The amplitude of these disturbances is a function of tooth design. The pulsations can be reduced markedly by the use of rotors with helical teeth. This in turn introduces end thrust, which can be eliminated by the use of double-helical or herringbone teeth.

**Screw Pumps** A modification of the helical gear pump is the screw pump. Both gear and screw pumps are positive-displacement pumps. Figure 10-33 illustrates a two-rotor version in which the liquid is fed to either the center or the ends, depending upon the direction of rotation, and progresses axially in the cavities formed by the meshing threads or teeth. In three-rotor versions, the center rotor is the driving member while the other two are driven. Figure 10-34 shows still another arrangement, in which a metal rotor of unique design rotates without clearance in an elastomeric stationary sleeve.

Screw pumps, because of multiple dams that reduce slip, are well adapted for producing higher pressure rises, for example, 6.895 MPa (1000 lbf/in<sup>2</sup>), especially when handling viscous liquids such as heavy oils. The all-metal pumps are generally subject to the same limitations on handling abrasive solids as conventional gear pumps. In addition, the wide bearing spans usually demand that the liquid have considerable lubricity to prevent metal-to-metal contact.

Among the liquids handled by rotary pumps are mineral oils, vegetable oils, animal oils, greases, glucose, viscose, molasses, paints, var-

nish, shellac, lacquers, alcohols, catsup, brine, mayonnaise, sizing, soap, tanning liquors, vinegar, and ink. Some screw-type units are specially designed for the gentle handling of large solids suspended in the liquid.

**Fluid-Displacement Pumps** In addition to pumps that depend on the mechanical action of pistons, plungers, or impellers to move the liquid, other devices for this purpose employ displacement by a secondary fluid. This group includes air lifts and acid eggs.

**The air lift** is a device for raising liquid by means of compressed air. In the past it was widely used for pumping wells, but it has been less widely used since the development of efficient centrifugal pumps. It operates by introducing compressed air into the liquid near the bottom of the well. The air-and-liquid mixture, being lighter than liquid alone, rises in the well casing. The advantage of this system of pumping lies in the fact that there are no moving parts in the well. The pumping equipment is an air compressor, which can be located on the surface.

A simplified sketch of an air lift for this purpose is shown in Fig. 10-35. Ingersoll-Rand has developed empirical information on air-lift performance which is available upon request.

An important application of the gas-lift principle involves the extraction of oil from wells. There are several references to both practical and theoretical work involving gas lift performance and related problems. Recommended sources are American Petroleum Institute, *Drilling and Production Practices*, 1952, pp. 257–317, and 1939, p. 266; *Trans. Am. Soc. Mining Metall. Eng.*, **92**, 296–313 (1931), **103**, 170–186 (1933), **118**, 56–70 (1936), **192**, 317–326 (1951), **189**, 73–82 (1950), and **198**, 271–278 (1953); *Trans. Am. Soc. Mining Metall.*, and *Pet. Eng.*, **213** (1958), and **207**, 17–24 (1956); and *Univ. Wisconsin Bull., Eng. Ser.*, **6**, no. 7 (1911, reprinted 1914).

An **acid egg**, or **blowcase**, consists of an egg-shaped container which can be filled with a charge of liquid that is to be pumped. This container is fitted with an inlet pipe for the charge, an outlet pipe for the discharge, and a pipe for the admission of compressed air or gas, as illustrated in Fig. 10-36. Pressure of air or gas on the surface of the liquid forces it out of the discharge pipe. Such pumps can be hand-operated or arranged for semiautomatic or automatic operation.

## CENTRIFUGAL PUMPS

The centrifugal pump is the type most widely used in the chemical industry for transferring liquids of all types—raw materials, materials in manufacture, and finished products—as well as for general services of water supply, boiler feed, condenser circulation, condensate return, etc. These pumps are available through a vast range of sizes, in capacities from 0.5 m<sup>3</sup>/h to 2 × 10<sup>4</sup> m<sup>3</sup>/h (2 gal/min to 10<sup>5</sup> gal/min), and for discharge heads (pressures) from a few meters to approximately 48 MPa (7000 lbf/in<sup>2</sup>). The size and type best suited to a particular application can be determined only by an engineering study of the problem.

The primary advantages of a centrifugal pump are simplicity, low first cost, uniform (nonpulsating) flow, small floor space, low maintenance expense, quiet operation, and adaptability for use with a motor or a turbine drive.

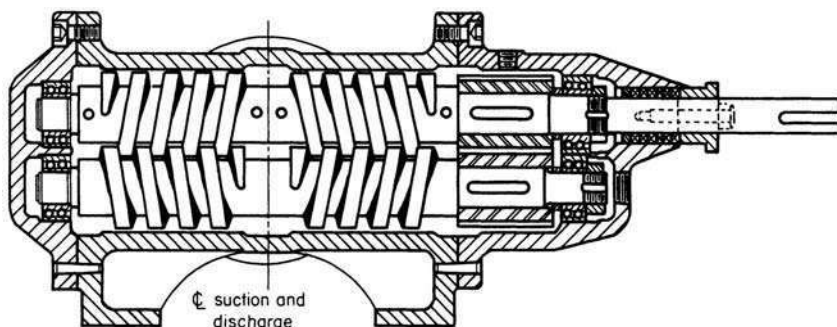


FIG. 10-33 Two-rotor screw pump. (Courtesy of Warren Quimby Pump Co.)

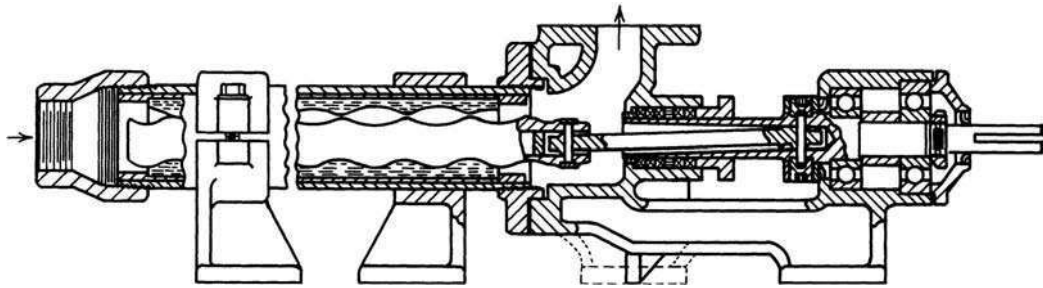


FIG. 10-34 Single-rotor screw pump with an elastomeric lining. (Courtesy of Moyno Pump Division, Robbins & Myers, Inc.)

A centrifugal pump, in its simplest form, consists of an impeller rotating within a casing. The **impeller** consists of a number of blades, either open or shrouded, mounted on a shaft that projects outside the casing. Its axis of rotation may be either horizontal or vertical, to suit the work to be done. **Closed-type**, or **shrouded**, impellers are generally the most efficient. **Open- or semiopen-type** impellers are used for viscous liquids or for liquids containing solid materials and on many small pumps for general service. Impellers may be of the **single-suction** or the **double-suction** type—single if the liquid enters from one side, double if it enters from both sides.

**Casings** There are three general types of casings, but each consists of a chamber in which the impeller rotates, provided with inlet and exit for the liquid being pumped. The simplest form is the **circular casing**, consisting of an annular chamber around the impeller; no attempt is made to overcome the losses that will arise from eddies and shock when the liquid leaving the impeller at relatively high velocities enters this chamber. Such casings are seldom used.

**Volute casings** take the form of a spiral increasing uniformly in cross-sectional area as the outlet is approached. The volute efficiently converts the velocity energy imparted to the liquid by the impeller into pressure energy.

A third type of casing is used in **diffuser-type** or turbine pumps. In this type, **guide vanes** or **diffusers** are interposed between the impeller discharge and the casing chamber. Losses are kept to a minimum in a well-designed pump of this type, and improved efficiency is obtained over a wider range of capacities. This construction is often used in multistage high-head pumps.

**Action of a Centrifugal Pump** Briefly, the action of a centrifugal pump may be shown by Fig. 10-37. Power from an outside source is applied to shaft A, rotating the impeller B within the stationary casing C. The blades of the impeller in revolving produce a reduction in

pressure at the entrance or eye of the impeller. This causes liquid to flow into the impeller from the suction pipe D. This liquid is forced outward along the blades at increasing tangential velocity. The velocity head it has acquired when it leaves the blade tips is changed to pressure head as the liquid passes into the volute chamber and thence out the discharge E.

**Centrifugal Pump Characteristics** Figure 10-38 shows a typical characteristic curve of a centrifugal pump. It is important to note that at any fixed speed the pump will operate along this curve and at no other points. For instance, on the curve shown, at  $45.5 \text{ m}^3/\text{h}$  (200 gal/min) the pump will generate 26.5-m (87-ft) head. If the head is increased to 30.48 m (100 ft),  $27.25 \text{ m}^3/\text{h}$  (120 gal/min) will be delivered. It is not possible to reduce the capacity to  $27.25 \text{ m}^3/\text{h}$  (120 gal/min) at 26.5-m (87-ft) head unless the discharge is throttled so that 30.48 m (100 ft) is actually generated within the pump. On pumps with variable-speed drivers such as steam turbines, it is possible to change the characteristic curve, as shown by Fig. 10-39.

As shown in Eq. (10-50), the head depends upon the velocity of the fluid, which in turn depends upon the capability of the impeller

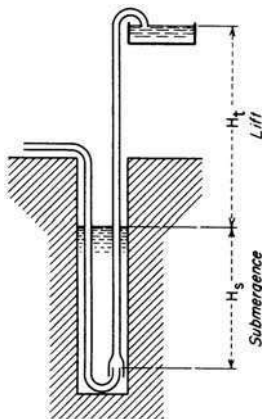


FIG. 10-35 Simplified sketch of an air lift, showing submergence and total head.

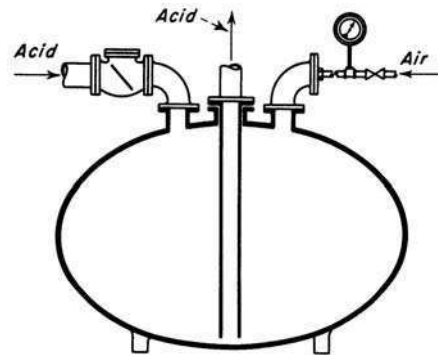


FIG. 10-36 A form of acid egg. External controls required for automatic operation are not shown.

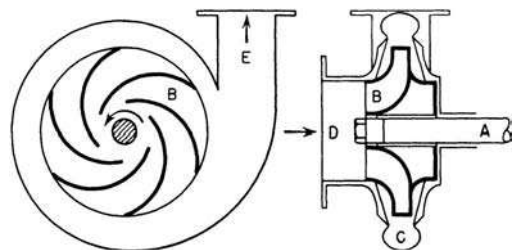
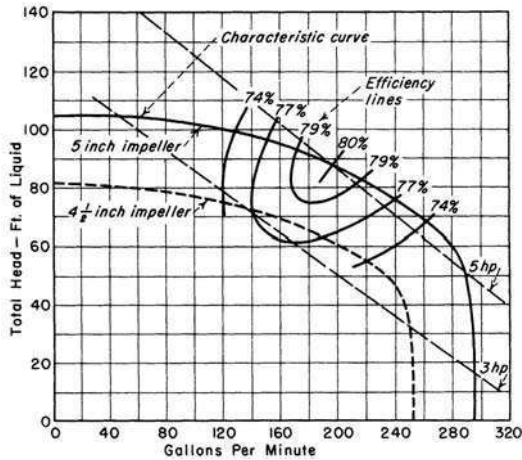


FIG. 10-37 A simple centrifugal pump.

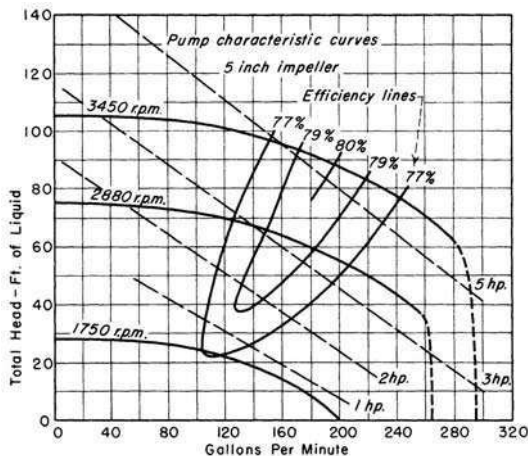


**FIG. 10-38** Characteristic curve of a centrifugal pump operating at a constant speed of 3450 r/min. To convert gallons per minute to cubic meters per hour, multiply by 0.2271; to convert feet to meters, multiply by 0.3048; to convert horsepower to kilowatts, multiply by 0.746; and to convert inches to centimeters, multiply by 2.54.

to transfer energy to the fluid. This is a function of the fluid viscosity and the impeller design. It is important to remember that the head produced will be the same for any liquid of the same viscosity. The pressure rise, however, will vary in proportion to the specific gravity.

For quick pump selection, manufacturers often give the most essential performance details for a whole range of pump sizes. Figure 10-40 shows typical performance data for a range of process pumps based on suction and discharge pipes and impeller diameters. The performance data consists of pump flow rate and head. Once a pump meets a required specification, then a more detailed performance data for the particular pump can be easily found based on the curve reference number. Figure 10-41 shows a more detailed pump performance curve that includes, in addition to pump head and flow, the brake horsepower required, NPSH required, number of vanes, and pump efficiency for a range of impeller diameters.

If detailed manufacturer-specified performance curves are not available for a different size of the pump or operating condition, a best



**FIG. 10-39** Characteristic curve of a centrifugal pump at various speeds. To convert gallons per minute to cubic meters per hour, multiply by 0.2271; to convert feet to meters, multiply by 0.3048; to convert horsepower to kilowatts, multiply by 0.746; and to convert inches to centimeters, multiply by 2.54.

estimate of the off-design performance of pumps can be obtained through similarity relationship or the affinity laws. These are:

1. Capacity ( $Q$ ) is proportional to impeller rotational speed ( $N$ ).
2. Head ( $h$ ) varies as square of the impeller rotational speed.
3. Brake horsepower (BHP) varies as the cube of the impeller rotational speed.

These equations can be expressed mathematically and appear in Table 10-13.

**System Curves** In addition to the pump design, the operational performance of a pump depends upon factors such as the downstream load characteristics, pipe friction, and valve performance. Typically, head and flow follow the following relationship:

$$\frac{(Q_2)^2}{(Q_1)^2} = \frac{h_2}{h_1} \quad (10-60)$$

where the subscript 1 refers to the design condition and 2 to the actual conditions. The above equation indicates that head will change as a square of the water flow rate.

Figure 10-42 shows the schematic of a pump, moving a fluid from tank A to tank B, both of which are at the same level. The only force that the pump has to overcome in this case is the pipe friction, variation of which with fluid flow rate is also shown in the figure. On the other for the use shown in Fig. 10-43, the pump in addition to pipe friction should overcome head due to difference in elevation between tanks A and B. In this case, elevation head is constant, whereas the head required to overcome friction depends on the flow rate. Figure 10-44 shows the pump performance requirement of a valve opening and closing.

**Pump Selection** One of the parameters that is extremely useful in selecting a pump for a particular application is specific speed  $N_s$ . Specific speed of a pump can be evaluated based on its design speed, flow, and head:

$$N_s = \frac{NQ^{1/2}}{H^{3/4}} \quad (10-61)$$

where  $N$  = rpm,  $Q$  is flow rate in gpm, and  $H$  is head in ft-lb/lbm.

Specific speed is a parameter that defines the speed at which impellers of geometrically similar design have to be run to discharge one gallon per minute against a one-foot head. In general, pumps with a low specific speed have a low capacity and high specific speed, high capacity. Specific speeds of different types of pumps are shown in Table 10-14 for comparison.

Another parameter that helps in evaluating the pump suction limitations, such as cavitation, is suction-specific speed.

$$S = \frac{NQ^{1/2}}{(\text{NPSH})^{3/4}} \quad (10-62)$$

Typically, for single-suction pumps, suction-specific speed above 11,000 is considered excellent. Below 7000 is poor and 7000–9000 is of an average design. Similarly, for double-suction pumps, suction-specific speed above 14,000 is considered excellent, below 7000 is poor, and 9000–11,000 is average.

Figure 10-45 shows the schematic of specific-speed variation for different types of pumps. The figure clearly indicates that, as the specific speed increases, the ratio of the impeller outer diameter  $D_1$  to inlet or eye diameter  $D_2$  decreases, tending to become unity for pumps of axial-flow type.

Typically, axial flow pumps are of high flow and low head type and have a high specific speed. On the other hand, purely radial pumps are of high head and low flow rate capability and have a low specific speed. Obviously, a pump with a moderate flow and head has an average specific speed.

A typical pump selection chart such as shown in Fig. 10-46 calculates the specific speed for a given flow, head, and speed requirements. Based on the calculated specific speed, the optimal pump design is indicated.

**Process Pumps** This term is usually applied to single-stage pedestal-mounted units with single-suction overhung impellers and with a single packing box. These pumps are ruggedly designed for

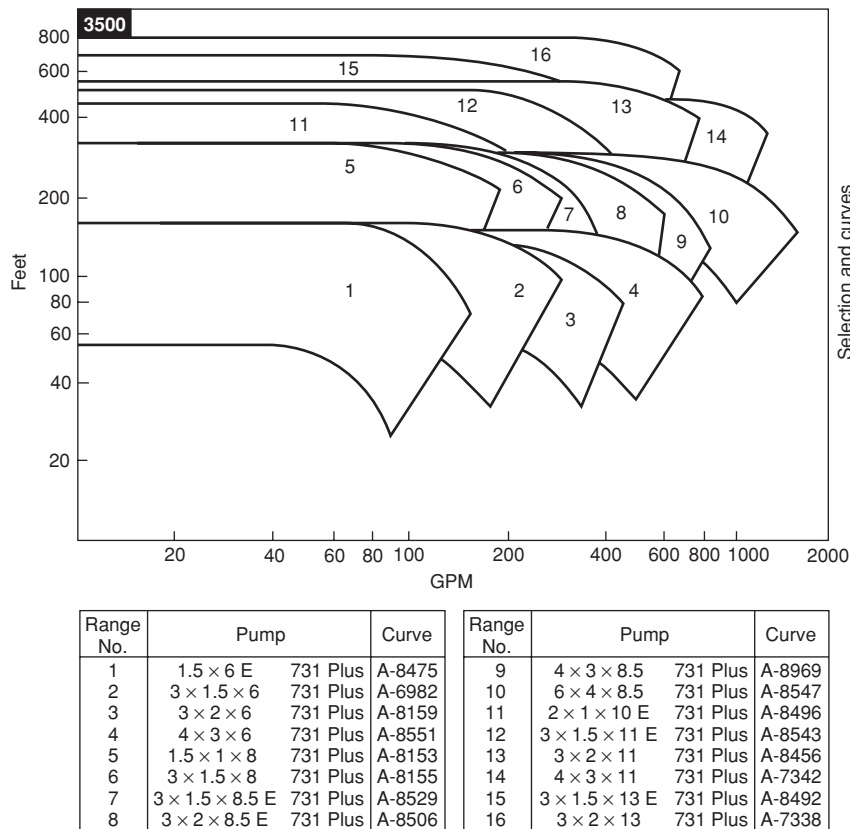


FIG. 10-40 Performance curves for a range of open impeller pumps.

ease in dismantling and accessibility, with mechanical seals or packing arrangements, and are built especially to handle corrosive or otherwise difficult-to-handle liquids.

Specifically but not exclusively for the chemical industry, most pump manufacturers now build to national standards **horizontal and vertical process pumps**. ASME Standards B73.1—2001 and B73.2—2003 apply to the horizontal (Fig. 10-47) and vertical in-line (Fig. 10-48) pumps, respectively.

The horizontal pumps are available for capacities up to 900 m<sup>3</sup>/h (4000 gal/min); the vertical in-line pumps, for capacities up to 320 m<sup>3</sup>/h (1400 gal/min). Both horizontal and vertical in-line pumps are available for heads up to 120 m (400 ft). The intent of each ANSI specification is that pumps from all vendors for a given nominal capacity and total dynamic head at a given rotative speed shall be dimensionally interchangeable with respect to mounting, size, and location of suction and discharge nozzles, input shaft, base plate, and foundation bolts.

The vertical in-line pumps, although relatively new additions, are finding considerable use in chemical and petrochemical plants in the United States. An inspection of the two designs will make clear the relative advantages and disadvantages of each.

Chemical pumps are available in a variety of materials. Metal pumps are the most widely used. Although they may be obtained in iron, bronze, and iron with bronze fittings, an increasing number of pumps of ductile-iron, steel, and nickel alloys are being used. Pumps are also available in glass, glass-lined iron, carbon, rubber, rubber-lined metal, ceramics, and a variety of plastics, such units usually being employed for special purposes.

**Sealing the Centrifugal Chemical Pump** Although detailed treatment of **shaft seals** is presented in the subsection “Sealing of

Rotating Shafts,” it is appropriate to mention here the special problems of sealing centrifugal chemical pumps. Current practice demands that packing boxes be designed to accommodate both packing and mechanical seals. With either type of seal, one consideration is of paramount importance in chemical service; the liquid present at the sealing surfaces must be free of solids. Consequently, it is necessary to provide a secondary compatible liquid to flush the seal or packing whenever the process liquid is not absolutely clean.

The use of **packing** requires the continuous escape of liquid past the seal to minimize and to carry away the frictional heat developed. If the effluent is toxic or corrosive, quench glands or catch pans are usually employed. Although packing can be adjusted with the pump operating, leaking mechanical seals require shutting down the pump to correct the leak. Properly applied and maintained **mechanical seals** usually show no visible leakage. In general, owing to the more effective performance of mechanical seals, they have gained almost universal acceptance.

**Double-Suction Single-Stage Pumps** These pumps are used for general water-supply and circulating service and for chemical service when liquids that are noncorrosive to iron or bronze are being handled. They are available for capacities from about 5.7 m<sup>3</sup>/h (25 gal/min) up to as high as  $1.136 \times 10^4$  m<sup>3</sup>/h (50,000 gal/min) and heads up to 304 m (1000 ft). Such units are available in iron, bronze, and iron with bronze fittings. Other materials increase the cost; when they are required, a standard chemical pump is usually more economical.

**Close-Coupled Pumps** (Fig. 10-49) Pumps equipped with a built-in electric motor or sometimes steam-turbine-driven (i.e., with pump

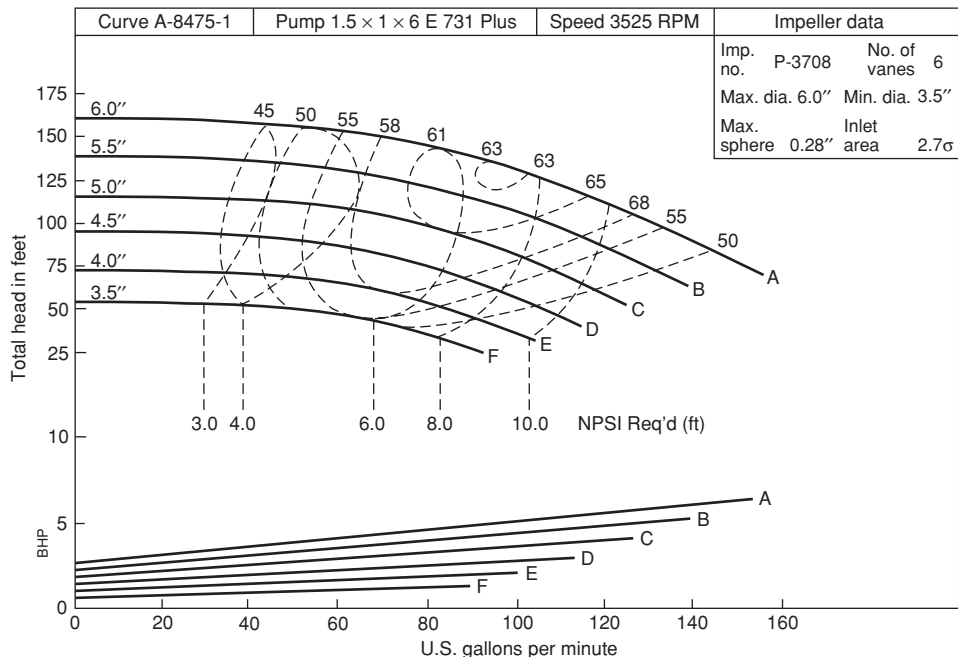


FIG. 10-41 Typical pump performance curve. The curve is shown for water at 85°F. If the specific gravity of the fluid is other than unity, BHP must be corrected.

impeller and driver on the same shaft) are known as close-coupled pumps. Such units are extremely compact and are suitable for a variety of services for which standard iron and bronze materials are satisfactory. They are available in capacities up to about 450 m<sup>3</sup>/h (2000 gal/min) for heads up to about 73 m (240 ft). Two-stage units in the smaller sizes are available for heads to around 150 m (500 ft).

**Canned-Motor Pumps** (Fig. 10-50) These pumps command considerable attention in the chemical industry. They are close-coupled units in which the cavity housing the motor rotor and the pump casing are interconnected. As a result, the motor bearings run in the process liquid and all seals are eliminated. Because the process liquid is the bearing lubricant, abrasive solids cannot be tolerated. Standard single-stage canned-motor pumps are available for flows up to 160 m<sup>3</sup>/h (700 gal/min) and heads up to 76 m (250 ft). Two-stage units are available for heads up to 183 m (600 ft). Canned-motor pumps are being widely used for handling organic solvents, organic heat-transfer liquids, and light oils as well as many clean toxic or hazardous liquids or for installations in which leakage is an economic problem.

**Vertical Pumps** In the chemical industry, the term **vertical process pump** (Fig. 10-51) generally applies to a pump with a vertical shaft having a length from drive end to impeller of approximately 1 m (3.1 ft) minimum to 20 m (66 ft) or more. Vertical pumps are used as either **wet-pit pumps** (immersed) or **dry-pit pumps** (externally mounted) in conjunction with stationary or mobile tanks containing

difficult-to-handle liquids. They have the following advantages: the liquid level is above the impeller, and the pump is thus self-priming; and the shaft seal is above the liquid level and is not wetted by the pumped liquid, which simplifies the sealing task. When no bottom connections are permitted on the tank (a safety consideration for highly corrosive or toxic liquid), the vertical wet-pit pump may be the only logical choice.

These pumps have the following disadvantages: intermediate or line bearings are generally required when the shaft length exceeds about 3 m (10 ft) in order to avoid shaft resonance problems; these

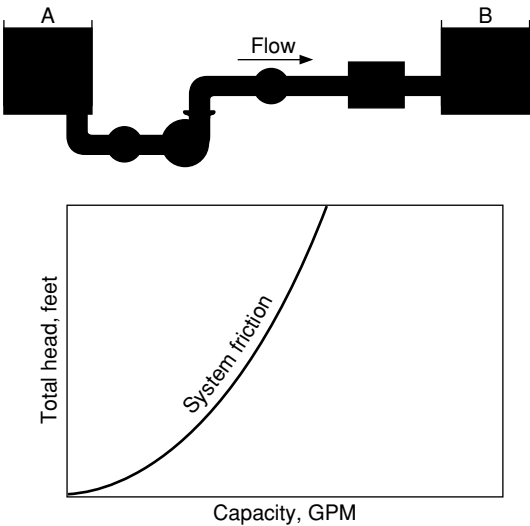
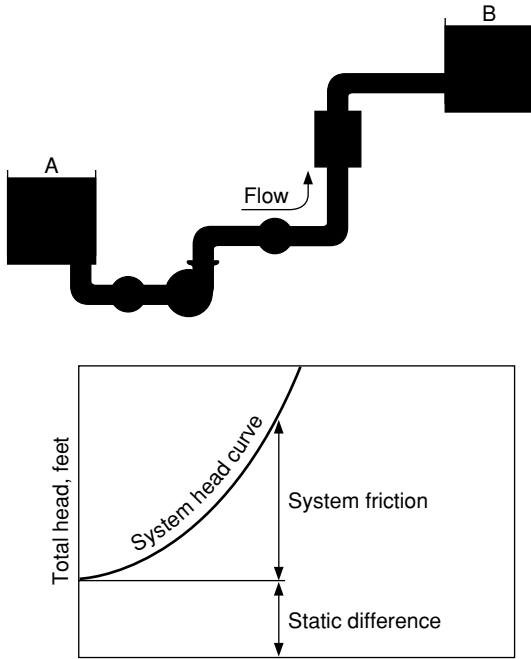


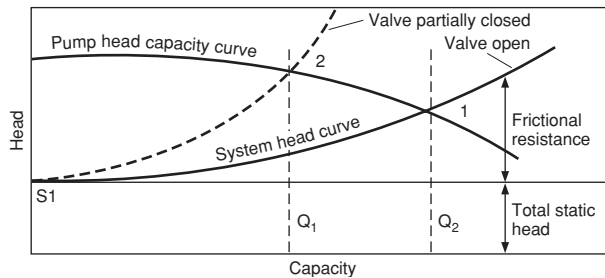
FIG. 10-42 Variation of total head versus flow rate to overcome friction.

TABLE 10-13 The Affinity Laws

|                  | Constant impeller diameter                             | Constant impeller speed                                |
|------------------|--------------------------------------------------------|--------------------------------------------------------|
| Capacity         | $\frac{Q_1}{Q_2} = \frac{N_1}{N_2}$                    | $\frac{Q_1}{Q_2} = \frac{D_1}{D_2}$                    |
| Head             | $\frac{H_1}{H_2} = \left(\frac{N_1}{N_2}\right)^2$     | $\frac{h_1}{h_2} = \left(\frac{D_1}{D_2}\right)^2$     |
| Brake horsepower | $\frac{BHP_1}{BHP_2} = \left(\frac{N_1}{N_2}\right)^3$ | $\frac{BHP_1}{BHP_2} = \left(\frac{P_1}{P_2}\right)^3$ |



**FIG. 10-43** Variation of total head as a function of flow rate to overcome both friction and static head.



**FIG. 10-44** Typical steady-state response of a pump system with a valve fully and partially open.

**TABLE 10-14 Specific Speeds of Different Types of Pumps**

| Pump type    | Specific speed range         |
|--------------|------------------------------|
| Below 2,000  | Process pumps and feed pumps |
| 2,000–5,000  | Turbine pumps                |
| 4,000–10,000 | Mixed-flow pumps             |
| 9,000–15,000 | Axial-flow pumps             |

bearings must be lubricated whenever the shaft is rotating. Since all wetted parts must be corrosion-resistant, low-cost materials may not be suitable for the shaft, column, etc. Maintenance is more costly since the pumps are larger and more difficult to handle.

For abrasive service, vertical cantilever designs requiring no line or foot bearings are available. Generally, these pumps are limited to about a 1-m (3.1-ft) maximum shaft length. Vertical pumps are also used to pump waters to reservoirs. One such application in the Los Angeles water basin has fourteen 4-stage pumps, each pump requiring 80,000 hp to drive them.

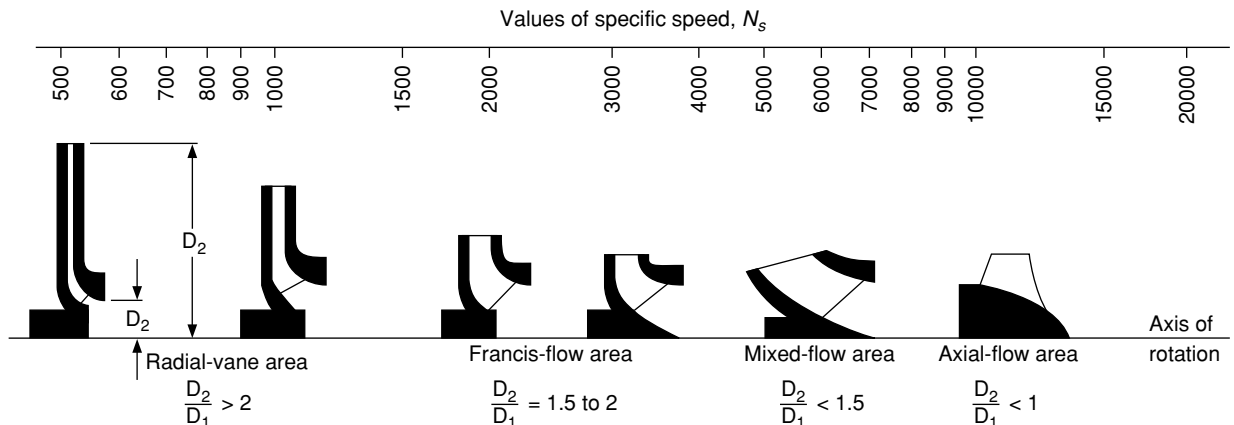
**Sump Pumps** These are small single-stage vertical pumps used to drain shallow pits or sumps. They are of the same general construction as vertical process pumps but are not designed for severe operating conditions.

**Multistage Centrifugal Pumps** These pumps are used for services requiring heads (pressures) higher than can be generated by a single impeller. All impellers are in series, the liquid passing from one impeller to the next and finally to the pump discharge. The total head then is the summation of the heads of the individual impellers. Deepwell pumps, high-pressure water-supply pumps, boiler-feed pumps, fire pumps, and charge pumps for refinery processes are examples of multistage pumps required for various services.

Multistage pumps may be of the **volute type** (Fig. 10-52), with single- or double-suction impellers (Fig. 10-53), or of the **diffuser type** (Fig. 10-54). They may have horizontally split casings or, for extremely high pressures, 20 to 40 MPa (3000 to 6000 lb/in<sup>2</sup>), vertically split barrel-type exterior casings with inner casings containing diffusers, interstage passages, etc.

## PROPELLER AND TURBINE PUMPS

**Axial-Flow (Propeller) Pumps** (Fig. 10-55) These pumps are essentially very-high-capacity low-head units. Normally they are designed for flows in excess of 450 m<sup>3</sup>/h (2000 gal/min) against heads of 15 m (50 ft) or less. They are used to great advantage in closed-loop circulation systems in which the pump casing becomes merely an



**FIG. 10-45** Specific speed variations of different types of pump.



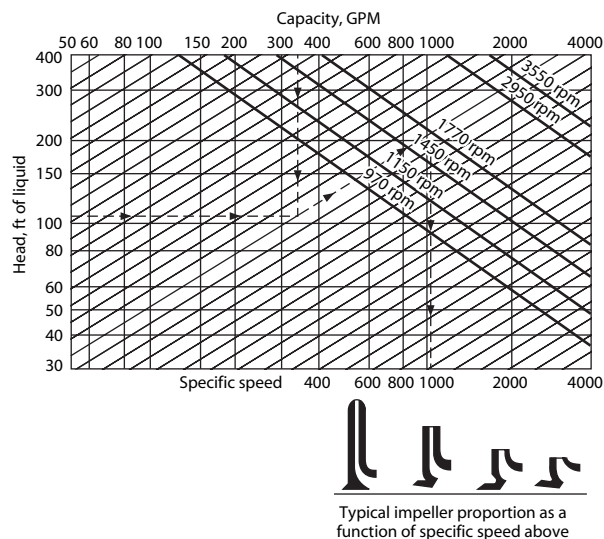


FIG. 10-46 Relationships between specific speed, rotative speed, and impeller proportions (Worthington Pump Inc., Pump World, vol. 4, no. 2, 1978).

elbow in the line. A common installation is for calandria circulation. A characteristic curve of an axial-flow pump is given in Fig. 10-56.

**Turbine Pumps** The term “turbine pump” is applied to units with mixed-flow (part axial and part centrifugal) impellers. Such units are available in capacities from 20 m<sup>3</sup>/h (100 gal/min) upward for heads up to about 30 m (100 ft) per stage. Turbine pumps are usually vertical.

A common form of turbine pump is the vertical pump, which has the pump element mounted at the bottom of a column that serves as the discharge pipe (see Fig. 10-57). Such units are immersed in the liquid to be pumped and are commonly used for wells, condenser circulating water, large-volume drainage, etc. Another form of the pump has a shell surrounding the pumping element which is connected to the intake pipe. In this form, the pump is used on condensate service in power plants and for process work in oil refineries.

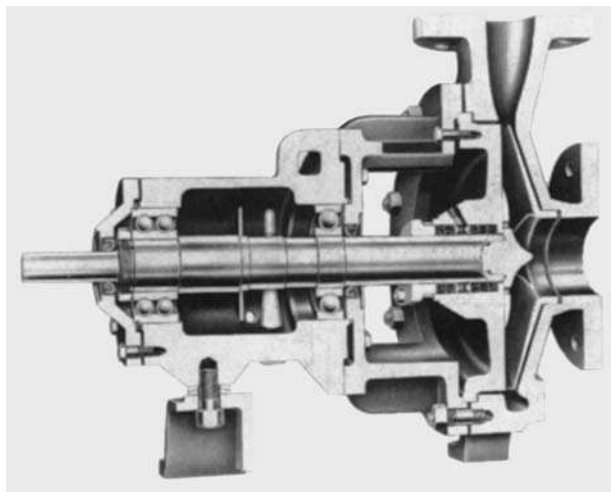


FIG. 10-47 Horizontal process pump conforming to American Society of Mechanical Engineers Standards B73.1—2001.

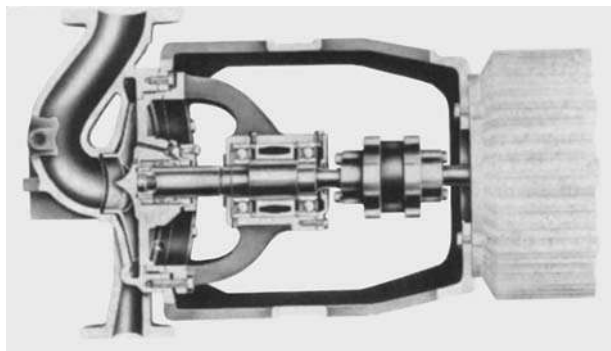


FIG. 10-48 Vertical in-line process pump conforming to ASME standard B73.2—2003. The pump shown is driven by a motor through flexible coupling. Not shown but also conforming to ASME Standard B73.2—2003 are vertical in-line pumps with rigid couplings and with no coupling (impeller-mounted on an extended motor shaft).

**Regenerative Pumps** Also referred to as turbine pumps because of the shape of the impeller, regenerative pumps employ a combination of mechanical impulse and centrifugal force to produce heads of several hundred meters (feet) at low volumes, usually less than 20 m<sup>3</sup>/h (100 gal/min). The impeller, which rotates at high speed with small clearances, has many short radial passages milled on each side at the periphery. Similar channels are milled in the mating surfaces of the casing. Upon entering, the liquid is directed into the impeller passages and proceeds in a spiral pattern around the periphery, passing alternately from the impeller to the casing and receiving successive impulses as it does so. Figure 10-58 illustrates a typical performance-characteristic curve.

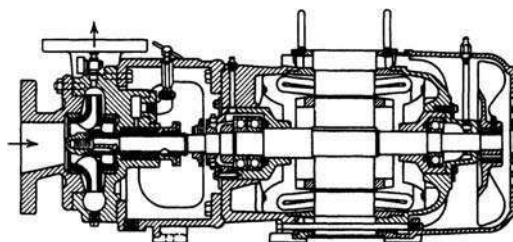


FIG. 10-49 Close-coupled pump.

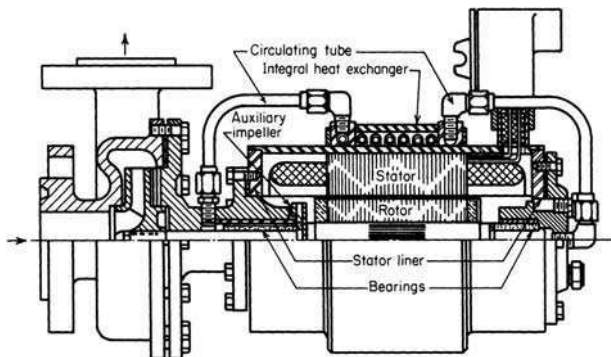
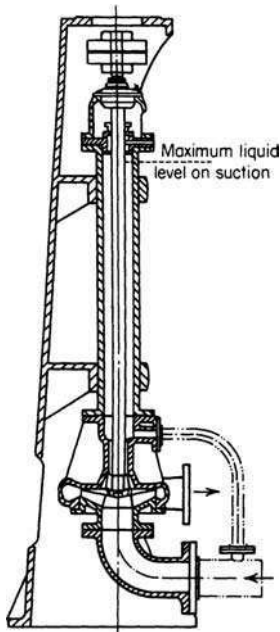


FIG. 10-50 Canned-motor pump. (Courtesy of Chempump Division, Crane Co.)



**FIG. 10-51** Vertical process pump for dry-pit mounting. (Courtesy of Lawrence Pumps, Inc.)

These pumps are particularly useful when low volumes of low-viscosity liquids must be handled at higher pressures than are normally available with centrifugal pumps. Close clearances limit their use to clean liquids. For very high heads, multistage units are available.

### JET PUMPS

Jet pumps are a class of liquid-handling device that makes use of the momentum of one fluid to move another.

**Ejectors and injectors** are the two types of jet pumps of interest to chemical engineers. The ejector, also called the siphon, exhauster, or eductor, is designed for use in operations in which the head pumped against is low and is less than the head of the fluid used for pumping. The injector is a special type of jet pump, operated by steam and used for boiler feed and similar services, in which the fluid being pumped is discharged into a space under the same pressure as that of the steam being used to operate the injector.

Figure 10-59 shows a simple design for a jet pump of the ejector type. The pumping fluid enters through the nozzle at the left and passes through the venturi nozzle at the center and out of the discharge opening at the right. As it passes into the venturi nozzle, it develops a suction that causes some of the fluid in the suction chamber to be entrained with the stream and delivered through this discharge.

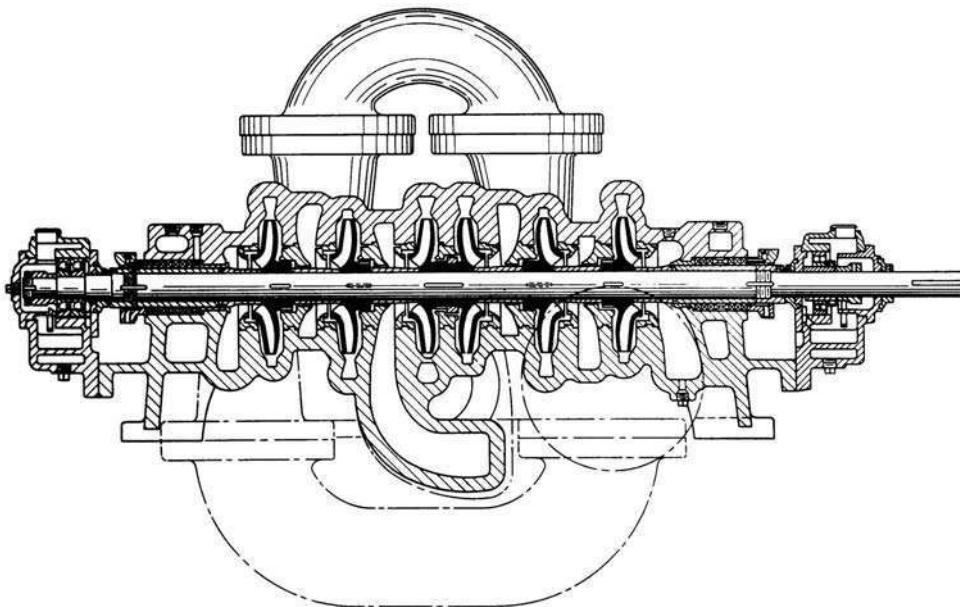
The efficiency of an ejector or jet pump is low, being only a few percent. The head developed by the ejector is also low except in special types. The device has the disadvantage of diluting the fluid pumped by mixing it with the pumping fluid. In steam injectors for boiler feed and similar services in which the heat of the steam is recovered, efficiency is close to 100 percent.

The simple ejector or siphon is widely used, in spite of its low efficiency, for transferring liquids from one tank to another, for lifting acids, alkalis, or solid-containing liquids of an abrasive nature, and for emptying sumps.

### ELECTROMAGNETIC PUMPS

The necessity of circulating liquid-metal heat-transfer media in nuclear-reactor systems has led to development of electromagnetic pumps. All electromagnetic pumps utilize the motor principle: a conductor in a magnetic field, carrying a current which flows at right angles to the direction of the field, has a force exerted on it, the force being mutually perpendicular to both the field and the current. In all electromagnetic pumps, the fluid is the conductor. This force, suitably directed in the fluid, manifests itself as a pressure if the fluid is suitably contained. The field and current can be produced in a number of different ways and the force utilized variously.

Both alternating- and direct-current units are available. While dc pumps (Fig. 10-60) are simpler, their high-current requirement is a definite limitation; ac pumps can readily obtain high currents by making use of transformers. Multipole induction ac pumps have been



**FIG. 10-52** Six-stage volute-type pump.



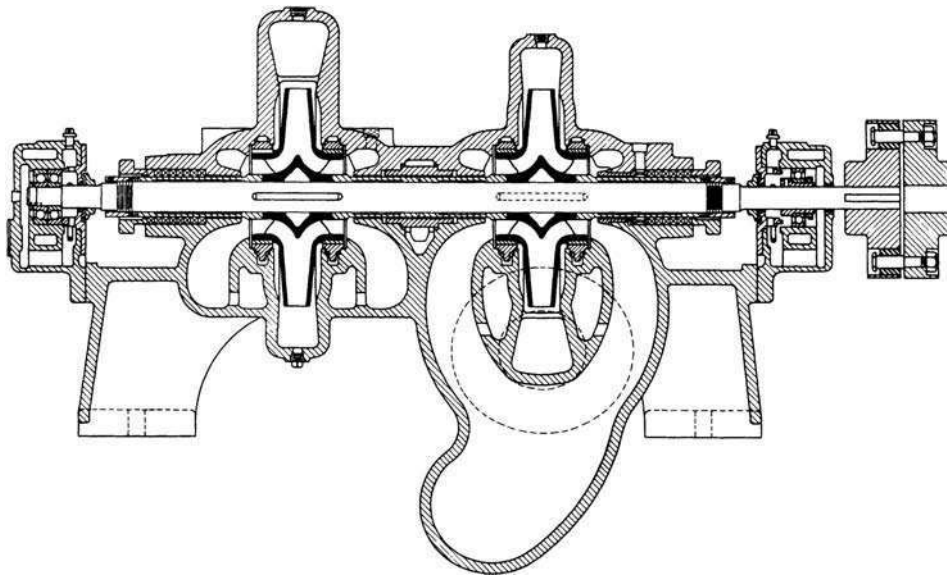


FIG. 10-53 Two-stage pump having double-suction impellers.

built in helical and linear configurations. Helical units are effective for relatively high heads and low flows, while linear induction pumps are best suited to large flows at moderate heads. Electromagnetic pumps are available for flow rates up to  $2.271 \times 10^3 \text{ m}^3/\text{h}$  (10,000 gal/min), and pressures up to 2 MPa (300 lbf/in<sup>2</sup>) are practical. Performance characteristics resemble those of centrifugal pumps.

### PUMP DIAGNOSTICS

Pump problems vary over a large range depending on the type of pumps and the usage of the pumps. They can be classified in the following manner by the pump type and the service:

1. *Positive-displacement pumps—reciprocating pumps* problems can be classified into the following categories:
  - a. Compressor valve problems: plate valves, feather valves, concentric disk valves, relief valves
  - b. Piston and rod assembly: piston rings, cylinder chatter, cylinder cooling, piston-rod packing
  - c. Lubrication system
2. *Positive displacement pumps—gear-type and roots-type* problems can be classified into the following categories:
  - a. Rotor dynamic problems: vibration problems, gear problems or roots rotor problems, bearing and seal problems
  - b. Lubrication systems

3. *Continuous flow pumps such as centrifugal pumps* problems can be classified into the following categories:

- a. Cavitation
- b. Capacity flow
- c. Motor overload
- d. Impeller
- e. Bearings and seals
- f. Lubrication systems

Table 10-15 classifies different types of centrifugal pump-related problems, their possible causes, and corrective actions that can be taken to solve some of the more common issues. These problems in the table are classified into three major categories for these type of pumps:

1. Cavitation
2. Capacity flow
3. Motor overload

The use of vibration monitoring to diagnose pump and compressor problems is discussed at the end of the subsection on compressor problems.

### COMPRESSORS

A compressor is a device which pressurizes a working fluid. One of the basic purposes of using a compressor is to compress the fluid and to

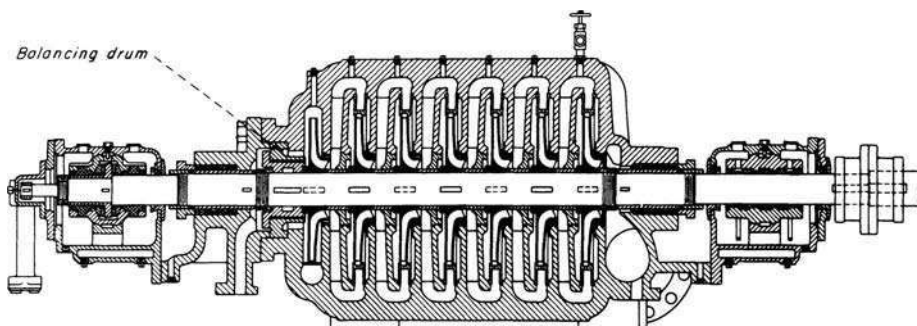
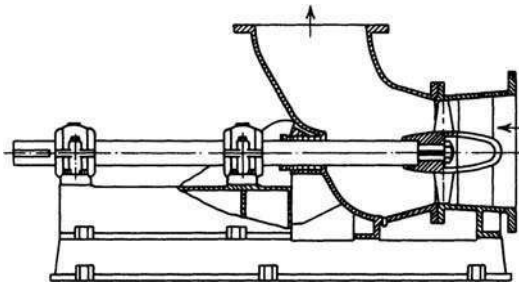


FIG. 10-54 Seven-stage diffuser-type pump.

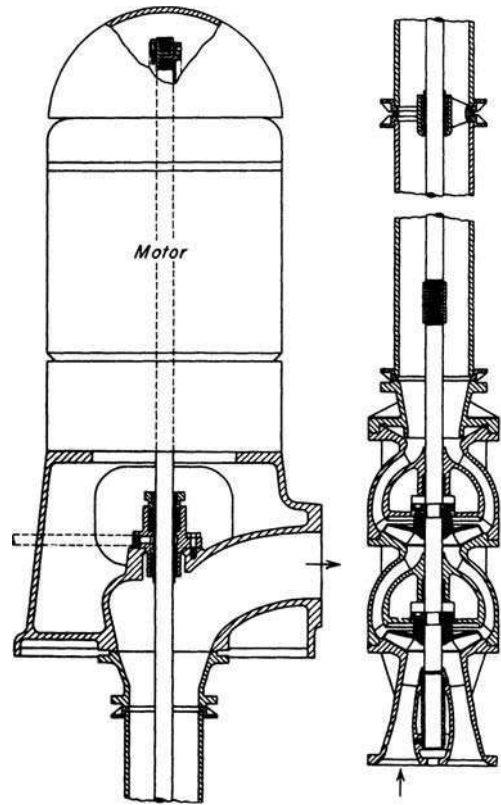


**FIG. 10-55** Axial-flow elbow-type propeller pump. (Courtesy of Lawrence Pumps, Inc.)

deliver it at a pressure higher than its original pressure. Compression is required for a variety of purposes, some of which are listed below:

1. To provide air for combustion
2. To transport process fluid through pipelines
3. To provide compressed air for driving pneumatic tools
4. To circulate process fluid within a process

Different types of compressors are shown in Fig. 10-61. Positive-displacement compressors are used for intermittent flow in which successive volumes of fluid are confined in a closed space to increase their pressures. Rotary compressors provide continuous flow. In rotary compressors, rapidly rotating parts (impellers) accelerate fluid to a high speed; this velocity is then converted to additional pressure by gradual deceleration in the diffuser or volute which surrounds the impeller. Positive-displacement compressors can be further classified as either reciprocating or rotary type, as shown in Fig. 10-61. The reciprocating compressor has a piston having a reciprocating motion within a cylinder. The rotary positive-displacement compressors have rotating elements whose positive action results in compression and displacement. The rotary positive-displacement compressors can be further subdivided into sliding vane, liquid piston, straight lobe, and helical lobe compressors. The continuous flow compressors (Fig. 10-61) can be classified as either dynamic compressors or ejectors. Ejectors entrain the in-flowing fluid by using a high-velocity gas or steam jet and then convert the velocity of the mixture to pressure in a diffuser. The dynamic compressors have rotating elements, which accelerate the in-flowing fluid, and convert the velocity head to pressure head, partially in the rotating elements and partially in the stationary diffusers or blade. The dynamic compressors can be further subdivided into centrifugal, axial-flow, and mixed-flow compressors. The main flow of gas in the centrifugal compressor is radial. The flow of gas in an axial com-

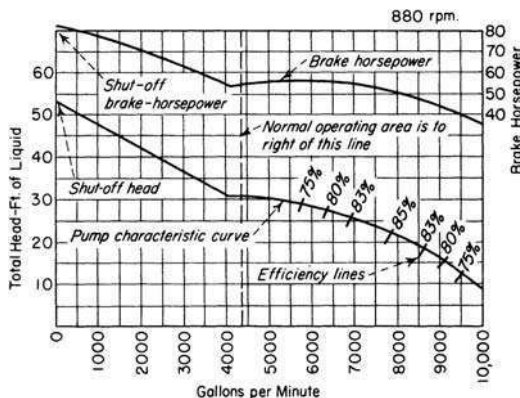


**FIG. 10-57** Vertical multistage turbine, or mixed-flow, pump.

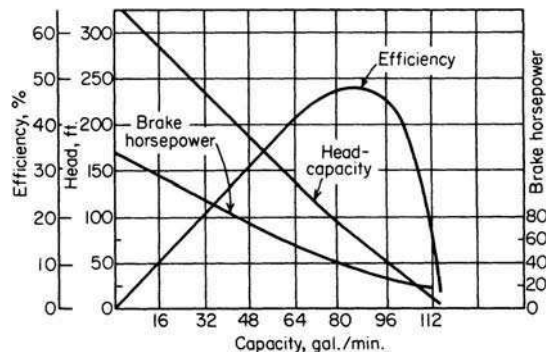
pressor is axial, and the mixed-flow compressor combines some characteristics of both centrifugal and axial compressors.

It is not always obvious what type of compressor is needed for an application. Of the many types of compressors used in the process industries, some of the more significant are the centrifugal, axial, rotary, and reciprocating compressors. They fall into three categories, as shown in Fig. 10-62.

For very high flows and low pressure ratios, an axial-flow compressor would be best. Axial-flow compressors usually have a higher efficiency, as seen in Fig. 10-63, but a smaller operating region than does



**FIG. 10-56** Characteristic curve of an axial-flow pump. To convert gallons per minute to cubic meters per hour, multiply by 0.2271; to convert feet to meters, multiply by 0.3048; and to convert horsepower to kilowatts, multiply by 0.746.



**FIG. 10-58** Characteristic curves of a regenerative pump. To convert gallons per minute to cubic meters per hour, multiply by 0.2271; to convert feet to meters, multiply by 0.3048; and to convert horsepower to kilowatts, multiply by 0.746.

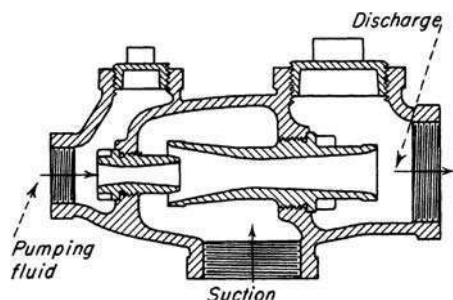


FIG. 10-59 Simple ejector using a liquid-motivating fluid.

a centrifugal machine. Centrifugal compressors operate most efficiently at medium flow rates and high pressure ratios. Rotary and reciprocating compressors (positive-displacement machines) are best used for low flow rates and high pressure ratios. The positive-displacement compressors and, in particular, reciprocating compressors were the most widely used in the process and pipeline industries up to and through the 1960s.

In turbomachinery the centrifugal flow and the axial-flow compressors are the ones used for compressing gases. Positive-displacement compressors such as reciprocating, gear-type, or lobe-type are widely used in the industry for many other applications such as slurry pumping.

The performance characteristics of a single stage of the three main types of compressors are given in Table 10-16. The pressure ratios of the axial and centrifugal compressors have been classified into three groups: industrial, aerospace, and research.

The industrial pressure ratio is low because the operating range needs to be large. The operating range is defined as the range between the surge point and the choke point. The surge point is the point at which the flow is reversed in the compressor. The choke point is the point at which the flow has reached  $Mach = 1.0$ , the point where no more flow can get through the unit, a "stone wall." When surge occurs, the flow is reversed, and so are all the forces acting on the compressor, especially the thrust forces. Surge can lead to total destruction of the compressor. Thus surge is a region that must be avoided. Choke conditions cause a large drop in efficiency, but do not lead to destruction of the unit. Note that with the increase in pressure ratio and the number of stages, the operating range is narrowed in axial-flow and centrifugal compressors.

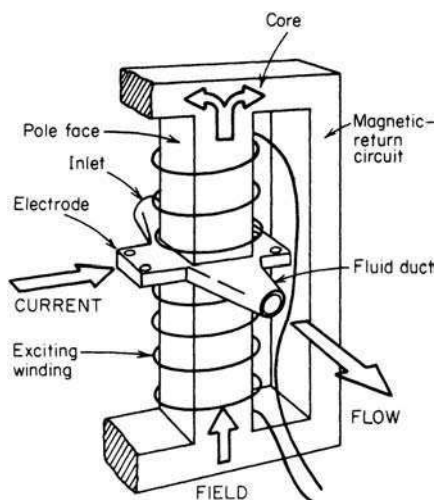


FIG. 10-60 Simplified diagram of a direct-current-operated electromagnetic pump.

**Compressor Selection** To select the most satisfactory compression equipment, engineers must consider a wide variety of types, each of which offers peculiar advantages for particular applications. Among the major factors to be considered are flow rate, head or pressure, temperature limitations, method of sealing, method of lubrication, power consumption, serviceability, and cost.

To be able to decide which compressor best fits the job, the engineer must analyze the flow characteristics of the units. The following dimensionless numbers describe the flow characteristics.

Reynolds number is the ratio of the inertia forces to the viscous forces

$$N_{Re} = \frac{\rho V D}{\mu} \quad (10-63)$$

where  $\rho$  is the density of the gas,  $V$  is the velocity of the gas,  $D$  is the diameter of the impeller, and  $\mu$  is the viscosity of the gas.

The specific speed compares the adiabatic head and flow rate in geometrically similar machines at various speeds.

$$N_s = \frac{N \sqrt{Q}}{H_{ad}^{3/4}} \quad (10-64)$$

where  $N$  is the speed of rotation of the compressor,  $Q$  is the volume flow rate, and  $H$  is the adiabatic head.

The specific diameter compares head and flow rates in geometrically similar machines at various diameters

$$D_s = \frac{D H^{1/4}}{\sqrt{Q}} \quad (10-65)$$

The flow coefficient is the capacity of the flow rate of the machine

$$\phi = \frac{Q^1}{N D^3} \quad (10-66)$$

The pressure coefficient is the pressure or the pressure rise of the machine

$$\Psi = \frac{H}{N^2 D^2} \quad (10-67)$$

In selecting the machines of choice, the use of specific speed and diameter best describe the flow. Figure 10-64 shows the characteristics of the three types of compressors. Other considerations in chemical plant service such as problems with gases which may be corrosive or have abrasive solids in suspension must be dealt with. Gases at elevated temperatures may create a potential explosion hazard, while air at the same temperatures may be handled quite normally; minute amounts of lubricating oil or water may contaminate the process gas and so may not be permissible, and for continuous-process use, a high degree of equipment reliability is required, since frequent shutdowns for inspection or maintenance cannot be tolerated.

## COMPRESSION OF GASES

**Theory of Compression** In any continuous compression process the relation of absolute pressure  $p$  to volume  $V$  is expressed by the formula

$$pV^n = C = \text{constant} \quad (10-68)$$

The plot of pressure versus volume for each value of exponent  $n$  is known as the **polytropic** curve. Since the work  $W$  performed in proceeding from  $p_1$  to  $p_2$  along any polytropic curve (Fig. 10-65) is

$$W = \int_1^2 p \, dV \quad (10-69)$$

it follows that the amount of work required is dependent upon the polytropic curve involved and increases with increasing values of  $n$ . The path requiring the least amount of input work is  $n = 1$ , which is equivalent to **isothermal** compression. For **adiabatic** compression (i.e., no heat is being added or taken away during the process),  $n = k =$  ratio of specific heat at constant pressure to that at constant volume.

TABLE 10-15 Pump Problems

| Possible causes                                                                                                                                                                                                                                                                                       | Corrective action                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cavitating-Type Problems</b>                                                                                                                                                                                                                                                                       |                                                                                                                                                       |
| Plugged suction screen.                                                                                                                                                                                                                                                                               | Check for indications of the presence of screen. Remove and clean screen.                                                                             |
| Piping gaskets with undersized IDs installed, a very common problem in small pumps.                                                                                                                                                                                                                   | Install proper-sized gaskets.                                                                                                                         |
| Column tray parts or ceramic packing lodged in the impeller eye.                                                                                                                                                                                                                                      | Remove suction piping and debris.                                                                                                                     |
| Deteriorated impeller eye due to corrosion.                                                                                                                                                                                                                                                           | Replace impeller and overhaul pump.                                                                                                                   |
| Flow rate is high enough above design that NPSH for flow rate has increased above NPSH.                                                                                                                                                                                                               | Reduce flow rate to that of design.                                                                                                                   |
| Lined pipe collapsed at gasket area or ID due to buildup of corrosion products between liner and carbon-steel pipe.                                                                                                                                                                                   | Replace deteriorated piping.                                                                                                                          |
| Poor suction piping layout, too many elbows in too many planes, a tee branch almost directly feeding the suction of the other pump, or not enough straight run before the suction flange of the pump.                                                                                                 | Redesign piping layout, using fewer elbows and laterals for tees, and have five or more straight pipe diameters before suction flange.                |
| Vertical pumps experience a vortex formation due to loss of submergence required by the pump. Observe the suction surface while the pump is in operation, if possible.                                                                                                                                | Review causes of vortexing. Consider installation of a vortex breaker such as a bell mouth umbrella or changes to sump design.                        |
| Spare pump begins to cavitate when attempt is made to switch it with the running pump. The spare is "backed off" by the running pump because its shutoff head is less than the head produced by the running pump. This is a frequent problem when one pump is turbine-driven and one is motor-driven. | Throttle discharge of running pump until spare can get in system. Slow down running pump if it is a turbine or variable-speed motor.                  |
| Suction piping configuration causes adverse fluid rotation when approaching impeller.                                                                                                                                                                                                                 | Install sufficient straight run of suction piping, or install vanes in piping to break up prerotation.                                                |
| Velocity of the liquid is too high as it approaches the impeller eye.                                                                                                                                                                                                                                 | Install larger suction piping or reduce flow through pump.                                                                                            |
| Pump is operating at a low-flow-producing suction recirculation in the impeller eye. This results in a cavitationlike sound.                                                                                                                                                                          | Install bypass piping back to suction vessel to increase flow through pump. Remember bypass flow may have to be as high as 50 percent of design flow. |
| <b>Capacity-Type Problems</b>                                                                                                                                                                                                                                                                         |                                                                                                                                                       |
| Check the discharge block valve opening first. It may be partially closed and thus the problem.                                                                                                                                                                                                       | Open block valve completely.                                                                                                                          |
| Wear-ring clearances are excessive (closed impeller design).                                                                                                                                                                                                                                          | Overhaul pump. Renew wear rings if clearance is about twice design value for energy and performance reasons.                                          |
| Impeller-to-case or head clearances are excessive (open impeller design).                                                                                                                                                                                                                             | Reposition impeller to obtain correct clearance.                                                                                                      |
| Air leaks into the system if the pump suction is below atmospheric pressure.                                                                                                                                                                                                                          | Take actions as needed to eliminate air leaks.                                                                                                        |
| Increase in piping friction to the discharge vessel due to the following:                                                                                                                                                                                                                             | Take the following actions:                                                                                                                           |
| 1. Gate has fallen off the discharge valve stem.                                                                                                                                                                                                                                                      | 1. Repair or replace gate valve.                                                                                                                      |
| 2. Spring is broken in the spring-type check valve.                                                                                                                                                                                                                                                   | 2. Repair valve by replacing spring.                                                                                                                  |
| 3. Check valve flapper pin is worn, and flapper will not swing open.                                                                                                                                                                                                                                  | 3. Overhaul check valve; restore proper clearance to pin and flapper bore.                                                                            |
| 4. Lined pipe collapsing.                                                                                                                                                                                                                                                                             | 4. Replace damaged pipe.                                                                                                                              |
| 5. Control valve stroke improperly set, causing too much pressure drop.                                                                                                                                                                                                                               | 5. Adjust control valve stroke as necessary.                                                                                                          |
| Suction and/or discharge vessel levels are not correct, a problem mostly seen in lower-speed pumps.                                                                                                                                                                                                   | Calibrate level controllers as necessary.                                                                                                             |
| Motor running backward or impeller of double suction design is mounted backward. Discharge pressure developed in both cases is about one-half design value.                                                                                                                                           | Check for proper rotation and mounting of impeller. Reverse motor leads if necessary.                                                                 |
| Entrained gas from the process lowering NPSH available.                                                                                                                                                                                                                                               | Reduce entrained gas in liquid by process changes as needed.                                                                                          |
| Polymer or scale buildup in discharge nozzle areas.                                                                                                                                                                                                                                                   | Shut down pump and remove scale or deposits.                                                                                                          |
| Mechanical seal in suction system under vacuum is leaking air into system, causing pump curve to drop.                                                                                                                                                                                                | Change percentage balance of seal faces or increase spring tension.                                                                                   |
| The pump may have formed a vortex at high flow rates or low liquid level. Does the vessel have a vortex breaker? Does the incoming flow cause the surface to swirl or be agitated?                                                                                                                    | Reduce flow to design rates. Raise liquid level in suction vessel. Install vortex breaker in suction vessel.                                          |
| Variable-speed motor running too slowly.                                                                                                                                                                                                                                                              | Adjust motor speed as needed.                                                                                                                         |
| Bypassing is occurring between volute channels in a double-volute pump casing due to a casting defect or extreme erosion.                                                                                                                                                                             | Overhaul pump; repair eroded area.                                                                                                                    |
| The positions of impellers are not centered with diffuser vanes. Several impellers will cause vibration and lower head output.                                                                                                                                                                        | Overhaul pump; reposition individual impellers as needed. Reposition whole rotor by changing thrust collar locator spacer.                            |
| When the suction system is under vacuum, the spare pump has difficulty getting into system.                                                                                                                                                                                                           | Install a positive-pressure steam (from running pump) to fill the suction line from the block valve through the check valve.                          |
| Certain pump designs use an internal bypass orifice port to alter head-flow curve. High liquid velocities often erode the orifice, causing the pump to go farther out on the pump curve. The system head curve increase corrects the flow back up the curve.                                          | Overhaul pump, restore orifice to correct size.                                                                                                       |
| Replacement impeller is not correct casting pattern; therefore NPSH required is different.                                                                                                                                                                                                            | Overhaul pump, replace impeller with correct pattern.                                                                                                 |
| Volute and cutwater area of casing is severely eroded.                                                                                                                                                                                                                                                | Overhaul pump; replace casing or repair by welding. Stress-relieve after welding as needed.                                                           |

TABLE 10-15 Pump Problems (Concluded)

| Possible causes                                                                                                                                     | Corrective action                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overload Problems                                                                                                                                   |                                                                                                                                                                                                                           |
| Polymer buildup between wear surfaces (rings or vanes).                                                                                             | Remove buildup to restore clearances.                                                                                                                                                                                     |
| Excessive wear ring (closed impeller) or cover-case clearance (open impeller).                                                                      | Replace wear rings or adjust axial clearance of open impeller. In severe cases, cover or case must be replaced.                                                                                                           |
| Pump circulating excessive liquid back to suction through a breakdown bushing or a diffuser gasket area.                                            | Overhaul pump, replacing parts as needed.                                                                                                                                                                                 |
| Minimum-flow loop left open at normal rates, or bypass around control valve is open.                                                                | Close minimum-flow loop or control valve bypass valve.                                                                                                                                                                    |
| Discharge piping leaking under liquid level in sump-type design.                                                                                    | Inspect piping for leakage. Replace as needed.                                                                                                                                                                            |
| Electrical switch gear problems cause one phase to have low amperage.                                                                               | Check out switch gear and repair as necessary.                                                                                                                                                                            |
| Specific gravity is higher than design specification.                                                                                               | Change process to adjust specific gravity to design value, or throttle pump to reduce horsepower requirements. This will not correct problem with some vertical turbine pumps that have a flat horsepower-required curve. |
| Pump motor not sized for end of curve operation.                                                                                                    | Replace motor with one of larger size, or reduce flow rate.                                                                                                                                                               |
| Open impeller has slight rub on casing. Most often occurs in operations from 250 to 400°F due to piping strain and differential growth in the pump. | Increase clearance of impeller to casing.                                                                                                                                                                                 |
| A replacement impeller was not trimmed to the correct diameter.                                                                                     | Remove impeller from pump and turn to correct diameter.                                                                                                                                                                   |

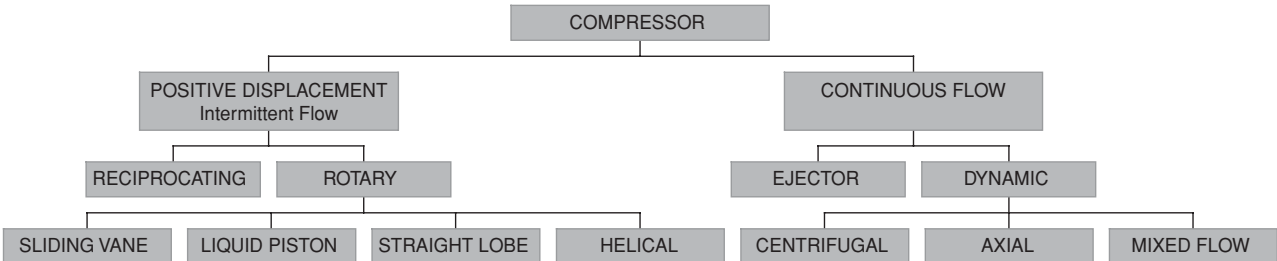


FIG. 10-61 Principal types of compressors.

Since most compressors operate along a polytropic path approaching the adiabatic, compressor calculations are generally based on the adiabatic curve.

Some formulas based upon the adiabatic equation and useful in compressor work are as follows:

**Pressure, volume, and temperature** relations for perfect gases:

$$p_2/p_1 = (V_1/V_2)^k \tag{10-70}$$

$$T_2/T_1 = (V_1/V_2)^{k-1} \tag{10-71}$$

$$p_2/p_1 = (T_2/T_1)^{k/(k-1)} \tag{10-72}$$

**Adiabatic Calculations** Adiabatic head is expressed as follows: In SI units,

$$H_{ad} = \frac{k \times ZRT_1}{k-1} \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \tag{10-73a}$$

where  $H_{ad}$  = adiabatic head, N·m/kg;  $R$  = gas constant, J/(kg·K) = 8314/molecular weight;  $T_1$  = inlet gas temperature, K;  $p_1$  = absolute inlet pressure, kPa; and  $p_2$  = absolute discharge pressure, kPa.

In U.S. customary units,

$$H_{ad} = \frac{k}{k-1} ZRT_1 \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \tag{10-73b}$$

where  $H_{ad}$  = adiabatic head, ft·lbf/lbm;  $R$  = gas constant, (ft·lbf)/(lbm·°R) = 1545/molecular weight;  $T_1$  = inlet gas temperature, °R;

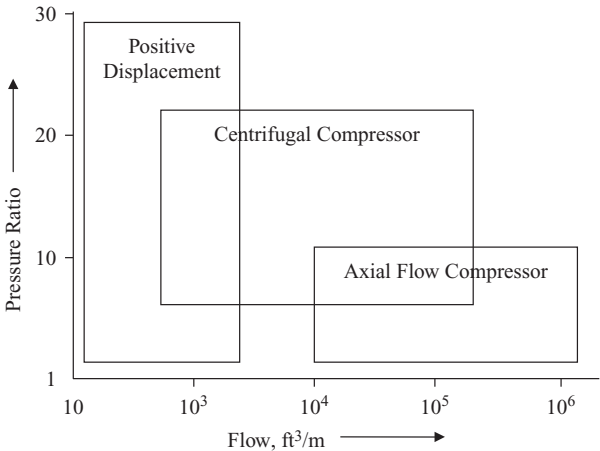
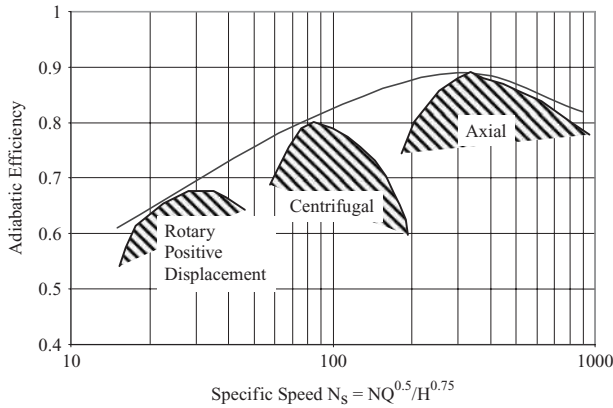


FIG. 10-62 Performance characteristics of different types of compressors.



**FIG. 10-63** Variation of adiabatic efficiency with specific speed for the three types of compressors.

$p_1$  = absolute inlet pressure, lbf/in<sup>2</sup>; and  $p_2$  = absolute discharge pressure, lbf/in<sup>2</sup>.

The **work** expended on the gas during compression is equal to the product of the adiabatic head and the mass flow of gas handled. Therefore, the adiabatic power is as follows:

In SI units,

$$kW_{ad} = \frac{WH_{ad}}{10^3} = \frac{k \times ZWRT_1}{k-1} \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \quad (10-74a)$$

$$\text{or} \quad kW_{ad} = 2.78 \times 10^{-4} \frac{k}{k-1} Q_1 p_1 \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \quad (10-75a)$$

where  $kW_{ad}$  = power, kW;  $W$  = mass flow, kg/s; and  $Q_1$  = volume rate of gas flow, m<sup>3</sup>/h, at compressor inlet conditions.

In U.S. Customary units,

$$hp_{ad} = \frac{WH_{ad}}{550} = \frac{k}{k-1} \frac{ZWRT_1}{550} \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \quad (10-74b)$$

$$\text{or} \quad hp_{ad} = \frac{k}{k-1} \frac{Q_1 p_1}{3600} \left[ \left( \frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] \quad (10-75b)$$

where  $hp_{ad}$  = power, hp;  $W$  = mass flow, lb/s; and  $Q_1$  = volume rate of gas flow, ft<sup>3</sup>/min.

Adiabatic discharge temperature is

$$T_2 = T_1 (p_2/p_1)^{(k-1)/k} \quad (10-76)$$

The work in a compressor under ideal conditions as previously shown occurs at constant entropy. The actual process is a polytropic process as shown in Fig. 10-65 and given by the equation of state  $Pv^n = \text{constant}$ .

Adiabatic efficiency is given by the following relationship:

$$\eta_{ad} = \frac{\text{ideal work}}{\text{actual work}} \quad (10-77)$$

In terms of the change in total temperatures the relationship can be written as:

$$\eta_{ad} = \frac{T_2 - T_1}{T_{2a} - T_1} \quad (10-78)$$

where  $T_{2a}$  is the total actual discharge temperature of the gas. The adiabatic efficiency can be represented in terms of total pressure change:

$$\eta_{ad} = \frac{\left( \frac{P_2}{P_1} \right)^{(k-1)/k} - 1}{\left( \frac{P_2}{P_1} \right)^{(n-1)/n} - 1} \quad (10-79)$$

Polytropic head can be expressed by the following relationship:

$$H_{poly} = \frac{n}{n-1} ZRT_1 \left[ \left( \frac{P_2}{P_1} \right)^{(n-1)/n} - 1 \right] \quad (10-80)$$

Likewise for polytropic efficiency, which is often considered as the small stage efficiency, or the hydraulic efficiency:

$$\eta_{pc} = \frac{(k-1)/k}{(n-1)/n} \quad (10-81)$$

Polytropic efficiency is the limited value of the isentropic efficiency as the pressure ratio approaches 1.0, and the value of the polytropic efficiency is higher than the corresponding adiabatic efficiency.

A characteristic of polytropic efficiency is that the polytropic efficiency of a multistage unit is equal to the stage efficiency if each stage has the same efficiency.

If the compression cycle approaches the isothermal condition,  $pV = \text{constant}$ , as is the case when several stages with intercoolers are used, a simple approximation of the power is obtained from the following formula:

In SI units,

$$kW = 2.78 \times 10^{-4} Q_1 p_1 \ln p_2/p_1 \quad (10-82a)$$

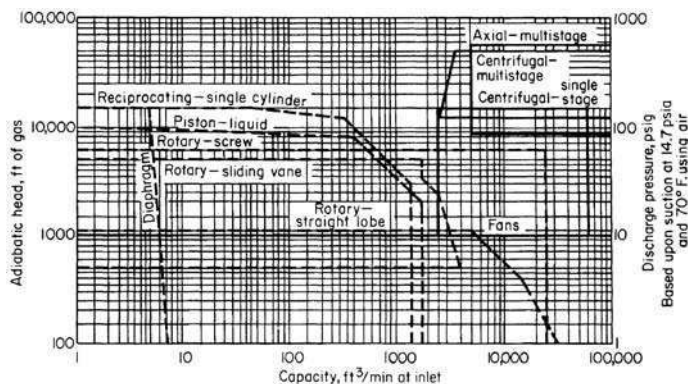
In U.S. customary units,

$$hp = 4.4 \times 10^{-3} Q_1 p_1 \ln p_2/p_1 \quad (10-82b)$$

**Reciprocating Compressors** Reciprocating compressors are used mainly when high-pressure head is required at a low flow. Reciprocating compressors are furnished in either single-stage or multistage types. The number of stages is determined by the required compressor ratio  $p_2/p_1$ . The compression ratio per stage is generally limited to 4, although low-capacity units are furnished with compression ratios of 8 and even higher. Generally, the maximum compression ratio is determined by the maximum allowable discharge-gas temperature.

**TABLE 10-16 Performance Characteristics of Compressors**

| Types of compressors  | Pressure ratio per stage |           |          | Efficiency, % | Operating range surge – choke, % |
|-----------------------|--------------------------|-----------|----------|---------------|----------------------------------|
|                       | Industrial               | Aerospace | Research |               |                                  |
| Positive displacement | Up to 30                 | —         | —        | 75–82         | —                                |
| Centrifugal           | 1.2–1.9                  | 2.0–7.0   | 13       | 75–87<br>25   | Large                            |
| Axial                 | 1.05–1.3                 | 1.1–1.45  | 2.1      | 80–91         | Narrow<br>3–10                   |



**FIG. 10-64** Compressor coverage chart based on the normal range of operation of commercially available types shown. Solid lines: use left ordinate, head. Broken lines: use right ordinate, pressure. To convert cubic feet per minute to cubic meters per hour, multiply by 1.699; to convert feet to meters, multiply by 0.3048; and to convert pounds-force per square inch to kilopascals, multiply by 6.895;  $(^{\circ}\text{F} - 32)/1.8 = ^{\circ}\text{C}$ .

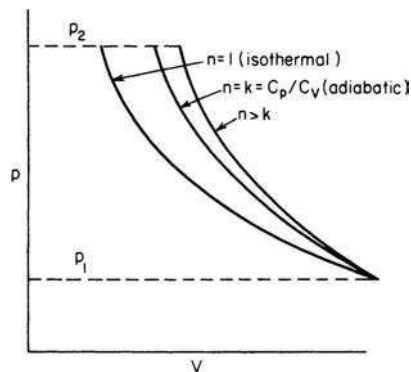
Single-acting air-cooled and water-cooled air compressors are available in sizes up to about 75 kW (100 hp). Such units are available in one, two, three, or four stages for pressure as high as 24 MPa (3500 lbf/in<sup>2</sup>). These machines are seldom used for gas compression because of the difficulty of preventing gas leakage and contamination of the lubricating oil.

The compressors most commonly used for compressing gases have a crosshead to which the connecting rod and piston rod are connected. This provides a straight-line motion for the piston rod and permits simple packing to be used. Figure 10-66 illustrates a simple single-stage machine of this type having a double-acting piston. Either single-acting (Fig. 10-67) or double-acting pistons (Fig. 10-68) may be used, depending on the size of the machine and the number of stages. In some machines double-acting pistons are used in the first stages and single-acting in the later stages.

On multistage machines, intercoolers are provided between stages. These heat exchangers remove the heat of compression from the gas and reduce its temperature to approximately the temperature existing at the compressor intake. Such cooling reduces the volume of gas going to the high-pressure cylinders, reduces the power required for compression, and keeps the temperature within safe operating limits.

Figure 10-69 illustrates a two-stage compressor end such as might be used on the compressor illustrated in Fig. 10-66.

Compressors with horizontal cylinders such as illustrated in Figs. 10-66 to 10-68 are most commonly used because of their accessibility. However, machines are also built with vertical cylinders and other arrangements such as right-angle (one horizontal and one vertical cylinder) and V-angle.



**Fig. 10-65** Polytropic compression curves.

Compressors up to around 75 kW (100 hp) usually have a single center-throw crank, as illustrated in Fig. 10-66. In larger sizes compressors are commonly of duplex construction with cranks on each end of the shaft (see Fig. 10-70). Some large synchronous motor-driven units are of four-corner construction; i.e., they are of double-duplex construction with two connecting rods from each of the two crank throws (see Fig. 10-71). Steam-driven compressors have one or more steam cylinders connected directly by piston rod or tie rods to the gas-cylinder piston or crosshead.

**Valve Losses** Above piston speeds of 2.5 m/s (500 ft/min), suction and discharge valve losses begin to exert significant effects on the actual internal compression ratio of most compressors, depending on the valve port area available. The obvious results are high temperature rise and higher power requirements than might be expected. These effects become more pronounced with higher-molecular-weight gases. Valve problems can be a very major contributor to down time experienced by these machines.

**Control Devices** In many installations the use of gas is intermittent, and some means of controlling the output of the compressor is therefore necessary. In other cases constant output is required despite variations in discharge pressure, and the control device must operate to maintain a constant compressor speed. Compressor capacity, speed, or pressure may be varied in accordance with requirements. The nature of the control device will depend on the function to be regulated. Regulation of pressure, volume, temperature, or some other factor determines the type of regulation required and the type of the compressor driver.

The most common control requirement is regulation of capacity. Many capacity controls, or unloading devices, as they are usually termed, are actuated by the pressure on the discharge side of the compressor. A falling pressure indicates that gas is being used faster than it is being compressed and that more gas is required. A rising pressure indicates that more gas is being compressed than is being used and that less gas is required.

An obvious method of controlling the capacity of a compressor is to vary the speed. This method is applicable to units driven by variable-speed drivers such as steam pistons, steam turbines, gas engines, diesel engines, etc. In these cases the regulator actuates the steam-admission or fuel-admission valve on the compressor driver and thus controls the speed.

Motor-driven compressors usually operate at constant speed, and other methods of controlling the capacity are necessary. On reciprocating compressors discharging into receivers, up to about 75 kW (100 hp), two types of control are usually available. These are automatic-start-and-stop control and constant-speed control.

Automatic-start-and-stop control, as its name implies, stops or starts the compressor by means of a pressure-actuated switch as the

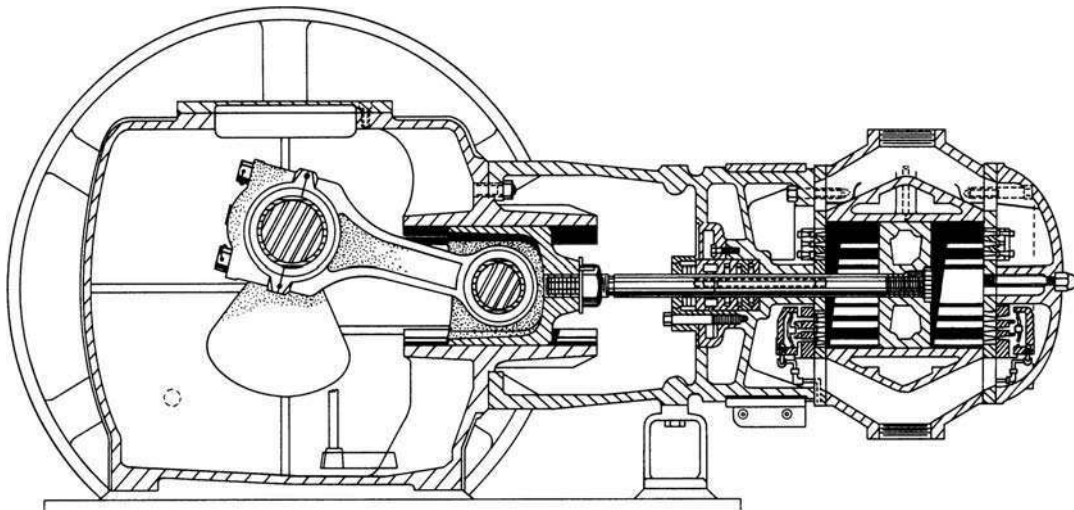


FIG. 10-66 Typical single-stage, double-acting water-cooled compressor.

gas demand varies. It should be used only when the demand for gas will be intermittent.

Constant-speed control should be used when gas demand is fairly constant. With this type of control, the compressor runs continuously but compresses only when gas is needed. Three methods of unloading the compressor with this type of control are in common use: (1) **closed suction unloaders**, (2) **open inlet-valve unloaders**, and (3) **clearance unloaders**. The closed suction unloader consists of a pressure-actuated valve which shuts off the compressor intake. Open inlet-valve unloaders (see Fig. 10-72) operate to hold the compressor inlet valves open and thereby prevent compression. Clearance unloaders (see Fig. 10-73) consist of pockets or small reservoirs which are opened when unloading is desired. The gas is compressed into them on the compression stroke and reexpands into the cylinder on the return stroke, thus preventing the compression of additional gas.

It is sometimes desirable to have a compressor equipped with both constant-speed and automatic-start-and-stop control. When this is done, a switch allows immediate selection of either type.

Motor-driven reciprocating compressors above about 75 kW (100 hp) in size are usually equipped with a step control. This is in reality a variation of constant-speed control in which unloading is accomplished in a series of steps, varying from full load down to no load.

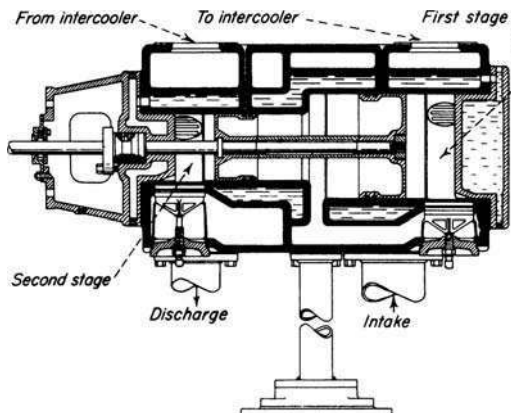


FIG. 10-67 Two-stage single-acting opposed piston in a single step-type cylinder.

**Three-step control** (full load, one-half load, and no load) is usually accomplished with inlet-valve unloaders. **Five-step control** (full load, three-fourths load, one-half load, one-fourth load, and no load) is accomplished by means of clearance pockets (see Fig. 10-74). On some machines, inlet-valve and clearance-control unloading are used in combination.

Although such control devices are usually automatically operated, manual operation is satisfactory for some services. When manual operation is provided, it often consists of a valve or valves to open and close clearance pockets. In some cases, a movable cylinder head is provided for variable clearance in the cylinder (see Fig. 10-75).

When no capacity control or unloading device is provided, it is necessary to provide bypasses between the inlet and discharge in order that the compressor can be started against no load (see Fig. 10-76).

**Nonlubricated Cylinders** Most compressors use oil to lubricate the cylinder. In some processes, however, the slightest oil contamination

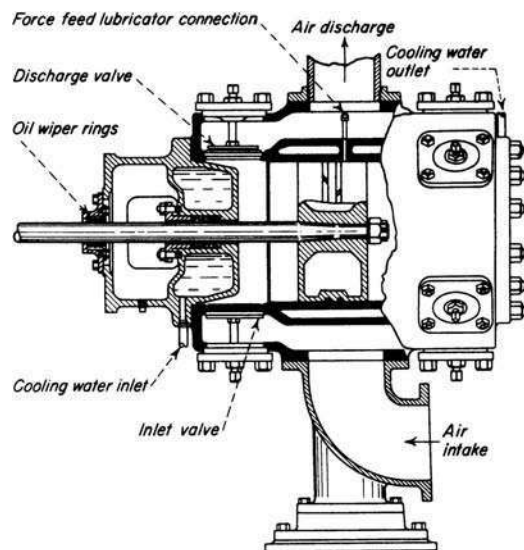


FIG. 10-68 Typical double-acting compressor piston and cylinder.



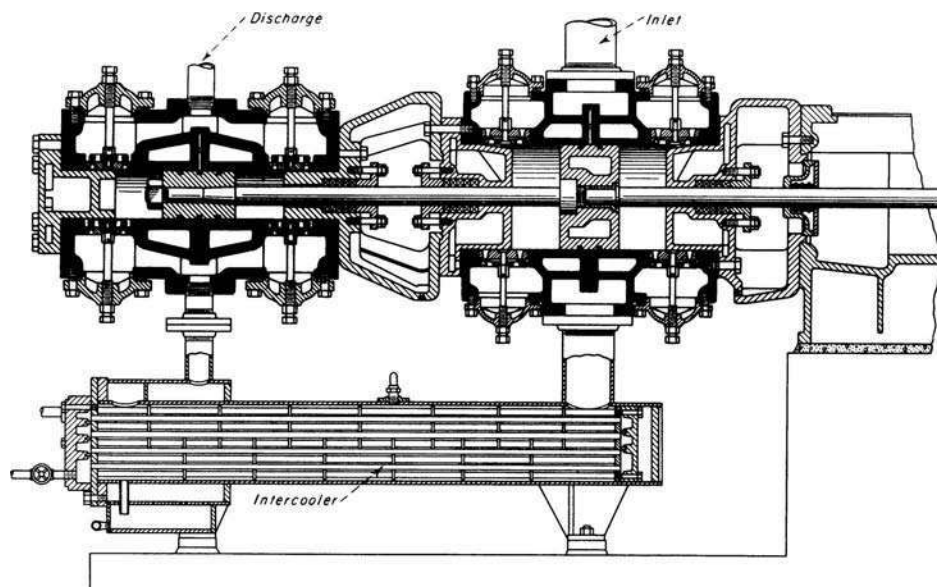


FIG. 10-69 Two-stage double-acting compressor cylinders with intercooler.

is objectionable. For such cases a number of manufacturers furnish a "nonlubricated" cylinder (see Fig. 10-77). The piston on these cylinders is equipped with piston rings of graphitic carbon or Teflon<sup>®</sup> as well as pads or rings of the same material to maintain proper clearance between the piston and the cylinder. Plastic packing of a type that requires no lubricant is used on the stuffing box. Although oil-wiper rings are used on the piston rod where it leaves the compressor frame, minute quantities of oil might conceivably enter the cylinder on the rod. If even such small amounts of oil are objectionable, an extended cylinder connecting piece can be furnished. This simply lengthens the piston rod enough so that no portion of the rod can alternately enter the frame and the cylinder.

In many cases, a small amount of gas leaking through the packing is objectionable. Special connecting pieces are furnished between the cylinder and the frame, which may be either single-compartment or double-compartment. These may be furnished gastight and

vented back to the suction or filled with a sealing gas or fluid and held under a slight pressure.

**High-Pressure Compressors** There is a definite trend in the chemical industry toward the use of high-pressure compressors with discharge pressures of from 34.5 to 172 MPa (5000 to 25,000 lbf/in<sup>2</sup>) and with capacities from  $8.5 \times 10^3$  to  $42.5 \times 10^3$  m<sup>3</sup>/h (5000 to 25,000 ft<sup>3</sup>/min). These require special design, and a complete knowledge of the characteristics of the gas is necessary. In most cases, these types of applications use the barrel-type centrifugal compressor.

The gas usually deviates considerably from the perfect-gas laws, and in many cases temperature or other limitations necessitate a thorough engineering study of the problem. These compressors usually have five, six, seven, or eight stages, and the cylinders must be properly proportioned to meet the various limitations involved and also to balance the load among the various stages. In many cases, scrubbing or other processing is carried on between stages. High-pressure cylinders are steel forgings with single-acting plungers (see Fig. 10-78). The compressors are usually designed so that the pressure load against the plunger is opposed by one or more single-acting pistons of the lower pressure stages. Piston-rod packing is usually of the segmental-ring metallic type. Accurate fitting and correct lubrication are very important. High-pressure compressor valves are designed for the conditions involved. Extremely high-grade engineering and skill are necessary.

**Piston-Rod Packing** Proper piston-rod packing is important. Many types are available, and the most suitable is determined by the gas handled and the operating conditions for a particular unit.

There are many types and compositions of soft packing, semimetallic packing, and metallic packing. In many cases, metallic packing is to be recommended. A typical low-pressure packing arrangement is shown in Fig. 10-79. A high-pressure packing arrangement is shown in Fig. 10-80.

When wet, volatile, or hazardous gases are handled or when the service is intermittent, an auxiliary packing gland and soft packing are usually employed (see Fig. 10-81).

**Metallic Diaphragm Compressors** (Fig. 10-82) These are available for small quantities [up to about 17 m<sup>3</sup>/h (10 ft<sup>3</sup>/min)] for compression ratios as high as 10:1 per stage. Temperature rise is not a serious problem, as the large wall area relative to the gas volume permits sufficient heat transfer to approach isothermal compression. These compressors possess the advantage of having no seals for the process gas. The diaphragm is actuated hydraulically by a plunger pump.

<sup>®</sup>Du Pont tetrafluoroethylene fluorocarbon resin.

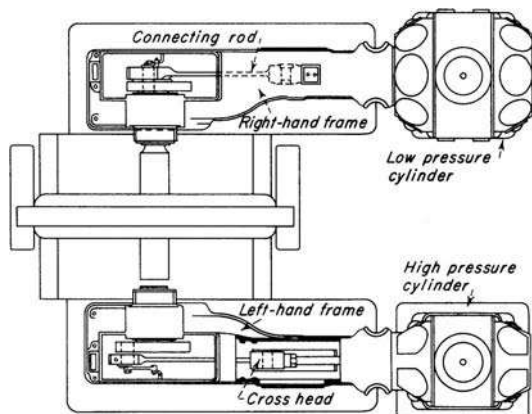


FIG. 10-70 Duplex two-stage compressor (plan view).

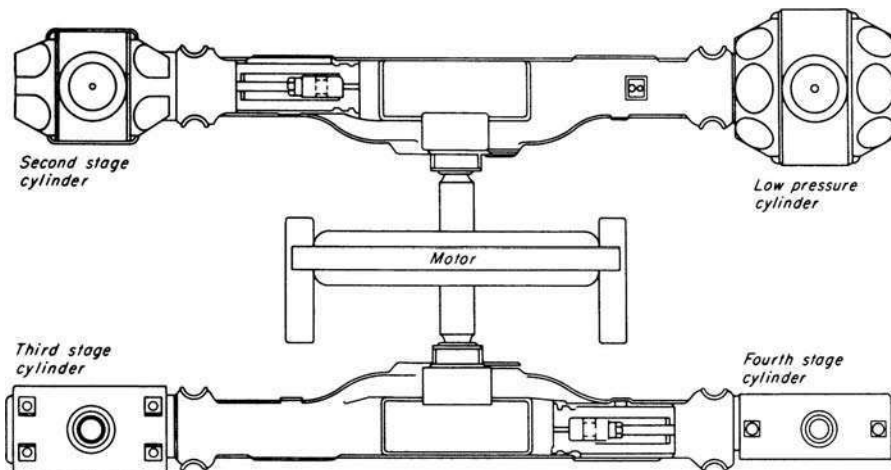


FIG. 10-71 Four-corner four-stage compressor (plan view).

## FANS AND BLOWERS

**Fans** are used for low pressures where generally the delivery pressure is less than 3.447 kPa (0.5 lb/in<sup>2</sup>), and **blowers** are used for higher pressures. However, they are usually below delivery pressures of 10.32 kPa (1.5 lb/in<sup>2</sup>). These units can either be centrifugal or the axial-flow type.

**Fans and blowers** are used for many types of ventilating work such as air-conditioning systems. In large buildings, blowers are often used due to the high delivery pressures needed to overcome the pressure drop in the ventilation system. Most of these blowers are of the centrifugal type. Blowers are also used to supply draft air to boilers and furnaces. Fans are used to move large volumes of air or gas through ducts, supplying air for drying, conveying material suspended in the gas stream, removing fumes, condensing towers and other high-flow, low-pressure applications.

**Axial-Flow Fans** These are designed to handle very high flow rates and low pressure. The disc-type fans are similar to those of a household fan. They are usually for general circulation or exhaust work without ducts.

The so-called propeller-type fans with blades that are aerodynamically designed (as seen in Fig. 10-83) can consist of two or more stages. The air in these fans enters in an axial direction and leaves in an axial direction. The fans usually have inlet guide vanes followed by a rotating blade, followed by a stationary (stator) blade.

**Centrifugal Blowers** These blowers have air or gases entering in the axial direction and being discharged in the radial direction. These blowers have three types of blades: radial or straight blades, forward-curved blades, and backward-curved blades (Figs. 10-84 to 10-86).

Radial blade blowers as seen in Fig. 10-84 are usually used in large-diameter or high-temperature applications. The blades being radial in direction have very low stresses as compared to the backward or forward

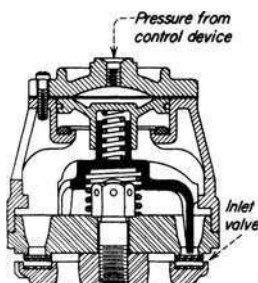


FIG. 10-72 Inlet-valve unloader.

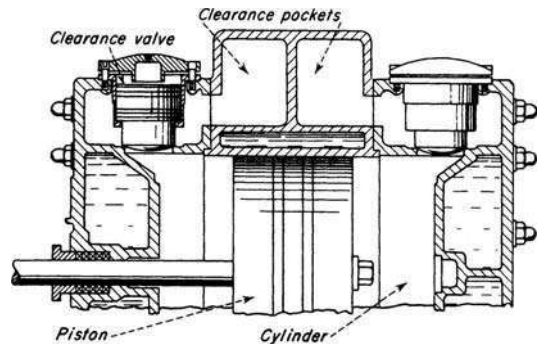


FIG. 10-73 Clearance-control cylinder. (Courtesy of Ingersoll-Rand.)

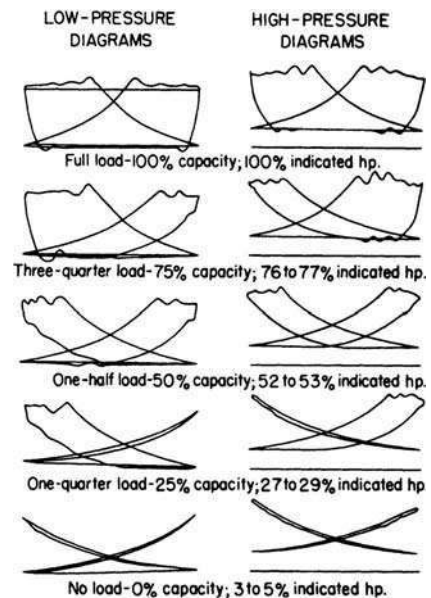


FIG. 10-74 Actual indicator diagram of a two-stage compressor showing the operation of clearance control at five load points.

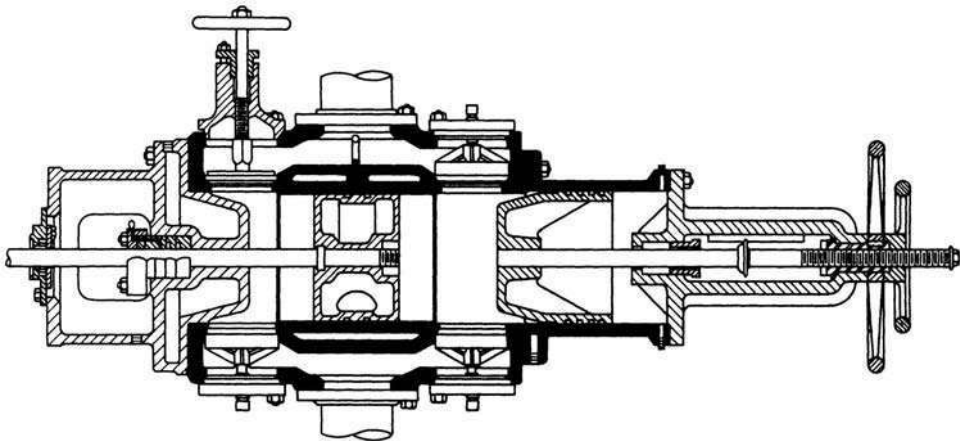


FIG. 10-75 Sectional view of a cylinder equipped with a hand-operated valve lifter on one end and a variable-volume clearance pocket at other end.

curve blades. The rotors have anywhere between 4 to 12 blades and usually operate at low speeds. These fans are used in exhaust work especially for gases at high temperature and with suspensions in the flow stream.

**Forward-Curved Blade Blowers** These blowers discharge the gas at a very high velocity. The pressure supplied by this blower is lower than that produced in the other two blade characteristics. The number of blades in such a rotor can be large—up to 50 blades—and the speed is high—usually 3600–1800 rpm in 60-cycle countries and 3000–1500 rpm in 50-cycle countries.

**Backward-Curved Blade Blowers** These blowers are used when a higher discharge pressure is needed. It is used over a wide range of applications. Both the forward and backward curved blades do have much higher stresses than the radial bladed blower.

| Starting compressor     | Stopping compressor |
|-------------------------|---------------------|
| Start with A and D open | Close ---- C        |
| Close ----- D           | Close --- B         |
| Close ----- A           | Open ---- A and D   |
| Open ----- B            |                     |
| Slowly open --- C       |                     |

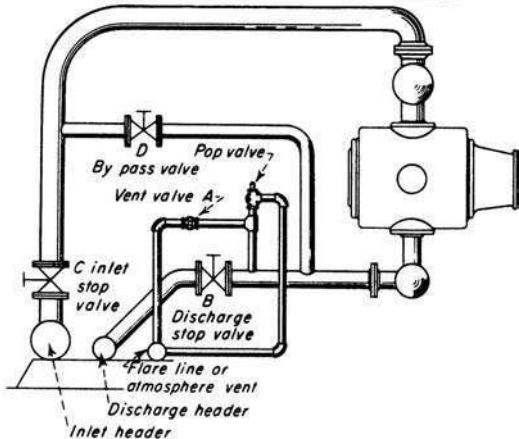


FIG. 10-76 Bypass arrangement for a single-stage compressor. On multistage machines, each stage is bypassed in a similar manner. Such an arrangement is necessary for no-load starting.

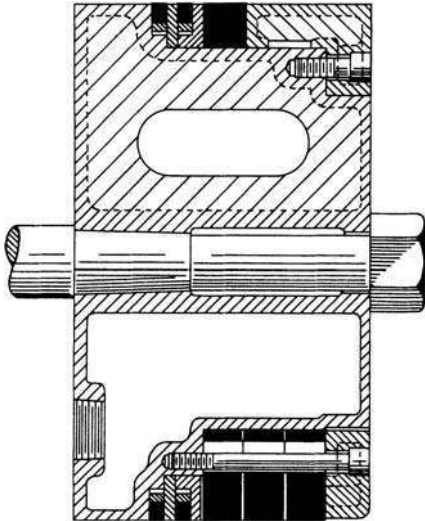


FIG. 10-77 Piston equipped with carbon piston and wearing rings for a non-lubricated cylinder.

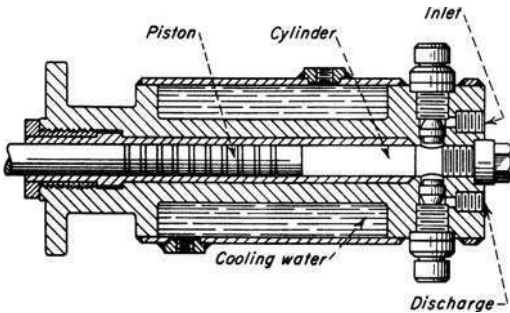


FIG. 10-78 Forged-steel single-acting high-pressure cylinder.

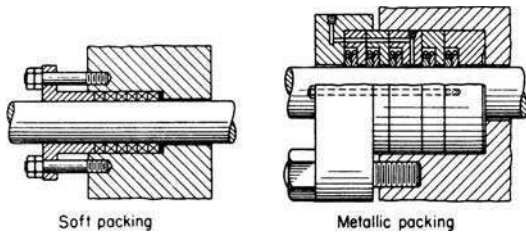


FIG. 10-79 Typical packing arrangements for low-pressure cylinders.

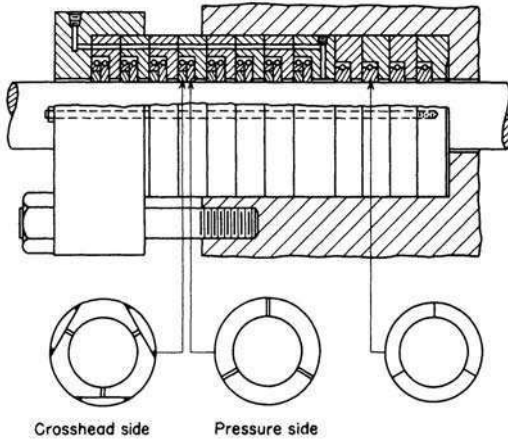


FIG. 10-80 Typical packing arrangement, using metallic packing, for high-pressure cylinders.

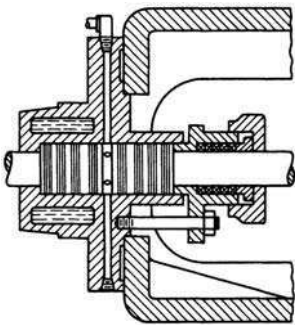


FIG. 10-81 Soft packing in an auxiliary stuffing box for handling gases.

The centrifugal blower produces energy in the air stream by the centrifugal force and imparts a velocity to the gas by the blades. Forward curved blades impart the most velocity to the gas. The scroll-shaped volute diffuses the air and creates an increase in the static pressure by reducing the gas velocity. The change in total pressure occurs in the impeller—this is usually a small change. The static pressure is increased in both the impeller and the diffuser section. Operating efficiencies of the fan range from 40 to 80 percent. The discharge total pressure is the summation of the static pressure and the velocity head.

The power needed to drive the fan can be computed as follows.

$$\text{Power (kW)} = 2.72 \times 10^{-5} QP \quad (10-83)$$

where  $Q$  is the fan volume ( $\text{m}^3/\text{h}$ ) and  $P$  is the total discharge pressure

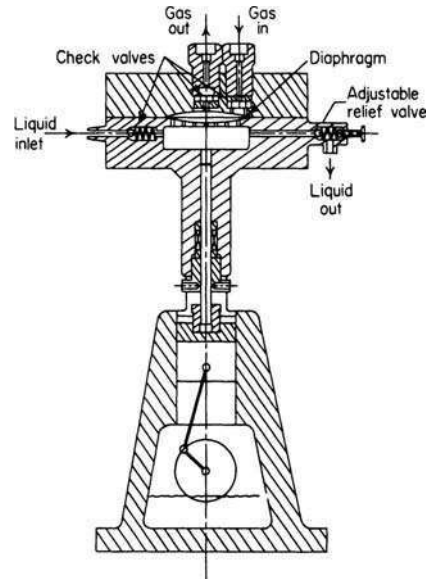


FIG. 10-82 High-pressure, low-capacity compressor having a hydraulically actuated diaphragm. (Pressure Products Industries.)

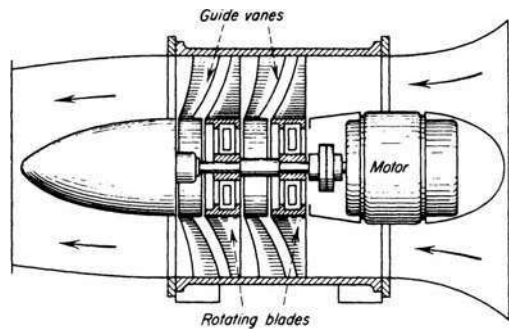


FIG. 10-83 Two-stage axial-flow fan.

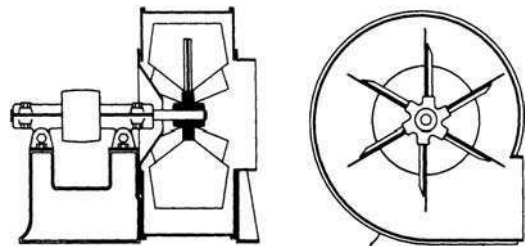


FIG. 10-84 Straight-blade, or steel-plate, fan.

in cm of water column.

In U.S. customary units,

$$\text{hp} = 1.57 \times 10^{-4} Qp \quad (10-84)$$

where  $\text{hp}$  is the fan power output,  $\text{hp}$ ;  $Q$  is the fan volume,  $\text{ft}^3/\text{min}$ ; and  $p$  is the fan-operating pressure, inches water column.

$$\text{Efficiency} = \frac{\text{air power output}}{\text{shaft power input}} \quad (10-85)$$

## 10-52 TRANSPORT AND STORAGE OF FLUIDS

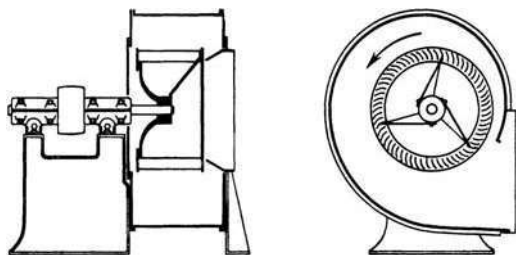


FIG. 10-85 Forward-curved blade, or "scirocco"-type, fan.

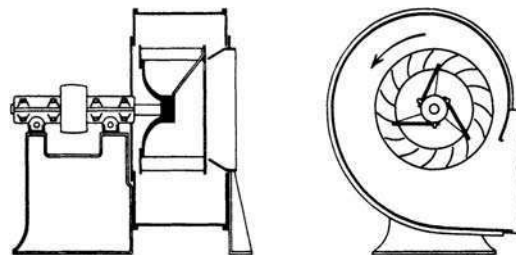


FIG. 10-86 Backward-curved-blade fan.

**Fan Performance** The performance of a centrifugal fan varies with changes in conditions such as temperature, speed, and density of the gas being handled. It is important to keep this in mind in using the catalog data of various fan manufacturers, since such data are usually based on stated standard conditions. Corrections must be made for variations from these standards. The usual variations are as follows:

When speed varies, (1) capacity varies directly as the speed ratio, (2) pressure varies as the square of the speed ratio, and (3) horsepower varies as the cube of the speed ratio.

When the temperature of air or gas varies, horsepower and pressure vary inversely as the absolute temperature, speed and capacity being constant. See Fig. 10-87.

When the density of air or gas varies, horsepower and pressure vary directly as the density, speed and capacity being constant.

### CONTINUOUS-FLOW COMPRESSORS

Continuous-flow compressors are machines where the flow is continuous, unlike positive-displacement machines where the flow is fluctuating. Continuous-flow compressors are also classified as turbomachines. These types of machines are widely used in the chemical and petroleum industry for many services. They are also used extensively in many other industries such as the iron and steel industry, pipeline boosters, and on offshore platforms for reinjection compressors. Continuous-flow machines are usually much smaller in size and produce much less vibration than their counterpart, positive-displacement units.

**Centrifugal Compressors** The flow in a centrifugal compressor enters the impeller in an axial direction and exits in a radial direction.

In a typical centrifugal compressor, the fluid is forced through the impeller by rapidly rotating impeller blades. The velocity of the fluid is converted to pressure, partially in the impeller and partially in the stationary diffusers. Most of the velocity leaving the impeller is converted into pressure energy in the diffuser as shown in Fig. 10-88. It is normal practice to design the compressor so that half the pressure rise takes place in the impeller and the other half in the diffuser. The diffuser consists of a vaneless space, a vane that is tangential to the impeller, or a combination of both. These vane passages diverge to convert the velocity head into pressure energy.

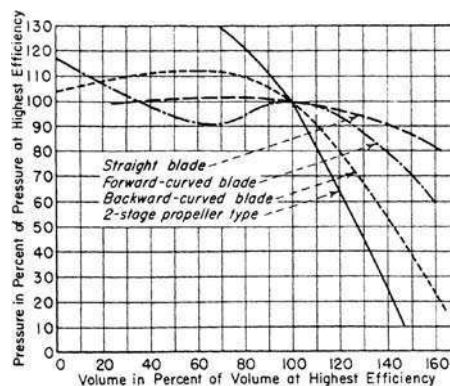


FIG. 10-87 Approximate characteristic curves of various types of fans.

Centrifugal compressors in general are used for higher pressure ratios and lower flow rates compared to lower-stage pressure ratios and higher flow rates in axial compressors. The pressure ratio in a single-stage centrifugal compressor varies depending on the industry and application. In the petrochemical industry the single stage pressure ratio is about 1.2:1. Centrifugal compressors used in the aerospace industry, usually as a compressor of a gas turbine, have pressure ratios between 3:1 to as high as 9:1 per stage.

In the petrochemical industry, the centrifugal compressors consist mainly of casings with multiple stages. In many instances, multiple casings are also used, and to reduce the power required to drive these multiple casings, there are intercoolers between them. Each casing can have up to 9 stages. In some cases, intercoolers are also used between single stages of compressor to reduce the power required for compression. These compressors are usually driven by gas turbines, steam turbines, and electric motors. Speed-increasing gears may be used in conjunction with these drivers to obtain the high speeds at which many of these units operate. Rotative speeds of as high as 50,000 rpm are not uncommon. Most of the petrochemical units run between 9,000 and 15,000 rpm.

The compressor's operating range is between two major regions as seen in Fig. 10-89, which is a performance map of a centrifugal compressor. These two regions are *surge*, which is the lower flow limit of stable operation, and *choke* or *stonewall*, which is the maximum flow through the compressor at a given operating speed. The centrifugal compressor's operating range between surge and choke is reduced as the pressure ratio per stage is increased or the number of stages are added.

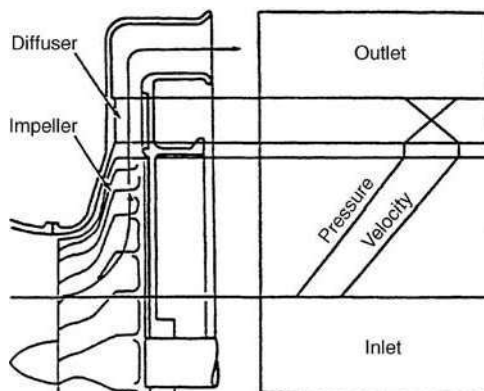


FIG. 10-88 Pressure and velocity through a centrifugal compressor.

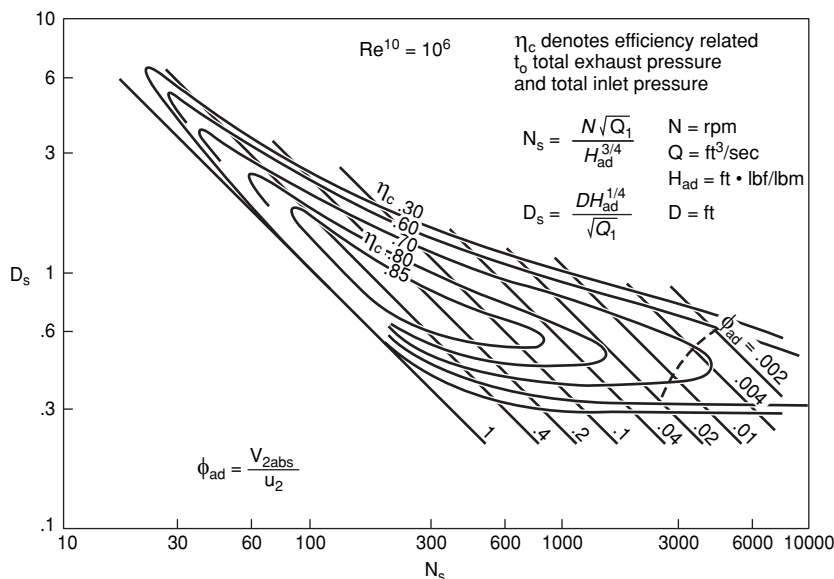


FIG. 10-89 Centrifugal compressor map. (Balje, O. E., "A Study of Reynolds Number Effects in Turbomachinery," Journal of Engineering for Power, ASME Trans., vol. 86, series A, p. 227.)

A compressor is said to be in surge when the main flow through the compressor reverses its direction. Surge is often symptomized by excessive vibration and a large audible sound. This flow reversal is accompanied with a very violent change in energy, which causes a reversal of the thrust force. The surge process is cyclic in nature and if allowed to cycle for some time, irreparable damage can occur to the compressor. In a centrifugal compressor, surge is usually initiated at the exit of the impeller or at the diffuser entrance for impellers producing a pressure ratio of less than 3:1. For higher pressure ratios, the initiation of surge can occur in the inducer.

A centrifugal compressor impeller can have three types of blades at the exit of the impeller. These are forward-curved, backward-curved, and radial blades. Forward-curved blades are not often used in a centrifugal compressor's impeller because of the very high-velocity discharge at the compressor that would require conversion of the high velocity to a pressure head in the diffuser, which is accompanied by high losses. Radial blades are used in impellers of high pressure ratio since the stress levels are minimal. Backward-curved blades give the highest efficiency and the largest operating margin of any of the various types of blades in an impeller. Most centrifugal compressors in the petrochemical industry use backward-curved impellers because of the higher efficiency and larger operating range.

Process compressors have impellers with very low pressure ratio impellers and thus large surge-to-choke margins. The common method of classifying process-type centrifugal compressors is based on the number of impellers and the casing design. Sectionalized casing types have impellers that are usually mounted on the extended motor shaft, and similar sections are bolted together to obtain the desired number of stages. Casing material is either steel or cast iron. These machines require minimum supervision and maintenance and are quite economic in their operating range. The sectionalized casing design is used extensively in supplying air for combustion in ovens and furnaces.

The horizontally split type have casings split horizontally at the mid-section and the top. The bottom halves are bolted and doweled together. This design type is preferred for large multistage units. The internal parts such as shaft, impellers, bearings, and seals are readily accessible for inspection and repairs by removing the top half. The casing material is cast iron or cast steel.

Barrel casings are used for high pressures in which the horizontally split joint is inadequate. This type of compressor consists of a barrel into which a compressor bundle of multiple stages is inserted. The bundle is itself a horizontally split casing compressor.

**Compressor Configuration** To properly design a centrifugal compressor, one must know the operating conditions—the type of gas, its pressure, temperature, and molecular weight. One must also know the corrosive properties of the gas so that proper metallurgical selection can be made. Gas fluctuations due to process instabilities must be pinpointed so that the compressor can operate without surging.

Centrifugal compressors for industrial applications have relatively low pressure ratios per stage. This condition is necessary so that the compressors can have a wide operating range while stress levels are kept at a minimum. Because of the low pressure ratios for each stage, a single machine may have a number of stages in one "barrel" to achieve the desired overall pressure ratio. Figure 10-90 shows some of the many configurations. Some factors to be considered when selecting a configuration to meet plant needs are:

1. Intercooling between stages can considerably reduce the power consumed.
2. Back-to-back impellers allow for a balanced rotor thrust and minimize overloading the thrust bearings.
3. Cold inlet or hot discharge at the middle of the case reduces oil-seal and lubrication problems.
4. Single inlet or single discharge reduces external piping problems.
5. Balance planes that are easily accessible in the field can appreciably reduce field-balancing times.
6. Balance piston with no external leakage will greatly reduce wear on the thrust bearings.
7. Hot and cold sections of the case that are adjacent to each other will reduce thermal gradients and thus reduce case distortion.
8. Horizontally split casings are easier to open for inspection than vertically split ones, reducing maintenance time.
9. Overhung rotors present an easier alignment problem because shaft-end alignment is necessary only at the coupling between the compressor and driver.
10. Smaller, high-pressure compressors that do the same job will reduce foundation problems but will have greatly reduced operational range.



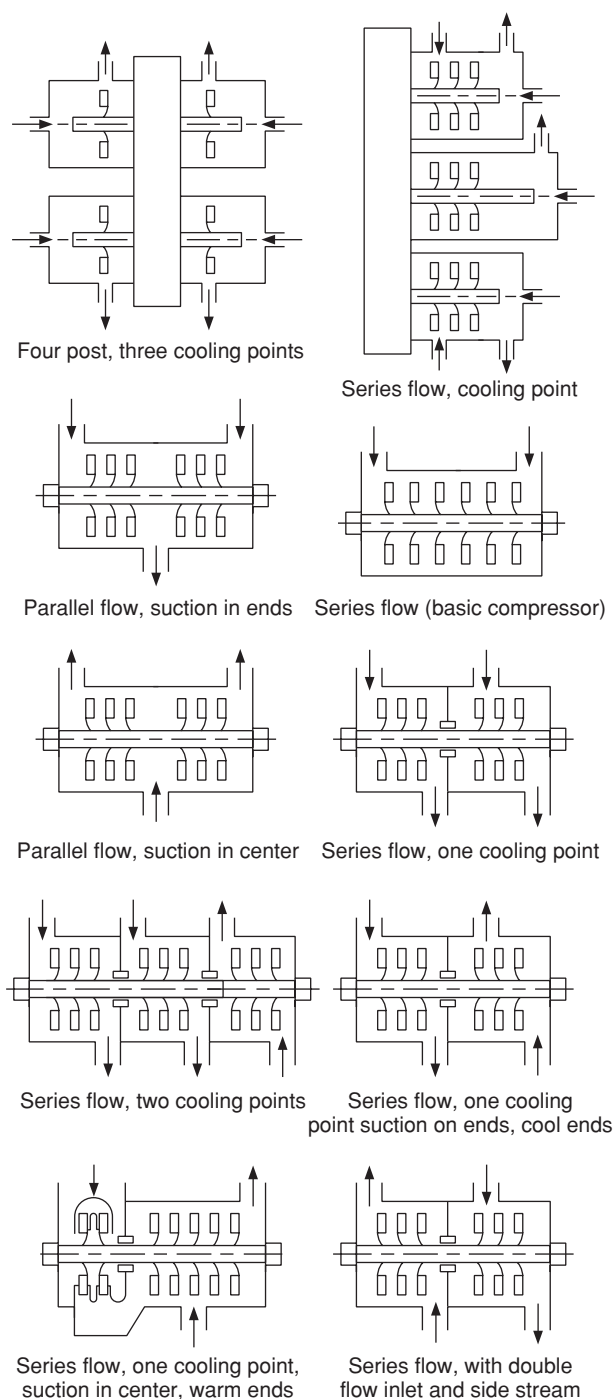


FIG. 10-90 Various configurations of centrifugal compressors.

**Impeller Fabrication** Centrifugal-compressor impellers are either shrouded or unshrouded. Open, shrouded impellers that are mainly used in single-stage applications are made by investment-casting techniques or by three-dimensional milling. Such impellers are used, in most cases, for the high-pressure-ratio stages. The

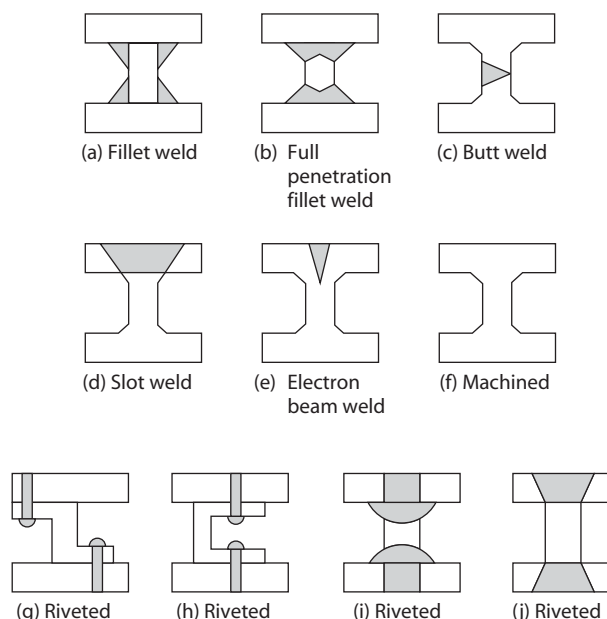


FIG. 10-91 Several fabrication techniques for centrifugal impellers.

shrouded impeller is commonly used in the process compressor because of its low pressure ratio stages. The low tip stresses in this application make it a feasible design. Figure 10-91 shows several fabrication techniques. The most common type of construction is seen in A and B where the blades are fillet-welded to the hub and shroud. In B, the welds are full penetration. The disadvantage in this type of construction is the obstruction of the aerodynamic passage. In C, the blades are partially machined with the covers and then butt-welded down the middle. For backward lean-angled blades, this technique has not been very successful, and there has been difficulty in achieving a smooth contour around the leading edge.

D illustrates a slot-welding technique and is used where blade-passage height is too small (or the backward lean-angle too high) to permit conventional fillet welding. In E, an electron-beam technique is shown. Its major disadvantage is that electron-beam welds should preferably be stressed in tension but, for the configuration of E, they are in shear. The configurations of G through J use rivets. Where the rivet heads protrude into the passage aerodynamic performance is reduced. Riveted impellers were used in the 1960s—they are very rarely used now. Elongation of these rivets occurs at certain critical surge conditions and can lead to major failures.

Materials for fabricating these impellers are usually low-alloy steels, such as AISI 4140 or AISI 4340. AISI 4140 is satisfactory for most applications; AISI 4340 is used for large impellers requiring higher strengths. For corrosive gases, AISI 410 stainless steel (about 12 percent chromium) is used. Monel K-500 is employed in halogen gas atmospheres and oxygen compressors because of its resistance to sparking. Titanium impellers have been applied to chlorine service. Aluminum-alloy impellers have been used in great numbers, especially at lower temperatures (below 300°F). With new developments in aluminum alloys, this range is increasing. Aluminum and titanium are sometimes selected because of their low density. This low density can cause a shift in the critical speed of the rotor, which may be advantageous.

**Axial-Flow Compressors** Axial-flow compressors are used mainly as compressors for gas turbines. They are also used in the steel industry as blast furnace blowers and in the chemical industry for large nitric acid plants. They are mainly used for applications where the head required is low and the flow large.

Figure 10-92 shows a typical axial-flow compressor. The rotating element consists of a single drum to which are attached several rows

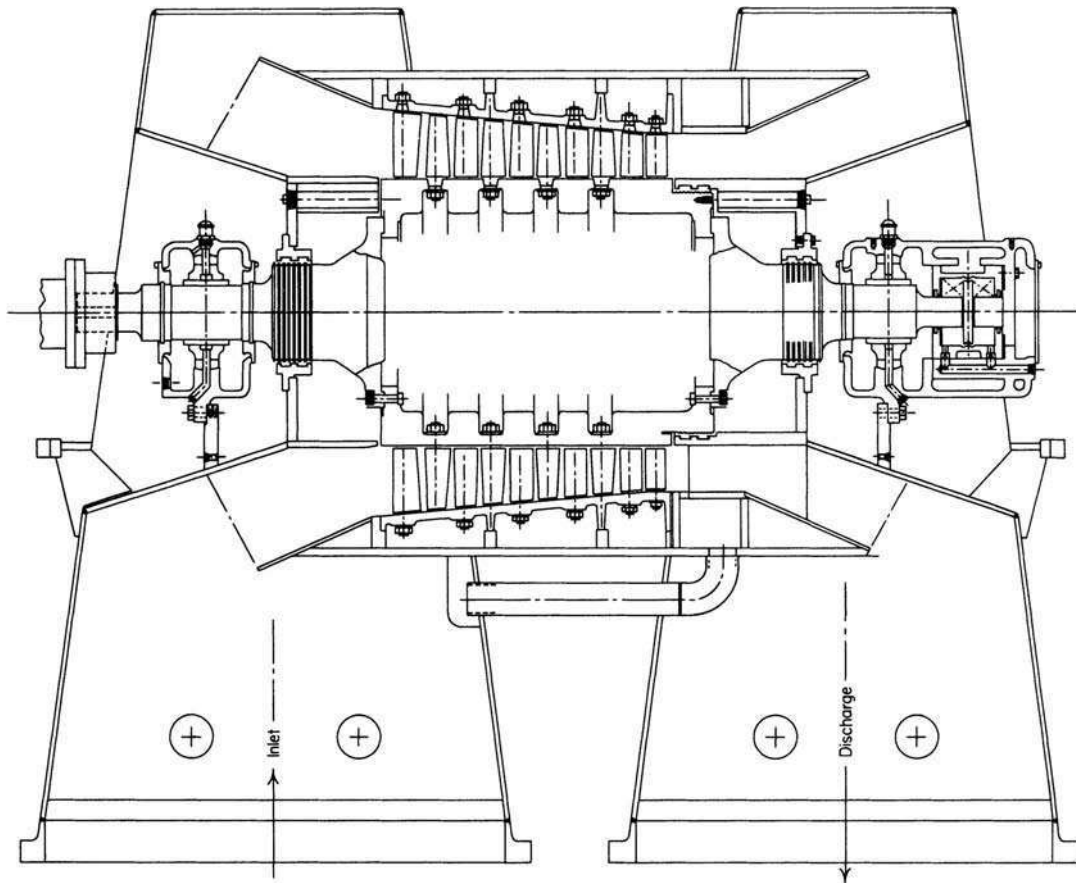


FIG. 10-92 Axial-flow compressor. (Courtesy of Allis-Chalmers Corporation.)

of decreasing-height blades having airfoil cross sections. Between each rotating blade row is a stationary blade row. All blade angles and areas are designed precisely for a given performance and high efficiency. The use of multiple stages permits overall pressure increases up to 30:1. The efficiency in an axial flow compressor is higher than the centrifugal compressor.

Pressure ratio per casing can be comparable with those of centrifugal equipment, although flow rates are considerably higher for a given casing diameter because of the greater area of the flow path. The pressure ratio per stage is less than in a centrifugal compressor. The pressure ratio per stage in industrial compressors is between 1:05 and 1:15, and for aeroturbines, 1.1 and 1.2.

The axial-flow compressors used in gas turbines vary depending on the type of turbines. The industrial-type gas turbine has an axial-flow compressor of a rugged construction. These units have blades that have low aspect ratio ( $R = \text{blade height/blade chord}$ ) with minimum streamline curvature, and the shafts are supported on sleeve-type bearings. The industrial gas turbine compressor has also a lower pressure ratio per stage (stage = rotor + stationary blade), giving a low blade loading. This also gives a larger operating range than its counterpart the aero axial gas turbine compressor but considerably less than the centrifugal compressor.

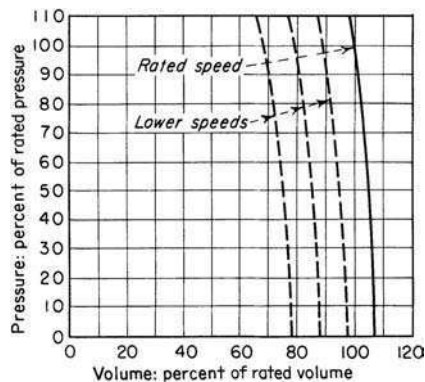
The axial-flow compressors in aero gas turbines are heavily loaded. The aspect ratio of the blades, especially the first few stages, can be as high as 4.0, and the effect of streamline curvature is substantial. The streamline configuration is a function of the annular passage area, the camber and thickness distribution of the blade, and the flow angles at the inlet and outlet of the blades. The

shafts on these units are supported on antifriction bearings (roller or ball bearings).

The operation of the axial-flow compressor is a function of the rotational speed of the blades and the turning of the flow in the rotor. The stationary blades (stator) are used to diffuse the flow and convert the velocity increased in the rotor to a pressure increase. One rotor and one stator make up a stage in a compressor. One additional row of fixed blades (inlet guide vanes) is frequently used at the compressor inlet to ensure that air enters the first stage rotors at the desired angle. In addition to the stators, another diffuser at the exit of the compressor further diffuses the gas and, in the case of gas turbines, controls its velocity entering the combustor. The axial-flow compressor has a much smaller operating range "Surge to Choke" than its counterpart in the centrifugal compressor. Because of the steep characteristics of the head/flow capacity curve, the surge point is usually within 10 percent of the design point.

The axial-flow compressor has three distinct stall phenomena. Rotating stall and individual blade stall are aerodynamic phenomena. Stall flutter is an aeroelastic phenomenon. Rotating stall (propagating stall) consists of large stall zones covering several blade passages and propagates in the direction of the rotor and at some fraction of rotor speed. The number of stall zones and the propagating rates vary considerably. Rotating stall is the most prevalent type of stall phenomenon. Individual blade stall occurs when all the blades around the compressor annulus stall simultaneously without the occurrence of the stall propagation mechanism. The phenomena of stall flutter is caused by self-excitation of the blade and is aeroelastic. It must be distinguished from classic flutter, since classic flutter is a coupled





**Fig. 10-93** Approximate performance curves for a rotary positive-displacement compressor. The safety valve in discharge line or bypass must be set to operate at a safe value determined by construction.

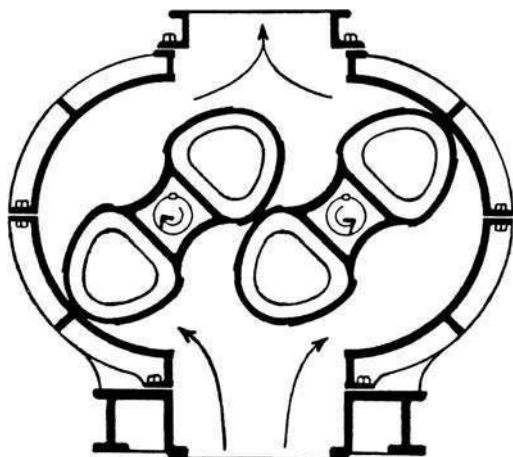
torsional-flexural vibration that occurs when the freestream velocity over an airfoil section reaches a certain critical velocity. Stall flutter, on the other hand, is a phenomenon that occurs due to the stalling of the flow around a blade. Blade stall causes Karman vortices in the airfoil wake. Whenever the frequency of the vortices coincides with the natural frequency of airfoil, flutter will occur. Stall flutter is a major cause of compressor-blade failure.

**Positive-Displacement Compressors** Positive-displacement compressors are machines that are essentially constant-volume machines with variable discharge pressures. These machines can be divided into two types:

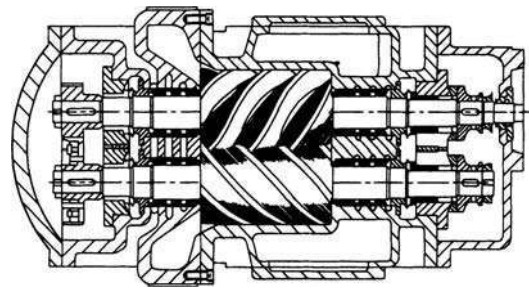
1. Rotary compressors
2. Reciprocating compressors

Many users consider rotary compressors, such as the "Rootes"-type blower, as turbomachines because their behavior in terms of the rotor dynamics is very close to centrifugal and axial flow machinery. Unlike the reciprocating machines, the rotary machines do not have a very high vibration problem but, like the reciprocating machines, they are positive-displacement machines.

**Rotary Compressors** Rotary compressors are machines of the positive-displacement type. Such units are essentially constant-volume machines with variable discharge pressure. The volume can be varied only by changing the speed or by bypassing or wasting some of the capacity of the machine. The discharge pressure will vary with the resistance on the discharge side of the system. A characteristic curve typical of the form produced by these rotary units is shown in Fig. 10-93.



**FIG. 10-94** Two-impeller type of rotary positive-displacement blower.



**FIG. 10-95** Screw-type rotary compressor.

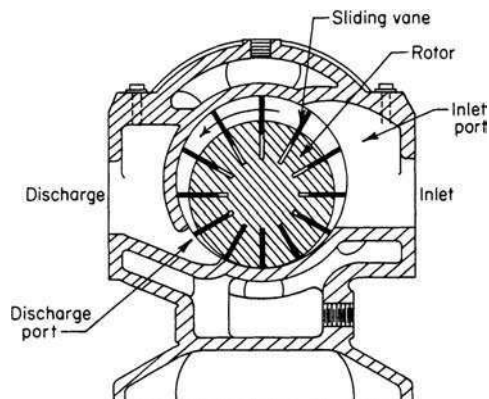
Rotary compressors are generally classified as of the straight-lobe type, screw type, sliding-vane type, and liquid-piston type.

**Straight-Lobe Type** This type is illustrated in Fig. 10-94. Such units are available for pressure differentials up to about 83 kPa (12 lbf/in<sup>2</sup>) and capacities up to  $2.549 \times 10^4$  m<sup>3</sup>/h (15,000 ft<sup>3</sup>/min). Sometimes multiple units are operated in series to produce higher pressures; individual-stage pressure differentials are limited by the shaft deflection, which must necessarily be kept small to maintain rotor and casing clearance.

**Screw Type** This type of rotary compressor, as shown in Fig. 10-95, is capable of handling capacities up to about  $4.248 \times 10^4$  m<sup>3</sup>/h (25,000 ft<sup>3</sup>/min) at pressure ratios of 4:1 and higher. Relatively small-diameter rotors allow rotative speeds of several thousand rev/min. Unlike the straight-lobe rotary machine, it has male and female rotors whose rotation causes the axial progression of successive sealed cavities. These machines are staged with intercoolers when such an arrangement is advisable. Their high-speed operation usually necessitates the use of suction- and discharge-noise suppressors. The bearings used are sleeve-type bearings. Due to the side pressures experienced, tilting pad bearings are highly recommended.

**Sliding-Vane Type** This type is illustrated in Fig. 10-96. These units are offered for operating pressures up to 0.86 MPa (125 lbf/in<sup>2</sup>) and in capacities up to  $3.4 \times 10^3$  m<sup>3</sup>/h (2000 ft<sup>3</sup>/min). Generally, pressure ratios per stage are limited to 4:1. Lubrication of the vanes is required, and the air or gas stream therefore contains lubricating oil.

**Liquid-Piston Type** This type is illustrated in Fig. 10-97. These compressors are offered as single-stage units for pressure differentials up to about 0.52 MPa (75 lbf/in<sup>2</sup>) in the smaller sizes and capacities up to  $6.8 \times 10^3$  m<sup>3</sup>/h (4000 ft<sup>3</sup>/min) when used with a lower pressure differential. Staging is employed for higher pressure differentials. These units have found wide application as vacuum pumps on wet-vacuum service. Inlet and discharge ports are located in the impeller hub. As the vaned impeller rotates, centrifugal force drives the sealing liquid



**FIG. 10-96** Sliding-vane type of rotary compressor.

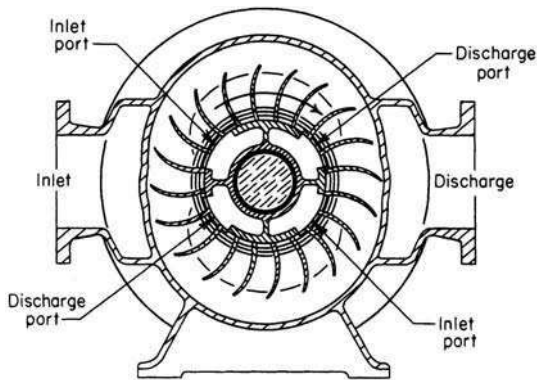


FIG. 10-97 Liquid-piston type of rotary compressor.

against the walls of the elliptical housing, causing the air to be successively drawn into the vane cavities and expelled against discharge pressure. The sealing liquid must be externally cooled unless it is used in a once-through system. A separator is usually employed in the discharge line to minimize carryover of entrained liquid. Compressor capacity can be considerably reduced if the gas is highly soluble in the sealing liquid.

The liquid-piston type of compressor has been of particular advantage when hazardous gases are being handled. Because of the gas-liquid contact and because of the much greater liquid specific heat, the gas-temperature rise is very small.

## EJECTORS

An ejector is a simplified type of vacuum pump or compressor which has no pistons, valves, rotors, or other moving parts. Figure 10-98 illustrates a steam-jet ejector. It consists essentially of a nozzle which discharges a high-velocity jet across a suction chamber that is con-

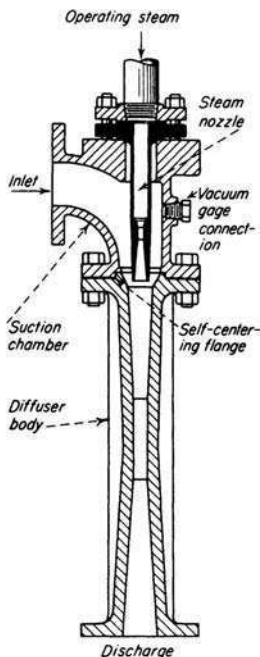


FIG. 10-98 Typical steam-jet ejector.

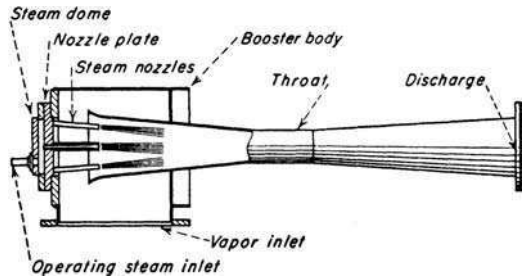


FIG. 10-99 Booster ejector with multiple steam nozzles.

nected to the equipment to be evacuated. The gas is entrained by the steam and carried into a venturi-shaped diffuser which converts the velocity energy into pressure energy. Figure 10-99 shows a large-sized ejector, sometimes called a booster ejector, with multiple nozzles. Nozzles are devices in subsonic flow that have a decreasing area and accelerate the flow. They convert pressure energy to velocity energy. A minimum area is reached when velocity reaches sonic flow. In supersonic flow, the nozzle is an increasing area device. A diffuser in subsonic flow has an increasing area and converts velocity energy into pressure energy. A diffuser in supersonic flow has a decreasing area.

Two or more ejectors may be connected in series or stages. Also, a number of ejectors may be connected in parallel to handle larger quantities of gas or vapor.

Liquid- or air-cooled condensers are usually used between stages. Liquid-cooled condensers may be of either the direct-contact (barometric) or the surface type. By condensing vapor the load on the following stage is reduced, thus minimizing its size and reducing consumption of motive gas. Likewise, a precondenser installed ahead of an ejector reduces its size and consumption if the suction gas contains vapors that are condensable at the temperature condition available. An **aftercondenser** is frequently used to condense vapors from the final stage, although this does not affect ejector performance.

**Ejector Performance** The performance of any ejector is a function of the area of the motive-gas nozzle and venturi throat, pressure of the motive gas, suction and discharge pressures, and ratios of specific heats, molecular weights, and temperatures. Figure 10-100, based on the assumption of **constant-area mixing**, is useful in evaluating single-stage-ejector performance for compression ratios up to 10 and area ratios up to 100 (see Fig. 10-101 for notation).

For example,\* assume that it is desired to evacuate air at 2.94 lbf/in<sup>2</sup> with a steam ejector discharging to 14.7 lbf/in<sup>2</sup> with available steam pressure of 100 lbf/in<sup>2</sup>. Entering the chart at  $p_{03}/p_{0b} = 5.0$ , at  $p_{0b}/p_{0a} = 2.94/100 = 0.0294$  the optimum area ratio is 12. Proceeding horizontally to the left,  $w_1/w_a$  is approximately 0.15 lb of air per 1 lb of steam. This value must be corrected for the temperature and molecular-weight differences of the two fluids by Eq. (10-86).

$$w/w_a = w_1/w_a \sqrt{T_{0a}M_b/T_{0b}M_a} \quad (10-86)$$

In addition, there are empirical correction factors which should be applied. Laboratory tests show that for ejectors with constant-area mixing the actual entrainment and compression ratios will be approximately 90 percent of the calculated values and even less at very small values of  $p_{0b}/p_{0a}$ . This compensates for ignoring wall friction in the mixing section and irreversibilities in the nozzle and diffuser. In theory, each point on a given design curve of Fig. 10-100 is associated with an optimum ejector for prevailing operating conditions. Adjacent points on the same curve represent theoretically different ejectors for the new conditions, the difference being that for each ratio of  $p_{0b}/p_{0a}$  there is an optimum area for the exit of the motive-gas nozzle. In practice, however, a segment of a given curve for constant  $A_2/A_1$  represents the

\*All data are given in U.S. customary units since the charts are in these units. Conversion factors to SI units are given on the charts.

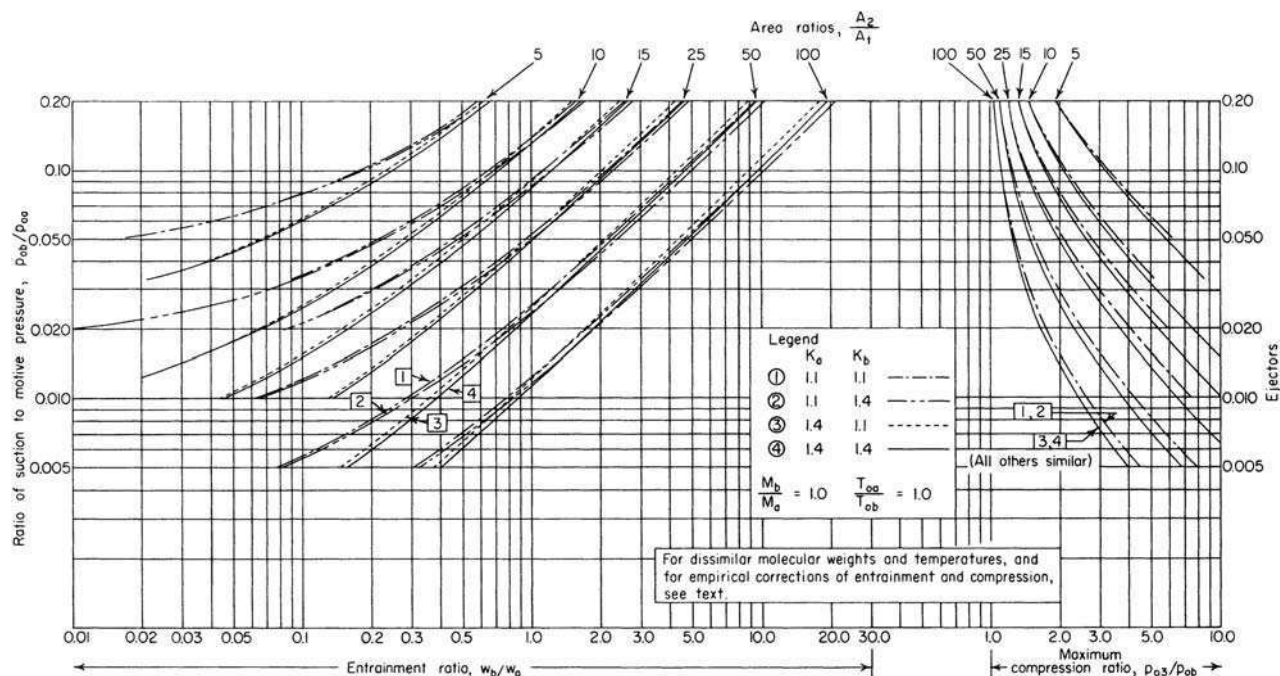


FIG. 10-100 Design curves for optimum single-stage ejectors. [DeFrate and Hoerl, Chem. Eng. Prog., 55, Symp. Ser. 21, 46 (1959).]

performance of a single ejector satisfactorily for estimating purposes, provided that the suction pressure lies within 20 to 130 percent of the design suction pressure and the motive pressure within 80 to 120 percent of design motive pressure. Thus the curves can be used to select an optimum ejector for the design point and to estimate its performance at off-design conditions within the limits noted. Final ejector selection should, of course, be made with the assistance of a manufacturer of such equipment.

**Uses of Ejectors** For the operating range of steam-jet ejectors in vacuum applications, see the subsection "Vacuum Systems."

The choice of the most suitable type of ejector for a given application depends upon the following factors:

1. *Steam pressure.* Ejector selection should be based upon the minimum pressure in the supply line selected to serve the unit.
2. *Water temperature.* Selection is based on the maximum water temperature.
3. *Suction pressure and temperature.* Overall process requirements should be considered. Selection is usually governed by the minimum suction pressure required (the highest vacuum).
4. *Capacity required.* Again overall process requirements should be considered, but selection is usually governed by the capacity required at the minimum process pressure.

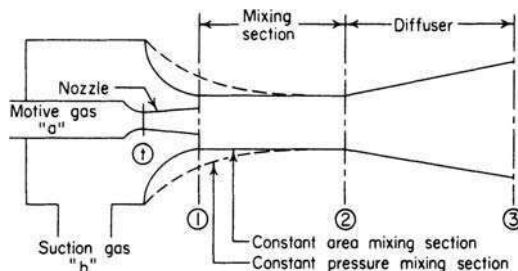


FIG. 10-101 Notation for Fig. 10-100.

Ejectors are easy to operate and require little maintenance. Installation costs are low. Since they have no moving parts, they have long life, sustained efficiency, and low maintenance cost. Ejectors are suitable for handling practically any type of gas or vapor. They are also suitable for handling wet or dry mixtures or gases containing sticky or solid matter such as chaff or dust.

Ejectors are available in many materials of construction to suit process requirements. If the gases or vapors are not corrosive, the diffuser is usually constructed of cast iron and the steam nozzle of stainless steel. For more corrosive gases and vapors, many combinations of materials such as bronze, various stainless-steel alloys, and other corrosion-resistant metals, carbon, and glass can be used.

## VACUUM SYSTEMS

Figure 10-102 illustrates the level of vacuum normally required to perform many of the common manufacturing processes. The attainment of various levels is related to available equipment in Fig. 10-103.

**Vacuum Equipment** The equipment shown in Fig. 10-103 has been discussed elsewhere in this section with the exception of the **diffusion pump**. Figure 10-104 depicts a typical design. A liquid of low absolute vapor pressure is boiled in the reservoir. The vapor is ejected at high velocity in a downward direction through multiple jets and is condensed on the walls, which are cooled by the surrounding coils. Molecules of the gas being pumped enter the vapor stream and are driven downward by collisions with the vapor molecules. The gas molecules are removed through the discharge line by a backing pump such as a rotary oil-sealed unit.

Diffusion pumps operate at very low pressures. The ultimate vacuum attainable depends somewhat upon the vapor pressure of the pump liquid at the temperature of the condensing surfaces. By providing a cold trap between the diffusion pump and the region being evacuated, pressures as low as  $10^{-7}$  mmHg absolute are achieved in this manner. Liquids used for diffusion pumps are mercury and oils of low vapor pressure. Silicone oils have excellent characteristics for this service.



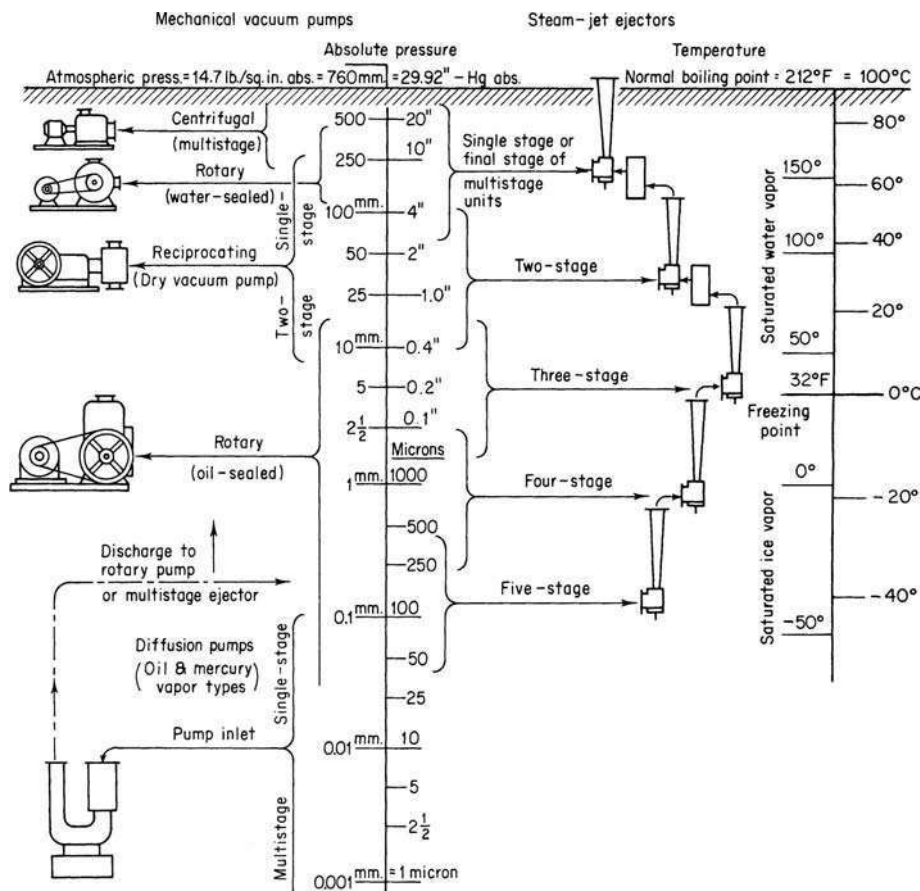


FIG. 10-103 Vacuum levels attainable with various types of equipment. (Courtesy of Compressed Air magazine.)

consumption, material selection flexibility, minimal effect on rotor dynamics, back diffusion reduction, integration of pressure, lack of pressure limitations, and tolerance to gross thermal variations. The major disadvantages are the high leakage, loss of machine efficiency, increased buffering costs, tolerance to ingestion of particulates with

resulting damage to other critical items such as bearings, the possibility of the cavity clogging due to low gas velocities or back diffusion, and the inability to provide a simple seal system that meets OSHA or EPA standards. Because of some of the foregoing disadvantages, many machines are being converted to other types of seals.

Labyrinth seals are simple to manufacture and can be made from conventional materials. Early designs of labyrinth seals used knife-edge seals and relatively large chambers or pockets between the knives. These relatively long knives are easily subject to damage. The modern, more functional, and more reliable labyrinth seals consist of sturdy, closely spaced lands. Some labyrinth seals are shown in Fig. 10-105. Figure 10-105a is the simplest form of the seal. Figure 10-105b shows a grooved seal; it is more difficult to manufacture but produces a tighter seal. Figure 10-105c and 10-105d is rotating labyrinth-type seals. Figure 10-105e shows a simple labyrinth seal with a buffered gas for which pressure must be maintained above the process gas pressure and the outlet pressure (which can be greater than or less than the atmospheric pressure). The buffered gas produces a fluid barrier to the process gas. The eductor sucks gas from the vent near the atmospheric end. Figure 10-105f shows a buffered, stepped labyrinth. The step labyrinth gives a tighter seal. The matching stationary seal is usually manufactured from soft materials such as bab-bitt or bronze, while the stationary or rotating labyrinth lands are made from steel. This composition enables the seal to be assembled with minimal clearance. The lands can therefore cut into the softer materials to provide the necessary running clearances for adjusting to the dynamic excursions of the rotor. To maintain maximum sealing efficiency, it is essential that the labyrinth lands maintain sharp edges in the direction of the flow.

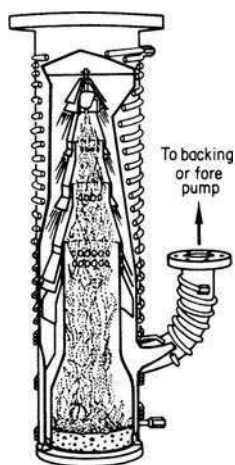


FIG. 10-104 Typical diffusion pump. (Courtesy of Compressed Air magazine.)

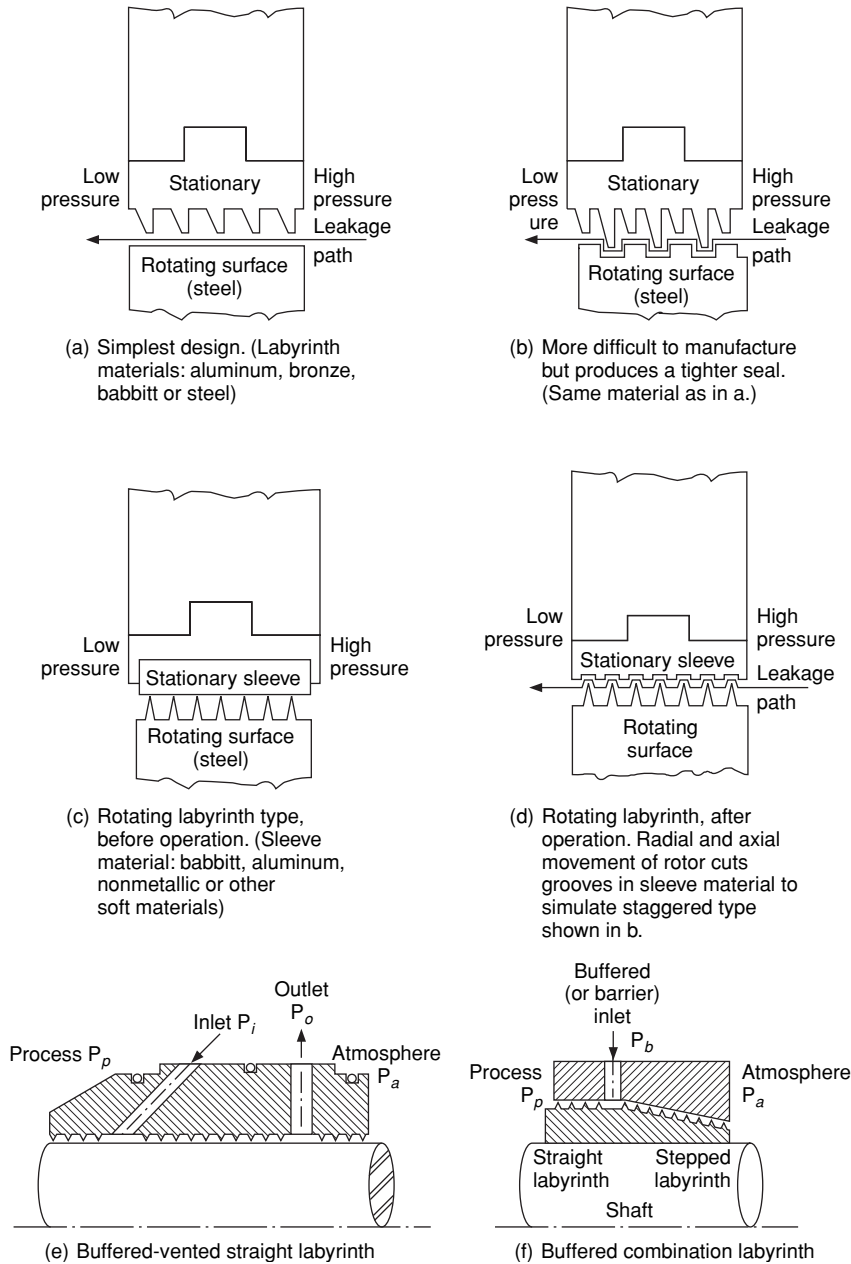


FIG. 10-105 Various configurations of labyrinth seals.

Leakage past these labyrinths is approximately inversely proportional to the square root of the number of labyrinth lands. This translates into the following relationship if leakage is to be cut in half in a four labyrinth seal: The number of labyrinths would have to be increased to 16. The Elgi leakage formula can be modified and written as:

$$m_l = AK \left[ \frac{(g/V_o)(P_o - P_n)}{n + \ln(P_n/P_o)} \right]^{1/2} \quad (10-87)$$

where  $m_l$  = leakage  
 $A$  = leakage area of single throttling

$K$  = labyrinth constant ( $K = .9$  for straight labyrinths,  $K = .75$  for staggered labyrinths)

$P_o$  = absolute pressure before the labyrinth

$P_n$  = absolute pressure after the last labyrinth

$V_o$  = specific volume before the labyrinth

$n$  = number of lands

The leakage of a labyrinth seal can be kept to a minimum by providing: (1) minimum clearance between the seal lands and the seal sleeve, (2) sharp edges on the lands to reduce the flow discharge coefficient, and (3) grooves or steps in the flow path for reducing dynamic head carryover from stage to stage.



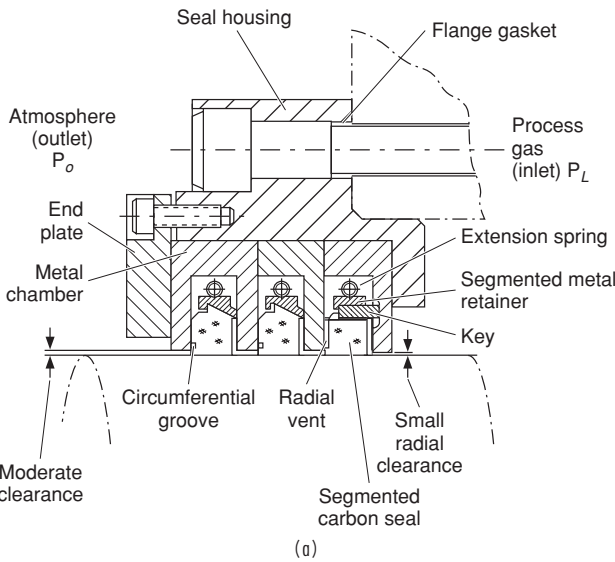


FIG. 10-106 Floating-type restrictive ring seal.

The labyrinth sleeve can be flexibly mounted to permit radial motion for self-aligning effects. In practice, a radial clearance of under 0.008 in is difficult to achieve.

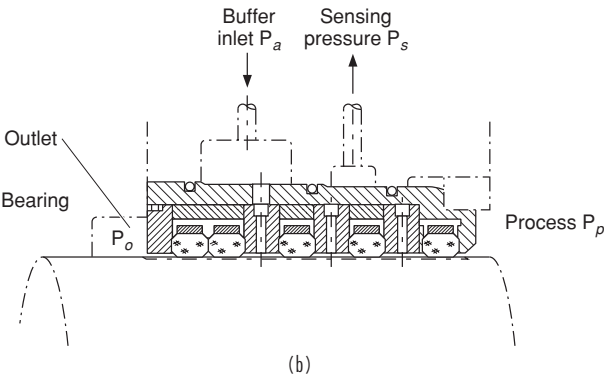
**Ring Seals** The restrictive ring seal is essentially a series of sleeves in which the bores form a small clearance around the shaft. Thus, the leakage is limited by the flow resistance in the restricted area and controlled by the laminar or turbulent friction. There are two types of ring seals: (1) fixed seal rings and (2) floating seal rings. The floating rings permit a much smaller leakage; they can be either the segmented type as shown in Fig. 10-106a or the rigid type as shown in Fig. 10-106b.

**Fixed Seal Rings** The fixed-seal ring consists of a long sleeve affixed to a housing in which the shaft rotates with small clearances. Long assemblies must be used to keep leakage within a reasonable limit. Long seal assemblies aggravate alignment and rubbing problems, thus requiring shafts to operate below their capacity. The fixed bushing seal operates with appreciable eccentricity and, combined with large clearances, produces large leakages, thus making this kind of seal impractical where leakage is undesirable.

**Floating Seal Rings** Clearance seals that are free to move in a radial direction are known as floating seals. The floating characteristics permit them to move freely, thus avoiding severe rubs. Due to differential thermal expansion between the shaft and bushing, the bushings should be made of material with a higher coefficient of thermal expansion. This is achieved by shrinking the carbon into a metallic retaining ring with a coefficient of expansion that equals or exceeds that of the shaft material. It is advisable in high shearing applications to lock the bushings against rotation.

Buildup of dirt and other foreign material lodged between the seal ring and seat will create an excessive spin and damage on the floating seal ring unit. It is therefore improper to use soft material such as babbit and silver as seal rings.

**Packing Seal** A common type of rotating shaft seal consists of packing composed of fibers which are first woven, twisted, or braided into strands and then formed into coils, spirals, or rings. To ensure initial lubrication and to facilitate installation, the basic materials are often impregnated. Common materials are braided and twisted rubber and duck, flax, jute, and metallic braids. The so-called plastic packings can be made up with varying amounts of fiber combined with a binder and lubricant for high-speed applications. Maximum temperatures that base materials of packings withstand and still give good service are as follows:



|                           | °C  | °F   |
|---------------------------|-----|------|
| Flax                      | 38  | 100  |
| Cotton                    | 93  | 200  |
| Duck and rubber           | 149 | 300  |
| Rubber                    | 177 | 350  |
| Metallic (lead-based)     | 218 | 425  |
| Metallic (aluminum-based) | 552 | 1025 |
| Metallic (copper-based)   | 829 | 1525 |

Packing may not provide a completely leak-free seal. With shaft surface speeds less than approximately 2.5 m/s (500 ft/min), the packing may be adjusted to seal completely. However, for higher speeds some leakage is required for lubrication, friction reduction, and cooling.

**Application of Packing** Coils and spirals are cut to form closed or nearly closed rings in the stuffing box. Clearance between ends should be sufficient to allow for fitting and possible expansion due to increased temperature or liquid absorption of the packing while in operation.

The correct form of the ring joint depends on materials and service requirements. Braided and flexible metallic packings usually have butt or square joints (Fig. 10-107a). With other packing material, service experience indicates that rings cut with bevel or skive joints (Fig. 10-107b) are more satisfactory. A slight advantage of the bevel joint over the butt joint is that the bevel permits a certain amount of sliding action, thus absorbing a portion of ring expansion.

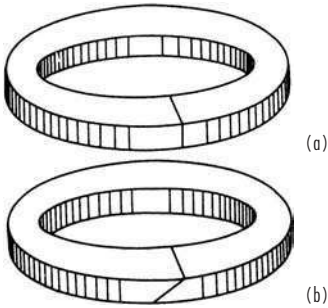


FIG. 10-107 Butt (a) and skive (b) joints for compression packing rings.

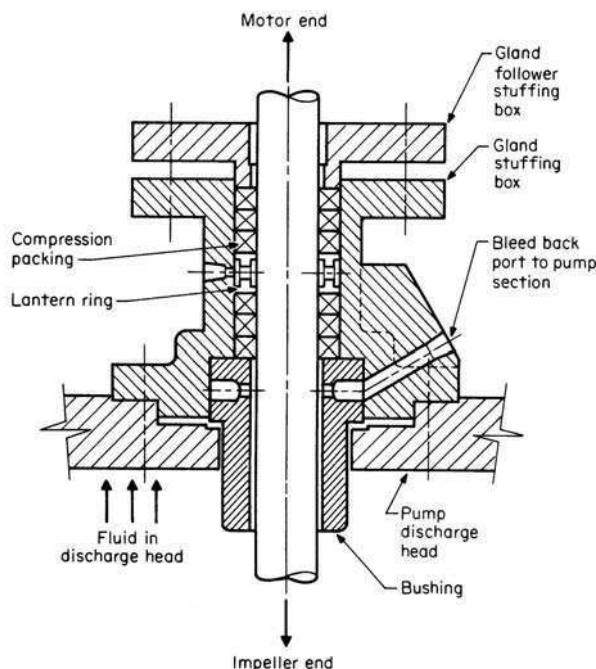


FIG. 10-108 Seal cage or lantern ring. (Courtesy of Crane Packing Co.)

In the manufacture of packings, the proper grade and type of **lubricant** is usually impregnated for each service for which the packing is recommended. However, it may be desirable to replenish the lubricant during the normal life of the packing. Lack of lubrication causes packing to become hard and lose its resiliency, thus increasing friction, shortening packing life, and increasing operating costs.

An effective auxiliary device frequently used with packing and rotary shafts is the **seal cage** (or **lantern ring**), shown in Fig. 10-108. The seal cage provides an annulus around the shaft for the introduction of a lubricant, oil, grease, etc. The seal cage is also used to introduce liquid for cooling, to prevent the entrance of atmospheric air, or to prevent the infiltration of abrasives from the process liquid.

The chief advantage of packing over other types of seals is the ease with which it can be adjusted or replaced. Most equipment is designed so that disassembly of major components is not required to remove or add packing rings. The major disadvantages of a packing-type seal are (1) short life, (2) requirement for frequent adjustment, and (3) need for some leakage to provide lubrication and cooling.

**Mechanical Face Seals** This type of seal forms a running seal between flat precision-finished surfaces. It is an excellent seal against leakages. The sealing surfaces are planes perpendicular to the rotating shaft, and the forces that hold the contact faces are parallel to the shaft axis. For a seal to function properly, there are four sealing points:

1. Stuffing box face
2. Leakage down the shaft
3. Mating ring in the gland plate
4. Dynamic faces

**Mechanical Seal Selection** There are many factors that govern the selection of seals. These factors apply to any type of seal:

1. Product
2. Seal environment
3. Seal arrangement
4. Equipment
5. Secondary packing
6. Seal face combinations
7. Seal gland plate
8. Main seal body

**Product** Physical and chemical properties of the liquid or gas being sealed places constraints on the type of material, design, and arrangement of the seal.

**Pressure.** Pressure affects the choice of material and whether balanced or unbalanced seal design can be used. Most unbalanced seals are good up to 100 psig stuffing box pressure. Over 100 psig, balanced seals should be used.

**Temperature.** The temperature of the liquid being pumped is important because it affects the seal face material selection as well as the wear life of the seal face.

**Lubricity.** In any mechanical seal design, there is rubbing motion between the dynamic seal faces. This rubbing motion is often lubricated by the fluid being pumped. Most seal manufacturers limit the speed of their seals to 90 ft/s (30 m/s). This is primarily due to centrifugal forces acting on the seal, which tends to restrict the seal's axial flexibility.

**Abrasion.** If there are entrained solids in the liquid, it is desirable to have a flushed single inside type with a face combination of very hard material.

**Corrosion.** This affects the type of seal body: what spring material, what face material, and what type of elastomer or gasket material. The corrosion rate will affect the decision of whether to use a single or multiple spring design because the spring can usually tolerate a greater amount of corrosion without weakening it appreciably.

**Seal Environment** The design of the seal environment is based on the product and the four general parameters that regulate it:

1. Pressure control
2. Temperature control
3. Fluid replacement
4. Atmospheric air elimination

**Seal Arrangement** There are four types of seal arrangements:

1. Double seals are standard with toxic and lethal products, but maintenance problems and seal design contribute to poor reliability. The double face-to-face seal may be a better solution.
2. Do not use a double seal in dirty service—the inside seal will hang up.
3. API standards for balanced and unbalanced seals are good guidelines; too low a pressure for a balanced seal may encourage face lift-off.
4. Arrangement of the seal will determine its success more than the vendor. Over 100 arrangements are available.

**Equipment** The geometry of the pump or compressor is very important in seal effectiveness. Different pumps with the same shaft diameter and the total differential head can present different sealing problems.

**Secondary Packing** Much more emphasis should be placed on secondary packing especially if Teflon is used. A wide variation in performance is seen between various seal vendors, depending on seal arrangement there can be differences in mating ring packing.

**Seal Face Combinations** The dynamics of seal faces is better understood today. Seal-face combinations have come a long way in the past 8–10 years. Stellite is being phased out of the petroleum and petrochemical applications. Better grades of ceramic are available, cost of tungsten has come down, and relapping of tungsten are available near most industrial areas. Silicon carbide is being used in abrasive service.

**Seal Gland Plate** The seal gland plate is caught in between the pump vendor and the seal vendor. Special glands should be furnished by seal vendors, especially if they require heating, quenching, and drain with a floating-throat bushing. Gland designs are complex and may have to be revisited, especially if seals are changed.

**Main Seal Body** The term *seal body* makes reference to all rotating parts on a pusher seal, excluding shaft packing and seal ring. In many cases it is the chief reason to avoid a particular design for a particular service.

Basically, most mechanical seals have the following components as seen in Fig. 10-109.

1. Rotating seal ring
2. Stationary seal ring
3. Spring devices to provide pressure
4. Static seals

A loading device such as a spring is needed to ensure that in the event of loss or hydraulic pressure the sealing surfaces are kept closed. The amount of the load on the sealing area is determined by the



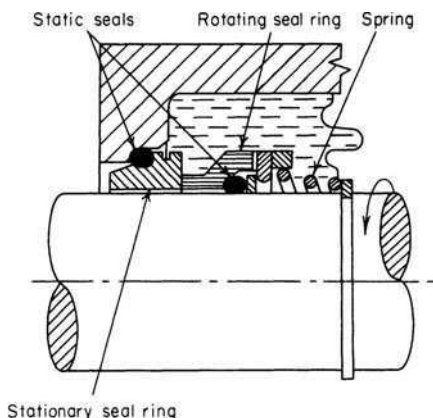


FIG. 10-109 Mechanical-seal components.

degree of “seal balance.” Figure 10-110 shows what seal balance means. A completely balanced seal is when the only force exerted on the sealing surfaces is the spring force; i.e., hydraulic pressure does not act on the sealing surface. The type of spring depends on the space available, loading characteristics, and the seal environment. Based on these considerations, either a single or multiple spring can be used. In small axial space, Belleville springs, finger washers, or curved washers can be used.

Shaft-sealing elements can be split up into two groups. The first type may be called pusher-type seals and includes the O-ring, V-ring, U-cup, and wedge configurations. Figure 10-111 shows some typical pusher-type seals. The second type is the bellows-type seals, which differ from the pusher-type seals in that they form a static seal between themselves and the shaft.

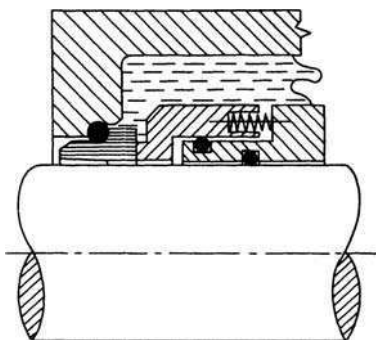


FIG. 10-110 Balanced internal mechanical seal.

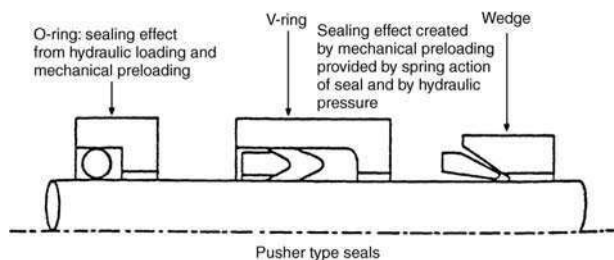


FIG. 10-111 Various types of shaft-sealing elements.

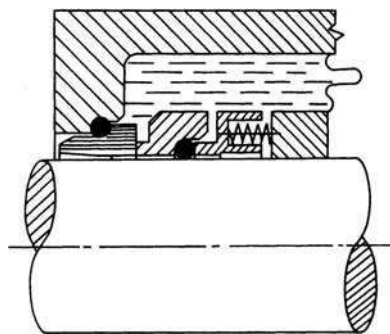


FIG. 10-112 Internal mechanical seal.

**Internal and External Seals** Mechanical seals are classified broadly as internal or external. **Internal seals** (Fig. 10-112) are installed with all seal components exposed to the fluid sealed. The advantages of this arrangement are (1) the ability to seal against high pressure, since the hydrostatic force is normally in the same direction as the spring force; (2) protection of seal parts from external mechanical damage; and (3) reduction in the shaft length required.

For high-pressure installations, it is possible to balance partially or fully the hydrostatic force on the rotating member of an internal seal by using a stepped shaft or shaft sleeve (Fig. 10-110). This method of relieving face pressure is an effective way of decreasing power consumption and extending seal life.

When abrasive solids are present and it is not permissible to introduce appreciable quantities of a secondary flushing fluid into the process, double internal seals are sometimes used (Fig. 10-113). Both sealing faces are protected by the flushing fluid injected between them even though the inward flow is negligible.

**External seals** (Fig. 10-114) are installed with all seal components protected from the process fluid. The advantages of this arrangement are that (1) fewer critical materials of construction are required, (2) installation and setting are somewhat simpler because of the exposed position of the parts, and (3) stuffing-box size is not a limiting factor. Hydraulic balancing is accomplished by proper proportioning of the seal face and secondary seal diameters.

**Throttle Bushings** These bushings (Fig. 10-115) are commonly used with single internal or external seals when solids are present in the fluid and the inflow of a flushing fluid is not objectionable. These close-clearance bushings are intended to serve as flow restrictions through which the maintenance of a small inward flow of flushing fluid prevents the entrance of a process fluid into the stuffing box.

A typical complex seal utilizes both the noncontact and mechanical aspects of sealing. Figure 10-116 shows such a seal with its two major elements. This type of seal will normally have buffering via a labyrinth

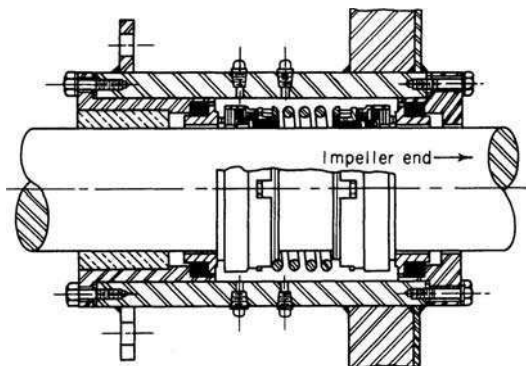


FIG. 10-113 Internal bellows-type double mechanical seal.

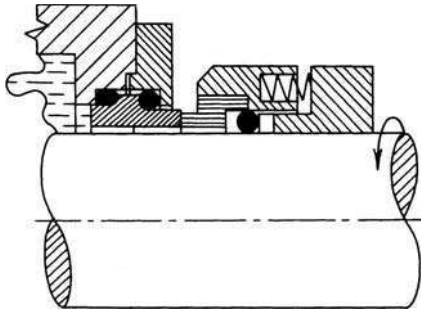


FIG. 10-114 External mechanical seal.

seal and a positive shutdown device. For shutdown, the carbon ring is tightly sandwiched between the rotating seal ring and the stationary sleeve with gas pressure to prevent gas from leaking out when no oil pressure is available.

In operation seal oil pressure is about 30–50 psi over the process gas pressure. The high-pressure oil enters the top and completely fills the seal cavity. A small percentage is forced across the carbon ring seal faces. The rotative speed of the carbon ring can be anywhere between zero and full rotational speed. Oil crossing the seal faces contacts the process gas and therefore is “contaminated oil.” The contaminated oil leaves through the contaminated oil drain to a degassifier for purification. The majority of the oil flows through the uncontaminated seal oil drain.

**Materials** Springs and other metallic components are available in a wide variety of alloys and are usually selected on the basis of temperature and corrosion conditions. The use of a particular mechanical seal is frequently restricted by the temperature limitations of the organic materials used in the static seals. Most elastomers are limited to about 121°C (250°F). Teflon will withstand temperatures of 260°C (500°F) but softens appreciably above 204°C (400°F). Glass-filled Teflon is dimensionally stable up to 232 to 260°C (450 to 500°F).

One of the most common elements used for seal faces is carbon. Although compatible with most process media, carbon is affected by strong oxidizing agents, including fuming nitric acid, hydrogen chloride, and high-temperature air [above 316°C (600°F)]. Normal mating-face materials for carbon are tungsten or chromium carbide, hard steel, stainless steel, or one of the cast irons.

Other sealing-face combinations that have been satisfactory in corrosive service are carbide against carbide, ceramic against ceramic, ceramic against carbon, and carbon against glass. The ceramics have also been mated with the various hard-facing alloys. When selecting seal materials the possibility of galvanic corrosion must also be considered.

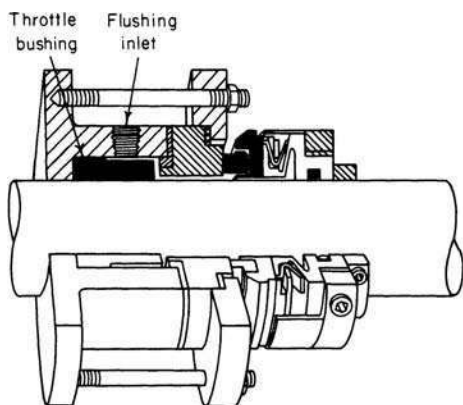
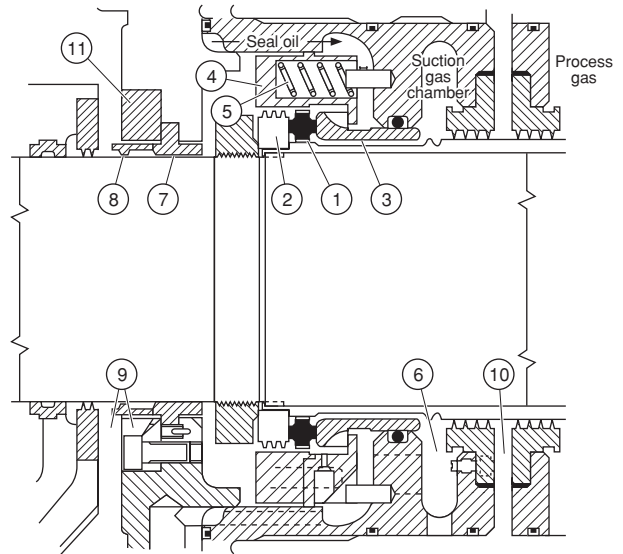


FIG. 10-115 External mechanical seal and throttle bushing.



- |                                   |                                      |
|-----------------------------------|--------------------------------------|
| 1. Rotating carbon ring           | 7. Floating babbitt-faced steel ring |
| 2. Rotating seal ring             | 8. Seal wiper ring                   |
| 3. Stationary sleeve              | 9. Seal oil drain line               |
| 4. Spring retainer                | 10. Buffer gas injection port        |
| 5. Spring                         | 11. Bypass orifice                   |
| 6. Gas and contaminated oil drain |                                      |

FIG. 10-116 Mechanical contact shaft seal.

## BEARINGS

Many factors enter into the selection of the proper design for bearings. Some of these factors are:

1. Shaft speed range.
2. Maximum shaft misalignment that can be tolerated.
3. Critical speed analysis and the influence of bearing stiffness on this analysis
4. Loading of the compressor impellers
5. Oil temperatures and viscosity
6. Foundation stiffness
7. Axial movement that can be tolerated
8. Type of lubrication system and its contamination
9. Maximum vibration levels that can be tolerated

**Types of Bearings** Figure 10-117 shows a number of different types of journal bearings. A description of a few of the pertinent types of journal bearings is given here:

1. *Plain journal.* Bearing is bored with equal amounts of clearance (on the order of one and one-half to two thousandths of an inch per inch of journal diameter) between the journal and bearing.
2. *Circumferential grooved bearing.* Normally the oil groove is half the bearing length. This configuration provides better cooling but reduces load capacity by dividing the bearing into two parts.
3. *Cylindrical bore bearings.* Another common bearing type used in turbines. It has a split construction with two axial oil-feed grooves at the split.
4. *Pressure or pressure dam.* Used in many places where bearing stability is required, this bearing is a plain journal bearing with a pressure pocket cut in the unloaded half. This pocket is approximately  $\frac{1}{32}$  of an inch deep with a width 50 percent of the bearing length. This groove or channel covers an arc of 135° and terminates abruptly in a sharp-edge dam. The direction of rotation is such that the oil is pumped down the channel toward the sharp edge. Pressure dam bearings are for one direction of rotation. They can be used in conjunction with cylindrical bore bearings as shown in Fig. 10-117.

| Bearing type                                                                                                             | Load capacity | Suitable direction of rotation                                                    | Resistance to half-speed whirl                                                                                                                                                                        | Stiffness and damping |
|--------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Cylindrical bore<br>                    | Good          |  | Worst<br><br>Increasing<br><br>Best | Moderate              |
| Cylindrical bore with dammed groove<br> | Good          |  |                                                                                                                                                                                                       | Moderate              |
| Lemon bore<br>                          | Good          |  |                                                                                                                                                                                                       | Moderate              |
| Three lobe<br>                          | Moderate      |  |                                                                                                                                                                                                       | Good                  |
| Offset halves<br>                       | Good          |  |                                                                                                                                                                                                       | Excellent             |
| Tilting pad<br>                         | Moderate      |  |                                                                                                                                                                                                       | Good                  |

FIG. 10-117 Comparison of general bearing types.

5. *Lemon bore or elliptical.* This bearing is bored with shims split line, which are removed before installation. The resulting shape approximates an ellipse with the major axis clearance approximately twice the minor axis clearance. Elliptical bearings are for both directions of rotation.

6. *Three-lobe bearing.* The three-lobe bearing is not commonly used in turbomachines. It has a moderate load-carrying capacity and can be operated in both directions.

7. *Offset halves.* In principle, this bearing acts very similar to a pressure dam bearing. Its load-carrying capacity is good. It is restricted to one direction of rotation.

8. *Tilt-pad bearings.* This bearing is the most common bearing type in today's machines. It consists of several bearing pads posed around the circumference of the shaft. Each pad is able to tilt to assume the most effective working position. This bearing also offers the greatest increase in fatigue life because of the following advantages:

- Thermal conductive backing material to dissipate heat developed in oil film.
- A thin babbitt layer can be centrifugally cast with a uniform thickness of about 0.005 inch. Thick babbitts greatly reduce bearing life. Babbitt thickness in the neighborhood of .01 reduce the bearing life by more than half.
- Oil film thickness is critical in bearing stiffness calculations. In a tilting-pad bearing, one can change this thickness in a number of ways:

(a) change the number of pads; (b) direct the load on or in between the pads; (c) change the axial length of pad.

The previous list contains some of the most common types of journal bearings. They are listed in the order of growing stability. All of the bearings designed for increased stability are obtained at higher manufacturing costs and reduced efficiency. The antiwhirl bearings all impose a parasitic load on the journal, which causes higher-power losses to the bearings and in turn requires higher oil flow to cool the bearing.

**Thrust Bearings** The most important function of a thrust bearing is to resist the unbalanced force in a machine's working fluid and to maintain the rotor in its position (within prescribed limits). A complete analysis of the thrust load must be conducted. As mentioned earlier, compressors with back-to-back rotors reduce this load greatly on thrust bearings. Figure 10-118 shows a number of thrust-bearing types. Plain, grooved thrust washers are rarely used with any continuous load, and their use tends to be confined to cases where the thrust load is very short duration or possibly occurs at standstill or low speed only. Occasionally, this type of bearing is used for light loads (less than 50 lb/in<sup>2</sup>), and in these circumstances the operation is probably hydrodynamic due to small distortions present in the nominally flat bearing surface.

When significant continuous loads have to be taken on a thrust washer, it is necessary to machine into the bearing surface a profile to generate a fluid film. This profile can be either a tapered wedge or occasionally a small step.

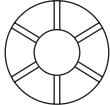
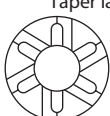
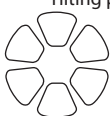
| Bearing type                                                                                                      | Load capacity | Suitable direction of rotation                                                    | Tolerance of changing load/speed | Tolerance of misalignment | Space requirement |
|-------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|----------------------------------|---------------------------|-------------------|
| Plain washer<br>                 | Poor          |  | Good                             | Moderate                  | Compact           |
| Taper land<br>Bidirectional<br>  | Moderate      |  | Poor                             | Poor                      | Compact           |
| Unidirectional                                                                                                    | Good          |  | Poor                             | Poor                      | Compact           |
| Tilting pad<br>Bidirectional<br> | Good          |  | Good                             | Good                      | Greater           |
| Unidirectional                                                                                                    | Good          |  | Good                             | Good                      | Greater           |

FIG. 10-118 Comparison of thrust-bearing types.

The tapered-land thrust bearing, when properly designed, can take and support a load equal to a tilting-pad thrust bearing. With perfect alignment, it can match the load of even a self-equalizing tilting-pad thrust bearing that pivots on the back of the pad along a radial line. For variable-speed operation, tilting-pad thrust bearings as shown in Fig. 10-119 are advantageous when compared to conventional taper-land bearings. The pads are free to pivot to form a proper angle for lubrication over a wide speed range. The self-leveling feature equalizes individual pad loadings and reduces the sensitivity to shaft misalignments that may occur during service. The major drawback of this bearing type is that standard designs require more axial space than a nonequalizing thrust bearing.

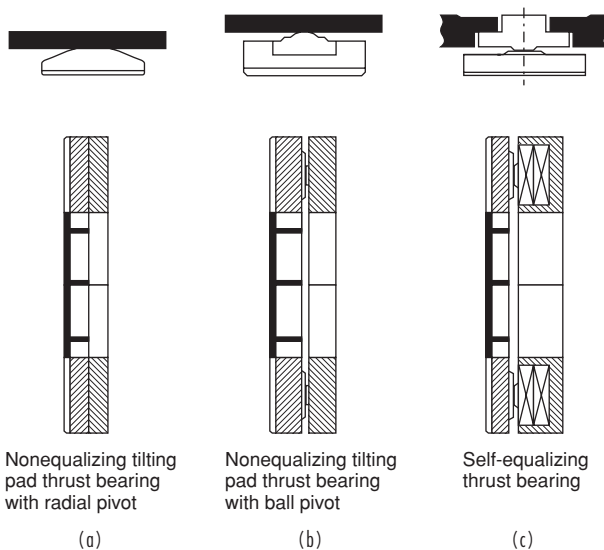


FIG. 10-119 Various types of thrust bearings.

The thrust-carrying capacity can be greatly improved by maintaining pad flatness and removing heat from the loaded zone. By the use of high thermal conductivity backing materials with proper thickness and proper support, the maximum continuous thrust limit can be increased to 1000 psi or more. This new limit can be used to increase either the factor of safety and improve the surge capacity of a given size bearing or reduce the thrust bearing size and consequently the losses generated for a given load.

Since the higher thermal conductivity material (copper or bronze) is a much better bearing material than the conventional steel backing, it is possible to reduce the babbitt thickness to .010–.030 in. Embedded thermocouples and RTDs will signal distress in the bearing if properly positioned. Temperature-monitoring systems have been found to be more accurate than axial-position indicators, which tend to have linearity problems at high temperatures.

In a change from steel-backing to copper-backing, a different set of temperature limiting criteria should be used. Figure 10-120 shows a typical set of curves for the two backing materials. This chart also shows that drain oil temperature is a poor indicator of bearing operating conditions because there is very little change in drain oil temperature from low load to failure load.

**Thrust-Bearing Power Loss** The power consumed by various thrust bearing types is an important consideration in any system. Power losses must be accurately predicted so that turbine efficiency can be computed and the oil supply system properly designed.

Figure 10-121 shows a typical power consumption in thrust bearings as a function of unit speed. The total power loss is usually about 0.8–10 percent of the total rate power of the unit. New vector lube bearings reduce the horsepower loss by as much as 30 percent. In large vertical pumps, thrust bearings take not only the load caused by the fluid but also the load caused by the weight of the entire assembly (shaft and impellers). In some large pumps these could be about 60 ft (20 m) high and weigh 16 tons. The thrust bearing for such a pump is over 5 ft (1.7 m) in diameter with each thrust pad weighing over 110 lb (50 kg). In such cases, the entire pump assembly is first floated before the unit is started.

## CENTRIFUGAL COMPRESSOR PROBLEMS

Compressors in process gas applications suffer from many problems. The following are some of the major categories in which these problems

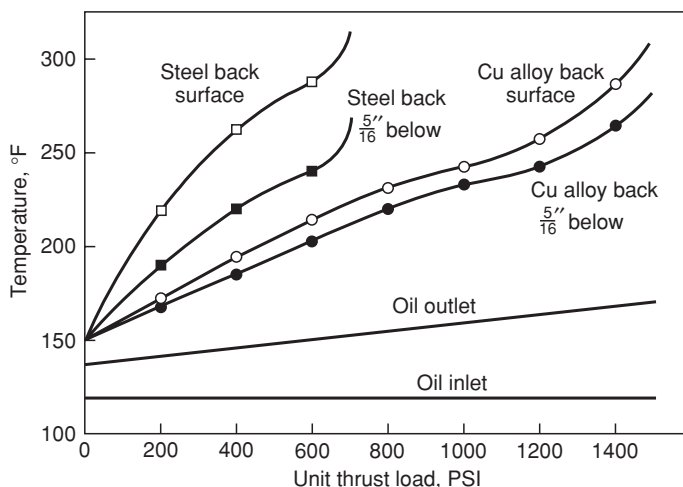


FIG. 10-120 Thrust-bearing temperature characteristics.

fall (see Meherwan P. Boyce, *Centrifugal Compressors: A Basic Guide*, PennWell, 2003):

1. Compressor fouling
2. Compressor failures
3. Impeller problems
4. Rotor thrust problems
5. Seals and bearings problems
6. Bearing maintenance

**Compressor Fouling** Centrifugal compressors, especially in the process gas applications, suffer greatly from fouling. Fouling is the deposit and nonuniform accumulation of debris in the gas on internal compressor surfaces. Fouling is due to the carryover of liquids and other debris from the suction knockout drums. Debris can roughen compressor surfaces. Polymerization can occur also due to changes in

process conditions. In wet gas compressors, ethylene plant cracked gas compressors, and polyethylene recycle compressors, the temperature of the gas must be kept below the threshold temperature that would initiate polymerization. The buildup will usually occur on the hub and the shroud with a larger buildup on the shroud at the elbow of the impeller on closed-face impellers. There is also a buildup on the blades, with the buildup usually more on the pressure side than on the suction side. Often the buildup is the heaviest on the pressure side at the blade exit where there is also flow separation.

#### Techniques to Prevent Fouling in Process Gas Compressors

1. *Condition monitoring of compressor aerodynamic and mechanical parameters.* Vibration monitoring could also alert the operator to fouling problems.

2. *Process control.* Accurate control of process conditions can prevent fouling in applications where polymers can form. Control of temperature is usually the most important. The following are examples of applications that can be affected by excessive process temperature:

- a. Ethylene cracked gas
- b. Linear low-density polyethylene
- c. High-density propylene
- d. Fluid catalytic cracker off-gas (wet gas)
- e. Thermal cracker off-gas (wet gas)
- f. Coker gas

The temperature below which fouling can be prevented varies with each process, compressor, and application. Monitoring of process conditions is necessary to establish a threshold temperature in each case. In some cases, fouling cannot be prevented with an existing compressor. It may be necessary to modify the aerodynamic design and/or add cooling.

3. *On-line solvent injection.* On-line solvent injection is very successful in various processes. The objective of this measure is to *continuously* inject a small amount of solvent to reduce the friction coefficient of the blade and impeller surface and thus prevent fouling of the surface. The injection should be done from the start; otherwise, the foulant could be dislodged and moved downstream, creating a major problem. The downstream areas are much smaller so foulant lodging there could create a blockage.

#### Air Compressors

1. *Inlet filter.* In air compressors filter selection is an important factor in preventing fouling. Most high-efficiency air filters have a triple-stage filtration system. Also these filters often have rain shades to prevent water from entering the filters. Site conditions play a very important part in the selection of the filters.

**Compressor Blade Coating** Coatings protect blades against oxidation, corrosion, and cracking problems. Coatings guard the base metal of the compressor from attack. Other benefits include reduced

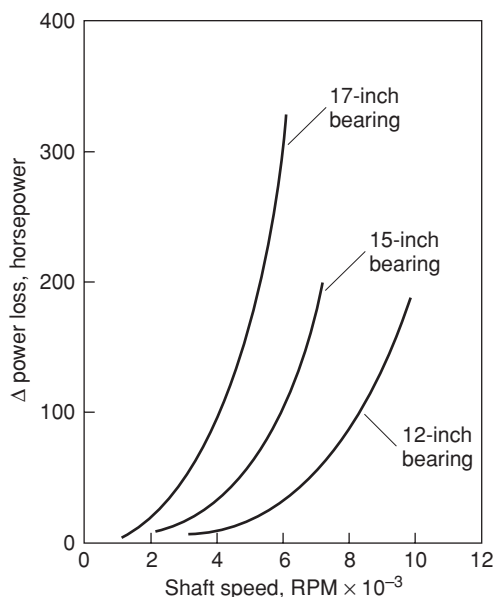


FIG. 10-121 Difference in total-power-loss data test minus catalog frictional losses versus shaft speed for 6 × 6 pad double-element thrust bearings.

thermal fatigue from cyclic operation, increased surface smoothness, reduced erosion, and reduced heat flux loading when one is considering thermal barriers. Coatings increase resistance to spalling from light impacts. Coatings also extend compressor life, better endure operational conditions, and serve as a sacrificial layer by allowing the coating to be restriped and recoated on the same base metal.

**Compressor Failures** In the process industry there are three types of compressors that have very different maintenance problems:

1. Barrel-type compressor
2. Horizontal split casing centrifugal compressor with closed-face impellers
3. Air integral gear-type compressor with open-face impellers

**Barrel-Type Compressors** Barrel-type compressors are being utilized in the process industry to an increased extent because the barrel design confines gases more effectively than horizontally split cases. This becomes a critical consideration in two areas: high-pressure and low-molecular-weight gas compression. API-617, *Centrifugal Compressors for General Refinery Services*, requires a barrel design based on the molecular percent of hydrogen contained in the process gas and the discharge pressure. API-617 defines high-pressure compressors as units in which the partial pressure of the hydrogen exceeds 200 psig, and specifies that these units be vertically (radially) split. Hydrogen partial pressure is given by the following relationship (in absolute pressures):

$$P_{H_2} = P(\%H_2/100) \quad (10-88)$$

Maximum casing working pressure for axially split compressors (psig) is

$$P_{\text{casing}} = \frac{200}{\%H_2/100} \quad (10-89)$$

The casings should include a minimum of a 1/8-in corrosion allowance. Casing strength and rigidity should limit change of alignment to 0.002 in if it is caused by the worst combination of pressure, torque, or allowable piping stress.

The barrel design is essentially a compressor placed inside a pressure vessel. For higher pressures some manufacturers have merely "beefed up" lower-pressure barrel designs, while others, have perfected unique designs such as the "shear ring" head design. All these designs make extensive use of elastomer O-rings as sealing devices. There are several inherent maintenance problems with barrel-type compressors:

**Handling.** Barrel-type machines must be removed from their foundations for total maintenance. Because barrel machines weigh up to 30 tons, the handling problems can be formidable.

**Inner casing alignment.** Since this type of compressor consists of a bundle contained within the pressure walls of the barrel, alignment and positive positioning are often very poor and the bundle is free to move to a certain extent. Bundle length is critical. Interstage leakage may occur if the bundle length is not correct. Assembly errors can be particularly detrimental in the case of a stacked diaphragm design, and care must be exercised to maintain proper impeller-diaphragm positioning. Since the bundle is subjected to discharge pressure on one end and suction pressure on the other, a force builds up that is transmitted from diaphragm to diaphragm, causing high loading on the inlet wall.

**Internal leakage.** The discharge and suction compartments of the inner bundle on a straight-through flow design are normally separated by a single O-ring. Compressors with side nozzles can have several bundles of O-rings. Excessive bundle-to-barrel clearance may cause leakage past the O-rings. O-rings are frequently pinched and cut across the suction nozzle opening in the barrel, a condition that is hard to prevent and doubly hard to detect if it occurs. Pressure differentials in excess of 400 to 500 psi even using good design practice can cause extrusion and failure of the O-rings. In many cases backup rings to the O-rings have been added to prevent failures. Grooves with O-ring ribbons have been added to the horizontal joints of the bundles of almost all the machines to prevent interstage leakage.

**Bearing bracket alignment.** In contrast to horizontally split compressors where the bearing brackets are normally an integral part of the lower case half, in barrel machines bearing brackets are bolted to

the barrel heads. Both the bearing brackets and the head are removed during the disassembly operation, thus requiring *all* internal alignment to be reestablished each time maintenance work is performed.

**Material problems.** To limit the physical size of the case or pressure vessel, to limit the rotor bearing span, and to maximize the number of stages within the heavy barrel, the gas path of a barrel is "squeezed" to a greater extent than in a horizontally split machine. This means the diaphragms and inlet guide vanes are intricate shapes with very small openings. Plain gray cast iron is normally used for these shapes because of casting ease and for other economic reasons. The gray iron is not strong enough in many instances to withstand the pressure differentials imposed on it, resulting in failures. Inlet guide vanes have been especially troublesome. On several occasions inlet guide vanes have been fabricated from wrought stainless and carbon steel materials. Replacement diaphragms and inlet guide vanes cast of nodular iron have also been used to alleviate some of these material problems.

**Impeller Problems** The high-speed rotation of the impeller of a centrifugal compressor imparts the vital aerodynamic velocity to the flow within the gas path. The buffeting effects of the gas flow can cause fatigue failures in the conventional fabricated shrouded impeller due to vibration-induced alternating stresses. These may be of the following types:

1. Resultant vibration in a principal mode
2. Forced-undamped vibration, associated with aerodynamic buffeting or high acoustic energy levels

The vibratory mode most frequently encountered is of the plate type and involves either the shroud or the disc. Fatigue failure generally originates at the impeller outside diameter, adjacent to a vane often due to the vibratory motion of the shroud or disc. The fatigue crack propagates inward along the nodal line, and finally a section of the shroud or disc tears out.

To eliminate failures of the covered impellers when operating at high density levels, the impellers are frequently scalloped between vanes at the outside diameter. The consequent reduction in disc friction also causes a small increase in impeller efficiency. However, there may be a slight reduction in overall efficiency due to higher losses in the diffuser. The advantages of scalloped impellers from a mechanical point of view are large. Several rotors have been salvaged by scalloping the wheels after a partial failure has occurred.

**Rotor Thrust Problems** Thrust loads in compressors due to aerodynamic forces are affected by impeller geometry, by pressure rise through the compressor, and by internal leakage due to labyrinth clearances. The impeller thrust is calculated by using correction factors to account for internal leakage, and a balance piston size is selected to compensate for the impeller thrust load. The common assumptions made in the calculation are

1. The radial pressure distribution along the outside of the disc cover is essentially balanced.
2. Only the "eye" area is effective in producing thrust.
3. The pressure differential applied to the "eye" area is equal to the difference between the static pressure at the impeller tip, corrected for the pumping action of the disc, and the total pressure at inlet.

These "common assumptions" are grossly erroneous and can be disastrous when applied to high-pressure, barrel-type compressors where a large part of the impeller-generated thrust is compensated by a balance piston. The actual thrust is about 50 percent more than the calculations indicate. The error is less when the thrust is compensated by opposed impellers, because the mistaken assumptions offset each other.

The magnitude of the thrust is considerably affected by leakage at the impeller labyrinth seals. Increased leakage here produces increased thrust independent of balancing piston labyrinth seal clearance or leakage.

The thrust errors are further compounded in the design of the balancing piston, labyrinths, and bleed line. API-617, *Centrifugal Compressors*, specifies that a separate pressure tap connection be provided to indicate the pressure in the balance chamber. It also specifies that the balance line be sized to handle balance piston labyrinth gas leakage at twice the initial clearance without exceeding the load ratings of the thrust bearing, and that thrust bearings for compressors be selected at no more than 50 percent of the bearing manufacturer's

rating. The leaks and the consequential pressure change across the balance piston destabilize the entire rotor system.

**Journal Bearing Failures** With high-speed machines, simple bearing failures are rare unless they are caused by faulty alignment, distortion, wrong clearance, or dirt. More common are failures caused by vibrations and rotor whirls. Some of these originate in the bearings; others can be amplified or attenuated by the bearings, the bearing cases, and the bearing support structure.

During inspection, all journal bearings should be closely inspected. If the machine has not suffered from excessive vibrations or lubrication problems, the bearings can be reinstalled and utilized.

Four places should be checked for wear during inspection periods:

1. Babbitted shoe surface
2. Pivoting shoe surface and seat in retaining ring
3. Seal ring bore or end plates
4. The shoe thickness at the pivot point or across ball and socket (all shoes should be within 0.0005 percent of the same thickness)

**Thrust Bearing Failures** Tilting pad-type thrust bearings are used in most major pieces of rotating equipment under the general term *Kingsbury type*. A thrust-bearing failure is one of the worst things that can happen to a machine, since it often wrecks the machine. To evaluate the reliability of a thrust bearing arrangement, one must first consider how a failure is initiated and evaluate the merits of the various designs.

Failures caused by bearing overload during normal operation (design error) are rare today, but still far more thrust failures occur than one would expect, considering all the precautions taken by the bearing designer. The causes in the following list are roughly in sequence of importance:

1. *Fluid slugging.* Passing a slug of fluid through a turbine or compressor can increase the thrust to many times its normal level.
2. *Buildup of solids in rotor and/or stator passages ("plugging" of turbine blades).* This problem should be noticed from performance or pressure distribution in the machine (first-stage pressure) long before the failure occurs.
3. *Off-design operation.* This arises especially from backpressure (vacuum), inlet pressure, extraction pressure, or moisture. Many failures are caused by overload, off-design speed, and flow fluctuations.
4. *Compressor surging.* This problem occurs especially in double-flow machines.
5. *Gear coupling thrust.* This is a frequent cause of failure, especially of upstream thrust bearings. The thrust is caused by friction in the loaded teeth that opposes thermal expansion. Therefore, thrust can get very high, since it has no relation to the normal thrust caused by pressure distribution inside the machine. The coupling thrust may act either way, adding to or subtracting from normal thrust. Much depends on tooth geometry and coupling quality.
6. *Dirt in oil.* This is a common cause of failures, especially when combined with other factors. The oil film at the end of the oil wedge is only a small fraction of a thousandth inch thick. If dirt goes through, it can cause the film to rupture, and the bearing may burn out. Therefore, very fine filtering of the oil is required.
7. *Momentary loss of oil pressure.* This type of failure is usually encountered while switching filters or coolers, or in some instances when the dc pump does not come on-line when the main pumps fail.

The thrust bearings must be closely maintained. This type of bearing consists of pivoted segments or pads (usually six) against which the thrust collar revolves, forming a wedge-shaped oil film. This film plus minute misalignment of the thrust collar and the bearing pads causes movement and wear of the various bearing parts. The erroneous thrust calculations discussed earlier cause the bearing to be loaded more heavily than desired. This accelerates the wear problem. There are seven wear points in the bearing. All these points must be checked for wear:

1. The soft babbitted shoe face
2. The hardened steel shoe insert face
3. The face of the hardened steel upper leveling plate
4. The outer edge of the upper leveling plate
5. The upper edge of lower leveling plate
6. The pivot point of the lower leveling plate
7. The inner face of the base ring

To protect thrust bearings, accurate and reliable instrumentation is now available to monitor thrust bearings well enough to ensure safe continuous operation and to prevent catastrophic failure in the event of an upset to the system.

Temperature sensors, such as RTDs (resistance temperature detectors), thermocouples, and thermistors, can be installed directly in the thrust bearing to measure metal temperature.

Axial proximity probes are another means of monitoring rotor position and the integrity of the thrust bearing. This method detects thrust collar runout and rotor movement. In most cases this ideal positioning of the probes is not possible. Many times the probes are indexed to the rotor or other convenient locations and thus do not truly show the movement of the rotor with respect to the thrust bearing.

A critical installation should have the metal temperature sensors in the thrust pad. Axial proximity probes may be used as a backup system. If metal temperatures are high and the rate of change of those temperatures begins to alter rapidly, thrust-bearing failure should be anticipated.

**Compressor Seal Problems** The extent of the leakage past the seals where the shaft comes through the casing frequently limits the running time of the compressor, yet the seals and the seal systems are not given adequate treatment in the maintenance manuals or in the operating instructions furnished by the compressor manufacturer.

Shaft seals are divided into the following categories by API Standard 617:

- Labyrinth
- Restrictive carbon rings
- Mechanical (contact) type
- Liquid film or floating bushing type
- Liquid film type with pump bushings

The first two seal categories are usually operated dry, and the last three categories require seal oil consoles either separately or as part of the lube system. Each of these seal designs has its own characteristics and maintenance difficulties.

Oil flows and critical clearances are not spelled out well in either the operating instructions or the maintenance manuals. Because of this, several maintenance technique improvements are needed:

1. *Radial clearances.* Radial clearance between the bushing and the shaft and the length of the bushing must be selected to obtain minimum leakage without exceeding fluid temperature limitations.
2. *Quality control.* The flatness, parallelism, and surface finish of the mating sleeve faces must be carefully controlled to obtain maximum seal effectiveness.
3. *Axial clearances.* Axial clearance between the bushing or sleeve and the housing is critical, there should be 12 to 15 mils of clearance per bushing between the bushing or sleeve and the housing; where the sleeves are mounted back-to-back, there will be 25 to 30 mils of clearance total for the seal.
4. *Seal design.* In higher-pressure seals, more than one outboard (i.e., high differential) sleeve may be used. Generally, it is desirable to use a single sleeve because the inboard sleeve operates with up to 80 percent of the total pressure drop across it. The outer sleeve with the lower differential causes lubrication and cooling problems that can shorten the life of one or both sleeves.
5. *Guidelines.* These should be explicit in indicating oil flow rates and the interaction of various components.
6. *Rules of thumb.* There are a few rules of thumb that help in understanding seal operation and maintenance.
  - a. The oil flow rate will vary: (1) *directly* with the differential pressure and the wetted perimeter of the sleeve; (2) with the *cube* of the radial clearance; (3) with the *square* of the eccentricity of the sleeve and shaft; (4) *inversely* with oil viscosity, temperature, and length of the sleeve.
  - b. Shear work done on the sealing fluid during its passage through the sleeve raises its temperature to a much higher level than may be expected.

## ROTOR DYNAMICS

The rotating elements consist of the impeller and the shaft. The shaft should be made of one-piece, heat-treated forged steel, with the shaft ends tapered for coupling fits. Interstage sleeves should be renewable and made of material which is corrosion-resistant in the



specified service. The rotor shaft sensing area observed by the non-contact probes should be concentric with the bearing journals and free of any scratches, marks, or other surface discontinuity. The surface finish should be 16 to 32  $\mu\text{in}$  root mean square, and the area should be demagnetized and treated. Electromechanical runout should not exceed 25 percent of the maximum allowed peak-to-peak vibration amplitude or 0.25 mil, whichever is greater. Although not mentioned in the standard, chrome plating of the shaft in the sensing area is unacceptable. Maximum vibration should not exceed 2.0 mils as given by

$$\text{Vib}_{\max} = \sqrt{\frac{12,000}{\text{rpm}}} + 0.25 \sqrt{\frac{12,000}{\text{rpm}}} \quad (10-90)$$

(Vibration) (runout)

At the trip speed of the driver (105 percent for a gas turbine), the vibration should not exceed this level by more than 0.5 mil.

The impellers can be an open-faced (stationary shroud) or closed-face (rotating shroud) design. As long as the tip velocities are below 1000 ft/s, closed-face impellers can be used. The standards allow the impellers to be welded, riveted, milled, or cast. Riveted impellers are unacceptable, especially if the impeller loading is high. Impellers are to be assembled on the shaft with a shrink fit with or without key. Shrink fits should be carefully done because excessive shrink fits can cause a problem known as hysteresis whirl. In compressors where the impellers require their thrust to be balanced, a balance drum is acceptable and preferred.

The high-speed pumps or compressors must operate in a region away from any critical speed. The amplification factor used to indicate the severity of the critical speed is given by the relationship

$$\text{AF} = \frac{\text{critical speed}}{\text{peak width of "half-power" point}} \quad (10-91)$$

$$\text{AF} = \frac{N_{C1}}{N_2 - N_1} \quad (10-92)$$

where  $N_2 - N_1$  is the rpm corresponding to the 0.707 peak critical amplitude.

The amplification factor should be below 8 and preferably above 5. A rotor response plot is shown in Fig. 10-122. The operational speed for units operating below their first critical speed should be at least 20 percent below the critical speed. For units operating above their first critical speed, the operational speed must be at least 15 percent above the critical speed and/or 20 percent below any critical speed. The preferred bearings for the various types of installation are tilting-shoe radial bearings and the self-equalizing tilting pad thrust bearings. Radial and thrust bearings should be equipped with embedded temperature sensors to detect pad surface temperatures.

## VIBRATION MONITORING

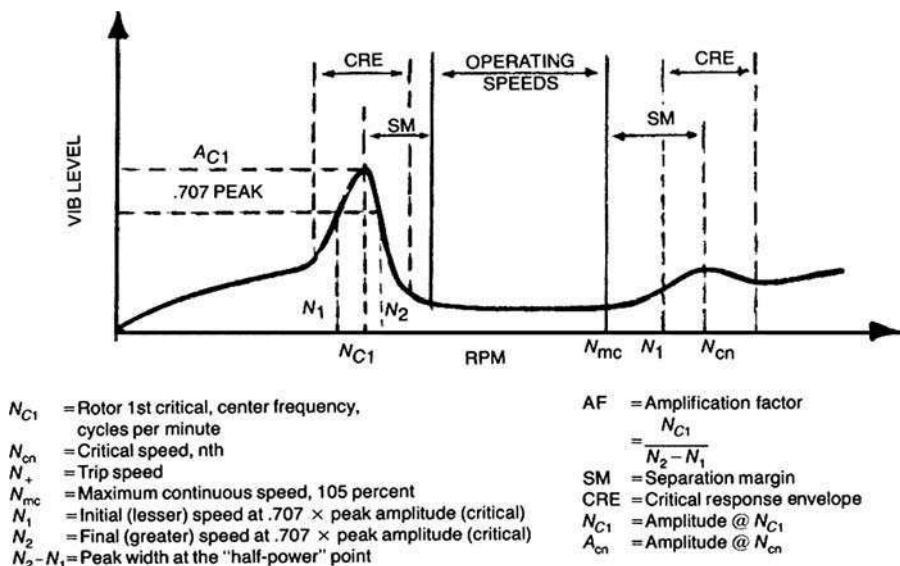
One of the major factors that causes pump failure is vibration, which usually causes seal damage and oil leakage. Vibration in pumps is caused by numerous factors such as cavitation, impeller unbalance, loose bearings, and pipe pulsations.

As the mechanical integrity of the pump system changes, the amplitude of vibration levels change. In some cases, in order to identify the source of vibration, pump speed may have to be varied, as these problems are frequency- or resonance-dependent. Pump impeller imbalance and cavitation are related to this category.

It is advisable in most of these cases to use accelerometers. Displacement probes will not give the high-frequency signals and velocity probes because their mechanical design is very directional and prone to deterioration. Figure 10-123 shows the signal from the various types of probes.

Typically, large-amplitude vibration occurs when the frequency of vibration coincides with that of the natural frequency of the pump system. This results in a catastrophic operating condition that should be avoided. If the natural frequency is close to the upper end of the operating speed range, then the pump system should be stiffened to reduce vibration. On the other hand, if the natural frequency is close to the lower end of the operating range, the unit should be made more flexible. During startup, the pump system may go through its system natural frequency, and vibration can occur. Continuous operation at this operating point should be avoided.

ASME recommends periodic monitoring of all pumps. Pump vibration level should fall within the prescribed limits. The reference vibration level is measured during acceptance testing. This level is specified by the manufacturer.



**FIG. 10-122** Rotor response plot. This plot is Figure 7 in API Standard 617, *Centrifugal Compressors for General Refinery Services*, 4th ed., 1979. (Courtesy American Petroleum Institute.)



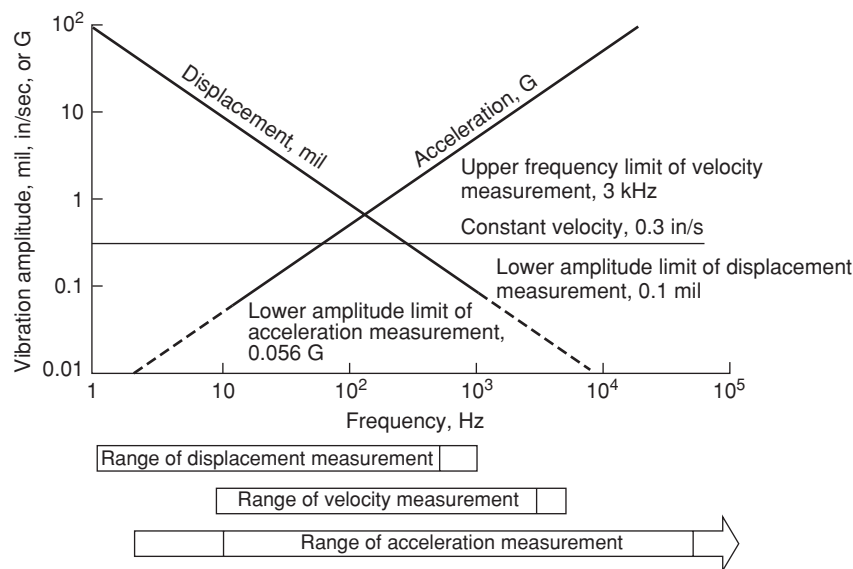


FIG. 10-123 Limitations on machinery vibrations analysis systems and transducers.

TABLE 10-17 Alert Levels

| Reference value mils. | Alert mils., $\mu\text{m}$ | Action required mils., $\mu\text{m}$ |
|-----------------------|----------------------------|--------------------------------------|
| $V_r < 0.5$           | 1.0                        | 1.5                                  |
| $0.5 < V_r < 2.0$     | $2V_r$                     | $3V_r$                               |
| $2.0 < V_r < 5.0$     | $2+V_r$                    | $4+V_r$                              |
| $5.0 < V_r$           | $1.4V_r$                   | $1.8V_r$                             |

During periodic maintenance, the vibration level should not exceed alert level (see Table 10-17). If the measured level exceeds the alert level then preventive maintenance should be performed, by diagnosing the cause of vibration and reducing the vibration level prior to continued operation.

Typical problems and their vibration frequency ranges are shown in Fig. 10-124.

Collection and analysis of vibration signatures is a complex procedure. By looking at a vibration spectrum, one can identify which components of the pump system are responsible for a particular frequency component. Comparison of vibration signatures at periodic intervals reveals if a particular component is deteriorating. The following example illustrates evaluation of the frequency composition of an electric motor gear pump system.

**Example 2: Vibration** Consider an electric motor rotating at 1800 rpm driving an 8-vane centrifugal pump rotating at 600 rpm. For this 3:1 speed reduction, assume a gear box having two gears of 100 and 300 teeth. Since 60 Hz is 1 rpm, Motor frequency =  $1800/60 = 30$  Hz  
Pump frequency =  $600/60 = 10$  Hz  
Gear mesh frequency =  $300 \text{ teeth} \times 600 \text{ rpm} = 3000$  Hz  
Vane frequency =  $8 \times 600 \text{ rpm} = 80$  Hz  
An ideal vibration spectra for this motor-gear pump assembly would appear as shown in Fig. 10-125.

Figure 10-126 shows an actual pump vibration spectra. In the figure, several amplitude peaks occur at several frequencies.

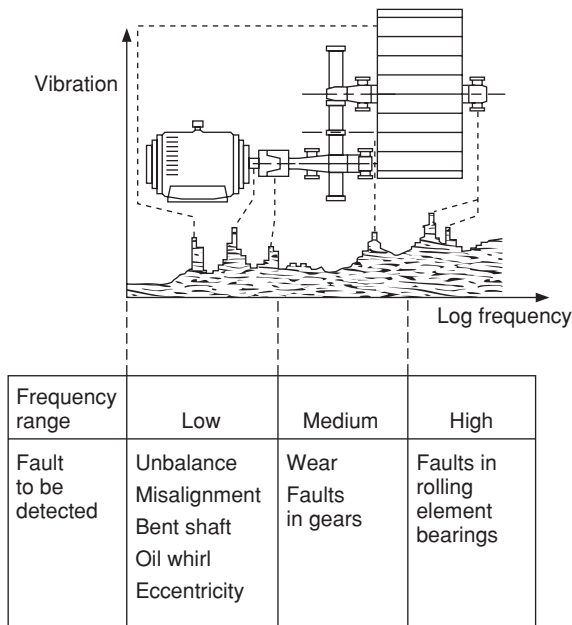


FIG. 10-124 Frequency range of typical machinery faults.

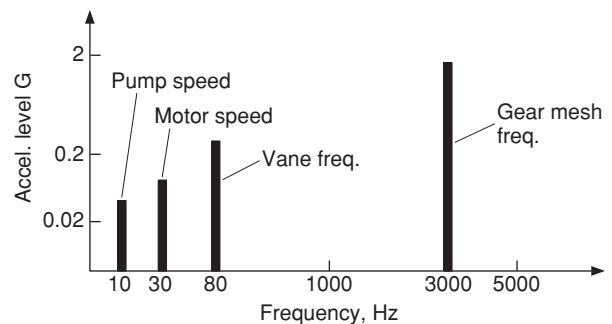


FIG. 10-125 An ideal vibration spectra from an electric motor pump assembly.

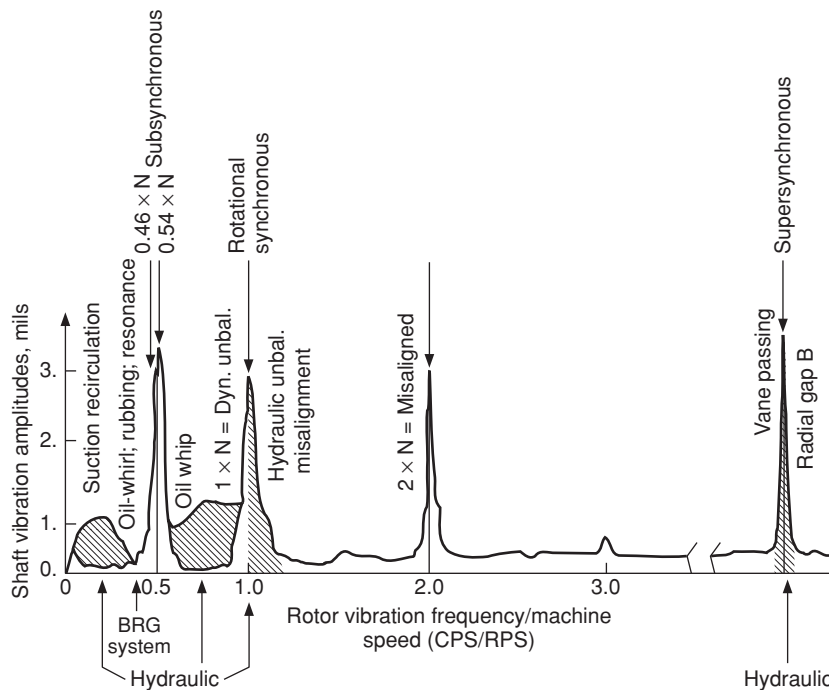


FIG. 10-126 An actual pump vibration spectrum.

## PROCESS PLANT PIPING

### INTRODUCTION

This section provides general comments that are pertinent to the design of process plant piping. It is intended to provide a convenient summary of commonly used information from various sources. It is not intended to serve as a comprehensive source of requirements or as a substitute for referenced codes, standards, and specifications. It is intended that qualified designers obtain copies of all applicable codes, standards, and specifications and thoroughly review all pertinent requirements of these documents prior to execution of work.

### CODES AND STANDARDS

**Units: Pipe and Tubing Sizes and Ratings** In this subsection pipe and tubing sizes are generally quoted in units of inches. To convert inches to millimeters, multiply by 25.4. Ratings are given in pounds. To convert pounds to kilograms, multiply by 0.454.

**Pressure-Piping Codes** The code for pressure piping (ASME B31) consists of a number of sections which collectively constitute the code. Table 10-18 shows the status of the B31 code as of July 2005. The sections are published as separate documents for simplicity and convenience. The sections differ extensively.

The Process Piping Code (ASME B31.3) is a subsection of the ASME code for Pressure Piping B31. It was derived from a merging of the code groups for chemical-plant (B31.6) and petroleum-refinery (B31.3) piping into a single committee. Some of the significant requirements of ASME B31.3, Process Piping (2004 edition) are summarized in the following presentation.

Where the word *code* is used in this subsection of the *Handbook* without other identification, it refers to the B31.3 section of ASME B31. The

code has been extensively quoted in this subsection of the *Handbook* with the permission of the publisher. The code is published by and copies are available from the American Society of Mechanical Engineers (ASME), Three Park Avenue, New York, New York 10016-5990.

**National Standards** The American Society of Mechanical Engineers (ASME) and the American Petroleum Institute (API) have established dimensional standards for the most widely used piping components. Lists of these standards as well as specifications for pipe and fitting materials and testing methods of the American Society for Testing and Materials (ASTM), American Welding Society (AWS) specifications, and standards of the Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) can be found in the ASME B31 code sections. Many of these standards contain pressure-temperature ratings which will be of assistance to engineers in their design function. The use of published standards does not eliminate the need for engineering judgment. For example, although the code calculation formulas recognize the need to provide an allowance for corrosion, the standard rating tables for valves, flanges, fittings, etc., do not incorporate a corresponding allowance. Judgments regarding the suitability of these components are left to the designer.

The introduction to the code sets forth engineering requirements deemed necessary for the safe design and construction of piping systems. While safety is the basic consideration of the code, this factor alone will not necessarily govern final specifications for any pressure piping system.

Designers are cautioned that the code is not a design handbook and does not do away with the need for competent engineering judgment.

**Government Regulations: OSHA** Sections of the ASME B31 code have been adopted with certain reservations or revisions by some state and local authorities as local codes.

**TABLE 10-18 Status of ASME B31 Code for Pressure Piping**

| ASME B31.3 section number, latest issue, and title                                                                                       | General scope and application                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31.1-2004, <i>Power Piping</i>                                                                                                          | Addresses piping typically found in electric power generating stations, industrial and institutional plants, geothermal heating systems, and central and district heating and cooling systems.                                 |
| B31.2-1968, <i>Fuel Gas Piping</i>                                                                                                       | Addresses fuel gas piping systems in and between buildings. The scope includes piping between the outlet of the consumer's meter and the outlet of the first pressure-containing valve upstream of the gas utilization device. |
| B31.3-2004, <i>Process Piping</i>                                                                                                        | Addresses piping typically found in petroleum refineries; chemical, pharmaceutical, textile, paper, semiconductor, and cryogenic plants; and related processing plants and terminals.                                          |
| B31.4-2002, <i>Pipeline Transportation Systems for Liquid Hydrocarbons and Other Liquids</i>                                             | Addresses piping transporting products which are predominately liquid between plants and terminals. Included are terminals, tank farms, and pumping, regulating, and metering stations.                                        |
| B31.5-2001, <i>Refrigeration Piping and Heat Transfer Components</i>                                                                     | Addresses refrigeration piping in packaged units and in commercial and public buildings.                                                                                                                                       |
| B31.7, <i>Nuclear Power Piping</i>                                                                                                       | Withdrawn; see <i>ASME Boiler and Pressure Vessel Code</i> , Sec. 3.                                                                                                                                                           |
| B31.8-2003, <i>Gas Transmission and Distribution Piping Systems</i>                                                                      | Addresses piping transporting products which are predominately gas between sources and terminals. Included are gas-gathering pipelines and compressor, regulating, and metering stations.                                      |
| B31.8S-2004, <i>Managing System Integrity of Gas Pipelines</i>                                                                           | Addresses development and implementation of integrity management systems for gas pipelines.                                                                                                                                    |
| B31.9-1996, <i>Building Services Piping</i>                                                                                              | Addresses piping outside the scope of ASME B31.1 that is typically found in multiunit residences and in industrial, institutional, commercial, and public buildings.                                                           |
| B31.11-2002, <i>Slurry Transportation Piping Systems</i>                                                                                 | Addresses piping transporting aqueous slurries between plants and terminals and within terminals, pumping, and regulating stations.                                                                                            |
| B31G-1991, <i>Manual for Determining the Remaining Strength of Corroded Pipelines: A Supplement to ASME B31 Code for Pressure Piping</i> | Addresses methods of determining the remaining strength of corroded pipelines that are within the scope of the <i>ASME B31 Code for Pressure Piping</i> .                                                                      |

Addenda are issued at intervals between publication of complete editions. Information on the latest issues can be obtained from the American Society of Mechanical Engineers, Three Park Avenue, New York, NY 10016-5990; or on the World Wide Web at ASME.org.

The specific requirements for piping systems in certain services have been promulgated as Occupational Safety and Health Administration (OSHA) regulations. These rules and regulations will presumably be revised and supplemented from time to time and may include specific requirements not addressed by the B31 sections.

**International Regulations** ASME piping codes have been widely used throughout the world for the design of facilities falling within their defined scopes. Although the use of ASME codes is widely acceptable in areas outside the United States, it is essential to identify additional local or national codes or standards that may apply. Such documents may require qualified third-party review and approval of project specifications, facility design, fabrication, material documentation, inspection, and testing. For example, within the European Community, such requirements are imposed by the Pressure Equipment Directive 97/23/EC (also known as the PED). These requirements must be recognized early in the project to avoid costly error.

## CODE CONTENTS AND SCOPE

The code prescribes minimum requirements for materials, design, fabrication, assembly, support, erection, examination, inspection, and testing of piping systems subject to pressure or vacuum. The scope of the piping covered by B31.3 is illustrated in Fig. 10-127. It applies to all fluids including fluidized solids and to all services except as noted in the figure.

The code also excludes piping systems designed for internal gauge pressures at or above zero but less than 0.105 MPa (15 lbf/in<sup>2</sup>) provided the fluid handled is nonflammable, nontoxic, and not damaging to human tissues, and its design temperature is from -29°C (-20°F) through 186°C (366°F). Refer to the code for definitions of nonflammable and nontoxic.

Some of the more significant requirements of ASME B31.3 (2004 edition) have been summarized and incorporated in this section of the *Handbook*. For a more comprehensive treatment of code require-

ments engineers are referred to the B31.3 code and the standards referenced therein.

## SELECTION OF PIPE SYSTEM MATERIALS

The selection of material to resist deterioration in service is outside the scope of the B31.3 code (see Sec. 25). Experience has, however, resulted in the following material considerations extracted from the code with the permission of the publisher, the American Society of Mechanical Engineers, New York.

**General Considerations\*** Following are some general considerations which should be evaluated when selecting and applying materials in piping:

1. The possibility of exposure of the piping to fire and the melting point, degradation temperature, loss of strength at elevated temperature, and combustibility of the piping material under such exposure.
2. The susceptibility to brittle failure or failure from thermal shock of the piping material when exposed to fire or to fire-fighting measures, and possible hazards from fragmentation of the material in the event of failure.
3. The ability of thermal insulation to protect piping against failure under fire exposure (e.g., its stability, fire resistance, and ability to remain in place during a fire).
4. The susceptibility of the piping material to crevice corrosion under backing rings, in threaded joints, in socket-welded joints, and in other stagnant, confined areas.
5. The possibility of adverse electrolytic effects if the metal is subject to contact with a dissimilar metal.
6. The compatibility of lubricants or sealants used on threads with the fluid service.

\*Extracted from ASME B31.3-2004, Section F323, with permission of the publisher, the American Society of Mechanical Engineers, New York.

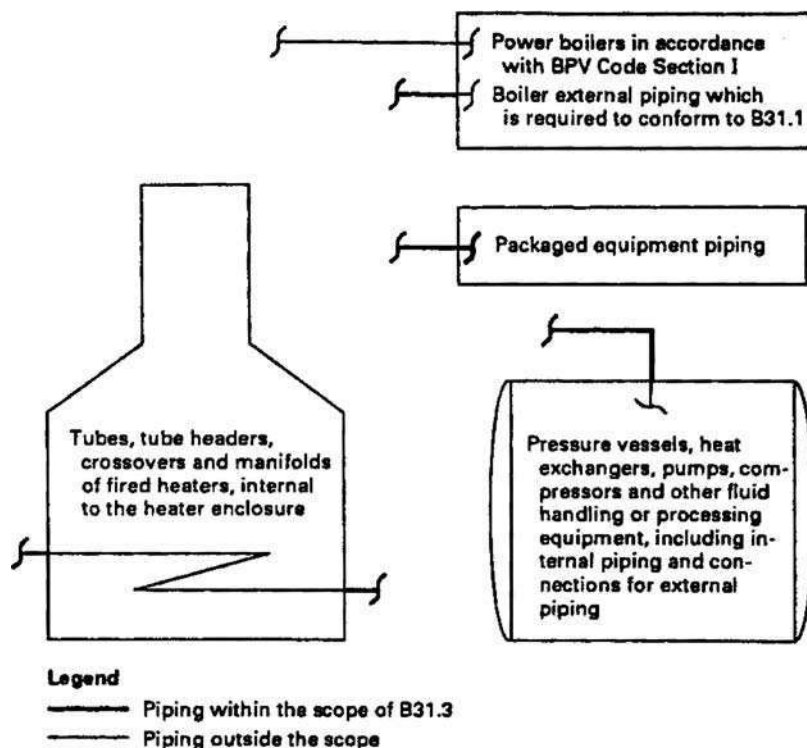


FIG. 10-127 Scope of work covered by process piping code ASME B31.3-2004.

7. The compatibility of packing, seals, and O-rings with the fluid service.

8. The compatibility of materials, such as cements, solvents, solders, and brazing materials, with the fluid service.

9. The chilling effect of sudden loss of pressure on highly volatile fluids as a factor in determining the lowest expected service temperature.

10. The possibility of pipe support failure resulting from exposure to low temperatures (which may embrittle the supports) or high temperatures (which may weaken them).

11. The compatibility of materials, including sealants, gaskets, lubricants, and insulation, used in strong oxidizer fluid service (e.g., oxygen or fluorine).

**Specific Material Considerations—Metals\*** Following are some specific considerations which should be evaluated when applying certain metals in piping.

1. *Irons—cast, malleable, and high silicon (14.5%).* Their lack of ductility and their sensitivity to thermal and mechanical shock.

2. *Carbon steel, and low and intermediate alloy steels.*

a. The possibility of embrittlement when handling alkaline or strong caustic fluids.

b. The possible conversion of carbides to graphite during long time exposure to temperatures above 427°C (800°F) of carbon steels, plain

nickel steel, carbon-manganese steel, manganese-vanadium steel, and carbon-silicon steel.

c. The possible conversion of carbides to graphite during long time exposure to temperatures above 468°C (875°F) of carbon-molybdenum steel, manganese-molybdenum-vanadium steel, and chromium-vanadium steel.

d. The advantages of silicon-killed carbon steel (0.1% silicon minimum) for temperatures above 482°C (900°F).

e. The possibility of damage due to hydrogen exposure at elevated temperature (see API RP941); hydrogen damage (blistering) may occur at lower temperatures under exposure to aqueous acid solutions.<sup>†</sup>

f. The possibility of stress corrosion cracking when exposed to cyanides, acids, acid salts, or wet hydrogen sulfide; a maximum hardness limit is usually specified (see NACE MR0175 and RP0472).<sup>†</sup>

g. The possibility of sulfidation in the presence of hydrogen sulfide at elevated temperatures.

3. *High-alloy (stainless) steels.*

a. The possibility of stress corrosion cracking of austenitic stainless steels exposed to media such as chlorides and other halides either internally or externally; the latter can result from improper selection or application of thermal insulation, or from use of marking inks, paints, labels, tapes, adhesives, and other accessory materials containing chlorides or other halides.

b. The susceptibility to intergranular corrosion of austenitic stainless steels sensitized by exposure to temperatures between 427 and 871°C (800 and 1600°F); as an example, stress corrosion cracking of sensitized metal at room temperature by polythionic acid (reaction of oxidizable sulfur compound, water, and air); stabilized or low-carbon grades may provide improved resistance (see NACE RP0170).<sup>†</sup>

c. The susceptibility to intercrystalline attack of austenitic stainless steels on contact with liquid metals (including aluminum, antimony, bismuth, cadmium, gallium, lead, magnesium, tin, and zinc) or their compounds.

\*Extracted from ASME B31.3-2004, Section F323, with permission of the publisher, the American Society of Mechanical Engineers, New York.

<sup>†</sup>Titles of referenced documents are:

API RP941, *Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants*

NACE MR0175, *Sulfide Stress-Cracking Resistant Metallic Materials for Oil Field Equipment*

NACE RP0472, *Methods and Controls to Prevent In-Service Cracking of Carbon Steel (P-1) Welds in Corrosive Petroleum Refining Environments*

NACE RP0170, *Protection of Austenitic Stainless Steel in Refineries Against Stress Corrosion Cracking by Use of Neutralizing Solutions During Shutdown*

d. The brittleness of ferritic stainless steels at room temperature after service at temperature above 371°C (700°F).

4. *Nickel and nickel-base alloys.*

a. The susceptibility to grain boundary attack of nickel and nickel-base alloys not containing chromium when exposed to small quantities of sulfur at temperatures above 316°C (600°F).

b. The susceptibility to grain boundary attack of nickel-base alloys containing chromium at temperatures above 593°C (1100°F) under reducing conditions and above 760°C (1400°F) under oxidizing conditions.

c. The possibility of stress corrosion cracking of nickel-copper Alloy 400 in hydrofluoric acid vapor in the presence of air, if the alloy is highly stressed (including residual stresses from forming or welding).

5. *Aluminum and aluminum alloys.*

a. The compatibility with aluminum of thread compounds used in aluminum threaded joints to prevent seizing and galling.

b. The possibility of corrosion from concrete, mortar, lime, plaster, or other alkaline materials used in buildings or structures.

c. The susceptibility of Alloy nos. 5083, 5086, 5154, and 5456 to exfoliation or intergranular attack; and the upper temperature limit of 66°C (150°F) shown in Appendix A to avoid such deterioration.

6. *Copper and copper alloys.*

a. The possibility of dezincification of brass alloys.

b. The susceptibility to stress corrosion cracking of copper-based alloys exposed to fluids such as ammonia or ammonium compounds.

c. The possibility of unstable acetylide formation when exposed to acetylene.

7. *Titanium and titanium alloys.* The possibility of deterioration of titanium and its alloys above 316°C (600°F).

8. *Zirconium and zirconium alloys.* The possibility of deterioration of zirconium and zirconium alloys above 316°C (600°F).

9. *Tantalum.* Above 299°C (570°F), the possibility of reactivity of tantalum with all gases except the inert gases. Below 299°C, the possibility of embrittlement of tantalum by nascent (monatomic) hydrogen (but not molecular hydrogen). Nascent hydrogen is produced by galvanic action, or as a product of corrosion by certain chemicals.

10. *Metals with enhanced properties.* The possible loss of strength, in a material whose properties have been enhanced by heat treatment, during long-continued exposure to temperatures above the tempering temperature.

11. The desirability of specifying some degree of production impact testing, in addition to the weld procedure qualification tests, when using materials with limited low-temperature service experience below the minimum temperature stated in ASME B31.3 Table A-1.

**Specific Material Considerations—Nonmetals** Following are some considerations to be evaluated when applying nonmetals in piping. Refer to Tables 10-19, 10-20, and 10-21 for typical temperature limits.

1. *Static charges.* Because of the possibility of producing hazardous electrostatic charges in nonmetallic piping and metallic piping lined with nonmetals, consideration should be given to grounding the metallic components of such systems conveying nonconductive fluids.

2. *Thermoplastics.* If thermoplastic piping is used aboveground for compressed air or other compressed gases, special precautions should be observed. In determining the needed safeguarding for such services, the energetics and the specific failure mechanism need to be evaluated. Encasement of the plastic piping in shatter-resistant material may be considered.

3. *Borosilicate glass.* Take into account its lack of ductility and its sensitivity to thermal and mechanical shock.

## METALLIC PIPING SYSTEM COMPONENTS

Metallic pipe systems comprise the majority of applications. Metallic pipe, tubing, and pipe fittings are divided into two main categories: seamless and welded. Both have advantages and disadvantages in terms of economy and function. Specifications governing the production of these products dictate the permissible mechanical and dimensional variations, and code design calculations account for these variations.

**Seamless Pipe and Tubing** Seamless pipe and tubing may be formed by various methods. A common technique involves piercing solid round forgings, followed by rolling and drawing. Other tech-

**TABLE 10-19 Recommended Temperature Limits for Thermoplastic Pipe\***

| ASTM spec. no.                                           | Material | Recommended temperature limits †,‡ |     |         |     |
|----------------------------------------------------------|----------|------------------------------------|-----|---------|-----|
|                                                          |          | Minimum                            |     | Maximum |     |
|                                                          |          | °C                                 | °F  | °C      | °F  |
| ...                                                      | ABS55535 | -40                                | -40 | 80      | 176 |
| ...                                                      | AP       | -18                                | 0   | 77      | 170 |
| D 2846<br>F 441<br>F 442                                 | CPVC4120 | -18                                | 0   | 99      | 210 |
| ...                                                      | ECTFE    | -40                                | -40 | 149     | 300 |
| ...                                                      | ETFE     | -40                                | -40 | 149     | 300 |
| D 2513<br>D 2662<br>D 2666<br>D 3000<br>D 3309           | PB2110   | -18                                | 0   | 99      | 210 |
| D 2104<br>D 2239<br>D 2447<br>D 2513<br>D 2737<br>D 3035 | PE3408   | -34                                | -30 | 82      | 180 |
| ...                                                      | PEEK     | -40                                | -40 | 250     | 450 |
| ...                                                      | PFA      | -40                                | -40 | 250     | 450 |
| ...                                                      | POP2125  | -1                                 | 30  | 99      | 210 |
| ...                                                      | PP       | -1                                 | 30  | 99      | 210 |
| D 1785                                                   | PVC1120  | -18                                | 0   | 66      | 150 |
| D 2241                                                   | PVC1220  | -18                                | 0   | 66      | 150 |
| D 2513                                                   | PVC2110  | -18                                | 0   | 54      | 130 |
| D 2672                                                   | PVC2120  | -18                                | 0   | 66      | 150 |
| ...                                                      | PVDC     | 4                                  | 40  | 71      | 160 |
| ...                                                      | PVDF     | -18                                | 0   | 135     | 275 |

\*Extracted from ASME B31.3-2004, Table B-1, with permission of the publisher, the American Society of Mechanical Engineers, New York.

†These recommended limits are for low-pressure applications with water and other fluids that do not significantly affect the properties of the thermoplastic. The upper temperature limits are reduced at higher pressures, depending on the combination of fluid and expected service life. Lower temperature limits are affected more by the environment, safeguarding, and installation conditions than by strength.

‡These recommended limits apply only to materials listed. Manufacturers should be consulted for temperature limits on specific types and kinds of materials not listed.

niques include forging and boring, extrusion, and static and centrifugal casting. Piercing frequently produces pipe with a less uniform wall thickness and concentricity of bore than is the case with products produced by other methods. Since seamless products have no weld joints, there is no reduction of strength due to weld joint efficiency.

**Welded Pipe and Tubing** These products are typically made by forming strips or plate into cylinders and seam-welding by various methods. Manufacturing by welding permits the production of larger diameter pipe than is possible with seamless manufacturing methods, as well as larger diameter/wall thickness ratios. While strip and plate thickness may be more closely controlled than is possible for some seamless products, the specifications governing production are not always more stringent for welded products.

Weld quality has the potential of making the weld weaker than the base material. Depending on the welding method and the degree of nondestructive examination required by the product specification or dictated by the designer, the code assigns a joint efficiency ranging from 60 to 100 percent of the strength of the base material. Although

**TABLE 10-20 Recommended Temperature Limits for Thermoplastics Used as Linings\***

| Materials | Minimum |      | Maximum |     |
|-----------|---------|------|---------|-----|
|           | °C      | °F   | °C      | °F  |
| PFA       | -198    | -325 | 260     | 500 |
| PTFE      | -198    | -325 | 260     | 500 |
| FEP       | -198    | -325 | 204     | 400 |
| ECTFE     | -198    | -325 | 171     | 340 |
| ETFE      | -198    | -325 | 149     | 300 |
| PVDF      | -18     | 0    | 135     | 275 |
| PP        | -18     | 0    | 107     | 225 |
| PVDC      | -18     | 0    | 79      | 175 |

\*Extracted from ASME B31.3-2004, Table A323.4.3, with permission of the publisher, the American Society of Mechanical Engineers, New York.

NOTE: These temperature limits are based on material tests and do not necessarily reflect evidence of successful use as piping component linings in specific fluid services at these temperatures. The designer should consult the manufacturer for specific applications, particularly as temperature limits are approached.

Abbreviations for plastics: ABS, acrylonitrile-butadiene-styrene; CPVC, chlorinated poly vinyl chloride; ECTFE, ethylene-chlorotrifluoroethylene; ETFE, ethylene-tetrafluoroethylene; PB, polybutylene; PE, polyethylene; PEEK, poly ether ether ketone; PFA, perfluoroalkoxy copolymer; POE, poly phenylene oxide; PP, polypropylene; PVC, poly vinyl chloride; PVDC, poly vinylidene chloride; PVDF, poly vinylidene fluoride.

some welding methods have the potential of producing short sections of partially fused joints that may develop into small leaks in corrosive conditions, proper matching of the weld method and the type and extent of examination will result in highly reliable joints that are suitable for use in critical services. Welds must be considered when developing specifications for bending, flaring or expanding welded pipe or tubing.

**Tubing** Tubing sizes typically reflect the actual outside diameter of the product. Pipe is manufactured to nominal diameters, which are not the same as the actual outside diameters for sizes 12 in and smaller. Facilities within the scope of the ASME B31 codes nearly exclusively use pipe, rather than tubing, for applications external to equipment. Tubing is commonly classified as suitable for either mechanical or pressure applications. Tubing is available in size and wall thickness combinations not normally produced as pipe. Tubing wall thickness (gauge) is specified as either average wall or minimum wall. Minimum wall is more costly than average wall, and because of closer tolerances on thickness and diameter, tubing of either gauge system is generally more costly than pipe. Tubing having outside diameters of 2 $\frac{3}{8}$ , 2 $\frac{1}{2}$ , 3 $\frac{1}{2}$ , and 4 $\frac{1}{2}$  in are commonly available; however, these sizes are generally considered to be nonstandard for typical piping applications.

Table 10-22 gives some of the more common standard size and wall-thickness combinations together with capacity and weight.

**Methods of Joining Pipe** Piping joints must be reliably leak-tight and provide adequate mechanical strength to resist external loads due to thermal expansion, weight, wind, seismic activity, and other factors. Joints for pipe buried in soil may be subjected to unique external loads resulting from thermal expansion and contraction, settlement, and other factors. Joint designs that permit rotation about an axis perpendicular to the longitudinal axis of the pipe may be advantageous in certain situations.

Disassembly frequency and ease should be considered when selecting joining methods. Ideally the method for joining piping system components provides minimum installed cost, maintains its integrity throughout the lifetime of the facility, provides restraint against axial thrust due to internal pressure, provides strength against external loads equal to that of the pipe, permits unrestricted flow with minimum pressure drop, and is free from crevices that may be detrimental to the product or contribute to corrosion or erosion problems.

Joint design and selection generally involves compromising between the ideal and practical. A number of manufacturers produce patented or "proprietary" joints that embody many ideal characteristics. Some are excellent products and are well suited to special applications. Valves and fittings are often available with proprietary joints that have gained wide acceptance; however, consideration should be given to the possible impact on product delivery time and cost.

**Welded Joints** The most widely used joint in piping systems is the **butt-weld joint** (Fig. 10-128). In all ductile pipe metals which can be welded, pipe, elbows, tees, laterals, reducers, caps, valves, flanges, and V-clamp joints are available in all sizes and wall thicknesses with ends prepared for butt welding. Joint strength equal to the original pipe (except for work-hardened pipes which are annealed by the welding), unimpaired flow pattern, and generally unimpaired corrosion resistance more than compensate for the necessary careful alignment, skilled labor, and equipment required.

Plain-end pipe used for socket-weld joints (Fig. 10-129) is available in all sizes, but fittings and valves with socket-weld ends are limited to sizes 3 in and smaller, for which the extra cost of the socket is outweighed by much easier alignment and less skill needed in welding.

Socket-welded joints are not as resistant to externally applied bending moments as are butt-welded joints, are not easily examined by volumetric nondestructive examination methods such as radiography and ultrasonic, and should not be used where crevice corrosion has been determined to be of concern. However, they are widely used in sizes 2 in and smaller and are quite satisfactory for most applications when used within the limits established by code restrictions and good engineering judgment. Components with socket-welded ends are generally specified as requiring compliance with ASME B16.11, *Forged-Fittings, Socket-Welding and Threaded*.

**Branch Connections** Branch connections may be made with manufactured tees, fabricated reinforced and nonreinforced branch connections (Fig. 10-130), or manufactured integrally reinforced branch connections. Butt-welded fittings offer the best opportunity

**TABLE 10-21 Recommended Temperature Limits for Reinforced Thermosetting Resin Pipe\***

| Materials   |             | Recommended temperature limits |     |         |     |
|-------------|-------------|--------------------------------|-----|---------|-----|
|             |             | Minimum                        |     | Maximum |     |
|             |             | °C                             | °F  | °C      | °F  |
| Resin       | Reinforcing |                                |     |         |     |
| Epoxy       | Glass fiber | -29                            | -20 | 149     | 300 |
| Phenolic    | Glass fiber | -29                            | -20 | 149     | 300 |
| Furan       | Carbon      | -29                            | -20 | 93      | 200 |
| Furan       | Glass fiber | -29                            | -20 | 93      | 200 |
| Polyester   | Glass fiber | -29                            | -20 | 93      | 200 |
| Vinyl ester | Glass fiber | -29                            | -20 | 93      | 200 |

\*Extracted from ASME B31.3-2004, Table A323.4.2C, with permission of the publisher, the American Society of Mechanical Engineers, New York.

NOTE: These temperature limits apply only to materials listed and do not reflect evidence of successful use in specific fluid services at these temperatures. The designer should consult the manufacturer for specific applications, particularly as the temperature limits are approached.

# 10-78 TRANSPORT AND STORAGE OF FLUIDS

**TABLE 10-22 Properties of Steel Pipe**

| Nominal pipe size, in | Outside diameter, in | Schedule no. | Wall thickness, in | Inside diameter, in | Cross-sectional area   |                       | Circumference, ft, or surface, ft <sup>2</sup> /ft of length |        | Capacity at 1-ft/s velocity |            | Weight of plain-end pipe, lb/ft |
|-----------------------|----------------------|--------------|--------------------|---------------------|------------------------|-----------------------|--------------------------------------------------------------|--------|-----------------------------|------------|---------------------------------|
|                       |                      |              |                    |                     | Metal, in <sup>2</sup> | Flow, ft <sup>2</sup> | Outside                                                      | Inside | U.S. gal/min                | lb/h water |                                 |
| 1/8                   | 0.405                | 10S          | 0.049              | 0.307               | 0.055                  | 0.00051               | 0.106                                                        | 0.0804 | 0.231                       | 115.5      | 0.19                            |
|                       |                      | 40ST, 40S    | .068               | .269                | .072                   | .00040                | .106                                                         | .0705  | .179                        | 89.5       | .24                             |
|                       |                      | 80XS, 80S    | .095               | .215                | .093                   | .00025                | .106                                                         | .0563  | .113                        | 56.5       | .31                             |
| 1/4                   | 0.540                | 10S          | .065               | .410                | .097                   | .00092                | .141                                                         | .107   | .412                        | 206.5      | .33                             |
|                       |                      | 40ST, 40S    | .088               | .364                | .125                   | .00072                | .141                                                         | .095   | .323                        | 161.5      | .42                             |
|                       |                      | 80XS, 80S    | .119               | .302                | .157                   | .00050                | .141                                                         | .079   | .224                        | 112.0      | .54                             |
| 3/8                   | 0.675                | 10S          | .065               | .545                | .125                   | .00162                | .177                                                         | .143   | .727                        | 363.5      | .42                             |
|                       |                      | 40ST, 40S    | .091               | .493                | .167                   | .00133                | .177                                                         | .129   | .596                        | 298.0      | .57                             |
|                       |                      | 80XS, 80S    | .126               | .423                | .217                   | .00098                | .177                                                         | .111   | .440                        | 220.0      | .74                             |
| 1/2                   | 0.840                | 5S           | .065               | .710                | .158                   | .00275                | .220                                                         | .186   | 1.234                       | 617.0      | .54                             |
|                       |                      | 10S          | .083               | .674                | .197                   | .00248                | .220                                                         | .176   | 1.112                       | 556.0      | .67                             |
|                       |                      | 40ST, 40S    | .109               | .622                | .250                   | .00211                | .220                                                         | .163   | 0.945                       | 472.0      | .85                             |
|                       |                      | 80XS, 80S    | .147               | .546                | .320                   | .00163                | .220                                                         | .143   | 0.730                       | 365.0      | 1.09                            |
|                       |                      | 160          | .188               | .464                | .385                   | .00117                | .220                                                         | .122   | 0.527                       | 263.5      | 1.31                            |
|                       |                      | XX           | .294               | .252                | .504                   | .00035                | .220                                                         | .066   | 0.155                       | 77.5       | 1.71                            |
| 3/4                   | 1.050                | 5S           | .065               | .920                | .201                   | .00461                | .275                                                         | .241   | 2.072                       | 1036.0     | 0.69                            |
|                       |                      | 10S          | .083               | .884                | .252                   | .00426                | .275                                                         | .231   | 1.903                       | 951.5      | 0.86                            |
|                       |                      | 40ST, 40S    | .113               | .824                | .333                   | .00371                | .275                                                         | .216   | 1.665                       | 832.5      | 1.13                            |
|                       |                      | 80XS, 80S    | .154               | .742                | .433                   | .00300                | .275                                                         | .194   | 1.345                       | 672.5      | 1.47                            |
|                       |                      | 160          | .219               | .612                | .572                   | .00204                | .275                                                         | .160   | 0.917                       | 458.5      | 1.94                            |
|                       |                      | XX           | .308               | .434                | .718                   | .00103                | .275                                                         | .114   | 0.461                       | 230.5      | 2.44                            |
| 1                     | 1.315                | 5S           | .065               | 1.185               | .255                   | .00768                | .344                                                         | .310   | 3.449                       | 1725       | 0.87                            |
|                       |                      | 10S          | .109               | 1.097               | .413                   | .00656                | .344                                                         | .287   | 2.946                       | 1473       | 1.40                            |
|                       |                      | 40ST, 40S    | .133               | 1.049               | .494                   | .00600                | .344                                                         | .275   | 2.690                       | 1345       | 1.68                            |
|                       |                      | 80XS, 80S    | .179               | 0.957               | .639                   | .00499                | .344                                                         | .250   | 2.240                       | 1120       | 2.17                            |
|                       |                      | 160          | .250               | 0.815               | .836                   | .00362                | .344                                                         | .213   | 1.625                       | 812.5      | 2.84                            |
|                       |                      | XX           | .358               | 0.599               | 1.076                  | .00196                | .344                                                         | .157   | 0.878                       | 439.0      | 3.66                            |
| 1 1/4                 | 1.660                | 5S           | .065               | 1.530               | 0.326                  | .01277                | .435                                                         | .401   | 5.73                        | 2865       | 1.11                            |
|                       |                      | 10S          | .109               | 1.442               | 0.531                  | .01134                | .435                                                         | .378   | 5.09                        | 2545       | 1.81                            |
|                       |                      | 40ST, 40S    | .140               | 1.380               | 0.668                  | .01040                | .435                                                         | .361   | 4.57                        | 2285       | 2.27                            |
|                       |                      | 80XS, 80S    | .191               | 1.278               | 0.881                  | .00891                | .435                                                         | .335   | 3.99                        | 1995       | 3.00                            |
|                       |                      | 160          | .250               | 1.160               | 1.107                  | .00734                | .435                                                         | .304   | 3.29                        | 1645       | 3.76                            |
|                       |                      | XX           | .382               | 0.896               | 1.534                  | .00438                | .435                                                         | .235   | 1.97                        | 985        | 5.21                            |
| 1 1/2                 | 1.900                | 5S           | .065               | 1.770               | 0.375                  | .01709                | .497                                                         | .463   | 7.67                        | 3835       | 1.28                            |
|                       |                      | 10S          | .109               | 1.682               | 0.614                  | .01543                | .497                                                         | .440   | 6.94                        | 3465       | 2.09                            |
|                       |                      | 40ST, 40S    | .145               | 1.610               | 0.800                  | .01414                | .497                                                         | .421   | 6.34                        | 3170       | 2.72                            |
|                       |                      | 80XS, 80S    | .200               | 1.500               | 1.069                  | .01225                | .497                                                         | .393   | 5.49                        | 2745       | 3.63                            |
|                       |                      | 160          | .281               | 1.338               | 1.429                  | .00976                | .497                                                         | .350   | 4.38                        | 2190       | 4.86                            |
|                       |                      | XX           | .400               | 1.100               | 1.885                  | .00660                | .497                                                         | .288   | 2.96                        | 1480       | 6.41                            |
| 2                     | 2.375                | 5S           | .065               | 2.245               | 0.472                  | .02749                | .622                                                         | .588   | 12.34                       | 6170       | 1.61                            |
|                       |                      | 10S          | .109               | 2.157               | 0.776                  | .02538                | .622                                                         | .565   | 11.39                       | 5695       | 2.64                            |
|                       |                      | 40ST, 40S    | .154               | 2.067               | 1.075                  | .02330                | .622                                                         | .541   | 10.45                       | 5225       | 3.65                            |
|                       |                      | 80ST, 80S    | .218               | 1.939               | 1.477                  | .02050                | .622                                                         | .508   | 9.20                        | 4600       | 5.02                            |
|                       |                      | 160          | .344               | 1.687               | 2.195                  | .01552                | .622                                                         | .436   | 6.97                        | 3485       | 7.46                            |
|                       |                      | XX           | .436               | 1.503               | 2.656                  | .01232                | .622                                                         | .393   | 5.53                        | 2765       | 9.03                            |
| 2 1/2                 | 2.875                | 5S           | .083               | 2.709               | 0.728                  | .04003                | .753                                                         | .709   | 17.97                       | 8985       | 2.48                            |
|                       |                      | 10S          | .120               | 2.635               | 1.039                  | .03787                | .753                                                         | .690   | 17.00                       | 8500       | 3.53                            |
|                       |                      | 40ST, 40S    | .203               | 2.469               | 1.704                  | .03322                | .753                                                         | .647   | 14.92                       | 7460       | 5.79                            |
|                       |                      | 80XS, 80S    | .276               | 2.323               | 2.254                  | .02942                | .753                                                         | .608   | 13.20                       | 6600       | 7.66                            |
|                       |                      | 160          | .375               | 2.125               | 2.945                  | .02463                | .753                                                         | .556   | 11.07                       | 5535       | 10.01                           |
|                       |                      | XX           | .552               | 1.771               | 4.028                  | .01711                | .753                                                         | .464   | 7.68                        | 3840       | 13.69                           |
| 3                     | 3.500                | 5S           | .083               | 3.334               | 0.891                  | .06063                | .916                                                         | .873   | 27.21                       | 13,605     | 3.03                            |
|                       |                      | 10S          | .120               | 3.260               | 1.274                  | .05796                | .916                                                         | .853   | 26.02                       | 13,010     | 4.33                            |
|                       |                      | 40ST, 40S    | .216               | 3.068               | 2.228                  | .05130                | .916                                                         | .803   | 23.00                       | 11,500     | 7.58                            |
|                       |                      | 80XS, 80S    | .300               | 2.900               | 3.016                  | .04587                | .916                                                         | .759   | 20.55                       | 10,275     | 10.25                           |
|                       |                      | 160          | .438               | 2.624               | 4.213                  | .03755                | .916                                                         | .687   | 16.86                       | 8430       | 14.32                           |
|                       |                      | XX           | .600               | 2.300               | 5.466                  | .02885                | .916                                                         | .602   | 12.95                       | 6475       | 18.58                           |
| 3 1/4                 | 4.0                  | 5S           | .083               | 3.834               | 1.021                  | .08017                | 1.047                                                        | 1.004  | 35.98                       | 17,990     | 3.48                            |
|                       |                      | 10S          | .120               | 3.760               | 1.463                  | .07711                | 1.047                                                        | 0.984  | 34.61                       | 17,305     | 4.97                            |
|                       |                      | 40ST, 40S    | .226               | 3.548               | 2.680                  | .06870                | 1.047                                                        | 0.929  | 30.80                       | 15,400     | 9.11                            |
|                       |                      | 80XS, 80S    | .318               | 3.364               | 3.678                  | .06170                | 1.047                                                        | 0.881  | 27.70                       | 13,850     | 12.50                           |
| 4                     | 4.5                  | 5S           | .083               | 4.334               | 1.152                  | .10245                | 1.178                                                        | 1.135  | 46.0                        | 23,000     | 3.92                            |
|                       |                      | 10S          | .120               | 4.260               | 1.651                  | .09898                | 1.178                                                        | 1.115  | 44.4                        | 22,200     | 5.61                            |
|                       |                      | 40ST, 40S    | .237               | 4.026               | 3.17                   | .08840                | 1.178                                                        | 1.054  | 39.6                        | 19,800     | 10.79                           |

TABLE 10-22 Properties of Steel Pipe (Continued)

| Nominal pipe size, in | Outside diameter, in | Schedule no. | Wall thickness, in | Inside diameter, in | Cross-sectional area   |                       | Circumference, ft, or surface, ft <sup>2</sup> of length |        | Capacity at 1-ft/s velocity |            | Weight of plain-end pipe, lb/ft |
|-----------------------|----------------------|--------------|--------------------|---------------------|------------------------|-----------------------|----------------------------------------------------------|--------|-----------------------------|------------|---------------------------------|
|                       |                      |              |                    |                     | Metal, in <sup>2</sup> | Flow, ft <sup>2</sup> | Outside                                                  | Inside | U.S. gal/min                | lb/h water |                                 |
| 5                     | 5.563                | 80XS, 80S    | .337               | 3.826               | 4.41                   | .07986                | 1.178                                                    | 1.002  | 35.8                        | 17,900     | 14.98                           |
|                       |                      | 120          | 0.438              | 3.624               | 5.58                   | 0.07170               | 1.178                                                    | 0.949  | 32.2                        | 16,100     | 19.00                           |
|                       |                      | 160          | .531               | 3.438               | 6.62                   | .06647                | 1.178                                                    | 0.900  | 28.9                        | 14,450     | 22.51                           |
|                       |                      | XX           | .674               | 3.152               | 8.10                   | .05419                | 1.178                                                    | 0.825  | 24.3                        | 12,150     | 27.54                           |
|                       |                      | 5S           | .109               | 5.345               | 1.87                   | .1558                 | 1.456                                                    | 1.399  | 69.9                        | 34,950     | 6.36                            |
|                       |                      | 10S          | .134               | 5.295               | 2.29                   | .1529                 | 1.456                                                    | 1.386  | 68.6                        | 34,300     | 7.77                            |
|                       |                      | 40ST, 40S    | .258               | 5.047               | 4.30                   | .1390                 | 1.456                                                    | 1.321  | 62.3                        | 31,150     | 14.62                           |
|                       |                      | 80XS, 80S    | .375               | 4.813               | 6.11                   | .1263                 | 1.456                                                    | 1.260  | 57.7                        | 28,850     | 20.78                           |
|                       |                      | 120          | .500               | 4.563               | 7.95                   | .1136                 | 1.456                                                    | 1.195  | 51.0                        | 25,500     | 27.04                           |
|                       |                      | 160          | .625               | 4.313               | 9.70                   | .1015                 | 1.456                                                    | 1.129  | 45.5                        | 22,750     | 32.96                           |
| 6                     | 6.625                | XX           | .750               | 4.063               | 11.34                  | .0900                 | 1.456                                                    | 1.064  | 40.4                        | 20,200     | 38.55                           |
|                       |                      | 5S           | .109               | 6.407               | 2.23                   | .2239                 | 1.734                                                    | 1.677  | 100.5                       | 50,250     | 7.60                            |
|                       |                      | 10S          | .134               | 6.357               | 2.73                   | .2204                 | 1.734                                                    | 1.664  | 98.9                        | 49,450     | 9.29                            |
|                       |                      | 40ST, 40S    | .280               | 6.065               | 5.58                   | .2006                 | 1.734                                                    | 1.588  | 90.0                        | 45,000     | 18.97                           |
|                       |                      | 80XS, 80S    | .432               | 5.761               | 8.40                   | .1810                 | 1.734                                                    | 1.508  | 81.1                        | 40,550     | 28.57                           |
|                       |                      | 120          | .562               | 5.501               | 10.70                  | .1650                 | 1.734                                                    | 1.440  | 73.9                        | 36,950     | 36.39                           |
|                       |                      | 160          | .719               | 5.187               | 13.34                  | .1467                 | 1.734                                                    | 1.358  | 65.9                        | 32,950     | 45.34                           |
|                       |                      | XX           | .864               | 4.897               | 15.64                  | .1308                 | 1.734                                                    | 1.282  | 58.7                        | 29,350     | 53.16                           |
|                       |                      | 5S           | .109               | 8.407               | 2.915                  | .3855                 | 2.258                                                    | 2.201  | 173.0                       | 86,500     | 9.93                            |
|                       |                      | 10S          | .148               | 8.329               | 3.941                  | .3784                 | 2.258                                                    | 2.180  | 169.8                       | 84,900     | 13.40                           |
| 8                     | 8.625                | 20           | .250               | 8.125               | 6.578                  | .3601                 | 2.258                                                    | 2.127  | 161.5                       | 80,750     | 22.36                           |
|                       |                      | 30           | .277               | 8.071               | 7.265                  | .3553                 | 2.258                                                    | 2.113  | 159.4                       | 79,700     | 24.70                           |
|                       |                      | 40ST, 40S    | .322               | 7.981               | 8.399                  | .3474                 | 2.258                                                    | 2.089  | 155.7                       | 77,850     | 28.55                           |
|                       |                      | 60           | .406               | 7.813               | 10.48                  | .3329                 | 2.258                                                    | 2.045  | 149.4                       | 74,700     | 35.64                           |
|                       |                      | 80XS, 80S    | .500               | 7.625               | 12.76                  | .3171                 | 2.258                                                    | 1.996  | 142.3                       | 71,150     | 43.39                           |
|                       |                      | 100          | .594               | 7.437               | 14.99                  | .3017                 | 2.258                                                    | 1.947  | 135.4                       | 67,700     | 50.95                           |
|                       |                      | 120          | .719               | 7.187               | 17.86                  | .2817                 | 2.258                                                    | 1.882  | 126.4                       | 63,200     | 60.71                           |
|                       |                      | 140          | .812               | 7.001               | 19.93                  | .2673                 | 2.258                                                    | 1.833  | 120.0                       | 60,000     | 67.76                           |
|                       |                      | XX           | .875               | 6.875               | 21.30                  | .2578                 | 2.258                                                    | 1.800  | 115.7                       | 57,850     | 72.42                           |
|                       |                      | 160          | .906               | 6.813               | 21.97                  | .2532                 | 2.258                                                    | 1.784  | 113.5                       | 56,750     | 74.69                           |
| 10                    | 10.75                | 5S           | .134               | 10.482              | 4.47                   | .5993                 | 2.814                                                    | 2.744  | 269.0                       | 134,500    | 15.19                           |
|                       |                      | 10S          | .165               | 10.420              | 5.49                   | .5922                 | 2.814                                                    | 2.728  | 265.8                       | 132,900    | 18.65                           |
|                       |                      | 20           | .250               | 10.250              | 8.25                   | .5731                 | 2.814                                                    | 2.685  | 257.0                       | 128,500    | 28.04                           |
|                       |                      | 30           | .307               | 10.136              | 10.07                  | .5603                 | 2.814                                                    | 2.655  | 252.0                       | 126,000    | 34.24                           |
|                       |                      | 40ST, 40S    | .365               | 10.020              | 11.91                  | .5475                 | 2.814                                                    | 2.620  | 246.0                       | 123,000    | 40.48                           |
|                       |                      | 80S, 60XS    | .500               | 9.750               | 16.10                  | .5185                 | 2.814                                                    | 2.550  | 233.0                       | 116,500    | 54.74                           |
|                       |                      | 80           | .594               | 9.562               | 18.95                  | .4987                 | 2.814                                                    | 2.503  | 223.4                       | 111,700    | 64.43                           |
|                       |                      | 100          | .719               | 9.312               | 22.66                  | .4729                 | 2.814                                                    | 2.438  | 212.3                       | 106,150    | 77.03                           |
|                       |                      | 120          | .844               | 9.062               | 26.27                  | .4479                 | 2.814                                                    | 2.372  | 201.0                       | 100,500    | 89.29                           |
|                       |                      | 140, XX      | 1.000              | 8.750               | 30.63                  | .4176                 | 2.814                                                    | 2.291  | 188.0                       | 94,000     | 104.13                          |
| 12                    | 12.75                | 160          | 1.125              | 8.500               | 34.02                  | .3941                 | 2.814                                                    | 2.225  | 177.0                       | 88,500     | 115.64                          |
|                       |                      | 5S           | 0.156              | 12.438              | 6.17                   | .8438                 | 3.338                                                    | 3.26   | 378.7                       | 189,350    | 20.98                           |
|                       |                      | 10S          | 0.180              | 12.390              | 7.11                   | .8373                 | 3.338                                                    | 3.24   | 375.8                       | 187,900    | 24.17                           |
|                       |                      | 20           | 0.250              | 12.250              | 9.82                   | .8185                 | 3.338                                                    | 3.21   | 367.0                       | 183,500    | 33.38                           |
|                       |                      | 30           | 0.330              | 12.090              | 12.88                  | .7972                 | 3.338                                                    | 3.17   | 358.0                       | 179,000    | 43.77                           |
|                       |                      | ST, 40S      | 0.375              | 12.000              | 14.58                  | .7854                 | 3.338                                                    | 3.14   | 352.5                       | 176,250    | 49.56                           |
|                       |                      | 40           | 0.406              | 11.938              | 15.74                  | .7773                 | 3.338                                                    | 3.13   | 349.0                       | 174,500    | 53.52                           |
|                       |                      | XS, 80S      | 0.500              | 11.750              | 19.24                  | .7530                 | 3.338                                                    | 3.08   | 338.0                       | 169,000    | 65.42                           |
|                       |                      | 60           | 0.562              | 11.626              | 21.52                  | .7372                 | 3.338                                                    | 3.04   | 331.0                       | 165,500    | 73.15                           |
|                       |                      | 80           | 0.688              | 11.374              | 26.07                  | .7056                 | 3.338                                                    | 2.98   | 316.7                       | 158,350    | 88.63                           |
| 14                    | 14                   | 100          | 0.844              | 11.062              | 31.57                  | .6674                 | 3.338                                                    | 2.90   | 299.6                       | 149,800    | 107.32                          |
|                       |                      | 120, XX      | 1.000              | 10.750              | 36.91                  | .6303                 | 3.338                                                    | 2.81   | 283.0                       | 141,500    | 125.49                          |
|                       |                      | 140          | 1.125              | 10.500              | 41.09                  | .6013                 | 3.338                                                    | 2.75   | 270.0                       | 135,000    | 139.67                          |
|                       |                      | 160          | 1.312              | 10.126              | 47.14                  | .5592                 | 3.338                                                    | 2.65   | 251.0                       | 125,500    | 160.27                          |
|                       |                      | 5S           | 0.156              | 13.688              | 6.78                   | 1.0219                | 3.665                                                    | 3.58   | 459                         | 229,500    | 23.07                           |
|                       |                      | 10S          | 0.188              | 13.624              | 8.16                   | 1.0125                | 3.665                                                    | 3.57   | 454                         | 227,000    | 27.73                           |
|                       |                      | 10           | 0.250              | 13.500              | 10.80                  | 0.9940                | 3.665                                                    | 3.53   | 446                         | 223,000    | 36.71                           |
|                       |                      | 20           | 0.312              | 13.376              | 13.42                  | 0.9750                | 3.665                                                    | 3.50   | 438                         | 219,000    | 45.61                           |
|                       |                      | 30, ST       | 0.375              | 13.250              | 16.05                  | 0.9575                | 3.665                                                    | 3.47   | 430                         | 215,000    | 54.57                           |
|                       |                      | 40           | 0.438              | 13.124              | 18.66                  | 0.9397                | 3.665                                                    | 3.44   | 422                         | 211,000    | 63.44                           |
| 16                    | 16                   | XS           | 0.500              | 13.000              | 21.21                  | 0.9218                | 3.665                                                    | 3.40   | 414                         | 207,000    | 72.09                           |
|                       |                      | 60           | 0.594              | 12.812              | 25.02                  | 0.8957                | 3.665                                                    | 3.35   | 402                         | 201,000    | 85.05                           |
|                       |                      | 80           | 0.750              | 12.500              | 31.22                  | 0.8522                | 3.665                                                    | 3.27   | 382                         | 191,000    | 106.13                          |
|                       |                      | 100          | 0.938              | 12.124              | 38.49                  | 0.8017                | 3.665                                                    | 3.17   | 360                         | 180,000    | 130.85                          |
|                       |                      | 120          | 1.094              | 11.812              | 44.36                  | 0.7610                | 3.665                                                    | 3.09   | 342                         | 171,000    | 150.79                          |
|                       |                      | 140          | 1.250              | 11.500              | 50.07                  | 0.7213                | 3.665                                                    | 3.01   | 324                         | 162,000    | 170.21                          |
|                       |                      | 160          | 1.406              | 11.188              | 55.63                  | 0.6827                | 3.665                                                    | 2.93   | 306                         | 153,000    | 189.11                          |
|                       |                      | 5S           | 0.165              | 15.670              | 8.21                   | 1.3393                | 4.189                                                    | 4.10   | 601                         | 300,500    | 27.90                           |
|                       |                      | 10S          | 0.188              | 15.624              | 9.34                   | 1.3314                | 4.189                                                    | 4.09   | 598                         | 299,000    | 31.75                           |
|                       |                      | 10           | 0.250              | 15.500              | 12.37                  | 1.3104                | 4.189                                                    | 4.06   | 587                         | 293,500    | 42.05                           |



# 10-80 TRANSPORT AND STORAGE OF FLUIDS

**TABLE 10-22 Properties of Steel Pipe** (Concluded)

| Nominal pipe size, in | Outside diameter, in | Schedule no. | Wall thickness, in | Inside diameter, in | Cross-sectional area   |                       | Circumference, ft, or surface, ft <sup>2</sup> /ft of length |        | Capacity at 1-ft/s velocity |            | Weight of plain-end pipe, lb/ft |
|-----------------------|----------------------|--------------|--------------------|---------------------|------------------------|-----------------------|--------------------------------------------------------------|--------|-----------------------------|------------|---------------------------------|
|                       |                      |              |                    |                     | Metal, in <sup>2</sup> | Flow, ft <sup>2</sup> | Outside                                                      | Inside | U.S. gal/min                | lb/h water |                                 |
| 18                    | 18                   | 20           | 0.312              | 15.376              | 15.38                  | 1.2985                | 4.189                                                        | 4.03   | 578                         | 289,000    | 52.27                           |
|                       |                      | 30, ST       | 0.375              | 15.250              | 18.41                  | 1.2680                | 4.189                                                        | 3.99   | 568                         | 284,000    | 62.58                           |
|                       |                      | 40, XS       | 0.500              | 15.000              | 24.35                  | 1.2272                | 4.189                                                        | 3.93   | 550                         | 275,000    | 82.77                           |
|                       |                      | 60           | 0.656              | 14.688              | 31.62                  | 1.1766                | 4.189                                                        | 3.85   | 528                         | 264,000    | 107.50                          |
|                       |                      | 80           | 0.844              | 14.312              | 40.19                  | 1.1171                | 4.189                                                        | 3.75   | 501                         | 250,500    | 136.61                          |
|                       |                      | 100          | 1.031              | 13.938              | 48.48                  | 1.0596                | 4.189                                                        | 3.65   | 474                         | 237,000    | 164.82                          |
|                       |                      | 120          | 1.219              | 13.562              | 56.61                  | 1.0032                | 4.189                                                        | 3.55   | 450                         | 225,000    | 192.43                          |
|                       |                      | 140          | 1.438              | 13.124              | 65.79                  | 0.9394                | 4.189                                                        | 3.44   | 422                         | 211,000    | 223.64                          |
|                       |                      | 160          | 1.594              | 12.812              | 72.14                  | 0.8953                | 4.189                                                        | 3.35   | 402                         | 201,000    | 245.25                          |
|                       |                      | 5S           | 0.165              | 17.670              | 9.25                   | 1.7029                | 4.712                                                        | 4.63   | 764                         | 382,000    | 31.43                           |
|                       |                      | 10S          | 0.188              | 17.624              | 10.52                  | 1.6941                | 4.712                                                        | 4.61   | 760                         | 379,400    | 35.76                           |
|                       |                      | 10           | 0.250              | 17.500              | 13.94                  | 1.6703                | 4.712                                                        | 4.58   | 750                         | 375,000    | 47.39                           |
|                       |                      | 20           | 0.312              | 17.376              | 17.34                  | 1.6468                | 4.712                                                        | 4.55   | 739                         | 369,500    | 58.94                           |
|                       |                      | ST           | 0.375              | 17.250              | 20.76                  | 1.6230                | 4.712                                                        | 4.52   | 728                         | 364,000    | 70.59                           |
|                       |                      | 30           | 0.438              | 17.124              | 24.16                  | 1.5993                | 4.712                                                        | 4.48   | 718                         | 359,000    | 82.15                           |
|                       |                      | XS           | 0.500              | 17.000              | 27.49                  | 1.5763                | 4.712                                                        | 4.45   | 707                         | 353,500    | 93.45                           |
|                       |                      | 40           | 0.562              | 16.876              | 30.79                  | 1.5533                | 4.712                                                        | 4.42   | 697                         | 348,500    | 104.67                          |
|                       |                      | 60           | 0.750              | 16.500              | 40.64                  | 1.4849                | 4.712                                                        | 4.32   | 666                         | 333,000    | 138.17                          |
|                       |                      | 80           | 0.938              | 16.124              | 50.28                  | 1.4180                | 4.712                                                        | 4.22   | 636                         | 318,000    | 170.92                          |
|                       |                      | 100          | 1.156              | 15.688              | 61.17                  | 1.3423                | 4.712                                                        | 4.11   | 602                         | 301,000    | 207.96                          |
|                       |                      | 120          | 1.375              | 15.250              | 71.82                  | 1.2684                | 4.712                                                        | 3.99   | 569                         | 284,500    | 244.14                          |
| 20                    | 20                   | 140          | 1.562              | 14.876              | 80.66                  | 1.2070                | 4.712                                                        | 3.89   | 540                         | 270,000    | 274.22                          |
|                       |                      | 160          | 1.781              | 14.438              | 90.75                  | 1.1370                | 4.712                                                        | 3.78   | 510                         | 255,000    | 308.50                          |
|                       |                      | 5S           | 0.188              | 19.624              | 11.70                  | 2.1004                | 5.236                                                        | 5.14   | 943                         | 471,500    | 39.78                           |
|                       |                      | 10S          | 0.218              | 19.564              | 13.55                  | 2.0878                | 5.236                                                        | 5.12   | 937                         | 467,500    | 46.06                           |
|                       |                      | 10           | 0.250              | 19.500              | 15.51                  | 2.0740                | 5.236                                                        | 5.11   | 930                         | 465,000    | 52.73                           |
|                       |                      | 20, ST       | 0.375              | 19.250              | 23.12                  | 2.0211                | 5.236                                                        | 5.04   | 902                         | 451,000    | 78.60                           |
|                       |                      | 30, XS       | 0.500              | 19.000              | 30.63                  | 1.9689                | 5.236                                                        | 4.97   | 883                         | 441,500    | 104.13                          |
|                       |                      | 40           | 0.594              | 18.812              | 36.21                  | 1.9302                | 5.236                                                        | 4.92   | 866                         | 433,000    | 123.11                          |
|                       |                      | 60           | 0.812              | 18.376              | 48.95                  | 1.8417                | 5.236                                                        | 4.81   | 826                         | 413,000    | 166.40                          |
|                       |                      | 80           | 1.031              | 17.938              | 61.44                  | 1.7550                | 5.236                                                        | 4.70   | 787                         | 393,500    | 208.87                          |
|                       |                      | 100          | 1.281              | 17.438              | 75.33                  | 1.6585                | 5.236                                                        | 4.57   | 744                         | 372,000    | 256.10                          |
|                       |                      | 120          | 1.500              | 17.000              | 87.18                  | 1.5763                | 5.236                                                        | 4.45   | 707                         | 353,500    | 296.37                          |
|                       |                      | 140          | 1.750              | 16.500              | 100.3                  | 1.4849                | 5.236                                                        | 4.32   | 665                         | 332,500    | 341.09                          |
|                       |                      | 160          | 1.969              | 16.062              | 111.5                  | 1.4071                | 5.236                                                        | 4.21   | 632                         | 316,000    | 397.17                          |
|                       |                      | 5S           | 0.218              | 23.564              | 16.29                  | 3.0285                | 6.283                                                        | 6.17   | 1359                        | 679,500    | 55.37                           |
|                       |                      | 10, 10S      | 0.250              | 23.500              | 18.65                  | 3.012                 | 6.283                                                        | 6.15   | 1350                        | 675,000    | 63.41                           |
|                       |                      | 20, ST       | 0.375              | 23.250              | 27.83                  | 2.948                 | 6.283                                                        | 6.09   | 1325                        | 662,500    | 94.62                           |
|                       |                      | XS           | 0.500              | 23.000              | 36.90                  | 2.885                 | 6.283                                                        | 6.02   | 1295                        | 642,500    | 125.49                          |
|                       |                      | 30           | 0.562              | 22.876              | 41.39                  | 2.854                 | 6.283                                                        | 5.99   | 1281                        | 640,500    | 140.68                          |
|                       |                      | 40           | 0.688              | 22.624              | 50.39                  | 2.792                 | 6.283                                                        | 5.92   | 1253                        | 626,500    | 171.29                          |
|                       |                      | 60           | 0.969              | 22.062              | 70.11                  | 2.655                 | 6.283                                                        | 5.78   | 1192                        | 596,000    | 238.35                          |
|                       |                      | 80           | 1.219              | 21.562              | 87.24                  | 2.536                 | 6.283                                                        | 5.64   | 1138                        | 569,000    | 296.58                          |
|                       |                      | 100          | 1.531              | 20.938              | 108.1                  | 2.391                 | 6.283                                                        | 5.48   | 1073                        | 536,500    | 367.39                          |
| 24                    | 24                   | 120          | 1.812              | 20.376              | 126.3                  | 2.264                 | 6.283                                                        | 5.33   | 1016                        | 508,000    | 429.39                          |
|                       |                      | 140          | 2.062              | 19.876              | 142.1                  | 2.155                 | 6.283                                                        | 5.20   | 965                         | 482,500    | 483.12                          |
|                       |                      | 160          | 2.344              | 19.312              | 159.5                  | 2.034                 | 6.283                                                        | 5.06   | 913                         | 456,500    | 542.13                          |
|                       |                      | 5S           | 0.250              | 29.500              | 23.37                  | 4.746                 | 7.854                                                        | 7.72   | 2130                        | 1,065,000  | 79.43                           |
|                       |                      | 10, 10S      | 0.312              | 29.376              | 29.10                  | 4.707                 | 7.854                                                        | 7.69   | 2110                        | 1,055,000  | 98.93                           |
|                       |                      | ST           | 0.375              | 29.250              | 34.90                  | 4.666                 | 7.854                                                        | 7.66   | 2094                        | 1,048,000  | 118.65                          |
|                       |                      | 20, XS       | 0.500              | 29.000              | 46.34                  | 4.587                 | 7.854                                                        | 7.59   | 2055                        | 1,027,500  | 157.53                          |
|                       |                      | 30           | 0.625              | 28.750              | 57.68                  | 4.508                 | 7.854                                                        | 7.53   | 2020                        | 1,010,000  | 196.08                          |
|                       |                      | 5S           | 0.250              | 29.500              | 23.37                  | 4.746                 | 7.854                                                        | 7.72   | 2130                        | 1,065,000  | 79.43                           |
|                       |                      | 10, 10S      | 0.312              | 29.376              | 29.10                  | 4.707                 | 7.854                                                        | 7.69   | 2110                        | 1,055,000  | 98.93                           |
|                       |                      | ST           | 0.375              | 29.250              | 34.90                  | 4.666                 | 7.854                                                        | 7.66   | 2094                        | 1,048,000  | 118.65                          |
|                       |                      | 20, XS       | 0.500              | 29.000              | 46.34                  | 4.587                 | 7.854                                                        | 7.59   | 2055                        | 1,027,500  | 157.53                          |
|                       |                      | 30           | 0.625              | 28.750              | 57.68                  | 4.508                 | 7.854                                                        | 7.53   | 2020                        | 1,010,000  | 196.08                          |
|                       |                      | 5S           | 0.250              | 29.500              | 23.37                  | 4.746                 | 7.854                                                        | 7.72   | 2130                        | 1,065,000  | 79.43                           |
|                       |                      | 10, 10S      | 0.312              | 29.376              | 29.10                  | 4.707                 | 7.854                                                        | 7.69   | 2110                        | 1,055,000  | 98.93                           |
|                       |                      | ST           | 0.375              | 29.250              | 34.90                  | 4.666                 | 7.854                                                        | 7.66   | 2094                        | 1,048,000  | 118.65                          |
|                       |                      | 20, XS       | 0.500              | 29.000              | 46.34                  | 4.587                 | 7.854                                                        | 7.59   | 2055                        | 1,027,500  | 157.53                          |
|                       |                      | 30           | 0.625              | 28.750              | 57.68                  | 4.508                 | 7.854                                                        | 7.53   | 2020                        | 1,010,000  | 196.08                          |
|                       |                      | 5S           | 0.250              | 29.500              | 23.37                  | 4.746                 | 7.854                                                        | 7.72   | 2130                        | 1,065,000  | 79.43                           |
|                       |                      | 10, 10S      | 0.312              | 29.376              | 29.10                  | 4.707                 | 7.854                                                        | 7.69   | 2110                        | 1,055,000  | 98.93                           |
|                       |                      | ST           | 0.375              | 29.250              | 34.90                  | 4.666                 | 7.854                                                        | 7.66   | 2094                        | 1,048,000  | 118.65                          |
|                       |                      | 20, XS       | 0.500              | 29.000              | 46.34                  | 4.587                 | 7.854                                                        | 7.59   | 2055                        | 1,027,500  | 157.53                          |
|                       |                      | 30           | 0.625              | 28.750              | 57.68                  | 4.508                 | 7.854                                                        | 7.53   | 2020                        | 1,010,000  | 196.08                          |

5S, 10S, 40S, and 80S are extracted from Stainless Steel Pipe, ASME B36.19M-1985, with permission of the publisher, the American Society of Mechanical Engineers, New York. ST = standard wall, XS = extra strong wall, XX = double extra strong wall, and Schedules 10 through 160 are extracted from Welded and Seamless Wrought Steel Pipe, ASME B36.10M-1996, with permission of the same publisher. Refer to these standards for a more comprehensive listing of material sizes and wall thicknesses. Decimal thicknesses for respective pipe sizes represent their nominal or average wall dimensions. Mill tolerances as high as  $\pm 12\frac{1}{2}$  percent are permitted.

Plain-end pipe is produced by a square cut. Pipe is also shipped from the mills threaded, with a threaded coupling on one end, or with the ends beveled for welding, or grooved or sized for patented couplings.

To convert inches to millimeters, multiply by 25.4; to convert square inches to square millimeters, multiply by 645; to convert feet to meters, multiply by 0.3048; to convert square feet to square meters, multiply by 0.0929; to convert pounds per foot to kilograms per meter, multiply by 1.49; to convert gallons to cubic meters, multiply by  $3.7854 \times 10^{-3}$ ; and to convert pounds to kilograms, multiply by 0.4536.



FIG. 10-128 Butt weld.

for nondestructive examination; however, branch connections are commonly specified for branches smaller than the header, and often best satisfy the design and economic requirements. Design of fabricated branch connections is addressed in the subsection "Pressure Design of Metallic Components: Wall Thickness." Integrally reinforced fittings are generally specified as requiring compliance with Manufacturer's Standardization Society specification MSS SP-97, *Integrally Reinforced Forged Branch Outlet Fittings—Socket Welding, Threaded, and Butt Welding Ends*.

**Threaded Joints** Pipe with **taper-pipe-thread** ends (Fig. 10-131), per ASME B1.20.1, is available 12 in and smaller, subject to minimum-wall limitations. Fittings and valves with taper-pipe-thread ends are available in most pipe metals.

Principal use of threaded joints is in sizes 2 in and smaller, in metals for which the most economically produced walls are thick enough to withstand pressure and corrosion after reduction in thickness due to threading. For threaded joints over 2 in, assembly difficulty and cost of tools increase rapidly. Careful alignment, required at the start of assembly and during rotation of the components, as well as variation in length produced by diametral tolerances in the threads, severely limits preassembly of the components. Threading is not a precise machining operation, and filler materials known as "pipe dope" are necessary to block the spiral leakage path.

Threads notch the pipe and cause loss of strength and fatigue resistance. Enlargement and contraction of the flow passage at threaded joints creates turbulence; sometimes corrosion and erosion are aggravated at the point where the pipe has already been thinned by threading. The tendency of pipe wrenches to crush pipe and fittings limits the torque available for tightening threaded joints. For low-pressure systems, a slight rotation in the joint may be used to impart flexibility to the system, but this same rotation, unwanted, may cause leaks to develop in higher-pressure systems. In some metals, galling occurs when threaded joints are disassembled.

**Straight Pipe Threads** These are confined to light-weight couplings in sizes 2 in and smaller (Fig. 10-132). Manufacturers of threaded pipe ship it with such couplings installed on one end of each pipe. The joint obtained is inferior to that obtained with taper threads. The code limits the joint shown in Fig. 10-129 to 1.0 MPa (150 lbf/in<sup>2</sup>) gauge maximum, 182°C (360°F) maximum, and to nonflammable, nontoxic fluids.

When both components of a threaded joint are of weldable metal, the joint may be **seal-welded** as shown in Fig. 10-133. Seal welds may be used only to prevent leakage of threaded joints. They are not considered as contributing any strength to the joint. This type of joint is limited to new construction and is not suitable as a repair procedure, since pipe dope in the threads would interfere with welding. Careful consideration should be given to the suitability of threaded joints when joining metals having significantly different coefficients of expansion. Thermal expansion and temperature cycling may eventually result in leakage.

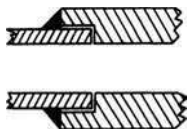


FIG. 10-129 Socket weld.

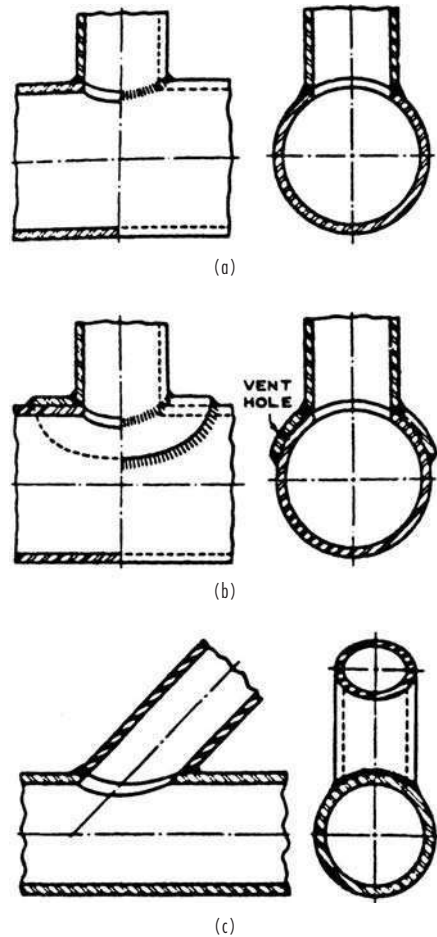


FIG. 10-130 Branch welds. (a) Without added reinforcement. (b) With added reinforcement. (c) Angular branch.

To assist in assembly and disassembly of both threaded and welded systems, **union joints** (Fig. 10-134) are used. They comprise metal-to-metal or gasketed seats drawn together by a shouldered straight thread nut and are available both in couplings for joining two lengths of pipe and on the ends of some fittings. On threaded piping systems in which disassembly is not contemplated, union joints installed at intervals permit future further tightening of threaded joints. Tightening of heavy unions yields tight joints even if the pipe is slightly misaligned at the start of tightening.

**Flanged Joints** For sizes larger than 2 in when disassembly is contemplated, the flanged joint (Fig. 10-135) is the most widely used. Figures 10-136 and 10-137 illustrate the wide variety of types and facings available. Though flanged joints consume a large volume of metal, precise machining is required only on the facing. Flanged joints do not impose severe diametral tolerances on the pipe. Alignment tolerances required for flanged joints are reasonably achieved with quality construction practices, and in comparison with taper threaded joints, required wrench sizes are smaller and sealing is more easily and reliably obtained.

Manufacturers offer **flanged-end pipe** in only a few metals. Otherwise, flanges are attached to pipe by various types of joints (Fig. 10-136). The lap joint involves a modification of the pipe which may be formed from the pipe itself or by welding a ring or a lap-joint stub end to it. **Flanged-end fittings** and valves are available in all sizes of most pipe metals.

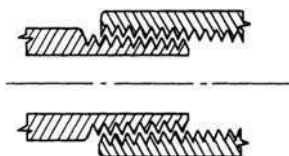


FIG. 10-131 Taper pipe thread.

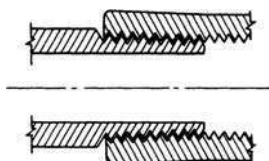


FIG. 10-132 Taper pipe to straight coupling thread.

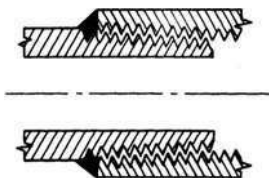


FIG. 10-133 Taper pipe thread seal-welded.

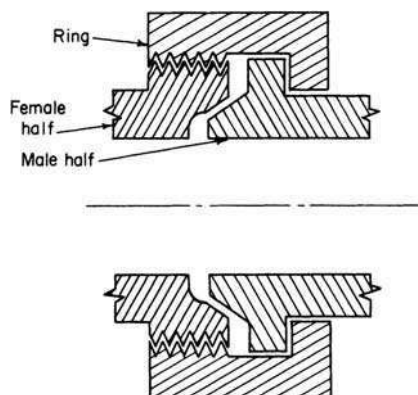


FIG. 10-134 Union.

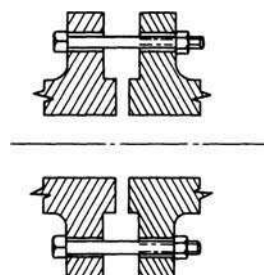


FIG. 10-135 Flanged joint.

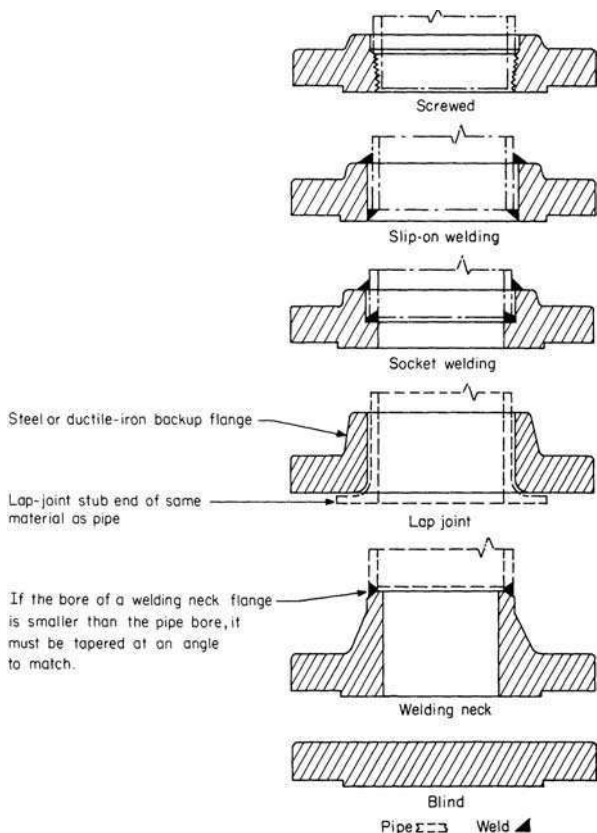


FIG. 10-136 Types of carbon and alloy steel flanges.

Of the flange types shown in Fig. 10-136, welding-neck flanges offer the highest mechanical strength and are the type most suitable for extreme temperatures and cyclic loading. Regardless of the type selected, designers must be aware that the flange's capability to resist external bending moments and maintain its seal does not necessarily match the bending moment capability of the attached pipe.

When selecting the flange type, the designer should review the usage restrictions contained in each section of the ASME B31 code. Each of the other types shown provides significant fabrication and economic advantages and is suitable for many of the routine applications. Lap-joint flanges permit adjustment of the bolt hole orientation and can greatly simplify construction when configurations are complex and bolt hole orientations are difficult to ensure.

Dimensions for alloy and carbon-steel pipe flanges are given in Tables 10-23 through 10-29. The dimensions were extracted from *Pipe Flanges and Flanged Fittings*, ASME B16.5-2003, with permission of the publisher, the American Society of Mechanical Engineers, New York. Dimensions for cast-iron flanges are provided in *Cast Iron Pipe Flanges and Flanged Fittings*, ASME B16.1. Bolt patterns and bolt sizes for Class 125 cast-iron flanges match the ASME B16.5 Class 150 flange dimensions, and bolt patterns for Class 250 cast-iron flanges match the ASME B16.5 Class 300 flange dimensions.

When mating with cast-iron flanged fittings or valves, steel pipe flanges are often preferred to cast-iron flanges because they permit welded rather than screwed assembly to the pipe and because cast-iron pipe flanges, not being reinforced by the pipe, are not so resistant to abuse as flanges cast integrally on cast-iron fittings.

Facing of flanges for alloy and carbon steel pipe and fittings is shown in Fig. 10-137; Class 125 cast-iron pipe and fitting flanges have flat faces, which with full-face gaskets minimize bending stresses; Class 250 cast-iron pipe and fitting flanges have 1.5-mm

**TABLE 10-23 Dimensions of ASME B16.5 Class 150 Flanges\***

All dimensions in inches

| Nominal pipe size | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |                              |
|-------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|------------------------------|
|                   |                            |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck | ANSI B16.1, screwed (125-lb) |
| 1/2               | 3.50                       | 0.44                         | 2.38                    | 1/2               | 4            | 0.62                            | 0.62      | 1.88         |                              |
| 3/4               | 3.88                       | 0.50                         | 2.75                    | 1/2               | 4            | 0.62                            | 0.62      | 2.06         |                              |
| 1                 | 4.25                       | 0.56                         | 3.12                    | 1/2               | 4            | 0.69                            | 0.69      | 2.19         | 0.69                         |
| 1 1/4             | 4.62                       | 0.62                         | 3.50                    | 1/2               | 4            | 0.81                            | 0.81      | 2.25         | 0.81                         |
| 1 1/2             | 5.00                       | 0.69                         | 3.88                    | 1/2               | 4            | 0.88                            | 0.88      | 2.44         | 0.88                         |
| 2                 | 6.00                       | 0.75                         | 4.75                    | 5/8               | 4            | 1.00                            | 1.00      | 2.50         | 1.00                         |
| 2 1/2             | 7.00                       | 0.88                         | 5.50                    | 5/8               | 4            | 1.12                            | 1.12      | 2.75         | 1.12                         |
| 3                 | 7.50                       | 0.94                         | 6.00                    | 5/8               | 4            | 1.19                            | 1.19      | 2.75         | 1.19                         |
| 3 1/2             | 8.50                       | 0.94                         | 7.00                    | 5/8               | 8            | 1.25                            | 1.25      | 2.81         | 1.25                         |
| 4                 | 9.00                       | 0.94                         | 7.50                    | 5/8               | 8            | 1.31                            | 1.31      | 3.00         | 1.31                         |
| 5                 | 10.00                      | 0.94                         | 8.50                    | 3/4               | 8            | 1.44                            | 1.44      | 3.50         | 1.44                         |
| 6                 | 11.00                      | 1.00                         | 9.50                    | 3/4               | 8            | 1.56                            | 1.56      | 3.50         | 1.56                         |
| 8                 | 13.50                      | 1.12                         | 11.75                   | 3/4               | 8            | 1.75                            | 1.75      | 4.00         | 1.75                         |
| 10                | 16.00                      | 1.19                         | 14.25                   | 7/8               | 12           | 1.94                            | 1.94      | 4.00         | 1.94                         |
| 12                | 19.00                      | 1.25                         | 17.00                   | 7/8               | 12           | 2.19                            | 2.19      | 4.50         | 2.19                         |
| 14                | 21.00                      | 1.38                         | 18.75                   | 1                 | 12           | 2.25                            | 3.12      | 5.00         | 2.25                         |
| 16                | 23.50                      | 1.44                         | 21.25                   | 1                 | 16           | 2.50                            | 3.44      | 5.00         | 2.50                         |
| 18                | 25.00                      | 1.56                         | 22.75                   | 1 1/8             | 16           | 2.69                            | 3.81      | 5.50         | 2.69                         |
| 20                | 27.50                      | 1.69                         | 25.00                   | 1 1/8             | 20           | 2.88                            | 4.06      | 5.69         | 2.88                         |
| 24                | 32.00                      | 1.88                         | 29.50                   | 1 1/4             | 20           | 3.25                            | 4.38      | 6.00         | 3.25                         |

\*Dimensions from ASME B16.5-2003, unless otherwise noted. To convert inches to millimeters, multiply by 25.4.

(1/16-in) raised faces (wider than on steel flanges) for the same purpose. Carbon steel and ductile- (nodular-) iron lap-joint flanges are widely used as backup flanges with stub ends in piping systems of austenitic stainless steel and other expensive materials to reduce costs (see Fig. 10-136). The code prohibits the use of ductile-iron flanges at temperatures above 343°C (650°F). When the type of facing affects the length through the hub dimension of flanges, correct dimensions for commonly used facings can be determined from the dimensional data in Fig. 10-137.

**Gaskets** Gaskets must resist corrosion by the fluids handled. The more expensive male-and-female or tongue-and-groove facings may be required to seat hard gaskets adequately. With these facings the gasket generally cannot blow out. Flanged joints, by placing the gasket material under heavy compression and permitting only edge attack by the fluid handled, can use gasket materials which in other joints might not satisfactorily resist the fluid handled.

Standards to which flanges are manufactured (e.g., ASME B16.1, ASME B16.5, ASME B16.47) typically specify a standard surface finish

**TABLE 10-24 Dimensions of ASME B16.5 Class 300 Flanges\***

All dimensions in inches

| Nominal pipe size | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |                                 |
|-------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|---------------------------------|
|                   |                            |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck | ANSI B16.1, screwed (Class 250) |
| 1/2               | 3.75                       | 0.56                         | 2.62                    | 1/2               | 4            | 0.88                            | 0.88      | 2.06         |                                 |
| 3/4               | 4.62                       | 0.62                         | 3.25                    | 5/8               | 4            | 1.00                            | 1.00      | 2.25         |                                 |
| 1                 | 4.88                       | 0.69                         | 3.50                    | 5/8               | 4            | 1.06                            | 1.06      | 2.44         | 0.88                            |
| 1 1/4             | 5.25                       | 0.75                         | 3.88                    | 5/8               | 4            | 1.06                            | 1.06      | 2.56         | 1.00                            |
| 1 1/2             | 6.12                       | 0.81                         | 4.50                    | 3/4               | 4            | 1.19                            | 1.19      | 2.69         | 1.12                            |
| 2                 | 6.50                       | 0.88                         | 5.00                    | 5/8               | 8            | 1.31                            | 1.31      | 2.75         | 1.25                            |
| 2 1/2             | 7.50                       | 1.00                         | 5.88                    | 3/4               | 8            | 1.50                            | 1.50      | 3.00         | 1.43                            |
| 3                 | 8.25                       | 1.12                         | 6.62                    | 3/4               | 8            | 1.69                            | 1.69      | 3.12         | 1.56                            |
| 3 1/2             | 9.00                       | 1.19                         | 7.25                    | 3/4               | 8            | 1.75                            | 1.75      | 3.19         | 1.62                            |
| 4                 | 10.00                      | 1.25                         | 7.88                    | 3/4               | 8            | 1.88                            | 1.88      | 3.38         | 1.75                            |
| 5                 | 11.00                      | 1.38                         | 9.25                    | 3/4               | 8            | 2.00                            | 2.00      | 3.88         | 1.88                            |
| 6                 | 12.50                      | 1.44                         | 10.62                   | 3/4               | 12           | 2.06                            | 2.06      | 3.88         | 1.94                            |
| 8                 | 15.00                      | 1.62                         | 13.00                   | 7/8               | 12           | 2.44                            | 2.44      | 4.38         | 2.19                            |
| 10                | 17.50                      | 1.88                         | 15.25                   | 1                 | 16           | 2.62                            | 3.75      | 4.62         | 2.38                            |
| 12                | 20.50                      | 2.00                         | 17.75                   | 1 1/8             | 16           | 2.88                            | 4.00      | 5.12         | 2.56                            |
| 14                | 23.00                      | 2.12                         | 20.25                   | 1 1/8             | 20           | 3.00                            | 4.38      | 5.62         | 2.69                            |
| 16                | 25.50                      | 2.25                         | 22.50                   | 1 1/4             | 20           | 3.25                            | 4.75      | 5.75         | 2.88                            |
| 18                | 28.00                      | 2.38                         | 24.75                   | 1 1/4             | 24           | 3.50                            | 5.12      | 6.25         |                                 |
| 20                | 30.50                      | 2.50                         | 27.00                   | 1 1/4             | 24           | 3.75                            | 5.50      | 6.38         |                                 |
| 24                | 36.00                      | 2.75                         | 32.00                   | 1 1/2             | 24           | 4.19                            | 6.00      | 6.62         |                                 |

\*Dimensions from ASME B16.5-2003, unless otherwise noted. To convert inches to millimeters, multiply by 25.4.

## 10-84 TRANSPORT AND STORAGE OF FLUIDS

**TABLE 10-25 Dimensions of ASME B16.5 Class 400 Flanges\***

All dimensions in inches

| Nominal pipe size                        | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |
|------------------------------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|
|                                          |                            |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck |
| 1/2                                      |                            |                              |                         |                   |              |                                 |           |              |
| 3/4                                      |                            |                              |                         |                   |              |                                 |           |              |
| 1                                        |                            |                              |                         |                   |              |                                 |           |              |
| 1 1/4                                    |                            |                              |                         |                   |              |                                 |           |              |
| 1 1/2                                    |                            |                              |                         |                   |              |                                 |           |              |
| 2                                        |                            |                              |                         |                   |              |                                 |           |              |
| 2 1/2                                    |                            |                              |                         |                   |              |                                 |           |              |
| 3                                        |                            |                              |                         |                   |              |                                 |           |              |
| 3 1/2                                    |                            |                              |                         |                   |              |                                 |           |              |
| Use Class 600 dimensions in these sizes. |                            |                              |                         |                   |              |                                 |           |              |
| 4                                        | 10.00                      | 1.38                         | 7.88                    | 7/8               | 8            | 2.00                            | 2.00      | 3.50         |
| 5                                        | 11.00                      | 1.50                         | 9.25                    | 7/8               | 8            | 2.12                            | 2.12      | 4.00         |
| 6                                        | 12.50                      | 1.62                         | 10.62                   | 7/8               | 12           | 2.25                            | 2.25      | 4.06         |
| 8                                        | 15.00                      | 1.88                         | 13.00                   | 1                 | 12           | 2.69                            | 2.69      | 4.62         |
| 10                                       | 17.50                      | 2.12                         | 15.25                   | 1 1/4             | 16           | 2.88                            | 4.00      | 4.88         |
| 12                                       | 20.50                      | 2.25                         | 17.75                   | 1 1/4             | 16           | 3.12                            | 4.25      | 5.38         |
| 14                                       | 23.00                      | 2.38                         | 20.25                   | 1 1/4             | 20           | 3.31                            | 4.62      | 5.88         |
| 16                                       | 25.50                      | 2.50                         | 22.50                   | 1 3/4             | 20           | 3.69                            | 5.00      | 6.00         |
| 18                                       | 28.00                      | 2.62                         | 24.75                   | 1 3/4             | 24           | 3.88                            | 5.38      | 6.50         |
| 20                                       | 30.50                      | 2.75                         | 27.00                   | 1 1/2             | 24           | 4.00                            | 5.75      | 6.62         |
| 24                                       | 36.00                      | 3.00                         | 32.00                   | 1 3/4             | 24           | 4.50                            | 6.25      | 6.88         |

\*Dimensions from ASME B16.5-2003. To convert inches to millimeters, multiply by 25.4.

for the gasket seal area. Flange rating and type, flange size, flange facing type, gasket style, commodity, design conditions, and bolting must all be considered to ensure proper seating of the gasket and reliable performance. Unless the user is familiar with gasket design and the particular application being considered, it is highly recommended that the gasket manufacturer be consulted regarding gasket selection. Upon request, gasket manufacturers typically provide assistance in determining the proper material selection and the proper gasket style to ensure an economical choice and a reliable system.

When appropriate for the commodity, elastomer sheet gaskets without fillers are generally the least expensive gasket type. They are

typically limited to Class 150 and temperatures below 120°C (250°F). Composition sheet gaskets are somewhat more expensive than elastomer sheet gaskets. They are generally composed of an elastomer binder with fiber filler. Their use is generally limited to Class 150 and Class 300, and depending on the filler selected the upper temperature limit typically ranges between 205°C (400°F) and 370°C (700°F). Nonelastomer sheet gaskets, such as graphite sheet gaskets, are generally somewhat more expensive than composition sheet gaskets. Their use is typically limited to Class 150 and Class 300, and the upper temperature limit may be 535°C (1000°F) or higher. Spiral-wound gaskets with graphite, PTFE, or other filler are generally appropriate

**TABLE 10-26 Dimensions of ASME B16.5 Class 600 Flanges\***

All dimensions in inches

| Nominal pipe size | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |
|-------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|
|                   |                            |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck |
| 1/2               | 3.75                       | 0.56                         | 2.62                    | 1/2               | 4            | 0.88                            | 0.88      | 2.06         |
| 3/4               | 4.62                       | 0.62                         | 3.25                    | 5/8               | 4            | 1.00                            | 1.00      | 2.25         |
| 1                 | 4.88                       | 0.69                         | 3.50                    | 5/8               | 4            | 1.06                            | 1.06      | 2.44         |
| 1 1/4             | 5.25                       | 0.81                         | 3.88                    | 5/8               | 4            | 1.12                            | 1.12      | 2.62         |
| 1 1/2             | 6.12                       | 0.88                         | 4.50                    | 3/4               | 4            | 1.25                            | 1.25      | 2.75         |
| 2                 | 6.50                       | 1.00                         | 5.00                    | 3/4               | 8            | 1.44                            | 1.44      | 2.88         |
| 2 1/2             | 7.50                       | 1.12                         | 5.88                    | 3/4               | 8            | 1.62                            | 1.62      | 3.12         |
| 3                 | 8.25                       | 1.25                         | 6.62                    | 3/4               | 8            | 1.81                            | 1.81      | 3.25         |
| 3 1/2             | 9.00                       | 1.38                         | 7.25                    | 7/8               | 8            | 1.94                            | 1.94      | 3.38         |
| 4                 | 10.75                      | 1.50                         | 8.50                    | 7/8               | 8            | 2.12                            | 2.12      | 4.00         |
| 5                 | 13.00                      | 1.75                         | 10.50                   | 1                 | 8            | 2.38                            | 2.38      | 4.50         |
| 6                 | 14.00                      | 1.88                         | 11.50                   | 1                 | 12           | 2.62                            | 2.62      | 4.62         |
| 8                 | 16.50                      | 2.19                         | 13.75                   | 1 1/4             | 12           | 3.00                            | 3.00      | 5.25         |
| 10                | 20.00                      | 2.50                         | 17.00                   | 1 1/4             | 16           | 3.38                            | 4.38      | 6.00         |
| 12                | 22.00                      | 2.62                         | 19.25                   | 1 1/4             | 20           | 3.62                            | 4.62      | 6.12         |
| 14                | 23.75                      | 2.75                         | 20.75                   | 1 3/4             | 20           | 3.69                            | 5.00      | 6.50         |
| 16                | 27.00                      | 3.00                         | 23.75                   | 1 1/2             | 20           | 4.19                            | 5.50      | 7.00         |
| 18                | 29.25                      | 3.25                         | 25.75                   | 1 3/4             | 20           | 4.62                            | 6.00      | 7.25         |
| 20                | 32.00                      | 3.50                         | 28.50                   | 1 3/4             | 24           | 5.00                            | 6.50      | 7.50         |
| 24                | 37.00                      | 4.00                         | 33.00                   | 1 3/4             | 24           | 5.50                            | 7.25      | 8.00         |

\*Dimensions from ASME B16.5-2003. To convert inches to millimeters, multiply by 25.4.

**TABLE 10-27 Dimensions of ASME B16.5 Class 900 Flanges\***

All dimensions in inches

| Nominal pipe size                                   | Outside diameter of flange                | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |
|-----------------------------------------------------|-------------------------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|
|                                                     |                                           |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck |
| 1/2<br>3/4<br>1<br>1 1/4<br><br>1 1/2<br>2<br>2 1/2 | Use Class 1500 dimensions in these sizes. |                              |                         |                   |              |                                 |           |              |
| 3                                                   | 9.50                                      | 1.50                         | 7.50                    | 7/8               | 8            | 2.12                            | 2.12      | 4.00         |
| 4                                                   | 11.50                                     | 1.75                         | 9.25                    | 1 1/8             | 8            | 2.75                            | 2.75      | 4.50         |
| 5                                                   | 13.75                                     | 2.00                         | 11.00                   | 1 1/4             | 8            | 3.12                            | 3.12      | 5.00         |
| 6                                                   | 15.00                                     | 2.19                         | 12.50                   | 1 1/4             | 12           | 3.38                            | 3.38      | 5.50         |
| 8                                                   | 18.50                                     | 2.50                         | 15.50                   | 1 3/4             | 12           | 4.00                            | 4.50      | 6.38         |
| 10                                                  | 21.50                                     | 2.75                         | 18.50                   | 1 3/4             | 16           | 4.25                            | 5.00      | 7.25         |
| 12                                                  | 24.00                                     | 3.12                         | 21.00                   | 1 3/4             | 20           | 4.62                            | 5.62      | 7.88         |
| 14                                                  | 25.25                                     | 3.38                         | 22.00                   | 1 1/2             | 20           | 5.12                            | 6.12      | 8.38         |
| 16                                                  | 27.75                                     | 3.50                         | 24.25                   | 1 3/4             | 20           | 5.25                            | 6.50      | 8.50         |
| 18                                                  | 31.00                                     | 4.00                         | 27.00                   | 1 3/4             | 20           | 6.00                            | 7.50      | 9.00         |
| 20                                                  | 33.75                                     | 4.25                         | 29.50                   | 2                 | 20           | 6.25                            | 8.25      | 9.75         |
| 24                                                  | 41.00                                     | 5.50                         | 35.50                   | 2 1/2             | 20           | 8.00                            | 10.50     | 11.50        |

\*Dimensions from ASME B16.5-2003. To convert inches to millimeters, multiply by 25.4.

for applications more demanding than those handled by sheet gaskets. They are generally more expensive than sheet gaskets and are commonly used in Class 150 through Class 1500 services (and higher) class ratings. Because of their breadth of capabilities and the advantages of standardizing, they are often used when less expensive gaskets would suffice. The solid metal outer ring on spiral-wound gaskets serves to center the gasket and provide blowout resistance. With the proper filler, spiral-wound gaskets and some sheet gaskets provide good sealing under fire conditions.

**Ring Joint Flanges** Ring joint (RTJ) flanges provide sealing capability for pressure-temperature combinations higher than those for which spiral-wound gaskets are typically used. Depending on the

service, use of RTJ flanges is often considered in Class 900 and higher applications. RTJ flange facings and gaskets are more expensive than the spiral-wound counterparts. The ring material must be softer than the flange seating surface, and corrosion-resistant to the service. They provide good resistance to leakage under fire conditions. RTJ flanges must be separated in the axial direction to permit insertion and removal of the gasket.

**Bolting** Bolt strength requirements are addressed to some extent by the code and by code-referenced flange standards. Bolts are categorized by the code as high strength, intermediate strength, and low strength. Bolting materials having allowable stresses meeting or exceeding those of ASTM A193 Grade B7 are categorized as high

**TABLE 10-28 Dimensions of ASME B16.5 Class 1500 Flanges\***

All dimensions in inches

| Nominal pipe size | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub              |           |              |
|-------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|---------------------------------|-----------|--------------|
|                   |                            |                              |                         |                   |              | Threaded slip-on socket welding | Lap joint | Welding neck |
| 1/2               | 4.75                       | 0.88                         | 3.25                    | 3/4               | 4            | 1.25                            | 1.25      | 2.38         |
| 3/4               | 5.12                       | 1.00                         | 3.50                    | 3/4               | 4            | 1.38                            | 1.38      | 2.75         |
| 1                 | 5.88                       | 1.12                         | 4.00                    | 7/8               | 4            | 1.62                            | 1.62      | 2.88         |
| 1 1/4             | 6.25                       | 1.12                         | 4.38                    | 7/8               | 4            | 1.62                            | 1.62      | 2.88         |
| 1 1/2             | 7.00                       | 1.25                         | 4.88                    | 1                 | 4            | 1.75                            | 1.75      | 3.25         |
| 2                 | 8.50                       | 1.50                         | 6.50                    | 7/8               | 8            | 2.25                            | 2.25      | 4.00         |
| 2 1/2             | 9.62                       | 1.62                         | 7.50                    | 1                 | 8            | 2.50                            | 2.50      | 4.12         |
| 3                 | 10.50                      | 1.88                         | 8.00                    | 1 1/8             | 8            | 2.88                            | 2.88      | 4.62         |
| 4                 | 12.25                      | 2.12                         | 9.50                    | 1 1/4             | 8            | 3.56                            | 3.56      | 4.88         |
| 5                 | 14.75                      | 2.88                         | 11.50                   | 1 1/2             | 8            | 4.12                            | 4.12      | 6.12         |
| 6                 | 15.50                      | 3.25                         | 12.50                   | 1 3/4             | 12           | 4.69                            | 4.69      | 6.75         |
| 8                 | 19.00                      | 3.62                         | 15.50                   | 1 3/4             | 12           | 5.62                            | 5.62      | 8.38         |
| 10                | 23.00                      | 4.25                         | 19.00                   | 1 3/4             | 12           | 6.25                            | 7.00      | 10.00        |
| 12                | 26.50                      | 4.88                         | 22.50                   | 2                 | 16           | 7.12                            | 8.62      | 11.12        |
| 14                | 29.50                      | 5.25                         | 25.00                   | 2 1/4             | 16           |                                 | 9.50      | 11.75        |
| 16                | 32.50                      | 5.75                         | 27.75                   | 2 1/2             | 16           |                                 | 10.25     | 12.25        |
| 18                | 36.00                      | 6.38                         | 30.50                   | 2 3/4             | 16           |                                 | 10.88     | 12.88        |
| 20                | 38.75                      | 7.00                         | 32.75                   | 3                 | 16           |                                 | 11.50     | 14.00        |
| 24                | 46.00                      | 8.00                         | 39.00                   | 3 1/2             | 16           |                                 | 13.00     | 16.00        |

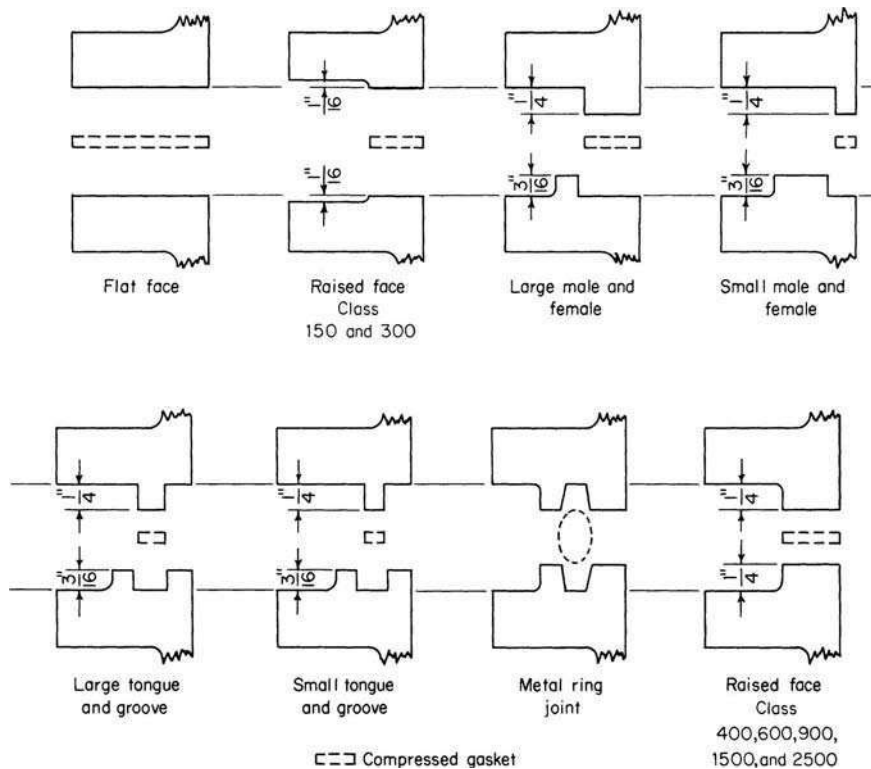
\*Dimensions from ASME B16.5-2003. To convert inches to millimeters, multiply by 25.4.

**TABLE 10-29 Dimensions of ASME B16.5 Class 2500\***

All dimensions in inches

| Nominal pipe size | Outside diameter of flange | Thickness of flange, minimum | Diameter of bolt circle | Diameter of bolts | No. of bolts | Length through hub |           |              |
|-------------------|----------------------------|------------------------------|-------------------------|-------------------|--------------|--------------------|-----------|--------------|
|                   |                            |                              |                         |                   |              | Threaded           | Lap joint | Welding neck |
| 1/2               | 5.25                       | 1.19                         | 3.50                    | 3/4               | 4            | 1.56               | 1.56      | 2.88         |
| 3/4               | 5.50                       | 1.25                         | 3.75                    | 3/4               | 4            | 1.69               | 1.69      | 3.12         |
| 1                 | 6.25                       | 1.38                         | 4.25                    | 7/8               | 4            | 1.88               | 1.88      | 3.50         |
| 1 1/4             | 7.25                       | 1.50                         | 5.12                    | 1                 | 4            | 2.06               | 2.06      | 3.75         |
| 1 1/2             | 8.00                       | 1.75                         | 5.75                    | 1 1/8             | 4            | 2.38               | 2.38      | 4.38         |
| 2                 | 9.25                       | 2.00                         | 6.75                    | 1                 | 8            | 2.75               | 2.75      | 5.00         |
| 2 1/2             | 10.50                      | 2.25                         | 7.75                    | 1 1/8             | 8            | 3.12               | 3.12      | 5.62         |
| 3                 | 12.00                      | 2.62                         | 9.00                    | 1 1/4             | 8            | 3.62               | 3.62      | 6.62         |
| 4                 | 14.00                      | 3.00                         | 10.75                   | 1 1/2             | 8            | 4.25               | 4.25      | 7.50         |
| 5                 | 16.50                      | 3.62                         | 12.75                   | 1 3/4             | 8            | 5.12               | 5.12      | 9.00         |
| 6                 | 19.00                      | 4.25                         | 14.50                   | 2                 | 8            | 6.00               | 6.00      | 10.75        |
| 8                 | 21.75                      | 5.00                         | 17.25                   | 2                 | 12           | 7.00               | 7.00      | 12.50        |
| 10                | 26.50                      | 6.50                         | 21.25                   | 2 1/2             | 12           | 9.00               | 9.00      | 16.50        |
| 12                | 30.00                      | 7.25                         | 24.38                   | 2 3/4             | 12           | 10.00              | 10.00     | 18.25        |

\*Dimensions from ASME B16.5-2003. To convert inches to millimeters, multiply by 25.4.



**FIG. 10-137** Flange facings, illustrated on welding-neck flanges. (On small male-and-female facings the outside diameter of the male face is less than the outside diameter of the pipe, so this facing does not apply to screwed or slip-on flanges. A similar joint can be made with screwed flanges and threaded pipe by projecting the pipe through one flange and recessing it in the other. However, pipe thicker than Schedule 40 is required to avoid crushing gaskets.) To convert inches to millimeters, multiply by 25.4.



strength. Bolting materials having specified minimum yield strengths of 207 MPa (30 ksi) or less are categorized as low strength. ASTM A307 Grade B is a commonly used specification for low-strength bolting.

The suitability of the strength of any bolting throughout the required temperature range should be verified by the designer. Verification of the suitability of intermediate-strength bolting for the intended joint design is required prior to its use. The code restricts the use of low-strength bolting to nonmetallic gaskets and flanges rated ASME B16.5 Class 300 and lower having bolt temperatures at  $-29$  to  $204^{\circ}\text{C}$  ( $-20$  to  $400^{\circ}\text{F}$ ) inclusive. Low-strength bolting is not permitted for use under severe cyclic conditions as defined by the code.

Except when bolting brittle flange materials such as gray cast iron, the code permits the use of high-strength bolting for any style of flanged joint and gasket type combination. Per the code, if either mating flange is specified in accordance with ASME B16.1, ASME B16.24, MSS SP-42, or MSS SP-51, the bolting material shall be no stronger than low-strength unless both mating flanges have a flat face and a full-face gasket is used. Exception to this requirement is permitted if sequence and torque limits for bolt-up are specified, with consideration given to sustained and occasional loads, and displacement strains. When both flanges are flat face and the gasket is full face extending to the outside diameter of the flange, intermediate-strength and high-strength bolts may be used.

### Miscellaneous Mechanical Joints

**Packed-Gland Joints** These joints (Fig. 10-138) require no special end preparation of pipe but do require careful control of the diameter of the pipe. Thus the supplier of the pipe should be notified when packed-gland joints are to be used. Cast- and ductile-iron pipe, fittings, and valves are available with the bell cast on one or more ends. Glands, bolts, and gaskets are shipped with the pipe. Couplings equipped with packed glands at each end, known as Dresser couplings, are available in several metals. The joints can be assembled with small wrenches and unskilled labor, in limited space, and if necessary, under water.

Packed-gland joints are designed to take the same hoop stress as the pipe. They do not resist bending moments or axial forces tending to separate the joints but yield to them to an extent indicated by the manufacturer's allowable-angular-deflection and end movement specifications. Further angular or end movement produces leakage, but end movement can be limited by harnessing or bridling with a combination of rods and welded clips or clamps, or by anchoring to existing or new structures. The crevice between the bell and the spigot may promote corrosion. The joints are widely used in underground lines. They are not affected by limited earth settlement, and friction of the earth is often adequate to prevent end separation. When disassembly by moving pipe axially is not practical, packed-joint couplings which can be slid entirely onto one of the two lengths joined are available.

**Poured Joints** Figure 10-139 illustrates a poured joint design. With regard to performance and ease of installation, most other joint

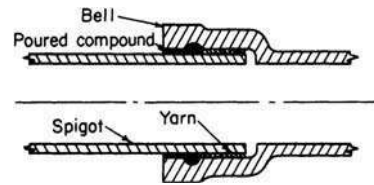


FIG. 10-139 Poured joint.

designs are preferable to poured joints, and their use can generally be avoided.

**Push-on Joints** These joints (Fig. 10-140) require diametral control of the end of the pipe. They are used for brittle and nonmetallic materials. Pipe, fittings, and valves are furnished with the bells on one or more ends.

Push-on joints do not resist bending moments or axial forces tending to separate the joints but yield to them to an extent limited by the manufacturer's allowable-angular-deflection and end-movement specifications. End movement can be limited by harnessing or bridling with a combination of rods and clamps, or by anchoring to existing or new structures. Some manufacturers offer O-rings with metallic embedments that grip the pipe and prevent axial separation under internal pressure loading. The joints are widely used on underground lines. They are not affected by limited earth settlement, and friction of the earth is often adequate to prevent end separation. A lubricant is used on the O ring during assembly. After this disappears, the O ring bonds somewhat to the spigot and disassembly is very difficult. Disassembly for maintenance is accomplished by cutting the pipe and reassembly by use of a coupling with a packed-gland joint on each end.

**Expanded Joints** These joints (Fig. 10-141) are confined to the smaller pipe sizes and ductile metals. Various proprietary designs are available in which either the pipe is expanded into the coupling or the

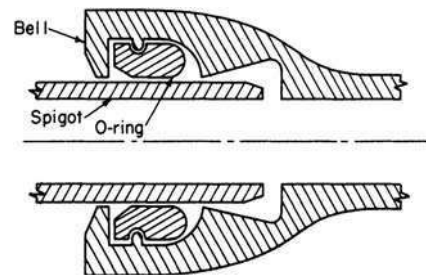


FIG. 10-140 Push-on joint.

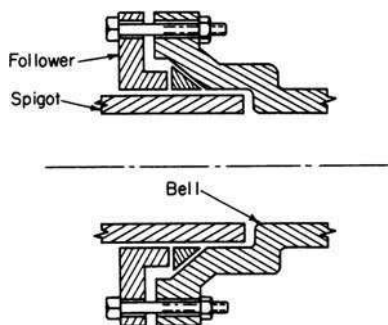


FIG. 10-138 Packed-gland joint.

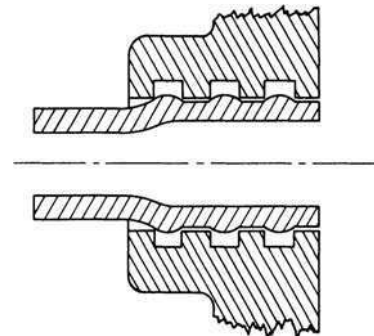


FIG. 10-141 Expanded joint.



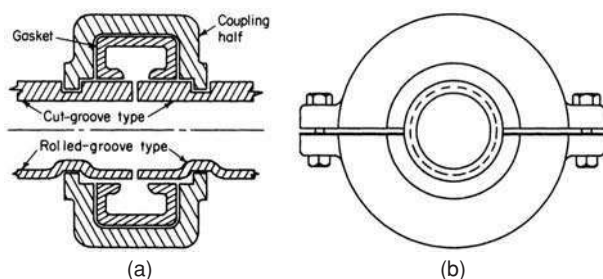


FIG. 10-142 Grooved joint. (a) Section. (b) End view.

coupling is crimped down onto the pipe. In some designs, the seal between the pipe and coupling is metal to metal, while in others elastomer O-rings are employed. Joints of these types typically are quickly and easily made with specialized equipment, and they may be particularly attractive in maintenance applications since no welding is involved. The designer should clearly understand the limitations of the joint design and should verify the success of its long-term service in similar applications.

**Grooved Joints** These joints (Fig. 10-142) are divided into two classes, cut grooves and rolled grooves. Rolled grooves are preferred because, compared with cut grooves, they are easier to form and reduce the metal wall less. However, they slightly reduce the flow area. They are limited to thin walls of ductile material, while cut grooves, because of their reduction of the pipe wall, are limited to thicker walls. In the larger pipe sizes, some commonly used wall thicknesses are too thick for rolled grooves but too thin for cut grooves. The thinning of the walls impairs resistance to corrosion and erosion but not to internal pressure, because the thinned area is reinforced by the coupling.

Control of outside diameter is important. Permissible minus tolerance is limited, since it impairs the grip of the couplings. Plus tolerance makes it necessary to cut the cut grooves more deeply, increasing the thinning of the wall. Plus tolerance is not a problem with rolled grooves, since they are confined to walls thin enough so that the couplings can compress the pipe. Pipe is available from vendors already grooved and also with heavier-wall grooved ends welded on.

Grooved joints resist axial forces tending to separate the joints. Angular deflection, up to the limit specified by the manufacturer, may be used to absorb thermal expansion and to permit the piping to be laid on uneven ground. Grooved joints provide quick and easy assembly and disassembly when compared with flanges, but may require more support than welded joints.

Gaskets are self-sealing against both internal and external pressure and are available in a wide variety of elastomers. However, successful performance of an elastomer as a flange gasket does not necessarily mean equally satisfactory performance in a grooved joint, since exposure to the fluid in the latter is much greater and hardening has a greater unfavorable effect. It is advisable to select coupling material that is suitably corrosion-resistant with respect to the service; but with proper gasket style it may be permissible to use a coupling material that might otherwise be unacceptable with respect to fluid contamination.

**V-Clamp Joints** These joints (Fig. 10-143) are attached to the pipe by butt-weld or expanded joints. Theoretically, there is only one relative position of the parts in which the conical surfaces of the clamp are completely in contact with the conical surfaces of the stub ends. In actual practice, there is considerable flexing of the stub ends and the clamp; also complete contact is not required. This permits use of elastomeric gaskets as well as metal gaskets. Fittings are also available with integral conical shouldered ends.

Conical ends vary from machined forgings to roll-formed tubing, and clamps vary from machined forgings to bands to which several roll-formed channels are attached at their centers by spot welding. A hinge may be inserted in the band as a substitute for one of the draw bolts. Latches may also be substituted for draw bolts.

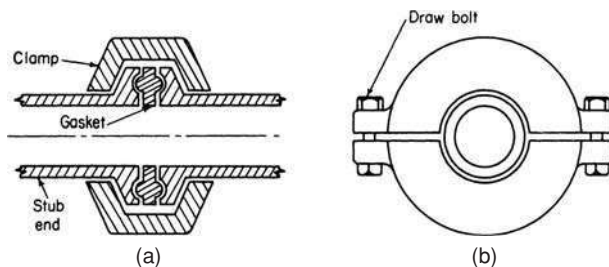


FIG. 10-143 V-clamp joint. (a) Section. (b) End view.

Compared with flanges, V-clamp joints use less metal, require less labor for assembly, and are less likely to leak under wide-range rapid temperature cycling. However, they are more susceptible to failure or damage from overtightening. They are widely used for high-alloy piping subject to periodic cleaning or relocation. Manufactured as forgings, they are used in carbon steel with metal gaskets for very high pressures. They resist both axial strain and bending moments. Each size of each type of joint is customarily rated by the vendor for both internal pressure and bending moment.

**Seal Ring Joints** These joints (Fig. 10-144) consist of hubs that are attached to pipe ends by welding. Joints of this type are proprietary, and their pressure/temperature ratings and external loading capabilities are established by the manufacturers. Variations of this design are offered by various manufacturers. Many of these designs have been widely used in critical high-pressure/temperature applications. They are particularly cost-effective in high-pressure alloy material applications.

**Pressure-Seal Joints** These joints (Fig. 10-145) are used for pressures of ASME Class 600 and higher. They use less metal than flanged joints but require much more machining of surfaces. There are several designs, in all of which increasing fluid pressure increases the force holding the sealing surfaces against each other. These joints are widely used as bonnet joints in carbon and alloy steel valves.

**Tubing Joints** **Flared-fitting joints** (see Fig. 10-146) are used for ductile tubing when the ratio of wall thickness to the diameter is small enough to permit flaring without cracking the inside surface. The tubing must have a smooth interior surface. The three-piece type avoids torsional strain on the tubing and minimizes vibration fatigue on the flared portion of the tubing. More labor is required for assembly, but the fitting is more resistant to temperature cycling than other tubing fittings and is less likely to be damaged by overtightening. Its efficiency is not impaired by repeated assembly and disassembly. Size is limited because of the large number of machined surfaces. The nut and, in the three-piece type, the sleeve need not be of the same material as the tubing. For these fittings, less control of tubing diameter is required.

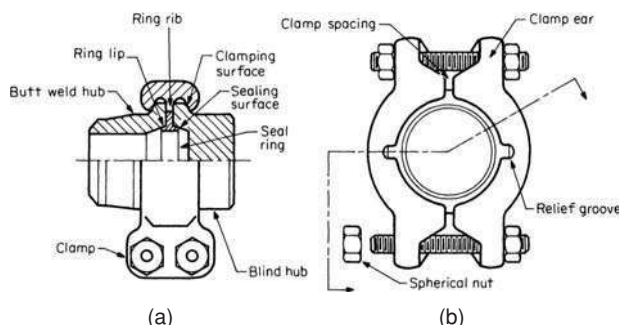


FIG. 10-144 Seal-ring joint. (Courtesy of Gray Tool Co.)

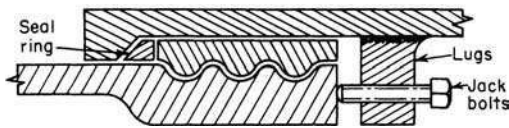


FIG. 10-145 Pressure-seal joint.

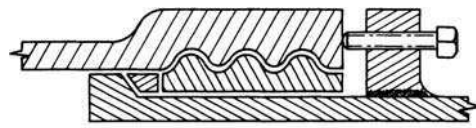
**Compression-Fitting Joints** These joints (Fig. 10-147) are used for ductile tubing with thin walls. The outside of the tubing must be clean and smooth. Assembly consists only of inserting the tubing and tightening the nut. These are the least costly tubing fittings but are not resistant to vibration or temperature cycling.

**Bite-Type-Fitting Joints** These joints (Fig. 10-148) are used when the tubing has too high a ratio of wall thickness to diameter for flaring, when the tubing lacks sufficient ductility for flaring, and for low assembly-labor cost. The outside of the tubing must be clean and smooth. Assembly consists in merely inserting the tubing and tightening the nut. The sleeve must be considerably harder than the tubing yet still ductile enough to be diametrically compressed and must be as resistant to corrosion by the fluid handled as the tubing. The fittings are resistant to vibration but not to wide-range rapid temperature cycling. Compared with flared fittings, they are less suited for repeated assembly and disassembly, require closer diametral control of the tubing, and are more susceptible to damage from overtightening. They are widely used for oil-filled hydraulic systems at all pressures.

**O-ring Seal Joints** These joints (Fig. 10-149) are also used for applications requiring heavy-wall tubing. The outside of the tubing must be clean and smooth. The joint may be assembled repeatedly, and as long as the tubing is not damaged, leaks can usually be corrected by replacing the O ring and the antiextrusion washer. This joint is used extensively in oil-filled hydraulic systems.

**Soldered Joints** These joints (Fig. 10-150) require precise control of the diameter of the pipe or tubing and of the cup or socket in the fitting in order to cause the solder to draw into the clearance between the cup and the tubing by capillary action (Fig. 10-150). Extrusion provides this diametral control, and the joints are most widely used in copper. A 50 percent lead, 50 percent tin solder is used for temperatures up to 93°C (200°F). Careful cleaning of the outside of the tubing and inside of the cup is required.

Heat for soldering is usually obtained from torches. The high conductivity of copper makes it necessary to use large flames for the larger sizes, and for this reason the location in which the joint will be made must be carefully considered. Soldered joints are most widely used in sizes 2 in and smaller for which heat requirements are less burdensome. Soldered joints should not be used in areas where plant fires are likely because exposure to fires results in rapid and complete failure of the joints. Properly made, the joints are completely impervious. The code permits the use of soldered joints only for Category D fluid service and then only if the system is not subject to severe cyclic conditions.



**Silver Brazed Joints** These are similar to soldered joints except that a temperature of about 600°C (1100°F) is required. A 15 percent silver, 80 percent copper, 5 percent phosphorus solder is used for copper and copper alloys, while 45 percent silver, 15 percent copper, 16 percent zinc, 24 percent cadmium solders are used for copper, copper alloys, carbon steel, and alloy steel. Silver-brazed joints are used for temperatures up to 200°C (400°F). Cast-bronze fittings and valves with preinserted rings of 15 percent silver, 80 percent copper, 5 percent phosphorus brazing alloy are available.

Silver-brazed joints are used when temperature or the combination of temperature and pressure is beyond the range of soldered joints. They are also more reliable in the event of plant fires and are more resistant to vibration. If they are used for fluids that are flammable, toxic, or damaging to human tissue, appropriate safeguarding is required by the code. There are OSHA regulations governing the use of silver brazing alloys containing cadmium and other toxic materials.

**Pipe Fittings and Bends** Directional changes in piping systems are typically made with bends or welded fittings. Bends are made as either hot bends or cold bends. Cold bending is done at temperatures below the material transformation temperature. Depending on the material and the amount of strain involved, annealing or stress relief may be required after bending. The bend radius that may be achieved for pipe of a given size, material, and thickness depends upon the bending machine capabilities and bending procedures used. When contemplating bending, the bending limitations should be reviewed with the pipe fabricators being considered for the project. Because bends are not generally made to radii as small as those of standard butt-weld or socket-weld fittings, the use of bends must be considered during piping layout. Wall thinning resulting from bending must also be considered when specifying the wall thickness of material that is to be bent. A detailed bending specification that addresses all aspects of bending, including requirements for bending procedure specifications, availability of bending procedure qualification records and bending operator qualification records, the range of bends covered by a single bending procedure qualification, in-process nondestructive examination requirements (including minimum wall thickness verification), dimensional tolerance requirements, etc., should be part of the bending agreement. Some bending operations and subsequent heat treatment can result in tenacious oxide formation on certain materials (such as 9Cr-1Mo-V). Removal of this oxide by conventional means such as abrasive blasting may be very difficult. Methods of avoiding this formation or of removing it should be discussed prior to bending when the application requires a high level of cleanliness, such as is the case with steam supply lines to turbines.

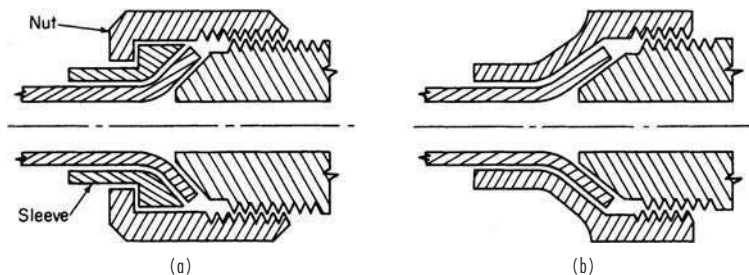


FIG. 10-146 Flared-fitting joint. (a) Three-piece. (b) Two-piece.

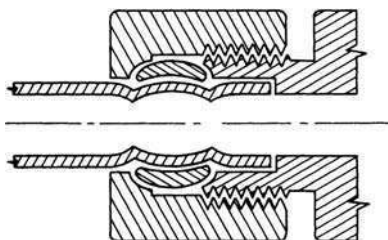


FIG. 10-147 Compression-fitting joint.

**Elbow Fittings** These fittings may be cast, forged, or hot- or cold-formed from short pieces of pipe or made by welding together pieces of miter-cut pipe. The thinning of pipe during the forming of elbows is compensated for by starting with heavier walls.

Flow in bends and elbow fittings is more turbulent than in straight pipe, thus increasing corrosion and erosion. This can be countered by selecting a component with greater radius of curvature, thicker wall, or smoother interior contour, but this is seldom economical in miter elbows.

Compared with elbow fittings, bends with a centerline radius of three or five nominal pipe diameters save the cost of joints and reduce pressure drop. It is sometimes difficult to nest bends of unequal pipe size when they lie in the same plane.

Flanged fittings are used when pipe is likely to be dismantled for frequent cleaning or extensive revision, or for lined piping systems. They are also used in areas where welding is not permitted. Cast fittings are usually flanged. Table 10-30 gives dimensions for flanged fittings.

Dimensions of carbon and alloy steel **butt-welding fittings** are shown in Table 10-31. Butt-welding fittings are available in the wall thicknesses shown in Table 10-22. Larger sizes and other wall thicknesses are also available. Schedule 5 and Schedule 10 stainless-steel butt-welding fittings are available with extensions for expanding into stainless-steel hubs mechanically locked in carbon steel ASME B16.5 dimension flanges. The use of expanded joints (Fig. 10-141) is restricted by the code.

Depending on the size, forged fittings are available with socket-weld (Fig. 10-129) or screwed ends in sizes 4 in and smaller; however, 2 in is the upper size limit normally used. ASME B16.11 gives minimum dimensions for Class 3000, 6000, and 9000 socket-weld fittings, and for Class 2000, 3000, and 6000 threaded fittings. The use of socket-weld and threaded fittings is restricted by the code.

Steel forged fittings with screwed ends may be installed without pipe dope in the threads and seal-welded (Fig. 10-133) to secure bubble-tight joints.

ASME B16.3–1998 gives pressure ratings and dimensions for Class 150 and Class 300 **malleable-iron threaded fittings**. Primary usage is 2 in and smaller; however, Class 150 fittings are available in 6 in and smaller, and Class 300 fittings are available in 3 in and smaller. Malleable-iron fittings are generally less expensive than forged carbon-

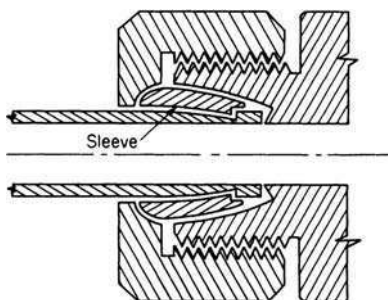


FIG. 10-148 Bite-type-fitting joint.

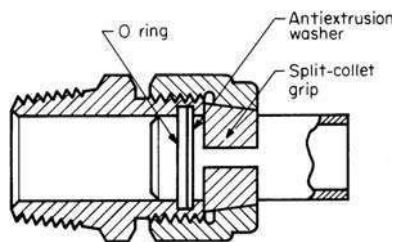


FIG. 10-149 O-ring seal joint. (Courtesy of the Lenz Co.)

steel fittings, but cannot be seal-welded. Threaded ends are typically female; however, male threads or a combination of male and female is available in some fittings. Among other restrictions, the code does not permit the use of malleable iron in severe cyclic conditions, in situations subject to thermal or mechanical shock, or in any fluid service below  $-29^{\circ}\text{C}$  ( $-20^{\circ}\text{F}$ ) or above  $343^{\circ}\text{C}$  ( $650^{\circ}\text{F}$ ). It also does not permit its use in flammable fluid service at temperatures above  $149^{\circ}\text{C}$  ( $300^{\circ}\text{F}$ ) or at gauge pressures above 2.76 MPa (400 lbf/in<sup>2</sup>). ASME B16.3 ratings for Class 150 fittings are 2.07 MPa (300 lbf/in<sup>2</sup>) at  $66^{\circ}\text{C}$  ( $150^{\circ}\text{F}$ ) and below and 1.03 MPa (150 lbf/in<sup>2</sup>) at  $186^{\circ}\text{C}$  ( $366^{\circ}\text{F}$ ). ASME B16.3 ratings for Class 300 fittings are size-dependent, but at least 6.89 MPa (1000 lbf/in<sup>2</sup>) at  $66^{\circ}\text{C}$  ( $150^{\circ}\text{F}$ ) and below and 2.07 MPa (300 lbf/in<sup>2</sup>) at  $288^{\circ}\text{C}$  ( $550^{\circ}\text{F}$ ).

ASME B16.4–1998 gives pressure ratings and dimensions for Class 125 and Class 250 **gray-iron (cast-iron) threaded fittings**. Threaded fittings in both classes are available in sizes 12 in and smaller; however, consideration should be given to other types of end connections prior to using threaded fittings in sizes larger than 2 in. Threaded ends are typically female. Cast-iron fittings are less expensive than forged carbon steel fittings, but cannot be seal-welded. The code places significant restrictions on the use of cast iron, and its use is typically limited to low-pressure, noncritical, nonflammable services. Its brittle nature should be considered before using it for compressed gas services. The minimum permissible design temperature is  $29^{\circ}\text{C}$  ( $-20^{\circ}\text{F}$ ). ASME B16.4 ratings for Class 125 fittings are 1.21 MPa (175 lbf/in<sup>2</sup>) at  $66^{\circ}\text{C}$  ( $150^{\circ}\text{F}$ ) and below and 0.86 MPa (125 lbf/in<sup>2</sup>) at  $178^{\circ}\text{C}$  ( $353^{\circ}\text{F}$ ). ASME B16.4 ratings for Class 250 fittings are 2.76 MPa (400 lbf/in<sup>2</sup>) at  $66^{\circ}\text{C}$  ( $150^{\circ}\text{F}$ ) and below and 1.72 MPa (250 lbf/in<sup>2</sup>) at  $208^{\circ}\text{C}$  ( $406^{\circ}\text{F}$ ).

**Tees** Tees may be cast, forged, or hot- or cold-formed from plate or pipe. Tees are typically stocked with both header (run) ends of the same size. In general, run ends of different sizes are not typically stocked or specified; however, occasionally run ends of different sizes are specified in threaded or socket-welded sizes. Branch connections may be full size or reducing sizes. Branch reductions two sizes smaller than the header are routinely stocked, and it is not typically difficult to purchase reducing tees with branches as small as those listed in ASME B16.9 (i.e., approximately one-half the header size). Economics, stress intensification factors, and nondestructive examination requirements typically dictate the branch connection type.

**Reducers** Reducers may be cast, forged, or hot- or cold-formed from pipe or plate. End connections may be concentric or eccentric, that is, tangent to the same plane at one point on their circumference. For pipe supported by hangers, concentric reducers permit maintenance

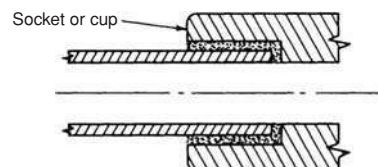
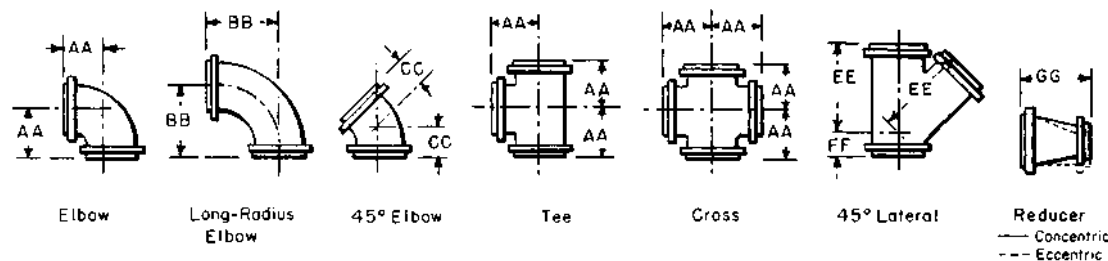


FIG. 10-150 Soldered, brazed, or cemented joint.

**TABLE 10-30 Dimensions of Flanged Fittings\***

All dimensions in inches



| Nominal<br>pipe size | ASME B16.5, Class 150<br>ASME B16.1, Class 125 |       |       |       |      |       | ASME B16.5, Class 300<br>ASME B16.1, Class 250 |       |       |       |       |       | ASME B16.5, Class 400                   |       |       |       |       | ASME B16.5, Class 600 |       |       |       |       |
|----------------------|------------------------------------------------|-------|-------|-------|------|-------|------------------------------------------------|-------|-------|-------|-------|-------|-----------------------------------------|-------|-------|-------|-------|-----------------------|-------|-------|-------|-------|
|                      | AA                                             | BB    | CC    | EE    | FF   | GG    | AA                                             | BB    | CC    | EE    | FF    | GG    | AA                                      | CC    | EE    | FF    | GG    | AA                    | CC    | EE    | FF    | GG    |
| ½                    |                                                |       |       |       |      |       |                                                |       |       |       |       |       | Use Class 600 dimensions in these sizes |       |       |       |       | 3.25                  | 2.00  | 5.75  | 1.75  | 5.00  |
| ¾                    |                                                |       |       |       |      |       |                                                |       |       |       |       |       |                                         |       |       |       |       | 3.75                  | 2.50  | 6.75  | 2.00  | 5.00  |
| 1                    | 3.50                                           | 5.00  | 1.75  | 5.75  | 1.75 | 4.50  | 4.00                                           | 5.00  | 2.25  | 6.50  | 2.00  | 4.50  |                                         |       |       |       |       | 4.25                  | 2.50  | 7.25  | 2.25  | 5.00  |
| 1¼                   | 3.75                                           | 5.50  | 2.00  | 6.25  | 1.75 | 4.50  | 4.25                                           | 5.50  | 2.50  | 7.25  | 2.25  | 4.50  |                                         |       |       |       |       | 4.50                  | 2.75  | 8.00  | 2.50  | 5.00  |
| 1½                   | 4.00                                           | 6.00  | 2.25  | 7.00  | 2.00 | 4.50  | 4.50                                           | 6.00  | 2.75  | 8.50  | 2.50  | 4.50  |                                         |       |       |       |       | 4.75                  | 3.00  | 9.00  | 2.75  | 5.00  |
| 2                    | 4.50                                           | 6.50  | 2.50  | 8.00  | 2.50 | 5.00  | 5.00                                           | 6.50  | 3.00  | 9.00  | 2.50  | 5.00  |                                         |       |       |       |       | 5.75                  | 4.25  | 10.25 | 3.50  | 6.00  |
| 2½                   | 5.00                                           | 7.00  | 3.00  | 9.50  | 2.50 | 5.50  | 5.50                                           | 7.00  | 3.50  | 10.50 | 2.50  | 5.50  |                                         |       |       |       |       | 6.50                  | 4.50  | 11.50 | 3.50  | 6.75  |
| 3                    | 5.50                                           | 7.75  | 3.00  | 10.00 | 3.00 | 6.00  | 6.00                                           | 7.75  | 3.50  | 11.00 | 3.00  | 6.00  |                                         |       |       |       |       | 7.00                  | 5.00  | 12.75 | 4.00  | 7.25  |
| 3½                   | 6.00                                           | 8.50  | 3.50  | 11.50 | 3.00 | 6.50  | 6.50                                           | 8.50  | 4.00  | 12.50 | 3.00  | 6.50  |                                         |       |       |       |       | 7.50                  | 5.50  | 14.00 | 4.50  | 7.75  |
| 4                    | 6.50                                           | 9.00  | 4.00  | 12.00 | 3.00 | 7.00  | 7.00                                           | 9.00  | 4.50  | 13.50 | 3.00  | 7.00  | 8.00                                    | 5.50  | 16.00 | 4.50  | 8.25  | 8.50                  | 6.00  | 16.50 | 4.50  | 8.75  |
| 5                    | 7.50                                           | 10.25 | 4.50  | 13.50 | 3.50 | 8.00  | 8.00                                           | 10.25 | 5.00  | 15.00 | 3.50  | 8.00  | 9.00                                    | 6.00  | 16.75 | 5.00  | 9.25  | 10.00                 | 7.00  | 19.50 | 6.00  | 10.25 |
| 6                    | 8.00                                           | 11.50 | 5.00  | 14.50 | 3.50 | 9.00  | 8.50                                           | 11.50 | 5.50  | 17.50 | 4.00  | 9.00  | 9.75                                    | 6.25  | 18.75 | 5.25  | 10.00 | 11.00                 | 7.50  | 21.00 | 6.50  | 11.25 |
| 8                    | 9.00                                           | 14.00 | 5.50  | 17.50 | 4.50 | 11.00 | 10.00                                          | 14.00 | 6.00  | 20.50 | 5.00  | 11.00 | 11.75                                   | 6.75  | 22.25 | 5.75  | 12.00 | 13.00                 | 8.50  | 24.50 | 7.00  | 13.25 |
| 10                   | 11.00                                          | 16.50 | 6.50  | 20.50 | 5.00 | 12.00 | 11.50                                          | 16.50 | 7.00  | 24.00 | 5.50  | 12.00 | 13.25                                   | 7.75  | 25.75 | 6.25  | 13.50 | 15.50                 | 9.50  | 29.50 | 8.00  | 15.75 |
| 12                   | 12.00                                          | 19.00 | 7.50  | 24.50 | 5.50 | 14.00 | 13.00                                          | 19.00 | 8.00  | 27.50 | 6.00  | 14.00 | 15.00                                   | 8.75  | 29.75 | 6.50  | 15.25 | 16.50                 | 10.00 | 31.50 | 8.50  | 16.75 |
| 14                   | 14.00                                          | 21.50 | 7.50  | 27.00 | 6.00 | 16.00 | 15.00                                          | 21.50 | 8.50  | 31.00 | 6.50  | 16.00 | 16.25                                   | 9.25  | 32.75 | 7.00  | 16.50 | 17.50                 | 10.75 | 34.25 | 9.00  | 17.75 |
| 16                   | 15.00                                          | 24.00 | 8.00  | 30.00 | 6.50 | 18.00 | 16.50                                          | 24.00 | 9.50  | 34.50 | 7.50  | 18.00 | 17.75                                   | 10.25 | 36.25 | 8.00  | 18.50 | 19.50                 | 11.75 | 38.50 | 10.00 | 19.75 |
| 18                   | 16.50                                          | 26.50 | 8.50  | 32.00 | 7.00 | 19.00 | 18.00                                          | 26.50 | 10.00 | 37.50 | 8.00  | 19.00 | 19.25                                   | 10.75 | 39.25 | 8.50  | 19.50 | 21.50                 | 12.25 | 42.00 | 10.50 | 21.75 |
| 20                   | 18.00                                          | 29.00 | 9.50  | 35.00 | 8.00 | 20.00 | 19.50                                          | 29.00 | 10.50 | 40.50 | 8.50  | 20.00 | 20.75                                   | 11.25 | 42.75 | 9.00  | 21.00 | 23.50                 | 13.00 | 45.50 | 11.00 | 23.75 |
| 24                   | 22.00                                          | 34.00 | 11.00 | 40.50 | 9.00 | 24.00 | 22.50                                          | 34.00 | 12.00 | 47.50 | 10.00 | 24.00 | 24.25                                   | 12.75 | 50.25 | 10.50 | 24.50 | 27.50                 | 14.75 | 53.00 | 13.00 | 27.75 |

**TABLE 10-30 Dimensions of Flanged Fittings\* (Concluded)**

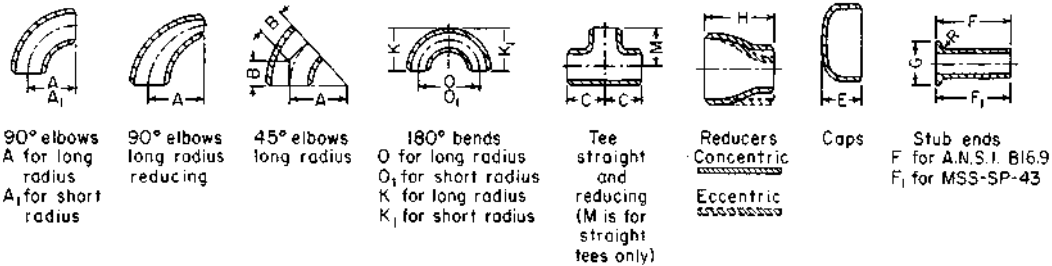
| Nominal<br>pipe size | ASME B16.5, Class 900                    |       |       |       |       | ASME B16.5, Class 1500 |       |       |       |       | ASME B16.5, Class 2500 |       |       |       |       |
|----------------------|------------------------------------------|-------|-------|-------|-------|------------------------|-------|-------|-------|-------|------------------------|-------|-------|-------|-------|
|                      | AA                                       | CC    | EE    | FF    | GG    | AA                     | CC    | EE    | FF    | GG    | AA                     | CC    | EE    | FF    | GG    |
| 1/2                  | Use Class 1500 dimensions in these sizes |       |       |       |       | 4.25                   | 3.00  |       |       |       | 5.19                   |       |       |       |       |
| 3/4                  |                                          |       |       |       |       | 4.50                   | 3.25  |       |       |       | 5.37                   |       |       |       |       |
| 1                    |                                          |       |       |       |       | 5.00                   | 3.50  | 9.00  | 2.50  | 5.00  | 6.06                   | 4.00  |       |       |       |
| 1 1/4                |                                          |       |       |       |       | 5.50                   | 4.00  | 10.00 | 3.00  | 5.75  | 6.87                   | 4.25  |       |       |       |
| 1 1/2                |                                          |       |       |       |       | 6.00                   | 4.25  | 11.00 | 3.50  | 6.25  | 7.56                   | 4.75  |       |       |       |
| 2                    |                                          |       |       |       |       | 7.25                   | 4.75  | 13.25 | 4.00  | 7.25  | 8.87                   | 5.75  | 15.25 | 5.25  | 9.50  |
| 2 1/2                |                                          |       |       |       |       | 8.25                   | 5.25  | 15.25 | 4.50  | 8.25  | 10.00                  | 6.25  | 17.25 | 5.75  | 10.50 |
| 3                    | 7.50                                     | 5.50  | 14.50 | 4.50  | 7.75  | 9.25                   | 5.75  | 17.25 | 5.00  | 9.25  | 11.37                  | 7.25  | 19.75 | 6.75  | 11.75 |
| 4                    | 9.00                                     | 6.50  | 17.50 | 5.50  | 9.25  | 10.75                  | 7.25  | 19.25 | 6.00  | 10.75 | 13.25                  | 8.50  | 23.00 | 7.75  | 13.50 |
| 5                    | 11.00                                    | 7.50  | 21.00 | 6.50  | 11.25 | 13.25                  | 8.75  | 23.25 | 7.50  | 13.75 | 15.62                  | 10.00 | 27.25 | 9.25  | 15.75 |
| 6                    | 12.00                                    | 8.00  | 22.50 | 6.50  | 12.25 | 13.88                  | 9.38  | 24.88 | 8.12  | 14.50 | 18.00                  | 11.50 | 31.25 | 10.50 | 18.00 |
| 8                    | 14.50                                    | 9.00  | 27.50 | 7.50  | 14.75 | 16.38                  | 10.88 | 29.88 | 9.12  | 17.00 | 20.12                  | 12.75 | 35.25 | 11.75 | 20.50 |
| 10                   | 16.50                                    | 10.00 | 31.50 | 8.50  | 16.75 | 19.50                  | 12.00 | 36.00 | 10.25 | 20.25 | 25.00                  | 16.00 | 43.25 | 14.75 | 25.50 |
| 12                   | 19.00                                    | 11.00 | 34.50 | 9.00  | 17.75 | 22.25                  | 13.25 | 40.75 | 12.00 | 23.00 | 28.00                  | 17.75 | 49.25 | 16.25 | 29.00 |
| 14                   | 20.25                                    | 11.50 | 36.50 | 9.50  | 19.00 | 24.75                  | 14.25 | 44.00 | 12.50 | 25.75 |                        |       |       |       |       |
| 16                   | 22.25                                    | 12.50 | 40.75 | 10.50 | 21.00 | 27.25                  | 16.25 | 48.25 | 14.75 | 28.25 |                        |       |       |       |       |
| 18                   | 24.00                                    | 13.25 | 45.50 | 12.00 | 24.50 | 30.25                  | 17.75 | 53.25 | 16.50 | 31.50 |                        |       |       |       |       |
| 20                   | 26.00                                    | 14.50 | 50.25 | 13.00 | 26.50 | 32.75                  | 18.75 | 57.75 | 17.75 | 34.00 |                        |       |       |       |       |
| 24                   | 30.50                                    | 18.00 | 60.00 | 15.50 | 30.50 | 38.25                  | 20.75 | 67.25 | 20.50 | 39.75 |                        |       |       |       |       |

\*Outline drawings show a 1/4-in (6.5-mm) raised face machined onto the flange for ASME Class 400 and higher. ASME B16.1 Class 250 and ASME B16.5 Classes 150 and 300 have a 1/16-in. (1.5-mm) raised face. ASME B16.1 Class 125 has no raised face. See Tables 10-23 through 10-29 for flange drillings. Dimensions for Class 400 and Class 600 fittings are identical for sizes 1/2 through 3 1/2 in. Dimensions for Class 900 and Class 1500 fittings are identical for sizes 1/2 through 2 1/2 in. To convert inches to millimeters, multiply by 25.4. The dimensions were extracted from *Cast Iron Pipe Flanges and Flanged Fittings*, ASME B16.1-1998, and *Pipe Flanges and Flanged Fittings*, ASME B16.5-2003, with permission of the publisher, the American Society of Mechanical Engineers, New York.



TABLE 10-31 Butt-Welding Fittings\*

All dimensions in inches



| Pipe size | A     | K     | A1    | K1    | B     | O     | O1    | M, C  | H     | E†    | G     | F     | F1   | R‡   |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
| ½         | 1.50  | 1.88  |       |       | 0.62  | 3.00  |       | 1.00  |       | 1.00  | 1.38  | 3.00  | 2.00 | 0.12 |
| ¾         | 1.50  | 2.00  |       |       | 0.75  | 3.00  |       | 1.12  | 1.50  | 1.00  | 1.69  | 3.00  | 2.00 | 0.12 |
| 1         | 1.50  | 2.19  | 1.00  | 1.62  | 0.88  | 3.00  | 2.00  | 1.50  | 2.00  | 1.50  | 2.00  | 4.00  | 2.00 | 0.12 |
| 1¼        | 1.88  | 2.75  | 1.25  | 2.06  | 1.00  | 3.75  | 2.50  | 1.88  | 2.00  | 1.50  | 2.50  | 4.00  | 2.00 | 0.19 |
| 1½        | 2.25  | 3.25  | 1.50  | 2.44  | 1.12  | 4.50  | 3.00  | 2.25  | 2.50  | 1.50  | 2.88  | 4.00  | 2.00 | 0.25 |
| 2         | 3.00  | 4.19  | 2.00  | 3.19  | 1.38  | 6.00  | 4.00  | 2.50  | 3.00  | 1.50  | 3.62  | 6.00  | 2.50 | 0.31 |
| 2½        | 3.75  | 5.19  | 2.50  | 3.94  | 1.75  | 7.50  | 5.00  | 3.00  | 3.50  | 1.50  | 4.12  | 6.00  | 2.50 | 0.31 |
| 3         | 4.50  | 6.25  | 3.00  | 4.75  | 2.00  | 9.00  | 6.00  | 3.38  | 3.50  | 2.00  | 5.00  | 6.00  | 2.50 | 0.38 |
| 3½        | 5.25  | 7.25  | 3.50  | 5.50  | 2.25  | 10.50 | 7.00  | 3.75  | 4.00  | 2.50  | 5.50  | 6.00  | 3.00 | 0.38 |
| 4         | 6.00  | 8.25  | 4.00  | 6.25  | 2.50  | 12.00 | 8.00  | 4.12  | 4.00  | 2.50  | 6.19  | 6.00  | 3.00 | 0.44 |
| 5         | 7.50  | 10.31 | 5.00  | 7.75  | 3.12  | 15.00 | 10.00 | 4.88  | 5.00  | 3.00  | 7.31  | 8.00  | 3.00 | 0.44 |
| 6         | 9.00  | 12.31 | 6.00  | 9.31  | 3.75  | 18.00 | 12.00 | 5.62  | 5.50  | 3.50  | 8.50  | 8.00  | 3.50 | 0.50 |
| 8         | 12.00 | 16.31 | 8.00  | 12.31 | 5.00  | 24.00 | 16.00 | 7.00  | 6.00  | 4.00  | 10.62 | 8.00  | 4.00 | 0.50 |
| 10        | 15.00 | 20.38 | 10.00 | 15.38 | 6.25  | 30.00 | 20.00 | 8.50  | 7.00  | 5.00  | 12.75 | 10.00 | 5.00 | 0.50 |
| 12        | 18.00 | 24.38 | 12.00 | 18.38 | 7.50  | 36.00 | 24.00 | 10.00 | 8.00  | 6.00  | 15.00 | 10.00 | 6.00 | 0.50 |
| 14        | 21.00 | 28.00 | 14.00 | 21.00 | 8.75  | 42.00 | 28.00 | 11.00 | 13.00 | 6.50  | 16.25 | 12.00 | 6.00 | 0.50 |
| 16        | 24.00 | 32.00 | 16.00 | 24.00 | 10.00 | 48.00 | 32.00 | 12.00 | 14.00 | 7.00  | 18.50 | 12.00 | 6.00 | 0.50 |
| 18        | 27.00 | 36.00 | 18.00 | 27.00 | 11.25 | 54.00 | 36.00 | 13.50 | 15.00 | 8.00  | 21.00 | 12.00 | 6.00 | 0.50 |
| 20        | 30.00 | 40.00 | 20.00 | 30.00 | 12.50 | 60.00 | 40.00 | 15.00 | 20.00 | 9.00  | 23.00 | 12.00 | 6.00 | 0.50 |
| 24        | 36.00 | 48.00 | 24.00 | 36.00 | 15.00 | 72.00 | 48.00 | 17.00 | 20.00 | 10.50 | 27.25 | 12.00 | 6.00 | 0.50 |

\*Extracted from *Factory-Made Wrought Butt-Welding Fittings*, ASME B16.9-2003, with permission of the publisher, the American Society of Mechanical Engineers, New York. O and K dimensions of 2.25 and 1.69 in respectively may be furnished for NPS ¾ at the manufacturer's option.

†For wall thicknesses greater than extra heavy, E is greater than shown here for sizes 2 in and larger.

‡For MSS SP-43 type B stub ends, which are designed to be backed up by slip-on flanges, R = ½ in for 4 in and smaller and ⅝ in for 6 through 12 in. To convert inches to millimeters multiply by 25.4.

of the same hanger length; for pipe laid on structural steel, eccentric reducers permit maintaining the same elevation of top of steel. Eccentric reducers with the common tangent plane on the bottom side permit complete drainage of branched horizontal piping systems. With the common tangent plane on the top side, they permit liquid flow in horizontal lines to sweep the line free of gas or vapor.

**Reducing Elbow Fittings** These permit change of direction and concentric size reduction in the same fitting.

**Valves** Valve bodies may be cast, forged, machined from bar stock, or fabricated from welded plate. Steel valves are available with screwed or socket-weld ends in the smaller sizes. Bronze and brass screwed-end valves are widely used for low-pressure service in steel systems. Table 10-32 gives contact-surface-of-face to contact-surface-of-face dimensions for flanged ferrous valves and end-to-end dimensions for butt-welding ferrous valves. Drilling of end flanges is shown in Tables 10-23 to 10-29. Bolt holes are located so that the stem is equidistant from the centerline of two bolt holes. Even if removal for maintenance is not anticipated, flanged valves are frequently used instead of butt-welding-end valves because they permit insertion of blanks for isolating sections of a loop piping system.

Ferrous valves are also available in nodular (ductile) iron, which has tensile strength and yield point approximately equal to cast carbon steel at temperatures of 343°C (650°F) and below and only slightly less elongation.

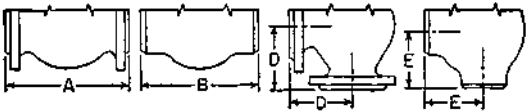
Valves serve not only to regulate the flow of fluids but also to isolate piping or equipment for maintenance without interrupting other connected units. Valve designers attempt to minimize body distortion due to pressure, changes in temperature, and applied loads. The sealing mechanisms of certain valve designs are inherently more tolerant of these factors than are others. The selection of valve type and materials of construction should provide a valve that functions reliably and that is acceptably tight across the sealing surfaces for the lowest lifetime cost. Valve manufacturers are a valuable source of information when evaluating the suitability of specific designs. The principal types are named, described, compared, and illustrated with line diagrams in subsequent subsections. In the line diagrams, the operating stem is shown in solid black, direction of flow by arrows on a thin solid line, and motion of valve parts by arrows on a dotted line. Moving parts are drawn with solid lines in the nearly closed position and with dotted lines in the fully open position. Packing is represented by an X in a square.

**Gate Valves** These valves are designed in two types (Fig. 10-151). The wedge-shaped-gate, **inclined-seat** type is most commonly used. The wedge gate may be solid or flexible (partly cut into halves by a plane at right angles to the pipe) or split (completely cleft by such a plane). Flexible and split wedges minimize galling of the sealing surfaces by distorting more easily to match angularly misaligned seats. In the double-disk **parallel-seat** type, an inclined-plane device mounted between the disks converts stem force to axial force, pressing the disks against the

10-94 TRANSPORT AND STORAGE OF FLUIDS

TABLE 10-32 Dimensions of Valves\*

All dimensions in inches



| Nominal valve size | Class 125 cast iron |               |                        |                        |               | Class 150 steel, MSS-SP-42 through 12-in size |                               |                              |                              |                     | Class 250 cast iron                |                                      |                        |
|--------------------|---------------------|---------------|------------------------|------------------------|---------------|-----------------------------------------------|-------------------------------|------------------------------|------------------------------|---------------------|------------------------------------|--------------------------------------|------------------------|
|                    | Flanged end         |               |                        |                        |               | Flanged end                                   | Welding end                   | Flanged end and welding end  |                              |                     | Flanged end                        |                                      |                        |
|                    | Gate                |               | Globe and lift check A | Angle and lift check D | Swing check A | Gate                                          | Gate                          | Globe and lift check A and B | Angle and lift check D and E | Swing check A and B | Gate Solid wedge and double disk A | Globe, lift check, and swing check A | Angle and lift check D |
|                    | Solid wedge A       | Double disk A |                        |                        |               | Solid wedge and double disk A                 | Solid wedge and double disk B |                              |                              |                     |                                    |                                      |                        |
| ¼                  |                     |               |                        |                        |               | 4                                             | 4                             | 4                            | 2                            | 4                   |                                    |                                      |                        |
| ⅜                  |                     |               |                        |                        |               | 4                                             | 4                             | 4                            | 2                            | 4                   |                                    |                                      |                        |
| ½                  |                     |               |                        |                        |               | 4¼                                            | 4¼                            | 4¼                           | 2¼                           | 4¼                  |                                    |                                      |                        |
| ¾                  |                     |               |                        |                        |               | 4½                                            | 4½                            | 4½                           | 2½                           | 4½                  |                                    |                                      |                        |
| 1                  |                     |               |                        |                        |               | 5                                             | 5                             | 5                            | 2¾                           | 5                   |                                    |                                      |                        |
| 1¼                 |                     |               |                        |                        |               | 5½                                            | 5½                            | 5½                           | 3                            | 5½                  |                                    |                                      |                        |
| 1½                 |                     |               |                        |                        |               | 6½                                            | 6½                            | 6½                           | 3¼                           | 6½                  |                                    |                                      |                        |
| 2                  | 7                   | 7             | 8                      | 4                      | 8             | 7                                             | 8½                            | 8                            | 4                            | 8                   | 8½                                 | 10½                                  | 5¼                     |
| 2½                 | 7½                  | 7½            | 8½                     | 4¼                     | 8½            | 7½                                            | 9½                            | 8½                           | 4¼                           | 8½                  | 9½                                 | 11½                                  | 5¾                     |
| 3                  | 8                   | 8             | 9½                     | 4¾                     | 9½            | 8                                             | 11¼                           | 9½                           | 4¾                           | 9½                  | 11¼                                | 12½                                  | 6¼                     |
| 3½                 | 8½                  | 8½            |                        |                        |               | †                                             |                               |                              |                              |                     | 12                                 | 14                                   | 7                      |
| 4                  | 9                   | 9             | 11½                    | 5¾                     | 11½           | 9                                             | 12                            | 11½                          | 5¾                           | 11½                 | 15                                 | 15¾                                  | 7½                     |
| 5                  | 10                  | 10            | 13                     | 6½                     | 13            | 10                                            | 15                            | 14                           | 7                            | 13                  | 15½                                | 17½                                  | 8¾                     |
| 6                  | 10½                 | 10½           | 14                     | 7                      | 14            | 10½                                           | 15½                           | 16                           | 8                            | 14                  | 16½                                | 21                                   | 10½                    |
| 8                  | 11½                 | 11½           | 19½                    | 9¾                     | 19½           | 11½                                           | 16½                           | 19½                          | 9¾                           | 19½                 | 18                                 | 24½                                  | 12¼                    |
| 10                 | 13                  | 13            | 24½                    | 12¼                    | 24½           | 13                                            | 18                            | 24½                          | 12¼                          | 24½                 | 19¾                                | 28                                   | 14                     |
| 12                 | 14                  | 14            | 27½                    | 13¾                    | 27½           | 14                                            | 19¾                           | 27½                          | 13¾                          | 27½                 | 22½                                | †                                    |                        |
| 14                 | 15                  | †             | 31                     | 15½                    | 31            | 15                                            | 22½                           | 31                           | 15½                          | 31                  | 24                                 | †                                    |                        |
| 16                 | 16                  | †             | 36                     | 18                     | †             | 16                                            | 24                            | 36                           | 18                           | †                   | 26                                 |                                      |                        |
| 18                 | 17                  | †             |                        |                        | †             | 17                                            | 26                            |                              |                              | †                   | 28                                 |                                      |                        |
| 20                 | 18                  | †             |                        |                        | †             | 18                                            | 28                            |                              |                              | †                   |                                    |                                      |                        |
| 24                 | 20                  | †             |                        |                        | †             | 20                                            | 32                            |                              |                              | †                   | 31                                 |                                      |                        |

| Nominal valve size | Class 300 steel                          |                              |                              |                     | Class 400 steel             |                     |                                            |                              | Class 600 steel             |                     |                  |                                                        |                                                  |                      |                 |
|--------------------|------------------------------------------|------------------------------|------------------------------|---------------------|-----------------------------|---------------------|--------------------------------------------|------------------------------|-----------------------------|---------------------|------------------|--------------------------------------------------------|--------------------------------------------------|----------------------|-----------------|
|                    | Flanged end and welding end              |                              |                              |                     | Flanged end and welding end |                     |                                            |                              | Flanged end and welding end |                     |                  |                                                        |                                                  |                      |                 |
|                    | Gate Solid wedge and double disk A and B | Globe and lift check A and B | Angle and lift check D and E | Swing check A and B | Gate                        |                     | Globe, lift check, and swing check A and B | Angle and lift check D and E | Gate                        |                     |                  | Regular globe, regular lift check, swing check A and B | Short pattern† globe, short pattern lift check B | Angle and lift check |                 |
|                    |                                          |                              |                              |                     | Solid wedge A and B         | Double disk A and B |                                            |                              | Solid wedge A and B         | Double disk A and B | Short pattern† B |                                                        |                                                  | Regular D and E      | Short pattern E |
| ½                  | 5½                                       | 6                            | 3                            |                     | 6½                          |                     | 6½                                         | 3¼                           | 6½                          |                     |                  | 6½                                                     |                                                  | 3¼                   |                 |
| ¾                  | 6                                        | 7                            | 3½                           |                     | 7½                          |                     | 7½                                         | 3¾                           | 7½                          |                     |                  | 7½                                                     |                                                  | 3¾                   |                 |
| 1                  | 6½                                       | 8                            | 4                            | 8½                  | 8½                          | 8½                  | 8½                                         | 4¼                           | 8½                          | 8½                  | 5¼               | 8½                                                     | 5¼                                               | 4¼                   |                 |
| 1¼                 | 7½                                       | 8½                           | 4¼                           | 9                   | 9                           | 9                   | 9                                          | 4½                           | 9                           | 9                   | 5¾               | 9                                                      | 5¾                                               | 4½                   |                 |
| 1½                 | 7½                                       | 9                            | 4½                           | 9½                  | 9½                          | 9½                  | 9½                                         | 4¾                           | 9½                          | 9½                  | 6                | 9½                                                     | 6                                                | 4¾                   |                 |
| 2                  | 8½                                       | 10½                          | 5¼                           | 10½                 | 11½                         | 11½                 | 11½                                        | 5¾                           | 11½                         | 11½                 | 7                | 11½                                                    | 7                                                | 5¾                   | 4¼              |
| 2½                 | 9½                                       | 11½                          | 5¾                           | 11½                 | 13                          | 13                  | 13                                         | 6½                           | 13                          | 13                  | 8½               | 13                                                     | 8½                                               | 6½                   | 5               |
| 3                  | 11½                                      | 12½                          | 6¼                           | 12½                 | 14                          | 14                  | 14                                         | 7                            | 14                          | 14                  | 10               | 14                                                     | 10                                               | 7                    | 6               |
| 4                  | 12                                       | 14                           | 7                            | 14                  | 16                          | 16                  | 16                                         | 8                            | 17                          | 17                  | 12               | 17                                                     | 12                                               | 8½                   | 7               |
| 5                  | 15                                       | 15¾                          | 7¾                           | 15¾                 | 18                          | 18                  | 18                                         | 9                            | 20                          | 20                  | 15               | 20                                                     | 15                                               | 10                   | 8½              |
| 6                  | 15½                                      | 17½                          | 8¾                           | 17½                 | 19½                         | 19½                 | 19½                                        | 9¾                           | 22                          | 22                  | 18               | 22                                                     | 18                                               | 11                   | 10              |
| 8                  | 16½                                      | 22                           | 11                           | 21                  | 23½                         | 23½                 | 23½                                        | 11¾                          | 26                          | 26                  | 23               | 26                                                     | 23                                               | 13                   |                 |
| 10                 | 18                                       | 24½                          | 12¼                          | 24½                 | 26½                         | 26½                 | 26½                                        | 13¼                          | 31                          | 31                  | 28               | 31                                                     | 28                                               | 15½                  |                 |
| 12                 | 19¾                                      | 28                           | 14                           | 28                  | 30                          | 30                  | 30                                         | 15                           | 33                          | 33                  | 32               | 33                                                     | 32                                               | 16½                  |                 |
| 14                 | 30                                       |                              |                              | †                   | 32½                         | 30½                 | †                                          |                              | 35                          | 35                  | 35               | †                                                      |                                                  |                      |                 |
| 16                 | 33                                       |                              |                              | †                   | 35½                         | 35½                 | †                                          |                              | 39                          | 39                  | 39               | †                                                      |                                                  |                      |                 |
| 18                 | 36                                       |                              |                              | †                   | 38½                         | 38½                 | †                                          |                              | 43                          | 43                  | 43               | †                                                      |                                                  |                      |                 |
| 20                 | 39                                       |                              |                              | †                   | 41½                         | 41½                 | †                                          |                              | 47                          | 47                  | 47               | †                                                      |                                                  |                      |                 |
| 22                 | 43                                       |                              |                              | †                   | 45                          | 45                  | †                                          |                              | 51                          | 51                  |                  | †                                                      |                                                  |                      |                 |
| 24                 | 45                                       |                              |                              | †                   | 48½                         | 48½                 | †                                          |                              | 55                          | 55                  | 55               | †                                                      |                                                  |                      |                 |

TABLE 10-32 Dimensions of Valves (Concluded)

| Nominal valve size | Class 900 steel             |                     |                  |                                                       |                                                  |                      | Class 1500 steel            |                     |                     |                  |                                        |                              |
|--------------------|-----------------------------|---------------------|------------------|-------------------------------------------------------|--------------------------------------------------|----------------------|-----------------------------|---------------------|---------------------|------------------|----------------------------------------|------------------------------|
|                    | Flanged end and welding end |                     |                  |                                                       |                                                  |                      | Flanged end and welding end |                     |                     |                  |                                        |                              |
|                    | Gate                        |                     |                  | Regular globe regular lift check, swing check A and B | Short pattern† globe, short pattern lift check B | Angle and lift check |                             | Gate                |                     |                  | Globe, lift check, swing check A and B | Angle and lift check D and E |
|                    | Solid wedge A and B         | Double disk A and B | Short pattern‡ B |                                                       |                                                  | Regular D and E      | Short pattern E             | Solid wedge A and B | Double disk A and B | Short pattern‡ B |                                        |                              |
| ¾                  |                             |                     |                  | 9                                                     |                                                  | 4½                   |                             |                     |                     |                  | 9                                      | 4½                           |
| 1                  | 10                          |                     | 5½               | 10                                                    |                                                  | 5                    |                             | 10                  |                     | 5½               | 10                                     | 5                            |
| 1¼                 | 11                          |                     | 6½               | 11                                                    |                                                  | 5½                   |                             | 11                  |                     | 6½               | 11                                     | 5½                           |
| 1½                 | 12                          |                     | 7                | 12                                                    |                                                  | 6                    |                             | 12                  |                     | 7                | 12                                     | 6                            |
| 2                  | 14½                         | 14½                 | 8½               | 14½                                                   |                                                  | 7¼                   |                             | 14½                 | 14½                 | 8½               | 14½                                    | 7¼                           |
| 2½                 | 16½                         | 16½                 | 10               | 16½                                                   |                                                  | 8¼                   |                             | 16½                 | 16½                 | 10               | 16½                                    | 8¼                           |
| 3                  | 15                          | 15                  | 12               | 15                                                    | 12                                               | 7½                   |                             | 18½                 | 18½                 | 12               | 18½                                    | 9¼                           |
| 4                  | 18                          | 18                  | 14               | 18                                                    | 14                                               | 9                    | 7                           | 21½                 | 21½                 | 16               | 21½                                    | 10¾                          |
| 5                  | 22                          | 22                  | 17               | 22                                                    | 17                                               | 11                   | 8½                          | 26½                 | 26½                 | 19               | 26½                                    | 13¼                          |
| 6                  | 24                          | 24                  | 20               | 24                                                    | 20                                               | 12                   | 10                          | 27¾                 | 27¾                 | 22               | 27¾                                    | 13¾                          |
| 8                  | 29                          | 29                  | 26               | 29                                                    | 26                                               | 14½                  | 13                          | 32¾                 | 32¾                 | 28               | 32¾                                    | 16¾                          |
| 10                 | 33                          | 33                  | 31               | 33                                                    | 31                                               | 16½                  | 15½                         | 39                  | 39                  | 34               | 39                                     | 19½                          |
| 12                 | 38                          | 38                  | 36               | 38                                                    | 36                                               | 19                   | 18                          | 44½                 | 44½                 | 39               | 44½                                    | 22¼                          |
| 14                 | 40½                         | 40½                 | 39               | 40½                                                   | 39                                               | 20¼                  | 19½                         | 49½                 | 49½                 | 42               | 49½                                    | 24¾                          |
| 16                 | 44½                         | 44½                 | 43               |                                                       |                                                  |                      |                             | 54½                 | 54½                 | 47               |                                        |                              |
| 18                 | 48                          | 48                  | †                |                                                       |                                                  |                      |                             | 60½                 | 60½                 | 53               |                                        |                              |
| 20                 | 52                          | 52                  | †                |                                                       |                                                  |                      |                             | 65½                 | 65½                 | 58               |                                        |                              |
| 24                 | 61                          | 61                  | †                |                                                       |                                                  |                      |                             | 76½                 | 76½                 |                  |                                        |                              |

| Nominal valve size | Class 2500 steel            |                     |                  |                                        |                        |
|--------------------|-----------------------------|---------------------|------------------|----------------------------------------|------------------------|
|                    | Flanged end and welding end |                     |                  |                                        |                        |
|                    | Gate                        |                     |                  | Globe, lift check, swing check A and B | Angle and lift check B |
|                    | Solid wedge A and B         | Double disk A and B | Short pattern‡ B |                                        |                        |
| ½                  | 10¾                         |                     |                  | 10¾                                    | 5¾ <sub>16</sub>       |
| ¾                  | 10¾                         |                     |                  | 10¾                                    | 5¾                     |
| 1                  | 12½                         |                     | 7¾ <sub>16</sub> | 12½                                    | 6¼ <sub>16</sub>       |
| 1¼                 | 13¾                         |                     | 9¾               | 13¾                                    | 6¾                     |
| 1½                 | 15¼                         |                     | 9¾               | 15¼                                    | 7¾ <sub>16</sub>       |
| 2                  | 17¾                         | 17¾                 | 11               | 17¾                                    | 8¾                     |
| 2½                 | 20                          | 20                  | 13               | 20                                     | 10                     |
| 3                  | 22¾                         | 22¾                 | 14½              | 22¾                                    | 11¾                    |
| 4                  | 26½                         | 26½                 | 18               | 26½                                    | 13¼                    |
| 5                  | 31¼                         | 31¼                 | 21               | 31¼                                    | 15¾                    |
| 6                  | 36                          | 36                  | 24               | 36                                     | 18                     |
| 8                  | 40¼                         | 40¼                 | 30               | 40¼                                    | 20¼                    |
| 10                 | 50                          | 50                  | 36               | 50                                     | 25                     |
| 12                 | 56                          | 56                  | 41               | 56                                     | 28                     |
| 14                 |                             |                     | 44               |                                        |                        |
| 16                 |                             |                     | 49               |                                        |                        |
| 18                 |                             |                     | 55               |                                        |                        |

NOTE: Outline drawings for flanged valves show ¼-in raised face machined onto flange, as for Class 400 cast-steel valves; Class 150 and 300 cast-steel valves and Class 250 cast-iron valves have ¼<sub>16</sub>-in raised faces; Class 125 cast-iron have no raised faces.

\*Extracted from Face-to-Face and End-to-End Dimensions of Valves, ASME B16.10, with permission of the publisher, the American Society of Mechanical Engineers, New York. To convert inches to millimeters, multiply by 25.4.

†Not shown in ANSI B16.10 but commercially available.

‡These dimensions apply to pressure-seal or flangeless bonnet valves only.

§Solid wedge only.



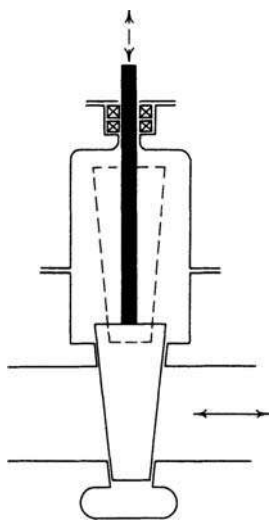


FIG. 10-151 Gate valve.

seats after the disks have been positioned for closing. This gate assembly distorts automatically to match both angular misalignment of the seats and longitudinal shrinkage of the valve body on cooling.

During opening and closing, some parallel-seat designs are more subject to vibration resulting from fluid flow than are wedge gates. Specific applications should be discussed with the manufacturer. In some applications it may be advisable to use a small bypass around the in-line valve to help lower opening and closing forces and to relieve binding between the gate and the seat due to high differential pressure or temperature. Double-disk parallel seat valves should be installed with the stem essentially vertical unless otherwise recommended by the manufacturer. All wedge gate valves are equipped with tongue-and-groove guides to keep the gate sealing surfaces from clattering on the seats and marring them during opening and closing. Depending on the velocity and density of the fluid stream being sheared, these guiding surfaces may be specified as cast, machined, or hard-surfaced and ground.

Gate valves may have nonrising stems, inside-screw rising stems, or outside-screw rising stems, listed in order of decreasing exposure of the stem threads to the fluid handled. Rising-stem valves require more space, but the position of the stem visually indicates the position of the gate. Indication is clearest on the outside-screw rising-stem valves, and on these the stem threads and thrust collars may be lubricated, reducing operating effort. The stem connection to the gate assembly prevents the stem from rotating.

Gate valves are used to minimize pressure drop in the open position and to stop the flow of fluid rather than to regulate it. The problem, when the valve is closed, of pressure buildup in the bonnet from cold liquids expanding or chemical action between fluid and bonnet should be solved by a relief valve or by notching the upstream seat ring.

**Globe Valves** (Fig. 10-152) These are designed as either inside-screw rising stem or outside-screw rising stem. In most designs the disk is free to rotate on the stem; this prevents galling between the disk and the seat. Various designs are used to maintain alignment between the disk and the seat, and to keep the fluid flow from vibrating or rotating the disk. Disks are typically guided either by the valve stem or against the valve body. Body guiding reduces side thrust loads on the stem. The suitability of each design can be determined by reviewing specific applications with valve manufacturers.

Disk shapes are commonly flat or conical. Conical designs provide either line or area contact between the seat and disk, and are generally more suitable than flat disks for high pressures and temperatures. Needle-type disks provide better flow control and are commonly available in valves 1 in and smaller.

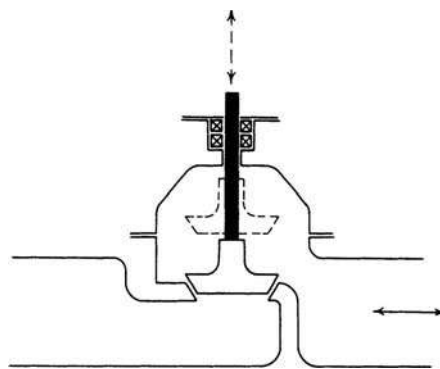


FIG. 10-152 Globe valve.

For certain valve designs, sizes, and applications, globe valves may be installed with the stem in the horizontal position; however, unless approved by the manufacturer, the stem orientation should be vertical. Globe valves are not symmetric with respect to flow. Generally globe valves are installed with pressure under the seat when the valve is in the closed position, and with the flow direction coming from under the seat. Opposite-flow direction provides pressure-assisted seating, lower seating torque requirements, and higher opening torques, and may result in blockage in dirty services. Consult the manufacturer before installing globe valves in the opposite-flow direction.

Pressure drop through globe valves is much greater than that for gate valves. In Y-type globe valves, the stem and seat are at about  $45^\circ$  to the pipe instead of  $90^\circ$ . This reduces pressure drop but presents design challenges with regard to disk alignment.

Globe valves in horizontal lines prevent complete drainage.

**Angle Valves** These valves are similar to globe valves; the same bonnet, stem, and disk are used for both (Fig. 10-153). They combine an elbow fitting and a globe valve into one component with a substantial saving in pressure drop.

**Diaphragm Valves** These valves are limited to pressures of approximately  $50 \text{ lbf/in}^2$  (Fig. 10-154). The fabric-reinforced diaphragms may be made from natural rubber, from a synthetic rubber, or from natural or synthetic rubbers faced with Teflon® fluorocarbon resin. The simple shape of the body makes lining it economical. Elastomers have shorter lives as diaphragms than as linings because of

\*Du Pont TFE fluorocarbon resin.

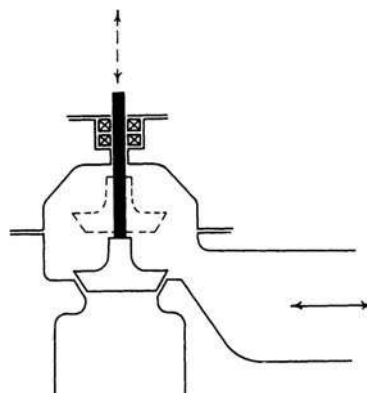


FIG. 10-153 Angle valve.

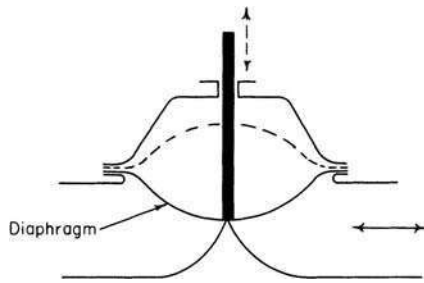


FIG. 10-154 Diaphragm valve.

flexing but still provide satisfactory service. Plastic bodies, which have low moduli of elasticity compared with metals, are practical in diaphragm valves since alignment and distortion are minor problems.

These valves are excellent for fluids containing suspended solids and can be installed in any position. Models in which the dam is very low, reducing pressure drop to a negligible quantity and permitting complete drainage in horizontal lines, are available. However, drainage can be obtained with any model simply by installing it with the stem horizontal. The only maintenance required is replacement of the diaphragm, which can be done very quickly without removing the valve from the line.

**Plug Valves** These valves (Fig. 10-155) are typically limited to temperatures below 260°C (500°F) since differential expansion between the plug and the body results in seizure. The size and shape of the port divide these valves into different types. In order of increasing cost they are short venturi, reduced rectangular port; long venturi, reduced rectangular port; full rectangular port; and full round port.

In lever-sealed plug valves, tapered plugs are used. The plugs are raised by turning one lever, rotated by another lever, and resealed by the first lever. **Lubricated** plug valves may use straight or tapered plugs. The tapered plugs may be raised slightly, to reduce turning effort, by injection of the lubricant, which also acts as a seal. Plastic is used in nonlubricated plug valves as a body liner, a plug coating, or port seals in the body or on the plug.

In plug valves other than lever-sealed plug valves, the contact area between plug and body is large, and gearing is usually used in sizes 6 in and larger to minimize operating effort. There are several lever-sealed plug valves incorporating mechanisms which convert the rotary motion of a handwheel into sequenced motion of the two levers.

For lubricated plug valves, the lubricant must have limited viscosity change over the range of operating temperature, must have low solubility in the fluid handled, and must be applied regularly. There must be no chemical reaction between the lubricant and the fluid which would harden or soften the lubricant or contaminate the fluid. For these reasons, lubricated plug valves are most often used when there are a large number handling the same or closely related fluids at approximately the same temperature.

Lever-sealed plug valves are used for throttling service. Because of the large contact area between plug and body, if a plug valve is operable, there is little likelihood of leakage when closed, and the handle position is a clearly visible indication of the valve position.

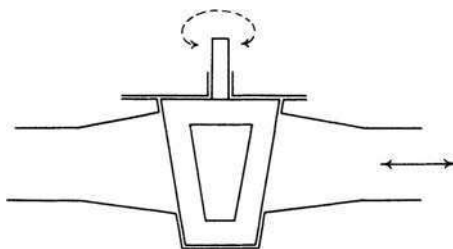


FIG. 10-155 Plug valve.

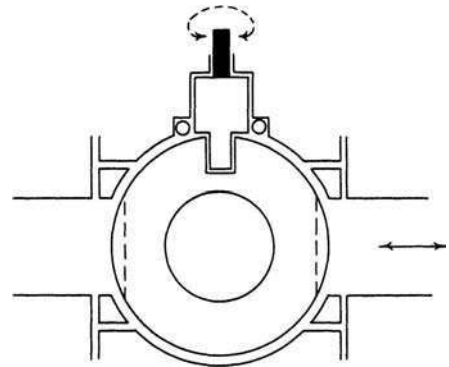


FIG. 10-156 Ball valve; floating ball.

**Ball Valves** Ball valves (Figs. 10-156 and 10-157) are of two primary designs, floating ball and trunnion-mounted ball. In floating ball designs, the ball is supported by the downstream seat. In trunnion ball designs, the ball is supported by the trunnion and the seat loads are less than those in floating ball valves. Because of operating torque and shutoff pressure ratings, trunnion ball valves are available in larger sizes and higher pressure ratings than floating ball valves.

Both floating and trunnion-mounted designs are available with other design variations that include metal seated valves, soft seated valves, top-entry valves, end-entry valves, and split body valves. Valves in all these design variations are available as either full port or reduced port. *Port* refers to the round-bore fluid flow area through the ball. Full port valves have a bore that is approximately equal to the inside diameter of the mating pipe. Reduced port valves have a bore that is approximately equal to the inside diameter of pipe one size smaller than full bore.

A variety of soft seat materials are available, including PTFE and nylon. Since the shutoff pressure capability of ball valves is limited by the load capabilities of the seat material, the upper temperature limit of soft seated valves is limited by the seat material selection. The shutoff pressure rating of soft seated valves typically declines rapidly with increasing temperature, and the shutoff rating is often less than the body pressure rating. Metal seated valves do not share this characteristic.

For equal port size, ball valves share the low pressure drop characteristics of gate valves. Also as is the case with gate valves, consideration must be given to venting the valve when the expansion of fluid trapped within the body cavity could overpressurize the valve body. Some seat designs are inherently self-venting to either the upstream or the downstream side of the valve. In floating ball valves, venting may result in a unidirectional valve that seats against flow in only one direction.

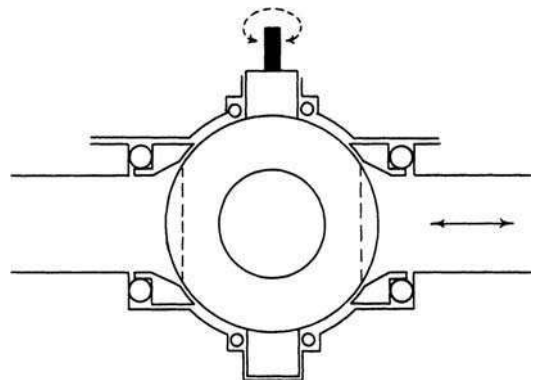


FIG. 10-157 Ball valve; trunnion-mounted ball.

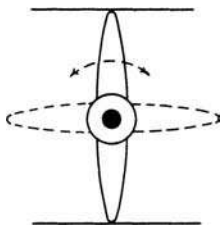


FIG. 10-158 Butterfly valve.

With split body designs, internals must be removed by separating the valve body in the axial direction of the mating pipe. Top-entry design permits removal of the internals through the top of the valve. When valves are butt-welded, top entry may be specified to permit repair without removing the valve from the piping. Top-entry valves are significantly more expensive than split body and end-entry valves, and full port valves are more expensive than reduced port in any body design. Metal seated valves are significantly more expensive than soft seated valves and are typically used only when other types of valves are unsuitable for the application.

**Butterfly Valves** These valves (Fig. 10-158) occupy less space, are much lighter than other types of block valves, and are available in body styles that include wafer, lugged (drilled-through or tapped), and flanged. They are available in ASME Class 900 and lower pressure ratings. The maximum size available varies with pressure rating. Valves in Class 150 are available in sizes exceeding 60-in diameter. Within limits, they may be used for throttling. Their relatively high pressure drop must be considered during design.

Like ball valves, butterfly valves are fully opened in one-quarter turn and are therefore well suited to automation. *Butterfly Valves: Double Flange, Lug- and Wafer-Type*, API 609, is one of the standards commonly used to specify butterfly valves. API 609 defines two major categories of butterfly valves: Category A and Category B. Category A valves are typically soft seated valves with shell pressure ratings that may be less than the flange rating of the valve. They are typically used for utility services and are commonly referred to as *utility valves*. Category B may be soft seated or metal seated, and must have shell pressure ratings equal to the full pressure rating of the valve flange, and seat ratings that essentially meet the shell rating within the temperature capability of the seat material. Within Category B, valves may be further divided into concentric shaft, double-offset shaft, and triple-offset shaft designs. *Offset* refers to the position of the shaft with respect to seat area. With minor exception, double- and triple-offset valve designs are metal-to-metal seated. They are distinguished from other designs by their exceptional seat tightness (often “zero” leakage) that is maintained throughout the life of the valve. Their tightness exceeds the seat tightness capability and reliability of wedge-type gate valves. Double-offset valves minimize rubbing between the disk and the seat, and triple-offset valves virtually eliminate rubbing. Although double- and triple-offset valves are more expensive than other butterfly valve designs, because of their weight they are often more economical than gate valves for some combinations of pressure class, size, and materials.

**Check Valves** These valves are used to prevent reversal of flow. They must be located where flow turbulence or instability does not result in chatter (high-frequency opening and closing of the valve) and in systems designed to prevent sudden high-velocity flow reversal which results in slamming upon closure. Many valve manufacturers can provide application advice.

**Swing Check Valves** These valves (Fig. 10-159) are normally designed for use in horizontal lines or in vertical lines with normally upward flow. Since their seating force is primarily due to pipeline pressure, they may not seal as tightly at low pressures as at higher pressures. When suitable, nonmetallic seats may be used to minimize this problem.

**Lift Check Valves** These valves (Figs. 10-160 through 10-162) are made in three styles. Vertical lift check valves are for installation in vertical lines with flow normally upward. Globe (or piston) valves with a 90° bonnet (Fig. 10-161) are for installation in horizontal lines, although inclined bonnet versions (approximately 45°) with spring

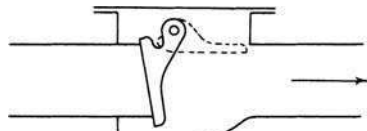


FIG. 10-159 Swing check valve.

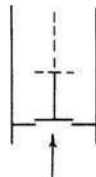


FIG. 10-160 Lift check valve, vertical.

assist may be used in vertical lines with normally upward flow. Globe and angle check valves often incorporate mechanisms to control the opening or closing rate of the piston, or to promote full opening under low-flow conditions. In some designs, spring-assisted closure is available, but this increases pressure drop. Lift check valves should not be used when the fluid contains suspended solids. Ball check valves having designs similar to those in Figs. 10-160 and 10-161 are available in sizes 2 in and smaller. They promote even wear of the seat area and are more suitable for viscous services, or services with limited solids.

**Tilting-Disk Check Valves** These valves (Fig. 10-163) may be installed in a horizontal line or in lines in which the flow is vertically upward. The pivot point is located so that the distribution of pressure in the fluid handled speeds the closing but arrests slamming. Compared with swing check valves of the same size, pressure drop is less at low velocities but greater at high velocities.

Closure at the instant of flow reversal is most nearly attained with tilting-disk, dual-plate, and specialty axial-flow check valves. However, quick closure is not the solution to all noise, shock, and water hammer problems. External dashpots are available when a controlled rate of closure is desired. Nonmetallic seats are also available.

**Dual-Plate Check Valves** These valves (Fig. 10-163) occupy less space, are much lighter than other types of check valves, and are available in body styles that include wafer, lugged (drilled-through or tapped), and flanged. They are available in all ASME pressure classes. The maximum size available varies with pressure rating. Valves in Class 150 are available in sizes 60 in or larger. They are available with either metallic or nonmetallic seats. Pressure drop is greater than that in a fully open swing check valve. Plate closure is spring-assisted, and the rate of closure can be controlled with proper spring selection. High-performance valves with fast closure rates are available to address water hammer problems. They typically weigh as little as 15 to 30 percent as much as swing check valves. Because of their weight they are often more economical than other types of check valves.

**Valve Trim** Various alloys are available for valve parts such as seats, disks, and stems which must retain smooth finish for successful operation. The problem in seat materials is fivefold: (1) resistance to corrosion by the fluid handled and to oxidation at high temperatures, (2) resistance to erosion by suspended solids in the fluid, (3) prevention of galling (seizure at point of contact) by differences in material or hardness or both, (4) maintenance of high strength at high temperature, and (5) avoidance of distortion.

Standard valve trims are defined by standards such as API 600 and API 602. Elastomer or plastic inserts may be specified to achieve bubble-tight shutoff. Valve manufacturers may be consulted for recommended trims.

## CAST IRON, DUCTILE IRON, AND HIGH-SILICON IRON PIPING SYSTEMS

**Cast Iron and Ductile Iron** Cast iron and ductile iron provide more metal for less cost than steel in piping systems and are widely used in low-pressure services in which internal and external corrosion

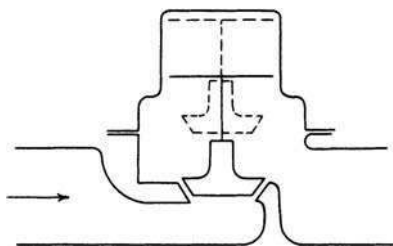


FIG. 10-161 Lift check valve, globe.

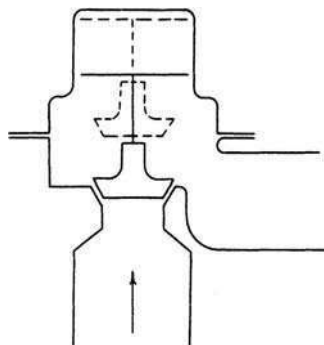


FIG. 10-162 Lift check valve, angle.

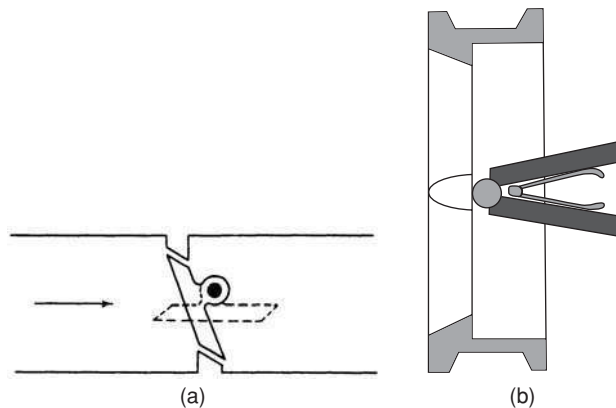


FIG. 10-163 (a) Tilting-disk check valve. (b) Dual-plate check valve.

may cause a considerable loss of metal. They are widely used for underground water distribution. Cement lining is available at a nominal cost for handling water causing tuberculation.

Ductile iron has an elongation of 10 percent or more compared with essentially nil elongation for cast iron and has for all practical purposes supplanted cast iron as a cast piping material. It is usually centrifugally cast. This manufacturing method improves tensile strength and reduces porosity. Ductile-iron pipe is manufactured to AWWA C151/A21.51-2002 and is available in nominal sizes from 3 through 64 in. Wall thicknesses are specified by seven standard thickness classes. Table 10-33 gives the outside diameter and standard thickness for various rated water working pressures for centrifugally cast ductile-iron pipe. The required wall thickness for underground installations increases with internal pressure, depth of laying, and weight of vehicles operating over the pipe. It is reduced by the degree to which the soil surrounding the pipe provides uniform support along the pipe and around the lower 180°.

Tables are provided in AWWA C151/A21.51 for determining wall-thickness-class recommendations for various installation conditions. The poured joint (Fig. 10-139) has been almost entirely superseded by the mechanical joint (Fig. 10-138) and the push-on joint (Fig. 10-140), which are better suited to wet trenches, bad weather, and unskilled labor. Such joints also minimize strain on the pipe from ground settlement. Lengths vary between 5 and 6 m (between 18 and 20 ft), depending on the supplier. Stock fittings are designed for 1.72-MPa (250-lbf/in<sup>2</sup>) cast iron or 2.41-MPa (350-lbf/in<sup>2</sup>) ductile iron in sizes through 12 in and for 1.0- and 1.72-MPa (150- and 250-lbf/in<sup>2</sup>) cast iron or 2.41-MPa (350-lbf/in<sup>2</sup>) ductile iron in sizes 14 in and larger. Stock fittings include 22½° and 11¼° bends. Ductile-iron pipe is also supplied with flanges that match the dimensions of Class 125 flanges shown in ASME B16.1 (see Table 10-23). These flanges are assembled to the pipe barrel by threaded joints.

**High-Silicon Iron** Pipe and fittings are cast products of material typically conforming to ASTM A518. Nominal silicon content is 14.5 percent, and nominal carbon content is approximately 0.85 percent. This material is corrosion-resistant to most chemicals, highly abrasion-resistant, and suitable for applications to 260°C (500°F). Applications are primarily gravity drain. Pipe and fittings are available under the trade name Duriron®.

Pipe and fitting sizes typically available are shown in Table 10-34. Pipe and fitting dimensions commonly conform to ASTM A861. Bell and spigot connections are sealed with chemical-resistant rope packing and molten lead. Mechanical joint connections are made with TFE gaskets and stainless steel clamps.

The coefficient of linear expansion of this alloy in the temperature range of 21 to 100°C (70 to 212°F) is  $12.2 \times 10^{-6}/^{\circ}\text{C}$  ( $6.8 \times 10^{-6}/^{\circ}\text{F}$ ), which is slightly above that of cast iron (National Bureau of Standards). Since this material has practically no elasticity, the need for expansion joints should be considered. Connections for flanged pipe, fittings, valves, and pumps are made to ASME B16.1, Class 125.

The use of high-silicon iron in flammable-fluid service or in Category M fluid service is prohibited by the code.

## NONFERROUS METAL PIPING SYSTEMS

**Aluminum** Seamless aluminum pipe and tube are produced by extrusion in essentially pure aluminum and in several alloys; 6-, 9-, and 12-m (20-, 30-, and 40-ft) lengths are available. Alloying and mill treatment improve physical properties, but welding reduces them. Essentially pure aluminum has an ultimate tensile strength of 58.6 MPa (8500 lbf/in<sup>2</sup>) subject to a slight increase by mill treatment which is lost during welding. Alloy 6061, which contains 0.25 percent copper, 0.6 percent silicon, 1 percent magnesium, and 0.25 percent chromium, has an ultimate tensile strength of 124 MPa (18,000 lbf/in<sup>2</sup>) in the annealed condition, 262 MPa (38,000 lbf/in<sup>2</sup>), mill-treated as 6061-T6, and 165 MPa (24,000 lbf/in<sup>2</sup>) at welded joints. Extensive use is made of alloy 1060, which is 99.6 percent pure aluminum, for hydrogen peroxide; of alloy 3003, which contains 1.2 percent manganese, for high-purity chemicals; and of alloys 6063 and 6061 for many other services. Alloy 6063 is the same as 6061 minus the chromium and has slightly lower mechanical properties.

Aluminum is not embrittled by low temperatures and is not subject to external corrosion when exposed to normal atmospheres. At 200°C (400°F) its strength is less than half that at room temperature. It is attacked by alkalis, by traces of copper, nickel, mercury, and other heavy-metal ions, and by prolonged contact with wet insulation. It suffers from galvanic corrosion when coupled to copper, nickel, or lead-base alloys but not when coupled to galvanized iron.

Aluminum pipe schedules conform to those in Table 10-22. Consult suppliers for available sizes and schedules.

Threaded aluminum fittings are seldom recommended for process piping. Wrought fittings with welding ends (see Table 10-31 for dimensions) and with grooved joint ends are available. Wrought 6061-T6 flanges with dimensions per Table 10-23 are also available. Consult suppliers on availability of cast flanges and fittings. Castings manufactured in accordance with ASTM B26 are available in several grades. Refer to Table 10-30 for dimensions. The low strength and

## 10-100 TRANSPORT AND STORAGE OF FLUIDS

**TABLE 10-33 Dimensions of Ductile-Iron Pipe\***

Standard thickness for internal pressure†

| Pipe size,<br>in | Outside<br>diameter, in | Rated water working pressure, lbf/in <sup>2</sup> ‡ |                    |                  |                    |                  |                    |                  |                    |                  |                    |
|------------------|-------------------------|-----------------------------------------------------|--------------------|------------------|--------------------|------------------|--------------------|------------------|--------------------|------------------|--------------------|
|                  |                         | 150                                                 |                    | 200              |                    | 250              |                    | 300              |                    | 350              |                    |
|                  |                         | Thickness,<br>in                                    | Thickness<br>class | Thickness,<br>in | Thickness<br>class | Thickness,<br>in | Thickness<br>class | Thickness,<br>in | Thickness<br>class | Thickness,<br>in | Thickness<br>class |
| 3                | 3.96                    | 0.25                                                | 51                 | 0.25             | 51                 | 0.25             | 51                 | 0.25             | 51                 | 0.25             | 51                 |
| 4                | 4.80                    | 0.26                                                | 51                 | 0.26             | 51                 | 0.26             | 51                 | 0.26             | 51                 | 0.26             | 51                 |
| 6                | 6.90                    | 0.25                                                | 50                 | 0.25             | 50                 | 0.25             | 50                 | 0.25             | 50                 | 0.25             | 50                 |
| 8                | 9.05                    | 0.27                                                | 50                 | 0.27             | 50                 | 0.27             | 50                 | 0.27             | 50                 | 0.27             | 50                 |
| 10               | 11.10                   | 0.29                                                | 50                 | 0.29             | 50                 | 0.29             | 50                 | 0.29             | 50                 | 0.29             | 50                 |
| 12               | 13.20                   | 0.31                                                | 50                 | 0.31             | 50                 | 0.31             | 50                 | 0.31             | 50                 | 0.31             | 50                 |
| 14               | 15.30                   | 0.33                                                | 50                 | 0.33             | 50                 | 0.33             | 50                 | 0.33             | 50                 | 0.33             | 50                 |
| 16               | 17.40                   | 0.34                                                | 50                 | 0.34             | 50                 | 0.34             | 50                 | 0.34             | 50                 | 0.34             | 50                 |
| 18               | 19.50                   | 0.35                                                | 50                 | 0.35             | 50                 | 0.35             | 50                 | 0.35             | 50                 | 0.35             | 50                 |
| 20               | 21.60                   | 0.36                                                | 50                 | 0.36             | 50                 | 0.36             | 50                 | 0.36             | 50                 | 0.39             | 51                 |
| 24               | 25.80                   | 0.38                                                | 50                 | 0.38             | 50                 | 0.38             | 50                 | 0.41             | 51                 | 0.44             | 52                 |
| 30               | 32.00                   | 0.39                                                | 50                 | 0.39             | 50                 | 0.43             | 51                 | 0.47             | 52                 | 0.51             | 53                 |
| 36               | 38.30                   | 0.43                                                | 50                 | 0.43             | 50                 | 0.48             | 51                 | 0.53             | 52                 | 0.58             | 53                 |
| 42               | 44.50                   | 0.47                                                | 50                 | 0.47             | 50                 | 0.53             | 51                 | 0.59             | 52                 | 0.65             | 53                 |
| 48               | 50.80                   | 0.51                                                | 50                 | 0.51             | 50                 | 0.58             | 51                 | 0.65             | 52                 | 0.72             | 53                 |
| 54               | 57.56                   | 0.57                                                | 50                 | 0.57             | 50                 | 0.65             | 51                 | 0.73             | 52                 | 0.81             | 53                 |

\*Extracted from *Ductile Iron Pipe, Centrifugally Cast*, AWWA C151/A21.51-2002, with permission of the publisher, the American Water Works Association, Denver, Colo.

†To convert from inches to millimeters, multiply by 25.4; to convert pounds-force per square inch to megapascals, multiply by 0.006895.

‡These pipe walls are adequate for the rated working pressure plus a surge allowance of 100 lbf/in<sup>2</sup>. For the effect of laying conditions and depth of bury, see ANSI A21.51.

modulus of elasticity of aluminum must be considered when using flanged connections.

Aluminum-body diaphragm and ball valves are used extensively.

**Copper and Copper Alloys** Seamless pipe and tubing manufactured from copper, bronze, brass, and copper-nickel alloys are produced by extrusion. The availability of pipe or tubing depends on the metallurgy and size. Copper tubing is widely used for plumbing, steam tracing, compressed air, instrument air, and inert gas applications. Copper tubing specifications are generally segregated into three types: water and general service, refrigeration service (characterized by cleanliness requirements), and drain/waste/vent (DWV) service. Tubing is available in the annealed or hard-drawn condition. Hard-drawn products are available only as straight lengths. Annealed products

are often available in coils or as straight lengths. Suppliers should be consulted regarding wall thickness availability and standard lengths; however, many products are available in 3.0-m (10-ft) and 6.1-m (20-ft) straight lengths. Coil lengths are generally 12.2 m (40 ft) to 30 m (100 ft).

ASTM copper tubing specifications for water and general-purpose applications include B75 and B88 (Table 10-35). Tubing, fittings, and solders specified for potable water services must always be approved by the appropriate authority, such as the National Sanitation Foundation (NSF). ASTM copper tubing specifications for refrigeration services include B68 and B280 (Table 10-36). ASTM specifications for DWV services include B306 (1¼ through 8-in OD). Pipe conforming to the diameters shown in Table 10-37 is available as ASTM B42 and B302. ASTM B302 has diameters matching ASTM B42, but is available with thinner walls (Table 10-38). Red brass pipe is available as ASTM B43.

Joints are typically soldered, silver-brazed, or mechanical. When using flare, compression, or other mechanical joint fittings, consideration must be given to the fitting manufacturer's recommendations regarding hardness, minimum and maximum wall thickness, finish requirements, and diameter tolerances. Flanges and flanged fittings are seldom used since soldered and brazed joints can be easily disassembled. Brass and bronze valves are available with female ends for soldering.

70 percent copper, 30 percent nickel and 90 percent copper, 10 percent nickel ASTM B466 are available as seamless pipe (ASTM B466) or welded pipe (ASTM B467) and welding fittings for handling brackish water in Schedule 10 and regular copper pipe thicknesses.

**Nickel and Nickel Alloys** A wide range of ferrous and nonferrous nickel and nickel-bearing alloys are available. They are usually selected because of their improved resistance to chemical attack or their superior resistance to the effects of high temperature. In general terms their cost and corrosion resistance are somewhat a function of their nickel content. The 300 Series stainless steels are the most generally used. Some other frequently used alloys are listed in Table 10-39 together with their nominal compositions. For metallurgical and corrosion resistance data, see Sec. 25.

**TABLE 10-34 High-Silicon Iron Pipe\***

| Size,<br>inside<br>diam.,<br>in | No hub (MJ) ends             |                               |                                 |                                       | Bell-and-spigot ends         |                               |                                 |                                       |
|---------------------------------|------------------------------|-------------------------------|---------------------------------|---------------------------------------|------------------------------|-------------------------------|---------------------------------|---------------------------------------|
|                                 | Out-<br>side<br>diam.,<br>in | Wall<br>thick-<br>ness,<br>in | Stand-<br>ard†<br>length,<br>ft | Est.<br>weight<br>per<br>piece,<br>lb | Out-<br>side<br>diam.,<br>in | Wall<br>thick-<br>ness,<br>in | Stand-<br>ard†<br>length,<br>ft | Est.<br>weight<br>per<br>piece,<br>lb |
| 1½                              | 2⅜                           | ⅝                             | 7                               | 46                                    |                              |                               |                                 |                                       |
| 2                               | 2⅞                           | ⅝                             | 7                               | 56                                    | 2⅞                           | ⅝                             | 7                               | 62                                    |
| 3                               | 3⅞                           | ⅝                             | 7                               | 79                                    | 3⅞                           | ⅝                             | 7                               | 90                                    |
| 4                               | 4⅞                           | ⅝                             | 7                               | 100                                   | 4⅞                           | ⅝                             | 7                               | 114                                   |
| 6                               |                              |                               |                                 |                                       | 6⅞                           | ⅝                             | 7                               | 169                                   |
| 8                               |                              |                               |                                 |                                       | 9                            | ⅝                             | 7                               | 234                                   |
| 10                              |                              |                               |                                 |                                       | 11¾                          | ⅝                             | 7                               | 340                                   |
| 12                              |                              |                               |                                 |                                       | 13¼                          | ⅝                             | 5                               | 470                                   |
| 15                              |                              |                               |                                 |                                       | 16¼                          | ⅝                             | 5                               | 800                                   |

\*Extracted from *Standard Specification for High-Silicon Iron Pipe and Fittings*, ASTM A861-2004, with permission of the American Society for Testing and Materials, West Conshohocken, Pa.

**TABLE 10-35 Dimensions, Weights, and Tolerances in Diameter and Wall Thickness for Nominal or Standard Copper Water Tube Sizes (ASTM B88-2003)\***

All tolerances are plus and minus except as otherwise noted.

| Nominal standard size, in | Outside diameter, in | Average outside diameter <sup>r</sup> tolerance, in |                  | Wall thickness and tolerances, in |                        |                |                        |                |                        | Theoretical weight, lb/ft |        |              |
|---------------------------|----------------------|-----------------------------------------------------|------------------|-----------------------------------|------------------------|----------------|------------------------|----------------|------------------------|---------------------------|--------|--------------|
|                           |                      |                                                     |                  | Type K                            |                        | Type L         |                        | Type M         |                        | Type K                    | Type L | Type M       |
|                           |                      | Annealed                                            | Drawn            | Wall thickness                    | Tolerance <sup>B</sup> | Wall thickness | Tolerance <sup>B</sup> | Wall thickness | Tolerance <sup>B</sup> |                           |        |              |
| ¼                         | 0.375                | 0.002                                               | 0.001            | 0.035                             | 0.0035                 | 0.030          | 0.003                  | <sup>c</sup>   | <sup>c</sup>           | 0.145                     | 0.126  | <sup>c</sup> |
| ⅜                         | 0.500                | 0.0025                                              | 0.001            | 0.049                             | 0.005                  | 0.035          | 0.004                  | 0.025          | 0.002                  | 0.269                     | 0.198  | 0.145        |
| ½                         | 0.625                | 0.0025                                              | 0.001            | 0.049                             | 0.005                  | 0.040          | 0.004                  | 0.028          | 0.003                  | 0.344                     | 0.285  | 0.204        |
| ⅝                         | 0.750                | 0.0025                                              | 0.001            | 0.049                             | 0.005                  | 0.042          | 0.004                  | <sup>c</sup>   | <sup>c</sup>           | 0.418                     | 0.362  | <sup>c</sup> |
| ¾                         | 0.875                | 0.003                                               | 0.001            | 0.065                             | 0.006                  | 0.045          | 0.004                  | 0.032          | 0.003                  | 0.641                     | 0.455  | 0.328        |
| 1                         | 1.125                | 0.0035                                              | 0.0015           | 0.065                             | 0.006                  | 0.050          | 0.005                  | 0.035          | 0.004                  | 0.839                     | 0.655  | 0.465        |
| 1¼                        | 1.375                | 0.004                                               | 0.0015           | 0.065                             | 0.006                  | 0.055          | 0.006                  | 0.042          | 0.004                  | 1.04                      | 0.884  | 0.682        |
| 1½                        | 1.625                | 0.0045                                              | 0.002            | 0.072                             | 0.007                  | 0.060          | 0.006                  | 0.049          | 0.005                  | 1.36                      | 1.14   | 0.940        |
| 2                         | 2.125                | 0.005                                               | 0.002            | 0.083                             | 0.008                  | 0.070          | 0.007                  | 0.058          | 0.006                  | 2.06                      | 1.75   | 1.46         |
| 2½                        | 2.625                | 0.005                                               | 0.002            | 0.095                             | 0.010                  | 0.080          | 0.008                  | 0.065          | 0.006                  | 2.93                      | 2.48   | 2.03         |
| 3                         | 3.125                | 0.005                                               | 0.002            | 0.109                             | 0.011                  | 0.090          | 0.009                  | 0.072          | 0.007                  | 4.00                      | 3.33   | 2.68         |
| 3½                        | 3.625                | 0.005                                               | 0.002            | 0.120                             | 0.012                  | 0.100          | 0.010                  | 0.083          | 0.008                  | 5.12                      | 4.29   | 3.58         |
| 4                         | 4.125                | 0.005                                               | 0.002            | 0.134                             | 0.013                  | 0.110          | 0.011                  | 0.095          | 0.010                  | 6.51                      | 5.38   | 4.66         |
| 5                         | 5.125                | 0.005                                               | 0.002            | 0.160                             | 0.016                  | 0.125          | 0.012                  | 0.109          | 0.011                  | 9.67                      | 7.61   | 6.66         |
| 6                         | 6.125                | 0.005                                               | 0.002            | 0.192                             | 0.019                  | 0.140          | 0.014                  | 0.122          | 0.012                  | 13.9                      | 10.2   | 8.92         |
| 8                         | 8.125                | 0.006                                               | +0.002<br>-0.004 | 0.271                             | 0.027                  | 0.200          | 0.020                  | 0.170          | 0.017                  | 25.9                      | 19.3   | 16.5         |
| 10                        | 10.125               | 0.008                                               | +0.002<br>-0.006 | 0.338                             | 0.034                  | 0.250          | 0.025                  | 0.212          | 0.021                  | 40.3                      | 30.1   | 25.6         |
| 12                        | 12.125               | 0.008                                               | +0.002<br>-0.006 | 0.405                             | 0.040                  | 0.280          | 0.028                  | 0.254          | 0.025                  | 57.8                      | 40.4   | 36.7         |

\* Extracted from ASTM B88-2003 with permission of the publisher, the American Society for Testing and Materials, West Conshohocken, Pa.

<sup>A</sup> The average outside diameter of a tube is the average of the maximum and minimum outside diameter, as determined at any one cross section of the tube.<sup>B</sup> Maximum deviation at any one point.<sup>C</sup> Indicates that the material is not generally available or that no tolerance has been established.

**Titanium** Seamless pipe is available as ASTM B861, and welded pipe is available as ASTM B862. Both standards offer numerous grades of unalloyed and alloyed materials. While the alloys often have higher tensile strengths, corrosion resistance may be sacrificed. Forged or wrought fittings and forged or cast valves are available. For many applications, elastomer-lined valves having carbon-steel or ductile-iron bodies and titanium trim offer an economical alternative to valves with titanium bodies. Titanium pipe is available with wall thicknesses conforming to many of those listed in Table 10-22, including Schedule 10S and Standard Weight. Properly selected and specified, titanium can be a good choice for seawater systems such as offshore fire

water systems. Seamless and welded tubing is manufactured to ASTM B338; however, availability may be limited.

**Flexible Metal Hose** Flexible hoses provide flexible connections for conveying gases or liquids, wherever rigid pipes are impractical. There are two basic types of flexible hose: corrugated hoses and interlocked hoses. These flexible hoses can absorb vibrations and noise. They can also provide some flexibility for misaligned rigid piping or equipment during construction. Corrugated or interlocked thin brass, bronze, Monel, aluminum, and steel tubes are covered with flexible braided-wire jackets to form flexible metal hose. Both tube and braid are brazed or welded to pipe-thread, union, or flanged ends. Failures are often the

**TABLE 10-36a Standard Dimensions and Weights, and Tolerances in Diameter and Wall Thickness for Coil Lengths (ASTM B280-2003\*)**

| Standard size, in | Outside diameter, in (mm) | Wall thickness, in (mm) | Weight, lb/ft (kg/m) | Tolerances                                                     |                                                      |
|-------------------|---------------------------|-------------------------|----------------------|----------------------------------------------------------------|------------------------------------------------------|
|                   |                           |                         |                      | Average <sup>A</sup> outside diameter, plus and minus, in (mm) | Wall <sup>B</sup> thickness, plus and minus, in (mm) |
| ¼                 | 0.125 (3.18)              | 0.030 (0.762)           | 0.0347 (0.0516)      | 0.002 (0.051)                                                  | 0.003 (0.08)                                         |
| ⅜                 | 0.187 (4.75)              | 0.030 (0.762)           | 0.0575 (0.0856)      | 0.002 (0.051)                                                  | 0.003 (0.08)                                         |
| ½                 | 0.250 (6.35)              | 0.030 (0.762)           | 0.0804 (0.120)       | 0.002 (0.051)                                                  | 0.0025 (0.08)                                        |
| ⅝                 | 0.312 (7.92)              | 0.032 (0.813)           | 0.109 (0.162)        | 0.002 (0.051)                                                  | 0.003 (0.08)                                         |
| ¾                 | 0.375 (9.52)              | 0.032 (0.813)           | 0.134 (0.199)        | 0.002 (0.051)                                                  | 0.003 (0.08)                                         |
| 1                 | 0.500 (12.7)              | 0.032 (0.813)           | 0.182 (0.271)        | 0.002 (0.051)                                                  | 0.003 (0.08)                                         |
| 1¼                | 0.625 (15.9)              | 0.035 (0.889)           | 0.251 (0.373)        | 0.002 (0.051)                                                  | 0.004 (0.011)                                        |
| 1½                | 0.750 (19.1)              | 0.042 (0.889)           | 0.305 (0.454)        | 0.0025 (0.064)                                                 | 0.004 (0.011)                                        |
| 2                 | 0.750 (19.1)              | 0.042 (1.07)            | 0.362 (0.539)        | 0.0025 (0.064)                                                 | 0.004 (0.011)                                        |
| 2½                | 0.875 (22.3)              | 0.045 (1.14)            | 0.455 (0.677)        | 0.003 (0.076)                                                  | 0.004 (0.11)                                         |
| 3                 | 1.125 (28.6)              | 0.050 (1.27)            | 0.665 (0.975)        | 0.0035 (0.089)                                                 | 0.005 (0.13)                                         |
| 3½                | 1.375 (34.9)              | 0.055 (1.40)            | 0.884 (1.32)         | 0.004 (0.10)                                                   | 0.006 (0.15)                                         |
| 4                 | 1.625 (41.3)              | 0.060 (1.52)            | 1.14 (1.70)          | 0.0045 (0.11)                                                  | 0.006 (0.15)                                         |

\* Extracted from ASTM B280-2003 with permission of the publisher, the American Society for Testing and Materials, West Conshohocken, Pa.

<sup>A</sup> The average outside diameter of a tube is the average of the maximum and minimum outside diameters as determined at any one cross section of the tube.<sup>B</sup> The tolerances listed represent the maximum deviation at any point.



**TABLE 10-36b Standard Dimensions and Weights, and Tolerances in Diameter and Wall Thickness for Straight Lengths (ASTM B280-2003\*)**

Note 1—Applicable to drawn temper tube only.

| Standard size, in | Outside diameter, in (mm) | Wall thickness, in (mm) | Weight, lb/ft (kg/m) | Tolerances                                                     |                                                      |
|-------------------|---------------------------|-------------------------|----------------------|----------------------------------------------------------------|------------------------------------------------------|
|                   |                           |                         |                      | Average <sup>A</sup> outside diameter, plus and minus, in (mm) | Wall <sup>B</sup> thickness, plus and minus, in (mm) |
| 3/8               | 0.375 (9.52)              | 0.030 (0.762)           | 0.126 (0.187)        | 0.001 (0.025)                                                  | 0.0035 (0.08)                                        |
| 1/2               | 0.500 (12.7)              | 0.035 (0.889)           | 0.198 (0.295)        | 0.001 (0.025)                                                  | 0.004 (0.09)                                         |
| 5/8               | 0.625 (15.9)              | 0.040 (1.02)            | 0.285 (0.424)        | 0.001 (0.025)                                                  | 0.004 (0.10)                                         |
| 3/4               | 0.750 (19.1)              | 0.042 (1.07)            | 0.362 (0.539)        | 0.001 (0.025)                                                  | 0.004 (0.11)                                         |
| 7/8               | 0.875 (22.3)              | 0.045 (1.14)            | 0.455 (0.677)        | 0.001 (0.025)                                                  | 0.004 (0.11)                                         |
| 1 1/8             | 1.125 (28.6)              | 0.050 (1.27)            | 0.655 (0.975)        | 0.0015 (0.038)                                                 | 0.004 (0.13)                                         |
| 1 1/4             | 1.375 (34.9)              | 0.055 (1.40)            | 0.884 (1.32)         | 0.0015 (0.038)                                                 | 0.006 (0.11)                                         |
| 1 3/8             | 1.625 (41.3)              | 0.060 (1.52)            | 1.14 (1.70)          | 0.002 (0.051)                                                  | 0.006 (0.11)                                         |
| 2 1/8             | 2.125 (54.0)              | 0.070 (1.78)            | 1.75 (2.60)          | 0.002 (0.051)                                                  | 0.007 (0.15)                                         |
| 2 3/8             | 2.625 (66.7)              | 0.080 (2.03)            | 2.48 (3.69)          | 0.002 (0.051)                                                  | 0.008 (0.15)                                         |
| 3 1/8             | 3.125 (79.4)              | 0.090 (2.29)            | 3.33 (4.96)          | 0.002 (0.051)                                                  | 0.009 (0.23)                                         |
| 3 3/8             | 3.625 (92.1)              | 0.100 (2.54)            | 4.29 (6.38)          | 0.002 (0.051)                                                  | 0.010 (0.25)                                         |
| 4 1/8             | 4.125 (105)               | 0.110 (2.79)            | 5.38 (8.01)          | 0.002 (0.051)                                                  | 0.011 (0.28)                                         |

\*Extracted from ASTM B280-2003, with permission of the publisher, the American Society for Testing and Materials, West Conshohocken, Pa.

<sup>A</sup> The average outside diameter of a tube is the average of the maximum and minimum outside diameters as determined at any one cross section of the tube.

<sup>B</sup> The tolerances listed represent the maximum deviation at any point.

**TABLE 10-37 Copper and Red-Brass Pipe (ASTM B42-2002 and B43-2004)\*: Standard Dimensions, Weights, and Tolerances**

| Standard pipe size, in | Nominal outside diameter, in (mm) | Average outside diameter tolerances, in (mm), all minus† | Nominal wall thickness, in (mm) | Tolerance, in (mm)‡ | Theoretical weight, lb/ft (kg/m) | Theoretical weight, lb/ft (kg/m) |
|------------------------|-----------------------------------|----------------------------------------------------------|---------------------------------|---------------------|----------------------------------|----------------------------------|
| Regular pipe           |                                   |                                                          |                                 | Red brass           |                                  | Copper                           |
| 1/8                    | 0.405 (10.3)                      | 0.004 (0.10)                                             | 0.062 (1.57)                    | 0.004 (0.10)        | 0.253 (0.376)                    | 0.259 (0.385)                    |
| 1/4                    | 0.540 (13.7)                      | 0.004 (0.10)                                             | 0.082 (2.08)                    | 0.005 (0.13)        | 0.447 (0.665)                    | 0.457 (0.680)                    |
| 3/8                    | 0.675 (17.1)                      | 0.005 (0.13)                                             | 0.090 (2.29)                    | 0.005 (0.13)        | 0.627 (0.933)                    | 0.641 (0.954)                    |
| 1/2                    | 0.840 (21.3)                      | 0.005 (0.13)                                             | 0.107 (2.72)                    | 0.006 (0.15)        | 0.934 (1.39)                     | 0.955 (1.42)                     |
| 3/4                    | 1.050 (26.7)                      | 0.006 (0.15)                                             | 0.114 (2.90)                    | 0.006 (0.15)        | 1.27 (1.89)                      | 1.30 (1.93)                      |
| 1                      | 1.315 (33.4)                      | 0.006 (0.15)                                             | 0.126 (3.20)                    | 0.007 (0.18)        | 1.78 (2.65)                      | 1.82 (2.71)                      |
| 1 1/4                  | 1.660 (42.2)                      | 0.006 (0.15)                                             | 0.146 (3.71)                    | 0.008 (0.20)        | 2.63 (3.91)                      | 2.69 (4.00)                      |
| 1 1/2                  | 1.900 (48.3)                      | 0.006 (0.15)                                             | 0.150 (3.81)                    | 0.008 (0.20)        | 3.13 (4.66)                      | 3.20 (4.76)                      |
| 2                      | 2.375 (60.3)                      | 0.008 (0.20)                                             | 0.156 (3.96)                    | 0.009 (0.23)        | 4.12 (6.13)                      | 4.22 (6.28)                      |
| 2 1/2                  | 2.875 (73.0)                      | 0.008 (0.20)                                             | 0.187 (4.75)                    | 0.010 (0.25)        | 5.99 (8.91)                      | 6.12 (9.11)                      |
| 3                      | 3.500 (88.9)                      | 0.010 (0.25)                                             | 0.219 (5.56)                    | 0.012 (0.30)        | 8.56 (12.7)                      | 8.76 (13.0)                      |
| 3 1/2                  | 4.000 (102)                       | 0.010 (0.25)                                             | 0.250 (6.35)                    | 0.013 (0.33)        | 11.2 (16.7)                      | 11.4 (17.0)                      |
| 4                      | 4.500 (114)                       | 0.012 (0.30)                                             | 0.250 (6.35)                    | 0.014 (0.36)        | 12.7 (18.9)                      | 12.9 (19.2)                      |
| 5                      | 5.562 (141)                       | 0.014 (0.36)                                             | 0.250 (6.35)                    | 0.014 (0.36)        | 15.8 (23.5)                      | 16.2 (24.1)                      |
| 6                      | 6.625 (168)                       | 0.016 (0.41)                                             | 0.250 (6.35)                    | 0.014 (0.36)        | 19.0 (28.3)                      | 19.4 (28.9)                      |
| 8                      | 8.625 (219)                       | 0.020 (0.51)                                             | 0.312 (7.92)                    | 0.022 (0.56)        | 30.9 (46.0)                      | 31.6 (47.0)                      |
| 10                     | 10.750 (273)                      | 0.022 (0.56)                                             | 0.365 (9.27)                    | 0.030 (0.76)        | 45.2 (67.3)                      | 46.2 (68.7)                      |
| 12                     | 12.750 (324)                      | 0.024 (0.61)                                             | 0.375 (9.52)                    | 0.030 (0.76)        | 55.3 (82.3)                      | 56.5 (84.1)                      |
| Extrastrong pipe       |                                   |                                                          |                                 | Red brass           |                                  | Copper                           |
| 1/8                    | 0.405 (10.3)                      | 0.004 (0.10)                                             | 0.100 (2.54)                    | 0.006 (0.15)        | 0.363 (0.540)                    | 0.371 (0.552)                    |
| 1/4                    | 0.540 (13.7)                      | 0.004 (0.10)                                             | 0.123 (3.12)                    | 0.007 (0.18)        | 0.611 (0.909)                    | 0.625 (0.930)                    |
| 3/8                    | 0.675 (17.1)                      | 0.005 (0.13)                                             | 0.127 (3.23)                    | 0.007 (0.18)        | 0.829 (1.23)                     | 0.847 (1.26)                     |
| 1/2                    | 0.840 (21.3)                      | 0.005 (0.13)                                             | 0.149 (3.78)                    | 0.008 (0.20)        | 1.23 (1.83)                      | 1.25 (1.86)                      |
| 3/4                    | 1.050 (26.7)                      | 0.006 (0.15)                                             | 0.157 (3.99)                    | 0.009 (0.23)        | 1.67 (2.48)                      | 1.71 (2.54)                      |
| 1                      | 1.315 (33.4)                      | 0.006 (0.15)                                             | 0.182 (4.62)                    | 0.010 (0.25)        | 2.46 (3.66)                      | 2.51 (3.73)                      |
| 1 1/4                  | 1.660 (42.2)                      | 0.006 (0.15)                                             | 0.194 (4.93)                    | 0.010 (0.25)        | 3.39 (5.04)                      | 3.46 (5.15)                      |
| 1 1/2                  | 1.900 (48.3)                      | 0.006 (0.15)                                             | 0.203 (5.16)                    | 0.011 (0.28)        | 4.10 (6.10)                      | 4.19 (6.23)                      |
| 2                      | 2.375 (60.3)                      | 0.008 (0.20)                                             | 0.221 (5.61)                    | 0.012 (0.30)        | 5.67 (8.44)                      | 5.80 (8.63)                      |
| 2 1/2                  | 2.875 (73.0)                      | 0.008 (0.20)                                             | 0.280 (7.11)                    | 0.015 (0.38)        | 8.66 (12.9)                      | 8.85 (13.2)                      |
| 3                      | 3.500 (88.9)                      | 0.010 (0.25)                                             | 0.304 (7.72)                    | 0.016 (0.41)        | 11.6 (17.3)                      | 11.8 (17.6)                      |
| 3 1/2                  | 4.000 (102)                       | 0.010 (0.25)                                             | 0.321 (8.15)                    | 0.017 (0.43)        | 14.1 (21.0)                      | 14.4 (21.4)                      |
| 4                      | 4.500 (114)                       | 0.012 (0.30)                                             | 0.341 (8.66)                    | 0.018 (0.46)        | 16.9 (25.1)                      | 17.3 (25.7)                      |
| 5                      | 5.562 (141)                       | 0.014 (0.36)                                             | 0.375 (9.52)                    | 0.019 (0.48)        | 23.2 (34.5)                      | 23.7 (35.3)                      |
| 6                      | 6.625 (168)                       | 0.016 (0.41)                                             | 0.437 (11.1)                    | 0.027 (0.69)        | 32.2 (47.9)                      | 32.9 (49.0)                      |
| 8                      | 8.625 (219)                       | 0.020 (0.51)                                             | 0.500 (12.7)                    | 0.035 (0.89)        | 48.4 (72.0)                      | 49.5 (73.7)                      |
| 10                     | 10.750 (273)                      | 0.022 (0.56)                                             | 0.500 (12.7)                    | 0.040 (1.0)         | 61.1 (90.9)                      | 62.4 (92.9)                      |

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†The average outside diameter of a tube is the average of the maximum and minimum outside diameters as determined at any one cross section of the tube.

‡Maximum deviation at any one point.

**TABLE 10-38 Hard-Drawn Copper Threadless Pipe (ASTM B302)\***

| Standard pipe size, in | Nominal dimensions, in (mm) |                 |                | Cross-sectional area of bore, in <sup>2</sup> (cm <sup>2</sup> ) | Nominal weight, lb/ft (kg/m) | Tolerances, in (mm)                  |                                |
|------------------------|-----------------------------|-----------------|----------------|------------------------------------------------------------------|------------------------------|--------------------------------------|--------------------------------|
|                        | Outside diameter            | Inside diameter | Wall thickness |                                                                  |                              | Average outside diameter, all minus† | Wall thickness, plus and minus |
| ¼                      | 0.540 (13.7)                | 0.410 (10.4)    | 0.065 (1.65)   | 0.132 (0.852)                                                    | 0.376 (0.559)                | 0.004 (0.10)                         | 0.0035 (0.089)                 |
| ⅜                      | 0.675 (17.1)                | 0.545 (13.8)    | 0.065 (1.65)   | 0.233 (1.50)                                                     | 0.483 (0.719)                | 0.004 (0.10)                         | 0.004 (0.10)                   |
| ½                      | 0.840 (21.3)                | 0.710 (18.0)    | 0.065 (1.65)   | 0.396 (2.55)                                                     | 0.613 (0.912)                | 0.005 (0.13)                         | 0.004 (0.10)                   |
| ¾                      | 1.050 (26.7)                | 0.920 (23.4)    | 0.065 (1.65)   | 0.665 (4.29)                                                     | 0.780 (1.16)                 | 0.005 (0.13)                         | 0.004 (0.10)                   |
| 1                      | 1.315 (33.4)                | 1.185 (30.1)    | 0.065 (1.65)   | 1.10 (7.10)                                                      | 0.989 (1.47)                 | 0.005 (0.13)                         | 0.004 (0.10)                   |
| 1¼                     | 1.660 (42.2)                | 1.530 (38.9)    | 0.065 (1.65)   | 1.84 (11.9)                                                      | 1.26 (1.87)                  | 0.006 (0.15)                         | 0.004 (0.10)                   |
| 1½                     | 1.900 (48.3)                | 1.770 (45.0)    | 0.065 (1.65)   | 2.46 (15.9)                                                      | 1.45 (2.16)                  | 0.006 (0.15)                         | 0.004 (0.10)                   |
| 2                      | 2.375 (60.3)                | 2.245 (57.0)    | 0.065 (1.65)   | 3.96 (25.5)                                                      | 1.83 (272)                   | 0.007 (0.18)                         | 0.006 (0.15)                   |
| 2½                     | 2.875 (73.0)                | 2.745 (69.7)    | 0.065 (1.65)   | 5.92 (38.2)                                                      | 2.22 (3.30)                  | 0.007 (0.18)                         | 0.006 (0.15)                   |
| 3                      | 3.500 (88.9)                | 3.334 (84.7)    | 0.083 (2.11)   | 8.73 (56.3)                                                      | 3.45 (5.13)                  | 0.008 (0.20)                         | 0.007 (0.18)                   |
| 3½                     | 4.000 (102)                 | 3.810 (96.8)    | 0.095 (2.41)   | 11.4 (73.5)                                                      | 4.52 (6.73)                  | 0.008 (0.20)                         | 0.007 (0.18)                   |
| 4                      | 4.500 (114)                 | 4.286 (109)     | 0.107 (2.72)   | 14.4 (92.9)                                                      | 5.72 (8.51)                  | 0.010 (0.25)                         | 0.009 (0.23)                   |
| 5                      | 5.562 (141)                 | 5.298 (135)     | 0.132 (3.40)   | 22.0 (142)                                                       | 8.73 (13.0)                  | 0.012 (0.30)                         | 0.010 (0.25)                   |
| 6                      | 6.625 (168)                 | 6.309 (160)     | 0.158 (4.01)   | 31.3 (202)                                                       | 12.4 (18.5)                  | 0.014 (0.36)                         | 0.010 (0.25)                   |
| 8                      | 8.625 (219)                 | 8.215 (209)     | 0.205 (5.21)   | 53.0 (342)                                                       | 21.0 (31.2)                  | 0.018 (0.46)                         | 0.014 (0.36)                   |
| 10                     | 10.750 (273)                | 10.238 (260)    | 0.256 (6.50)   | 82.3 (531)                                                       | 32.7 (48.7)                  | 0.018 (0.46)                         | 0.016 (0.41)                   |
| 12                     | 12.750 (324)                | 12.124 (308)    | 0.313 (7.95)   | 115 (742)                                                        | 47.4 (70.5)                  | 0.018 (0.46)                         | 0.020 (0.51)                   |

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†The average outside diameter of a tube is the average of the maximum and minimum outside diameters, as determined at any one cross section of the tube.

result of corrosion of the braided-wire jacket or of a poor jacket-to-fitting weld. Maximum recommended temperature for bronze hose is approximately 230°C (450°F). Metal thickness is much less than for straight tube for the same pressure-temperature conditions; so accurate data on corrosion and erosion are required to make proper selection.

## NONMETALLIC PIPE AND METALLIC PIPING SYSTEMS WITH NONMETALLIC LININGS

**Cement-Lined Carbon-Steel Pipe** Cement-lined carbon-steel pipe is made by lining steel pipe with special cement. The cement lin-

ing prevents pickup of iron by the fluid handled, corrosion of the metal by brackish or saline water, and growth of tuberculation. Various grades of cement are available, and the proper grade should be selected to match the application.

Cement-lined pipe in sizes smaller than 1½ in is not generally recommended. Cement-lined carbon steel pipe can be supplied with butt-weld or flanged ends. Butt-welded construction involves the use of special joint grouts at the weld joint and controlled welding procedures. See Table 10-40.

American Water Works Association Standard C205 addresses shop-applied cement mortar lining of pipe sizes 4 in and larger. Fittings are

**TABLE 10-39 Common Nickel and Nickel-Bearing Alloys**

| Common trade name or registered trademark | Code designation             | Alloy no. | ASTM specification (pipe) | Nominal composition, % |      |     |      |                |                 |     |      |                  |                  |                |
|-------------------------------------------|------------------------------|-----------|---------------------------|------------------------|------|-----|------|----------------|-----------------|-----|------|------------------|------------------|----------------|
|                                           |                              |           |                           | Ni                     | Cr   | Mo  | Fe   | C <sup>a</sup> | Si <sup>a</sup> | Mn  | Cu   | Cb               | Co               | W              |
| Type 304 stainless steel                  |                              | S30400    | A312                      | 8                      | 18   |     | BAL  | 0.08           | 1.00            | 2.0 |      |                  |                  |                |
| Type 304L stainless steel                 |                              | S30403    | A312                      | 8                      | 18   |     | BAL  | 0.03           | 1.00            | 2.0 |      |                  |                  |                |
| Type 316 stainless steel                  |                              | S31600    | A312                      | 12                     | 16   | 2   | BAL  | 0.08           | 1.00            | 2.0 |      |                  |                  |                |
| Type 316L stainless steel                 |                              | S31603    | A312                      | 12                     | 16   | 2   | BAL  | 0.03           | 1.00            | 2.0 |      |                  |                  |                |
| Carpenter 20cb <sup>b</sup>               | Ni-Cr-Fe-Mo-Cu-Cb stabilized | N08020    | B464                      | 33                     | 20   | 2.5 | 38.5 | 0.06           |                 | 2.0 | 3    | 1                |                  |                |
| Incoloy 800 <sup>c</sup>                  | Ni-Fe-Cr                     | N08800    | B407                      | 32.5                   | 21   |     | 46   | 0.05           | 0.5             | 0.8 | 0.4  |                  |                  |                |
| Incoloy 825 <sup>c</sup>                  | Ni-Fe-Cr-Mo-Cu               | N08825    | B423                      | 42                     | 21.5 | 3   | 30   | 0.03           | 0.2             | 0.5 | 2.2  |                  |                  |                |
| Hastelloy C-276 <sup>d</sup>              | Ni-Mo-Cr low carbon          | N10276    | B575 <sup>e</sup>         | 54                     | 15   | 16  | 5    | 0.02           | 0.08            | 1   |      |                  | 2.5              | 4              |
| Hastelloy B-2 <sup>d</sup>                | Ni-Mo                        | N10001    | B333 <sup>e</sup>         | 64                     | 1    | 28  | 2    | 0.02           | 0.1             | 1   |      |                  |                  |                |
| Inconel 625 <sup>e</sup>                  | Ni-Cr-Mo-Cb                  | N06625    | B444                      | 61                     | 21.5 | 9   | 2.5  | 0.05           | 0.2             | 0.2 |      | 4                |                  |                |
| Inconel 600 <sup>e</sup>                  | Ni-Cr-Fe                     | N06600    | B167                      | 76                     | 15.5 |     | 8    | 0.08           | 0.2             | 0.5 | 0.2  |                  |                  |                |
| Monel 400 <sup>e</sup>                    | Ni-Cu                        | N04400    | B165                      | 66                     |      |     | 1.2  | 0.20           | 0.2             | 1   | 31.5 |                  |                  |                |
| Nickel 200 <sup>e</sup>                   | Ni                           | N02200    | B161                      | 99+                    |      |     | 0.2  | 0.08           |                 |     |      |                  |                  |                |
| Hastelloy G <sup>d</sup>                  | Ni-Cr-Fe-Mo-Cu               | N06007    | B622                      | 42                     | 22.2 | 6.5 | 19.5 | 0.05           | 1               | 1.5 | 2    | 2.2 <sup>f</sup> | 2.5 <sup>e</sup> | 1 <sup>a</sup> |

<sup>a</sup> Maximum.

<sup>b</sup> Registered trademark, Carpenter Technology Corp.

<sup>c</sup> Registered trademark, Huntington Alloys, Inc.

<sup>d</sup> Registered trademark, Cabot Corp.

<sup>e</sup> Plate.

<sup>f</sup> Cb + Ta.



TABLE 10-40 Cement-Lined Carbon-Steel Pipe\*

| Stand-<br>ard pipe<br>size, in | Inside<br>diam.<br>after<br>lining, in | Typical<br>thickness<br>of lining,<br>in | Weight,<br>per ft,<br>lb | Stand-<br>ard pipe<br>size, in | Inside<br>diam.<br>after<br>lining, in | Typical<br>thickness<br>of lining,<br>in | Weight<br>per ft,<br>lb |
|--------------------------------|----------------------------------------|------------------------------------------|--------------------------|--------------------------------|----------------------------------------|------------------------------------------|-------------------------|
|                                |                                        |                                          |                          | 3                              | 2.70                                   | 0.13                                     | 8.3                     |
|                                |                                        |                                          |                          | 4                              | 3.60                                   | .16                                      | 12.0                    |
|                                |                                        |                                          |                          | 6                              | 5.40                                   | .25                                      | 24.0                    |
| 1½                             | 1.40                                   | .09                                      | 3.0                      | 8                              | 7.40                                   | .25                                      | 32.0                    |
| 2                              | 1.80                                   | .10                                      | 4.1                      | 10                             | 9.40                                   | .30                                      | 43.0                    |
| 2½                             | 2.20                                   | .10                                      | 6.6                      | 12                             | 11.40                                  | .30                                      | 55.0                    |

\*To convert inches to millimeters, multiply by 25.4; to convert pounds per foot to kilograms per meter, multiply by 1.49.

available as cement mortar lined butt-weld or flanged carbon steel, flanged cast iron, or flanged ductile iron. AWWA C602 addresses in-place (i.e., in situ or field) application of cement mortar lining for pipe sizes 4 in and larger.

**Concrete Pipe** Concrete piping and nonmetallic piping such as PVC, RTR, and HDPE are commonly used for buried gravity drain and pressurized applications. Common applications for both include construction culverts and forced water mains, sewage, industrial waste, and storm water systems. Some of the factors to be considered when deciding whether to use concrete or nonmetallic piping include local code requirements, pipe size, soil and commodity corrosivity, commodity temperature and pressure, resistance to tuberculin growth, traffic and burial loads, soil conditions and bedding requirements, groundwater level and buoyancy issues, suitability of available joining methods, ability of joints to resist internal pressure thrust without the use of thrust blocks, availability of pressure-rated and non-pressure-rated fittings, shipping weight, load capacity of available construction equipment, requirements for special equipment such as fusion bonding machines, contractor's experience and labor skill level requirements, and final installed cost.

Nonreinforced concrete culvert pipe for gravity drain applications is manufactured to ASTM C14 in strength Classes 1, 2, and 3. It is available with internal diameters 4 through 36 in. Reinforced concrete culvert for gravity drain applications is manufactured to ASTM C76 with internal diameters of 12 through 144 in. Metric sizes are manufactured to ASTM C14M and C76M. Joints are typically bell and spigot (or a similar variation) with rubber gaskets.

Concrete pressure pipe is typically custom-designed to three different specifications. Each design provides a cement mortar lining or concrete interior. It is advisable to consult manufacturers regarding the most appropriate specification for a given application and the availability of fittings. The names of some manufacturers can be obtained through the American Concrete Pressure Pipe Association. American Water Works Association standard AWWA C300 addresses steel cylinder reinforced concrete pressure pipe in inside-diameter sizes 30 through 144 in. AWWA C301 addresses prestressed reinforced pipe with a steel cylinder wrapped with steel wire. Inside-diameter sizes are 16 through 144 in. AWWA C302 addresses circumferentially reinforced pipe without a steel cylinder or prestress. Inside diameter sizes are 12 through 144 in, with continuous pressure ratings to 0.38 MPa (55 lbf/in<sup>2</sup>) and total pressure (including surge) to 0.45 MPa (65 lbf/in<sup>2</sup>). AWWA C303 addresses reinforced pipe with a steel cylinder helically wrapped with steel bar. Inside-diameter sizes are 10 through 60 in, with pressure ratings to 2.7-MPa (400 lbf/in<sup>2</sup>) working pressure. Joints are typically bell and spigot (or a similar variation) with rubber gaskets. In addition to the gasket, grouting is used on the exterior and interior of the joint to seal otherwise exposed steel.

**Glass Pipe and Fittings** These are made from heat- and chemical-resistant borosilicate glass in accordance with ASTM E-438 Type 1 Class A. This glass is resistant to chemical attack by almost all products, which makes its resistance much more comprehensive than that of other well-known materials. It is highly resistant to water, saline solutions, organic substances, halogens such as chlorine and bromine, and many acids. There are only a few chemicals that can cause noticeable corrosion of the glass surface, such as hydrofluoric acid, concen-

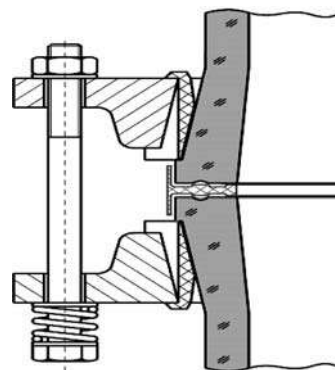


FIG. 10-164 Flanged joint with conical ends. (Adapted with permission of De Dietrich Process Systems, Mountainside, N.J.)

trated phosphoric acid, and strong caustic solutions at elevated temperatures. Some important physical properties are as follows:

Mean linear expansion coefficient between 20 and 300°C:  $[(3.3 \pm 0.1) \times 10^{-6}]/\text{K}$

Mean thermal conductivity between 20 and 200°C: 1.2 W/(m K)

Mean specific heat capacity between 20 and 100°C: 0.8 kJ/kg(K)

Mean specific heat capacity between 101 and 200°C: 0.9 kJ/kg(K)

Density at 20°C: 2.23 kg/dm<sup>3</sup>

Flanged glass pipe with conical ends (Fig. 10-164) should be used in applications requiring a pressure rating or that are expected to see thermal cycling. Flanged ends are normally required to mate with other glass components (vessels, coil heat exchangers, etc). The flanges are specially designed plastic backing flanges that are cushioned from the glass by either molded plastic or fiber inserts. The liquid seal is provided by means of a gasket that is gripped between the grooved pipe ends.

Glass pipe is made in the sizes shown in Table 10-41. Depending on the nominal diameter, lengths range from 0.075 to 3 m. Design pressure ranges from -0.10 MPa (-14.5 lbf/in<sup>2</sup>) vacuum to 0.40 MPa (58 lbf/in<sup>2</sup>) for nominal diameters of 15 through 50 mm, 3 MPa (43 lbf/in<sup>2</sup>) for nominal diameter of 80 mm, 0.20 MPa (29 lbf/in<sup>2</sup>) for nominal diameters of 100 and 150 mm, and 0.10 MPa (14.5 lbf/in<sup>2</sup>) for nominal diameters of 200 and 300 mm. Maximum permissible thermal shock as a general guide is 120 K. Maximum operating temperature is 200°C (248°F). A complete line of fittings is available, and special parts can be made to order. Thermal expansion stresses should be completely relieved by tied PTFE, corrugated expansion joints and offsets. Temperature rating may be limited by joint design and materials.

Beaded pipe is used for process waste lines and vent lines. Some applications have also been made in low-pressure and vacuum lines. Beaded end pipe is available in nominal diameters of 1½ through 6 in. For operating conditions manufacturers should be consulted. In this system the ends of the pipe and fittings are formed into a bead, as

TABLE 10-41 Dimensions for Glass Pipe and Flanged Joints (see Fig. 10-165)\*

| Nominal pipe<br>size, mm | D1    | D2    | D3    | D4         | Type |
|--------------------------|-------|-------|-------|------------|------|
| 15                       | 16.8  | 28.6  | 23    | 15.5–17.5  | A    |
| 25                       | 26.5  | 42.2  | 34    | 25–27      | A    |
| 40                       | 38.5  | 57.4  | 48    | 36.5–39.75 | A    |
| 50                       | 50.5  | 70    | 60.5  | 48–52      | A    |
| 80                       | 76    | 99.2  | 88    | 72–78      | A    |
| 100                      | 104.5 | 132.6 | 120.5 | 97.6–110   | A    |
| 150                      | 154   | 185   | 172   | 150–156    | A    |
| 200                      | 203   | 235   | 220   | 197–205    | B    |
| 300                      | 300   | 340   | 321   | 299–303    | B    |

\*Adapted with permission of De Dietrich Process Systems, Mountainside, N.J.

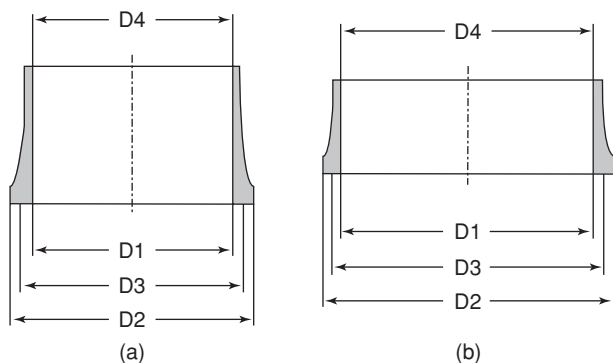


FIG. 10-165 Flanged pipe ends. (Adapted with permission of De Dietrich Process Systems, Mountainside, N.J.)

shown in Fig. 10-166. The coupling consists of a stainless steel outer shell, a rubber collar, and a TFE liner-gasket. When the single coupling nut is tightened, the thick rubber sleeve forces the TFE gasket against the glass to make the seal.

**Glass-Lined Steel Pipe and Fittings** This pipe is fully resistant to all acids except hydrofluoric and concentrated phosphoric acids at temperatures up to 120°C (248°F). It is also resistant to alkaline solutions at moderate temperatures. Glass-lined steel pipe can be used at temperatures up to 220°C (428°F) under some exposure conditions provided there are no excessive temperature changes. The operating pressure rating of commonly available systems is 1 MPa (145 lbf/in<sup>2</sup>). The glass lining is approximately 1.6 mm (1/16 in) thick. It is made by lining Schedule 40 steel pipe. Fittings are available in glass-lined cast steel. Standard nominal pipe sizes available are 1½ through 8 in. Larger-diameter pipe up to 48 in is available on a custom-order basis. A range of lengths are generally available. See Table 10-42 for dimensional data. Steel split flanges drilled to ANSI B16.5 Class 150 dimensions along with PTFE envelope gaskets are used for the assembly of the system.

**Fused Silica or Fused Quartz** Containing 99.8 percent silicon dioxide, fused silica or fused quartz can be obtained as opaque or transparent pipe and tubing. The melting point is 1710°C (3100°F). Tensile strength is approximately 48 MPa (7000 lbf/in<sup>2</sup>); specific gravity is about 2.2. The pipe and tubing can be used continuously at temperatures up to 1000°C (1830°F) and intermittently up to 1500°C (2730°F). The material's chief assets are noncontamination of most chemicals in high-temperature service, thermal-shock resistance, and high-temperature electrical insulating characteristics.

Transparent tubing is available in inside diameters from 1 to 125 mm in a range of wall thicknesses. Satin-surface tubing is available in inside diameters from 1/8 to 2 in, and sand-surface pipe and tubing are

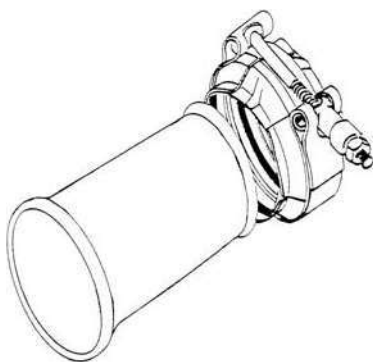


FIG. 10-166 Joint with beaded pipe ends. (Adapted with permission of De Dietrich Process Systems, Mountainside, N.J.)

TABLE 10-42 Glass-Lined Steel Pipe\*

| Size, in. | Outside diameter, in | Range of standard lengths, in |         |
|-----------|----------------------|-------------------------------|---------|
|           |                      | Minimum                       | Maximum |
| 1½        | 1.900                | 6                             | 60      |
| 2         | 2.375                | 6                             | 84      |
| 3         | 3.500                | 6                             | 120     |
| 4         | 4.500                | 6                             | 120     |
| 6         | 6.625                | 6                             | 120     |
| 8         | 8.625                | 6                             | 120     |

\*Adapted with permission of De Dietrich Process Systems, Mountainside, N.J. Other manufacturers may offer different standard lengths.

available in ½- to 24-in inside diameters and lengths up to 6 m (20 ft). Sand-surface pipe and tubing are obtainable in wall thicknesses varying from 1/8 to 1 in. Pipe and tubing sections in both opaque and transparent fused silica or fused quartz can be readily machine-ground to special tolerances for pressure joints or other purposes. Also, fused-silica piping and tubing can be reprocessed to meet special-design requirements. Manufacturers should be consulted for specific details.

**Plastic-Lined Steel Pipe** Use of a variety of polymeric materials as liners for steel pipe rather than as piping systems solves problems which the relatively low tensile strength of the polymer at elevated temperature and high thermal expansion, compared with steel, would produce. The steel outer shell permits much wider spacing of supports, reliable flanged joints, and higher pressure and temperature in the piping. The size range is 1 through 12 in. The systems are flanged with 125-lb cast-iron, 150-lb ductile-iron, and 150- and 300-lb steel flanges. The linings are factory-installed in both pipe and fittings. Lengths are available up to 6 m (20 ft). Lined ball, diaphragm, and check valves and plug cocks are available.

One method of manufacture consists of inserting the liner into an oversize, approximately Schedule 40 steel tube and swaging the assembly to produce iron-pipe-size outside diameter, firmly engaging the liner which projects from both ends of the pipe. Flanges are then screwed onto the pipe, and the projecting liner is hot-flared over the flange faces nearly to the bolt holes. In another method, the liner is pushed into steel pipe having cold-flared laps backed up by flanges at the ends and then hot-flared over the faces of the laps. Pipe lengths made by either method may be shortened in the field and reflared with special procedures and tools. Square and tapered spacers are furnished to adjust for small discrepancies in assembly.

Liner types available are suitable for a wide variety of chemical services, including acids, alkalis, and various solvents. All liners are permeable to some degree, and manufacturers use various methods to vent gas out of the interspace between the liner and casing. All plastics are subject to environmental stress cracking (ESC). ESC can occur even when the liner is chemically resistant to the service. Lined pipe manufacturers should always be consulted regarding liner selections and service applications. Also consult manufacturers regarding vacuum service limits.

**Polyvinylidene Chloride Liners** Polyvinylidene chloride liners have excellent resistance to hydrochloric acid. Maximum temperature is 80°C (175°F). Polyvinylidene chloride is also known as Saran, a product of the Dow Chemical Co.

**Polypropylene Liners** Polypropylene liners are used in sulfuric acid service. At 10 to 30 percent concentration the upper temperature limit is 93°C (200°F). Polypropylene is also suitable for higher concentrations at lower temperatures.

**Kynar Liners** Kynar (Pennwalt Chemicals Corp.) polyvinylidene fluoride liners are used for many chemicals, including bromine and 50 percent hydrochloric acid.

**PTFE and PFA Lined Steel Pipe** These are available in sizes from 1 through 12 in and in lengths through 6 m (20 ft). Experience has determined that practical upper temperature limits are approximately 204°C (400°F) for PTFE (polytetrafluoroethylene) and PFA (perfluoroalkoxy) and 149°C (300°F) for FEP (fluoroethylene polymer); Class 150 and 300 ductile-iron or steel flanged lined fittings and valves are used. The nonadhesive properties of the liner make it ideal for handling sticky or viscous substances. Thickness of the lining varies from 1.5 to 3.8 mm (60 to 150 mil), depending on pipe size. Only flanged joints are used.

**Rubber-Lined Steel Pipe** This pipe is made in lengths up to 6 m (20 ft) with seamless, straight seam-welded and some types of spiral-welded pipe using various types of natural and synthetic adhering rubber. The type of rubber is selected to provide the most suitable lining for the specific service. In general, soft rubber is used for abrasion resistance, semihard for general service, and hard for the more severe service conditions. Multiple-ply lining and combinations of hard and soft rubber are available. The thickness of lining ranges from 3.2 to 6.4 mm ( $\frac{1}{8}$  to  $\frac{1}{4}$  in) depending on the service, the type of rubber, and the method of lining. Cast-steel, ductile-iron, and cast-iron flanged fittings are available rubber-lined. The fittings are usually purchased by the vendor since absence of porosity on the inner surface is essential. Pipe is flanged before rubber lining, and welding elbows and tees may be incorporated at one end of the length of pipe, subject to the conditions that the size of the pipe and the location of the fittings are such that the operator doing the lining can place a hand on any point on the interior surface of the fitting. Welds must be ground smooth on the inside, and a radius is required at the inner edge of the flange face.

The rubber lining is extended out over the face of flanges. With hard-rubber lining, a gasket is required. With soft-rubber lining, a release coating or a polyethylene sheet is required in place of a gasket to avoid bonding of the lining of one flange to the lining on the other and to permit disassembly of the flanged joint. Also, for pressures over 0.86 MPa (125 lbf/in<sup>2</sup>), the tendency of soft-rubber linings to extrude out between the flanges may be prevented by terminating the lining inside the bolt holes and filling the balance of the space between the flange faces with a spacer of the proper thickness. Hard-rubber-lined gate, diaphragm, and swing check valves are available. In gate valves, the stem, wedge assembly, and seat rings, and in the check valves, the hinge pin, flapper arm, disk, and seat ring must be made of metal resistant to the solution handled.

**Plastic Pipe** In contrast to other piping materials, plastic pipe is free from internal and external corrosion, is easily cut and joined, and does not cause galvanic corrosion when coupled to other materials. Allowable stresses and upper temperature limits are low. Normal operation is in the creep range. Fluids for which a plastic is not suited penetrate and soften it rather than dissolve surface layers. Coefficients of thermal expansion are high.

Plastic pipe or tubing may be used for a wide variety of services. As with all nonmetallic materials, code restrictions limit the applications in which their use is permitted. In general, their use in flammable or toxic service is limited. Plastic tubing of various types may be used for instrument air-signal connections; however, as is the case with all non-metallic applications, the need for fire resistance must be considered. When used in specialized applications such as potable water or underground fire water, care should be taken to ensure that the specified products are certified by appropriate agencies such as the National Sanitation Foundation and Factory Mutual.

Support spacing must be much closer than for carbon steel. As temperature increases, the allowable stress for many plastic pipes decreases very rapidly, and heat from sunlight or adjacent hot uninsulated equipment has a marked effect. Many plastics deteriorate with exposure to ultraviolet light if not provided with a UV-resistant coating or other surface barrier. Successful economical underground use of plastic pipe does not necessarily indicate similar economies outdoors aboveground.

Methods of joining include threaded joints with IPS dimensions, solvent-welded joints, heat-fused joints, and insert fittings. Schedules 40 and 80 (see Table 10-22) have been used as a source for standardized dimensions at joints. Some plastics are available in several grades with allowable stresses varying by a factor of 2 to 1. For the same plastic,  $\frac{1}{2}$ -in Schedule 40 pipe of the strongest grade may have 4 times the allowable internal pressure of the weakest grade of a 2-in Schedule 40 pipe. For this reason, the plastic-pipe industry is shifting to standard dimension ratios (approximately the same ratio of diameter to wall thickness over a wide range of pipe sizes).

ASTM and the Plastics Pipe Institute have established identifications for plastic pipe in which the first group of letters identifies the plastic, the two following numbers identify the grade of that plastic, and the last two numbers represent the design stress in the nearest lower 0.7-MPa (100-lbf/in<sup>2</sup>) unit at 23°C (73.4°F).

**Polyethylene** Plastics Pipe Institute ([www.plasticpipe.org](http://www.plasticpipe.org)) is an excellent source of information regarding specification, design, fabrication, and testing of polyethylene piping. Polyethylene (PE) pipe and tubing are available in sizes 48 in and smaller. They have excellent resistance at room temperature to salts, sodium and ammonium hydroxides, and sulfuric, nitric, and hydrochloric acids. High-density polyethylene (HDPE) is often used for underground fire water. Pipe and tubing are produced by extrusion from resins whose density varies with the manufacturing process. Physical properties and therefore wall thickness depend on the particular resin used. In some products, about 3 percent carbon black is added to provide resistance to ultraviolet light. Use of higher-density resin reduces splitting and pinholing in service and increases the strength of the material and the maximum service temperature.

ASTM D2104 covers PE pipe in sizes  $\frac{1}{2}$  through 6 in with IPS Schedule 40 outside and inside diameters for insert-fitting joints. ASTM D2239 covers six standard dimension ratios of pipe diameter to wall thickness in sizes  $\frac{1}{2}$  through 6 in. ASTM D2447 covers sizes  $\frac{1}{2}$  through 12 in with IPS Schedule 40 and 80 outside and inside diameters for use with heat-fusion socket-type and butt-type fittings. ASTM D3035 covers 10 standard dimension ratios of pipe sizes from  $\frac{1}{2}$  through 24 in with IPS outside diameters. All these specifications cover various PE materials. Hydrostatic design stresses within the recommended temperature limits are given in Appendix B, Table B-1, of the code. The hydrostatic design stress is the maximum tensile hoop stress due to internal hydrostatic water pressure that can be applied continuously with a high degree of certainty that failure of the pipe will not occur. Both manufacturers and the Plastic Pipe Institute publish literature describing design calculations required to determine the required wall thickness.

Polyethylene water piping is not damaged by freezing. Pipe and tubing 2 in and smaller are shipped in coils several hundred feet in length.

Clamped-insert joints (Fig. 10-167) are used for flexible plastic pipe up through the 2-in size. Friction between the pipe and the spud is developed both by forcing the spud into the pipe and by tightening the clamp. For the larger sizes, which have thicker walls, these methods cannot develop adequate friction. Insert joints also have high pressure drop. Stainless-steel bands are available. Inserts are available in nylon, polypropylene, and a variety of metals.

Joints of all sizes may be made with heat fusion techniques. Fused joints may be made with either electrofusion or conventional heat fusion. Electrofusion joints are made with fittings which have embedded heating wires. Conventional fusion joints are made with special machines which trim the pipe ends, apply heat, and then force them together to form a bond. Consult the manufacturer regarding the sizes for which electrofusion fittings are available.

A significant use for PE and PP pipe is the technique of rehabilitating deteriorated pipe lines by lining them with plastic pipe. Lining an existing pipe with plastic pipe has a large cost advantage over replacing the line, particularly if replacement of the old line would require excavation.

**Polyvinyl Chloride** Polyvinyl chloride (PVC) pipe and tubing are available with socket fittings for solvent-cemented joints in sizes 24 in and smaller. PVC with gasketed bell and spigot joints is available in sizes 4 through 48 in. Chlorinated polyvinyl chloride (CPVC) pipe and tubing are available with socket fittings for solvent-cemented joints in sizes 4 in and smaller. PVC and CPVC are suitable

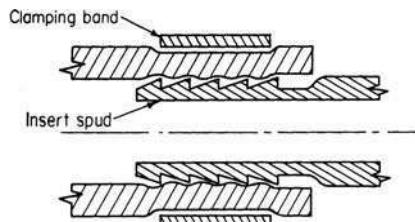


FIG. 10-167 Clamped-insert joint.

for a variety of chemical services and are commonly used for potable water. Consult manufacturers or the Plastics Pipe Institute for chemical resistance data. Hydrostatic design stresses within the temperature limits are given in Appendix B, Table B-1, of the code.

ASTM D1785 covers sizes from 1/4 through 12 in of PVC pipe in IPS Schedules 40, 80, and 120, except that Schedule 120 starts at 1/2 in and is not IPS for sizes from 1/2 through 3 in. ASTM D2241 covers sizes 1/4 through 36 in but with IPS outside diameter and seven standard dimension ratios: 13.5, 17, 21, 26, 32.5, 41, and 64.

ASTM D2513 specifies requirements for thermoplastic materials for buried fuel gas and relining applications. Materials addressed include PE, PVC, and crosslinked polyethylene (PEX). Tubing sizes covered are 1/4 through 1 3/4 in. IPS sizes covered are 1/2 through 12 in. Size availability depends upon the material.

Solvent-cemented joints (Fig. 10-150) are standard, but screwed joints are sometimes used with Schedule 80 pipe. Cemented joints must not be disturbed for 5 min and achieve full strength in 1 day. A wide variety of valve types are available in PVC and CPVC.

**Polypropylene** Polypropylene (PP) pipe and fittings have excellent resistance to most common organic and mineral acids and their salts, strong and weak alkalis, and many organic chemicals. They are available in sizes 1/2 through 6 in, in Schedules 40 and 80, but are not covered as such by ASTM specifications.

**Reinforced-Thermosetting-Resin (RTR) Pipe** Glass-reinforced epoxy resin has good resistance to nonoxidizing acids, alkalis, salt water, and corrosive gases. The glass reinforcement is many times stronger at room temperature than plastics, does not lose strength with increasing temperature, and reinforces the resin effectively up to 149°C (300°F). (See Table 10-21 for temperature limits.) The glass reinforcement is located near the outside wall, protected from the contents by a thick wall of resin and protected from the atmosphere by a thin wall of resin. Stock sizes are 2 through 12 in.

Pipe is supplied in 6- and 12-m (20- and 40-ft) lengths. It is more economical for long, straight runs than for systems containing numerous fittings. When the pipe is sawed to nonfactory lengths, it must be sawed very carefully to avoid cracking the interior plastic zone. A two-component cement may be used to bond lengths into socket couplings or flanges or cemented-joint fittings. Curing of the cement is temperature-sensitive; it sets to full strength in 45 min at 93°C (200°F), in 12 h at 38°C (100°F), and in 24 h at 10°C (50°F). Extensive use is made of shop-fabricated flanged preassemblies. Only flanged joints are used to connect to metallic piping systems. Compared with that of other plastics, the ratio of fitting cost to pipe cost is high. Cemented-joint fittings and flanged fittings are available. Internally lined flanged metallic valves are used.

RTR is more flexible than metallic pipe and consequently requires closer support spacing. While the recommended spacing varies among manufacturers and with the type of product, Table 10-43 gives typical hanger-spacing ranges. The pipe fabricator should be consulted for recommended hanger spacing on the specific pipe-wall construction being used.

Epoxy resin has a higher strength at elevated temperatures than polyester resins but is not as resistant to attack by some fluids. Some glass-reinforced epoxy-resin pipe is made with a polyester-resin liner. The coefficient of thermal expansion of glass-reinforced resin pipe is higher than that for carbon steel but much less than that for plastics.

Glass-reinforced polyester is the most widely used reinforced-resin system. A wide choice of polyester resins is available. The bisphenol resins resist strong acids as well as alkaline solutions. The size range is 2 through at least 36 in; the temperature range is shown

in Table 10-21. Diameters are not standardized. Adhesive-cemented socket joints and hand-lay-up reinforced butt joints are used. For the latter, reinforcement consists of layers of glass cloth saturated with adhesive cement.

## DESIGN OF PIPING-SYSTEMS

**Safeguarding** Safeguarding may be defined as the provision of protective measures as required to ensure the safe operation of a proposed piping system. General considerations to be evaluated should include (1) the hazardous properties of the fluid, (2) the quantity of fluid which could be released by a piping failure, (3) the effect of a failure (such as possible loss of cooling water) on overall plant safety, (4) evaluation of effects on a reaction with the environment (i.e., possibility of a nearby source of ignition), (5) the probable extent of exposure of operating or maintenance personnel, and (6) the relative inherent safety of the piping by virtue of materials of construction, methods of joining, and history of service reliability.

Evaluation of safeguarding requirements might include engineered protection against possible failures such as thermal insulation, armor, guards, barricades, and damping for protection against severe vibration, water hammer, or cyclic operating conditions. Simple means to protect people and property such as shields for valve bonnets, flanged joints, and sight glasses should not be overlooked. The necessity for means to shut off or control flow in the event of a piping failure such as block valves or excess-flow valves should be examined.

**Classification of Fluid Services** The code applies to piping systems as illustrated in Fig. 10-127, but two categories of fluid services are segregated for special consideration as follows:

**Category D** Category D fluid service is defined as "a fluid service to which all the following apply: (1) the fluid handled is non-flammable and nontoxic; (2) the design gage pressure does not exceed 150 psi (1.0 MPa); and (3) the design temperature is between -20°F (-29°C) and 366°F (186°C)."

**Category M** Category M fluid service is defined as "a fluid service in which a single exposure to a very small quantity of a toxic fluid, caused by leakage, can produce serious irreversible harm to persons on breathing or bodily contact, even when prompt restorative measures are taken."

The code assigns to the owner the responsibility for identifying those fluid services which are in Categories D and M. The design and fabrication requirements for Class M toxic-service piping are beyond the scope of this *Handbook*. See ASME B31.3—2004, chap. VIII.

**Design Conditions** Definitions of the temperatures, pressures, and various forces applicable to the design of piping systems are as follows:

**Design Pressure** The design pressure of a piping system shall not be less than the pressure at the most severe condition of coincident internal or external pressure and temperature resulting in the greatest required component thickness or rating.

**Design Temperature** The design temperature is the material temperature representing the most severe condition of coincident pressure and temperature. For uninsulated metallic pipe with fluid below 65°C (150°F), the metal temperature is taken as the fluid temperature.

With fluid at or above 65°C (150°F) and without external insulation, the metal temperature is taken as a percentage of the fluid temperature unless a lower temperature is determined by test or calculation. For pipe, threaded and welding-end valves, fittings, and other components with a wall thickness comparable with that of the pipe, the percentage is 95 percent; for flanges and flanged valves and fittings, 90 percent; for lap-joint flanges, 85 percent; and for bolting, 80 percent.

With external insulation, the metal temperature is taken as the fluid temperature unless service data, tests, or calculations justify lower values. For internally insulated pipe, the design metal temperature shall be calculated or obtained from tests.

**Ambient Influences** If cooling results in a vacuum, the design must provide for external pressure or a vacuum breaker installed; also provision must be made for thermal expansion of contents trapped between or in closed valves. Nonmetallic or nonmetallic-lined pipe may require protection when ambient temperature exceeds design temperature.

**TABLE 10-43 Typical Hanger-Spacing Ranges Recommended for Reinforced-Thermosetting-Resin Pipe**

| Nominal pipe size, in    | 2   | 3   | 4    | 6    | 8    | 10    | 12    |
|--------------------------|-----|-----|------|------|------|-------|-------|
| Hanger-spacing range, ft | 5-8 | 6-9 | 6-10 | 8-11 | 9-13 | 10-14 | 11-15 |

NOTE: Consult pipe manufacturer for recommended hanger spacing for the specific RTR pipe being used. Tabulated values are based on a specific gravity of 1.25 for the contents of the pipe. To convert feet to meters, multiply by 0.3048.

Occasional variations of pressure or temperature, or both, above operating levels are characteristic of certain services. If the following criteria are met, such variations need not be considered in determining pressure-temperature design conditions. Otherwise, the most severe conditions of coincident pressure and temperature during the variation shall be used to determine design conditions. (Application of pressures exceeding pressure-temperature ratings of valves may under certain conditions cause loss of seat tightness or difficulty of operation. Such an application is the owner's responsibility.)

All the following criteria must be met:

1. The piping system shall have no pressure-containing components of cast iron or other nonductile metal.
2. Nominal pressure stresses shall not exceed the yield strength at temperature (see  $S_y$  data in BPV Code, Sec. II, Part D).
3. Combined longitudinal stresses  $S_L$  shall not exceed the limits established in the code (see pressure design of piping components for  $S_L$  limitations).
4. The total number of pressure-temperature variations above the design condition shall not exceed 1000 during the life of the piping system.
5. Occasional variations above design conditions shall remain within one of the following limits for pressure design:
  - When the variation lasts no more than 10 h at any one time and no more than 100 h per year, it is permissible to exceed the pressure rating or the allowable stress for pressure design at the temperature of the increased condition by not more than 33 percent.
  - When the variation lasts no more than 50 h at any one time and not more than 500 h per year, it is permissible to exceed the pressure rating or the allowable stress for pressure design at the temperature of the increased condition by not more than 20 percent.

**Dynamic Effects** Design must provide for impact (hydraulic shock, etc.), wind (exposed piping), earthquake, discharge reactions, and vibrations (of piping arrangement and support).

**Weight Effects** Weight considerations include (1) live loads (contents, ice, and snow), (2) dead loads (pipe, valves, insulation, etc.), and (3) test loads (test fluid).

**Thermal Expansion and Contraction Effects** Thermal-expansion and -contraction loads occur when a piping system is prevented from free thermal expansion or contraction as a result of anchors and restraints or undergoes large, rapid temperature changes or unequal temperature distribution because of an injection of cold liquid striking the wall of a pipe carrying hot gas.

**Effects of Support, Anchor, and Terminal Movements** The effects of movements of piping supports, anchors, and connected equipment shall be taken into account in the design of piping. These movements may result from the flexibility and/or thermal expansion of equipment, supports, or anchors; and from settlement, tidal movements, or wind sway.

**Reduced Ductility** The harmful effects of reduced ductility shall be taken into account in the design of piping. The effects may, e.g., result from welding, heat treatment, forming, bending, or low operating temperatures, including the chilling effect of sudden loss of pressure on highly volatile fluids. Low ambient temperatures expected during operation shall be considered.

**Cyclic Effects** Fatigue due to pressure cycling, thermal cycling, and other cyclic loading shall be considered in the design.

**Air Condensation Effects** At operating temperatures below  $-191^{\circ}\text{C}$  ( $-312^{\circ}\text{F}$ ) in ambient air, condensation and oxygen enrichment occur. These shall be considered in selecting materials, including insulation, and adequate shielding and/or disposal shall be provided.

**Design Criteria: Metallic Pipe** The code uses three different approaches to design, as follows:

1. It provides for the use of dimensionally standardized components at their published pressure-temperature ratings.
2. It provides design formulas and maximum stresses.
3. It prohibits the use of materials, components, or assembly methods in certain conditions.

**Components Having Specific Ratings** These are listed in ASME, API, and industry standards. These ratings are acceptable for design pressures and temperatures unless limited in the code. A list of component standards is given in Appendix E of the ASME B31.3 code. Table 10-44 lists pressure-temperature ratings for

**TABLE 10-44a Pressure-Temperature Ratings for Group 1.1 Materials (Carbon Steel)\***

| Nominal designation                | Forgings                |     |     | Castings          |      | Plates                |      |
|------------------------------------|-------------------------|-----|-----|-------------------|------|-----------------------|------|
| C-Si                               | A 105 (1)               |     |     | A 216 Gr. WCB (1) |      | A 515 Gr. 70 (1)      |      |
| C-Mn-Si                            | A 350 Gr. LF2 (1)       |     |     |                   |      | A 516 Gr. 70 (1), (2) |      |
| C-Mn-Si-V                          | A 350 Gr. LF6 Cl. 1 (4) |     |     |                   |      | A 537 Cl. 1 (3)       |      |
| 3½ NI                              | A 350 Gr. LF 3          |     |     |                   |      |                       |      |
| Working pressures by classes, psig |                         |     |     |                   |      |                       |      |
| Class temp., °F                    | 150                     | 300 | 400 | 600               | 900  | 1500                  | 2500 |
| -20 to 100                         | 285                     | 740 | 985 | 1480              | 2220 | 3705                  | 6170 |
| 200                                | 260                     | 680 | 905 | 1360              | 2035 | 3395                  | 5655 |
| 300                                | 230                     | 655 | 870 | 1310              | 1965 | 3270                  | 5450 |
| 400                                | 200                     | 635 | 845 | 1265              | 1900 | 3170                  | 5280 |
| 500                                | 170                     | 605 | 805 | 1205              | 1810 | 3015                  | 5025 |
| 600                                | 140                     | 570 | 755 | 1135              | 1705 | 2840                  | 4730 |
| 650                                | 125                     | 550 | 730 | 1100              | 1650 | 2745                  | 4575 |
| 700                                | 110                     | 530 | 710 | 1060              | 1590 | 2655                  | 4425 |
| 750                                | 95                      | 505 | 675 | 1015              | 1520 | 2535                  | 4230 |
| 800                                | 80                      | 410 | 550 | 825               | 1235 | 2055                  | 3430 |
| 850                                | 65                      | 320 | 425 | 640               | 955  | 1595                  | 2655 |
| 900                                | 50                      | 230 | 305 | 460               | 690  | 1150                  | 1915 |
| 950                                | 35                      | 135 | 185 | 275               | 410  | 685                   | 1145 |
| 1000                               | 20                      | 85  | 115 | 170               | 255  | 430                   | 715  |

NOTES:

- (1) Upon prolonged exposure to temperatures above  $800^{\circ}\text{F}$ , the carbide phase of steel may be converted to graphite. Permissible, but not recommended for prolonged use above  $800^{\circ}\text{F}$ .
- (2) Not to be used over  $850^{\circ}\text{F}$ .
- (3) Not to be used over  $700^{\circ}\text{F}$ .
- (4) Not to be used over  $500^{\circ}\text{F}$ .

**TABLE 10-44b Pressure-Temperature Ratings for Group 1.5 Materials (Carbon, ½Mo Steel)**

| Nominal designation                | Forgings         |     |     | Castings |      | Plates          |                 |
|------------------------------------|------------------|-----|-----|----------|------|-----------------|-----------------|
| C-1/2Mo                            | A 182 Gr. F1 (1) |     |     |          |      | A 204 Gr. A (1) | A 204 Gr. B (1) |
| Working pressures by classes, psig |                  |     |     |          |      |                 |                 |
| Class temp., °F                    | 150              | 300 | 400 | 600      | 900  | 1500            | 2500            |
| -20 to 100                         | 265              | 695 | 930 | 1395     | 2090 | 3480            | 5805            |
| 200                                | 260              | 695 | 930 | 1395     | 2090 | 3480            | 5805            |
| 300                                | 230              | 685 | 915 | 1375     | 2060 | 3435            | 5725            |
| 400                                | 200              | 660 | 885 | 1325     | 1985 | 3310            | 5520            |
| 500                                | 170              | 640 | 855 | 1285     | 1925 | 3210            | 5350            |
| 600                                | 140              | 605 | 805 | 1210     | 1815 | 3025            | 5040            |
| 650                                | 125              | 590 | 785 | 1175     | 1765 | 2940            | 4905            |
| 700                                | 110              | 570 | 755 | 1135     | 1705 | 2840            | 4730            |
| 750                                | 95               | 530 | 710 | 1065     | 1595 | 2660            | 4430            |
| 800                                | 80               | 510 | 675 | 1015     | 1525 | 2540            | 4230            |
| 850                                | 65               | 485 | 650 | 975      | 1460 | 2435            | 4060            |
| 900                                | 50               | 450 | 600 | 900      | 1350 | 2245            | 3745            |
| 950                                | 35               | 280 | 375 | 560      | 845  | 1405            | 2345            |
| 1000                               | 20               | 165 | 220 | 330      | 495  | 825             | 1370            |

NOTE:

(1) Upon prolonged exposure to temperatures above 875°F, the carbide phase of carbon-molybdenum steel may be converted to graphite. Permissible, but not recommended for prolonged use above 875°F.

**TABLE 10-44c Pressure-Temperature Ratings for Group 2.1 Materials (Type 304 Stainless Steel)**

| Nominal designation                | Forgings           |     |     | Castings          |      | Plates            |      |
|------------------------------------|--------------------|-----|-----|-------------------|------|-------------------|------|
| 18Cr–8Ni                           | A 182 Gr. F304 (1) |     |     | A 351 Gr. CF3 (2) |      | A 240 Gr. 304 (1) |      |
|                                    | A 182 Gr. F304H    |     |     | A 351 Gr. CF8 (1) |      | A 240 Gr. 304H    |      |
| Working pressures by classes, psig |                    |     |     |                   |      |                   |      |
| Class temp., °F                    | 150                | 300 | 400 | 600               | 900  | 1500              | 2500 |
| –20 to 100                         | 275                | 720 | 960 | 1440              | 2160 | 3600              | 6000 |
| 200                                | 230                | 600 | 800 | 1200              | 1800 | 3000              | 5000 |
| 300                                | 205                | 540 | 715 | 1075              | 1615 | 2690              | 4480 |
| 400                                | 190                | 495 | 660 | 995               | 1490 | 2485              | 4140 |
| 500                                | 170                | 465 | 620 | 930               | 1395 | 2330              | 3880 |
| 600                                | 140                | 440 | 590 | 885               | 1325 | 2210              | 3680 |
| 650                                | 125                | 430 | 575 | 865               | 1295 | 2160              | 3600 |
| 700                                | 110                | 420 | 565 | 845               | 1265 | 2110              | 3520 |
| 750                                | 95                 | 415 | 550 | 825               | 1240 | 2065              | 3440 |
| 800                                | 80                 | 405 | 540 | 810               | 1215 | 2030              | 3380 |
| 850                                | 65                 | 395 | 530 | 790               | 1190 | 1980              | 3300 |
| 900                                | 50                 | 390 | 520 | 780               | 1165 | 1945              | 3240 |
| 950                                | 35                 | 380 | 510 | 765               | 1145 | 1910              | 3180 |
| 1000                               | 20                 | 355 | 470 | 710               | 1065 | 1770              | 2950 |
| 1050                               | ...                | 325 | 435 | 650               | 975  | 1630              | 2715 |
| 1100                               | ...                | 255 | 345 | 515               | 770  | 1285              | 2145 |
| 1150                               | ...                | 205 | 275 | 410               | 615  | 1030              | 1715 |
| 1200                               | ...                | 165 | 220 | 330               | 495  | 825               | 1370 |
| 1250                               | ...                | 135 | 180 | 265               | 400  | 670               | 1115 |
| 1300                               | ...                | 115 | 150 | 225               | 340  | 565               | 945  |
| 1350                               | ...                | 95  | 125 | 185               | 280  | 465               | 770  |
| 1400                               | ...                | 75  | 100 | 150               | 225  | 380               | 630  |
| 1450                               | ...                | 60  | 80  | 115               | 175  | 290               | 485  |
| 1500                               | ...                | 40  | 55  | 85                | 125  | 205               | 345  |

NOTES:

(1) At temperatures over 1000°F, use only when the carbon content is 0.04% or higher.

(2) Not to be used over 800°F.

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**TABLE 10-44d Pressure-Temperature Ratings for Group 2.2 Materials (Type 316 Stainless Steel)**

| Nominal designation                | Forgings           |     | Castings           |      | Plates            |      |      |
|------------------------------------|--------------------|-----|--------------------|------|-------------------|------|------|
| 16Cr-12Ni-2Mo                      | A 182 Gr. F316 (1) |     | A 351 Gr. CF3M (2) |      | A 240 Gr. 316 (1) |      |      |
|                                    | A 182 Gr. F316H    |     | A 351 Gr. CF8M (1) |      | A 240 Gr. 316H    |      |      |
| 18Cr-13Ni-3Mo                      | A 182 Gr. F317 (1) |     |                    |      | A 240 Gr. 317 (1) |      |      |
| 19Cr-10Ni-3Mo                      |                    |     | A 351 Gr. CG8M (3) |      |                   |      |      |
| Working pressures by classes, psig |                    |     |                    |      |                   |      |      |
| Class temp., °F                    | 150                | 300 | 400                | 600  | 900               | 1500 | 2500 |
| -20 to 100                         | 275                | 720 | 960                | 1440 | 2160              | 3600 | 6000 |
| 200                                | 235                | 620 | 825                | 1240 | 1860              | 3095 | 5160 |
| 300                                | 215                | 560 | 745                | 1120 | 1680              | 2795 | 4660 |
| 400                                | 195                | 515 | 685                | 1025 | 1540              | 2570 | 4280 |
| 500                                | 170                | 480 | 635                | 955  | 1435              | 2390 | 3980 |
| 600                                | 140                | 450 | 600                | 900  | 1355              | 2255 | 3760 |
| 650                                | 125                | 440 | 590                | 885  | 1325              | 2210 | 3680 |
| 700                                | 110                | 435 | 580                | 870  | 1305              | 2170 | 3620 |
| 750                                | 95                 | 425 | 570                | 855  | 1280              | 2135 | 3560 |
| 800                                | 80                 | 420 | 565                | 845  | 1265              | 2110 | 3520 |
| 850                                | 65                 | 420 | 555                | 835  | 1255              | 2090 | 3480 |
| 900                                | 50                 | 415 | 555                | 830  | 1245              | 2075 | 3460 |
| 950                                | 35                 | 385 | 515                | 775  | 1160              | 1930 | 3220 |
| 1000                               | 20                 | 365 | 485                | 725  | 1090              | 1820 | 3030 |
| 1050                               | ...                | 360 | 480                | 720  | 1080              | 1800 | 3000 |
| 1100                               | ...                | 305 | 405                | 610  | 915               | 1525 | 2545 |
| 1150                               | ...                | 235 | 315                | 475  | 710               | 1185 | 1970 |
| 1200                               | ...                | 185 | 245                | 370  | 555               | 925  | 1545 |
| 1250                               | ...                | 145 | 195                | 295  | 440               | 735  | 1230 |
| 1300                               | ...                | 115 | 155                | 235  | 350               | 585  | 970  |
| 1350                               | ...                | 95  | 130                | 190  | 290               | 480  | 800  |
| 1400                               | ...                | 75  | 100                | 150  | 225               | 380  | 630  |
| 1450                               | ...                | 60  | 80                 | 115  | 175               | 290  | 485  |
| 1500                               | ...                | 40  | 55                 | 85   | 125               | 205  | 345  |

NOTES:

(1) At temperatures over 1000°F, use only when the carbon content is 0.04% or higher.

(2) Not to be used over 850°F.

(3) Not to be used over 1000°F.

**TABLE 10-44e Pressure-Temperature Ratings for Group 2.2 Materials (Type 304L and 316L Stainless Steel)**

| Nominal designation                | Forgings            |     |     | Castings |      | Plates             |      |
|------------------------------------|---------------------|-----|-----|----------|------|--------------------|------|
| 16Cr–12Ni–2Mo                      | A 182 Gr. F316L     |     |     |          |      | A 240 Gr. 316L     |      |
| 18Cr–8Ni                           | A 182 Gr. F304L (1) |     |     |          |      | A 240 Gr. 304L (1) |      |
| Working pressures by classes, psig |                     |     |     |          |      |                    |      |
| Class temp., °F                    | 150                 | 300 | 400 | 600      | 900  | 1500               | 2500 |
| –20 to 100                         | 230                 | 600 | 800 | 1200     | 1800 | 3000               | 5000 |
| 200                                | 195                 | 510 | 680 | 1020     | 1535 | 2555               | 4260 |
| 300                                | 175                 | 455 | 610 | 910      | 1370 | 2280               | 3800 |
| 400                                | 160                 | 420 | 560 | 840      | 1260 | 2100               | 3500 |
| 500                                | 150                 | 395 | 525 | 785      | 1180 | 1970               | 3280 |
| 600                                | 140                 | 370 | 495 | 745      | 1115 | 1860               | 3100 |
| 650                                | 125                 | 365 | 485 | 730      | 1095 | 1825               | 3040 |
| 700                                | 110                 | 360 | 480 | 720      | 1080 | 1800               | 3000 |
| 750                                | 95                  | 355 | 470 | 705      | 1060 | 1765               | 2940 |
| 800                                | 80                  | 345 | 460 | 690      | 1035 | 1730               | 2880 |
| 850                                | 65                  | 340 | 450 | 675      | 1015 | 1690               | 2820 |

NOTE:

(1) Not to be used over 800°F.

flanges, flanged fittings, and flanged valves, and has been extracted from ASME B16.5 with permission of the publisher, the American Society of Mechanical Engineers, New York. Only a few of the more common materials of construction of piping are reproduced here. See ASME B16.5 for other materials. Flanged joints, flanged valves in the open position, and flanged fittings may be subjected to system hydrostatic tests at a pressure not to exceed the hydrostatic-shell test pressure. Flanged valves in the closed position may be subjected to a system hydrostatic test at a pressure not to exceed 110 percent of the 100°F rating of the valve unless otherwise limited by the manufacturer.

Pressure-temperature ratings for soldered tubing joints are given in ASME B16.22—2001.

**Components without Specific Ratings** Components such as pipe and butt-welding fittings are generally furnished in nominal thicknesses. Fittings are rated for the same allowable pressures as pipe of the same nominal thickness and, along with pipe, are rated by the rules for pressure design and other provisions of the code.

#### Limits of Calculated Stresses due to Sustained Loads and Displacement Strains

1. *Internal pressure stresses.* Stresses due to internal pressure shall be considered safe when the wall thickness of the piping component, including any reinforcement, meets the requirements of the pressure design of components defined by the ASME B31.3 code.

2. *External pressure stresses.* Stresses due to external pressure shall be considered safe when the wall thickness of the piping component, and its means of stiffening meet the requirements of the pressure design of components defined by the ASME B31.3 code.

3. *Longitudinal stresses  $S_L$ .* The sum of longitudinal stresses  $S_L$  in any component in a piping system, due to sustained loads such as pressure and weight, shall not exceed the product  $S_h W$ , where  $S_h$  is the basic allowable stress at maximum metal temperature expected during the displacement cycle under analysis, and  $W$  is the weld joint strength reduction factor.

4. *Allowable displacement stress range  $S_A$ .* The computed displacement stress range  $S_E$  in a piping system shall not exceed the allowable displacement stress range  $S_A$ .

$$S_A = f(1.25S_c + 0.25S_h) \quad (10-93)$$

When  $S_h$  is greater than  $S_L$ , the difference between them may be added to the term  $0.25S_h$  in Eq. (10-93). In that case, the allowable stress range is calculated by

$$S_A = f[1.25(S_c + S_h) - S_L] \quad (10-94)$$

where  $S_A$  = allowable displacement stress range

$f$  = stress range factor (see Fig. 10-168)

$S_c$  = basic allowable stress at minimum metal temperature expected during displacement cycle under analysis

$S_h$  = basic allowable stress at maximum metal temperature expected during displacement cycle under analysis

$S_L$  = longitudinal stresses, including pressure and weight

5. *Weld joint strength reduction factor  $W$ .* It is very important to include the weld joint strength reduction factor  $W$  in the design consideration. Especially, at elevated temperatures, the long-term strength of weld joints may be lower than the long-term strength of the base material. The weld joint strength reduction factor only applies at weld locations.

**Pressure Design of Metallic Components** External-pressure stress evaluation of piping is the same as for pressure vessels. But an important difference exists when one is establishing design pressure and wall thickness for internal pressure as a result of the ASME Boiler and Pressure Vessel Code's requirement that the relief-valve setting be not higher than the design pressure. For vessels this means

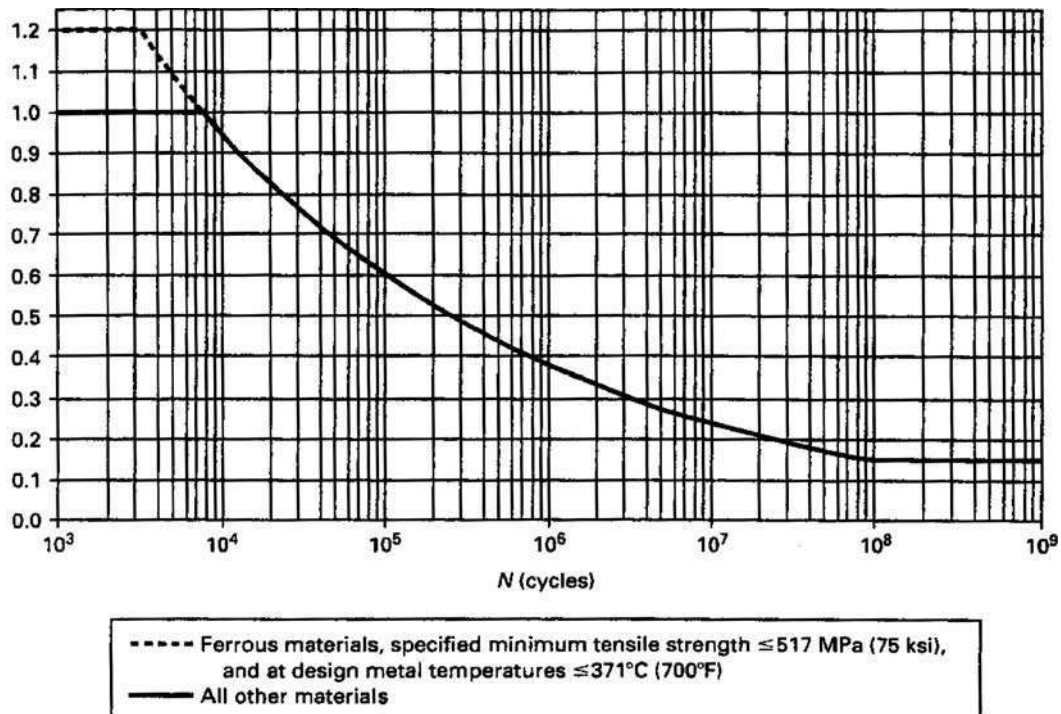


FIG. 10-168 Stress range reduction factor,  $f$ . (Reproduced from ASME B31.3-2004 with permission of the publisher, the American Society of Mechanical Engineers, New York.)



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that the design is for a pressure 10 percent more or less above the intended maximum operating pressure to avoid popping or leakage from the valve during normal operation. However, on piping the design pressure and temperature are taken as the maximum intended operating pressure and coincident temperature combination which results in the maximum thickness. The temporary increased operating conditions listed under "Design Criteria" cover temporary operation at pressures that cause relief valves to leak or open fully. Allowable stresses for nearly 1000 materials are contained in the code. For convenience, the allowable stresses for commonly used carbon and stainless steels have been extracted from the code and listed in Table 10-44.

For **straight metal pipe under internal pressure** the formula for minimum required wall thickness  $t_m$  is applicable for:

1.  $t < D/6$ . The internal pressure design thickness for straight pipe shall be not less than that calculated in accordance with the equation below. The more conservative Barlow and Lamé equations may also be used. Equation (10-95) includes a factor  $Y$  varying with material and temperature to account for the redistribution of circumferential

stress which occurs under steady-state creep at high temperature and permits slightly lesser thickness at this range.

$$t_m = \frac{PD_o}{2(SE + PY)} \div C \quad (10-95)$$

where (in consistent units)

$P$  = design pressure

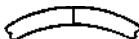
$D_o$  = outside diameter of pipe

$C$  = sum of allowances for corrosion, erosion, and any thread or groove depth. For threaded components the depth is  $h$  of ASME B1.20.1, and for grooved components the depth is the depth removed (0.02 in when no tolerance is specified).

$SE$  = allowable stress (see Table 10-44)

$S$  = basic allowable stress for materials, excluding casting, joint, or structural-grade quality factors

$E$  = quality factor. The quality factor  $E$  is one or the product of more than one of the following quality factors: casting quality factor  $E_c$ , joint quality factor  $E_j$  (see Fig. 10-169).

| No. | Type of Joint                                                                  |                                                                                     | Type of Seam                                             | Examination                                                                    | Factor $E_j$       |
|-----|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|--------------------|
| 1   | Furnace but weld, continuous weld                                              |    | Straight                                                 | As required by listed specification                                            | 0.60<br>[Note (1)] |
| 2   | Electric resistance weld                                                       |    | Straight or spiral                                       | As required by listed specification                                            | 0.85<br>[Note (1)] |
| 3   | Electric fusion weld                                                           |                                                                                     |                                                          |                                                                                |                    |
|     | (a) Single butt weld                                                           |    | Straight or spiral                                       | As required by listed specification or this code                               | 0.80               |
|     | (with or without filler metal)                                                 |                                                                                     |                                                          | Additionally spot radiographed per para. 341.5.1, ASME B31.3                   | 0.90               |
|     |                                                                                |                                                                                     |                                                          | Additionally 100% radiographed per para. 344.5.1 and Table 341.3.2, ASME B31.3 | 1.00               |
|     | (b) Double butt weld                                                           |  | Straight or spiral<br>[except as provided in 4(a) below] | As required by listed specification or this code                               | 0.85               |
|     | (with or without filler metal)                                                 |                                                                                     |                                                          | Additionally spot radiographed per para. 341.5.1, ASME B31.3                   | 0.90               |
|     | Additionally 100% radiographed per para. 344.5.1 and Table 341.3.2, ASME B31.3 |                                                                                     |                                                          | 1.00                                                                           |                    |
| 4   | Per specific specification                                                     |                                                                                     |                                                          |                                                                                |                    |
|     | (a) API 5L                                                                     | Submerged arc weld (SAW)<br>Gas metal arc weld (GMAW)<br>Combined GMAW, SAW         | Straight with one or two seams<br>Spiral                 | As required by specification                                                   | 0.95               |
|     |                                                                                |  |                                                          |                                                                                |                    |

**FIG. 10-169** Longitudinal weld joint quality factor,  $E_j$ . [NOTE (1): It is not permitted to increase the joint quality factor by additional examination for joint 1 or 2.] (Reproduced from ASME B31.3 with permission of the publisher, the American Society of Mechanical Engineers, New York.)

$Y$  = coefficient having value in Table 10-45 valid for  $t < D/6$  and for materials shown

$t_m$  = minimum required thickness, in, including mechanical, corrosion, and erosion allowances

2.  $t \geq D/6$  or  $P/SE > 0.385$ . Calculation of pressure design thickness for straight pipe requires special consideration of factors such as theory of failure, effects of fatigue, and thermal stress.

For flanges of nonstandard dimensions or for sizes beyond the scope of the approved standards, design shall be in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Sec. VIII, except that requirements for fabrication, assembly, inspection testing, and the pressure and temperature limits for materials of the Piping Code are to prevail. Countermoment flanges of flat face or otherwise providing a reaction outside the bolt circle are permitted if designed or tested in accordance with code requirements under pressure-containing components "not covered by standards and for which design formulas or procedures are not given."

**Test Conditions** The shell pressure test for flanged fittings shall be at a pressure no less than 1.5 times the 38°C (100°F) pressure rating rounded off to the next higher 1-bar (psi) increment.

In accordance with listed standards, **blind flanges** may be used at their pressure-temperature ratings. The minimum thickness of non-standard blind flanges shall be the same as for a bolted flat cover, in accordance with the rules of the ASME Boiler and Pressure Vessel Code, Sec. VIII.

**Operational blanks** shall be of the same thickness as blind flanges or may be calculated by the following formula (use consistent units):

$$t = d\sqrt{3P/16SE + C} \quad (10-96)$$

where  $d$  = inside diameter of gasket for raised- or flat (plain)-face flanges, or the gasket pitch diameter for retained gasketed flanges

$P$  = internal design pressure or external design pressure

$S$  = applicable allowable stress

$E$  = quality factor

$C$  = sum of the mechanical allowances

**Valves** must comply with the applicable standards listed in Appendix E of the code and with the allowable pressure-temperature limits established thereby but not beyond the code-established service or materials limitations. Special valves must meet the same requirements as for countermoment flanges.

The code contains no specific rules for the design of **fittings** other than as branch openings. Ratings established by recognized standards are acceptable, however. ASME Standard B16.5 for steel-flanged fittings incorporates a 1.5 shape factor and thus requires the entire fitting to be 50 percent heavier than a simple cylinder in order to provide reinforcement for openings and/or general shape. ASME B16.9 for butt-welding fittings, on the other hand, requires only that the fittings be able to withstand the calculated bursting strength of the straight pipe with which they are to be used.

The thickness of **pipe bends** shall be determined as for straight pipe, provided the bending operation does not result in a difference between maximum and minimum diameters greater than 8 and 3 percent of the nominal outside diameter of the pipe for internal and external pressure, respectively.

**TABLE 10-45 Values of Coefficient  $Y$  for  $t < D/6$**

| Materials            | Temperature, °C (°F)       |              |               |               |               |                          |
|----------------------|----------------------------|--------------|---------------|---------------|---------------|--------------------------|
|                      | ≤482<br>(900<br>and lower) | 510<br>(950) | 538<br>(1000) | 566<br>(1050) | 593<br>(1100) | ≥621<br>(1150<br>and up) |
| Ferritic steels      | 0.4                        | 0.5          | 0.7           | 0.7           | 0.7           | 0.7                      |
| Austenitic steels    | 0.4                        | 0.4          | 0.4           | 0.4           | 0.5           | 0.7                      |
| Other ductile metals | 0.4                        | 0.4          | 0.4           | 0.4           | 0.4           | 0.4                      |
| Cast iron            | 0.0                        | ...          | ...           | ...           | ...           | ...                      |

The maximum allowable internal pressure for multiple miter bends shall be the lesser value calculated from Eqs. (10-97) and (10-98). These equations are not applicable when  $\theta$  exceeds 22.5°.

$$P_m = \frac{SEW(T - C)}{r_2} = \left[ \frac{T - C}{(T - C) + 0.643 \tan \theta \sqrt{r_2(T - C)}} \right] \quad (10-97)$$

$$P_m = \frac{SEW(T - C)}{r_2} = \left( \frac{R_1 - r_2}{R_1 - 0.5r_2} \right) \quad (10-98)$$

where nomenclature is the same as for straight pipe except as follows (see Fig. 10-170):

$S$  = stress value for material

$C$  = sum of mechanical allowances

$r_2$  = mean radius of pipe

$R_1$  = effective radius of miter bend, defined as the shortest distance from the pipe centerline to the intersection of the planes of adjacent miter joints

$\theta$  = angle of miter cut, °

$\alpha$  = angle of change in direction at miter joint  
=  $2\theta$ , °

$T$  = pipe wall thickness

$W$  = weld joint strength reduction factor

$E$  = quality factor

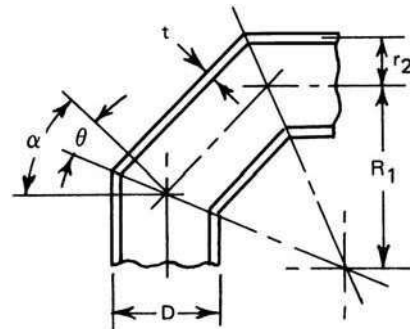
For compliance with the code, the value of  $R_1$  shall not be less than that given by Eq. (10-99):

$$R_1 = A/\tan \theta + D/2 \quad (10-99)$$

where  $A$  has the following empirical values:

| $t$ , in         | $A$                   |
|------------------|-----------------------|
| ≤ 0.5            | 1.0                   |
| 0.5 < $t$ < 0.88 | $2(T - C)$            |
| ≥ 0.88           | $[2(T - C)/3] + 1.17$ |

**Piping branch connections** involve the same considerations as pressure-vessel nozzles. However, outlet size in proportion to piping header size is unavoidably much greater for piping. The current Piping Code rules for calculation of branch-connection reinforcement are similar to those of the ASME Boiler and Pressure Vessel Code, Sec. VIII, Division I—2004 for a branch with axis at right angles to the header axis. If the branch connection makes an angle  $\beta$  with the header axis from 45 to 90°, the Piping Code requires that the area to be replaced be increased by dividing it by  $\sin \beta$ . In such cases the half width of the reinforcing zone measured along the header axis is similarly increased, except that it may not exceed the outside diameter of the header. Some details of commonly used reinforced branch connections are given in Fig. 10-171.



**FIG. 10-170** Nomenclature for miter bends. (Extracted from the Process Piping Code, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.)

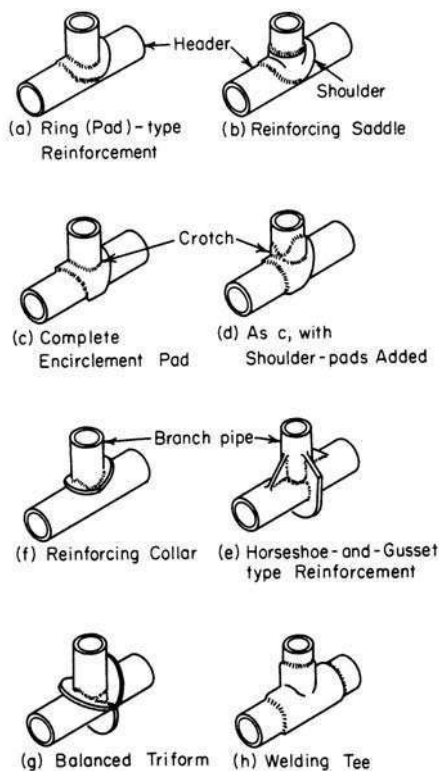


FIG. 10-171 Types of reinforcement for branch connections. (From Kellogg, *Design of Piping Systems*, Wiley, New York, 1965.)

The rules provide that a branch connection has adequate strength for pressure if a fitting (tee, lateral, or cross) is in accordance with an approved standard and is used within the pressure-temperature limitations or if the connection is made by welding a coupling or half coupling (wall thickness not less than the branch anywhere in reinforcement zone or less than extra heavy or 3000 lb) to the run and provided the ratio of branch to run diameters is not greater than one-fourth and that the branch is not greater than 2 in nominal diameter.

Dimensions of extra-heavy couplings are given in the *Steel Products Manual* published by the American Iron and Steel Institute. In ASME B16.11—2001, 2000-lb couplings were superseded by 3000-lb couplings.

ASME B31.3 states that the reinforcement area for resistance to external pressure is to be at least one-half of that required to resist internal pressure.

The code provides no guidance for analysis but requires that external and internal **attachments** be designed to avoid flattening of the pipe, excessive localized bending stresses, or harmful thermal gradients, with further emphasis on minimizing stress concentrations in cyclic service.

The code provides design requirements for **closures** which are flat, ellipsoidal, spherically dished, hemispherical, conical (without transition knuckles), conical convex to pressure, toriconical concave to pressure, and toriconical convex to pressure.

**Openings in closures** over 50 percent in diameter are designed as flanges in flat closures and as reducers in other closures. Openings of not over one-half of the diameter are to be reinforced as branch connections.

**Thermal Expansion and Flexibility: Metallic Piping** ASME B31.3 requires that piping systems have sufficient flexibility to prevent thermal expansion or contraction or the movement of piping supports or terminals from causing (1) failure of piping supports from overstress or fatigue; (2) leakage at joints; or (3) detrimental stresses or distortions in piping or in connected equipment (pumps, turbines, or valves, for example), resulting from excessive thrusts or movements in the piping.

To assure that a system meets these requirements, the computed displacement-stress range  $S_E$  shall not exceed the allowable stress range  $S_A$  [Eqs. (10-93) and (10-94)], the reaction forces  $R_m$  [Eq. (10-106)] shall not be detrimental to supports or connected equipment, and movement of the piping shall be within any prescribed limits.

**Displacement Strains** Strains result from piping being displaced from its unrestrained position:

1. **Thermal displacements.** A piping system will undergo dimensional changes with any change in temperature. If it is constrained from free movement by terminals, guides, and anchors, it will be displaced from its unrestrained position.

2. **Reaction displacements.** If the restraints are not considered rigid and there is a predictable movement of the restraint under load, this may be treated as a compensating displacement.

3. **Externally imposed displacements.** Externally caused movement of restraints will impose displacements on the piping in addition to those related to thermal effects. Such movements may result from causes such as wind sway or temperature changes in connected equipment.

**Total Displacement Strains** Thermal displacements, reaction displacements, and externally imposed displacements all have equivalent effects on the piping system and must be considered together in determining total displacement strains in a piping system.

Expansion strains may be taken up in three ways: by bending, by torsion, or by axial compression. In the first two cases maximum stress occurs at the extreme fibers of the cross section at the critical location. In the third case the entire cross-sectional area over the entire length is for practical purposes equally stressed.

Bending or torsional flexibility may be provided by bends, loops, or offsets; by corrugated pipe or expansion joints of the bellows type; or by other devices permitting rotational movement. These devices must be anchored or otherwise suitably connected to resist end forces from fluid pressure, frictional resistance to pipe movement, and other causes.

Axial flexibility may be provided by expansion joints of the slipjoint or bellows types, suitably anchored and guided to resist end forces from fluid pressure, frictional resistance to movement, and other causes.

**Displacement Stresses** Stresses may be considered proportional to the total displacement strain only if the strains are well distributed and not excessive at any point. The methods outlined here and in the code are applicable only to such a system. Poor distribution of strains (unbalanced systems) may result from:

1. Highly stressed small-size pipe runs in series with large and relatively stiff pipe runs

2. Local reduction in size or wall thickness or local use of a material having reduced yield strength (for example, girth welds of substantially lower strength than the base metal)

3. A line configuration in a system of uniform size in which expansion or contraction must be absorbed largely in a short offset from the major portion of the run

If unbalanced layouts cannot be avoided, appropriate analytical methods must be applied to assure adequate flexibility. If the designer determines that a piping system does not have adequate inherent flexibility, additional flexibility may be provided by adding bends, loops, offsets, swivel joints, corrugated pipe, expansion joints of the bellows or slip-joint type, or other devices. Suitable anchoring must be provided.

As contrasted with stress from sustained loads such as internal pressure or weight, displacement stresses may be permitted to cause limited overstrain in various portions of a piping system. When the system is operated initially at its greatest displacement condition, any yielding reduces stress. When the system is returned to its original condition, there occurs a redistribution of stresses which is referred to as self-springing. It is similar to cold springing in its effects.

While stresses resulting from thermal strain tend to diminish with time, the algebraic difference in displacement condition and in either the original (as-installed) condition or any anticipated condition with a greater opposite effect than the extreme displacement condition remains substantially constant during any one cycle of operation. This difference is defined as the displacement-stress range, and it is a determining factor in the design of piping for flexibility. See Eqs. (10-93) and (10-94) for the allowable stress range  $S_A$  and Eq. (10-101) for the computed stress range  $S_E$ .

**Cold Spring** Cold spring is the intentional deformation of piping during assembly to produce a desired initial displacement and stress. For pipe operating at a temperature higher than that at which it was installed, cold spring is accomplished by fabricating it slightly shorter than design length. Cold spring is beneficial in that it serves to balance the magnitude of stress under initial and extreme displacement conditions. When cold spring is properly applied, there is less likelihood of overstrain during initial operation; hence, it is recommended especially for piping materials of limited ductility. There is also less deviation from as-installed dimensions during initial operation, so that hangers will not be displaced as far from their original settings.

Inasmuch as the service life of a system is affected more by the range of stress variation than by the magnitude of stress at a given time, no credit for cold spring is permitted in stress-range calculations. However, in calculating the thrusts and moments when actual reactions as well as their range of variations are significant, credit is given for cold spring.

Values of thermal-expansion coefficients to be used in determining total displacement strains for computing the stress range are determined from Table 10-46 as the algebraic difference between the value at design maximum temperature and that at the design minimum temperature for the thermal cycle under analysis.

**Values for Reactions** Values of thermal displacements to be used in determining total displacement strains for the computation of reactions on supports and connected equipment shall be determined as the algebraic difference between the value at design maximum (or minimum) temperature for the thermal cycle under analysis and the value at the temperature expected during installation.

The as-installed and maximum or minimum moduli of elasticity,  $E_a$  and  $E_m$  respectively, shall be taken as the values shown in Table 10-47.

Poisson's ratio may be taken as 0.3 at all temperatures for all metals. The allowable stress range for displacement stresses  $S_A$  and permissible additive stresses shall be as specified in Eqs. (10-93) and (10-94) for systems primarily stressed in bending and/or torsion. For pipe or piping components containing longitudinal welds the basic allowable stress  $S$  may be used to determine  $S_A$ .

Nominal thicknesses and outside diameters of pipe and fittings shall be used in flexibility calculations.

In the absence of more directly applicable data, the flexibility factor  $k$  and stress-intensification factor  $i$  shown in Table 10-48 may be used in flexibility calculations in Eq. (10-102). For piping components or attachments (such as valves, strainers, anchor rings, and bands) not covered in the table, suitable stress-intensification factors may be assumed by comparison of their significant geometry with that of the components shown.

**Requirements for Analysis** No formal analysis of adequate flexibility is required in systems which (1) are duplicates of successfully operating installations or replacements without significant change of systems with a satisfactory service record; (2) can readily be judged adequate by comparison with previously analyzed systems; or (3) are of uniform size, have no more than two points of fixation, have no intermediate restraints, and fall within the limitations of empirical Eq. (10-100):\*

$$\frac{Dy}{(L-U)^2} \leq K_1 \quad (10-100)$$

where  $D$  = outside diameter of pipe, in (mm)  
 $y$  = resultant of total displacement strains, in (mm), to be absorbed by the piping system  
 $L$  = developed length of piping between anchors, ft (m)  
 $U$  = anchor distance, straight line between anchors, ft (m)  
 $K_1$  = 0.03 for U.S. customary units listed  
 = 208.3 for SI units listed in parentheses

\*WARNING: No general proof can be offered that this equation will yield accurate or consistently conservative results. It is not applicable to systems used under severe cyclic conditions. It should be used with caution in configurations such as unequal leg U bends ( $L/U > 2.5$ ) or near-straight sawtooth runs, or for large thin-wall pipe ( $i \geq 5$ ), or when extraneous displacements (not in the direction connecting anchor points) constitute a large part of the total displacement. There is no assurance that terminal reactions will be acceptably low even if a piping system falls within the limitations of Eq. (10-100).

1. All systems not meeting these criteria shall be analyzed by simplified, approximate, or comprehensive methods of analysis appropriate for the specific case.

2. Approximate or simplified methods may be applied only if they are used in the range of configurations for which their adequacy has been demonstrated.

3. Acceptable comprehensive methods of analysis include analytical and chart methods which provide an evaluation of the forces, moments, and stresses caused by displacement strains.

4. Comprehensive analysis shall take into account stress-intensification factors for any component other than straight pipe. Credit may be taken for the extra flexibility of such a component.

In calculating the flexibility of a piping system between anchor points, the system shall be treated as a whole. The significance of all parts of the line and of all restraints introduced for the purpose of reducing moments and forces on equipment or small branch lines and also the restraint introduced by support friction shall be recognized. Consider all displacements over the temperature range defined by operating and shutdown conditions.

**Flexibility Stresses** Bending and torsional stresses shall be computed using the as-installed modulus of elasticity  $E_a$  and then combined in accordance with Eq. (10-101) to determine the computed displacement stress range  $S_E$ , which shall not exceed the allowable stress range  $S_A$  [Eqs. (10-93) and (10-94)]:

$$S_E = \sqrt{S_b^2 + 4S_t^2} \quad (10-101)$$

where  $S_b$  = resultant bending stress, lbf/in<sup>2</sup> (MPa)  
 $S_t = M_t/2Z$  = torsional stress, lbf/in<sup>2</sup> (MPa)  
 $M_t$  = torsional moment, in-lbf (N-mm)  
 $Z$  = section modulus of pipe, in<sup>3</sup> (mm<sup>3</sup>)

The resultant bending stresses  $S_b$  to be used in Eq. (10-101) for elbows and miter bends shall be calculated in accordance with Eq. (10-102), with moments as shown in Fig. (10-172):

$$S_b = \frac{\sqrt{(i_t M_t)^2 + (i_o M_o)^2}}{Z} \quad (10-102)$$

where  $S_b$  = resultant bending stress, lbf/in<sup>2</sup> (MPa)  
 $i_t$  = in-plane stress-intensification factor from Table 10-48  
 $i_o$  = out-plane stress-intensification factor from Table 10-48  
 $M_t$  = in-plane bending moment, in-lbf (N-mm)  
 $M_o$  = out-plane bending moment, in-lbf (N-mm)  
 $Z$  = section modulus of pipe, in<sup>3</sup> (mm<sup>3</sup>)

The resultant bending stresses  $S_b$  to be used in Eq. (10-101) for branch connections shall be calculated in accordance with Eqs. (10-103) and (10-104), with moments as shown in Fig. 10-173.

For header (legs 1 and 2):

$$S_b = \frac{\sqrt{(i_t M_t)^2 + (i_o M_o)^2}}{Z} \quad (10-103)$$

For branch (leg 3):

$$S_b = \frac{\sqrt{(i_t M_t)^2 + (i_o M_o)^2}}{Z_e} \quad (10-104)$$

where  $S_b$  = resultant bending stress, lbf/in<sup>2</sup> (MPa)  
 $Z_e$  = effective section modulus for branch, in<sup>3</sup> (mm<sup>3</sup>)

$$Z_e = \pi r_s^2 T_s \quad (10-105)$$

$r_s$  = mean branch cross-sectional radius, in (mm)  
 $T_s$  = effective branch wall thickness, in (mm) [lesser of  $\bar{T}_h$  and  $(i_o) (\bar{T}_b)$ ]  
 $\bar{T}_h$  = thickness of pipe matching run of tee or header exclusive of reinforcing elements, in (mm)

**TABLE 10-46 Thermal Coefficients, U.S. Units, for Metals**Mean coefficient of linear thermal expansion between 70°F and indicated temperature,  $\mu\text{in}/(\text{in}\cdot^{\circ}\text{F})$ 

| Temp., °F | Material                                                       |                             |                                               |                        |           |                                     |      |                                   |          |                      |        |       |           |                                     |                                     |                 |
|-----------|----------------------------------------------------------------|-----------------------------|-----------------------------------------------|------------------------|-----------|-------------------------------------|------|-----------------------------------|----------|----------------------|--------|-------|-----------|-------------------------------------|-------------------------------------|-----------------|
|           | Carbon steel<br>carbon-moly-<br>low-chrome<br>(through 3Cr-Mo) | 5Cr-Mo<br>through<br>9Cr-Mo | Austenitic<br>stainless<br>steels<br>18Cr-8Ni | 12Cr,<br>17Cr,<br>27Cr | 25Cr-20Ni | UNS<br>N04400<br>Monel<br>67Ni-30Cu | 3½Ni | Copper<br>and<br>copper<br>alloys | Aluminum | Gray<br>cast<br>iron | Bronze | Brass | 70Cu-30Ni | UNS<br>N08XXX<br>series<br>Ni-Fe-Cr | UNS<br>N06XXX<br>series<br>Ni-Cr-Fe | Ductile<br>iron |
| -450      | ...                                                            | ...                         | ...                                           | ...                    | ...       | ...                                 | ...  | 6.30                              | ...      | ...                  | ...    | ...   | ...       | ...                                 | ...                                 | ...             |
| -425      | ...                                                            | ...                         | ...                                           | ...                    | ...       | ...                                 | ...  | 6.61                              | ...      | ...                  | ...    | ...   | ...       | ...                                 | ...                                 | ...             |
| -400      | ...                                                            | ...                         | ...                                           | ...                    | ...       | ...                                 | ...  | 6.93                              | ...      | ...                  | ...    | ...   | ...       | ...                                 | ...                                 | ...             |
| -375      | ...                                                            | ...                         | ...                                           | ...                    | ...       | ...                                 | ...  | 7.24                              | ...      | ...                  | ...    | ...   | ...       | ...                                 | ...                                 | ...             |
| -350      | ...                                                            | ...                         | ...                                           | ...                    | ...       | ...                                 | ...  | 7.51                              | ...      | ...                  | ...    | ...   | ...       | ...                                 | ...                                 | ...             |
| -325      | 5.00                                                           | 4.70                        | 8.15                                          | 4.30                   | ...       | 5.55                                | 4.76 | 7.74                              | 9.90     | ...                  | 8.40   | 8.20  | 6.65      | ...                                 | ...                                 | ...             |
| -300      | 5.07                                                           | 4.77                        | 8.21                                          | 4.36                   | ...       | 5.72                                | 4.90 | 7.94                              | 10.04    | ...                  | 8.45   | 8.24  | 6.76      | ...                                 | ...                                 | ...             |
| -275      | 5.14                                                           | 4.84                        | 8.28                                          | 4.41                   | ...       | 5.89                                | 5.01 | 8.11                              | 10.18    | ...                  | 8.50   | 8.29  | 6.86      | ...                                 | ...                                 | ...             |
| -250      | 5.21                                                           | 4.91                        | 8.34                                          | 4.47                   | ...       | 6.06                                | 5.15 | 8.26                              | 10.33    | ...                  | 8.55   | 8.33  | 6.97      | ...                                 | ...                                 | ...             |
| -225      | 5.28                                                           | 4.98                        | 8.41                                          | 4.53                   | ...       | 6.23                                | 5.30 | 8.40                              | 10.47    | ...                  | 8.60   | 8.37  | 7.08      | ...                                 | ...                                 | ...             |
| -200      | 5.35                                                           | 5.05                        | 8.47                                          | 4.59                   | ...       | 6.40                                | 5.45 | 8.51                              | 10.61    | ...                  | 8.65   | 8.41  | 7.19      | ...                                 | ...                                 | 4.65            |
| -175      | 5.42                                                           | 5.12                        | 8.54                                          | 4.64                   | ...       | 6.57                                | 5.52 | 8.62                              | 10.76    | ...                  | 8.70   | 8.46  | 7.29      | ...                                 | ...                                 | 4.76            |
| -150      | 5.50                                                           | 5.20                        | 8.60                                          | 4.70                   | ...       | 6.75                                | 5.59 | 8.72                              | 10.90    | ...                  | 8.75   | 8.50  | 7.40      | ...                                 | ...                                 | 4.87            |
| -125      | 5.57                                                           | 5.26                        | 8.66                                          | 4.78                   | ...       | 6.85                                | 5.67 | 8.81                              | 11.08    | ...                  | 8.85   | 8.61  | 7.50      | ...                                 | ...                                 | 4.98            |
| -100      | 5.65                                                           | 5.32                        | 8.75                                          | 4.85                   | ...       | 6.95                                | 5.78 | 8.89                              | 11.25    | ...                  | 8.95   | 8.73  | 7.60      | ...                                 | ...                                 | 5.10            |
| -75       | 5.72                                                           | 5.38                        | 8.83                                          | 4.93                   | ...       | 7.05                                | 5.83 | 8.97                              | 11.43    | ...                  | 9.05   | 8.84  | 7.70      | ...                                 | ...                                 | 5.20            |
| -50       | 5.80                                                           | 5.45                        | 8.90                                          | 5.00                   | ...       | 7.15                                | 5.88 | 9.04                              | 11.60    | ...                  | 9.15   | 8.95  | 7.80      | ...                                 | ...                                 | 5.30            |
| -25       | 5.85                                                           | 5.51                        | 8.94                                          | 5.05                   | ...       | 7.22                                | 5.94 | 9.11                              | 11.73    | ...                  | 9.23   | 9.03  | 7.87      | ...                                 | ...                                 | 5.40            |
| 0         | 5.90                                                           | 5.56                        | 8.98                                          | 5.10                   | ...       | 7.28                                | 6.00 | 9.17                              | 11.86    | ...                  | 9.32   | 9.11  | 7.94      | ...                                 | ...                                 | 5.50            |
| 25        | 5.96                                                           | 5.62                        | 9.03                                          | 5.14                   | ...       | 7.35                                | 6.08 | 9.23                              | 11.99    | ...                  | 9.40   | 9.18  | 8.02      | ...                                 | ...                                 | 5.58            |
| 50        | 6.01                                                           | 5.67                        | 9.07                                          | 5.19                   | ...       | 7.41                                | 6.16 | 9.28                              | 12.12    | ...                  | 9.49   | 9.26  | 8.09      | ...                                 | ...                                 | 5.66            |
| 70        | 6.07                                                           | 5.73                        | 9.11                                          | 5.24                   | ...       | 7.48                                | 6.25 | 9.32                              | 12.25    | ...                  | 9.57   | 9.34  | 8.16      | ...                                 | 7.13                                | 5.74            |
| 100       | 6.13                                                           | 5.79                        | 9.16                                          | 5.29                   | ...       | 7.55                                | 6.33 | 9.39                              | 12.39    | ...                  | 9.66   | 9.42  | 8.24      | ...                                 | 7.20                                | 5.82            |
| 125       | 6.19                                                           | 5.85                        | 9.20                                          | 5.34                   | ...       | 7.62                                | 6.36 | 9.43                              | 12.53    | ...                  | 9.75   | 9.51  | 8.31      | ...                                 | 7.25                                | 5.87            |
| 150       | 6.25                                                           | 5.92                        | 9.25                                          | 5.40                   | ...       | 7.70                                | 6.39 | 9.48                              | 12.67    | ...                  | 9.85   | 9.59  | 8.39      | ...                                 | 7.30                                | 5.92            |
| 175       | 6.31                                                           | 5.98                        | 9.29                                          | 5.45                   | ...       | 7.77                                | 6.42 | 9.52                              | 12.81    | ...                  | 9.93   | 9.68  | 8.46      | ...                                 | 7.35                                | 5.97            |
| 200       | 6.38                                                           | 6.04                        | 9.34                                          | 5.50                   | 8.79      | 7.84                                | 6.45 | 9.56                              | 12.95    | 5.75                 | 10.03  | 9.76  | 8.54      | 7.90                                | 7.40                                | 6.02            |
| 225       | 6.43                                                           | 6.08                        | 9.37                                          | 5.54                   | 8.81      | 7.89                                | 6.50 | 9.60                              | 13.03    | 5.80                 | 10.05  | 9.82  | 8.58      | 8.01                                | 7.44                                | 6.08            |
| 250       | 6.49                                                           | 6.12                        | 9.41                                          | 5.58                   | 8.83      | 7.93                                | 6.55 | 9.64                              | 13.12    | 5.84                 | 10.08  | 9.88  | 8.63      | 8.12                                | 7.48                                | 6.14            |
| 275       | 6.54                                                           | 6.15                        | 9.44                                          | 5.62                   | 8.85      | 7.98                                | 6.60 | 9.68                              | 13.20    | 5.89                 | 10.10  | 9.94  | 8.67      | 8.24                                | 7.52                                | 6.20            |
| 300       | 6.60                                                           | 6.19                        | 9.47                                          | 5.66                   | 8.87      | 8.02                                | 6.65 | 9.71                              | 13.28    | 5.93                 | 10.12  | 10.00 | 8.71      | 8.35                                | 7.56                                | 6.25            |
| 325       | 6.65                                                           | 6.23                        | 9.50                                          | 5.70                   | 8.89      | 8.07                                | 6.69 | 9.74                              | 13.36    | 5.97                 | 10.15  | 10.06 | 8.76      | 8.46                                | 7.60                                | 6.31            |
| 350       | 6.71                                                           | 6.27                        | 9.53                                          | 5.74                   | 8.90      | 8.11                                | 6.73 | 9.78                              | 13.44    | 6.02                 | 10.18  | 10.11 | 8.81      | 8.57                                | 7.63                                | 6.37            |
| 375       | 6.76                                                           | 6.30                        | 9.56                                          | 5.77                   | 8.91      | 8.16                                | 6.77 | 9.81                              | 13.52    | 6.06                 | 10.20  | 10.17 | 8.85      | 8.69                                | 7.67                                | 6.43            |
| 400       | 6.82                                                           | 6.34                        | 9.59                                          | 5.81                   | 8.92      | 8.20                                | 6.80 | 9.84                              | 13.60    | 6.10                 | 10.23  | 10.23 | 8.90      | 8.80                                | 7.70                                | 6.48            |
| 425       | 6.87                                                           | 6.38                        | 9.62                                          | 5.85                   | 8.92      | 8.25                                | 6.83 | 9.86                              | 13.68    | 6.15                 | 10.25  | 10.29 | ...       | 8.82                                | 7.72                                | 6.57            |
| 450       | 6.92                                                           | 6.42                        | 9.65                                          | 5.89                   | 8.92      | 8.30                                | 6.86 | 9.89                              | 13.75    | 6.19                 | 10.28  | 10.35 | ...       | 8.85                                | 7.75                                | 6.66            |
| 475       | 6.97                                                           | 6.46                        | 9.67                                          | 5.92                   | 8.92      | 8.35                                | 6.89 | 9.92                              | 13.83    | 6.24                 | 10.30  | 10.41 | ...       | 8.87                                | 7.77                                | 6.75            |
| 500       | 7.02                                                           | 6.50                        | 9.70                                          | 5.96                   | 8.93      | 8.40                                | 6.93 | 9.94                              | 13.90    | 6.28                 | 10.32  | 10.47 | ...       | 8.90                                | 7.80                                | 6.85            |
| 525       | 7.07                                                           | 6.54                        | 9.73                                          | 6.00                   | 8.93      | 8.45                                | 6.97 | 9.97                              | 13.98    | 6.33                 | 10.35  | 10.53 | ...       | 8.92                                | 7.82                                | 6.88            |
| 550       | 7.12                                                           | 6.58                        | 9.76                                          | 6.05                   | 8.93      | 8.49                                | 7.01 | 9.99                              | 14.05    | 6.38                 | 10.38  | 10.58 | ...       | 8.95                                | 7.85                                | 6.92            |
| 575       | 7.17                                                           | 6.62                        | 9.79                                          | 6.09                   | 8.93      | 8.54                                | 7.04 | 10.1                              | 14.13    | 6.42                 | 10.41  | 10.64 | ...       | 8.97                                | 7.88                                | 6.95            |
| 600       | 7.23                                                           | 6.66                        | 9.82                                          | 6.13                   | 8.94      | 8.58                                | 7.08 | 10.04                             | 14.20    | 6.47                 | 10.44  | 10.69 | ...       | 9.00                                | 7.90                                | 6.98            |
| 625       | 7.28                                                           | 6.70                        | 9.85                                          | 6.17                   | 8.94      | 8.63                                | 7.12 | ...                               | ...      | 6.52                 | 10.46  | 10.75 | ...       | 9.02                                | 7.92                                | 7.02            |
| 650       | 7.33                                                           | 6.73                        | 9.87                                          | 6.20                   | 8.95      | 8.68                                | 7.16 | ...                               | ...      | 6.56                 | 10.48  | 10.81 | ...       | 9.05                                | 7.95                                | 7.04            |
| 675       | 7.38                                                           | 6.77                        | 9.90                                          | 6.23                   | 8.95      | 8.73                                | 7.19 | ...                               | ...      | 6.61                 | 10.50  | 10.86 | ...       | 9.07                                | 7.98                                | 7.08            |
| 700       | 7.44                                                           | 6.80                        | 9.92                                          | 6.26                   | 8.96      | 8.78                                | 7.22 | ...                               | ...      | 6.65                 | 10.52  | 10.92 | ...       | 9.10                                | 8.00                                | 7.11            |
| 725       | 7.49                                                           | 6.84                        | 9.95                                          | 6.29                   | 8.96      | 8.83                                | 7.25 | ...                               | ...      | 6.70                 | 10.55  | 10.98 | ...       | 9.12                                | 8.02                                | 7.14            |
| 750       | 7.54                                                           | 6.88                        | 9.99                                          | 6.33                   | 8.96      | 8.87                                | 7.29 | ...                               | ...      | 6.74                 | 10.57  | 11.04 | ...       | 9.15                                | 8.05                                | 7.18            |
| 775       | 7.59                                                           | 6.92                        | 10.02                                         | 6.36                   | 8.96      | 8.92                                | 7.31 | ...                               | ...      | 6.79                 | 10.60  | 11.10 | ...       | 9.17                                | 8.08                                | 7.22            |
| 800       | 7.65                                                           | 6.96                        | 10.05                                         | 6.39                   | 8.97      | 8.96                                | 7.34 | ...                               | ...      | 6.83                 | 10.62  | 11.16 | ...       | 9.20                                | 8.10                                | 7.25            |
| 825       | 7.70                                                           | 7.00                        | 10.08                                         | 6.42                   | 8.97      | 9.01                                | 7.37 | ...                               | ...      | 6.87                 | 10.65  | 11.22 | ...       | 9.22                                | ...                                 | 7.27            |

|      |      |      |       |      |      |       |      |     |     |      |       |       |     |       |     |      |
|------|------|------|-------|------|------|-------|------|-----|-----|------|-------|-------|-----|-------|-----|------|
| 850  | 7.75 | 7.03 | 10.11 | 6.46 | 8.98 | 9.06  | 7.40 | ... | ... | 6.92 | 10.67 | 11.28 | ... | 9.25  | ... | 7.31 |
| 875  | 7.79 | 7.07 | 10.13 | 6.49 | 8.99 | 9.11  | 7.43 | ... | ... | 6.96 | 10.70 | 11.34 | ... | 9.27  | ... | 7.34 |
| 900  | 7.84 | 7.10 | 10.16 | 6.52 | 9.00 | 9.16  | 7.45 | ... | ... | 7.00 | 10.72 | 11.40 | ... | 9.30  | ... | 7.37 |
| 925  | 7.87 | 7.13 | 10.19 | 6.55 | 9.05 | 9.21  | 7.47 | ... | ... | 7.05 | 10.74 | 11.46 | ... | 9.32  | ... | 7.41 |
| 950  | 7.91 | 7.16 | 10.23 | 6.58 | 9.10 | 9.25  | 7.49 | ... | ... | 7.10 | 10.76 | 11.52 | ... | 9.35  | ... | 7.44 |
| 975  | 7.94 | 7.19 | 10.26 | 6.60 | 9.15 | 9.30  | 7.52 | ... | ... | 7.14 | 10.78 | 11.57 | ... | 9.37  | ... | 7.47 |
| 1000 | 7.97 | 7.22 | 10.29 | 6.63 | 9.18 | 9.34  | 7.55 | ... | ... | 7.19 | 10.80 | 11.63 | ... | 9.40  | ... | 7.50 |
| 1025 | 8.01 | 7.25 | 10.32 | 6.65 | 9.20 | 9.39  | ...  | ... | ... | ...  | 10.83 | 11.69 | ... | 9.42  | ... | ...  |
| 1050 | 8.05 | 7.27 | 10.34 | 6.68 | 9.22 | 9.43  | ...  | ... | ... | ...  | 10.85 | 11.74 | ... | 9.45  | ... | ...  |
| 1075 | 8.08 | 7.30 | 10.37 | 6.70 | 9.24 | 9.48  | ...  | ... | ... | ...  | 10.88 | 11.80 | ... | 9.47  | ... | ...  |
| 1100 | 8.12 | 7.32 | 10.39 | 6.72 | 9.25 | 9.52  | ...  | ... | ... | ...  | 10.90 | 11.85 | ... | 9.50  | ... | ...  |
| 1125 | 8.14 | 7.34 | 10.41 | 6.74 | 9.29 | 9.57  | ...  | ... | ... | ...  | 10.93 | 11.91 | ... | 9.52  | ... | ...  |
| 1150 | 8.16 | 7.37 | 10.44 | 6.75 | 9.33 | 9.61  | ...  | ... | ... | ...  | 10.95 | 11.97 | ... | 9.55  | ... | ...  |
| 1175 | 8.17 | 7.39 | 10.46 | 6.77 | 9.36 | 9.66  | ...  | ... | ... | ...  | 10.98 | 12.03 | ... | 9.57  | ... | ...  |
| 1200 | 8.19 | 7.41 | 10.48 | 6.78 | 9.39 | 9.70  | ...  | ... | ... | ...  | 11.00 | 12.09 | ... | 9.60  | ... | ...  |
| 1225 | 8.21 | 7.43 | 10.50 | 6.80 | 9.43 | 9.75  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.64  | ... | ...  |
| 1250 | 8.24 | 7.45 | 10.51 | 6.82 | 9.47 | 9.79  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.68  | ... | ...  |
| 1275 | 8.26 | 7.47 | 10.53 | 6.83 | 9.50 | 9.84  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.71  | ... | ...  |
| 1300 | 8.28 | 7.49 | 10.54 | 6.85 | 9.53 | 9.88  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.75  | ... | ...  |
| 1325 | 8.30 | 7.51 | 10.56 | 6.86 | 9.53 | 9.92  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.79  | ... | ...  |
| 1350 | 8.32 | 7.52 | 10.57 | 6.88 | 9.54 | 9.96  | ...  | ... | ... | ...  | ...   | ...   | ... | 9.83  | ... | ...  |
| 1375 | 8.34 | 7.54 | 10.59 | 6.89 | 9.55 | 10.00 | ...  | ... | ... | ...  | ...   | ...   | ... | 9.86  | ... | ...  |
| 1400 | 8.36 | 7.55 | 10.60 | 6.90 | 9.56 | 10.04 | ...  | ... | ... | ...  | ...   | ...   | ... | 9.90  | ... | ...  |
| 1425 | ...  | ...  | 10.64 | ...  | ...  | ...   | ...  | ... | ... | ...  | ...   | ...   | ... | 9.94  | ... | ...  |
| 1450 | ...  | ...  | 10.68 | ...  | ...  | ...   | ...  | ... | ... | ...  | ...   | ...   | ... | 9.98  | ... | ...  |
| 1475 | ...  | ...  | 10.72 | ...  | ...  | ...   | ...  | ... | ... | ...  | ...   | ...   | ... | 10.01 | ... | ...  |
| 1500 | ...  | ...  | 10.77 | ...  | ...  | ...   | ...  | ... | ... | ...  | ...   | ...   | ... | 10.05 | ... | ...  |

GENERAL NOTE: For Code references to this table, see para. 319.3.1, ASME B31.3. These data are for use in the absence of more applicable data. It is the designer's responsibility to verify that materials are suitable for the intended service at the temperatures shown.

|                                                                                                         | <i>E</i> = modulus of elasticity, Msi (millions of psi), at temperature, °F |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|--|
| Material                                                                                                | −425                                                                        | −400 | −350 | −325 | −200 | −100 | 70   | 200  | 300  | 400  | 500  | 600  | 700  | 800  | 900  | 1000 | 1100 | 1200 | 1300 | 1400 | 1500 |  |
| <b>Ferrous Metals</b>                                                                                   |                                                                             |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Gray cast iron                                                                                          | ...                                                                         | ...  | ...  | ...  | ...  | ...  | 13.4 | 13.2 | 12.9 | 12.6 | 12.2 | 11.7 | 11.0 | 10.2 | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| Carbon steels, <i>C</i> ≤ 0.3%                                                                          | 31.9                                                                        | ...  | ...  | 31.4 | 30.8 | 30.2 | 29.5 | 28.8 | 28.3 | 27.7 | 27.3 | 26.7 | 25.5 | 24.2 | 22.4 | 20.4 | 18.0 | ...  | ...  | ...  | ...  |  |
| Carbon steels, <i>C</i> ≤ 0.3%                                                                          | 31.7                                                                        | ...  | ...  | 31.2 | 30.6 | 30.0 | 29.3 | 28.6 | 28.1 | 27.5 | 27.1 | 26.5 | 25.3 | 24.0 | 22.2 | 20.2 | 17.9 | 15.4 | ...  | ...  | ...  |  |
| Carbon-moly steels                                                                                      | 31.7                                                                        | ...  | ...  | 31.1 | 30.5 | 29.9 | 29.2 | 28.5 | 28.0 | 27.4 | 27.0 | 26.4 | 25.3 | 23.9 | 22.2 | 20.1 | 17.8 | 15.3 | ...  | ...  | ...  |  |
| Nickel steels, Ni 2%–9%                                                                                 | 30.1                                                                        | ...  | ...  | 29.6 | 29.1 | 28.5 | 27.8 | 27.1 | 26.7 | 26.1 | 25.7 | 25.2 | 24.6 | 23.0 | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| Cr-Mo steels, Cr ½%–2%                                                                                  | 32.1                                                                        | ...  | ...  | 31.6 | 31.0 | 30.4 | 29.7 | 29.0 | 28.5 | 27.9 | 27.5 | 26.9 | 26.3 | 25.5 | 24.8 | 23.9 | 23.0 | 21.8 | 20.5 | 18.9 | ...  |  |
| Cr-Mo steels, Cr 2¼%–3%                                                                                 | 33.1                                                                        | ...  | ...  | 32.6 | 32.0 | 31.4 | 30.6 | 29.8 | 29.4 | 28.8 | 28.3 | 27.7 | 27.1 | 26.3 | 25.6 | 24.6 | 23.7 | 22.5 | 21.1 | 19.4 | ...  |  |
| Cr-Mo steels, Cr 5%–9%                                                                                  | 33.4                                                                        | ...  | ...  | 32.9 | 32.3 | 31.7 | 30.9 | 30.1 | 29.7 | 29.0 | 28.6 | 28.0 | 27.3 | 26.1 | 24.7 | 22.7 | 20.4 | 18.2 | 15.5 | 12.7 | ...  |  |
| Chromium steels, Cr 12%, 17%, 27%                                                                       | 31.8                                                                        | ...  | ...  | 31.2 | 30.7 | 30.1 | 29.2 | 28.5 | 27.9 | 27.3 | 26.7 | 26.1 | 25.6 | 24.7 | 23.2 | 21.5 | 19.1 | 16.6 | ...  | ...  | ...  |  |
| Austenitic steels (TP304, 310, 316, 321, 347)                                                           | 30.8                                                                        | ...  | ...  | 30.3 | 29.7 | 29.0 | 28.3 | 27.6 | 27.0 | 26.5 | 25.8 | 25.3 | 24.8 | 24.1 | 23.5 | 22.8 | 22.1 | 21.2 | 20.2 | 19.2 | 18.1 |  |
| <b>Copper and Copper Alloys (UNS Nos.)</b>                                                              |                                                                             |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Comp. and leaded Sn-bronze (C83600, C92200)                                                             | ...                                                                         | ...  | ...  | 14.8 | 14.6 | 14.4 | 14.0 | 13.7 | 13.4 | 13.2 |      |      |      |      |      |      |      |      |      |      |      |  |
| Naval brass, Si- & Al-bronze (C46400, C65500, C95200, C95400)                                           | ...                                                                         | ...  | ...  | 15.9 | 15.6 | 15.4 | 15.0 | 14.6 | 14.4 | 14.1 | 12.9 | 12.5 | 12.0 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| Copper (C11000)                                                                                         | ...                                                                         | ...  | ...  | 16.9 | 16.6 | 16.5 | 16.0 | 15.6 | 15.4 | 15.0 | 13.8 | 13.4 | 12.8 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| Copper, red brass, Al-bronze (C10200, C12000, C12200, C12500, C14200, C23000, C61400)                   | ...                                                                         | ...  | ...  | 18.0 | 17.7 | 17.5 | 17.0 | 16.6 | 16.3 | 16.0 | 14.7 | 14.2 | 13.7 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| 90Cu-10Ni (C70600)                                                                                      | ...                                                                         | ...  | ...  | 19.0 | 18.7 | 18.5 | 18.0 | 17.6 | 17.3 | 16.9 | 15.6 | 15.1 | 14.5 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| Leaded Ni-bronze                                                                                        | ...                                                                         | ...  | ...  | 20.1 | 19.8 | 19.6 | 19.0 | 18.5 | 18.2 | 17.9 | 16.6 | 16.0 | 15.4 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| 80Cu-20Ni (C71000)                                                                                      | ...                                                                         | ...  | ...  | 21.2 | 20.8 | 20.6 | 20.0 | 19.5 | 19.2 | 18.8 | 17.5 | 16.9 | 16.2 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| 70Cu-30Ni (C71500)                                                                                      | ...                                                                         | ...  | ...  | 23.3 | 22.9 | 22.7 | 22.0 | 21.5 | 21.1 | 20.7 | 18.4 | 17.8 | 17.1 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
|                                                                                                         |                                                                             |      |      |      |      |      |      |      |      |      | 20.2 | 19.6 | 18.8 | ...  | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| <b>Nickel and Nickel Alloys (UNS Nos.)</b>                                                              |                                                                             |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Monel 400 N04400                                                                                        | 28.3                                                                        | ...  | ...  | 27.8 | 27.3 | 26.8 | 26.0 | 25.4 | 25.0 | 24.7 | 24.3 | 24.1 | 23.7 | 23.1 | 22.6 | 22.1 | 21.7 | 21.2 | ...  | ...  | ...  |  |
| Alloys N06007, N08320                                                                                   | 30.3                                                                        | ...  | ...  | 29.5 | 29.2 | 28.6 | 27.8 | 27.1 | 26.7 | 26.4 | 26.0 | 25.7 | 25.3 | 24.7 | 24.2 | 23.6 | 23.2 | 22.7 | ...  | ...  | ...  |  |
| Alloys N08900, N08810, N06002                                                                           | 31.1                                                                        | ...  | ...  | 30.5 | 29.9 | 29.4 | 28.5 | 27.8 | 27.4 | 27.1 | 26.6 | 26.4 | 25.9 | 25.4 | 24.8 | 24.2 | 23.8 | 23.2 | ...  | ...  | ...  |  |
| Alloys N06455, N10276                                                                                   | 32.5                                                                        | ...  | ...  | 31.6 | 31.3 | 30.6 | 29.8 | 29.1 | 28.6 | 28.3 | 27.9 | 27.6 | 27.1 | 26.5 | 25.9 | 25.3 | 24.9 | 24.3 | ...  | ...  | ...  |  |
| Alloys N02200, N02201, N06625                                                                           | 32.7                                                                        | ...  | ...  | 32.1 | 31.5 | 30.9 | 30.0 | 29.3 | 28.8 | 28.5 | 28.1 | 27.8 | 27.3 | 26.7 | 26.1 | 25.5 | 25.1 | 24.5 | ...  | ...  | ...  |  |
| Alloy N06600                                                                                            | 33.8                                                                        | ...  | ...  | 33.2 | 32.6 | 31.9 | 31.0 | 30.2 | 29.9 | 29.5 | 29.0 | 28.7 | 28.2 | 27.6 | 27.0 | 26.4 | 25.9 | 25.3 | ...  | ...  | ...  |  |
| Alloy N10001                                                                                            | 33.9                                                                        | ...  | ...  | 33.3 | 32.7 | 32.0 | 31.1 | 30.3 | 29.9 | 29.5 | 29.1 | 28.8 | 28.3 | 27.7 | 27.1 | 26.4 | 26.0 | 25.3 | ...  | ...  | ...  |  |
| Alloy N10665                                                                                            | 34.2                                                                        | ...  | ...  | 33.3 | 33.0 | 32.3 | 31.4 | 30.6 | 30.1 | 29.8 | 29.4 | 29.0 | 28.6 | 27.9 | 27.3 | 26.7 | 26.2 | 25.6 | ...  | ...  | ...  |  |
| <b>Unalloyed Titanium</b>                                                                               |                                                                             |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Grades 1, 2, 3, and 7                                                                                   | ...                                                                         | ...  | ...  | ...  | ...  | ...  | 15.5 | 15.0 | 14.6 | 14.0 | 13.3 | 12.6 | 11.9 | 11.2 | ...  | ...  | ...  | ...  | ...  | ...  | ...  |  |
| <b>Aluminum and Aluminum Alloys (UNS Nos.)</b>                                                          |                                                                             |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Grades 443, 1060, 1100, 3003, 3004, 6061, 6063 (A24430, A91060, A91100, A93003, A93004, A96061, A96063) | 11.4                                                                        | ...  | ...  | 11.1 | 10.8 | 10.5 | 10.0 | 9.6  | 9.2  | 8.7  |      |      |      |      |      |      |      |      |      |      |      |  |
| Grades 5052, 5154, 5454, 5652 (A95052, A95154, A95454, A95652)                                          | 11.6                                                                        | ...  | ...  | 11.3 | 11.0 | 10.7 | 10.2 | 9.7  | 9.4  | 8.9  |      |      |      |      |      |      |      |      |      |      |      |  |
| Grades 356, 5083, 5086, 5456 (A03560, A95083, A95086, A95456)                                           | 11.7                                                                        | ...  | ...  | 11.4 | 11.1 | 10.8 | 10.3 | 9.8  | 9.5  | 9.0  |      |      |      |      |      |      |      |      |      |      |      |  |

**TABLE 10-48 Flexibility Factor  $k$  and Stress Intensification Factor  $i$** 

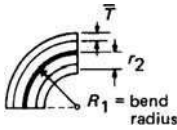
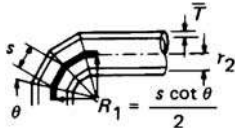
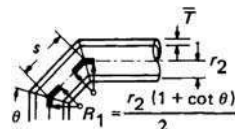
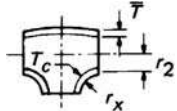
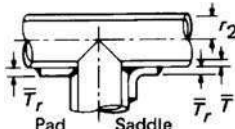
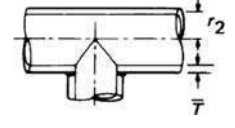
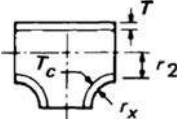
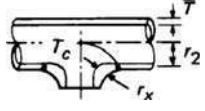
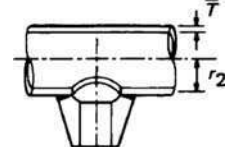
| Description                                                                                              | Flexibility factor $k$ | Stress intensification factor [Notes (2), (3)] |                                 | Flexibility characteristic $h$                                      | Sketch                                                                                |
|----------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------|---------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                                                                                          |                        | Out-of-plane, $i_o$                            | In-plane, $i_i$                 |                                                                     |                                                                                       |
| Welding elbow or pipe bend<br>[Notes (2)], (4)-(7)]                                                      | $\frac{1.65}{h}$       | $\frac{0.75}{h^{2/3}}$                         | $\frac{0.9}{h^{2/3}}$           | $\frac{\bar{T}R_1}{r_2^2}$                                          |    |
| Closely spaced miter bend<br>$s < r_2 (1 + \tan \theta)$<br>[Notes (2), (4), (5), (7)]                   | $\frac{1.52}{h^{5/6}}$ | $\frac{0.9}{h^{2/3}}$                          | $\frac{0.9}{h^{2/3}}$           | $\frac{\cot \theta}{2} \left( \frac{s\bar{T}}{r_2^2} \right)$       |    |
| Single miter bend or widely spaced miter bend<br>$s \geq r_2 (1 + \tan \theta)$<br>[Notes (2), (4), (7)] | $\frac{1.52}{h^{5/6}}$ | $\frac{0.9}{h^{2/3}}$                          | $\frac{0.9}{h^{2/3}}$           | $\frac{1 + \cot \theta}{2} \left( \frac{\bar{T}}{r_2} \right)$      |    |
| Welding tee per ASME B16.9<br>[Notes (2), (4), (6), (11), (13)]                                          | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{3}{4} i_o + \frac{1}{4}$ | $3.1 \frac{\bar{T}}{r_2}$                                           |    |
| Reinforced fabricated tee with pad or saddle<br>[Notes (2), (4), (8), (12), (13)]                        | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{3}{4} i_o + \frac{1}{4}$ | $\frac{(\bar{T} + \frac{1}{2} \bar{T}_r)^{2.5}}{\bar{T}^{1.5} r_2}$ |   |
| Unreinforced fabricated tee<br>[Notes (2), (4), (12), (13)]                                              | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{3}{4} i_o + \frac{1}{4}$ | $\frac{\bar{T}}{r_2}$                                               |  |
| Extruded welding tee with<br>$r_s \geq 0.05 D_b$<br>$T_c < 1.5 \bar{T}$<br>[Notes (2), (4), (13)]        | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{3}{4} i_c + \frac{1}{4}$ | $\left( 1 + \frac{r_s}{r_2} \right) \frac{\bar{T}}{r_2}$            |  |
| Welded-in contour insert<br>[Notes (2), (4), (11), (13)]                                                 | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{3}{4} i_c + \frac{1}{4}$ | $3.1 \frac{\bar{T}}{r_2}$                                           |  |
| Branch welded-on fitting (integrally reinforced)<br>[Notes (2), (4), (9), (12)]                          | 1                      | $\frac{0.9}{h^{2/3}}$                          | $\frac{0.9}{h^{2/3}}$           | $3.3 \frac{\bar{T}}{r_2}$                                           |  |



TABLE 10-48 Flexibility Factor  $k$  and Stress-Intensification Factor  $i$  (Concluded)

| Description                                                         | Flexibility factor $k$ | Stress intensification factor $i$ [Note (1)] |
|---------------------------------------------------------------------|------------------------|----------------------------------------------|
| Butt-welded joint, reducer, or weld-neck flange                     | 1                      | 1.0                                          |
| Double-welded slip-on flange                                        | 1                      | 1.2                                          |
| Fillet welded joint, or socket weld flange or fitting               | 1                      | Note (14)                                    |
| Lap joint flange (with ASME B16.9 lap joint stub)                   | 1                      | 1.6                                          |
| Threaded pipe joint or threaded flange                              | 1                      | 2.3                                          |
| Corrugated straight pipe, or corrugated or creased bend [Note (10)] | 5                      | 2.5                                          |

## NOTES:

(1) Stress intensification and flexibility factor data in this table are for use in the absence of more directly applicable data (see para. 319.3.6). Their validity has been demonstrated for  $D/T \leq 100$ .

(2) The flexibility factor,  $k$ , in this table applies to bending in any plane. The flexibility factors,  $k$ , and stress intensification factors,  $i$ , shall not be less than unity; factors for torsion equal unity. Both factors apply over the effective arc length (shown by heavy centerlines in the sketches) for curved and miter bends, and to the intersection point for tees.

(3) A single intensification factor equal to  $0.9/h^{2/3}$  may be used for both  $i_i$  and  $i_o$  if desired.

(4) The values of  $k$  and  $i$  can be read directly from Chart A by entering with the characteristic  $h$  computed from the formulas given above. Nomenclature is as follows:

$D_o$  = outside diameter of branch

$R_1$  = bend radius of welding elbow or pipe bend

$r_s$  = see definition in para. 304.3.4(c) and Table 10-48

$r_2$  = mean radius of matching pipe

$s$  = miter spacing at centerline

$\bar{T}$  = for elbows and miter bends, the nominal wall thickness of the fitting,

= for tees, the nominal wall thickness of the matching pipe

$T_c$  = crotch thickness of branch connections measured at the center of the crotch where shown in the sketches

$\bar{T}_r$  = pad or saddle thickness

$\theta$  = one-half angle between adjacent miter axes

(5) Where flanges are attached to one or both ends, the values of  $k$  and  $i$  in this table shall be corrected by the factors  $C_1$ , which can be read directly from Chart B, entering with the computed  $h$ .

(6) The designer is cautioned that cast butt-welded fittings may have considerably heavier walls than that of the pipe with which they are used. Large errors may be introduced unless the effect of these greater thicknesses is considered.

(7) In large diameter thin-wall elbows and bends, pressure can significantly affect the magnitudes of  $k$  and  $i$ . To correct values from the table, divide  $k$  by

$$1 + 6 \left( \frac{P}{E} \right) \left( \frac{r_2}{\bar{T}} \right)^{2/3} \left( \frac{R_1}{r_2} \right)^{1/3}$$

divide  $i$  by

$$1 + 3.25 \left( \frac{P}{E} \right) \left( \frac{r_2}{\bar{T}} \right)^{5/3} \left( \frac{R_1}{r_2} \right)^{2/3}$$

For consistency, use kPa and mm for SI metric, and psi and in. for U.S. customary notation.

(8) When  $\bar{T}_r$  is  $> 1\frac{1}{2} \bar{T}$ , use  $h = 4 \bar{T}/r_2$ .

(9) The designer must be satisfied that this fabrication has a pressure rating equivalent to straight pipe.

(10) Factors shown apply to bending. Flexibility factor for torsion equals 0.9.

(11) If  $r_s \geq \frac{1}{8} D_o$  and  $T_c \geq 1.5 \bar{T}$  a flexibility characteristic of  $4.4 \bar{T}/r_2$  may be used.

(12) The out-of-plane stress intensification factor (SIF) for a reducing branch connection with branch-to-run diameter ratio of  $0.5 < d/D < 1.0$  may be nonconservative. A smooth concave weld contour has been shown to reduce the SIF. Selection of the appropriate SIF is the designer's responsibility.

(13) Stress intensification factors for branch connections are based on tests with at least two diameters of straight run pipe on each side of the branch centerline. More closely loaded branches may require special consideration.

(14) 2.1 max. or  $2.1 \bar{T}/C_x$ , but not less than 1.3.  $C_x$  is the fillet weld leg length (see Fig. 328.5.2C). For unequal leg lengths, use the smaller leg for  $C_x$ .

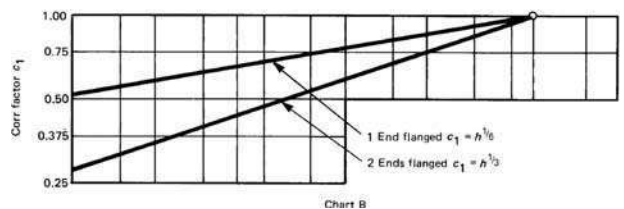
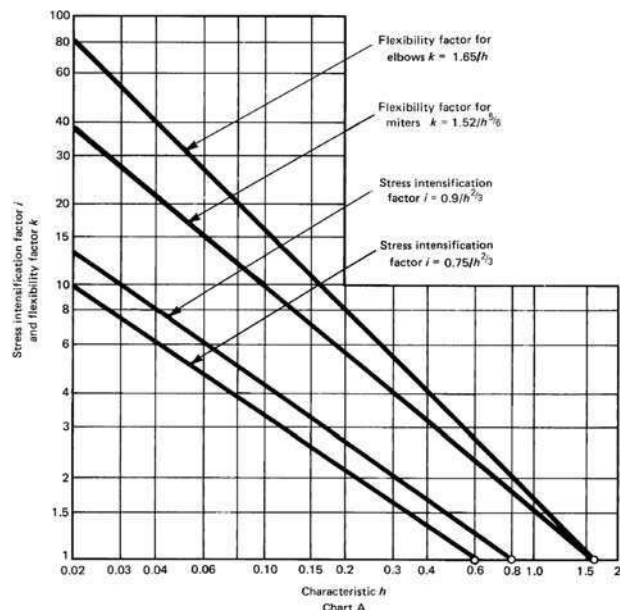
$\bar{T}_b$  = thickness of pipe matching branch, in (mm)

$i_o$  = out-plane stress-intensification factor (Table 10-48)

$i_i$  = in-plane stress-intensification factor (Table 10-48)

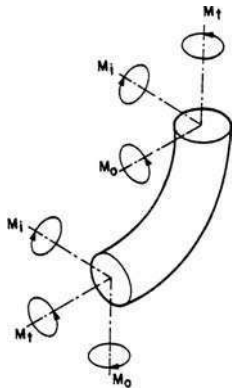
Allowable stress range  $S_A$  and permissible additive stresses shall be computed in accordance with Eqs. (10-93) and (10-94).

**Required Weld Quality Assurance** Any weld at which  $S_E$  exceeds  $0.8S_A$  for any portion of a piping system and the equivalent number of cycles  $N$  exceeds 7000 shall be fully examined in accordance with the requirements for severe cyclic service (presented later in this section).



**Reactions: Metallic Piping** Reaction forces and moments to be used in the design of restraints and supports and in evaluating the effects of piping displacements on connected equipment shall be based on the reaction range  $R$  for the extreme displacement conditions, considering the range previously defined for reactions and using  $E_a$ . The designer shall consider instantaneous maximum values of forces and moments in the original and extreme displacement conditions as well as the reaction range in making these evaluations.

**Maximum Reactions for Simple Systems** For two-anchor systems without intermediate restraints, the maximum instantaneous values



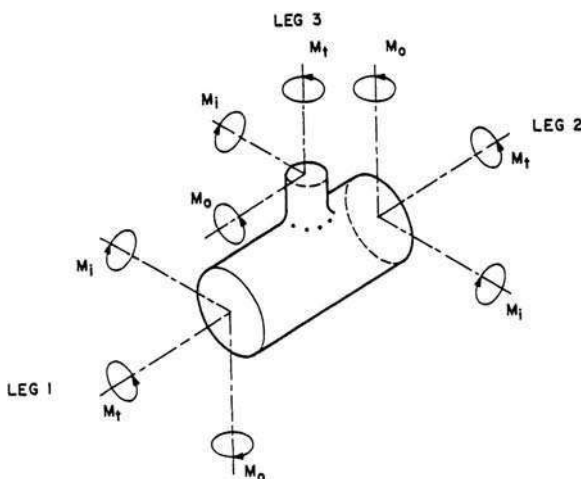
**FIG. 10-172** Moments in bends. (Extracted from the *Process Piping Code*, B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.)

of reaction forces and moments may be estimated from Eqs. (10-106) and (10-107).

1. For extreme displacement conditions,  $R_m$ . The temperature for this computation is the design maximum or design minimum temperature as previously defined for reactions, whichever produces the larger reaction:

$$R_m = R \left( 1 - \frac{2C}{3} \right) \frac{E_m}{E_a} \quad (10-106)$$

where  $C$  = cold-spring factor varying from zero for no cold spring to 1.0 for 100 percent cold spring. (The factor  $\frac{2}{3}$  is based on experience, which shows that specified cold spring cannot be fully assured even with elaborate precautions.)  
 $E_a$  = modulus of elasticity at installation temperature, lbf/in<sup>2</sup> (MPa)  
 $E_m$  = modulus of elasticity at design maximum or design minimum temperature, lbf/in<sup>2</sup> (MPa)  
 $R$  = range of reaction forces or moments (derived from flexibility analysis) corresponding to the full displacement-stress range and based on  $E_a$ , lbf or in·lbf (N or N·mm)



**FIG. 10-173** Moments in branch connections. (Extracted from the *Process Piping Code*, B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.)

$R_m$  = estimated instantaneous maximum reaction force or moment at design maximum or design minimum temperature, lbf or in·lbf (N or N·mm)

2. For original condition,  $R_a$ . The temperature for this computation is the expected temperature at which the piping is to be assembled.

$$R_a = CR \quad \text{or} \quad C_1 R, \text{ whichever is greater} \quad (10-107)$$

where nomenclature is as for Eq. (10-106) and

$C_1 = 1 - (S_h E_a / S_E E_m)$   
 = estimated self-spring or relaxation factor (use zero if value of  $C_1$  is negative)

$R_a$  = estimated instantaneous reaction force or moment at installation temperature, lbf or in·lbf (N or N·mm)

$S_E$  = computed displacement-stress range, lbf/in<sup>2</sup> (MPa). See Eq. (10-101).

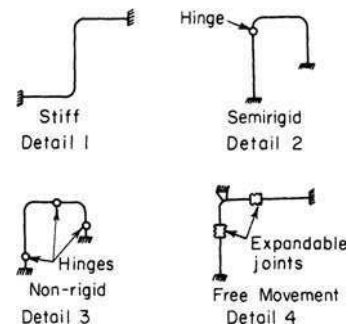
$S_h$  = See Eq. (10-93).

**Maximum Reactions for Complex Systems** For multianchor systems and for two-anchor systems with intermediate restraints, Eqs. (10-106) and (10-107) are not applicable. Each case must be studied to estimate the location, nature, and extent of local overstrain and its effect on stress distribution and reactions.

Acceptable comprehensive methods of analysis are analytical, model-test, and chart methods, which evaluate for the entire piping system under consideration the forces, moments, and stresses caused by bending and torsion from a simultaneous consideration of terminal and intermediate restraints to thermal expansion and include all external movements transmitted under thermal change to the piping by its terminal and intermediate attachments. Correction factors, as provided by the details of these rules, must be applied for the stress intensification of curved pipe and branch connections and may be applied for the increased flexibility of such component parts.

**Expansion Joints** All the foregoing applies to "stiff piping systems," i.e., systems without expansion joints (see detail 1 of Fig. 10-174). When space limitations, process requirements, or other considerations result in configurations of insufficient flexibility, capacity for deflection within allowable stress range limits may be increased successively by the use of one or more hinged bellows expansion joints, viz., semirigid (detail 2) and nonrigid (detail 3) systems, and expansion effects essentially eliminated by a free-movement joint (detail 4) system. Expansion joints for semirigid and nonrigid systems are restrained against longitudinal and lateral movement by the hinges with the expansion element under bending movement only and are known as "rotation" or "hinged" joints (see Fig. 10-175). Semirigid systems require a minimum of three joints for two-dimensional and five joints for three-dimensional expansion movement.

Joints similar to that shown in Fig. 10-175, except with two pairs of hinge pins equally spaced around a gimbal ring, achieve similar results with a lesser number of joints.



**FIG. 10-174** Flexibility classification for piping systems. (From Kellogg, *Design of Piping Systems*, Wiley, New York, 1965.)

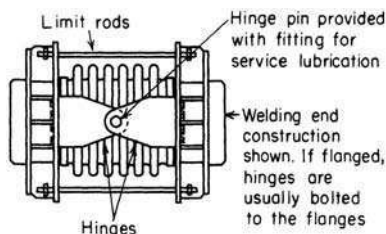


FIG. 10-175 Hinged expansion joint. (From Kellogg, Design of Piping Systems, Wiley, New York, 1965.)

Expansion joints for free-movement systems can be designed for axial or offset movement alone, or for combined axial and offset movements (see Fig. 10-176). For offset movement alone, the end load due to pressure and weight can be transferred across the joint by tie rods or structural members (see Fig. 10-177). For axial or combined movements, anchors must be provided to absorb the unbalanced pressure load and force bellows to deflect.

Commercial bellows elements are usually light-gauge (of the order of 0.05 to 0.10 in thick and are available in stainless and other alloy steels, copper, and other nonferrous materials. Multi-ply bellows, bellows with external reinforcing rings, and toroidal contour bellows are available for higher pressures. Since bellows elements are ordinarily rated for strain ranges which involve repetitive yielding, predictable performance is assured only by adequate fabrication controls and knowledge of the potential fatigue performance of each design. The attendant cold work can affect corrosion resistance and promote susceptibility to corrosion fatigue or stress corrosion; joints in a horizontal position cannot be drained and have frequently undergone pitting or cracking due to the presence of condensate during operation or offstream. For low-pressure essentially nonhazardous service, nonmetallic bellows of fabric-reinforced rubber or special materials are sometimes used. For corrosive service PTFE bellows may be used.

Because of the inherently greater susceptibility of expansion bellows to failure from unexpected corrosion, failure of guides to control joint movements, etc., it is advisable to examine critically their design choice in comparison with a stiff system.

Slip-type expansion joints (Fig. 10-178) substitute packing (ring or plastic) for bellows. Their performance is sensitive to adequate design

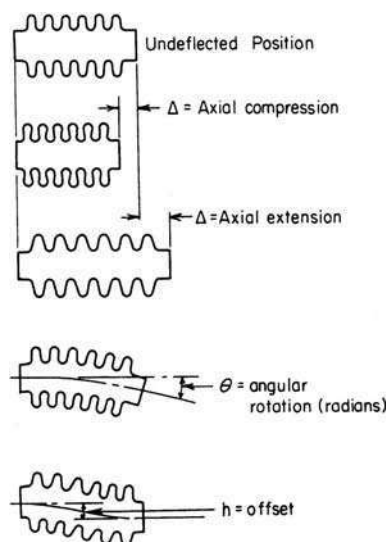


FIG. 10-176 Action of expansion bellows under various movements. (From Kellogg, Design of Piping Systems, Wiley, New York, 1965.)

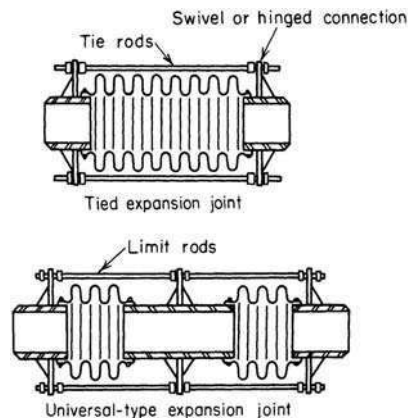


FIG. 10-177 Constrained-bellows expansion joints. (From Kellogg, Design of Piping Systems, Wiley, New York, 1965.)

with respect to guiding to prevent binding and the adequacy of stuffing boxes and attendant packing, sealant, and lubrication. Anchors must be provided for the unbalanced pressure force and for the friction forces to move the joint. The latter can be much higher than the elastic force required to deflect a bellows joint. Rotary packed joints, ball joints, and other special joints can absorb end load.

Corrugated pipe and corrugated and creased bends are also used to decrease stiffness.

**Pipe Supports** Loads transmitted by piping to attached equipment and supporting elements include weight, temperature- and pressure-induced effects, vibration, wind, earthquake, shock, and thermal expansion and contraction. The design of supports and restraints is based on concurrently acting loads (if it is assumed that wind and earthquake do not act simultaneously).

Resilient and constant-effort-type supports shall be designed for maximum loading conditions including test unless temporary supports are provided.

Though not specified in the code, supports for discharge piping from relief valves must be adequate to withstand the jet reaction produced by their discharge.

The code states further that pipe-supporting elements shall (1) avoid excessive interference with thermal expansion and contraction of pipe which is otherwise adequately flexible; (2) be such that they do not contribute to leakage at joints or excessive sag in piping requiring drainage; (3) be designed to prevent overstress, resonance, or disengagement due to variation of load with temperature; also, so that combined longitudinal stresses in the piping shall not exceed the code allowable limits; (4) be such that a complete release of the piping load will be prevented in the event of spring failure or misalignment, weight transfer, or added load due to test during erection; (5) be of steel or wrought iron; (6) be of alloy steel or protected from temperature when the temperature limit for carbon steel may be exceeded; (7) not be cast

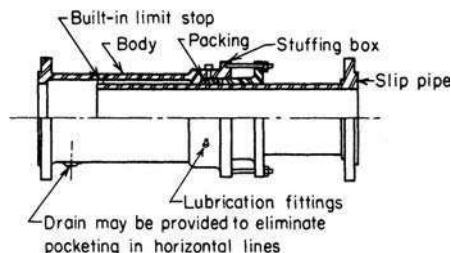


FIG. 10-178 Slip-type expansion joint. (From Kellogg, Design of Piping Systems, Wiley, New York, 1965.)

iron except for roller bases, rollers, anchor bases, etc., under mainly compression loading; (8) not be malleable or nodular iron except for pipe clamps, beam clamps, hanger flanges, clips, bases, and swivel rings; (9) not be wood except for supports mainly in compression when the pipe temperature is at or below ambient; and (10) have threads for screw adjustment which shall conform to ASME B1.1.

A supporting element used as an anchor shall be designed to maintain an essentially fixed position.

To protect terminal equipment or other (weaker) portions of the system, restraints (such as anchors and guides) shall be provided where necessary to control movement or to direct expansion into those portions of the system that are adequate to absorb them. The design, arrangement, and location of restraints shall ensure that expansion-joint movements occur in the directions for which the joint is designed. In addition to the other thermal forces and moments, the effects of friction in other supports of the system shall be considered in the design of such anchors and guides.

**Anchors for Expansion Joints** Anchors (such as those of the corrugated, omega, disk, or slip type) shall be designed to withstand the algebraic sum of the forces at the maximum pressure and temperature at which the joint is to be used. These forces are:

1. Pressure thrust, which is the product of the effective thrust area times the maximum pressure to which the joint will be subjected during normal operation. (For slip joints the effective thrust area shall be computed by using the outside diameter of the pipe. For corrugated, omega, or disk-type joints, the effective thrust area shall be that area recommended by the joint manufacturer. If this information is unobtainable, the effective area shall be computed by using the maximum inside diameter of the expansion-joint bellows.)

2. The force required to compress or extend the joint in an amount equal to the calculated expansion movement.

3. The force required to overcome the static friction of the pipe in expanding or contracting on its supports, from installed to operating position. The length of pipe considered should be that located between the anchor and the expansion joint.

**Support Fixtures** Hanger rods may be pipe straps, chains, bars, or threaded rods which permit free movement for thermal expansion or contraction. Sliding supports shall be designed for friction and bearing loads. Brackets shall be designed to withstand movements due to friction in addition to other loads. Spring-type supports shall be designed for weight load at the point of attachment and to prevent misalignment, buckling, or eccentric loading of springs, and provided with stops to prevent spring overtravel. Compensating-type spring hangers are recommended for high-temperature and critical-service piping to make the supporting force uniform with appreciable movement. Counterweight supports shall have stops to limit travel. Hydraulic supports shall be provided with safety devices and stops to support load in the event of loss of pressure. Vibration dampers or sway braces may be used to limit vibration amplitude.

The code requires that the safe load for threaded hanger rods be based on the root area of the threads. This, however, assumes concentric loading. When hanger rods move to a nonvertical position so that the load is transferred from the rod to the supporting structure via the edge of one flat of the nut on the rod, it is necessary to consider the root area to be reduced by one-third. If a clamp is connected to a vertical line to support its weight, it is recommended that shear lugs be welded to the pipe, or that the clamp be located below a fitting or flange, to prevent slippage. Consideration shall be given to the localized stresses induced in the piping by the integral attachment. Typical pipe supports are shown in Fig. 10-179.

Much piping is supported from structures installed for other purposes. It is common practice to use beam formulas for tubular sections to determine stress, maximum deflection, and maximum slope of piping in **spans between supports**. When piping is supported from structures installed for that sole purpose and those structures rest on driven piles, detailed calculations are usually made to determine maximum permissible spans. Limits imposed on maximum slope to make the contents of the line drain to the lower end require calculations made on the weight per foot of the empty line. To avoid interference with other components, maximum deflection should be limited to 25.4 mm (1 in).

Pipe hangers are essentially frictionless but require taller pipe-support structures which cost more than structures on which pipe is laid. Devices that reduce friction between laid pipe subject to thermal movement and its supports are used to accomplish the following:

1. Reduce loads on anchors or on equipment acting as anchors.
2. Reduce the tendency of pipe acting as a column loaded by friction at supports to buckle sideways off supports.
3. Reduce nonvertical loads imposed by piping on its supports so as to minimize cost of support foundations.
4. Reduce longitudinal stress in pipe.

Linear bearing surfaces made of fluorinated hydrocarbons or of graphite and also rollers are used for this purpose.

**Design Criteria: Nonmetallic Pipe** In using a nonmetallic material, designers must satisfy themselves as to the adequacy of the material and its manufacture, considering such factors as strength at design temperature, impact- and thermal-shock properties, toxicity, methods of making connections, and possible deterioration in service. Rating information, based usually on ASTM standards or specifications, is generally available from the manufacturers of these materials. Particular attention should be given to provisions for the thermal expansion of nonmetallic piping materials, which may be as much as 5 to 10 times that of steel (Table 10-49). Special consideration should be given to the strength of small pipe connections to piping and equipment and to the need for extra flexibility at the junction of metallic and nonmetallic systems.

Table 10-50 gives values for the modulus of elasticity for nonmetals; however, no specific stress-limiting criteria or methods of stress analysis are presented. Stress-strain behavior of most nonmetals differs considerably from that of metals and is less well-defined for mathematic analysis. The piping system should be designed and laid out so that flexural stresses resulting from displacement due to expansion, contraction, and other movement are minimized. This concept requires special attention to supports, terminals, and other restraints.

**Displacement Strains** The concepts of strain imposed by restraint of thermal expansion or contraction and by external movement described for metallic piping apply in principle to nonmetals. Nevertheless, the assumption that stresses throughout the piping system can be predicted from these strains because of fully elastic behavior of the piping materials is not generally valid for nonmetals.

In thermoplastics and some thermosetting resins, displacement strains are not likely to produce immediate failure of the piping but may result in detrimental distortion. Especially in thermoplastics, progressive deformation may occur upon repeated thermal cycling or on prolonged exposure to elevated temperature.

In brittle nonmetallics (such as porcelain, glass, impregnated graphite, etc.) and some thermosetting resins, the materials show rigid behavior and develop high displacement stresses up to the point of sudden breakage due to overstrain.

**Elastic Behavior** The assumption that displacement strains will produce proportional stress over a sufficiently wide range to justify an elastic-stress analysis often is not valid for nonmetals. In brittle nonmetallic piping, strains initially will produce relatively large elastic stresses. The total displacement strain must be kept small, however, since overstrain results in failure rather than plastic deformation. In plastic and resin nonmetallic piping strains generally will produce stresses of the overstressed (plastic) type even at relatively low values of total displacement strain.

## FABRICATION, ASSEMBLY, AND ERECTION

**Welding, Brazing, or Soldering** Code requirements dealing with fabrication are more detailed for welding than for other methods of joining, since welding is the predominant method of construction and the method used for the most demanding applications. The code requirements for welding processes and operators are essentially the same as covered in the subsection on pressure vessels (i.e., qualification to Sec. IX of the ASME Boiler and Pressure Vessel Code) except that welding processes are not restricted, the material grouping (P number) must be in accordance with ASME B31.3 Appendix A-1, and welding positions are related to pipe position. The code also permits one fabricator to accept welders or welding operators qualified by another employer

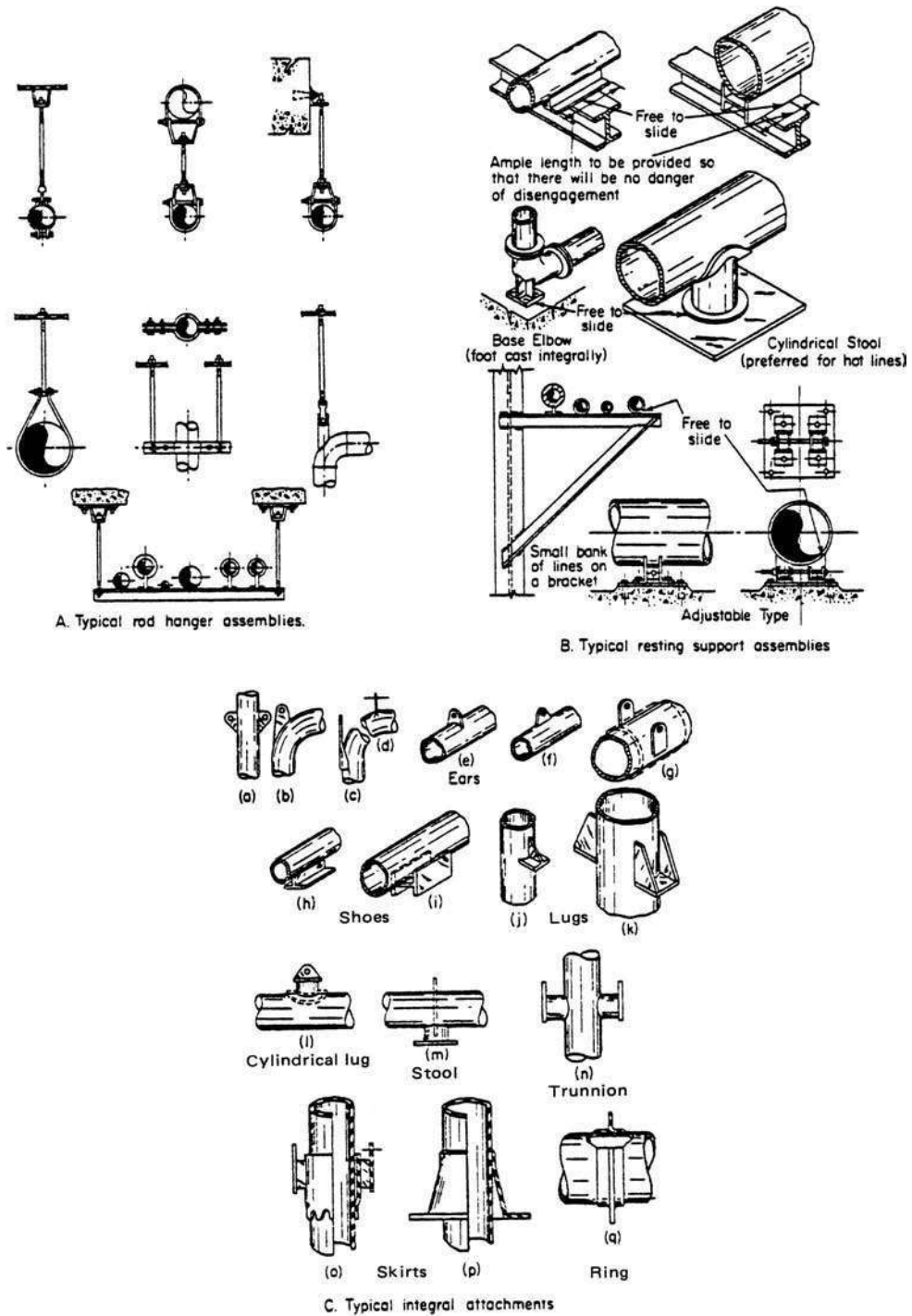


FIG. 10-179 Typical pipe supports and attachments. (From Kellogg, Design of Piping Systems, Wiley, New York, 1965.)

without requalification when welding pipe by the same or equivalent procedure. The code may require that the welding procedure qualification include low-temperature toughness testing (see. Table 10-51).

Filler metal is required to conform with the requirements of Sec. IX. Backing rings (of ferrous material), when used, shall be of weldable quality with sulfur limited to 0.05 percent. Backing rings of nonferrous and nonmetallic materials may be used provided they are proved satisfactory by procedure-qualification tests and provided their use has been approved by the designer.

The code requires internal alignment within the dimensional limits specified in the welding procedure and the engineering design without specific dimensional limitations. Internal trimming is permitted

for correcting internal misalignment provided such trimming does not result in a finished wall thickness before welding of less than required minimum wall thickness  $t_m$ . When necessary, weld metal may be deposited on the inside or outside of the component to provide alignment or sufficient material for trimming.

Table 10-53 is a summary of the code acceptance criteria (limits on imperfections) for welds. The defects referred to are illustrated in Fig. 10-180.

Brazing procedures, brazers, and brazing operators must be qualified in accordance with the requirements of Part QB, Sec. IX, ASME Code. At the owner's option, qualification is not required for Category D fluid service not exceeding 93°C (200°F). The clearance between

**TABLE 10-49 Thermal Expansion Coefficients, Nonmetals**

| Material description                                                  | Mean coefficients (divide table values by 10 <sup>6</sup> ) |           |           |           |
|-----------------------------------------------------------------------|-------------------------------------------------------------|-----------|-----------|-----------|
|                                                                       | in/in, °F                                                   | Range, °F | mm/mm, °C | Range, °C |
| <b>Thermoplastics</b>                                                 |                                                             |           |           |           |
| Acetal AP2012                                                         | 2                                                           | ...       | 3.6       | ...       |
| Acrylonitrile-butadiene-styrene                                       |                                                             |           |           |           |
| ABS 1208                                                              | 60                                                          | ...       | 108       | ...       |
| ABS 1210                                                              | 55                                                          | 45–55     | 99        | 7–13      |
| ABS 1316                                                              | 40                                                          | ...       | 72        | ...       |
| ABS 2112                                                              | 40                                                          | ...       | 72        | ...       |
| Cellulose acetate butyrate                                            |                                                             |           |           |           |
| CAB MH08                                                              | 80                                                          | ...       | 144       | ...       |
| CAB 5004                                                              | 95                                                          | ...       | 171       | ...       |
| Chlorinated poly(vinyl chloride)                                      |                                                             |           |           |           |
| CPVC 4120                                                             | 35                                                          | ...       | 63        | ...       |
| Polybutylene PB 2110                                                  | 72                                                          | ...       | 130       | ...       |
| Polyether, chlorinated                                                | 45                                                          | ...       | 81        | ...       |
| Polyethylene                                                          |                                                             |           |           |           |
| PE 1404                                                               | 100                                                         | 46–100    | 180       | 8–38      |
| PE 2305                                                               | 90                                                          | 46–100    | 162       | 8–38      |
| PE 2306                                                               | 80                                                          | 46–100    | 144       | 8–38      |
| PE 3306                                                               | 70                                                          | 46–100    | 126       | 8–38      |
| PE 3406                                                               | 60                                                          | 46–100    | 108       | 8–38      |
| Polyphenylene POP 2125                                                | 30                                                          | ...       | 54        | ...       |
| Polypropylene                                                         |                                                             |           |           |           |
| PP1110                                                                | 48                                                          | 33–67     | 86        | 1–19      |
| PP1208                                                                | 43                                                          | ...       | 77        | ...       |
| PP2105                                                                | 40                                                          | ...       | 72        | ...       |
| Poly(vinyl chloride)                                                  |                                                             |           |           |           |
| PVC 1120                                                              | 30                                                          | 23–37     | 54        | –5 to +3  |
| PVC 1220                                                              | 35                                                          | 34–40     | 63        | 1–4       |
| PVC 2110                                                              | 50                                                          | ...       | 90        | ...       |
| PVC 2112                                                              | 45                                                          | ...       | 81        | ...       |
| PVC 2116                                                              | 40                                                          | 37–45     | 72        | 3–7       |
| PVC 2120                                                              | 30                                                          | ...       | 54        | ...       |
| Poly(vinylidene fluoride)                                             | 79                                                          | ...       | 142       | ...       |
| Poly(vinylidene chloride)                                             | 100                                                         | ...       | 180       | ...       |
| Polytetrafluoroethylene                                               | 55                                                          | 73–140    | 99        | 23–60     |
| Poly(fluorinated ethylenepropylene)                                   | 46–58                                                       | 73–140    | 83–104    | 23–60     |
| Poly(perfluoroalkoxy alkane)                                          | 67                                                          | 70–212    | 121       | 21–100    |
| Poly(perfluoroalkoxy alkane)                                          | 94                                                          | 212–300   | 169       | 100–149   |
| Poly(perfluoroalkoxy alkane)                                          | 111                                                         | 300–408   | 200       | 149–209   |
| <b>Reinforced Thermosetting Resins and Reinforced Plastic Mortars</b> |                                                             |           |           |           |
| Glass-epoxy, centrifugally cast                                       | 9–13                                                        | ...       | 16–23.5   | ...       |
| Glass-polyester, centrifugally cast                                   | 9–15                                                        | ...       | 16–27     | ...       |
| Glass-polyester, filament-wound                                       | 9–11                                                        | ...       | 16–20     | ...       |
| Glass-polyester, hand lay-up                                          | 12–15                                                       | ...       | 21.5–27   | ...       |
| Glass-epoxy, filament-wound                                           | 9–13                                                        | ...       | 16–23.5   | ...       |
| <b>Other Nonmetallic Materials</b>                                    |                                                             |           |           |           |
| Borosilicate glass                                                    | 1.8                                                         | ...       | 3.25      | ...       |

GENERAL NOTES:

(a) For Code references to this table, see para. A319.3.1, ASME B31.3—2004. These data are for use in the absence of more applicable data. It is the designer's responsibility to verify that materials are suitable for the intended service at the temperatures shown.

(b) Individual compounds may vary from the values shown. Consult manufacturer for specific values for products.

TABLE 10-50 Modulus of Elasticity, Nonmetals

| Material description                            | E, ksi (73.4°F) | E, MPa (23°C) |
|-------------------------------------------------|-----------------|---------------|
| <b>Thermoplastics [Note (1)]</b>                |                 |               |
| Acetal                                          | 410             | 2,830         |
| ABS, Type 1210                                  | 250             | 1,725         |
| ABS, Type 1316                                  | 340             | 2,345         |
| CAB                                             | 120             | 825           |
| PVC, Type 1120                                  | 420             | 2,895         |
| PVC, Type 1220                                  | 410             | 2,825         |
| PVC, Type 2110                                  | 340             | 2,345         |
| PVC, Type 2116                                  | 380             | 2,620         |
| Chlorinated PVC                                 | 420             | 2,895         |
| Chlorinated polyether                           | 160             | 1,105         |
| PE, Type 2306                                   | 90              | 620           |
| PE, Type 3306                                   | 130             | 895           |
| PE, Type 3406                                   | 150             | 1,035         |
| Polypropylene                                   | 120             | 825           |
| Poly(vinylidene chloride)                       | 100             | 690           |
| Poly(vinylidene fluoride)                       | 194             | 1,340         |
| Poly(tetrafluorethylene)                        | 57              | 395           |
| Poly(fluorinated ethylenepropylene)             | 67              | 460           |
| Poly(perfluoroalkoxy alkane)                    | 100             | 690           |
| <b>Thermosetting Resins, Axially Reinforced</b> |                 |               |
| Epoxy-glass, centrifugally cast                 | 1,200–1,900     | 8,275–13,100  |
| Epoxy-glass, filament-wound                     | 1,100–2,000     | 7,585–13,790  |
| Polyester-glass, centrifugally cast             | 1,200–1,900     | 8,275–13,100  |
| Polyester-glass, hand lay-up                    | 800–1,000       | 5,515–6,895   |
| <b>Other</b>                                    |                 |               |
| Borosilicate glass                              | 9,800           | 67,570        |

GENERAL NOTE: For Code references to this table, see para. A319.3.2, ASME B31.3—2004. These data are for use in the absence of more applicable data. It is the designer's responsibility to verify that materials are suitable for the intended service at the temperatures shown.

## NOTE:

(1) The modulus of elasticity data shown for thermoplastics are based on short-term tests. The manufacturer should be consulted to obtain values for use under long-term loading.

surfaces to be joined by brazing or soldering shall be no larger than is necessary to allow complete capillary distribution of the filler metal.

The only requirement for solderers is that they follow the procedure in the *Copper Tube Handbook* of the Copper Development Association.

**Bending and Forming** Pipe may be bent to any radius for which the bend-arc surface will be free of cracks and substantially free of buckles. The use of qualified bends which are creased or corrugated is permitted. Bending may be done by any hot or cold method that produces a product meeting code and service requirements, and that does not have an adverse effect on the essential characteristics of the material. Hot bending and hot forming must be done within a temperature range consistent with material characteristics, end use, and postoperation heat treatment. Postbend heat treatment may be required for bends in some materials; its necessity is dependent upon the type of bending operation and the severity of the bend. Postbend heat treatment requirements are defined in the code.

Piping components may be formed by any suitable hot or cold pressing, rolling, forging, hammering, spinning, drawing, or other method. Thickness after forming shall not be less than required by design. Special rules cover the forming and pressure design verification of flared laps.

The development of fabrication facilities for bending pipe to the radius of commercial butt-welding long-radius elbows and forming flared metallic (Van Stone) laps on pipe are important techniques in reducing welded-piping costs. These techniques save both the cost of the ell or stub end and the welding operation required to attach the fitting to the pipe.

**Preheating and Heat Treatment** Preheating and postoperation heat treatment are used to avert or relieve the detrimental effects of the high temperature and severe thermal gradients inherent in the

welding of metals. In addition, heat treatment may be needed to relieve residual stresses created during the bending or forming of metals. The code provisions shown in Tables 10-54 and 10-55 represent basic practices which are acceptable for most applications of welding, bending, and forming, but they are not necessarily suitable for all service conditions. The specification of more or less stringent preheating and heat-treating requirements is a function of those responsible for the engineering design.

Refer to the code for rules establishing the thickness to be used in determining PWHT requirements for configurations other than butt welds (e.g., fabricated branch connections and socket welds).

**Joining Nonmetallic Pipe** All joints should be made in accordance with procedures complying with the manufacturer's recommendations and code requirements. General welding and heat fusion procedures are described in ASTM D-2657. ASTM D2855 describes general solvent-cementing procedures.

Depending on size, thermoplastic piping can be joined with mechanical joints, solvent-cemented joints, hot-gas welding, or heat fusion procedures. Mechanical joints are frequently bell-and-spigot joints which employ an elastomer O-ring gasket. Joints of this type are generally not self-restrained against internal pressure thrust.

Thermosetting resin pipe can be joined with mechanical joints or adhesive-bonded joints. Mechanical joints are generally a variation of gasketed bell-and-spigot joints and may be either nonrestrained or self-restrained. Adhesive-bonded joints are typically bell-and-spigot or butt-and-strap. Butt-and-strap joints join piping components with multiple layers of resin-saturated glass reinforcement.

**Assembly and Erection** Flanged-joint faces shall be aligned to the design plane to within  $\frac{1}{10}$  in/ft (0.5 percent) maximum measured across any diameter, and flange bolt holes shall be aligned to within 3.2-mm ( $\frac{1}{8}$ -in) maximum offset. Flanged joints involving flanges with widely differing mechanical properties shall be assembled with extra care, and tightening to a predetermined torque is recommended.

The use of flat washers under bolt heads and nuts is a code requirement when assembling nonmetallic flanges. It is preferred that the bolts extend completely through their nuts; however, a lack of complete thread engagement not exceeding one thread is permitted by the code. The assembly of cast-iron bell-and-spigot piping is covered in AWWA Standard C600.

Screwed joints which are intended to be seal-welded shall be made up without any thread compound.

When installing conductive nonmetallic piping and metallic pipe with nonmetallic linings, consideration should be given to the need to provide electrical continuity throughout the system and to grounding requirements. This is particularly critical in areas with potentially explosive atmospheres.

## EXAMINATION, INSPECTION, AND TESTING

This subsection provides a general synopsis of code requirements. It should not be viewed as comprehensive.

**Examination and Inspection** The code differentiates between examination and inspection. "Examination" applies to quality-control functions performed by personnel of the piping manufacturer, fabricator, or erector. "Inspection" applies to functions performed for the owner by the authorized inspector.

The authorized inspector shall be designated by the owner and shall be the owner, an employee of the owner, an employee of an engineering or scientific organization, or an employee of a recognized insurance or inspection company acting as the owner's agent. The inspector shall not represent or be an employee of the piping erector, the manufacturer, or the fabricator unless the owner is also the erector, the manufacturer, or the fabricator.

The authorized inspector shall have a minimum of 10 years' experience in the design, fabrication, or inspection of industrial pressure piping. Each 20 percent of satisfactory work toward an engineering degree accredited by the Engineers' Council for Professional Development shall be considered equivalent to 1 year's experience, up to 5 years total.

**TABLE 10-51 Requirements for Low-Temperature Toughness Tests for Metals\***

These toughness requirements are in addition to tests required by the material specification.

|                    | Type of material                                                                                                                                                                                                                                                                                                                     | Column A<br>Design minimum temperature at or above min. temp. in Table A-1 of<br>ASME B31.3—2004 or Table 10-52                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Column B<br>Design minimum temperature below<br>min. temp. in Table A-1 of ASME<br>B31.3—2004 or Table 10-52                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Listed Materials   | 1 Gray cast iron                                                                                                                                                                                                                                                                                                                     | A-1 No additional requirements                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | B-1 No additional requirements                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                    | 2 Malleable and ductile cast iron:<br>carbon steel per Note (1)                                                                                                                                                                                                                                                                      | A-2 No additional requirements                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | B-2 Materials designated in Box 2 shall<br>not be used.                                                                                                                                                                                                                                                                                                                                                                                                         |
|                    |                                                                                                                                                                                                                                                                                                                                      | (a) Base metal                                                                                                                                                                                                                                                    | (b) Weld metal and heat affected<br>zone (HAZ) [Note (2)]                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                    | 3 Other carbon steels, low<br>and intermediate alloy<br>steels, high alloy ferritic<br>steels, duplex stainless<br>steels                                                                                                                                                                                                            | A-3 (a) No additional requirements                                                                                                                                                                                                                                | A-3 (b) Weld metal deposits shall be<br>impact tested per para. 323.3. <sup>a</sup><br>If design min. temp. < -29°C<br>(-20°F), except as provided in<br>Notes (3) and (5), and except as<br>follows: for materials listed for<br>Curves C and D of Table 10-52,<br>where corresponding welding con-<br>sumables are qualified by impact<br>testing at the design minimum tem-<br>perature or lower in accordance<br>with the applicable AWS specification,<br>additional testing is not required. | B-3 Except as provided in Notes (3)<br>and (5), heat treat base metal per<br>applicable ASTM specification listed<br>in para. 323.3.2 <sup>a</sup> ; then impact test base<br>metal, weld deposits, and HAZ<br>per para. 323.3 <sup>a</sup> [see Note (2)]. When<br>materials are used at design min. temp.<br>below the assigned curve as permitted<br>by Notes (2) and (3) of Table 10-52,<br>weld deposits and HAZ shall be<br>impact tested [see Note (2)]. |
|                    | 4 Austenitic stainless steels                                                                                                                                                                                                                                                                                                        | A-4 (a) If:<br>(1) carbon content by analysis<br>>0.1%; or<br>(2) material is not in solution<br>heat treated condition;<br>then, impact test per para.<br>323.3 <sup>a</sup> for design min. temp.<br>< -29°C (-20°F) except as<br>provided in Notes (3) and (6) | A-4 (b) Weld metal deposits shall be<br>impact tested per para. 323.3. <sup>a</sup> If design<br>min. temp. < -29°C (-20°F)<br>except as provided in para.<br>323.2.2 <sup>b</sup> and in Notes (3) and (6)                                                                                                                                                                                                                                                                                        | B-4 Base metal and weld metal<br>deposits shall be impact tested<br>per para. 323.3 <sup>a</sup> . See Notes (2),(3),<br>and (6).                                                                                                                                                                                                                                                                                                                               |
|                    | 5 Austenitic ductile iron,<br>ASTM A 571                                                                                                                                                                                                                                                                                             | A-5 (a) No additional requirements                                                                                                                                                                                                                                | A-5 (b) Welding is not permitted                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | B-5 Base metal shall be impact tested<br>per para. 323.3. <sup>a</sup> Do not use<br>< -196°C (-320°F). Welding is not<br>permitted.                                                                                                                                                                                                                                                                                                                            |
| Unlisted Materials | 6 Aluminum, copper,<br>nickel, and their alloys;<br>unalloyed titanium                                                                                                                                                                                                                                                               | A-6 (a) No additional requirements                                                                                                                                                                                                                                | A-6 (b) No additional requirements<br>unless filler metal composition is<br>outside the range for base metal<br>composition; then test per column<br>B-6                                                                                                                                                                                                                                                                                                                                           | B-6 Designer shall be assured by<br>suitable tests [see Note (4)] that base<br>metal, weld deposits, and HAZ are<br>suitable at the design min. temp.                                                                                                                                                                                                                                                                                                           |
|                    | 7 An unlisted material shall conform to a published specification. Where composition, heat treatment, and product form are comparable to those of a listed material, requirements for the corresponding listed material shall be met. Other unlisted materials shall be qualified as required in the applicable section of column B. |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## NOTES:

- (1) Carbon steels conforming to the following are subject to the limitations in Box B-2; plates per ASTM A 36, A 283, and A 570; pipe per ASTM A 134 when made from these plates; and pipe per ASTM A 53 Type F and API 5L Gr. A25 butt weld.
- (2) Impact tests that meet the requirements of Table 323.3.1,<sup>a</sup> which are performed as part of the weld procedure qualification, will satisfy all requirements of para. 323.2.2.<sup>b</sup> and need not be repeated for production welds.
- (3) Impact testing is not required if the design minimum temperature is below -29°C (-20°F) but at or above -104°C (-155°F) and the Stress Ratio defined in Fig. 323.2.2B<sup>a</sup> does not exceed 0.3.
- (4) Tests may include tensile elongation, sharp-notch tensile strength (to be compared with unnotched tensile strength), and/or other tests, conducted at or below design minimum temperature. See also para. 323.3.4.<sup>b</sup>
- (5) Impact tests are not required when the maximum obtainable Charpy specimen has a width along the notch of less than 2.5 mm (0.098 in). Under these conditions, the design minimum temperature shall not be less than the lower of -48°C (-55°F) or the minimum temperature for the material in Table A-1.
- (6) Impact tests are not required when the maximum obtainable Charpy specimen has a width along the notch of less than 2.5 mm (0.098 in).

<sup>a</sup>Refer to the referenced code paragraph for impact testing methods and acceptance.

<sup>b</sup>Refer to the referenced code paragraph for comments regarding circumstances under which impact testing can be excluded.

\*Table 10-51 and notes have been extracted (with minor modifications) from *Process Piping*, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.



**TABLE 10-52 Tabular Values for Minimum Temperatures without Impact Testing for Carbon Steel Materials\***

| Nominal thickness<br>[Note (6)] |        | Design minimum temperature |     |                       |     |                       |     |         |     |
|---------------------------------|--------|----------------------------|-----|-----------------------|-----|-----------------------|-----|---------|-----|
|                                 |        | Curve A<br>[Note (2)]      |     | Curve B<br>[Note (3)] |     | Curve C<br>[Note (3)] |     | Curve D |     |
|                                 |        | °C                         | °F  | °C                    | °F  | °C                    | °F  | °C      | °F  |
| 6.4                             | 0.25   | -9.4                       | 15  | -28.9                 | -20 | -48.3                 | -55 | -48.3   | -55 |
| 7.9                             | 0.3125 | -9.4                       | 15  | -28.9                 | -20 | -48.3                 | -55 | -48.3   | -55 |
| 9.5                             | 0.375  | -9.4                       | 15  | -28.9                 | -20 | -48.3                 | -55 | -48.3   | -55 |
| 10.0                            | 0.394  | -9.4                       | 15  | -28.9                 | -20 | -48.3                 | -55 | -48.3   | -55 |
| 11.1                            | 0.4375 | -6.7                       | 20  | -28.9                 | -20 | -41.7                 | -43 | -48.3   | -55 |
| 12.7                            | 0.5    | -1.1                       | 30  | -28.9                 | -20 | -37.8                 | -36 | -48.3   | -55 |
| 14.3                            | 0.5625 | 2.8                        | 37  | -21.7                 | -7  | -35.0                 | -31 | -45.6   | -50 |
| 15.9                            | 0.625  | 6.1                        | 43  | -16.7                 | 2   | -32.2                 | -26 | -43.9   | -47 |
| 17.5                            | 0.6875 | 8.9                        | 48  | -12.8                 | 9   | -29.4                 | -21 | -41.7   | -43 |
| 19.1                            | 0.75   | 11.7                       | 53  | -9.4                  | 15  | -27.2                 | -17 | -40.0   | -40 |
| 20.6                            | 0.8125 | 14.4                       | 58  | -6.7                  | 20  | -25.0                 | -13 | -38.3   | -37 |
| 22.2                            | 0.875  | 16.7                       | 62  | -3.9                  | 25  | -23.3                 | -10 | -36.7   | -34 |
| 23.8                            | 0.9375 | 18.3                       | 65  | -1.7                  | 29  | -21.7                 | -7  | -35.6   | -32 |
| 25.4                            | 1.0    | 20.0                       | 68  | 0.6                   | 33  | -19.4                 | -3  | -34.4   | -30 |
| 27.0                            | 1.0625 | 22.2                       | 72  | 2.2                   | 36  | -18.3                 | -1  | -33.3   | -28 |
| 28.6                            | 1.125  | 23.9                       | 75  | 3.9                   | 39  | -16.7                 | 2   | -32.2   | -26 |
| 30.2                            | 1.1875 | 25.0                       | 77  | 5.6                   | 42  | -15.6                 | 4   | -30.6   | -23 |
| 31.8                            | 1.25   | 26.7                       | 80  | 6.7                   | 44  | -14.4                 | 6   | -29.4   | -21 |
| 33.3                            | 1.3125 | 27.8                       | 82  | 7.8                   | 46  | -13.3                 | 8   | -28.3   | -19 |
| 34.9                            | 1.375  | 28.9                       | 84  | 8.9                   | 48  | -12.2                 | 10  | -27.8   | -18 |
| 36.5                            | 1.4375 | 30.0                       | 86  | 9.4                   | 49  | -11.1                 | 12  | -26.7   | -16 |
| 38.1                            | 1.5    | 31.1                       | 88  | 10.6                  | 51  | -10.0                 | 14  | -25.6   | -14 |
| 39.7                            | 1.5625 | 32.2                       | 90  | 11.7                  | 53  | -8.9                  | 16  | -25.0   | -13 |
| 41.3                            | 1.625  | 33.3                       | 92  | 12.8                  | 55  | -8.3                  | 17  | -23.9   | -11 |
| 42.9                            | 1.6875 | 33.9                       | 93  | 13.9                  | 57  | -7.2                  | 19  | -23.3   | -10 |
| 44.5                            | 1.75   | 34.4                       | 94  | 14.4                  | 58  | -6.7                  | 20  | -22.2   | -8  |
| 46.0                            | 1.8125 | 35.6                       | 96  | 15.0                  | 59  | -5.6                  | 22  | -21.7   | -7  |
| 47.6                            | 1.875  | 36.1                       | 97  | 16.1                  | 61  | -5.0                  | 23  | -21.1   | -6  |
| 49.2                            | 1.9375 | 36.7                       | 98  | 16.7                  | 62  | -4.4                  | 24  | -20.6   | -5  |
| 50.8                            | 2.0    | 37.2                       | 99  | 17.2                  | 63  | -3.3                  | 26  | -20.0   | -4  |
| 51.6                            | 2.0325 | 37.8                       | 100 | 17.8                  | 64  | -2.8                  | 27  | -19.4   | -3  |
| 54.0                            | 2.125  | 38.3                       | 101 | 18.3                  | 65  | -2.2                  | 28  | -18.9   | -2  |
| 55.6                            | 2.1875 | 38.9                       | 102 | 18.9                  | 66  | -1.7                  | 29  | -18.3   | -1  |
| 57.2                            | 2.25   | 38.9                       | 102 | 19.4                  | 67  | -1.1                  | 30  | -17.8   | 0   |
| 58.7                            | 2.3125 | 39.4                       | 103 | 20.0                  | 68  | -0.6                  | 31  | -17.2   | 1   |
| 60.3                            | 2.375  | 40.0                       | 104 | 20.6                  | 69  | 0.0                   | 32  | -16.7   | 2   |
| 61.9                            | 2.4375 | 40.6                       | 105 | 21.1                  | 70  | 0.6                   | 33  | -16.1   | 3   |
| 63.5                            | 2.5    | 40.6                       | 105 | 21.7                  | 71  | 1.1                   | 34  | -15.6   | 4   |
| 65.1                            | 2.5625 | 41.1                       | 106 | 21.7                  | 71  | 1.7                   | 35  | -15.0   | 5   |
| 66.7                            | 2.625  | 41.7                       | 107 | 22.8                  | 73  | 2.2                   | 36  | -14.4   | 6   |
| 68.3                            | 2.6875 | 41.7                       | 107 | 22.8                  | 73  | 2.8                   | 37  | -13.9   | 7   |
| 69.9                            | 2.75   | 42.2                       | 108 | 23.3                  | 74  | 3.3                   | 38  | -13.3   | 8   |
| 71.4                            | 2.8125 | 42.2                       | 108 | 23.9                  | 75  | 3.9                   | 39  | -13.3   | 8   |
| 73.0                            | 2.875  | 42.8                       | 109 | 24.4                  | 76  | 4.4                   | 40  | -12.8   | 9   |
| 74.6                            | 2.9375 | 42.8                       | 109 | 25.0                  | 77  | 4.4                   | 40  | -12.2   | 10  |
| 76.2                            | 3.0    | 43.3                       | 110 | 25.0                  | 77  | 5.0                   | 41  | -11.7   | 11  |

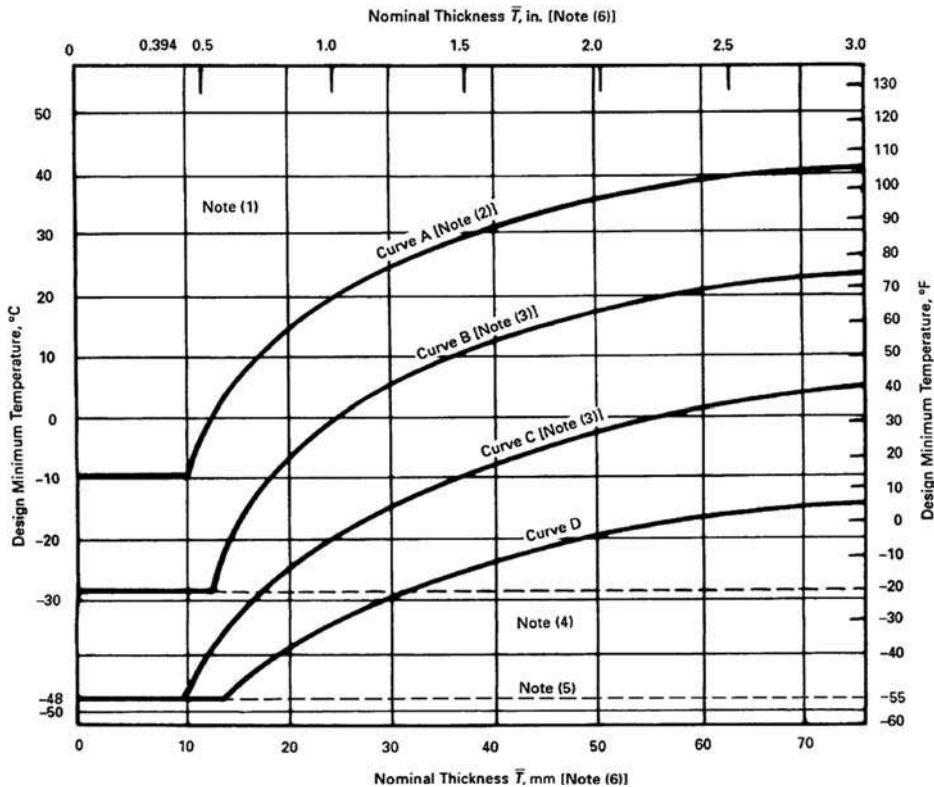
It is the owner's responsibility, exercised through the authorized inspector, to verify that all required examinations and testing have been completed and to inspect the piping to the extent necessary to be satisfied that it conforms to all applicable requirements of the code and the engineering design. This verification may include certifications and records pertaining to materials, components, heat treatment, examination and testing, and qualifications of operators and procedures. The authorized inspector may delegate the performance of inspection to a qualified person.

Inspection does not relieve the manufacturer, the fabricator, or the erector of responsibility for providing materials, components, and skill in accordance with requirements of the code and the engineering design, performing all required examinations, and preparing records of examinations and tests for the inspector's use.

**Examination Methods** The code establishes the types of examinations for evaluating various types of imperfections (see Table 10-56).

Personnel performing examinations other than visual shall be qualified in accordance with applicable portions of SNT TC-1A, *Recommended Practice for Nondestructive Testing Personnel Qualification and Certification*. Procedures shall be qualified as required in Par. T-150, Art. 1, Sec. V of the ASME Code. Limitations on imperfections shall be in accordance with the engineering design but shall at least meet the requirements of the code (see Tables 10-54 and 10-55) for the specific type of examination. Repairs shall be made as applicable.

**Visual Examination** This consists of observation of the portion of components, joints, and other piping elements that are or can be exposed to view before, during, or after manufacture, fabrication, assembly, erection, inspection, or testing.

**TABLE 10-52 Tabular Values for Minimum Temperatures without Impact Testing for Carbon Steel Materials\*** (Concluded)

## NOTES:

- (1) Any carbon steel material may be used to a minimum temperature of  $-29^{\circ}\text{C}$  ( $-20^{\circ}\text{F}$ ) for Category D Fluid Service.
  - (2) X Grades of API SL, and ASTM A 381 materials, may be used in accordance with Curve B if normalized or quenched and tempered.
  - (3) The following materials may be used in accordance with Curve D if normalized:
    - (a) ASTM A 516 Plate, all grades
    - (b) ASTM A 671 Pipe, Grades CE55, CE60, and all grades made with A 516 plate
    - (c) ASTM A 672 Pipe, Grades E55, E60, and all grades made with A 516 plate
  - (4) A welding procedure for the manufacture of pipe or components shall include impact testing of welds and HAZ for any design minimum temperature below  $-29^{\circ}\text{C}$  ( $-20^{\circ}\text{F}$ ), except as provided in Table 10-51, A-3(b).
  - (5) Impact testing in accordance with para. 323.3<sup>b</sup> is required for any design minimum temperature below  $-48^{\circ}\text{C}$  ( $-55^{\circ}\text{F}$ ), except as permitted by Note (3) in Table 10-51.
  - (6) For blind flanges and blanks,  $T$  shall be  $1/4$  of the flange thickness.
- <sup>a</sup>Refer to the referenced code paragraph for impact testing methods and acceptance.
- <sup>b</sup>Refer to the referenced code paragraph for comments regarding circumstances under which impact testing can be excluded.
- \*Table 10-52 and notes have been extracted (with minor modifications) from *Process Piping*, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.

The examination includes verification of code and engineering design requirements for materials and components, dimensions, joint preparation, alignment, welding, bonding, brazing, bolting, threading and other joining methods, supports, assembly, and erection.

Visual examination shall be performed in accordance with Art. 9, Sec. V of the ASME Code.

**Magnetic-Particle Examination** This examination shall be performed in accordance with Art. 7, Sec. V of the ASME Code.

**Liquid-Penetrant Examination** This examination shall be performed in accordance with Art. 6, Sec. V of the ASME Code.

**Radiographic Examination** The following definitions apply to radiography required by the code or by the engineering design:

1. "Random radiography" applies only to girth and miter groove welds. It is radiographic examination of the complete circumference of a specified percentage of the girth butt welds in a designated lot of piping.

2. Unless otherwise specified in engineering design, 100 percent radiography applies only to girth welds, miter groove welds, and fabricated branch connections that utilize butt-type welds to join the header and the branch. The design engineer may, however, elect to designate other types of welds as requiring 100 percent radiography. By definition, 100 percent radiography requires radiographic examination of the complete length of all such welds in a designated lot of piping.

# 10-130 TRANSPORT AND STORAGE OF FLUIDS

**TABLE 10-53 Acceptance Criteria for Welds and Examination Methods for Evaluating Weld Imperfections\***

| Criteria (A to M) for types of welds and for service conditions [Note (1)] |                                |                   |                                                    |                                |                   |                          |                                |                   |                              |                                                                      | Examination methods |             |                   |                  |
|----------------------------------------------------------------------------|--------------------------------|-------------------|----------------------------------------------------|--------------------------------|-------------------|--------------------------|--------------------------------|-------------------|------------------------------|----------------------------------------------------------------------|---------------------|-------------|-------------------|------------------|
| Normal and Category M fluid service                                        |                                |                   | Severe cyclic conditions                           |                                |                   | Category D fluid service |                                |                   |                              | Weld imperfection                                                    | Visual              | Radiography | Magnetic particle | Liquid penetrant |
| Type of weld                                                               |                                |                   | Type of weld                                       |                                |                   | Type of weld             |                                |                   |                              |                                                                      |                     |             |                   |                  |
| Girth, miter groove & branch connection [Note (4)]                         | Longitudinal groove [Note (2)] | Fillet [Note (3)] | Girth, miter groove & branch connection [Note (4)] | Longitudinal groove [Note (2)] | Fillet [Note (3)] | Girth and miter groove   | Longitudinal groove [Note (2)] | Fillet [Note (3)] | Branch connection [Note (4)] |                                                                      |                     |             |                   |                  |
| A                                                                          | A                              | A                 | A                                                  | A                              | A                 | A                        | A                              | A                 | A                            | Crack                                                                | ✓                   | ✓           | ✓                 | ✓                |
| A                                                                          | A                              | A                 | A                                                  | A                              | A                 | C                        | A                              | N/A               | A                            | Lack of fusion                                                       | ✓                   | ✓           | ...               | ...              |
| B                                                                          | A                              | N/A               | A                                                  | A                              | N/A               | C                        | A                              | N/A               | B                            | Incomplete penetration                                               | ✓                   | ✓           | ...               | ...              |
| E                                                                          | E                              | N/A               | D                                                  | D                              | N/A               | N/A                      | N/A                            | N/A               | N/A                          | Internal porosity                                                    | ...                 | ✓           | ...               | ...              |
| G                                                                          | G                              | N/A               | F                                                  | F                              | N/A               | N/A                      | N/A                            | N/A               | N/A                          | Internal slag inclusion, tungsten inclusion, or elongated indication | ...                 | ✓           | ...               | ...              |
| H                                                                          | A                              | H                 | A                                                  | A                              | A                 | I                        | A                              | H                 | H                            | Undercutting                                                         | ...                 | ✓           | ...               | ...              |
| A                                                                          | A                              | A                 | A                                                  | A                              | A                 | A                        | A                              | A                 | A                            | Surface porosity or exposed slag inclusion [Note (5)]                | ✓                   | ...         | ...               | ...              |
| N/A                                                                        | N/A                            | N/A               | J                                                  | J                              | J                 | N/A                      | N/A                            | N/A               | N/A                          | Surface finish                                                       | ✓                   | ...         | ...               | ...              |
| K                                                                          | K                              | N/A               | K                                                  | K                              | N/A               | K                        | K                              | N/A               | K                            | Concave root surface (suck up)                                       | ✓                   | ✓           | ...               | ...              |
| L                                                                          | L                              | L                 | L                                                  | L                              | L                 | M                        | M                              | M                 | M                            | Weld reinforcement or internal protrusion                            | ✓                   | ...         | ...               | ...              |

## Criterion value notes

| Criterion |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symbol    | Measure                                                                                                                                                                                                                                                                | Acceptable value limits [Note (6)]                                                                                                                                                                                                                                                                                                                                                |
| A         | Extent of imperfection                                                                                                                                                                                                                                                 | Zero (no evident imperfection)                                                                                                                                                                                                                                                                                                                                                    |
| B         | Depth of incomplete penetration                                                                                                                                                                                                                                        | $\leq 1 \text{ mm } (1/32 \text{ in})$ and $\leq 0.2\bar{T}_w$                                                                                                                                                                                                                                                                                                                    |
|           | Cumulative length of incomplete penetration                                                                                                                                                                                                                            | $\leq 38 \text{ mm } (1.5 \text{ in})$ in any 150 mm (6 in) weld length                                                                                                                                                                                                                                                                                                           |
| C         | Depth of lack of fusion and incomplete penetration                                                                                                                                                                                                                     | $\leq 0.2\bar{T}_w$                                                                                                                                                                                                                                                                                                                                                               |
|           | Cumulative length of lack of fusion and incomplete penetration [Note (7)]                                                                                                                                                                                              | $\leq 38 \text{ mm } (1.5 \text{ in})$ in any 150 mm (6 in) weld length                                                                                                                                                                                                                                                                                                           |
| D         | Size and distribution of internal porosity                                                                                                                                                                                                                             | See BPV Code, Section VIII, Division 1, Appendix 4                                                                                                                                                                                                                                                                                                                                |
| E         | Size and distribution of internal porosity                                                                                                                                                                                                                             | For $\bar{T}_w \leq 6 \text{ mm } (1/4 \text{ in})$ , limit is same as D<br>For $\bar{T}_w > 6 \text{ mm } (1/4 \text{ in})$ , limit is $1.5 \times D$                                                                                                                                                                                                                            |
| F         | Slag inclusion, tungsten inclusion, or elongated indication                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                   |
|           | Individual length                                                                                                                                                                                                                                                      | $\leq \bar{T}_w/3$                                                                                                                                                                                                                                                                                                                                                                |
|           | Individual width                                                                                                                                                                                                                                                       | $\leq 2.5 \text{ mm } (3/32 \text{ in})$ and $\leq \bar{T}_w/3$                                                                                                                                                                                                                                                                                                                   |
|           | Cumulative length                                                                                                                                                                                                                                                      | $\leq \bar{T}_w$ in any 12 $\bar{T}_w$ weld length                                                                                                                                                                                                                                                                                                                                |
| G         | Slag inclusion, tungsten inclusion, or elongated indication                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                   |
|           | Individual length                                                                                                                                                                                                                                                      | $\leq 2\bar{T}_w$                                                                                                                                                                                                                                                                                                                                                                 |
|           | Individual width                                                                                                                                                                                                                                                       | $\leq 3 \text{ mm } (1/8 \text{ in})$ and $\leq \bar{T}_w/2$                                                                                                                                                                                                                                                                                                                      |
|           | Cumulative length                                                                                                                                                                                                                                                      | $\leq 4\bar{T}_w$ in any 150 mm (6 in) weld length                                                                                                                                                                                                                                                                                                                                |
| H         | Depth of undercut                                                                                                                                                                                                                                                      | $\leq 1 \text{ mm } (1/32 \text{ in})$ and $\leq \bar{T}_w/4$                                                                                                                                                                                                                                                                                                                     |
| I         | Depth of undercut                                                                                                                                                                                                                                                      | $\leq 1.5 \text{ mm } (1/16 \text{ in})$ and $\leq [\bar{T}_w/4 \text{ or } 1 \text{ mm } (1/32 \text{ in})]$                                                                                                                                                                                                                                                                     |
| J         | Surface roughness                                                                                                                                                                                                                                                      | $\leq 500 \text{ min. } Ra$ per ASME B46.1                                                                                                                                                                                                                                                                                                                                        |
| K         | Depth of root surface concavity                                                                                                                                                                                                                                        | Total joint thickness, incl. weld reinf., $\geq \bar{T}_w$                                                                                                                                                                                                                                                                                                                        |
| L         | Height of reinforcement or internal protrusion [Note (8)] in any plane through the weld shall be within limits of the applicable height value in the tabulation at right, except as provided in Note (9). Weld metal shall merge smoothly into the component surfaces. | <div>For <math>\bar{T}_w</math> mm (in)</div> <div> <math>\leq 6 (1/4)</math><br/> <math>&gt; 6 (1/4), \leq 13 (1/2)</math><br/> <math>&gt; 13 (1/2), \leq 25 (1)</math><br/> <math>&gt; 25 (1)</math> </div> <div>Height, mm (in)</div> <div> <math>\leq 1.5 (1/16)</math><br/> <math>\leq 3 (1/8)</math><br/> <math>\leq 4 (5/32)</math><br/> <math>\leq 5 (3/16)</math> </div> |
| M         | Height of reinforcement or internal protrusion [Note (8)] as described in L. Note (9) does not apply.                                                                                                                                                                  | Limit is twice the value applicable for L above                                                                                                                                                                                                                                                                                                                                   |

**TABLE 10-53 Acceptance Criteria for Welds and Examination Methods for Evaluating Weld Imperfections\*** (Concluded)

## GENERAL NOTES:

(a) Weld imperfections are evaluated by one or more of the types of examination methods. The types and extent of examination shall comply with code requirements and those specified by engineering design.

(b) "N/A" indicates the code does not establish acceptance criteria or does not require evaluation of this kind of imperfection for this type of weld.

(c) Check (✓) indicates examination method generally used for evaluating this kind of weld imperfection.

(d) Ellipsis (...) indicates examination method not generally used for evaluating this kind of weld imperfection.

## NOTES:

(1) Criteria given are for required examination. More stringent criteria may be specified in the engineering design. Other methods of examination may be specified by engineering design to supplement the examination required by the code. The extent of supplementary examination and any acceptance criteria differing from those specified by the code shall be specified by engineering design. Any examination method recognized by the code may be used to resolve doubtful indications. Acceptance criteria shall be those established by the code for the required examination.

(2) Longitudinal groove weld includes straight and spiral seam. Criteria are not intended to apply to welds made in accordance with component standards recognized by the code (ref. ASME B31.3 Table A1 and Table 326.1); however, alternative leak test requirements dictate that all component welds be examined (see code for specific requirements).

(3) Fillet weld includes socket and seal welds, and attachment welds for slip-on flanges, branch reinforcement, and supports.

(4) Branch connection weld includes pressure containing welds in branches and fabricated laps.

(5) These imperfections are evaluated only for welds  $\leq 5$  mm ( $\frac{3}{16}$  in) in nominal thickness.

(6) Where two limiting values are separated by *and* the lesser of the values determines acceptance. Where two sets of values are separated by *or* the larger value is acceptable.  $T_w$  is the nominal wall thickness of the thinner of two components joined by a butt weld.

(7) Tightly butted unfused root faces are unacceptable.

(8) For groove welds, height is the lesser of the measurements made from the surfaces of the adjacent components; both reinforcement and internal protrusion are permitted in a weld. For fillet welds, height is measured from the theoretical throat defined by the code; internal protrusion does not apply.

(9) For welds in aluminum alloy only, internal protrusion shall not exceed the following values:

(a) for thickness  $\leq 2$  mm ( $\frac{3}{64}$  in): 1.5 mm ( $\frac{1}{16}$  in)

(b) for thickness  $> 2$  mm and  $\leq 6$  mm ( $\frac{1}{4}$  in): 2.5 mm ( $\frac{3}{32}$  in)

For external reinforcement and for greater thicknesses, see the tabulation for Symbol L.

\*Table 10-53 and notes have been extracted (with minor modifications) from Section 341 of *Process Piping*, ASME B31.3–2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.

3. "Spot radiography" is the practice of making a single-exposure radiograph at a point within a specified extent of welding. Required coverage for a single spot radiograph is as follows:

- For longitudinal welds, at least 150 mm (6 in) of weld length.
- For girth, miter, and branch welds in piping  $2\frac{1}{2}$  in NPS and smaller, a single elliptical exposure which encompasses the entire weld circumference, and in piping larger than  $2\frac{1}{2}$  in NPS, at least 25 percent of the inside circumference or 150 mm (6 in), whichever is less.

Radiography of components other than castings and of welds shall be in accordance with Art. 2, Sec. V of the ASME Code. Limitations on imperfections in components other than castings and welds shall be as stated in Table 10-53 for the degree of radiography involved.

**Ultrasonic Examination** Ultrasonic examination of welds shall be in accordance with Art. 5, Sec. V of the ASME Code, except that the modifications stated in Par. 336.4.6 of the code shall be substituted for T542.2.1 and T542.8.1.1. Refer to the code for additional requirements.

**Type and Extent of Required Examination** The intent of examinations is to provide the examiner and the inspector with reasonable assurance that the requirements of the code and the engineering design have been met. For P-number 3, 4, and 5 materials, examination shall be performed after any heat treatment has been completed.

**Examination Normally Required\*** Piping in normal fluid service shall be examined to the extent specified herein or to any greater extent specified in the engineering design. Acceptance criteria are as stated in the code for Normal Fluid Service unless more stringent requirements are specified.

1. **Visual examination.** At least the following shall be examined in accordance with code requirements:

a. Sufficient materials and components, selected at random, to satisfy the examiner that they conform to specifications and are free from defects.

b. At least 5 percent of fabrication. For welds, each welder's and welding operator's work shall be represented.

c. 100 percent of fabrication for longitudinal welds, except those in components made in accordance with a listed specification. Longi-

tudinal welds required to have a joint efficiency of 0.9 must be spot-radiographed to the extent of 300 mm (12 in) in each 30 m (100 ft) of weld for each welder or welding operator. Acceptance criteria shall comply with code radiography acceptance criteria for Normal Fluid Service.

d. Random examination of the assembly of threaded, bolted, and other joints to satisfy the examiner that they conform to the applicable code requirements for erection and assembly. When pneumatic testing is to be performed, all threaded, bolted, and other mechanical joints shall be examined.

e. Random examination during erection of piping, including checking of alignment, supports, and cold spring.

f. Examination of erected piping for evidence of defects that would require repair or replacement, and for other evident deviations from the intent of the design.

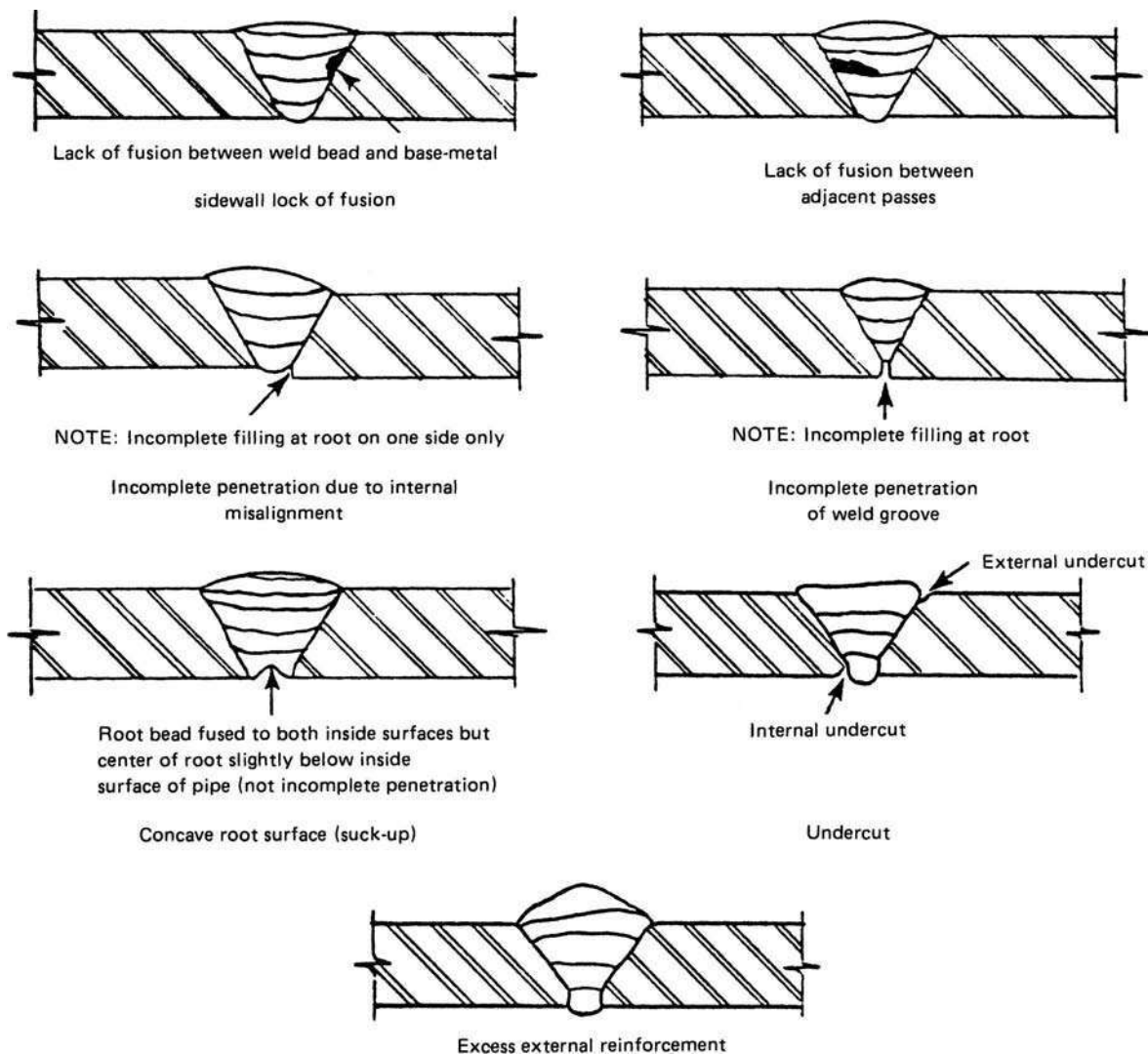
2. **Other examination**

a. Not less than 5 percent of circumferential butt and miter groove welds shall be examined fully by random radiography or random ultrasonic examination in accordance with code requirements established for these methods. The welds to be examined shall be selected to ensure that the work product of each welder or welding operator doing the production welding is included. They shall also be selected to maximize coverage of intersections with longitudinal joints. When a circumferential weld with intersecting longitudinal weld(s) is examined, at least the adjacent 38 mm ( $1\frac{1}{2}$  in) of each intersecting weld shall be examined. In-process examination in accordance with code requirements may be substituted for all or part of the radiographic or ultrasonic examination on a weld-for-weld basis if specified in the engineering design or specifically authorized by the Inspector.

b. Not less than 5 percent of all brazed joints shall be examined by in-process examination in accordance with the code definition of in-process examination, the joints to be examined being selected to ensure that the work of each brazer making the production joints is included.

3. **Certifications and records.** The examiner shall be assured, by examination of certifications, records, and other evidence, that the materials and components are of the specified grades and that they have received required heat treatment, examination, and testing. The examiner shall provide the Inspector with a certification that all the quality control requirements of the code and of the engineering design have been carried out.

\*Extracted (with minor editing) from *Process Piping*, ASME B31.3—2004, paragraph 341, with permission of the publisher, the American Society of Mechanical Engineers, New York.



**FIG. 10-180** Typical weld imperfections. (Extracted from *Process Piping*, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.)

**Examination—Category D Fluid Service\*** Piping and piping elements for Category D fluid service as designated in the engineering design shall be visually examined in accordance with code requirements for visual examination to the extent necessary to satisfy the examiner that components, materials, and workmanship conform to the requirements of this code and the engineering design. Acceptance criteria shall be in accordance with code requirements and criteria in Table 10-53, for Category D fluid service, unless otherwise specified.

**Examination—Severe Cyclic Conditions\*** Piping to be used under severe cyclic conditions shall be examined to the extent specified herein or to any greater extent specified in the engineering design. Acceptance criteria shall be in accordance with code requirements and criteria in Table 10-53, for severe cyclic conditions, unless otherwise specified.

\*Extracted (with minor editing) from *Process Piping*, ASME B31.3—2004, paragraph 341, with permission of the publisher, the American Society of Mechanical Engineers, New York.

1. **Visual examination.** The requirements for Normal Fluid Service apply with the following exceptions.

- a. All fabrication shall be examined.
- b. All threaded, bolted, and other joints shall be examined.
- c. All piping erection shall be examined to verify dimensions and alignment. Supports, guides, and points of cold spring shall be checked to ensure that movement of the piping under all conditions of startup, operation, and shutdown will be accommodated without undue binding or unanticipated constraint.

2. **Other examination.** All circumferential butt and miter groove welds and all fabricated branch connection welds comparable to those recognized by the code (ref. Fig. 10-130) shall be examined by 100% radiography or 100% ultrasonic (if specified in engineering design) in accordance with code requirements. Socket welds and branch connection welds which are not radiographed shall be examined by magnetic particle or liquid penetrant methods in accordance with code requirements.

3. In-process examination in accordance with the code definition, supplemented by appropriate nondestructive examination, may be

**TABLE 10-54 Preheat Temperatures\***

| Base metal<br>P-no. or<br>S-no.<br>[Note (1)] | Weld metal<br>analysis<br>A-no.<br>[Note (2)] | Base metal group                   | Nominal wall<br>thickness |      | Specified min.<br>tensile strength,<br>base metal |     | Min. temperature |                  |                  |                  |
|-----------------------------------------------|-----------------------------------------------|------------------------------------|---------------------------|------|---------------------------------------------------|-----|------------------|------------------|------------------|------------------|
|                                               |                                               |                                    | mm                        | in.  | MPa                                               | ksi | Required         |                  | Recommended      |                  |
|                                               |                                               |                                    |                           |      |                                                   |     | °C               | °F               | °C               | °F               |
| 1                                             | 1                                             | Carbon steel                       | <25                       | <1   | ≤490                                              | ≤71 | ...              | ...              | 10               | 50               |
|                                               |                                               |                                    | ≥25                       | ≥1   | All                                               | All | ...              | ...              | 79               | 175              |
|                                               |                                               |                                    | All                       | All  | >490                                              | >71 | ...              | ...              | 79               | 175              |
| 3                                             | 2,11                                          | Alloy steels,<br>Cr ≤ 1/2%         | <13                       | <1/2 | ≤490                                              | ≤71 | ...              | ...              | 10               | 50               |
|                                               |                                               |                                    | ≥13                       | ≥1/2 | All                                               | All | ...              | ...              | 79               | 175              |
|                                               |                                               |                                    | All                       | All  | >490                                              | >71 | ...              | ...              | 79               | 175              |
| 4                                             | 3                                             | Alloy steels,<br>1/2% ≤ Cr ≤ 2%    | All                       | All  | All                                               | All | 149              | 300              | ...              | ...              |
| 5A, 5B, 5C                                    | 4, 5                                          | Alloy steels,<br>2 1/4% ≤ Cr ≤ 10% | All                       | All  | All                                               | All | 177              | 350              | ...              | ...              |
| 6                                             | 6                                             | High alloy steels<br>martensitic   | All                       | All  | All                                               | All | ...              | ...              | 149 <sup>4</sup> | 300 <sup>4</sup> |
| 7                                             | 7                                             | High alloy steels<br>ferritic      | All                       | All  | All                                               | All | ...              | ...              | 10               | 50               |
| 8                                             | 8, 9                                          | High alloy steels<br>austenitic    | All                       | All  | All                                               | All | ...              | ...              | 10               | 50               |
| 9A, 9B                                        | 10                                            | Nickel alloy steels                | All                       | All  | All                                               | All | ...              | ...              | 93               | 200              |
| 10                                            | ...                                           | Cr-Cu steel                        | All                       | All  | All                                               | All | 149–204          | 300–400          | ...              | ...              |
| 101                                           | ...                                           | 27Cr steel                         | All                       | All  | All                                               | All | 149 <sup>3</sup> | 300 <sup>3</sup> | ...              | ...              |
| 11A SG 1                                      | ...                                           | 8Ni, 9Ni steel                     | All                       | All  | All                                               | All | ...              | ...              | 10               | 50               |
| 11A SG 2                                      | ...                                           | 5Ni steel                          | All                       | All  | All                                               | All | 10               | 50               | ...              | ...              |
| 21–52                                         | ...                                           | ...                                | All                       | All  | All                                               | All | ...              | ...              | 10               | 50               |

\*Extracted from *Process Piping*, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.

## NOTES:

(1) P-number or S-number from BPV Code, Section IX, QW/QB-442.

(2) A-number from Section IX, QW-442.

(3) Maintain interpass temperature between 177°C and 232°C (350°F and 450°F).

(4) Maximum interpass temperature 316°C (600°F).

substituted for the examination required in 2 above on a weld-for-weld basis if specified in the engineering design or specifically authorized by the Inspector.

4. *Certification and records.* The requirements established by the code for Normal Fluid Service apply.

**Impact Testing** In specifying materials, it is critical that the low-temperature limits of materials and impact testing requirements of the applicable code edition be clearly understood. In the recent past, code criteria governing low-temperature limits and requirements for impact testing have undergone extensive revision. The code contains extensive criteria detailing when impact testing is required and describing how it is to be performed. Because of the potentially changing requirements and the complexity of the code requirements, this handbook does not attempt to provide a comprehensive treatment of this subject or a comprehensive presentation of the requirements of the current code edition. Some of the general guidelines are provided here; however, the designer should consult the code to clearly understand additional requirements and special circumstances under which impact testing may be omitted. These exclusions permitted by the code may be significant in selecting materials or establishing material requirements.

In general, materials conforming to ASTM specifications listed by the code may be used at temperatures down to the lowest temperature listed for that material in ASME B31.3 Table A-10. When welding or other operations are performed on these materials, additional low-temperature toughness tests (impact testing) may be required. Refer to Table 10-51 for a general summary of these requirements.

**Pressure Testing** Prior to initial operation, installed piping shall be pressure-tested to assure tightness except as permitted for Category D fluid service described later. The pressure test shall be main-

tained for a sufficient time to determine the presence of any leaks but not less than 10 min.

If repairs or additions are made following the pressure tests, the affected piping shall be retested except that, in the case of minor repairs or additions, the owner may waive retest requirements when precautionary measures are taken to assure sound construction.

When pressure tests are conducted at metal temperatures near the ductile-to-brittle transition temperature of the material, the possibility of brittle fracture shall be considered.

The test shall be hydrostatic, using water, with the following exceptions. If there is a possibility of damage due to freezing or if the operating fluid or piping material would be adversely affected by water, any other suitable nontoxic liquid may be used. If a flammable liquid is used, its flash point shall not be less than 50°C (120°F), and consideration shall be given to the test environment.

The hydrostatic-test pressure at any point in the system shall be as follows:

1. Not less than 1 1/2 times the design pressure.

2. For a design temperature above the test temperature, the minimum test pressure shall be as calculated by the following formula:

$$P_T = 1.5PS_T/S \quad (10-108)$$

where  $P_T$  = test hydrostatic gauge pressure, MPa (lbf/in<sup>2</sup>)

$P$  = internal design pressure, MPa (lbf/in<sup>2</sup>)

$S_T$  = allowable stress at test temperature, MPa (lbf/in<sup>2</sup>)

$S$  = allowable stress at design temperature, MPa (lbf/in<sup>2</sup>)

$S_T/S$  shall not exceed 6.5

If the test pressure as so defined would produce a stress in excess of the yield strength at test temperature, the test pressure may be reduced

TABLE 10-55 Requirements for Heat Treatment\*

| Base metal<br>P-no.<br>or S-no.<br>[Note (1)] | Weld-metal<br>analysis<br>A-no.<br>[Note (2)] | Base metal<br>group                                                                                        | Nominal wall<br>thickness |                   | Specified min.<br>tensile<br>strength,<br>base metal |                     | Metal<br>temperature range    |                                 | Holding time               |               | Brinell<br>hardness,<br>max.<br>[Note (4)] |                    |
|-----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------|-------------------|------------------------------------------------------|---------------------|-------------------------------|---------------------------------|----------------------------|---------------|--------------------------------------------|--------------------|
|                                               |                                               |                                                                                                            | mm                        | in                | MPa                                                  | ksi                 | °C                            | °F                              | Nominal wall<br>[Note (3)] |               |                                            | Min.<br>time,<br>h |
|                                               |                                               |                                                                                                            |                           |                   |                                                      |                     |                               |                                 | min/mm                     | h/in          |                                            |                    |
| 1                                             | 1                                             | Carbon steel                                                                                               | ≤ 19<br>> 19              | ≤ ¾<br>> ¾        | All<br>All                                           | All<br>All          | None<br>593–649               | None<br>1100–1200               | ...<br>2.4                 | ...<br>1      | ...<br>1                                   | ...<br>...         |
| 3                                             | 2, 11                                         | Alloy steels<br>Cr ½%                                                                                      | ≤ 19<br>> 19<br>All       | ≤ ¾<br>> ¾<br>All | ≤ 490<br>All<br>> 490                                | ≤ 71<br>All<br>> 71 | None<br>593–718<br>593–718    | None<br>1100–1325<br>1100–1325  | ...<br>2.4<br>2.4          | ...<br>1<br>1 | ...<br>1<br>1                              | ...<br>225<br>225  |
| 4 [Note (5)]                                  | 3                                             | Alloy steels,<br>½% < Cr ≤ 2%                                                                              | ≤ 13<br>> 13<br>All       | ≤ ½<br>> ½<br>All | ≤ 490<br>All<br>> 490                                | ≤ 71<br>All<br>> 71 | None<br>704–746<br>704–746    | None<br>1300–1375<br>1300–1375  | ...<br>2.4<br>2.4          | ...<br>1<br>1 | ...<br>2<br>2                              | ...<br>225<br>225  |
| 5A, 5B, 5C<br>[Note (5)]                      | 4, 5                                          | Alloy steels 2¼%<br>≤ Cr ≤ 10% ≤ 3%<br>Cr and ≤ 0.15%<br>C ≤ 3% Cr and ≤<br>0.15% C > 3%<br>Cr or ≤ 0.15%C | ≤ 13<br>> 13<br>All       | ≤ ½<br>> ½<br>All | All<br>All<br>All                                    | All<br>All<br>All   | None<br>704–760<br>704–760    | None<br>1300–1400<br>1300–1400  | ...<br>2.4<br>2.4          | ...<br>1<br>1 | ...<br>2<br>2                              | ...<br>241<br>241  |
| 6                                             | 6                                             | High-alloy steels:<br>martensitic<br>A 240, Gr. 429                                                        | All<br>All                | All<br>All        | All<br>All                                           | All<br>All          | 732–788<br>621–663            | 1350–1450<br>1150–1225          | 2.4<br>2.4                 | 1<br>1        | 2<br>2                                     | 241<br>241         |
| 7                                             | 7                                             | High-alloy steels:<br>ferritic                                                                             | All                       | All               | All                                                  | All                 | None                          | None                            | ...                        | ...           | ...                                        | ...                |
| 8                                             | 8, 9                                          | High-alloy steels:<br>austenitic                                                                           | All                       | All               | All                                                  | All                 | None                          | None                            | ...                        | ...           | ...                                        | ...                |
| 9A, 9B                                        | 10                                            | Nickel alloy steels                                                                                        | ≤ 19<br>> 19              | ≤ ¾<br>> ¾        | All<br>All                                           | All<br>All          | None<br>593–635               | None<br>1100–1175               | ...<br>1.2                 | ...<br>½      | ...<br>1                                   | ...<br>...         |
| 10                                            | ...                                           | Cr–Cu steel                                                                                                | All                       | All               | All                                                  | All                 | 760–816<br>[Note (6)]         | 1400–1500<br>[Note (6)]         | 1.2                        | ½             | ½                                          | ...                |
| 10H                                           | ...                                           | Duplex stainless<br>steel                                                                                  | All                       | All               | All                                                  | All                 | Note (7)                      | Note (7)                        | 1.2                        | ½             | ½                                          | ...                |
| 10I                                           | ...                                           | 27Cr steel                                                                                                 | All                       | All               | All                                                  | All                 | 663–704<br>[Note (8)]         | 1225–1300<br>[Note (8)]         | 2.4                        | 1             | 1                                          | ...                |
| 11A SG 1                                      | ...                                           | 8Ni, 9Ni steel                                                                                             | ≤ 51<br>> 51              | ≤ 2<br>> 2        | All<br>All                                           | All<br>All          | None<br>552–585<br>[Note (9)] | None<br>1025–1085<br>[Note (9)] | ...<br>2.4                 | ...<br>1      | ...<br>1                                   | ...<br>...         |
| 11A SG 2                                      | ...                                           | 5Ni steel                                                                                                  | > 51                      | > 2               | All                                                  | All                 | 552–585<br>[Note (9)]         | 1025–1085<br>[Note (9)]         | 2.4                        | 1             | 1                                          | ...                |
| 62                                            | ...                                           | Zr R60705                                                                                                  | All                       | All               | All                                                  | All                 | 538–593<br>[Note (10)]        | 1000–1100<br>[Note (10)]        | Note<br>(10)               | Note<br>(10)  | 1                                          | ...                |

\* Extracted from *Process Piping*, ASME B31.3—2004, with permission of the publisher, the American Society of Mechanical Engineers, New York.

## NOTES:

(1) P-number or S-number from BPV Code, Section IX, QW/QB-422.

(2) A-number from Section IX, QW-442.

(3) For holding time in SI metric units, use min/mm (minutes per mm thickness). For U.S. units, use h/in thickness.

(4) See para. 331.1.7.<sup>a</sup>

(5) See Appendix F, para. F331.1.<sup>a</sup>

(6) Cool as rapidly as possible after the hold period.

(7) Postweld heat treatment is neither required nor prohibited, but any heat treatment applied shall be as required in the material specification.

(8) Cooling rate to 649°C (1200°F) shall be less than 56°C (100°F)/h; thereafter, the cooling rate shall be fast enough to prevent embrittlement.

(9) Cooling rate shall be > 167°C (300°F)/h to 316°C (600°F).

(10) Heat treat within 14 days after welding. Hold time shall be increased by ½ h for each 25 mm (1 in) over 25 mm thickness. Cool to 427°C (800°F) at a rate ≤ 278°C (500°F)/h, per 25 mm (1 in) nominal thickness, 278°C (500°F)/h max. Cool in still air from 427°C (800°F).

<sup>a</sup> Refer to the referenced code paragraph for details.

to the maximum pressure that will not exceed the yield strength at test temperature. If the test liquid in the system is subject to thermal expansion, precautions shall be taken to avoid excessive pressure.

A preliminary air test at not more than 0.17-MPa (25-lbf/in<sup>2</sup>) gauge pressure may be made prior to hydrostatic test in order to locate major leaks.

If hydrostatic testing is not considered practicable by the owner, a pneumatic test in accordance with the following procedure may be substituted, using air or another nonflammable gas.

If the piping is tested pneumatically, the test pressure shall be 110 percent of the design pressure. When pneumatically testing nonmetallic

materials, assure that the materials are suitable for compressed gas. Pneumatic testing involves a hazard due to the possible release of energy stored in compressed gas. Therefore, particular care must be taken to minimize the chance of brittle failure of metals and thermoplastics. The test temperature is important in this regard and must be considered when material is chosen in the original design. Any pneumatic test shall include a preliminary check at not more than 0.17-MPa (25-lbf/in<sup>2</sup>) gauge pressure. The pressure shall be increased gradually in steps providing sufficient time to allow the piping to equalize strains during test and to check for leaks. Once test pressure has been achieved, the pressure shall be reduced to design pressure prior to examining for leakage.

**TABLE 10-56 Types of Examination for Evaluating Imperfections\***

| Type of imperfection     | Type of examination |                                       |                            |      |
|--------------------------|---------------------|---------------------------------------|----------------------------|------|
|                          | Visual              | Liquid-penetrant or magnetic-particle | Ultrasonic or radiographic |      |
|                          |                     |                                       | Random                     | 100% |
| Crack                    | X                   | X                                     | X                          | X    |
| Incomplete penetration   | X                   |                                       | X                          | X    |
| Lack of fusion           | X                   |                                       | X                          | X    |
| Weld undercutting        | X                   |                                       |                            |      |
| Weld reinforcement       | X                   |                                       |                            |      |
| Internal porosity        |                     |                                       | X                          | X    |
| External porosity        | X                   |                                       |                            |      |
| Internal slag inclusions |                     |                                       | X                          | X    |
| External slag inclusions | X                   |                                       |                            |      |
| Concave root surface     | X                   |                                       | X                          | X    |

\*Extracted from the Chemical Plant and Petroleum Refinery Piping Code, ANSI B31.3—1980, with permission of the publisher, the American Society of Mechanical Engineers, New York. For code acceptance criteria (limits on imperfections) for welds see Table 10-53.

At the owner's option, a piping system used only for Category D fluid service as defined in the subsection "Classification of Fluid Service" may be tested at the normal operating conditions of the system during or prior to initial operation by examining for leaks at every joint not previously tested. A preliminary check shall be made at not more than 0.17-MPa (25-lbf/in<sup>2</sup>) gauge pressure when the contained fluid is a gas or a vapor. The pressure shall be increased gradually in steps providing sufficient time to allow the piping to equalize strains during testing and to check for leaks.

Tests alternative to those required by these provisions may be applied under certain conditions described in the code.

Piping required to have a sensitive leak test shall be tested by the gas- and bubble-formation testing method specified in Art. 10, Sec. V of the ASME Boiler and Pressure Vessel Code or by another method demonstrated to have equal or greater sensitivity. The sensitivity of the test shall be at least (100 Pa-mL/s) [(10<sup>3</sup> atm-mL/s)] under test conditions.

Records shall be kept of each piping installation during the testing.

## COST COMPARISON OF PIPING SYSTEMS

Piping may represent as much as 25 percent of the cost of a chemical-process plant. The installed cost of piping systems varies widely with the materials of construction and the complexity of the system. A study of piping costs shows that the most economical choice of material for a simple straight piping run may not be the most economical for a complex installation made up of many short runs involving numerous fittings and valves. The economics also depends heavily on the pipe size and fabrication techniques employed. Fabrication methods such as bending to standard long-radius-elbow dimensions and machine-flaring lap joints have a large effect on the cost of fabricating pipe from ductile materials suited to these techniques. Cost reductions of as high as 35 percent are quoted by some custom fabricators utilizing advanced techniques; however, the basis for pricing comparisons should be carefully reviewed.

Figure 10-181 is based on data extracted from a comparison of the installed cost of piping systems of various materials published by the Dow Chemical Co. The chart shows the relative cost ratios for systems of various materials based on two installations, one consisting of 152 m (500 ft) of 2-in pipe in a complex piping arrangement and the other of 305 m (1000 ft) of 2-in pipe in a straight-run piping arrangement. Figure 10-181 is based on field-fabrication construction techniques using welding stubs, the method commonly used by contractors. A considerably different ranking would result from using other construction methods such as machine-formed lap joints and bends in place of welding elbows. Piping-cost experience shows that it is difficult to generalize and reflect accurate piping-cost comparisons. Although the prices for many of the metallurgies shown in Fig. 10-181 are very volatile even over short periods, Fig. 10-181 may still be used as a reasonable initial estimate of the relative cost of metallic materials. The cost of nonmetallic materials and lined metallic materials versus

solid alloy materials should be carefully reviewed prior to material selection. For an accurate comparison the cost for each type of material must be estimated individually on the basis of the actual fabrication and installation methods that will be used, pipe sizes and the conditions anticipated for the proposed installation.

## FORCES OF PIPING ON PROCESS MACHINERY AND PIPING VIBRATION

The reliability of process rotating machinery is affected by the quality of the process piping installation. Excessive external forces and moments upset casing alignment and can reduce clearance between motor and casing. Further, the bearings, seals, and coupling can be adversely affected, resulting in repeated failures that may be correctly diagnosed as misalignment, and may have excessive piping forces as the root causes. Most turbine and compressor manufacturers have prescribed specification or will follow NEMA standards for allowable nozzle loading. For most of the pumps, API or ANSI pump standards will be followed when evaluating the pump nozzle loads. Pipe support restraints need to be placed at the proper locations to protect the machinery nozzles during operation.

Prior to any machinery alignment procedure, it is imperative to check for machine pipe strain. This is accomplished by the placement of dial indicators on the shaft and then loosening the hold-down bolts. Movements of greater than 1 mil are considered indication of a pipe strain condition.

This is an important practical problem area, as piping vibration can cause considerable downtime or even pipe failure.

Pipe vibration is caused by:

1. Internal flow (pulsation)
2. Plant machinery (such as compressors, pumps)

Pulsation can be problematic and difficult to predict. Pulsations are also dependent on acoustic resonance characteristics. For reciprocating equipment, such as reciprocating compressors and pumps, in some cases, an analog (digital) study needs to be performed to identify the deficiency in the piping and pipe support systems as well as to evaluate the performance of the machine during operation. The study will also provide recommendations on how to improve the machine and piping system's performance.

When a pulsation frequency coincides with a mechanical or acoustic resonance, severe vibration can result. A common cause for pulsation is the presence of flow control valves or pressure regulators. These often operate with high pressure drops (i.e., high flow velocities), which can result in the generation of severe pulsation. Flashing and cavitation can also contribute.

Modern-day piping design codes can model the vibration situation, and problems can thus be resolved in the design phases.

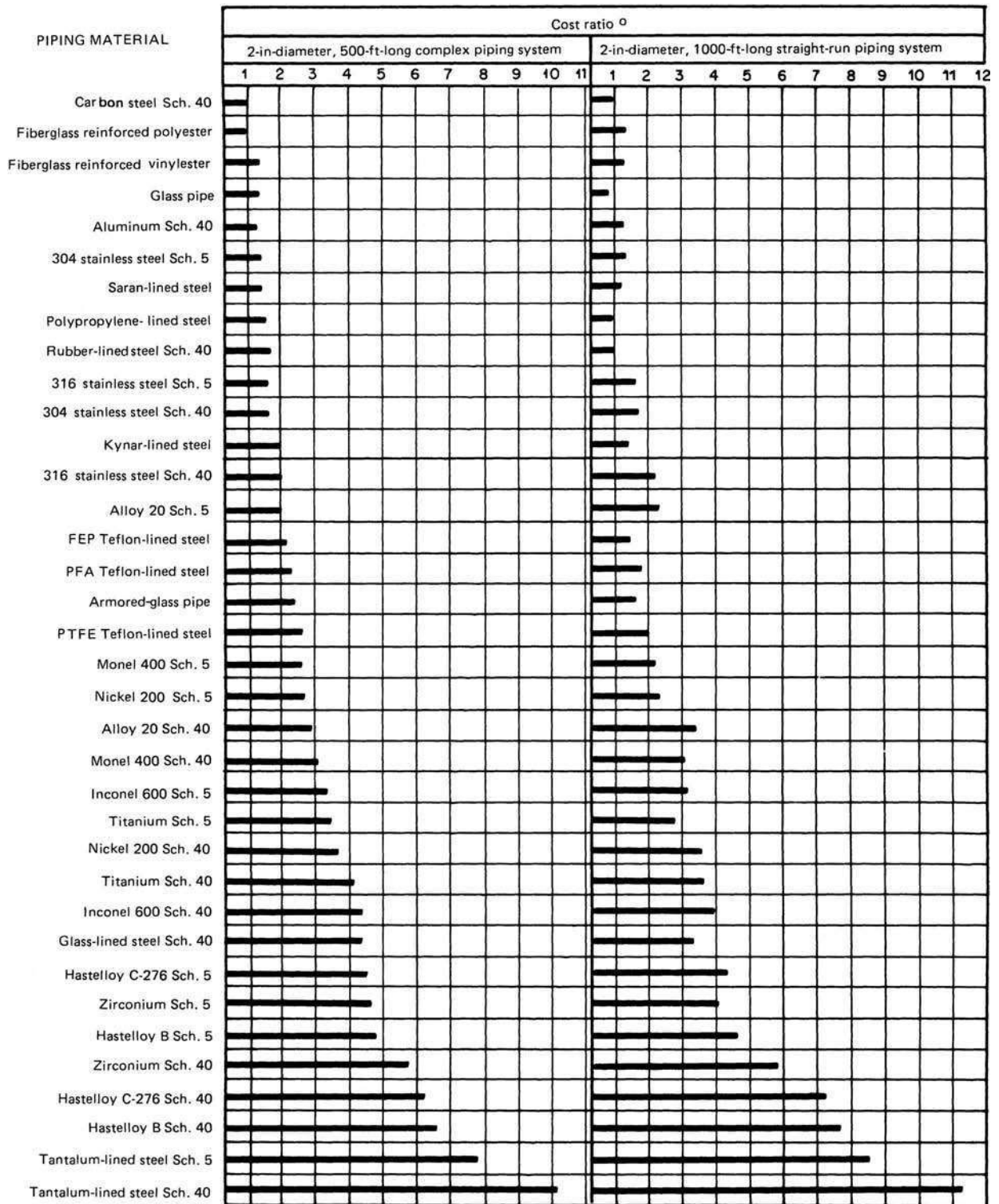
## HEAT TRACING OF PIPING SYSTEMS

Heat tracing is used to maintain pipes and the material that pipes contain at temperatures above the ambient temperature. Two common uses of heat tracing are preventing water pipes from freezing and maintaining fuel oil pipes at high enough temperatures such that the viscosity of the fuel oil will allow easy pumping. Heat tracing is also used to prevent the condensation of a liquid from a gas and to prevent the solidification of a liquid commodity.

A heat-tracing system is often more expensive on an installed cost basis than the piping system it is protecting, and it will also have significant operating costs. A study on heat-tracing costs by a major chemical company showed installed costs of \$31/ft to \$142/ft and yearly operating costs of \$1.40/ft to \$16.66/ft. In addition to being a major cost, the heat-tracing system is an important component of the reliability of a piping system. A failure in the heat-tracing system will often render the piping system inoperable. For example, with a water freeze protection system, the piping system may be destroyed by the expansion of water as it freezes if the heat-tracing system fails.

The vast majority of heat-traced pipes are insulated to minimize heat loss to the environment. A heat input of 2 to 10 watts per foot is generally required to prevent an insulated pipe from freezing. With high wind speeds, an uninsulated pipe could require well over 100





**FIG. 10-181** Cost rankings and cost ratios for various piping materials. This figure is based on field-fabrication construction techniques using welding stubs, as this is the method most often employed by contractors. A considerably different ranking would result from using other construction methods, such as machined-formed lap joints, for the alloy pipe. <sup>a</sup>Cost ratio = (cost of listed item)/(cost of Schedule 40 carbon steel piping system, field-fabricated by using welding stubs). (Extracted with permission from *Installed Cost of Corrosion Resistant Piping*, copyright 1977, Dow Chemical Co.)

watts per foot to prevent freezing. Such a high heat input would be very expensive.

Heat tracing for insulated pipes is generally only required for the period when the material in the pipe is not flowing. The heat loss of an insulated pipe is very small compared to the heat capacity of a flowing fluid. Unless the pipe is extremely long (several thousands of feet), the temperature drop of a flowing fluid will not be significant.

The three major methods of avoiding heat tracing are:

1. Changing the ambient temperature around the pipe to a temperature that will avoid low-temperature problems. Burying water pipes below the frost line or running them through a heated building are the two most common examples of this method.

2. Emptying a pipe after it is used. Arranging the piping such that it drains itself when not in use, can be an effective method of avoiding the need for heat tracing. Some infrequently used lines can be pigged or blown out with compressed air. This technique is not recommended for commonly used lines due to the high labor requirement.

3. Arranging a process such that some lines have continuous flow can eliminate the need for tracing these lines. This technique is generally not recommended because a failure that causes a flow stoppage can lead to blocked or broken pipes.

Some combination of these techniques may be used to minimize the quantity of traced pipes. However, the majority of pipes containing fluids that must be kept above the minimum ambient temperature are generally going to require heat tracing.

**Types of Heat-Tracing Systems** Industrial heat-tracing systems are generally fluid systems or electrical systems. In fluid systems, a pipe or tube called the *tracer* is attached to the pipe being traced, and a warm fluid is put through it. The tracer is placed under the insulation. Steam is by far the most common fluid used in the tracer, although ethylene glycol and more exotic heat-transfer fluids are used. In electrical systems, an electrical heating cable is placed against the pipe under the insulation.

**Fluid Tracing Systems** Steam tracing is the most common type of industrial pipe tracing. In 1960, over 95 percent of industrial tracing systems were steam traced. By 1995, improvements in electric heating technology increased the electric share to 30 to 40 percent. Fluid systems other than steam are rather uncommon and account for less than 5 percent of tracing systems.

Half-inch copper tubing is commonly used for steam tracing. Three-eighths-inch tubing is also used, but the effective circuit length is then decreased from 150 ft to about 60 ft. In some corrosive environments, stainless steel tubing is used.

In addition to the tracer, a steam tracing system (Fig. 10-182) consists of steam supply lines to transport steam from the existing steam lines to the traced pipe, a steam trap to remove the condensate and hold back the steam, and in most cases a condensate return system to return the condensate to the existing condensate return system. In the past, a significant percentage of condensate from steam tracing was simply dumped to drains, but increased energy costs and environmental rules have caused almost all condensate from new steam tracing systems to be returned. This has significantly increased the initial cost of steam tracing systems.

Applications requiring accurate temperature control are generally limited to electric tracing. For example, chocolate lines cannot be exposed to steam temperatures or the product will degrade and if caustic soda is heated above 65°C (150°F) it becomes extremely corrosive to carbon steel pipes.

For some applications, either steam or electricity is simply not available and this makes the decision. It is rarely economic to install a steam boiler just for tracing. Steam tracing is generally considered only when a boiler already exists or is going to be installed for some other primary purpose. Additional electric capacity can be provided in most situations for reasonable costs. It is considerably more expensive to supply steam from a long distance than it is to provide electricity. Unless steam is available close to the pipes being traced, the automatic choice is usually electric tracing.

For most applications, particularly in processing plants, either steam tracing or electric tracing could be used, and the correct choice is dependent on the installed costs and the operating costs of the competing systems.

**Economics of Steam Tracing versus Electric Tracing** The question of the economics of various tracing systems has been examined thoroughly. All of these papers have concluded that electric tracing is generally less expensive to install and significantly less expensive to operate. Electric tracing has significant cost advantages in terms of installation because less labor is required than steam tracing. However, it is clear that there are some special cases where steam tracing is more economical.

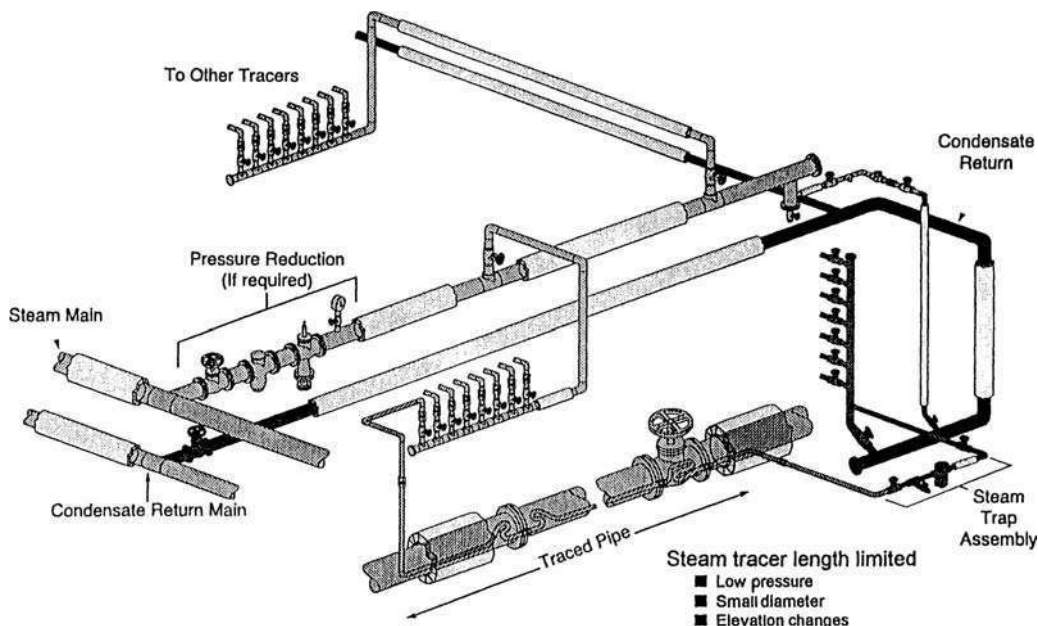


FIG. 10-182 Steam tracing system.

TABLE 10-57 Steam versus Electric Tracing\*

| Temperature maintained | TIC    |          | Ratio S/E | TOC   |          | Ratio S/E |
|------------------------|--------|----------|-----------|-------|----------|-----------|
|                        | Steam  | Electric |           | Steam | Electric |           |
| 50°F                   | 22,265 | 7,733    | 2.88      | 1,671 | 334      | 5.00      |
| 150°F                  | 22,265 | 13,113   | 1.70      | 4,356 | 1,892    | 2.30      |
| 250°F                  | 22,807 | 17,624   | 1.29      | 5,348 | 2,114    | 2.53      |
| 400°F                  | 26,924 | 14,056   | 1.92      | 6,724 | 3,942    | 1.71      |

\*Specifications: 400 feet of four-inch pipe, \$25/hr labor, \$0.07/kWh, \$4.00/1,000# steam, 100-foot supply lines. TIC = total installed cost; TOC = total operating costs.

The two key variables in the decision to use steam tracing or electric tracing are the temperature at which the pipe must be maintained and the distance to the supply of steam and a source of electric power.

Table 10-57 shows the installed costs and operating costs for 400 ft of 4-in pipe, maintained at four different temperatures, with supply lengths of 100 ft for both electricity and steam and \$25/h labor.

The major advantages of a steam tracing system are:

1. **High heat output.** Due to its high temperature, a steam tracing system provides a large amount of heat to the pipe. There is a very high heat transfer rate between the metallic tracer and a metallic pipe. Even with damage to the insulation system, there is very little chance of a low temperature failure with a steam-tracing system.

2. **High reliability.** Many things can go wrong with a steam-tracing system but very few of the potential problems lead to a heat-tracing failure. Steam traps fail, but they usually fail in the open position, allowing for a continuous flow of steam to the tracer. Other problems such as steam leaks that can cause wet insulation are generally prevented from becoming heat-tracing failures by the extremely high heat output of a steam tracer. Also, a tracing tube is capable of withstanding a large amount of mechanical abuse without failure.

3. **Safety.** While steam burns are fairly common, there are generally fewer safety concerns than with electric tracing.

4. **Common usage.** Steam tracing has been around for many years and many operators are familiar with the system. Because of this familiarity, failures due to operator error are not very common.

The weaknesses of a steam-tracing system are:

1. **High installed costs.** The incremental piping required for the steam supply system and the condensate return system must be installed, insulated, and, in the case of the supply system, additional

steam traps are often required. The tracer itself is not expensive, but the labor required for installation is relatively high. Studies have shown that steam tracing systems typically cost from 50 to 150 percent more than a comparable electric-tracing system.

2. **Energy inefficiency.** A steam-tracing system's total energy use is often more than 20 times the actual energy requirement to keep the pipe at the desired temperature. The steam tracer itself puts out significantly more energy than required. The steam traps use energy even when they are properly operating and waste large amounts of energy when they fail in the open position, which is the common failure mode. Steam leaks waste large amounts of energy, and both the steam supply system and the condensate return system use significant amounts of energy.

3. **Poor temperature control.** A steam-tracing system offers very little temperature control capability. The steam is at a constant temperature (50 psig steam is 300°F) usually well above that desired for the pipe. The pipe will reach an equilibrium temperature somewhere between the steam temperature and the ambient temperature. However, the section of pipe against the steam tracer will effectively be at the steam temperature. This is a serious problem for temperature-sensitive fluids such as food products. It also represents a problem with fluids such as bases and acids, which are not damaged by high temperatures but often become extremely corrosive to piping systems at higher temperatures.

4. **High maintenance costs.** Leaks must be repaired and steam traps must be checked and replaced if they have failed. Numerous studies have shown that, due to the energy lost through leaks and failed steam traps, an extensive maintenance program is an excellent investment. Steam maintenance costs are so high that for low-temperature maintenance applications, total steam operating costs are sometimes greater than electric operating costs, even if no value is placed on the steam.

**Electric Tracing** An electric-tracing system (see Fig. 10-183) consists of an electric heater placed against the pipe under the thermal insulation, the supply of electricity to the tracer, and any control or monitoring system that may be used (optional). The supply of electricity to the tracer usually consists of an electrical panel and electrical conduit or cable trays. Depending on the size of the tracing system and the capacity of the existing electrical system, an additional transformer may be required.

#### Advantages of Electric Tracing

1. **Lower installed and operating costs.** Most studies have shown that electric tracing is less expensive to install and less expensive to

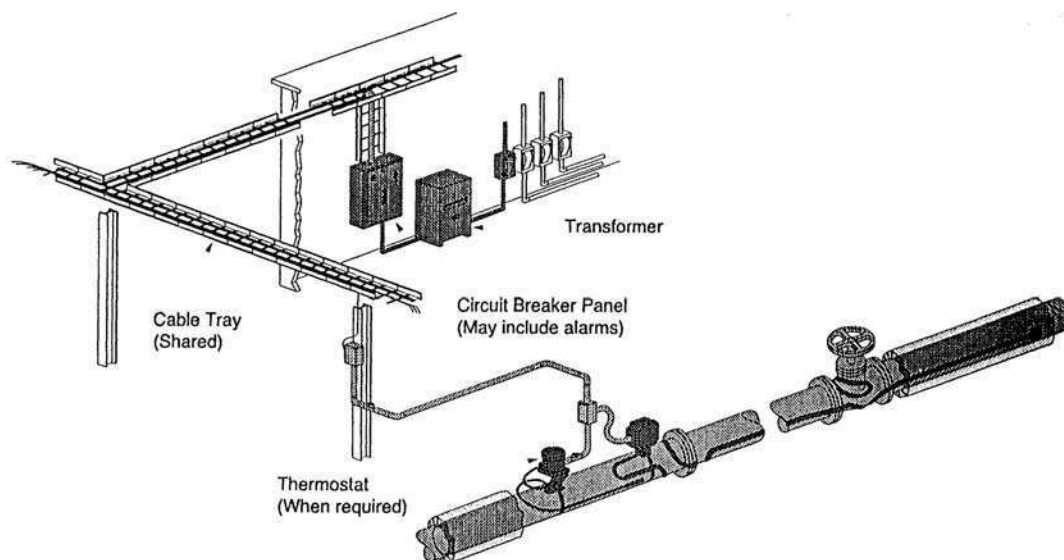


FIG. 10-183 Electrical heat tracing system.

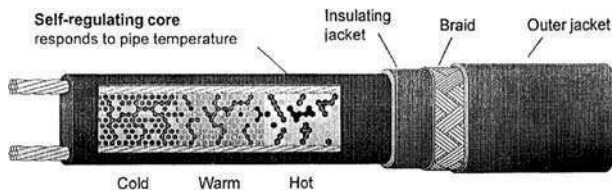


FIG. 10-184 Self-regulating heating cable.

operate. This is true for most applications. However, for some applications, the installed costs of steam tracing are equal to or less than electric tracing.

2. **Reliability.** In the past, electric heat tracing had a well-deserved reputation for poor reliability. However, since the introduction of self-regulating heaters in 1971, the reliability of electric heat tracing has improved dramatically. Self-regulating heaters cannot destroy themselves with their own heat output. This eliminates the most common failure mode of polymer-insulated constant wattage heaters. Also, the technology used to manufacture mineral-insulated cables, high-temperature electric heat tracing, has improved significantly, and this has improved their reliability.

3. **Temperature control.** Even without a thermostat or any control system, an electric tracing system usually provides better temperature control than a steam tracing system. With thermostatic or electronic control, very accurate temperature control can be achieved.

4. **Safety.** The use of self-regulating heaters and ground leakage circuit breakers has answered the safety concerns of most engineers considering electric tracing. Self-regulating heaters eliminate the problems from high-temperature failures, and ground leakage circuit breakers minimize the danger of an electrical fault to ground, causing injury or death.

5. **Monitoring capability.** One question often asked about any heat-tracing system is, "How do I know it is working?" Electric tracing now has available almost any level of monitoring desired. The temperature at any point can be monitored with both high and low alarm capability. This capability has allowed many users to switch to electric tracing with a high degree of confidence.

6. **Energy efficiency.** Electric heat tracing can accurately provide the energy required for each application without the large additional energy use of a steam system. Unlike steam tracing systems, other parts of the system do not use significant amounts of energy.

#### Disadvantages of Electric Tracing

1. **Poor reputation.** In the past, electric tracing has been less than reliable. Due to past failures, some operating personnel are unwilling to take a chance on any electric tracing.

2. **Design requirements.** A slightly higher level of design expertise is required for electric tracing than for steam tracing.

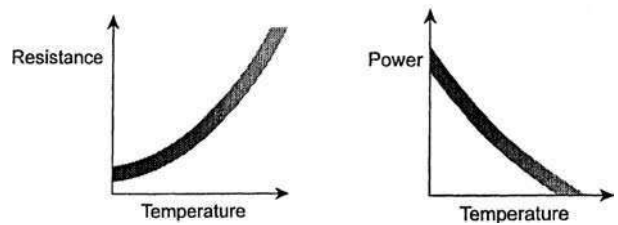


FIG. 10-185 Self-regulation.

3. **Lower power output.** Since electric tracing does not provide a large multiple of the required energy, it is less forgiving to problems such as damaged insulation or below design ambient temperatures. Most designers include a 10 to 20 percent safety factor in the heat loss calculation to cover these potential problems. Also, a somewhat higher than required design temperature is often specified to provide an additional safety margin. For example, many water systems are designed to maintain 50°F to prevent freezing.

**Types of Electric Tracing** Self-regulating electric tracing (see Fig. 10-184) is by far the most popular type of electric tracing. The heating element in a self-regulating heater is a conductive polymer between the bus wires. This conductive polymer increases its resistance as its temperature increases. The increase in resistance with temperature causes the heater to lower its heat output at any point where its temperature increases (Fig. 10-185). This self-regulating effect eliminates the most common failure mode of constant wattage electric heaters, which is destruction of the heater by its own heat output.

Because self-regulating heaters are parallel heaters, they may be cut to length at any point without changing their power output per unit of length. This makes them much easier to deal with in the field. They may be terminated, teed, or spliced in the field with hazardous-area-approved components.

**MI Cables** Mineral insulated cables (Fig. 10-186) are the electric heat tracers of choice for high-temperature applications. High-temperature applications are generally considered to maintain temperatures above 250°F or exposure temperatures above 420°F where self-regulating heaters cannot be used. MI cable consists of one or two heating wires, magnesium oxide insulation (from whence it gets its name), and an outer metal sheath. Today the metal sheath is generally inconel. This eliminates both the corrosion problems with copper sheaths and the stress cracking problems with stainless steel.

MI cables can maintain temperatures up to 1200°F and withstand exposure to up to 1500°F. The major disadvantage of MI cable is that it must be factory-fabricated to length. It is very difficult to terminate

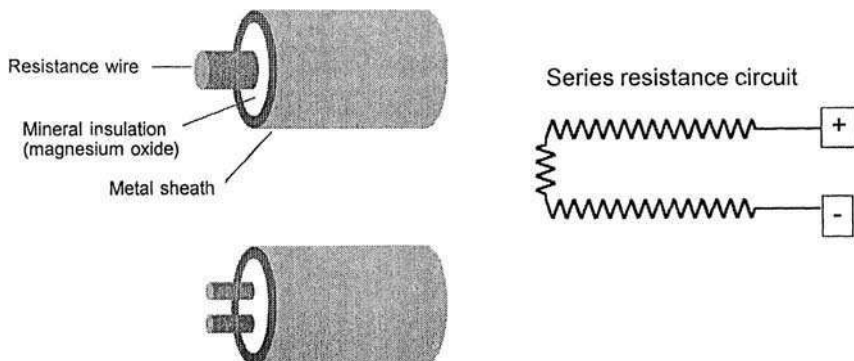


FIG. 10-186 Mineral insulated cable (MI cable).

or splice the heater in the field. This means pipe measurements are necessary before the heaters are ordered. Also, any damage to an MI cable generally requires a complete new heater. It's not as easy to splice in a good section as with self-regulating heaters.

**Polymer-Insulated Constant Wattage Electric Heaters** These are slightly cheaper than self-regulating heaters, but they are generally being replaced with self-regulating heaters due to inferior reliability. These heaters tend to destroy themselves with their own heat output when they are overlapped at valves or flanges. Since overlapping self-regulating heaters is the standard installation technique, it is difficult to prevent this technique from being used on the similar-looking constant-wattage heaters.

**SECT (Skin-Effect Current Tracing)** This is a special type of electric tracing employing a tracing pipe, usually welded to the pipe being traced, that is used for extremely long lines. With SECT tracing circuits, up to 10 miles can be powered from one power point. All SECT systems are specially designed by heat-tracing vendors.

**Impedance Tracing** This uses the pipe being traced to carry the current and generate the heat. Less than 1 percent of electric heat-tracing systems use this method. Low voltages and special electrical isolation techniques are used. Impedance heating is useful when extremely high heat densities are required, like when a pipe containing aluminum metal must be melted from room temperature on a regular basis. Most impedance systems are specially designed by heat tracing vendors.

**Choosing the Best Tracing System** Some applications require either steam tracing or electric tracing regardless of the relative economics. For example, a large line that is regularly allowed to cool and needs to be quickly heated would require steam tracing because of its much higher heat output capability. In most heat-up applications, steam tracing is used with heat-transfer cement, and the heat output is increased by a factor of up to 10. This is much more heat than would be

practical to provide with electric tracing. For example, a half-inch copper tube containing 50 psig steam with heat transfer cement would provide over 1100 Btu/hr/ft to a pipe at 50°F. This is over 300 watts per foot or more than 15 times the output of a high-powered electric tracer.

Table 10-57 shows that electric tracing has a large advantage in terms of cost at low temperatures and smaller but still significant advantages at higher temperatures. Steam tracing does relatively better at higher temperatures because steam tracing supplies significantly more power than is necessary to maintain a pipe at low temperatures. Table 10-57 indicates that there is very little difference between the steam tracing system at 50°F and the system at 250°F. However, the electric system more than doubles in cost between these two temperatures because more heaters, higher powered heaters, and higher temperature heaters are required.

The effect of supply lengths on a 150°F system can be seen from Table 10-58. Steam supply pipe is much more expensive to run than electrical conduit. With each system having relatively short supply lines (40 ft each) the electric system has only a small cost advantage (10 percent, or a ratio of 1.1). This ratio is 2.1 at 50°F and 0.8 at 250°F. However, as the supply lengths increase, electric tracing has a large cost advantage.

**TABLE 10-58 Effect of Supply Lengths**

Ratio of steam TIC to electric TIC maintained at 150°F

| Steam supply length | Electric supply length |          |          |
|---------------------|------------------------|----------|----------|
|                     | 40 feet                | 100 feet | 300 feet |
| 40 feet             | 1.1                    | 1.0      | 0.7      |
| 100 feet            | 1.9                    | 1.7      | 1.1      |
| 300 feet            | 4.9                    | 4.2      | 2.9      |

## STORAGE AND PROCESS VESSELS

### STORAGE OF LIQUIDS

**Atmospheric Tanks** The term *atmospheric tank* as used here applies to any tank that is designed to operate at pressures from atmospheric through 3.45 kPag (0.5 psig). It may be either open to the atmosphere or enclosed. Atmospheric tanks may be either shop-fabricated or field-erected. The most common application is storage of fuel for transportation and power generation. See the subsections "Pressure Tanks" and "Pressure Vessels" later in this section.

**Shop-Fabricated Storage Tanks** A shop-fabricated storage tank is a storage tank constructed at a tank manufacturer's plant and transported to a facility for installation. In general, tanks 190 m<sup>3</sup> (50,000 gal) and under can be shop-fabricated and shipped in one piece to an installation site. Shop-fabricated storage tanks may be either underground storage tanks (USTs) or aboveground storage tanks (ASTs).

**USTs versus ASTs** For decades, USTs were the standard means of storing petroleum and chemicals in quantities of 190 m<sup>3</sup> (50,000 gal) or less. However, during the 1990s, many industrial and commercial facilities shifted to ASTs for hazardous product storage. Reasons include the ability to visually monitor the storage tank as well as to avoid perceived risks and the myriad regulations surrounding underground storage tanks. Nonetheless, AST installations are also subject to certain regulations and codes, particularly fire codes.

The choice of utilizing USTs or ASTs is driven by numerous factors. Local authorities having jurisdiction may allow only underground storage. Total required storage capacity factored against available real estate may also preclude the use of ASTs. In addition, ASTs are subject to minimum distance separations from one another and from buildings, property lines, and public ways. ASTs are visually monitorable, yet that same attribute may be aesthetically undesirable. Other considerations are adequate protection against spills, vandal-

ism, and vehicular damage. Additionally, central design elements regarding product transfer and system functionality must be taken into account.

USTs are subject to myriad EPA regulations as well as fire codes. Further, a commonly cited drawback is the potential for unseen leaks and subsequent environmental damage and cleanup. However, advances in technology have addressed these concerns. Corrosion protection and leak detection are now standard in all UST systems. Sophisticated tank and pipe secondary containment systems have been developed to meet the EPA's secondary containment mandate for underground storage of nonpetroleum chemicals. Computerized in-tank monitoring devices for tracking inventory, evolutionary tank integrity testing equipment, statistical inventory reconciliation analysis, leak-free dry-disconnect pipe and hose joints for loading/unloading, and in-tank fill shutoff valves are just a few of the many pieces of equipment which have surfaced as marketplace solutions.

Properly designed and installed in accordance with industry standards, regulations, and codes, both UST and AST systems are reliable and safe. Because of space limitations and the prevalence of ASTs at plant sites, only ASTs will be discussed further.

**Aboveground Storage Tanks** Aboveground storage tanks are classified as either field-erected or shop-fabricated. The latter is typically 190-m<sup>3</sup> (50,000-gal) capacity or less and is able to be shipped over the highway, while larger tanks are more economically erected in the field. Whereas field-erected tanks likely constitute the majority of total AST storage capacity, shop-fabricated ASTs constitute the clear majority of the total number of ASTs in existence today. Most of these shop-fabricated ASTs store flammable or combustible liquids at atmospheric pressure and have 45-m<sup>3</sup> (12,000-gal) capacity or smaller.

Shop-fabricated ASTs can be designed and fabricated as pressure vessels, but are more typically vented to atmosphere. They are oriented

for either horizontal or vertical installation and are made in either cylindrical or rectangular form. Tanks are often secondarily contained and may also include insulation for fire safety or temperature control. Compartmented ASTs are also available. Over 90 percent of the atmospheric tank applications store some sort of hydrocarbon. Within that, a majority are used to store fuels for motor vehicle dispensing.

**Fire Codes** Many chemicals are classified as either flammable or combustible or subject to fire codes. All hydrocarbon tanks are classified as flammable or combustible. Notably, with the tremendous increase in the use of ASTs at private fleet fueling facilities in the 1990s, fire codes were rapidly modified to address the understandable safety concerns. In the United States, two principal organizations publish fire codes pertinent to underground storage tanks, with each state adopting all or part of the respective codes.

The **National Fire Protection Association (NFPA)** has developed several principal codes pertaining to the storage of flammable and combustible liquids:

**NFPA 30, Flammable and Combustible Liquids Code (2003)**

**NFPA 30A, Code for Motor Fuel Dispensing Facilities and Repair Garages (2003)**

**NFPA 31, Standard for the Installation of Oil Burning Equipment**

The **International Code Council (ICC)** was formed by the consolidation of three formerly separate fire code organizations: International Conference of Building Officials (ICBO), which had published the Uniform Fire Code under its fire service arm, the International Fire Code Institute (IFCI); Building Officials and Code Administrators (BOCA), which had published the National Fire Prevention Code; and Southern Building Congress Code International (SBCCI), which had published the Standard Fire Prevention Code. When the three groups merged in 2000, in part to develop a common fire code, the individual codes became obsolete; however, they are noted above since references to them may periodically surface. The consolidated code is **IFC-2006, International Fire Code**.

The **Canadian Commission on Building and Fire Codes (CCBFC)** developed a recommended model code to permit adoption by various regional authorities. The National Research Council of Canada publishes the model code document **National Fire Code of Canada (1995)**.

**Standards** Third-party standards for AST fabrication have evolved significantly over the past two decades, as have various recommended practice guidelines for the installation and operations of AST systems. Standards developed by Underwriters Laboratories (UL) have been perhaps the most predominant guidelines—in fact, ASTs are often categorized according to the UL standard that they meet, such as “a UL 142 tank.”

**Underwriters Laboratories Inc.** standards for steel ASTs storing flammable and combustible liquids include the following:

**UL 142, Steel Aboveground Tanks for Flammable and Combustible Liquids**, 9th ed., covers steel atmospheric tanks intended for aboveground storage of noncorrosive, stable, flammable, and combustible liquids that have a specific gravity not exceeding that of water. UL 142 applies to single-wall and double-wall horizontal carbon-steel and stainless steel tanks up to 190 m<sup>3</sup> (50,000 gal), with a maximum diameter of 3.66 m (12 ft) and a maximum length-to-diameter ratio of 8 : 1. A formula from *Roark's Formulas for Stress and Strain* has been incorporated within UL 58 to calculate minimum steel wall thicknesses. UL 142 also applies to vertical tanks up to 10.7 m (35 ft) in height. UL 142 has been the primary AST standard since its development in 1922.

Tanks covered by these requirements can be fabricated in cylindrical or rectangular configurations. The standard covers secondary contained tanks, either of dual-wall construction or a tank in a steel dike or bund. It also provides listings for special AST constructions, such as those used under generators for backup power. These tanks are fabricated, inspected, and tested for leakage before shipment from the factory as completely assembled vessels.

UL 142 provides details for steel type, wall thickness, compartments and bulkheads, manways, and other fittings and appurtenances. Issues relating to leakage, venting, and the ability of the tank to with-

stand the development of internal pressures encountered during operation and production leak testing are also addressed.

These requirements do not apply to large field-erected storage tanks covered by the Standard for Welded Steel Tanks for Oil Storage, ANSI/API 650, or the Specification for Field-Welded Tanks for Storage of Production Liquids, API 12D; or to the Specification for Shop-Welded Tanks for Storage of Production Liquids, API 12F.

**UL 2085, Protected Aboveground Tanks for Flammable and Combustible Liquids**, covers shop-fabricated, aboveground atmospheric protected tanks intended for storage of stable, flammable or combustible liquids that have a specific gravity not greater than 1.0 and that are compatible with the material and construction of the tank.

The UL 2085 tank construction is intended to limit the heat transferred to the primary tank when the AST is exposed to a 2-h hydrocarbon pool fire of 1093°C (2000°F). The tank must be insulated to withstand the test without leakage and with an average maximum temperature rise on the primary tank not exceeding 127°C (260°F). Temperatures on the inside surface of the primary tank cannot exceed 204°C (400°F).

UL 2085 also provides criteria for resistance against vehicle impact, ballistic impact, and fire hose impact. These tanks are also provided with integral secondary containment intended to prevent any leakage from the primary tank from entering the environment.

**UL 2080, Fire Resistant Tanks for Flammable and Combustible Liquids**, is similar to UL 2085 tanks, except with an average maximum temperature rise on the primary tank limited to 427°C (800°F) during a 2-h pool fire. Temperatures on the inside surface of the primary tank cannot exceed 538°C (1000°F).

**UL 2244, Standard for Aboveground Flammable Liquid Tank Systems**, covers factory-fabricated, preengineered aboveground atmospheric tank systems intended for dispensing flammable or combustible liquids, such as gasoline or diesel fuel, into motor vehicles, generators, or aircraft.

**UL 80, Steel Tanks for Oil-Burner Fuel**, covers the design and construction of welded steel tanks of the atmospheric type with a maximum capacity of 0.23 to 2.5 m<sup>3</sup> (60 to 660 gal) intended for unenclosed installation inside of buildings or for outside aboveground applications as permitted by the Standard for Installation of Oil-Burning Equipment, NFPA 31, primarily for the storage and supply of fuel oil for oil burners.

**UL 2245, Below-Grade Vaults for Flammable Liquid Storage Tanks**, covers below-grade vaults intended for the storage of flammable or combustible liquids in an aboveground atmospheric tank. Below-grade vaults, constructed of a minimum of 150 mm (6 in) of reinforced concrete or other equivalent noncombustible material, are designed to contain one aboveground tank, which can be a compartment tank. Adjacent vaults may share a common wall. The lid of the vault may be at grade or below. Vaults provide a safe means to install hazardous tanks so that the system is accessible to the operator without unduly exposing the public.

**Southwest Research Institute (SwRI)** standards for steel ASTs storing flammable and combustible liquids include the following:

**SwRI 93-01, Testing Requirements for Protected Aboveground Flammable Liquid/Fuel Storage Tanks**, includes tests to evaluate the performance of ASTs under fire, hose stream, ballistics, heavy vehicular impact, and different environments. This standard requires pool-fire resistance similar to that of UL 2085.

**SwRI 97-04, Standard for Fire Resistant Tanks**, includes tests to evaluate the performance of ASTs under fire and hose stream. This standard is similar to UL 2080 in that the construction is exposed to a 2-h hydrocarbon pool fire of 1093°C (2000°F). However, SwRI 97-04 is concerned only with the integrity of the tank after the 2-h test and not concerned with the temperature inside the tank due to the heat transfer. As a result, UL 142 tanks have been tested to the SwRI standard and passed. Secondary containment with insulation is not necessarily an integral component of the system.

**Underwriters Laboratories of Canada (ULC)** publishes a number of standards for aboveground tanks and accessories. All the following pertain to the aboveground storage of flammable liquids and combustible liquids such as gasoline, fuel oil, or similar products with a relative density not greater than 1.0:

**ULC S601, Shop Fabricated Steel Aboveground Horizontal Tanks for Flammable and Combustible Liquids**, covers single- and double-wall cylindrical horizontal steel nonpressure tanks. These requirements do not cover tanks of capacities greater than 200 m<sup>3</sup>.

**ULC S630, Shop Fabricated Steel Aboveground Vertical Tanks for Flammable and Combustible Liquids**, covers single- and double-wall cylindrical vertical steel nonpressure tanks.

**ULC S643, Shop Fabricated Steel Utility Tanks for Flammable and Combustible Liquids**, covers single- and double-wall cylindrical horizontal steel nonpressure tanks.

**ULC S653-94, Aboveground Steel Contained Tank Assemblies for Flammable and Combustible Liquids**, covers steel contained tank assemblies.

**ULC S655, Aboveground Protected Tank Assemblies for Flammable and Combustible Liquids**, covers shop-fabricated primary tanks that provided with secondary containment and protective encasement and are intended for stationary installation and use in accordance with

1. National Fire Code of Canada, Part 4
2. CAN/CSA-B139, Installation Code for Oil Burning Equipment
3. The Environmental Code of Practice for Aboveground Storage Tank Systems Containing Petroleum Products
4. The requirements of the authority having jurisdiction

**ULC/ORD C142.20, Secondary Containment for Aboveground Flammable and Combustible Liquid Storage Tanks**, covers secondary containments for aboveground primary tanks.

**ULC S602, Aboveground Steel Tanks for the Storage of Combustible Liquids Intended to Be Used as Heating and/or Generator Fuels**, covers the design and construction of tanks of the nonpressure type, intended for installation inside or outside buildings. This standard covers single-wall tanks and tanks with secondary containment, having a maximum capacity of 2.5 m<sup>3</sup>.

The **Petroleum Equipment Institute (PEI)** has developed a recommended practice for AST system installation: **PEI-RP200, RP 200, Recommended Practices for Installation of Aboveground Storage Systems for Motor Vehicle Fueling**.

The **American Petroleum Institute (API)** has developed a series of standards and specifications involving ASTs:

**API 12F, Shop Welded Tanks for Storage of Production Liquids**  
**RP 12R1, Setting, Maintenance, Inspection, Operation, and Repair of Tanks in Production Service**

**RP 575, Inspection of Atmospheric and Low Pressure Storage Tanks**

**RP 579, Fitness-For-Service**

**API 650, Welded Steel Tanks for Oil Storage**

**API 652, Lining of Aboveground Storage Tank Bottoms**

**API 653, Tank Inspection, Repair, Alteration, and Reconstruction**

**API 2350, Overfill Protection for Storage Tanks in Petroleum Facilities** (overfill is the primary cause of AST releases)

The **American Society of Mechanical Engineers (ASME)** has developed a standard for welded aluminum-alloy storage tanks: **B96.1, Welded Aluminum-Alloy Storage Tanks** (1999), covers the design, materials, fabrication, erection, inspection, and testing requirements for welded aluminum alloy (field-erected or shop-fabricated) aboveground, vertical, cylindrical, flat bottom, open or closed-top tanks storing liquids under pressures approximating atmospheric pressure at ambient temperatures.

The **American Water Works Association (AWWA)** has many standards dealing with water handling and storage. A list of its publications is given in the *AWWA Handbook* (annually). *AWWA D100, Standard for Steel Tanks—Standpipes, Reservoirs, and Elevated Tanks for Water Storage*, contains rules for design and fabrication. Although AWWA tanks are intended for water, they could be used for the storage of other liquids.

The **Steel Tank Institute (STI)** publishes construction standards and recommended installation practices pertaining to ASTs fabricated to STI technologies. STI's recommended installation practices are notable for their applicability to similar respective technologies:

**SP001-06, Standard for Inspection of In-Service Shop Fabricated Aboveground Tanks for Storage of Combustible and Flammable Liquids**

**R912-00, Installation Instructions for Shop Fabricated Aboveground Storage Tanks for Flammable, Combustible Liquids**

**F921, Standard for Aboveground Tanks with Integral Secondary Containment**

**Standard for Fire Resistant Tanks (Flameshield)**

**Standard for Fireguard Thermally Insulated Aboveground Storage Tanks**

**F911, Standard for Diked Aboveground Storage Tanks**

The **National Association of Corrosion Engineers (NACE International)** has developed the following to protect the soil side of bottoms of on-grade carbon-steel storage tanks: **NACE RP0193-01, Standard Recommended Practice—External Cathodic Protection of On-Grade Metallic Storage Tank Bottoms**.

**Environmental Regulations** A key **United States Environmental Protection Agency (US EPA)** requirement for certain aboveground storage facilities is the development and submittal of Spill Prevention Control and Countermeasure (SPCC) Plans within 40 CFR 112, the Oil Pollution Prevention regulation, which in turn is part of the Clean Water Act (CWA). SPCC Plans and Facility Response Plans pertain to facilities which may discharge oil into groundwater or storm runoff, which in turn may flow into navigable waters. Enacted in 1973, these requirements were principally used by owners of large, field-fabricated aboveground tanks predominant at that time, although the regulation applied to all bulk containers larger than 2.5 m<sup>3</sup> (660 gal) and included the requirement for a Professional Engineer to certify the spill plan.

In July 2002, the US EPA issued a final rule amending 40 CFR 112 which included differentiation of shop-fabricated from field-fabricated ASTs. The rule also includes new subparts outlining the requirements for various classes of oil, revises the applicability of the regulation, amends the requirements for completing SPCC Plans, and makes other modifications.

The revised rule also states that all bulk storage container installations must provide a secondary means of containment for the entire capacity of the largest single container, with sufficient freeboard to contain precipitation, and that such containment is sufficiently impervious to contain discharged oil. US EPA encourages the use of industry standards to comply with the rules. Many owners of shop-fabricated tanks have opted for double-wall tanks built to STI or UL standards as a means to comply with this requirement.

**State and Local Jurisdictions** Due to the manner in which aboveground storage tank legislation was promulgated in 1972 for protection of surface waters from oil pollution, state environmental agencies did not receive similar jurisdiction as they did within the underground storage tank rules. Nonetheless, many state environmental agencies, state fire marshals, or Weights and Measures departments—including Minnesota, Florida, Wisconsin, Virginia, Oklahoma, Missouri, Maryland, Delaware, and Michigan—are presently regulating aboveground storage tanks through other means. Other regulations exist for hazardous chemicals and should be consulted for specific requirements.

**Aboveground Storage Tank Types and Options** Most hydrocarbon storage applications use carbon steel as the most economical and available material that provides suitable strength and compatibility for the specific storage application. For vertical tanks installed on grade, corrosion protection can be given to exterior tank bottoms in contact with soil. The interior of the tank can incorporate special coatings and linings (e.g., polymer, glass, or other metals). Some chemical storage applications will dictate that the storage tank be made from a stainless steel or nickel alloy. Fiberglass-reinforced plastic (FRP), polyethylene, or polypropylene may be used for nonflammable storage in smaller sizes. Suppliers can be contacted to verify the appropriate material to be used.

As stated earlier, shop-fabricated ASTs are often categorized according to the standards to which the tanks are fabricated, e.g., a UL 142 or UL 2085 tank. That said, however, there are defined categories such as diked tanks, protected tanks, fire-resistant tanks, and insulated tanks. It's critical, then, for the tank buyer or specifier to know precisely what is required or desired given the application, code requirements, and/or owner/operator preferences—and to discuss this with the tank contractor and/or manufacturer.



Cylindrical or rectangular tanks storing flammable and combustible liquids (UL 142 ASTs) will normally comply with UL 142. The Seventh Edition published in 1993 was particularly notable, as it incorporated secondary containment designs (diking or steel secondary containment tanks) and rectangular tank designs.

Rectangular tanks became a desirable option for small tanks, typically less than 7.6 m<sup>3</sup> (2000 gal), as operators liked the accessibility of the flat top to perform operations and maintenance without the need for special ladders or catwalks.

**Posttensioned concrete** is frequently used for tanks to about 57,000 m<sup>3</sup> (15 × 10<sup>6</sup> gal), usually containing water. Their design is treated in detail by Creasy (*Prestressed Concrete Cylindrical Tanks*, Wiley, New York, 1961). For the most economical design of large open tanks at ground levels, he recommends limiting vertical height to 6 m (20 ft). Seepage can be a problem if unlined concrete is used with some liquids (e.g., gasoline).

**Elevated tanks** can supply a large flow when required, but pump capacities need be only for average flow. Thus, they may save on pump and piping investment. They also provide flow after pump failure, an important consideration for fire systems.

**Open tanks** may be used to store materials that will not be harmed by water, weather, or atmospheric pollution. Otherwise, a roof, either fixed or floating, is required. **Fixed roofs** are usually either domed or coned. Large tanks have coned roofs with intermediate supports. Since negligible pressure is involved, snow and wind are the principal design loads. Local building codes often give required values.

Fixed-roof atmospheric tanks require **vents** to prevent pressure changes which would otherwise result from temperature changes and withdrawal or addition of liquid. API Standard 2000, Venting Atmospheric and Low Pressure Storage Tanks, gives practical rules for conservative vent design. The principles of this standard can be applied to fluids other than petroleum products. Excessive losses of volatile liquids, particularly those with flash points below 38°C (100°F), may result from the use of open vents on fixed-roof tanks. Sometimes vents are manifolded and led to a vent tank, or the vapor may be extracted by a recovery system.

An effective way of preventing vent loss is to use one of the many types of **variable-volume tanks**. These are built under API Standard 650. They may have floating roofs of the double-deck or the single-deck type. There are lifter-roof types in which the roof either has a skirt moving up and down in an annular liquid seal or is connected to the tank shell by a flexible membrane. A fabric expansion chamber housed in a compartment on top of the tank roof also permits variation in volume.

**Floating roofs** must have a seal between the roof and the tank shell. If not protected by a fixed roof, they must have drains for the removal of water, and the tank shell must have a "wind girder" to avoid distortion. An industry has developed to retrofit existing tanks with floating roofs. Much detail on the various types of tank roofs is given in manufacturers' literature. Figure 10-187 shows types. These roofs cause less condensation buildup and are highly recommended.

**Fire-Rated or Insulated ASTs: Protected and Fire-Resistant** These ASTs have received much attention within the fire regulatory community, particularly for motor fuel storage and dispensing appli-

cations and generator base tanks. National model codes have been revised to allow this type of storage aboveground.

An insulated tank can be a protected tank, built to third-party standards UL 2085 and/or SwRI 93-01, or a fire-resistant tank built to UL 2080 or SwRI 97-04. Protected tanks were developed in line with NFPA requirements and terminology, while fire-resistant ASTs were developed in line with Uniform Fire Code (now International Fire Code) requirements and terminology. Both protected tanks and fire-resistant tanks must pass a 1093°C (2000°F), 2-h fire test.

The insulation properties of many fire-rated ASTs marketed today are typically provided by concrete; i.e., the primary steel tank is surrounded by concrete. Due to the weight of concrete, this design is normally limited to small tanks. Another popular AST technology meeting all applicable code requirements for insulated tanks and fabricated to UL 2085 is a tank that utilizes a lightweight monolithic thermal insulation in between two walls of steel to minimize heat transfer from the outer tank to the inner tank and to make tank handling more palatable.

A **secondary containment AST** to prevent contamination of our environment has become a necessity for all hazardous liquid storage, regardless of its chemical nature, in order to minimize liability and protect neighboring property. A number of different regulations exist, but the regulations with the greatest impact are fire codes and the US EPA SPCC rules for oil storage.

In 1991, the Spill Prevention Control and Countermeasure (SPCC) rule proposed a revision to require secondary containment that was impermeable for at least 72 h after a release occurred. The 2003 promulgated EPA SPCC rule no longer mandates a 72-h containment requirement, instead opting to require means to contain releases until they can be detected and removed. Nonetheless, the need for impermeable containment continues to position steel as a material of choice for shop-fabricated tanks. However, release prevention barriers made from plastic or concrete can also meet US EPA requirements when frequently inspected for releases.

**Diked ASTs** Fire codes dictate flammable and combustible liquid tanks have spill control in the form of dike, remote impounding, or small secondary containment tanks. The dike must contain the content of the largest tank to prevent hazardous liquids from endangering the public and property. Traditional bulk storage systems will include multiple tanks within a concrete or earthen dike wall.

From a shop-fabricated tank manufacturer's perspective, a diked AST generally refers to a steel tank within a factory-fabricated steel box, or dike. An example of a diked AST is the STI F911 standard, providing an open-topped steel rectangular dike and floor as support and secondary containment of a UL 142 steel tank. The dike will contain 110 percent of the tank capacity; as rainwater may already have collected in the dike, the additional 10 percent acts as freeboard should a catastrophic failure dump a full tank's contents into a dike. Many fabricators offer steel dikes with rain shields to prevent precipitation from collecting.

A **double-wall AST** of steel fulfills the same function as a diked AST with rain shield. Double-wall designs consist of a steel wrap over a horizontal or vertical steel storage tank. The steel wrap provides an intimate, secondary containment over the primary tank. One such design is the Steel Tank Institute's F921 standard, based upon

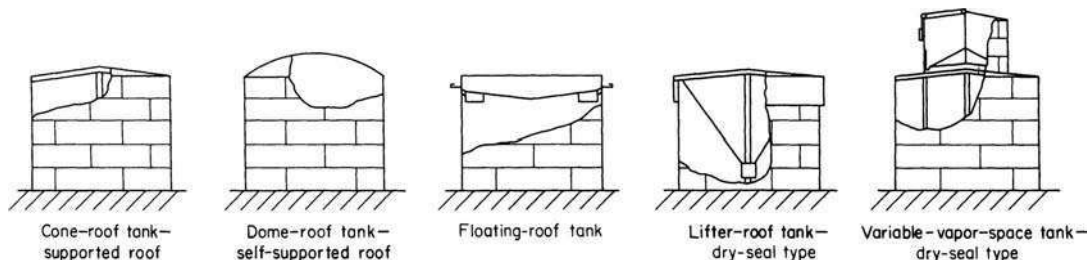


FIG. 10-187 Some types of atmospheric storage tanks.



UL 142-listed construction for the primary tank, outer containment, associated tank supports or skids.

**Venting** of ASTs is critical, since they are exposed to greater ambient temperature fluctuations than are USTs. Properly designed and sized venting, both normal (atmospheric) and emergency, is required. Normal vents permit the flow of air or inert gas into and out of the tank. The vent line must be large enough to accommodate the maximum filling or withdrawal rates without exceeding the allowable stress for the tank.

Fire codes' recommended installation procedures also detail specifics on pressure/vacuum venting devices and vent line flame arresters. For example, codes mandate different ventilation requirements for Class I-A liquids versus Class I-B or I-C liquids. Tank vent piping is generally not manifolded unless required for special purposes such as air pollution control or vapor recovery. As always, local codes must be consulted.

Emergency venting prevents an explosion during a fire or other emergency. All third-party laboratory standards except UL 80 include emergency relief provisions, since these tanks are designed for atmospheric pressure conditions.

**Separation distances** are also important. Aboveground storage tanks must be separated from buildings, property lines, fuel dispensers and delivery trucks in accordance with the level of safety the tank design provides, depending upon whether they are constructed of traditional steel or are vault/fire-resistant.

For most chemical storage tanks, codes such as NFPA 30 and the International Fire Code give specific separation distances. For motor vehicle fueling applications, the codes are more stringent on separation requirements due to a greater exposure of the public to the hazards. Hence codes such as NFPA 30A establish variable separation distances depending upon whether the facility is private or public.

Separation distance requirements may dictate whether a tank buyer purchases a traditional steel UL 142 tank, a fire-resistant tank, or a tank in a vault. For example, NFPA 30, NFPA 30A, and the IFC codes allow UL 2085 tanks to be installed closer to buildings and property lines, thereby reducing the real estate necessary to meet fire codes.

Under NFPA 30A, dispensers may be installed directly over vaults or upon fire-resistant tanks at fleet-type installations, whereas a 7.6- to 15.2-m (25- to 50-ft) separation distance is required at retail-type service station installations. The IFC only allows gasoline and diesel to be dispensed from ASTs, designed with a 2-h fire rating. Non-2-h fire-rated UL 142 tanks dispensing diesel can be installed upon approval of the local authority.

**Maintenance and Operations** Water in any storage system can cause myriad problems from product quality to system degradation due to corrosion from trace contaminants and microbial action. Subsequently, any operations and maintenance program must include a proactive program of monitoring for water—and removing it if found. Other operations and maintenance procedures would include periodic integrity testing and corrosion control for vertical tank bottoms. Additional guidance is available from organizations such as the American Petroleum Institute (API), Petroleum Equipment Institute (PEI), ASTM International, National Oilheat Research Alliance (NORA), and Steel Tank Institute (STI). Also see the Steel Tank Institute document *Keeping Water Out of Your Storage System* (<http://www.steel-tank.com/library/pubs/waterinfuel tanks.pdf>).

Integrity testing and visual inspection requirements are discussed in the SPCC requirements, Subpart B, Para. 112.8 c (6). Chemical tanks storing toluene and benzene, e.g., are subject to the rule in addition to traditional fuels. But the need for a good inspection program is recommended regardless of applicable regulations. Both visual inspection and another testing technique are required. Comparison records must be kept, and frequent inspections must be made of the outside of the tank and system components for signs of deterioration, discharges, or accumulation of oil inside diked areas.

For inspection of large field-erected tanks, API 653, *Tank Inspection, Repair, Alteration, and Reconstruction*, is referenced by the US EPA. A certified inspector must inspect tanks. US EPA references the Steel Tank Institute Standard SP001-06, *Standard for Inspection of In-Service Shop Fabricated Aboveground Tanks for Storage of Combustible and Flammable Liquids*, as an industry standard that may assist an owner or operator with the integrity testing and inspection of shop-fabricated tanks. The STI SP001-06 standard includes inspec-

tion techniques for all types of shop-fabricated tanks—horizontal cylindrical, vertical, and rectangular. SP001-06 also addresses tanks that rest directly on the ground or on release prevention barriers, tanks that are elevated on supports, and tanks that are either single- or double-wall using a risk-based approach.

**Pressure Tanks** Vertical cylindrical tanks constructed with domed or coned roofs, which operate at pressures above several hundred pascals (a few pounds per square foot) but which are still relatively close to atmospheric pressure, can be built according to API Standard 650. The pressure force acting against the roof is transmitted to the shell, which may have sufficient weight to resist it. If not, the uplift will act on the tank bottom. The strength of the bottom, however, is limited, and if it is not sufficient, an anchor ring or a heavy foundation must be used. In the larger sizes uplift forces limit this style of tank to very low pressures.

As the size or the pressure goes up, curvature on all surfaces becomes necessary. Tanks in this category, up to and including a pressure of 103.4 kPa (15 lbf/in<sup>2</sup>), can be built according to API Standard 620. Shapes used are spheres, ellipsoids, toroidal structures, and circular cylinders with torispherical, ellipsoidal, or hemispherical heads. The ASME Pressure Vessel Code (Sec. VIII of the ASME Boiler and Pressure Vessel Code), although not required below 103.4 kPa (15 lbf/in<sup>2</sup>), is also useful for designing such tanks.

Tanks that could be subjected to vacuum should be provided with vacuum-breaking valves or be designed for vacuum (external pressure). The ASME Pressure Vessel Code contains design procedures.

**Calculation of Tank Volume** A tank may be a single geometrical element, such as a cylinder, a sphere, or an ellipsoid. It may also have a compound form, such as a cylinder with hemispherical ends or a combination of a toroid and a sphere. To determine the volume, each geometrical element usually must be calculated separately. Calculations for a full tank are usually simple, but calculations for partially filled tanks may be complicated.

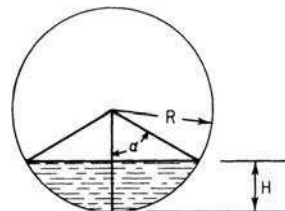
To calculate the volume of a **partially filled horizontal cylinder** refer to Fig. 10-188. Calculate the angle  $\alpha$  in degrees. Any units of length can be used, but they must be the same for  $H$ ,  $R$ , and  $L$ . The liquid volume

$$V = LR^2 \left( \frac{\alpha}{57.30} - \sin \alpha \cos \alpha \right) \quad (10-109)$$

This formula may be used for any depth of liquid between zero and the full tank, provided the algebraic signs are observed. If  $H$  is greater than  $R$ ,  $\sin \alpha \cos \alpha$  will be negative and thus will add numerically to  $\alpha/57.30$ . Table 10-59 gives liquid volume, for a partially filled horizontal cylinder, as a fraction of the total volume, for the dimensionless ratio  $H/D$  or  $H/2R$ .

The **volumes of heads** must be calculated separately and added to the volume of the cylindrical portion of the tank. The four types of heads most frequently used are the standard dished head,\* torispherical or ASME head, ellipsoidal head, and hemispherical head. Dimensions and volumes for all four of these types are given in *Lukens Spun*

\*The standard dished head does not comply with the ASME Pressure Vessel Code.



**FIG. 10-188** Calculation of partially filled horizontal tanks.  $H$  = depth of liquid;  $R$  = radius;  $D$  = diameter;  $L$  = length;  $\alpha$  = half of the included angle; and  $\cos \alpha = 1 - H/R = 1 - 2H/D$ .

**TABLE 10-59 Volume of Partially Filled Horizontal Cylinders**

| H/D  | Fraction of volume | H/D  | Fraction of volume | H/D  | Fraction of volume | H/D  | Fraction of volume |
|------|--------------------|------|--------------------|------|--------------------|------|--------------------|
| 0.01 | 0.00169            | 0.26 | 0.20660            | 0.51 | 0.51273            | 0.76 | 0.81545            |
| .02  | .00477             | .27  | .21784             | .52  | .52546             | .77  | .82625             |
| .03  | .00874             | .28  | .22921             | .53  | .53818             | .78  | .83688             |
| .04  | .01342             | .29  | .24070             | .54  | .55088             | .79  | .84734             |
| .05  | .01869             | .30  | .25231             | .55  | .56356             | .80  | .85762             |
| .06  | .02450             | .31  | .26348             | .56  | .57621             | .81  | .86771             |
| .07  | .03077             | .32  | .27587             | .57  | .58884             | .82  | .87760             |
| .08  | .03748             | .33  | .28779             | .58  | .60142             | .83  | .88727             |
| .09  | .04458             | .34  | .29981             | .59  | .61397             | .84  | .89673             |
| .10  | .05204             | .35  | .31192             | .60  | .62647             | .85  | .90594             |
| .11  | .05985             | .36  | .32410             | .61  | .63892             | .86  | .91491             |
| .12  | .06797             | .37  | .33636             | .62  | .65131             | .87  | .92361             |
| .13  | .07639             | .38  | .34869             | .63  | .66364             | .88  | .93203             |
| .14  | .08509             | .39  | .36108             | .64  | .67590             | .89  | .94015             |
| .15  | .09406             | .40  | .37353             | .65  | .68808             | .90  | .94796             |
| .16  | .10327             | .41  | .38603             | .66  | .70019             | .91  | .95542             |
| .17  | .11273             | .42  | .39858             | .67  | .71221             | .92  | .96252             |
| .18  | .12240             | .43  | .41116             | .68  | .72413             | .93  | .96923             |
| .19  | .13229             | .44  | .42379             | .69  | .73652             | .94  | .97550             |
| .20  | .14238             | .45  | .43644             | .70  | .74769             | .95  | .98131             |
| .21  | .15266             | .46  | .44912             | .71  | .75930             | .96  | .98658             |
| .22  | .16312             | .47  | .46182             | .72  | .77079             | .97  | .99126             |
| .23  | .17375             | .48  | .47454             | .73  | .78216             | .98  | .99523             |
| .24  | .18455             | .49  | .48727             | .74  | .79340             | .99  | .99831             |
| .25  | .19550             | .50  | .50000             | .75  | .80450             | 1.00 | 1.00000            |

*Heads*, Lukens Inc., Coatesville, Pennsylvania. Approximate volumes can also be calculated by the formulas in Table 10-60. Consistent units must be used in these formulas.

A partially filled horizontal tank requires the determination of the partial volume of the heads. The Lukens catalog gives approximate volumes for partially filled (axis horizontal) standard ASME and ellipsoidal heads. A formula for **partially filled heads** (excluding conical), by Doolittle [*Ind. Eng. Chem.*, **21**, 322-323 (1928)], is

$$V = 0.215H^2(3R - H) \quad (10-110)$$

where in consistent units  $V$  = volume,  $R$  = radius, and  $H$  = depth of liquid. Doolittle made some simplifying assumptions which affect the

volume given by the equation, but the equation is satisfactory for determining the volume as a fraction of the entire head. This fraction, calculated by Doolittle's formula, is given in Table 10-61 as a function of  $H/D_i$  ( $H$  is the depth of liquid, and  $D_i$  is the inside diameter). Table 10-61 can be used for standard dished, torispherical, ellipsoidal, and hemispherical heads with an error of less than 2 percent of the volume of the entire head. The error is zero when  $H/D_i = 0, 0.5$ , and  $1.0$ . Table 10-61 cannot be used for conical heads.

When a tank volume cannot be calculated or when greater precision is required, **calibration** may be necessary. This is done by draining (or filling) the tank and measuring the volume of liquid. The measurement may be made by weighing, by a calibrated fluid meter, or by repeatedly filling small measuring tanks which have been calibrated by weight.

**Container Materials and Safety** Storage tanks are made of almost any structural material. Steel and reinforced concrete are most widely used. Plastics and glass-reinforced plastics are used for tanks up to about 230 m<sup>3</sup> (60,000 gal). Resistance to corrosion, light weight, and lower cost are their advantages. Plastic and glass coatings are also applied to steel tanks. Aluminum and other nonferrous metals are used when their special properties are required. When expensive metals such as tantalum are required, they may be applied as tank linings or as clad metals.

Some grades of steel listed by API and AWWA Standards are of lower quality than is customarily used for pressure vessels. The stresses allowed by these standards are also higher than those allowed by the ASME Pressure Vessel Code. Small tanks containing nontoxic substances are not particularly hazardous and can tolerate a reduced factor of safety. Tanks containing highly toxic substances and very large tanks containing any substance can be hazardous. The designer must consider the magnitude of the hazard. The possibility of brittle behavior of ferrous metal should be taken into account in specifying materials (see subsection "Safety in Design").

Volume 1 of National Fire Codes (National Fire Protection Association, Quincy, Massachusetts) contains recommendations (Code 30) for venting, drainage, and dike construction of tanks for **flammable liquids**.

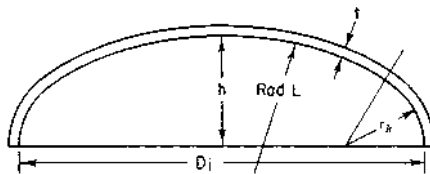
**Container Insulation** Tanks containing materials above atmospheric temperature may require insulation to reduce loss of heat. Almost any of the commonly used insulating materials can be employed. Calcium silicate, glass fiber, mineral wool, cellular glass, and plastic foams are among those used. Tanks exposed to weather must have jackets or protective coatings, usually asphalt, to keep water out of the insulation.

Tanks with contents at lower than atmospheric temperature may require insulation to minimize heat absorption. The insulation must have

**TABLE 10-60 Volumes of Heads\***

| Type of head              | Knuckle radius, $r_k$ | $h$     | $L$           | Volume                            | % Error   | Remarks                                      |
|---------------------------|-----------------------|---------|---------------|-----------------------------------|-----------|----------------------------------------------|
| Standard dished           | Approx. $3t$          |         | Approx. $D_i$ | Approx. $0.050D_i^3 + 1.65tD_i^2$ | $\pm 10$  | $h$ varies with $t$                          |
| Torispherical or A.S.M.E. | $0.06L$               |         | $D_i$         | $0.0809D_i^3$                     | $\pm 0.1$ | $r_k$ must be the larger of $0.06L$ and $3t$ |
| Torispherical or A.S.M.E. | $3t$                  |         | $D_i$         | Approx. $0.513hD_i^2$             | $\pm 8$   |                                              |
| Ellipsoidal               |                       | $D_i/4$ | $D_i/2$       | $\pi D_i^2 h/6$                   | 0         | Standard proportions                         |
| Ellipsoidal               |                       | $D_i/2$ |               | $\pi D_i^3/24$                    | 0         |                                              |
| Hemispherical             |                       | $D_i/2$ |               | $\pi D_i^3/12$                    | 0         |                                              |
| Conical                   |                       |         |               | $\pi h(D_i^2 + D_i d + d^2)/12$   | 0         |                                              |

\*Use consistent units.



**TABLE 10-61 Volume of Partially Filled Heads on Horizontal Tanks\***

| H/D <sub>i</sub> | Fraction of volume | H/D <sub>i</sub> | Fraction of volume | H/D <sub>i</sub> | Fraction of volume | H/D <sub>i</sub> | Fraction of volume |
|------------------|--------------------|------------------|--------------------|------------------|--------------------|------------------|--------------------|
| 0.02             | 0.0012             | 0.28             | 0.1913             | 0.52             | 0.530              | 0.78             | 0.8761             |
| .04              | .0047              | .30              | .216               | .54              | .560               | .80              | .8960              |
| .06              | .0104              | .32              | .242               | .56              | .590               | .82              | .9145              |
| .08              | .0182              | .34              | .268               | .58              | .619               | .84              | .9314              |
| .10              | .0280              | .36              | .295               | .60              | .648               | .86              | .9467              |
| .12              | .0397              | .38              | .323               | .62              | .677               | .88              | .9603              |
| .14              | .0533              | .40              | .352               | .64              | .705               | .90              | .9720              |
| .16              | .0686              | .42              | .381               | .66              | .732               | .92              | .9818              |
| .18              | .0855              | .44              | .410               | .68              | .758               | .94              | .9896              |
| .20              | .1040              | .46              | .440               | .70              | .784               | .96              | .9953              |
| .22              | .1239              | .48              | .470               | .72              | .8087              | .98              | .9988              |
| .24              | .1451              | .50              | .500               | .74              | .8324              | 1.00             | 1.0000             |
| .26              | .1676              |                  |                    | .76              | .8549              |                  |                    |

\*Based on Eq. (10-110).

a vapor barrier at the outside to prevent condensation of atmospheric moisture from reducing its effectiveness. An insulation not damaged by moisture is preferable. The insulation techniques presently used for refrigerated systems can be applied (see subsection "Low-Temperature and Cryogenic Storage").

**Tank Supports** Large vertical atmospheric steel tanks may be built on a base of about 150 cm (6 in) of sand, gravel, or crushed stone if the subsoil has adequate bearing strength. It can be level or slightly coned, depending on the shape of the tank bottom. The porous base provides drainage in case of leaks. A few feet beyond the tank perimeter the surface should drop about 1 m (3 ft) to assure proper drainage of the subsoil. API Standard 650, Appendix B, and API Standard 620, Appendix C, give recommendations for tank foundations.

The bearing pressure of the tank and contents must not exceed the **bearing strength** of the soil. Local building codes usually specify allowable soil loading. Some approximate bearing values are:

|                                             | kPa       | Tons/ft <sup>2</sup> |
|---------------------------------------------|-----------|----------------------|
| Soft clay (can be crumbled between fingers) | 100       | 1                    |
| Dry fine sand                               | 200       | 2                    |
| Dry fine sand with clay                     | 300       | 3                    |
| Coarse sand                                 | 300       | 3                    |
| Dry hard clay (requires a pick to dig it)   | 350       | 3.5                  |
| Gravel                                      | 400       | 4                    |
| Rock                                        | 1000–4000 | 10–40                |

For high, heavy tanks, a foundation ring may be needed. Prestressed concrete tanks are sufficiently heavy to require foundation rings. Foundations must extend below the frost line. Some tanks that are not flat-bottomed may also be supported by soil if it is suitably graded and drained. When soil does not have adequate bearing strength, it may be excavated and backfilled with a suitable soil, or piles capped with a concrete mat may be required.

Spheres, spheroids, and toroids use steel or concrete saddles or are supported by columns. Some may rest directly on soil. Horizontal cylindrical tanks should have two rather than multiple saddles to avoid indeterminate load distribution. Small horizontal tanks are sometimes supported by legs. Most tanks must be designed to resist the reactions of the saddles or legs, and they may require reinforcing. Neglect of this can cause collapse. Tanks without stiffeners usually need to make contact with the saddles on at least 2.1 rad (120°) of their circumference. An elevated steel tank may have either a circle of steel columns or a large central steel standpipe. Concrete tanks usually have concrete columns. Tanks are often supported by buildings.

**Pond and Underground Storage** Low-cost liquid materials, if they will not be damaged by rain or atmospheric pollution, may be stored in **ponds**. A pond may be excavated or formed by damming a ravine. To prevent loss by seepage, the soil which will be submerged may require

treatment to make it sufficiently impervious. This can also be accomplished by lining the pond with concrete, plastic film, or some other barrier. Prevention of seepage is especially necessary if the pond contains material that could contaminate present or future water supplies.

**Underground Cavern Storage** Large volumes of liquids are often stored below ground in artificial caverns as an economical alternative to aboveground tanks and other modes of storage. The liquid to be stored must tolerate water, brine, and other contaminants that are usually present to some degree in the cavern. The liquids that are most commonly stored are natural gas liquids (NGLs), LPGs, crude oil, and refined petroleum products. If the liquid is suitable for cavern storage, this method may be less expensive, safer, and more secure than other storage modes.

There are two types of caverns used for storing liquids. *Hard rock (mined)* caverns are constructed by mining rock formations such as shale, granite, limestone, and many other types of rock. *Solution-mined* caverns are constructed by dissolution processes, i.e., solution mining or leaching a mineral deposit, most often salt (sodium chloride). The salt deposit may take the form of a massive salt dome or thinner layers of bedded salt that are stratified between layers of rock. Hard rock and solution-mined caverns have been constructed in the United States and many other parts of the world.

**Mined Caverns** Caverns mined in hard rock are generally situated 100 to 150 m (325 to 500 ft) below ground level. These caverns are constructed by excavating rock with conventional drill-and-blast mining methods. The excavated cavern consists of a grouping of interconnecting tunnels or storage "galleries." Mined caverns have been constructed for volumes ranging from as little as 3200 to 800,000 m<sup>3</sup> [20,000 API barrels\* (bbl) to 5 million bbl]. Figure 10-189 illustrates a typical mined cavern for liquid storage.

Hard rock caverns are designed so that the internal storage pressure is at all times less than the hydrostatic head of the water contained in the surrounding rock matrix. Thus, the depth of a cavern determines its maximum allowable operating pressure. Groundwater that continuously seeps into hard rock caverns in permeable formations is periodically pumped out of the cavern. The maximum operating pressure of the cavern is established after a thorough geological and hydrogeological evaluation is made of the rock formation and the completed cavern is pressure-tested.

**Salt Caverns** Salt caverns are constructed in both *domal salt*, more commonly referred to as "salt domes," and *bedded salt*, which consists of a body of salt sandwiched between layers of rock. The greatest total volume of underground liquid storage in the United States is stored in salt dome caverns. A salt dome is a large body, mostly consisting of sodium chloride, which over geologic time moved upward thousands of feet from extensive halite deposits deep below the earth's crust. There are numerous salt domes in the United States and other parts of the world [see Harben, P. W., and R. L. Bates, "Industrial Minerals Geology and World Deposits," *Metal Bulletin Plc*, UK, pp. 229–234 (1990)]. An individual salt dome may exceed 1 mi in diameter and contain many storage caverns. The depth to the top of a salt dome may range from a few hundred to several thousand feet, although depths to about 1070 m (3500 ft) are commercially viable for cavern development. The extent of many salt domes allows for caverns of many different sizes and depths to be developed. The extensive nature of salt domes has allowed the development of caverns as large as 5.7 × 10<sup>6</sup> m<sup>3</sup> (36 million bbl) (US DOE Bryan Mound Strategic Petroleum Reserve) and larger; however, cavern volumes of 159,000 to 1.59 × 10<sup>6</sup> m<sup>3</sup> (1 to 10 million bbl) are more common for liquid storage.

The benefits of salt are its high compressive strength of 13.8 to 27.6 MPa (2000 to 4000 psi), its impermeability to hydrocarbon liquids and gases, and its non-chemically reactive (inert) nature. Due to the impervious nature of salt, the maximum allowed storage pressure gradient in this type of cavern is greater than that of a mined cavern. A typical storage pressure gradient for liquids is about 18 kPa/m of depth (0.80 psi/ft) to the bottom of the well casing. Actual maximum and minimum allowable operating pressure gradients are determined from geologic evaluations and rock mechanics studies. Typical depths to the top of a salt cavern may range from 500 to 4000 ft (about 150 to 1200 m).

\*One API barrel = 42 US gal = 5.615 ft<sup>3</sup> = 0.159 m<sup>3</sup>.

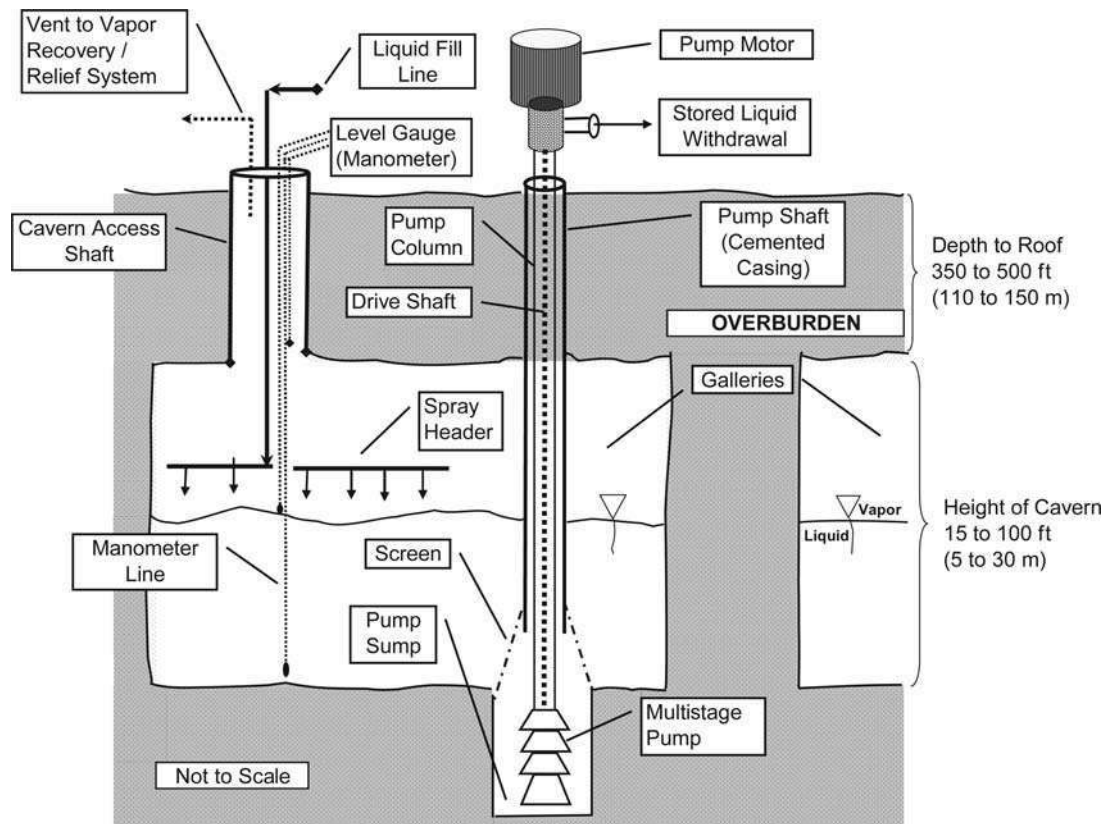


FIG. 10-189 Mined cavern.

Therefore, the maximum storage pressure (2760 to 32,060 kPa, or 400 to 3200 psig) usually exceeds the vapor pressure of all commonly stored hydrocarbon liquids. Higher-vapor-pressure products such as ethylene or ethane cannot be stored in relatively shallow caverns.

Salt caverns are developed by solution mining, a process (leaching) in which water is injected to dissolve the salt. Approximately 7 to 10 units of fresh water are required to leach 1 unit of cavern volume. Figure 10-190 illustrates the leaching process for two caverns. Modern salt dome caverns are shaped as relatively tall, slender cylinders. The leaching process produces nearly saturated brine from the cavern. Brine may be disposed into nearby disposal wells or offshore disposal fields, or it may be supplied to nearby plants as a feedstock for manufacturing of caustic (NaOH) and chlorine (Cl<sub>2</sub>). The final portion of the produced brine is retained and stored in artificial surface ponds or tanks to be used to displace the stored liquid from the cavern.

Salt caverns are usually developed using a single well, although some employ two or more wells. The well consists of a series of concentric casings that protect the water table and layers of rock and sediments (overburden) that lie above the salt dome. The innermost well casing is referred to as the last cemented or well "production" casing and is cemented in place, sealing the cavern and protecting the surrounding geology. Once the last cemented casing is in place, a borehole is drilled from the bottom of the well, through the salt to the design cavern depth. For single-well leaching, two concentric tubing strings are then suspended in the well. A liquid, such as diesel, or a gas is then injected through the outer annular space and into the top of the cavern to act as a "blanket" to prevent undesired leaching of the top of the cavern. Water is then injected into one of the suspended tubing strings, and brine is withdrawn from the other. During the leaching process, the flow path for the injected water is alternated between the innermost

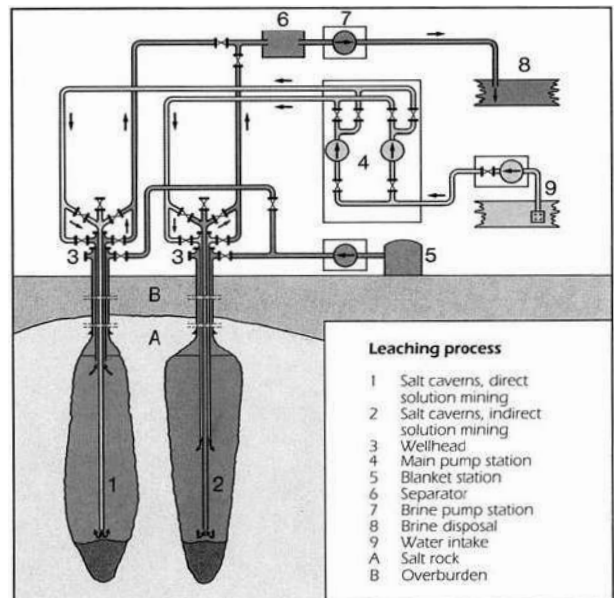


FIG. 10-190 Cavern leaching process.

tubing and the inner annulus, and these strings are periodically raised upward to control the cavern shape. A typical salt dome cavern may require 18 to 30 months of leaching time, whereas smaller, bedded salt caverns may be developed in a shorter time frame.

**Brine-Compensated Storage** As the stored product is pumped into the cavern, brine is displaced into an aboveground brine storage reservoir. To withdraw the product from the cavern, brine is pumped back into the cavern, displacing the stored liquid. This method of product transfer is termed *brine-compensated*, and caverns that operate in this fashion remain liquid-filled at all times. Figure 10-191 illustrates brine-compensated storage operations.

**Uncompensated Storage** Hard rock caverns and a few bedded salt caverns do not use brine for product displacement. This type of storage operation is referred to as pumpout or *uncompensated* storage operations. When the cavern is partially empty of liquid, the void space is filled with the vapor that is in equilibrium with the stored liquid. When liquid is introduced into the cavern, it compresses and condenses this saturated vapor phase. In some cases, vapor may be vented to the surface where it may be refrigerated and recycled to the cavern.

Submersible pumps or vertical line shaft pumps are used for withdrawing the stored liquid. Vertical line shaft pumps are suited for depths of no more than several hundred feet. Figure 10-189 illustrates an example of uncompensated storage operations.

Water is also stored underground when suitable formations are available. When an excess of surface water is available part of the time, the excess is treated, if required, and pumped into the ground to be retrieved when needed. Sometimes pumping is unnecessary, and it will seep into the ground.

Underground chambers are also constructed in frozen earth (see subsection "Low-Temperature and Cryogenic Storage"). Underground tunnel or tank storage is often the most practical way of storing hazardous or radioactive materials, such as proposed at Yucca Mountain, Nevada. A cover of 30 m (100 ft) of rock or dense earth can exert a pressure of about 690 kPa (100 lbf/in<sup>2</sup>).

## STORAGE OF GASES

**Gas Holders** Gas is sometimes stored in expandable gas holders of either the liquid-seal or dry-seal type. The liquid-seal holder is a familiar sight. It has a cylindrical container, closed at the top, and

varies its volume by moving it up and down in an annular water-filled seal tank. The seal tank may be staged in several lifts (as many as five). Seal tanks have been built in sizes up to 280,000 m<sup>3</sup> (10 × 10<sup>6</sup> ft<sup>3</sup>). The dry-seal holder has a rigid top attached to the sidewalls by a flexible fabric diaphragm which permits it to move up and down. It does not involve the weight and foundation costs of the liquid-seal holder. Additional information on gas holders can be found in *Gas Engineers Handbook*, Industrial Press, New York, 1966.

**Solution of Gases in Liquids** Certain gases will dissolve readily in liquids. In some cases in which the quantities are not large, this may be a practical storage procedure. Examples of gases that can be handled in this way are ammonia in water, acetylene in acetone, and hydrogen chloride in water. Whether or not this method is used depends mainly on whether the end use requires the anhydrous or the liquid state. Pressure may be either atmospheric or elevated. The solution of acetylene in acetone is also a safety feature because of the instability of acetylene.

**Storage in Pressure Vessels, Bottles, and Pipe Lines** The distinction between pressure vessels, bottles, and pipes is arbitrary. They can all be used for storing gases under pressure. A storage pressure vessel is usually a permanent installation. Storing a gas under pressure not only reduces its volume but also in many cases liquefies it at ambient temperature. Some gases in this category are carbon dioxide, several petroleum gases, chlorine, ammonia, sulfur dioxide, and some types of Freon. Pressure tanks are frequently installed underground.

Liquefied petroleum gas (LPG) is the subject of API Standard 2510, The Design and Construction of Liquefied Petroleum Gas Installations at Marine and Pipeline Terminals, Natural Gas Processing Plants, Refineries, and Tank Farms. This standard in turn refers to:

1. National Fire Protection Association (NFPA) Standard 58, Standard for the Storage and Handling of Liquefied Petroleum Gases
2. NFPA Standard 59, Standard for the Storage and Handling of Liquefied Petroleum Gases at Utility Gas Plants
3. NFPA Standard 59A, Standard for the Production, Storage, and Handling of Liquefied Natural Gas (LNG)

The API Standard gives considerable information on the construction and safety features of such installations. It also recommends minimum distances from property lines. The user may wish to obtain added safety by increasing these distances.

The term **bottle** is usually applied to a pressure vessel that is small enough to be conveniently portable. Bottles range from about 57 L (2 ft<sup>3</sup>) down to CO<sub>2</sub> capsules of about 16.4 mL (1 in<sup>3</sup>). Bottles are convenient for small quantities of many gases, including air, hydrogen, nitrogen, oxygen, argon, acetylene, Freon, and petroleum gas. Some are one-time-use disposable containers.

**Pipe Lines** A pipe line is not ordinarily a storage device. Pipes, however, have been buried in a series of connected parallel lines and used for storage. This avoids the necessity of providing foundations, and the earth protects the pipe from extremes of temperature. The economics of such an installation would be doubtful if it were designed to the same stresses as a pressure vessel. Storage is also obtained by increasing the pressure in operating pipe lines and thus using the pipe volume as a tank.

**Low-Temperature and Cryogenic Storage** This type is used for gases that liquefy under pressure at atmospheric temperature. In cryogenic storage the gas is at, or near to, atmospheric pressure and remains liquid because of low temperature. A system may also operate with a combination of pressure and reduced temperature. The term "cryogenic" usually refers to temperatures below -101°C (-150°F). Some gases, however, liquefy between -101°C and ambient temperatures. The principle is the same, but cryogenic temperatures create different problems with insulation and construction materials.

The liquefied gas must be maintained at or below its boiling point. Refrigeration can be used, but the usual practice is to cool by evaporation. The quantity of liquid evaporated is minimized by insulation. The vapor may be vented to the atmosphere (wasteful), it may be compressed and reliquefied, or it may be used.

At very low temperatures with liquid air and similar substances, the tank may have double walls with the interspace evacuated. The well-known Dewar flask is an example. Large tanks and even pipe

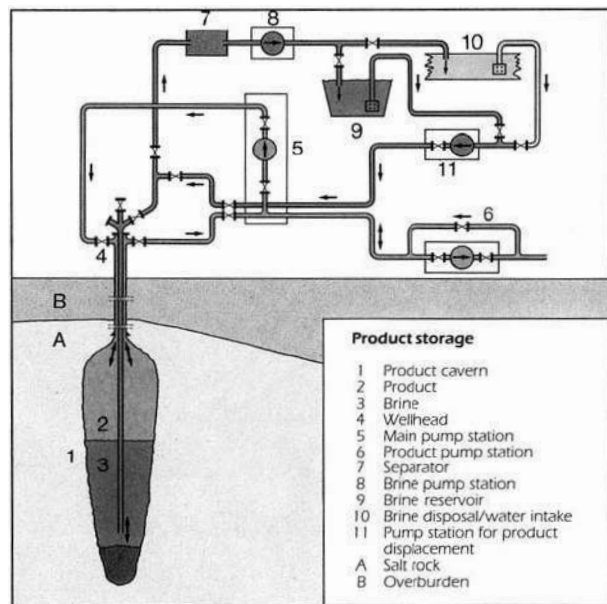


FIG. 10-191 Brine-compensated storage.

lines are now built this way. An alternative is to use double walls without vacuum but with an insulating material in the interspace. Perlite and plastic foams are two insulating materials employed in this way. Sometimes both insulation and vacuum are used.

**Materials** Materials for liquefied-gas containers must be suitable for the temperatures, and they must not be brittle. Some carbon steels can be used down to  $-59^{\circ}\text{C}$  ( $-75^{\circ}\text{F}$ ), and low-alloy steels to  $-101^{\circ}\text{C}$  ( $-150^{\circ}\text{F}$ ) and sometimes  $-129^{\circ}\text{C}$  ( $-200^{\circ}\text{F}$ ). Below these temperatures austenitic stainless steel (AISI 300 series) and aluminum are the principal materials. (See discussion of brittle fracture on p.10-160.)

Low temperatures involve problems of **differential thermal expansion**. With the outer wall at ambient temperature and the inner wall at the liquid boiling point, relative movement must be accommodated. Some systems for accomplishing this are patented. The Gaz Transport of France reduces dimensional change by using a thin inner liner of Invar. Another patented French system accommodates this change by means of the flexibility of thin metal which is creased. The creases run in two directions, and the form of the crossings of the creases is a feature of the system.

Low-temperature tanks may be installed underground to take advantage of the insulating value of the earth. Frozen-earth storage is also used. The frozen earth forms the tank. Some installations using this technique have been unsuccessful because of excessive heat absorption.

**Cavern Storage** Gases are also stored below ground in salt caverns. The most common type of gas stored in caverns is natural gas, although hydrogen and air have also been stored. Hydrogen storage requires special consideration in selecting metallurgy for the wellhead and the tubular goods. Air is stored for the purpose of providing compressed air energy for peak shaving power plants. Two such plants are in operation, one in the United States (Alabama), the other in Germany. A discussion of the Alabama plant is presented in *History of First U.S. Compressed Air Energy Storage (CAES) Plant (110-MW-26 h)*, vol. 1, *Early CAES Development*, Electric Power Research Institute (EPRI), Palo Alto, Calif. (1992).

Since salt caverns contain brine and other contaminants, the type of gas to be stored should not be sensitive to the presence of contaminants. If the gas is determined suitable for cavern storage, then cavern storage may not offer only economic benefits and enhanced safety and security; salt caverns also offer relatively high rates of deliverability compared to reservoir and aquifer storage fields. Solution-mined gas storage caverns in salt formations operate as *uncompensated* storage—no fluid is injected into the well to displace the compressed gas.

Surface gas handling facilities for storage caverns typically include receipt and delivery measurement stations, gas compressors, and gas processing equipment. When compressors are required for cavern injection and/or withdrawal, banks of positive-displacement-type compressors are used, since this compressor type is well suited for handling the highly variable compression ratios and flow rates associated with cavern injection and withdrawal operations. Cavern withdrawal operations typically involve single or multiple pressure reduction stations and full or partial gas dehydration. Large pressure throttling requirements often require heating the gas upon withdrawal, and injection of methanol to help control hydrate formation is also often necessary.

An in-depth discussion on natural gas storage in underground caverns may be found in *Gas Engineering and Operating Practices, Supply*, Book S-1, Part 1, *Underground Storage of Natural Gas*, and Part 2, Chapter 2, "Leached Caverns," American Gas Association, Arlington, Va. (1990).

**Additional References** API Recommended Practice 1114, *Design of Solution-Mined Underground Storage Facilities*, 1st ed., Washington, June, 1994. API 1115, *Operation of Solution-Mined Underground Storage Facilities*, 1st ed., Washington, September 1994. LeFond, Stanley J., *Handbook of World Salt Resources, Monographs in Geoscience*, Department of Geology, Columbia University, New York, 1969. SME *Mining Engineering Handbook*, 2d ed., vol. 2, 1992.

## COST OF STORAGE FACILITIES

Contractors' bids offer the most reliable information on cost. Order-of-magnitude costs, however, may be required for preliminary studies. One way of estimating them is to obtain cost information from similar

facilities and scale it to the proposed installation. Costs of steel storage tanks and vessels have been found to vary approximately as the 0.6 to 0.7 power of their weight [see Happel, *Chemical Process Economics*, Wiley, 1958, p. 267; also Williams, *Chem. Eng.*, **54**(12), 124 (1947)]. All estimates based on the costs of existing equipment must be corrected for changes in the price index from the date when the equipment was built. Considerable uncertainty is involved in adjusting data more than a few years old.

Based on a survey in 1994 for storage tanks, the prices for field-erected tanks are for multiple-tank installations erected by the contractor on foundations provided by the owner. Some cost information on tanks is given in various references cited in Sec. 9. Cost data vary considerably from one reference to another.

Prestressed (posttensioned) concrete tanks cost about 20 percent more than steel tanks of the same capacity. Once installed, however, concrete tanks require very little maintenance. A true comparison with steel would, therefore, require evaluating the maintenance cost of both types.

## BULK TRANSPORT OF FLUIDS

Transportation is often an important part of product cost. Bulk transportation may provide significant savings. When there is a choice between two or more forms of transportation, the competition may result in rate reduction. Transportation is subject to considerable regulation, which will be discussed in some detail under specific headings.

**Pipe Lines** For quantities of fluid which an economic investigation indicates are sufficiently large and continuous to justify the investment, pipe lines are one of the lowest-cost means of transportation. They have been built up to 1.22 m (48 in) or more in diameter and about 3200 km (2000 mi) in length for oil, gas, and other products. Water is usually not transported more than 160 to 320 km (100 to 200 miles), but the conduits may be much greater than 1.22 m (48 in) in diameter. Open canals are also used for water transportation.

Petroleum pipe lines before 1969 were built to ASA (now ANSI) Standard B31.4 for liquids and Standard B31.8 for gas. These standards were seldom mandatory because few states adopted them. The U.S. Department of Transportation (DOT), which now has responsibility for pipe-line regulation, issued Title 49, Part 192—Transportation of Natural Gas and Other Gas by Pipeline: Minimum Safety Standards, and Part 195—Transportation of Liquids by Pipeline. These contain considerable material from B31.4 and B31.8. They allow generally higher stresses than the ASME Pressure Vessel Code would allow for steels of comparable strength. The enforcement of their regulations is presently left to the states and is therefore somewhat uncertain.

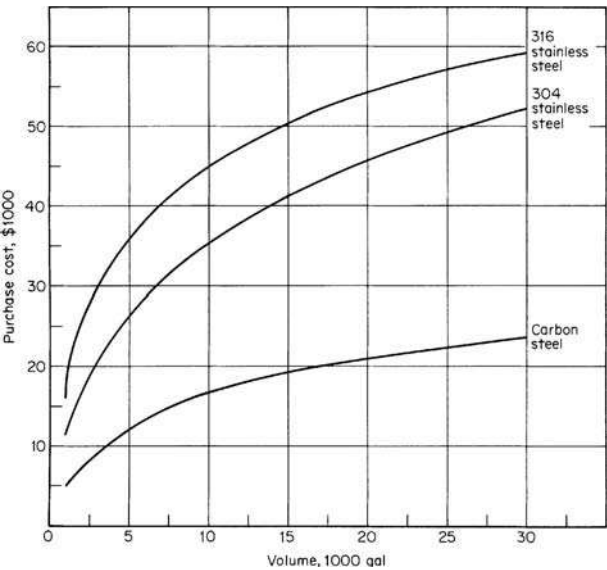
Pipe-line pumping stations usually range from 16 to 160 km (10 to 100 miles) apart, with maximum pressures up to 6900 kPa (1000 lbf/in<sup>2</sup>) and velocities up to 3 m/s (10 ft/s) for liquid. Gas pipe lines have higher velocities and may have greater spacing of stations.

**Tanks** Tank cars (single and multiple tank), tank trucks, portable tanks, drums, barrels, carboys, and cans are used to transport fluids (see Figs. 10-192 to 10-194). Interstate transportation is regulated by the DOT. There are other regulating agencies—state, local, and private. Railroads make rules determining what they will accept, some states require compliance with DOT specifications on intrastate movements, and tunnel authorities as well as fire chiefs apply restrictions. Water shipments involve regulations of the U.S. Coast Guard. The American Bureau of Shipping sets rules for design and construction which are recognized by insurance underwriters.

The most pertinent **DOT regulations** (*Code of Federal Regulations*, Title 49, Parts 171–179 and 397) were published by R. M. Graziano (then agent and attorney for carriers and freight forwarders) in his tariff titled *Hazardous Materials Regulations of the Department of Transportation* (1978). New tariffs identified by number are issued at intervals, and interim revisions are sent out. Agents change at intervals.

Graziano's tariff lists many regulated (dangerous) commodities (Part 172, DOT regulations) for transportation. This includes those that are poisonous, flammable, oxidizing, corrosive, explosive, radioactive, and compressed gases. Part 178 covers specifications for





**FIG. 10-192** Cost of shop-fabricated tanks in mid-1980 with ¼-in walls. Multiplying factors on carbon steel costs for other materials are: carbon steel, 1.0; rubber-lined carbon steel, 1.5; aluminum, 1.6; glass-lined carbon steel, 4.5; and fiber-reinforced plastic, 0.75 to 1.5. Multiplying factors on type 316 stainless-steel costs for other materials are: 316 stainless steel, 1.0; Monel, 2.0; Inconel, 2.0; nickel, 2.0; titanium, 3.2; and Hastelloy C, 3.8. Multiplying factors for wall thicknesses different from ¼ in are:

| Thickness, in | Carbon steel | 304 stainless steel | 316 stainless steel |
|---------------|--------------|---------------------|---------------------|
| ½             | 1.4          | 1.8                 | 1.8                 |
| ¾             | 2.1          | 2.5                 | 2.6                 |
| 1             | 2.7          | 3.3                 | 3.5                 |

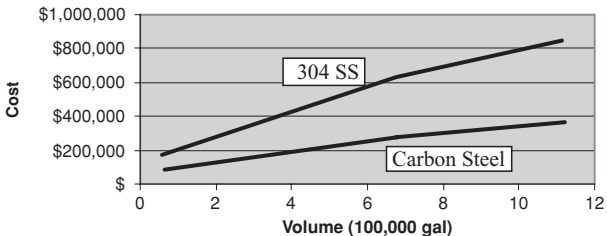
To convert gallons to cubic meters, multiply by  $3.785 \times 10^{-3}$ .

all types of containers from carboys to large portable tanks and tank trucks. Part 179 deals with tank-car construction.

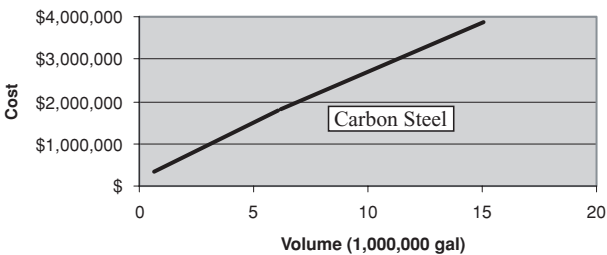
An Association of American Railroads (AAR) publication, *Specifications for Tank Cars*, covers many requirements beyond the DOT regulations.

Some additional details are given later. Because of frequent changes, it is always necessary to check the latest rules. The **shipper**, not the carrier, has the ultimate responsibility for shipping in the correct container.

**Tank Cars** These range in size from about 7.6 to 182 m<sup>3</sup> (2000 to 48,000 gal), and a car may be single or multiunit. The DOT now limits them to 130 m<sup>3</sup> (34,500 gal) and 120,000 kg (263,000 lb) gross mass. Large cars usually result in lower investment per cubic meter



**FIG. 10-193** Cost ( $\pm 30$  percent) of field-erected, domed, flat-bottom API 650 tanks, October 2005, includes concrete foundation and typical nozzles, ladders, and platforms. 1 gal = 0.003785 m<sup>3</sup>.



**FIG. 10-194** Cost ( $\pm 30$  percent) of field-erected, floating roof tanks, October 2005, includes concrete foundation and typical nozzles, ladders, and platforms. 1 gal = 0.003785 m<sup>3</sup>.

and take lower shipping rates. Cars may be insulated to reduce heating or cooling of the contents. Certain liquefied gases may be carried in insulated cars; temperatures are maintained by evaporation (see subsection “Low-Temperature and Cryogenic Storage”). Cars may be heated by steam coils or by electricity. Some products are loaded hot, solidify in transport, and are melted for removal. Some low-temperature cargoes must be unloaded within a given time (usually 30 days) to prevent pressure buildup.

Tank cars are classified as pressure or general-purpose. Pressure cars have relief-valve settings of 517 kPa (75 lbf/in<sup>2</sup>) and above. Those designated as general-purpose cars are, nevertheless, pressure vessels and may have relief valves or rupture disks. The DOT specification code number indicates the type of car. For instance, 105A500W indicates a pressure car with a test pressure of 3447 kPa (500 lbf/in<sup>2</sup>) and a relief-valve setting of 2585 kPa (375 lbf/in<sup>2</sup>). In most cases, loading and unloading valves, safety valves, and vent valves must be in a dome or an enclosure.

Companies shipping dangerous materials sometimes build tank cars with metal thicker than required by the specifications in order to reduce the possibility of leakage during a wreck or fire. The punching of couplers or rail ends into heads of tanks is a hazard.

Older tank cars have a center sill or beam running the entire length of the car. Most modern cars have no continuous sill, only short stub sills at each end. Cars with full sills have tanks anchored longitudinally at the center of the sill. The anchor is designed to be weaker than either the tank shell or the doubler plate between anchor and shell. Cars with stub sills have similar safeguards. Anchors and other parts are designed to meet AAR requirements.

The impact forces on car couplers put high stresses in sills, anchors, and doublers. This may start fatigue cracks in the shell, particularly at the corners of welded doubler plates. With brittle steel in cold weather, such cracks sometimes cause complete rupture of the tank. Large end radii on the doublers and tougher steels will reduce this hazard. Inspection of older cars can reveal cracks before failure.

A difference between tank cars and most pressure vessels is that tank cars are designed in terms of the theoretical ultimate or bursting strength of the tank. The test pressure is usually 40 percent of the bursting pressure (sometimes less). The safety valves are set at 75 percent of the test pressure. Thus, the maximum operating pressure is usually 30 percent of the bursting pressure. This gives a nominal factor of safety of 3.3, compared with 3.5 for Division 1 of the ASME Pressure Vessel Code.

The DOT rules require that pressure cars have relief valves designed to limit pressure to 82.5 percent (with certain exceptions) of test pressure (110 percent of maximum operating pressure) when exposed to fire. Appendix A of AAR Specifications deals with the flow capacity of relief devices. The formulas apply to cars in the upright position with the device discharging vapor. They may not protect the car adequately when it is overturned and the device is discharging liquid.

Appendix B of AAR Specifications deals with the certification of facilities. Fabrication, repairing, testing, and specialty work on tank cars must be done in certified facilities. The AAR certifies shops to build cars of certain materials, to do test work on cars, or to make certain repairs and alterations.

**Tank Trucks** These trucks may have single, compartmented, or multiple tanks. Many of their requirements are similar to those for tank cars, except that thinner shells are permitted in most cases. Trucks for nonhazardous materials are subject to few regulations other than the normal highway laws governing all motor vehicles. But trucks carrying hazardous materials must comply with DOT regulations, Parts 173, 177, 178, and 397. Maximum weight, axle loading, and length are governed by state highway regulations. Many states have limits in the vicinity of 31,750 kg (70,000 lb) total mass, 14,500 kg (32,000 lb) for tandem axles, and 18.3 m (60 ft) or less overall length. Some allow tandem trailers.

Truck cargo tanks (for dangerous materials) are built under Part 173 and Subpart J of Part 178, DOT regulations. This includes Specifications MC-306, MC-307, MC-312, and MC-331. MC-331 is required for compressed gas. Subpart J requires tanks for pressures above 345 kPa (50 lbf/in<sup>2</sup>) in one case and 103 kPa (15 lbf/in<sup>2</sup>) in another to be built according to the ASME Pressure Vessel Code. A particular issue of the code is specified.

Because of the demands of highway service, the DOT specifications have a number of requirements in addition to the ASME Code. These include design for impact forces and rollover protection for fittings.

**Portable tanks, drums, or bottles** are shipped by rail, ship, air, or truck. Portable tanks containing hazardous materials must conform to DOT regulations, Parts 173 and 178, Subpart H.

Some tanks are designed to be shipped by trailer and transferred to railcars or ships (see following discussion).

**Marine Transportation** Seagoing **tankers** are for high tonnage. The traditional tanker uses the ship structure as a tank. It is subdivided into a number of tanks by means of transverse bulkheads and a center-line bulkhead. More than one product can be carried. An elaborate piping system connects the tanks to a pumping plant which can discharge or transfer the cargo. Harbor and docking facilities appear to be the only limit to tanker size. The largest crude oil tanker size to date is about 560,000 deadweight tons. In the United States, tankers are built to specifications of the American Bureau of Shipping and the U.S. Coast Guard.

Low-temperature liquefied gases are shipped in special ships with insulation between the hull and an inner tank. The largest LNG carrier's capacity is about 145,000 m<sup>3</sup>. Poisonous materials are shipped in separate tanks built into the ship. This prevents tank leakage from contaminating harbors. Separate tanks are also used to transport pressurized gases.

Barges are used on inland waterways. Popular sizes are up to 16 m (52½ ft) wide by 76 m (250 ft) long, with 2.6 m (8½ ft) to 4.3 m (14 ft) draft. Cargo requirements and waterway limitations determine design. Use of barges of uniform size facilitates rafting them together.

**Portable tanks** may be stowed in the holds of conventional cargo ships or special container ships, or they may be fastened on deck.

**Container ships** have guides in the hold and on deck which hold boxlike containers or tanks. The tank is latched to a trailer chassis and hauled to shipside. A movable gantry, sometimes permanently installed on the ship, hoists the tank from the trailer and lowers it into the guides on the ship. This system achieves large savings in labor, but its application is sometimes limited by lack of agreement between ship operators and unions.

Portable tanks for regulated commodities in marine transportation must be designed and built under Coast Guard regulations (see discussion under "Pressure Vessels").

**Materials of Construction for Bulk Transport** Because of the more severe service, construction materials for transportation usually are more restricted than for storage. Most large pipe lines are constructed of steel conforming to API Specification 5L or 5LX. Most tanks (cars, etc.) are built of pressure-vessel steels or AAR specification steels, with a few of aluminum or stainless steel. Carbon steel tanks may be lined with rubber, plastic, nickel, glass, or other materials. In many cases this is practical and cheaper than using a stainless-steel tank. Other materials for tank construction may be proposed and used if approved by the appropriate authorities (AAR and DOT).

## PRESSURE VESSELS

This discussion of pressure vessels is intended as an overview of the codes most frequently used for the design and construction of pressure

vessels. Chemical engineers who design or specify pressure vessels should determine the federal and local laws relevant to the problem and then refer to the most recent issue of the pertinent code or standard before proceeding. Laws, codes, and standards are frequently changed.

A pressure vessel is a closed container of limited length (in contrast to the indefinite length of piping). Its smallest dimension is considerably larger than the connecting piping, and it is subject to pressures above 7 or 14 kPa (1 or 2 lbf/in<sup>2</sup>). It is distinguished from a boiler, which in most cases is used to generate steam for use external to itself.

**Code Administration** The American Society of Mechanical Engineers has written the ASME Boiler and Pressure Vessel Code, which contains rules for the design, fabrication, and inspection of boilers and pressure vessels. The ASME Code is an American National Standard. Most states in the United States and all Canadian provinces have passed legislation which makes the ASME Code or certain parts of it their legal requirement. Only a few jurisdictions have adopted the code for all vessels. The others apply it to certain types of vessels or to boilers. States employ inspectors (usually under a chief boiler inspector) to enforce code provisions. The authorities also depend a great deal on insurance company inspectors to see that boilers and pressure vessels are maintained in a safe condition.

The ASME Code is written by a large committee and many subcommittees, composed of engineers appointed by the ASME. The Code Committee meets regularly to review the code and consider requests for its revision, interpretation, or extension. **Interpretation and extension** are accomplished through "code cases." The decisions are published in *Mechanical Engineering*. Code cases are also mailed to those who subscribe to the service. A typical code case might be the approval of the use of a metal which is not presently on the list of approved code materials. Inquiries relative to code cases should be addressed to the secretary of the ASME Boiler and Pressure Vessel Committee, American Society of Mechanical Engineers, New York.

A new edition of the code is issued every 3 years. Between editions, alterations are handled by issuing semiannual addenda, which may be purchased by subscription. The ASME considers any issue of the code to be adequate and safe, but some government authorities specify certain issues of the code as their legal requirement.

**Inspection Authority** The National Board of Boiler and Pressure Vessel Inspectors is composed of the chief inspectors of states and municipalities in the United States and Canadian provinces which have made any part of the Boiler and Pressure Vessel Code a legal requirement. This board promotes uniform enforcement of boiler and pressure-vessel rules. One of the board's important activities is providing examinations for, and commissioning of, inspectors. Inspectors so qualified and employed by an insurance company, state, municipality, or Canadian province may inspect a pressure vessel and permit it to be stamped ASME—NB (National Board). An inspector employed by a vessel user may authorize the use of only the ASME stamp. The ASME Code Committee authorizes fabricators to use the various ASME stamps. The stamps, however, may be applied to a vessel only with the approval of the inspector.

The ASME Boiler and Pressure Vessel Code consists of eleven sections as follows:

- I. Power Boilers
- II. Materials
  - a. Ferrous
  - b. Nonferrous
  - c. Welding rods, electrodes, and filler metals
  - d. Properties
- III. Rules for Construction of Nuclear Power Plant Components
- IV. Heating Boilers
- V. Nondestructive Examination
- VI. Rules for Care and Operation of Heating Boilers
- VII. Guidelines for the Care of Power Boilers
- VIII. Pressure Vessels
  - IX. Welding and Brazing Qualifications
  - X. Fiber-Reinforced Plastic Pressure Vessels
  - XI. Rules for Inservice Inspection of Nuclear Power Plant Components

Pressure vessels (as distinguished from boilers) are involved with Secs. II, III, V, VIII, IX, X, and XI. Section VIII, Division 1, is the



Pressure Vessel Code as it existed in the past (and will continue). Division 1, was brought out as a means of permitting higher design stresses while ensuring at least as great a degree of safety as in Division 1. These two divisions plus Secs. III and X will be discussed briefly here. They refer to Secs. II and IX.

**ASME Code Section VIII, Division 1** Most pressure vessels used in the process industry in the United States are designed and constructed in accordance with Sec. VIII, Division 1 (see Fig. 10-195). This division is divided into three subsections followed by appendices.

**Introduction** The Introduction contains the scope of the division and defines the responsibilities of the user, the manufacturer, and the inspector. The scope defines pressure vessels as containers for the containment of pressure. It specifically excludes vessels having an internal pressure not exceeding 103 kPa (15 lbf/in<sup>2</sup>) and further states that the rules are applicable for pressures not exceeding 20,670 kPa (3000 lbf/in<sup>2</sup>). For higher pressures it is usually necessary to deviate from the rules in this division.

The scope covers many other less basic exclusions, and inasmuch as the scope is occasionally revised, except for the most obvious cases, it is prudent to review the current issue before specifying or designing pressure vessels to this division. Any vessel which meets all the requirements of this division may be stamped with the code *U* symbol even though exempted from such stamping.

**Subsection A** This subsection contains the general requirements applicable to all materials and methods of construction. Design temperature and pressure are defined here, and the loadings to be considered in design are specified. For stress failure and yielding, this section of the code uses the maximum-stress theory of failure as its criterion.

This subsection refers to the tables elsewhere in the division in which the maximum allowable tensile-stress values are tabulated. The basis for the establishment of these allowable stresses is defined in detail in Appendix P; however, as the safety factors used were very important in establishing the various rules of this division, it is noted that the safety factors for internal-pressure loads are 3.5 on ultimate strength and 1.6 or 1.5 on yield strength, depending on the material. For external-pressure loads on cylindrical shells, the safety factors are 3 for both elastic buckling and plastic collapse. For other shapes subject to external pressure and for longitudinal shell compression, the safety factors are 3.5 for both elastic buckling and plastic collapse. Longitudinal compressive stress in cylindrical elements is limited in this subsection by the lower of either stress failure or buckling failure.

Internal-pressure design rules and formulas are given for cylindrical and spherical shells and for ellipsoidal, torispherical (often called ASME heads), hemispherical, and conical heads. The formulas given assume membrane-stress failure, although the rules for heads include consideration for buckling failure in the transition area from cylinder to head (knuckle area).

Longitudinal joints in cylinders are more highly stressed than circumferential joints, and the code takes this fact into account. When forming heads, there is usually some thinning from the original plate thickness in the knuckle area, and it is prudent to specify the minimum allowable thickness at this point.

Unstayed flat heads and covers can be designed by very specific rules and formulas given in this subsection. The stresses caused by pressure on these members are bending stresses, and the formulas include an allowance for additional edge moments induced when the head, cover, or blind flange is attached by bolts. Rules are provided for quick-opening closures because of the risk of incomplete attachment or opening while the vessel is pressurized. Rules for braced and stayed surfaces are also provided.

External-pressure failure of shells can result from overstress at one extreme or from elastic instability at the other or at some intermediate loading. The code provides the solution for most shells by using a number of charts. One chart is used for cylinders where the shell diameter-to-thickness ratio and the length-to-diameter ratio are the variables. The rest of the charts depict curves relating the geometry of cylinders and spheres to allowable stress by curves which are determined from the modulus of elasticity, tangent modulus, and yield strength at temperatures for various materials or classes of materials. The text of this subsection explains how the allowable stress is deter-

mined from the charts for cylinders, spheres, and hemispherical, ellipsoidal, torispherical, and conical heads.

Frequently cost savings for cylindrical shells can result from reducing the effective length-to-diameter ratio and thereby reducing shell thickness. This can be accomplished by adding circumferential stiffeners to the shell. Rules are included for designing and locating the stiffeners.

Openings are always required in pressure-vessel shells and heads. Stress intensification is created by the existence of a hole in an otherwise symmetrical section. The code compensates for this by an area-replacement method. It takes a cross section through the opening, and it measures the area of the metal of the required shell that is removed and replaces it in the cross section by additional material (shell wall, nozzle wall, reinforcing plate, or weld) within certain distances of the opening centerline. These rules and formulas for calculation are included in Subsec. A.

When a cylindrical shell is drilled for the insertion of multiple tubes, the shell is significantly weakened and the code provides rules for tube-hole patterns and the reduction in strength that must be accommodated.

Fabrication tolerances are covered in this subsection. The tolerances permitted for shells for external pressure are much closer than those for internal pressure because the stability of the structure is dependent on the symmetry. Other paragraphs cover repair of defects during fabrication, material identification, heat treatment, and impact testing.

Inspection and testing requirements are covered in detail. Most vessels are required to be hydrostatic-tested (generally with water) at 1.3 times the maximum allowable working pressure. Some enameled (glass-lined) vessels are permitted to be hydrostatic-tested at lower pressures. Pneumatic tests are permitted and are carried to at least 1¼ times the maximum allowable working pressure, and there is provision for proof testing when the strength of the vessel or any of its parts cannot be computed with satisfactory assurance of accuracy. Pneumatic or proof tests are rarely conducted.

Pressure-relief-device requirements are defined in Subsec. A. Set point and maximum pressure during relief are defined according to the service, the cause of overpressure, and the number of relief devices. Safety, safety relief, relief valves, rupture disk, breaking pin, and rules on tolerances for the relieving point are given.

Testing, certification, and installation rules for relieving devices are extensive. Every chemical engineer responsible for the design or operation of process units should become very familiar with these rules. The pressure-relief-device paragraphs are the only parts of Sec. VIII, Division 1, that are concerned with the installation and ongoing operation of the facility; all other rules apply only to the design and manufacture of the vessel.

**Subsection B** This subsection contains rules pertaining to the methods of fabrication of pressure vessels. Part UW is applicable to welded vessels. Service restrictions are defined. Lethal service is for "lethal substances," which are defined as poisonous gases or liquids of such a nature that a very small amount of the gas or the vapor of the liquid mixed or unmixed with air is dangerous to life when inhaled. It is stated that it is the user's responsibility to advise the designer or manufacturer if the service is lethal. All vessels in lethal service shall have all butt-welded joints fully radiographed, and when practical, joints shall be butt-welded. All vessels fabricated of carbon or low-alloy steel shall be postweld-heat-treated.

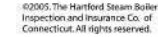
Low-temperature service is defined as being below -29°C (-20°F), and impact testing of many materials is required. The code is restrictive in the type of welding permitted.

Unfired steam boilers with design pressures exceeding 345 kPa (50 lbf/in<sup>2</sup>) have restrictive rules on welded-joint design, and all butt joints require full radiography.

Pressure vessels subject to direct firing have special requirements relative to welded-joint design and postweld heat treatment.

This subsection includes rules governing welded-joint designs and the degree of radiography, with efficiencies for welded joints specified as functions of the quality of joint. These efficiencies are used in the formulas in Subsec. A for determining vessel thicknesses.

Details are provided for head-to-shell welds, tube sheet-to-shell welds, and nozzle-to-shell welds. Acceptable forms of welded stay-bolts and plug and slot welds for staying plates are given here.



**FIG. 10-195** Quick reference guide to ASME Boiler and Pressure Vessel Code Section VIII, Division 1 (2004 edition). (Reprinted with permission of publisher, HSB Global Standards, Hartford, Conn.)

| ORGANIZATION                                                        |  |  |  | GENERAL NOTES                                                                                               |  |  |  |
|---------------------------------------------------------------------|--|--|--|-------------------------------------------------------------------------------------------------------------|--|--|--|
| Introduction                                                        |  |  |  | Code Jurisdiction for Piping.....U-1                                                                        |  |  |  |
| Subsection A                                                        |  |  |  | Design Pressure.....UG-21; UG-98; UCD-3                                                                     |  |  |  |
| Part UG-General requirements for all construction and all materials |  |  |  | Design Temperature.....UG-20; UCL-24; UCD-3                                                                 |  |  |  |
| Subsection B                                                        |  |  |  | Dimpled or Embossed Assemblies.....Appendix 17                                                              |  |  |  |
| Part UW                                                             |  |  |  | Inspector's Responsibility.....U-2; UG-90                                                                   |  |  |  |
| Part UF                                                             |  |  |  | Loadings.....UG-22; Appendix G, H                                                                           |  |  |  |
| Part UB                                                             |  |  |  | Low Temperature Service.....UW-2; UCS-66-68; UHA-51; Part ULT                                               |  |  |  |
| Subsection C                                                        |  |  |  | Manufacturer's Responsibility.....U-2; UG-90                                                                |  |  |  |
| Part UCS                                                            |  |  |  | Material, General.....UG-4, 10, 11, 15                                                                      |  |  |  |
| Part UNF                                                            |  |  |  | a. Bolts and Studs.....UG-12                                                                                |  |  |  |
| Part UHA                                                            |  |  |  | b. Castings.....UG-7                                                                                        |  |  |  |
| Part UCI                                                            |  |  |  | c. Forgings.....UG-6                                                                                        |  |  |  |
| Part UCL                                                            |  |  |  | d. Nuts and Washers.....UG-13                                                                               |  |  |  |
| Part UCD                                                            |  |  |  | e. Pipes and Tubes.....UG-8                                                                                 |  |  |  |
| Part UHT                                                            |  |  |  | f. Plates.....UG-5                                                                                          |  |  |  |
| Part ULW                                                            |  |  |  | g. Rod and Bar.....UG-14                                                                                    |  |  |  |
| Part ULT                                                            |  |  |  | h. Standard Parts.....UG-11, 44                                                                             |  |  |  |
| Part UHX                                                            |  |  |  | i. Welding.....UG-9                                                                                         |  |  |  |
| Mandatory Appendices 1-33                                           |  |  |  | Material Identification, Marking and Certification.....UG-77, 93, 94                                        |  |  |  |
| Nonmandatory Appendices A-Y, DD, EE, FF, GG                         |  |  |  | Material Tolerances.....UG-16                                                                               |  |  |  |
|                                                                     |  |  |  | Nameplates, Stamping and Reports.....UG-115-120; UHT-115; ULW-115; ULT-115; App. W & 18                     |  |  |  |
|                                                                     |  |  |  | Nondestructive Examination                                                                                  |  |  |  |
|                                                                     |  |  |  | a. Liquid Penetrant.....Appendix 8                                                                          |  |  |  |
|                                                                     |  |  |  | b. Magnetic Particle.....Appendix 6                                                                         |  |  |  |
|                                                                     |  |  |  | c. Radiography.....UW-51, 52                                                                                |  |  |  |
|                                                                     |  |  |  | d. Ultrasonic.....Appendix 12                                                                               |  |  |  |
|                                                                     |  |  |  | Porosity Charts.....Appendix 4                                                                              |  |  |  |
|                                                                     |  |  |  | Pressure Tests.....UG-99, 100, 101; UW-50; UCL-99; UCD-99; ULT-99                                           |  |  |  |
|                                                                     |  |  |  | Quality Control System.....U-2; Appendix 10                                                                 |  |  |  |
|                                                                     |  |  |  | Quick Actuating Closures.....U-1; UG-35; App. FF                                                            |  |  |  |
|                                                                     |  |  |  | Service Restrictions.....UW-2; UB-3; UCL-3; UCD-2                                                           |  |  |  |
|                                                                     |  |  |  | Stress, Maximum Allowable.....UG-23; App. P; UCS-23; UNF-23; UHA-23; UCD-23; UHT-23; ULT-23; UCL-23; UCI-23 |  |  |  |
|                                                                     |  |  |  | User's Responsibility.....UG-125; U-2                                                                       |  |  |  |
|                                                                     |  |  |  | Welding Preheat Recommendations.....Appendix R                                                              |  |  |  |
|                                                                     |  |  |  | Welding Qualifications.....UW-26 thru 29; ULT-82; UHT-82; UHA-52; UNF-95                                    |  |  |  |

| PARA.  | REQUIREMENTS/REMARKS                                                                                                         | PT | MT | UT | RT | PARA.  | REQUIREMENTS/REMARKS                                                                                                                                                                   | PT | MT | UT |
|--------|------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|
| UG-24  | General requirements for castings.....X                                                                                      |    | X  | X  | X  | UCL-35 | Vessels or parts of vessels constructed of clad plate and those having applied corrosion resistant linings.....                                                                        |    |    |    |
| UG-93  | General requirements for the inspection of all materials.....X                                                               |    | X  |    |    | UCL-36 | Chromium stainless steel cladding or lining.....                                                                                                                                       |    |    |    |
| UW-11  | Radiographic and ultrasonic examination required for pressure vessels and vessel parts (also see UW-51, UW-52).....          |    |    | X  | X  | UCD-78 | Repair of defects in cast ductile iron pressure vessels and vessel parts.....                                                                                                          |    |    |    |
| UW-42  | Surface weld metal buildup.....X                                                                                             |    | X  |    | X  | UHT-57 | Pressure vessels or vessel parts constructed of ferritic steels having tensile properties enhanced by heat treatment.....X                                                             |    | X  |    |
| UW-50  | Welds on pneumatically tested vessels.....X                                                                                  |    | X  |    |    | UHT-83 | Metal removal accomplished by methods involving melting on pressure vessels and vessel parts constructed of ferritic steels having tensile properties enhanced by heat treatment.....X |    | X  |    |
| UF-31  | Vessels fabricated from SA-372 forging material to be liquid quenched and tempered.....X                                     |    | X  |    |    | UHT-85 | Removal of temporary welds on pressure vessels and vessel parts constructed of ferritic steels having tensile properties enhanced by heat treatment.....X                              |    | X  |    |
| UF-32  | Finished welds after postweld heat treatment.....X                                                                           |    | X  |    |    | ULW-51 | Inner shells and inner heads of layered pressure vessels.....                                                                                                                          |    |    |    |
| UF-37  | Repair welds in forgings.....X                                                                                               |    | X  |    | X  | ULW-52 | Welded joints in the layers of layered pressure vessels.....X                                                                                                                          |    | X  |    |
| UF-55  | Vessels constructed of SA-372 Class VIII material.....X                                                                      |    |    | X  |    | ULW-53 | Step welded girth joints in the layers of layered pressure vessels.....X                                                                                                               |    | X  |    |
| UCS-57 | Examination in addition to UW-11 for butt welded joints on carbon and low alloy steel pressure vessels and vessel parts..... |    |    | X  |    | ULW-54 | Butt welded joints in layered pressure vessels.....                                                                                                                                    |    |    |    |
| UNF-57 | Examination in addition to UW-11 for pressure vessels and vessel parts constructed of nonferrous materials.....              |    |    |    | X  | ULW-55 | Flat head and tube sheet welded joints in layered pressure vessels.....X                                                                                                               |    | X  |    |
| UNF-58 | All groove and fillet welds in vessels constructed of certain nonferrous materials.....X                                     |    |    |    |    | ULW-56 | Nozzle and communicating chamber welded joints in layered pressure vessels.....X                                                                                                       |    | X  |    |
| UHA-33 | Exceptions for radiographic examinations of high alloy steel vessels.....                                                    |    |    |    | X  | ULW-57 | Random spot examinations and repairs of welds in layered pressure vessels.....X                                                                                                        |    | X  |    |
| UHA-34 | Austenitic chromium-nickel alloy steel butt and fillet welds.....X                                                           |    |    |    |    | ULT-57 | Welds in pressure vessels and vessel parts constructed of materials having increased design stress values due to low temperature applications.....X                                    |    | X  | X  |
| UCI-78 | Repair of defects in cast iron pressure vessels and vessel parts.....X                                                       |    | X  |    | X  |        |                                                                                                                                                                                        |    |    |    |

| Qualifications of Personnel Performing Section VIII, Division 1 Nondestructive Examinations |            |             |             |
|---------------------------------------------------------------------------------------------|------------|-------------|-------------|
| Examination                                                                                 | Reference  | Examination | Reference   |
| PT                                                                                          | Appendix 8 | UT          | Appendix 12 |
| MT                                                                                          | Appendix 6 | RT          | UW-51       |

FIG. 10-195 (Continued)

Rules for the welded fabrication of pressure vessels cover welding processes, manufacturer's record keeping on welding procedures, welder qualification, cleaning, fit-up alignment tolerances, and repair of weld defects. Procedures for postweld heat treatment are detailed. Checking the procedures and welders and radiographic and ultrasonic examination of welded joints are covered.

Requirements for vessels fabricated by forging in Part UF include unique design requirements with particular concern for stress risers, fabrication, heat treatment, repair of defects, and inspection. Vessels fabricated by brazing are covered in Part UB. Brazed vessels cannot be used in lethal service, for unfired steam boilers, or for direct firing. Permitted brazing processes as well as testing of brazed joints for strength are covered. Fabrication and inspection rules are also included.

**Subsection C** This subsection contains requirements pertaining to classes of materials. Carbon and low-alloy steels are governed by Part UCS, nonferrous materials by Part UNF, high-alloy steels by Part UHA, and steels with tensile properties enhanced by heat treatment by Part UHT. Each of these parts includes tables of maximum allowable stress values for all code materials for a range of metal temperatures. These stress values include appropriate safety factors. Rules governing the application, fabrication, and heat treatment of the vessels are included in each part.

Part UHT also contains more stringent details for nozzle welding that are required for some of these high-strength materials. Part UCI has rules for cast-iron construction, Part UCL has rules for welded vessels of clad plate as lined vessels, and Part UCD has rules for ductile-iron pressure vessels.

A relatively recent addition to the code is Part ULW, which contains requirements for vessels fabricated by layered construction. This type of construction is most frequently used for high pressures, usually in excess of 13,800 kPa (2000 lbf/in<sup>2</sup>).

There are several methods of layering in common use: (1) thick layers shrunk together; (2) thin layers, each wrapped over the other and the longitudinal seam welded by using the prior layer as backup; and (3) thin layers spirally wrapped. The code rules are written for either thick or thin layers. Rules and details are provided for all the usual welded joints and nozzle reinforcement. Supports for layered vessels require special consideration, in that only the outer layer could contribute to the support. For lethal service only the inner shell and inner heads need comply with the requirements in Subsec. B. Inasmuch as radiography would not be practical for inspection of many of the welds, extensive use is made of magnetic-particle and ultrasonic inspection. When radiography is required, the code warns the inspector

that indications sufficient for rejection in single-wall vessels may be acceptable. Vent holes are specified through each layer down to the inner shell to prevent buildup of pressure between layers in the event of leakage at the inner shell.

**Mandatory Appendixes** These include a section on supplementary design formulas for shells not covered in Subsec. A. Formulas are given for thick shells, heads, and dished covers. Another appendix gives very specific rules, formulas, and charts for the design of bolted-flange connections. The nature of these rules is such that they are readily programmable for a digital computer, and most flanges now are designed by using computers. One appendix includes only the charts used for calculating shells for external pressure discussed previously. Jacketed vessels are covered in a separate appendix in which very specific rules are given, particularly for the attachment of the jacket to the inner shell. Other appendixes cover inspection and quality control.

**Nonmandatory Appendixes** These cover a number of subjects, primarily suggested good practices and other aids in understanding the code and in designing with the code. Several current nonmandatory appendixes will probably become mandatory.

Figure 10-195 illustrates a pressure vessel with the applicable code paragraphs noted for the various elements. Additional important paragraphs are referenced at the bottom of the figure.

**ASME Code Section VIII, Division 2** Paragraph AG-100e of Division 2 states: "In relation to the rules of Division 1 of Section VIII, these rules of Division 2 are more restrictive in the choice of materials which may be used but permit higher design stress intensity values to be employed in the range of temperatures over which the design stress intensity value is controlled by the ultimate strength or the yield strength; more precise design procedures are required and some common design details are prohibited; permissible fabrication procedures are specifically delineated and more complete testing and inspection are required." Most Division 2 vessels fabricated to date have been large or intended for high pressure and, therefore, expensive when the material and labor savings resulting from smaller safety factors have been greater than the additional engineering, administrative, and inspection costs.

The organization of Division 2 differs from that of Division 1.

**Part AG** This part gives the scope of the division, establishes its jurisdiction, and sets forth the responsibilities of the user and the manufacturer. Of particular importance is the fact that no upper limitation in pressure is specified and that a user's design specification is required. The user or the user's agent shall provide requirements for intended operating conditions in such detail as to constitute an adequate basis for selecting materials and designing, fabricating, and inspecting the vessel. The user's design specification shall include the method of supporting the vessel and any requirement for a fatigue analysis. If a fatigue analysis is required, the user must provide information in sufficient detail so that an analysis for cyclic operation can be made.

**Part AM** This part lists permitted individual construction materials, applicable specifications, special requirements, design stress-intensity values, and other property information. Of particular importance are the ultrasonic-test and toughness requirements. Among the properties for which data are included are thermal conductivity and diffusivity, coefficient of thermal expansion, modulus of elasticity, and yield strength. The design stress-intensity values include a safety factor of 3 on ultimate strength at temperature or 1.5 on yield strength at temperature.

**Part AD** This part contains requirements for the design of vessels. The rules of Division 2 are based on the maximum-shear theory of failure for stress failure and yielding. Higher stresses are permitted when wind or earthquake loads are considered. Any rules for determining the need for fatigue analysis are given here.

Rules for the design of shells of revolution under internal pressure differ from the Division 1 rules, particularly the rules for formed heads when plastic deformation in the knuckle area is the failure criterion. Shells of revolution for external pressure are determined on the same criterion, including safety factors, as in Division 1. Reinforcement for openings uses the same area-replacement method as Division 1; however, in many cases the reinforcement metal must be closer to the opening centerline.

The rest of the rules in Part AD for flat heads, bolted and studed connections, quick-actuating closures, and layered vessels essentially

duplicate Division 1. The rules for support skirts are more definitive in Division 2.

**Part AF** This part contains requirements governing the fabrication of vessels and vessel parts.

**Part AR** This part contains rules for pressure-relieving devices.

**Part AI** This part contains requirements controlling inspection of vessel.

**Part AT** This part contains testing requirements and procedures.

**Part AS** This part contains requirements for stamping and certifying the vessel and vessel parts.

**Appendixes** Appendix 1 defines the basis used for defining stress-intensity values. Appendix 2 contains external-pressure charts, and Appendix 3 has the rules for bolted-flange connections; these two are exact duplicates of the equivalent appendixes in Division 1.

Appendix 4 gives definitions and rules for stress analysis for shells, flat and formed heads, and tube sheets, layered vessels, and nozzles including discontinuity stresses. Of particular importance are Table 4-120.1, "Classification of Stresses for Some Typical Cases," and Fig. 4-130.1, "Stress Categories and Limits of Stress Intensity." These are very useful in that they clarify a number of paragraphs and simplify stress analysis.

Appendix 5 contains rules and data for stress analysis for cyclic operation. Except in short-cycle batch processes, pressure vessels are usually subject to few cycles in their projected lifetime, and the endurance-limit data used in the machinery industries are not applicable. Curves are given for a broad spectrum of materials, covering a range from 10 to 1 million cycles with allowable stress values as high as 650,000 lbf/in<sup>2</sup>. This low-cycle fatigue has been developed from strain-fatigue work in which stress values are obtained by multiplying the strains by the modulus of elasticity. Stresses of this magnitude cannot occur, but strains do. The curves given have a factor of safety of 2 on stress or 20 on cycles.

Appendix 6 contains requirements of experimental stress analysis, Appendix 8 has acceptance standards for radiographic examination, Appendix 9 covers nondestructive examination, Appendix 10 gives rules for capacity conversions for safety valves, and Appendix 18 details quality-control-system requirements.

The remaining appendixes are nonmandatory but useful to engineers working with the code.

**General Considerations** Most pressure vessels for the chemical-process industry will continue to be designed and built to the rules of Sec. VIII, Division 1. While the rules of Sec. VIII, Division 2, will frequently provide thinner elements, the cost of the engineering analysis, stress analysis and higher-quality construction, material control, and inspection required by these rules frequently exceeds the savings from the use of thinner walls.

### Additional ASME Code Considerations

**ASME Code Sec. III: Nuclear Power Plant Components** This section of the code includes vessels, storage tanks, and concrete containment vessels as well as other nonvessel items.

**ASME Code Sec. X: Fiberglass-Reinforced-Plastic Pressure Vessels** This section is limited to four types of vessels: bag-molded and centrifugally cast, each limited to 1000 kPa (150 lbf/in<sup>2</sup>); filament-wound with cut filaments limited to 10,000 kPa (1500 lbf/in<sup>2</sup>); and filament-wound with uncut filaments limited to 21,000 kPa (3000 lbf/in<sup>2</sup>). Operating temperatures are limited to the range from +66°C (150°F) to -54°C (-65°F). Low modulus of elasticity and other property differences between metal and plastic required that many of the procedures in Sec. X be different from those in the sections governing metal vessels. The requirement that at least one vessel of a particular design and fabrication shall be tested to destruction has prevented this section from being widely used. The results from the combined fatigue and burst test must give the design pressure a safety factor of 6 to the burst pressure.

**Safety in Design** Designing a pressure vessel in accordance with the code will, under most circumstances, provide adequate safety. In the code's own words, however, the rules "cover minimum construction requirements for the design, fabrication, inspection, and certification of pressure vessels." The significant word is "minimum." The **ultimate responsibility** for safety rests with the user and the designer. They must decide whether anything beyond code requirements is necessary. The code cannot foresee and provide for all the

unusual conditions to which a pressure vessel might be exposed. If it tried to do so, the majority of pressure vessels would be unnecessarily restricted. Some of the conditions that a vessel might encounter are unusually low temperatures, unusual thermal stresses, stress ratcheting caused by thermal cycling, vibration of tall vessels excited by von Karman vortices caused by wind, very high pressures, runaway chemical reactions, repeated local overheating, explosions, exposure to fire, exposure to materials that rapidly attack the metal, containment of extremely toxic materials, and very large sizes of vessels. Large vessels, although they may contain nonhazardous materials, could, by their very size, create a serious hazard if they burst. The failure of the Boston molasses tank in 1919 killed 12 people. For pressure vessels which are outside code jurisdiction, there are sometimes special hazards in very-high-strength materials and plastics. There may be many others which the designers should recognize if they encounter them.

**Metal fatigue**, when it is present, is a serious hazard. Section VIII, Division 1, mentions rapidly fluctuating pressures. Division 2 and Sec. III do require a fatigue analysis. In extreme cases vessel contents may affect the fatigue strength (endurance limit) of the material. This is corrosion fatigue. Although most ASME Code materials are not particularly sensitive to corrosion fatigue, even they may suffer an endurance limit loss of 50 percent in some environments. High-strength heat-treated steels, on the other hand, are very sensitive to corrosion fatigue. It is not unusual to find some of these which lose 75 percent of their endurance in corrosive environments. In fact, in corrosion fatigue many steels do not have an endurance limit. The curve of stress versus cycles to failure (*S/N* curve) continues to slope downward regardless of the number of cycles.

**Brittle fracture** is probably the most insidious type of pressure-vessel failure. Without brittle fracture, a pressure vessel could be pressurized approximately to its ultimate strength before failure. With brittle behavior some vessels have failed well below their design pressures (which are about 25 percent of the theoretical bursting pressures). In order to reduce the possibility of brittle behavior, Division 2 and Sec. III require impact tests.

The subject of brittle fracture has been understood only since about 1950, and knowledge of some of its aspects is still inadequate. A notched or cracked plate of pressure-vessel steel, stressed at 66°C (150°F), would elongate and absorb considerable energy before breaking. It would have a ductile or plastic fracture. As the temperature is lowered, a point is reached at which the plate would fail in a brittle manner with a flat fracture surface and almost no elongation. The transition from ductile to brittle fracture actually takes place over a temperature range, but a point in this range is selected as the **transition temperature**. One of the ways of determining this temperature is the Charpy impact test (see ASTM Specification E-23). After the transition temperature has been determined by laboratory impact tests, it must be correlated with service experience on full-size plates. The literature on brittle fracture contains information on the relation of impact tests to service experience on some carbon steels.

A more precise but more elaborate method of dealing with the ductile-brittle transition is the **fracture-analysis diagram**. This uses a transition known as the **nil-ductility temperature** (NDT), which is determined by the drop-weight test (ASTM Standard E208) or the drop-weight tear test (ASTM Standard E436). The application of this diagram is explained in two papers by Pellini and Puzak [*Trans. Am. Soc. Mech. Eng.*, 429 (October 1964); *Welding Res. Coun. Bull.* 88, 1963].

Section VIII, Division 1, is rather lax with respect to brittle fracture. It allows the use of many steels down to -29°C (-20°F) without a check on toughness. Occasional brittle failures show that some vessels are operating below the nil-ductility temperature, i.e., the lower limit of ductility. Division 2 has resolved this problem by requiring impact tests in certain cases. Tougher grades of steel, such as the SA516 steels (in preference to SA515 steel), are available for a small price premium. Stress relief, steel made to fine-grain practice, and normalizing all reduce the hazard of brittle fracture.

**Nondestructive testing** of both the plate and the finished vessel is important to safety. In the analysis of fracture hazards, it is important to know the size of the flaws that may be present in the completed vessel. The four most widely used methods of examination are radiographic, magnetic-particle, liquid-penetrant, and ultrasonic.

**Radiographic examination** is either by **x-rays** or by **gamma radiation**. The former has greater penetrating power, but the latter is more portable. Few x-ray machines can penetrate beyond 300-mm (12-in) thickness.

**Ultrasonic techniques** use vibrations with a frequency between 0.5 and 20 MHz transmitted to the metal by a transducer. The instrument sends out a series of pulses. These show on a cathode-ray screen as they are sent out and again when they return after being reflected from the opposite side of the member. If there is a crack or an inclusion along the way, it will reflect part of the beam. The initial pulse and its reflection from the back of the member are separated on the screen by a distance which represents the thickness. The reflection from a flaw will fall between these signals and indicate its magnitude and position. Ultrasonic examination can be used for almost any thickness of material from a fraction of an inch to several feet. Its use is dependent upon the shape of the body because irregular surfaces may give confusing reflections. Ultrasonic transducers can transmit pulses normal to the surface or at an angle. Transducers transmitting pulses that are oblique to the surface can solve a number of special inspection problems.

**Magnetic-particle examination** is used only on magnetic materials. Magnetic flux is passed through the part in a path parallel to the surface. Fine magnetic particles, when dusted over the surface, will concentrate near the edges of a crack. The sensitivity of magnetic-particle examination is proportional to the sine of the angle between the direction of the magnetic flux and the direction of the crack. To be sure of picking up all cracks, it is necessary to probe the area in two directions.

**Liquid-penetrant examination** involves wetting the surface with a fluid which penetrates open cracks. After the excess liquid has been wiped off, the surface is coated with a material which will reveal any liquid that has penetrated the cracks. In some systems a colored dye will seep out of cracks and stain whitewash. Another system uses a penetrant that becomes fluorescent under ultraviolet light.

Each of these four popular methods has its advantages. Frequently, best results are obtained by using more than one method. Magnetic particles or liquid penetrants are effective on surface cracks. Radiography and ultrasonics are necessary for subsurface flaws. *No known method of nondestructive testing can guarantee the absence of flaws.* There are other less widely used methods of examination. Among these are eddy-current, electrical-resistance, acoustics, and thermal testing. *Nondestructive Testing Handbook* [Robert C. McMaster (ed.), Ronald, New York, 1959] gives information on many testing techniques.

The **eddy-current technique** involves an alternating-current coil along and close to the surface being examined. The electrical impedance of the coil is affected by flaws in the structure or changes in composition. Commercially, the principal use of eddy-current testing is for the examination of tubing. It could, however, be used for testing other things.

The **electrical-resistance method** involves passing an electric current through the structure and exploring the surface with voltage probes. Flaws, cracks, or inclusions will cause a disturbance in the voltage gradient on the surface. Railroads have used this method for many years to locate transverse cracks in rails.

The **hydrostatic test** is, in one sense, a method of examination of a vessel. It can reveal gross flaws, inadequate design, and flange leaks. Many believe that a hydrostatic test guarantees the safety of a vessel. This is not necessarily so. A vessel that has passed a hydrostatic test is probably safer than one that has not been tested. It can, however, still fail in service, even on the next application of pressure. Care in material selection, examination, and fabrication do more to guarantee vessel integrity than the hydrostatic test.

The ASME Codes recommend that hydrostatic tests be run at a temperature that is usually above the nil-ductility temperature of the material. This is, in effect, a pressure-temperature treatment of the vessel. When tested in the relatively ductile condition above the nil-ductility temperature, the material will yield at the tips of cracks and flaws and at points of high residual weld stress. This procedure will actually reduce the residual stresses and cause a redistribution at crack tips. The vessel will then be in a safer condition for subsequent operation. This procedure is sometimes referred to as **notch nullification**.

It is possible to design a hydrostatic test in such a way that it probably will be a proof test of the vessel. This usually requires, among other things, that the test be run at a temperature as low as and

preferably lower than the minimum operating temperature of the vessel. Proof tests of this type are run on vessels built of ultrahigh-strength steel to operate at cryogenic temperatures.

**Other Regulations and Standards** Pressure vessels may come under many types of regulation, depending on where they are and what they contain. Although many states have adopted the ASME Boiler and Pressure Vessel Code, either in total or in part, any state or municipality may enact its own requirements. The federal government regulates some pressure vessels through the Department of Transportation, which includes the Coast Guard. If pressure vessels are shipped into foreign countries, they may face additional regulations.

Pressure vessels carried aboard United States-registered ships must conform to rules of the **U.S. Coast Guard**. Subchapter F of Title 46, *Code of Federal Regulations*, covers marine engineering. Of this, Parts 50 through 61 and 98 include pressure vessels. Many of the rules are similar to those in the ASME Code, but there are differences.

The **American Bureau of Shipping** (ABS) has rules that insurance underwriters require for the design and construction of pressure vessels which are a permanent part of a ship. Pressure cargo tanks may be permanently attached and come under these rules. Such tanks supported at several points are independent of the ship's structure and are distinguished from "integral cargo tanks" such as those in a tanker. ABS has pressure vessel rules in two of its publications. Most of them are in *Rules for Building and Classing Steel Vessels*.

**Standards of Tubular Exchanger Manufacturers Association** (TEMA) give recommendations for the construction of tubular heat exchangers. Although TEMA is not a regulatory body and there is no legal requirement for the use of its standards, they are widely accepted as a good basis for design. By specifying TEMA standards, one can obtain adequate equipment without having to write detailed specifications for each piece. TEMA gives formulas for the thickness of tube sheets. Such formulas are not in ASME Codes. (See further discussion of TEMA in Sec. 11.)

**Vessels with Unusual Construction** High pressures create design problems. The ASME Code Sec. VIII, Division 1, applies to vessels rated for pressures up to 20,670 kPa (3000 lbf/in<sup>2</sup>). Division 2 is unlimited. At high pressures, special designs not necessarily in accordance with the code are sometimes used. At such pressures, a vessel designed for ordinary low-carbon-steel plate, particularly in large diameters, would become too thick for practical fabrication by ordinary methods. The alternatives are to make the vessel of high-strength plate, use a solid forging, or use multilayer construction.

**High-strength steels** with tensile strengths over 1380 MPa (200,000 lbf/in<sup>2</sup>) are limited largely to applications for which weight is very important. Welding procedures are carefully controlled, and preheat is used. These materials are brittle at almost any temperature, and vessels must be designed to prevent brittle fracture. Flat spots and variations in curvature are avoided. Openings and changes in shape require appropriate design. The maximum permissible size of flaws is determined by fracture mechanics, and the method of examination must assure as much as possible that larger flaws are not present. All methods of nondestructive testing may be used. Such vessels require the most sophisticated techniques in design, fabrication, and operation.

**Solid forgings** are frequently used in construction for pressure vessels above 20,670 kPa (3000 lbf/in<sup>2</sup>) and even lower. Almost any shell thickness can be obtained, but most of them range between 50 and 300 mm (2 and 12 in). The ASME Code lists forging materials with tensile strengths from 414 to 930 MPa (from 60,000 to 135,000 lbf/in<sup>2</sup>). Brittle fracture is a possibility, and the hazard increases with thickness. Furthermore, some forging alloys have nil-ductility temperatures as high as 121°C (250°F). A forged vessel should have an NDT at least 17°C (30°F) below the design temperature. In operation, it should be slowly and uniformly heated at least to NDT before it is subjected to pressure. During construction, nondestructive testing should be used to detect dangerous cracks or flaws. Section VIII of the ASME Code, particularly Division 2, gives design and testing techniques.

As the size of a forged vessel increases, the sizes of ingot and handling equipment become larger. The cost may increase faster than the weight. The problems of getting sound material and avoiding brittle fracture also become more difficult. Some of these problems are avoided by use of **multilayer construction**. In this type of vessel, the

heads and flanges are made of forgings, and the cylindrical portion is built up by a series of layers of thin material. The thickness of these layers may be between 3 and 50 mm ( $\frac{1}{8}$  and 2 in), depending on the type of construction. There is an inner lining which may be different from the outer layers.

Although there are multilayer vessels as small as 380-mm (15-in) inside diameter and 2400 mm (8 ft) long, their principal advantage applies to the larger sizes. When properly made, a multilayer vessel is probably safer than a vessel with a solid wall. The layers of thin material are tougher and less susceptible to brittle fracture, have less probability of defects, and have the statistical advantage of a number of small elements instead of a single large one. The heads, flanges, and welds, of course, have the same hazards as other thick members. Proper attention is necessary to avoid cracks in these members.

There are several assembly techniques. One frequently used is to form successive layers in half cylinders and butt-weld them over the previous layers. In doing this, the welds are staggered so that they do not fall together. This type of construction usually uses plates from 6 to 12 mm ( $\frac{1}{4}$  to  $\frac{1}{2}$  in) thick. Another method is to weld each layer separately to form a cylinder and then shrink it over the previous layers. Layers up to about 50-mm (2-in) thickness are assembled in this way. A third method of fabrication is to wind the layers as a continuous sheet. This technique is used in Japan. The Wickel construction, fabricated in Germany, uses helical winding of interlocking metal strip. Each method has its advantages and disadvantages, and choice will depend upon circumstances.

Because of the possibility of voids between layers, it is preferable not to use multilayer vessels in applications where they will be subjected to fatigue. Inward thermal gradients (inside temperature lower than outside temperature) are also undesirable.

Articles on these vessels have been written by Fratcher [*Pet. Refiner*, 34(11), 137 (1954)] and by Strelzoff, Pan, and Miller [*Chem. Eng.*, 75(21), 143-150 (1968)].

**Vessels for high-temperature service** may be beyond the temperature limits of the stress tables in the ASME Codes. Section VIII, Division 1, makes provision for construction of pressure vessels up to 650°C (1200°F) for carbon and low-alloy steel and up to 815°C (1500°F) for stainless steels (300 series). If a vessel is required for temperatures above these values and above 103 kPa (15 lbf/in<sup>2</sup>), it would be necessary, in a code state, to get permission from the state authorities to build it as a special project. Above 815°C (1500°F), even the 300 series stainless steels are weak, and creep rates increase rapidly. If the metal which resists the pressure operates at these temperatures, the vessel pressure and size will be limited. The vessel must also be expendable because its life will be short. Long exposure to high temperature may cause the metal to deteriorate and become brittle. Sometimes, however, economics favor this type of operation.

One way to circumvent the problem of low metal strength is to use a metal inner liner surrounded by insulating material, which in turn is confined by a pressure vessel. The liner, in some cases, may have perforations which will allow pressure to pass through the insulation and act on the outer shell, which is kept cool to obtain normal strength. The liner has no pressure differential acting on it and, therefore, does not need much strength. Ceramic linings are also useful for high-temperature work.

**Lined vessels** are used for many applications. Any type of lining can be used in an ASME Code vessel, provided it is compatible with the metal of the vessel and the contents. Glass, rubber, plastics, rare metals, and ceramics are a few types. The lining may be installed separately, or if a metal is used, it may be in the form of clad plate. The cladding on plate can sometimes be considered as a stress-carrying part of the vessel.

A **ceramic lining** when used with high temperature acts as an insulator so that the steel outer shell is at a moderate temperature while the temperature at the inside of the lining may be very high. Ceramic linings may be of unstressed brick, or prestressed brick, or cast in place. Cast ceramic linings or unstressed brick may develop cracks and are used when the contents of the vessel will not damage the outer shell. They are usually designed so that the high temperature at the inside will expand them sufficiently to make them tight in the outer (and cooler) shell. This, however, is not usually sufficient to prevent some penetration by the product.

Prestressed-brick linings can be used to protect the outer shell. In this case, the bricks are installed with a special thermosetting-resin mortar. After lining, the vessel is subjected to internal pressure and heat. This expands the steel vessel shell, and the mortar expands to take up the space. The pressure and temperature must be at least as high as the maximum that will be encountered in service. After the mortar has set, reduction of pressure and temperature will allow the vessel to contract, putting the brick in compression. The upper temperature limit for this construction is about 190°C (375°F). The installation of such linings is highly specialized work done by a few companies. Great care is usually exercised in operation to protect the vessel from exposure to unsymmetrical temperature gradients. Side nozzles and other unsymmetrical designs are avoided insofar as possible.

**Concrete pressure vessels** may be used in applications that require large sizes. Such vessels, if made of steel, would be too large and heavy to ship. Through the use of posttensioned (prestressed) concrete, the vessel is fabricated on the site. In this construction, the reinforcing steel is placed in tubes or plastic covers, which are cast into the concrete. Tension is applied to the steel after the concrete has acquired most of its strength.

Concrete nuclear reactor vessels, of the order of magnitude of 15-m (50-ft) inside diameter and length, have inner linings of steel which confine the pressure. After fabrication of the liner, the tubes for the cables or wires are put in place and the concrete is poured. High-strength reinforcing steel is used. Because there are thousands of reinforcing tendons in the concrete vessel, there is a statistical factor of safety. The failure of 1 or even 10 tendons would have little effect on the overall structure.

**Plastic pressure vessels** have the *advantages of chemical resistance* and light weight. Above 103 kPa (15 lbf/in<sup>2</sup>), with certain exceptions, they must be designed according to the ASME Code Section X (see "Storage of Gases") and are confined to the three types of approved code construction. Below 103 kPa (15 lbf/in<sup>2</sup>), any construction may be used. Even in this pressure range, however, the code should be used for guidance. Solid plastics, because of low strength and creep, can be used only for the lowest pressures and sizes. A stress of a few hundred pounds-force per square inch is the maximum for most plastics. To obtain higher strength, the filled plastics or filament-wound vessels, specified by the code, must be used. Solid-plastic parts, however, are often employed inside a steel shell, particularly for heat exchangers.

**Graphite and ceramic vessels** are used fully armored; that is, they are enclosed within metal pressure vessels. These materials are also used for boxlike vessels with backing plates on the sides. The plates are drawn together by tie bolts, thus putting the material in compression so that it can withstand low pressure.

**ASME Code Developments** At the time of this publication, ASME Section VIII is currently being rewritten and reorganized into three classes. Class 1 will be for low-pressure vessels employing spot radiography. Class 2 will be for vessels requiring full radiography. Class 3 will be for vessels experiencing fatigue. This rewriting effort is also using material stress levels similar to those of competing vessel codes from Europe and Asia.

**Vessel Codes Other than ASME** Different design and construction rules are used in other countries. Chemical engineers concerned with pressure vessels outside the United States must become familiar with local pressure-vessel laws and regulations. *Boilers and Pressure Vessels*, an international survey of design and approval requirements published by the British Standards Institution, Maylands Avenue, Hemel Hempstead, Hertfordshire, England, in 1975, gives pertinent information for 76 political jurisdictions.

The British Code (British Standards) and the German Code (*A. D. Merkblätter*) in addition to the ASME Code are most commonly permitted, although Netherlands, Sweden, and France also have codes. The major difference between the codes lies in factors of safety and in whether or not ultimate strength is considered. ASME Code, Sec. VIII, Division 1, vessels are generally heavier than vessels built to the other codes; however, the differences in allowable stress for a given material are less in the higher temperature (creep) range.

Engineers and metallurgists have developed alloys to comply economically with individual codes. In Germany, where design stress is determined from yield strength and creep-rupture strength and no

allowance is made for ultimate strength, steels which have a very high yield-strength-to-ultimate-strength ratio are used.

Other differences between codes include different bases for the design of reinforcement for openings and the design of flanges and heads. Some codes include rules for the design of heat-exchanger tube sheets, while others (ASME Code) do not. The Dutch Code (*Grondslagen*) includes very specific rules for calculation of wind loads, while the ASME Code leaves this entirely to the designer.

There are also significant differences in construction and inspection rules. Unless engineers make a detailed study of the individual codes and keep current, they will be well advised to make use of responsible experts for any of the codes.

**Vessel Design and Construction** The ASME Code lists a number of loads that must be considered in designing a pressure vessel. Among them are impact, weight of the vessel under operating and test conditions, superimposed loads from other equipment and piping, wind and earthquake loads, temperature-gradient stresses, and localized loadings from internal and external supports. In general, the code gives no values for these loads or methods for determining them, and no formulas are given for determining the stresses from these loads. Engineers must be knowledgeable in mechanics and strength of materials to solve these problems.

Some of the problems are treated by Brownell and Young, *Process Equipment Design*, Wiley, New York, 1959. ASME papers treat others, and a number of books published by the ASME are collections of papers on pressure-vessel design: *Pressure Vessels and Piping Design: Collected Papers, 1927-1959*; *Pressure Vessels and Piping Design and Analysis*, four volumes; and *International Conference: Pressure Vessel Technology*, published annually.

Throughout the year the Welding Research Council publishes bulletins which are final reports from projects sponsored by the council, important papers presented before engineering societies, and other reports of current interest which are not published in *Welding Research*. A large number of the published bulletins are pertinent for vessel designers.

**Care of Pressure Vessels** Protection against **excessive pressure** is largely taken care of by code requirements for relief devices. Exposure to fire is also covered by the code. The code, however, does not provide for the possibility of local overheating and weakening of a vessel in a fire. Insulation reduces the required relieving capacity and also reduces the possibility of local overheating.

A pressure-reducing valve in a line leading to a pressure vessel is not adequate protection against overpressure. Its failure will subject the vessel to full line pressure.

Vessels that have an operating cycle which involves the solidification and remelting of solids can develop excessive pressures. A solid plug of material may seal off one end of the vessel. If heat is applied at that end to cause melting, the expansion of the liquid can build up a high pressure and possibly result in yielding or rupture. Solidification in connecting piping can create similar problems.

Some vessels may be exposed to a runaway chemical reaction or even an explosion. This requires relief valves, rupture disks, or, in extreme cases, a frangible roof design or barricade (the vessel is expendable). A vessel with a large rupture disk needs anchors designed for the jet thrust when the disk blows.

**Vacuum** must be considered. It is nearly always possible that the contents of a vessel might contract or condense sufficiently to subject it to an internal vacuum. If the vessel cannot withstand the vacuum, it must have vacuum-breaking valves.

Improper operation of a process may result in the vessel's **exceeding design temperature**. Proper control is the only solution to this problem. Maintenance procedures can also cause excessive temperatures. Sometimes the contents of a vessel may be burned out with torches. If the flame impinges on the vessel shell, overheating and damage may occur.

**Excessively low temperature** may involve the hazard of brittle fracture. A vessel that is out of use in cold weather could be at a sub-zero temperature and well below its nil-ductility temperature. In startup, the vessel should be warmed slowly and uniformly until it is above the NDT. A safe value is 38°C (100°F) for plate if the NDT is unknown. The vessel should not be pressurized until this temperature is exceeded. Even after the NDT has been passed, excessively rapid heating or cooling can cause high thermal stresses.



**Corrosion** is probably the greatest threat to vessel life. Partially filled vessels frequently have severe pitting at the liquid-vapor interface. Vessels usually do not have a corrosion allowance on the outside. Lack of protection against the weather or against the drip of corrosive chemicals can reduce vessel life. Insulation may contain damaging substances. Chlorides in insulating materials can cause cracking of stainless steels. Water used for hydrotesting should be free of chlorides.

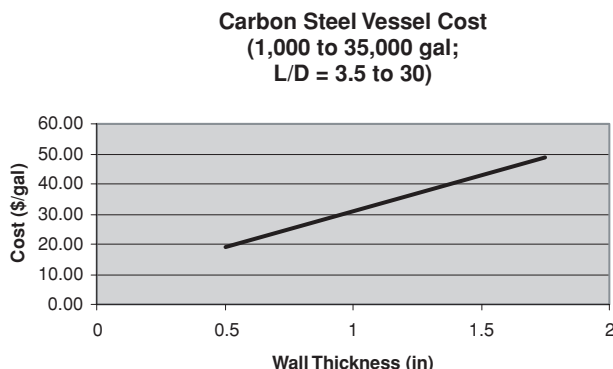
Pressure vessels should be **inspected periodically**. No rule can be given for the frequency of these inspections. Frequency depends on operating conditions. If the early inspections of a vessel indicate a low corrosion rate, intervals between inspections may be lengthened. Some vessels are inspected at 5-year intervals; others, as frequently as once a year. Measurement of corrosion is an important inspection item. One of the most convenient ways of measuring thickness (and corrosion) is to use an ultrasonic gauge. The location of the corrosion and whether it is uniform or localized in deep pits should be observed and reported. Cracks, any type of distortion, and leaks should be observed. Cracks are particularly dangerous because they can lead to sudden failure. Insulation is usually left in place during inspection of insulated vessels. If, however, severe external corrosion is suspected, the insulation should be removed. All forms of nondestructive testing are useful for examinations.

There are many ways in which a pressure vessel can suffer **mechanical damage**. The shells can be dented or even punctured, they can be dropped or have hoisting cables improperly attached, bolts can be broken, flanges are bent by excessive bolt tightening, gasket contact faces can be scratched and dented, rotating paddles can drag against the shell and cause wear, and a flange can be bolted up with a gasket half in the groove and half out. Most of these forms of damage can be prevented by care and common sense. If damage is repaired by straightening, as with a dented shell, it may be necessary to stress-relieve the repaired area. Some steels are susceptible to embrittlement by aging after severe straining. A safer procedure is to cut out the damaged area and replace it.

The National Board Inspection Code, published by the National Board of Boiler and Pressure Vessel Inspectors, Columbus, Ohio, is helpful. Any repair, however, is acceptable if it is made in accordance with the rules of the Pressure Vessel Code.

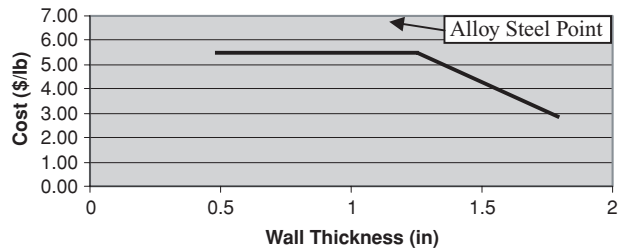
Care in **reassembling** the vessel is particularly important. Gaskets should be properly located, particularly if they are in grooves. Bolts should be tightened in proper sequence. In some critical cases and with large bolts, it is necessary to control bolt tightening by torque wrenches, micrometers, patented bolt-tightening devices, or heating bolts. After assembly, vessels are sometimes given a hydrostatic test.

**Pressure-Vessel Cost and Weight** Figure 10-196 can be used for estimating carbon-steel vessel cost when a weight estimate is not available and Fig. 10-197 with a weight estimate. Weight and cost include skirts and other supports. The cost is based on several 2005 pressure-vessels. Costs are for vessels not of unusual design. Complicated vessels could cost considerably more. Guthrie [*Chem. Eng.*, 76(6), 114-142 (1969)] also gives pressure-vessel cost data.



**FIG. 10-196** Carbon-steel pressure-vessel cost as a function of wall thickness. 1 gal = 0.003875 cm<sup>3</sup>; 1 in = 0.0254 m. (Courtesy of E. S. Fox, Ltd.)

### Carbon Steel Vessel Cost (1,000 to 35,000 gal; L/D = 3.5 to 30)



**FIG. 10-197** Carbon-steel pressure-vessel cost as a function of wall thickness. 1 gal = 0.003875 cm<sup>3</sup>; 1 in = 0.0254 m; 1 lb = 0.4536 kg. (Courtesy of E. S. Fox, Ltd.)

When vessels have complicated construction (large, heavy bolted connections, support skirts, etc.), it is preferable to estimate their weight and apply a unit cost in dollars per pound.

Pressure-vessel weights are obtained by calculating the cylindrical shell and heads separately and then adding the weights of nozzles and attachments. Steel weighs 7817 kg/m<sup>3</sup> (488 lb/ft<sup>3</sup>). Metal in heads can be approximated by calculating the area of the blank (disk) used for forming the head. The required diameter of blank can be calculated by multiplying the head outside diameter by the approximate factors given in Table 10-62. These factors make no allowance for the straight flange which is a cylindrical extension that is formed on the head. The blank diameter obtained from these factors must be increased by twice the length of straight flange, which is usually 1½ to 2 in but can be up to several inches in length. Manufacturers' catalogs give weights of heads.

Forming a head thins it in certain areas. To obtain the required minimum thickness of a head, it is necessary to use a plate that is initially thicker. Table 10-63 gives allowances for additional thickness.

Nozzles and flanges may add considerably to the weight of a vessel. Their weights can be obtained from manufacturers' catalogs (Taylor Forge Division of Gulf & Western Industries, Inc., Tube Turns Inc., Ladish Co., Lenape Forge, and others). Other parts such as skirts, legs, support brackets, and other details must be calculated.

**TABLE 10-62 Factors for Estimating Diameters of Blanks for Formed Heads**

|                    | Ratio $d/t$ | Blank diameter factor |
|--------------------|-------------|-----------------------|
| ASME head          | Over 50     | 1.09                  |
|                    | 30-50       | 1.11                  |
|                    | 20-30       | 1.15                  |
| Ellipsoidal head   | Over 20     | 1.24                  |
|                    | 10-20       | 1.30                  |
| Hemispherical head | Over 30     | 1.60                  |
|                    | 18-30       | 1.65                  |
|                    | 10-18       | 1.70                  |

$d$  = head diameter

$t$  = nominal minimum head thickness

**TABLE 10-63 Extra Thickness Allowances for Formed Heads\***

| Minimum head thickness, in | Extra thickness, in          |                       |               |
|----------------------------|------------------------------|-----------------------|---------------|
|                            | ASME and ellipsoidal         |                       | Hemispherical |
|                            | Head o.d. up to 150 in incl. | Head o.d. over 150 in |               |
| Up to 0.99                 | 1/16                         | 1/8                   | 3/16          |
| 1 to 1.99                  | 1/8                          | 1/4                   | 3/8           |
| 2 to 2.99                  | 1/4                          | 1/2                   | 3/4           |

\*Lukens, Inc.



SECTION 11

Heat-Transfer  
Equipment

# PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



RICHARD L. SHILLING, PATRICK M. BERNHAGEN  
VICTOR M. GOLDSCHMIDT, PREDRAG S. HRNJAK  
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# THERMAL DESIGN OF HEAT-TRANSFER EQUIPMENT

## INTRODUCTION TO THERMAL DESIGN

Designers commonly use computer software to design heat exchangers. The best sources of such software are Heat Transfer Research, Inc. (HTRI), and Heat Transfer and Fluid Flow Services (HTFS), a division of ASPENTECH. These are companies that develop proprietary correlations based on their research and provide software that utilizes these correlations. However, it is important that engineers understand the fundamental principles that lie beneath the framework of the software. Therefore, design methods for several important classes of process heat-transfer equipment are presented in the following portions of Sec. 11. Mechanical descriptions and specifications of equipment are given in this section and should be read in conjunction with the use of this material. It is impossible to present here a comprehensive treatment of heat-exchanger selection, design, and application. The best general references in this field are Hewitt, Shires, and Bott, *Process Heat Transfer*, CRC Press, Boca Raton, FL, 1994; and Schlönder (ed.), *Heat Exchanger Design Handbook*, Begell House, New York, 2002.

**Approach to Heat-Exchanger Design** The proper use of basic heat-transfer knowledge in the design of practical heat-transfer equipment is an art. Designers must be constantly aware of the differences between the idealized conditions for and under which the basic knowledge was obtained and the real conditions of the mechanical expression of their design and its environment. The result must satisfy process and operational requirements (such as availability, flexibility, and maintainability) and do so economically. An important part of any design process is to consider and offset the consequences of error in the basic knowledge, in its subsequent incorporation into a design method, in the translation of design into equipment, or in the operation of the equipment and the process. Heat-exchanger design is not a highly accurate art under the best of conditions.

The design of a process heat exchanger usually proceeds through the following steps:

1. Process conditions (stream compositions, flow rates, temperatures, pressures) must be specified.
2. Required physical properties over the temperature and pressure ranges of interest must be obtained.
3. The type of heat exchanger to be employed is chosen.
4. A preliminary estimate of the size of the exchanger is made, using a heat-transfer coefficient appropriate to the fluids, the process, and the equipment.
5. A first design is chosen, complete in all details necessary to carry out the design calculations.
6. The design chosen in step 5 is evaluated, or *rated*, as to its ability to meet the process specifications with respect to both heat transfer and pressure drop.
7. On the basis of the result of step 6, a new configuration is chosen if necessary and step 6 is repeated. If the first design was inadequate to meet the required heat load, it is usually necessary to increase the size of the exchanger while still remaining within specified or feasible limits of pressure drop, tube length, shell diameter, etc. This will sometimes mean going to multiple-exchanger configurations. If the first design more than meets heat-load requirements or does not use all the allowable pressure drop, a less expensive exchanger can usually be designed to fulfill process requirements.
8. The final design should meet process requirements (within reasonable expectations of error) at lowest cost. The lowest cost should include operation and maintenance costs and credit for ability to meet long-term process changes, as well as installed (capital) cost. Exchangers should not be selected entirely on a lowest-first-cost basis, which frequently results in future penalties.

**Overall Heat-Transfer Coefficient** The basic design equation for a heat exchanger is

$$dA = dQ/U \Delta T \quad (11-1)$$

where  $dA$  is the element of surface area required to transfer an amount of heat  $dQ$  at a point in the exchanger where the overall heat-transfer coefficient is  $U$  and where the overall bulk temperature difference between the two streams is  $\Delta T$ . The overall heat-transfer coefficient is related to the individual film heat-transfer coefficients and fouling and wall resistances by Eq. (11-2). Basing  $U_o$  on the outside surface area  $A_o$  results in

$$U_o = \frac{1}{1/h_o + R_{do} + xA_o/k_w A_{wm} + (1/h_i + R_{di})A_o/A_i} \quad (11-2)$$

Equation (11-1) can be formally integrated to give the outside area required to transfer the total heat load  $Q_T$ :

$$A_o = \int_0^{Q_T} \frac{dQ}{U_o \Delta T} \quad (11-3)$$

To integrate Eq. (11-3),  $U_o$  and  $\Delta T$  must be known as functions of  $Q$ . For some problems,  $U_o$  varies strongly and nonlinearly throughout the exchanger. In these cases, it is necessary to evaluate  $U_o$  and  $\Delta T$  at several intermediate values and numerically or graphically integrate. For many practical cases, it is possible to calculate a constant mean overall coefficient  $U_{om}$  from Eq. (11-2) and define a corresponding mean value of  $\Delta T_m$ , such that

$$A_o = Q_T/U_{om} \Delta T_m \quad (11-4)$$

Care must be taken that  $U_o$  does not vary too strongly, that the proper equations and conditions are chosen for calculating the individual coefficients, and that the mean temperature difference is the correct one for the specified exchanger configuration.

**Mean Temperature Difference** The temperature difference between the two fluids in the heat exchanger will, in general, vary from point to point. The mean temperature difference ( $\Delta T_m$  or MTD) can be calculated from the terminal temperatures of the two streams if the following assumptions are valid:

1. All elements of a given fluid stream have the same thermal history in passing through the exchanger.\*
2. The exchanger operates at steady state.
3. The specific heat is constant for each stream (or if either stream undergoes an isothermal phase transition).
4. The overall heat-transfer coefficient is constant.
5. Heat losses are negligible.

**Countercurrent or Cocurrent Flow** If the flow of the streams is either *completely* countercurrent or *completely* cocurrent or if one or both streams are isothermal (condensing or vaporizing a pure component with negligible pressure change), the correct MTD is the logarithmic-mean temperature difference (LMTD), defined as

$$\text{LMTD} = \Delta T_m = \frac{(t'_1 - t''_2) - (t'_2 - t''_1)}{\ln \left( \frac{t'_1 - t''_2}{t'_2 - t''_1} \right)} \quad (11-5a)$$

for *countercurrent flow* (Fig. 11-1a) and

$$\text{LMTD} = \Delta T_m = \frac{(t'_1 - t''_1) - (t'_2 - t''_2)}{\ln \left( \frac{t'_1 - t''_1}{t'_2 - t''_2} \right)} \quad (11-5b)$$

for *cocurrent flow* (Fig. 11-1b)

\*This assumption is vital but is usually omitted or less satisfactorily stated as "each stream is well mixed at each point." In a heat exchanger with substantial bypassing of the heat-transfer surface, e.g., a typical baffled shell-and-tube exchanger, this condition is not satisfied. However, the error is in some degree offset if the same MTD formulation used in reducing experimental heat-transfer data to obtain the basic correlation is used in applying the correlation to design a heat exchanger. The compensation is not in general exact, and insight and judgment are required in the use of the MTD formulations. Particularly, in the design of an exchanger with a very close temperature approach, bypassing may result in an exchanger that is inefficient and even thermodynamically incapable of meeting specified outlet temperatures.

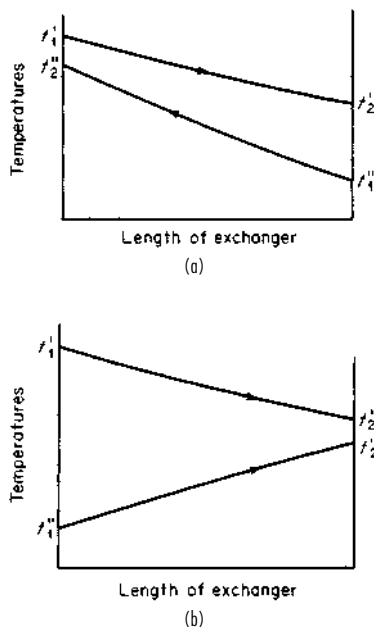


FIG. 11-1 Temperature profiles in heat exchangers. (a) Countercurrent. (b) Cocurrent.

If  $U$  is not constant but a linear function of  $\Delta T$ , the correct value of  $U_{om}\Delta T_m$  to use in Eq. (11-4) is [Colburn, *Ind. Eng. Chem.*, **25**, 873 (1933)]

$$U_{om}\Delta T_m = \frac{U_o''(t_1' - t_2'') - U_o'(t_2' - t_1'')}{\ln \left( \frac{U_o''(t_1' - t_2'')}{U_o'(t_2' - t_1'')} \right)} \quad (11-6a)$$

for countercurrent flow, where  $U_o''$  is the overall coefficient evaluated when the stream temperatures are  $t_1'$  and  $t_2''$  and  $U_o'$  is evaluated at  $t_2'$  and  $t_1''$ . The corresponding equation for cocurrent flow is

$$U_{om}\Delta T_m = \frac{U_o''(t_1' - t_1'') - U_o'(t_2' - t_2'')}{\ln \left( \frac{U_o''(t_1' - t_1'')}{U_o'(t_2' - t_2'')} \right)} \quad (11-6b)$$

where  $U_o'$  is evaluated at  $t_2'$  and  $t_2''$  and  $U_o''$  is evaluated at  $t_1'$  and  $t_1''$ . To use these equations, it is necessary to calculate two values of  $U_o$ .

The use of Eq. (11-6) will frequently give satisfactory results even if  $U_o$  is not strictly linear with temperature difference.

**Reversed, Mixed, or Cross-Flow** If the flow pattern in the exchanger is not completely countercurrent or cocurrent, it is necessary to apply a **correction factor**  $F_T$  by which the LMTD is multiplied to obtain the appropriate MTD. These corrections have been mathematically derived for flow patterns of interest, still by making assumptions 1 to 5 [see Bowman, Mueller, and Nagle, *Trans. Am. Soc. Mech. Eng.*, **62**, 283 (1940) or Hewitt, et al. op. cit.]. For a common flow pattern, the 1-2 exchanger (Fig. 11-2), the correction factor  $F_T$  is given in Fig. 11-4a, which is also valid for finding  $F_T$  for a 1-2 exchanger in which the shell-side flow direction is reversed from that shown in Fig. 11-2. Figure 11-4a is also applicable with negligible error to exchangers with one shell pass and any number of tube passes. Values of  $F_T$  less than 0.8 (0.75 at the very lowest) are generally unacceptable because the exchanger configuration chosen is inefficient; the chart is difficult to read accurately; and even a small violation of the first assumption

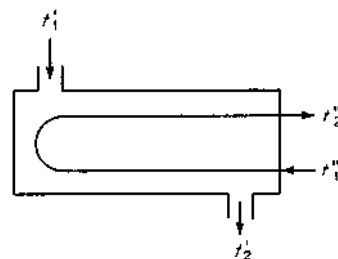


FIG. 11-2 Diagram of a 1-2 exchanger (one well-baffled shell pass and two tube passes with an equal number of tubes in each pass).

underlying the MTD will invalidate the mathematical derivation and lead to a thermodynamically inoperable exchanger.

Correction-factor charts are also available for exchangers with more than one shell pass provided by a longitudinal shell-side baffle. However, these exchangers are seldom used in practice because of mechanical complications in their construction. Also thermal and physical leakages across the longitudinal baffle further reduce the mean temperature difference and are not properly incorporated into the correction-factor charts. Such charts are useful, however, when it is necessary to construct a multiple-shell exchanger train such as that shown in Fig. 11-3 and are included here for two, three, four, and six separate identical shells and two or more tube passes per shell in Fig. 11-4b, c, d, and e. If only one tube pass per shell is required, the piping can and should be arranged to provide pure countercurrent flow, in which case the LMTD is used with no correction.

Cross-flow exchangers of various kinds are also important and require correction to be applied to the LMTD calculated by assuming countercurrent flow. Several cases are given in Fig. 11-4f, g, h, i, and j.

Many other MTD correction-factor charts have been prepared for various configurations. The  $F_T$  charts are often employed to make approximate corrections for configurations even in cases for which they are not completely valid.

## THERMAL DESIGN FOR SINGLE-PHASE HEAT TRANSFER

**Double-Pipe Heat Exchangers** The design of double-pipe heat exchangers is straightforward. It is generally conservative to

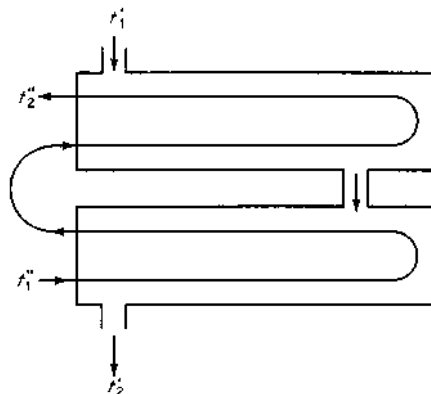
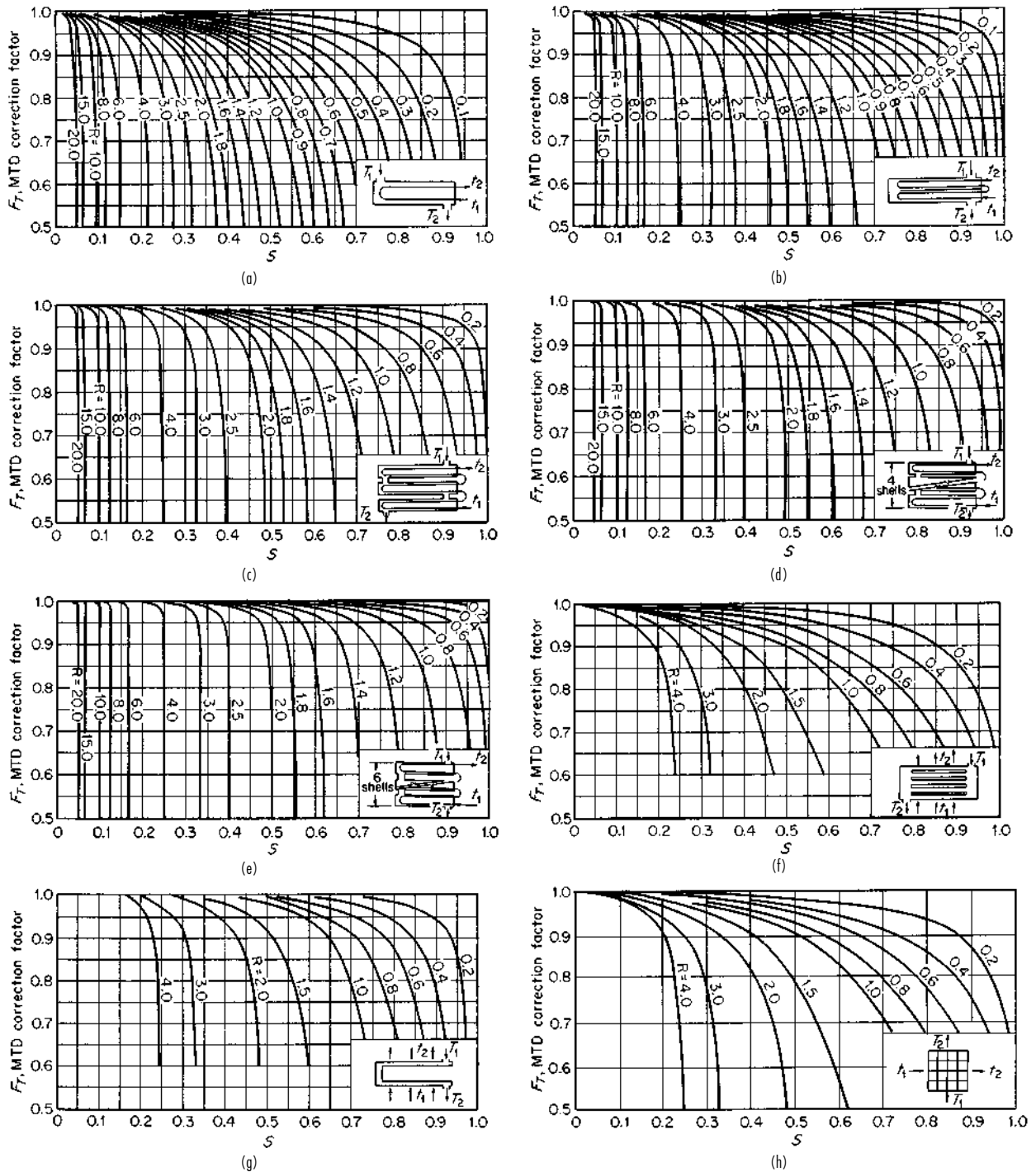


FIG. 11-3 Diagram of a 2-4 exchanger (two separate identical well-baffled shells and four or more tube passes).

\*This task can be avoided if a hydrocarbon stream is the limiting resistance by the use of the caloric temperature charts developed by Colburn [*Ind. Eng. Chem.*, **25**, 873 (1933)].



**FIG. 11-4** LMTD correction factors for heat exchangers. In all charts,  $R = (T_1 - T_2)/(t_2 - t_1)$  and  $S = (t_2 - t_1)/(T_1 - t_1)$ . (a) One shell pass, two or more tube passes. (b) Two shell passes, four or more tube passes. (c) Three shell passes, six or more tube passes. (d) Four shell passes, eight or more tube passes. (e) Six shell passes, twelve or more tube passes. (f) Cross-flow, one shell pass, one or more parallel rows of tubes. (g) Cross-flow, two passes, two rows of tubes; for more than two passes, use  $F_T = 1.0$ . (h) Cross-flow, one shell pass, one tube pass, both fluids unmixed. (i) Cross-flow (drip type), two horizontal passes with U-bend connections (trombone type). (j) Cross-flow (drip type), helical coils with two turns.



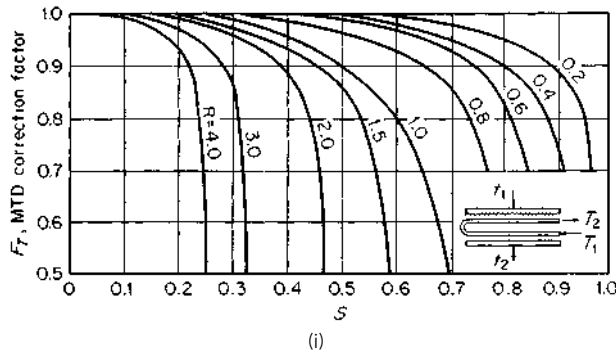
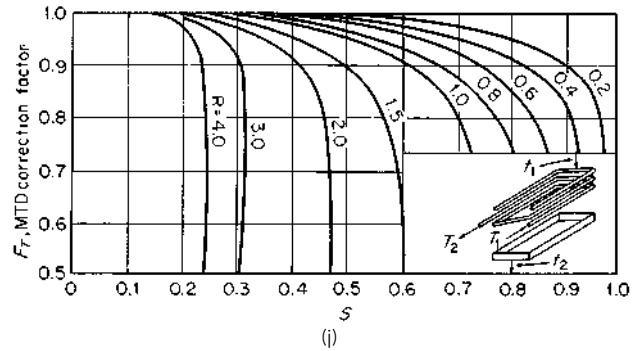


FIG. 11-4 (Continued)



neglect natural-convection and entrance effects in turbulent flow. In laminar flow, natural convection effects can increase the theoretical Graetz prediction by a factor of 3 or 4 for fully developed flows. Pressure drop is calculated by using the correlations given in Sec. 6.

If the inner tube is longitudinally finned on the outside surface, the equivalent diameter is used as the characteristic length in both the Reynolds-number and the heat-transfer correlations. The fin efficiency must also be known to calculate an *effective* outside area to use in Eq. (11-2).

Fittings contribute strongly to the pressure drop on the annulus side. General methods for predicting this are not reliable, and manufacturer's data should be used when available.

Double-pipe exchangers are often piped in complex series-parallel arrangements on both sides. The MTD to be used has been derived for some of these arrangements and is reported in Kern (*Process Heat Transfer*, McGraw-Hill, New York, 1950). More complex cases may require trial-and-error balancing of the heat loads and rate equations for subsections or even for individual exchangers in the bank.

**Baffled Shell-and-Tube Exchangers** The method given here is based on the research summarized in Final Report, Cooperative Research Program on Shell and Tube Heat Exchangers, Univ. Del. Eng. Exp. Sta. Bull. 5 (June 1963). The method assumes that the shell-side heat transfer and pressure-drop characteristics are equal to those of the ideal tube bank corresponding to the cross-flow sections of the exchanger, modified for the distortion of flow pattern introduced by the baffles and the presence of leakage and bypass flow through the various clearances required by mechanical construction.

It is assumed that process conditions and physical properties are known and the following are known or specified: tube outside diameter  $D_o$ , tube geometrical arrangement (unit cell), shell inside diameter  $D_s$ , shell outer tube limit  $D_{ot}$ , baffle cut  $l_c$ , baffle spacing  $l_s$ , and number of sealing strips  $N_s$ . The effective tube length between tube sheets  $L$  may be either specified or calculated after the heat-transfer coefficient has been determined. If additional specific information (e.g., tube-baffle clearance) is available, the exact values (instead of estimates) of certain parameters may be used in the calculation with some improvement in accuracy. To complete the rating, it is necessary to know also the tube material and wall thickness or inside diameter.

This rating method, though apparently generally the best in the open literature, is not extremely accurate. An exhaustive study by Palen and Taborek [*Chem. Eng. Prog. Symp. Ser. 92*, 65, 53 (1969)] showed that this method predicted shell-side coefficients from about 50 percent low to 100 percent high, while the pressure-drop range was from about 50 percent low to 200 percent high. The mean error for heat transfer was about 15 percent low (safe) for all Reynolds numbers, while the mean error for pressure drop was from about 5 percent low (unsafe) at Reynolds numbers above 1000 to about 100 percent high at Reynolds numbers below 10.

#### Calculation of Shell-Side Geometrical Parameters

1. *Total number of tubes in exchanger  $N_t$ .* If not known by direct count, estimate using Eq. (11-74) or (11-75).

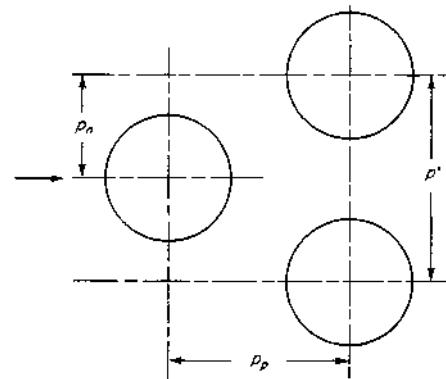
2. *Tube pitch parallel to flow  $p_p$  and normal to flow  $p_n$ .* These quantities are needed only for estimating other parameters. If a detailed drawing of the exchanger is available, it is better to obtain these other parameters by direct count or calculation. The pitches are described by Fig. 11-5 and read therefrom for common tube layouts.

3. *Number of tube rows crossed in one cross-flow section  $N_c$ .* Count from exchanger drawing or estimate from

$$N_c = \frac{D_s [1 - 2(l_c/D_s)]}{p_p} \quad (11-7)$$

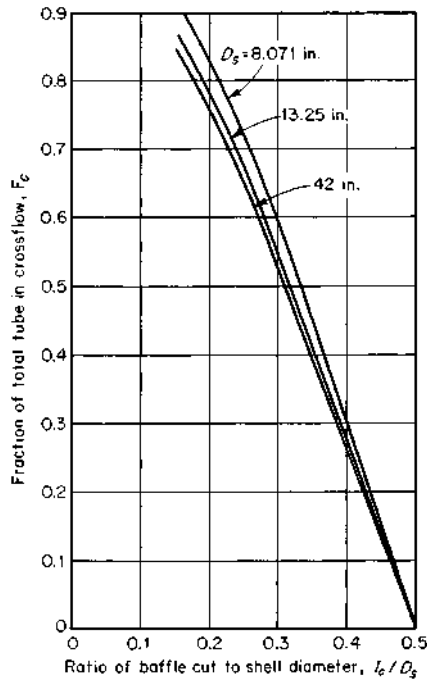
4. *Fraction of total tubes in cross-flow  $F_c$*

$$F_c = \frac{1}{\pi} \left[ \pi + 2 \frac{D_s - 2l_c}{D_{ot}} \sin \left( \cos^{-1} \frac{D_s - 2l_c}{D_{ot}} \right) - 2 \cos^{-1} \frac{D_s - 2l_c}{D_{ot}} \right] \quad (11-8)$$



| Tube O.D.<br>$D_o$ , in. | Tube pitch<br>$p'$ , in. | Layout | $p_p$ , in. | $p_n$ , in. |
|--------------------------|--------------------------|--------|-------------|-------------|
| 0.625                    | 0.812                    |        | 0.704       | 0.406       |
| 0.750                    | 0.938                    |        | 0.814       | 0.469       |
| 0.750                    | 1                        |        | 1.000       | 1.000       |
| 0.750                    | 1                        |        | 0.707       | 0.707       |
| 0.750                    | 1                        |        | 0.866       | 0.500       |
| 1.000                    | 1.250                    |        | 1.250       | 1.250       |
| 1.000                    | 1.250                    |        | 0.884       | 0.884       |
| 1.000                    | 1.250                    |        | 1.082       | 0.625       |

FIG. 11-5 Values of tube pitch for common tube layouts. To convert inches to meters, multiply by 0.0254. Note that  $D_o$ ,  $p'$ ,  $p_p$ , and  $p_n$  have units of inches.



**FIG. 11-6** Estimation of fraction of tubes in cross-flow  $F_c$  [Eq. (11-8)]. To convert inches to meters, multiply by 0.0254. Note that  $l_c$  and  $D_s$  have units of inches.

$F_c$  is plotted in Fig. 11-6. This figure is strictly applicable only to splitting, floating-head construction but may be used for other situations with minor error.

5. Number of effective cross-flow rows in each window  $N_{cw}$

$$N_{cw} = \frac{0.8l_c}{p_p} \quad (11-9)$$

6. Cross-flow area at or near centerline for one cross-flow section  $S_m$

a. For rotated and in-line square layouts:

$$S_m = l_s \left[ D_s - D_{otl} + \frac{D_{otl} - D_o}{p_n} (p' - D_o) \right] \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-10a)$$

b. For triangular layouts:

$$S_m = l_s \left[ D_s - D_{otl} + \frac{D_{otl} - D_o}{p'} (p' - D_o) \right] \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-10b)$$

7. Fraction of cross-flow area available for bypass flow  $F_{bp}$

$$F_{bp} = \frac{(D_s - D_{otl})l_s}{S_m} \quad (11-11)$$

8. Tube-to-baffle leakage area for one baffle  $S_{tb}$ . Estimate from

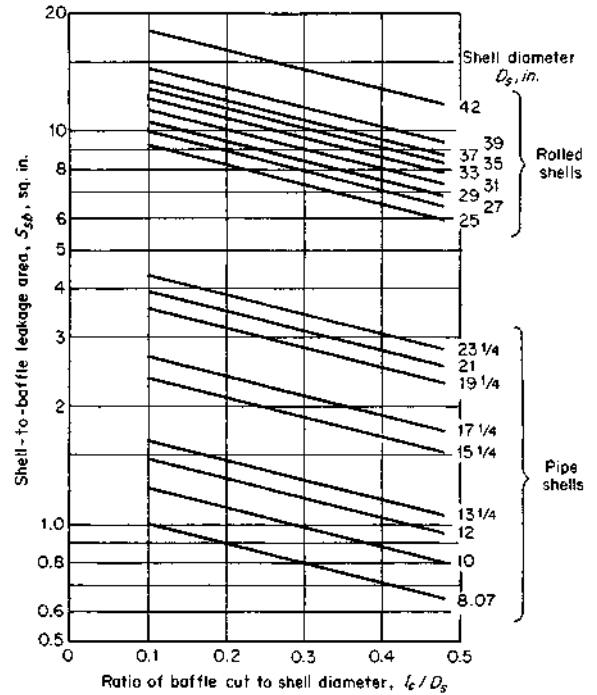
$$S_{tb} = bD_oN_T(1 + F_c) \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-12)$$

where  $b = (6.223)(10^{-4})$  (SI) or  $(1.701)(10^{-4})$  (U.S. customary). These values are based on Tubular Exchanger Manufacturers Association (TEMA) Class R construction which specifies  $1/32$ -in diametral clearance between tube and baffle. Values should be modified if extra tight or loose construction is specified or if clogging by dirt is anticipated.

9. Shell-to-baffle leakage area for one baffle  $S_{sb}$ . If diametral shell-baffle clearance  $\delta_{sb}$  is known,  $S_{sb}$  can be calculated from

$$S_{sb} = \frac{D_s \delta_{sb}}{2} \left[ \pi - \cos^{-1} \left( 1 - \frac{2l_c}{D_s} \right) \right] \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-13)$$

where the value of the term  $\cos^{-1} (1 - 2l_c/D_s)$  is in radians and is between 0 and  $\pi/2$ .  $S_{sb}$  is plotted in Fig. 11-7, based on TEMA Class R standards. Since pipe shells are generally limited to diameters



**FIG. 11-7** Estimation of shell-to-baffle leakage area [Eq. (11-13)]. To convert inches to meters, multiply by 0.0254; to convert square inches to square meters, multiply by  $(6.45)(10^{-4})$ . Note that  $l_c$  and  $D_s$  have units of inches.

below 24 in, the larger sizes are shown by using the rolled-shell specification. Allowance should be made for especially tight or loose construction.

10. Area for flow through window  $S_w$ . This area is obtained as the difference between the gross window area  $S_{wg}$  and the window area occupied by tubes  $S_{wt}$ :

$$S_w = S_{wg} - S_{wt} \quad (11-14)$$

$$S_{wg} = \frac{D_s^2}{4} \left[ \cos^{-1} \left( 1 - 2 \frac{l_c}{D_s} \right) - \left( 1 - 2 \frac{l_c}{D_s} \right) \sqrt{1 - \left( 1 - 2 \frac{l_c}{D_s} \right)^2} \right] \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-15)$$

$S_{wg}$  is plotted in Fig. 11-8.  $S_{wt}$  can be calculated from

$$S_{wt} = (N_T/8)(1 - F_c)\pi D_o^2 \text{ m}^2 \text{ (ft}^2\text{)} \quad (11-16)$$

11. Equivalent diameter of window  $D_w$  [required only if laminar flow, defined as  $(N_{Re})_s \leq 100$ , exists]

$$D_w = \frac{4S_w}{(\pi/2)N_T(1 - F_c)D_o + D_s\theta_b} \text{ m (ft)} \quad (11-17)$$

where  $\theta_b$  is the baffle-cut angle given by

$$\theta_b = 2 \cos^{-1} \left( 1 - \frac{2l_c}{D_s} \right) \text{ rad} \quad (11-18)$$

12. Number of baffles  $N_b$

$$N_b = \frac{L - 2le}{l_s} + 1 \quad (11-19)$$

where  $le$  is the entrance/exit baffle spacing, often different from the central baffle spacing. The effective tube length  $L$  must be known to calculate  $N_b$ , which is needed to calculate shell-side pressure drop. In designing an exchanger, the shell-side coefficient may be calculated and the required exchanger length for heat transfer obtained before  $N_b$  is calculated.

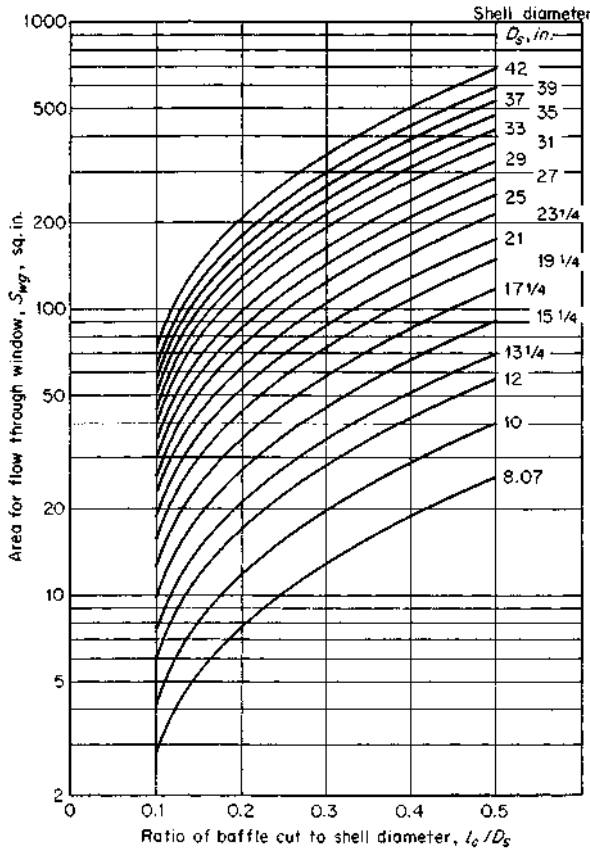


FIG. 11-8 Estimation of window cross-flow area [Eq. (11-15)]. To convert inches to meters, multiply by 0.0254. Note that  $l_c$  and  $D_s$  have units of inches.

### Shell-Side Heat-Transfer Coefficient Calculation

1. Calculate the shell-side Reynolds number  $(N_{Re})_s$ .

$$(N_{Re})_s = D_s W / \mu_b S_m \quad (11-20)$$

where  $W$  = mass flow rate and  $\mu_b$  = viscosity at bulk temperature. The arithmetic mean bulk shell-side fluid temperature is usually adequate to evaluate all bulk properties of the shell-side fluid. For large temperature ranges or for viscosity that is very sensitive to temperature change, special care must be taken, such as using Eq. (11-6).

2. Find  $j_k$  from the ideal-tube bank curve for a given tube layout at the calculated value of  $(N_{Re})_s$  using Fig. 11-9, which is adapted from

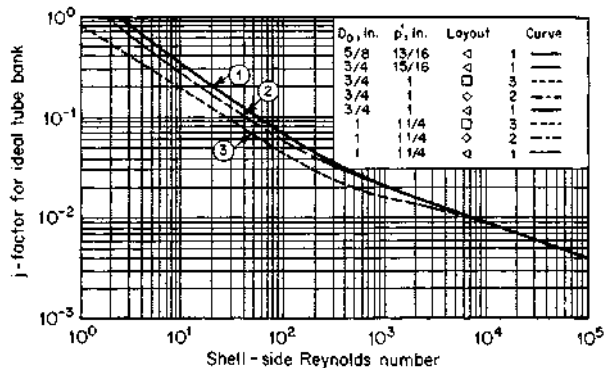


FIG. 11-9 Correlation of  $j$  factor for ideal tube bank. To convert inches to meters, multiply by 0.0254. Note that  $p'$  and  $D_s$  have units of inches.

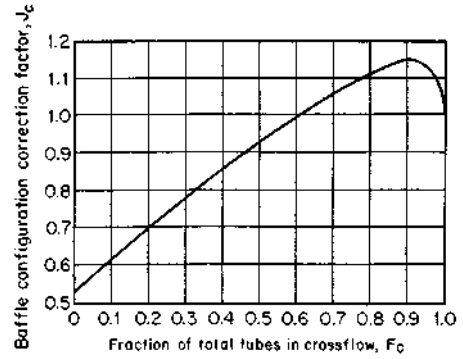


FIG. 11-10 Correction factor for baffle-configuration effects.

ideal-tube-bank data obtained at Delaware by Bergelin et al. [Trans. Am. Soc. Mech. Eng., **74**, 953 (1952) and the Grimson correlation [Trans. Am. Soc. Mech. Eng., **59**, 583 (1937)].

3. Calculate the shell-side heat-transfer coefficient for an ideal tube bank  $h_k$ .

$$h_k = j_k c \frac{W}{S_m} \left( \frac{k}{c\mu} \right)^{2/3} \left( \frac{\mu_b}{\mu_w} \right)^{0.14} \quad (11-21)$$

where  $c$  is the specific heat,  $k$  is the thermal conductivity, and  $\mu_w$  is the viscosity evaluated at the mean surface temperature.

4. Find the correction factor for baffle-configuration effects  $J_c$  from Fig. 11-10.

5. Find the correction factor for baffle-leakage effects  $J_l$  from Fig. 11-11.

6. Find the correction factor for bundle-bypassing effects  $J_b$  from Fig. 11-12

7. Find the correction factor for adverse temperature-gradient buildup at low Reynolds number  $J_r$ :

- a. If  $(N_{Re})_s < 100$ , find  $J_r^*$  from Fig. 11-13, knowing  $N_b$  and  $(N_c + N_{ce})$ .

- b. If  $(N_{Re})_s \leq 20$ ,  $J_r = J_r^*$ .

- c. If  $20 < (N_{Re})_s < 100$ , find  $J_r$  from Fig. 11-14, knowing  $J_r^*$  and  $(N_{Re})_s$ .

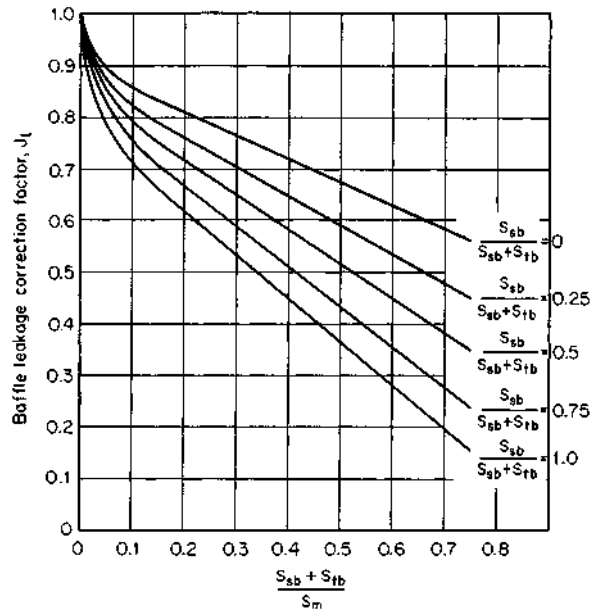


FIG. 11-11 Correction factor for baffle-leakage effects.

## 11-10 HEAT-TRANSFER EQUIPMENT

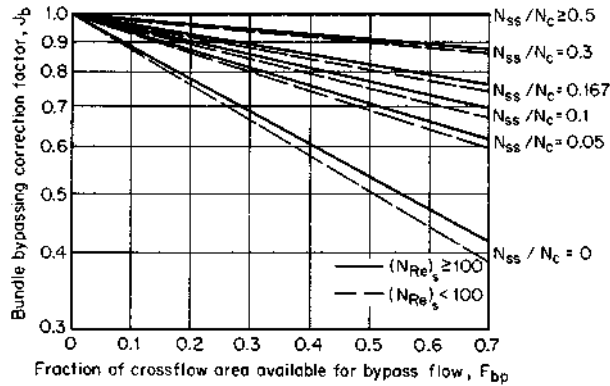


FIG. 11-12 Correction factor for bypass flow.

8. Calculate the *shell-side heat-transfer coefficient for the exchanger*  $h_s$  from

$$h_s = h_k J_c J_b J_r \quad (11-22)$$

### Shell-Side Pressure-Drop Calculation

1. Find  $f_k$  from the ideal-tube-bank friction-factor curve for the given tube layout at the calculated value of  $(N_{Re})_o$  using Fig. 11-15a for triangular and rotated square arrays and Fig. 11-15b for in-line square arrays. These curves are adapted from Bergelin et al. and Grimson (loc. cit.).

2. Calculate the *pressure drop for an ideal cross-flow section*.

$$\Delta P_{bk} = b \frac{f_k W^2 N_c}{\rho S_m^2} \left( \frac{\mu_w}{\mu_b} \right)^{0.14} \quad (11-23)$$

where  $b = (2.0)(10^{-3})$  (SI) or  $(9.9)(10^{-5})$  (U.S. customary).

3. Calculate the *pressure drop for an ideal window section*. If  $(N_{Re})_s \geq 100$ ,

$$\Delta P_{wk} = b \frac{W^2 (2 + 0.6 N_{cw})}{S_m S_w \rho} \quad (11-24a)$$

where  $b = (5)(10^{-4})$  (SI) or  $(2.49)(10^{-5})$  (U.S. customary).

If  $(N_{Re})_s < 100$ ,

$$\Delta P_{wk} = b_1 \frac{\mu_b W}{S_m S_w \rho} \left( \frac{N_{cw}}{p' - D_o} + \frac{l_s}{D_w^2} \right) + b_2 \frac{W^2}{S_m S_w \rho} \quad (11-24b)$$

where  $b_1 = (1.681)(10^{-5})$  (SI) or  $(1.08)(10^{-4})$  (U.S. customary), and  $b_2 = (9.99)(10^{-4})$  (SI) or  $(4.97)(10^{-5})$  (U.S. customary).

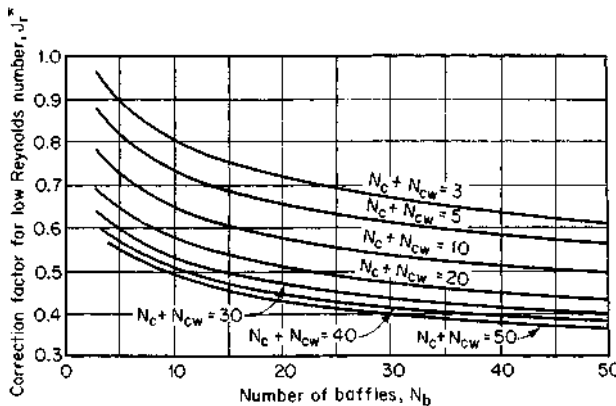


FIG. 11-13 Basic correction factor for adverse temperature gradient at low Reynolds numbers.

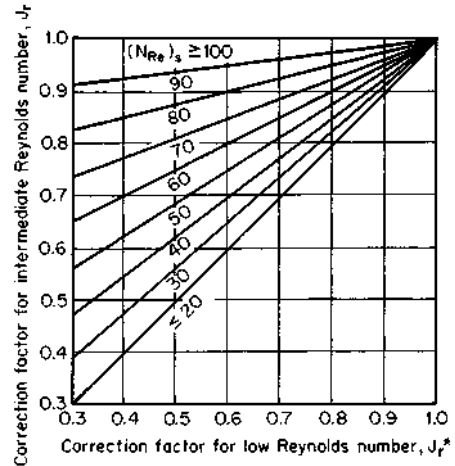
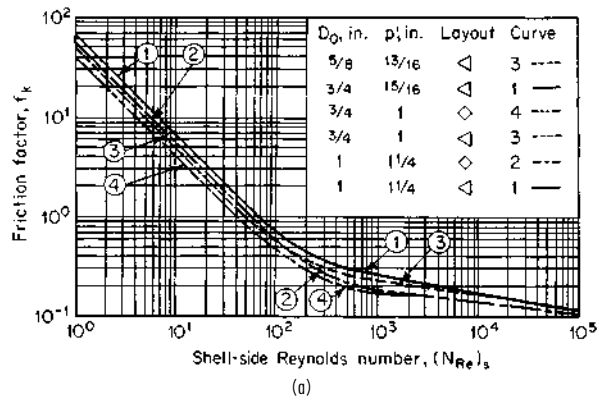


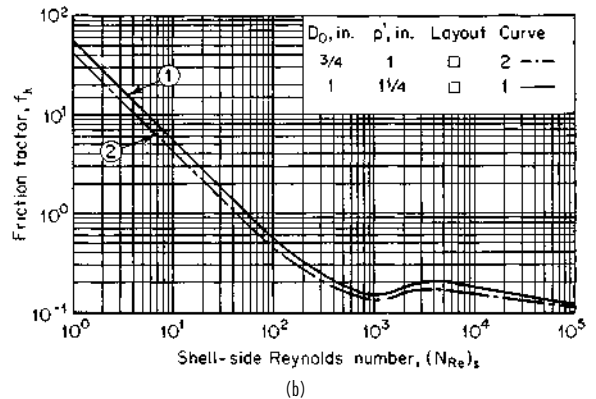
FIG. 11-14 Correction factor for adverse temperature gradient at intermediate Reynolds numbers.

4. Find the correction factor for the effect of baffle leakage on pressure drop  $R_l$  from Fig. 11-16. Curves shown are not to be extrapolated beyond the points shown.

5. Find the correction factor for bundle bypass  $R_b$  from Fig. 11-17.



(a)



(b)

FIG. 11-15 Correction of friction factors for ideal tube banks. (a) Triangular and rotated square arrays. (b) In-line square arrays.

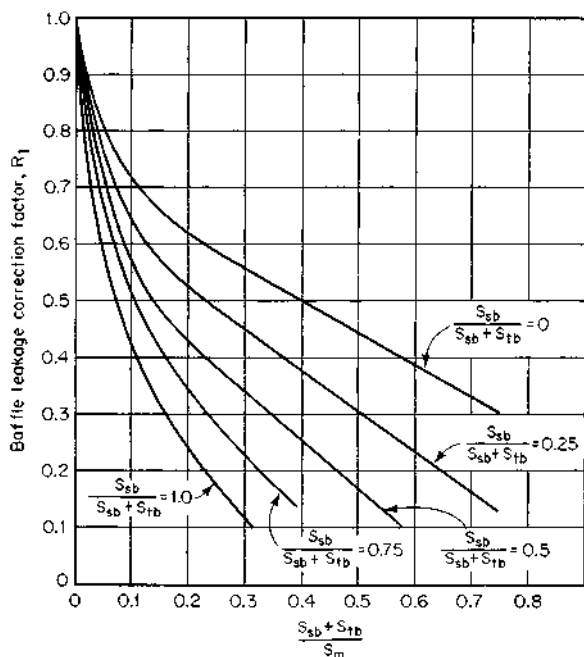


FIG. 11-16 Correction factor for baffle-leakage effect on pressure drop.

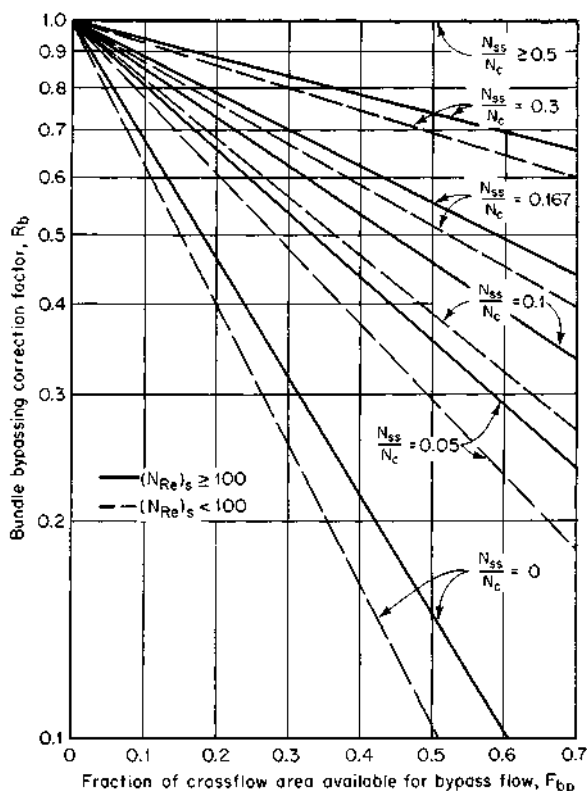


FIG. 11-17 Correction factor on pressure drop for bypass flow.

6. Calculate the *pressure drop across the shell side* (excluding nozzles). Units for pressure drop are  $\text{lb}/\text{ft}^2$ .

$$\Delta P_s = [(N_b - 1)(\Delta P_{bk})R_b + N_b \Delta P_{ak}]R_l + 2 \Delta P_{bh}R_b \left(1 + \frac{N_{cw}}{N_c}\right) \quad (11-25)$$

The values of  $h_s$  and  $\Delta P_s$  calculated by this procedure are for clean exchangers and are intended to be as accurate as possible, not conservative. A fouled exchanger will generally give lower heat-transfer rates, as reflected by the dirt resistances incorporated into Eq. (11-2), and higher pressure drops. Some estimate of **fouling effects** on pressure drop may be made by using the methods just given by assuming that the fouling deposit blocks the leakage and possibly the bypass areas. The fouling may also decrease the clearance between tubes and significantly increase the pressure drop in cross-flow.

## THERMAL DESIGN OF CONDENSERS

### Single-Component Condensers

**Mean Temperature Difference** In condensing a single component at its saturation temperature, the entire resistance to heat transfer on the condensing side is generally assumed to be in the layer of condensate. A mean condensing coefficient is calculated from the appropriate correlation and combined with the other resistances in Eq. (11-2). The overall coefficient is then used with the LMTD (no  $F_T$  correction is necessary for isothermal condensation) to give the required area, even though the condensing coefficient and hence  $U$  are not constant throughout the condenser.

If the vapor is **superheated** at the inlet, the vapor may first be desuperheated by sensible heat transfer from the vapor. This occurs if the surface temperature is above the saturation temperature, and a single-phase heat-transfer correlation is used. If the surface is below the saturation temperature, condensation will occur directly from the superheated vapor, and the effective coefficient is determined from the appropriate condensation correlation, using the saturation temperature in the LMTD. To determine whether or not condensation will occur directly from the superheated vapor, calculate the surface temperature by assuming single-phase heat transfer.

$$T_{\text{surface}} = T_{\text{vapor}} - \frac{U}{h} (T_{\text{vapor}} - T_{\text{coolant}}) \quad (11-26)$$

where  $h$  is the sensible heat-transfer coefficient for the vapor,  $U$  is calculated by using  $h$ , and both are on the same area basis. If  $T_{\text{surface}} > T_{\text{saturation}}$ , no condensation occurs at that point and the heat flux is actually higher than if  $T_{\text{surface}} \leq T_{\text{saturation}}$  and condensation did occur. It is generally conservative to design a pure-component desuperheater-condenser as if the entire heat load were transferred by condensation, using the saturation temperature in the LMTD.

The design of an integral **condensate subcooling section** is more difficult, especially if close temperature approach is required. The condensate layer on the surface is on the average subcooled by one-third to one-half of the temperature drop across the film, and this is often sufficient if the condensate is not reheated by raining through the vapor. If the condensing-subcooling process is carried out inside tubes or in the shell of a vertical condenser, the single-phase subcooling section can be treated separately, giving an area that is added onto that needed for condensation. If the subcooling is achieved on the shell side of a horizontal condenser by flooding some of the bottom tubes with a weir or level controller, the rate and heat-balance equations must be solved for each section to obtain the area required.

**Pressure drop** on the condensing side reduces the final condensing temperature and the MTD and should always be checked. In designs requiring close approach between inlet coolant and exit condensate (subcooled or not), underestimation of pressure drop on the condensing side can lead to an exchanger that cannot meet specified terminal temperatures. Since pressure-drop calculations in two-phase flows such as condensation are relatively inaccurate, designers must consider carefully the consequences of a larger-than-calculated pressure drop.

**Horizontal In-Shell Condensers** The mean **condensing coefficient** for the outside of a bank of horizontal tubes is calculated from Eq. (5-93) for a single tube, corrected for the number of tubes

in a vertical row. For undisturbed laminar flow over all the tubes, Eq. (5-97) is, for realistic condenser sizes, overly conservative because of rippling, splashing, and turbulent flow (*Process Heat Transfer*, McGraw-Hill, New York, 1950). Kern proposed an exponent of  $-1/6$  on the basis of experience, while Freon-11 data of Short and Brown (*General Discussion on Heat Transfer*, Institute of Mechanical Engineers, London, 1951) indicate independence of the number of tube rows. It seems reasonable to use no correction for inviscid liquids and Kern's correction for viscous condensates. For a cylindrical tube bundle, where  $N$  varies, it is customary to take  $N$  equal to two-thirds of the maximum or centerline value.

Baffles in a horizontal in-shell condenser are oriented with the cuts vertical to facilitate drainage and eliminate the possibility of flooding in the upward cross-flow sections. **Pressure drop** on the vapor side can be estimated by the data and method of Diehl and Unruh [*Pet. Refiner*, **36**(10), 147 (1957); **37**(10), 124 (1958)].

High vapor velocities across the tubes enhance the condensing coefficient. There is no correlation in the open literature to permit designers to take advantage of this. Since the vapor flow rate varies along the length, an incremental calculation procedure would be required in any case. In general, the pressure drops required to gain significant benefit are above those allowed in most process applications.

**Vertical In-Shell Condensers** Condensers are often designed so that condensation occurs on the outside of vertical tubes. Equation (5-88) is valid as long as the condensate film is laminar. When it becomes turbulent, Fig. 5-10 or Colburn's equation [*Trans. Am. Inst. Chem. Eng.*, **30**, 187 (1933-1934)] may be used.

Some judgment is required in the use of these correlations because of construction features of the condenser. The tubes must be supported by baffles, usually with maximum cut (45 percent of the shell diameter) and maximum spacing to minimize pressure drop. The flow of the condensate is interrupted by the baffles, which may draw off or redistribute the liquid and which will also cause some splashing of free-falling drops onto the tubes.

For **subcooling**, a liquid inventory may be maintained in the bottom end of the shell by means of a weir or a liquid-level-controller. The subcooling heat-transfer coefficient is given by the correlations for natural convection on a vertical surface [Eqs. (5-33a), (5-33b)], with the pool assumed to be well mixed (isothermal) at the subcooled condensate exit temperature. Pressure drop may be estimated by the shell-side procedure.

**Horizontal In-Tube Condensers** Condensation of a vapor inside horizontal tubes occurs in kettle and horizontal thermosiphon reboilers and in air-cooled condensers. In-tube condensation also offers certain advantages for condensation of multicomponent mixtures, discussed in the subsection "Multicomponent Condensers." The various in-tube correlations are closely connected to the **two-phase flow pattern** in the tube [*Chem. Eng. Prog. Symp. Ser.*, **66**(102), 150 (1970)]. At low flow rates, when gravity dominates the flow pattern, Eq. (5-101) may be used. At high flow rates, the flow and heat transfer are governed by vapor shear on the condensate film, and Eq. (5-100a) is valid. A simple and generally conservative procedure is to calculate the coefficient for a given case by both correlations and use the *larger* one.

Pressure drop during condensation inside horizontal tubes can be computed by using the correlations for two-phase flow given in Sec. 6 and neglecting the pressure recovery due to deceleration of the flow.

**Vertical In-Tube Condensation** Vertical-tube condensers are generally designed so that vapor and liquid flow cocurrently downward; if pressure drop is not a limiting consideration, this configuration can result in higher heat-transfer coefficients than shell-side condensation and has particular advantages for multicomponent condensation. If gravity controls, the mean heat-transfer coefficient for condensation is given by Figs. 5-9 and 5-10. If vapor shear controls, Eq. (5-99a) is applicable. It is generally conservative to calculate the coefficients by both methods and choose the *higher* value. The pressure drop can be calculated by using the Lockhart-Martinelli method [*Chem. Eng. Prog.*, **45**, 39 (1945)] for friction loss, neglecting momentum and hydrostatic effects.

Vertical in-tube condensers are often designed for **reflux or knock-back application** in reactors or distillation columns. In this

case, vapor flow is upward, countercurrent to the liquid flow on the tube wall; the vapor shear acts to thicken and retard the drainage of the condensate film, reducing the coefficient. Neither the fluid dynamics nor the heat transfer is well understood in this case, but Soliman, Schuster, and Berenson [*J. Heat Transfer*, **90**, 267-276 (1968)] discuss the problem and suggest a computational method. The Diehl-Koppány correlation [*Chem. Eng. Prog. Symp. Ser.* **92**, **65** (1969)] may be used to estimate the maximum allowable vapor velocity at the tube inlet. If the vapor velocity is great enough, the liquid film will be carried upward; this design has been employed in a few cases in which only part of the stream is to be condensed. This velocity cannot be accurately computed, and a very conservative (high) outlet velocity must be used if unstable flow and flooding are to be avoided; 3 times the vapor velocity given by the Diehl-Koppány correlation for incipient flooding has been suggested as the design value for completely stable operation.

## Multicomponent Condensers

**Thermodynamic and Mass-Transfer Considerations** *Multicomponent vapor mixture* includes several different cases: all the components may be liquids at the lowest temperature reached in the condensing side, or there may be components which dissolve substantially in the condensate even though their boiling points are below the exit temperature, or one or more components may be both noncondensable and nearly insoluble.

Multicomponent condensation always involves sensible-heat changes in the vapor and liquid along with the latent-heat load. Compositions of both phases in general change through the condenser, and **concentration gradients** exist in both phases. Temperature and concentration profiles and transport rates at a point in the condenser usually cannot be calculated, but the binary cases have been treated: condensation of one component in the presence of a completely insoluble gas [Colburn and Hougen, *Ind. Eng. Chem.*, **26**, 1178-1182 (1934); and Colburn and Edison, *Ind. Eng. Chem.*, **33**, 457-458 (1941)] and condensation of a binary vapor [Colburn and Drew, *Trans. Am. Inst. Chem. Eng.*, **33**, 196-215 (1937)]. It is necessary to know or calculate diffusion coefficients for the system, and a reasonable approximate method to avoid this difficulty and the reiterative calculations is desirable. To integrate the point conditions over the total condensation requires the temperature, composition enthalpy, and flow-rate profiles as functions of the heat removed. These are calculated from component thermodynamic data if the vapor and liquid are assumed to be in equilibrium at the local vapor temperature. This assumption is not exactly true, since the condensate and the liquid-vapor interface (where equilibrium does exist) are intermediate in temperature between the coolant and the vapor.

In calculating the condensing curve, it is generally assumed that the vapor and liquid flow collinearly and in intimate contact so that composition equilibrium is maintained between the total streams at all points. If, however, the condensate drops out of the vapor (as can happen in horizontal shell-side condensation) and flows to the exit without further interaction, the remaining vapor becomes excessively enriched in light components with a decrease in condensing temperature and in the temperature difference between vapor and coolant. The result may be not only a small reduction in the amount of heat transferred in the condenser but also an inability to condense totally the light ends even at reduced throughput or with the addition of more surface. To prevent the liquid from segregating, in-tube condensation is preferred in critical cases.

**Thermal Design** If the controlling resistance for heat and mass transfer in the vapor is sensible-heat removal from the cooling vapor, the following design equation is obtained:

$$A = \int_0^{Q_T} \frac{1 + U'Z_H/h_{se}}{U'(T_c - T_v)} dQ \quad (11-27)$$

$U'$  is the overall heat-transfer coefficient between the vapor-liquid interface and the coolant, including condensate film, dirt and wall resistances, and coolant. The condensate film coefficient is calculated from the appropriate equation or correlation for pure vapor condensation for the geometry and flow regime involved, using mean liquid properties.  $Z_H$  is the ratio of the sensible heat removed from the

vapor-gas stream to the total heat transferred; this quantity is obtained from thermodynamic calculations and may vary substantially from one end of the condenser to the other, especially when removing vapor from a noncondensable gas. The sensible-heat-transfer coefficient for the vapor-gas stream  $h_{vg}$  is calculated by using the appropriate correlation or design method for the geometry involved, neglecting the presence of the liquid. As the vapor condenses, this coefficient decreases and must be calculated at several points in the process.  $T_v$  and  $T_c$  are temperatures of the vapor and of the coolant respectively. This procedure is similar in principle to that of Ward [*Petro/Chem. Eng.*, **32**(11), 42-48 (1960)]. It may be nonconservative for condensing steam and other high-latent-heat substances, in which case it may be necessary to increase the calculated area by 25 to 50 percent.

Pressure drop on the condensing side may be estimated by judicious application of the methods suggested for pure-component condensation, taking into account the generally nonlinear decrease of vapor-gas flow rate with heat removal.

## THERMAL DESIGN OF REBOILERS

For a **single-component reboiler design**, attention is focused upon the mechanism of heat and momentum transfer at the hot surface. In *multicomponent systems*, the light components are preferentially vaporized at the surface, and the process becomes limited by their rate of diffusion. The net effect is to decrease the effective temperature difference between the hot surface and the bulk of the boiling liquid. If one attempts to vaporize too high a fraction of the feed liquid to the reboiler, the temperature difference between surface and liquid is reduced to the point that nucleation and vapor generation on the surface are suppressed and heat transfer to the liquid proceeds at the lower rate associated with single-phase natural convection. The only safe procedure in design for wide-boiling-range mixtures is to vaporize such a limited fraction of the feed that the boiling point of the remaining liquid mixture is still at least 5.5°C (10°F) below the surface temperature. Positive flow of the unvaporized liquid through and out of the reboiler should be provided.

**Kettle Reboilers** It has been generally assumed that kettle reboilers operate in the pool boiling mode, but with a lower peak heat flux because of vapor binding and blanketing of the upper tubes in the bundle. There is some evidence that vapor generation in the bundle causes a high circulation rate through the bundle. The result is that, at the lower heat fluxes, the kettle reboiler actually gives higher heat-transfer coefficients than a single tube. Present understanding of the recirculation phenomenon is insufficient to take advantage of this in design. Available nucleate pool boiling correlations are only very approximate, failing to account for differences in the nucleation characteristics of different surfaces. The Mostinski correlation [Eq. (5-102)] and the McNelly correlation [Eq. (5-103)] are generally the best for single components or narrow-boiling-range mixtures at low fluxes, though they may give errors of 40 to 50 percent. Experimental heat-transfer coefficients for pool boiling of a given liquid on a given surface should be used if available. The bundle **peak heat flux** is a function of tube-bundle geometry, especially of tube-packing density; in the absence of better information, the Palen-Small modification [Eq. (5-108)] of the Zuber maximum-heat-flux correlation is recommended.

A general method for analyzing kettle reboiler performance is by Fair and Klip, *Chem. Eng. Prog.* **79**(3), 86 (1983). It is effectively limited to computer application.

Kettle reboilers are generally assumed to require negligible pressure drop. It is important to provide good longitudinal liquid flow paths within the shell so that the liquid is uniformly distributed along the entire length of the tubes and excessive local vaporization and vapor binding are avoided.

This method may also be used for the thermal design of **horizontal thermosiphon reboilers**. The recirculation rate and pressure profile of the thermosiphon loop can be calculated by the methods of Fair [*Pet. Refiner*, **39**(2), 105-123 (1960)].

**Vertical Thermosiphon Reboilers** Vertical thermosiphon reboilers operate by natural circulation of the liquid from the still through the downcomer to the reboiler and of the two-phase mixture

from the reboiler through the return piping. The flow is induced by the hydrostatic pressure imbalance between the liquid in the downcomer and the two-phase mixture in the reboiler tubes. Thermosiphons do not require any pump for recirculation and are generally regarded as less likely to foul in service because of the relatively high two-phase velocities obtained in the tubes. Heavy components are not likely to accumulate in the thermosiphon, but they are more difficult to design satisfactorily than kettle reboilers, especially in vacuum operation. Several shortcut methods have been suggested for thermosiphon design, but they must generally be used with caution. The method due to Fair (loc. cit.), based upon two-phase flow correlations, is the most complete in the open literature but requires a computer for practical use. Fair also suggests a shortcut method that is satisfactory for preliminary design and can be reasonably done by hand.

**Forced-Recirculation Reboilers** In forced-recirculation reboilers, a pump is used to ensure circulation of the liquid past the heattransfer surface. Force-recirculation reboilers may be designed so that boiling occurs inside vertical tubes, inside horizontal tubes, or on the shell side. For forced boiling inside vertical tubes, Fair's method (loc. cit.) may be employed, making only the minor modification that the recirculation rate is fixed and does not need to be balanced against the pressure available in the downcomer. Excess pressure required to circulate the two-phase fluid through the tubes and back into the column is supplied by the pump, which must develop a positive pressure increase in the liquid.

Fair's method may also be modified to design forced-recirculation reboilers with horizontal tubes. In this case the hydrostatic-head-pressure effect through the tubes is zero but must be considered in the two-phase return lines to the column.

The same procedure may be applied in principle to design of forced-recirculation reboilers with shell-side vapor generation. Little is known about two-phase flow on the shell side, but a reasonable estimate of the friction pressure drop can be made from the data of Diehl and Unruh [*Pet. Refiner*, **36**(10), 147 (1957); **37**(10), 124 (1958)]. No void-fraction data are available to permit accurate estimation of the hydrostatic or acceleration terms. These may be roughly estimated by assuming homogeneous flow.

## THERMAL DESIGN OF EVAPORATORS

Heat duties of evaporator heating surfaces are usually determined by conventional heat and material balance calculations. Heating surface areas are normally, but not always taken as those in contact with the material being evaporated. It is the heat transfer  $\Delta T$  that presents the most difficulty in deriving or applying heat-transfer coefficients. The total  $\Delta T$  between heat source and heat sink is never all available for heat transfer. Since energy usually is carried to and from an evaporator body or effect by condensible vapors, loss in pressure represents a loss in  $\Delta T$ . Such losses include pressure drop through entrainment separators, friction in vapor piping, and acceleration losses into and out of the piping. The latter loss has often been overlooked, even though it can be many times greater than the friction loss. Similarly, friction and acceleration losses past the heating surface, such as in a falling film evaporator, cause a loss of  $\Delta T$  that may or may not have been included in the heat transfer  $\Delta T$  when reporting experimental results. Boiling-point rise, the difference between the boiling point of the solution and the condensing point of the solvent at the same pressure, is another loss. Experimental data are almost always corrected for boiling-point rise, but plant data are suspect when based on temperature measurements because vapor at the point of measurement may still contain some superheat, which represents but a very small fraction of the heat given up when the vapor condenses but may represent a substantial fraction of the actual net  $\Delta T$  available for heat transfer. A  $\Delta T$  loss that must be considered in forced-circulation evaporators is that due to temperature rise through the heater, a consequence of the heat being absorbed there as sensible heat. A further loss may occur when the heater effluent flashes as it enters the vapor-liquid separator. Some of the liquid may not reach the surface and flash to equilibrium with the vapor pressure in the separator, instead of recirculating to the heater, raising the average temperature at which heat is absorbed and further reducing the net  $\Delta T$ . Whether or not these  $\Delta T$  losses are allowed for in the heat-transfer coefficients reported

depends on the method of measurement. Simply basing the liquid temperature on the measured vapor head pressure may ignore both—or only the latter if temperature rise through the heater is estimated separately from known heat input and circulation rate. In general, when calculating overall heat-transfer coefficients from individual-film coefficients, all of these losses must be allowed for, while when using reported overall coefficients care must be exercised to determine which losses may already have been included in the heat transfer  $\Delta T$ .

**Forced-Circulation Evaporators** In evaporators of this type in which hydrostatic head prevents boiling at the heating surface, **heat-transfer coefficients** can be predicted from the usual correlations for condensing steam (Fig. 5-10) and forced-convection sensible heating [Eq. (5-50)]. The liquid film coefficient is improved if boiling is not completely suppressed. When only the film next to the wall is above the boiling point, Boarts, Badger, and Meisenberg [*Ind. Eng. Chem.*, **29**, 912 (1937)] found that results could be correlated by Eq. (5-50) by using a constant of 0.0278 instead of 0.023. In such cases, the course of the liquid temperature can still be calculated from known circulation rate and heat input.

When the bulk of the liquid is boiling in part of the tube length, the film coefficient is even higher. However, the liquid temperature starts dropping as soon as full boiling develops, and it is difficult to estimate the course of the temperature curve. It is certainly safe to estimate heat transfer on the basis that no bulk boiling occurs. Fragen and Badger [*Ind. Eng. Chem.*, **28**, 534 (1936)] obtained an **empirical correlation** of overall heat-transfer coefficients in this type of evaporator, based on the  $\Delta T$  at the heater inlet:

In U.S. customary units

$$U = 2020D^{0.57}(V_s)^{3.6/L}/\mu^{0.25}\Delta T^{0.1} \quad (11-28)$$

where  $D$  = mean tube diameter,  $V_s$  = inlet velocity,  $L$  = tube length, and  $\mu$  = liquid viscosity. This equation is based primarily on experiments with copper tubes of 0.022 m (8/8 in) outside diameter, 0.00165 m (16 gauge), 2.44 m (8 ft) long, but it includes some work with 0.0127-m (1/2-in) tubes 2.44 m (8 ft) long and 0.0254-m (1-in) tubes 3.66 m (12 ft) long.

**Long-Tube Vertical Evaporators** In the rising-film version of this type of evaporator, there is usually a nonboiling zone in the bottom section and a boiling zone in the top section. The length of the nonboiling zone depends on heat-transfer characteristics in the two zones and on pressure drop during two-phase flow in the boiling zone. The work of Martinelli and coworkers [Lockhart and Martinelli, *Chem. Eng. Prog.*, **45**, 39–48 (January 1949); and Martinelli and Nelson, *Trans. Am. Soc. Mech. Eng.*, **70**, 695–702 (August 1948)] permits a prediction of pressure drop, and a number of correlations are available for estimating film coefficients of heat transfer in the two zones. In estimating pressure drop, integrated curves similar to those presented by Martinelli and Nelson are the easiest to use. The curves for pure water are shown in Figs. 11-18 and 11-19, based on the assumption that the flow of both vapor and liquid would be turbulent if each were flowing alone in the tube. Similar curves can be prepared if one or both flows are laminar or if the properties of the liquid differ appreciably from the properties of pure water. The **acceleration pressure drop**  $\Delta P_a$  is calculated from the equation

$$\Delta P_a = br_2G^2/32.2 \quad (11-29)$$

where  $b = (2.6)(10^7)$  (SI) and 1.0 (U.S. customary) and using  $r_2$  from Fig. 11-18. The frictional pressure drop is derived from Fig. 11-19, which shows the ratio of two-phase pressure drop to that of the entering liquid flowing alone.

Pressure drop due to hydrostatic head can be calculated from liquid holdup  $R_1$ . For nonfoaming dilute aqueous solutions,  $R_1$  can be estimated from  $R_1 = 1/[1 + 2.5(V/L)(\rho_1/\rho_v)^{1/2}]$ . Liquid holdup, which represents the ratio of liquid-only velocity to actual liquid velocity, also appears to be the principal determinant of the convective coefficient in the boiling zone (Dengler, Sc.D. thesis, MIT, 1952). In other words, the convective coefficient is that calculated from Eq. (5-50) by using the liquid-only velocity divided by  $R_1$  in the Reynolds number. Nucleate boiling augments convective heat transfer, primarily when  $\Delta T$ 's are high and the convective coefficient is low [Chen, *Ind. Eng. Chem. Process Des. Dev.*, **5**, 322 (1966)].

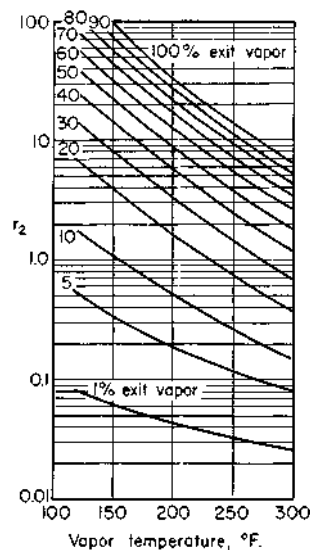


FIG. 11-18 Acceleration losses in boiling flow.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ .

Film coefficients for the **boiling of liquids other than water** have been investigated. Coulson and McNelly [*Trans. Inst. Chem. Eng.*, **34**, 247 (1956)] derived the following relation, which also correlated the data of Badger and coworkers [*Chem. Metall. Eng.*, **46**, 640 (1939); *Chem. Eng.*, **61**(2), 183 (1954); and *Trans. Am. Inst. Chem. Eng.*, **33**, 392 (1937); **35**, 17 (1939); **36**, 759 (1940)] on water:

$$N_{Nu} = (1.3 + bD)(N_{Pr})^{0.9}(N_{Re})^{0.23}(N_{Re_g})^{0.34}\left(\frac{\rho_l}{\rho_g}\right)^{0.25}\left(\frac{\mu_g}{\mu_l}\right) \quad (11-30)$$

where  $b = 128$  (SI) or 39 (U.S. customary),  $N_{Nu}$  = Nusselt number based on liquid thermal conductivity,  $D$  = tube diameter, and the remaining terms are dimensionless groupings of liquid Prandtl number, liquid Reynolds number, vapor Reynolds number, and ratios of densities and viscosities. The Reynolds numbers are calculated on the basis of each fluid flowing by itself in the tube.

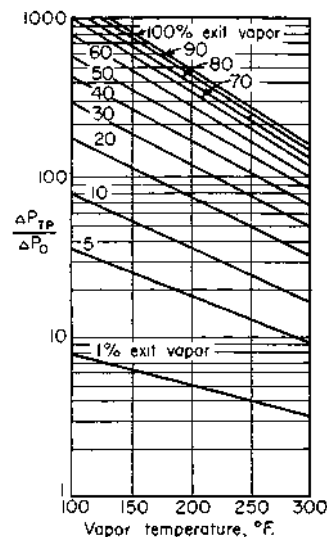


FIG. 11-19 Friction pressure drop in boiling flow.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ .



Additional corrections must be applied when the fraction of vapor is so high that the remaining liquid does not wet the tube wall or when the velocity of the mixture at the tube exits approaches sonic velocity. McAdams, Woods, and Bryan (*Trans. Am. Soc. Mech. Eng.*, 1940), Dengler and Addoms (loc. cit.), and Stroebe, Baker, and Badger [*Ind. Eng. Chem.*, **31**, 200 (1939)] encountered dry-wall conditions and reduced coefficients when the weight fraction of vapor exceeded about 80 percent. Schweppe and Foust [*Chem. Eng. Prog.*, **49**, *Symp. Ser.* 5, 77 (1953)] and Harvey and Foust (ibid., p. 91) found that "sonic choking" occurred at surprisingly low flow rates.

The simplified method of calculation outlined includes no allowance for the **effect of surface tension**. Stroebe, Baker, and Badger (loc. cit.) found that by adding a small amount of surface-active agent the boiling-film coefficient varied inversely as the square of the surface tension. Coulson and Mehta [*Trans. Inst. Chem. Eng.*, **31**, 208 (1953)] found the exponent to be -1.4. The higher coefficients at low surface tension are offset to some extent by a higher pressure drop, probably because the more intimate mixture existing at low surface tension causes the liquid fraction to be accelerated to a velocity closer to that of the vapor. The pressure drop due to acceleration  $\Delta P_a$  derived from Fig. 11-18 allows for some slippage. In the limiting case, such as might be approached at low surface tension, the acceleration pressure drop in which "fog" flow is assumed (no slippage) can be determined from the equation

$$\Delta P'_a = \frac{y(V_g - V_l)G^2}{g_c} \quad (11-31)$$

where  $y$  = fraction vapor by weight  
 $V_g, V_l$  = specific volume gas, liquid  
 $G$  = mass velocity

While the foregoing methods are valuable for detailed evaporator design or for evaluating the effect of changes in conditions on performance, they are cumbersome to use when making preliminary designs or cost estimates. Figure 11-20 gives the general range of **overall long-tube vertical- (LTV) evaporator heat-transfer coefficients** usually encountered in commercial practice. The higher coefficients are encountered when evaporating dilute solutions and the lower range when evaporating viscous liquids. The dashed curve represents the approximate lower limit, for liquids with viscosities of about 0.1 Pa·s (100 cP). The LTV evaporator does not work well at low temperature differences, as indicated by the results shown in Fig. 11-21 for seawater in 0.051-m (2-in.), 0.0028-m (12-gauge) brass tubes 7.32 m (24 ft) long (W. L. Badger Associates, Inc., U.S. Department of the Interior, Office of Saline Water Rep. 26, December 1959, OTS

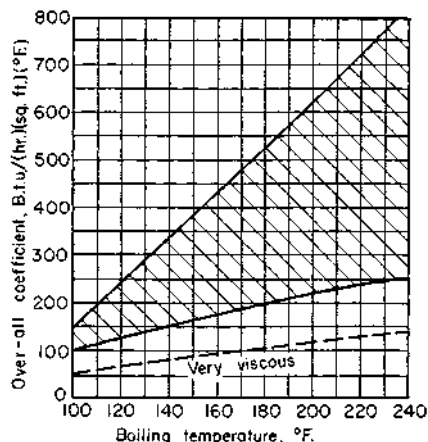


FIG. 11-20 General range of long-tube vertical- (LTV) evaporator coefficients.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

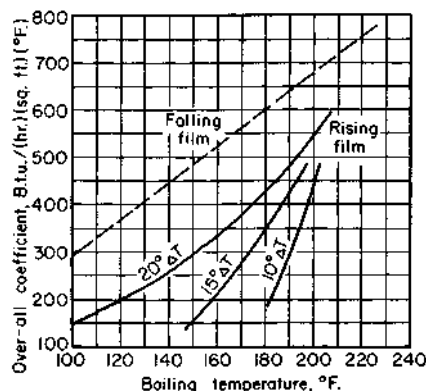


FIG. 11-21 Heat-transfer coefficients in LTV seawater evaporators.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

Publ. PB 161290). The feed was at its boiling point at the vapor-head pressure, and feed rates varied from 0.025 to 0.050 kg/(s-tube) [200 to 400 lb/(h-tube)] at the higher temperature to 0.038 to 0.125 kg/(s-tube) [300 to 1000 lb/(h-tube)] at the lowest temperature.

Falling film evaporators find their widest use at low temperature differences—also at low temperatures. Under most operating conditions encountered, heat transfer is almost all by pure convection, with a negligible contribution from nucleate boiling. Film coefficients on the condensing side may be estimated from Dukler's correlation, [*Chem. Eng. Prog.*, **55**, 62 1950]. The same Dukler correlation presents curves covering falling film heat transfer to non-boiling liquids that are equally applicable to the falling film evaporator [Sinek and Young, *Chem. Eng. Prog.*, **58**, No. 12, 74 (1962)]. Kunz and Yerazunis [*J. Heat Transfer* **8**, 413 (1969)] have since extended the range of physical properties covered, as shown in Fig. 11-22. The boiling point in the tubes of such an evaporator is higher than in the vapor head because of both frictional-pressure drop and the head needed to accelerate the vapor to the tube-exit velocity. These factors, which can easily be predicted, make the overall apparent coefficients somewhat lower than those for nonboiling conditions. Figure 11-21 shows overall apparent heat-transfer coefficients determined in a falling-film seawater evaporator using the same tubes and flow rates as for the rising-film tests (W. L. Badger Associates, Inc., loc. cit.).

**Short-Tube Vertical Evaporators** Coefficients can be estimated by the same detailed method described for recirculating LTV evaporators. Performance is primarily a function of temperature level, temperature difference, and viscosity. While liquid level can also have an important influence, this is usually encountered only at levels lower than considered safe in commercial operation. **Overall heat-transfer coefficients** are shown in Fig. 11-23 for a basket-type evaporator (one with an annular downtake) when boiling water with 0.051-m (2-in.) outside-diameter 0.0028-m-wall (12-gauge), 1.22-m-(4-ft-) long steel tubes [Badger and Shepard, *Chem. Metall. Eng.*, **23**, 281 (1920)]. Liquid level was maintained at the top tube sheet. Foust, Baker, and Badger [*Ind. Eng. Chem.*, **31**, 206 (1939)] measured recirculating velocities and heat-transfer coefficients in the same evaporator except with 0.064-m (2.5-in.) 0.0034-m-wall (10-gauge), 1.22-m-(4-ft-) long tubes and temperature differences from 7 to 26°C (12 to 46°F). In the normal range of liquid levels, their results can be expressed as

$$U_c = \frac{b(\Delta T_c)^{0.22} N_{Pr}^{0.4}}{(V_g - V_l)^{0.37}} \quad (11-32)$$

where  $b = 153$  (SI) or 375 (U.S. customary) and the subscript  $c$  refers to true liquid temperature, which under these conditions was about 0.56°C (1°F) above the vapor-head temperature. This work was done with water.

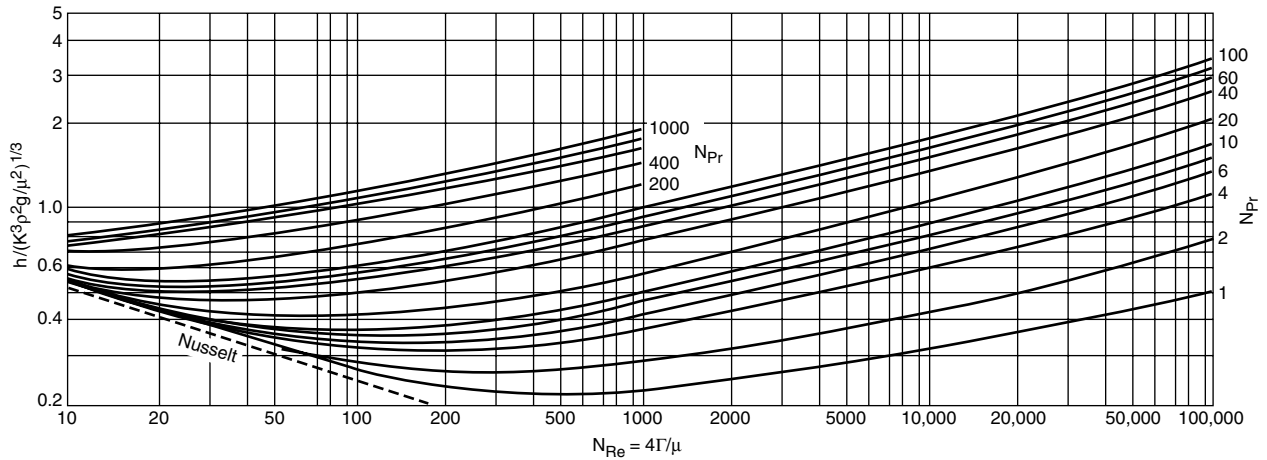


FIG. 11-22 Kunz and Yerazunis Correlation for falling-film heat transfer.

No detailed tests have been reported for the performance of propeller calandrias. Not enough is known regarding the performance of the propellers themselves under the cavitating conditions usually encountered to permit predicting circulation rates. In many cases, it appears that the propeller does no good in accelerating heat transfer over the transfer for natural circulation (Fig. 11-23).

**Miscellaneous Evaporator Types** **Horizontal-tube evaporators** operating with partially or fully submerged heating surfaces behave in much the same way as short-tube verticals, and heat-transfer coefficients are of the same order of magnitude. Some test results for water were published by Badger [*Trans. Am. Inst. Chem. Eng.*, **13**, 139 (1921)]. When operating unsubmerged, their heat transfer performance is roughly comparable to the falling-film vertical tube evaporator. Condensing coefficients inside the tubes can be derived from Nusselt's theory which, based on a constant-heat flux rather than a constant film  $\Delta T$ , gives:

$$\frac{h}{(k^3 p^2 g / \mu^2)^{1/3}} = 1.59 (4\Gamma / \mu)^{-1/3} \quad (11-33a)$$

For the boiling side, a correlation based on seawater tests gave:

$$\frac{h}{(k^3 p^2 g / \mu^2)^{1/3}} = 0.0147 (4\Gamma / \mu)^{1/3} (D)^{-1/3} \quad (11-33b)$$

where  $\Gamma$  is based on feed-rate per unit length of the top tube in each vertical row of tubes and  $D$  is in meters.

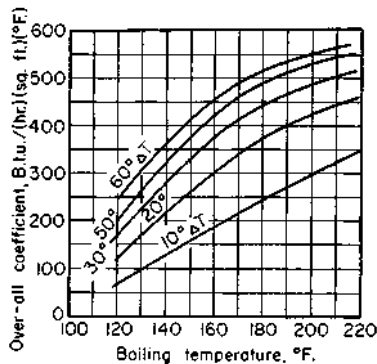


FIG. 11-23 Heat-transfer coefficients for water in short-tube evaporators.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

Heat-transfer coefficients in clean coiled-tube evaporators for seawater are shown in Fig. 11-24 [Hillier, *Proc. Inst. Mech. Eng. (London)*, **1B**(7), 295 (1953)]. The tubes were of copper.

Heat-transfer coefficients in **agitated-film evaporators** depend primarily on liquid viscosity. This type is usually justifiable only for very viscous materials. Figure 11-25 shows general ranges of overall coefficients [Hauschild, *Chem. Ing. Tech.*, **25**, 573 (1953); Lindsey, *Chem. Eng.*, **60**(4), 227 (1953); and Leniger and Veldstra, *Chem. Ing. Tech.*, **31**, 493 (1959)]. When used with nonviscous fluids, a wiped-film evaporator having fluted external surfaces can exhibit very high coefficients (Lustenader et al., *Trans. Am. Soc. Mech. Eng.*, Paper 59-SA-30, 1959), although at a probably unwarranted first cost.

**Heat Transfer from Various Metal Surfaces** In an early work, Pridgeon and Badger [*Ind. Eng. Chem.*, **16**, 474 (1924)] published test results on copper and iron tubes in a horizontal-tube evaporator that indicated an extreme **effect of surface cleanliness** on heat-transfer coefficients. However, the high degree of cleanliness needed for high coefficients was difficult to achieve, and the tube layout and liquid level were changed during the course of the tests so as to make direct comparison of results difficult. Other workers have found little or no effect of conditions of surface or tube material on boiling-film

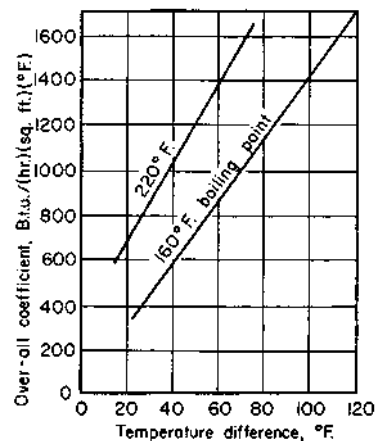


FIG. 11-24 Heat-transfer coefficients for seawater in coil-tube evaporators.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

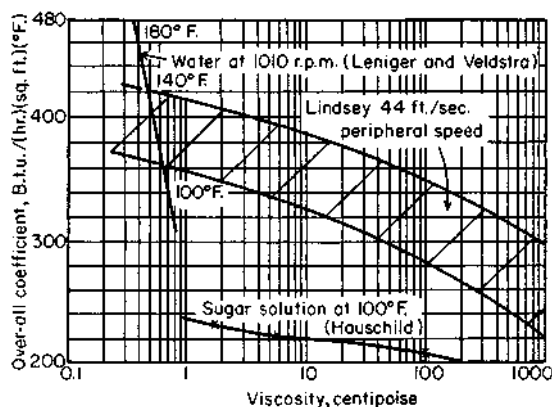


FIG. 11-25 Overall heat-transfer coefficients in agitated-film evaporators.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; to convert centipoises to pascal-seconds, multiply by  $10^{-3}$ .

coefficients in the range of commercial operating conditions [Averin, *Izv. Akad. Nauk SSSR Otd. Tekh. Nauk*, no. 3, p. 116, 1954; and Coulson and McNelly, *Trans. Inst. Chem. Eng.*, **34**, 247 (1956)].

Work in connection with desalination of seawater has shown that **specially modified surfaces** can have a profound effect on heat-transfer coefficients in evaporators. Figure 11-26 (Alexander and Hoffman, Oak Ridge National Laboratory TM-2203) compares overall coefficients for some of these surfaces when boiling fresh water in 0.051-m (2-in) tubes 2.44-m (8-ft) long at atmospheric pressure in both upflow and downflow. The area basis used was the nominal outside area. Tube 20 was a smooth 0.0016-m- (0.062-in-) wall aluminum brass tube that had accumulated about 6 years of fouling in seawater

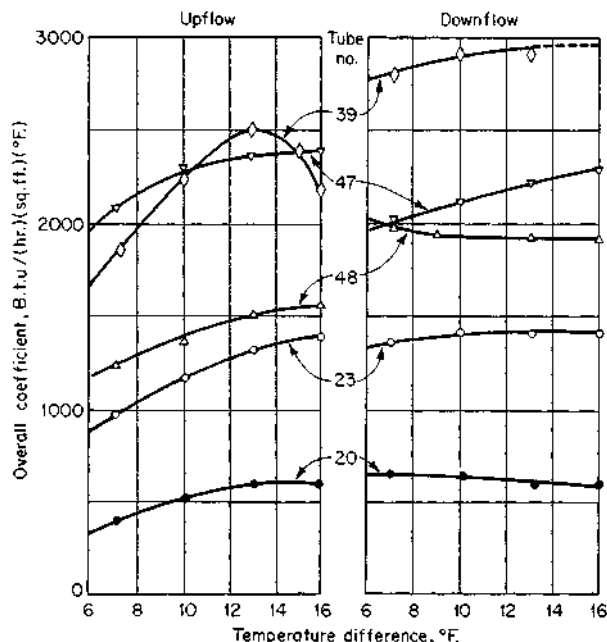


FIG. 11-26 Heat-transfer coefficients for enhanced surfaces.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783. (By permission from Oak Ridge National Laboratory TM-2203.)

service and exhibited a fouling resistance of about  $(2.6)/(10^{-5})$  ( $\text{m}^2 \cdot \text{s} \cdot \text{K}/\text{J}$ ) [ $0.00015$  ( $\text{ft}^2 \cdot \text{h} \cdot ^{\circ}\text{F}/\text{Btu}$ )]. Tube 23 was a clean aluminum tube with 20 spiral corrugations of 0.0032-m ( $1/8$ -in) radius on a 0.254-m (10-in) pitch indented into the tube. Tube 48 was a clean copper tube that had 50 longitudinal flutes pressed into the wall (General Electric double-flute profile, Diedrich, U.S. Patent 3,244,601, Apr. 5, 1966). Tubes 47 and 39 had a specially patterned porous sintered-metal deposit on the boiling side to promote nucleate boiling (Minton, U.S. Patent 3,384,154, May 21, 1968). Both of these tubes also had steam-side coatings to promote dropwise condensation—parylene for tube 47 and gold plating for tube 39.

Of these special surfaces, only the **double-fluted tube** has seen extended services. Most of the gain in heat-transfer coefficient is due to the condensing side; the flutes tend to collect the condensate and leave the lands bare [Carnavos, *Proc. First Int. Symp. Water Desalination*, **2**, 205 (1965)]. The condensing-film coefficient (based on the actual outside area, which is 28 percent greater than the nominal area) may be approximated from the equation

$$h = b \left( \frac{k^3 \rho^2 g}{\mu^2} \right)^{1/3} \left( \frac{\mu L}{q} \right)^{1/3} \left( \frac{q}{A} \right)^{-0.833} \quad (11-34a)$$

where  $b = 2100$  (SI) or 1180 (U.S. customary). The boiling-side coefficient (based on actual inside area) for salt water in downflow may be approximated from the equation

$$h = 0.035(k^3 \rho^2 g / \mu^2)^{1/3} (4\Gamma / \mu)^{1/3} \quad (11-34b)$$

The boiling-film coefficient is about 30 percent lower for pure water than it is for salt water or seawater. There is as yet no accepted explanation for the superior performance in salt water. This phenomenon is also seen in evaporation from smooth tubes.

**Effect of Fluid Properties on Heat Transfer** Most of the heat-transfer data reported in the preceding paragraphs were obtained with water or with dilute solutions having properties close to those of water. Heat transfer with other materials will depend on the type of evaporator used. For forced-circulation evaporators, methods have been presented to calculate the effect of changes in fluid properties. For natural-circulation evaporators, **viscosity** is the most important variable as far as aqueous solutions are concerned. Badger (*Heat Transfer and Evaporation*, Chemical Catalog, New York, 1926, pp. 133–134) found that, as a rough rule, overall heat-transfer coefficients varied in inverse proportion to viscosity if the boiling film was the main resistance to heat transfer. When handling molasses solutions in a forced-circulation evaporator in which boiling was allowed to occur in the tubes, Coates and Badger [*Trans. Am. Inst. Chem. Eng.*, **32**, 49 (1936)] found that from 0.005 to 0.03 Pa·s (5 to 30 cP) the overall heat-transfer coefficient could be represented by  $U = b/\mu_j^{1/24}$ , where  $b = 2.55$  (SI) or 7043 (U.S. customary). Fragen and Badger [*Ind. Eng. Chem.*, **28**, 534 (1936)] correlated overall coefficients on sugar and sulfite liquor in the same evaporator for viscosities to 0.242 Pa·s (242 cP) and found a relationship that included the viscosity raised only to the 0.25 power.

Little work has been published on the effect of viscosity on heat transfer in the long-tube vertical evaporator. Cessna, Leintz, and Badger [*Trans. Am. Inst. Chem. Eng.*, **36**, 759 (1940)] found that the overall coefficient in the nonboiling zone varied inversely as the 0.7 power of viscosity (with sugar solutions). Coulson and Mehta [*Trans. Inst. Chem. Eng.*, **31**, 208 (1953)] found the exponent to be  $-0.44$ , and Stroebel, Baker, and Badger (loc. cit.) arrived at an exponent of  $-0.3$  for the effect of viscosity on the film coefficient in the boiling zone.

Kerr (Louisiana Agr. Exp. Sta. Bull. 149) obtained plant data shown in Fig. 11-27 on various types of full-sized evaporators for cane sugar. These are invariably forward-feed evaporators concentrating to about 50° Brix, corresponding to a viscosity on the order of 0.005 Pa·s (5 cP) in the last effect. In Fig. 11-27 curve A is for short-tube verticals with central downtake, B is for standard horizontal tube evaporators, C is for Lillie evaporators (which were horizontal-tube machines with no liquor level but having recirculating liquor showered over the tubes), and D is for long-tube vertical evaporators. These curves show apparent coefficients, but sugar solutions have boiling-point rises low enough not to affect the results noticeably. Kerr also obtained the data

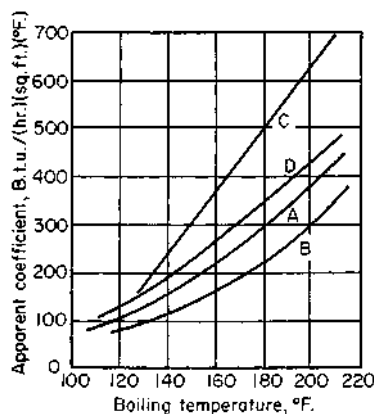


FIG. 11-27 Kerr's tests with full-sized sugar evaporators.  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

shown in Fig. 11-28 on a laboratory short-tube vertical evaporator with 0.44- by 0.61-m (1¾- by 24-in) tubes. This work was done with sugar juices boiling at 57°C (135°F) and an 11°C (20°F) temperature difference.

**Effect of Noncondensables on Heat Transfer** Most of the heat transfer in evaporators does not occur from pure steam but from vapor evolved in a preceding effect. This vapor usually contains inert gases—from air leakage if the preceding effect was under vacuum, from air entrained or dissolved in the feed, or from gases liberated by decomposition reactions. To prevent these inerts from seriously impeding heat transfer, the gases must be channeled past the heating surface and vented from the system while the gas concentration is still quite low. The influence of inert gases on heat transfer is due partially to the effect on  $\Delta T$  of lowering the partial pressure and hence condensing temperature of the steam. The primary effect, however, results from the formation at the heating surface of an insulating blanket of gas through which the steam must diffuse before it can condense. The latter effect can be treated as an added resistance or fouling factor equal to  $6.5 \times 10^{-5}$  times the local mole percent inert gas (in  $\text{J}^{-1}\text{-s-m}^2\text{-K}$ ) [Standiford, *Chem. Eng. Prog.*, **75**, 59–62 (July 1979)]. The effect on  $\Delta T$  is readily calculated from Dalton's law. Inert-gas concentrations may vary by a factor of 100 or more between vapor inlet and vent outlet, so these relationships should be integrated through the tube bundle.

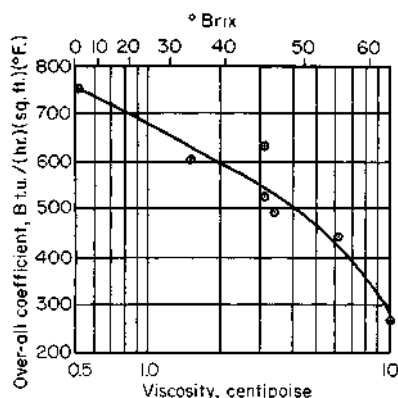


FIG. 11-28 Effect of viscosity on heat transfer in short-tube vertical evaporator. To convert centipoises to pascal-seconds, multiply by  $10^{-3}$ ; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

## BATCH OPERATIONS: HEATING AND COOLING OF VESSELS

**Nomenclature** (Use consistent units.)  $A$  = heat-transfer surface;  $C, c$  = specific heats of hot and cold fluids respectively;  $L_0$  = flow rate of liquid added to tank;  $M$  = mass of fluid in tank;  $T, t$  = temperature of hot and cold fluids respectively;  $T_1, t_1$  = temperatures at beginning of heating or cooling period or at inlet;  $T_2, t_2$  = temperature at end of period or at outlet;  $T_0, t_0$  = temperature of liquid added to tank;  $U$  = coefficient of heat transfer; and  $\dot{W}, w$  = flow rate through external exchanger of hot and cold fluids respectively.

**Applications** One typical application in heat transfer with batch operations is the heating of a reactor mix, maintaining temperature during a reaction period, and then cooling the products after the reaction is complete. This subsection is concerned with the heating and cooling of such systems in either unknown or specified periods.

The technique for deriving expressions relating time for heating or cooling agitated batches to coil or jacket area, heat-transfer coefficients, and the heat capacity of the vessel contents was developed by Bowman, Mueller, and Nagle [*Trans. Am. Soc. Mech. Eng.*, **62**, 283–294 (1940)] and extended by Fisher [*Ind. Eng. Chem.*, **36**, 939–942 (1944)] and Chaddock and Sanders [*Trans. Am. Inst. Chem. Eng.*, **40**, 203–210 (1944)] to external heat exchangers. Kern (*Process Heat Transfer*; McGraw-Hill, New York, 1950, Chap. 18) collected and published the results of these investigators.

The assumptions made were that (1)  $U$  is constant for the process and over the entire surface, (2) liquid flow rates are constant, (3) specific heats are constant for the process, (4) the heating or cooling medium has a constant inlet temperature, (5) agitation produces a uniform batch fluid temperature, (6) no partial phase changes occur, and (7) heat losses are negligible. The developed equations are as follows. If any of the assumptions do not apply to a system being designed, new equations should be developed or appropriate corrections made. Heat exchangers are counterflow except for the 1-2 exchangers, which are one-shell-pass, two-tube-pass, parallel-flow counterflow.

### Coil-in-Tank or Jacketed Vessel: Isothermal Heating Medium

$$\ln (T_1 - t_1)/(T_1 - t_2) = UA\theta/Mc \quad (11-35)$$

### Cooling-in-Tank or Jacketed Vessel: Isothermal Cooling Medium

$$\ln (T_1 - t_1)/(T_2 - t_1) = UA\theta/MC \quad (11-35a)$$

### Coil-in-Tank or Jacketed Vessel: Nonisothermal Heating Medium

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{WC}{Mc} \left( \frac{K_1 - 1}{K_1} \right) \theta \quad (11-35b)$$

where  $K_1 = e^{UA/MC}$

### Coil-in-Tank: Nonisothermal Cooling Medium

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{wc}{MC} \left( \frac{K_2 - 1}{K_2} \right) \theta \quad (11-35c)$$

where  $K_2 = e^{UA/wc}$

### External Heat Exchanger: Isothermal Heating Medium

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \frac{wc}{Mc} \left( \frac{K_2 - 1}{K_2} \right) \theta \quad (11-35d)$$

### External Exchanger: Isothermal Cooling Medium

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \frac{WC}{MC} \left( \frac{K_1 - 1}{K_1} \right) \theta \quad (11-35e)$$

### External Exchanger: Nonisothermal Heating Medium

$$\ln \frac{T_1 - t_1}{T_1 - t_2} = \left( \frac{K_3 - 1}{M} \right) \left( \frac{wWC}{K_3wc - WC} \right) \theta \quad (11-35f)$$

where  $K_3 = e^{UA/(1/WC - 1/wc)}$

### External Exchanger: Nonisothermal Cooling Medium

$$\ln \frac{T_1 - t_1}{T_2 - t_1} = \left( \frac{K_4 - 1}{M} \right) \left( \frac{Wwc}{K_4wc - WC} \right) \theta \quad (11-35g)$$

where  $K_4 = e^{UA/(1/WC - 1/wc)}$

**External Exchanger with Liquid Continuously Added to Tank: Isothermal Heating Medium**

$$\ln \frac{t_1 - t_0 - \frac{w}{L_0} \left( \frac{K_2 - 1}{K_2} \right) (T_1 - t_1)}{t_2 - t_0 - \frac{w}{L_0} \left( \frac{K_2 - 1}{K_2} \right) (T_1 - t_2)} = \left[ \frac{w}{L_0} \left( \frac{K_2 - 1}{K_2} \right) + 1 \right] \ln \frac{M + L_0 \theta}{M} \quad (11-35h)$$

If the addition of liquid to the tank causes an average endothermic or exothermic heat of solution,  $\pm q_s/\text{kg}$  (Btu/lb) of makeup, it may be included by adding  $\pm q_s/c_0$  to both the numerator and the denominator of the left side. The subscript 0 refers to the makeup.

**External Exchanger with Liquid Continuously Added to Tank: Isothermal Cooling Medium**

$$\ln \frac{T_0 - T_1 - \frac{W}{L_0} \left( \frac{K_1 - 1}{K_1} \right) (T_1 - t_1)}{T_0 - T_2 - \frac{W}{L_0} \left( \frac{K_1 - 1}{K_1} \right) (T_2 - t_1)} = \left[ 1 - \frac{W}{L_0} \left( \frac{K_1 - 1}{K_1} \right) \right] \ln \frac{M + L_0 \theta}{M} \quad (11-35i)$$

The heat-of-solution effects can be included by adding  $\pm q_s/C_0$  to both the numerator and the denominator of the left side.

**External Exchanger with Liquid Continuously Added to Tank: Nonisothermal Heating Medium**

$$\ln \frac{t_0 - t_1 + \frac{wWC(K_5 - 1)(T_1 - t_1)}{L_0(K_5WC - wc)}}{t_0 - t_2 + \frac{wWC(K_5 - 1)(T_2 - t_2)}{L_0(K_5WC - wc)}} = \left[ \frac{wWC(K_5 - 1)}{L_0(K_5WC - wc)} + 1 \right] \ln \frac{M + L_0 \theta}{M} \quad (11-35j)$$

where  $K_5 = e^{(UA/wc)(1 - wc/WC)}$

The heat-of-solution effects can be included by adding  $\pm q_s/c_0$  to both the numerator and the denominator of the left side.

**External Exchanger with Liquid Continuously Added to Tank: Nonisothermal Cooling Medium**

$$\ln \frac{T_0 - T_1 - \frac{Wwc(K_6 - 1)(T_1 - t_1)}{L_0(K_6wc - WC)}}{T_0 - T_2 - \frac{Wwc(K_6 - 1)(T_2 - t_1)}{L_0(K_6wc - WC)}} = \left[ \frac{Wwc(K_6 - 1)}{L_0(K_6wc - WC)} + 1 \right] \ln \frac{M + L_0 \theta}{M} \quad (11-35k)$$

where  $K_6 = e^{(UA/WC)(1 - WC/wc)}$

The heat-of-solution effects can be included by adding  $\pm q_s/C_0$  to both the numerator and the denominator of the left side.

**Heating and Cooling Agitated Batches: 1-2 Parallel Flow-Counterflow**

$$\frac{UA}{wc} = \frac{1}{\sqrt{R^2 + 1}} \ln \frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} \quad (11-35l)$$

$$R = \frac{T_1 - T_2}{t' - t} = \frac{wc}{WC} \quad \text{and} \quad S = \frac{t' - t}{T_1 - t}$$

$$\frac{2 - S(R + 1 - \sqrt{R^2 + 1})}{2 - S(R + 1 + \sqrt{R^2 + 1})} = e^{(UA/wc)\sqrt{R^2 + 1}} = K_7 \quad (11-35m)$$

$$S = \frac{2(K_7 - 1)}{K_7(R + 1 + R^2 + 1) - (R + 1 - R^2 + 1)}$$

**External 1-2 Exchanger: Heating**

$$\ln (T_1 - t_1)/(T_1 - t_2) = (Sw/M)\theta \quad (11-35n)$$

**External 1-2 Exchanger: Cooling**

$$\ln (T_1 - t_1)/(T_2 - t_1) = S(wc/MC)\theta \quad (11-35o)$$

The cases of multipass exchangers with liquid continuously added to the tank are covered by Kern, as cited earlier. An alternative method for all multipass-exchanger gases, including those presented as well as cases with two or more shells in series, is as follows:

1. Determine  $UA$  for using the applicable equations for counter-flow heat exchangers.
2. Use the initial batch temperature  $T_1$  or  $t_1$ .
3. Calculate the outlet temperature from the exchanger of each fluid. (This will require trial-and-error methods.)
4. Note the  $F_T$  correction factor for the corrected mean temperature difference. (See Fig. 11-4.)
5. Repeat steps 2, 3, and 4 by using the final batch temperature  $T_2$  and  $t_2$ .
6. Use the average of the two values for  $F$ , then increase the required multipass  $UA$  as follows:

$$UA(\text{multipass}) = UA(\text{counterflow})/F_T$$

In general, values of  $F_T$  below 0.8 are uneconomical and should be avoided.  $F_T$  can be raised by increasing the flow rate of either or both of the flow streams. Increasing flow rates to give values well above 0.8 is a matter of economic justification.

If  $F_T$  varies widely from one end of the range to the other,  $F_T$  should be determined for one or more intermediate points. The average should then be determined for each step which has been established and the average of these taken for use in step 6.

**Effect of External Heat Loss or Gain** If heat loss or gain through the vessel walls cannot be neglected, equations which include this heat transfer can be developed by using energy balances similar to those used for the derivations of equations given previously. Basically, these equations must be modified by adding a heat-loss or heat-gain term.

A simpler procedure, which is probably acceptable for most practical cases, is to ratio  $UA$  or  $\theta$  either up or down in accordance with the required modification in total heat load over time  $\theta$ .

Another procedure, which is more accurate for the external-heat-exchanger cases, is to use an equivalent value for  $MC$  (for a vessel being heated) derived from the following energy balance:

$$Q = (Mc)_s(t_2 - t_1) = Mc(t_2 - t_1) + U'A'(MTD')\theta \quad (11-35p)$$

where  $Q$  is the total heat transferred over time  $\theta$ ,  $U'A'$  is the heat-transfer coefficient for heat loss times the area for heat loss, and  $MTD'$  is the mean temperature difference for the heat loss.

A similar energy balance would apply to a vessel being cooled.

**Internal Coil or Jacket Plus External Heat Exchanger** This case can be most simply handled by treating it as two separate problems.  $M$  is divided into two separate masses  $M_1$  and  $(M - M_1)$ , and the appropriate equations given earlier are applied to each part of the system. Time  $\theta$ , of course, must be the same for both parts.

**Equivalent-Area Concept** The preceding equations for batch operations, particularly Eq. 11-35 can be applied for the calculation of heat loss from tanks which are allowed to cool over an extended period of time. However, different surfaces of a tank, such as the top (which would not be in contact with the tank contents) and the bottom, may have coefficients of heat transfer which are different from those of the vertical tank walls. The simplest way to resolve this difficulty is to use an equivalent area  $A_e$  in the appropriate equations where

$$A_e = A_b U_b / U_s + A_t U_t / U_s + A_s \quad (11-35q)$$

TABLE 11-1 Typical Values for Use with Eqs. (11-36) to (11-44)\*

| Application                                   | Fluid           | $U_s$ | $A_s$                       |
|-----------------------------------------------|-----------------|-------|-----------------------------|
| Tanks on legs, outdoors, not insulated        | Oil             | 3.7   | $0.22 A_t + A_b + A_s$      |
|                                               | Water at 150°F. | 5.1   | $0.16 A_t + A_b + A_s$      |
| Tanks on legs, outdoors, insulated 1 in.      | Oil             | 0.45  | $0.7 A_t + A_b + A_s$       |
|                                               | Water           | 0.43  | $0.67 A_t + A_b + A_s$      |
| Tanks on legs, indoors, not insulated         | Oil             | 1.5   | $0.53 A_t + A_b + A_s$      |
|                                               | Water           | 1.8   | $0.35 A_t + A_b + A_s$      |
| Tanks on legs, indoors, insulated 1 in.       | Oil             | 0.36  | $0.8 A_t + A_b + A_s$       |
|                                               | Water           | 0.37  | $0.73 A_t + A_b + A_s$      |
| Flat-bottom tanks,† outdoors, not insulated   | Oil             | 3.7   | $0.22 A_t + A_s + 0.43 D_t$ |
|                                               | Water           | 5.1   | $0.16 A_t + A_s + 0.31 D_t$ |
| Flat-bottom tanks,† outdoors, insulated 1 in. | Oil             | 0.36  | $0.7 A_t + A_s + 3.9 D_t$   |
|                                               | Water           | 0.37  | $0.16 A_t + A_s + 3.7 D_t$  |
| Flat-bottom tanks, indoors, not insulated     | Oil             | 1.5   | $0.53 A_t + A_s + 1.1 D_t$  |
|                                               | Water           | 1.8   | $0.35 A_t + A_s + 0.9 D_t$  |
| Flat-bottom tanks, indoors, insulated 1 in.   | Oil             | 0.36  | $0.8 A_t + A_s + 4.4 D_t$   |
|                                               | Water           | 0.37  | $0.73 A_t + A_s + 4.5 D_t$  |

\*Based on typical coefficients.

†The ratio  $(t - t_o)/(t - t')$  assumed at 0.85 for outdoor tanks. °C = (°F - 32)/1.8; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

and the subscripts  $b$ ,  $s$ , and  $t$  refer to the bottom, sides, and top respectively.  $U$  is usually taken as  $U_s$ . Table 11-1 lists typical values for  $U_s$  and expressions for  $A_s$  for various tank configurations.

**Nonagitated Batches** Cases in which vessel contents are vertically stratified, rather than uniform in temperature, have been treated by Kern (op. cit.). These are of little practical importance except for tall, slender vessels heated or cooled with external exchangers. The result is that a smaller exchanger is required than for an equivalent agitated batch system that is uniform.

**Storage Tanks** The equations for batch operations with agitation may be applied to storage tanks even though the tanks are not agitated. This approach gives conservative results. The important cases (nonsteady state) are:

1. *Tanks cool; contents remain liquid.* This case is relatively simple and can easily be handled by the equations given earlier.
2. *Tanks cool, contents partially freeze, and solids drop to bottom or rise to top.* This case requires a two-step calculation. The first step is handled as in case 1. The second step is calculated by assuming an isothermal system at the freezing point. It is possible, given time and a sufficiently low ambient temperature, for tank contents to freeze solid.
3. *Tanks cool and partially freeze; solids form a layer of self-insulation.* This complex case, which has been known to occur with heavy hydrocarbons and mixtures of hydrocarbons, has been discussed by Stuhlberg [*Pet. Refiner*, **38**, 143 (Apr. 1, 1959)]. The contents in the center of such tanks have been known to remain warm and liquid even after several years of cooling.

It is very important that a melt-out riser be installed whenever tank contents are expected to freeze on prolonged shutdown. The purpose is to provide a molten chimney through the crust for relief of thermal expansion or cavitation if fluids are to be pumped out or recirculated through an external exchanger. An external heat tracer, properly located, will serve the same purpose but may require more remelt time before pumping can be started.

## THERMAL DESIGN OF TANK COILS

The thermal design of tank coils involves the determination of the area of heat-transfer surface required to maintain the contents of the tank at a constant temperature or to raise or lower the temperature of the contents by a specified magnitude over a fixed time.

**Nomenclature**  $A$  = area;  $A_b$  = area of tank bottom;  $A_s$  = area of coil;  $A_t$  = equivalent area;  $A_s$  = area of sides;  $A_t$  = area of top;  $A_1$  = equivalent area receiving heat from external coils;  $A_2$  = equivalent area not covered with external coils;  $D_t$  = diameter of tank;  $F$  = design (safety) factor;  $h$  = film coefficient;  $h_a$  = coefficient of ambient air;  $h_c$  = coefficient of coil;  $h_h$  = coefficient of heating medium;  $h_l$  = coefficient of liquid phase of tank contents or tube-side coefficient referred to

outside of coil;  $h_s$  = coefficient of insulation;  $k$  = thermal conductivity;  $k_g$  = thermal conductivity of ground below tank;  $M$  = mass of tank contents when full;  $t$  = temperature;  $t_a$  = temperature of ambient air;  $t_d$  = temperature of dead-air space;  $t_f$  = temperature of contents at end of heating;  $t_g$  = temperature of ground below tank;  $t_h$  = temperature of heating medium;  $t_o$  = temperature of contents at beginning of heating;  $U$  = overall coefficient;  $U_b$  = coefficient at tank bottom;  $U_c$  = coefficient of coil;  $U_d$  = coefficient of dead air to the tank contents;  $U_i$  = coefficient through insulation;  $U_s$  = coefficient at sides;  $U_t$  = coefficient at top; and  $U_2$  = coefficient at area  $A_2$ .

Typical coil coefficients are listed in Table 11-2. More exact values can be calculated by using the methods for natural convection or forced convection given elsewhere in this section.

**Maintenance of Temperature** Tanks are often maintained at temperature with internal coils if the following equations are assumed to be applicable:

$$q = U_s A_s (T - t') \quad (11-36)$$

and

$$A_c = q/U_c (MTD) \quad (11-36a)$$

These make no allowance for unexpected shutdowns. One method of allowing for shutdown is to add a safety factor to Eq. 11-36a.

In the case of a tank maintained at temperature with internal coils, the coils are usually designed to cover only a portion of the tank. The temperature  $t_d$  of the dead-air space between the coils and the tank is obtained from

$$U_d A_1 (t_d - t) = U_2 A_2 (t - t') \quad (11-37)$$

The heat load is

$$q = U_d A_1 (t_d - t) + A_1 U_i (t_d - t') \quad (11-38)$$

The coil area is

$$A_c = \frac{qF}{U_c (t_h - t_d)_m} \quad (11-39)$$

where  $F$  is a safety factor.

## Heating

**Heating with Internal Coil from Initial Temperature for Specified Time**

$$Q = Wc(t_f - t_o) \quad (11-40)$$

$$A_c = \left[ \frac{Q}{\theta_h} + U_s A_s \left( \frac{t_f + t_o}{2} - t' \right) \right] \left[ \frac{1}{U_c [t_h - (t_f + t_o)/2]} \right] (F) \quad (11-41)$$

where  $\theta_h$  is the length of heating period. This equation may also be used when the tank contents have cooled from  $t_f$  to  $t_o$  and must be

**TABLE 11-2 Overall Heat-Transfer Coefficients for Coils Immersed in Liquids**  
*U* Expressed as Btu/(h · ft<sup>2</sup> · °F)

| Substance inside coil                  | Substance outside coil          | Coil material    | Agitation                               | <i>U</i> |
|----------------------------------------|---------------------------------|------------------|-----------------------------------------|----------|
| Steam                                  | Water                           | Lead             | Agitated                                | 70       |
| Steam                                  | Sugar and molasses solutions    | Copper           | None                                    | 50–240   |
| Steam                                  | Boiling aqueous solution        |                  |                                         | 600      |
| Cold water                             | Dilute organic dye intermediate | Lead             | Turboagitator at 95 r.p.m.              | 300      |
| Cold water                             | Warm water                      | Wrought iron     | Air bubbled into water surrounding coil | 150–300  |
| Cold water                             | Hot water                       | Lead             | 0.40 r.p.m. paddle stirrer              | 90–360   |
| Brine                                  | Amino acids                     |                  | 30 r.p.m.                               | 100      |
| Cold water                             | 25% oleum at 60°C.              | Wrought iron     | Agitated                                | 20       |
| Water                                  | Aqueous solution                | Lead             | 500 r.p.m. sleeve propeller             | 250      |
| Water                                  | 8% NaOH                         |                  | 22 r.p.m.                               | 155      |
| Steam                                  | Fatty acid                      | Copper (pancake) | None                                    | 96–100   |
| Milk                                   | Water                           |                  | Agitation                               | 300      |
| Cold water                             | Hot water                       | Copper           | None                                    | 105–180  |
| 60°F. water                            | 50% aqueous sugar solution      | Lead             | Mild                                    | 50–60    |
| Steam and hydrogen at 1500 lb./sq. in. | 60°F. water                     | Steel            |                                         | 100–165  |
| Steam 110–146 lb./sq. in. gage         | Vegetable oil                   | Steel            | None                                    | 23–29    |
| Steam                                  | Vegetable oil                   | Steel            | Various                                 | 39–72    |
| Cold water                             | Vegetable oil                   | Steel            | Various                                 | 29–72    |

NOTES: Chilton, Drew, and Jebens [*Ind. Eng. Chem.*, **36**, 510 (1944)] give film coefficients for heating and cooling agitated fluids using a coil in a jacketed vessel. Because of the many factors affecting heat transfer, such as viscosity, temperature difference, and coil size, the values in this table should be used primarily for preliminary design estimates and checking calculated coefficients.

°C = (°F – 32)/1.8; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

reheated to  $t_f$ . If the contents cool during a time  $\theta_c$ , the temperature at the end of this cooling period is obtained from

$$\ln \left( \frac{t_f - t'}{t_o - t'} \right) = \frac{U_s A_s \theta_c}{Wc} \quad (11-42)$$

**Heating with External Coil from Initial Temperature for Specified Time** The temperature of the dead-air space is obtained from

$$U_d A_1 [t_d - 0.5(t_f - t_o)] = U_2 A_2 [0.5(t_f - t_o) - t'] + Q/\theta_h \quad (11-43)$$

The heat load is

$$q = U_1 A_1 (t_d - t') + U_2 A_2 [0.5(t_f - t_o) - t'] + Q/\theta_h \quad (11-44)$$

The coil area is obtained from Eq. 11-39.

The safety factor used in the calculations is a matter of judgment based on confidence in the design. A value of 1.10 is normally not considered excessive. Typical design parameters are shown in Tables 11-1 and 11-2.

## HEATING AND COOLING OF TANKS

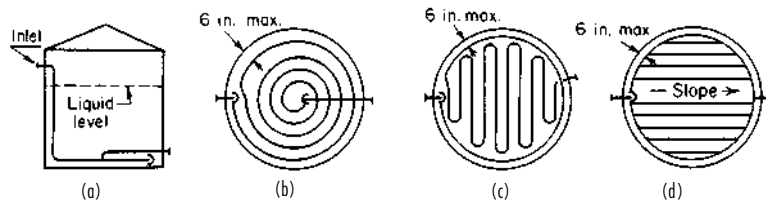
**Tank Coils** Pipe tank coils are made in a wide variety of configurations, depending upon the application and shape of the vessel. **Helical** and **spiral** coils are most commonly shop-fabricated, while the **hairpin** pattern is generally field-fabricated. The helical coils are used principally in process tanks and pressure vessels when large areas

for rapid heating or cooling are required. In general, heating coils are placed low in the tank, and cooling coils are placed high or distributed uniformly through the vertical height.

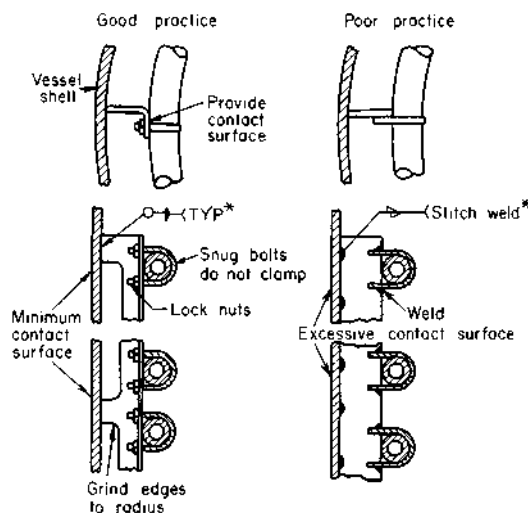
Stocks which tend to solidify on cooling require uniform coverage of the bottom or agitation. A maximum spacing of 0.6 m (2 ft) between turns of 50.8-mm (2-in) and larger pipe and a close approach to the tank wall are recommended. For smaller pipe or for low-temperature heating media, closer spacing should be used. In the case of the common hairpin coils in vertical cylindrical tanks, this means adding an encircling ring within 152 mm (6 in) of the tank wall (see Fig. 11-29a for this and other typical coil layouts). The coils should be set directly on the bottom or raised not more than 50.8 to 152 mm (2 to 6 in), depending upon the difficulty of remelting the solids, in order to permit free movement of product within the vessel. The coil inlet should be above the liquid level (or an internal melt-out riser installed) to provide a molten path for liquid expansion or venting of vapors.

Coils may be sloped to facilitate drainage. When it is impossible to do so and remain close enough to the bottom to get proper remelting, the coils should be blown out after usage in cold weather to avoid damage by freezing.

Most coils are firmly clamped (but not welded) to supports. **Supports** should allow expansion but be rigid enough to prevent uncontrolled motion (see Fig. 11-29b). Nuts and bolts should be securely fastened. Reinforcement of the inlet and outlet connections through the tank wall is recommended, since bending stresses due to thermal expansion are usually high at such points.



**FIG. 11-29a** Typical coil designs for good bottom coverage. (a) Elevated inlet on spiral coil. (b) Spiral with recirculating ring. (c) Hairpin with encircling ring. (d) Ring header type.



\* See Amer. Standards Assn. Standard Y32.3-1959

FIG. 11-29b Right and wrong ways to support coils. [*Chem. Eng.*, 172 (May 16, 1960).]

In general, 50.8- and 63.4-mm (2- and 2½-in) coils are the most economical for shop fabrication and 38.1- and 50.8-mm (1½- and 2-in) for field fabrication. The tube-side heat-transfer coefficient, high-pressure, or layout problems may lead to the use of smaller-size pipe.

The wall thickness selected varies with the service and material. Carbon steel coils are often made from schedule 80 or heavier pipe to allow for corrosion. When stainless-steel or other high-alloy coils are not subject to corrosion or excessive pressure, they may be of schedule 5 or 10 pipe to keep costs at a minimum, although high-quality welding is required for these thin walls to assure trouble-free service.

Methods for calculating heat loss from tanks and the sizing of tank coils have been published by Stuhlbarg [*Pet. Refiner*, **38**, 143 (April 1959)].

**Fin-tube coils** are used for fluids which have poor heat-transfer characteristics to provide more surface for the same configuration at reduced cost or when temperature-driven fouling is to be minimized. Fin tubing is not generally used when bottom coverage is important.

**Fin-tube tank heaters** are compact prefabricated bundles which can be brought into tanks through manholes. These are normally installed vertically with longitudinal fins to produce good convection currents. To keep the heaters low in the tank, they can be installed horizontally with helical fins or with perforated longitudinal fins to prevent entrapment. Fin tubing is often used for heat-sensitive material because of the lower surface temperature for the same heating medium, resulting in a lesser tendency to foul.

**Plate or panel coils** made from two metal sheets with one or both embossed to form passages for a heating or cooling medium can be used in lieu of pipe coils. Panel coils are relatively light in weight, easy to install, and easily removed for cleaning. They are available in a range of standard sizes and in both flat and curved patterns. Process tanks have been built by using panel coils for the sides or bottom. A serpentine construction is generally utilized when liquid flows through the unit. Header-type construction is used with steam or other condensing media.

Standard **glass coils** with 0.18 to 11.1 m<sup>2</sup> (2 to 120 ft<sup>2</sup>) of heat-transfer surface are available. Also available are plate-type units made of **impervious graphite**.

**Teflon Immersion Coils** Immersion coils made of Teflon fluorocarbon resin are available with 2.5-mm (0.10-in) ID tubes to increase overall heat-transfer efficiency. The flexible bundles are available with 100, 160, 280, 500, and 650 tubes with standard lengths

varying in 0.6-m (2-ft) increments between 1.2 and 4.8 m (4 and 16 ft). These coils are most commonly used in metal-finishing baths and are adaptable to service in reaction vessels, crystallizers, and tanks where corrosive fluids are used.

**Bayonet Heaters** A bayonet-tube element consists of an outer and an inner tube. These elements are inserted into tanks and process vessels for heating and cooling purposes. Often the outer tube is of expensive alloy or nonmetallic (e.g., glass, impervious graphite), while the inner tube is of carbon steel. In glass construction, elements with 50.8- or 76.2-mm (2- or 3-in) glass pipe [with lengths to 2.7 m (9 ft)] are in contact with the external fluid, with an inner tube of metal.

**External Coils and Tracers** Tanks, vessels, and pipe lines can be equipped for heating or cooling purposes with external coils. These are generally 9.8 to 19 mm (¾ to ¾ in) so as to provide good distribution over the surface and are often of soft copper or aluminum, which can be bent by hand to the contour of the tank or line. When necessary to avoid "hot spots," the tracer is so mounted that it does not touch the tank.

External coils spaced away from the tank wall exhibit a coefficient of around 5.7 W/(m<sup>2</sup>·°C) [1 Btu/(h·ft<sup>2</sup> of coil surface·°F)]. Direct contact with the tank wall produces higher coefficients, but these are difficult to predict since they are strongly dependent upon the degree of contact. The use of **heat-transfer cements** does improve performance. These puttylike materials of high thermal conductivity are troweled or caulked into the space between the coil and the tank or pipe surface.

Costs of the cements (in 1960) varied from 37 to 63 cents per pound, with requirements running from about 0.27 lb/ft of ¾-in outside-diameter tubing to 1.48 lb/ft of 1-in pipe. Panel coils require ½ to 1 lb/ft<sup>2</sup>. A rule of thumb for preliminary estimating is that the per-foot installed cost of tracer with cement is about double that of the tracer alone.

**Jacketed Vessels** Jacketing is often used for vessels needing frequent cleaning and for glass-lined vessels which are difficult to equip with internal coils. The jacket eliminates the need for the coil yet gives a better overall coefficient than external coils. However, only a limited heat-transfer area is available. The conventional jacket is of simple construction and is frequently used. It is most effective with a condensing vapor. A liquid heat-transfer fluid does not maintain uniform flow characteristics in such a jacket. Nozzles, which set up a swirling motion in the jacket, are effective in improving heat transfer. Wall thicknesses are often high unless reinforcement rings are installed.

**Spiral baffles**, which are sometimes installed for liquid services to improve heat transfer and prevent channeling, can be designed to serve as reinforcements. A spiral-wound channel welded to the vessel wall is an alternative to the spiral baffle which is more predictable in performance, since cross-baffle leakage is eliminated, and is reportedly lower in cost [Feichtinger, *Chem. Eng.*, **67**, 197 (Sept. 5, 1960)].

The half-pipe jacket is used when high jacket pressures are required. The flow pattern of a liquid heat-transfer fluid can be controlled and designed for effective heat transfer. The dimple jacket offers structural advantages and is the most economical for high jacket pressures. The low volumetric capacity produces a fast response to temperature changes.

## EXTENDED OR FINNED SURFACES

**Finned-Surface Application** Extended or finned surfaces are often used when one film coefficient is substantially lower than the other, the goal being to make  $h_o A_{oe} \approx h_i A_i$ . A few typical fin configurations are shown in Fig. 11-30a. Longitudinal fins are used in double-pipe exchangers. Transverse fins are used in cross-flow and shell-and-tube configurations. High transverse fins are used mainly with low-pressure gases; low fins are used for boiling and condensation of nonaqueous streams as well as for sensible-heat transfer. Finned surfaces have been proven to be a successful means of controlling temperature driven fouling such as coking and scaling. Fin spacing should be great enough to avoid entrapment of particulate matter in the fluid stream (5 mm minimum spacing).

The area added by the fin is not as efficient for heat transfer as bare tube surface owing to resistance to conduction through the fin. The



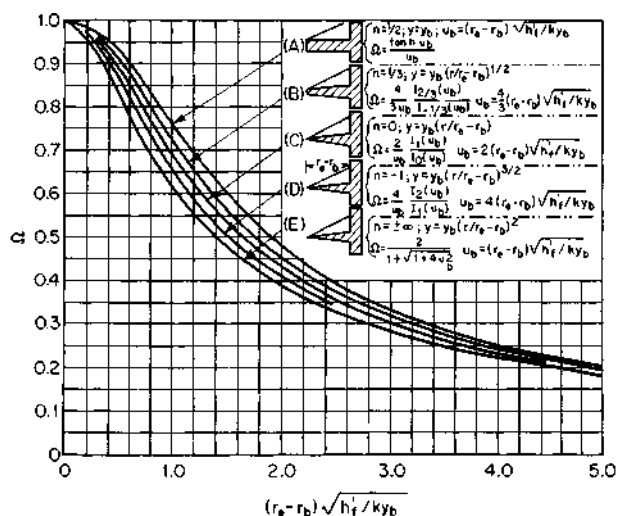


FIG. 11-30a Efficiencies for several longitudinal fin configurations.

effective heat-transfer area is

$$A_{oe} = A_{uf} + A_f \Omega \quad (11-45)$$

The fin efficiency is found from mathematically derived relations, in which the film heat-transfer coefficient is assumed to be constant over the entire fin and temperature gradients across the thickness of the fin have been neglected (see Kraus, *Extended Surfaces*, Spartan Books, Baltimore, 1963). The efficiency curves for some common fin configurations are given in Figs. 11-30a and 11-30b.

**High Fins** To calculate heat-transfer coefficients for cross-flow to a transversely finned surface, it is best to use a correlation based on experimental data for that surface. Such data are not often available, and a more general correlation must be used, making allowance for the possible error. Probably the best general correlation for bundles of finned tubes is given by Schmidt [Kaltetechnik, 15, 98–102, 370–378 (1963)]:

$$hD_r/k = K(D_r \rho V'_{\max}/\mu)^{0.625} R_f^{-0.375} N_{Pr}^{1/3} \quad (11-46)$$

where  $K = 0.45$  for staggered tube arrays and  $0.30$  for in-line tube

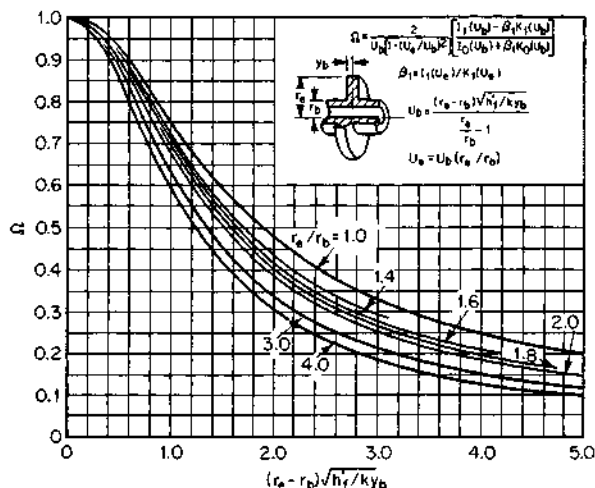


FIG. 11-30b Efficiencies for annular fins of constant thickness.

arrays:  $D_r$  is the root or base diameter of the tube;  $V'_{\max}$  is the maximum velocity through the tube bank, i.e., the velocity through the minimum flow area between adjacent tubes; and  $R_f$  is the ratio of the total outside surface area of the tube (including fins) to the surface of a tube having the same root diameter but without fins.

**Pressure drop** is particularly sensitive to geometrical parameters, and available correlations should be extrapolated to geometries different from those on which the correlation is based only with great caution and conservatism. The best correlation is that of Robinson and Briggs [Chem. Eng. Prog., 62, Symp. Ser. 64, 177–184 (1966)].

**Low Fins** Low-finned tubing is generally used in shell-and-tube configurations. For sensible-heat transfer, only minor modifications are needed to permit the shell-side method given earlier to be used for both heat transfer and pressure [see Briggs, Katz, and Young, Chem. Eng. Prog., 59(11), 49–59 (1963)]. For condensing on low-finned tubes in horizontal bundles, the Nusselt correlation is generally satisfactory for low-surface-tension [ $\sigma < (3/10^{-6})\text{N/m}$  (30 dyn/cm)] condensates fins of finned surfaces should not be closely spaced for high-surface-tension condensates (notably water), which do not drain easily.

The modified Palen-Small method can be employed for reboiler design using finned tubes, but the maximum flux is calculated from  $A_o$ , the total outside heat-transfer area including fins. The resulting value of  $q_{\max}$  refers to  $A_o$ .

## FOULING AND SCALING

Fouling refers to any change in the solid boundary separating two heat transfer fluids, whether by dirt accumulation or other means, which results in a decrease in the rate of heat transfer occurring across that boundary. Fouling may be classified by mechanism into six basic categories:

1. **Corrosion fouling.** The heat transfer surface reacts chemically with elements of the fluid stream producing a less conductive, corrosion layer on all or part of the surface.
2. **Biofouling.** Organisms present in the fluid stream are attracted to the warm heat-transfer surface where they attach, grow, and reproduce. The two subgroups are microbiofouls such as slime and algae and macrobiofouls such as snails and barnacles.
3. **Particulate fouling.** Particles held in suspension in the flow stream will deposit out on the heat-transfer surface in areas of sufficiently lower velocity.
4. **Chemical reaction fouling** (ex.—Coking). Chemical reaction of the fluid takes place on the heat-transfer surface producing an adhering solid product of reaction.
5. **Precipitation fouling** (ex.—Scaling). A fluid containing some dissolved material becomes supersaturated with respect to this material at the temperatures seen at the heat-transfer surface. This results in a crystallization of the material which “plates out” on the warmer surface.
6. **Freezing fouling.** Overcooling of a fluid below the fluid’s freezing point at the heat-transfer surface causes solidification and coating of the heat-transfer surface.

**Control of Fouling** Once the combination of mechanisms contributing to a particular fouling problem are recognized, methods to substantially reduce the fouling rate may be implemented. For the case of **corrosion fouling**, the common solution is to choose a less corrosive material of construction balancing material cost with equipment life. In cases of **biofouling**, the use of copper alloys and/or chemical treatment of the fluid stream to control organism growth and reproduction are the most common solutions.

In the case of **particulate fouling**, one of the more common types, insuring a sufficient flow velocity and minimizing areas of lower velocities and stagnant flows to help keep particles in suspension is the most common means of dealing with the problem. For water, the recommended tubeside minimum velocity is about 0.9 to 1.0 m/s. This may not always be possible for moderate to high-viscosity fluids where the resulting pressure drop can be prohibitive.

Special care should be taken in the application of any velocity requirement to the shellside of segmental-baffled bundles due to the many different flow streams and velocities present during operation, the unavoidable existence of high-fouling areas of flow stagnation, and

the danger of flow-induced tube vibration. In general, shellside-particulate fouling will be greatest for segmentally baffled bundles in the regions of low velocity and the TEMA-fouling factors (which are based upon the use of this bundle type) should be used. However, since the 1940's, there have been a host of successful, low-fouling exchangers developed, some tubular and some not, which have in common the elimination of the cross-flow plate baffle and provide practically no regions of flow stagnation at the heat-transfer surface. Some examples are the plate and frame exchanger, the spiral plate exchanger, and the twisted tube exchanger, all of which have dispensed with baffles altogether and use the heat-transfer surface itself for bundle support. The general rule for these designs is to provide between 25 and 30 percent excess surface to compensate for potential fouling, although this can vary in special applications.

For the remaining classifications—**polymerization, precipitation, and freezing**—fouling is the direct result of temperature extremes at the heat-transfer surface and is reduced by reducing the temperature difference between the heat-transfer surface and the bulk-fluid stream. Conventional wisdom says to increase velocity, thus increasing the local heat-transfer coefficient to bring the heat-transfer surface temperature closer to the bulk-fluid temperature. However, due to a practical limit on the amount of heat-transfer coefficient increase available by increasing velocity, this approach, although better than nothing, is often not satisfactory by itself.

A more effective means of reducing the temperature difference is by using, in concert with adequate velocities, some form of extended surface. As discussed by Shilling (*Proceedings of the 10th International Heat Transfer Conference*, Brighton, U.K., 4, p. 423), this will tend to reduce the temperature extremes between fluid and heat transfer surface and not only reduce the rate of fouling but make the heat exchanger generally less sensitive to the effects of any fouling that does occur. In cases where unfinned tubing in a triangular tube layout would not be acceptable because fouling buildup and eventual mechanical cleaning are inevitable, extended surface should be used only when the exchanger construction allows access for cleaning.

**Fouling Transients and Operating Periods** Three common behaviors are noted in the development of a fouling film over a period of time. One is the so-called *asymptotic fouling* in which the speed of fouling resistance increase decreases over time as it approaches some asymptotic value beyond which no further fouling can occur. This is commonly found in temperature-driven fouling. A second is *linear fouling* in which the increase in fouling resistance follows a straight line over the time of operation. This could be experienced in a case of severe particulate fouling where the accumulation of dirt during the time of operation did not appreciably increase velocities to mitigate the problem. The third, *falling rate fouling*, is neither linear nor asymptotic but instead lies somewhere between these two extremes. The rate of fouling decreases with time but does not appear to approach an asymptotic maximum during the time of operation. This is the most common type of fouling in the process industry and is usually the result of a combination of different fouling mechanisms occurring together.

The optimum operating period between cleanings depends upon the rate and type of fouling, the heat exchanger used (i.e. baffle type, use of extended surface, and velocity and pressure drop design constraints), and the ease with which the heat exchanger may be removed from service for cleaning. As noted above, care must be taken in the use of fouling factors for exchanger design, especially if the exchanger configuration has been selected specifically to minimize fouling accumulation. An oversurfaced heat exchanger which will not foul enough to operate properly can be almost as much a problem as an undersized exchanger. This is especially true in steam-heated exchangers where the ratio of design MTD to minimum achievable MTD is less than  $U_{\text{clean}}/U_{\text{fouled}}$ .

**Removal of Fouling Deposits** Chemical removal of fouling can be achieved in some cases by weak acid, special solvents, and so on. Other deposits adhere weakly and can be washed off by periodic operation at very high velocities or by flushing with a high-velocity steam or water jet or using a sand-water slurry. These methods may be applied to both the shell side and tube side without pulling the bundle. Many fouling deposits, however, must be removed by positive

mechanical action such as rodding, turbing, or scraping the surface. These techniques may be applied inside of tubes without pulling the bundle but can be applied on the shellside only after bundle removal. Even then there is limited access because of the tube pitch and rotated square or large triangular layouts are recommended. In many cases, it has been found that designs developed to minimize fouling often develop a fouling layer which is more easily removed.

**Fouling Resistances** There are no published methods for predicting fouling resistances a priori. The accumulated experience of exchanger designers and users was assembled more than 40 years ago based primarily upon segmental-baffled exchanger bundles and may be found in the *Standards of Tubular Exchanger Manufacturers Association* (TEMA). In the absence of other information, the fouling resistances contained therein may be used.

## TYPICAL HEAT-TRANSFER COEFFICIENTS

Typical overall heat-transfer coefficients are given in Tables 11-3 through 11-8. Values from these tables may be used for preliminary estimating purposes. They should not be used in place of the design methods described elsewhere in this section, although they may serve as a useful check on the results obtained by those design methods.

## THERMAL DESIGN FOR SOLIDS PROCESSING

Solids in divided form, such as powders, pellets, and lumps, are heated and/or cooled in chemical processing for a variety of objectives such as solidification or fusing (Sec. 11), drying and water removal (Sec. 20), solvent recovery (Secs. 13 and 20), sublimation (Sec. 17), chemical reactions (Sec. 20), and oxidation. For process and mechanical-design considerations, see the referenced sections.

**Thermal design** concerns itself with sizing the equipment to effect the heat transfer necessary to carry on the process. The design equation is the familiar one basic to all modes of heat transfer, namely,

$$A = Q/U \Delta t \quad (11-47)$$

where  $A$  = effective heat-transfer surface,  $Q$  = quantity of heat required to be transferred,  $\Delta t$  = temperature difference of the process, and  $U$  = overall heat-transfer coefficient. It is helpful to define the modes of heat transfer and the corresponding overall coefficient as  $U_{co}$  = overall heat-transfer coefficient for (indirect through-a-wall) conduction,  $U_{cw}$  = overall heat-transfer coefficient for the little-used convection mechanism,  $U_{ct}$  = heat-transfer coefficient for the *contactive* mechanism in which the gaseous-phase heat carrier passes directly through the solids bed, and  $U_{ra}$  = heat-transfer coefficient for radiation.

There are two general methods for determining numerical values for  $U_{co}$ ,  $U_{cw}$ ,  $U_{ct}$ , and  $U_{ra}$ . One is by analysis of actual operating data. Values so obtained are used on geometrically similar systems of a size not too different from the equipment from which the data were obtained. The second method is predictive and is based on the material properties and certain operating parameters. Relative values of the coefficients for the various modes of heat transfer at temperatures up to 980°C (1800°F) are as follows (Holt, Paper 11, Fourth National Heat Transfer Conference, Buffalo, 1960):

|            |     |
|------------|-----|
| Convective | 1   |
| Radiant    | 2   |
| Conductive | 20  |
| Contactive | 200 |

Because heat-transfer equipment for solids is generally an adaptation of a primarily material-handling device, the area of heat transfer is often small in relation to the overall size of the equipment. Also peculiar to solids heat transfer is that the  $\Delta t$  varies for the different heat-transfer mechanisms. With a knowledge of these mechanisms, the  $\Delta t$  term generally is readily estimated from temperature limitations imposed by the burden characteristics and/or the construction.

**Conductive Heat Transfer** Heat-transfer equipment in which heat is transferred by conduction is so constructed that the solids load (burden) is separated from the heating medium by a wall.

For a high proportion of applications,  $\Delta t$  is the log-mean temperature difference. Values of  $U_{co}$  are reported in Secs. 11, 15, 17, and 19.

**TABLE 11-3 Typical Overall Heat-Transfer Coefficients in Tubular Heat Exchangers**

$$U = \text{Btu}/(^{\circ}\text{F} \cdot \text{ft}^2 \cdot \text{h})$$

| Shell side                                  | Tube side                       | Design $U$ | Includes total dirt | Shell side                                   | Tube side                                | Design $U$ | Includes total dirt |
|---------------------------------------------|---------------------------------|------------|---------------------|----------------------------------------------|------------------------------------------|------------|---------------------|
| <b>Liquid-liquid media</b>                  |                                 |            |                     |                                              |                                          |            |                     |
| Aroclor 1248                                | Jet fuels                       | 100–150    | 0.0015              | Dowtherm vapor                               | Dowtherm liquid                          | 80–120     | .0015               |
| Cutback asphalt                             | Water                           | 10–20      | .01                 | Gas-plant tar                                | Steam                                    | 40–50      | .0055               |
| Demineralized water                         | Water                           | 300–500    | .001                | High-boiling hydrocarbons V                  | Water                                    | 20–50      | .003                |
| Ethanol amine (MEA or DEA) 10–25% solutions | Water or DEA, or MEA solutions  | 140–200    | .003                | Low-boiling hydrocarbons A                   | Water                                    | 80–200     | .003                |
| Fuel oil                                    | Water                           | 15–25      | .007                | Hydrocarbon vapors (partial condenser)       | Oil                                      | 25–40      | .004                |
| Fuel oil                                    | Oil                             | 10–15      | .008                | Organic solvents A                           | Water                                    | 100–200    | .003                |
| Gasoline                                    | Water                           | 60–100     | .003                | Organic solvents high NC, A                  | Water or brine                           | 20–60      | .003                |
| Heavy oils                                  | Heavy oils                      | 10–40      | .004                | Organic solvents low NC, V                   | Water or brine                           | 50–120     | .003                |
| Heavy oils                                  | Water                           | 15–50      | .005                | Kerosene                                     | Water                                    | 30–65      | .004                |
| Hydrogen-rich reformer stream               | Hydrogen-rich reformer stream   | 90–120     | .002                | Kerosene                                     | Oil                                      | 20–30      | .005                |
| Kerosene or gas oil                         | Water                           | 25–50      | .005                | Naphtha                                      | Water                                    | 50–75      | .005                |
| Kerosene or gas oil                         | Oil                             | 20–35      | .005                | Naphtha                                      | Oil                                      | 20–30      | .005                |
| Kerosene or jet fuels                       | Trichlorethylene                | 40–50      | .0015               | Stabilizer reflux vapors                     | Water                                    | 80–120     | .003                |
| Jacket water                                | Water                           | 230–300    | .002                | Steam                                        | Feed water                               | 400–1000   | .0005               |
| Lube oil (low viscosity)                    | Water                           | 25–50      | .002                | Steam                                        | No. 6 fuel oil                           | 15–25      | .0055               |
| Lube oil (high viscosity)                   | Water                           | 40–80      | .003                | Steam                                        | No. 2 fuel oil                           | 60–90      | .0025               |
| Lube oil                                    | Oil                             | 11–20      | .006                | Sulfur dioxide                               | Water                                    | 150–200    | .003                |
| Naphtha                                     | Water                           | 50–70      | .005                | Tall-oil derivatives, vegetable oils (vapor) | Water                                    | 20–50      | .004                |
| Naphtha                                     | Oil                             | 25–35      | .005                | Water                                        | Aromatic vapor-stream azeotrope          | 40–80      | .005                |
| Organic solvents                            | Water                           | 50–150     | .003                | <b>Gas-liquid media</b>                      |                                          |            |                     |
| Organic solvents                            | Brine                           | 35–90      | .003                | Air, N <sub>2</sub> , etc. (compressed)      | Water or brine                           | 40–80      | .005                |
| Organic solvents                            | Organic solvents                | 20–60      | .002                | Air, N <sub>2</sub> , etc., A                | Water or brine                           | 10–50      | .005                |
| Tall oil derivatives, vegetable oil, etc.   | Water                           | 20–50      | .004                | Water or brine                               | Air, N <sub>2</sub> (compressed)         | 20–40      | .005                |
| Water                                       | Caustic soda solutions (10–30%) | 100–250    | .003                | Water or brine                               | Air, N <sub>2</sub> , etc., A            | 5–20       | .005                |
| Water                                       | Water                           | 200–250    | .003                | Water                                        | Hydrogen containing natural-gas mixtures | 80–125     | .003                |
| Wax distillate                              | Water                           | 15–25      | .005                | <b>Vaporizers</b>                            |                                          |            |                     |
| Wax distillate                              | Oil                             | 13–23      | .005                | Anhydrous ammonia                            | Steam condensing                         | 150–300    | .0015               |
| <b>Condensing vapor-liquid media</b>        |                                 |            |                     | Chlorine                                     | Steam condensing                         | 150–300    | .0015               |
| Alcohol vapor                               | Water                           | 100–200    | .002                | Chlorine                                     | Light heat-transfer oil                  | 40–60      | .0015               |
| Asphalt (450°F.)                            | Dowtherm vapor                  | 40–60      | .006                | Propane, butane, etc.                        | Steam condensing                         | 200–300    | .0015               |
| Dowtherm vapor                              | Tall oil and derivatives        | 60–80      | .004                | Water                                        | Steam condensing                         | 250–400    | .0015               |

NC = noncondensable gas present.

V = vacuum.

A = atmospheric pressure.

Dirt (or fouling factor) units are (h · ft<sup>2</sup> · °F)/Btu.

To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; to convert hours per square foot-degree Fahrenheit-British thermal units to square meters per second-kelvin-joules, multiply by 0.1761.

**TABLE 11-4 Typical Overall Heat-Transfer Coefficients in Refinery Service**

$$\text{Btu}/(^{\circ}\text{F} \cdot \text{ft}^2 \cdot \text{h})$$

| Fluid                       | API gravity | Fouling factor (one stream) | Reboiler, steam-heated | Condenser, water-cooled* | Exchangers, liquid to liquid (tube-side fluid designation appears below) |    |    | Reboiler (heating liquid designated below) |    |    | Condenser (cooling liquid designated below) |    |    |    |
|-----------------------------|-------------|-----------------------------|------------------------|--------------------------|--------------------------------------------------------------------------|----|----|--------------------------------------------|----|----|---------------------------------------------|----|----|----|
|                             |             |                             |                        |                          | C                                                                        | G  | H  | C                                          | G† | K  | D                                           | F  | G  | J  |
| A Propane                   |             | 0.001                       | 160                    | 95                       | 85                                                                       | 85 | 80 | 110                                        | 95 | 35 |                                             |    |    |    |
| B Butane                    |             | .001                        | 155                    | 90                       | 80                                                                       | 75 | 75 | 105                                        | 90 | 35 | 80                                          | 55 | 40 | 30 |
| C 400°F. end-point gasoline | 50          | .001                        | 120                    | 80                       | 70                                                                       | 65 | 60 | 65                                         | 50 | 30 |                                             |    |    |    |
| D Virgin light naphtha      | 70          | .001                        | 140                    | 85                       | 70                                                                       | 55 | 55 | 75                                         | 60 | 35 | 75                                          |    |    |    |
| E Virgin heavy naphtha      | 45          | .001                        | 95                     | 75                       | 65                                                                       | 55 | 50 | 55                                         | 45 | 30 | 70                                          | 50 | 35 | 30 |
| F Kerosene                  | 40          | .001                        | 85                     | 60                       | 60                                                                       | 55 | 50 |                                            | 45 | 25 |                                             | 50 | 35 | 30 |
| G Light gas oil             | 30          | .002                        | 70                     | 50                       | 60                                                                       | 50 | 50 |                                            | 40 | 25 | 70                                          | 45 | 30 | 30 |
| H Heavy gas oil             | 22          | .003                        | 60                     | 45                       | 55                                                                       | 50 | 45 | 50                                         | 40 | 20 | 70                                          | 40 | 30 | 20 |
| J Reduced crude             | 17          | .005                        |                        |                          | 55                                                                       | 45 | 40 |                                            |    |    |                                             |    |    |    |
| K Heavy fuel oil (tar)      | 10          | .005                        |                        |                          | 50                                                                       | 40 | 35 |                                            |    |    |                                             |    |    |    |

Fouling factor, water side 0.0002; heating or cooling streams are shown at top of columns as C, D, F, G, etc.; to convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; to convert hours per square foot-degree Fahrenheit-British thermal units to square meters per second-kelvin-joules, multiply by 0.1761.

\*Cooler, water-cooled, rates are about 5 percent lower.

†With heavy gas oil (H) as heating medium, rates are about 5 percent lower.

**TABLE 11-5 Overall Coefficients for Air-Cooled Exchangers on Bare-Tube Basis**

| Btu/(°F · ft <sup>2</sup> · h) |                                      |                            |             |
|--------------------------------|--------------------------------------|----------------------------|-------------|
| Condensing                     | Coefficient                          | Liquid cooling             | Coefficient |
| Ammonia                        | 110                                  | Engine-jacket water        | 125         |
| Freon-12                       | 70                                   | Fuel oil                   | 25          |
| Gasoline                       | 80                                   | Light gas oil              | 65          |
| Light hydrocarbons             | 90                                   | Light hydrocarbons         | 85          |
| Light naphtha                  | 75                                   | Light naphtha              | 70          |
| Heavy naphtha                  | 65                                   | Reformer liquid streams    | 70          |
| Reformer reactor effluent      | 70                                   | Residuum                   | 15          |
| Low-pressure steam             | 135                                  | Tar                        | 7           |
| Overhead vapors                | 65                                   |                            |             |
| Gas cooling                    | Operating pressure, lb./sq. in. gage | Pressure drop, lb./sq. in. | Coefficient |
| Air or flue gas                | 50                                   | 0.1 to 0.5                 | 10          |
|                                | 100                                  | 2                          | 20          |
|                                | 100                                  | 5                          | 30          |
| Hydrocarbon gas                | 35                                   | 1                          | 35          |
|                                | 125                                  | 3                          | 55          |
|                                | 1000                                 | 5                          | 80          |
| Ammonia reactor stream         |                                      |                            | 85          |

Bare-tube external surface is 0.262 ft<sup>2</sup>/ft.

Fin-tube surface/bare-tube surface ratio is 16.9.

To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; to convert pounds-force per square inch to kilopascals, multiply by 6.895.

A predictive equation for  $U_{co}$  is

$$U_{co} = \left( \frac{h}{h - 2ca/d_m} \right) \left( \frac{2ca}{d_m} \right) \quad (11-48)$$

where  $h$  = wall film coefficient,  $c$  = volumetric heat capacity,  $d_m$  = depth of the burden, and  $\alpha$  = thermal diffusivity. Relevant thermal

properties of various materials are given in Table 11-9. For details of terminology, equation development, numerical values of terms in typical equipment and use, see Holt [*Chem. Eng.*, **69**, 107 (Jan. 8, 1962)].

Equation (11-48) is applicable to burdens in the solid, liquid, or gaseous phase, either static or in laminar motion; it is applicable to solidification equipment and to divided-solids equipment such as metal belts, moving trays, stationary vertical tubes, and stationary-shell fluidizers.

**Fixed (or packed) bed operation** occurs when the fluid velocity is low or the particle size is large so that fluidization does not occur. For such operation, Jakob (*Heat Transfer*, vol. 2, Wiley, New York, 1957) gives

$$hD_t/k = b_1 b D_t^{0.17} (D_p G/\mu)^{0.83} (c\mu/k) \quad (11-49a)$$

where  $b_1 = 1.22$  (SI) or 1.0 (U.S. customary),  $h = U_{co}$  = overall coefficient between the inner container surface and the fluid stream,

$$b = .2366 + .0092 \left( \frac{D_p}{D_t} \right) - 4.0672 \left( \frac{D_p}{D_t} \right)^2 + 18.229 \left( \frac{D_p}{D_t} \right)^3 - 11.837 \left( \frac{D_p}{D_t} \right)^4 \quad (11-49b)$$

$D_p$  = particle diameter,  $D_t$  = vessel diameter, (note that  $D_p/D_t$  has units of foot per foot in the equation),  $G$  = superficial mass velocity,  $k$  = fluid thermal conductivity,  $\mu$  = fluid viscosity, and  $c$  = fluid specific heat. Other correlations are those of Leva [*Ind. Eng. Chem.*, **42**, 2498 (1950)]:

$$h = 0.813 \frac{k}{D_t} e^{-6D_p/D_t} \left( \frac{D_p G}{\mu} \right)^{0.90} \quad \text{for } \frac{D_p}{D_t} < 0.35 \quad (11-50a)$$

$$h = 0.125 \frac{k}{D_t} \left( \frac{D_p G}{\mu} \right)^{0.75} \quad \text{for } 0.35 < \frac{D_p}{D_t} < 0.60 \quad (11-50b)$$

and Calderbank and Pogerski [*Trans. Inst. Chem. Eng. (London)*, **35**, 195 (1957)]:

$$hD_p/k = 3.6 (D_p G/\mu \epsilon_e)^{0.365} \quad (11-51)$$

where  $\epsilon_e$  = fraction voids in the bed.

**TABLE 11-6 Panel Coils Immersed in Liquid: Overall Average Heat-Transfer Coefficients\***

$U$  expressed in Btu/(h · ft<sup>2</sup> · °F)

| Hot side                           | Cold side                  | Clean-surface coefficients |                   | Design coefficients, considering usual fouling in this service |                   |
|------------------------------------|----------------------------|----------------------------|-------------------|----------------------------------------------------------------|-------------------|
|                                    |                            | Natural convection         | Forced convection | Natural convection                                             | Forced convection |
| Heating applications:              |                            |                            |                   |                                                                |                   |
| Steam                              | Watery solution            | 250–500                    | 300–550           | 100–200                                                        | 150–275           |
| Steam                              | Light oils                 | 50–70                      | 110–140           | 40–45                                                          | 60–110            |
| Steam                              | Medium lube oil            | 40–60                      | 100–130           | 35–40                                                          | 50–100            |
| Steam                              | Bunker C or No. 6 fuel oil | 20–40                      | 70–90             | 15–30                                                          | 60–80             |
| Steam                              | Tar or asphalt             | 15–35                      | 50–70             | 15–25                                                          | 40–60             |
| Steam                              | Molten sulfur              | 35–45                      | 45–55             | 20–35                                                          | 35–45             |
| Steam                              | Molten paraffin            | 35–45                      | 45–55             | 25–35                                                          | 40–50             |
| Steam                              | Air or gases               | 2–4                        | 5–10              | 1–3                                                            | 4–8               |
| Steam                              | Molasses or corn sirup     | 20–40                      | 70–90             | 15–30                                                          | 60–80             |
| High temperature hot water         | Watery solutions           | 115–140                    | 200–250           | 70–100                                                         | 110–160           |
| High temperature heat-transfer oil | Tar or asphalt             | 12–30                      | 45–65             | 10–20                                                          | 30–50             |
| Dowtherm or Aroclor                | Tar or asphalt             | 15–30                      | 50–60             | 12–20                                                          | 30–50             |
| Cooling applications:              |                            |                            |                   |                                                                |                   |
| Water                              | Watery solution            | 110–135                    | 195–245           | 65–95                                                          | 105–155           |
| Water                              | Quench oil                 | 10–15                      | 25–45             | 7–10                                                           | 15–25             |
| Water                              | Medium lube oil            | 8–12                       | 20–30             | 5–8                                                            | 10–20             |
| Water                              | Molasses or corn sirup     | 7–10                       | 18–26             | 4–7                                                            | 8–15              |
| Water                              | Air or gases               | 2–4                        | 5–10              | 1–3                                                            | 4–8               |
| Freon or ammonia                   | Watery solution            | 35–45                      | 60–90             | 20–35                                                          | 40–60             |
| Calcium or sodium brine            | Watery solution            | 100–120                    | 175–200           | 50–75                                                          | 80–125            |

\*Tranter Manufacturing, Inc.

NOTE: To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783.

TABLE 11-7 Jacketed Vessels: Overall Coefficients

| Jacket fluid      | Fluid in vessel  | Wall material   | Overall $U^*$                  |                            |
|-------------------|------------------|-----------------|--------------------------------|----------------------------|
|                   |                  |                 | Btu/(h · ft <sup>2</sup> · °F) | J/(m <sup>2</sup> · s · K) |
| Steam             | Water            | Stainless steel | 150–300                        | 850–1700                   |
| Steam             | Aqueous solution | Stainless steel | 80–200                         | 450–1140                   |
| Steam             | Organics         | Stainless steel | 50–150                         | 285–850                    |
| Steam             | Light oil        | Stainless steel | 60–160                         | 340–910                    |
| Steam             | Heavy oil        | Stainless steel | 10–50                          | 57–285                     |
| Brine             | Water            | Stainless steel | 40–180                         | 230–1625                   |
| Brine             | Aqueous solution | Stainless steel | 35–150                         | 200–850                    |
| Brine             | Organics         | Stainless steel | 30–120                         | 170–680                    |
| Brine             | Light oil        | Stainless steel | 35–130                         | 200–740                    |
| Brine             | Heavy oil        | Stainless steel | 10–30                          | 57–170                     |
| Heat-transfer oil | Water            | Stainless steel | 50–200                         | 285–1140                   |
| Heat-transfer oil | Aqueous solution | Stainless steel | 40–170                         | 230–965                    |
| Heat-transfer oil | Organics         | Stainless steel | 30–120                         | 170–680                    |
| Heat-transfer oil | Light oil        | Stainless steel | 35–130                         | 200–740                    |
| Heat-transfer oil | Heavy oil        | Stainless steel | 10–40                          | 57–230                     |
| Steam             | Water            | Glass-lined CS  | 70–100                         | 400–570                    |
| Steam             | Aqueous solution | Glass-lined CS  | 50–85                          | 285–480                    |
| Steam             | Organics         | Glass-lined CS  | 30–70                          | 170–400                    |
| Steam             | Light oil        | Glass-lined CS  | 40–75                          | 230–425                    |
| Steam             | Heavy oil        | Glass-lined CS  | 10–40                          | 57–230                     |
| Brine             | Water            | Glass-lined CS  | 30–80                          | 170–450                    |
| Brine             | Aqueous solution | Glass-lined CS  | 25–70                          | 140–400                    |
| Brine             | Organics         | Glass-lined CS  | 20–60                          | 115–340                    |
| Brine             | Light oil        | Glass-lined CS  | 25–65                          | 140–370                    |
| Brine             | Heavy oil        | Glass-lined CS  | 10–30                          | 57–170                     |
| Heat-transfer oil | Water            | Glass-lined CS  | 30–80                          | 170–450                    |
| Heat-transfer oil | Aqueous solution | Glass-lined CS  | 25–70                          | 140–400                    |
| Heat-transfer oil | Organics         | Glass-lined CS  | 25–65                          | 140–370                    |
| Heat-transfer oil | Light oil        | Glass-lined CS  | 20–70                          | 115–400                    |
| Heat-transfer oil | Heavy oil        | Glass-lined CS  | 10–35                          | 57–200                     |

\*Values listed are for moderate nonproximity agitation. CS = carbon steel.

A technique for calculating radial temperature gradients in a packed bed is given by Smith (*Chemical Engineering Kinetics*, McGraw-Hill, New York, 1956).

**Fluidization** occurs when the fluid flow rate is great enough so that the pressure drop across the bed equals the weight of the bed. As stated previously, the solids film thickness adjacent to the wall  $d_m$  is difficult to measure and/or predict. Wen and Fau [*Chem. Eng.*, **64**(7),

254 (1957)] give for *external walls*:

$$h = bk(c_p \rho_s)^{0.4} (G \eta / \mu N_f)^{0.36} \quad (11-51a)$$

where  $b = 0.29$  (SI) or 11.6 (U.S. customary),  $c_p$  = heat capacity of solid,  $\rho_s$  = particle density,  $\eta$  = fluidization efficiency (Fig. 11-31) and  $N_f$  = bed expansion ratio (Fig. 11-32). For *internal walls*, Wen

TABLE 11-8 External Coils; Typical Overall Coefficients\*

$U$  expressed in Btu/(h · ft<sup>2</sup> · °F)

| Type of coil                                                   | Coil spacing, in.† | Fluid in coil                  | Fluid in vessel                      | Temp. range, °F. | $U$ ‡ without cement | $U$ with heat-transfer cement |
|----------------------------------------------------------------|--------------------|--------------------------------|--------------------------------------|------------------|----------------------|-------------------------------|
| ¾ in. o.d. copper tubing attached with bands at 24-in. spacing | 2                  | 5 to 50 lb./sq. in. gage steam | Water under light agitation          | 158–210          | 1–5                  | 42–46                         |
|                                                                | 3½                 |                                |                                      | 158–210          | 1–5                  | 50–53                         |
|                                                                | 6¼                 |                                |                                      | 158–210          | 1–5                  | 60–64                         |
|                                                                | 12½ or greater     |                                |                                      | 158–210          | 1–5                  | 69–72                         |
| ¾ in. o.d. copper tubing attached with bands at 24-in. spacing | 2                  | 50 lb./sq. in. gage steam      | No. 6 fuel oil under light agitation | 158–258          | 1–5                  | 20–30                         |
|                                                                | 3½                 |                                |                                      | 158–258          | 1–5                  | 25–38                         |
|                                                                | 6¼                 |                                |                                      | 158–240          | 1–5                  | 30–40                         |
|                                                                | 12½ or greater     |                                |                                      | 158–238          | 1–5                  | 35–46                         |
| Panel coils                                                    |                    | 50 lb./sq. in. gage steam      | Boiling water                        | 212              | 29                   | 48–54                         |
|                                                                |                    |                                | Water                                | 158–212          | 8–30                 | 19–48                         |
|                                                                |                    |                                | No. 6 fuel oil                       | 228–278          | 6–15                 | 24–56                         |
|                                                                |                    |                                | Water                                | 130–150          | 7                    | 15                            |
|                                                                |                    |                                | No. 6 fuel oil                       | 130–150          | 4                    | 9–19                          |

\*Data courtesy of Thermon Manufacturing Co.

†External surface of tubing or side of panel coil facing tank.

‡For tubing, the coefficients are more dependent upon tightness of the coil against the tank than upon either fluid. The low end of the range is recommended.

NOTE: To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; to convert inches to meters, multiply by 0.0254; and to convert pounds-force per square inch to kilopascals, multiply by 6.895.

**TABLE 11-9 Thermal Properties of Various Materials as Affecting Conductive Heat Transfer**

| Material                          | Thermal conductivity,<br>B.t.u./(hr.)(sq. ft.)(°F./ft.) | Volume specific heat,<br>B.t.u./(cu. ft.)(°F.) | Thermal diffusivity,<br>sq. ft./hr. |
|-----------------------------------|---------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Air                               | 0.0183                                                  | 0.016                                          | 1.143                               |
| Water                             | 0.3766                                                  | 62.5                                           | 0.0755                              |
| Double steel plate, sand divider  | 0.207                                                   | 19.1                                           | 0.0108                              |
| Sand                              | 0.207                                                   | 19.1                                           | 0.0108                              |
| Powdered iron                     | 0.0533                                                  | 12.1                                           | 0.0044                              |
| Magnetite iron ore                | 0.212                                                   | 63                                             | 0.0033                              |
| Aerocat catalysts                 | 0.163                                                   | 20                                             | 0.0062                              |
| Table salt                        | 0.168                                                   | 12.6                                           | 0.0133                              |
| Bone char                         | 0.0877                                                  | 16.9                                           | 0.0051                              |
| Pitch coke                        | 0.333                                                   | 16.2                                           | 0.0198                              |
| Phenolformaldehyde resin granules | 0.0416                                                  | 10.5                                           | 0.0042                              |
| Phenolformaldehyde resin powder   | 0.070                                                   | 10                                             | 0.0070                              |
| Powdered coal                     | 0.070                                                   | 15                                             | 0.0047                              |

To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per meter-second-kelvins, multiply by 1.7307; to convert British thermal units per cubic foot-degrees Fahrenheit to joules per cubic meter-kelvins, multiply by (6.707)(10<sup>4</sup>); and to convert square feet per hour to square meters per second, multiply by (2.581)(10<sup>-5</sup>).

and Fau give

$$h_i = bhG^{-0.37} \quad (11-51b)$$

where  $b = 0.78$  (SI) or 9 (U.S. customary),  $h_i$  is the coefficient for internal walls, and  $h$  is calculated from Eq. (11-51a).  $G_{mf}$ , the minimum fluidizing velocity, is defined by

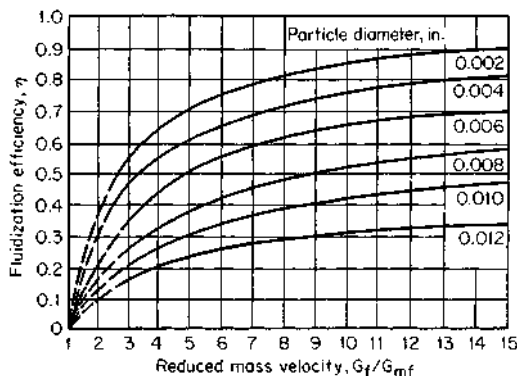
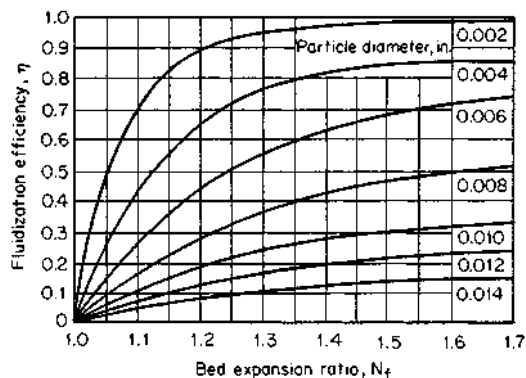
$$G_{mf} = \frac{b\rho_g^{1.1}(\rho_s - \rho_g)^{0.9}D_p^2}{\mu} \quad (11-51c)$$

where  $b = (1.23)(10^{-2})$  (SI) or (5.23)(10<sup>5</sup>) (U.S. customary).

Wender and Cooper [*Am. Inst. Chem. Eng. J.*, **4**, 15 (1958)] developed an empirical correlation for *internal walls*:

$$\frac{hD_p/k}{1 - \epsilon_v} \left( \frac{k}{c_p} \right)^{0.43} = bC_R \left( \frac{D_p G}{\mu} \right)^{0.23} \left( \frac{c_s}{c_g} \right)^{0.80} \left( \frac{\rho_s}{\rho_g} \right)^{0.66} \quad (11-52a)$$

where  $b = (3.51)(10^{-4})$  (SI) or 0.033 (U.S. customary) and  $C_R$  = correction for displacement of the immersed tube from the axis of the

**FIG. 11-31** Fluidization efficiency.**FIG. 11-32** Bed expansion ratio.

vessel (see the reference). For external walls:

$$\frac{hD_p}{k_g(1 - \epsilon_v)(c_s\rho_s/c_g\rho_g)} = f(1 + 7.5e^{-x}) \quad (11-52b)$$

where  $x = 0.44L_H c_s/D_p c_g$  and  $f$  is given by Fig. 11-33. An important feature of this equation is inclusion of the ratio of bed depth to vessel diameter  $L_H/D_p$ .

For **dilute fluidized beds** on the shell side of an unbaffled tubular bundle Genetti and Knudsen [*Inst. Chem. Eng. (London) Symp. Ser.* 3,172 (1968)] obtained the relation:

$$\frac{hD_p}{k} = \frac{5\phi(1 - \epsilon_v)}{\left[ 1 + \frac{580}{N_{Re}} \left( \frac{k_s}{D_p^{1.5} c_s \rho_g^{0.5}} \right) \left( \frac{\rho_s}{\rho_g} \right)^{1.1} \left( \frac{G_{mf}}{G} \right)^{7/3} \right]} \quad (11-53a)$$

where  $\phi$  = particle surface area per area of sphere of same diameter. When particle transport occurred through the bundle, the heat-transfer coefficients could be predicted by

$$j_H = 0.14(N_{Re}/\phi)^{-0.68} \quad (11-53b)$$

In Eqs. (11-53a) and (11-53b),  $N_{Re}$  is based on particle diameter and superficial fluid velocity.

Zenz and Othmer (see "Introduction: General References") give an excellent summary of fluidized bed-to-wall heat-transfer investigations.

**Solidification** involves heavy heat loads transferred essentially at a steady temperature difference. It also involves the varying values of liquid- and solid-phase thickness and thermal diffusivity. When these are substantial and/or in the case of a liquid flowing over a changing solid layer interface, Siegel and Savino (ASME Paper 67-WA/Ht-34, November 1967) offer equations and charts for prediction of the layer-growth time. For solidification (or melting) of a slab or a semi-infinite bar, initially at its transition temperature, the position of the interface is given by the one-dimensional Newmann's solution given in Carslaw and Jaeger (*Conduction of Heat in Solids*, Clarendon Press, Oxford, 1959).

Later work by Hashem and Sliepcevich [*Chem. Eng. Prog.*, **63**, Symp. Ser. 79, 35, 42 (1967)] offers more accurate second-order finite-difference equations.

The heat-transfer rate is found to be substantially higher under conditions of **agitation**. The heat transfer is usually said to occur by combined conductive and convective modes. A discussion and explanation are given by Holt [*Chem. Eng.*, **69**(1), 110 (1962)]. Prediction of  $U_{co}$  by Eq. (11-48) can be accomplished by replacing  $\alpha$  by  $\alpha_e$ , the effective thermal diffusivity of the bed. To date so little work has been performed in evaluating the effect of mixing parameters that few predictions can be made. However, for agitated liquid-phase devices Eq. (18-19) is applicable. Holt (loc. cit.) shows that this equation can be converted for solids heat transfer to yield

$$U_{co} = a' c_s D_i^{-0.3} N^{0.7} (\cos \omega)^{0.2} \quad (11-54)$$

where  $D_i$  = agitator or vessel diameter;  $N$  = turning speed, r/min;  $\omega$  = effective angle of repose of the burden; and  $a'$  is a proportionality con-

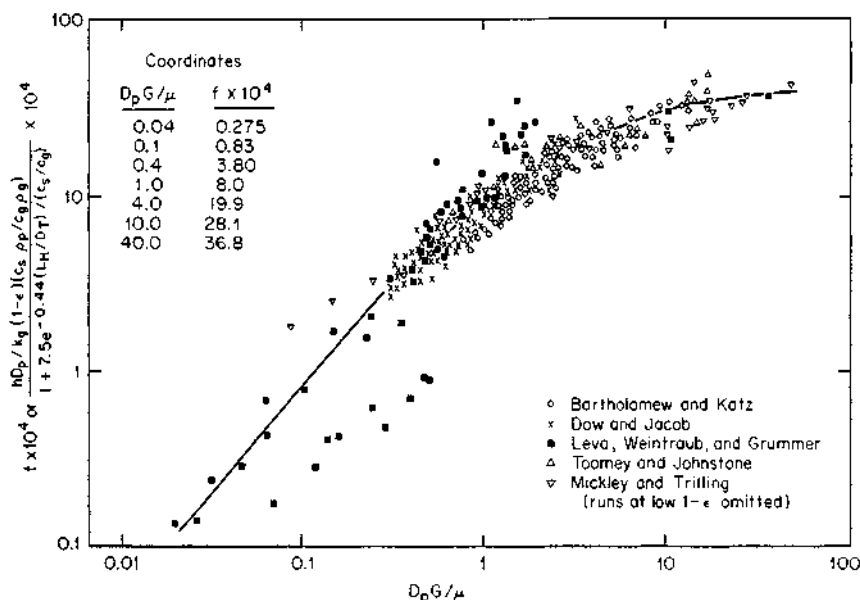


FIG. 11-33  $f$  factor for Eq. (11-52b).

stant. This is applicable for such devices as agitated pans, agitated kettles, spiral conveyors, and rotating shells.

The solids passage time through **rotary devices** is given by Saemann [Chem. Eng. Prog., 47, 508, (1951)]:

$$\theta = 0.318L \sin \omega / S_r N D_i \quad (11-55a)$$

and by Marshall and Friedman [Chem. Eng. Prog., 45, 482-493, 573-588 (1949)]:

$$\theta = (0.23L / S_r N^{0.9} D_i) \pm (0.6BLG / F_a) \quad (11-55b)$$

where the second term of Eq. (11-55b) is positive for counterflow of air, negative for concurrent flow, and zero for indirect rotary shells. From these equations a predictive equation is developed for rotary-shell devices, which is analogous to Eq. (11-54):

$$U_{co} = \frac{b' c_s D_i N^{0.9} Y}{(\Delta t) L \sin \omega} \quad (11-56)$$

where  $\theta$  = solids-bed passage time through the shell, min;  $S_r$  = shell slope;  $L$  = shell length;  $Y$  = percent fill; and  $b'$  is a proportionality constant.

**Vibratory devices** which constantly agitate the solids bed maintain a relatively constant value for  $U_{co}$  such that

$$U_{co} = a' c_s \alpha_e \quad (11-57)$$

with  $U_{co}$  having a nominal value of 114 J/(m<sup>2</sup>·s·K) [20 Btu/(h·ft<sup>2</sup>·°F)].

**Contactive (Direct) Heat Transfer** Contactive heat-transfer equipment is so constructed that the particulate burden in solid phase is directly exposed to and permeated by the heating or cooling medium (Sec. 20). The carrier may either heat or cool the solids. A large amount of the industrial heat processing of solids is effected by this mechanism. Physically, these can be classified into packed beds and various degrees of agitated beds from dilute to dense fluidized beds.

The temperature difference for heat transfer is the log-mean temperature difference when the particles are large and/or the beds packed, or the difference between the inlet fluid temperature  $t_3$  and average exhausting fluid temperature  $t_4$ , expressed  $\Delta t_4$ , for small particles. The use of the log mean for packed beds has been confirmed by Thodos and Wilkins (Second American Institute of Chemical Engineers-IIQPR Meeting, Paper 30D, Tampa, May 1968). When fluid and solid flow directions are axially concurrent and particle size is

small, as in a vertical-shell fluid bed, the temperature of the exiting solids  $t_2$  (which is also that of exiting gas  $t_4$ ) is used as  $\Delta_3 t_2$ , as shown by Levenspiel, Olson, and Walton [Ind. Eng. Chem., 44, 1478 (1952)], Marshall [Chem. Eng. Prog., 50, Monogr. Ser. 2, 77 (1954)], Leva (Fluidization, McGraw-Hill, New York, 1959), and Holt (Fourth Int. Heat Transfer Conf. Paper 11, American Institute of Chemical Engineers-American Society of Mechanical Engineers, Buffalo, 1960). This temperature difference is also applicable for well-fluidized beds of small particles in cross-flow as in various vibratory carriers.

The **packed-bed-to-fluid heat-transfer coefficient** has been investigated by Baumeister and Bennett [Am. Inst. Chem. Eng. J., 4, 69 (1958)], who proposed the equation

$$j_H = (h/cG)(c\mu/k)^{2/3} = a N_{Re}^m \quad (11-58)$$

where  $N_{Re}$  is based on particle diameter and superficial fluid velocity. Values of  $a$  and  $m$  are as follows:

| $D_i/D_p$<br>(dimensionless) | $a$  | $m$   |
|------------------------------|------|-------|
| 10.7                         | 1.58 | -0.40 |
| 16.0                         | 0.95 | -0.30 |
| 25.7                         | 0.92 | -0.28 |
| >30                          | 0.90 | -0.28 |

Glaser and Thodos [Am. Inst. Chem. Eng. J., 4, 63 (1958)] give a correlation involving individual particle shape and bed porosity. Kunii and Suzuki [Int. J. Heat Mass Transfer, 10, 845 (1967)] discuss heat and mass transfer in packed beds of fine particles.

**Particle-to-fluid** heat-transfer coefficients in gas **fluidized beds** are predicted by the relation (Zenz and Othmer, op. cit.)

$$\frac{hD_p}{k} = 0.017(D_p G_{mf}/\mu)^{1.21} \quad (11-59a)$$

where  $G_{mf}$  is the superficial mass velocity at incipient fluidization.

A more general equation is given by Frantz [Chem. Eng., 69(20), 89 (1962)]:

$$hD_p/k = 0.015(D_p G/\mu)^{1.6}(c\mu/k)^{0.67} \quad (11-59b)$$

where  $h$  is based on true gas temperature.

**Bed-to-wall coefficients in dilute-phase transport** generally can be predicted by an equation of the form of Eq. (5-50). For example,

Bonilla et al. (American Institute of Chemical Engineers Heat Transfer Symp., Atlantic City, N.J., December 1951) found for 1- to 2- $\mu\text{m}$  chalk particles in water up to 8 percent by volume that the coefficient on Eq. (5-50) is 0.029 where  $k$ ,  $\rho$ , and  $c$  were arithmetic weighted averages and the viscosity was taken equal to the coefficient of rigidity. Farber and Morley [*Ind. Eng. Chem.*, **49**, 1143 (1957)] found the coefficient on Eq. (5-50) to be 0.025 for the upward flow of air transporting silica-alumina catalyst particles at rates less than 2 kg solids/kg air (2 lb solids/lb air). Physical properties used were those of the transporting gas. See Zenz and Othmer (op. cit.) for additional details covering wider porosity ranges.

The thermal performance of **cylindrical rotating shell** units is based upon a volumetric heat-transfer coefficient

$$U_{ct} = \frac{Q}{V_r(\Delta T)} \quad (11-60a)$$

where  $V_r$  = volume. This term indirectly includes an area factor so that thermal performance is governed by a cross-sectional area rather than by a heated area. Use of the heated area is possible, however:

$$U_{ct} = \frac{Q}{(\Delta_3 t_2)A} \quad \text{or} \quad \frac{Q}{(\Delta_3 t_4)A} \quad (11-60b)$$

For **heat transfer directly to solids**, predictive equations give directly the volume  $V$  or the heat-transfer area  $A$ , as determined by heat balance and airflow rate. For devices with gas flow normal to a fluidized-solids bed,

$$A = \frac{Q}{\Delta t_p (c p_g)(F_g)} \quad (11-61)$$

where  $\Delta t_p = \Delta_3 t_4$  as explained above,  $c p$  = volumetric specific heat, and  $F_g$  = gas flow rate. For air,  $c p$  at normal temperature and pressure is about 1100 J/(m<sup>3</sup>·K) [0.0167 Btu/(ft<sup>3</sup>·°F)]; so

$$A = \frac{bQ}{(\Delta_3 t_4)F_g} \quad (11-62)$$

where  $b = 0.0009$  (SI) or 60 (U.S. customary). Another such equation, for stationary vertical-shell and some horizontal rotary-shell and pneumatic-transport devices in which the gas flow is parallel with and directionally concurrent with the fluidized bed, is the same as Eq. (11-62) with  $\Delta_3 t_4$  replaced by  $\Delta_3 t_2$ . If the operation involves drying or chemical reaction, the heat load  $Q$  is much greater than for sensible-heat transfer only. Also, the gas flow rate to provide moisture carry-off and stoichiometric requirements must be considered and simultaneously provided. A good treatise on the latter is given by Pinkey and Plint (*Miner. Process.*, June 1968, p. 17).

**Evaporative cooling** is a special patented technique that often can be advantageously employed in cooling solids by contactive heat transfer. The drying operation is terminated *before* the desired final moisture content is reached, and solids temperature is at a moderate value. The cooling operation involves contacting the burden (preferably fluidized) with air at normal temperature and pressure. The air adiabatically absorbs and carries off a large part of the moisture and, in doing so, picks up heat from the warm (or hot) solids particles to supply the latent heat demand of evaporation. For entering solids at temperatures of 180°C (350°F) and less with normal heat-capacity values of 0.85 to 1.0 kJ/(kg·K) [0.2 to 0.25 Btu/(lb·°F)], the effect can be calculated by:

1. Using 285 m<sup>3</sup> (1000 ft<sup>3</sup>) of airflow at normal temperature and pressure at 40 percent relative humidity to carry off 0.45 kg (1 lb) of water [latent heat 2326 kJ/kg (1000 Btu/lb)] and to lower temperature by 22 to 28°C (40 to 50°F).

2. Using the lowered solids temperature as  $t_3$  and calculating the remainder of the heat to be removed in the regular manner by Eq. (11-62). The required air quantity for (2) must be equal to or greater than that for (1).

When the solids heat capacity is higher (as is the case for most organic materials), the temperature reduction is inversely proportional to the heat capacity.

A nominal result of this technique is that the required airflow rate and equipment size is about two-thirds of that when evaporative cooling is not used. See Sec. 20 for equipment available.

**Convective Heat Transfer** Equipment using the true convective mechanism when the heated particles are mixed with (and remain with) the cold particles is used so infrequently that performance and sizing equations are not available. Such a device is the pebble heater as described by Norton (*Chem. Metall. Eng.*, July 1946). For operation data, see Sec. 9.

Convective heat transfer is often used as an adjunct to other modes, particularly to the conductive mode. It is often more convenient to consider the agitative effect a performance-improvement influence on the thermal diffusivity factor  $\alpha$ , modifying it to  $\alpha_e$ , the effective value.

A *pseudo-convective heat-transfer* operation is one in which the heating gas (generally air) is passed over a bed of solids. Its use is almost exclusively limited to drying operations (see Sec. 12, tray and shelf dryers). The operation, sometimes termed direct, is more akin to the conductive mechanism. For this operation, Tsao and Wheelock [*Chem. Eng.*, **74**(13), 201 (1967)] predict the heat-transfer coefficient when radiative and conductive effects are absent by

$$h = bG^{0.8} \quad (11-63)$$

where  $b = 14.31$  (SI) or 0.0128 (U.S. customary),  $h$  = convective heat transfer, and  $G$  = gas flow rate.

The **drying rate** is given by

$$K_{cv} = \frac{h(T_d - T_w)}{\lambda} \quad (11-64)$$

where  $K_{cv}$  = drying rate, for constant-rate period, kg/(m<sup>2</sup>·s) [lb/(h·ft<sup>2</sup>)];  $T_d$  and  $T_w$  = respective dry-bulb and wet-bulb temperatures of the air; and  $\lambda$  = latent heat of evaporation at temperature  $T_w$ . Note here that the temperature-difference determination of the operation is a simple linear one and of a steady-state nature. Also note that the operation is a function of the airflow rate. Further, the solids are granular with a fairly uniform size, have reasonable capillary voids, are of a firm texture, and have the particle surface wetted.

The coefficient  $h$  is also used to predict (in the constant-rate period) the total overall air-to-solids heat-transfer coefficient  $U_{cv}$  by

$$1/U_{cv} = 1/h + x/k \quad (11-65)$$

where  $k$  = solids thermal conductivity and  $x$  is evaluated from

$$x = \frac{z(X_c - X_o)}{X_c - X_e} \quad (11-65a)$$

where  $z$  = bed (or slab) thickness and is the total thickness when drying and/or heat transfer is from one side only but is one-half of the thickness when drying and/or heat transfer is simultaneously from both sides;  $X_o$ ,  $X_c$ , and  $X_e$  are respectively the initial (or feed-stock), critical, and equilibrium (with the drying air) moisture contents of the solids, all in kg H<sub>2</sub>O/kg dry solids (lb H<sub>2</sub>O/lb dry solids). This coefficient is used to predict the *instantaneous* drying rate

$$-\frac{W}{A} \frac{dX}{d\theta} = \frac{U_{cv}(T_d - T_w)}{\lambda} \quad (11-66)$$

By rearrangement, this can be made into a design equation as follows:

$$A = -\frac{W\lambda(dX/d\theta)}{U_{cv}(T_d - T_w)} \quad (11-67)$$

where  $W$  = weight of dry solids in the equipment,  $\lambda$  = latent heat of evaporation, and  $\theta$  = drying time. The reader should refer to the full reference article by Tsao and Wheelock (loc. cit.) for other solids conditions qualifying the use of these equations.

**Radiative Heat Transfer** Heat-transfer equipment using the radiative mechanism for divided solids is constructed as a "table" which is stationary, as with trays, or moving, as with a belt, and/or agitated, as with a vibrated pan, to distribute and expose the burden in a plane parallel to (but not in contact with) the plane of the radiant-heat sources. Presence of air is not necessary (see Sec. 12 for vacuum-shelf dryers and Sec. 22 for resublimation). In fact, if air in the intervening space has a high humidity or CO<sub>2</sub> content, it acts as an energy absorber, thereby depressing the performance.

For the radiative mechanism, the temperature difference is evaluated as

$$\Delta t = T_e^4 - T_r^4 \quad (11-68)$$



where  $T_e$  = absolute temperature of the radiant-heat source, K (°R); and  $T_r$  = absolute temperature of the bed of divided solids, K (°R).

Numerical values for  $U_{ra}$  for use in the general design equation may be calculated from experimental data by

$$U_{ra} = \frac{Q}{A(T_e^4 - T_r^4)} \quad (11-69)$$

The literature to date offers practically no such values. However, enough proprietary work has been performed to present a reliable evaluation for the comparison of mechanisms (see "Introduction: Modes of Heat Transfer").

For the radiative mechanism of heat transfer to solids, the rate equation for parallel-surface operations is

$$q_{ra} = b(T_e^4 - T_r^4)i_f \quad (11-70)$$

where  $b = (5.67)(10^{-8})$  (SI) or  $(0.172)(10^{-8})$  (U.S. customary),  $q_{ra}$  = radiative heat flux, and  $i_f$  = an interchange factor which is evaluated from

$$1/i_f = 1/e_s + 1/e_r - 1 \quad (11-70a)$$

where  $e_s$  = coefficient of emissivity of the source and  $e_r$  = "emissivity" (or "absorptivity") of the receiver, which is the divided-solids bed. For the emissivity values, particularly of the heat source  $e_s$ , an important consideration is the wavelength at which the radiant source emits as well as the flux density of the emission. Data for these values are available from Polentz [*Chem. Eng.*, **65**(7), 137; (8), 151 (1958)] and Adlam [*Radiant Heating*, Industrial Press, New York, p. 40]. Both give radiated flux density versus wavelength at varying temperatures. Often, the seemingly cooler but longer wavelength source is the better selection.

Emitting sources are (1) pipes, tubes, and platters carrying steam, 2100 kPa (300 lbf/in<sup>2</sup>); (2) electrical-conducting glass plates, 150 to 315°C (300 to 600°F) range; (3) light-bulb type (tungsten-filament resistance heater); (4) modules of refractory brick for gas burning at high temperatures and high fluxes; and (5) modules of quartz tubes, also operable at high temperatures and fluxes. For some emissivity values see Table 11-10.

For predictive work, where  $U_{ra}$  is desired for sizing, this can be obtained by dividing the flux rate  $q_{ra}$  by  $\Delta t$ :

$$U_{ra} = q_{ra}/(T_e^4 - T_r^4) = i_f b \quad (11-71)$$

where  $b = (5.67)(10^{-8})$  (SI) or  $(0.172)(10^{-8})$  (U.S. customary). Hence:

$$A = \frac{Q}{U_{ra}(T_e^4 - T_r^4)} \quad (11-72)$$

where  $A$  = bed area of solids in the equipment.

Important considerations in the application of the foregoing equations are:

1. Since the temperature of the emitter is generally known (pre-selected or readily determined in an actual operation), the absorptivity value  $e_r$  is the unknown. This absorptivity is partly a measure of the ability of radiant heat to penetrate the body of a solid particle (or a moisture film) instantly, as compared with diffusional heat transfer by conduction. Such instant penetration greatly reduces processing time and case-hardening effects. Moisture release and other mass transfer, however, still progress by diffusional means.

2. In one of the major applications of radiative devices (drying), the surface-held moisture is a good heat absorber in the 2- to 7- $\mu$ m wavelength range. Therefore, the absorptivity, color, and nature of the solids are of little importance.

3. For drying, it is important to provide a small amount of venting air to carry away the water vapor. This is needed for two reasons. First, water vapor is a good absorber of 2- to 7- $\mu$ m energy. Second, water-vapor accumulation depresses further vapor release by the solids. If the air over the solids is kept fairly dry by venting, very little heat is carried off, because dry air does not absorb radiant heat.

4. For some of the devices, when the overall conversion efficiency has been determined, the application is primarily a matter of computing the required heat load. It should be kept in mind, however, that

there are two conversion efficiencies that must be differentiated. One measure of efficiency is that with which the source converts input energy to output radiated energy. The other is the overall efficiency that measures the proportion of input energy that is actually absorbed by the solids. This latter is, of course, the one that really matters.

Other applications of radiant-heat processing of solids are the toasting, puffing, and baking of foods and the low-temperature roasting and preheating of plastic powder or pellets. Since the determination of heat loads for these operations is not well established, bench and pilot tests are generally necessary. Such processes require a fast input of heat and higher heat fluxes than can generally be provided by indirect equipment. Because of this, infrared-equipment size and space requirements are often much lower.

Although direct contact heat transfer can provide high temperatures and heat concentrations and at the same time be small in size, its use may not always be preferable because of undesired side effects such as drying, contamination, case hardening, shrinkage, off color, and dusting.

When radiating and receiving surfaces are not in parallel, as in rotary-kiln devices, and the solids burden bed may be only intermittently exposed and/or agitated, the calculation and procedures become very complex, with photometric methods of optics requiring consideration. The following equation for heat transfer, which allows for convective effects, is commonly used by designers of **high-temperature furnaces**:

$$q_{ra} = Q/A = b\sigma [(T_g/100)^4 - (T_s/100)^4] \quad (11-73)$$

where  $b = 5.67$  (SI) or  $0.172$  (U.S. customary);  $Q$  = total furnace heat transfer;  $\sigma$  = an emissivity factor with recommended values of 0.74 for gas, 0.75 for oil, and 0.81 for coal;  $A$  = effective area for absorbing heat (here the solids burden exposed area);  $T_g$  = exiting-combustion-gas absolute temperature; and  $T_s$  = absorbing surface temperature.

In rotary devices, reradiation from the exposed shell surface to the solids bed is a major design consideration. A treatise on furnaces, including radiative heat-transfer effects, is given by Ellwood and Danatos [*Chem. Eng.*, **73**(8), 174 (1966)]. For discussion of radiation heat-transfer computational methods, heat fluxes obtainable, and emissivity values, see Schornshort and Viskanta (ASME Paper 68-H 7-32), Sherman (ASME Paper 56-A-111), and the following subsection.

## SCRAPED-SURFACE EXCHANGERS

Scraped-surface exchangers have a rotating element with spring-loaded scraper blades to scrape the inside surface (Fig. 11-34). Generally a

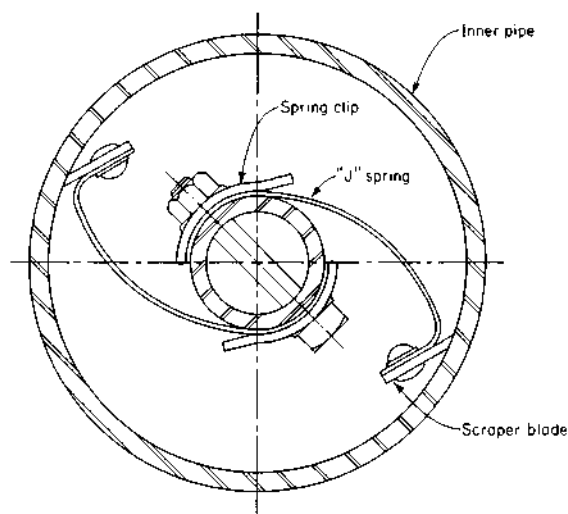


FIG. 11-34 Scraper blade of scraped-surface exchanger. (Henry Vogt Machine Co., Inc.)

# 11-32 HEAT-TRANSFER EQUIPMENT

**TABLE 11-10 Normal Total Emissivity of Various Surfaces**

| A. Metals and Their Oxides                                     |                            |                         |                                                                                                                            |                            |                         |
|----------------------------------------------------------------|----------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------|
| Surface                                                        | $t, ^\circ\text{F.}^\circ$ | Emissivity <sup>o</sup> | Surface                                                                                                                    | $t, ^\circ\text{F.}^\circ$ | Emissivity <sup>o</sup> |
| Aluminum                                                       |                            |                         | Sheet steel, strong rough oxide layer                                                                                      | 75                         | 0.80                    |
| Highly polished plate, 98.3% pure                              | 440–1070                   | 0.039–0.057             | Dense shiny oxide layer                                                                                                    | 75                         | 0.82                    |
| Polished plate                                                 | 73                         | 0.040                   | Cast plate:                                                                                                                |                            |                         |
| Rough plate                                                    | 78                         | 0.055                   | Smooth                                                                                                                     | 73                         | 0.80                    |
| Oxidized at 1110°F                                             | 390–1110                   | 0.11–0.19               | Rough                                                                                                                      | 73                         | 0.82                    |
| Aluminum-surfaced roofing                                      | 100                        | 0.216                   | Cast iron, rough, strongly oxidized                                                                                        | 100–480                    | 0.95                    |
| Calorized surfaces, heated at 1110°F.                          |                            |                         | Wrought iron, dull oxidized                                                                                                | 70–680                     | 0.94                    |
| Copper                                                         | 390–1110                   | 0.18–0.19               | Steel plate, rough                                                                                                         | 100–700                    | 0.94–0.97               |
| Steel                                                          | 390–1110                   | 0.52–0.57               | High temperature alloy steels (see Nickel Alloys)                                                                          |                            |                         |
| Brass                                                          |                            |                         | Molten metal                                                                                                               |                            |                         |
| Highly polished:                                               |                            |                         | Cast iron                                                                                                                  | 2370–2550                  | 0.29                    |
| 73.2% Cu, 26.7% Zn                                             | 476–674                    | 0.028–0.031             | Mild steel                                                                                                                 | 2910–3270                  | 0.28                    |
| 62.4% Cu, 36.8% Zn, 0.4% Pb, 0.3% Al                           | 494–710                    | 0.033–0.037             | Lead                                                                                                                       |                            |                         |
| 82.9% Cu, 17.0% Zn                                             | 530                        | 0.030                   | Pure (99.96%), unoxidized                                                                                                  | 260–440                    | 0.057–0.075             |
| Hard rolled, polished:                                         |                            |                         | Gray oxidized                                                                                                              | 75                         | 0.281                   |
| But direction of polishing visible                             | 70                         | 0.038                   | Oxidized at 390°F                                                                                                          | 390                        | 0.63                    |
| But somewhat attacked                                          | 73                         | 0.043                   | Mercury                                                                                                                    | 32–212                     | 0.09–0.12               |
| But traces of stearin from polish left on                      | 75                         | 0.053                   | Molybdenum filament                                                                                                        | 1340–4700                  | 0.096–0.292             |
| Polished                                                       | 100–600                    | 0.096                   | Monel metal, oxidized at 1110°F                                                                                            | 390–1110                   | 0.41–0.46               |
| Rolled plate, natural surface                                  | 72                         | 0.06                    | Nickel                                                                                                                     |                            |                         |
| Rubbed with coarse emery                                       | 72                         | 0.20                    | Electroplated on polished iron, then polished                                                                              | 74                         | 0.045                   |
| Dull plate                                                     | 120–660                    | 0.22                    | Technically pure (98.9% Ni, + Mn), polished                                                                                | 440–710                    | 0.07–0.087              |
| Oxidized by heating at 1110°F                                  | 390–1110                   | 0.61–0.59               | Electroplated on pickled iron, not polished                                                                                | 68                         | 0.11                    |
| Chromium; see Nickel Alloys for Ni-Cr steels                   | 100–1000                   | 0.08–0.26               | Wire                                                                                                                       | 368–1844                   | 0.096–0.186             |
| Copper                                                         |                            |                         | Plate, oxidized by heating at 1110°F                                                                                       | 390–1110                   | 0.37–0.48               |
| Carefully polished electrolytic copper                         | 176                        | 0.018                   | Nickel oxide                                                                                                               | 1200–2290                  | 0.59–0.86               |
| Commercial, emiered, polished, but pits remaining              | 66                         | 0.030                   | Nickel alloys                                                                                                              |                            |                         |
| Commercial, scraped shiny but not mirror-like                  | 72                         | 0.072                   | Chromnickel                                                                                                                | 125–1894                   | 0.64–0.76               |
| Polished                                                       | 242                        | 0.023                   | Nickelin (18–32 Ni; 55–68 Cu; 20 Zn), gray oxidized                                                                        | 70                         | 0.262                   |
| Plate, heated long time, covered with thick oxide layer        | 77                         | 0.78                    | KA-2S alloy steel (8% Ni; 18% Cr), light silvery, rough, brown, after heating                                              | 420–914                    | 0.44–0.36               |
| Plate heated at 1110°F                                         | 390–1110                   | 0.57                    | After 42 hr. heating at 980°F                                                                                              | 420–980                    | 0.62–0.73               |
| Cuprous oxide                                                  | 1470–2010                  | 0.66–0.54               | NCT-3 alloy (20% Ni; 25% Cr), brown, splotted, oxidized from service                                                       | 420–980                    | 0.90–0.97               |
| Molten copper                                                  | 1970–2330                  | 0.16–0.13               | NCT-6 alloy (60% Ni; 12% Cr), smooth, black, firm adhesive oxide coat from service                                         | 520–1045                   | 0.89–0.82               |
| Gold                                                           |                            |                         | Platinum                                                                                                                   |                            |                         |
| Pure, highly polished                                          | 440–1160                   | 0.018–0.035             | Pure, polished plate                                                                                                       | 440–1160                   | 0.054–0.104             |
| Iron and steel                                                 |                            |                         | Strip                                                                                                                      | 1700–2960                  | 0.12–0.17               |
| Metallic surfaces (or very thin oxide layer):                  |                            |                         | Filament                                                                                                                   | 80–2240                    | 0.036–0.192             |
| Electrolytic iron, highly polished                             | 350–440                    | 0.052–0.064             | Wire                                                                                                                       | 440–2510                   | 0.073–0.182             |
| Polished iron                                                  | 800–1880                   | 0.144–0.377             | Silver                                                                                                                     |                            |                         |
| Iron freshly emiered                                           | 68                         | 0.242                   | Polished, pure                                                                                                             | 440–1160                   | 0.0198–0.0324           |
| Cast iron, polished                                            | 392                        | 0.21                    | Polished                                                                                                                   | 100–700                    | 0.0221–0.0312           |
| Wrought iron, highly polished                                  | 100–480                    | 0.28                    | Steel, see Iron                                                                                                            |                            |                         |
| Cast iron, newly turned                                        | 72                         | 0.435                   | Tantalum filament                                                                                                          | 2420–5430                  | 0.194–0.31              |
| Polished steel casting                                         | 1420–1900                  | 0.52–0.56               | Tin—bright tinned iron sheet                                                                                               | 76                         | 0.043 and 0.064         |
| Ground sheet steel                                             | 1720–2010                  | 0.55–0.61               | Tungsten                                                                                                                   |                            |                         |
| Smooth sheet iron                                              | 1650–1900                  | 0.55–0.60               | Filament, aged                                                                                                             | 80–6000                    | 0.032–0.35              |
| Cast iron, turned on lathe                                     | 1620–1810                  | 0.60–0.70               | Filament                                                                                                                   | 6000                       | 0.39                    |
| Oxidized surfaces:                                             |                            |                         | Zinc                                                                                                                       |                            |                         |
| Iron plate, pickled, then rusted red                           | 68                         | 0.612                   | Commercial, 99.1% pure, polished                                                                                           | 440–620                    | 0.045–0.053             |
| Completely rusted                                              | 67                         | 0.685                   | Oxidized by heating at 750°F                                                                                               | 750                        | 0.11                    |
| Rolled sheet steel                                             | 70                         | 0.657                   | Galvanized sheet iron, fairly bright                                                                                       | 82                         | 0.228                   |
| Oxidized iron                                                  | 212                        | 0.736                   | Galvanized sheet iron, gray oxidized                                                                                       | 75                         | 0.276                   |
| Cast iron, oxidized at 1100°F                                  | 390–1110                   | 0.64–0.78               |                                                                                                                            |                            |                         |
| Steel, oxidized at 1100°F                                      | 390–1110                   | 0.79                    |                                                                                                                            |                            |                         |
| Smooth oxidized electrolytic iron                              | 260–980                    | 0.78–0.82               |                                                                                                                            |                            |                         |
| Iron oxide                                                     | 930–2190                   | 0.85–0.89               |                                                                                                                            |                            |                         |
| Rough ingot iron                                               | 1700–2040                  | 0.87–0.95               |                                                                                                                            |                            |                         |
| B. Refractories, Building Materials, Paints, and Miscellaneous |                            |                         |                                                                                                                            |                            |                         |
| Asbestos                                                       |                            |                         | Carbon                                                                                                                     |                            |                         |
| Board                                                          | 74                         | 0.96                    | T-carbon (Gebr. Siemens) 0.9% ash (this started with emissivity at 260°F. of 0.72, but on heating changed to values given) | 260–1160                   | 0.81–0.79               |
| Paper                                                          | 100–700                    | 0.93–0.945              | Carbon filament                                                                                                            | 1900–2560                  | 0.526                   |
| Brick                                                          |                            |                         | Candle soot                                                                                                                | 206–520                    | 0.952                   |
| Red, rough, but no gross irregularities                        | 70                         | 0.93                    | Lampblack-waterglass coating                                                                                               | 209–362                    | 0.959–0.947             |
| Silica, unglazed, rough                                        | 1832                       | 0.80                    |                                                                                                                            |                            |                         |
| Silica, glazed, rough                                          | 2012                       | 0.85                    |                                                                                                                            |                            |                         |
| Grog brick, glazed                                             | 2012                       | 0.75                    |                                                                                                                            |                            |                         |
| See Refractory Materials below.                                |                            |                         |                                                                                                                            |                            |                         |

TABLE 11-10 Normal Total Emissivity of Various Surfaces (Concluded)

| A. Metals and Their Oxides                          |                 |             |                                                      |                 |                                                                                           |
|-----------------------------------------------------|-----------------|-------------|------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------|
| Surface                                             | <i>t</i> , °F.° | Emissivity° | Surface                                              | <i>t</i> , °F.° | Emissivity°                                                                               |
| Same                                                | 260–440         | 0.957–0.952 | Oil paints, sixteen different, all colors            | 212             | 0.92–0.96                                                                                 |
| Thin layer on iron plate                            | 69              | 0.927       | Aluminum paints and lacquers                         |                 |                                                                                           |
| Thick coat                                          | 68              | 0.967       | 10% Al, 22% lacquer body, on rough or smooth surface | 212             | 0.52                                                                                      |
| Lampblack, 0.003 in. or thicker                     | 100–700         | 0.945       | 26% Al, 27% lacquer body, on rough or smooth surface | 212             | 0.3                                                                                       |
| Enamel, white fused, on iron                        | 66              | 0.897       | Other Al paints, varying age and Al content          | 212             | 0.27–0.67                                                                                 |
| Glass, smooth                                       | 72              | 0.937       | Al lacquer, varnish binder, on rough plate           | 70              | 0.39                                                                                      |
| Gypsum, 0.02 in. thick on smooth or blackened plate | 70              | 0.903       | Al paint, after heating to 620°F                     | 300–600         | 0.35                                                                                      |
| Marble, light gray, polished                        | 72              | 0.931       | Paper, thin                                          |                 |                                                                                           |
| Oak, planed                                         | 70              | 0.895       | Pasted on tinned iron plate                          | 66              | 0.924                                                                                     |
| Oil layers on polished nickel (lube oil)            | 68              |             | On rough iron plate                                  | 66              | 0.929                                                                                     |
| Polished surface, alone                             |                 | 0.045       | On black lacquered plate                             | 66              | 0.944                                                                                     |
| +0.001-in. oil                                      |                 | 0.27        | Plaster, rough lime                                  | 50–190          | 0.91                                                                                      |
| +0.002-in. oil                                      |                 | 0.46        | Porcelain, glazed                                    | 72              | 0.924                                                                                     |
| +0.005-in. oil                                      |                 | 0.72        | Quartz, rough, fused                                 | 70              | 0.932                                                                                     |
| Infinitely thick oil layer                          |                 | 0.82        | Refractory materials, 40 different poor radiators    | 1110–1830       | $\begin{bmatrix} 0.65 \\ 0.70 \end{bmatrix} - \begin{bmatrix} 0.75 \\ 0.85 \end{bmatrix}$ |
| Oil layers on aluminum foil (linseed oil)           |                 |             | good radiators                                       |                 | 0.91                                                                                      |
| Al foil                                             | 212             | 0.087†      | Roofing paper                                        | 69              |                                                                                           |
| +1 coat oil                                         | 212             | 0.561       | Rubber                                               |                 |                                                                                           |
| +2 coats oil                                        | 212             | 0.574       | Hard, glossy plate                                   | 74              | 0.945                                                                                     |
| Paints, lacquers, varnishes                         |                 |             | Soft, gray, rough (reclaimed)                        | 76              | 0.859                                                                                     |
| Snowwhite enamel varnish or rough iron plate        | 73              | 0.906       | Serpentine, polished                                 | 74              | 0.900                                                                                     |
| Black shiny lacquer, sprayed on iron                | 76              | 0.875       | Water                                                | 32–212          | 0.95–0.963                                                                                |
| Black shiny shellac on tinned iron sheet            | 70              | 0.821       |                                                      |                 |                                                                                           |
| Black matte shellac                                 | 170–295         | 0.91        |                                                      |                 |                                                                                           |
| Black lacquer                                       | 100–200         | 0.80–0.95   |                                                      |                 |                                                                                           |
| Flat black lacquer                                  | 100–200         | 0.96–0.98   |                                                      |                 |                                                                                           |
| White lacquer                                       | 100–200         | 0.80–0.95   |                                                      |                 |                                                                                           |

\*When two temperatures and two emissivities are given, they correspond, first to first and second to second, and linear interpolation is permissible. °C = (°F – 32)/1.8.

†Although this value is probably high, it is given for comparison with the data by the same investigator to show the effect of oil layers. See Aluminum, Part A of this table.

double-pipe construction is used; the scraping mechanism is in the inner pipe, where the process fluid flows; and the cooling or heating medium is in the outer pipe. The most common size has 6-in inside and 8-in outside pipes. Also available are 3- by 4-in, 8- by 10-in, and 12- by 14-in sizes (in  $\times$  25.4 = mm). These double-pipe units are commonly connected in series and arranged in double stands.

For **chilling** and **crystallizing** with an evaporating refrigerant, a 27-in shell with seven 6-in pipes is available (Henry Vogt Machine Co.). In direct contact with the scraped surface is the process fluid which may

deposit crystals upon chilling or be extremely fouling or of very high viscosity. Motors, chain drives, appropriate guards, and so on are required for the rotating element. For chilling service with a refrigerant in the outer shell, an accumulator drum is mounted on top of the unit.

Scraped-surface exchangers are particularly suitable for heat transfer with crystallization, heat transfer with severe fouling of surfaces, heat transfer with solvent extraction, and heat transfer of high-viscosity fluids. They are extensively used in paraffin-wax plants and in petrochemical plants for crystallization.

## TEMA-STYLE SHELL-AND-TUBE HEAT EXCHANGERS

### TYPES AND DEFINITIONS

TEMA-style shell-and-tube-type exchangers constitute the bulk of the unfired heat-transfer equipment in chemical-process plants, although increasing emphasis has been developing in other designs. These exchangers are illustrated in Fig. 11-35, and their features are summarized in Table 11-11.

**TEMA Numbering and Type Designation** Recommended practice for the designation of TEMA-style shell-and-tube heat exchangers by numbers and letters has been established by the Tubular Exchanger Manufacturers Association (TEMA). This information from the sixth edition of the TEMA Standards is reproduced in the following paragraphs.

It is recommended that heat-exchanger size and type be designated by numbers and letters.

1. *Size.* Sizes of shells (and tube bundles) shall be designated by numbers describing shell (and tube-bundle) diameters and tube lengths as follows:

2. *Diameter.* The nominal diameter shall be the inside diameter of the shell in inches, rounded off to the nearest integer. For kettle reboilers the nominal

diameter shall be the port diameter followed by the shell diameter, each rounded off to the nearest integer.

3. *Length.* The nominal length shall be the tube length in inches. Tube length for straight tubes shall be taken as the actual overall length. For U tubes the length shall be taken as the straight length from end of tube to bend tangent.

4. *Type.* Type designation shall be by letters describing stationary head, shell (omitted for bundles only), and rear head, in that order, as indicated in Fig. 11-1.

**Typical Examples** (A) Split-ring floating-heat exchanger with removable channel and cover, single-pass shell, 591-mm (23¼-in) inside diameter with tubes 4.9 m (16 ft) long. SIZE 23–192 TYPE AES.

(B) U-tube exchanger with bonnet-type stationary head, split-flow shell, 483-mm (19-in) inside diameter with tubes 21-m (7-ft) straight length. SIZE 19–84 TYPE GBU.

(C) Pull-through floating-heat-kettle-type reboiler having stationary head integral with tube sheet, 584-mm (23-in) port diameter and 940-mm (37-in) inside shell diameter with tubes 4.9-m (16-ft) long. SIZE 23/37–192 TYPE CKT.

(D) Fixed-tube sheet exchanger with removable channel and cover, bonnet-type rear head, two-pass shell, 841-mm (33½-in) diameter with tubes 2.4 m (8-ft) long. SIZE 33–96 TYPE AFM.

(E) Fixed-tube sheet exchanger having stationary and rear heads integral with tube sheets, single-pass shell, 432-mm (17-in) inside diameter with tubes 4.9-m (16-ft) long. SIZE 17–192 TYPE CEN.

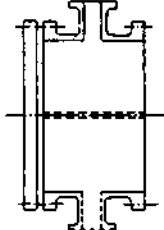
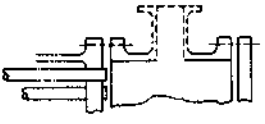
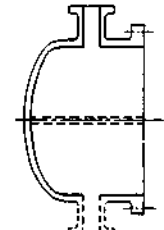
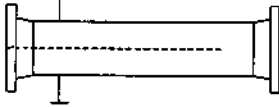
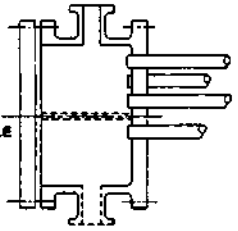
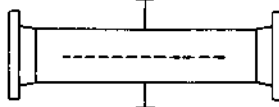
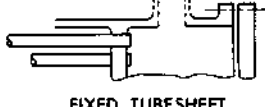
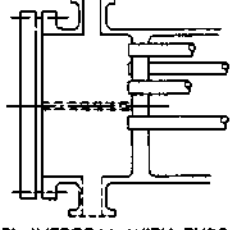
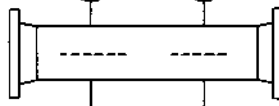
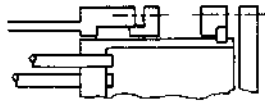
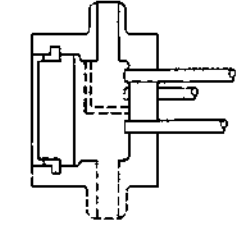
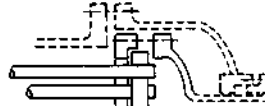
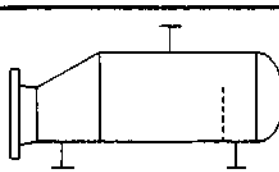
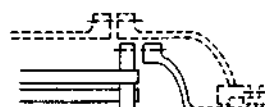
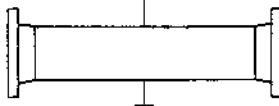
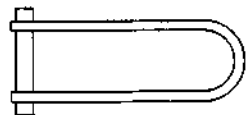
| FRONT END<br>STATIONARY HEAD TYPES |                                                                                                 | SHELL TYPES |                                                                                     | REAR END<br>HEAD TYPES |                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------|
| <b>A</b>                           |                | <b>E</b>    |    | <b>L</b>               |    |
|                                    | CHANNEL<br>AND REMOVABLE COVER                                                                  |             | ONE PASS SHELL                                                                      |                        | FIXED TUBESHEET<br>LIKE "A" STATIONARY HEAD                                           |
|                                    |                |             |    |                        |    |
|                                    | BONNET (INTEGRAL COVER)                                                                         |             | TWO PASS SHELL<br>WITH LONGITUDINAL BAFFLE                                          |                        | FIXED TUBESHEET<br>LIKE "B" STATIONARY HEAD                                           |
|                                    |               |             |    |                        |    |
| <b>C</b>                           | REMOVABLE<br>TUBE<br>BUNDLE<br>ONLY<br>CHANNEL INTEGRAL WITH TUBE-<br>SHEET AND REMOVABLE COVER | <b>G</b>    | SPLIT FLOW                                                                          | <b>N</b>               | FIXED TUBESHEET<br>LIKE "N" STATIONARY HEAD                                           |
| <b>N</b>                           |              | <b>H</b>    |    | <b>P</b>               |    |
|                                    | CHANNEL INTEGRAL WITH TUBE-<br>SHEET AND REMOVABLE COVER                                        |             | DOUBLE SPLIT FLOW                                                                   |                        | OUTSIDE PACKED FLOATING HEAD                                                          |
| <b>D</b>                           |              | <b>J</b>    |  | <b>S</b>               |   |
|                                    | SPECIAL HIGH PRESSURE CLOSURE                                                                   | <b>K</b>    |  |                        |  |
|                                    |                                                                                                 |             | KETTLE TYPE REBOILER                                                                |                        | PULL THROUGH FLOATING HEAD                                                            |
|                                    |                                                                                                 | <b>X</b>    |  | <b>U</b>               |  |
|                                    |                                                                                                 |             | CROSS FLOW                                                                          | <b>W</b>               |  |
|                                    |                                                                                                 |             |                                                                                     |                        | EXTERNALLY SEALED<br>FLOATING TUBESHEET                                               |

FIG. 11-35 TEMA-type designations for shell-and-tube heat exchangers. (Standards of Tubular Exchanger Manufacturers Association, 6th ed., 1978.)

TABLE 11-11 Features of TEMA Shell-and-Tube-Type Exchangers\*

| Type of design                                                              | Fixed tube sheet         | U-tube                          | Packed lantern-ring floating head | Internal floating head (split backing ring) | Outside-packed floating head | Pull-through floating head |
|-----------------------------------------------------------------------------|--------------------------|---------------------------------|-----------------------------------|---------------------------------------------|------------------------------|----------------------------|
| T.E.M.A. rear-head type                                                     | L or M or N              | U                               | W                                 | S                                           | P                            | T                          |
| Relative cost increases from A (least expensive) through E (most expensive) | B                        | A                               | C                                 | E                                           | D                            | E                          |
| Provision for differential expansion                                        | Expansion joint in shell | Individual tubes free to expand | Floating head                     | Floating head                               | Floating head                | Floating head              |
| Removable bundle                                                            | No                       | Yes                             | Yes                               | Yes                                         | Yes                          | Yes                        |
| Replacement bundle possible                                                 | No                       | Yes                             | Yes                               | Yes                                         | Yes                          | Yes                        |
| Individual tubes replaceable                                                | Yes                      | Only those in outside row†      | Yes                               | Yes                                         | Yes                          | Yes                        |
| Tube cleaning by chemicals inside and outside                               | Yes                      | Yes                             | Yes                               | Yes                                         | Yes                          | Yes                        |
| Interior tube cleaning mechanically                                         | Yes                      | Special tools required          | Yes                               | Yes                                         | Yes                          | Yes                        |
| Exterior tube cleaning mechanically:                                        |                          |                                 |                                   |                                             |                              |                            |
| Triangular pitch                                                            | No                       | No‡                             | No‡                               | No‡                                         | No‡                          | No‡                        |
| Square pitch                                                                | No                       | Yes                             | Yes                               | Yes                                         | Yes                          | Yes                        |
| Hydraulic-jet cleaning:                                                     |                          |                                 |                                   |                                             |                              |                            |
| Tube interior                                                               | Yes                      | Special tools required          | Yes                               | Yes                                         | Yes                          | Yes                        |
| Tube exterior                                                               | No                       | Yes                             | Yes                               | Yes                                         | Yes                          | Yes                        |
| Double tube sheet feasible                                                  | Yes                      | Yes                             | No                                | No                                          | Yes                          | No                         |
| Number of tube passes                                                       | No practical limitations | Any even number possible        | Limited to one or two passes      | No practical limitations§                   | No practical limitations     | No practical limitations§  |
| Internal gaskets eliminated                                                 | Yes                      | Yes                             | Yes                               | No                                          | Yes                          | No                         |

NOTE: Relative costs A and B are not significantly different and interchange for long lengths of tubing.

\*Modified from page a-8 of the Patterson-Kelley Co. Manual No. 700A, Heat Exchangers.

†U-tube bundles have been built with tube supports which permit the U-bends to be spread apart and tubes inside of the bundle replaced.

‡Normal triangular pitch does not permit mechanical cleaning. With a wide triangular pitch, which is equal to 2 (tube diameter plus cleaning lane)/ $\sqrt{3}$ , mechanical cleaning is possible on removable bundles. This wide spacing is infrequently used.

§For odd number of tube side passes, floating head requires packed joint or expansion joint.

**Functional Definitions** Heat-transfer equipment can be designated by type (e.g., fixed tube sheet, outside packed head, etc.) or by function (chiller, condenser, cooler, etc.). Almost any type of unit can be used to perform any or all of the listed functions. Many of these terms have been defined by Donahue [*Pet. Process.*, 103 (March 1956)].

| Equipment                   | Function                                                                                                                                                                                                                             |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chiller                     | Cools a fluid to a temperature below that obtainable if water only were used as a coolant. It uses a refrigerant such as ammonia or Freon.                                                                                           |
| Condenser                   | Condenses a vapor or mixture of vapors, either alone or in the presence of a noncondensable gas.                                                                                                                                     |
| Partial condenser           | Condenses vapors at a point high enough to provide a temperature difference sufficient to preheat a cold stream of process fluid. This saves heat and eliminates the need for providing a separate preheater (using flame or steam). |
| Final condenser             | Condenses the vapors to a final storage temperature of approximately 37.8°C (100°F). It uses water cooling, which means that the transferred heat is lost to the process.                                                            |
| Cooler                      | Cools liquids or gases by means of water.                                                                                                                                                                                            |
| Exchanger                   | Performs a double function: (1) heats a cold fluid by (2) using a hot fluid which it cools. None of the transferred heat is lost.                                                                                                    |
| Heater                      | Imparts sensible heat to a liquid or a gas by means of condensing steam or Dowtherm.                                                                                                                                                 |
| Reboiler                    | Connected to the bottom of a fractionating tower, it provides the reboil heat necessary for distillation. The heating medium may be either steam or a hot-process fluid.                                                             |
| Thermosiphon reboiler       | Natural circulation of the boiling medium is obtained by maintaining sufficient liquid head to provide for circulation.                                                                                                              |
| Forced-circulation reboiler | A pump is used to force liquid through the reboiler.                                                                                                                                                                                 |
| Steam generator             | Generates steam for use elsewhere in the plant by using the available high-level heat in tar or a heavy oil.                                                                                                                         |

Superheater  
Vaporizer  
Waste-heat boiler

Heats a vapor above the saturation temperature.  
A heater which vaporizes part of the liquid.  
Produces steam; similar to steam generator, except that the heating medium is a hot gas or liquid produced in a chemical reaction.

## GENERAL DESIGN CONSIDERATIONS

**Selection of Flow Path** In selecting the flow path for two fluids through an exchanger, several general approaches are used. The tube-side fluid is more corrosive or dirtier or at a higher pressure. The shell-side fluid is a liquid of high viscosity or a gas.

When alloy construction for one of the two fluids is required, a carbon steel shell combined with alloy tube-side parts is less expensive than alloy in contact with the shell-side fluid combined with carbon steel headers.

Cleaning of the inside of tubes is more readily done than cleaning of exterior surfaces.

For gauge pressures in excess of 2068 kPa (300 lbf/in<sup>2</sup>) for one of the fluids, the less expensive construction has the high-pressure fluid in the tubes.

For a given pressure drop, higher heat-transfer coefficients are obtained on the shell side than on the tube side.

Heat-exchanger shutdowns are most often caused by fouling, corrosion, and erosion.

**Construction Codes** "Rules for Construction of Pressure Vessels, Division 1," which is part of Section VIII of the ASME Boiler and Pressure Vessel Code (American Society of Mechanical Engineers), serves as a construction code by providing minimum standards. New editions of the code are usually issued every 3 years. Interim revisions are made semiannually in the form of addenda. Compliance with ASME Code requirements is mandatory in much of the United States and Canada. Originally these rules were not prepared for heat exchangers. However, the welded joint between tube sheet and shell of the fixed-tube-sheet heat exchanger is now included. A nonmandatory appendix on tube-to-tube-sheet joints is also included. Additional rules for heat exchangers are being developed.

*Standards of Tubular Exchanger Manufacturers Association*, 6th ed., 1978 (commonly referred to as the TEMA Standards), serve to supplement and define the ASME Code for all shell-and-tube-type heat-exchanger applications (other than double-pipe construction). TEMA Class R design is "for the generally severe requirements of petroleum and related processing applications. Equipment fabricated in accordance with these standards is designed for safety and durability under the rigorous service and maintenance conditions in such applications." TEMA Class C design is "for the generally moderate requirements of commercial and general process applications," while TEMA Class B is "for chemical process service."

The mechanical-design requirements are identical for all three classes of construction. The differences between the TEMA classes are minor and were listed by Rubin [*Hydrocarbon Process.*, **59**, 92 (June 1980)].

Among the topics of the TEMA Standards are nomenclature, fabrication tolerances, inspection, guarantees, tubes, shells, baffles and support plates, floating heads, gaskets, tube sheets, channels, nozzles, end flanges and bolting, material specifications, and fouling resistances.

*Shell and Tube Heat Exchangers for General Refinery Services*, API Standard 660, 4th ed., 1982, is published by the American Petroleum Institute to supplement both the TEMA Standards and the ASME Code. Many companies in the chemical and petroleum processing fields have their own standards to supplement these various requirements. *The Interrelationships between Codes, Standards, and Customer Specifications for Process Heat Transfer Equipment* is a symposium volume which was edited by F. L. Rubin and published by ASME in December 1979. (See discussion of pressure-vessel codes in Sec. 6.)

Design pressures and temperatures for exchangers usually are specified with a margin of safety beyond the conditions expected in service. Design pressure is generally about 172 kPa (25 lbf/in<sup>2</sup>) greater than the maximum expected during operation or at pump shutoff. Design temperature is commonly 14°C (25°F) greater than the maximum temperature in service.

**Tube Bundle Vibration** Damage from tube vibration has become an increasing problem as plate baffled heat exchangers are designed for higher flow rates and pressure drops. The most effective method of dealing with this problem is the avoidance of cross flow by use of tube support baffles which promote only longitudinal flow. However, even then, strict attention must be given the bundle area under the shell inlet nozzle where flow is introduced through the side of the shell. TEMA has devoted an entire section in its standards to this topic. In general, the mechanisms of tube vibration are as follows:

**Vortex Shedding** The vortex-shedding frequency of the fluid in cross-flow over the tubes may coincide with a natural frequency of the tubes and excite large resonant vibration amplitudes.

**Fluid-Elastic Coupling** Fluid flowing over tubes causes them to vibrate with a whirling motion. The mechanism of fluid-elastic coupling occurs when a "critical" velocity is exceeded and the vibration then becomes self-excited and grows in amplitude. This mechanism frequently occurs in process heat exchangers which suffer vibration damage.

**Pressure Fluctuation** Turbulent pressure fluctuations which develop in the wake of a cylinder or are carried to the cylinder from upstream may provide a potential mechanism for tube vibration. The tubes respond to the portion of the energy spectrum that is close to their natural frequency.

**Acoustic Coupling** When the shell-side fluid is a low-density gas, acoustic resonance or coupling develops when the standing waves in the shell are in phase with vortex shedding from the tubes. The standing waves are perpendicular to the axis of the tubes and to the direction of cross-flow. Damage to the tubes is rare. However, the noise can be extremely painful.

**Testing** Upon completion of shop fabrication and also during maintenance operations it is desirable hydrostatically to test the shell side of tubular exchangers so that visual examination of tube ends can be made. Leaking tubes can be readily located and serviced. When leaks are determined without access to the tube ends, it is necessary to reroll or reweld all the tube-to-tube-sheet joints with possible damage to the satisfactory joints.

Testing for leaks in heat exchangers was discussed by Rubin [*Chem. Eng.*, **68**, 160-166 (July 24, 1961)].

**Performance** testing of heat exchangers is described in the American Institute of Chemical Engineers' *Standard Testing Procedure for Heat Exchangers*, Sec. 1. "Sensible Heat Transfer in Shell-and-Tube-Type Equipment."

## PRINCIPAL TYPES OF CONSTRUCTION

Figure 11-36 shows details of the construction of the TEMA types of shell-and-tube heat exchangers. These and other types are discussed in the following paragraphs.

**Fixed-Tube-Sheet Heat Exchangers** Fixed-tube-sheet exchangers (Fig. 11-36b) are used more often than any other type, and the frequency of use has been increasing in recent years. The tube sheets are welded to the shell. Usually these extend beyond the shell and serve as flanges to which the tube-side headers are bolted. This construction requires that the shell and tube-sheet materials be weldable to each other.

When such welding is not possible, a "blind"-gasket type of construction is utilized. The blind gasket is not accessible for maintenance or replacement once the unit has been constructed. This construction is used for steam surface condensers, which operate under vacuum.

The tube-side header (or channel) may be welded to the tube sheet, as shown in Fig. 11-35 for type C and N heads. This type of construction is less costly than types B and M or A and L and still offers the advantage that tubes may be examined and replaced without disturbing the tube-side piping connections.

There is no limitation on the number of tube-side passes. Shell-side passes can be one or more, although shells with more than two shell-side passes are rarely used.

Tubes can completely fill the heat-exchanger shell. Clearance between the outermost tubes and the shell is only the minimum necessary for fabrication. Between the inside of the shell and the baffles some clearance must be provided so that baffles can slide into the shell. Fabrication tolerances then require some additional clearance between the outside of the baffles and the outermost tubes. The edge distance between the outer tube limit (OTL) and the baffle diameter must be sufficient to prevent vibration of the tubes from breaking through the baffle holes. The outermost tube must be contained within the OTL. Clearances between the inside shell diameter and OTL are 13 mm (½ in) for 635-mm-(25-in)-inside-diameter shells and up, 11 mm (⅞ in) for 254- through 610-mm (10- through 24-in) pipe shells, and slightly less for smaller-diameter pipe shells.

Tubes can be replaced. Tube-side headers, channel covers, gaskets, etc., are accessible for maintenance and replacement. Neither the shell-side baffle structure nor the blind gasket is accessible. During tube removal, a tube may break within the shell. When this occurs, it is most difficult to remove or to replace the tube. The usual procedure is to plug the appropriate holes in the tube sheets.

Differential expansion between the shell and the tubes can develop because of differences in length caused by thermal expansion. Various types of expansion joints are used to eliminate excessive stresses caused by expansion. The need for an expansion joint is a function of both the amount of differential expansion and the cycling conditions to be expected during operation. A number of types of expansion joints are available (Fig. 11-37).

a. *Flat plates.* Two concentric flat plates with a bar at the outer edges. The flat plates can flex to make some allowance for differential expansion. This design is generally used for vacuum service and gauge pressures below 103 kPa (15 lbf/in<sup>2</sup>). All are subject to severe stress during differential expansion.

b. *Flanged-only heads.* The flat plates are flanged (or curved). The diameter of these heads is generally 203 mm (8 in) or more greater than the shell diameter. The welded joint at the shell is subject to the stress referred to before, but the joint connecting the heads is subjected to less stress during expansion because of the curved shape.

c. *Flared shell or pipe segments.* The shell may be flared to connect with a pipe section, or a pipe may be halved and quartered to produce a ring.

d. *Formed heads.* A pair of dished-only or elliptical or flanged and dished heads can be used. These are welded together or connected by a ring. This type of joint is similar to the flanged-only-head type but apparently is subject to less stress.

e. *Flanged and flued heads.* A pair of flanged-only heads is provided with concentric reverse flue holes. These heads are relatively expensive because of

the cost of the fluing operation. The curved shape of the heads reduces the amount of stress at the welds to the shell and also connecting the heads.

*f. Toroidal.* The toroidal joint has a mathematically predictable smooth stress pattern of low magnitude, with maximum stresses at sidewalls of the corrugation and minimum stresses at top and bottom.

The foregoing designs were discussed as ring expansion joints by Kopp and Sayre, "Expansion Joints for Heat Exchangers" (ASME Misc. Pap., vol. 6, no. 211). All are statically indeterminate but are subjected to analysis by introducing various simplifying assumptions. Some joints in current industrial use are of lighter wall construction than is indicated by the method of this paper.

*g. Bellows.* Thin-wall bellows joints are produced by various manufacturers. These are designed for differential expansion and are tested for axial and transverse movement as well as for cyclical life. Bellows may be of stainless steel, nickel alloys, or copper. (Aluminum, Monel, phosphor bronze, and titanium bellows have been manufactured.) Welding nipples of the same composition as the heat-exchanger shell are generally furnished. The bellows may be hydraulically formed from a single piece of metal or may consist of welded pieces. External insulation covers of carbon steel are often provided to protect the light-gauge bellows from damage. The cover also prevents insulation from interfering with movement of the bellows (see *h*).

*h. Toroidal bellows.* For high-pressure service the bellows type of joint has been modified so that movement is taken up by thin-wall small-diameter bellows of a toroidal shape. Thickness of parts under high pressure is reduced considerably (see *f*).

Improper handling during manufacture, transit, installation, or maintenance of the heat exchanger equipped with the thin-wall-bellows type

or toroidal type of expansion joint can damage the joint. In larger units these light-wall joints are particularly susceptible to damage, and some designers prefer the use of the heavier walls of formed heads.

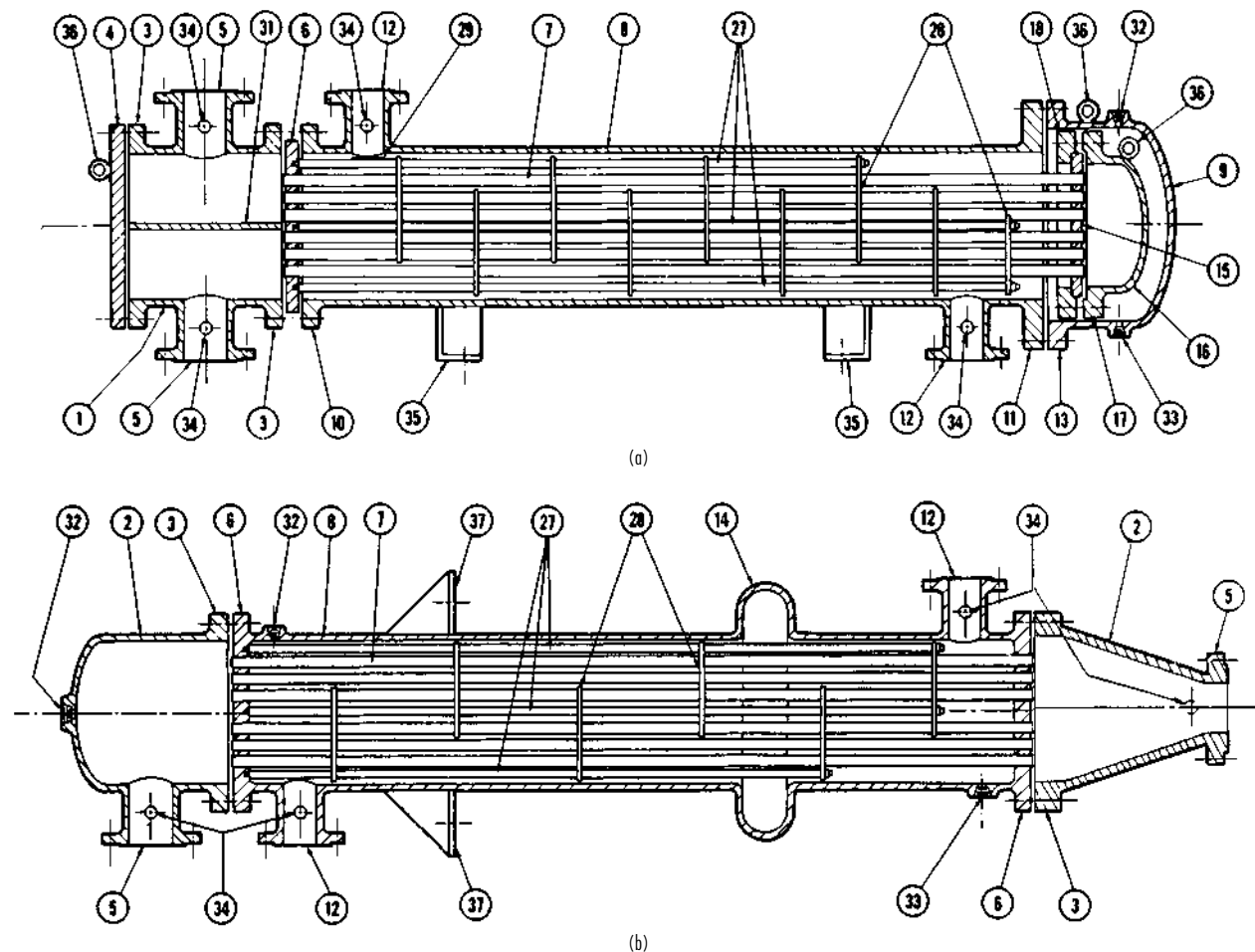
Chemical-plant exchangers requiring expansion joints most commonly have used the flanged-and-flued-head type. There is a trend toward more common use of the light-wall-bellows type.

**U-Tube Heat Exchanger (Fig. 11-36*d*)** The tube bundle consists of a stationary tube sheet, U tubes (or hairpin tubes), baffles or support plates, and appropriate tie rods and spacers. The tube bundle can be removed from the heat-exchanger shell. A tube-side header (stationary head) and a shell with integral shell cover, which is welded to the shell, are provided. Each tube is free to expand or contract without any limitation being placed upon it by the other tubes.

The U-tube bundle has the advantage of providing minimum clearance between the outer tube limit and the inside of the shell for any of the removable-tube-bundle constructions. Clearances are of the same magnitude as for fixed-tube-sheet heat exchangers.

The number of tube holes in a given shell is less than that for a fixed-tube-sheet exchanger because of limitations on bending tubes of a very short radius.

The U-tube design offers the advantage of reducing the number of joints. In high-pressure construction this feature becomes of considerable importance in reducing both initial and maintenance costs. The use of U-tube construction has increased significantly with the development



**FIG. 11-36** Heat-exchanger-component nomenclature. (*a*) Internal-floating-head exchanger (with floating-head backing device). Type AES. (*b*) Fixed-tube-sheet exchanger. Type BEM. (Standards of the Tubular Exchanger Manufacturers Association, 6th ed., 1978.)

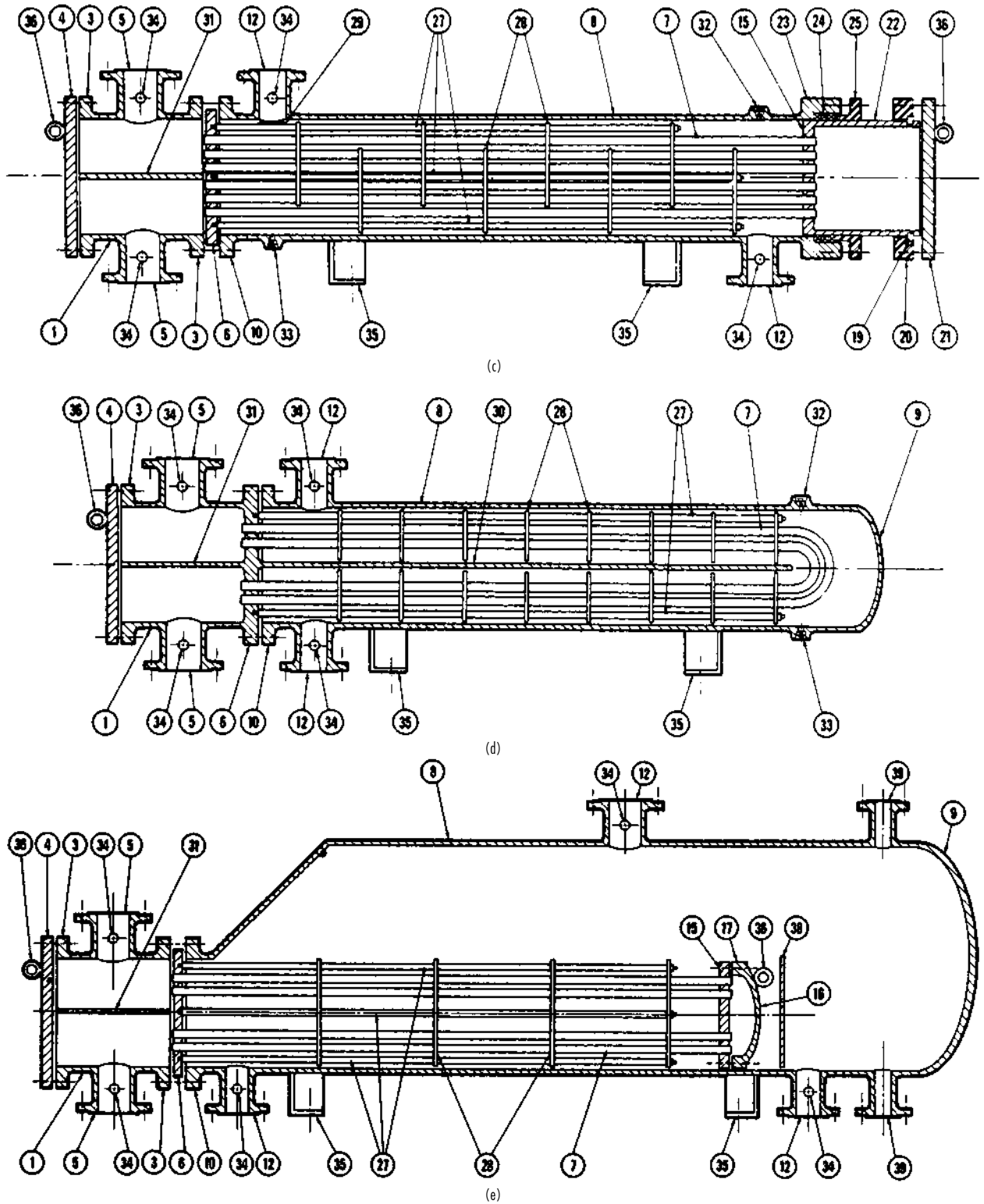
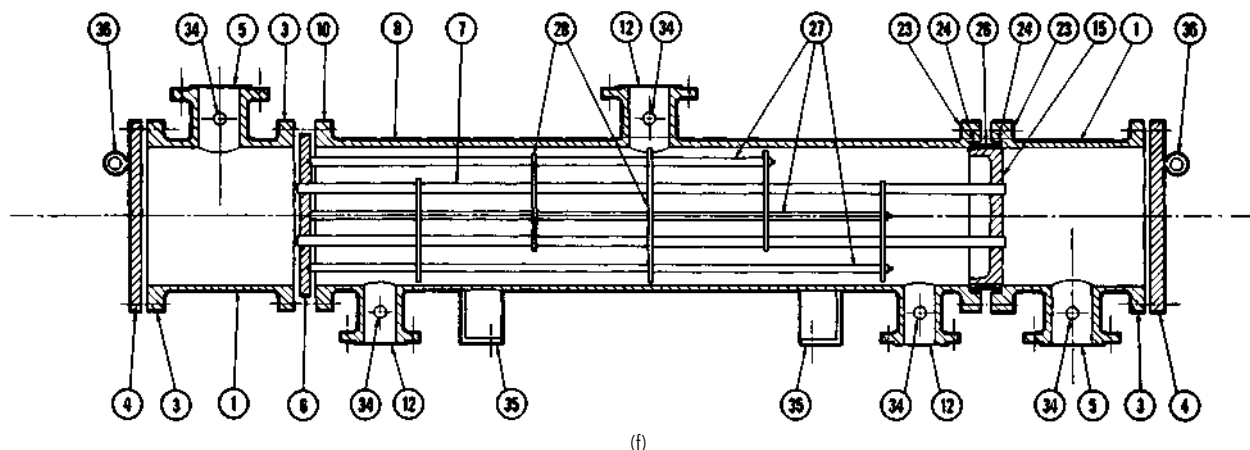


FIG. 11-36 (Continued) Heat-exchanger-component nomenclature, (c) Outside-packed floating-head exchanger. Type AEP. (d) U-tube heat exchanger. Type CFU. (e) Kettle-type floating-head reboiler. Type AKT. (Standards of the Tubular Exchanger Manufacturers Association, 6th ed., 1978.)





- |                                             |                                          |
|---------------------------------------------|------------------------------------------|
| 1. Stationary Head—Channel                  | 20. Slip-on Backing Flange               |
| 2. Stationary Head—Bonnet                   | 21. Floating Head Cover—External         |
| 3. Stationary Head Flange—Channel or Bonnet | 22. Floating Tubesheet Skirt             |
| 4. Channel Cover                            | 23. Packing Box Flange                   |
| 5. Stationary Head Nozzle                   | 24. Packing                              |
| 6. Stationary Tubesheet                     | 25. Packing Gland                        |
| 7. Tubes                                    | 26. Lantern Ring                         |
| 8. Shell                                    | 27. Tie Rods and Spacers                 |
| 9. Shell Cover                              | 28. Transverse Baffles or Support Plates |
| 10. Shell Flange—Stationary Head End        | 29. Impingement Plate                    |
| 11. Shell Flange—Rear Head End              | 30. Longitudinal Baffle                  |
| 12. Shell Nozzle                            | 31. Pass Partition                       |
| 13. Shell Cover Flange                      | 32. Vent Connection                      |
| 14. Expansion Joint                         | 33. Drain Connection                     |
| 15. Floating Tubesheet                      | 34. Instrument Connection                |
| 16. Floating Head Cover                     | 35. Support Saddle                       |
| 17. Floating Head Flange                    | 36. Lifting Lug                          |
| 18. Floating Head Backing Device            | 37. Support Bracket                      |
| 19. Split Shear Ring                        | 38. Weir                                 |
|                                             | 39. Liquid Level Connection              |

**FIG. 11-36** (Continued) Heat-exchanger-component nomenclature. (f) Exchanger with packed floating tube sheet and lantern ring. Type AJW. (Standards of the Tubular Exchanger Manufacturers Association, 6th ed., 1978.)

of hydraulic tube cleaners, which can remove fouling residues from both the straight and the U-bend portions of the tubes.

Mechanical cleaning of the inside of the tubes was described by John [*Chem. Eng.*, **66**, 187–192 (Dec. 14, 1959)]. Rods and conventional mechanical tube cleaners cannot pass from one end of the U tube to the other. Power-driven tube cleaners, which can clean both the straight legs of the tubes and the bends, are available.

Hydraulic jetting with water forced through spray nozzles at high pressure for cleaning tube interiors and exteriors of removal bundles is reported by Canada ("Hydraulic Jetting Tools for Cleaning Heat Exchangers," ASME Pap. 58-A-217, unpublished).

The tank suction heater, as illustrated in Fig. 11-38, contains a U-tube bundle. This design is often used with outdoor storage tanks for heavy fuel oils, tar, molasses, and similar fluids whose viscosity must be lowered to permit easy pumping. Usually the tube-side heating medium is steam. One end of the heater shell is open, and the liquid being heated passes across the outside of the tubes. Pumping costs can be reduced without heating the entire contents of the tank. Bare tube and integral low-fin tubes are provided with baffles. Longitudinal

fin-tube heaters are not baffled. Fins are most often used to minimize the fouling potential in these fluids.

Kettle-type reboilers, evaporators, etc., are often U-tube exchangers with enlarged shell sections for vapor-liquid separation. The U-tube bundle replaces the floating-heat bundle of Fig. 11-36e.

The U-tube exchanger with copper tubes, cast-iron header, and other parts of carbon steel is used for water and steam services in office buildings, schools, hospitals, hotels, etc. Nonferrous tube sheets and admiralty or 90-10 copper-nickel tubes are the most frequently used substitute materials. These standard exchangers are available from a number of manufacturers at costs far below those of custom-built process-industry equipment.

**Packed-Lantern-Ring Exchanger (Fig. 11-36f)** This construction is the least costly of the straight-tube removable bundle types. The shell- and tube-side fluids are each contained by separate rings of packing separated by a lantern ring and are installed at the floating tube sheet. The lantern ring is provided with weep holes. Any leakage passing the packing goes through the weep holes and then drops to the ground. Leakage at the packing will not result in mixing within the exchanger of the two fluids.

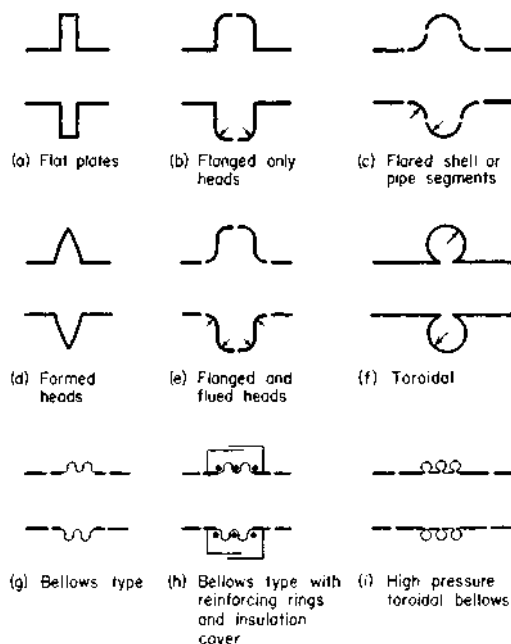


FIG. 11-37 Expansion joints.

The width of the floating tube sheet must be great enough to allow for the packings, the lantern ring, and differential expansion. Sometimes a small skirt is attached to a thin tube sheet to provide the required bearing surface for packings and lantern ring.

The clearance between the outer tube limit and the inside of the shell is slightly larger than that for fixed-tube-sheet and U-tube exchangers. The use of a floating-tube-sheet skirt increases this clearance. Without the skirt the clearance must make allowance for tube-hole distortion during tube rolling near the outside edge of the tube sheet or for tube-end welding at the floating tube sheet.

The packed-lantern-ring construction is generally limited to design temperatures below 191°C (375°F) and to the mild services of water, steam, air, lubricating oil, etc. Design gauge pressure does not exceed 2068 kPa (300 lbf/in<sup>2</sup>) for pipe shell exchangers and is limited to 1034 kPa (150 lbf/in<sup>2</sup>) for 610- to 1067-mm- (24- to 42-in-) diameter shells.

**Outside-Packed Floating-Head Exchanger (Fig. 11-36c)** The shell-side fluid is contained by rings of packing, which are compressed within a stuffing box by a packing follower ring. This construction was frequently used in the chemical industry, but in recent years usage has decreased. The removable-bundle construction accommodates differential expansion between shell and tubes and is used for shell-side

service up to 4137 kPa gauge pressure (600 lbf/in<sup>2</sup>) at 316°C (600°F). There are no limitations upon the number of tube-side passes or upon the tube-side design pressure and temperature. The outside-packed floating-head exchanger was the most commonly used type of removable-bundle construction in chemical-plant service.

The floating-tube-sheet skirt, where in contact with the rings of packing, has fine machine finish. A split shear ring is inserted into a groove in the floating-tube-sheet skirt. A slip-on backing flange, which in service is held in place by the shear ring, bolts to the external floating-head cover.

The floating-head cover is usually a circular disk. With an odd number of tube-side passes, an axial nozzle can be installed in such a floating-head cover. If a side nozzle is required, the circular disk is replaced by either a dished head or a channel barrel (similar to Fig. 11-36f) bolted between floating-head cover and floating-tube-sheet skirt.

The outer tube limit approaches the inside of the skirt but is farther removed from the inside of the shell than for any of the previously discussed constructions. Clearances between shell diameter and bundle OTL are 22 mm (7/8 in) for small-diameter pipe shells, 44 mm (1 3/4 in) for large-diameter pipe shells, and 58 mm (2 1/8 in) for moderate-diameter plate shells.

**Internal Floating-Head Exchanger (Fig. 11-36a)** The internal floating-head design is used extensively in petroleum-refinery service, but in recent years there has been a decline in usage.

The tube bundle is removable, and the floating tube sheet moves (or floats) to accommodate differential expansion between shell and tubes. The outer tube limit approaches the inside diameter of the gasket at the floating tube sheet. Clearances (between shell and OTL) are 29 mm (1 1/4 in) for pipe shells and 37 mm (1 7/8 in) for moderate-diameter plate shells.

A split backing ring and bolting usually hold the floating-head cover at the floating tube sheet. These are located beyond the end of the shell and within the larger-diameter shell cover. Shell cover, split backing ring, and floating-head cover must be removed before the tube bundle can pass through the exchanger shell.

With an even number of tube-side passes the floating-head cover serves as return cover for the tube-side fluid. With an odd number of passes a nozzle pipe must extend from the floating-head cover through the shell cover. Provision for both differential expansion and tube-bundle removal must be made.

**Pull-Through Floating-Head Exchanger (Fig. 11-36e)** Construction is similar to that of the internal-floating-head split-backing-ring exchanger except that the floating-head cover bolts directly to the floating tube sheet. The tube bundle can be withdrawn from the shell without removing either shell cover or floating-head cover. This feature reduces maintenance time during inspection and repair.

The large clearance between the tubes and the shell must provide for both the gasket and the bolting at the floating-head cover. This clearance is about 2 to 2 1/2 times that required by the split-ring design. Sealing strips or dummy tubes are often installed to reduce bypassing of the tube bundle.

**Falling-Film Exchangers** Falling-film shell-and-tube heat exchangers have been developed for a wide variety of services and are described by Sack [*Chem. Eng. Prog.*, **63**, 55 (July 1967)]. The fluid enters at the top of the vertical tubes. Distributors or slotted tubes put the liquid in film flow in the inside surface of the tubes, and the film adheres to the tube surface while falling to the bottom of the tubes. The film can be cooled, heated, evaporated, or frozen by means of the proper heat-transfer medium outside the tubes. Tube distributors have been developed for a wide range of applications. Fixed tube sheets, with or without expansion joints, and outside-packed-head designs are used.

Principal advantages are high rate of heat transfer, no internal pressure drop, short time of contact (very important for heat-sensitive materials), easy accessibility to tubes for cleaning, and, in some cases, prevention of leakage from one side to another.

These falling-film exchangers are used in various services as described in the following paragraphs.

**Liquid Coolers and Condensers** Dirty water can be used as the cooling medium. The top of the cooler is open to the atmosphere for

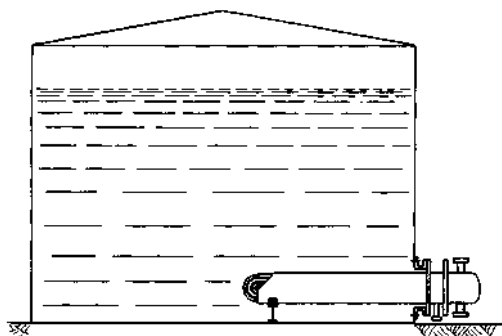


FIG. 11-38 Tank suction heater.

access to tubes. These can be cleaned without shutting down the cooler by removing the distributors one at a time and scrubbing the tubes.

**Evaporators** These are used extensively for the concentration of ammonium nitrate, urea, and other chemicals sensitive to heat when minimum contact time is desirable. Air is sometimes introduced in the tubes to lower the partial pressure of liquids whose boiling points are high. These evaporators are built for pressure or vacuum and with top or bottom vapor removal.

**Absorbers** These have a two-phase flow system. The absorbing medium is put in film flow during its fall downward on the tubes as it is cooled by a cooling medium outside the tubes. The film absorbs the gas which is introduced into the tubes. This operation can be cocurrent or countercurrent.

**Freezers** By cooling the falling film to its freezing point, these exchangers convert a variety of chemicals to the solid phase. The most common application is the production of sized ice and paradichlorobenzene. Selective freezing is used for isolating isomers. By melting the solid material and refreezing in several stages, a higher degree of purity of product can be obtained.

## TUBE-SIDE CONSTRUCTION

**Tube-Side Header** The tube-side header (or stationary head) contains one or more flow nozzles.

The **bonnet** (Fig. 11-35B) bolts to the shell. It is necessary to remove the bonnet in order to examine the tube ends. The fixed-tube-sheet exchanger of Fig. 11-36b has bonnets at both ends of the shell.

The **channel** (Fig. 11-35A) has a removable channel cover. The tube ends can be examined by removing this cover without disturbing the piping connections to the channel nozzles. The channel can bolt to the shell as shown in Fig. 11-36a and c. The Type C and Type N channels of Fig. 11-35 are welded to the tube sheet. This design is comparable in cost with the bonnet but has the advantages of permitting access to the tubes without disturbing the piping connections and of eliminating a gasketed joint.

**Special High-Pressure Closures (Fig. 11-35D)** The channel barrel and the tube sheet are generally forged. The removable channel cover is seated in place by hydrostatic pressure, while a shear ring subjected to shearing stress absorbs the end force. For pressures above 6205 kPa (900 lbf/in<sup>2</sup>) these designs are generally more economical than bolted constructions, which require larger flanges and bolting as pressure increases in order to contain the end force with bolts in tension. Relatively light-gauge internal pass partitions are provided to direct the flow of tube-side fluids but are designed only for the differential pressure across the tube bundle.

**Tube-Side Passes** Most exchangers have an even number of tube-side passes. The fixed-tube-sheet exchanger (which has no shell cover) usually has a return cover without any flow nozzles as shown in Fig. 11-35M; Types L and N are also used. All removable-bundle designs (except for the U tube) have a floating-head cover directing the flow of tube-side fluid at the floating tube sheet.

**Tubes** Standard heat-exchanger tubing is  $\frac{1}{4}$ ,  $\frac{3}{8}$ ,  $\frac{1}{2}$ ,  $\frac{5}{8}$ ,  $\frac{3}{4}$ , 1, 1 $\frac{1}{4}$ , and 1 $\frac{1}{2}$  in in outside diameter (in  $\times$  25.4 = mm). Wall thickness is measured in Birmingham wire gauge (BWG) units. A comprehensive list of tubing characteristics and sizes is given in Table 11-12. The most commonly used tubes in chemical plants and petroleum refineries are 19- and 25-mm ( $\frac{3}{4}$ - and 1-in) outside diameter. Standard tube lengths are 8, 10, 12, 16, and 20 ft, with 20 ft now the most common (ft  $\times$  0.3048 = m).

**Manufacturing tolerances** for steel, stainless-steel, and nickel-alloy tubes are such that the tubing is produced to either average or minimum wall thickness. Seamless carbon steel tube of minimum wall thickness may vary from 0 to 20 percent above the nominal wall thickness. Average-wall seamless tubing has an allowable variation of plus or minus 10 percent. Welded carbon steel tube is produced to closer tolerances (0 to plus 18 percent on minimum wall; plus or minus 9 percent on average wall). Tubing of aluminum, copper, and their alloys can be drawn easily and usually is made to minimum wall specifications.

Common practice is to specify **exchanger surface** in terms of total external square feet of tubing. The effective outside heat-transfer surface is based on the length of tubes measured between the inner faces

of tube sheets. In most heat exchangers there is little difference between the total and the effective surface. Significant differences are usually found in high-pressure and double-tube-sheet designs.

**Integrally finned tube**, which is available in a variety of alloys and sizes, is being used in shell-and-tube heat exchangers. The fins are radially extruded from thick-walled tube to a height of 1.6 mm ( $\frac{1}{16}$  in) spaced at 1.33 mm (19 fins per inch) or to a height of 3.2 mm ( $\frac{1}{8}$  in) spaced at 2.3 mm (11 fins per inch). External surface is approximately  $2\frac{1}{2}$  times the outside surface of a bare tube with the same outside diameter. Also available are 0.93-mm- (0.037-in-) high fins spaced 0.91 mm (28 fins per inch) with an external surface about 3.5 times the surface of the bare tube. Bare ends of nominal tube diameter are provided, while the fin height is slightly less than this diameter. The tube can be inserted into a conventional tube bundle and rolled or welded to the tube sheet by the same means, used for bare tubes. An integrally finned tube rolled into a tube sheet with double serrations and flared at the inlet is shown in Fig. 11-39. Internally finned tubes have been manufactured but have limited application.

**Longitudinal fins** are commonly used in double-pipe exchangers upon the outside of the inner tube. U-tube and conventional removable tube bundles are also made from such tubing. The ratio of external to internal surface generally is about 10 or 15:1.

**Transverse fins** upon tubes are used in low-pressure gas services. The primary application is in air-cooled heat exchangers (as discussed under that heading), but shell-and-tube exchangers with these tubes are in service.

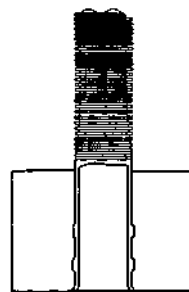
**Rolled Tube Joints** Expanded tube-to-tube-sheet joints are standard. Properly rolled joints have uniform tightness to minimize tube fractures, stress corrosion, tube-sheet ligament pushover and enlargement, and dishing of the tube sheet. Tubes are expanded into the tube sheet for a length of two tube diameters, or 50 mm (2 in), or tube-sheet thickness minus 3 mm ( $\frac{1}{8}$  in). Generally tubes are rolled for the last of these alternatives. The expanded portion should never extend beyond the shell-side face of the tube sheet, since removing such a tube is extremely difficult. Methods and tools for tube removal and tube rolling were discussed by John [*Chem. Eng.*, **66**, 77-80 (Dec. 28, 1959)], and rolling techniques by Bach [*Pet. Refiner*, **39**, 8, 104 (1960)].

Tube ends may be projecting, flush, flared, or beaded (listed in order of usage). The flare or bell-mouth tube end is usually restricted to water service in condensers and serves to reduce erosion near the tube inlet.

For moderate general process requirements at gauge pressures less than 2058 kPa (300 lbf/in<sup>2</sup>) and less than 177°C (350°F), tube-sheet holes without grooves are standard. For all other services with expanded tubes at least two grooves in each tube hole are common. The number of grooves is sometimes changed to one or three in proportion to tube-sheet thickness.

Expanding the tube into the **grooved tube holes** provides a stronger joint but results in greater difficulties during tube removal.

**Welded Tube Joints** When suitable materials of construction are used, the tube ends may be welded to the tube sheets. Welded joints may be seal-welded "for additional tightness beyond that of tube



**FIG. 11-39** Integrally finned tube rolled into tube sheet with double serrations and flared inlet. (Woverine Division, UOP, Inc.)

**TABLE 11-12 Characteristics of Tubing (From *Standards of the Tubular Exchanger Manufacturers Association*, 8th Ed., 1999; 25 North Broadway, Tarrytown, N.Y.)**

| Tube<br>O.D.,<br>in. | B.W.G.<br>gage | Thickness,<br>in. | Internal<br>area,<br>in. <sup>2</sup> | Sq. ft.<br>external<br>surface<br>per foot<br>length | Sq. ft.<br>internal<br>surface<br>per foot<br>length | Weight<br>per ft.<br>length<br>steel,<br>lb* | Tube<br>I.D.,<br>in. | Moment<br>of inertia,<br>in. <sup>4</sup> | Section<br>modulus,<br>in. <sup>3</sup> | Radius of<br>gyration,<br>in. | Constant<br>C† | O.D.<br>I.D. | Transverse<br>metal area,<br>in. <sup>2</sup> |
|----------------------|----------------|-------------------|---------------------------------------|------------------------------------------------------|------------------------------------------------------|----------------------------------------------|----------------------|-------------------------------------------|-----------------------------------------|-------------------------------|----------------|--------------|-----------------------------------------------|
| 1/4                  | 22             | 0.028             | 0.0296                                | 0.0654                                               | 0.0508                                               | 0.066                                        | 0.194                | 0.00012                                   | 0.00098                                 | 0.0791                        | 46             | 1.289        | 0.0195                                        |
|                      | 24             | 0.022             | 0.0333                                | 0.0654                                               | 0.0539                                               | 0.054                                        | 0.206                | 0.00010                                   | 0.00083                                 | 0.0810                        | 52             | 1.214        | 0.0158                                        |
|                      | 26             | 0.018             | 0.0360                                | 0.0654                                               | 0.0560                                               | 0.045                                        | 0.214                | 0.00009                                   | 0.00071                                 | 0.0823                        | 56             | 1.168        | 0.0131                                        |
|                      | 27             | 0.016             | 0.0373                                | 0.0654                                               | 0.0571                                               | 0.040                                        | 0.218                | 0.00008                                   | 0.00065                                 | 0.0829                        | 58             | 1.147        | 0.0118                                        |
| 3/8                  | 18             | 0.049             | 0.0603                                | 0.0982                                               | 0.0725                                               | 0.171                                        | 0.277                | 0.00068                                   | 0.0036                                  | 0.1166                        | 94             | 1.354        | 0.0502                                        |
|                      | 20             | 0.035             | 0.0731                                | 0.0982                                               | 0.0798                                               | 0.127                                        | 0.305                | 0.00055                                   | 0.0029                                  | 0.1208                        | 114            | 1.230        | 0.0374                                        |
|                      | 22             | 0.028             | 0.0799                                | 0.0982                                               | 0.0835                                               | 0.104                                        | 0.319                | 0.00046                                   | 0.0025                                  | 0.1231                        | 125            | 1.176        | 0.0305                                        |
|                      | 24             | 0.022             | 0.0860                                | 0.0982                                               | 0.0867                                               | 0.083                                        | 0.331                | 0.00038                                   | 0.0020                                  | 0.1250                        | 134            | 1.133        | 0.0244                                        |
| 1/2                  | 16             | 0.065             | 0.1075                                | 0.1309                                               | 0.0969                                               | 0.302                                        | 0.370                | 0.0021                                    | 0.0086                                  | 0.1555                        | 168            | 1.351        | 0.0888                                        |
|                      | 18             | 0.049             | 0.1269                                | 0.1309                                               | 0.1052                                               | 0.236                                        | 0.402                | 0.0018                                    | 0.0071                                  | 0.1604                        | 198            | 1.244        | 0.0694                                        |
|                      | 20             | 0.035             | 0.1452                                | 0.1309                                               | 0.1126                                               | 0.174                                        | 0.430                | 0.0014                                    | 0.0056                                  | 0.1649                        | 227            | 1.163        | 0.0511                                        |
|                      | 22             | 0.028             | 0.1548                                | 0.1309                                               | 0.1162                                               | 0.141                                        | 0.444                | 0.0012                                    | 0.0046                                  | 0.1672                        | 241            | 1.126        | 0.0415                                        |
| 5/8                  | 12             | 0.109             | 0.1301                                | 0.1636                                               | 0.1066                                               | 0.601                                        | 0.407                | 0.0061                                    | 0.0197                                  | 0.1865                        | 203            | 1.536        | 0.177                                         |
|                      | 13             | 0.095             | 0.1486                                | 0.1636                                               | 0.1139                                               | 0.538                                        | 0.435                | 0.0057                                    | 0.0183                                  | 0.1904                        | 232            | 1.437        | 0.158                                         |
|                      | 14             | 0.083             | 0.1655                                | 0.1636                                               | 0.1202                                               | 0.481                                        | 0.459                | 0.0053                                    | 0.0170                                  | 0.1939                        | 258            | 1.362        | 0.141                                         |
|                      | 15             | 0.072             | 0.1817                                | 0.1636                                               | 0.1259                                               | 0.426                                        | 0.481                | 0.0049                                    | 0.0156                                  | 0.1972                        | 283            | 1.299        | 0.125                                         |
|                      | 16             | 0.065             | 0.1924                                | 0.1636                                               | 0.1296                                               | 0.389                                        | 0.495                | 0.0045                                    | 0.0145                                  | 0.1993                        | 300            | 1.263        | 0.114                                         |
|                      | 17             | 0.058             | 0.2035                                | 0.1636                                               | 0.1333                                               | 0.352                                        | 0.509                | 0.0042                                    | 0.0134                                  | 0.2015                        | 317            | 1.228        | 0.103                                         |
|                      | 18             | 0.049             | 0.2181                                | 0.1636                                               | 0.1380                                               | 0.302                                        | 0.527                | 0.0037                                    | 0.0119                                  | 0.2044                        | 340            | 1.186        | 0.089                                         |
|                      | 19             | 0.042             | 0.2299                                | 0.1636                                               | 0.1416                                               | 0.262                                        | 0.541                | 0.0033                                    | 0.0105                                  | 0.2067                        | 359            | 1.155        | 0.077                                         |
|                      | 20             | 0.035             | 0.2419                                | 0.1636                                               | 0.1453                                               | 0.221                                        | 0.555                | 0.0028                                    | 0.0091                                  | 0.2090                        | 377            | 1.126        | 0.065                                         |
|                      | 20             | 0.035             | 0.2419                                | 0.1636                                               | 0.1453                                               | 0.221                                        | 0.555                | 0.0028                                    | 0.0091                                  | 0.2090                        | 377            | 1.126        | 0.065                                         |
| 3/4                  | 10             | 0.134             | 0.1825                                | 0.1963                                               | 0.1262                                               | 0.833                                        | 0.482                | 0.0129                                    | 0.0344                                  | 0.2229                        | 285            | 1.556        | 0.259                                         |
|                      | 11             | 0.120             | 0.2043                                | 0.1963                                               | 0.1335                                               | 0.808                                        | 0.510                | 0.0122                                    | 0.0326                                  | 0.2267                        | 319            | 1.471        | 0.238                                         |
|                      | 12             | 0.109             | 0.2223                                | 0.1963                                               | 0.1393                                               | 0.747                                        | 0.532                | 0.0116                                    | 0.0309                                  | 0.2299                        | 347            | 1.410        | 0.219                                         |
|                      | 13             | 0.095             | 0.2463                                | 0.1963                                               | 0.1466                                               | 0.665                                        | 0.560                | 0.0107                                    | 0.0285                                  | 0.2340                        | 384            | 1.339        | 0.195                                         |
|                      | 14             | 0.083             | 0.2679                                | 0.1963                                               | 0.1529                                               | 0.592                                        | 0.584                | 0.0098                                    | 0.0262                                  | 0.2376                        | 418            | 1.284        | 0.174                                         |
|                      | 15             | 0.072             | 0.2884                                | 0.1963                                               | 0.1587                                               | 0.522                                        | 0.606                | 0.0089                                    | 0.0238                                  | 0.2411                        | 450            | 1.238        | 0.153                                         |
|                      | 16             | 0.065             | 0.3019                                | 0.1963                                               | 0.1623                                               | 0.476                                        | 0.620                | 0.0083                                    | 0.0221                                  | 0.2433                        | 471            | 1.210        | 0.140                                         |
|                      | 17             | 0.058             | 0.3157                                | 0.1963                                               | 0.1660                                               | 0.429                                        | 0.634                | 0.0076                                    | 0.0203                                  | 0.2455                        | 492            | 1.183        | 0.126                                         |
|                      | 18             | 0.049             | 0.3339                                | 0.1963                                               | 0.1707                                               | 0.367                                        | 0.652                | 0.0067                                    | 0.0178                                  | 0.2484                        | 521            | 1.150        | 0.108                                         |
|                      | 20             | 0.035             | 0.3632                                | 0.1963                                               | 0.1780                                               | 0.268                                        | 0.680                | 0.0050                                    | 0.0134                                  | 0.2531                        | 567            | 1.103        | 0.079                                         |
| 7/8                  | 10             | 0.134             | 0.2894                                | 0.2291                                               | 0.1589                                               | 1.062                                        | 0.607                | 0.0221                                    | 0.0505                                  | 0.2662                        | 451            | 1.442        | 0.312                                         |
|                      | 11             | 0.120             | 0.3167                                | 0.2291                                               | 0.1662                                               | 0.969                                        | 0.635                | 0.0208                                    | 0.0475                                  | 0.2703                        | 494            | 1.378        | 0.285                                         |
|                      | 12             | 0.109             | 0.3390                                | 0.2291                                               | 0.1720                                               | 0.893                                        | 0.657                | 0.0196                                    | 0.0449                                  | 0.2736                        | 529            | 1.332        | 0.262                                         |
|                      | 13             | 0.095             | 0.3685                                | 0.2291                                               | 0.1793                                               | 0.792                                        | 0.685                | 0.0180                                    | 0.0411                                  | 0.2778                        | 575            | 1.277        | 0.233                                         |
|                      | 14             | 0.083             | 0.3948                                | 0.2291                                               | 0.1856                                               | 0.703                                        | 0.709                | 0.0164                                    | 0.0374                                  | 0.2815                        | 616            | 1.234        | 0.207                                         |
|                      | 15             | 0.072             | 0.4197                                | 0.2291                                               | 0.1914                                               | 0.618                                        | 0.731                | 0.0148                                    | 0.0337                                  | 0.2850                        | 655            | 1.197        | 0.182                                         |
|                      | 16             | 0.065             | 0.4359                                | 0.2291                                               | 0.1950                                               | 0.563                                        | 0.745                | 0.0137                                    | 0.0312                                  | 0.2873                        | 680            | 1.174        | 0.165                                         |
|                      | 17             | 0.058             | 0.4525                                | 0.2291                                               | 0.1987                                               | 0.507                                        | 0.759                | 0.0125                                    | 0.0285                                  | 0.2896                        | 706            | 1.153        | 0.149                                         |
|                      | 18             | 0.049             | 0.4742                                | 0.2291                                               | 0.2034                                               | 0.433                                        | 0.777                | 0.0109                                    | 0.0249                                  | 0.2925                        | 740            | 1.126        | 0.127                                         |
|                      | 20             | 0.035             | 0.5090                                | 0.2291                                               | 0.2107                                               | 0.314                                        | 0.805                | 0.0082                                    | 0.0187                                  | 0.2972                        | 794            | 1.087        | 0.092                                         |
| 1                    | 8              | 0.165             | 0.3526                                | 0.2618                                               | 0.1754                                               | 1.473                                        | 0.670                | 0.0392                                    | 0.0784                                  | 0.3009                        | 550            | 1.493        | 0.433                                         |
|                      | 10             | 0.134             | 0.4208                                | 0.2618                                               | 0.1916                                               | 1.241                                        | 0.732                | 0.0350                                    | 0.0700                                  | 0.3098                        | 656            | 1.366        | 0.365                                         |
|                      | 11             | 0.120             | 0.4536                                | 0.2618                                               | 0.1990                                               | 1.129                                        | 0.760                | 0.0327                                    | 0.0654                                  | 0.3140                        | 708            | 1.316        | 0.332                                         |
|                      | 12             | 0.109             | 0.4803                                | 0.2618                                               | 0.2047                                               | 1.038                                        | 0.782                | 0.0307                                    | 0.0615                                  | 0.3174                        | 749            | 1.279        | 0.305                                         |
|                      | 13             | 0.095             | 0.5153                                | 0.2618                                               | 0.2121                                               | 0.919                                        | 0.810                | 0.0280                                    | 0.0559                                  | 0.3217                        | 804            | 1.235        | 0.270                                         |
|                      | 14             | 0.083             | 0.5463                                | 0.2618                                               | 0.2183                                               | 0.814                                        | 0.834                | 0.0253                                    | 0.0507                                  | 0.3255                        | 852            | 1.199        | 0.239                                         |
|                      | 15             | 0.072             | 0.5755                                | 0.2618                                               | 0.2241                                               | 0.714                                        | 0.856                | 0.0227                                    | 0.0455                                  | 0.3291                        | 898            | 1.168        | 0.210                                         |
|                      | 16             | 0.065             | 0.5945                                | 0.2618                                               | 0.2278                                               | 0.650                                        | 0.870                | 0.0210                                    | 0.0419                                  | 0.3314                        | 927            | 1.149        | 0.191                                         |
|                      | 18             | 0.049             | 0.6390                                | 0.2618                                               | 0.2361                                               | 0.498                                        | 0.902                | 0.0166                                    | 0.0332                                  | 0.3367                        | 997            | 1.109        | 0.146                                         |
|                      | 20             | 0.035             | 0.6793                                | 0.2618                                               | 0.2435                                               | 0.361                                        | 0.930                | 0.0124                                    | 0.0247                                  | 0.3414                        | 1060           | 1.075        | 0.106                                         |
| 1-1/4                | 7              | 0.180             | 0.6221                                | 0.3272                                               | 0.2330                                               | 2.059                                        | 0.890                | 0.0890                                    | 0.1425                                  | 0.3836                        | 970            | 1.404        | 0.605                                         |
|                      | 8              | 0.165             | 0.6648                                | 0.3272                                               | 0.2409                                               | 1.914                                        | 0.920                | 0.0847                                    | 0.1355                                  | 0.3880                        | 1037           | 1.359        | 0.562                                         |
|                      | 10             | 0.134             | 0.7574                                | 0.3272                                               | 0.2571                                               | 1.599                                        | 0.982                | 0.0742                                    | 0.1187                                  | 0.3974                        | 1182           | 1.273        | 0.470                                         |
|                      | 11             | 0.120             | 0.8012                                | 0.3272                                               | 0.2644                                               | 1.450                                        | 1.010                | 0.0688                                    | 0.1100                                  | 0.4018                        | 1250           | 1.238        | 0.426                                         |
|                      | 12             | 0.109             | 0.8365                                | 0.3272                                               | 0.2702                                               | 1.330                                        | 1.032                | 0.0642                                    | 0.1027                                  | 0.4052                        | 1305           | 1.211        | 0.391                                         |
|                      | 13             | 0.095             | 0.8825                                | 0.3272                                               | 0.2775                                               | 1.173                                        | 1.060                | 0.0579                                    | 0.0926                                  | 0.4097                        | 1377           | 1.179        | 0.345                                         |
|                      | 14             | 0.083             | 0.9229                                | 0.3272                                               | 0.2838                                               | 1.036                                        | 1.084                | 0.0521                                    | 0.0833                                  | 0.4136                        | 1440           | 1.153        | 0.304                                         |
|                      | 16             | 0.065             | 0.9852                                | 0.3272                                               | 0.2932                                               | 0.824                                        | 1.120                | 0.0426                                    | 0.0682                                  | 0.4196                        | 1537           | 1.116        | 0.242                                         |
|                      | 18             | 0.049             | 1.0423                                | 0.3272                                               | 0.3016                                               | 0.629                                        | 1.152                | 0.0334                                    | 0.0534                                  | 0.4250                        | 1626           | 1.085        | 0.185                                         |
|                      | 20             | 0.035             | 1.0936                                | 0.3272                                               | 0.3089                                               | 0.455                                        | 1.180                | 0.0247                                    | 0.0395                                  | 0.4297                        | 1706           | 1.059        | 0.134                                         |
| 1-1/2                | 10             | 0.134             | 1.1921                                | 0.3927                                               | 0.3225                                               | 1.957                                        | 1.232                | 0.1354                                    | 0.1806                                  | 0.4853                        | 1860           | 1.218        | 0.575                                         |
|                      | 12             | 0.109             | 1.2908                                | 0.3927                                               | 0.3356                                               | 1.621                                        | 1.282                | 0.1159                                    | 0.1545                                  | 0.4933                        | 2014           | 1.170        | 0.476                                         |
|                      | 14             | 0.083             | 1.3977                                | 0.3927                                               | 0.3492                                               | 1.257                                        | 1.334                | 0.0931                                    | 0.1241                                  | 0.5018                        | 2180           | 1.124        | 0.369                                         |
|                      | 16             | 0.065             | 1.4741                                | 0.3927                                               | 0.3587                                               | 0.997                                        | 1.370                | 0.0756                                    | 0.1008                                  | 0.5079                        | 2300           | 1.095        | 0.293                                         |
| 2                    | 11             | 0.120             | 2.4328                                | 0.5236                                               | 0.4608                                               | 2.412                                        | 1.760                | 0.3144                                    | 0.3144                                  | 0.6660                        | 3795           | 1.136        | 0.709                                         |
|                      | 12             | 0.109             | 2.4941                                | 0.5236                                               | 0.4665                                               | 2.204                                        | 1.782                | 0.2904                                    | 0.2904                                  | 0.6697                        | 3891           | 1.122        | 0.648                                         |
|                      | 13             | 0.095             | 2.5730                                | 0.5236                                               | 0.4739                                               | 1.935                                        | 1.810                | 0.2586                                    | 0.2588                                  | 0.6744                        | 4014           | 1.105        | 0.569                                         |
|                      | 14             | 0.083             | 2.6417                                | 0.5236                                               | 0.4801                                               | 1.701                                        | 1.834                | 0.2300                                    | 0.2300                                  | 0.6784                        | 4121           | 1.091        | 0.500                                         |

\*Weights are based on low-carbon steel with a density of 0.2836 lb/cu. in. For other metals multiply by the following factors: aluminum, 0.35; titanium, 0.58; A.I.S.I. 400 Series S/steels, 0.99; A.I.S.I. 300 Series S/steels, 1.02; aluminum bronze, 1.04; aluminum brass, 1.06; nickel-chrome-iron, 1.07; Admiralty, 1.09; nickel, 1.13; nickel-copper, 1.12; copper and cupro-nickels, 1.14.

$$\dagger \text{Liquid velocity} = \frac{\text{lb per tube hour}}{C \times \text{sp gr of liquid}} \quad \text{ft/s (sp gr of water at 60°F = 1.0)}$$

rolling" or may be strength-welded. Strength-welded joints have been found satisfactory in very severe services. Welded joints may or may not be rolled before or after welding.

The variables in tube-end welding were discussed in two unpublished papers (Emhardt, "Heat Exchanger Tube-to-Tubesheet Joints," ASME Pap. 69-WA/HT-47; and Reynolds, "Tube Welding for Conventional and Nuclear Power Plant Heat Exchangers," ASME Pap. 69-WA/HT-24), which were presented at the November 1969 meeting of the American Society of Mechanical Engineers.

Tube-end rolling before welding may leave lubricant from the tube expander in the tube hole. Fouling during normal operation followed by maintenance operations will leave various impurities in and near the tube ends. Satisfactory welds are rarely possible under such conditions, since tube-end welding requires extreme cleanliness in the area to be welded.

Tube expansion after welding has been found useful for low and moderate pressures. In high-pressure service tube rolling has not been able to prevent leakage after weld failure.

**Double-Tube-Sheet Joints** This design prevents the passage of either fluid into the other because of leakage at the tube-to-tubesheet joints, which are generally the weakest points in heat exchangers. Any leakage at these joints admits the fluid to the gap between the tube sheets. Mechanical design, fabrication, and maintenance of double-tube-sheet designs require special consideration.

## SHELL-SIDE CONSTRUCTION

**Shell Sizes** Heat-exchanger shells are generally made from standard-wall steel pipe in sizes up to 305-mm (12-in) diameter; from 9.5-mm (3/8-in) wall pipe in sizes from 356 to 610 mm (14 to 24 in); and from steel plate rolled at discrete intervals in larger sizes. Clearances between the outer tube limit and the shell are discussed elsewhere in connection with the different types of construction.

The following formulae may be used to estimate tube counts for various bundle sizes and tube passes. The estimated values include the removal of tubes to provide an entrance area for shell nozzle sizes of one-fifth the shell diameter. Due to the large effect from other parameters such as design pressure/corrosion allowance, baffle cuts, seal strips, and so on, these are to be used as estimates only. Exact tube counts are part of the design package of most reputable exchanger design software and are normally used for the final design.

**Triangular** tube layouts with pitch equal to 1.25 times the tube outside diameter:

$$C = 0.75 (D/d) - 36; \text{ where } D = \text{Bundle O.D. } d = \text{Tube O.D.}$$

$$\text{Range of accuracy: } -24 \leq C \leq 24.$$

$$1 \text{ Tube Pass: } N_t = 1298. + 74.86C + 1.283C^2 - .0078C^3 - .0006C^4 \quad (11-74a)$$

$$2 \text{ Tube Pass: } N_t = 1266. + 73.58C + 1.234C^2 - .0071C^3 - .0005C^4 \quad (11-74b)$$

$$4 \text{ Tube Pass: } N_t = 1196. + 70.79C + 1.180C^2 - .0059C^3 - .0004C^4 \quad (11-74c)$$

$$6 \text{ Tube Pass: } N_t = 1166. + 70.72C + 1.269C^2 - .0074C^3 - .0006C^4 \quad (11-74d)$$

**Square** tube layouts with pitch equal to 1.25 times the tube outside diameter:

$$C = (D/d) - 36.; \text{ where } D = \text{Bundle O.D. } d = \text{Tube O.D.}$$

$$\text{Range of accuracy: } -24 \leq C \leq 24.$$

$$1 \text{ Tube Pass: } N_t = 593.6 + 33.52C + .3782C^2 - .0012C^3 + .0001C^4 \quad (11-75a)$$

$$2 \text{ Tube Pass: } N_t = 578.8 + 33.36C + .3847C^2 - .0013C^3 + .0001C^4 \quad (11-75b)$$

$$4 \text{ Tube Pass: } N_t = 562.0 + 33.04C + .3661C^2 - .0016C^3 + .0002C^4 \quad (11-75c)$$

$$6 \text{ Tube Pass: } N_t = 550.4 + 32.49C + .3873C^2 - .0013C^3 + .0001C^4 \quad (11-75d)$$

**Shell-Side Arrangements** The **one-pass shell** (Fig. 11-35E) is the most commonly used arrangement. Condensers from single component vapors often have the nozzles moved to the center of the shell for vacuum and steam services.

A solid longitudinal baffle is provided to form a two-pass shell (Fig. 11-35F). It may be insulated to improve thermal efficiency. (See further discussion on baffles). A two-pass shell can improve thermal effectiveness at a cost lower than for two shells in series.

For **split flow** (Fig. 11-35G), the longitudinal baffle may be solid or perforated. The latter feature is used with condensing vapors.

A **double-split-flow** design is shown in Fig. 11-35H. The longitudinal baffles may be solid or perforated.

The **divided flow** design (Fig. 11-35J), mechanically is like the one-pass shell except for the addition of a nozzle. Divided flow is used to meet low-pressure-drop requirements.

The **kettle reboiler** is shown in Fig. 11-35K. When nucleate boiling is to be done on the shell-side, this common design provides adequate dome space for separation of vapor and liquid above the tube bundle and surge capacity beyond the weir near the shell cover.

## BAFFLES AND TUBE BUNDLES

The **tube bundle** is the most important part of a tubular heat exchanger. The tubes generally constitute the most expensive component of the exchanger and are the one most likely to corrode. Tube sheets, baffles, or support plates, tie rods, and usually spacers complete the bundle.

**Minimum baffle spacing** is generally one-fifth of the shell diameter and not less than 50.8 mm (2 in). Maximum baffle spacing is limited by the requirement to provide adequate support for the tubes. The maximum unsupported tube span in inches equals  $74 d^{0.75}$  (where  $d$  is the outside tube diameter in inches). The unsupported tube span is reduced by about 12 percent for aluminum, copper, and their alloys.

Baffles are provided for heat-transfer purposes. When shell-side baffles are not required for heat-transfer purposes, as may be the case in condensers or reboilers, tube supports are installed.

**Segmental Baffles** Segmental or cross-flow baffles are standard. Single, double, and triple segmental baffles are used. Baffle cuts are illustrated in Fig. 11-40. The double segmental baffle reduces cross-flow velocity for a given baffle spacing. The triple segmental baffle reduces both cross-flow and long-flow velocities and has been identified as the "window-cut" baffle.

Baffle cuts are expressed as the ratio of segment opening height to shell inside diameter. Cross-flow baffles with horizontal cut are shown in Fig. 11-36a, c, and f. This arrangement is not satisfactory for horizontal condensers, since the condensate can be trapped between baffles, or for dirty fluids in which the dirt might settle out. Vertical-cut baffles are used for side-to-side flow in horizontal exchangers with condensing fluids or dirty fluids. Baffles are notched to assure complete drainage when the units are taken out of service. (These notches permit some bypassing of the tube bundle during normal operation.)

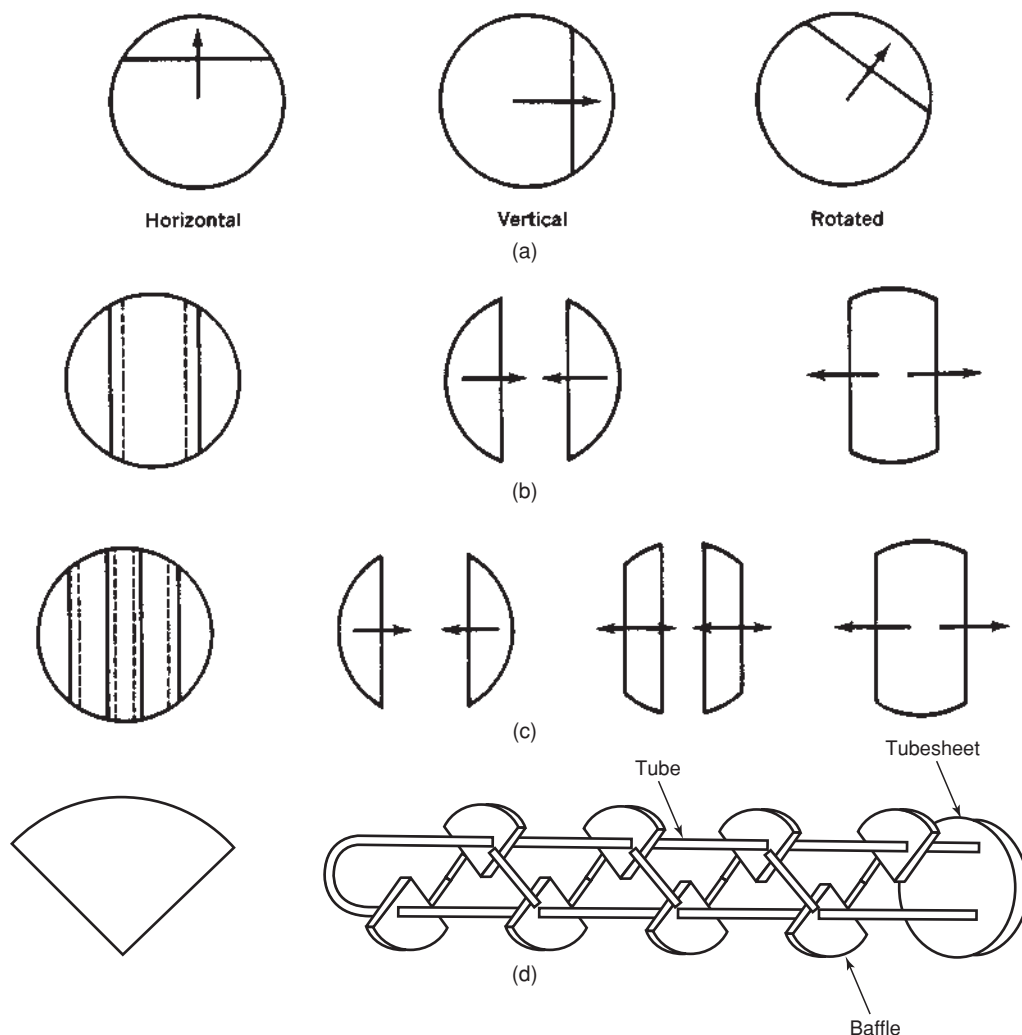
Tubes are most commonly arranged on an equilateral triangular pitch. Tubes are arranged on a square pitch primarily for mechanical cleaning purposes in removable-bundle exchangers.

**Maximum baffle cut** is limited to about 45 percent for single segmental baffles so that every pair of baffles will support each tube. Tube bundles are generally provided with baffles cut so that at least one row of tubes passes through all the baffles or support plates. These tubes hold the entire bundle together. In pipe-shell exchangers with a horizontal baffle cut and a horizontal pass rib for directing tube-side flow in the channel, the maximum baffle cut, which permits a minimum of one row of tubes to pass through all baffles, is approximately 33 percent in small shells and 40 percent in larger pipe shells.

Maximum shell-side heat-transfer rates in forced convection are apparently obtained by cross-flow of the fluid at right angles to the tubes. In order to maximize this type of flow some heat exchangers are built with segmental-cut baffles and with "no tubes in the window" (or the baffle cutout). Maximum baffle spacing may thus equal maximum unsupported-tube span, while conventional baffle spacing is limited to one-half of this span.

The maximum baffle spacing for no tubes in the window of single segmental baffles is unlimited when intermediate supports are provided. These are cut on both sides of the baffle and therefore do not affect the flow of the shell-side fluid. Each support engages all the tubes; the supports are spaced to provide adequate support for the tubes.

**Rod Baffles** Rod or bar baffles have either rods or bars extending through the lanes between rows of tubes. A baffle set can consist of a baffle with rods in all the vertical lanes and another baffle with rods in all the horizontal lanes between the tubes. The shell-side flow is uniform and parallel to the tubes. Stagnant areas do not exist.



**FIG. 11-40** Plate baffles. (a) Baffle cuts for single segmental baffles. (b) Baffle cuts for double segmental baffles. (c) Baffle cuts for triple segmental baffles. (d) Helical baffle construction.

One device uses four baffles in a baffle set. Only half of either the vertical or the horizontal tube lanes in a baffle have rods. The new design apparently provides a maximum shell-side heat-transfer coefficient for a given pressure drop.

**Tie Rods and Spacers** Tie rods are used to hold the baffles in place with spacers, which are pieces of tubing or pipe placed on the rods to locate the baffles. Occasionally baffles are welded to the tie rods, and spacers are eliminated. Properly located tie rods and spacers serve both to hold the bundle together and to reduce bypassing of the tubes.

In very large fixed-tube-sheet units, in which concentricity of shells decreases, baffles are occasionally welded to the shell to eliminate bypassing between the baffle and the shell.

Metal baffles are standard. Occasionally plastic baffles are used either to reduce corrosion or in vibratory service, in which metal baffles may cut the tubes.

**Impingement Baffle** The tube bundle is customarily protected against impingement by the incoming fluid at the shell inlet nozzle when the shell-side fluid is at a high velocity, is condensing, or is a two-phase fluid. Minimum entrance area about the nozzle is generally equal to the inlet nozzle area. Exit nozzles also require adequate area

between the tubes and the nozzles. A full bundle without any provision for shell inlet nozzle area can increase the velocity of the inlet fluid by as much as 300 percent with a consequent loss in pressure.

Impingement baffles are generally made of rectangular plate, although circular plates are more desirable. Rods and other devices are sometimes used to protect the tubes from impingement. In order to maintain a maximum tube count the impingement plate is often placed in a conical nozzle opening or in a dome cap above the shell.

Impingement baffles or flow-distribution devices are recommended for axial tube-side nozzles when entrance velocity is high.

**Vapor Distribution** Relatively large shell inlet nozzles, which may be used in condensers under low pressure or vacuum, require provision for uniform vapor distribution.

**Tube-Bundle Bypassing** Shell-side heat-transfer rates are maximized when bypassing of the tube bundle is at a minimum. The most significant bypass stream is generally between the outer tube limit and the inside of the shell. The clearance between tubes and shell is at a minimum for fixed-tube-sheet construction and is greatest for straight-tube removable bundles.

Arrangements to reduce tube-bundle bypassing include:

**Dummy tubes.** These tubes do not pass through the tube sheets and can be located close to the inside of the shell.

**Tie rods with spacers.** These hold the baffles in place but can be located to prevent bypassing.

**Sealing strips.** These longitudinal strips either extend from baffle to baffle or may be inserted in slots cut into the baffles.

Dummy tubes or tie rods with spacers may be located within the pass partition lanes (and between the baffle cuts) in order to ensure maximum bundle penetration by the shell-side fluid.

When tubes are omitted from the tube layout to provide entrance area about an impingement plate, the need for sealing strips or other devices to cause proper bundle penetration by the shell-side fluid is increased.

**Helical Baffles** An increasingly popular variant to the segmental baffle is the helical baffle. These are quadrant-shaped plate baffles installed at an angle to the axial bundle centerline to produce a pseudo-spiraling flow down the length of the tube bundle (Fig. 11-40d). This baffle has the advantage of producing shell-side heat-transfer coefficients similar to those of the segmental baffle with much less shell-side pressure loss for the same size of shell. In the case of equal pressure drops, the helical baffle exchanger will be smaller than the segmental baffle exchanger; or, for identical shell sizes, the helical baffle exchanger will permit a much higher throughput of flow for the same process inlet/outlet temperatures.

A great amount of proprietary research has been conducted by a few companies into the workings of helical baffled heat exchangers. The only known open literature method for estimating helical baffle performance has been "Comparison of Correction Factors for Shell-and-Tube Heat Exchangers with Segmental or Helical Baffles" by Stehlik, Nemcansky, Kral, and Swanson [*Heat Transfer Engineering*, 15(1), 55-65].

Unique design variables for helical baffles include the baffle angle, adjacent baffle contact diameter (which sets the baffle spacing and is usually about half of the shell I.D.), and the number of baffle starts (i.e., number of intermediate baffle starts). Of course, consideration is also given to the tube layout, tube pitch, use of seal strips, and all the other configuration characteristics common to any plate baffled bundle.

A helical baffle bundle built in this way produces two distinct flow regions. The area outside of the adjacent baffle contact diameter tends to produce a stable helical cross flow. However, inside the diameter where adjacent baffles touch is a second region where vortical flow is induced but in which the intensity of the rotational component tends to decrease as one approaches the center of the bundle. For a fixed flow rate and helix angle, this tendency may be minimized by the proper selection of the baffle contact diameter. With the correct selection, stream temperatures may be made to be close to uniform across the bundle cross section through the shell. However, below a critical velocity (for the baffle configuration and fluid state), the tendency for nonuniformity of temperatures increases as velocity decreases until ever-increasing portions of the central core surface area pinch out with respect to temperature and become ineffective for further heat transfer.

The design approach involves varying the baffle spacing for the primary purpose of balancing the flows in the two regions and maximizing the effectiveness of the total surface area. In many cases, a shallower helix angle is chosen in conjunction with the baffle spacing in order to minimize the central core component while still achieving a reduced overall bundle pressure drop.

**Longitudinal Flow Baffles** In fixed-tube-sheet construction with multipass shells, the baffle is usually welded to the shell and positive assurance against bypassing results. Removable tube bundles have a sealing device between the shell and the longitudinal baffle. Flexible light-gauge sealing strips and various packing devices have been used. Removable U-tube bundles with four tube-side passes and two shell-side passes can be installed in shells with the longitudinal baffle welded in place.

In split-flow shells the longitudinal baffle may be installed without a positive seal at the edges if design conditions are not seriously affected by a limited amount of bypassing.

Fouling in petroleum-refinery service has necessitated rough treatment of tube bundles during cleaning operations. Many refineries

avoid the use of longitudinal baffles, since the sealing devices are subject to damage during cleaning and maintenance operations.

## CORROSION IN HEAT EXCHANGERS

Some of the special considerations in regard to heat-exchanger corrosion are discussed in this subsection. A more extended presentation in Sec. 23 covers corrosion and its various forms as well as materials of construction.

**Materials of Construction** The most common material of construction for heat exchangers is carbon steel. Stainless-steel construction throughout is sometimes used in chemical-plant service and on rare occasions in petroleum refining. Many exchangers are constructed from dissimilar metals. Such combinations are functioning satisfactorily in certain services. Extreme care in their selection is required since electrolytic attack can develop.

Carbon steel and alloy combinations appear in Table 11-13 "Alloys" in chemical- and petrochemical-plant service in approximate order of use are stainless-steel series 300, nickel, Monel, copper alloy, aluminum, Inconel, stainless-steel series 400, and other alloys. In petroleum-refinery service the frequency order shifts, with copper alloy (for water-cooled units) in first place and low-alloy steel in second place. In some segments of the petroleum industry copper alloy, stainless series 400, low-alloy steel, and aluminum are becoming the most commonly used alloys.

Copper-alloy tubing, particularly inhibited admiralty, is generally used with cooling water. Copper-alloy tube sheets and baffles are generally of naval brass.

Aluminum alloy (and in particular alclad aluminum) tubing is sometimes used in water service. The alclad alloy has a sacrificial aluminum-alloy layer metallurgically bonded to a core alloy.

Tube-side headers for water service are made in a wide variety of materials: carbon steel, copper alloy, cast iron, and lead-lined or plastic-lined or specially painted carbon steel.

**Bimetallic Tubes** When corrosive requirements or temperature conditions do not permit the use of a single alloy for the tubes, bimetallic (or duplex) tubes may be used. These can be made from almost any possible combination of metals. Tube sizes and gauges can be varied. For thin gauges the wall thickness is generally divided equally between the two components. In heavier gauges the more expensive component may comprise from a fifth to a third of the total thickness.

The component materials comply with applicable ASTM specifications, but after manufacture the outer component may increase in hardness beyond specification limits, and special care is required during the tube-rolling operation. When the harder material is on the outside, precautions must be exercised to expand the tube properly. When the inner material is considerably softer, rolling may not be practical unless ferrules of the soft material are used.

In order to eliminate galvanic action the outer tube material may be stripped from the tube ends and replaced with ferrules of the inner tube material. When the end of a tube with a ferrule is expanded or welded to a tube sheet, the tube-side fluid can contact only the inner tube material, while the outer material is exposed to the shell-side fluid.

Bimetallic tubes are available from a small number of tube mills and are manufactured only on special order and in large quantities.

**TABLE 11-13 Dissimilar Materials in Heat-Exchanger Construction**

| Part              | Relative use  | 1 | 2 | 3 | 4 | 5 | 6 |
|-------------------|---------------|---|---|---|---|---|---|
|                   | Relative cost | A | B | C | D | C | E |
| Tubes             |               | ● | ● | ● | ● | ● | ● |
| Tube sheets       |               |   | ● | ● | ● | ● | ● |
| Tube-side headers |               |   |   | ● | ● |   |   |
| Baffles           |               |   |   |   | ● | ● | ● |
| Shell             |               |   |   |   |   |   | ● |

Carbon steel replaced by an alloy when ● appears.

Relative use: from 1 (most popular) through 6 (least popular) combinations.

Relative cost: from A (least expensive) to E (most expensive).

**Clad Tube Sheets** Usually tube sheets and other exchanger parts are of a solid metal. Clad or bimetallic tube sheets are used to reduce costs or because no single metal is satisfactory for the corrosive conditions. The alloy material (e.g., stainless steel, Monel) is generally bonded or clad to a carbon steel backing material. In fixed-tube-sheet construction a copper-alloy-clad tube sheet can be welded to a steel shell, while most copper-alloy tube sheets cannot be welded to steel in a manner acceptable to ASME Code authorities.

Clad tube sheets in service with carbon steel backer material include stainless-steel types 304, 304L, 316, 316L, and 317, Monel, Inconel, nickel, naval rolled brass, copper, admiralty, silicon bronze, and titanium. Naval rolled brass and Monel clad on stainless steel are also in service.

Ferrous-alloy-clad tube sheets are generally prepared by a weld overlay process in which the alloy material is deposited by welding upon the face of the tube sheet. Precautions are required to produce a weld deposit free of defects, since these may permit the process fluid to attack the base metal below the alloy. Copper-alloy-clad tube sheets are prepared by brazing the alloy to the carbon steel backing material.

Clad materials can be prepared by bonding techniques, which involve rolling, heat treatment, explosive bonding, etc. When properly manufactured, the two metals do not separate because of thermal-expansion differences encountered in service. Applied tube-sheet facings prepared by tack welding at the outer edges of alloy and base metal or by bolting together the two metals are in limited use.

**Nonmetallic Construction** Shell-and-tube exchangers with glass tubes 14 mm (0.551 in) in diameter and 1 mm (0.039 in) thick with tube lengths from 2.015 m (79.3 in) to 4.015 m (158 in) are available. Steel shell exchangers have a maximum design pressure of 517 kPa (75 lbf/in<sup>2</sup>). Glass shell exchangers have a maximum design gauge pressure of 103 kPa (15 lbf/in<sup>2</sup>). Shell diameters are 229 mm (9 in), 305 mm (12 in), and 457 mm (18 in). Heat-transfer surface ranges from 3.16 to 51 m<sup>2</sup> (34 to 550 ft<sup>2</sup>). Each tube is free to expand, since a Teflon sealer sheet is used at the tube-to-tube-sheet joint.

**Impervious graphite** heat-exchanger equipment is made in a variety of forms, including outside-packed-head shell-and-tube exchangers. They are fabricated with impervious graphite tubes and tube-side head-

ers and metallic shells. Single units containing up to 1300 m<sup>2</sup> (14,000 ft<sup>2</sup>) of heat-transfer surface are available.

Teflon heat exchangers of special construction are described later in this section.

**Fabrication** Expanding the tube into the tube sheet reduces the tube wall thickness and work-hardens the metal. The induced stresses can lead to **stress corrosion**. Differential expansion between tubes and shell in fixed-tube-sheet exchangers can develop stresses, which lead to stress corrosion.

When austenitic stainless-steel tubes are used for corrosion resistance, a close fit between the tube and the tube hole is recommended in order to minimize work hardening and the resulting loss of corrosion resistance.

In order to facilitate removal and replacement of tubes it is customary to roller-expand the tubes to within 3 mm ( $\frac{1}{8}$  in) of the shell-side face of the tube sheet. A 3-mm- ( $\frac{1}{8}$ -in-) long gap is thus created between the tube and the tube hole at this tube-sheet face. In some services this gap has been found to be a focal point for corrosion.

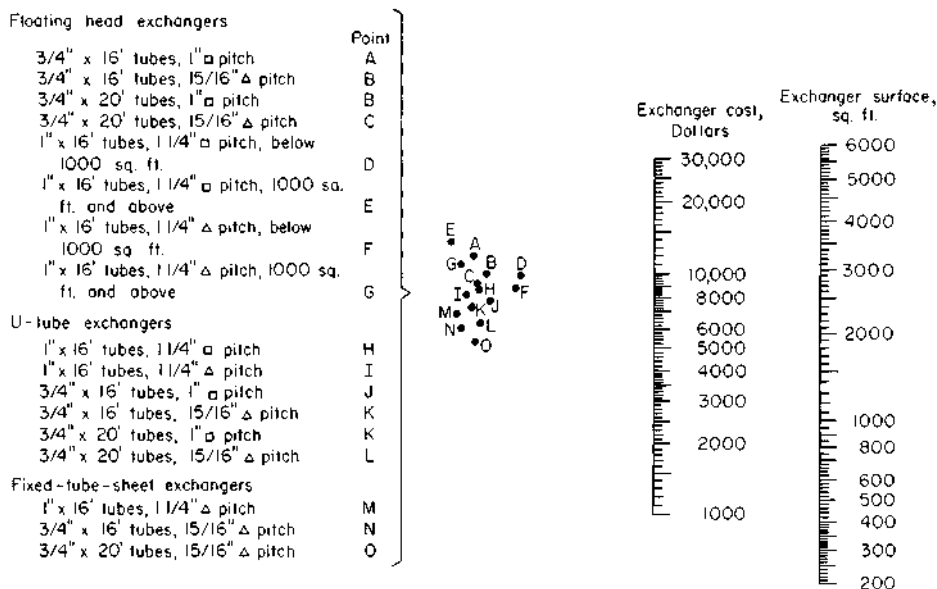
It is standard practice to provide a chamfer at the inside edges of tube holes in tube sheets to prevent cutting of the tubes and to remove burrs produced by drilling or reaming the tube sheet. In the lower tube sheet of vertical units this chamfer serves as a pocket to collect material, dirt, etc., and to serve as a corrosion center.

Adequate venting of exchangers is required both for proper operation and to reduce corrosion. Improper venting of the water side of exchangers can cause alternate wetting and drying and accompanying chloride concentration, which is particularly destructive to the series 300 stainless steels.

Certain corrosive conditions require that special consideration be given to complete drainage when the unit is taken out of service. Particular consideration is required for the upper surfaces of tube sheets in vertical heat exchangers, for sagging tubes, and for shell-side baffles in horizontal units.

## SHELL-AND-TUBE EXCHANGER COSTS

Basic costs of shell-and-tube heat exchangers made in the United States of carbon steel construction in 1958 are shown in Fig. 11-41.



**FIG. 11-41** Costs of basic exchangers—all steel, TEMA Class R, 150 lbf/in<sup>2</sup>, 1958. To convert pounds-force per square inch to kilopascals, multiply by 6.895; to convert square feet to square meters, multiply by 0.0929; to convert inches to millimeters, multiply by 25.4; and to convert feet to meters, multiply by 0.3048.



**TABLE 11-14 Extras for Pressure and Alloy Construction and Surface and Weights\***Percent of steel base price, 1500-lbf/in<sup>2</sup> working pressure

|                                                                                                    | Shell diameters, in |      |      |      |      |      |      |        |        |        |        |        |        |
|----------------------------------------------------------------------------------------------------|---------------------|------|------|------|------|------|------|--------|--------|--------|--------|--------|--------|
|                                                                                                    | 12                  | 14   | 16   | 18   | 20   | 22   | 24   | 27     | 30     | 33     | 36     | 39     | 42     |
| <b>Pressure†</b>                                                                                   |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| 300 lbf/in <sup>2</sup>                                                                            | 7                   | 7    | 8    | 8    | 9    | 9    | 10   | 11     | 11     | 12     | 13     | 14     | 15     |
| 450 lbf/in <sup>2</sup>                                                                            | 18                  | 19   | 20   | 21   | 22   | 23   | 24   | 27     | 29     | 31     | 32     | 33     | 35     |
| 600 lbf/in <sup>2</sup>                                                                            | 28                  | 29   | 31   | 33   | 35   | 37   | 39   | 40     | 41     | 32     | 44     | 45     | 50     |
| <b>Alloy</b>                                                                                       |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| All-steel heat exchanger                                                                           | 100                 | 100  | 100  | 100  | 100  | 100  | 100  | 100    | 100    | 100    | 100    | 100    | 100    |
| Tube sheets and baffles                                                                            |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| Naval rolled brass                                                                                 | 14                  | 17   | 19   | 21   | 22   | 22   | 22   | 22     | 22     | 23     | 24     | 24     | 25     |
| Monel                                                                                              | 24                  | 31   | 35   | 37   | 39   | 39   | 40   | 40     | 41     | 41     | 41     | 41     | 42     |
| 1½ Cr, ½ Mo                                                                                        | 6                   | 7    | 7    | 7    | 8    | 8    | 8    | 8      | 9      | 10     | 10     | 10     | 11     |
| 4-6 Cr, ½ Mo                                                                                       | 19                  | 22   | 24   | 25   | 26   | 26   | 26   | 25     | 25     | 25     | 26     | 26     | 26     |
| 11-13 Cr (stainless 410)                                                                           | 21                  | 24   | 26   | 27   | 27   | 27   | 27   | 27     | 27     | 27     | 27     | 27     | 28     |
| Stainless 304                                                                                      | 22                  | 27   | 29   | 30   | 31   | 31   | 31   | 31     | 30     | 30     | 30     | 31     | 31     |
| Shell and shell cover                                                                              |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| Monel                                                                                              | 45                  | 48   | 51   | 52   | 53   | 52   | 52   | 51     | 49     | 47     | 45     | 44     | 44     |
| 1½ Cr, ½ Mo                                                                                        | 20                  | 22   | 24   | 25   | 25   | 25   | 24   | 22     | 20     | 19     | 18     | 17     | 17     |
| 4-6 Cr, ½ Mo                                                                                       | 28                  | 31   | 33   | 35   | 35   | 35   | 34   | 32     | 30     | 28     | 27     | 26     | 26     |
| 11-13 Cr (stainless 410)                                                                           | 29                  | 33   | 35   | 36   | 36   | 36   | 35   | 34     | 32     | 30     | 29     | 27     | 27     |
| Stainless 304                                                                                      | 32                  | 34   | 36   | 37   | 38   | 37   | 37   | 35     | 33     | 31     | 30     | 29     | 28     |
| Channel and floating-head cover                                                                    |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| Monel                                                                                              | 40                  | 42   | 42   | 43   | 42   | 41   | 40   | 37     | 34     | 32     | 31     | 40     | 30     |
| 1½ Cr, ½ Mo                                                                                        | 23                  | 24   | 24   | 25   | 24   | 24   | 23   | 22     | 21     | 21     | 21     | 20     | 20     |
| 4-6 Cr, ½ Mo                                                                                       | 36                  | 37   | 38   | 38   | 37   | 36   | 34   | 31     | 29     | 27     | 26     | 25     | 24     |
| 11-13 Cr (stainless 410)                                                                           | 37                  | 38   | 39   | 39   | 38   | 37   | 35   | 32     | 30     | 28     | 27     | 26     | 25     |
| Stainless 304                                                                                      | 37                  | 39   | 39   | 39   | 38   | 37   | 36   | 33     | 31     | 29     | 28     | 26     | 26     |
| <b>Surface</b>                                                                                     |                     |      |      |      |      |      |      |        |        |        |        |        |        |
| Surface, ft <sup>2</sup> , internal floating head, ¾-in OD by 1-in square pitch, 16 ft 0 in, tube† | 251                 | 302  | 438  | 565  | 726  | 890  | 1040 | 1470   | 1820   | 2270   | 2740   | 3220   | 3700   |
| 1-in OD by 1¼-in square pitch, 16-ft 0-in tube§                                                    | 218                 | 252  | 352  | 470  | 620  | 755  | 876  | 1260   | 1560   | 1860   | 2360   | 2770   | 3200   |
| Weight, lb, internal floating head, 1-in OD, 14 BWG tube                                           | 2750                | 3150 | 4200 | 5300 | 6600 | 7800 | 9400 | 11,500 | 14,300 | 17,600 | 20,500 | 24,000 | 29,000 |

\*Modified from E. N. Sieder and G. H. Elliot, *Pet. Refiner*, **39**(5), 223 (1960).†Total extra is  $0.7 \times$  pressure extra on shell side plus  $0.3 \times$  pressure extra on tube side.

‡Fixed-tube-sheet construction with ¾-in OD tube on 1½-in triangular pitch provides 36 percent more surface.

§Fixed-tube-sheet construction with 1-in OD tube on 1¼-in triangular pitch provides 18 percent more surface.

For an all-steel heat exchanger with mixed design pressures the total extra for pressure is  $0.7 \times$  pressure extra on shell side plus  $0.3 \times$  pressure extra tube side.

For an exchanger with alloy parts and a design pressure of 150 lbf/in<sup>2</sup>, the alloy extras are added. For shell and shell cover the combined alloy-pressure extra is the alloy extra times the shell-side pressure extra/100. For tube sheets and baffles the combined alloy-pressure extra is the alloy extra times the higher-pressure extra times 0.9/100. (The 0.9 factor is included since baffle thickness does not increase because of pressure.)

NOTE: To convert pounds-force per square inch to kilopascals, multiply by 6.895; to convert square feet to square meters, multiply by 0.0929; and to convert inches to millimeters, multiply by 25.4.

Cost data for shell-and-tube exchangers from 15 sources were correlated and found to be consistent when scaled by the Marshall and Swift index [Woods et al., *Can. J. Chem. Eng.*, **54**, 469–489 (December 1976)].

Costs of shell-and-tube heat exchangers can be estimated from Fig. 11-41 and Tables 11-14 and 11-15. These 1960 costs should be updated by use of the Marshall and Swift Index, which appears in each issue of *Chemical Engineering*. Note that during periods of high and low demand for heat exchangers the prices in the marketplace may vary significantly from those determined by this method.

Small heat exchangers and exchangers bought in small quantities are likely to be more costly than indicated.

Standard heat exchangers (which are in some instances off-the-shelf items) are available in sizes ranging from 1.9 to 37 m<sup>2</sup> (20 to 400 ft<sup>2</sup>) at costs lower than for custom-built units. Steel costs are approximately one-half, admiralty tube-side costs are two-thirds, and stainless costs are three-fourths of those for equivalent custom-built exchangers.

**Kettle-type-reboiler** costs are 15 to 25 percent greater than for equivalent internal-floating-head or U-tube exchangers. The higher extra is applicable with relatively large kettle-to-port-diameter ratios

and with increased internals (e.g., vapor-liquid separators, foam breakers, sight glasses).

To estimate exchanger costs for varying construction details and alloys, first determine the base cost of a similar heat exchanger of basic construction (carbon steel, Class R, 150 lbf/in<sup>2</sup>) from Fig. 11-41. From Table 11-14, select appropriate extras for higher pressure rating and for alloy construction of tube sheets and baffles, shell and shell cover, and channel and floating-head cover. Compute these extras in accordance with the notes below the table. For tubes other than welded carbon steel, compute the extra by multiplying the exchanger surface by the appropriate cost per square foot from Table 11-15.

When points for 20-ft-long tubes do not appear in Fig. 11-41, use 0.95 times the cost of the equivalent 16-ft-long exchanger. Length variation of steel heat exchangers affects costs by approximately \$1 per square foot. Shell diameters for a given surface are approximately equal for U-tube and floating-head construction.

Low-fin tubes (1/16-in-high fins) provide 2.5 times the surface per lineal foot. Surface required should be divided by 2.5; then use Fig. 11-41 to determine basic cost of the heat exchanger. Actual surface times extra costs (from Table 11-15) should then be added to determine cost of fin-tube exchanger.

**TABLE 11-15 Base Quantity Extra Cost for Tube Gauge and Alloy**  
Dollars per square foot

|                        | ¾-in OD tubes |        |        | 1-in OD tubes |        |        |
|------------------------|---------------|--------|--------|---------------|--------|--------|
|                        | 16 BWG        | 14 BWG | 12 BWG | 16 BWG        | 14 BWG | 12 BWG |
| Carbon steel           | 0             | 0.02   | 0.06   | 0             | 0.01   | 0.07   |
| Admiralty              | 0.78          | 1.20   | 1.81   | 0.94          | 1.39   | 2.03   |
| (T-11) 1¼ Cr, ½ Mo     | 1.01          | 1.04   | 1.11   | 0.79          | 0.82   | 0.95   |
| (T-5) 4-6 Cr           | 1.61          | 1.65   | 1.74   | 1.28          | 1.32   | 1.48   |
| Stainless 410 welded   | 2.62          | 3.16   | 4.12   | 2.40          | 2.89   | 3.96   |
| Stainless 410 seamless | 3.10          | 3.58   | 4.63   | 2.84          | 3.31   | 4.47   |
| Stainless 304 welded   | 2.50          | 3.05   | 3.99   | 2.32          | 2.83   | 3.88   |
| Stainless 304 seamless | 3.86          | 4.43   | 5.69   | 3.53          | 4.08   | 5.46   |
| Stainless 316 welded   | 3.40          | 4.17   | 5.41   | 3.25          | 3.99   | 5.36   |
| Stainless 316 seamless | 7.02          | 7.95   | 10.01  | 6.37          | 7.27   | 9.53   |
| 90-10 cupronickel      | 1.33          | 1.89   | 2.67   | 1.50          | 2.09   | 2.90   |
| Monel                  | 4.25          | 5.22   | 6.68   | 4.01          | 4.97   | 6.47   |
| Low fin                |               |        |        |               |        |        |
| Carbon steel           | 0.22          | 0.23   |        | 0.18          | 0.19   |        |
| Admiralty              | 0.58          | 0.75   |        | 0.70          | 0.87   |        |
| 90-10 cupronickel      | 0.72          | 0.96   |        | 0.86          | 1.06   |        |

NOTE: To convert inches to millimeters, multiply by 25.4.

## HAIRPIN/DOUBLE-PIPE HEAT EXCHANGERS

### PRINCIPLES OF CONSTRUCTION

Hairpin heat exchangers (often also referred to as “double pipes”) are characterized by a construction form which imparts a U-shaped appearance to the heat exchanger. In its classical sense, the term *double pipe* refers to a heat exchanger consisting of a pipe within a pipe, usually of a straight-leg construction with no bends. However, due to the need for removable bundle construction and the ability to handle differential thermal expansion while avoiding the use of expansion joints (often the weak point of the exchanger), the current U-shaped configuration has become the standard in the industry (Fig. 11-42). A further departure from the classical definition comes when more than one pipe or tube is used to make a tube bundle, complete with tubesheets and tube supports similar to the TEMA style exchanger.

Hairpin heat exchangers consist of two shell assemblies housing a common set of tubes and interconnected by a return-bend cover referred to as the *bonnet*. The shell is supported by means of bracket assemblies designed to cradle both shells simultaneously. These brackets are configured to permit the modular assembly of many hairpin sections into an exchanger bank for inexpensive future-expansion capability and for providing the very long thermal lengths demanded by special process applications.

The bracket construction permits support of the exchanger without fixing the supports to the shell. This provides for thermal movement of the shells within the brackets and prevents the transfer of thermal stresses into the process piping. In special cases the brackets may be welded to the shell. However, this is usually avoided due to the resulting loss of flexibility in field installation and equipment reuse at other sites and an increase in piping stresses.

The hairpin heat exchanger, unlike the removable bundle TEMA styles, is designed for bundle insertion and removal from the return end rather than the tubesheet end. This is accomplished by means of removable split rings which slide into grooves machined around the outside of each tubesheet and lock the tubesheets to the external closure flanges. This provides a distinct advantage in maintenance since bundle removal

takes place at the exchanger end furthest from the plant process piping without disturbing any gasketed joints of this piping.

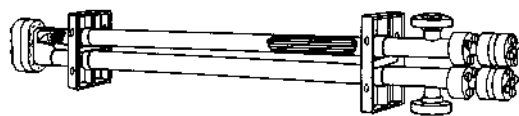
### FINNED DOUBLE PIPES

The design of the classical single-tube double-pipe heat exchanger is an exercise in pure longitudinal flow with the shellside and tubeside coefficients differing primarily due to variations in flow areas. Adding longitudinal fins gives the more common double-pipe configuration (Table 11-16). Increasing the number of tubes yields the *multitube* hairpin.

### MULTITUBE HAIRPINS

For years, the slightly higher mechanical-design complexity of the hairpin heat exchanger relegated it to only the smallest process requirements with shell sizes not exceeding 100 mm. In the early 1970s the maximum available sizes were increased to between 300 and 400 mm depending upon the manufacturer. At the present time, due to recent advances in design technology, hairpin exchangers are routinely produced in shell sizes between 50 (2 in) and 800 mm (30 in) for a wide range of pressures and temperatures and have been made in larger sizes as well. Table 11-17 gives common hairpin tube counts and areas for 19 mm (¾ in) O.D. tubes arranged on a 24 mm (1½ in) triangular tube layout.

The hairpin width and the centerline distance of the two legs (shells) of the hairpin heat exchanger are limited by the outside diameter of the closure flanges at the tubesheets. This diameter, in turn, is a function of the design pressures. As a general rule, for low-to-moderate design pressures (less than 15 bar), the center-to-center distance is approximately

**FIG. 11-42** Double-pipe-exchanger section with longitudinal fins. (Brown Fin-tube Co.)**TABLE 11-16 Double-Pipe Hairpin Section Data**

| Shell pipe O.D. |       | Inner pipe O.D. |       | Fin height |        | Fin count | Surface-area-per-unit length |          |
|-----------------|-------|-----------------|-------|------------|--------|-----------|------------------------------|----------|
| mm              | in    | mm              | in    | mm         | in     | (max)     | sq m/m                       | sq ft/ft |
| 60.33           | 2.375 | 25.4            | 1.000 | 12.7       | 0.50   | 24        | 0.692                        | 2.27     |
| 88.9            | 3.500 | 48.26           | 1.900 | 12.7       | 0.50   | 36        | 1.07                         | 3.51     |
| 114.3           | 4.500 | 48.26           | 1.900 | 25.4       | 1.00   | 36        | 1.98                         | 6.51     |
| 114.3           | 4.500 | 60.33           | 2.375 | 19.05      | 0.75   | 40        | 1.72                         | 5.63     |
| 114.3           | 4.500 | 73.03           | 2.875 | 12.70      | 0.50   | 48        | 1.45                         | 4.76     |
| 141.3           | 5.563 | 88.9            | 3.500 | 17.46      | 0.6875 | 56        | 2.24                         | 7.34     |
| 168.3           | 6.625 | 114.3           | 4.500 | 17.46      | 0.6875 | 72        | 2.88                         | 9.44     |

TABLE 11-17 Multitube Hairpin Section Data

| Size  | Shell O.D. |       | Shell thickness |        | Tube count | Surface area for 6.1 m (20 ft.) nominal length |       |
|-------|------------|-------|-----------------|--------|------------|------------------------------------------------|-------|
|       | mm         | in    | mm              | in     |            | sq m                                           | sq ft |
| 03-MT | 88.9       | 3.500 | 5.49            | 0.216  | 5          | 3.75                                           | 40.4  |
| 04-MT | 114.3      | 4.500 | 6.02            | 0.237  | 9          | 6.73                                           | 72.4  |
| 05-MT | 141.3      | 5.563 | 6.55            | 0.258  | 14         | 10.5                                           | 113.2 |
| 06-MT | 168.3      | 6.625 | 7.11            | 0.280  | 22         | 16.7                                           | 179.6 |
| 08-MT | 219.1      | 8.625 | 8.18            | 0.322  | 42         | 32.0                                           | 344.3 |
| 10-MT | 273.1      | 10.75 | 9.27            | 0.365  | 68         | 52.5                                           | 564.7 |
| 12-MT | 323.9      | 12.75 | 9.53            | 0.375  | 109        | 84.7                                           | 912.1 |
| 14-MT | 355.6      | 14.00 | 9.53            | 0.375  | 136        | 107.                                           | 1159. |
| 16-MT | 406.4      | 16.00 | 9.53            | 0.375  | 187        | 148.                                           | 1594. |
| 18-MT | 457.2      | 18.00 | 9.53            | 0.375  | 241        | 191.                                           | 2054. |
| 20-MT | 508.0      | 20.00 | 9.53            | 0.375  | 304        | 244.                                           | 2622. |
| 22-MT | 558.8      | 22.00 | 9.53            | 0.375  | 380        | 307.                                           | 3307. |
| 24-MT | 609.6      | 24.00 | 9.53            | 0.375  | 463        | 378.                                           | 4065. |
| 26-MT | 660.4      | 26.00 | 9.53            | 0.375  | 559        | 453.                                           | 4879. |
| 28-MT | 711.2      | 28.00 | 9.53            | 0.375  | 649        | 529.                                           | 5698. |
| 30-MT | 762.0      | 30.00 | 11.11           | 0.4375 | 752        | 630.                                           | 6776. |

1.5 to 1.8 times the shell outside diameter, with this ratio decreasing slightly for the larger sizes.

One interesting consequence of this fact is the inability to construct a hairpin tube bundle having the smallest radius bends common to a conventional U-tube, TEMA shell, and tube bundle. In fact, in the larger hairpin sizes the tubes might be better described as curved rather than bent. The smallest U-bend diameters are greater than the outside diameter of shells less than 300 mm in size. The U-bend diameters are greater than 300 mm in larger shells. As a general rule, mechanical tube cleaning around the radius of a U-bend may be accomplished with a flexible shaft-cleaning tool for bend diameters greater than ten times the tube's inside diameter. This permits the tool to pass around the curve of the tube bend without binding.

In all of these configurations, maintaining longitudinal flow on both the shellside and tubeside allows the decision for placement of a fluid

stream on either one side or the other to be based upon design efficiency (mass flow rates, fluid properties, pressure drops, and velocities) and not because there is any greater tendency to foul on one side than the other. Experience has shown that, in cases where fouling is influenced by flow velocity, overall fouling in tube bundles is less in properly designed longitudinal flow bundles where areas of low velocity can be avoided without flow-induced tube vibration.

This same freedom of stream choice is not as readily applied when a segmental baffle is used. In those designs, the baffle's creation of low velocities and stagnant flow areas on the outside of the bundle can result in increased shellside fouling at various locations of the bundle. The basis for choosing the stream side in those cases will be similar to the common shell and tube heat exchanger. At times a specific selection of stream side must be made regardless of tube-support mechanism in expectation of an unresolvable fouling problem. However, this is often the exception rather than the rule.

## DESIGN APPLICATIONS

One benefit of the hairpin exchanger is its ability to handle high tube-side pressures at a lower cost than other removable-bundle exchangers. This is due in part to the lack of pass partitions at the tubesheets which complicate the gasketing design process. Present mechanical design technology has allowed the building of dependable, removable-bundle, hairpin multitubes at tubeside pressures of 825 bar (12,000 psi).

The best known use of the hairpin is its operation in true counter-current flow which yields the most efficient design for processes that have a close temperature approach or temperature cross. However, maintaining countercurrent flow in a tubular heat exchanger usually implies one tube pass for each shell pass. As recently as 30 years ago, the lack of inexpensive, multiple-tube pass capability often diluted the advantages gained from countercurrent flow.

The early attempts to solve this problem led to investigations into the area of heat transfer augmentation. This familiarity with augmentation techniques inevitably led to improvements in the efficiency and capacity of the small heat exchangers. The result has been the application of the hairpin heat exchanger to the solution of unique process problems, such as dependable, once-through, convective boilers offering high-exit qualities, especially in cases of process-temperature crosses.

## AIR-COOLED HEAT EXCHANGERS

### AIR-COOLED HEAT EXCHANGERS

Atmospheric air has been used for many years to cool and condense fluids in areas of water scarcity. During the 1960s the use of air-cooled heat exchangers grew rapidly in the United States and elsewhere. In Europe, where seasonal variations in ambient temperatures are relatively small, air-cooled exchangers are used for the greater part of process cooling. In some new plants all cooling is done with air. Increased use of air-cooled heat exchangers has resulted from lack of available water, significant increases in water costs, and concern for water pollution.

Air-cooled heat exchangers include a tube bundle, which generally has spiral-wound fins upon the tubes, and a fan, which moves air across the tubes and is provided with a driver. Electric motors are the most commonly used drivers; typical drive arrangements require a V belt or a direct right-angle gear. A plenum and structural supports are basic components. Louvers are often used.

A bay generally has two tube bundles installed in parallel. These may be in the same or different services. Each bay is usually served by two (or more) fans and is furnished with a structure, a plenum, and other attendant equipment.

The location of air-cooled heat exchangers must consider the large space requirements and the possible recirculation of heated air because of the effect of prevailing winds upon buildings, fired heaters, towers, various items of equipment, and other air-cooled exchangers.

Inlet air temperature at the exchanger can be significantly higher than the ambient air temperature at a nearby weather station. See *Air-Cooled Heat Exchangers for General Refinery Services*, API Standard 661, 2d ed., January 1978, for information on refinery-process air-cooled heat exchangers.

**Forced and Induced Draft** The forced-draft unit, which is illustrated in Fig. 11-43 pushes air across the finned tube surface. The fans are located below the tube bundles. The induced-draft design has the fan above the bundle, and the air is pulled across the finned tube surface. In theory, a primary advantage of the forced-draft unit is that less power is required. This is true when the air-temperature rise exceeds 30°C (54°F).

Air-cooled heat exchangers are generally arranged in banks with several exchangers installed side by side. The height of the bundle aboveground must be one-half of the tube length to produce an inlet velocity equal to the face velocity. This requirement applies both to ground-mounted exchangers and to those pipe-rack-installed exchangers which have a fire deck above the pipe rack.

The forced-draft design offers better accessibility to the fan for on-stream maintenance and fan-blade adjustment. The design also provides a fan and V-belt assembly, which are not exposed to the hot-air stream that exits from the unit. Structural costs are less, and mechanical life is longer.

Induced-draft design provides more even distribution of air across the bundle, since air velocity approaching the bundle is relatively low.

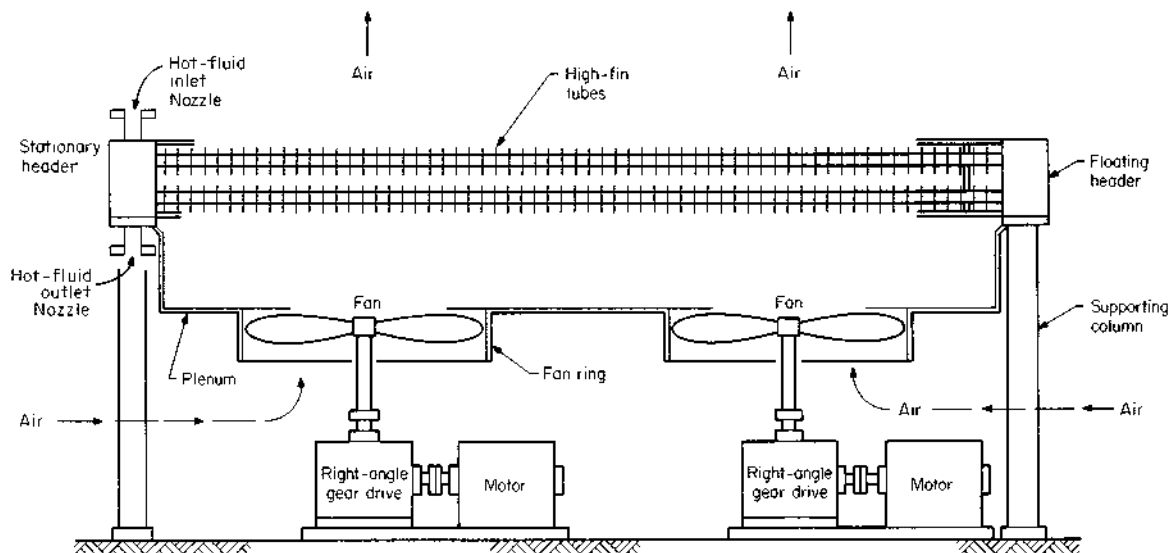


FIG. 11-43 Forced-draft air-cooled heat exchanger. [Chem. Eng., 114 (Mar. 27, 1978).]

This design is better suited for exchangers designed for a close approach of product outlet temperature to ambient-air temperature.

Induced-draft units are less likely to recirculate the hot exhaust air, since the exit air velocity is several times that of the forced-draft unit. Induced-draft design more readily permits the installation of the air-cooled equipment above other mechanical equipment such as pipe racks or shell-and-tube exchangers.

In a service in which sudden temperature change would cause upset and loss of product, the induced-draft unit gives more protection in that only a fraction of the surface (as compared with the forced-draft unit) is exposed to rainfall, sleet, or snow.

**Tube Bundle** The principal parts of the tube bundle are the finned tubes and the header. Most commonly used is the plug header, which is a welded box that is illustrated in Fig. 11-44. The finned tubes are described in a subsequent paragraph. The components of a tube bundle are identified in the figure.

The second most commonly used header is a cover-plate header. The cover plate is bolted to the top, bottom, and end plates of the

header. Removing the cover plate provides direct access to the tubes without the necessity of removing individual threaded plugs.

Other types of headers include the bonnet-type header, which is constructed similarly to the bonnet construction of shell-and-tube heat exchangers; manifold-type headers, which are made from pipe and have tubes welded into the manifold; and billet-type headers, made from a solid piece of material with machined channels for distributing the fluid. Serpentine-type tube bundles are sometimes used for very viscous fluids. A single continuous flow path through pipe is provided.

Tube bundles are designed to be rigid and self-contained and are mounted so that they expand independently of the supporting structure.

The face area of the tube bundle is its length times width. The net free area for air flow through the bundle is about 50 percent of the face area of the bundle.

The standard air face velocity (FV) is the velocity of standard air passing through the tube bundle and generally ranges from 1.5 to 3.6 m/s (300 to 700 ft/min).

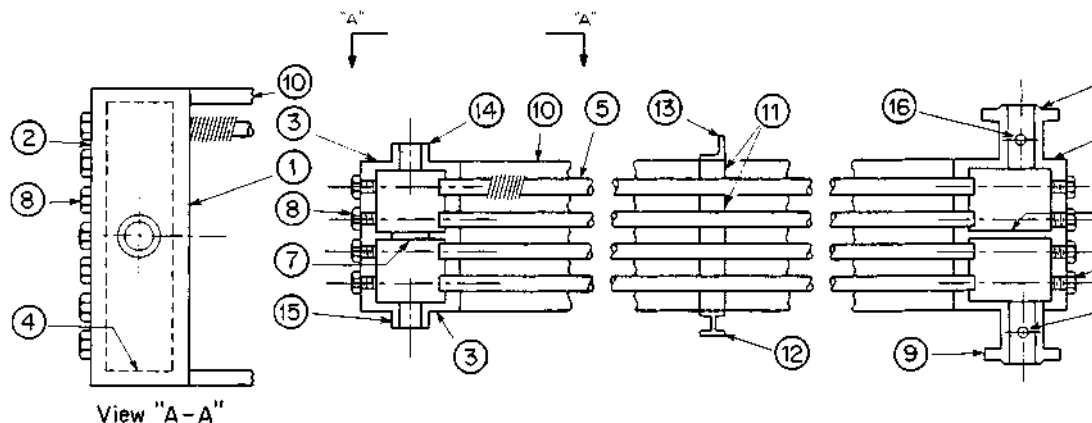


FIG. 11-44 Typical construction of a tube bundle with plug headers: (1) tube sheet; (2) plug sheet; (3) top and bottom plates; (4) end plate; (5) tube; (6) pass partition; (7) stiffener; (8) plug; (9) nozzle; (10) side frame; (11) tube spacer; (12) tube-support cross member; (13) tube keeper; (14) vent; (15) drain; (16) instrument connection. (API Standard 661.)

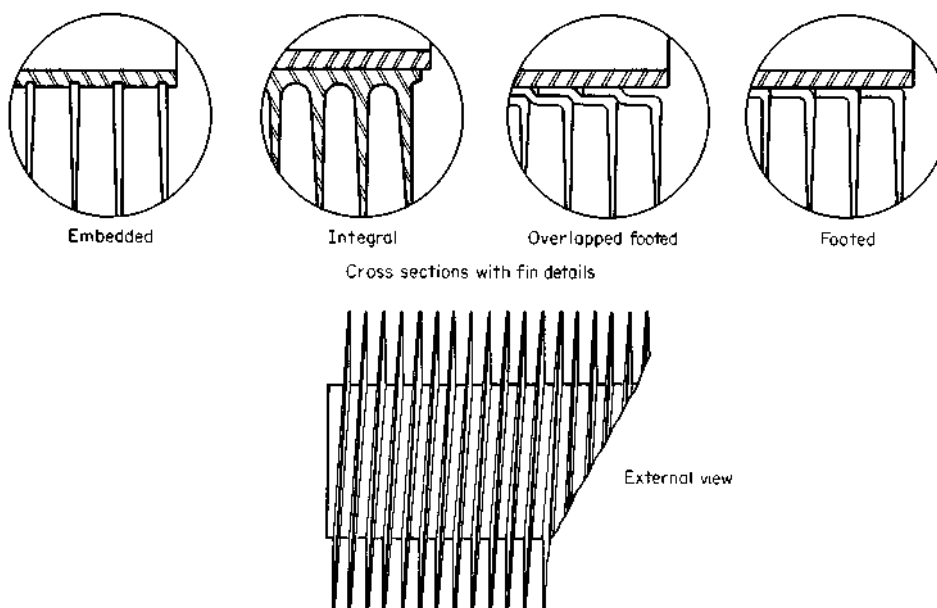


FIG. 11-45 Finned-tube construction.

**Tube** The 25.4-mm (1-in) outside-diameter tube is most commonly used. Fin heights vary from 12.7 to 15.9 mm (0.5 to 0.625 in), fin spacing from 3.6 to 2.3 mm (7 to 11 per linear inch), and tube triangular pitch from 50.8 to 63.5 mm (2.0 to 2.5 in). Ratio of extended surface to bare-tube outside surface varies from about 7 to 20. The 38-mm (1½-in) tube has been used for flue-gas and viscous-oil service. Tube size, fin heights, and fin spacing can be further varied.

Tube lengths vary and may be as great as 18.3 m (60 ft). When tube length exceeds 12.2 m (40 ft), three fans are generally installed in each bay. Frequently used tube lengths vary from 6.1 to 12.2 m (20 to 40 ft).

**Finned-Tube Construction** The following are descriptions of commonly used finned-tube constructions (Fig. 11-45).

1. **Embedded.** Rectangular-cross-section aluminum fin which is wrapped under tension and mechanically embedded in a groove  $0.25 \pm 0.05$  mm ( $0.010 \pm 0.002$  in) deep, spirally cut into the outside surface of a tube.

2. **Integral (or extruded).** An aluminum outer tube from which fins have been formed by extrusion, mechanically bonded to an inner tube or liner.

3. **Overlapped footed.** L-shaped aluminum fin wrapped under tension over the outside surface of a tube, with the tube fully covered by the overlapped feet under and between the fins.

4. **Footed.** L-shaped aluminum fin wrapped under tension over the outside surface of a tube with the tube fully covered by the feet between the fins.

5. **Bonded.** Tubes on which fins are bonded to the outside surface by hot-dip galvanizing, brazing, or welding.

Typical metal design temperatures for these finned-tube constructions are 399°C (750°F) embedded, 288°C (550°F) integral, 232°C (450°F) overlapped footed, and 177°C (350°F) footed.

Tube ends are left bare to permit insertion of the tubes into appropriate holes in the headers or tube sheets. Tube ends are usually roller-expanded into these tube holes.

**Fans** Axial-flow fans are large-volume, low-pressure devices. Fan diameters are selected to give velocity pressures of approximately 2.5 mm (0.1 in) of water. Total fan efficiency (fan, driver, and transmission device) is about 75 percent, and fan drives usually have a minimum of 95 percent mechanical efficiency.

Usually fans are provided with four or six blades. Larger fans may have more blades. Fan diameter is generally slightly less than the width of the bay.

At the fan-tip speeds required for economical performance, a large amount of noise is produced. The predominant source of noise is vortex shedding at the trailing edge of the fan blade. Noise control of air-cooled exchangers is required by the Occupational Safety and Health Act (OSHA). API Standard 661 (*Air-Cooled Heat Exchangers for General Refinery Services*, 2d ed., January 1978) has the purchaser specifying sound-pressure-level (SPL) values per fan at a location designated by the purchaser and also specifying sound-power-level (PWL) values per fan. These are designated at the following octave-band-center frequencies: 63, 125, 250, 1000, 2000, 4000, 8000, and also the dBA value (the dBA is a weighted single-value sound-pressure level).

Reducing the fan-tip speed results in a straight-line reduction in air flow while the noise level decreases. The API Standard limits fan-tip speed to 61 m/s (12,000 ft/min) for typical constructions. Fan-design changes which reduce noise include increasing the number of fan blades, increasing the width of the fan blades, and reducing the clearance between fan tip and fan ring.

Both the quantity of air and the developed static pressure of fans in air-cooled heat exchangers are lower than indicated by fan manufacturers' test data, which are applicable to testing-facility tolerances and not to heat-exchanger constructions.

The axial-flow fan is inherently a device for moving a consistent volume of air when blade setting and speed of rotation are constant. Variation in the amount of air flow can be obtained by adjusting the blade angle of the fan and the speed of rotation. The blade angle can be either (1) permanently fixed, (2) hand-adjustable, or (3) automatically adjusted. Air delivery and power are a direct function of blade pitch angle.

Fan mounting should provide a minimum of one-half to three-fourths diameter between fan and ground on a forced-draft heat exchanger and one-half diameter between tubes and fan on an induced-draft cooler.

Fan blades can be made of aluminum, molded plastic, laminated plastic, carbon steel, stainless steel, and Monel.

**Fan Drivers** Electric motors or steam turbines are most commonly used. These connect with gears or V belts. (Gas engines connected through gears and hydraulic motors either direct-connected or connected through gears are in use. Fans may be driven by a prime mover such as a compressor with a V-belt takeoff from the flywheel to a jack shaft and then through a gear or V belt to the fan. Direct motor drive is generally limited to small-diameter fans.)

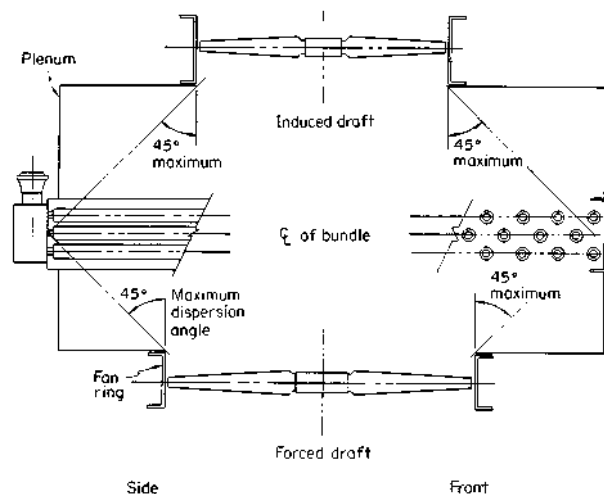


FIG. 11-46 Fan dispersion angle. (API Standard 661.)

V-belt drive assemblies are generally used with fans 3 m (10 ft) and less in diameter and motors of 22.4 kW (30 hp) and less.

Right-angle gear drive is preferred for fans over 3 m (10 ft) in diameter, for electric motors over 22.4 kW (30 hp), and with steam-turbine drives.

**Fan Ring and Plenum Chambers** The air must be distributed from the circular fan to the rectangular face of the tube bundle. The air velocity at the fan is between 3.8 and 10.2 m/s (750 and 2000 ft/in). The plenum-chamber depth (from fan to tube bundle) is dependent upon the fan dispersion angle (Fig. 11-46), which should have a maximum value of 45°.

The fan ring is made to commercial tolerances for the relatively large diameter fan. These tolerances are greater than those upon closely machined fan rings used for small-diameter laboratory-performance testing. Fan performance is directly affected by this increased clearance between the blade tip and the ring, and adequate provision in design must be made for the reduction in air flow. API Standard 661 requires that fan-tip clearance be a maximum of 0.5 percent of the fan diameter for diameters between 1.9 and 3.8 m (6.25 and 12.5 ft). Maximum clearance is 9.5 mm (3/8 in) for smaller fans and 19 mm (3/4 in) for larger fans.

The depth of the fan ring is critical. Worsham (ASME Pap. 59-PET-27, Petroleum Mechanical Engineering Conference, Houston, 1959) reports an increase in flow varying from 5 to 15 percent with the same power consumption when the depth of a fan ring was doubled. The percentage increase was proportional to the volume of air and static pressure against which the fan was operating.

When making a selection, the stall-out condition, which develops when the fan cannot produce any more air regardless of power input, should be considered.

**Air-Flow Control** Process operating requirements and weather conditions are considered in determining the method of controlling air flow. The most common methods include simple on-off control, on-off step control (in the case of multiple-driver units), two-speed-motor control, variable-speed drivers, controllable fan pitch, manually or automatically adjustable louvers, and air recirculation.

Winterization is the provision of design features, procedures, or systems for air-cooled heat exchangers to avoid process-fluid operating problems resulting from low-temperature inlet air. These include fluid freezing, pour point, wax formation, hydrate formation, laminar flow, and condensation at the dew point (which may initiate corrosion). Freezing points for some commonly encountered fluids in refinery service include: benzene, 5.6°C (42°F); *p*-xylene 15.5°C (55.9°F); cyclohexane, 6.6°C (43.8°F); phenol, 40.9°C (105.6°F); monoethanolamine, 10.3°C (50.5°F); and diethanolamine, 25.1°C (77.2°F). Water solutions

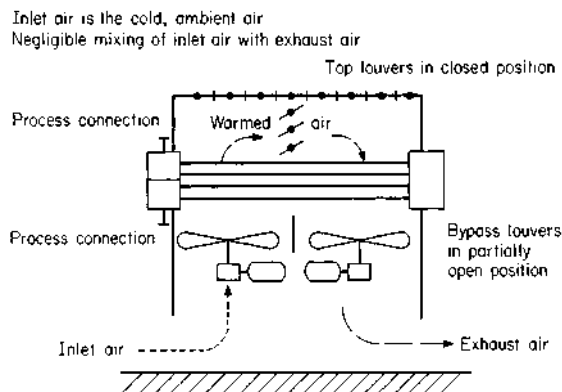


FIG. 11-47 Contained internal recirculation (with internal louvers). [*Hydrocarbon Process*, 59, 148-149 (October 1980).]

of these organic compounds are likely to freeze in air-cooled exchangers during winter service. Paraffinic and olefinic gases ( $C_1$  through  $C_4$ ) saturated with water vapor form hydrates when cooled. These hydrates are solid crystals which can collect and plug exchanger tubes.

Air-flow control in some services can prevent these problems. Cocurrent flow of air and process fluid during winter may be adequate to prevent problems. (Normal design has countercurrent flow of air and process fluid.) In some services when the hottest process fluid is in the bottom tubes, which are exposed to the lowest-temperature air, winterization problems may be eliminated.

Following are references which deal with problems in low-temperature environments: Brown and Benkley, "Heat Exchangers in Cold Service—A Contractor's View," *Chem. Eng. Prog.*, 70, 59-62 (July 1974); Franklin and Munn, "Problems with Heat Exchangers in Low Temperature Environments," *Chem. Eng. Prog.*, 70, 63-67 (July 1974); Newell, "Air-Cooled Heat Exchangers in Low Temperature Environments: A Critique," *Chem. Eng. Prog.*, 70, 86-91 (October 1974); Rubin, "Winterizing Air Cooled Heat Exchangers," *Hydrocarbon Process.*, 59, 147-149 (October 1980); Shipes, "Air-Cooled Heat Exchangers in Cold Climates," *Chem. Eng. Prog.*, 70, 53-58 (July 1974).

**Air Recirculation** Recirculation of air which has been heated as it crosses the tube bundle provides the best means of preventing operating problems due to low-temperature inlet air. Internal recirculation is the movement of air within a bay so that the heated air which has crossed the bundle is directed by a fan with reverse flow across another part of the bundle. Wind skirts and louvers are generally provided to minimize the entry of low-temperature air from the surroundings. Contained internal recirculation uses louvers within the bay to control the flow of warm air in the bay as illustrated in Fig. 11-47. Note that low-temperature inlet air has access to the tube bundle.

External recirculation is the movement of the heated air within the bay to an external duct, where this air mixes with inlet air, and the mixture serves as the cooling fluid within the bay. Inlet air does not have direct access to the tube bundle; an adequate mixing chamber is essential. Recirculation over the end of the exchanger is illustrated in Fig. 11-48. Over-the-side recirculation also is used. External recirculation systems maintain the desired low temperature of the air crossing the tube bundle.

**Trim Coolers** Conventional air-cooled heat exchangers can cool the process fluid to within 8.3°C (15°F) of the design dry-bulb temperature. When a lower process outlet temperature is required, a trim cooler is installed in series with the air-cooled heat exchanger. The water-cooled trim cooler can be designed for a 5.6 to 11.1°C (10 to 20°F) approach to the wet-bulb temperature (which in the United States is about 8.3°C (15°F) less than the dry-bulb temperature). In arid areas the difference between dry- and wet-bulb temperatures is much greater.

**Humidification Chambers** The air-cooled heat exchanger is provided with humidification chambers in which the air is cooled to a

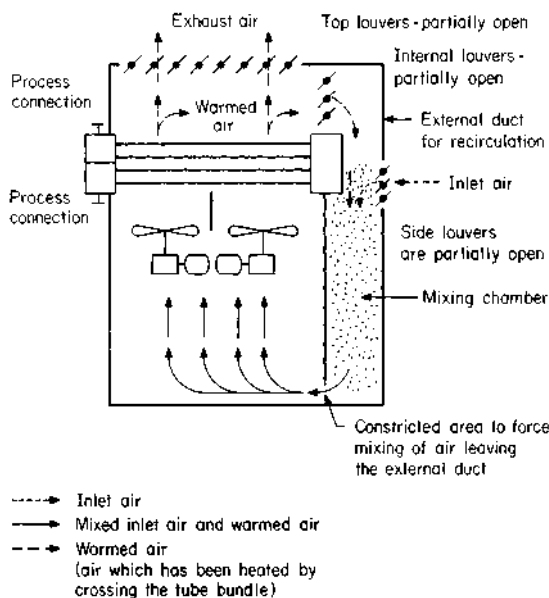


FIG. 11-48 External recirculation with adequate mixing chamber. [*Hydrocarbon Process*, 59, 148-149 (October 1980).]

close approach to the wet-bulb temperature before entering the finned-tube bundle of the heat exchanger.

**Evaporative Cooling** The process fluid can be cooled by using evaporative cooling with the sink temperature approaching the wet-bulb temperature.

**Steam Condensers** Air-cooled steam condensers have been fabricated with a single tube-side pass and several rows of tubes. The bottom row has a higher temperature difference than the top row, since the air has been heated as it crosses the rows of tubes. The bottom row condenses all the entering steam before the steam has traversed the length of the tube. The top row, with a lower temperature driving force, does not condense all the entering steam. At the exit header, uncondensed steam flows from the top row into the bottom row. Since noncondensable gases are always present in steam, these accumulate within the bottom row because steam is entering from both ends of the tube. Performance suffers.

Various solutions have been used. These include orifices to regulate the flow into each tube, a "blow-through steam" technique with a vent condenser, complete separation of each row of tubes, and inclined tubes.

**Air-Cooled Overhead Condensers** Air-cooled overhead condensers (AOC) have been designed and installed above distillation columns as integral parts of distillation systems. The condensers generally have inclined tubes, with air flow over the finned surfaces induced by a fan. Prevailing wind affects both structural design and performance.

AOC provide the additional advantages of reducing ground-space requirements and piping and pumping requirements and of providing smoother column operation.

The downflow condenser is used mainly for nonisothermal condensation. Vapors enter through a header at the top and flow downward. The reflux condenser is used for isothermal and small-temperature-change conditions. Vapors enter at the bottom of the tubes.

AOC usage first developed in Europe but became more prevalent in the United States during the 1960s. A state-of-the-art article was published by Dehne [*Chem. Eng. Prog.*, 64, 51 (July 1969)].

**Air-Cooled Heat-Exchanger Costs** The cost data that appear in Table 11-18 are unchanged from those published in the 1963 edition of this *Handbook*. In 1969 Guthrie [*Chem. Eng.*, 75, 114 (Mar. 24, 1969)] presented cost data for field-erected air-cooled

TABLE 11-18 Air-Cooled Heat-Exchanger Costs (1970)

| Surface (bare tube), sq. ft.                     | 500  | 1000 | 2000 | 3000 | 5000 |
|--------------------------------------------------|------|------|------|------|------|
| Cost for 12-row-deep bundle, dollars/square foot | 9.0  | 7.6  | 6.8  | 5.7  | 5.3  |
| Factor for bundle depth:                         |      |      |      |      |      |
| 6 rows                                           | 1.07 | 1.07 | 1.07 | 1.12 | 1.12 |
| 4 rows                                           | 1.2  | 1.2  | 1.2  | 1.3  | 1.3  |
| 3 rows                                           | 1.25 | 1.25 | 1.25 | 1.5  | 1.5  |

Base: Bare-tube external surface 1 in. o.d. by 12 B.W.G. by 24 ft. 0 in. steel tube with 8 aluminum fins per inch  $\frac{5}{8}$ -in. high. Steel headers. 150 lb./sq. in. design pressure. V-belt drive and explosion-proof motor. Bare-tube surface 0.262 sq. ft./ft. Fin-tube surface/bare-tube surface ratio is 16.9.

Factors: 20 ft. tube length 1.05  
30 ft. tube length 0.95  
18 B.W.G. admiralty tube 1.04  
16 B.W.G. admiralty tube 1.12

NOTE: To convert feet to meters, multiply by 0.3048; to convert square feet to square meters, multiply by 0.0929; and to convert inches to millimeters, multiply by 25.4.

exchangers. These costs are only 25 percent greater than those of Table 11-18 and include the costs of steel stairways, indirect subcontractor charges, and field-erection charges. Since minimal field costs would be this high (i.e., 25 percent of purchase price), the basic costs appear to be unchanged. (Guthrie indicated a cost band of plus or minus 25 percent.) Preliminary design and cost estimating of air-cooled heat exchangers have been discussed by J. E. Lerner [*Simplified Air Cooler Estimating*, *Hydrocarbon Process*, 52, 93-100 (February 1972)].

### Design Considerations

1. *Design dry-bulb temperature.* The typically selected value is the temperature which is equalled or exceeded  $2\frac{1}{2}$  percent of the time during the warmest consecutive 4 months. Since air temperatures at industrial sites are frequently higher than those used for these weather-data reports, it is good practice to add 1 to 3°C (2 to 6°F) to the tabulated value.

2. *Air recirculation.* Prevailing winds and the locations and elevations of buildings, equipment, fired heaters, etc., require consideration. All air-cooled heat exchangers in a bank are of one type, i.e., all forced-draft or all induced-draft. Banks of air-cooled exchangers must be placed far enough apart to minimize air recirculation.

3. *Wintertime operations.* In addition to the previously discussed problems of winterization, provision must be made for heavy rain, strong winds, freezing of moisture upon the fins, etc.

4. *Noise.* Two identical fans have a noise level 3 dBA higher than one fan, while eight identical fans have a noise level 9 dBA higher than a single fan. Noise level at the plant site is affected by the exchanger position, the reflective surfaces near the fan, the hardness of these surfaces, and noise from adjacent equipment. The extensive use of air-cooled heat exchangers contributes significantly to plant noise level.

5. *Ground area and space requirements.* Comparisons of the overall space requirements for plants using air cooling versus water cooling are not consistent. Some air-cooled units are installed above other equipment—pipe racks, shell-and-tube exchangers, etc. Some plants avoid such installations because of safety considerations, as discussed later.

6. *Safety.* Leaks in air-cooled units are directly to the atmosphere and can cause fire hazards or toxic-fume hazards. However, the large air flow through an air-cooled exchanger greatly reduces any concentration of toxic fluids. Segal [*Pet. Refiner*, 38, 106 (April 1959)] reports that air-fin coolers "are not located over pumps, compressors, electrical switchgear, control houses and, in general, the amount of equipment such as drums and shell-and-tube exchangers located beneath them are minimized."

Pipe-rack-mounted air-cooled heat exchangers with flammable fluids generally have concrete fire decks which isolate the exchangers from the piping.

7. *Atmospheric corrosion.* Air-cooled heat exchangers should not be located where corrosive vapors and fumes from vent stacks will pass through them.

8. *Air-side fouling.* Air-side fouling is generally negligible.
9. *Process-side cleaning.* Either chemical or mechanical cleaning on the inside of the tubes can readily be accomplished.
10. *Process-side design pressure.* The high-pressure process fluid is always in the tubes. Tube-side headers are relatively small as compared with water-cooled units when the high pressure is generally on the shell side. High-pressure design of rectangular headers is complicated. The plug-type header is normally used for design gauge pressures to 13,790 kPa (2000 lbf/in<sup>2</sup>) and has been used to 62,000 kPa (9000 lbf/in<sup>2</sup>). The use of threaded plugs at these pressures creates problems. Removable cover plate headers are generally limited to gauge pressures of 2068 kPa (300 lbf/in<sup>2</sup>). The expensive billet-type header is used for high-pressure service.
11. *Bond resistance.* Vibration and thermal cycling affect the bond resistance of the various types of tubes in different manners and thus affect the amount of heat transfer through the fin tube.
12. *Approach temperature.* The approach temperature, which is the difference between the process-fluid outlet temperature and the design dry-bulb air temperature, has a practical minimum of 8 to 14°C (15 to 25°F). When a lower process-fluid outlet temperature is required, an air-humidification chamber can be provided to reduce the inlet air temperature toward the wet-bulb temperature. A 5.6°C (10°F) approach is feasible. Since typical summer wet-bulb design temperatures in the United States are 8.3°C (15°F) lower than dry-bulb temperatures, the outlet process-fluid temperature can be 3°C (5°F) below the dry-bulb temperature.

13. *Mean-temperature-difference (MTD) correction factor.* When the outlet temperatures of both fluids are identical, the MTD correction factor for a 1:2 shell-and-tube exchanger (one pass shell side, two or more passes tube side) is approximately 0.8. For a single-pass air-cooled heat exchanger the factor is 0.91. A two-pass exchanger has a factor of 0.96, while a three-pass exchanger has a factor of 0.99 when passes are arranged for counterflow.

14. *Maintenance cost.* Maintenance for air-cooled equipment as compared with shell-and-tube coolers (complete with cooling-tower costs) indicates that air-cooling maintenance costs are approximately 0.3 to 0.5 those for water-cooled equipment.

15. *Operating costs.* Power requirements for air-cooled heat exchangers can be lower than at the summer design condition provided that an adequate means of air-flow control is used. The annual power requirement for an exchanger is a function of the means of air-flow control, the exchanger service, the air-temperature rise, and the approach temperature.

When the mean annual temperature is 16.7°C (30°F) lower than the design dry-bulb temperature and when both fans in a bay have automatically controllable pitch of fan blades, annual power required has been found to be 22, 36, and 54 percent respectively of that needed at the design condition for three process services [Frank L. Rubin, "Power Requirements Are Lower for Air-Cooled Heat Exchangers with AV Fans," *Oil Gas J.*, 165-167 (Oct. 11, 1982)]. Alternatively, when fans have two-speed motors, these deliver one-half of the design flow of air at half speed and use only one-eighth of the power of the full-speed condition.

## COMPACT AND NONTUBULAR HEAT EXCHANGERS

### COMPACT HEAT EXCHANGERS

With equipment costs rising and limited available plot space, compact heat exchangers are gaining a larger portion of the heat exchange market. Numerous types use special enhancement techniques to achieve the required heat transfer in smaller plot areas and, in many cases, less initial investment. As with all items that afford a benefit there is a series of restrictions that limit the effectiveness or application of these special heat exchanger products. In most products discussed some of these considerations are presented, but a thorough review with reputable suppliers of these products is the only positive way to select a compact heat exchanger. The following guidelines will assist in prequalifying one of these.

### PLATE-AND-FRAME EXCHANGERS

There are two major types gasketed and welded-plate heat exchangers. Each shall be discussed individually.

#### GASKETED-PLATE EXCHANGERS (G. PHE)

**Description** This type is the fastest growing of the compact exchangers and the most recognized (see Fig. 11-49). A series of corrugated alloy material channel plates, bounded by elastomeric gaskets are hung off and guided by longitudinal carrying bars, then compressed by large-diameter tightening bolts between two pressure retaining frame plates (cover plates). The frame and channel plates have portholes which allow the process fluids to enter alternating flow passages (the space between two adjacent-channel plates). Gaskets around the periphery of the channel plate prevent leakage to the atmosphere and also prevent process fluids from coming in contact with the frame plates. No interfluid leakage is possible in the port area due to a dual-gasket seal.

The frame plates are typically epoxy-painted carbon-steel material and can be designed per most pressure vessel codes. Design limitations are in the Table 11-19. The channel plates are always an alloy material with 304SS as a minimum (see Table 11-19 for other materials).

Channel plates are typically 0.4 to 0.8 mm thick and have corrugation depths of 2 to 10 mm. Special Wide Gap (WG PHE) plates are available, in limited sizes, for slurry applications with depths of approximately 16 mm. The channel plates are compressed to achieve metal-to-metal contact for pressure-retaining integrity. These narrow gaps and high number of contact points which change fluid flow direction, combine to create a very high turbulence between the plates. This means high individual-heat-transfer coefficients (up to 14,200 W/m<sup>2</sup> °C), but also very high pressure drops per length as well. To compensate, the channel plate lengths are usually short, most under 2 and few over 3 meters in length. In general, the same pressure drops as conventional exchangers are used without loss of the enhanced heat transfer.

Expansion of the initial unit is easily performed in the field without special considerations. The original frame length typically has an additional capacity of 15-20 percent more channel plates (i.e., surface area). In fact, if a known future capacity is available during fabrication stages, a longer carrying bar could be installed, and later, increasing the surface area would be easily handled. When the expansion is needed, simply untighten the carrying bolts, pull back the frame plate, add the additional channel plates, and tighten the frame plate.

**Applications** Most PHE applications are liquid-liquid services but there are numerous steam heater and evaporator uses from their heritage in the food industry. Industrial users typically have chevron style channel plates while some food applications are washboard style.

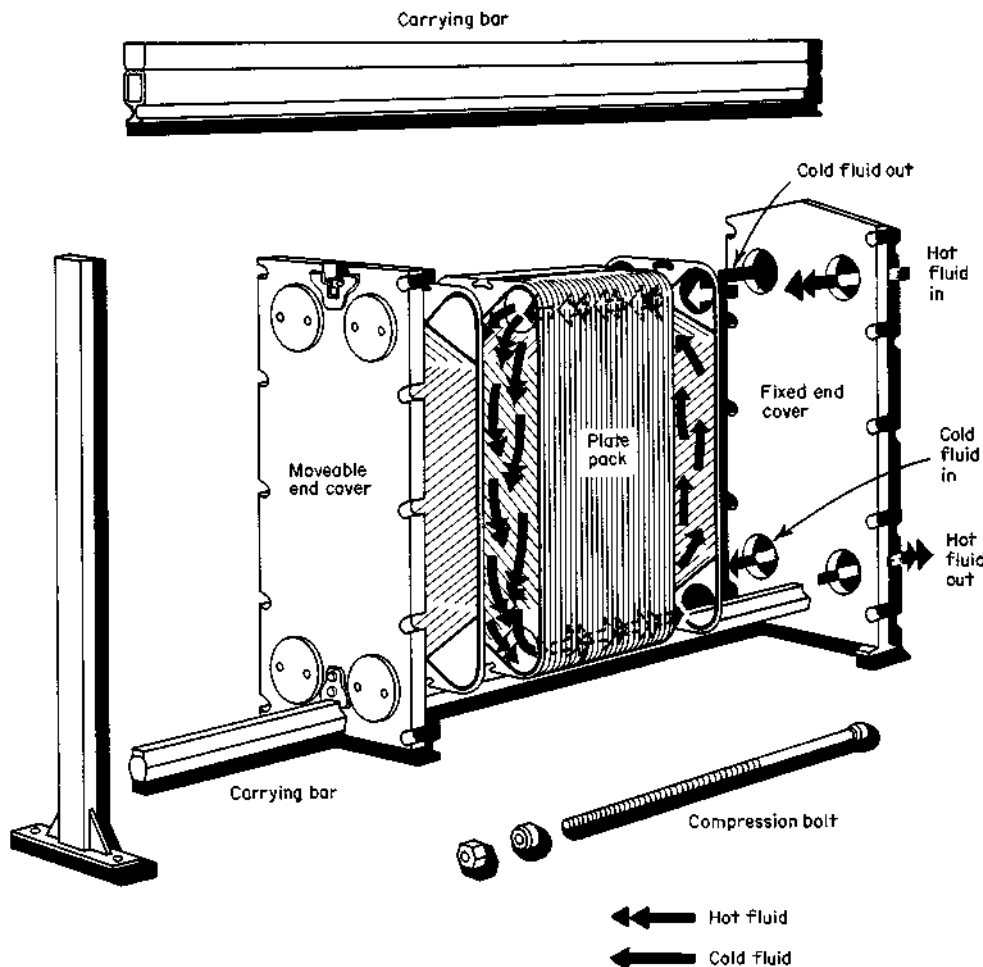
Fine particulate slurries in concentrations up to 70 percent by weight are possible with standard channel spacings. Wide-gap units are used with larger particle sizes. Typical particle size should not exceed 75 percent of the single plate (not total channel) gap.

Close temperature approaches and tight temperature control possible with PHE's and the ability to sanitize the entire heat transfer surface easily were a major benefit in the food industry.

Multiple services in a single frame are possible.

Gasket selection is one of the most critical and limiting factors in PHE usage. Table 11-20 gives some guidelines for fluid compatibility. Even trace fluid components need to be considered. The higher the operating temperature and pressure, the shorter the anticipated gasket life. *Always* consult the supplier on gasket selection and obtain an estimated or guaranteed lifetime.





**FIG. 11-49** Plate-and-frame heat exchanger. Hot fluid flows down between alternate plates, and cold fluid flows up between alternate plates. (Thermal Division, Alfa-Laval, Inc.)

The major applications are, but not limited to, as follows:

|                                |                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------|
| Temperature cross applications | (lean/rich solvent)                                                                   |
| Close approaches               | (fresh water/seawater)                                                                |
| Viscous fluids                 | (emulsions)                                                                           |
| Sterilized surface required    | (food, pharmaceutical)                                                                |
| Polished surface required      | (latex, pharmaceutical)                                                               |
| Future expansion required      |                                                                                       |
| Space restrictions             |                                                                                       |
| Barrier coolant services       | (closed-loop coolers)                                                                 |
| Slurry applications            | (TiO <sub>2</sub> , Kaolin, precipitated calcium carbonate, and beet sugar raw juice) |

**Design** Plate exchangers are becoming so commonplace that there is now an API 662 document available for the specification of these products. In addition, commercial computer programs are available from HTRI among others. Standard channel-plate designs, unique to each manufacturer, are developed with limited modifications of each plates' corrugation depths and included angles. Manufacturers combine their different style plates to custom-fit each service. Due to the possible combinations, it is impossible to present a way to exactly size PHEs. However, it is possible to estimate areas for new units and to predict performance of existing

units with different conditions (chevron-type channel plates are presented).

The fixed length and limited corrugation included angles on channel plates makes the NTU method of sizing practical. (Waterlike fluids are assumed for the following examples).

$$NTU = \frac{\Delta t \text{ of either side}}{LMTD} \quad (11-76)$$

Most plates have NTU values of 0.5 to 4.0, with 2.0 to 3.0 as the most common, (multipass shell and tube exchangers are typically less than 0.75). The more closely the fluid profile matches that of the channel plate, the smaller the required surface area. Attempting to increase the service NTU beyond the plate's NTU capability causes oversurfacing (inefficiency).

True sizing from scratch is impractical since a pressure balance on a channel-to-channel basis, from channel closest to inlet to furthest, must be achieved and when mixed plate angles are used; this is quite a challenge. Computer sizing is not just a benefit, it is a necessity for supplier's selection. Averaging methods are recommended to perform any sizing calculations.

From the APV heat-transfer handbook—*Design & Application of Paraflow-Plate Heat Exchangers* and J. Marriott's article, "Where and How To Use Plate Heat Exchangers," Chemical Engineering, April 5,

TABLE 11-19 Compact Exchanger Applications Guide

| Design conditions                | G. PHE | W. PHE  | WG. PHE | BHE  | DBL  | MLT  | STE     | CP      | SHE     | THE  |
|----------------------------------|--------|---------|---------|------|------|------|---------|---------|---------|------|
| Design temperature °C            | 165    | 150     | 150     | 185  | +500 | +500 | +500    | 450     | +400    | +500 |
| Minimum metal temp °C            | -30    | -30     | -30     | -160 | -160 | -160 | -160    | -160    | -160    | -160 |
| Design pressure MPa              | 2.5    | 2.5     | 0.7     | 3.1  | +20  | +20  | +20     | 3.1     | 2.0     | +20  |
| Inspect for leakage              | Yes    | Partial | Yes     | No   | Yes  | Yes  | Yes     | Partial | Yes     | Yes  |
| Mechanical cleaning              | Yes    | Yes/no  | Yes     | No   | Yes  | Yes  | Yes/no  | Yes     | Yes     | Yes  |
| Chemical cleaning                | Yes    | Yes     | Yes     | Yes  | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Expansion capability             | Yes    | Yes     | Yes     | No   | No   | No   | No      | No      | No      | No   |
| Repair                           | Yes    | Yes/no  | Yes     | No   | Yes  | Yes  | Partial | Partial | Partial | Yes  |
| Temperature cross                | Yes    | Yes     | Yes     | Yes  | Yes  | Yes  | Yes     | Yes     | Yes     | No°  |
| Surface area/unit m <sup>2</sup> | 1850   | 900     | 250     | 50   | 10   | 150  | 60      | 275     | 450     | High |
| Holdup volume                    | Low    | Low     | Low     | Low  | Med  | Med  | Low     | Low     | Med     | High |
| Materials                        | G. PHE | W. PHE  | WG. PHE | BHE  | DBL  | MLT  | STE     | CP      | SHE     | THE  |
| Mild steel                       | No     | No      | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Stainless                        | Yes    | Yes     | Yes     | Yes  | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Titanium                         | Yes    | Yes     | Yes     | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Hastalloy                        | Yes    | Yes     | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Nickel                           | Yes    | Yes     | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Alloy 20                         | Yes    | Yes     | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Incoloy 825                      | Yes    | Yes     | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Monel                            | Yes    | Yes     | No      | No   | Yes  | Yes  | Yes     | Yes     | Yes     | Yes  |
| Impervious graphite              | Yes    | No      | No      | No   | No   | No   | No      | No      | No      | Yes  |
| Service                          | G. PHE | W. PHE  | WG. PHE | BHE  | DBL  | MLT  | STE     | CP      | SHE     | THE  |
| Clean fluids                     | A      | A       | A       | A    | A    | A    | A       | A       | A       | A    |
| Gasket incompatibility           | D      | A/D     | D       | A    | A    | A    | A       | A       | A       | A    |
| Medium viscosity                 | A/B    | A/B     | A/B     | B    | A    | A    | A/B     | A/B     | A       | A    |
| High viscosity                   | A/B    | A/B     | A/B     | D    | A    | A    | A/B     | A/B     | A       | A    |
| Slurries & pulp (fine)           | B/D    | D       | A/B     | C    | A    | A/B  | C       | B       | A       | A/D  |
| Slurries & pulp (coarse)         | D      | D       | B       | D    | A    | B/C  | D       | B       | A       | A/D  |
| Refrigerants                     | D      | A       | D       | A    | A    | A    | B/C     | A       | A       | A    |
| Thermal fluids                   | D      | A/B     | D       | A/B  | A    | A    | C       | A       | A       | A    |
| Vent condensers                  | D      | D       | D       | D    | A/D  | A    | A       | B/C     | A       | A    |
| Process condenser                | D      | C       | D       | D    | A/D  | A    | A       | B/C     | B       | A    |
| Vacuum reboil/cond               | D      | D       | B       | D    | A/D  | B    | A       | B/C     | B       | A/C  |
| Evaporator                       | D      | C       | C       | A    | B    | B    | A       | B/C     | C       | A    |
| Tight temp control               | A      | A       | A       | A    | A    | A    | A       | B       | A       | C    |
| High scaling                     | B      | B       | A       | D    | A    | A/B  | B/C     | B       | B       | A/D  |

°Multipass

Adapted from Alfa-Laval and Vicarb literature

A—Very good C—Fair

B—Good D—Poor

TABLE 11-20 Elastomer Selection Guide

|                           | Uses                                                                                                             | Avoid                                                                                                                                                               |
|---------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitrile (NBR)             | Oil resistant<br>Fat resistant<br>Food stuffs<br>Mineral oil<br>Water                                            | Oxidants<br>Acids<br>Aromatics<br>Alkalies<br>Alcohols                                                                                                              |
| Resin cured butyl (IIR)   | Acids<br>Lyes<br>Strong alkalies<br>Strong phosphoric acid<br>Dilute mineral acids<br>Ketones<br>Amines<br>Water | Fats and fatty acids<br>Petroleum oils<br>Chlorinated hydrocarbons<br>Liquids with dissolved chlorine<br>Mineral oil<br>Oxygen rich demin. water<br>Strong oxidants |
| Ethylene-propylene (EPDM) | Oxidizing agents<br>Dilute acids<br>Amines<br>Water<br>(Mostly any IIR fluid)                                    | Oils<br>Hot & conc. acids<br>Very strong oxidants<br>Fats & fatty acids<br>Chlorinated hydrocarbons                                                                 |
| Viton (FKM, FPM)          | Water<br>Petroleum oils<br>Many inorganic acids<br>(Most all NBR fluids)                                         | Amines<br>Ketones<br>Esters<br>Organic acids<br>Liquid ammonia                                                                                                      |

1971, there are the following equations for plate heat transfer.

$$Nu = \frac{hDe}{k} = 0.28 * (Re)^{0.65} * (Pr)^{0.4} \quad (11-77)$$

where  $De = 2 \times \text{depth of single-plate corrugation}$ 

$$G = \frac{W}{Np * w * De} \quad (11-78)$$

Width of the plate ( $w$ ) is measured from inside to inside of the channel gasket. If not available, use the tear-sheet drawing width and subtract two times the bolt diameter and subtract another 50 mm. For depth of corrugation ask supplier, or take the compressed plate pack dimension, divide by the number of plates and subtract the plate thickness from the result. The number of passages ( $Np$ ) is the number of plates minus 1 then divided by 2.

Typical overall coefficients to start a rough sizing are as below. Use these in conjunction with the NTU calculated for the process. The closer the NTU matches the plate (say between 2.0 and 3.0), the higher the range of listed coefficients can be used. The narrower (smaller) the depth of corrugation, the higher the coefficient (and pressure drop), but also the lower the ability to carry through any particulate.

|                |                                 |
|----------------|---------------------------------|
| Water-water    | 5700–7400 W/(m <sup>2</sup> °C) |
| Steam-water    | 5700–7400 W/(m <sup>2</sup> °C) |
| Glycol/Glycol  | 2300–4000 W/(m <sup>2</sup> °C) |
| Amine/Amine    | 3400–5000 W/(m <sup>2</sup> °C) |
| Crude/Emulsion | 400–1700 W/(m <sup>2</sup> °C)  |

Pressure drops typically can match conventional tubular exchangers. Again from the APV handbook an average correlation is as follows:

$$\Delta P = \frac{2fG^2L}{g\rho De} \quad (11-79)$$

where  $f = 2.5 (GDe/\mu)^{-0.3}$   
 $g$  = gravitational constant

Fouling factors are typically  $\frac{1}{10}$  of TEMA values or a percent over-surfacing of 10–20 percent is used (J. Kerner, "Sizing Plate Exchangers," *Chemical Engineering*, November 1993).

LMTD is calculated like a 1 pass-1 pass shell and tube with no  $F$  correction factor required in most cases.

Overall coefficients are determined like shell and tube exchangers; that is, sum all the resistances, then invert. The resistances include the hot-side coefficient, the cold-side coefficient, the fouling factor (usually only a total value not individual values per fluid side) and the wall resistance.

## WELDED- AND BRAZED-PLATE EXCHANGERS (W. PHE & BHE)

The title of this group of plate exchangers has been used for a great variety of designs for various applications from normal gasketed-plate exchanger services to air-preheater services on fired heaters or boilers. The intent here is to discuss more traditional heat-exchanger designs, not the heat-recovery designs on fired equipment flue-gas streams. Many similarities exist between these products but the manufacturing techniques are quite different due to the normal operating conditions these units experience.

To overcome the gasket limitations, PHE manufacturers have developed welded-plate exchangers. There are numerous approaches to this solution: weld plate pairs together with the other fluid-side conventionally gasketed, weld up both sides but use a horizontal stacking of plates method of assembly, entirely braze the plates together with copper or nickel brazing, diffusion bond then pressure form plates and bond etched, passage plates.

Most methods of welded-plate manufacturing do not allow for inspection of the heat-transfer surface, mechanical cleaning of that surface, and have limited ability to repair or plug off damage channels. Consider these limitations when the fluid is heavily fouling, has solids, or in general the repair or plugging ability for severe services.

One of the previous types has an additional consideration of the brazing material to consider for fluid compatibility. The brazing compound entirely coats both fluid's heat-transfer surfaces.

The second type, a Compabloc (CP) from Alpha-Laval Thermal, has the advantage of removable cover plates, similar to air-cooled exchanger headers, to observe both fluids surface area. The fluids flow at 90° angles to each other on a horizontal plane. LMTD correction factors approach 1.0 for Compabloc just like the other welded and gasketed PHEs. Hydroblasting of Compabloc surfaces is also possible. The Compabloc has higher operating conditions than PHE's or W-PHE.

The performances and estimating methods of welded PHEs match those of gasketed PHEs in most cases, but normally the Compabloc, with larger depth of corrugations, can be lower in overall coefficient. Some extensions of the design operating conditions are possible with welded PHEs, most notably is that cryogenic applications are possible. Pressure vessel code acceptance is available on most units.

## COMBINATION WELDED-PLATE EXCHANGERS

Plate exchangers are well known for their high efficiency but suffer from limitations on operating pressure. Several companies have rectified this limitation by placing the welded plate exchanger inside a pressure vessel to withstand the pressure. One popular application is the feed effluent exchange in a catalytic reforming plant for oil refineries. Large volumes of gases with some liquids require cross-exchange to feed a reactor system. Close temperature approaches and lower pressure drops are required. These combined units provide an economic alternative to shell-and-tube exchangers.

## SPIRAL-PLATE EXCHANGERS (SHE)

**Description** The *spiral-plate heat exchanger* (SHE) may be one exchanger selected primarily on its virtues and not on its initial cost. SHEs offer high reliability and on-line performance in many severely fouling services such as slurries.

The SHE is formed by rolling two strips of plate, with welded-on spacer studs, upon each other into clock-spring shape. This forms two passages. Passages are sealed off on one end of the SHE by welding a bar to the plates; hot and cold fluid passages are sealed off on opposite ends of the SHE. A single rectangular flow passage is now formed for each fluid, producing very high shear rates compared to tubular designs. Removable covers are provided on each end to access and clean the entire heat transfer surface. Pure countercurrent flow is achieved and LMTD correction factor is essentially = 1.0.

Since there are no dead spaces in a SHE, the helical flow pattern combines to entrain any solids and create high turbulence creating a self-cleaning flow passage.

There are no thermal-expansion problems in spirals. Since the center of the unit is not fixed, it can torque to relieve stress.

The SHE can be expensive when only one fluid requires a high-alloy material. Since the heat-transfer plate contacts both fluids, it is required to be fabricated out of the higher alloy. SHEs can be fabricated out of any material that can be cold-worked and welded.

The channel spacings can be different on each side to match the flow rates and pressure drops of the process design. The spacer studs are also adjusted in their pitch to match the fluid characteristics.

As the coiled plate spirals outward, the plate thickness increases from a minimum of 2 mm to a maximum (as required by pressure) up to 10 mm. This means relatively thick material separates the two fluids compared to tubing of conventional exchangers. Pressure vessel code conformance is a common request.

**Applications** The most common applications that fit SHE are slurries. The rectangular channel provides high shear and turbulence to sweep the surface clear of blockage and causes no distribution problems associated with other exchanger types. A localized restriction causes an increase in local velocity which aids in keeping the unit free flowing. Only fibers that are long and stringy cause SHE to have a blockage it cannot clear itself.

As an additional antifoulant measure, SHEs have been coated with a phenolic lining. This provides some degree of corrosion protection as well, but this is not guaranteed due to pinholes in the lining process.

There are three types of SHE to fit different applications:

**Type I** is the spiral-spiral flow pattern. It is used for all heating and cooling services and can accommodate temperature crosses such as lean/rich services in one unit. The removable covers on each end allow access to one side at a time to perform maintenance on that fluid side. Never remove a cover with one side under pressure as the unit will telescope out like a collapsible cup.

**Type II** units are the condenser and reboiler designs. One side is spiral flow and the other side is in cross flow. These SHEs provide very stable designs for vacuum condensing and reboiling services. A SHE can be fitted with special mounting connections for reflux-type vent-condenser applications. The vertically mounted SHE directly attaches on the column or tank.

**Type III** units are a combination of the Type I and Type II where part is in spiral flow and part is in cross flow. This SHE can condense and subcool in a single unit.

The unique channel arrangement has been used to provide on-line cleaning, by switching fluid sides to clean the fouling (caused by the fluid that previously flowed there) off the surface. Phosphoric acid coolers use pond water for cooling and both sides foul; water, as you expect, and phosphoric acid deposit crystals. By reversing the flow sides, the water dissolves the acid crystals and the acid clears up the organic fouling. SHEs are also used as oleum coolers, sludge coolers/heaters, slop oil heaters, and in other services where multiple-flow-passage designs have not performed well.

**Design** A thorough article by P. E. Minton of Union Carbide called "Designing Spiral-Plate Heat Exchangers," appeared in *Chemical Engineering*, May 4, 1970. It covers the design in detail. Also an

article in *Chemical Engineering Progress* titled "Applications of Spiral Plate Heat Exchangers" by A. Hargis, A. Beckman, and J. Loicano appeared in July 1967, provides formulae for heat-transfer and pressure-drop calculations.

Spacings are from 6.35 to 31.75 mm (in 6.35 mm increments) with 9.5 mm the most common. Stud densities are 60 × 60 to 110 × 110 mm, the former the most common. The width (measured to the spiral flow passage), is from 150 to 2500 mm (in 150 mm increments). By varying the spacing and the width, separately for each fluid, velocities can be maintained at optimum rates to reduce fouling tendencies or utilize the allowable pressure drop most effectively. Diameters can reach 1500 mm. The total surface areas exceed 465 sqm. Materials that work harder are not suitable for spirals since hot-forming is not possible and heat treatment after forming is impractical.

$$Nu = \frac{HDe}{k} = 0.0315 (Re)^{0.8} (Pr)^{0.25} (\mu/\mu_w)^{0.17} \quad (11-80)$$

where  $De = 2 \times \text{spacing}$   
Flow area = width × spacing

$$\Delta P = \frac{LV^2 p}{1.705E03} * 1.45 \quad (1.45 \text{ for } 60 \times 60 \text{ mm studs}) \quad (11-81)$$

LMTD and overall coefficient are calculated like in PHE section above.

### BRAZED-PLATE-FIN HEAT EXCHANGER

Brazed-aluminum-plate-fin heat exchangers (or core exchangers or cold boxes) as they are sometimes called, were first manufactured for the aircraft industry during World War II. In 1950, the first tonnage air-separation plant with these compact, lightweight, reversing heat exchangers began producing oxygen for a steel mill. Aluminum-plate-fin exchangers are used in the process and gas-separation industries, particularly for services below -45°C.

Core exchangers are made up of a stack of rectangular sheets of aluminum separated by a wavy, usually perforated, aluminum fin. Two ends are sealed off to form a passage (see Fig. 11-50). The layers have the wavy fins and sealed ends alternating at 90° to each. Aluminum half-pipe-type headers are attached to the open ends to route the fluids into the alternating passages. Fluids usually flow at this same 90° angle to each other. Variations in the fin height, number of passages, and the length and width of the prime sheet allow for the core exchanger to match the needs of the intended service.

Design conditions range in pressures from full vacuum to 96.5 bar g and in temperatures from -269°C to 200°C. This is accomplished meeting the quality standards of most pressure vessel codes.

**Design and Application** Brazed plate heat exchangers have two design standards that are available. One is ALPEMA, the Brazed Aluminum Plate-Fin Heat Exchanger Manufacturers' Association, and the other is the API 662 document for plate heat exchangers.

Applications are varied for this highly efficient, compact exchanger. Mainly it is seen in the cryogenic fluid services of air-separation plants, refrigeration trains like in ethylene plants, and in natural-gas processing plants. Fluids can be all vapor, liquid, condensing, or vaporizing. Multifluid exchangers and multiservice cores, that is one exchanger with up to 10 different fluids, are common for this type of product.

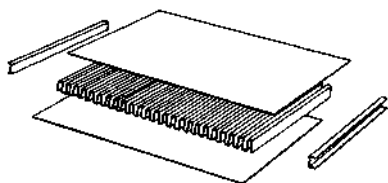


FIG. 11-50 Exploded view of a typical plate-fin arrangement. (Trane Co.)

Cold boxes are a group of cores assembled into a single structure or module, prepiped for minimum field connections. (Data obtained from ALTEC INTERNATIONAL, now Chart Industries. For detailed information refer to *GPSA Engineering Handbook* Section 9.)

### PLATE-FIN TUBULAR EXCHANGERS (PFE)

**Description** These shell and tube exchangers are designed to use a group of tightly spaced plate fins to increase the shellside heat transfer performance as fins do on double-pipe exchangers. In this design, a series of very thin plates (fins), usually of copper or aluminum material, are punched to the same pattern as the tube layout, spaced very close together, and mechanically bonded to the tube. Fin spacing is 315–785 FPM (Fins Per Meter) with 550 FPM most common. The fin thicknesses are 0.24 mm for aluminum and 0.19 mm for copper. Surface-area ratios over bare prime-tube units can be 20:1 to 30:1. The cost of the additional plate-fin material, without a reduction in shell diameter in many cases, and increased fabrication has to be offset by the total reduction of plot space and prime tube-surface area. The more costly the prime tube or plot space cost, the better the pay-out for this design. A rectangular tube layout is normally used, *no tubes in the window* (NTIW). The window area (where no tubes are) of the plate-fins are cut out. This causes a larger shell diameter for a given tube count compared to conventional tubular units. A dome area on top and bottom of the inside of the shell has been created for the fluid to flow along the tube length. In order to exit the unit the fluid must flow across the plate-finned tube bundle with extremely low pressure loss. The units from the outside and from the tubeside appear like any conventional shell and tube exchanger.

**Applications** Two principal applications are rotating equipment oil coolers and compressor inter- and after-coolers. Although seemingly different applications, both rely on the shellside finning to enhance the heat transfer of low heat-transfer characteristic fluids, viscous oils, and gases. By nature of the fluids and their applications, both are clean servicing. The tightly spaced fins would be a maintenance problem otherwise.

**Design** The economics usually work out in the favor of gas coolers when the centrifugal machine's flow rate reaches about 5000 scfm. The pressure loss can be kept to 7.0 kPa in most cases. When the ratio of  $A_{h1}$  to  $A_{h2}$  is 20:1, is another point to consider these plate-fin designs. Vibration is practically impossible with this design, and uses in reciprocating compressors are possible due to this.

Marine and hydraulic-oil coolers use these characteristics to enhance the coefficient of otherwise poorly performing fluids. The higher metallurgies in marine applications like 90/10 Cu-Ni afford the higher cost of plate-fin design to be offset by the less amount of alloy material being used. On small hydraulic coolers, these fins usually allow one to two size smaller coolers for the package and save skid space and initial cost.

Always check on metallurgy compatibility and cleanliness of the shellside fluid! (Data provided by Bos-Hatten and ITT-Standard.)

### PRINTED-CIRCUIT HEAT EXCHANGERS

These are a variation of the welded or brazed plate heat exchangers but using a chemical etching process to form the flow channels and diffusion bonding technique to secure the plates together. These units have the high heat-transfer characteristics and extended operating conditions that welded or brazed units have, but the diffusion process makes the bond the same strength as that of the prime plate material. The chemical etching, similar to that used in printed circuitry, allows greater flexibility in flow channel patterns than any other heat exchanger. This type of heat exchanger is perhaps the most compact design of all due to the infinite variations in passage size, layout, and direction.

Headers are welded on the core block to direct the fluids into the appropriate passages. The all-metal design allows very high operating conditions for both temperature and pressure. The diffusion bonding provides a near-homogeneous material for fluids that are corrosive or require high purity.

These exchangers can handle gases, liquids, and two-phase applications. It has the greatest potential in cryogenic, refrigeration, gas

processing, and corrosive chemical applications. Other applications are possible with the exception of fluids containing solids: the narrow passages, like most plate exchangers, are conducive to plugging.

### SPIRAL-TUBE EXCHANGERS (STE)

**Description** These exchangers are typically a series of stacked helical-coiled tubes connected to manifolds, then inserted into a casing or shell. They have many advantages like spiral-plate designs, such as avoiding differential expansion problems, acceleration effects of the helical flow increasing the heat transfer coefficient, and compactness of plot area. They are typically selected because of their economical design.

The most common form has both sides in helical flow patterns, pure countercurrent flow is followed and the LMTD correction factor approaches 1.0. Temperature crosses are possible in single units. Like the spiral-plate unit, different configurations are possible for special applications.

Tube material includes any that can be formed into a coil, but usually copper, copper alloys, and stainless steel are most common. The casing or shell material can be cast iron, cast steel, cast bronze, fabricated steel, stainless, and other high-alloy materials. Units are available with pressure vessel code conformance.

The data provided herein has been supplied by Graham Mfg. for their units called Heliflow.

**Applications** The common Heliflow applications are tank-vent condensers, sample coolers, pump-seal coolers, and steam-jet vacuum condensers. Instant water heaters, glycol/water services, and cryogenic vaporizers use the spiral tube's ability to reduce thermally induced stresses caused in these applications.

Many other applications are well suited for spiral tube units but many believe only small surface areas are possible with these units. Graham Mfg. states units are available to 60 m<sup>2</sup>. Their ability to polish the surfaces, double-wall the coil, use finned coil, and insert static mixers, among others configurations in design, make them quite flexible. Tubeside design pressures can be up to 69000 kPa. A cross-flow design on the external surface of the coil is particularly useful in steam-jet ejector condensing service. These Heliflow units, can be made very cost-effective, especially in small units. The main differences, compared to spiral plate, is that the tubeside cannot be cleaned except chemically and that multiple flow passages make tubeside slurry applications (or fouling) impractical.

**Design** The fluid flow is similar to the spiral-plate exchangers, but through parallel tube passages. Graham Mfg. has a liquid-liquid sizing pamphlet available from their local distributor. An article by M.A. Noble, J.S. Kamlani, and J.J. McKetta "Heat Transfer in Spiral Coils", was published in *Petroleum Engineer*, April 1952 p. 723, discussing sizing techniques.

The tubeside fluid must be clean or at least chemically cleanable. With a large number of tubes in the coil, cleaning of inside surfaces is not totally reliable. Fluids that attack stressed materials such as chlorides should be reviewed as to proper coil-material selection. Fluids that contain solids can be a problem due to erosion of relatively thin coil materials unlike the thick plates in spiral-plate units and multiple, parallel, fluid passages compared to a single passage in spiral-plate units.

### GRAPHITE HEAT EXCHANGERS

Impervious graphite exchangers now come in a variety of geometries to suit the particular requirements of the service. They include cubic block form, drilled cylinder block, shell and tube, and plate and frame.

**Description** Graphite is one of three crystalline forms of carbon. The other two are diamond and charcoal. Graphite has a hexagonal crystal structure, diamond is cubic, and charcoal is amorphous. Graphite is inert to most chemicals and resists corrosion attack. It is however porous and to be used, it must be impregnated with a resin sealer. Two main resins used are phenolic and PTFE with furan (one currently being phased out of production). Selection of resins include chemical compatibility, operating temperatures, and type of unit to be used. For proper selection, consult with a graphite supplier.

Shell-and-tube units in graphite were started by Karbate in 1939. The European market started using block design in the 1940s. Both technologies utilize the high thermal conductivity of the graphite material to compensate for the poor mechanical strength. The thicker materials needed to sustain pressure do not adversely impede the heat transfer. Maximum design pressures range from 0.35 to 1.0 kPa depending on type and size of exchanger. Design temperature is dependent on the fluids and resin selection, the maximum is 230 °C.

In all situations, the graphite heat transfer surface is contained within a metal structure or a shell (graphite lined on process side) to maintain the design pressure. For shell and tube units, the design is a packed floating tubesheet at both ends within a shell and channel. For stacked block design, the standardize blocks are glued together with special adhesives and compressed within a framework that includes manifold connections for each fluid. The cylindrical block unit is a combination of the above two with blocks glued together and surrounded by a pressure retaining shell. Pressure vessel code conformance of the units is possible due to the metallic components of these designs. Since welding of graphite is not possible, the selection and application of the adhesives used are critical to the proper operating of these units. Tube to tubesheet joints are glued since rolling of tubes into tubesheet is not possible. The packed channels and gasketed manifold connections are two areas of additional concern when selecting sealants for these units.

**Applications and Design** The major applications for these units are in the acid-related industries. Sulfuric, phosphoric, and hydrochloric acids require either very costly metals or impervious graphite. Usually graphite is the more cost-effective material to be used. Applications are increasing in the herbicide and pharmaceutical industries as new products with chlorine and fluorine compounds expand. Services are coolers, condensers, and evaporators, basically all services requiring this material. Types of units are shell-and-tube, block-type (circular and rectangular), and plate-and-frame-type exchangers. The design of the shell-and-tube units are the same as any but the design characteristics of tubes, spacing, and thickness are unique to the graphite design. The block and plate and frame also can be evaluated using techniques previously addressed but again, the unique characteristics of the graphite materials require input from a reputable supplier. Most designs will need the supplier to provide the most cost-effective design for the immediate and future operation of the exchangers. Also, consider the entire system design as some condensers and/or evaporators can be integral with their associated column.

### CASCADE COOLERS

Cascade coolers are a series of standard pipes, usually manifolded in parallel, and connected in series by vertically or horizontally oriented U-bends. Process fluid flows inside the pipe entering at the bottom and water trickles from the top downward over the external pipe surface. The water is collected from a trough under the pipe sections, cooled, and recirculated over the pipe sections. The pipe material can be any of the metallic and also glass, impervious graphite, and ceramics. The tubeside coefficient and pressure drop is as in any circular duct. The water coefficient (with Re number less than 2100) is calculated from the following equation by W.H. McAdams, T.B. Drew, and G.S. Bays Jr., from the ASME trans. **62**, 627-631 (1940).

$$h = 218 * (G'/D_o)^{1/3} \text{ (W/m}^2 \text{ °C)} \text{ (11-82)}$$

$$G' = m/(2L)$$

$$m = \text{water rate (kg/hr)}$$

$$L = \text{length of each pipe section (meter)}$$

$$D_o = \text{outside diameter of pipe (meter)}$$

LMTD corrections are per Fig. 11-4 *i* or *j* depending on U-bend orientation.

### BAYONET-TUBE EXCHANGERS

This type of exchanger gets its name from its design which is similar to a bayonet sword and its associated scabbard or sheath. The bayonet tube is a smaller-diameter tube inserted into a larger-diameter tube

that has been capped at one end. The fluid flow is typically entering the inner tube, exiting, hitting the cap of the larger tube, and returning the opposite direction in the annular area. The design eliminates any thermal expansion problems. It also creates a unique nonfreeze-type tube-side for steam heating of cryogenic fluids, the inner tube steam keeps the annulus condensate from freezing against the cold shellside fluid. This design can be expensive on a surface-area basis due to the need of a double channel design and only the outer tube surface is used to transfer heat. LMTD calculations for nonisothermal fluid are quite extensive and those applications are far too few to attempt to define it. The heat transfer is like the annular calculation of a double-pipe unit. The shellside is a conventional-baffled shell-and-tube design.

### ATMOSPHERIC SECTIONS

These consist of a rectangular bundle of tubes in similar fashion to air cooler bundles, placed just under the cooled water distribution section of a cooling tower. It, in essence, combines the exchanger and cooling tower into a single piece of equipment. This design is only practical for single-service cooler/condenser applications, and expansion capabilities are not provided. The process fluid flows inside the tubes and the cooling tower provides cool water that flows over the outside of the tube bundle. Water quality is critical for these applications to prevent fouling or corrosive attack on the outside of the tube surfaces and to prevent blockage of the spray nozzles. The initial and operating costs are lower than separate cooling tower and exchanger. Principal applications now are in the HVAC, Refrigeration and Industrial systems. Sometimes these are called "Wet Surface Air Coolers."

$$h = 1729 [(m^2/hr)/\text{face area } m^2]^{1/3} \quad (11-83)$$

### NONMETALLIC HEAT EXCHANGERS

Another growing field is that of nonmetallic heat exchanger designs which typically are of the shell and tube or coiled-tubing type. The graphite units were previously discussed but numerous other materials are available. The materials include Teflon, PVDF, glass, ceramic, and others as the need arises.

When using these types of products, consider the following topics and discuss the application openly with experienced suppliers.

1. The tube-to-tubesheet joint, how is it made? Many use "O" rings to add another material to the selection process. Preference should be given to a fusing technique of similar material.
2. What size tube or flow passage is available? Small tubes plug unless filtration is installed. Size of filtering is needed from the supplier.
3. These materials are very sensitive to temperature and pressure. Thermal or pressure shocks must be avoided.
4. Thermal conductivity of these materials is very low and affects the overall coefficient. When several materials are compatible, explore all of them, as final cost is not always the same as raw material costs.

### PVDF HEAT EXCHANGERS

These shell-and-tube-type exchangers are similar to the Teflon designs but have some mechanical advantages over Teflon units. First the tubes are available in 9.5 mm sizes which reduces the chances of plugging that are found in Teflon units with unfiltered fluids. Second, the material has higher strength even at lower temperatures almost double. Larger units are possible with PVDF materials.

Tube to tubesheet joints, a weakness of most nonmetallic units, are fused by special techniques that do not severely affect the chemical suitability of the unit. Some nonmetallics use Teflon or "O" rings that add an extra consideration to material selection.

The shell is usually a steel design and, like the graphite units before, can obtain pressure-vessel certification.

### CERAMIC HEAT EXCHANGERS

These include glass, silicon carbide, and similar variations. Even larger tubes are available in these materials, up to 19-mm diameter. They have high thermal conductivities and are usually very smooth surfaces to resist fouling. Very high material/fluid compatibility is seen for these products, not many fluids are excluded. Brittleness is a consideration of these materials and a complete discussion of the service with an experienced supplier is warranted. The major selection criteria to explore is the use of "O" rings and other associated joints at tubesheet. The shell is steel in most cases.

### TEFLON HEAT EXCHANGERS

Teflon tube shell-and-tube heat exchangers (Ametek) made with tubes of chemically inert Teflon fluorocarbon resin are available. The tubes are 0.25-in OD by 0.20-in ID, 0.175-in OD by 0.160-in ID, or 0.125-in OD by 0.100-in ID (in  $\times$  25.4 equal mm). The larger tubes are primarily used when pressure-drop limitations or particles reduce the effectiveness of smaller tubes. These heat exchangers generally operate at higher pressure drops than conventional units and are best suited for relatively clean fluids. Being chemically inert, the tubing has many applications in which other materials corrode. Fouling is negligible because of the antistick properties of Teflon.

The heat exchangers are of single-pass, countercurrent-flow design with removable tube bundles. Tube bundles are made of straight flexible tubes of Teflon joined together in integral honeycomb tube sheets. Baffles and O-ring gaskets are made of Teflon. Standard shell diameters are 102, 204, and 254 mm (4, 8, and 10 in). Tube counts range from 105 to 2000. Surface varies from 1.9 to 87 m<sup>2</sup> (20 to 940 ft<sup>2</sup>). Tube lengths vary from 0.9 to 4.9 m (3 to 16 ft). At 37.8°C (100°F) maximum operating gauge pressures are 690 kPa (100 lbf/in<sup>2</sup>) internal and 379 kPa (55 lbf/in<sup>2</sup>) external. At 149°C (300°F) the maximum pressures are 207 kPa (30 lbf/in<sup>2</sup>) internal and 124 kPa (18 lbf/in<sup>2</sup>) external.

## HEAT EXCHANGERS FOR SOLIDS

This section describes equipment for heat transfer to or from solids by the indirect mode. Such equipment is so constructed that the solids load (burden) is separated from the heat-carrier medium by a wall; the two phases are never in direct contact. Heat transfer is by conduction based on diffusion laws. Equipment in which the phases are in direct contact is covered in other sections of this *Handbook*, principally in Sec. 20.

Some of the devices covered here handle the solids burden in a static or laminar-flowing bed. Other devices can be considered as continuously agitated kettles in their heat-transfer aspect. For the latter, unit-area performance rates are higher.

Computational and graphical methods for predicting performance are given for both major heat-transfer aspects in Sec. 10. In solids heat processing with indirect equipment, the engineer should remember that the heat-transfer capability of the wall is many times that of the

solids burden. Hence the solids properties and bed geometry govern the rate of heat transfer. This is more fully explained earlier in this section. Only limited resultant (not predictive) and "experience" data are given here.

### EQUIPMENT FOR SOLIDIFICATION

A frequent operation in the chemical field is the removal of heat from a material in a molten state to effect its conversion to the solid state. When the operation is carried on batchwise, it is termed casting, but when done continuously, it is termed flaking. Because of rapid heat transfer and temperature variations, jacketed types are limited to an initial melt temperature of 232°C (450°F). Higher temperatures [to 316°C (600°F)] require extreme care in jacket design and cooling-liquid flow pattern. Best performance and greatest capacity are

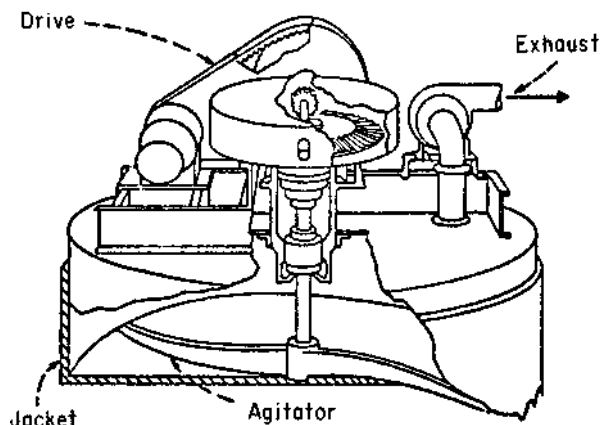


FIG. 11-51 Heat-transfer equipment for solidification (with agitation); agitated-pan type.

obtained by (1) holding precooling to the minimum and (2) optimizing the cake thickness. The latter cannot always be done from the heat-transfer standpoint, as size specifications for the end product may dictate thickness.

**Table Type** This is a simple flat metal sheet with slightly up-turned edges and jacketed on the underside for coolant flow. For many years this was the mainstay of food processors. Table types are still widely used when production is in small batches, when considerable batch-to-batch variation occurs, for pilot investigation, and when the cost of continuous devices is unjustifiable. Slab thicknesses are usually in the range of 13 to 25 mm ( $\frac{1}{2}$  to 1 in). These units are home-made, with no standards available. Initial cost is low, but operating labor is high.

**Agitated-Pan Type** A natural evolution from the table type is a circular flat surface with jacketing on the underside for coolant flow and the added feature of a stirring means to sweep over the heat-transfer surface. This device is the agitated-pan type (Fig. 11-51). It is a batch-operation device. Because of its age and versatility it still serves a variety of heat-transfer operations for the chemical-process industries. While the most prevalent designation is agitated-pan dryer (in this mode, the burden is heated rather than cooled), considerable use is made of it for solidification applications. In this field, it is particularly suitable for processing burdens that change phase (1)

slowly, by "thickening," (2) over a wide temperature range, (3) to an amorphous solid form, or (4) to a soft semigummy form (versus the usual hard crystalline structure).

The stirring produces the end product in the desired divided-solids form. Hence, it is frequently termed a "granulator" or a "crystallizer." A variety of factory-made sizes in various materials of construction are available. Initial cost is modest, while operating cost is rather high (as is true of all batch devices), but the ability to process "gummy" burdens and/or simultaneously effect two unit operations often yields an economical application.

**Vibratory Type** This construction (Fig. 11-52) takes advantage of the burden's special needs and the characteristic of vibratory actuation. A flammable burden requires the use of an inert atmosphere over it and a suitable nonhazardous fluid in the jacket. The vibratory action permits construction of rigid self-cleaning chambers with simple flexible connections. When solidification has been completed and vibrators started, the intense vibratory motion of the whole deck structure (as a rigid unit) breaks free the friable cake [up to 76 mm (3 in) thick], shatters it into lumps, and conveys it up over the dam to discharge. Heat-transfer performance is good, with overall coefficient  $U$  of about  $68 \text{ W/(m}^2\cdot^\circ\text{C)}$  [ $12 \text{ Btu/(h}\cdot\text{ft}^2\cdot^\circ\text{F)}$ ] and values of heat flux  $q$  in the order of  $11,670 \text{ W/m}^2$  [ $3700 \text{ Btu/(h}\cdot\text{ft}^2)$ ]. Application of timing-cycle controls and a surge hopper for the discharge solids facilitates automatic operation of the caster and continuous operation of subsequent equipment.

**Belt Types** The patented metal-belt type (Fig. 11-53a), termed the "water-bed" conveyor, features a thin wall, a well-agitated fluid side for a thin water film (there are no rigid welded jackets to fail), a stainless-steel or Swedish-iron conveyor belt "floated" on the water with the aid of guides, no removal knife, and cleanliness. It is mostly used for cake thicknesses of 3.2 to 15.9 mm ( $\frac{1}{8}$  to  $\frac{3}{8}$  in) at speeds up to 15 m/min (50 ft/min), with 45.7-m (150-ft) pulley centers common. For 25- to 32-mm (1- to  $1\frac{1}{4}$ -in) cake, another belt on top to give two-sided cooling is frequently used. Applications are in food operations for cooling to harden candies, cheeses, gelatins, margarines, gums, etc.; and in chemical operations for solidification of sulfur, greases, resins, soaps, waxes, chloride salts, and some insecticides. Heat transfer is good, with sulfur solidification showing values of  $q = 5800 \text{ W/m}^2$  [ $1850 \text{ Btu/(h}\cdot\text{ft}^2)$ ] and  $U = 96 \text{ W/(m}^2\cdot^\circ\text{C)}$  [ $17 \text{ Btu/(h}\cdot\text{ft}^2\cdot^\circ\text{F)}$ ] for a 7.9-mm ( $\frac{5}{16}$ -in) cake.

The submerged metal belt (Fig. 11-53b) is a special version of the metal belt to meet the peculiar handling properties of pitch in its solidification process. Although adhesive to a dry metal wall, pitch will not stick to the submerged wetted belt or rubber edge strips. Submergence helps to offset the very poor thermal conductivity through two-sided heat transfer.

A fairly recent application of the water-cooled metal belt to solidification duty is shown in Fig. 11-54. The operation is termed pastillizing

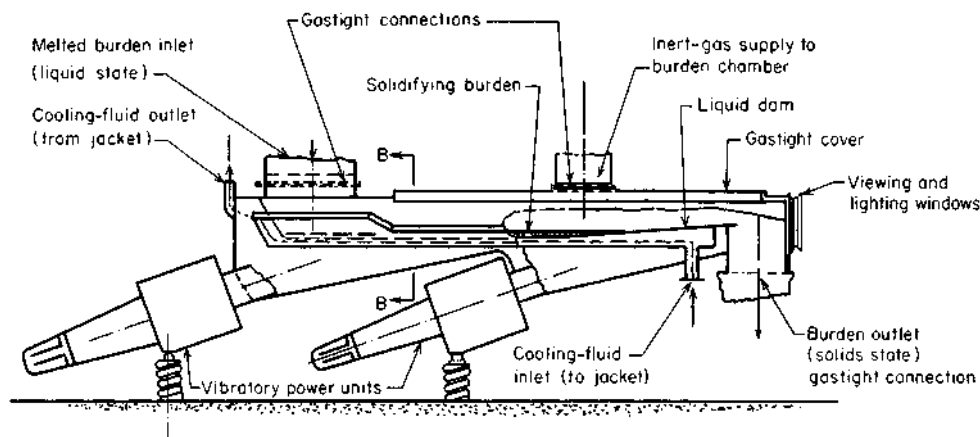


FIG. 11-52 Heat-transfer equipment for batch solidification; vibrating-conveyor type. (Courtesy of Jeffrey Mfg. Co.)

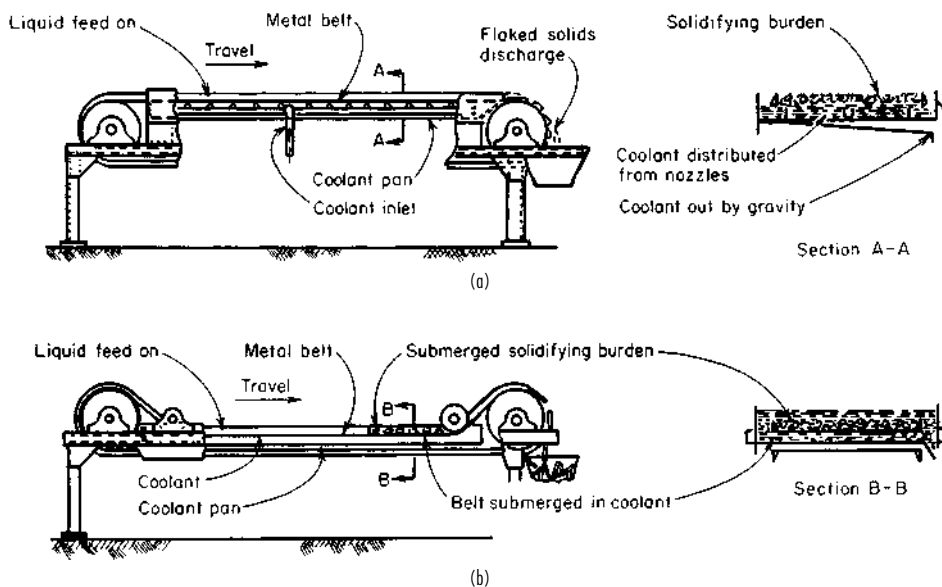


FIG. 11-53 Heat-transfer equipment for continuous solidification. (a) Cooled metal belt. (Courtesy of Sandvik, Inc.) (b) Submerged metal belt. (Courtesy of Sandvik, Inc.)

from the form of the solidified end product, termed "pastilles." The novel feature is a one-step operation from the molten liquid to a fairly uniformly sized and shaped product without intermediate operations on the solid phase.

Another development features a nonmetallic belt [*Plast. Des. Process.*, **13** (July 1968)]. When rapid heat transfer is the objective, a glass-fiber, Teflon-coated construction in a thickness as little as 0.08 mm (0.003 in) is selected for use. No performance data are available, but presumably the thin belt permits rapid heat transfer while taking advantage of the nonsticking property of Teflon. Another development [*Food Process. Mark.*, **69** (March 1969)] is extending the capability of belt solidification by providing use of subzero temperatures.

**Rotating-Drum Type** This type (Fig. 11-55a and b) is not an adaptation of a material-handling device (though volumetric material throughput is a first consideration) but is designed specifically for heat-transfer service. It is well engineered, established, and widely used. The twin-drum type (Fig. 11-55b) is best suited to thin [0.4- to 6-mm ( $\frac{1}{64}$  to  $\frac{1}{4}$ -in)] cake production. For temperatures to 149°C (300°F) the coolant water is piped in and siphoned out. Spray application of coolant water to the inside is employed for high-temperature work, permitting feed temperatures to at least 538°C (1000°F), or double those for jacketed equipment. Vaporizing refrigerants are readily applicable for very low temperature work.

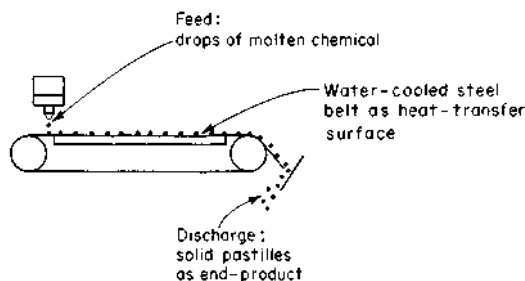


FIG. 11-54 Heat-transfer equipment for solidification; belt type for the operation of pastillization. (Courtesy of Sandvik, Inc.)

The burden must have a definite solidification temperature to assure proper pickup from the feed pan. This limitation can be overcome by side feeding through an auxiliary rotating spreader roll. Application limits are further extended by special feed devices for burdens having oxidation-sensitive and/or supercooling characteristics. The standard double-drum model turns downward, with adjustable roll spacing to control sheet thickness. The newer twin-drum model (Fig. 11-55b) turns upward and, though subject to variable cake thickness, handles viscous and indefinite solidification-temperature-point burden materials well.

Drums have been successfully applied to a wide range of chemical products, both inorganic and organic, pharmaceutical compounds, waxes, soaps, insecticides, food products to a limited extent (including lard cooling), and even flake-ice production. A novel application is that of using a water-cooled roll to pick up from a molten-lead bath and turn out a 1.2-m- (4-ft-) wide continuous sheet, weighing 4.9 kg/m<sup>2</sup> (1 lb/ft<sup>2</sup>), which is ideal for a sound barrier. This technique is more economical than other sheeting methods [*Mech. Eng.*, **631** (March 1968)].

**Heat-transfer performance** of drums, in terms of reported heat flux is: for an 80°C (176°F) melting-point wax, 7880 W/m<sup>2</sup> [2500 Btu/(h·ft<sup>2</sup>)]; for a 130°C (266°F) melting-point organic chemical, 20,000 W/m<sup>2</sup> [6500 Btu/(h·ft<sup>2</sup>)]; and for high- [318°C (604°F)] melting-point caustic soda (water-sprayed in drum), 95,000 to 125,000 W/m<sup>2</sup> [30,000 to 40,000 Btu/(h·ft<sup>2</sup>)], with overall coefficients of 340 to 450 W/(m<sup>2</sup>·°C) [60 to 80 Btu/(h·ft<sup>2</sup>·°F)]. An innovation that is claimed often to increase these performance values by as much as 300 percent is the addition of hoods to apply impinging streams of heated air to the solidifying and drying solids surface as the drums carry it upward [*Chem. Eng.*, **74**, 152 (June 19, 1967)]. Similar rotating-drum indirect heat-transfer equipment is also extensively used for drying duty on liquids and thick slurries of solids (see Sec. 20).

**Rotating-Shelf Type** The patented **Roto-shelf** type (Fig. 11-55c) features (1) a large heat-transfer surface provided over a small floor space and in a small building volume, (2) easy floor cleaning, (3) non-hazardous machinery, (4) stainless-steel surfaces, (5) good control range, and (6) substantial capacity by providing as needed 1 to 10 shelves operated in parallel. It is best suited for thick-cake production and burden materials having an indefinite solidification temperature. Solidification of liquid sulfur into 13- to 19-mm- ( $\frac{1}{2}$ - to  $\frac{3}{4}$ -in-) thick



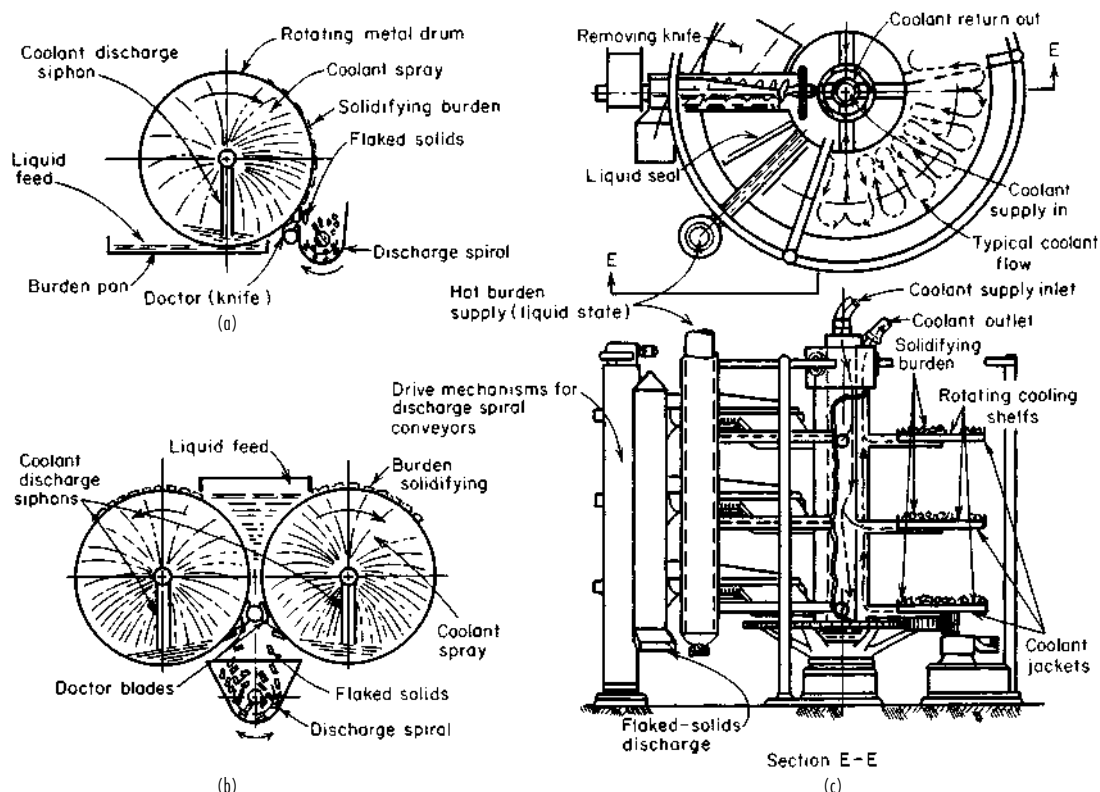


FIG. 11-55 Heat-transfer equipment for continuous solidification. (a) Single drum. (b) Twin drum. (c) Roto-shelf. (Courtesy of Buflovak Division, Blaw-Knox Food & Chemical Equipment, Inc.)

lumps is a successful application. Heat transfer, by liquid-coolant circulation through jackets, limits feed temperatures to 204°C (400°F). Heat-transfer rate, controlled by the thick cake rather than by equipment construction, should be equivalent to the belt type. Thermal performance is aided by applying water sprayed directly to the burden top to obtain two-sided cooling.

### EQUIPMENT FOR FUSION OF SOLIDS

The thermal duty here is the opposite of solidification operations. The indirect heat-transfer equipment suitable for one operation is not suitable for the other because of the material-handling rather than the thermal aspects. Whether the temperature of transformation is a definite or a ranging one is of little importance in the selection of equipment for fusion. The burden is much agitated, but the beds are deep. Only fair overall coefficient values may be expected, although heat-flux values are good.

**Horizontal-Tank Type** This type (Fig. 11-56a) is used to transfer heat for melting or cooking dry powdered solids, rendering lard from meat-scrap solids, and drying divided solids. Heat-transfer coefficients are 17 to 85 W/(m<sup>2</sup>·°C) [3 to 15 Btu/(h·ft<sup>2</sup>·°F)] for drying and 28 to 140 W/(m<sup>2</sup>·°C) [5 to 25 Btu/(h·ft<sup>2</sup>·°F)] for vacuum and/or solvent recovery.

**Vertical Agitated-Kettle Type** Shown in Fig. 11-56b this type is used to cook, melt to the liquid state, and provide or remove reaction heat for solids that vary greatly in "body" during the process so that material handling is a real problem. The virtues are simplicity and 100 percent cleanability. These often outweigh the poor heat-transfer aspect. These devices are available from the small jacketed type illustrated to huge cast-iron direct-underfired bowls for calcining gypsum.

Temperature limits vary with construction; the simpler jackets allow temperatures to 371°C (700°F) (as with Dowtherm), which is not true of all jacketed equipment.

**Mill Type** Figure 11-56c shows one model of roll construction used. Note the ruggedness, as it is a *power device* as well as one for *indirect* heat transfer, employed to knead and heat a mixture of dry powdered-solid ingredients with the objective of reacting and reforming via fusion to a consolidated product. In this compounding operation, frictional heat generated by the kneading may require heat-flow reversal (by cooling). Heat-flow control and temperature-level considerations often predominate over heat-transfer performance. Power and mixing considerations, rather than heat transfer, govern. The two-roll mill shown is employed in compounding raw plastic, rubber, and rubberlike elastomer stocks. Multiple-roll mills less knives (termed calenders) are used for continuous sheet or film production in widths up to 2.3 m (7.7 ft). Similar equipment is employed in the chemical compounding of inks, dyes, paint pigments, and the like.

### HEAT-TRANSFER EQUIPMENT FOR SHEETED SOLIDS

**Cylinder Heat-Transfer Units** Sometimes called "can" dryers or drying rolls, these devices are differentiated from drum dryers in that they are used for solids in flexible continuous-sheet form, whereas drum dryers are used for liquid or paste forms. The construction of the individual cylinders, or drums, is similar in most respects to that of drum dryers. Special designs are used to obtain uniform distribution of steam within large drums when uniform heating across the drum surface is critical.

A cylinder dryer may consist of one large cylindrical drum, such as the so-called Yankee dryer, but more often it comprises a number of

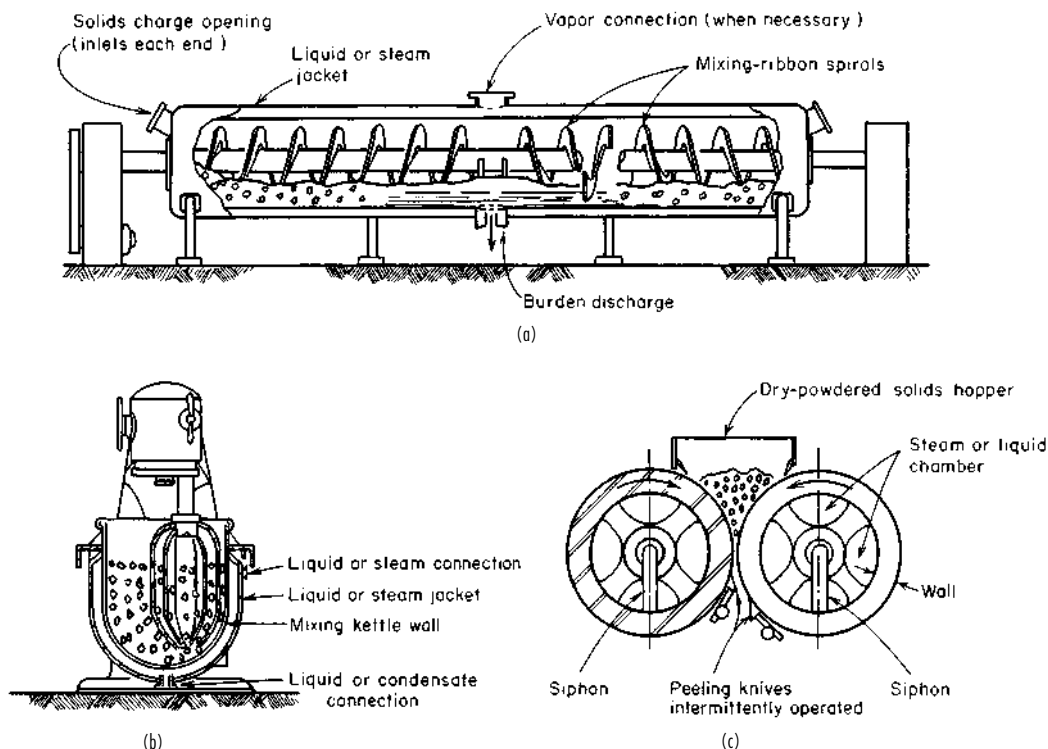


FIG. 11-56 Heat-transfer equipment for fusion of solids. (a) Horizontal-tank type. (Courtesy of Struthers Wells Corp.) (b) Agitated kettle. (Courtesy of Read-Standard Division, Capital Products Co.) (c) Double-drum mill. (Courtesy of Farrel-Birmingham Co.)

drums arranged so that a continuous sheet of material may pass over them in series. Typical of this arrangement are Fourdrinier-paper-machine dryers, cellophane dryers, slashers for textile piece goods and fibers, etc. The multiple cylinders are arranged in various ways. Generally they are staggered in two horizontal rows. In any one row, the cylinders are placed close together. The sheet material contacts the undersurface of the lower rolls and passes over the upper rolls, contacting 60 to 70 percent of the cylinder surface. The cylinders may also be arranged in a single horizontal row, in more than two horizontal rows, or in one or more vertical rows. When it is desired to contact only one side of the sheet with the cylinder surface, unheated guide rolls are used to conduct the sheeting from one cylinder to the next. For sheet materials that shrink on processing, it is frequently necessary to drive the cylinders at progressively slower speeds through the dryer. This requires elaborate individual electric drives on each cylinder.

Cylinder dryers usually operate at atmospheric pressure. However, the Minton paper dryer is designed for operation under vacuum. The drying cylinders are usually heated by steam, but occasionally single cylinders may be gas-heated, as in the case of the Pease blueprinting machine. Upon contacting the cylinder surface, wet sheet material is first heated to an equilibrium temperature somewhere between the wet-bulb temperature of the surrounding air and the boiling point of the liquid under the prevailing total pressure. The heat-transfer resistance of the vapor layer between the sheet and the cylinder surface may be significant.

These cylinder units are applicable to almost any form of sheet material that is not injuriously affected by contact with steam-heated metal surfaces. They are used chiefly when the sheet possesses certain properties such as a tendency to shrink or lacks the mechanical strength necessary for most types of continuous-sheeting air dryers. Applications are to dry films of various sorts, paper pulp in sheet form, paper sheets, paperboard, textile piece goods and fibers, etc. In some cases, imparting a special finish to the surface of the sheet may be an objective.

The **heat-transfer performance capacity** of cylinder dryers is not easy to estimate without a knowledge of the sheet temperature, which, in turn, is difficult to predict. According to published data, steam temperature is the largest single factor affecting capacity. Overall evaporation rates based on the total surface area of the dryers cover a range of 3.4 to 23 kg water/(h·m<sup>2</sup>) [0.7 to 4.8 lb water/(h·ft<sup>2</sup>)].

The value of the **coefficient of heat transfer** from steam to sheet is determined by the conditions prevailing on the inside and on the surface of the dryers. Low coefficients may be caused by (1) poor removal of air or other noncondensables from the steam in the cylinders, (2) poor removal of condensate, (3) accumulation of oil or rust on the interior of the drums, and (4) accumulation of a fiber lint on the outer surface of the drums. In a test reported by Lewis et al. [*Pulp Pap. Mag. Can.*, 22 (February 1927)] on a sulfite-paper dryer, in which the actual sheet temperatures were measured, a value of 187 W/(m<sup>2</sup>·°C) [33 Btu/(h·ft<sup>2</sup>·°F)] was obtained for the coefficient of heat flow between the steam and the paper sheet.

Operating-cost data for these units are meager. Power costs may be estimated by assuming 1 hp per cylinder for diameters of 1.2 to 1.8 m (4 to 6 ft). Data on labor and maintenance costs are also lacking.

The size of commercial cylinder dryers covers a wide range. The individual rolls may vary in diameter from 0.6 to 1.8 m (2 to 6 ft) and up to 8.5 m (28 ft) in width. In some cases, the width of rolls decreases throughout the dryer in order to conform to the shrinkage of the sheet. A single-cylinder dryer, such as the Yankee dryer, generally has a diameter between 2.7 and 4.6 m (9 and 15 ft).

#### HEAT-TRANSFER EQUIPMENT FOR DIVIDED SOLIDS

Most equipment for this service is some adaptation of a *material-handling* device whether or not the transport ability is desired. The old vertical tube and the vertical shell (fluidizer) are exceptions. Material-handling problems, plant transport needs, power, and maintenance are

prime considerations in equipment selection and frequently overshadow heat-transfer and capital-cost considerations. Material handling is generally the most important aspect. Material-handling characteristics of the divided solids may vary during heat processing. The body changes are usually important in drying, occasionally significant for heating, and only on occasion important for cooling. The ability to minimize effects of changes is a major consideration in equipment selection. Dehydration operations are better performed on contactive apparatus (see Sec. 12) that provides air to carry off released water vapor before a semiliquid form develops.

Some types of equipment are convertible from heat removal to heat supply by simply changing the temperature level of the fluid or air. Other types require an auxiliary change. Others require constructional changes. Temperature limits for the equipment generally vary with the thermal operation. The kind of thermal operation has a major effect on heat-transfer values. For drying, overall coefficients are substantially higher in the presence of substantial moisture for the constant-rate period than in finishing. However, a stiff "body" occurrence due to moisture can prevent a normal "mixing" with an adverse effect on the coefficient.

**Fluidized-Bed Type** Known as the cylindrical fluidizer, this operates with a bed of **fluidized solids** (Fig. 11-57). It is an indirect heat-transfer version of the contactive type in Sec. 17. An application disadvantage is the need for batch operation unless some short circuiting can be tolerated. Solids-cooling applications are few, as they can be more effectively accomplished by the fluidizing gas via the contactive mechanism that is referred to in Sec. 11. Heating applications are many and varied. These are subject to one shortcoming, which is the dissipation of the heat input by carry-off in the fluidizing gas. Heat-transfer performance for the indirect mode to solids has been outstanding, with overall coefficients in the range of 570 to 850 W/(m<sup>2</sup>·°C) [100 to 150 Btu/(h·ft<sup>2</sup>·°F)]. This device with its thin film does for solids what the falling-film and other thin-film techniques do for fluids, as shown by Holt (Pap. 11, 4th National Heat-Transfer Conference, August 1960). In a design innovation with high heat-transfer capability, heat is supplied indirectly to the fluidized solids through the walls of in-bed, horizontally placed, finned tubes [Petrie, Freeby, and Buckham, *Chem. Eng. Prog.*, **64**(7), 45 (1968)].

**Moving-Bed Type** This concept uses a single-pass tube bundle in a vertical shell with the divided solids flowing by gravity in the tubes. It is little used for solids. A major difficulty in divided-solids applications is the problem of charging and discharging with uniformity. A second is poor heat-transfer rates. Because of these limitations, this tube-bundle type is not the workhorse for solids that it is for liquid and gas-phase heat exchange.

However, there are applications in which the nature of a specific chemical reactor system requires indirect heating or cooling of a

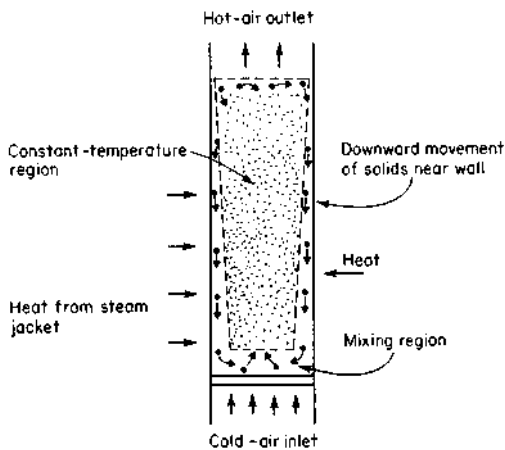


FIG. 11-57 Heat-transfer equipment for divided solids; stationary vertical-shell type. The indirect fluidizer.

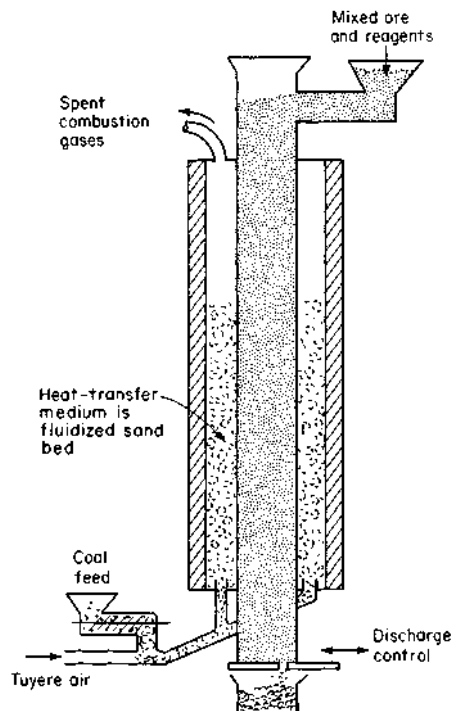


FIG. 11-58 Stationary vertical-tube type of indirect heat-transfer equipment with divided solids inside tubes, laminar solids flow and steady-state heat conditions.

moving bed of divided solids. One of these is the segregation process which through a gaseous reaction frees chemically combined copper in an ore to a free copper form which permits easy, efficient subsequent recovery [Pinkey and Plint, *Miner. Process.*, 17-30 (June 1968)]. The apparatus construction and principle of operation are shown in Fig. 11-58. The functioning is abetted by a novel heat-exchange provision of a fluidized sand bed in the jacket. This provides a much higher unit heat-input rate (coefficient value) than would the usual low-density hot-combustion-gas flow.

**Agitated-Pan Type** This device (Fig. 11-52) is not an adaptation of a material-handling device but was developed many years ago primarily for heat-transfer purposes. As such, it has found wide application. In spite of its batch operation with high attendant labor costs, it is still used for processing divided solids when no phase change is occurring. Simplicity and easy cleanout make the unit a wise selection for handling small, experimental, and even some production runs when quite a variety of burden materials are heat-processed. Both heating and cooling are feasible with it, but greatest use has been for drying [see Sec. 12 and Uhl and Root, *Chem. Eng. Prog.*, **63**(7), 8 (1967)]. This device, because it can be readily covered (as shown in the illustration) and a vacuum drawn or special atmosphere provided, features versatility to widen its use. For drying granular solids, the heat-transfer rate ranges from 28 to 227 W/(m<sup>2</sup>·°C) [5 to 40 Btu/(h·ft<sup>2</sup>·°F)]. For atmospheric applications, thermal efficiency ranges from 65 to 75 percent. For vacuum applications, it is about 70 to 80 percent. These devices are available from several sources, fabricated of various metals used in chemical processes.

**Kneading Devices** These are closely related to the agitated pan but differ as being primarily mixing devices with heat transfer a secondary consideration. Heat transfer is provided by jacketed construction of the main body and is effected by a coolant, hot water, or steam. These devices are applicable for the compounding of divided solids by mechanical rather than chemical action. Application is largely in the pharmaceutical and food-processing industries. For a

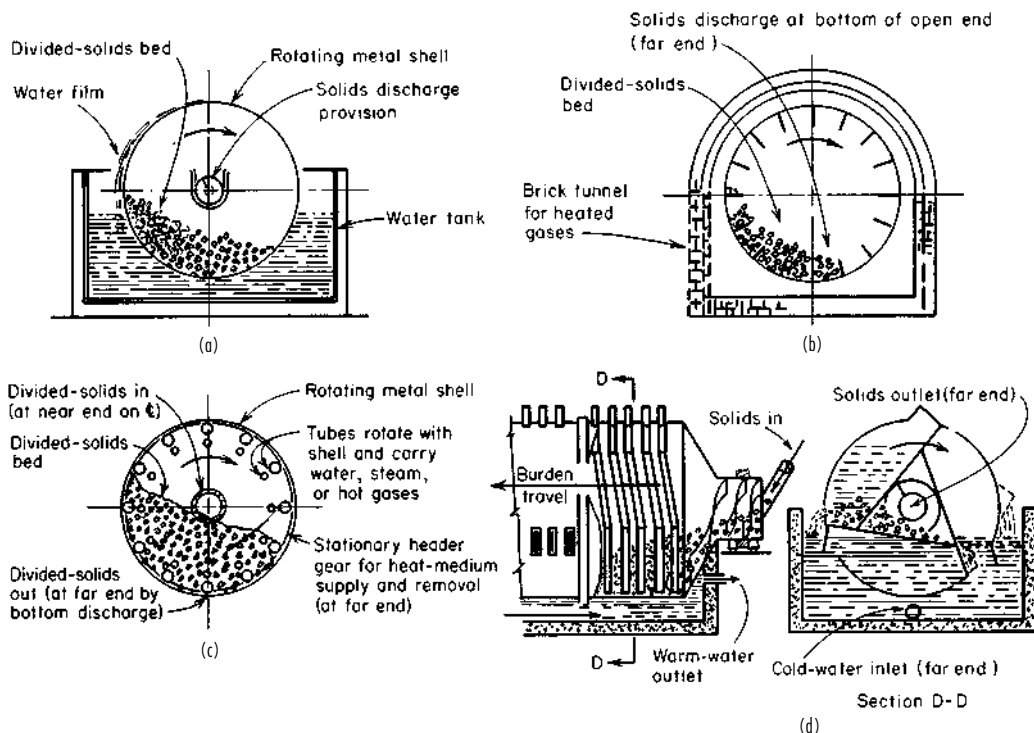


FIG. 11-59 Rotating shells as indirect heat-transfer equipment. (a) Plain. (Courtesy of BSP Corp.) (b) Flighted. (Courtesy of BSP Corp.) (c) Tubed. (d) Deep-finned type. (Courtesy of Link-Belt Co.)

more complete description, illustrations, performance, and power requirements, refer to Sec. 19.

**Shelf Devices** Equipment having heated and/or cooled shelves is available but is little used for divided-solids heat processing. Most extensive use of stationary shelves is freezing of packaged solids for food industries and for freeze drying by sublimation (see Sec. 22).

**Rotating-Shell Devices** These (see Fig. 11-59) are installed horizontally, whereas stationary-shell installations are vertical. Material-handling aspects are of greater importance than thermal performance. Thermal results are customarily given in terms of overall coefficient on the basis of the total area provided, which varies greatly with the design. The effective use, chiefly percent fill factor, varies widely, affecting the reliability of stated coefficient values. For performance calculations see Sec. 10 on heat-processing theory for solids. These devices are variously used for cooling, heating, and drying and are the workhorses for heat-processing divided solids in the large-capacity range. Different modifications are used for each of the three operations.

The **plain** type (Fig. 11-59a) features simplicity and yet versatility through various end-construction modifications enabling wide and varied applications. Thermal performance is strongly affected by the "body" characteristics of the burden because of its dependency for material handling on frictional contact. Hence, performance ranges from well-agitated beds with good thin-film heat-transfer rates to poorly agitated beds with poor thick-film heat-transfer rates. Temperature limits in application are (1) low-range cooling with shell dipped in water, 400°C (750°F) and less; (2) intermediate cooling with forced circulation of tank water, to 760°C (1400°F); (3) primary cooling, above 760°C (1400°F), water copiously sprayed and loading kept light; (4) low-range heating, below steam temperature, hot-water dip; and (5) high-range heating by tempered combustion gases or ribbon radiant-gas burners.

The **flighted** type (Fig. 11-59b) is a first-step modification of the plain type. The simple flight addition improves heat-transfer performance. This type is most effective on semifluid burdens which slide readily. Flighted models are restricted from applications in which soft-cake sticking occurs, breakage must be minimized, and abrasion is severe. A special flighting is one having the cross section compartmented into four lesser areas with ducts between. Hot gases are drawn through the ducts en route from the outer oven to the stack to provide about 75 percent more heating surface, improving efficiency and capacity with a modest cost increase. Another similar unit has the flights made in a triangular-duct cross section with hot gases drawn through.

The **tubed-shell** type (Fig. 11-59c) is basically the same device more commonly known as a "steam-tube rotary dryer" (see Sec. 20). The rotation, combined with slight inclination from the horizontal, moves the shell-side solids through it continuously. This type features good mixing with the objective of increased heat-transfer performance. Tube-side fluid may be water, steam, or combustion gas. Bottom discharge slots in the shell are used so that heat-transfer-medium supply and removal can be made through the ends; these restrict wide-range loading and make the tubed type inapplicable for floody materials. These units are seldom applicable for sticky, soft-caking, scaling, or heat-sensitive burdens. They are not recommended for abrasive materials. This type has high thermal efficiency because heat loss is minimized. **Heat-transfer coefficient** values are: water, 34 W/(m<sup>2</sup>·°C) [6 Btu/(h·ft<sup>2</sup>·°F)]; steam, same, with heat flux reliably constant at 3800 W/m<sup>2</sup> [1200 Btu/(h·ft<sup>2</sup>)]; and gas, 17 W/(m<sup>2</sup>·°C) [3 Btu/(h·ft<sup>2</sup>·°F)], with a high temperature difference. Although from the preceding discussion the device may seem rather limited, it is nevertheless widely used for drying, with condensing steam predominating as the heat-carrying fluid. But with water or refrigerants flowing in the tubes, it is also effective for cooling operations. The units are custom-built by several manufacturers in a wide range of sizes and materials.

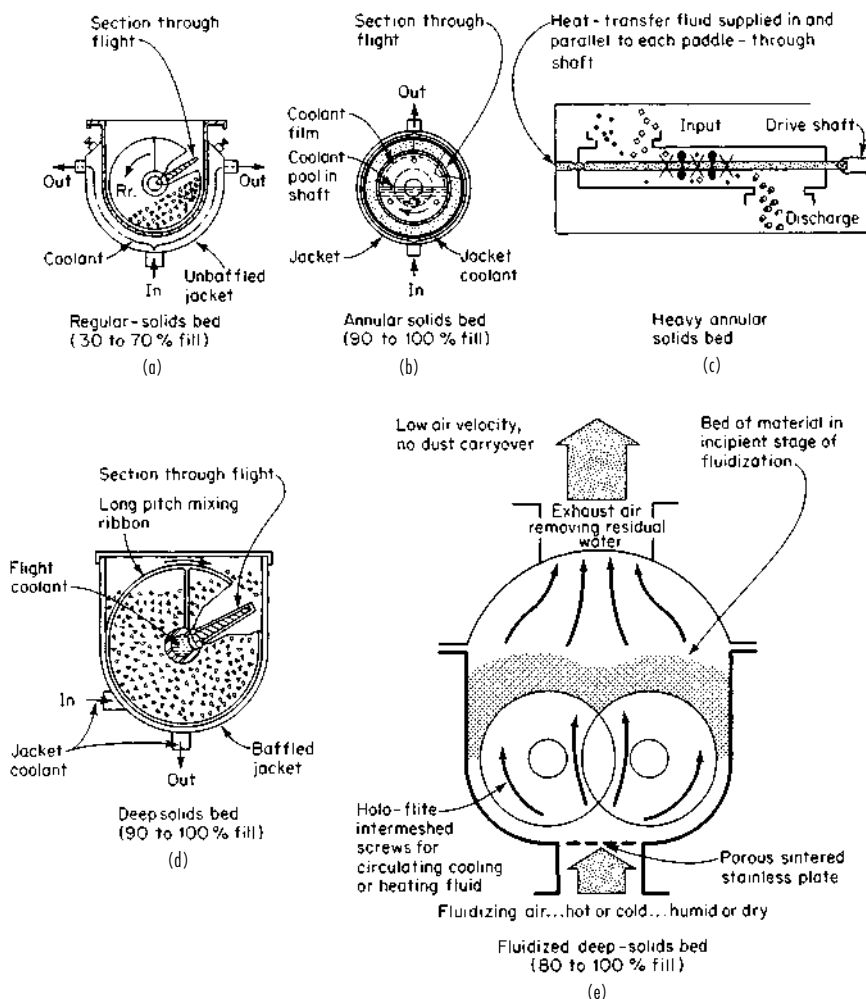
A few fabricators that specialize in this type of equipment have accumulated a vast store of data for determining application sizing.

The patented **deep-finned** type in Fig. 11-59d is named the "Rotofin cooler." It features loading with a small layer thickness, excellent mixing to give a good effective diffusivity value, and a thin fluid-side film. Unlike other rotating-shell types, it is installed horizontally, and the burden is moved positively by the fins acting as an Archimedes spiral. Rotational speed and spiral pitch determine travel time. For cooling, this type is applicable to both secondary and intermediate cooling duties. Applications include solids in small lumps [9 mm ( $\frac{3}{4}$  in.)] and granular size [6 mm and less ( $\frac{1}{4}$  to 0 in.)] with no larger pieces to plug the fins, solids that have a free-flowing body characteristic with no sticking or caking tendencies, and drying of solids that have a low moisture and powder content unless special modifications are made for substantial vapor and dust handling. Thermal performance is very good, with overall coefficients to 110  $W/(m^2 \cdot ^\circ C)$  [20  $Btu/(h \cdot ft^2 \cdot ^\circ F)$ ], with one-half of these coefficients nominal for cooling based on the total area provided (nearly double those reported for other indirect rotaries).

**Conveyor-Belt Devices** The metal-belt type (Fig. 11-55) is the only device in this classification of material-handling equipment that has had serious effort expended on it to adapt it to indirect heat-transfer service with divided solids. It features a lightweight construction of a large area with a thin metal wall. Indirect-cooling applications have been made with poor thermal performance, as could be expected with a static layer. Auxiliary plowlike mixing devices, which are considered an absolute necessity to secure any worthwhile results for this service, restrict applications.

**Spiral-Conveyor Devices** Figure 11-60 illustrates the major adaptations of this widely used class of material-handling equipment to indirect heat-transfer purposes. These conveyors can be considered for heat-transfer purposes as continuously agitated kettles. The adaptation of Fig. 11-60d offers a batch-operated version for evaporation duty. For this service, all are package-priced and package-shipped items requiring few, if any, auxiliaries.

The **jacketed solid-flight** type (Fig. 11-60a) is the standard low-cost (parts-basis-priced) material-handling device, with a simple jacket added and employed for secondary-range heat transfer of an



**FIG. 11-60** Spiral-conveyor adaptations as heat-transfer equipment. (a) Standard jacketed solid flight. (Courtesy of Jeffrey Mfg. Co.) (b) Small spiral, large shaft. (Courtesy of Fuller Co.) (c) "Porcupine" medium shaft. (Courtesy of Bethlehem Corp.) (d) Large spiral, hollow flight. (Courtesy of Rietz Mfg. Co.) (e) Fluidized-bed large spiral, helical flight. (Courtesy of Western Precipitation Division, Joy Mfg. Co.)

incidental nature. Heat-transfer coefficients are as low as 11 to 34 W/(m<sup>2</sup>·°C) [2 to 6 Btu/(h·ft<sup>2</sup>·°F)] on sensible heat transfer and 11 to 68 W/(m<sup>2</sup>·°C) [2 to 12 Btu/(h·ft<sup>2</sup>·°F)] on drying because of substantial static solids-side film.

The **small-spiral-large-shaft** type (Fig. 11-60b) is inserted in a solids-product line as pipe banks are in a fluid line, solely as a heat-transfer device. It features a thin burden ring carried at a high rotative speed and subjected to two-sided conductance to yield an estimated heat-transfer coefficient of 285 W/(m<sup>2</sup>·°C) [50 Btu/(h·ft<sup>2</sup>·°F)], thereby ranking thermally next to the shell-fluidizer type. This device for powdered solids is comparable with the Votator of the fluid field.

Figure 11-60c shows a fairly new spiral device with a medium-heavy annular solids bed and having the combination of a jacketed, stationary outer shell with moving paddles that carry the heat-transfer fluid. A unique feature of this device to increase volumetric throughput, by providing an overall greater temperature drop, is that the heat medium is supplied to and withdrawn from the rotor paddles by a parallel piping arrangement in the rotor shaft. This is a unique flow arrangement compared with the usual series flow. In addition, the rotor carries burden-agitating spikes which give it the trade name of Porcupine Heat-Processor (*Chem. Equip. News*, April 1966; and Uhl and Root, AIChE Prepr. 21, 11th National Heat-Transfer Conference, August 1967).

The **large-spiral hollow-flight** type (Fig. 11-60d) is an adaptation, with external bearings, full fill, and salient construction points as shown, that is highly versatile in application. Heat-transfer coefficients are 34 to 57 W/(m<sup>2</sup>·°C) [6 to 10 Btu/(h·ft<sup>2</sup>·°F)] for poor, 45 to 85 W/(m<sup>2</sup>·°C) [8 to 15 Btu/(h·ft<sup>2</sup>·°F)] for fair, and 57 to 114 W/(m<sup>2</sup>·°C) [10 to 20 Btu/(h·ft<sup>2</sup>·°F)] for wet conductors. A popular version of this employs two such spirals in one material-handling chamber for a pug-mill agitation of the deep solids bed. The spirals are seldom heated. The shaft and shell are heated.

Another deep-bed spiral-activated solids-transport device is shown by Fig. 11-60e. The flights carry a heat-transfer medium as well as the jacket. A unique feature of this device which is purported to increase heat-transfer capability in a given equipment space and cost is the dense-phase fluidization of the deep bed that promotes agitation and moisture removal on drying operations.

**Double-Cone Blending Devices** The original purpose of these devices was mixing (see Sec. 19). Adaptations have been made; so many models now are primarily for indirect heat-transfer processing. A jacket on the shell carries the heat-transfer medium. The mixing action, which breaks up agglomerates (but also causes some degradation), provides very effective burden exposure to the heat-transfer surface. On drying operations, the vapor release (which in a static bed is a slow diffusional process) takes place relatively quickly. To provide vapor removal from the burden chamber, a hollow shaft is used. Many of these devices carry the hollow-shaft feature a step further by adding a rotating seal and drawing a vacuum. This increases thermal performance notably and makes the device a natural for solvent-recovery operations.

These devices are replacing the older tank and spiral-conveyor devices. Better provisions for speed and ease of fill and discharge (without powered rotation) minimize downtime to make this batch-operated device attractive. Heat-transfer coefficients ranging from 28 to 200 W/(m<sup>2</sup>·°C) [5 to 35 Btu/(h·ft<sup>2</sup>·°F)] are obtained. However, if caking on the heat-transfer walls is serious, then values may drop to 5.5 or 11 W/(m<sup>2</sup>·°C) [1 or 2 Btu/(h·ft<sup>2</sup>·°F)], constituting a misapplication. The double cone is available in a fairly wide range of sizes and construction materials. The users are the fine-chemical, pharmaceutical, and biological-preparation industries.

A novel variation is a cylindrical model equipped with a tube bundle to resemble a shell-and-tube heat exchanger with a bloated shell [*Chem. Process.*, 20 (Nov. 15, 1968)]. Conical ends provide for redistribution of burden between passes. The improved heat-transfer performance is shown by Fig. 11-61.

**Vibratory-Conveyor Devices** Figure 11-62 shows the various adaptations of vibratory material-handling equipment for indirect heat-transfer service on divided solids. The basic vibratory-equipment data are given in Sec. 21. These indirect heat-transfer adaptations feature simplicity, nonhazardous construction, nondegradation, nondusting, no wear, ready conveying-rate variation [1.5 to 4.5 m/min (5 to 15 ft/min)], and good heat-transfer coefficient—115 W/(m<sup>2</sup>·°C)

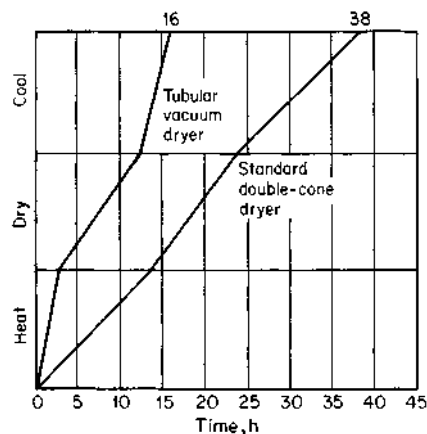


FIG. 11-61 Performance of tubed, blender heat-transfer device.

[20 Btu/(h·ft<sup>2</sup>·°F)] for sand. They usually require feed-rate and distribution auxiliaries. They are suited for heating and cooling of divided solids in powdered, granular, or moist forms but no sticky, liquefying, or floody ones. Terminal-temperature differences less than 11°C (20°F) on cooling and 17°C (30°F) on heating or drying operations are seldom practical. These devices are for medium and light capacities.

The **heavy-duty jacketed** type (Fig. 11-62a) is a special custom-built adaptation of a heavy-duty vibratory conveyor shown in Fig. 11-60. Its application is continuously to cool the crushed material [from about 177°C (350°F)] produced by the vibratory-type "caster" of Fig. 11-53. It does not have the liquid dam and is made in longer lengths that employ L, switchback, and S arrangements on one floor. The capacity rate is 27,200 to 31,700 kg/h (30 to 35 tons/h) with heat-transfer coefficients in the order of 142 to 170 W/(m<sup>2</sup>·°C) [25 to 30 Btu/(h·ft<sup>2</sup>·°F)]. For heating or drying applications, it employs steam to 414 kPa (60 lbf/in<sup>2</sup>).

The **jacketed or coolant-spraying** type (Fig. 11-62b) is designed to assure a very thin, highly agitated liquid-side film and the same initial coolant temperature over the entire length. It is frequently employed for transporting substantial quantities of hot solids, with cooling as an incidental consideration. For heating or drying applications, hot water or steam at a gauge pressure of 7 kPa (1 lbf/in<sup>2</sup>) may be employed. This type is widely used because of its versatility, simplicity, cleanability, and good thermal performance.

The **light-duty jacketed** type (Fig. 11-62c) is designed for use of air as a heat carrier. The flow through the jacket is highly turbulent and is usually counterflow. On long installations, the air flow is parallel to every two sections for more heat-carrying capacity and a fairly uniform surface temperature. The outstanding feature is that a wide range of temperature control is obtained by merely changing the heat-carrier temperature level from as low as atmospheric moisture condensation will allow to 204°C (400°F). On heating operations, a very good thermal efficiency can be obtained by insulating the machine and recycling the air. While heat-transfer rating is good, the heat-removal capacity is limited. Cooler units are often used in series with like units operated as dryers or when clean water is unavailable. Drying applications are for heat-sensitive [49 to 132°C (120° to 270°F)] products; when temperatures higher than steam at a gauge pressure of 7 kPa (1 lbf/in<sup>2</sup>) can provide are wanted but heavy-duty equipment is too costly; when the jacket-corrosion hazard of steam is unwanted; when headroom space is at a premium; and for highly abrasive burden materials such as fritted or crushed glasses and porcelains.

The **tiered arrangement** (Fig. 11-62d) employs the units of Fig. 11-62 with either air or steam at a gauge pressure of 7 kPa (1 lbf/in<sup>2</sup>) as a heat medium. These are custom-designed and built to provide a large amount of heat-transfer surface in a small space with the minimum of transport and to provide a complete processing system. These

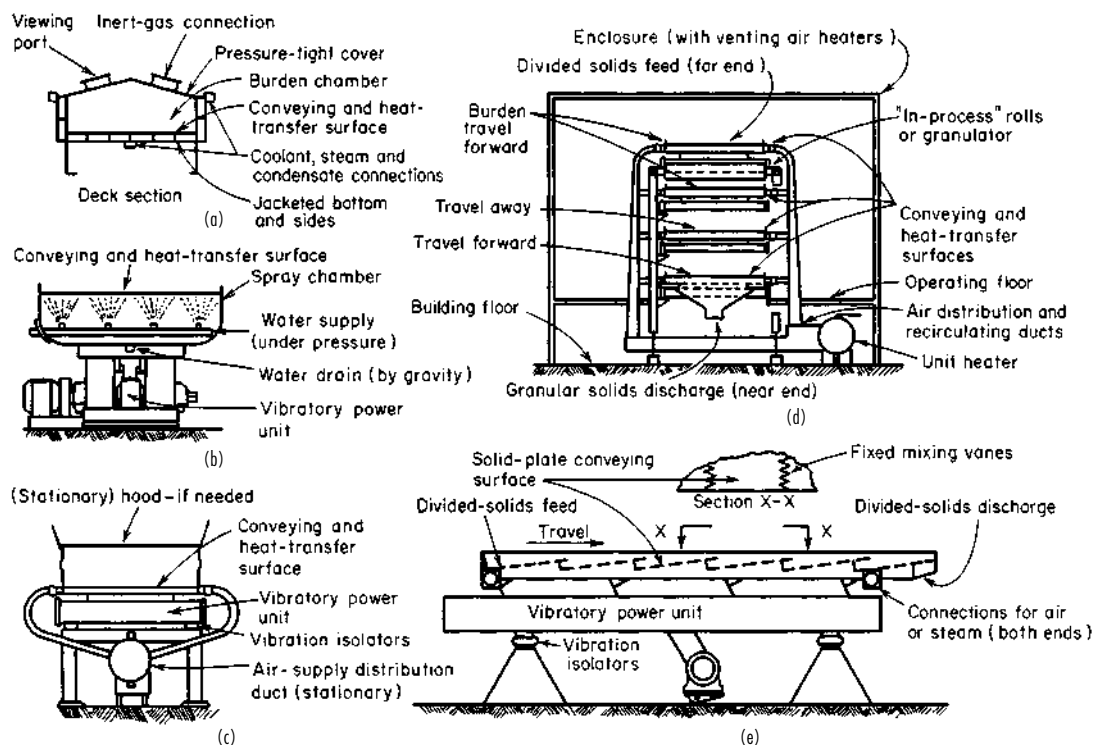


FIG. 11-62 Vibratory-conveyor adaptations as indirect heat-transfer equipment. (a) Heavy-duty jacketed for liquid coolant or high-pressure steam. (b) Jacketed for coolant spraying. (c) Light-duty jacketed construction. (d) Jacketed for air or steam in tiered arrangement. (e) Jacketed for air or steam with Mix-R-Step surface. (Courtesy of Jeffrey Mfg. Co.)

receive a damp material, resize while in process by granulators or rolls, finish dry, cool, and deliver to packaging or tableting. The applications are primarily in the fine-chemical, food, and pharmaceutical manufacturing fields.

The **Mix-R-Step** type in Fig. 11-62e is an adaptation of a vibratory conveyor. It features better heat-transfer rates, practically doubling the coefficient values of the standard flat surface and trebling heat-flux values, as the layer depth can be increased from the normal 13 to 25 and 32 mm ( $\frac{1}{2}$  to 1 and  $1\frac{1}{4}$  in). It may be provided on decks jacketed for air, steam, or water spray. It is also often applicable when an infrared heat source is mounted overhead to supplement the indirect or as the sole heat source.

**Elevator Devices** The **vibratory elevating-spiral** type (Fig. 11-63) adapts divided-solids-elevating material-handling equipment to heat-transfer service. It features a large heat-transfer area over a small floor space and employs a reciprocating shaker motion to effect transport. Applications, layer depth, and capacities are restricted, as burdens must be of such "body" character as to convey uphill by the microhopping-transport principle. The type lacks self-emptying ability. Complete washdown and cleaning is a feature not inherent in any other elevating device. A typical application is the cooling of a low-density plastic powder at the rate of 544 kg/h (1200 lb/h).

Another elevator adaptation is that for a **spiral-type elevating** device developed for ground cement and thus limited to fine powdery burdens. The spiral operates inside a cylindrical shell, which is externally cooled by a falling film of water. The spiral not only elevates the material in a thin layer against the wall but keeps it agitated to achieve high heat-transfer rates. Specific operating data are not available [*Chem. Eng. Prog.*, 68(7), 113 (1968)]. The falling-water film, besides being ideal thermally, by virtue of no jacket pressure very greatly reduces the hazard that the cooling water may contact the water-sensitive burden in process. Surfaces wet by water are accessible for

cleaning. A fair range of sizes is available, with material-handling capacities to 60 tons/h.

**Pneumatic-Conveying Devices** See Sec. 21 for descriptions, ratings, and design factors on these devices. Use is primarily for transport purposes, and heat transfer is a very secondary consideration. Applications have largely been for plastics in powder and pellet forms.

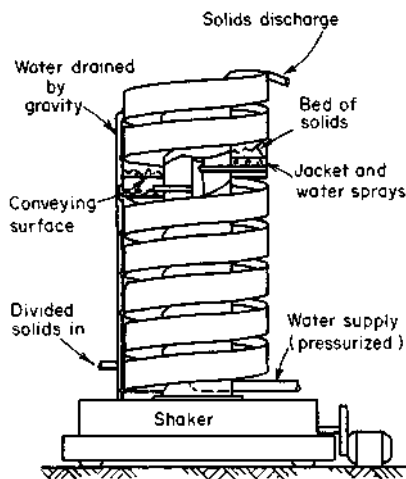


FIG. 11-63 Elevator type as heat-transfer equipment. (Courtesy of Carrier Conveyor Corp.)

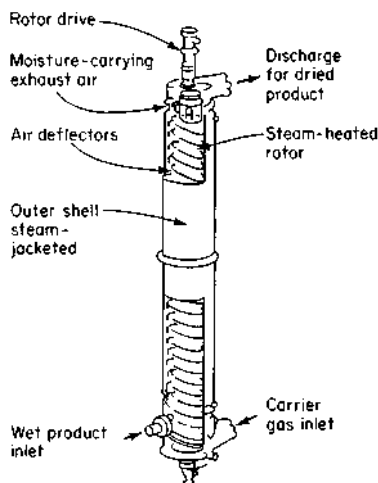


FIG. 11-64 A pneumatic-transport adaptation for heat-transfer duty. (Courtesy of Werner & Pfleiderer Corp.)

By modifications, needed cooling operations have been simultaneously effected with transport to stock storage [*Plast. Des. Process.*, **28** (December 1968)].

Heat-transfer aspects and performance were studied and reported on by Depew and Farbar (ASME Pap. 62-HT-14, September 1962). Heat-transfer coefficient characteristics are similar to those shown in Sec. 11 for the indirectly heated fluid bed. Another frequent application on plastics is a small, rather incidental but necessary amount of drying required for plastic pellets and powders on receipt when

shipped in bulk to the users. Pneumatic conveyors modified for heat transfer can handle this readily.

A pneumatic-transport device designed primarily for heat-sensitive products is shown in Fig. 11-64. This was introduced into the United States after 5 years' use in Europe [*Chem. Eng.*, **76**, 54 (June 16, 1969)].

Both the shell and the rotor carry steam as a heating medium to effect indirect transfer as the burden briefly contacts those surfaces rather than from the transport air, as is normally the case. The rotor turns slowly (1 to 10 r/min) to control, by deflectors, product distribution and prevent caking on walls. The carrier gas can be inert, as nitrogen, and also recycled through appropriate auxiliaries for solvent recovery. Application is limited to burdens that (1) are fine and uniformly grained for the pneumatic transport, (2) dry very fast, and (3) have very little, if any, sticking or decomposition characteristics. Feeds can carry 5 to 100 percent moisture (dry basis) and discharge at 0.1 to 2 percent. Wall temperatures range from 100 to 170°C (212 to 340°F) for steam and lower for a hot-water-heat source. Pressure drops are in order of 500 to 1500 mmH<sub>2</sub>O (20 to 60 inH<sub>2</sub>O). Steam consumption approaches that of a contractive-mechanism dryer down to a low value of 2.9 kg steam/kg water (2.9 lb steam/lb water). Available burden capacities are 91 to 5900 kg/h (200 to 13,000 lb/h).

**Vacuum-Shelf Types** These are very old devices, being a version of the table type. Early-day use was for drying (see Sec. 12). Heat transfer is slow even when supplemented by vacuum, which is 90 percent or more of present-day use. The newer vacuum blender and cone devices are taking over many applications. The slow heat-transfer rate is quite satisfactory in a major application, freeze drying, which is a sublimation operation (see Sec. 22 for description) in which the water must be retained in the solid state during its removal. Then slow diffusional processes govern. Another extensive application is freezing packaged foods for preservation purposes.

Available sizes range from shelf areas of 0.4 to 67 m<sup>2</sup> (4 to 726 ft<sup>2</sup>). These are available in several manufacturers' standards, either as system components or with auxiliary gear as packaged systems.

## THERMAL INSULATION

Materials or combinations of materials which have air- or gas-filled pockets or void spaces that retard the transfer of heat with reasonable effectiveness are thermal insulators. Such materials may be particulate and/or fibrous, with or without binders, or may be assembled, such as multiple heat-reflecting surfaces that incorporate air- or gas-filled void spaces.

The ability of a material to retard the flow of heat is expressed by its thermal **conductivity** (for unit thickness) or **conductance** (for a specific thickness). Low values for thermal conductivity or conductance (or high thermal resistivity or resistance value) are characteristics of thermal insulation.

Heat is transferred by radiation, conduction, and convection. Radiation is the primary mode and can occur even in a vacuum. The amount of heat transferred for a given area is relative to the temperature differential and emissivity from the radiating to the absorbing surface. Conduction is due to molecular motion and occurs within gases, liquids, and solids. The tighter the molecular structure, the higher the rate of transfer. As an example, steel conducts heat at a rate approximately 600 times that of typical thermal-insulation materials. Convection is due to mass motion and occurs only in fluids. The prime purpose of a thermal-insulation system is to minimize the amount of heat transferred.

### INSULATION MATERIALS

**Materials** Thermal insulations are produced from many materials or combinations of materials in various forms, sizes, shapes, and thickness. The most commonly available materials fall within the following categories:

*Fibrous or cellular—mineral.* Alumina, asbestos, glass, perlite, rock, silica, slag, or vermiculite.

*Fibrous or cellular—organic.* Cane, cotton, wood, and wood bark (cork).

*Cellular organic plastics.* Elastomer, polystyrene, polyisocyanate, polyisocyanurate, and polyvinyl acetate.

*Cements.* Insulating and/or finishing.

*Heat-reflecting metals (reflective).* Aluminum, nickel, stainless steel.

*Available forms.* Blanket (felt and batt), block, cements, loose fill, foil and sheet, formed or foamed in place, flexible, rigid, and semi-rigid.

The actual thicknesses of piping insulation differ from the nominal values. Dimensional data of ASTM Standard C585 appear in Table 11-21.

**Thermal Conductivity (K Factor)** Depending on the type of insulation, the thermal conductivity (K factor) can vary with age, manufacturer, moisture content, and temperature. Typical published values are shown in Fig. 11-65. Mean temperature is equal to the arithmetic average of the temperatures on both sides of the insulating material.

**Actual system heat loss** (or gain) will normally exceed calculated values because of projections, axial and longitudinal seams, expansion-contraction openings, moisture, workers' skill, and physical abuse.

**Finishes** Thermal insulations require an external covering (finish) to provide protection against entry of water or process fluids, mechanical damage, and ultraviolet degradation of foamed materials. In some cases the finish can reduce the flame-spread rating and/or provide fire protection.

The finish may be a coating (paint, asphaltic, resinous, or polymeric), a membrane (coated felt or paper, metal foil, or laminate of plastic, paper, foil or coatings), or sheet material (fabric, metal, or plastic).



TABLE 11-21 Thicknesses of Piping Insulation

| in<br>mm                              | Insulation, nominal thickness |     |         |    |          |    |         |    |          |    |         |    |          |     |          |     |
|---------------------------------------|-------------------------------|-----|---------|----|----------|----|---------|----|----------|----|---------|----|----------|-----|----------|-----|
|                                       | Outer<br>diameter             |     | 1<br>25 |    | 1½<br>38 |    | 2<br>51 |    | 2½<br>64 |    | 3<br>76 |    | 3½<br>89 |     | 4<br>102 |     |
| Nominal iron-<br>pipe size, in        | Approximate wall thickness    |     |         |    |          |    |         |    |          |    |         |    |          |     |          |     |
|                                       | in                            | mm  | in      | mm | in       | mm | in      | mm | in       | mm | in      | mm | in       | mm  | in       | mm  |
| ½                                     | 0.84                          | 21  | 1.01    | 26 | 1.57     | 40 | 2.07    | 53 | 2.88     | 73 | 3.38    | 86 | 3.88     | 99  | 4.38     | 111 |
| ¾                                     | 1.05                          | 27  | 0.90    | 23 | 1.46     | 37 | 1.96    | 50 | 2.78     | 71 | 3.28    | 83 | 3.78     | 96  | 4.28     | 109 |
| 1                                     | 1.32                          | 33  | 1.08    | 27 | 1.58     | 40 | 2.12    | 54 | 2.64     | 67 | 3.14    | 80 | 3.64     | 92  | 4.14     | 105 |
| 1¼                                    | 1.66                          | 42  | 0.91    | 23 | 1.66     | 42 | 1.94    | 49 | 2.47     | 63 | 2.97    | 75 | 3.47     | 88  | 3.97     | 101 |
| 1½                                    | 1.90                          | 48  | 1.04    | 26 | 1.54     | 39 | 2.35    | 60 | 2.85     | 72 | 3.35    | 85 | 3.85     | 98  | 4.42     | 112 |
| 2                                     | 2.38                          | 60  | 1.04    | 26 | 1.58     | 40 | 2.10    | 53 | 2.60     | 66 | 3.10    | 79 | 3.60     | 91  | 4.17     | 106 |
| 2½                                    | 2.88                          | 73  | 1.04    | 26 | 1.86     | 47 | 2.36    | 60 | 2.86     | 73 | 3.36    | 85 | 3.92     | 100 | 4.42     | 112 |
| 3                                     | 3.50                          | 89  | 1.02    | 26 | 1.54     | 39 | 2.04    | 52 | 2.54     | 65 | 3.04    | 77 | 3.61     | 92  | 4.11     | 104 |
| 3½                                    | 4.00                          | 102 | 1.30    | 33 | 1.80     | 46 | 2.30    | 58 | 2.80     | 71 | 3.36    | 85 | 3.86     | 98  | 4.36     | 111 |
| 4                                     | 4.50                          | 114 | 1.04    | 26 | 1.54     | 39 | 2.04    | 52 | 2.54     | 65 | 3.11    | 79 | 3.61     | 92  | 4.11     | 104 |
| 4½                                    | 5.00                          | 127 | 1.30    | 33 | 1.80     | 46 | 2.30    | 58 | 2.86     | 73 | 3.36    | 85 | 3.86     | 98  | 4.48     | 114 |
| 5                                     | 5.56                          | 141 | 0.99    | 25 | 1.49     | 38 | 1.99    | 51 | 2.56     | 65 | 3.06    | 78 | 3.56     | 90  | 4.18     | 106 |
| 6                                     | 6.62                          | 168 | 0.96    | 24 | 1.46     | 37 | 2.02    | 51 | 2.52     | 64 | 3.02    | 77 | 3.65     | 93  | 4.15     | 105 |
| 7                                     | 7.62                          | 194 |         |    | 1.52     | 39 | 2.02    | 51 | 2.52     | 64 | 3.15    | 80 | 3.65     | 93  | 4.15     | 105 |
| 8                                     | 8.62                          | 219 |         |    | 1.52     | 39 | 2.02    | 51 | 2.65     | 67 | 3.15    | 80 | 3.65     | 93  | 4.15     | 105 |
| 9                                     | 9.62                          | 244 |         |    | 1.52     | 39 | 2.15    | 55 | 2.65     | 67 | 3.15    | 80 | 3.65     | 93  | 4.15     | 105 |
| 10                                    | 10.75                         | 273 |         |    | 1.58     | 40 | 2.08    | 53 | 2.58     | 66 | 3.08    | 78 | 3.58     | 91  | 4.08     | 104 |
| 11                                    | 11.75                         | 298 |         |    | 1.58     | 40 | 2.08    | 53 | 2.58     | 66 | 3.08    | 78 | 3.58     | 91  | 4.08     | 104 |
| 12                                    | 12.75                         | 324 |         |    | 1.58     | 40 | 2.08    | 53 | 2.58     | 66 | 3.08    | 78 | 3.58     | 91  | 4.08     | 104 |
| 14                                    | 14.00                         | 356 |         |    | 1.46     | 37 | 1.96    | 50 | 2.46     | 62 | 2.96    | 75 | 3.46     | 88  | 3.96     | 101 |
| Over 14, up to<br>and including<br>36 |                               |     |         |    | 1.46     | 37 | 1.96    | 50 | 2.46     | 62 | 2.96    | 75 | 3.46     | 88  | 3.96     | 101 |

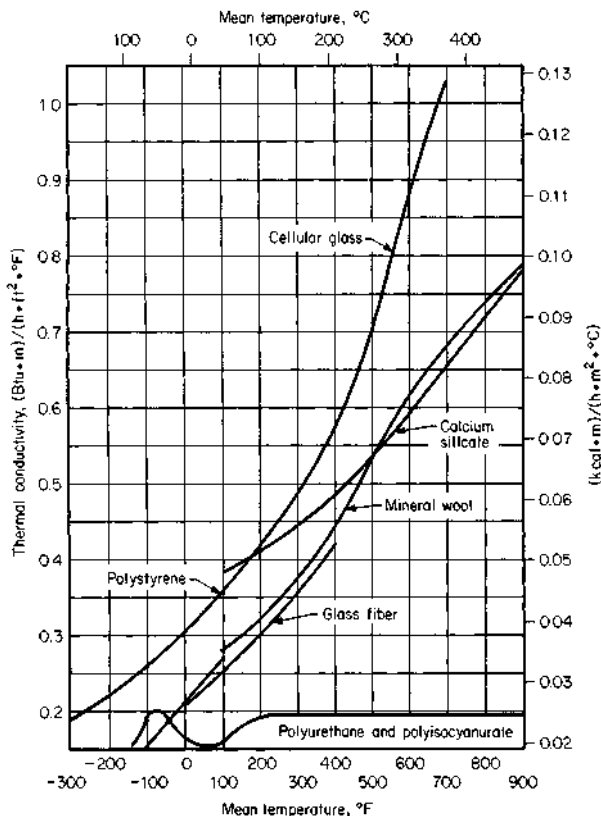


FIG. 11-65 Thermal conductivity of insulating materials.

Finishes for systems operating below 2°C (35°F) must be sealed and retard vapor transmission. Those from 2°C (35°F) through 27°C (80°F) should retard vapor transmission (to prevent surface condensation), and those above 27°C (80°F) should prevent water entry and allow moisture to escape.

Metal finishes are more durable, require less maintenance, reduce heat loss, and, if uncoated, increase the surface temperature on hot systems.

## SYSTEM SELECTION

A combination of insulation and finish produces the thermal-insulation system. Selection of these components depends on the purpose for which the system is to be used. No single system performs satisfactorily from the cryogenic through the elevated-temperature range. Systems operating below freezing have a low vapor pressure, and atmospheric moisture is pushed into the insulation system, while the reverse is true for hot systems. Some general guidelines for system selection follow.

**Cryogenic [−273 to −101°C (−459 to −150°F)] High Vacuum** This technique is based on the Dewar flask, which is a double-walled vessel with reflective surfaces on the evacuated side to reduce radiation losses. Figure 11-66 shows a typical laboratory-size Dewar. Figure 11-67 shows a semiportable type. Radiation losses can be further reduced by filling the cavity with powders such as perlite or silica prior to pulling the vacuum.

**Multilayer** Multilayer systems consist of series of radiation-reflective shields of low emittance separated by fillers or spacers of very low conductance and exposed to a high vacuum.

**Foamed or Cellular** Cellular plastics such as polyurethane and polystyrene do not hold up or perform well in the cryogenic temperature range because of permeation of the cell structure by water vapor, which in turn increases the heat-transfer rate. Cellular glass holds up better and is less permeable.

**Low Temperature [−101 to −1°C (−150 to +30°F)]** Cellular glass, glass fiber, polyurethane foam, and polystyrene foam are frequently used for this service range. A vapor-retarder finish with a

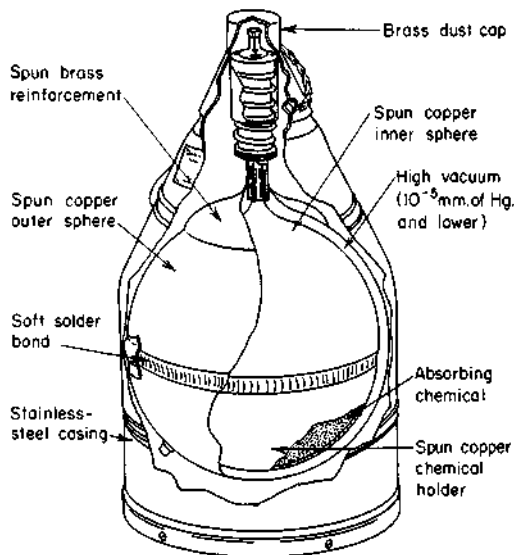


FIG. 11-66 Dewar flask.

perm rating less than 0.02 is required. In addition, it is good practice to coat all contact surfaces of the insulation with a vapor-retardant mastic to prevent moisture migration when the finish is damaged or is not properly maintained. Closed-cell insulation should not be relied on as the vapor retarder. Hairline cracks can develop, cells can break

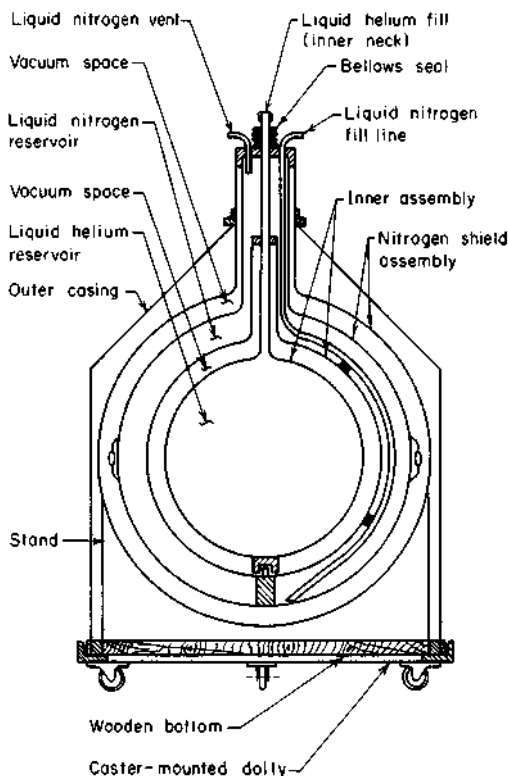


FIG. 11-67 Hydrogen bottle.

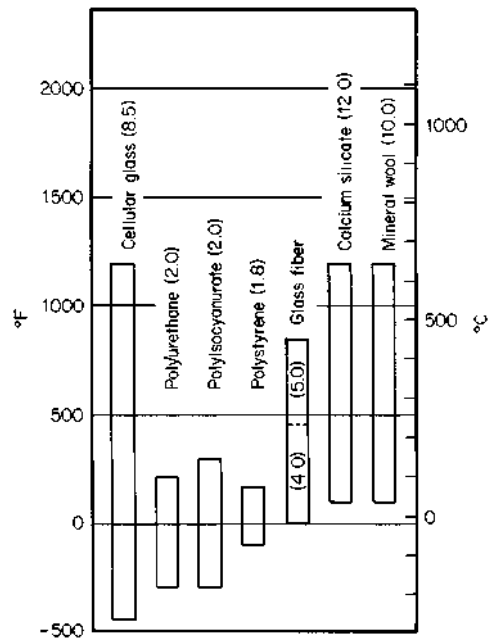


FIG. 11-68 Insulating materials and applicable temperature ranges.

down, glass-fiber binders are absorbent, and moisture can enter at joints between all materials.

**Moderate and High Temperature [over 2°C (36°F)]** Cellular or fibrous materials are normally used. See Fig. 11-68 for nominal temperature range. Nonwicking insulation is desirable for systems operating below 100°C (212°F).

**Other Considerations** Autoignition can occur if combustible fluids are absorbed by wicking-type insulations. **Chloride stress corrosion** of austenitic stainless steel can occur when chlorides are concentrated on metal surfaces at or above approximately 60°C (140°F). The chlorides can come from sources other than the insulation. Some calcium silicates are formulated to exceed the requirements of the MIL-I-24244A specification. **Fire resistance** of insulations varies widely. Calcium silicate, cellular glass, glass fiber, and mineral wool are fire-resistant but do not perform equally under actual fire conditions. A steel jacket provides protection, but aluminum does not.

Traced pipe performs better with a nonwicking insulation which has low thermal conductivity. Underground systems are very difficult to keep dry permanently. Methods of insulation include factory-preinsulated pouring types and conventionally applied types. Corrosion can occur under wet insulation. A protective coating, applied directly to the metal surface, may be required.

### ECONOMIC THICKNESS OF INSULATION

Optimal economic insulation thickness may be determined by various methods. Two of these are the minimum-total-cost method and the incremental-cost method (or marginal-cost method). The minimum-total-cost method involves the actual calculations of lost energy and insulation costs for each insulation thickness. The thickness producing the lowest total cost is the optimal economic solution. The optimum thickness is determined to be the point where the last dollar invested in insulation results in exactly \$1 in energy-cost savings ("ETI—Economic Thickness for Industrial Insulation," Conservation Pap. 46, Federal Energy Administration, August 1976). The incremental-cost method provides a simplified and direct solution for the least-cost thickness.

The total-cost method does *not* in general provide a satisfactory means for making most insulation investment decisions, since an

economic return on investment is required by investors and the method does not properly consider this factor. Return on investment is considered by Rubin ("Piping Insulation—Economics and Profits," in *Practical Considerations in Piping Analysis*, ASME Symposium, vol. 69, 1982, pp. 27–46). The incremental method used in this reference requires that each incremental ½ in of insulation provide the predetermined return on investment. The minimum thickness of installed insulation is used as a base for calculations. The incremental installed capital cost for each additional ½ in of insulation is determined. The energy saved for each increment is then determined. The value of this energy varies directly with the temperature level [e.g., steam at 538°C

(1000°F) has a greater value than condensate at 100°C (212°F)]. The final increment selected for use is required either to provide a satisfactory return on investment or to have a suitable payback period.

**Recommended Thickness of Insulation** Indoor insulation thickness appears in Table 11-22, and outdoor thickness appears in Table 11-23. These selections were based upon calcium silicate insulation with a suitable aluminum jacket. However, the variation in thickness for fiberglass, cellular glass, and rock wool is minimal. Fiberglass is available for maximum temperatures of 260, 343, and 454°C (500, 650, and 850°F). Rock wool, cellular glass, and calcium silicate are used up to 649°C (1200°F).

**TABLE 11-22 Indoor Insulation Thickness, 80°F Still Ambient Air\***

| Pipe size, in | Insulation thickness, in | Minimum pipe temperature, °F |      |      |      |      |      |      |      |
|---------------|--------------------------|------------------------------|------|------|------|------|------|------|------|
|               |                          | Energy cost, \$/million Btu  |      |      |      |      |      |      |      |
|               |                          | 1                            | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
| ¾             | 1½                       | 950                          | 600  | 550  | 400  | 350  | 300  | 250  | 250  |
|               | 2                        |                              |      |      | 1100 | 1000 | 900  | 800  | 750  |
|               | 2½                       |                              |      |      | 1750 | 1050 | 950  | 850  | 800  |
|               | 3                        |                              |      |      |      |      |      |      | 1200 |
| 1             | 1½                       | 1200                         | 800  | 600  | 500  | 450  | 400  | 350  | 300  |
|               | 2                        |                              |      | 1200 | 1000 | 900  | 800  | 700  | 700  |
|               | 2½                       |                              |      |      |      | 1200 | 1050 | 1000 | 900  |
|               | 3                        |                              |      |      |      |      | 1100 | 1150 | 950  |
| 1½            | 1½                       | 1100                         | 750  | 550  | 450  | 400  | 400  | 350  | 300  |
|               | 2                        |                              |      | 1000 | 850  | 700  | 650  | 600  | 500  |
|               | 2½                       |                              |      |      | 1050 | 900  | 800  | 750  | 650  |
|               | 3                        |                              |      |      |      |      | 1150 | 1100 | 1000 |
| 2             | 1½                       | 1050                         | 700  | 500  | 450  | 400  | 350  | 300  | 300  |
|               | 2                        |                              |      | 1050 | 850  | 750  | 700  | 600  | 600  |
|               | 2½                       |                              |      | 1100 | 950  | 1000 | 750  | 700  | 650  |
|               | 3                        |                              |      |      | 1200 | 1050 | 950  | 850  | 800  |
| 3             | 1½                       | 950                          | 650  | 500  | 400  | 350  | 300  | 300  | 250  |
|               | 2                        |                              |      | 900  | 700  | 600  | 550  | 500  | 450  |
|               | 2½                       |                              |      | 1050 | 850  | 750  | 650  | 500  | 500  |
|               | 3                        |                              |      |      | 1050 | 950  | 800  | 750  | 700  |
| 4             | 1½                       | 950                          | 600  | 500  | 400  | 350  | 300  | 300  | 250  |
|               | 2                        |                              |      | 850  | 700  | 600  | 550  | 500  | 450  |
|               | 2½                       |                              |      | 1200 | 1000 | 850  | 750  | 700  | 650  |
|               | 3                        |                              |      |      | 1050 | 900  | 800  | 750  | 700  |
|               | 3½                       |                              |      |      |      |      |      | 1150 | 1050 |
|               | 4                        |                              |      |      |      |      |      |      |      |
| 6             | 1½                       | 600                          | 350  | 300  | 250  | 250  | 200  | 200  | 200  |
|               | 2                        |                              |      | 850  | 700  | 600  | 550  | 500  | 500  |
|               | 2½                       |                              |      | 900  | 800  | 650  | 600  | 550  | 550  |
|               | 3                        |                              |      | 1150 | 1000 | 850  | 750  | 700  | 600  |
|               | 3½                       |                              |      |      |      |      | 1100 | 1000 | 900  |
|               | 4                        |                              |      |      |      |      |      |      | 1200 |
| 8             | 2                        |                              | 1000 | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 2½                       |                              |      | 850  | 700  | 600  | 550  | 500  | 450  |
|               | 3                        |                              |      |      | 1050 | 900  | 800  | 750  | 700  |
|               | 3½                       |                              |      |      |      | 1200 | 1100 | 1000 | 900  |
|               | 4                        |                              |      |      |      |      |      | 1150 | 1100 |
| 10            | 2                        |                              | 1100 | 850  | 700  | 650  | 550  | 500  | 450  |
|               | 2½                       |                              |      | 900  | 750  | 700  | 600  | 550  | 500  |
|               | 3                        |                              |      | 1050 | 900  | 750  | 700  | 600  | 550  |
|               | 3½                       |                              |      |      |      | 1200 | 1050 | 950  | 900  |
|               | 4                        |                              |      |      |      |      |      |      | 1200 |
| 12            | 2                        | 1150                         | 750  | 600  | 500  | 400  | 400  | 350  | 300  |
|               | 2½                       |                              |      | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 3                        |                              |      | 1200 | 1000 | 900  | 800  | 700  | 650  |
|               | 3½                       |                              |      |      |      | 1200 | 1100 | 1000 | 900  |
|               | 4                        |                              |      |      |      |      | 1150 | 1050 | 950  |
|               | 4½                       |                              |      |      |      |      | 1200 | 1100 | 1000 |
| 14            | 2                        | 1050                         | 650  | 550  | 450  | 400  | 350  | 300  | 300  |
|               | 2½                       |                              |      | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 3                        |                              |      | 1100 | 950  | 800  | 700  | 650  | 600  |
|               | 3½                       |                              |      |      |      | 1150 | 1000 | 950  | 850  |
|               | 4                        |                              |      |      |      | 1200 | 1050 | 1000 | 900  |
|               | 4½                       |                              |      |      |      |      | 1200 | 1100 | 1000 |

**TABLE 11-22 Indoor Insulation Thickness, 80°F Still Ambient Air\* (Concluded)**

| Pipe size, in | Insulation thickness, in | Minimum pipe temperature, °F |      |      |     |      |      |      |      |
|---------------|--------------------------|------------------------------|------|------|-----|------|------|------|------|
|               |                          | Energy cost, \$/million Btu  |      |      |     |      |      |      |      |
|               |                          | 1                            | 2    | 3    | 4   | 5    | 6    | 7    | 8    |
| 16            | 2                        | 950                          | 650  | 500  | 400 | 350  | 300  | 300  | 300  |
|               | 2½                       |                              | 1000 | 800  | 700 | 600  | 550  | 500  | 450  |
|               | 3                        |                              | 1200 | 950  | 800 | 700  | 600  | 550  | 500  |
|               | 3½                       |                              |      |      |     | 1150 | 1050 | 950  | 850  |
|               | 4                        |                              |      |      |     | 1200 | 1100 | 1000 | 900  |
|               | 4½                       |                              |      |      |     |      | 1150 | 1050 | 950  |
| 18            | 2                        | 1000                         | 650  | 500  | 400 | 350  | 350  | 300  | 300  |
|               | 2½                       |                              | 950  | 750  | 600 | 550  | 500  | 450  | 400  |
|               | 3                        |                              | 1150 | 900  | 750 | 650  | 550  | 500  | 500  |
|               | 3½                       |                              |      |      |     | 1200 | 1100 | 1000 | 900  |
|               | 4                        |                              |      |      |     |      | 1150 | 1050 | 950  |
|               | 4½                       |                              |      |      |     |      | 1200 | 1100 | 1000 |
| 20            | 2                        | 1050                         | 700  | 550  | 450 | 400  | 350  | 350  | 300  |
|               | 2½                       |                              | 1000 | 800  | 600 | 550  | 500  | 450  | 400  |
|               | 3                        |                              | 1150 | 900  | 750 | 650  | 550  | 500  | 500  |
|               | 3½                       |                              |      |      |     |      | 1100 | 1000 | 950  |
|               | 4                        |                              |      |      |     |      | 1150 | 1050 | 1000 |
|               | 4½                       |                              |      |      |     |      |      | 1200 | 1100 |
| 24            | 2                        | 950                          | 600  | 500  | 400 | 350  | 300  | 300  | 250  |
|               | 2½                       |                              | 1150 | 900  | 750 | 650  | 550  | 500  | 450  |
|               | 3                        |                              |      | 1050 | 900 | 750  | 700  | 600  | 550  |
|               | 3½                       |                              |      |      |     | 1100 | 1000 | 900  | 800  |
|               | 4                        |                              |      |      |     | 1150 | 1050 | 950  | 850  |
|               | 4½                       |                              |      |      |     |      | 1150 | 1050 | 950  |

\*Aluminum-jacketed calcium silicate insulation with an emissivity factor of 0.05. To convert inches to millimeters, multiply by 25.4, to convert dollars per 1 million British thermal units to dollars per 1 million kilojoules, multiply by 0.948, °C = 5/9 (°F – 32).

**TABLE 11-23 Outdoor Insulation Thickness, 7.5-mi/h Wind, 60°F Air\***

| Pipe size, in | Thickness, in | Minimum pipe temperature, °F |      |      |      |      |      |      |      |
|---------------|---------------|------------------------------|------|------|------|------|------|------|------|
|               |               | Energy cost, \$/million Btu  |      |      |      |      |      |      |      |
|               |               | 1                            | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
| ¾             | 1             | 450                          | 300  | 250  | 250  | 200  | 200  | 150  | 150  |
|               | 1½            | 800                          | 500  | 400  | 300  | 250  | 250  | 200  | 200  |
|               | 2             |                              |      | 1150 | 950  | 850  | 750  | 700  | 650  |
|               | 2½            |                              |      | 1100 | 1000 | 900  | 800  | 750  | 700  |
| 1             | 1             | 400                          | 300  | 250  | 200  | 200  | 150  | 150  | 150  |
|               | 1½            | 1000                         | 650  | 500  | 400  | 350  | 300  | 300  | 250  |
|               | 2             |                              |      | 1100 | 900  | 800  | 700  | 600  | 600  |
|               | 2½            |                              |      |      | 1200 | 1050 | 950  | 850  | 800  |
|               | 3             |                              |      |      |      | 1100 | 1000 | 900  | 850  |
| 1½            | 1             | 350                          | 250  | 200  | 200  | 150  | 150  | 150  | 150  |
|               | 1½            | 900                          | 600  | 450  | 350  | 300  | 300  | 250  | 250  |
|               | 2             |                              | 100  | 850  | 700  | 600  | 550  | 500  | 450  |
|               | 2½            |                              |      | 1150 | 950  | 800  | 750  | 700  | 600  |
|               | 3             |                              |      |      |      | 1200 | 1050 | 1000 | 900  |
| 2             | 1             | 350                          | 250  | 200  | 150  | 150  | 150  | 150  | 150  |
|               | 1½            | 900                          | 550  | 450  | 400  | 300  | 300  | 250  | 250  |
|               | 2             |                              | 1150 | 900  | 750  | 650  | 600  | 550  | 500  |
|               | 2½            |                              |      | 1000 | 850  | 750  | 650  | 600  | 550  |
|               | 3             |                              |      |      | 1050 | 950  | 850  | 750  | 700  |
| 3             | 1             | 300                          | 200  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 1½            | 750                          | 500  | 400  | 300  | 250  | 250  | 250  | 200  |
|               | 2             |                              | 950  | 750  | 600  | 500  | 450  | 400  | 350  |
|               | 2½            |                              | 1150 | 950  | 750  | 650  | 600  | 500  | 500  |
|               | 3             |                              |      | 1150 | 1000 | 850  | 750  | 650  | 600  |
|               | 3½            |                              |      |      |      |      |      |      | 1150 |
| 4             | 1             | 250                          | 200  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 1½            | 750                          | 500  | 350  | 300  | 250  | 250  | 200  | 200  |
|               | 2             |                              | 950  | 750  | 600  | 500  | 450  | 400  | 350  |
|               | 2½            |                              |      | 1050 | 900  | 700  | 650  | 600  | 550  |
|               | 3             |                              |      | 1100 | 950  | 750  | 700  | 650  | 600  |
|               | 3½            |                              |      |      |      |      | 1200 | 1100 | 1000 |

**TABLE 11-23 Outdoor Insulation Thickness, 7.5-mi/h Wind, 60°F Air\* (Concluded)**

| Pipe size, in | Thickness, in | Minimum pipe temperature, °F |      |      |      |      |      |      |      |
|---------------|---------------|------------------------------|------|------|------|------|------|------|------|
|               |               | Energy cost, \$/million Btu  |      |      |      |      |      |      |      |
|               |               | 1                            | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
| 6             | 1             | 250                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 1½            | 450                          | 300  | 200  | 200  | 150  | 150  | 150  | 150  |
|               | 2             |                              | 900  | 700  | 600  | 500  | 450  | 400  | 350  |
|               | 2½            |                              | 1050 | 800  | 650  | 600  | 500  | 450  | 400  |
|               | 3             |                              |      | 1050 | 900  | 750  | 700  | 600  | 550  |
|               | 3½            |                              |      |      |      | 1150 | 1050 | 950  | 850  |
|               | 4             |                              |      |      |      |      |      | 1200 | 1150 |
| 8             | 1             | 250                          | 200  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             |                              | 850  | 650  | 550  | 450  | 400  | 350  | 350  |
|               | 2½            |                              | 900  | 700  | 600  | 500  | 450  | 400  | 400  |
|               | 3             |                              |      | 1100 | 950  | 800  | 750  | 700  | 600  |
|               | 3½            |                              |      |      |      | 1150 | 1000 | 950  | 850  |
|               | 4             |                              |      |      |      |      |      | 1050 | 1000 |
|               | 4½            |                              |      |      |      |      |      |      |      |
| 10            | 2             | 200                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2½            |                              | 1000 | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 3             |                              | 1200 | 950  | 800  | 700  | 600  | 550  | 500  |
|               | 3½            |                              |      |      |      | 1100 | 1000 | 900  | 800  |
|               | 4             |                              |      |      |      |      |      | 1150 | 1050 |
|               | 4½            |                              |      |      |      |      |      | 1200 | 1100 |
|               |               |                              |      |      |      |      |      |      |      |
| 12            | 1½            | 250                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 950                          | 600  | 500  | 400  | 350  | 300  | 250  | 250  |
|               | 2½            |                              | 900  | 700  | 550  | 500  | 400  | 400  | 350  |
|               | 3             |                              |      | 1100 | 900  | 800  | 700  | 650  | 550  |
|               | 3½            |                              |      |      |      | 1100 | 1000 | 900  | 850  |
|               | 4             |                              |      |      |      | 1150 | 1050 | 950  | 900  |
|               | 4½            |                              |      |      |      | 1200 | 1100 | 1000 | 950  |
| 14            | 1½            | 250                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 850                          | 550  | 400  | 350  | 300  | 250  | 250  | 250  |
|               | 2½            |                              | 850  | 650  | 550  | 500  | 400  | 400  | 400  |
|               | 3             |                              |      | 1000 | 850  | 700  | 650  | 550  | 500  |
|               | 3½            |                              |      |      | 1200 | 1000 | 950  | 850  | 800  |
|               | 4             |                              |      |      |      | 1050 | 1000 | 900  | 850  |
|               | 4½            |                              |      |      |      |      | 1100 | 1000 | 950  |
| 16            | 1½            | 250                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 800                          | 500  | 350  | 300  | 300  | 250  | 250  | 200  |
|               | 2½            |                              | 900  | 700  | 550  | 500  | 450  | 400  | 350  |
|               | 3             |                              | 1000 | 850  | 700  | 600  | 500  | 450  | 400  |
|               | 3½            |                              |      |      | 1200 | 1000 | 950  | 850  | 800  |
|               | 4             |                              |      |      |      | 1100 | 1000 | 900  | 850  |
|               | 4½            |                              |      |      |      | 1150 | 1000 | 950  | 900  |
| 18            | 1½            | 250                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 850                          | 550  | 400  | 350  | 300  | 250  | 250  | 200  |
|               | 2½            |                              | 800  | 650  | 500  | 450  | 400  | 350  | 350  |
|               | 3             |                              | 1000 | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 3½            |                              |      |      |      | 1100 | 1000 | 900  | 850  |
| 20            | 1½            | 150                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 900                          | 550  | 450  | 350  | 300  | 300  | 250  | 250  |
|               | 2½            |                              | 850  | 650  | 550  | 450  | 400  | 350  | 350  |
|               | 3             |                              | 1000 | 800  | 650  | 550  | 500  | 450  | 400  |
|               | 3½            |                              |      |      |      | 1150 | 1050 | 950  | 900  |
|               | 4             |                              |      |      |      | 1200 | 1100 | 1000 | 950  |
| 24            | 1½            | 150                          | 150  | 150  | 150  | 150  | 150  | 150  | 150  |
|               | 2             | 800                          | 500  | 400  | 300  | 250  | 250  | 200  | 200  |
|               | 2½            |                              | 950  | 750  | 650  | 550  | 500  | 450  | 400  |
|               | 3             |                              | 1150 | 950  | 750  | 650  | 600  | 550  | 500  |
|               | 3½            |                              |      |      | 1150 | 1000 | 900  | 800  | 750  |
|               | 4             |                              |      |      | 1200 | 1050 | 950  | 850  | 800  |
|               | 4½            |                              |      |      |      |      | 1050 | 950  | 850  |
|               | 5             |                              |      |      |      |      |      |      |      |

\*Aluminum-jacketed calcium silicate insulation with an emissivity factor of 0.05. To convert inches to millimeters, multiply by 25.4; to convert miles per hour to kilometers per hour, multiply by 1.609; and to convert dollars per 1 million British thermal units to dollars per 1 million kilojoules, multiply by 0.948; °C = 5/9 (°F – 32).

The tables were based upon the cost of energy at the end of the first year, a 10 percent inflation rate on energy costs, a 15 percent interest cost, and a present-worth pretax profit of 40 percent per annum on the last increment of insulation thickness. Dual-layer insulation was used for 3½-in and greater thicknesses. The tables and a full explanation of their derivation appear in a paper by F. L. Rubin (op. cit.). Alternatively, the selected thicknesses have a payback period on the last nominal ½-in increment of 1.44 years as presented in a later paper by Rubin ["Can You Justify More Piping Insulation?" *Hydrocarbon Process.*, 152–155 (July 1982)].

**Example 1** For 24-in pipe at 371°C (700°F) with an energy cost of \$4/million Btu, select 2-in thickness for indoor and 2½-in for outdoor locations. [A 2½-in thickness would be chosen at 399°C (750°F) indoors and 3½-in outdoors.]

**Example 2** For 16-in pipe at 343°C (650°F) with energy valued at \$5/million Btu, select 2½-in insulation indoors [use 3-in thickness at 371°C (700°F)]. Outdoors choose 3-in insulation [use 3½-in dual-layer insulation at 535°C (1000°F)].

**Example 3** For 12-in pipe at 593°C (1100°F) with an energy cost of \$6/million Btu, select 3½-in thickness for an indoor installation and 4½-in thickness for an outdoor installation.

## INSTALLATION PRACTICE

**Pipe** Depending on diameter, pipe is insulated with cylindrical half, third, or quarter sections or with flat segmental insulation. Fittings and valves are insulated with preformed insulation covers or with individual pieces cut from sectional straight pipe insulation.

**Method of Securing** Insulation with factory-applied jacketing may be secured with adhesive on the overlap, staples, tape, or wire, depending on the type of jacket and the outside diameter. Insulation which has a separate jacket is wired or banded in place before the jacket (finish) is applied.

**Double Layer** Pipe expansion is a significant factor at temperatures above 600°F (316°C). Above this temperature, insulation should be applied in a double layer with all joints staggered to prevent excessive heat loss and high surface temperature at joints opened by pipe

expansion. This procedure also minimizes thermal stresses in the insulation.

**Finish** Covering for cylindrical surfaces ranges from asphalt-saturated or saturated and coated organic and asbestos paper, through laminates of such papers and plastic films or aluminum foil, to medium-gauge aluminum, galvanized steel, or stainless steel. Fittings and irregular surfaces may be covered with fabric-reinforced mastics or preformed metal or plastic covers. Finish selection depends on function and location. Vapor-barrier finishes may be in sheet form or a mastic, which may or may not require reinforcing, depending on the method of application, and additional protection may be required to prevent mechanical abuse and/or provide fire resistance. Criteria for selecting other finishes should include protection of insulation against water entry, mechanical abuse, or chemical attack. Appearance, life-cycle cost, and fire resistance may also be determining factors. Finish may be secured with tape, adhesive, bands, or screws. Fasteners which will penetrate vapor-retarder finishes should not be used.

**Tanks, Vessels, and Equipment** Flat, curved, and irregular surfaces such as tanks, vessels, boilers, and breechings are normally insulated with flat blocks, beveled lags, curved segments, blankets, or spray-applied insulation. Since no general procedure can apply to all materials and conditions, it is important that manufacturers' specifications and instructions be followed for specific insulation applications.

**Method of Securing** On small-diameter cylindrical vessels, the insulation may be secured by banding around the circumference. On larger cylindrical vessels, banding may be supplemented with angle-iron ledges to support the insulation and prevent slipping. On large flat and cylindrical surfaces, banding or wiring may be supplemented with various types of welded studs or pins. Breather springs may be required with bands to accommodate expansion and contraction.

**Finish** The materials are the same as for pipe and should satisfy the same criteria. Breather springs may be required with bands.

**ADDITIONAL REFERENCES:** ASHRAE *Handbook and Product Directory: Fundamentals*, American Society of Heating, Refrigerating and Air Conditioning Engineers, Atlanta, 1981. Turner and Malloy, *Handbook of Thermal Insulation Design Economics for Pipes and Equipment*, Krieger, New York, 1980. Turner and Malloy, *Thermal Insulation Handbook*, McGraw-Hill, New York, 1981.

## AIR CONDITIONING

### INTRODUCTION

Air Conditioning is the process of treating air so as to control simultaneously its temperature, humidity, cleanliness, and distribution to meet the requirements of the conditioned space. The portions relating only to temperature and humidity control will be discussed here. For detailed discussions of air cleanliness and distribution, refer, for example, to the current edition of the *HVAC Applications* volume of the A.S.H.R.A.E. *Handbooks* (ASHRAE, 1791 Tullie Circle, N.E., Atlanta, Ga.). Applications of air conditioning include the promotion of human comfort and the maintenance of proper conditions for the manufacture, processing, and preserving of material and equipment. Also, in industrial environments where, for economical or other reasons, conditions cannot be made entirely comfortable, air conditioning may be used for maintaining the efficiency and health of workers at safe tolerance limits.

### COMFORT AIR CONDITIONING

Comfort is influenced by temperature, humidity, air velocity, radiant heat, clothing, and work intensity. Psychological factors may also influence comfort, but their discussion is beyond the scope of this handbook. The reader is referred to Chap. 42 of the *HVAC Applications* volume of the A.S.H.R.A.E. *Handbooks* for a full discussion of the control of noise, which must also be considered in air-conditioning design. Figure 5 in Chap. 8 of the *HVAC Fundamentals* volume of the A.S.H.R.A.E. *Handbooks* relates the variables of ambient temperature, dew point temperature (or humidity ratio) to comfort under clothing and activity conditions typical for office space occupancy. It also shows boundary values for ET\*, the effective temperature index. This index combines temperature and moisture conditions into a

single index representing the temperature of an environment at 50 percent relative humidity resulting in the same heat transfer from the skin as for the actual case. Hence, the ET\* for 50 percent relative humidity is equal in value to the ambient dry-bulb temperature.

### INDUSTRIAL AIR CONDITIONING

Industrial buildings have to be designed according to their intended use. For instance, the manufacture of hygroscopic materials (paper, textiles, foods, etc.) will require relatively tight controls of relative humidity. On the other hand, the storage of furs will demand relatively low temperatures, while the ambient in a facility manufacturing refractories might be acceptable at notably higher temperatures. Chapter 12 of the *HVAC Applications* volume of the A.S.H.R.A.E. *Handbooks* provides extensive tables of suggested temperatures and humidities for industrial air conditioning.

### VENTILATION

In the design of comfort air-conditioning systems, odors arising from occupants, cooking, or other sources must be controlled. This is accomplished by introducing fresh air or purified recirculated air in sufficient quantities to reduce odor concentrations to an acceptable level by dilution. Recommended fresh-air requirements for different types of buildings are called for in A.S.H.R.A.E. standard 62-1989 "Ventilation for Acceptable Indoor Air Quality." These values range in the order of 15 to 30 cfm per person, according to application.

In industrial air-conditioning systems, harmful environmental gases, vapors, dusts, and fumes are often encountered. These contaminants can be controlled by exhaust systems at the source, by dilution

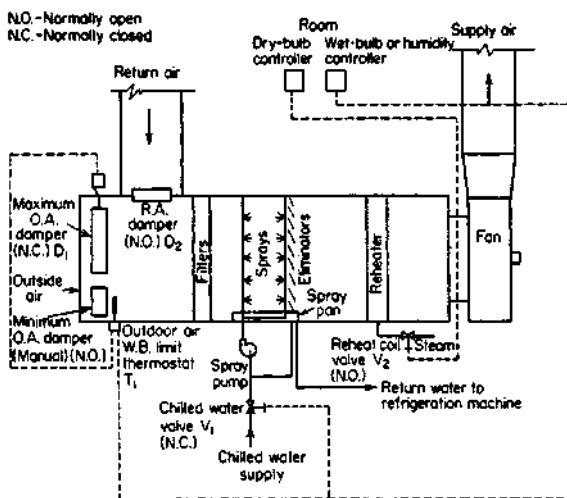
ventilation, or by a combination of the two methods. When exhaust systems are used, it is necessary to introduce sufficient fresh air into the air-conditioned area to make up for that exhausted. Generally, low exhaust systems are used where the contaminant sources are concentrated and/or where the contaminant may be highly toxic. Where the contaminant comes from widely dispersed points, however, dilution ventilation is usually employed. Combinations of the two systems sometimes provide the least expensive installation. Dilution ventilation is not appropriate for cases where large volumes of contaminant are released and cases where the employees must work near the contaminant source. The selection of dilution ventilation for cases with potential fire or smoke should be accompanied by careful study. Details for design of dilution ventilation systems are given in Chap. 25 of the *HVAC Applications* volume of the A.S.H.R.A.E. *Handbooks*. Chapter 27 of the same volume discusses industrial exhaust systems. Exhaust stacks should be high enough to adequately disperse the contaminated air and to prevent recirculation into fresh air intakes (Chap. 14 of the 1993 *HVAC Fundamentals* volume of the A.S.H.R.A.E. *Handbooks*). Depending on the contaminant and air pollution legislation, it may be necessary to reduce the contaminant emission rate by such methods as filtering, scrubbing, catalytic oxidation, or incineration.

## AIR-CONDITIONING EQUIPMENT

Basically, an air-conditioning system consists of a fan unit which forces a mixture of fresh outdoor air and room air through a series of devices which act upon the air to clean it, to increase or decrease its temperature, and to increase or decrease its water-vapor content or humidity. In general, air conditioning equipment can be classified into two broad types: central (sometimes called field erected) and unitary (or packaged).

## CENTRAL SYSTEMS

Figure 11-69 describes a typical central system. Either water or direct-expansion refrigerant coils or air washers may be used for cooling. Steam or hot-water coils are available for heating. Humidification may be provided by target-type water nozzles, pan humidifiers, air washers,



**FIG. 11-69** Typical central-station air-conditioning unit and control system. On a rising room wet-bulb temperature, the wet-bulb branch-line air pressure increases through the reverse-acting outdoor-air wet-bulb temperature-limit thermostat  $T_1$  to open gradually the maximum outdoor-air damper  $D_1$  and simultaneously close return-air damper  $D_2$ , then gradually open chilled-water valve  $V_1$ . On a rising room dry-bulb temperature, the dry-bulb branch-line air pressure gradually increases to close reheat steam valve  $V_2$ . When outdoor wet-bulb temperature exceeds the set point of the outdoor-air wet-bulb-limit thermostat  $T_1$ , which is set at the return-air wet-bulb temperature, this thermostat decreases branch-line pressure to close gradually maximum outdoor damper  $D_1$  and simultaneously open return-air damper  $D_2$ . The reverse sequences are followed during the heating season.

or sprayed coils. Air cleaning is usually provided by cleanable or throw-away filters. Central-station air-conditioning units in capacities up to about 50,000 cu ft/min are available in prefabricated units.

The principle types of refrigeration equipment used in large central systems are: Reciprocating (up to 300 hp); helical rotary (up to 750 tons); absorption (up to 2000 tons); and centrifugal (up to 10,000 tons). The drives for the reciprocating, rotary, and centrifugal compressors may be electric motors, gas or steam turbines, or gas or diesel engines. The heat rejected from the condensers usually calls for cooling towers or air-cooled condensers; in some cases evaporative cooling might be practical.

## UNITARY REFRIGERANT-BASED AIR-CONDITIONING SYSTEMS

These systems include window-mounted air conditioners and heat pumps, outdoor unitary equipment, indoor unitary equipment, unitary self-contained systems, and commercial self-contained systems. These are described in detail in the *HVAC Systems and Equipment* volume of the *A.S.H.R.A.E. Handbooks*. A detailed analysis of the proposed installation is usually necessary to select the air conditioning equipment which is best in overall performance. Each type of air conditioner has its own particular advantages and disadvantages. Important factors to be considered in the selection of air conditioning equipment are degree of temperature and humidity control required, investment, owning, and operating costs, and space requirements. Another important factor is the building itself, that is, whether it is new or existing construction. For example, for existing buildings where it may be inadvisable to install air-supply ducts, the self-contained or unit-type air conditioner may offer the greatest advantages in reduced installation costs. For large industrial processes where close temperature and humidity control are required, a central station system is usually employed.

## LOAD CALCULATION

First step in the solution of an air-conditioning problem is to determine the proper design temperature conditions. Since both outdoor and indoor temperatures greatly influence the size of the equipment, the designer must exercise good judgment in selecting the proper conditions for his/her particular case. Table 11-24 lists winter and summer outdoor temperature conditions in common use for comfort applications for various United States cities. For critical-process air conditioning, it may be desirable to use a different set of outdoor temperature conditions. However, it is seldom good practice to design for the extreme maximum or minimum outside conditions. (See the 1993 *HVAC Fundamentals* volume of the *A.S.H.R.A.E. Handbooks*).

After the proper inside and summer outside temperature conditions for comfort and temperature conditions for process air conditioning have been selected, the next step is to calculate the space cooling load which is made up of sensible heat and latent heat loads. The sensible heat load consists of (1) transmission through walls, roofs, floors, ceilings, and window glass, (2) solar and sky radiation, (3) heat gains from infiltration of outside air, (4) heat gains from people, lights, appliances, and power equipment (including the supply-air fan motor), and (5) heat to be removed from materials or products brought in at higher than room temperature. The latent heat load includes loads due to moisture (1) given off from people, appliances, and products and (2) from infiltration of outside air. The space total heat load is the sum of the sensible heat load and latent heat load of the space. The total refrigeration load consists of the total space load plus the sensible and latent heat loads from the outside air introduced at the conditioning unit.

The procedure for load calculation in nonresidential buildings should account for thermal mass (storage) effects as well as occupancy and other uses affecting the load. The load can in turn be strongly dependent on the nature of the building utilization; as an example, lighting might be a major component in the thermal load for a high-rise office building causing a need for cooling even in winter days. There are various approaches to load calculation, some requiring elaborate computer models. Chapter 26 of the 1993 *HVAC Fundamentals* volume of the A.S.H.R.A.E. *Handbooks* presents a step-by-step outline of the current methods in practice for load calculation.

TABLE 11-24 Outdoor Design Temperatures\*

| City-state          | Winter           | Summer           |                  | City-state            | Winter           | Summer           |                  |
|---------------------|------------------|------------------|------------------|-----------------------|------------------|------------------|------------------|
|                     | Dry bulb,<br>°F. | Dry bulb,<br>°F. | Wet bulb,<br>°F. |                       | Dry bulb,<br>°F. | Dry bulb,<br>°F. | Wet bulb,<br>°F. |
| Akron, Ohio         | -5               | 95               | 75               | Milwaukee, Wis.       | -15              | 95               | 75               |
| Albany, N.Y.        | -10              | 93               | 75               | Minneapolis, Minn.    | -20              | 95               | 75               |
| Albuquerque, N.M.   | 0                | 95               | 70               | Nashville, Tenn.      | 0                | 95               | 78               |
| Atlanta, Ga.        | 10               | 95               | 76               | New Haven, Conn.      | 0                | 95               | 75               |
| Baltimore, Md.      | 0                | 95               | 78               | New Orleans, La.      | 20               | 95               | 80               |
| Billings, Mont.     | -25              | 90               | 66               | New York, N.Y.        | 0                | 95               | 75               |
| Birmingham, Ala.    | 10               | 95               | 78               | Newark, N.J.          | 0                | 95               | 75               |
| Bloomfield, N.J.    | 0                | 95               | 75               | Norfolk, Va.          | 15               | 95               | 78               |
| Boise, Idaho        | -10              | 95               | 65               | Oakland, Calif.       | 30               | 85               | 65               |
| Boston, Mass.       | 0                | 92               | 75               | Oklahoma City, Okla.  | 0                | 101              | 77               |
| Bridgeport, Conn.   | 0                | 95               | 75               | Omaha, Nebr.          | -10              | 95               | 78               |
| Buffalo, N.Y.       | -5               | 93               | 73               | Peoria, Ill.          | -10              | 96               | 76               |
| Charleston, S.C.    | 15               | 95               | 78               | Philadelphia, Pa.     | 0                | 95               | 78               |
| Chattanooga, Tenn.  | 10               | 95               | 76               | Phoenix, Ariz.        | 0                | 105              | 76               |
| Chicago, Ill.       | -10              | 95               | 75               | Pittsburgh, Pa.       | -5               | 95               | 75               |
| Cincinnati, Ohio    | 0                | 95               | 78               | Portland, Me.         | -5               | 90               | 73               |
| Cleveland, Ohio     | 0                | 95               | 75               | Portland, Ore.        | 10               | 90               | 68               |
| Columbus, Ohio      | -10              | 95               | 76               | Providence, R.I.      | 0                | 93               | 75               |
| Dallas, Tex.        | 0                | 100              | 78               | Reno, Nev.            | -5               | 95               | 65               |
| Dayton, Ohio        | 0                | 95               | 78               | Richmond, Va.         | 15               | 95               | 78               |
| Denver, Colo.       | -10              | 95               | 64               | Roanoke, Va.          | 0                | 95               | 76               |
| Des Moines, Iowa    | -15              | 95               | 78               | Rochester, N.Y.       | -5               | 95               | 75               |
| Detroit, Mich.      | -10              | 95               | 75               | St. Louis, Mo.        | 0                | 95               | 78               |
| Duluth, Minn.       | -25              | 93               | 73               | St. Paul, Minn.       | -20              | 95               | 75               |
| East Orange, N.J.   | 0                | 95               | 75               | Salt Lake City, Utah  | -10              | 95               | 65               |
| El Paso, Tex.       | 10               | 100              | 69               | San Antonio, Tex.     | 20               | 100              | 78               |
| Erie, Pa.           | -5               | 93               | 75               | San Francisco, Calif. | 35               | 85               | 65               |
| Fitchburg, Mass.    | -10              | 93               | 75               | Schenectady, N.Y.     | -10              | 93               | 75               |
| Flint, Mich.        | -10              | 95               | 75               | Scranton, Pa.         | -5               | 95               | 75               |
| Fort Wayne, Ind.    | -10              | 95               | 75               | Seattle, Wash.        | 15               | 85               | 65               |
| Fort Worth, Tex.    | 10               | 100              | 78               | Shreveport, La.       | 20               | 100              | 78               |
| Grand Rapids, Mich. | -10              | 95               | 75               | Sioux City, Iowa      | -20              | 95               | 78               |
| Hartford, Conn.     | 0                | 93               | 75               | Spokane, Wash.        | -15              | 93               | 65               |
| Houston, Tex.       | 20               | 95               | 80               | Springfield, Mass.    | -10              | 93               | 75               |
| Indianapolis, Ind.  | -10              | 95               | 76               | Syracuse, N.Y.        | -10              | 93               | 75               |
| Jacksonville, Fla.  | 25               | 95               | 78               | Tampa, Fla.           | 30               | 95               | 78               |
| Jersey City, N.J.   | 0                | 95               | 75               | Toledo, Ohio          | -10              | 95               | 75               |
| Kansas City, Mo.    | -10              | 100              | 76               | Tucson, Ariz.         | 25               | 105              | 72               |
| Lincoln, Nebr.      | -10              | 95               | 78               | Tulsa, Okla.          | 0                | 101              | 77               |
| Little Rock, Ark.   | 5                | 95               | 78               | Washington, D.C.      | 0                | 95               | 78               |
| Long Beach, Calif.  | 35               | 90               | 70               | Wichita, Kans.        | -10              | 100              | 75               |
| Los Angeles, Calif. | 35               | 90               | 70               | Wilmington, Del.      | 0                | 95               | 78               |
| Louisville, Ky.     | 0                | 95               | 78               | Worcester, Mass.      | 0                | 93               | 75               |
| Memphis, Tenn.      | 0                | 95               | 78               | Youngstown, Ohio      | -5               | 95               | 75               |
| Miami, Fla.         | 35               | 91               | 79               |                       |                  |                  |                  |

\*Carrier, Cherne, Grant, and Roberts, *Modern Air Conditioning, Heating, and Ventilating*, 3d ed., p. 531, Pitman, New York, 1959.

## REFRIGERATION

### INTRODUCTION

Refrigeration is a process where heat is transferred from a lower- to a higher-temperature level by doing work on a system. In some systems heat transfer is used to provide the energy to drive the refrigeration cycle. All refrigeration systems are heat pumps ("pumps energy from a lower to a higher potential"). The term heat pump is mostly used to describe refrigeration system applications where heat rejected to the condenser is of primary interest.

There are many means to obtain refrigerating effect, but here three will be discussed: mechanical vapor refrigeration cycles, absorption and steam jet cycles due to their significance for industry.

**Basic Principles** Since refrigeration is the practical application of the thermodynamics, comprehending the basic principles of thermodynamics is crucial for full understanding of refrigeration. Section 4 includes a thorough approach to the theory of thermodynamics. Since our goal is to understand refrigeration processes, cycles are of the crucial interest.

The Carnot refrigeration cycle is reversible and consists of adiabatic (isentropic due to reversible character) compression (1-2), isothermal rejection of heat (2-3), adiabatic expansion (3-4) and isothermal addition of heat (4-1). The temperature-entropy diagram is shown in Fig. 11-70. The Carnot cycle is an unattainable ideal which serves as a standard of comparison and it provides a convenient guide to the temperatures that should be maintained to achieve maximum effectiveness.

The measure of the system performance is *coefficient of performance* (COP). For refrigeration applications COP is the ratio of heat removed from the low-temperature level ( $Q_{low}$ ) to the energy input ( $W$ ):

$$COP_R = \frac{Q_{low}}{W} \quad (11-84)$$

For the heat pump (HP) operation, heat rejected at the high temperature ( $Q_{high}$ ) is the objective, thus:

$$COP_{HP} = \frac{Q_{high}}{W} = \frac{Q + W}{W} = COP_R + 1 \quad (11-85)$$



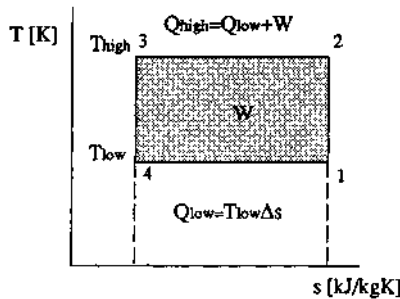


FIG. 11-70 Temperature-entropy diagram of the Carnot cycle.

For a Carnot cycle (where  $\Delta Q = T\Delta s$ ), the COP for the refrigeration application becomes (note that  $T$  is absolute temperature [K]):

$$\text{COP}_R = \frac{T_{\text{low}}}{T_{\text{high}} - T_{\text{low}}} \quad (11-86)$$

and for heat pump application:

$$\text{COP}_{\text{HP}} = \frac{T_{\text{high}}}{T_{\text{high}} - T_{\text{low}}} \quad (11-87)$$

The COP in real refrigeration cycles is always less than for the ideal (Carnot) cycle and there is constant effort to achieve this ideal value.

**Basic Refrigeration Methods** Three basic methods of refrigeration (mentioned above) use similar processes for obtaining refrigeration effect: evaporation in the evaporator, condensation in the condenser where heat is rejected to the environment, and expansion in a flow restrictor. The main difference is in the way compression is being done (Fig. 11-71): using mechanical work (in compressor), thermal energy (for absorption and desorption), or pressure difference (in ejector).

In the next figure (Fig. 11-72) basic refrigeration systems are displayed more detailed. More elaborated approach is presented in the text.

## MECHANICAL REFRIGERATION (VAPOR-COMPRESSION SYSTEMS)

**Vapor-Compression Cycles** The most widely used refrigeration principle is vapor compression. Isothermal processes are realized through isobaric evaporation and condensation in the tubes. Standard vapor compression refrigeration cycle (counterclockwise Rankine cycle) is marked in Fig. 11-72a) by 1, 2, 3, 4.

Work that could be obtained in turbine is small, and iturbine is substituted for an expansion valve. For the reasons of proper compressor

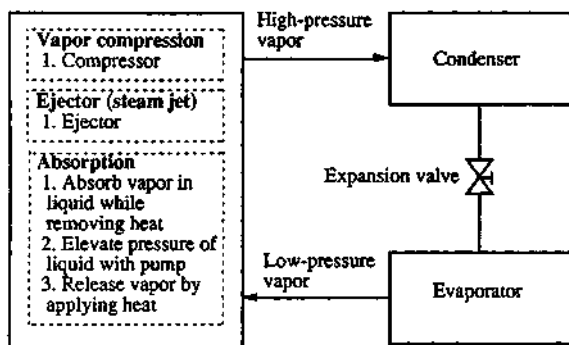


FIG. 11-71 Methods of transforming low-pressure vapor into high-pressure vapor in refrigeration systems (Stoecker, Refrigeration and Air Conditioning).

function, wet compression is substituted for an compression of dry vapor.

Although the  $T$ - $s$  diagram is very useful for thermodynamic analysis, the pressure enthalpy diagram is used much more in refrigeration practice due to the fact that both evaporation and condensation are isobaric processes so that heat exchanged is equal to enthalpy difference  $\Delta Q = \Delta h$ . For the ideal, isentropic compression, the work could be also presented as enthalpy difference  $\Delta W = \Delta h$ . The vapor compression cycle (Rankine) is presented in Fig. 11-73 in  $p$ - $h$  coordinates.

Figure 11-74 presents actual versus standard vapor-compression cycle. In reality, flow through the condenser and evaporator must be accompanied by pressure drop. There is always some subcooling in the condenser and superheating of the vapor entering the compressor-suction line, both due to continuing process in the heat exchangers and the influence of the environment. Subcooling and superheating are usually desirable to ensure only liquid enters the expansion device. Superheating is recommended as a precaution against droplets of liquid being carried over into the compressor.

There are many ways to increase cycle efficiency (COP). Some of them are better suited to one, but not for the other refrigerant. Sometimes, for the same refrigerant, the impact on COP could be different for various temperatures. One typical example is the use of a liquid-to-suction heat exchanger (Fig. 11-75).

The suction vapor coming from the evaporator could be used to subcool the liquid from the condenser. Graphic interpretation in  $T$ - $s$  diagram for such a process is shown in Fig. 11-76. The result of the use of suction line heat exchanger is to increase the refrigeration effect  $\Delta Q$  and to increase the work by  $\Delta W$ . The change in COP is then:

$$\Delta \text{COP} = \text{COP}' - \text{COP} = \frac{Q + \Delta Q}{(P + \Delta P) - Q/P} \quad (11-88)$$

When dry, or superheated, vapor is used to subcool the liquid, the COP in R12 systems will increase, and decrease the COP in  $\text{NH}_3$  systems. For R22 systems it could have both effects, depending on the operating regime. Generally, this measure is advantageous (COP is improved) for fluids with high, specific heat of liquid (less-inclined saturated-liquid line on the  $p$ - $h$  diagram), small heat of evaporation  $h_{fg}$ , when vapor-specific heat is low (isobars in superheated regions are steep), and when the difference between evaporation and condensation temperature is high. Measures to increase COP should be studied for every refrigerant. Sometimes the purpose of the suction-line heat exchanger is not only to improve the COP, but to ensure that only the vapor reaches the compressor, particularly in the case of a malfunctioning expansion valve.

The system shown in Fig. 11-75 is direct expansion where dry or slightly superheated vapor leaves the evaporator. Such systems are predominantly used in small applications because of their simplicity and light weight. For the systems where efficiency is crucial (large industrial systems), recirculating systems (Fig. 11-77) are more appropriate.

Ammonia refrigeration plants are almost exclusively built as recirculating systems. The main advantage of recirculating versus direct expansion systems is better utilization of evaporator surface area. The diagram showing influence of quality on the local heat-transfer coefficients is shown in figure 11-90. It is clear that heat-transfer characteristics will be better if the outlet quality is lower than 1. Circulation could be achieved either by pumping (mechanical or gas) or using gravity (thermosiphon effect: density of pure liquid at the evaporator entrance is higher than density of the vapor-liquid mixture leaving the evaporator). The circulation ratio (ratio of actual mass flow rate to the evaporated mass flow rate) is higher than 1 and up to 5. Higher values are not recommended due to a small increase in heat-transfer rate for a significant increase in pumping costs.

**Multistage Systems** When the evaporation and condensing pressure (or temperature) difference is large, it is prudent to separate compression in two stages. The use of multistage systems opens up the opportunity to use flash-gas removal and intercooling as measures to improve performance of the system. One typical two-stage system with two evaporating temperatures and both flash-gas removal and intercooling is shown in figure 11-78. The purpose of the flash-tank intercooler is to: (1) separate vapor created in the expansion process,

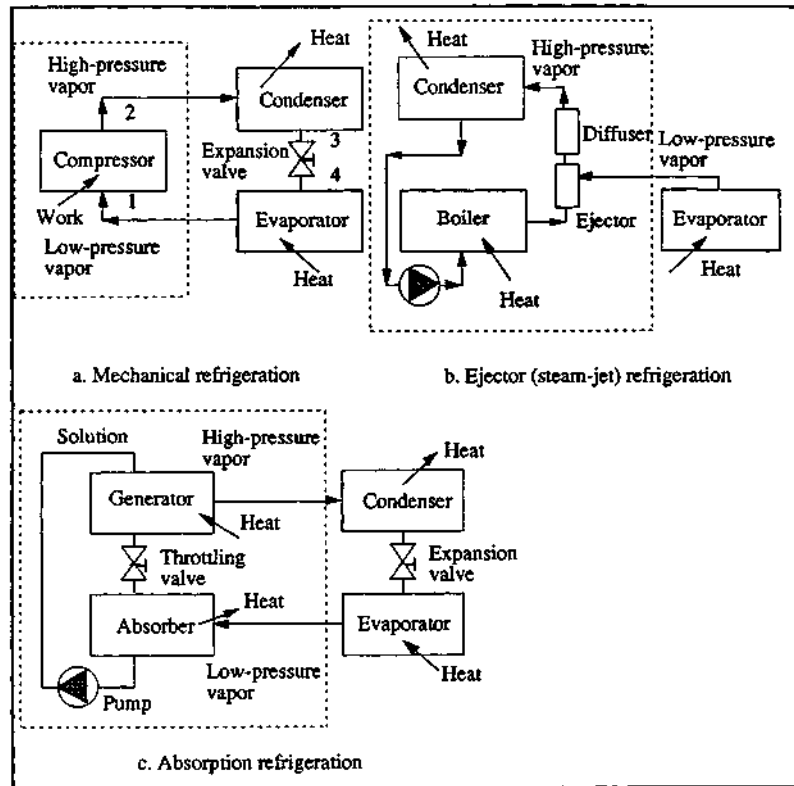


FIG. 11-72 Basic refrigeration systems.

(2) cool superheated vapor from compressor discharge, and (3) to eventually separate existing droplets at the exit of the medium-temperature evaporator. The first measure will decrease the size of the low-stage compressor because it will not wastefully compress the portion of the flow which cannot perform the refrigeration and second will decrease the size of the high-stage compressor due to lowering

the specific volume of the vapor from the low-stage compressor discharge, positively affecting operating temperatures of the high-stage compressor due to cooling effect.

If the refrigerating requirement at a low-evaporating temperature is  $Q_l$  and at the medium level is  $Q_m$ , then mass flow rates ( $m_l$  and  $m_m$

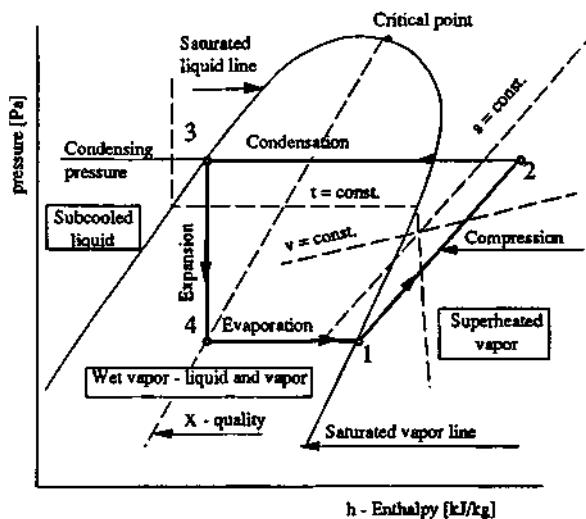


FIG. 11-73  $p$ - $h$  diagram for vapor-compression cycle.

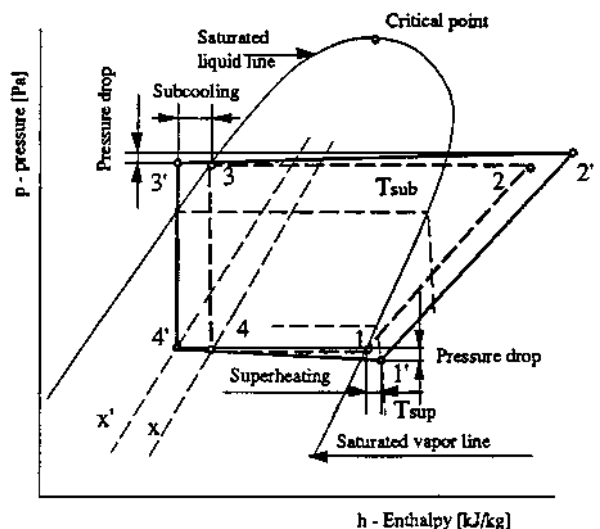


FIG. 11-74 Actual vapor-compression cycle compared with standard cycle.

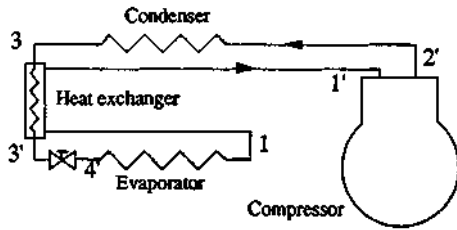


FIG. 11-75 Refrigeration system with a heat exchanger to subcool the liquid from the condenser.

respectively) needed are:

$$m_1 = \frac{Q_l}{h_1 - h_8} = \frac{Q_l}{h_1 - h_7} \quad (11-89)$$

$$m_m = \frac{Q_m}{h_3 - h_6} \quad (11-90)$$

The mass flow rate at the flash-tank inlet  $m_i$  consists of three components ( $m_i = m_1 + m_{sup} + m_{flash}$ ):

$m_1$  = liquid at  $p_m$  feeding low temperature evaporator,

$m_{sup}$  = liquid at  $p_m$  to evaporate in flash tank to cool superheated discharge,

$m_{flash}$  = flashed refrigerant, used to cool remaining liquid.

Vapor component is:

$$m_{flash} = x_m * m_i \quad (11-91)$$

and liquid component is:

$$(1 - x_m) * m_i = m_1 + m_{sup} \quad (11-92)$$

Liquid part of flow to cool superheated compressor discharge is determined by:

$$m_{sup} = \frac{Q_l}{h_1 - h_8} * \frac{h_2 - h_3}{h_3 - h_7} = m_1 * \frac{h_2 - h_3}{h_{igm}} \quad (11-93)$$

Since the quality  $x_m$  is:

$$x_m = \frac{h_6 - h_7}{h_3 - h_7} \quad (11-94)$$

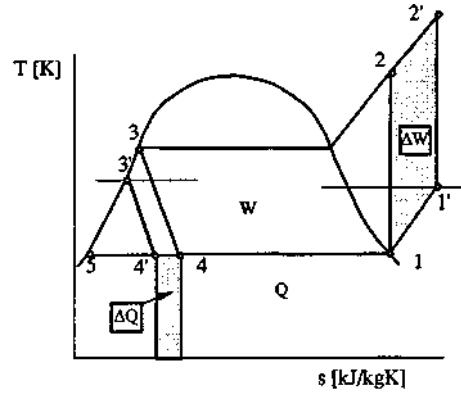


FIG. 11-76 Refrigeration system with a heat exchanger to subcool the liquid from the condenser.

mass flow rate through condenser and high-stage compressor  $m_h$  is finally:

$$m_h = m_m + m_i \quad (11-95)$$

The optimum intermediate pressure for the two-stage refrigeration cycles is determined as the geometric mean between evaporation

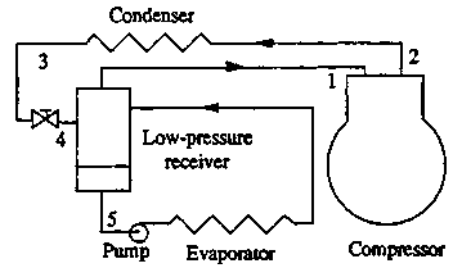


FIG. 11-77 Recirculation system.

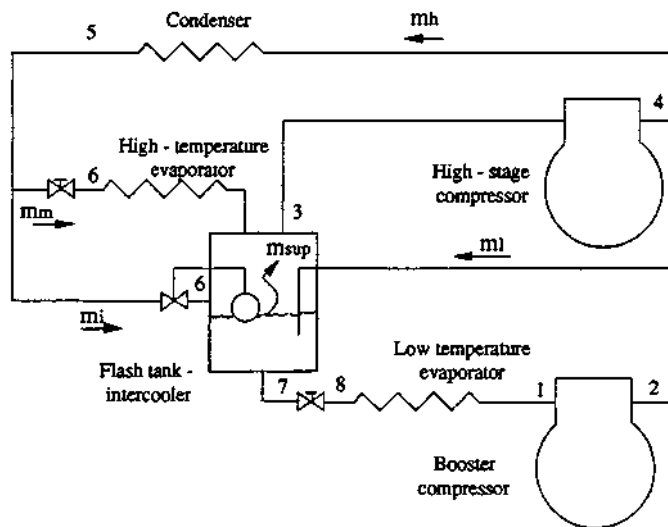


FIG. 11-78 Typical two-stage system with two evaporating temperatures, flash-gas removal, and intercooling.

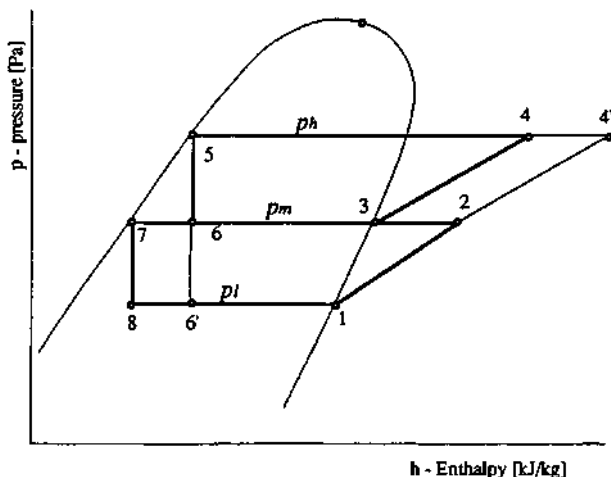


FIG. 11-79 Pressure enthalpy diagram for typical two-stage system with two evaporating temperatures, flash-gas removal, and intercooling.

pressure ( $p_l$ ) and condensing pressure ( $p_h$ , Fig. 11-79):

$$p_m = (\text{sqrt}) \left( \frac{p_h}{p_l} \right)$$

based on equal pressure ratios for low- and high-stage compressors. Optimum interstage pressure is slightly higher than the geometric mean of the suction and the discharge pressures, but, due to very flat optimum of power versus interstage pressure relation geometric mean, it is widely accepted for determining the intermediate pressure. Required pressure of intermediate-level evaporator may dictate interstage pressure other than determined as optimal.

Two-stage systems should be seriously considered when the evaporating temperature is below  $-20^\circ\text{C}$ . Such designs will save on power and reduce compressor discharge temperatures, but will increase initial cost.

**Cascade System** This is a reasonable choice in cases when the evaporating temperature is very low (below  $-60^\circ\text{C}$ ). When condensing pressures are to be in the rational limits, the same refrigerant has a high, specific volume at very low temperatures, requiring a large compressor. The evaporating pressure may be below atmospheric, which could cause moisture and air infiltration into the system if there is a leak. In other words, when the temperature difference between the medium that must be cooled and the environment is too high to be served with one refrigerant, it is wise to use different refrigerants in the high and low stages. Figure 11-80 shows a cascade system

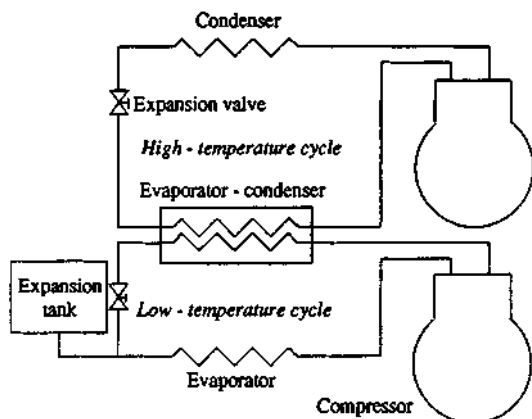


FIG. 11-80 Cascade system.

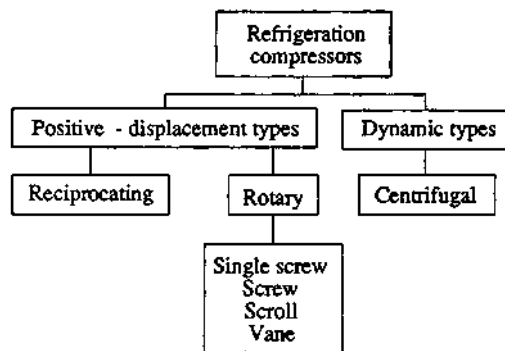


FIG. 11-81 Types of refrigeration compressors.

schematic diagram. There are basically two independent systems linked via a heat exchanger: the evaporator of the high-stage and the condenser of the low-stage system.

## EQUIPMENT

**Compressors** These could be classified by one criteria (the way the increase in pressure is obtained) as positive-displacement and dynamic types as shown in Fig. 11-81 (see Sec. 10 for drawings and mechanical description of the various types of compressors). *Positive-displacement compressors* (PDC) are the machines that increase the pressure of the vapor by reducing the volume of the chamber. Typical PDC are reciprocating (in a variety of types) or rotary as screw (with one and two rotors), vane, scroll, and so on. Centrifugal or turbocompressors are machines where the pressure is raised converting some of kinetic energy obtained by a rotating mechanical element which continuously adds angular momentum to a steadily flowing fluid, similar to a fan or pump.

Generally, reciprocating compressors dominate in the range up to 300 kW refrigeration capacity. Centrifugal compressors are more accepted for the range over 500 kW, while screw compressors are in between with a tendency to go toward smaller capacities. The vane and the scroll compressors are finding their places primarily in very low capacity range (domestic refrigerators and the air conditioners), although vane compressors could be found in industrial compressors. Frequently, screw compressors operate as boosters, for the base load, while reciprocating compressors accommodate the variation of capacity, in the high stage. The major reason is for such design is the advantageous operation of screw compressors near full load and in design conditions, while reciprocating compressors seem to have better efficiencies at part-load operation than screw.

Using other criteria, compressors are classified as *open*, *semihermetic* (*accessible*), or *hermetic*. Open type is characterized by shaft extension out of compressor where it is coupled to the driving motor. When the electric motor is in the same housing with the compressor mechanism, it could be either hermetic or accessible (semihermetic). Hermetic compressors have welded enclosures, not designed to be repaired, and are generally manufactured for smaller capacities (seldom over 30 kW), while semihermetic or an accessible type is located in the housing which is tightened by screws. Semihermetic compressors have all the advantages of hermetic (no sealing of moving parts, e.g., no refrigerant leakage at the seal shaft, no external motor mounting, no coupling alignment) and could be serviced, but it is more expensive.

Compared to other applications, refrigeration capacities in the chemical industry are usually high. That leads to wide usage of either centrifugal, screw, or high-capacity rotary compressors. Most centrifugal and screw compressors use economizers to minimize power and suction volume requirements. Generally, there is far greater use of open-drive type compressors in the chemical plants than in air-conditioning, commercial, or food refrigeration. Very frequently, compressor lube oil systems are provided with auxiliary oil pumps, filters, coolers, and other equipment to permit maintenance and repair without shut down.

**Positive-Displacement Compressors** *Reciprocating compressors* are built in different sizes (up to about one megawatt refrigeration capacity per unit). Modern compressors are high-speed, mostly direct-coupled, single-acting, from one to mostly eight, and occasionally up to sixteen cylinders.

Two characteristics of compressors for refrigeration are the most important: refrigerating capacity and power. Typical characteristics are as presented in the Fig. 11-82.

Refrigerating capacity  $Q$ , is the product of mass flow rate of refrigerant  $m$  and refrigerating effect  $R$  which is (for isobaric evaporation)  $R = h_{\text{evaporator outlet}} - h_{\text{evaporator inlet}}$ . Power  $P$  required for the compression, necessary for the motor selection, is the product of mass flow rate  $m$  and work of compression  $W$ . The latter is, for the isentropic compression,  $W = h_{\text{discharge}} - h_{\text{suction}}$ . Both of these characteristics could be calculated for the ideal (without losses) and for the actual compressor. Ideally, the mass flow rate is equal to the product of the compressor displacement  $V_i$  per unit time and the gas density  $\rho$ :  $m = V_i \cdot \rho$ . The compressor displacement rate is volume swept through by the pistons (product of the cylinder number  $n$ , and volume of cylinder  $V = \text{stroke} \cdot d^2 \pi / 4$ ) per second. In reality, the actual compressor delivers less refrigerant.

Ratio of the actual flow rate (entering compressor) to the displacement rate is the volumetric efficiency  $\eta_{va}$ . The volumetric efficiency is less than unity due to: reexpansion of the compressed vapor in clearance volume, pressure drop (through suction and discharge valves, strainers, manifolds, etc.), internal gas leakage (through the clearance between piston rings and cylinder walls, etc.), valve inefficiencies, and due to expansion of the vapor in the suction cycle caused by the heat exchanged (hot cylinder walls, oil, motor, etc.).

Similar to volumetric efficiency, isentropic (adiabatic) efficiency  $\eta_a$  is the ratio of the work required for isentropic compression of the gas to work input to the compressor shaft. The adiabatic efficiency is less than one mainly due to pressure drop through the valve ports and other restricted passages and the heating of the gas during compression.

Figure 11-83 presents the compression on a pressure-volume diagram for an ideal compressor with clearance volume (thin lines) and

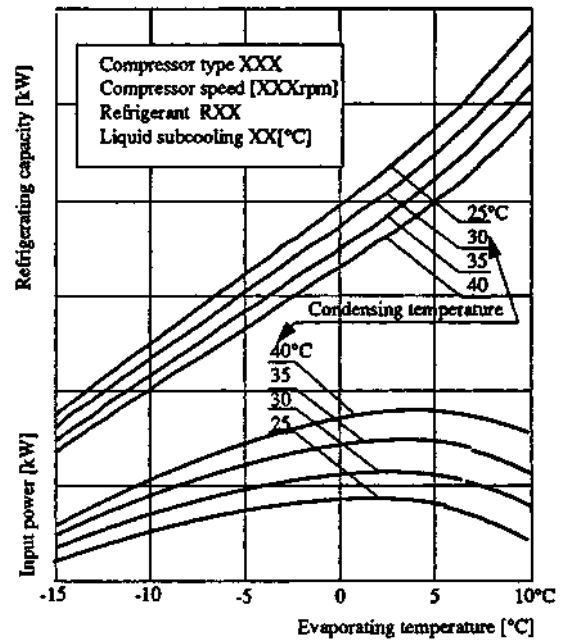


FIG. 11-82 Typical capacity and power-input curves for reciprocating compressor.

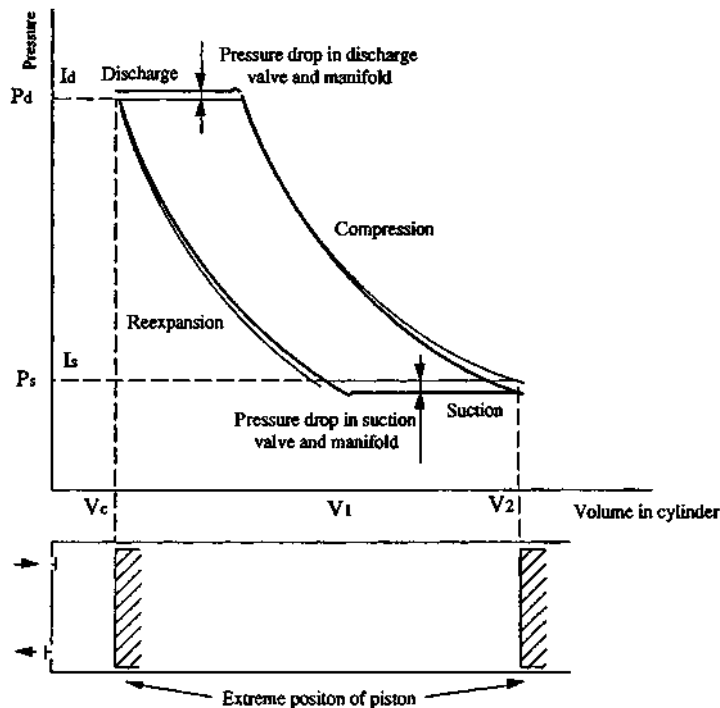


FIG. 11-83 Pressure-volume diagram of an ideal (thin line) and actual (thick line) reciprocating compressor.

actual (thick lines). Compression in an ideal compressor without clearance is extended using dashed lines to the points  $I_d$  (end of discharge), line  $I_d - I_s$  (suction), and  $I_s$  (beginning of suction). The area surrounded by the lines of compression, discharge, reexpansion and intake presents the work needed for compression. Actual compressor only appears to demand less work for compression due to smaller area in the  $p$ - $V$  diagram. Mass flow rate for an ideal compressor is higher, which cannot be seen in the diagram. In reality, an actual compressor will have diabatic compression and reexpansion and higher-discharge and lower-suction pressures due to pressure drops in valves and lines. The slight increase in the pressure at the beginning of the discharge and suction is due to forces needed to initially open valves.

When the suction pressure is lowered, the influence of the clearance will increase, causing in the extreme cases the entire volume to be used for reexpansion, which drives the volumetric efficiency to zero.

There are various options for capacity control of reciprocating refrigeration compressors:

1. Opening the suction valves by some external force (oil from the lubricating system, discharge gas, electromagnets . . .).
2. Gas bypassing—returning discharge gas to suction (within the compressor or outside the compressor).
3. Controlling suction pressure by throttling in the suction line.
4. Controlling discharge pressure.
5. Adding reexpansion volume.
6. Changing the stroke.
7. Changing the compressor speed.

The first method is used most frequently. The next preference is for the last method, mostly used in small compressors due to problems with speed control of electrical motors. Other means of capacity control are very seldom utilized due to thermodynamic inefficiencies and design difficulties. Energy losses in a compressor, when capacity regulation is provided by lifting the suction valves, are due to friction of gas flowing in and out the unloaded cylinder. This is shown in Fig. 11-84 where the comparison is made for ideal partial load operation, reciprocating, and screw compressors.

Rotary compressors are also PDC types, but where refrigerant flow rotates during compression. Unlike the reciprocating type, rotary compressors have a built-in volume ratio which is defined as volume in cavity when the suction port is closed ( $V_s = m \cdot v_s$ ) over the volume in the cavity when the discharge port is uncovered ( $V_d = m \cdot v_d$ ). Built-in volume ratio determines for a given refrigerant and conditions the pressure ratio which is:

$$\frac{p_d}{p_s} = \left( \frac{v_s}{v_d} \right)^n \quad (11-96)$$

where  $n$  represents the polytropic exponent of compression.

In other words, in a reciprocating compressor the discharge valve opens when the pressure in the cylinder is slightly higher than the pressure in the high-pressure side of the system, while in rotary compressors the discharge pressure will be established only by inlet conditions and built-in volume ratio regardless of the system discharge pressure. Very seldom are the discharge and system (condensing)

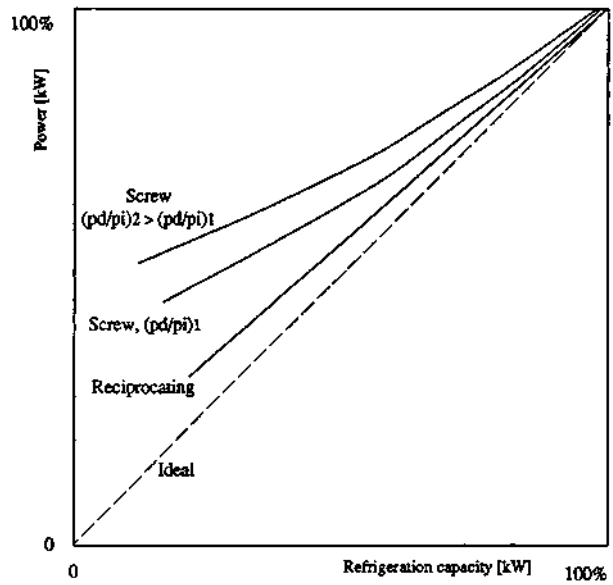


FIG. 11-84 Typical power-refrigeration capacity data for different types of compressors during partial, unloaded operation.

pressure equal, causing the situation shown in Fig. 11-85. When condensing pressure ( $p$ ) is lower than discharge ( $p_2$ ), shown as case (a), "over compression" will cause energy losses presented by the horn on the diagram. If the condensing pressure is higher, in the moment when the discharge port uncovers there will be flow of refrigerant backwards into the compressor, causing losses shown in Fig. 11-85b and the last stage will be only discharge without compression. The case when the compressor discharge pressure is equal to the condensing pressure is shown in the Fig. 11-85c.

Double helical rotary (twin) screw compressors consist of two mating helically grooved rotors (male and female) with asymmetric profile, in a housing formed by two overlapped cylinders, with inlet and outlet ports. Developed relatively recently (in 1930s) the first twin screw compressors were used for air, and later (1950s) became popular for refrigeration. Screw compressors have some advantages over reciprocating compressors (fewer moving parts and more compact) but also some drawbacks (lower efficiency at off-design conditions, as discussed above, higher manufacturing cost due to complicated screw geometry, large separators and coolers for oil which is important as a sealant). Figure 11-86 shows the oil circuit of a screw compressor. Oil cooling could be provided by water, glycol, or refrigerant either in the heat-exchanger-utilizing-thermosiphon effect or the using-direct-expansion concept.

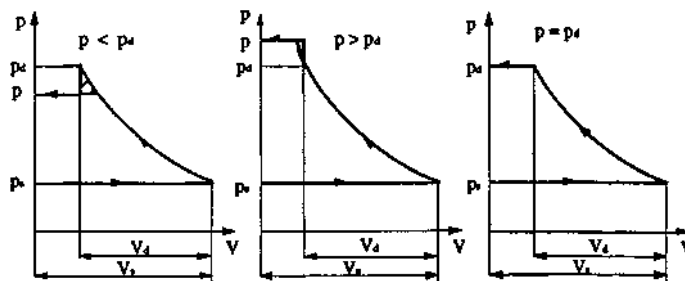


FIG. 11-85 Matching compressor built-in pressure ratio with actual pressure difference.

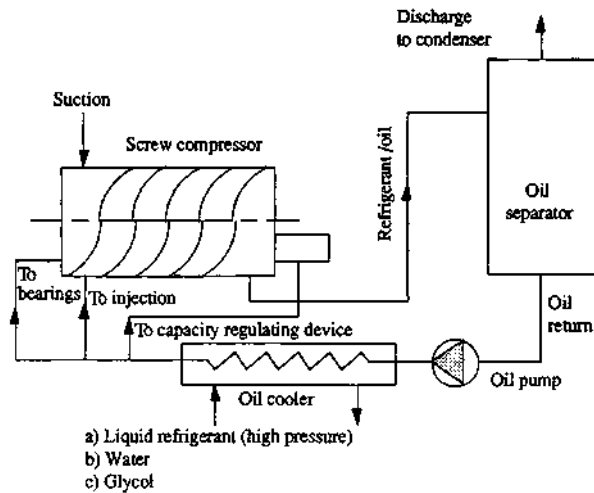


FIG. 11-86 Oil cooling in a screw compressor.

In order to overcome some inherent disadvantages, screw compressors have been initially used predominantly as booster (low-stage) compressors, and following development in capacity control and decreasing prices, they are widely used for high-stage applications. There are several methods for capacity regulation of screw compressors. One is variable speed drive, but a more economical first-cost concept is a slide valve that is used in some form by practically all screw compressors.

The slide is located in the compressor casting below the rotors, allowing internal gas recirculation without compression. Slide valve is operated by a piston located in a hydraulic cylinder and actuated by high-pressure oil from both sides. When the compressor is started, the slide valve is fully open and the compressor is unloaded. To increase capacity, a solenoid valve on the hydraulic line opens, moving the piston in the direction of increasing capacity. In order to increase part-load efficiency, the slide valve is designed to consist of two parts, one traditional slide valve for capacity regulation and other for built-in volume adjustment.

**Single screw compressors** are a newer design (early 1960s) compared to twin screw compressors, and are manufactured in the range of capacity from 100 kW to 4 MW. The compressor screw is cylindrical with helical grooves mated with two star wheels (gaterotors) rotating in opposite direction from one another. Each tooth acts as the piston in the rotating "cylinder" formed by screw flute and cylindrical main-rotor casting.

As compression occurs concurrently in both halves of the compressor, radial forces are oppositely directed, resulting in negligible net-radial loads on the rotor bearings (unlike twin screw compressors), but there are some loads on the star wheel shafts.

**Scroll compressors** are currently used in relatively small-sized installations, predominantly for residential air-conditioning (up to 50 kW). They are recognized for low-noise operation. Two scrolls (free-standing, involute spirals bounded on one side by a flat plate) facing each other form a closed volume while one moves in a controlled orbit around a fixed point on the other, fixed scroll.

The suction gas which enters from the periphery is trapped by the scrolls. The closed volumes move radially inward until the discharge port is reached, when vapor is pressed out. The orbiting scroll is driven by a short-throw crank mechanism. Similar to screw compressors, internal leakage should be kept low, and is occurring in gaps between cylindrical surfaces and between the tips of the involute and the opposing scroll base plate.

Similar to the screw compressor, the scroll compressor is a constant-volume-ratio machine. Losses occur when operating conditions of the compressor do not match the built-in volume ratio (see Fig. 11-85).

**Vane compressors** are used in small, hermetic units, but sometimes as booster compressors in industrial applications. Two basic types are

*fixed* (roller) or single-vane type and the *rotating* or multiple-vane type. In the single-vane type the rotor (called roller) is eccentrically placed in the cylinder so these two are always in contact. The contact line make the first separation between the suction and discharge chambers while the vane (spring-loaded divider) makes the second. In the multiple-vane compressors the rotor and the cylinder are not in the contact. The rotor has two or more sliding vanes which are held against the cylinder by centrifugal force. In the vane rotary compressors, no suction valves are needed. Since the gas enters the compressor continuously, gas pulsations are at minimum. Vane compressors have a high volumetric efficiency because of the small clearance volume and consequent low reexpansion losses. Rotary vane compressors have a low weight-to-displacement ratio, which makes them suitable for transport applications.

**Centrifugal Compressors** These are sometimes called turbo-compressors and mostly serve refrigeration systems in the capacity range 200 to 10,000 kW. The main component is a spinning impeller wheel, backwards curved, which imparts energy to the gas being compressed. Some of the kinetic energy converts into pressure in a volute. Refrigerating centrifugal compressors are predominantly multistage, compared to other turbocompressors, that produce high-pressure ratios.

The torque  $T$  (Nm) the impeller ideally imparts to the gas is:

$$T = m (u_{\text{tang,out}} r_{\text{out}} - u_{\text{tang,in}} r_{\text{in}}) \quad (11-97)$$

where:

$m$  (kg/s) = mass flow rate

$r_{\text{out}}$  (m) = radius of exit of impeller

$r_{\text{in}}$  (m) = radius of exit of impeller

$u_{\text{tang,out}}$  (m/s) = tangential velocity of refrigerant leaving impeller

$u_{\text{tang,in}}$  (m/s) = tangential velocity of refrigerant entering impeller

When refrigerant enters essentially radially,  $u_{\text{tang,in}} = 0$  and torque becomes:

$$T = m * u_{\text{tang,out}} * r_{\text{out}} \quad (11-98)$$

The power  $P$  (W), is the product of torque and rotative speed  $\omega$  [1/s] so is

$$P = T * \omega = m * u_{\text{tang,out}} * r_{\text{out}} * \omega \quad (11-99)$$

which for  $u_{\text{tang,out}} = r_{\text{out}} * \omega$  becomes

$$P = m * u_{\text{tang,out}}^2 \quad (11-100)$$

or for isentropic compression

$$P = m * \Delta h \quad (11-101)$$

The performance of a centrifugal compressor (discharge to suction-pressure ratio vs. the flow rate) for different speeds is shown in Fig. 11-87. Lines of constant efficiencies show the maximum efficiency. Unstable operation sequence, called surging, occurs when compressors fails to operate in the range left of the surge envelope. It is characterized by noise and wide fluctuations of load on the compressor and the motor. The period of the cycle is usually 2 to 5 s, depending upon the size of the installation.

The capacity could be controlled by: (1) adjusting the prerotation vanes at the impeller inlet, (2) varying the speed, (3) varying the condenser pressure, and (4) bypassing discharge gas. The first two methods are predominantly used.

**Condensers** These are heat exchangers that convert refrigerant vapor to a liquid. Heat is transferred in three main phases: (1) desuperheating, (2) condensing, and (3) subcooling. In reality condensation occurs even in the superheated region and subcooling occurs in the condensation region. Three main types of refrigeration condensers are: air cooled, water cooled, and evaporative.

Air-cooled condensers are used mostly in air-conditioning and for smaller-refrigeration capacities. The main advantage is availability of cooling medium (air) but heat-transfer rates for the air side are far below values when water is used as a cooling medium. Condensation always occurs inside tubes, while the air side uses extended surface (fins).

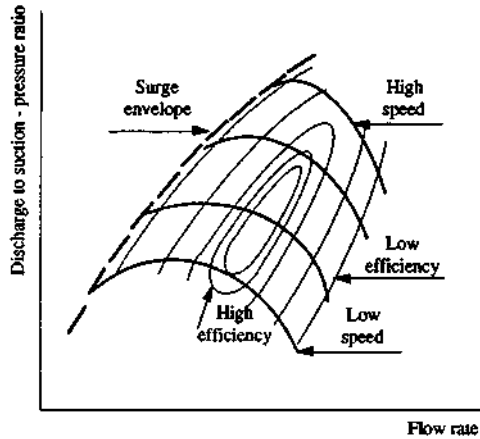


FIG. 11-87 Performance of the centrifugal compressor.

The most common types of water-cooled refrigerant condensers are: (1) shell-and-tube, (2) shell-and-coil, (3) tube-in-tube, and (4) brazed-plate. *Shell-and-tube* condensers are built up to 30 MW capacity. Cooling water flows through the tubes in a single or multipass circuit. Fixed-tube sheet and straight-tube construction are common. Horizontal layout is typical, but sometimes vertical is used. Heat-transfer coefficients for the vertical types are lower due to poor condensate drainage, but less water of lower purity can be utilized. Condensation always occurs on the tubes, and often the lower portion of the shell is used as a receiver. In *shell-and-coil* condensers water circulates through one or more continuous or assembled coils contained within the shell while refrigerant condenses outside. The tubes cannot be mechanically cleaned nor replaced. *Tube-in-tube* condensers could be found in versions where condensation occurs either in the inner tube or in the annulus. Condensing coefficients are more difficult to predict, especially in the cases where tubes are formed in spiral. Mechanical cleaning is more complicated, sometimes impossible, and tubes are not replaceable. *Brazed-plate* condensers are constructed of plates brazed together to make up an assembly of separate channels. The plates are typically stainless steel, wave-style corrugated, enabling high heat-transfer rates. Performance calculation is difficult, with very few correlations available. The main advantage is the highest performance/volume (mass) ratio and the lowest refrigerant charge. The last mentioned advantage seems to be the most important feature for many applications where minimization of charge inventory is crucial.

*Evaporative condensers* (Fig. 11-88) are widely used due to lower condensing temperatures than in the air-cooled condensers and also lower than the water-cooled condenser combined with the cooling tower. Water demands are far lower than for water-cooled condensers. The chemical industry uses shell-and-tube condensers widely, although the use of air-cooled condensing equipment and evaporative condensers is on the increase.

Generally, cooling water is of a lower quality than normal, having also higher mud and silt content. Sometimes even replaceable copper tubes in shell-and-tube heat exchangers are required. It is advisable to use cupronickel instead of copper tubes (when water is high in chlorides) and to use conservative water side velocities (less than 2 m/s for copper tubes).

Evaporative condensers are used quite extensively. In most cases commercial evaporative condensers are not totally suitable for chemical plants due to the hostile atmosphere which usually abounds in vapor and dusts which can cause either chemical (corrosion) or mechanical problems (plugging of spray nozzles).

Air-cooled condensers are similar to evaporative in that the service dictates either the use of more expensive alloys in the tube construction or conventional materials of greater wall thickness.

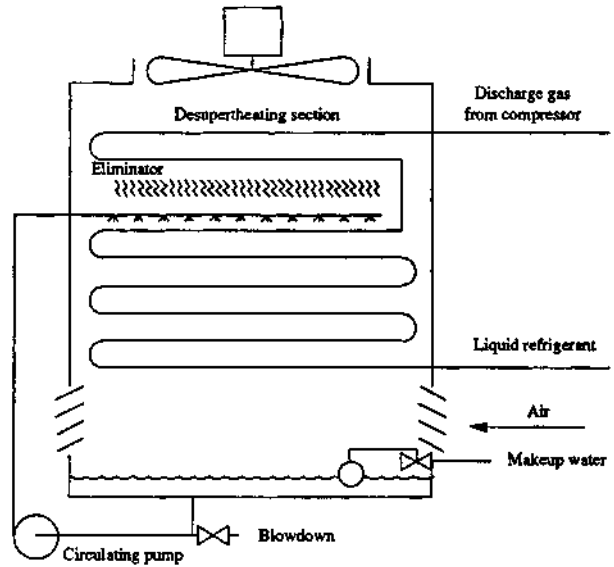


FIG. 11-88 Evaporative condenser with desuperheating coil.

Heat rejected in the condenser  $Q_{Cd}$  consists of heat absorbed in the evaporator  $Q_{Evap}$  and energy  $W$  supplied by the compressor:

$$Q_{Cd} = Q_{Evap} + W \quad (11-102)$$

For the actual systems, compressor work will be higher than for ideal for the isentropic efficiency and other losses. In the case of hermetic or accessible compressors where an electrical motor is cooled by the refrigerant, condenser capacity should be:

$$Q_{Cd} = Q_{Evap} + P_{EM} \quad (11-103)$$

It is common that compressor manufacturers provide data for the ratio of the heat rejected at the condenser to the refrigeration capacity as shown in Fig. 11-89. The solid line represents data for the open compressors while the dotted line represents the hermetic and accessible compressors. The difference between solid and dotted line is due

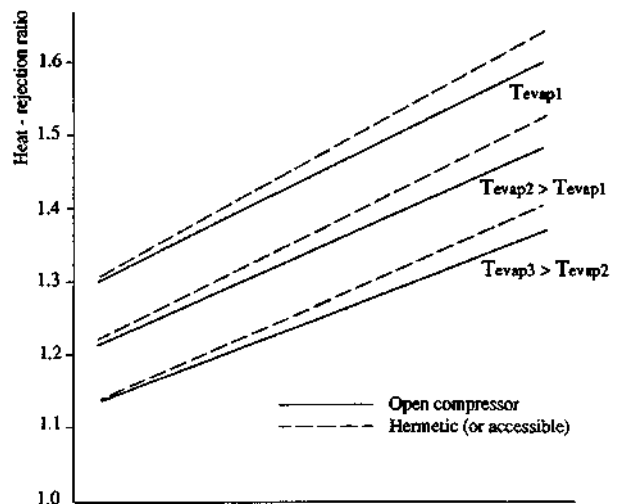


FIG. 11-89 Typical values of the heat-rejection ratio of the heat rejected at the condenser to the refrigerating capacity.



to all losses (mechanical and electrical in the electrical motor). Condenser design is based on the value:

$$Q_{Cd} = Q_{Evap} * \text{heat-rejection ratio} \quad (11-104)$$

Thermal and mechanical design of heat exchangers (condensers and evaporators) is presented earlier in this section.

**Evaporators** These are heat exchangers where refrigerant is evaporated while cooling the product, fluid, or body. Refrigerant could be in direct contact with the body that is being cooled, or some other medium could be used as secondary fluid. Mostly air, water, or antifreeze are fluids that are cooled. Design is strongly influenced by the application. Evaporators for air cooling will have in-tube evaporation of the refrigerant, while liquid chillers could have refrigerant evaporation inside or outside the tube. The heat-transfer coefficient for evaporation inside the tube (vs. length or quality) is shown in the Fig. 11-90. Fundamentals of the heat transfer in evaporators, as well as design aspects, are presented in Sec. 11. We will point out only some specific aspects of refrigeration applications.

Refrigeration evaporators could be classified according to the method of feed as either direct (dry) expansion or flooded (liquid overfeed). In dry-expansion the evaporator's outlet is dry or slightly superheated vapor. This limits the liquid feed to the amount that can be completely vaporized by the time it reaches the end of evaporator. In the liquid overfeed evaporator, the amount of liquid refrigerant circulating exceeds the amount evaporated by the circulation number. Decision on the type of the system to be used is one of the first in the design process. Direct-expansion evaporator is generally applied in smaller systems where compact design and the low first costs are crucial. Control of the refrigerant mass flow is then obtained by either a thermostatic expansion valve or a capillary tube. Figure 11-90 suggests that the evaporator surface is the most effective in the regions with quality which is neither low nor high. In dry-expansion evaporators, inlet qualities are 10–20 percent, but when controlled by the thermostatic expansion valve, vapor at the outlet is not only dry, but even superheated.

In recirculating systems saturated liquid ( $x = 0$ ) is entering the evaporator. Either the pump or gravity will deliver more refrigerant liquid than will evaporate, so outlet quality could be lower than one. The ratio of refrigerant flow rate supplied to the evaporator overflow rate of refrigerant vaporized is the circulation ratio,  $n$ . When  $n$  increases, the coefficient of heat transfer will increase due to the wetted outlet of the evaporator and the increased velocity at the inlet (Fig. 11-91). In the range of  $n = 2$  to 4, the overall  $U$  value for air cooler increases roughly by 20 to 30 percent compared to the direct-expansion case. Circulation rates higher than four are not efficient.

The price for an increase in heat-transfer characteristics is a more complex system with more auxiliary equipment: low-pressure receivers, refrigerant pumps, valves, and controls. Liquid refrigerant is predominantly pumped by mechanical pumps, however, sometimes gas at condensing pressure is used for pumping, in the variety of concepts.

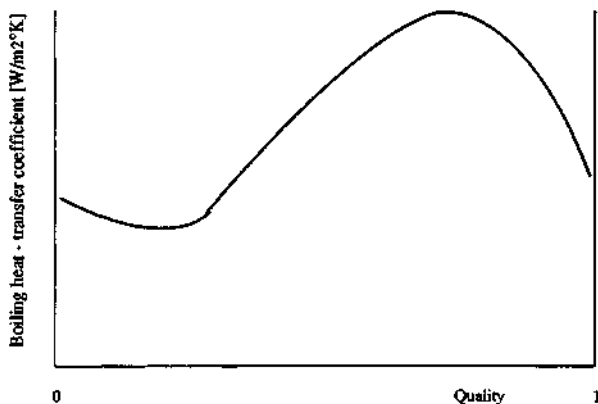


FIG. 11-90 Heat-transfer coefficient for boiling inside the tube.

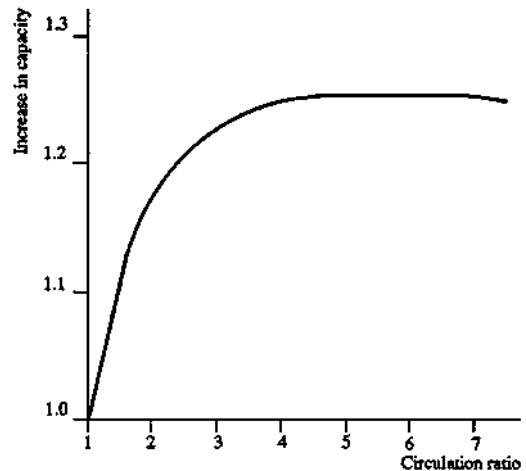


FIG. 11-91 Effect of circulation ratio on the overall heat-transfer coefficient of an air-cooling coil.

The important characteristics of the refrigeration evaporators is the presence of the oil. The system is contaminated with oil in the compressor, in spite of reasonably efficient oil separators. Some systems will recirculate oil, when miscible with refrigerant, returning it to the compressor crankcase. This is found mostly in the systems using halocarbon refrigerants. Although oils that are miscible with ammonia exist, immiscibles are predominantly used. This inhibits the ammonia systems from recirculating the oil. In systems with oil recirculation when halocarbons are used special consideration should be given to proper sizing and layout of the pipes. Proper pipeline configuration, slopes, and velocities (to ensure oil circulation under all operating loads) are essential for good system operation. When refrigerant is lighter than the oil in systems with no oil recirculation, oil will be at the bottom of every volume with a top outlet. Then oil must be drained periodically to avoid decreasing the performance of the equipment.

It is essential for proper design to have the data for refrigerant-oil miscibility under all operating conditions. Some refrigerant-oil combinations will always be miscible, some always immiscible, but some will have both characteristics, depending on temperatures and pressures applied. Defrosting is the important issue for evaporators which are cooling air below freezing. Defrosting is done periodically, actuated predominantly by time relays, but other frost indicators are used (temperature, visual, or pressure-drop sensors). Defrost technique is determined mostly by fluids available and tolerable complexity of the system. Defrosting is done by the following mechanisms when the system is off:

- Hot (or cool) refrigerant gas (the predominant method in industrial applications)
- Water (defrosting from the outside, unlike hot gas defrost)
- Air (only when room temperature is above freezing)
- Electricity (for small systems where hot-gas defrost will be too complex and water is not available)
- Combinations of above.

**System Analysis** Design calculations are made on the basis of the close to the highest refrigeration load, however the system operates at the design conditions very seldom. The purpose of regulating devices is to adjust the system performance to cooling demands by decreasing the effect or performance of some component. Refrigeration systems have inherent self-regulating control which the engineer could rely on to a certain extent. When the refrigeration load starts to decrease, less refrigerant will evaporate. This causes a drop in evaporation temperature (as long as compressor capacity is unchanged) due to the imbalance in vapor being taken by the compressor and produced by evaporation in evaporator. With a drop in evaporation pressure, the compressor capacity will decrease due to: (1) lower vapor density

(lower mass flow for the same volumetric flow rate) and (2) decrease in volumetric efficiency. On the other hand, when the evaporation temperature drops, for the unchanged temperature of the medium being cooled, the evaporator capacity will increase due to increase in the mean-temperature difference between refrigerant and cooled medium, causing a positive effect (increase) on the cooling load. With a decrease in the evaporation temperature the heat-rejection factor will increase causing an increase of heat rejected to the condenser, but refrigerant mass flow rate will decrease due to compressor characteristics. These will have an opposite effect on condenser load. Even a simplified analysis demonstrates the necessity for better understanding of system performance under different operating conditions. Two methods could be used for more accurate analysis. The traditional method of refrigeration-system analysis is through determination of balance points, while in recent years, system analysis is performed by system simulation or mathematical modeling, using mathematical (equation solving) rather than graphical (intersection of two curves) procedures. Systems with a small number of components such as the vapor-compression refrigeration system could be analyzed both ways. Graphical presentation, better suited for understanding trends is not appropriate for more complex systems, more detailed component description, and frequent change of parameters. There is a variety of different mathematical models tailored to fit specific systems, refrigerants, resources available, demands, and complexity. Although limited in its applications, graphical representation is valuable as the starting tool and for clear understanding of the system performance.

Refrigeration capacity  $q_e$  and power  $P$  curves for the reciprocating compressor are shown in Fig. 11-92. They are functions of temperatures of evaporation and condensation:

$$q_e = q_e(t_{\text{evap}}, t_{\text{cd}}) \quad (11-105a)$$

$$P = P(t_{\text{evap}}, t_{\text{cd}}) \quad (11-105b)$$

where  $q_e$  (kW) = refrigerating capacity  
 $P$  (kW) = power required by the compressor  
 $t_{\text{evap}}$  (°C) = evaporating temperature  
 $t_{\text{cd}}$  (°C) = condensing temperature.

A more detailed description of compressor performance is shown in the section on the refrigeration compressors.

Condenser performance, shown in figure 11-93, could be simplified as:

$$q_{\text{cd}} = F(t_{\text{cd}} - t_{\text{amb}}) \quad (11-105c)$$

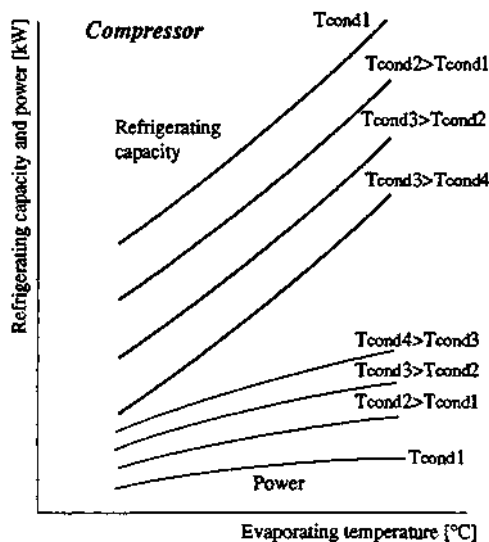


FIG. 11-92 Refrigerating capacity and power requirement for the reciprocating compressor.

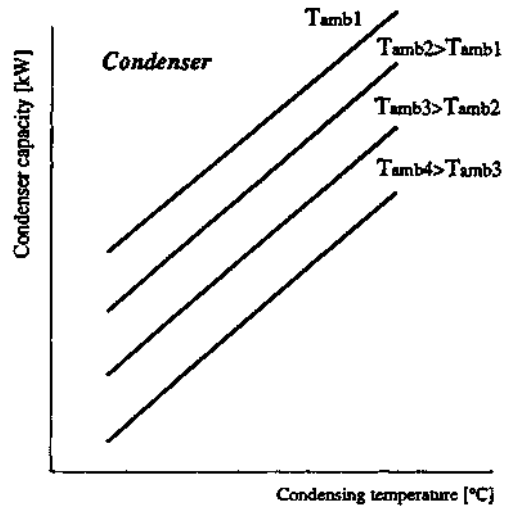


FIG. 11-93 Condenser performance.

where  $q_{\text{cd}}$  (kW) = capacity of condenser;  
 $F$  (kW/°C) = capacity of condenser per unit inlet temperature difference ( $F = U * A$ );  
 $t_{\text{amb}}$  (°C) = ambient temperature (or temperature of condenser cooling medium).

In this analysis  $F$  will be constant but it could be described more accurately as a function of parameters influencing heat transfer in the condenser (temperature, pressure, flow rate, fluid thermodynamical, and thermophysical characteristics . . .).

Condenser performance should be expressed as “evaporating effect” to enable matching with compressor and evaporator performance. Condenser “evaporating effect” is the refrigeration capacity of an evaporator served by a particular condenser. It is the function of the cycle, evaporating temperature, and the compressor. The “evaporating effect” could be calculated from the heat-rejection ratio  $q_{\text{cd}}/q_e$ :

$$q_e = \frac{q_{\text{cd}}}{\text{heat-rejection ratio}} \quad (11-105d)$$

The heat-rejection rate is presented in Fig. 11-94 (or Fig. 11-89).

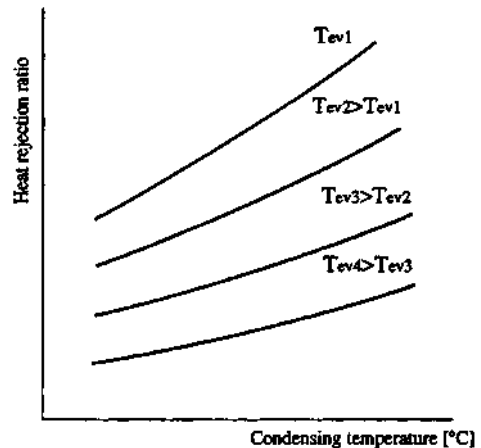


FIG. 11-94 Heat-rejection ratio.

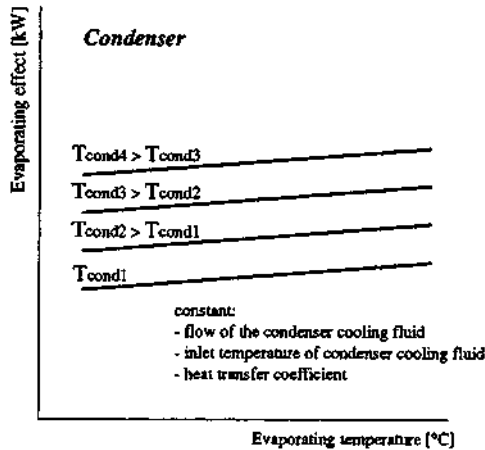


FIG. 11-95 Condenser evaporating effect.

Finally, the evaporating effect of the condenser is shown in Fig. 11-95.

The performance of the condensing-unit (compressor and condenser) subsystem could be developed as shown in Fig. 11-96 by superimposing two graphs, one for compressor performance and the other for condenser evaporating effect.

Evaporator performance could be simplified as:

$$q_e = F_{evap}(t_{amb} - t_{evap}) \quad (11-106)$$

where  $q_e$  (kW) = evaporator capacity  
 $F_e$  (kW/°C) = evaporator capacity per unit inlet temperature difference  
 $t_{amb}$  (°C) = ambient temperature (or temperature of cooled body or fluid).

The diagram of the evaporator performance is shown in the Fig. 11-97. The character of the curvature of the lines (variable heat-transfer rate) indicates that the evaporator is cooling air. Influences of the flow rate of cooled fluid are also shown in this diagram; i.e., higher flow rate will increase heat transfer. The same effect could be shown in the condenser-performance curve. It is omitted only for the reasons of simplicity.

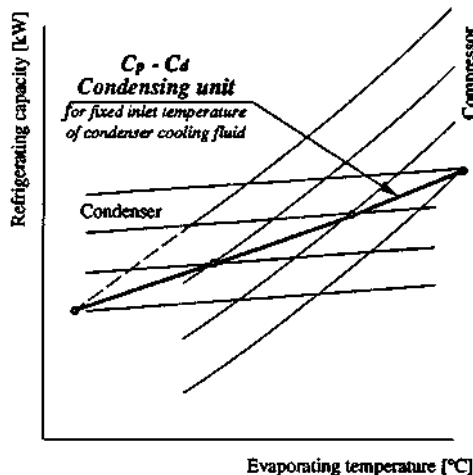


FIG. 11-96 Balance points of compressor and condenser determines performance of condensing unit for fixed temperature of condenser cooling fluid (flow rate and heat-transfer coefficient are constant).

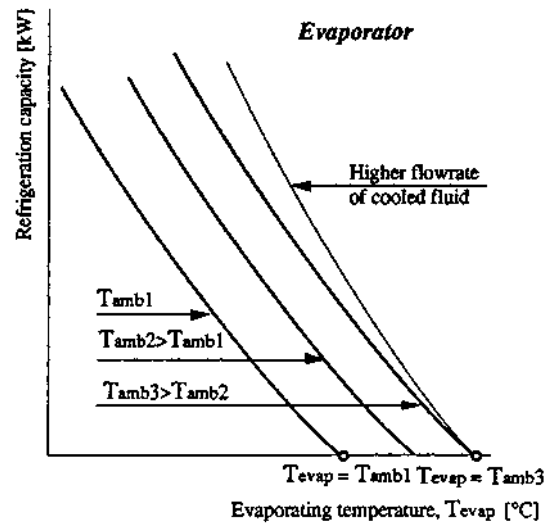


FIG. 11-97 Refrigerating capacity of evaporator.

The performance of the complete system could be predicted by superimposing the diagrams for the condensing unit and the evaporator, as shown in Fig. 11-98. Point 1 reveals a balance for constant flows and inlet temperatures of chilled fluid and fluid for condenser cooling. When this point is transferred in the diagram for the condensing unit in the Figs. 11-95 or 11-96, the condensing temperature could be determined. When the temperature of entering fluid in the evaporator  $t_{amb1}$  is lowered to  $t_{amb2}$  the new operating conditions will be determined by the state at point 2. Evaporation temperature drops from  $t_{evap1}$  to  $t_{evap2}$ . If the evaporation temperature should be unchanged, the same reduction of inlet temperature could be achieved by reducing the capacity of the condensing unit from  $C_p$  to  $C_p^*$ . The new operating point 3 shows reduction in capacity for  $\Delta$  due to the reduction in the compressor or the condenser capacity.

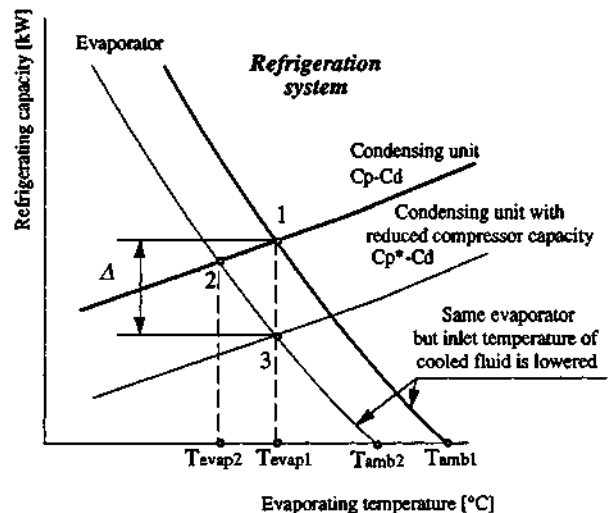


FIG. 11-98 Performance of complete refrigeration system (1), when there is reduction in heat load (2), and when for the same ambient (or inlet in evaporator) evaporation temperature is maintained constant by reducing capacity of compressor/condenser part (3).

Mathematical modeling is essentially the same process, but the description of the component performance is generally much more complex and detailed. This approach enables a user to vary more parameters easier, look into various possibilities for intervention, and predict the response of the system on different influences. Equation-solving does not necessarily have to be done by successive substitution or iteration as this procedure could suggest.

**System, Equipment, and Refrigerant Selection** There is no universal rule which can be used to decide which system, equipment type, or refrigerant is the most appropriate for a given application. A number of variables influence the final-design decision:

- Refrigeration load
- Type of installation
- Temperature level of medium to be cooled
- Condensing media characteristics: type (water, air, . . . ), temperature level, available quantities
- Energy source for driving the refrigeration unit (electricity, natural gas, steam, waste heat)
- Location and space available (urban areas, sensitive equipment around, limited space . . . )
- Funds available (i.e. initial vs. run-cost ratio)
- Safety requirements (explosive environment, aggressive fluids, . . . )
- Other demands (compatibility with existing systems, type of load, compactness, level of automatization, operating life, possibility to use process fluid as refrigerant)

Generally, vapor *compression systems* are considered first. They can be used for almost every task. Whenever it is possible, prefabricated elements or complete units are recommended. Reciprocating compressors are widely used for lower rates, more uneven heat loads (when frequent and wider range of capacity reduction is required). They ask for more space and have higher maintenance costs than centrifugal compressors, but are often the most economical in first costs. Centrifugal compressors are considered for huge capacities, when the evaporating temperature is not too low. Screw compressors are considered first when space in the machine room is limited, when system is operating long hours, and when periods between service should be longer.

Direct-expansions are more appropriate for smaller systems which should be compact, and where there are just one or few evaporators. Overfeed (recirculation) systems should be considered for all applications where first cost for additional equipment (surge drums, low-pressure receivers, refrigerant pumps, and accessories) is lower than the savings for the evaporator surface.

Choice of refrigerant is complex and not straightforward. For industrial applications, advantages of ammonia (thermodynamical and economical) overcome drawbacks which are mostly related to possible low-toxic and panics created by accidental leaks when used in urban areas. Halocarbons have many advantages (not toxic, not explosive, odorless . . . ), but environmental issues and slightly inferior thermodynamical and thermophysical properties compared to ammonia or hydrocarbons as well as rising prices are giving the chance to other options. When this text was written the ozone-depletion issue was not resolved, R22 was still used but facing phase-out, and R134a was considered to be the best alternative for CFCs and HCFCs, having similar characteristics to the already banned R12. Very often, fluid to be cooled is used as a refrigerant in the chemical industry. Use of secondary refrigerants in combination with the ammonia central-refrigeration unit is becoming a viable alternative in many applications.

Absorption systems will be considered when there is low-cost low-pressure steam or waste heat available and evaporation temperature and refrigeration load are relatively high. Typical application range is for water chilling at 7–10°C, and capacities from 300 kW to 5 MW in a single unit. The main drawback is the difficulty in maintaining a tight system with the highly corrosive lithium bromide, and an operating pressure in the evaporator and the absorber below atmospheric.

Ejector (steam-jet) refrigeration systems are used for similar applications, when chilled water-outlet temperature is relatively high, when relatively cool condensing water and cheap steam at 7 bar are available, and for similar high duties (0.3–5 MW). Even though these systems usually have low first and maintenance costs, there are not many steam-jet systems running.

## OTHER REFRIGERATION SYSTEMS APPLIED IN THE INDUSTRY

**Absorption Refrigeration Systems** Two main absorption systems are used in industrial application: lithium bromide-water and ammonia-water. Lithium bromide-water systems are limited to evaporation temperatures above freezing because water is used as the refrigerant, while the refrigerant in an ammonia-water system is ammonia and consequently it can be applied for the lower-temperature requirements.

Single-effect indirect-fired lithium bromide cycle is shown in Fig. 11-99. The machine consists of five major components:

*Evaporator* is the heat exchanger where refrigerant (water) evaporates (being sprayed over the tubes) due to low pressure in the vessel. Evaporation chills water flow inside the tubes that bring heat from the external system to be cooled.

*Absorber* is a component where strong absorber solution is used to absorb the water vapor flashed in the evaporator. A solution pump sprays the lithium bromide over the absorber tube section. Cool water is passing through the tubes taking refrigeration load, heat of dilution, heat to cool condensed water, and sensible heat for solution cooling.

*Heat exchanger* is used to improve efficiency of the cycle, reducing consumption of steam and condenser water.

*Generator* is a component where heat brought to a system in a tube section is used to restore the solution concentration by boiling off the water vapor absorbed in the absorber.

*Condenser* is an element where water vapor, boiled in the generator, is condensed, preparing pure water (refrigerant) for discharge to an evaporator.

Heat supplied to the generator is boiling weak (dilute) absorbent solution on the outside of the tubes. Evaporated water is condensed on the outside of the condenser tubes. Water utilized to cool the condenser is usually cooled in the cooling tower. Both condenser and generator are located in the same vessel, being at the absolute pressure of about 6 kPa. The water condensate passes through a liquid trap and enters the evaporator. Refrigerant (water) boils on the evaporator tubes and cools the water flow that brings the refrigeration load. Refrigerant that is not evaporated flows to the recirculation pump to be sprayed over the evaporator tubes. Solution with high water concentration that enters the generator increases in concentration as water evaporates. The resulting strong, absorbent solution (solution with low water concentration) leaves the generator on its way to the heat exchanger. There the stream of high water concentration that flows to the generator cools the stream of solution with low water concentration that flows to the second vessel. The solution with low water concentration is distributed over the absorber tubes. Absorber and evaporator are located in the same vessel, so the refrigerant evaporated on the evaporator tubes is readily absorbed into the absorbent solution. The pressure in the second vessel during the operation is 7 kPa (absolute). Heat of absorption and dilution are removed by cooling water (usually from the cooling tower). The resulting solution with high water concentration is pumped through the heat exchanger to the generator, completing the cycle. Heat exchanger increases the efficiency of the system by preheating, that is, reducing the amount of heat that must be added to the high water solution before it begins to evaporate in the generator.

The absorption machine operation is analyzed by the use of a lithium bromide-water equilibrium diagram, as shown in Fig. 11-100. Vapor pressure is plotted against the mass concentration of lithium bromide in the solution. The corresponding saturation temperature for a given vapor pressure is shown on the left-hand side of the diagram. The line in the lower right corner of the diagram is the crystallization line. It indicates the point at which the solution will begin to change from liquid to solid, and this is the limit of the cycle. If the solution becomes overconcentrated, the absorption cycle will be interrupted owing to solidification, and capacity will not be restored until the unit is desolidified. This normally requires the addition of heat to the outside of the solution heat exchanger and the solution pump.

The diagram in Fig. 11-101 presents enthalpy data for LiBr-water solutions. It is needed for the thermal calculation of the cycle. Enthalpies for water and water vapor can be determined from the table of properties of water. The data in Fig. 11-101 are applicable to

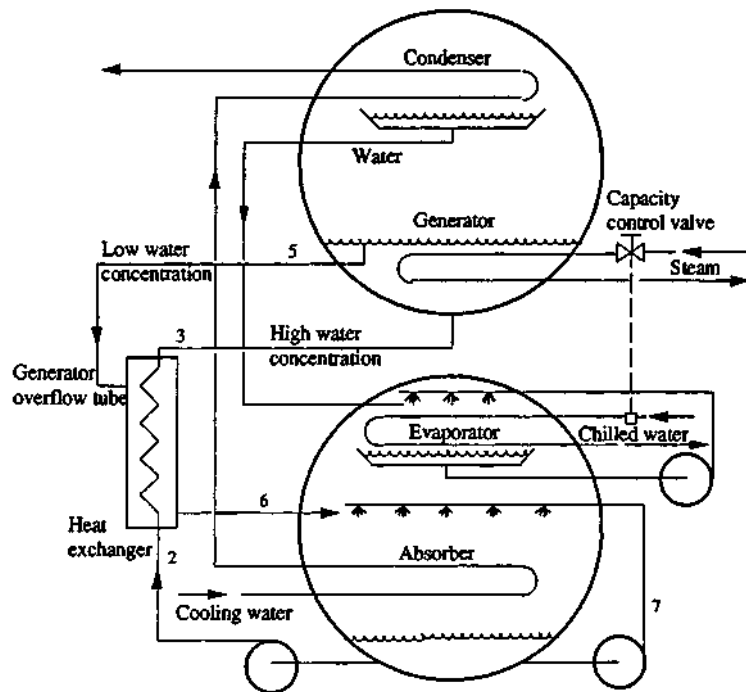


FIG. 11-99 Two-shell lithium bromide-water cycle chiller.

saturated or subcooled solutions and are based on a zero enthalpy of liquid water at 0°C and a zero enthalpy of solid LiBr at 25°C. Since the zero enthalpy for the water in the solution is the same as that in conventional tables of properties of water, the water property tables can be used in conjunction with diagram in Fig. 11-100.

Coefficient of performance of the absorption cycle is defined on the same principle as for the mechanical refrigeration:

$$\text{COP}_{\text{abs}} = \frac{\text{useful effect}}{\text{heat input}} = \frac{\text{refrigeration rate}}{\text{heat input at generator}}$$

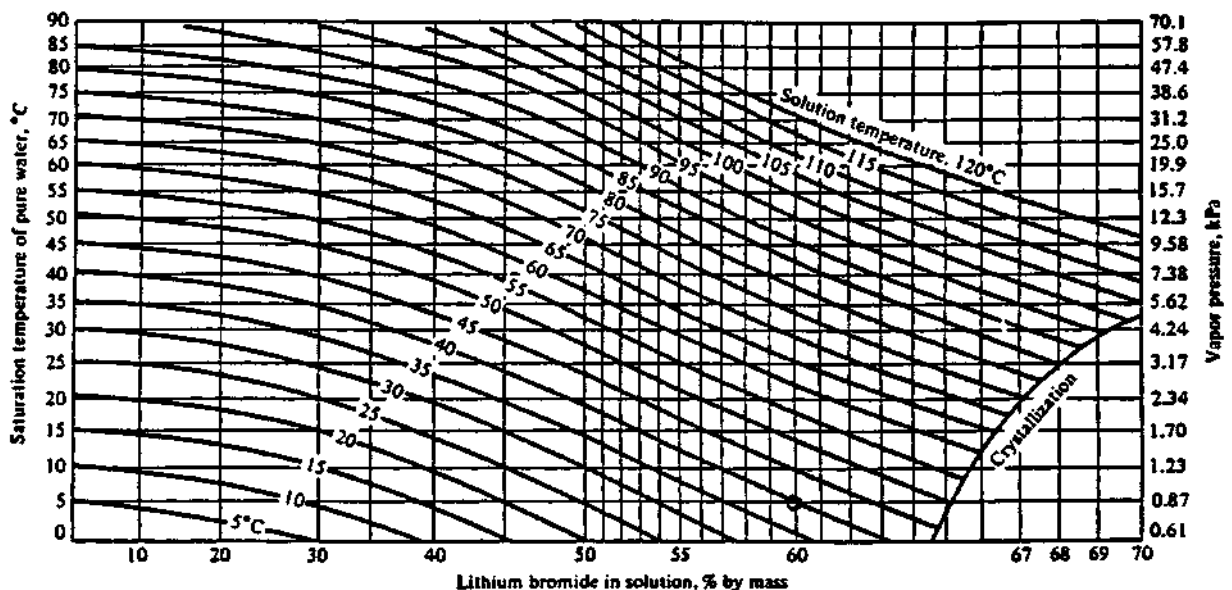


FIG. 11-100 Temperature-pressure-concentration diagram of saturated LiBr-water solutions (W. F. Stoecker and J. W. Jones: Refrigeration and Air-Conditioning)

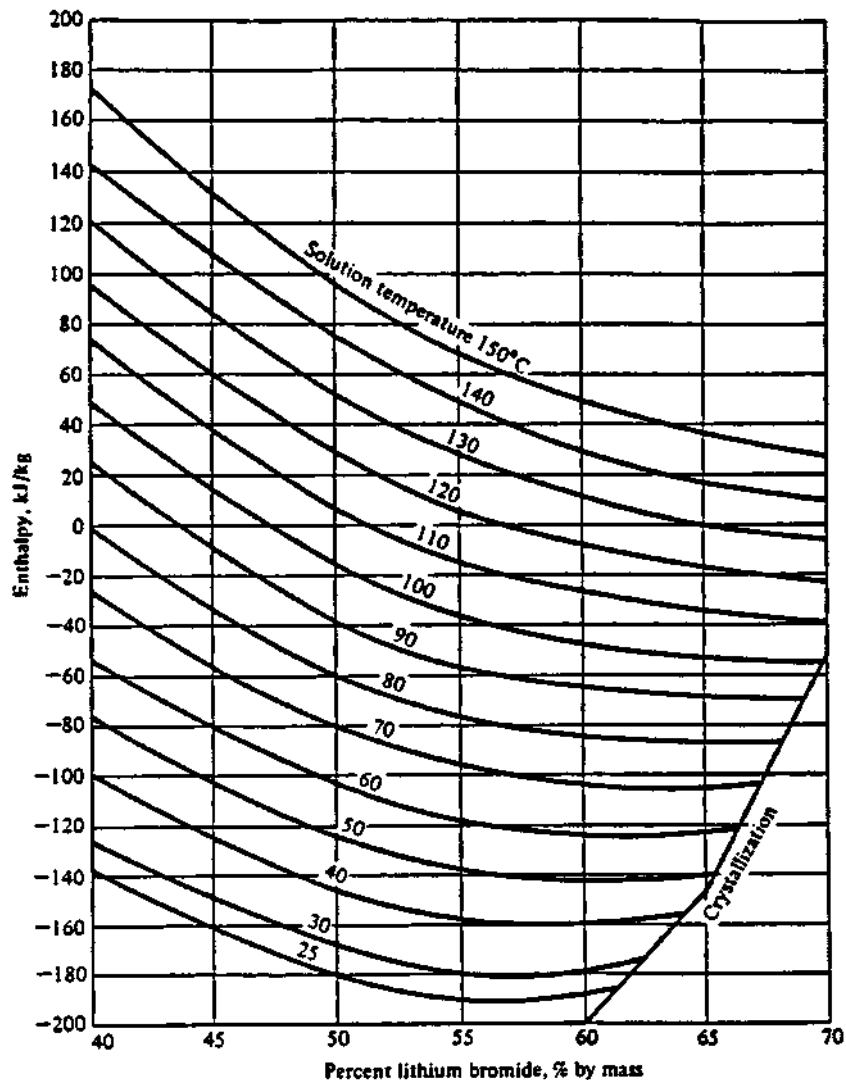


FIG. 11-101 Enthalpy of LiBr-water solutions (W. F. Stoecker and J. W. Jones: Refrigeration and Air-Conditioning).

but it should be noted that here denominator for the  $COP_{abs}$  is heat while for the mechanical refrigeration cycle it is work. Since these two forms of energy are not equal,  $COP_{abs}$  is not as low (0.6–0.8) as it appears compared to COP for mechanical system (2.5–3.5).

The double-effect absorption unit is shown in Fig. 11-102. All major components and operation of the double-effect absorption machine is similar to that for the single-effect machine. The primary generator, located in the vessel 1, is using an external heat source to evaporate water from dilute-absorbent (high water concentration) solution. Water vapor readily flows to the generator II where it is condensed on the tubes. The absorbent (LiBr) intermediate solution from generator I will pass through the heat exchanger on the way to the generator II where it is heated by the condensing water vapor. The throttling valve reduces pressure from vessel 1 (about 103 kPa absolute) to that of vessel 2. Following the reduction of pressure some water in the solution flashes to vapor, which is liquefied at the condenser. In the high temperature heat exchanger intermediate solution heats the weak (high water concentration) solution stream coming

from the low temperature heat exchanger. In the low temperature heat exchanger strong solution is being cooled before entering the absorber. The absorber is on the same pressure as the evaporator. The double-effect absorption units achieve higher COPs than the single stage.

The ammonia-water absorption system was extensively used until the fifties when the LiBr-water combination became popular. Figure 11-103 shows a simplified ammonia-water absorption cycle. The refrigerant is ammonia, and the absorbent is dilute aqueous solution of ammonia. Ammonia-water systems differ from water-lithium bromide equipment to accommodate major differences: Water (here absorbent) is also volatile, so the regeneration of weak water solution to strong water solution is a fractional distillation. Different refrigerant (ammonia) causes different, much higher pressures: about 1100–2100 kPa absolute in condenser.

Ammonia vapor from the evaporator and the weak water solution from the generator are producing strong water solution in the absorber. Strong water solution is then separated in the rectifier producing (1)

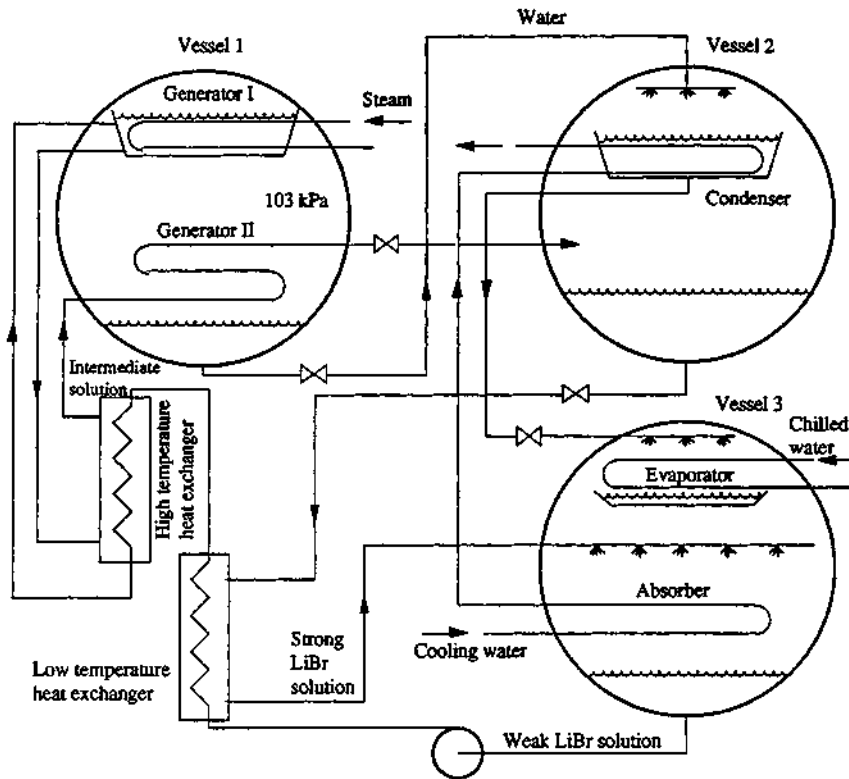


FIG. 11-102 Double-effect absorption unit.

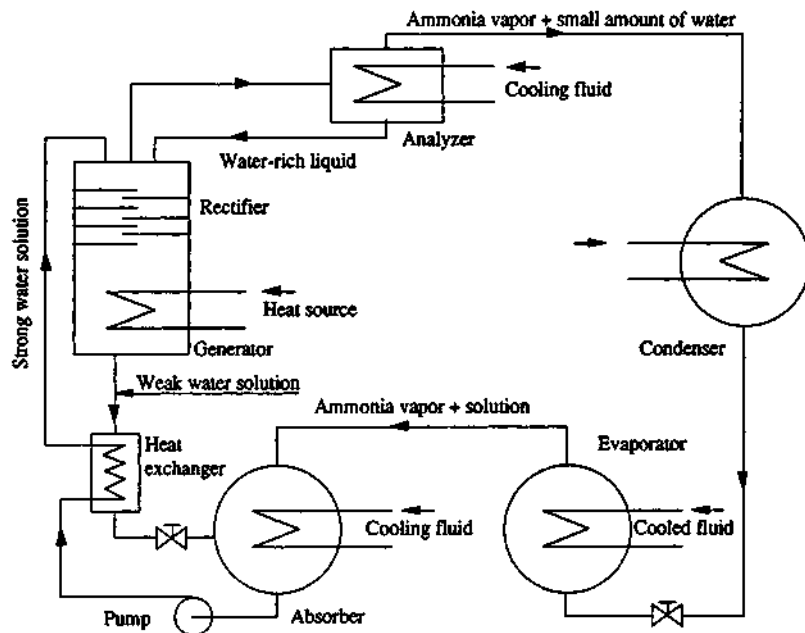


FIG. 11-103 Simplified ammonia-water absorption cycle.

ammonia with some water vapor content and (2) very strong water solution at the bottom, in the generator. Heat in the generator vaporizes ammonia and the weak solution returns to absorber. On its way to the absorber the weak solution stream passes through the heat exchanger, heating strong solution from the absorber on the way to the rectifier. The other stream, mostly ammonia vapor but with some water-vapor content flows to the condenser. To remove water as much as possible, the vapor from the rectifier passes through the analyzer where it is additionally cooled. The remaining water escaped from the analyzer pass as liquid through the condenser and the evaporator to the absorber.

Ammonia-water units can be arranged for single-stage or cascaded two-stage operation. The advantage of two-staging is it creates the possibility of utilizing only part of the heat on the higher and the rest on the lower temperature level but the price is increase in first cost and heat required.

Ammonia-water and lithium bromide-water systems operate under comparable COP. The ammonia-water system is capable achieving evaporating temperatures below 0°C because the refrigerant is ammonia. Water as the refrigerant limits evaporating temperatures to no lower than freezing, better to 3°C. Advantage of the lithium bromide-water system is that it requires less equipment and operates at lower pressures. But this is also a drawback, because pressures are below atmospheric, causing air infiltration in the system which must be purged periodically. Due to corrosion problems, special inhibitors must be used in the lithium-bromide-water system. The infiltration of air in the ammonia-water system is also possible, but when evaporating temperature is below -33°C. This can result in formation of corrosive ammonium carbonate.

**Further readings:** ASHRAE Handbook 1994 Refrigeration Systems and Applications; Bogart, M., 1981: *Ammonia Absorption Refrigeration in Industrial Processes*, Gulf Publishing Co. Houston; Stoecker, W. F., and Jones, J. W., 1982: *Refrigeration and Air-Conditioning*, McGraw-Hill Book Company, New York.

**Steam-Jet (Ejector) Systems** These systems substitute an ejector for a mechanical compressor in a vapor compression system. Since refrigerant is water, maintaining temperatures lower than the environment requires that the pressure of water in the evaporator must be below atmospheric. A typical arrangement for the steam-jet refrigeration cycle is shown in Fig. 11-104.

**Main Components** The main components of steam-jet refrigeration systems are:

1. **Primary steam ejector:** A kinetic device that utilizes the momentum of a high-velocity jet to entrain and accelerate a slower-moving medium into which it is directed. High-pressure steam is delivered to the nozzle of the ejector. The steam expands while flowing through the nozzle where the velocity increases rapidly. The velocity of steam leaving the nozzle is around 1200 m/s. Because of this high velocity, flash vapor from the tank is continually aspirated into the moving steam. The mixture of steam and flash vapor then enters the diffuser section where the velocity is gradually reduced because of increasing cross-sectional area. The energy of the high-velocity steam compresses the vapor during its passage through the diffuser, raising its temperature above the temperature of the condenser cooling water.

2. **Condenser:** The component of the system where the vapor condenses and where the heat is rejected. The rate of heat rejected is:

$$Q_{\text{cond}} = (W_s + W_w) h_{fg} \quad (11-107)$$

where:  $Q_{\text{cond}}$  = heat rejection (kW)

$W_s$  = primary booster steam rate (kg/s)

$W_w$  = flash vapor rate (kg/s)

$h_{fg}$  = latent heat (kJ/kg)

The condenser design, surface area, and condenser cooling water quantity should be based on the highest cooling water temperature likely to be encountered. If the inlet cooling water temperature becomes hotter than the design, the primary booster (ejector) may cease functioning because of the increase in condenser pressure.

Two types of condensers could be used: the surface condenser (shown in Fig. 11-104) and the barometric or jet condenser (Fig. 11-105). The surface condenser is of shell-and-tube design with water flowing through the tubes and steam condensed on the outside surface. In the jet condenser, condensing water and the steam being condensed are mixed directly, and no tubes are provided. The jet condenser can be barometric or a low-level type. The barometric condenser requires a height of ~10 m above the level of the water in the hot well. A tailpipe of this length is needed so that condenser water and condensate can drain by gravity. In the low-level jet type, the tailpipe is eliminated, and it becomes necessary to remove the condenser water and condensate

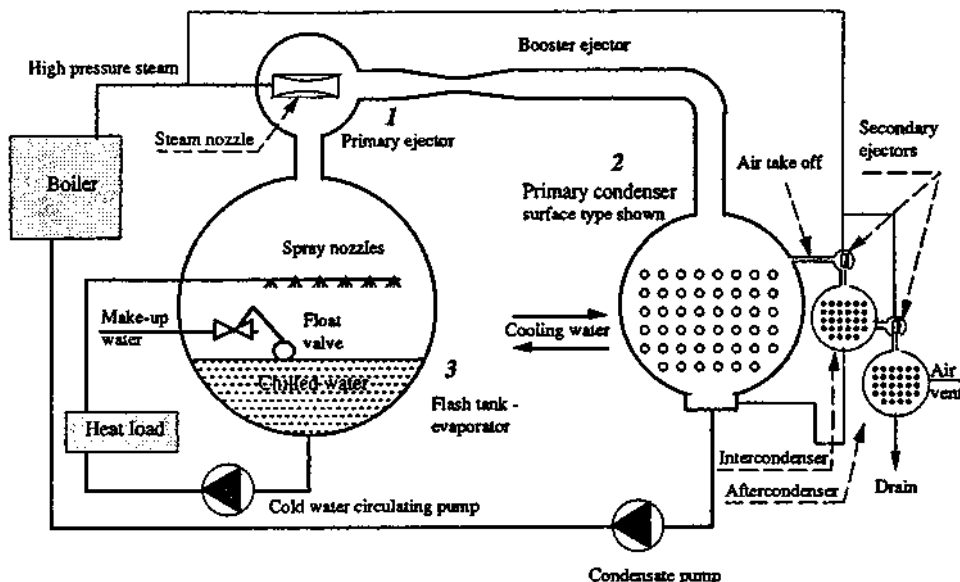


FIG. 11-104 Ejector (steam-jet) refrigeration cycle (with surface-type condenser).



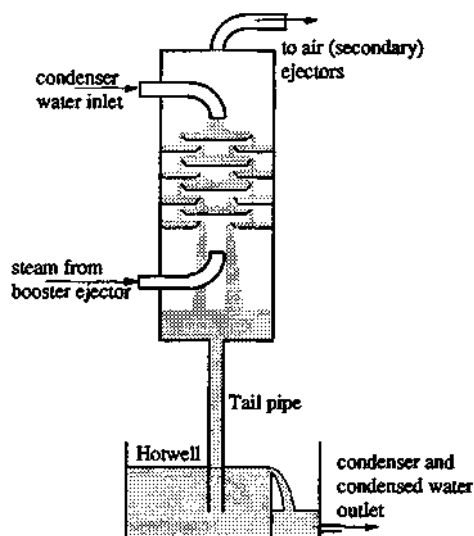


FIG. 11-105 Barometric condenser for steam-jet system.

by pumping from the condenser to the hot well. The main advantages of the jet condenser are low maintenance with the absence of tubes and the fact that condenser water of varying degrees of cleanliness may be used.

3. **Flash tank.** This is the evaporator of the ejector system and is usually a large-volume vessel where large water-surface area is needed for efficient evaporative cooling action. Warm water returning from the process is sprayed into the flash chamber through nozzles (sometimes cascades are used for maximizing the contact surface, being less susceptible to carryover problems) and the cooled effluent is pumped from the bottom of the flash tank.

When the steam supply to one ejector of a group is closed, some means must be provided to for preventing the pressure in the condenser and flash tank from equalizing through that ejector. A compartmental flash tank is frequently used for such purposes. With this arrangement, partitions are provided so that each booster is operating on its own flash tank. When the steam is shut off to any booster, the valve to the inlet spray water to that compartment also is closed.

A float valve is provided to control the supply of makeup water to replace the water vapor that has flashed off. The flash tank should be insulated.

**Applications** The steam-jet refrigeration is suited for:

1. Processes where direct vaporization is used for concentration or drying of heat-sensitive foods and chemicals, where, besides elimination of the heat exchanger, preservation of the product quality is an important advantage.

2. Enabling the use of hard or even sea water for heat rejection e.g. for absorption of gases ( $\text{CO}_2$ ,  $\text{SO}_2$ ,  $\text{ClO}_2$  . . . ) in chilled water (desorption is provided simultaneously with chilling) when a direct contact barometric condenser is used.

Despite being simple, rugged, reliable, requiring low maintenance, low cost, and vibration free, steam-jet systems are not widely accepted in water chilling for air-conditioning due to characteristics of the cycle.

**Factors Affecting Capacity** Ejector (steam-jet) units become attractive when cooling relatively high-temperature chilled water with a source of about 7 bar gauge steam and relatively cool condensing water. The factors involved with steam-jet capacity include the following:

1. **Steam pressure.** The main boosters can operate on steam pressures from as low as 0.15 bar up to 7 bar gauge. The quantity of steam required increases rapidly as the steam pressure drops (Fig. 11-106). The best steam rates are obtained with about 7 bar. Above this pressure the change in quantity of steam required is practically

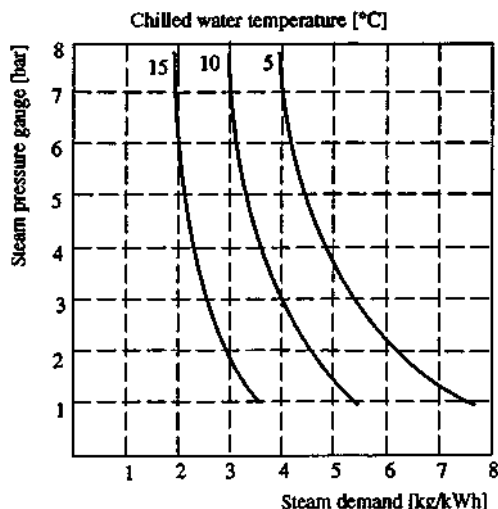


FIG. 11-106 Effect of steam pressure on steam demand at 38°C condenser temperature (ASHRAE 1983 Equipment Handbook).

negligible. Ejectors must be designed for the highest available steam pressure, to take advantage of the lower steam consumption for various steam-inlet pressures.

The secondary ejector systems used for removing air require steam pressures of 2.5 bar or greater. When the available steam pressure is lower than this, an electrically driven vacuum pump is used for either the final secondary ejector or for the entire secondary group. The secondary ejectors normally require 0.2–0.3 kg/h of steam per kW of refrigeration capacity.

2. **Condenser water temperature.** In comparison with other vapor-compression systems, steam-jet machines require relatively large water quantities for condensation. The higher the inlet-water temperature, the higher are the water requirements (Fig. 11-107). The condensing water temperature has an important effect on steam rate per refrigeration effect, rapidly decreasing with colder condenser cooling water. Figure 11-108 presents data on steam rate versus condenser water inlet for given chilled-water outlet temperatures and steam pressure.

3. **Chilled-water temperature.** As the chilled-water outlet temperature decreases, the ratio of steam/refrigeration effect decreases, thus increasing condensing temperatures and/or increasing the condensing-water requirements.

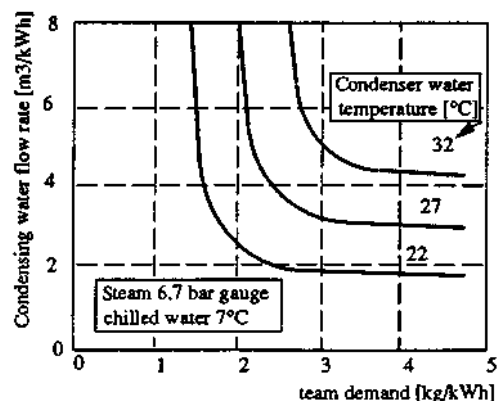


FIG. 11-107 Steam demand versus condenser-water flow rate.

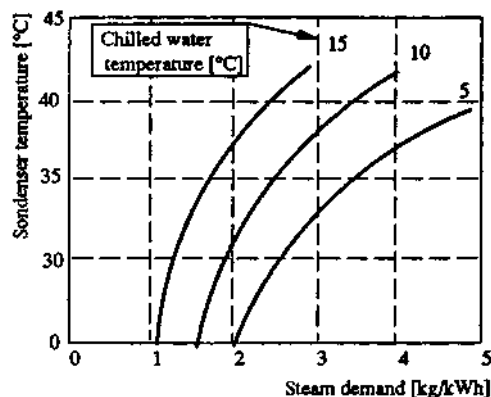


FIG. 11-108 Steam demand versus chilled-water temperature for typical steam-jet system (ASHRAE 1983 Equipment Handbook).

Unlike other refrigeration systems, the chilled-water flow rate is of no particular importance in steam-jet system design, because there is, due to direct heat exchange, no influence of evaporator tube velocities and related temperature differences on heat-transfer rates. Widely varying return chilled-water temperatures have little effect on steam-jet equipment.

**Multistage Systems** The majority of steam-jet systems being currently installed are multistage. Up to five stage systems are in commercial operation.

**Capacity Control** The simplest way to regulate the capacity of most steam vacuum refrigeration systems is to furnish several primary boosters in parallel and operate only those required to handle the heat load. It is not uncommon to have as many as four main boosters on larger units for capacity variation. A simple automatic on-off type of control may be used for this purpose. By sensing the chilled-water temperature leaving the flash tank, a controller can turn steam on and off to each ejector as required.

Additionally, two other control systems which will regulate steam flow or condenser-water flow to the machine are available. As the condenser-water temperature decreases during various periods of the year, the absolute condenser pressure will decrease. This will permit the ejectors to operate on less steam because of the reduced discharge pressure. Either the steam flow or the condenser water quantities can be reduced in order to lower operating costs at other than design periods. The arrangement selected depends on cost considerations between the two flow quantities. Some systems have been arranged for a combination of the two, automatically reducing steam flow down to a point, followed by a reduction in condenser-water flow. For maximum operating efficiency, automatic control systems are usually justifiable in keeping operating cost to a minimum without excessive operator attention. In general, steam savings of about 10 percent of rated booster flow are realized for each 2.5°C reduction in condensing-water temperature below the design point.

In some cases, with relatively cold inlet condenser water it has been possible to adjust automatically the steam inlet pressure in response to

chilled-water outlet temperatures. In general, however, this type of control is not possible because of the differences in temperature between the flash tank and the condenser. Under usual conditions of warm condenser-water temperatures, the main ejectors must compress water vapor over a relatively high ratio, requiring an ejector with entirely different operating characteristics. In most cases, when the ejector steam pressure is throttled, the capacity of the jet remains almost constant until the steam pressure is reduced to a point at which there is a sharp capacity decrease. At this point, the ejectors are unstable, and the capacity is severely curtailed. With a sufficient increase in steam pressure, the ejectors will once again become stable and operate at their design capacity. In effect, steam jets have a vapor-handling capacity fixed by the pressure at the suction inlet. In order for the ejector to operate along its characteristic pumping curve, it requires a certain minimum steam flow rate which is fixed for any particular pressure in the condenser. (For further information on the design of ejectors, see Sec. 6)

**Further reading and reference:** ASHRAE 1983 *Equipment Handbook*; Spencer, E., 1961, *New Development in Steam Vacuum Refrigeration*, ASHRAE Transactions Vol. 67, p. 339.

**Refrigerants** A refrigerant is any body or substance which acts as a cooling agent by absorbing heat from another body or substance which has to be cooled. Primary refrigerants are those that are used in the refrigeration systems, where they alternately vaporize and condense as they absorb or give off heat respectively. Secondary refrigerants are heat transfer fluids or heat carriers. Refrigerant pairs in absorption systems are ammonia-water and lithium bromide-water, while steam (water) is used as a refrigerant in ejector systems. Refrigerants used in the mechanical refrigeration systems are far more numerous.

A list of the most significant refrigerants is presented in the *ASHRAE Handbook Fundamentals*. More data are shown in the Chap. 3 of this handbook—"Physical and Chemical Data." Due to rapid changes in refrigerant, issue readers are advised to consult the most recent data and publications at the time of application.

The first refrigerants were natural: air, ammonia, CO<sub>2</sub>, SO<sub>2</sub>, and so on. Fast expansion of refrigeration in the second and third quarters of the 20th century is marked by the new refrigerants, chlorofluorocarbons (CFC) and hydrochlorofluorocarbons (HCFC). They are halocarbons which contain one or more of three halogens chlorine, fluorine, and bromine (Fig. 11-109). These refrigerants introduced many advantageous qualities compared to most of the existing refrigerants: odorless, nonflammable, nonexplosive, compatible with the most engineering materials, reasonably high COP, and nontoxic.

In the last decade, the refrigerant issue is extensively discussed due to the accepted hypothesis that the chlorine and bromine atoms from halocarbons released to the environment were using up ozone in the stratosphere, depleting it specially above the polar regions. Montreal Protocol and later agreements ban use of certain CFCs and halon compounds. It seems that all CFCs and most of the HCFCs will be out of production by the time this text will be published.

Chemical companies are trying to develop safe and efficient refrigerant for the refrigeration industry and application, but uncertainty in CFC and HCFC substitutes is still high. When this text was written HFCs were a promising solution. That is true especially for the R134a which seems to be the best alternative for R12. Substitutes for R22 and R502 are still under debate. Numerous ecologists and chemists are for

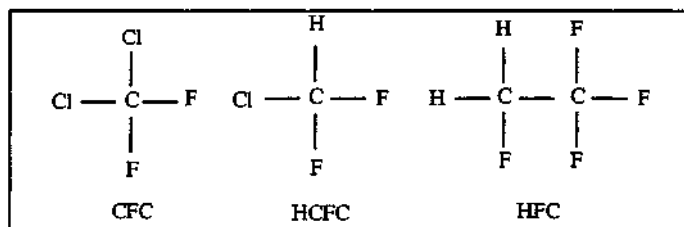


FIG. 11-109 Halocarbon refrigerants.

an extended ban on HFCs as well, mostly due to significant use of CFCs in production of HFCs. Extensive research is ongoing to find new refrigerants. Many projects are aimed to design and study refrigerant mixtures, both azeotropic (mixture which behaves physically as a single, pure compound) and zeotropic having desirable qualities for the processes with temperature glides in the evaporator and the condenser.

Ammonia (R717) is the single natural refrigerant being used extensively (beside halocarbons). It is significant in industrial applications for its excellent thermodynamic and thermophysical characteristics. Many engineers are considering ammonia as a CFC substitute for various applications. Significant work is being done on reducing the refrigerant inventory and consequently problems related to leaks of this fluid with strong odor. There is growing interest in hydrocarbons in some countries, particularly in Europe. Indirect cooling (secondary refrigeration) is under reconsideration for many applications.

Due to the vibrant refrigerant issue it will be a challenge for every engineer to find the best solution for the particular application, but basic principles are the same. Good refrigerant should be:

- Safe: nontoxic, nonflammable, and nonexplosive
- Environmentally friendly
- Compatible with materials normally used in refrigeration: oils, metals, elastomers, etc.
- Desirable thermodynamic and thermophysical characteristics:
  - High latent heat
  - Low specific volume of vapor
  - Low compression ratio
  - Low viscosity
  - Reasonably low pressures for operating temperatures
  - Low specific heat of liquid
  - High specific heat of vapor
  - High conductivity and other heat transfer related characteristics
  - Reasonably low compressor discharge temperatures
  - Easily detected if leaking
  - High dielectric constant
  - Good stability

**Secondary Refrigerants (Antifreezes or Brines)** These are mostly liquids used for transporting heat energy from the remote heat

source (process heat exchanger) to the evaporator of the refrigeration system. Antifreezes or brines do not change state in the process, but there are examples where some secondary refrigerants are either changing state themselves, or just particles which are carried in them.

Indirect refrigeration systems are more prevalent in the chemical industry than in the food industry, commercial refrigeration, or comfort air-conditioning. This is even more evident in the cases where a large amount of heat is to be removed or where a low temperature level is involved. Advantage of an indirect system is centralization of refrigeration equipment, which is specially important for relocation of refrigeration equipment in a nonhazardous area, both for people and equipment.

**Salt Brines** The typical curve of freezing point is shown in Fig. 11-110. Brine of concentration  $x$  (water concentration is  $1-x$ ) will not solidify at  $0^\circ\text{C}$  (freezing temperature for water, point A). When the temperature drops to B, the first crystal of ice is formed. As the temperature decreases to C, ice crystals continue to form and their mixture with the brine solution forms the slush. At the point C there will be part ice in the mixture  $l_2/(l_1 + l_2)$ , and liquid (brine)  $l_1/(l_1 + l_2)$ . At point D there is mixture of  $m_1$  parts eutectic brine solution  $D_1$  [concentration  $m_1/(m_1 + m_2)$ ], and  $m_2$  parts of ice [concentration  $m_2/(m_1 + m_2)$ ]. Cooling the mixture below D solidifies the entire solution at the eutectic temperature. Eutectic temperature is the lowest temperature that can be reached with no solidification.

It is obvious that further strengthening of brine has no effect, and can cause a different reaction—salt sometimes freezes out in the installations where concentration is too high.

Sodium chloride, an ordinary salt ( $\text{NaCl}$ ), is the least expensive per volume of any brine available. It can be used in contact with food and in open systems because of its low toxicity. Heat transfer coefficients are relatively high. However, its drawbacks are it has a relatively high freezing point and is highly corrosive (requires inhibitors thus must be checked on a regular schedule).

Calcium chloride ( $\text{CaCl}_2$ ) is similar to  $\text{NaCl}$ . It is the second lowest-cost brine, with a somewhat lower freezing point (used for temperatures as low as  $-37^\circ\text{C}$ ). Highly corrosive and not appropriate for direct contact with food. Heat transfer coefficients are rapidly reduced at

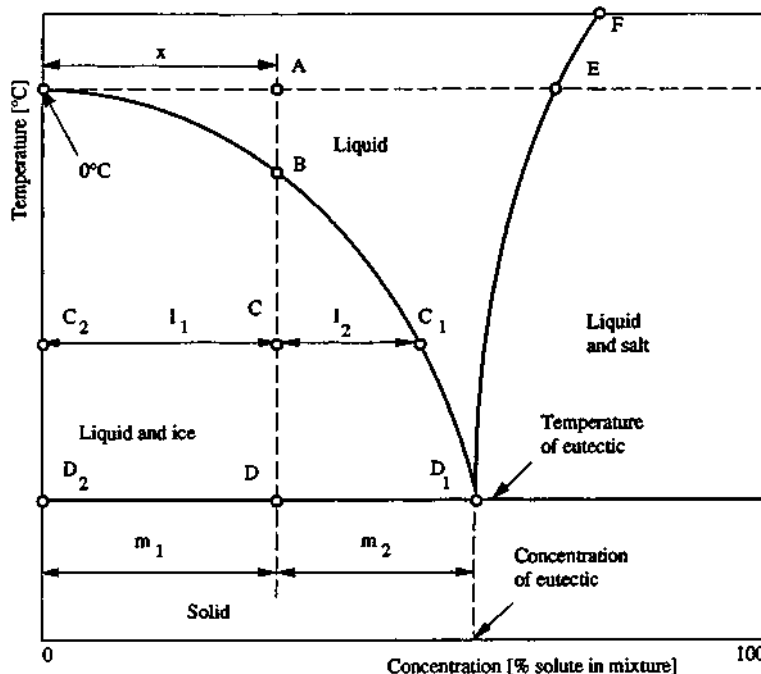


FIG. 11-110 Phase diagram of the brine.

temperatures below  $-20^{\circ}\text{C}$ . The presence of magnesium salts in either sodium or calcium chloride is undesirable because they tend to form sludge. Air and carbon dioxide are contaminants and excessive aeration of the brine should be prevented by use of close systems. Oxygen, required for corrosion, normally comes from the atmosphere and dissolves in the brine solution. Dilute brines dissolve oxygen more readily and are generally more corrosive than concentrated brines. It is believed that even a closed brine system will not prevent the infiltration of oxygen.

To adjust alkaline condition to pH 7.0–8.5 use caustic soda (to correct up to 7.0) or sodium dichromate (to reduce excessive alkalinity below pH 8.5). Such slightly alkaline brines are generally less corrosive than neutral or acid ones, although with high alkalinity the activity may increase.

If the untreated brine has the proper pH value, the acidifying effect of the dichromate may be neutralized by adding commercial flake caustic soda (76 percent pure) in quantity that corresponds to 27 percent of sodium dichromate used. Caustic soda must be thoroughly dissolved in warm water before it is added to the brine.

Recommended inhibitor (sodium dichromate) concentrations are  $2\text{ kg/m}^3$  of  $\text{CaCl}_2$  and  $3.2\text{ kg/m}^3$  of  $\text{NaCl}$  brine. Sodium dichromate when dissolved in water or brine makes the solution acid. Steel, iron, copper, or red brass can be used with brine circulating systems. Calcium chloride systems are generally equipped with all-iron-and-steel pumps and valves to prevent electrolysis in event of acidity. Copper and red brass tubing are used for calcium chloride evaporators. Sodium chloride systems are using all-iron or all-bronze pumps.

**Organic Compounds (Inhibited Glycols)** *Ethylene glycol* is colorless and practically odorless and completely miscible with water. Advantages are low volatility and relatively low corrosivity when properly inhibited. Main drawbacks are relatively low heat-transfer coefficients at lower temperatures due to high viscosities (even higher than for propylene glycol). It is somewhat toxic, but less harmful than methanol water solutions. It is not appropriate for food industry and should not stand in open containers. Preferably waters that are classified as soft and are low in chloride and sulfate ions should be used for preparation of ethylene glycol solution.

Pure ethylene glycol freezes at  $-12.7^{\circ}\text{C}$ . Exact composition and temperature for eutectic point are unknown, since solutions in this region turn to viscous, glassy mass that makes it difficult to determine the true freezing point. For the concentrations lower than eutectic, ice forms on freezing, while on the concentrated, solid glycol separates from the solution.

Ethylene glycol normally has pH of 8.8 to 9.2 and should not be used below 7.5. Addition of more inhibitor can not restore the solution to original condition. Once inhibitor has been depleted, it is recommended that the old glycol be removed from the system and the new charge be installed.

*Propylene glycol* is very similar to ethylene glycol, but it is not toxic and is used in direct contact with food. It is more expensive and, having higher viscosity, shows lower heat transfer coefficients.

*Methanol water* is an alcohol-base compound. It is less expensive than other organic compounds and, due to lower viscosity, has better heat transfer and pressure drop characteristics. It is used up to  $-35^{\circ}\text{C}$ . Disadvantages are (1) considered more toxic than ethylene glycol and thus more suitable for outdoor applications (2) flammable and could be assumed to be a potential fire hazard.

For ethylene glycol systems copper tubing is often used (up to 3 in), while pumps, cooler tubes, or coils are made of iron, steel, brass, copper, or aluminum. Galvanized tubes should not be used in ethylene glycol systems because of reaction of the inhibitor with the zinc.

Methanol water solutions are compatible with most materials but in sufficient concentration will badly corrode aluminum.

*Ethanol water* is a solution of denatured grain alcohol. Its main advantage is that it is nontoxic and thus is widely used in the food and chemical industry. By using corrosion inhibitors it could be made noncorrosive for brine service. It is more expensive than methanol water and has somewhat lower heat transfer coefficients. As an alcohol derivative it is flammable.

Secondary refrigerants shown below, listed under their generic names, are sold under different trade names. Some other secondary refrigerants appropriate for various refrigeration application will be

listed under their trade names. More data could be obtained from the manufacturer.

*Syltherm XLT (Dow Corning Corporation)*. A silicone polymer (Dimethyl Polysiloxane); recommended temperature range  $-70^{\circ}\text{C}$  to  $250^{\circ}\text{C}$ ; odorless; low in acute oral toxicity; noncorrosive toward metals and alloys commonly found in heat transfer systems.

*Syltherm 800 (Dow Corning Corporation)*. A silicone polymer (Dimethyl Polysiloxane); recommended temperature range  $-40^{\circ}\text{C}$  to  $400^{\circ}\text{C}$ ; similar to Syltherm XLT, more appropriate for somewhat higher temperatures; flash point is  $160^{\circ}\text{C}$ .

*D-limonene (Florida Chemicals)*. A compound of optically active terpene ( $\text{C}_{10}\text{H}_{16}$ ) derived as an extract from orange and lemon oils; limited data shows very low viscosity at low temperatures—only one centipoise at  $-50^{\circ}\text{C}$ ; natural substance having questionable stability.

*Therminol D-12 (Monsanto)*. A synthetic hydrocarbon; clear liquid; recommended range  $-40^{\circ}\text{C}$  to  $250^{\circ}\text{C}$ ; not appropriate for contact with food; precautions against ignitions and fires should be taken with this product; could be found under trade names Santotherm or Gilotherm.

*Therminol LT (Monsanto)*. Akybenzene, synthetic aromatic ( $\text{C}_{10}\text{H}_{14}$ ); recommended range  $-70^{\circ}\text{C}$  to  $-180^{\circ}\text{C}$ ; not appropriate for contact with food; precautions against ignitions and fire should be taken dealing with this product.

*Dowtherm J (Dow Corning Corporation)*. A mixture of isomers of an alkylated aromatic; recommended temperature range  $-70^{\circ}\text{C}$  to  $300^{\circ}\text{C}$ ; noncorrosive toward steel, common metals and alloys; combustible material; flash point  $58^{\circ}\text{C}$ ; low toxic; prolonged and repeated exposure to vapors should be limited 10 ppm for daily exposures of eight hours.

*Dowtherm Q (Dow Corning Corporation)*. A mixture of dyphenylene-hane and alkylated aromatics; recommended temperature range  $-30^{\circ}\text{C}$  to  $330^{\circ}\text{C}$ ; combustible material; flash point  $120^{\circ}\text{C}$ ; considered low toxic, similar to Dowtherm J.

**Safety in Refrigeration Systems** This is of paramount importance and should be considered at every stage of installation.

The design engineer should have safety as the primary concern by choosing suitable system and refrigerant; selecting components, choosing materials and thicknesses of vessels, pipes, and relief valves of pressure vessels, proper venting of machine rooms, and arranging the equipment for convenient access for service and maintenance (piping arrangements, valve location, machine room layout, etc.). He or she should conform to the stipulation of the safety codes, which is also important for the purpose of professional liability.

During construction and installation, the installer's good decisions and judgments are crucial for safety, because design documentation never specifies all details. This is especially important when there is reconstruction or repair while the facility has been charged.

During operation the plant is in the hands of the operating personnel. They should be properly trained and familiar with the installation. Very often, accidents are caused by an improper practice, such as making an attempt to repair when proper preparation is not made. Operators should be trained in first-aid procedures and how to respond to emergencies.

Most frequently needed standards and codes are listed below, and the reader can find comments in: W. F. Stoecker: *Industrial Refrigeration*, Vol. 2. Ch. 12, Business News Publishing Co., Troy, MI, 1995; *ASHRAE Handbook Refrigeration System and Applications*, 1994, Ch. 51. Some important standards and codes on safety that a refrigeration engineer should consult are: ANSI/ASHRAE Standard 15-92—Safety Code for Mechanical Refrigeration, ASHRAE, Atlanta GA, 1992; ANSI/ASHRAE Standard 34-92—Number Designation of Refrigerants, ASHRAE, Atlanta GA, 1992; ANSI/ASME Boiler and Pressure Vessel Code, ASME, New York, 1989; ANSI/ASME Code for Pressure Piping, B31, B31.5-1987, ASME, New York, 1987; ANSI/IIAR 2—1984, Equipment, Design and Installation of Ammonia Mechanical Refrigeration Systems, IIAR, Chicago, 1984; IIAR Minimum Safety Criteria for a Safe Ammonia Refrigeration Systems, Bulletin 109; IIAR, IIAR Start-up, Inspection, and Maintenance of Ammonia Mechanical Refrigeration Systems, Bulletin 110, Chicago, 1988; IIAR Recommended Procedures in Event of Ammonia Spills, Bulletin No. 106, IIAR, Chicago, 1977; A Guide to Good Practices for the Operation of an Ammonia Refrigeration System, IIAR Bulletin R1, 1983.

## CRYOGENIC PROCESSES

### INTRODUCTION

Cryogenics, the production of low temperatures, is a major business in the United States with an annual market in 2003 of almost 14 billion dollars. It is a very diverse supporting technology, a means to an end and not an end in itself. For example, the combined production of oxygen and nitrogen, obtained by the cryogenic separation of air, accounts for 15 percent of the total annual U.S. production of  $2.9 \times 10^{11}$  kg (639 billion lb) of organics and inorganics (2003 *C & EN Annual Report*). Liquid hydrogen production, in the last four decades, has risen from laboratory quantities to a level of over 2.2 kg/s, first spurred by nuclear weapons development and later by the United States space program. Similarly, the space age increased the need for liquid helium by more than a factor of ten, requiring the construction of large plants to separate helium from natural gas by cryogenic means. The demands for energy have likewise accelerated the construction of large base-load liquefied natural gas (LNG) plants around the world and have been responsible for the associated domestic LNG industry of today with its use of peak shaving plants.

Freezing as a means of preserving food dates back to 1840. However, today the food industry uses large quantities of liquid nitrogen for this purpose and uses the refrigerant in frozen-food transport systems. In biological applications liquid-nitrogen cooled containers are routinely used to preserve whole blood, tissue, bone marrow, and animal semen for extended periods of time. Cryogenic surgery has become accepted in curing such involuntary disorders as Parkinson's disease. Medical analysis of patients has increased in sophistication with the use of *magnetic resonance imaging* (MRI) which utilizes cryogenically cooled superconducting magnets. Finally, one must recognize the role that cryogenics plays in the chemical-processing industry with the recovery of valuable feedstocks from natural gas streams, upgrading the heat content of fuel gas, purification of various process and waste streams, production of ethylene, as well as other chemical processes.

### PROPERTIES OF CRYOGENIC FLUIDS

There are presently several database programs of thermodynamic properties data developed specifically for fluids commonly associated with low temperature processing including helium, hydrogen, neon, nitrogen, oxygen, argon, and methane. For example, the NIST Standard Reference Database 12, Version 3.0 includes a total of 34 thermophysical properties for seventeen fluids in the database. Cryodata Inc. provides a similar computer version for 28 pure fluids as well as for mixtures incorporating many of these fluids. To fully appreciate the major effort in developing these thermophysical properties, consult the presentation by Jacobsen and coworkers (*Thermodynamics Properties of Cryogenic Fluids*, Plenum Press, New York, 1997). A few peculiarities associated with the fluids of helium, hydrogen, oxygen, and air need to be noted below.

Liquid helium-4 can exist in two different liquid phases: liquid helium I, the normal liquid, and liquid helium II, the superfluid, since under certain conditions the latter fluid acts as if it had no viscosity. The phase transition between the two liquid phases is identified as the lambda line and where this transition intersects the vapor-pressure curve is designated as the lambda point. Thus, there is no triple point for this fluid as for other fluids. In fact, solid helium can only exist under a pressure of 2.5 MPa or more.

A unique property of hydrogen is that it can exist in two different molecular forms: orthohydrogen and parahydrogen. (This is also true for deuterium, an isotope of hydrogen with an atomic mass of 2.) The thermodynamic equilibrium composition of the ortho- and para-varieties is temperature dependent. The equilibrium mixture of 75 percent orthohydrogen and 25 percent parahydrogen at ambient temperatures is recognized as normal hydrogen.

In contrast to other cryogenic fluids, liquid oxygen is slightly magnetic. It is also chemically reactive, particularly with hydrocarbon materials. Oxygen thus presents a safety problem and requires extra precautions in handling.

Since air is a mixture of predominantly nitrogen, oxygen, and a host of lesser impurities, there has been less interest in developing precise thermodynamic properties. A widely used correlation of thermodynamic properties is that published by NIST (NIST Database 72, NIST AIRPROPS, Version 1.0).

### PROPERTIES OF SOLIDS

A knowledge of the properties and behavior of materials used in any cryogenic system is essential for proper design considerations. Often the choice of materials for the construction of cryogenic equipment will be dictated by other considerations besides mechanical properties as, for example, thermal conductivity (heat transfer along a structural member), thermal expansivity (expansion and contraction during cycling between ambient and low temperatures), and density (mass of the system). Since properties at low temperatures are often significantly different from those at ambient temperature, there is no substitute for test data on a truly representative sample specimen when designing for the limit of effectiveness of a cryogenic material or structure. For example, some metals including elements, intermetallic compounds and alloys exhibit the phenomenon of superconductivity at very low temperatures. The properties that are affected when a material becomes superconducting include specific heat, thermal conductivity, electrical resistance, magnetic permeability, and thermoelectric effect. As a result, the use of superconducting metals in the construction of equipment for temperatures lower than 10 K needs to be evaluated carefully. (High temperature superconductors because of their brittle ceramic structure are generally not considered as construction materials.)

**Structural Properties at Low Temperatures** It is most convenient to classify metals by their lattice symmetry for low temperature mechanical properties considerations. The *face-centered-cubic* (fcc) metals and their alloys are most often used in the construction of cryogenic equipment. Al, Cu Ni, their alloys, and the austenitic stainless steels of the 18-8 type are fcc and do not exhibit an impact ductile-to-brittle transition at low temperatures. As a general rule, the mechanical properties of these metals with the exception of 2024-T4 aluminum, improve as the temperature is reduced. Since annealing of these metals and alloys can affect both the ultimate and yield strengths, care must be exercised under these conditions.

The *body-centered-cubic* (bcc) metals and alloys are normally classified as undesirable for low temperature construction. This class includes Fe, the martensitic steels (low carbon and the 400-series stainless steels), Mo, and Nb. If not brittle at room temperature, these materials exhibit a ductile-to-brittle transition at low temperatures. Cold working of some steels, in particular, can induce the austenite-to-martensite transition.

The *hexagonal-close-packed* (hcp) metals generally exhibit mechanical properties intermediate between those of the fcc and bcc metals. For example Zn encounters a ductile-to-brittle transition whereas Zr and pure Ti do not. The latter and their alloys with a hcp structure remain reasonably ductile at low temperatures and have been used for many applications where weight reduction and reduced heat leakage through the material have been important. However, small impurities of O, N, H, and C can have a detrimental effect on the low temperature ductility properties of Ti and its alloys.

Plastics increase in strength as the temperature is decreased, but this is also accompanied by a rapid decrease in elongation in a tensile test and a decrease in impact resistance. Teflon and glass-reinforced plastics retain appreciable impact resistance as the temperature is lowered. The glass-reinforced plastics also have high strength-to-weight and strength-to-thermal conductivity ratios. All elastomers, on the other hand, become brittle at low temperatures. Nevertheless, many of these materials including rubber, Mylar, and nylon can be used for static seal gaskets provided they are highly compressed at room temperature prior to cooling.

The strength of glass under constant loading also increases with decrease in temperature. Since failure occurs at a lower stress when

the glass surface contains surface defects, the strength can be improved by tempering the surface.

**Thermal Properties at Low Temperatures** The thermodynamic properties of most gases at low temperature approximate quite closely those of an ideal gas, and thus the ideal equation of state is adequate for most cryogenic engineering designs. However, the heat capacity of an ideal gas depends on its molecular composition. In most cryogenic gases, electronic and vibrational degrees of freedom are not excited, while rotational contributions to the specific heat at constant-volume conditions have reached their full classical value. Under these conditions the heat capacity depends only on whether the gas is composed of isolated atoms, linear molecules, or nonlinear molecules. The values for  $C_v$ , the heat capacity at constant volume are  $3/2R$ ,  $5/2R$  and  $3R$ , respectively, where  $R$  is the universal gas constant. For  $C_p$ , the heat capacity at constant pressure, the corresponding values are  $5/2R$ ,  $7/2R$ , and  $4R$ , respectively. The  $C_v$  data for a number of real gases presented by Barron and White (*Heat Capacity and Thermal Expansion at Low Temperatures*, Kluwer Academic/Plenum Publisher, New York, 1999, p. 132) show that these values provide a good approximation over wide ranges in temperatures, although there are large deviations for hydrogen below about 200 K and for methane at higher temperatures due to vibrational excitation. For solids, the Debye model developed with the aid of statistical mechanics and quantum theory gives a satisfactory representation of the specific heat with temperature. Procedures for calculating values of  $\Theta_D$ , the Debye characteristic temperature, using either elastic constants, the compressibility, the melting point, or the temperature dependence of the expansion coefficient are outlined by Barron (*Cryogenic Systems*, 2d ed., Oxford University Press, 1985, pp. 24–29).

Adequate prediction of the thermal conductivity for pure metals can be made by means of the Wiedeman-Franz law which states that the ratio of the thermal conductivity to the product of the electrical conductivity and the absolute temperature is a constant. This ratio for high-conductivity metals extrapolates essentially to the Sommerfeld value of  $2.449 \times 10^{-8} \text{ W } \Omega/\text{K}^2$  at 0 K, but falls considerably below it at higher temperatures. High-purity aluminum and copper exhibit peaks in thermal conductivity between 20 to 50 K, but these peaks are rapidly suppressed with increased impurity levels and cold work of the metal. Some metals including Monel, Inconel, stainless steel, and structural and aluminum alloys show a steady decrease in thermal conductivity with a decrease in temperature.

All cryogenic liquids except hydrogen and helium have thermal conductivities that increase as the temperature is decreased. For these two exceptions, the thermal conductivity decreases with a decrease in temperature. The kinetic theory of gases correctly predicts the decrease in thermal conductivity of all gases when the temperature is lowered.

The expansion coefficient of a solid can be estimated with the aid of an approximate thermodynamic equation of state for solids which equates the thermal expansion coefficient  $\beta$  with the quantity  $\gamma C_p \rho/B$  where  $\gamma$  is the Grüneisen dimensionless ratio,  $C_p$  is the specific heat of the solid,  $\rho$  is the density of the material, and  $B$  is the bulk modulus. For fcc metals the average value of the Grüneisen constant is near 2.3. However, there is a tendency for this constant to increase with atomic number.

**Electrical Properties at Low Temperatures** The electrical resistivity of most pure metallic elements at ambient and moderately low temperatures is approximately proportional to the absolute temperature. At very low temperatures, however, the resistivity (with the exception of superconductors) approaches a residual value almost independent of temperature. Alloys, on the other hand, have resistivities much higher than those of their constituent elements and resistance-temperature coefficients that are quite low. The electrical resistivity of alloys as a consequence is largely independent of temperature and may often be of the same magnitude as the room temperature value.

**Superconductivity** The physical state in which all resistance to the flow of direct-current electricity disappears is defined as superconductivity. The Bardeen-Cooper-Schrieffer (BCS) theory has been reasonably successful in accounting for most of the basic features observed of the superconducting state for low-temperature superconductors (LTS) operating below 23 K. The advent of the ceramic high-temperature superconductors (HTS) by Bednorz and Miller (*Z. Phys.*

**TABLE 11-25 Composition and Critical Temperature  $T_c$  for HTS Materials**

| Formula*      | Accepted notation | Forms reported |      | Critical $T_c$ , K |
|---------------|-------------------|----------------|------|--------------------|
| Y-Ba-Cu-O†    | YBCO              | 123            | 124  | 80–92              |
| Bi-Sr-Ca-Cu-O | BSCCO             | 2212           | 2223 | 80–110             |
| Tl-Ba-Ca-Cu-O | TBCCO             | Several        |      | to 125             |
| Hg-Ba-Ca-Cu-O | HBCCO             | 1201           | 1223 | 95–155‡            |

\*Subscripts for compounds are not listed since there are generally several forms that can be produced (see column 3).

†Other rare earths may be substituted for Y (yttrium) providing new compounds with somewhat different properties.

‡Highest  $T_c$  obtained while subjecting sample to external pressure.

B64, 189, 1989) has called for modifications to existing theories which are still actively being debated. The massive interest in the new superconductors that can be cooled with liquid nitrogen is presently underway with many new applications.

Three important characteristics of the superconducting state are the critical temperature, the critical magnetic field, and the critical current. These parameters many times can be varied by using different materials or giving them special metallurgical treatments.

The alloy *niobium titanium* (NbTi) and the intermetallic compound of *niobium and tin* (Nb<sub>3</sub>Sn) are the most technologically advanced LTS materials presently available. Even though NbTi has a lower critical field and critical current density, it is often selected because its metallurgical properties favor convenient wire fabrication. In contrast, Nb<sub>3</sub>Sn is a very brittle material and requires wire fabrication under very well-defined temperature conditions. However, the recent discovery by Akimitsu and coworkers (*Nature* **10**, 63, 2001) of the superconducting property of MgBr<sub>2</sub> has spurred new activity in the development of still another LTS conductor for magnetic coils.

There are essentially four families of high-temperature superconductors under investigation for practical magnet applications. Table 11-25 shows that all HTS are copper oxide ceramics even though the oxygen content may vary. However, this variation generally has little effect on the physical properties of importance to superconductivity.

The most widely used development in HTS wire production is the powder-in-tube procedure with BSCCO ceramic materials. In this procedure very fine HTS powder, placed inside of a hollow silver tube, is fused as the tube length is mechanically increased to form a wire. Very high magnetic fields with this wire have been reported at 4 K; however, performance degrades substantially above 20 to 30 K.

HTS materials, because of their ceramic nature, are quite brittle. This has introduced problems relative to the winding of superconducting magnets. One solution is to first wind the magnet with the powder-in-tube wire before the ceramic powder has been bonded and then heat treat the desired configuration to form the final product. Another solution is to form the superconductor into such fine filaments that they remain sufficiently flexible even after the powder has been heat treated.

## REFRIGERATION AND LIQUEFACTION

A process for producing refrigeration or liquefaction at cryogenic temperatures usually involves ambient compression of a process fluid with heat rejection to a coolant. During the compression process, the enthalpy and entropy of the fluid are decreased. At the cryogenic temperature where heat is absorbed, the enthalpy and entropy are increased. The reduction in temperature of the process fluid is usually accomplished by heat exchange with a colder fluid and then followed by an expansion. This expansion may take place using either a throttling device (isenthalpic expansion) with only a reduction in temperature or a work-producing device (isentropic expansion) in which both temperature and enthalpy are decreased. Because of liquid withdrawal, a liquefaction system experiences an unbalanced flow in the heat exchanger while a refrigeration system with no liquid withdrawal system usually operates with a balanced flow in the heat exchanger, except where a portion of the flow is diverted through the work-producing expander.

**Principles** The performance of a real refrigerator is measured by the coefficient of performance, COP, defined as

$$\text{COP} = \frac{Q}{W} = \frac{\text{heat removed at low temperature}}{\text{net work input}} \quad (11-108)$$

Another means of comparing the performance of a practical refrigerator is by the use of the figure of merit, FOM, defined as

$$\text{FOM} = \frac{\text{COP}}{\text{COP}_i} \quad (11-109)$$

where COP is the coefficient of performance of the actual refrigerator system and  $\text{COP}_i$  is the coefficient of performance for the thermodynamically ideal system. For a liquefier, the FOM is generally specified as

$$\text{FOM} = \frac{W_i/\dot{m}_f}{W/\dot{m}_f} \quad (11-110)$$

where  $W_i$  is the work of compression for the ideal cycle,  $W$  is the work of compression for the actual cycle, and  $\dot{m}_f$  is the mass rate liquefied in the ideal or actual cycle.

The methods of refrigeration and/or liquefaction generally used include (1) vaporization of a liquid, (2) application of the Joule-Thomson effect in a gas, and (3) expansion of a gas in a work-producing engine. Normal commercial refrigeration generally is accomplished in a vapor-compression process. Temperatures to about 200 K can be obtained by cascading vapor-compression processes in which refrigeration is accomplished by liquid evaporation. Below this temperature, isenthalpic or isentropic expansions are generally used either singly or in combination. With few exceptions, refrigerators using these methods also absorb heat by vaporization of the liquid.

If refrigeration is to be accomplished at a temperature range where no suitable liquid exists to absorb heat by evaporation, then a cold gas must be available to absorb the heat. This is generally accomplished by using a work-producing expansion engine.

**Expansion Types of Refrigerators** A thermodynamic process utilizing isenthalpic expansion to obtain cryogenic temperatures, and commonly referred to as the simple Linde or J-T cycle, is shown schematically with its corresponding temperature-entropy diagram in Fig. 11-111. The gaseous refrigerant is compressed at ambient temperature while essentially rejecting heat isothermally to a coolant. The compressed refrigerant is cooled countercurrently in a heat exchanger by the cold gas stream leaving the liquid reservoir before it enters the throttling valve. Upon expansion, Joule-Thomson cooling further reduces the temperature until, at steady state, a portion of the refrigerant is liquefied. For a refrigerator, the unliquefied fraction and the vapor formed by liquid evaporation from the absorbed heat  $Q$  are warmed in the heat exchanger before returning to the intake of the compressor. Assuming no heat leaks, as well as negligible kinetic and potential energy changes in the fluid, the refrigeration duty  $Q$  is equivalent to  $\dot{m}(h_1 - h_2)$ , where the subscripts refer to the locations

shown on Fig. 11-111. Applying Eq. 11-108, the coefficient of performance for the ideal J-T refrigerator is given by

$$\text{COP} = \frac{h_1 - h_2}{T_1[s_1 - s_2 - (h_1 - h_2)]} \quad (11-111)$$

For a simple J-T liquefier, the liquefied portion is continuously withdrawn from the reservoir and only the unliquefied portion of the fluid is warmed in the countercurrent heat exchanger and returned to the compressor. The fraction  $y$  that is liquefied is obtained by applying the first law to the heat exchanger, J-T valve, and liquid reservoir. This results in

$$y = \frac{h_1 - h_2}{h_1 - h_f} \quad (11-112)$$

where  $h_f$  is the specific enthalpy of the liquid being withdrawn. Maximum liquefaction occurs when the difference between  $h_1$  and  $h_2$  is maximized. To account for heat leak,  $q_L$ , the relation needs to be modified to

$$y = \frac{h_1 - h_2 - q_L}{h_1 - h_f} \quad (11-113)$$

with a resultant decrease in the fraction liquefied.

Refrigerants used in this process have a critical temperature well below ambient; consequently liquefaction by direct compression is not possible. In addition, the inversion temperature of the refrigerant must be above ambient temperature to provide initial cooling by the J-T process. Auxiliary refrigeration is required if the simple J-T cycle is to be used to liquefy neon, hydrogen, or helium whose inversion temperatures are below ambient. Liquid nitrogen is the optimum refrigerant for hydrogen and neon liquefaction systems, while liquid hydrogen is the normal refrigerant for helium liquefaction systems.

To reduce the work of compression in this cycle a two-stage or dual-pressure process may be used whereby the pressure is reduced by two successive isenthalpic expansions. Since the isothermal work of compression is approximately proportional to the logarithm of the pressure ratio, and the Joule-Thomson cooling is roughly proportional to the pressure difference, there is a greater reduction in compressor work than in refrigerating performance for this dual-pressure process.

In a work-producing expansion, the temperature of the process fluid is always reduced; hence, cooling does not depend on being below the inversion temperature prior to expansion. Additionally, the work-producing expansion results in a larger amount of cooling than in an isenthalpic expansion over the same pressure difference.

In large systems utilizing expanders, the work produced during expansion is conserved. In small refrigerators, the energy from the expansion is usually expended in a gas or hydraulic pump, or other suitable device. A schematic of a simple cold-gas refrigerator using this expansion principle and the corresponding temperature-entropy diagram is shown in Fig. 11-112. Gas compressed isothermally at ambient temperature is cooled in a heat exchanger by gas being warmed on its return to the compressor intake. Further cooling takes

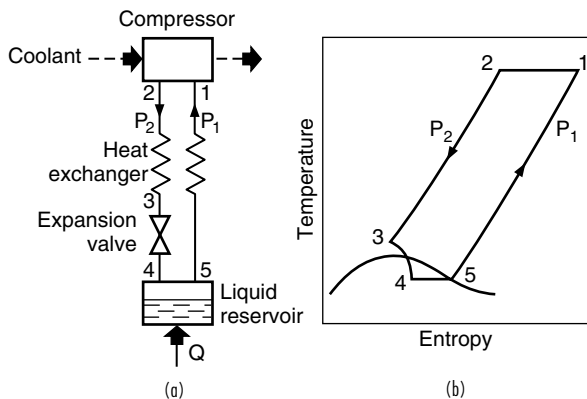


FIG. 11-111 Refrigerator using simple J-T cycle.

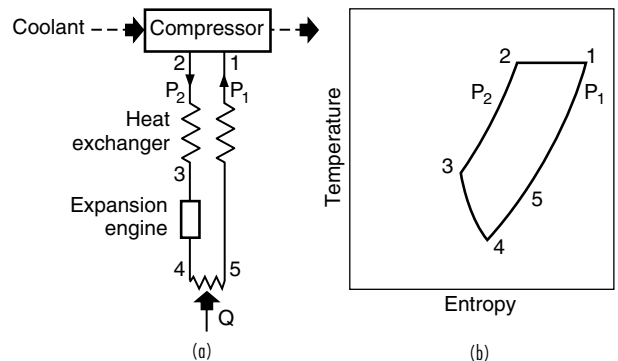


FIG. 11-112 Cold-gas refrigerator.

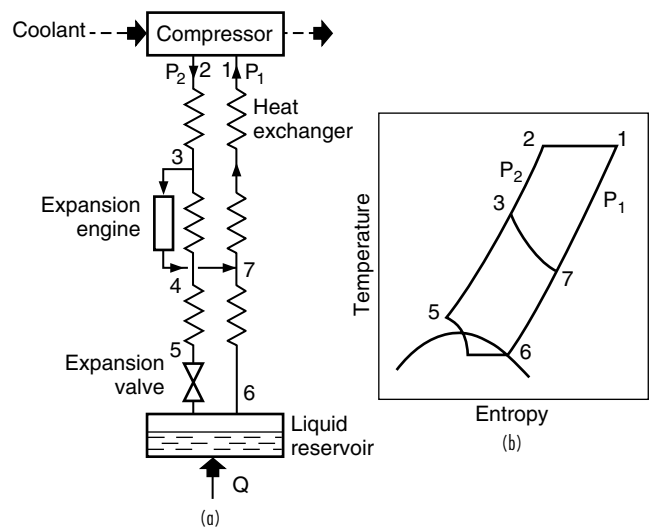


FIG. 11-113 Claude cycle refrigerator utilizing both expansion processes.

place during the engine expansion. In practice this expansion is never truly isentropic, and is reflected by path 3-4 on the temperature-entropy diagram. This specific refrigerant produces a cold gas which absorbs heat from 4-5 and provides a method of refrigeration that can be used to obtain temperatures between those of the boiling points of the lower-boiling cryogenes.

It is not uncommon to utilize both the isentropic and isenthalpic expansions in a cycle. This is done to avoid the technical difficulties associated with the formation of liquid in the expander. The Claude or expansion engine cycle is an example of a combination of these methods and is shown in Fig. 11-113 along with the corresponding temperature-entropy diagram.

The mixed refrigerant cycle was developed to meet the need for liquefying large quantities of natural gas to minimize transportation costs of this fuel. This cycle resembles the classic cascade cycle in principle and may best be understood by referring to that cycle. In the latter, the natural gas stream after purification is cooled successively by vaporization of propane, ethylene, and methane. Each refrigerant may be vaporized at two or three pressure levels to increase the natural gas cooling efficiency, but at a cost of considerable increased process complexity.

Cooling curves for natural gas liquefaction by the cascade process are shown in Fig. 11-114. It is evident that the cascade cycle efficiency can be improved by increasing the number of refrigerants employed.

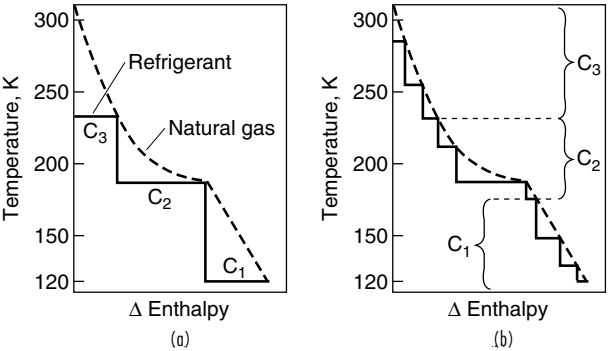


FIG. 11-114 Three- and nine-level cascade cycle cooling curve for natural gas.

For the same refrigeration capacity, the actual work required for the nine-level cascade cycle depicted is approximately 80 percent of that required by the three-level cascade cycle. This increase in efficiency is achieved by minimizing the temperature difference between the refrigerant and the natural gas stream throughout each increment of the cooling curve.

The mixed refrigerant cycle is a variation of the cascade cycle and involves the circulation of a single multicomponent refrigerant stream. The simplification of the compression and heat exchange services in such a cycle can offer potential for reduced capital expenditure over the conventional cascade cycle.

Figure 11-115 shows the basic concepts for a mixed refrigerant cycle (Gaumer, *Advances in Cryogenic Engineering* Vol. 31, Plenum, New York, 1986, p. 1095). Variations of the cycle are proprietary with those cryogenic engineering firms that have developed the technology. However, all of the mixed refrigerant processes use a carefully prepared refrigerant mix which is repeatedly condensed, vaporized, separated, and expanded. Thus, these processes require

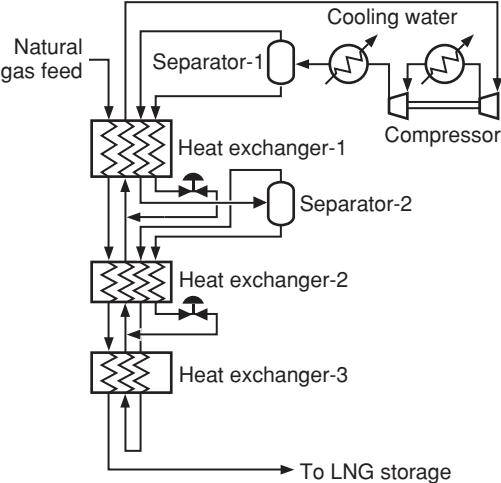


FIG. 11-115 Mixed-refrigerant cycle.



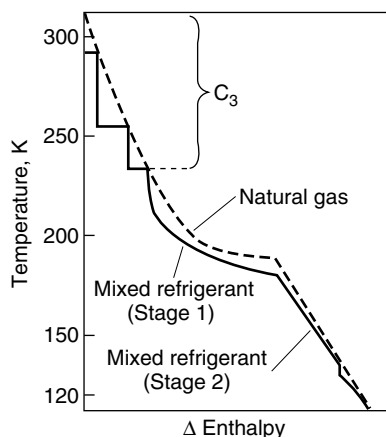


FIG. 11-116 Propane precooled mixed-refrigerant cycle cooling curve for natural gas.

more complete knowledge of the thermodynamic properties of gaseous mixtures than those required in the expander or classical cascade cycles. This is particularly evident when cooling curves similar to the one shown in Fig. 11-116 are desired. An inspection of the mixed refrigerant cycle also shows that these processes must routinely handle two-phase flows in the heat exchangers.

**Miniature Refrigerators** Expanded space and science projects have provided a need for miniature cryogenic coolers designated as cryocoolers. Such coolers provide useful cooling from a fraction of a watt to several watts at temperature levels from 1 to 90 K. These coolers are used to increase the sensitivity and signal-to-noise ratio of detectors by providing the required cryogenic operating temperatures as well as cooling the optical components to decrease the detector background radiation. The types of coolers developed to meet various specific requirements include solid cryogen coolers, radiative coolers, mechanical coolers,  $^3\text{He}$  adsorption coolers, adiabatic demagnetization refrigerators, and liquid helium storage systems. Mechanical coolers are generally classified as regenerative or recuperative. Regenerative coolers use reciprocating components that move the working fluid back and forth through a regenerator. The recuperative coolers, on the other hand, use countercurrent heat exchangers to accomplish the heat transfer operation. The Stirling, Gifford-McMahon, and pulse tube cycles are typically regenerative coolers while the Joule-Thomson and Brayton cycles are associated with recuperative coolers.

The miniature split single-stage Stirling cooler developed by Philips Laboratories produces 5 W of cooling at 65 K with the aid of linear motors and magnetically moving parts. A smaller split Stirling cycle cooler that uses a stacked diaphragm spring rather than magnetic means to levitate the piston and displacer has been developed at Oxford University (Bradshaw, et al., *Advances in Cryogenic Engineering*, Vol. 31, Plenum, New York, 1986, p. 801). The promise of higher reliability has spurred interest in the *pulse-tube refrigerator* (PTR). In the latest version of this device, an orifice and reservoir have been added to the warm end of the pulse tube (OPTR) to permit control of the phase shift required for optimum resonance in the system.

The Joule-Thomson cycle has also benefited from creative thinking. For example, Little (6th *International Cryocooler Conference*, Naval Postgraduate School, Monterey, CA, 1989, p. 3) has introduced a new method of fabricating J-T refrigerators using a photolithographic manufacturing process in which gas channels for the heat exchangers, expansion capillary, and liquid reservoir are etched on planar, glass substrates that are fused together to form the sealed refrigerator. These microminiature refrigerators have been made in a wide range of sizes and capacities.

Mixtures of highly polar gases are receiving considerable attention for J-T cycles since the magnitude of the Joule-Thomson coefficient

increases with nonideality of the gas. For example, for pure nitrogen with a high pressure of 5 MPa, a low pressure of 0.1 MPa, and an ambient temperature of 300 K, the ideal efficiency at 77 K is only 8.6 percent of Carnot. For a mixture of 40% nitrogen, 17% methane, 15% ethane, and 28% propane, the ideal efficiency for the same operating conditions is 52 percent of Carnot. The higher efficiency possible permits the use of compressors for domestic and commercial refrigeration, thereby reducing the overall costs of the cryogenic unit.

**Thermodynamic Analyses of Cycles** The thermodynamic quality measure of either a piece of equipment or an entire process is its reversibility. The second law, or more precisely the entropy increase, is an effective guide to this degree of irreversibility. However, to obtain a clearer picture of what these entropy increases mean, it has become convenient to relate such an analysis to the additional work that is required to overcome these irreversibilities. The fundamental equation for such an analysis is

$$W = W_{\text{rev}} + T_o \sum \dot{m} \Delta s \quad (11-114)$$

where the total work,  $W$ , is the sum of the reversible work,  $W_{\text{rev}}$ , plus a summation of the losses in availability for various unit operations in the analysis. Application of this method has been demonstrated numerically by Timmerhaus and Flynn (*Cryogenic Process Engineering*, Plenum Press, 1989, p. 175).

Numerous analyses and comparisons of refrigeration and liquefaction cycles are available in the literature. Great care must be exercised in accepting these comparisons since it is quite difficult to put all processes on a strictly comparable basis. Many assumptions need to be made in the course of the calculations, and these can have considerable effect on the conclusions. Major factors upon which assumptions generally have to be made include heat leak, temperature differences in the exchangers, efficiencies of compressors and expanders, number of stages of compression, fraction of expander work recovered, state of expander exhaust, purity and condition of inlet gases, pressure drop due to fluid flow, and so on. In view of this fact, differences in power requirements of 10 to 20 percent can readily be due to differences in assumed variables and can negate the advantage of one cycle over another. Barron (*Cryogenic Systems*, 2d ed., Oxford University Press, New York, 1985, p. 94) has made an analysis of some of the more common liquefaction systems described earlier that emphasize this point rather well.

## PROCESS EQUIPMENT

The equipment normally associated with cryogenic systems includes heat exchangers, compressors, expanders, throttling valves, and storage containers. Since the reciprocal or centrifugal compressors used generally operate at ambient temperatures, their operating principles are not covered here but in Sec. 10. Storage containers, are discussed later in Sec. 11.

**Heat Exchangers** Since most cryogens, with the exception of helium II behave as classical fluids, well-established principles of mechanics and thermodynamics at ambient temperature also apply for cryogens. Thus, similar conventional heat transfer correlations have been formulated for simple low-temperature heat exchangers. These correlations are described in terms of well-known dimensionless quantities such as the Nusselt, Reynolds, Prandtl, and Grashof numbers.

Because of the need to operate more efficiently at low temperatures, the simple heat exchangers have generally been replaced with more sophisticated types. Guidance for the development of such units for low-temperature service include the following factors:

1. Small temperature differences between inlet and exit streams to enhance efficiency.
2. Large surface area-to-volume ratio to minimize heat leak.
3. High heat transfer to reduce surface area.
4. Low mass to minimize start-up time.
5. Multichannel capability to minimize the number of units.
6. High-pressure capability to provide design flexibility.
7. Minimum pressure drop to reduce compression costs.
8. High reliability with minimal maintenance to reduce shutdowns.

Problems sometimes occur in trying to minimize the temperature difference at the cold end of the heat exchanger, particularly if the specific heat of the warm fluid decreases with decreasing temperature as is the case with gaseous hydrogen.

The selection of an exchanger for low-temperature operation is usually determined by the process-design requirements, mechanical-design limitations, and economic considerations. Laboratory needs are generally met by concentric tube and extended surface exchangers, while industrial needs are most often met by the coiled-tube, plate-fin, reversing, and regenerator types of exchangers.

The coiled-tube heat exchanger offers unique advantages, especially when dealing with low-temperature design conditions where (1) simultaneous heat transfer between more than two streams is desired, (2) a large number of heat transfer units is required, and (3) high operating pressures are involved. Heat transfer for single-phase flow of either gas or liquid on the tubeside is generally well represented by either the Colburn correlation or modified forms of the Dittus-Boelter relationship.

The shape of the cooling and warming curves in coiled-tube heat exchangers is affected by the pressure drop in both the tube and shell-sides of the heat exchanger. This is particularly important for two-phase flows of multicomponent systems. For example, an increase in pressure drop on the shellside causes boiling to occur at a higher temperature, while an increase in pressure drop on the tubeside will cause condensation to occur at a lower temperature. The net result is both a decrease in the effective temperature difference between the two streams and a requirement for additional heat transfer area to compensate for these losses.

Plate-fin heat exchangers are about nine times as compact as conventional shell-and-tube heat exchangers with the same amount of surface area, weigh less than conventional heat exchangers, and withstand design pressures up to 6 MPa for temperatures between 4 and 340 K. Flow instability frequently becomes a limiting design parameter for plate-fin heat exchangers handling either boiling or condensing two-phase flows. This results in lower optimum economic mass flow velocities for plate-fin heat exchangers when compared with coiled-tube heat exchangers. The use of fins or extended surfaces in plate-fin or similar exchangers greatly increases the heat transfer area. Calculations using finned surfaces are outlined earlier in Sec. 11.

There are two basic approaches to heat-exchanger design for low temperatures: (1) the effectiveness-NTU approach and (2) the *log-mean-temperature-difference* (LMTD) approach. The LMTD approach is used most frequently when all the required mass flows are known and the area of the exchanger is to be determined. The effectiveness-NTU approach is used more often when the inlet temperatures and the flow rates are specified for an exchanger with fixed area and the outlet temperatures are to be determined. Both methods are described earlier in Sec. 11.

System performance in cryogenic liquefiers and refrigerators is directly related to the effectiveness of the heat exchangers used in the system. For example, the liquid yield for a simple J-T cycle as given by Eq. 11-112 needs to be modified to

$$y = \frac{(h_1 - h_2) - (1 - \epsilon)(h_1 - h_g)}{(h_1 - h_f) - (1 - \epsilon)(h_1 - h_g)} \quad (11-115)$$

if the heat exchanger is less than 100 percent effective. Likewise, the heat exchanger ineffectiveness increases the work required for the system by an amount of

$$\Delta \dot{W} = \dot{m}(h_1 - h_g)(1 - \epsilon) \quad (11-116)$$

Uninterrupted operation of heat exchangers at low temperatures requires removal of essentially all impurities present in the streams that are to be cooled. Equipment is readily available for the satisfactory removal of these impurities by both chemical and physical methods, but at increased operating expense. Another effective method for also accomplishing this impurity removal utilizes reversing heat exchangers. Proper functioning of the reversing heat exchanger is dependent upon the relationship between the pressures and temperatures of the two streams. Since the pressures are generally fixed by other factors, the purification function of the heat exchanger is normally controlled by selecting the right temperature difference

throughout the heat exchanger. To assure that reevaporation takes place, these differences must be such that the vapor pressure of the impurity is greater than the partial pressure of the impurity in the purging stream.

Another type of reversing heat exchanger is the regenerator as detailed by Ackerman (*Cryogenic Regenerative Heat Exchangers*, Plenum Press, New York, 1997). As with all reversing heat exchangers, regenerators provide the simultaneous cooling and purification of gases in low-temperature processes. As noted earlier, reversing heat exchangers usually operate continuously. Regenerators do not operate continuously; instead, they operate by periodically storing heat in a packing during the first half of the cycle and then giving up this stored heat to the fluid during the second half of the cycle. Typically, a regenerator consists of two identical columns, which are packed with a porous solid material with a high heat capacity such as metal ribbon, through which the gases flow.

The low cost of the packing material, its large surface-area-per-unit volume, and the low-pressure drops encountered provide compelling arguments for utilizing regenerators. However, the intercontamination of fluid streams, caused by the periodic flow reversals, and the problems associated with designing regenerators to handle three or more fluids, has restricted their use to simple fluids, and favored adoption of plate-fin reversing heat exchangers.

**Expanders** The primary function of cryogenic expansion equipment is the reduction of the temperature of the gas being expanded to provide needed refrigeration. The expansion of a fluid to produce refrigeration may be carried out in two distinct ways: (1) in an expander where mechanical work is produced, and (2) in a Joule-Thomson valve where no work is produced.

**Mechanical Expanders** Reciprocating expanders are very similar in concept and design to reciprocating compressors. Generally these units are used with inlet pressures of 4 to 20 MPa. These machines operate at speeds up to 500 rpm. The thermal efficiencies (actual enthalpy difference/maximum possible enthalpy difference) range from about 75 percent for small units to 85 percent for large machines.

Turboexpanders have replaced reciprocating expanders in high-power installations as well as in small helium liquefiers. Sizes range from 0.75 to 7500 kW with flow rates up to 28 million m<sup>3</sup>/day. Today's large-tonnage air-separation plants are a reality due to the development of highly reliable and efficient turboexpanders. These expanders are being selected over other cryogenic equipment because of their ability to handle condensed ethane and heavier hydrocarbons. This type of expander usually weighs and costs less and requires less space and operating personnel.

Turboexpanders can be classified as either axial or radial. Axial flow expanders have either impulse or reaction type blades and are suitable for multistage expanders because they permit a much easier flow path from one stage to the next. However, radial turboexpanders have lower stresses at a given tip speed, which permits them to run at higher speeds. This results in higher efficiencies with correspondingly lower energy requirements. As a consequence, most turboexpanders built today are of the radial type.

**Joule-Thomson Valves** The principal function of a J-T valve is to obtain isenthalpic cooling of the gas flowing through the valve. These valves generally are needle-type valves modified for cryogenic operation. They are an important component in most refrigeration systems, particularly in the last stage of the liquefaction process. Joule-Thomson valves also offer an attractive alternative to turboexpanders for small-scale gas-recovery applications.

## SEPARATION AND PURIFICATION SYSTEMS

The energy required to reversibly separate gas mixtures is the same as that necessary to isothermally compress each component in the mixture from the partial pressure of the gas in the mixture to the final pressure of the mixture. This reversible isothermal work is given by the familiar relation

$$\frac{-\dot{W}_i}{\dot{m}} = T(s_1 - s_2) - (h_1 - h_2) \quad (11-117)$$

where  $s_1$  and  $h_1$  refer to conditions before the separation and  $s_2$  and  $h_2$  refer to conditions after the separation. For a binary system, and assuming a perfect gas for both components, Eq. (11-117) simplifies to

$$\frac{-W_i}{\dot{m}} = RT \left( n_A \ln \frac{P_T}{p_A} + n_B \ln \frac{P_T}{p_B} \right) \quad (11-118)$$

where  $n_A$  and  $n_B$  are the moles of A and B in the mixture,  $p_A$  and  $p_B$  are the partial pressures of these two components in the mixture, and  $P_T$  is the total pressure of the mixture. The figure of merit for a separation system is defined similar to that for a liquefaction system; see Eq. (11-110).

If the mixture to be separated is essentially a binary, both the McCabe-Thiele and the Ponchon-Savarit methods outlined in Sec. 13, with the appropriate cryogenic properties, can be used to obtain the ideal number of stages required. It should be noted, however, that it is not satisfactory in the separation of air to treat it as a binary mixture of oxygen and nitrogen if high purity (99 percent or better) oxygen is desired. The separation of oxygen from argon is a more difficult separation than oxygen from nitrogen and would require correspondingly many more plates. In fact, if the argon is not extracted from air, only 95 percent oxygen would be produced. The other rare gas constituents of air (helium, neon, krypton, and xenon) are present in such small quantities and have boiling points so far removed from those of oxygen and nitrogen that they introduce no important complications.

**Air-Separation Systems** Of the various separation schemes available today, the simplest is known as the Linde single-column system shown in Fig. 11-117 and first introduced in 1902. In it, purified compressed air passes through a precooling heat exchanger (if oxygen gas is the desired product, a three-channel exchanger for air, waste nitrogen, and oxygen gas is used; if liquid oxygen is to be recovered from the bottom of the column, a two-channel exchanger for air and waste nitrogen is employed), then through a coil in the boiler of the rectifying column where it is further cooled (acting at the same time as the boiler heat source); following this, it expands essentially to atmospheric pressure through a J-T valve and reenters the top of the column with the liquid providing the required reflux. Rectification occurs as liquid and vapor on each plate establish equilibrium. If oxygen gas is to be the product, purified air need only be compressed to a pressure of 3 to 6 MPa; if the product is to be liquid oxygen, a compressor outlet pressure of 20 MPa is necessary. Note that the Linde single-column separation system is simply a J-T liquefaction system

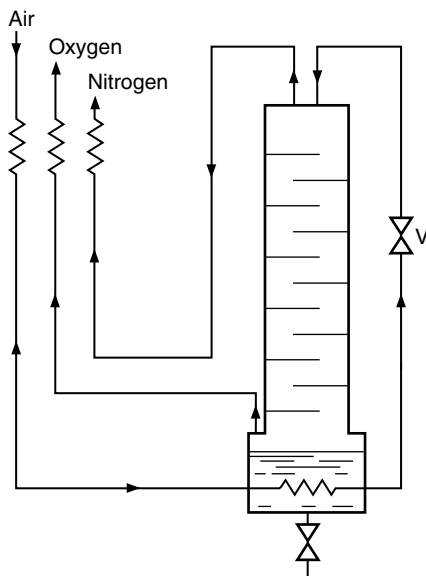


FIG. 11-117 Linde single-column air separator.

with a substitution of a rectification column for the liquid reservoir. However, any of the other liquefaction systems discussed earlier could have been used to furnish the liquid for the column.

In a simple single-column process, although the oxygen purity is high, the nitrogen effluent stream is impure. The equilibrium vapor concentration of the overhead nitrogen effluent for an initial liquid mixture of 21 percent oxygen/79 percent nitrogen at 100 kPa is about 6 to 7 percent oxygen. Thus, the nitrogen waste gas stream with such an impurity may only be usable as a purge gas for certain conditions.

The impurity problem noted in the previous paragraph was solved by the introduction of the Linde double-column system shown in Fig. 11-118. Two rectification columns are placed one on top of the other (hence the name double-column system).

In this system the liquid air is introduced at an intermediate point B into the lower column, and a condenser-evaporator at the top of the lower column makes the arrangement a complete reflux distillation column which delivers almost pure nitrogen at E. In order for the column simultaneously to deliver pure oxygen, the oxygen-rich liquid (about 45 percent  $O_2$ ) from the bottom boiler is introduced at an intermediate level C in the upper column. The reflux and rectification in the upper column produce pure oxygen at the bottom and pure nitrogen at the top provided all major impurities are first removed from the column. More than enough liquid nitrogen is produced in the lower column for the needed reflux in both columns. Since the condenser must condense nitrogen vapor by evaporating liquid oxygen, it is necessary to operate the lower column at a higher pressure, about 500 kPa, while the upper column is operated at approximately 110 kPa. This requires a reduction in the pressure of the fluids from the lower column as they are admitted to the upper column.

In the cycle shown, gaseous oxygen and nitrogen are withdrawn at room temperature. Liquid oxygen could be withdrawn from point A and liquid nitrogen from point E, but in this case more refrigeration would be needed.

Even the best modern low-temperature air separation plant has an efficiency only a small fraction of the theoretical optimum, that is, about 15 to 20 percent. The principal sources of inefficiency are threefold: (1) the nonideality of the refrigerating process, (2) the imperfection of the heat exchangers, and (3) losses of refrigeration through heat leak.

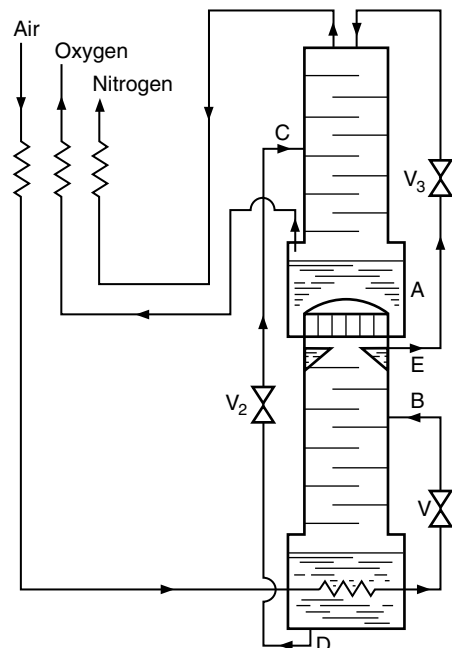
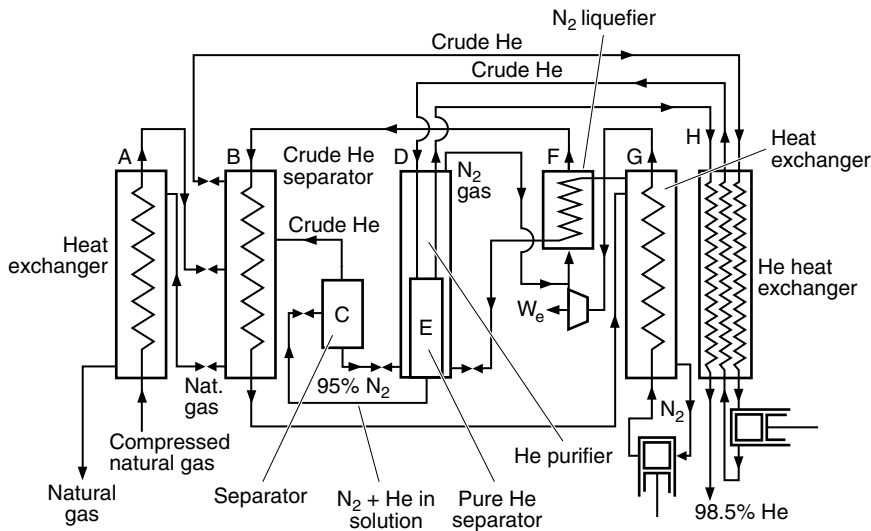


FIG. 11-118 Linde double-column air separator.



**FIG. 11-119** Typical helium-separation plant as operated by the U.S. Bureau of Mines.

**Helium and Natural-Gas Systems Separation** Helium is produced primarily by separation of helium-rich natural gas. The helium content of the natural gas from plants operated by the U.S. Bureau of Mines normally has varied from 1 to 2 percent while the nitrogen content of the natural gas has varied from 12 to 80 percent. The remainder of the natural gas is methane, ethane, and heavier hydrocarbons.

A Bureau of Mines system for the separation of helium from natural gas is shown in Fig. 11-119. Since the major constituents of natural gas have boiling points very much different from that of helium, a distillation column is not necessary and the separation can be accomplished with condenser-evaporators.

The need to obtain greater recoveries of the  $C_2$ ,  $C_3$ , and  $C_4$ 's in natural gas has resulted in the expanded use of low-temperature processing of these streams. The majority of the natural gas processing at low temperatures to recover light hydrocarbons is now accomplished using the turboexpander cycle. Feed gas is normally available from 1 to 10 MPa. The gas is first dehydrated to a dew point of 200 K and lower. After dehydration the feed is cooled with cold residue gas. Liquid produced at this point is separated before entering the expander and sent to the condensate stabilizer. The gas from the separator is expanded in a turboexpander where the exit stream can contain as much as 20 wt % liquid. This two-phase mixture is sent to the top section of the stabilizer which separates the two phases. The liquid is used as reflux in this unit while the cold gas exchanges heat with the fresh feed and is recompressed by the expander-driven compressor. Many variations to this cycle are possible and have been used in actual plants.

**Gas Purification** The nature and concentration of impurities to be removed depends on the type of process involved. For example, in the production of large quantities of oxygen, various impurities must be removed to avoid plugging of the cold process lines or to avoid buildup of hazardous contaminants. The impurities in air that would contribute most to plugging would be water and carbon dioxide. Helium, hydrogen, and neon, on the other hand, will accumulate on the condensing side of the condenser-reboiler located between the two separation columns and will reduce the rate of heat transfer unless removed by intermittent purging. The buildup of acetylene, however, can prove to be dangerous even though the feed concentration in the air is no greater than 0.04 ppm.

Refrigeration purification is a relatively simple method for removing water, carbon dioxide, and certain other contaminants from a process stream by condensation or freezing. (Either regenerators or reversing heat exchangers may be used for this purpose since a flow

reversal is periodically necessary to reevaporate and remove the solid deposits.) The effectiveness of this method depends upon the vapor pressure of the impurities relative to that of the major components of the process stream at the refrigeration temperature. Thus, assuming ideal gas behavior, the maximum impurity content in a gas stream after refrigeration would be inversely proportional to its vapor pressure. However, due to the departure from ideality at higher pressures, the impurity content will be considerably higher than predicted for the ideal situation. For example, the actual water vapor content in air will be over four times that predicted by ideal gas behavior at a temperature of 228 K and a pressure of 20 MPa.

Purification by a solid adsorbent is one of the most common low-temperature methods for removing impurities. Materials such as silica gel, carbon, and synthetic zeolites (molecular sieves) are widely used as adsorbents because of their extremely large effective surface areas. Most of the gels and carbon have pores of varying sizes in a given sample, but the synthetic zeolites are manufactured with closely controlled pore-size openings ranging from about 0.4 to 1.3 nm. This makes them even more selective than other adsorbents since it permits separation of gases on the basis of molecular size.

Information needed in the design of low-temperature adsorbers includes the equilibrium between the solid and gas and the rate of adsorption. Equilibrium data for the common systems generally are available from the suppliers of such material. The rate of adsorption is usually very rapid and the adsorption is essentially complete in a relatively narrow zone of the adsorber. If the concentration of the adsorbed gas is more than a trace, then heat of adsorption may also be a factor of importance in the design. (The heat of adsorption is usually of the same order or larger than the normal heat associated with the phase change.) Under such situations it is generally advisable to design the purification in two steps, that is, first removing a significant portion of the impurity either by condensation or chemical reaction and then completing the purification with a low-temperature adsorption system. A scheme combining the condensation and adsorption is shown in Fig. 11-120.

In normal plant operation at least two adsorption purifiers are employed—one in service while the other is desorbed of its impurities. Often there is an advantage in using an additional purifier by placing this unit in series with the adsorption unit to provide a backup if impurities are not trapped by the first unit. Cooling of the purifier must utilize some of the purified gas to avoid adsorption during this period.

Experience in air separation plant operations and other cryogenic processing plants has shown that local freeze-out of impurities such as

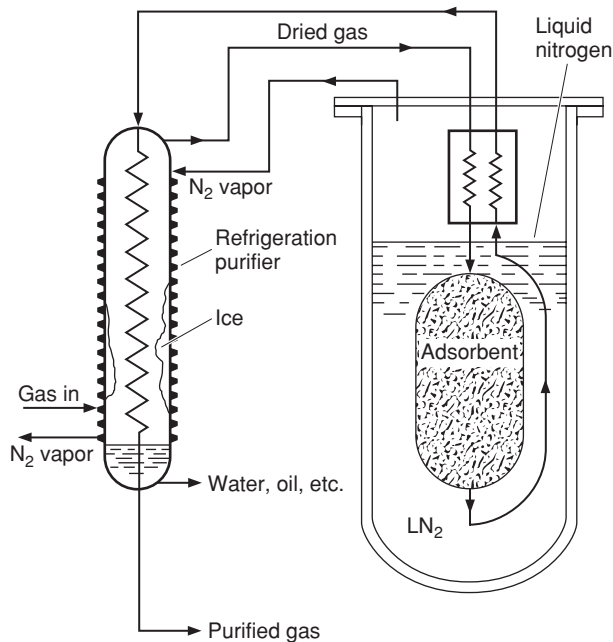


FIG. 11-120 Purifier using refrigeration and adsorption schemes in series.

carbon dioxide can occur at concentrations well below the solubility limit. For this reason, the carbon dioxide content of the feed gas subject to the minimum operating temperature is usually kept below 50 ppm. The amine process and the molecular sieve adsorption process are the most widely used methods for carbon dioxide removal. The amine process involves adsorption of the impurity by a lean aqueous organic amine solution. With sufficient amine recirculation rate, the carbon dioxide in the treated gas can be reduced to less than 25 ppm. Oxygen is removed by a catalytic reaction with hydrogen to form water.

## STORAGE AND TRANSFER SYSTEMS

Storage vessels range in type from low-performance containers, insulated by rigid foam or fibrous insulation where the liquid in the container evaporates in a few hours, up to high-performance containers, insulated with multilayer insulations where less than 0.1 percent of the fluid contents is evaporated per day. In the more effective units, the storage container consists of an inner vessel which encloses the cryogenic fluid to be stored and an outer vessel or vacuum jacket. The latter maintains the vacuum necessary to make the insulation effective and at the same time serves as a vapor barrier to the migration of water and other condensibles to the cold surface of the inner vessel. Improvements have been made in the insulation used in these containers, but the vacuum-insulated double-walled Dewar is still the basic idea for high-performance cryogenic-fluid container design.

**Insulation Principles** The effectiveness of a liquefier or refrigerator is highly dependent upon the heat leak entering such a system. Since heat removal becomes more costly with a lowering in temperature as demonstrated by the Carnot limitation, most cryogenic systems employ some form of insulation to minimize the effect. The insulation strategy is to minimize radiative heat transfer, minimize convective heat transfer, and use only a minimum of solid conductance media. Factors considered in the selection of the most suitable insulation include its ruggedness, convenience, volume, weight, ease of fabrication and handling, and thermal effectiveness and cost. It is common practice to use an experimentally obtained apparent thermal conductivity to characterize the thermal effectiveness of various insulations. Typical  $k_a$  values for insulations used in cryogenic service are listed in Table 11-26.

TABLE 11-26 Representative Apparent Thermal Conductivity Values

| Type of insulation                                        | $k_a$ , mW/m·K (77–300K)       |
|-----------------------------------------------------------|--------------------------------|
| Pure vacuum, $1.3 \times 10^{-4}$ Pa                      | 5                              |
| Foam insulation                                           | 26–35                          |
| Nonevacuated powders (perlite, silica aerogel)            | 19–44                          |
| Evacuated powders and fibers ( $1.3 \times 10^{-1}$ Pa)   | 1–2                            |
| Opacified powdered insulations ( $1.3 \times 10^{-1}$ Pa) | $3.5 \times 10^{-1}$           |
| Multilayer insulations ( $1.3 \times 10^{-4}$ Pa)         | $1.7\text{--}4 \times 10^{-2}$ |

**Types of Insulation** Cryogenic insulations have generally been divided into five general categories: high vacuum, multilayer insulation, powder, foam, and special insulations. Each is discussed in turn in the following sections.

**Vacuum Insulation** Heat transport across an evacuated space ( $1.3 \times 10^{-4}$  Pa or lower), is by radiation and by conduction through the residual gas. The heat transfer by radiation generally is predominant and can be approximated by

$$\frac{\dot{Q}_r}{A_1} = \sigma (T_2^4 - T_1^4) \left[ \frac{1}{e_1} + \frac{A_1}{A_2} \left( \frac{1}{e_2} - 1 \right) \right]^{-1} \quad (11-119)$$

where  $\dot{Q}_r/A_1$  is the heat transfer by radiation-per-unit area,  $\sigma$  is the Stefan-Boltzmann constant, and  $e$  is the emissivity of the surfaces. Subscript 2 refers to the hot surface and subscript 1 refers to the cold surface. The bracketed term on the right-hand side of this relation is designated as the overall emissivity factor,  $F_e$ .

The insertion of low-emissivity floating shields within the evacuated space can effectively reduce the heat transport by radiation. The effect of the shields is to greatly reduce the emissivity factor. For example, for  $N$  shields or  $(N+2)$  surfaces, an emissivity of the outer and inner surface of  $e_o$ , and an emissivity of the shields of  $e_s$ , the emissivity factor reduces to

$$\left[ 2 \left( \frac{1}{e_o} + \frac{1}{e_s} - 1 \right) + \frac{(N-1)(2-e_s)}{e_s} \right]^{-1} \quad (11-120)$$

In essence, one properly located low-emissivity shield can reduce the radiant heat transfer to around one-half of the rate without the shield, two shields can reduce this to around one-fourth of the rate without the shield, and so on.

**Multilayer Insulation** Multilayer insulation consists of alternating layers of highly reflecting material, such as aluminum foil or aluminized Mylar, and a low-conductivity spacer material or insulator, such as fiberglass mat or paper, glass fabric, or nylon net, all under high vacuum. When properly applied at the optimum density, this type of insulation can have an apparent thermal conductivity as low as 10 to 50  $\mu$ W/m·K between 20 and 300 K.

For a highly evacuated (on the order of  $1.3 \times 10^{-4}$  Pa) multilayer insulation, heat is transferred primarily by radiation and solid conduction through the spacer material. The apparent thermal conductivity of the insulation material under these conditions may be determined from

$$k_a = \frac{1}{N/\Delta x} \left\{ h_s + \frac{\sigma e T_2^3}{2-e} \left[ 1 + \left( \frac{T_1}{T_2} \right)^2 \right] \left( 1 + \frac{T_1}{T_2} \right) \right\} \quad (11-121)$$

where  $N/\Delta x$  is the number of complete layers (reflecting shield plus spacer) of the insulation-per-unit thickness,  $h_s$  is the solid conductance of the spacer material,  $\sigma$  is the Stefan-Boltzmann constant,  $e$  is the effective emissivity of the reflecting shield, and  $T_2$  and  $T_1$  are the temperatures of the warm and cold sides of the insulation, respectively. It is evident that the apparent thermal conductivity can be reduced by increasing the layer density up to a certain point. It is not obvious from the above relation that a compressive load affects the apparent thermal conductivity and thus the performance of a multilayer insulation. However, under a compressive load the solid conductance increases much more rapidly than  $N/\Delta x$  resulting in an overall increase in  $k_a$ . Plots of heat flux versus compressive load on a logarithmic scale result in straight lines with slopes between 0.5 and 0.67.

The effective thermal conductivity values generally obtained in practice are at least a factor of two greater than the one-dimensional thermal conductivity values measured in the laboratory with carefully

controlled techniques. This degradation in insulation thermal performance is caused by the combined presence of edge exposure to isothermal boundaries, gaps, joints, or penetrations in the insulation blanket required for structural supports, fill and vent lines, and high lateral thermal conductivity of these insulation systems.

**Powder Insulation** A method of realizing some of the benefits of multiple floating shields without incurring the difficulties of awkward structural complexities is to use evacuated powder insulation. The penalty incurred in the use of this type of insulation, however, is a tenfold reduction in the overall thermal effectiveness of the insulation system over that obtained for multilayer insulation. In applications where this is not a serious factor, such as LNG storage facilities, and investment cost is of major concern, even unevacuated powder-insulation systems have found useful applications. The variation in apparent mean thermal conductivity of several powders as a function of interstitial gas pressure is shown in the familiar S-shaped curves of Fig. 11-121.

The apparent thermal conductivity of powder insulation at cryogenic temperatures is generally obtained from

$$k_a = \frac{k_g}{1 - V_r(1 - k_g/k_s)} \quad (11-122)$$

where  $k_g$  is the thermal conductivity of the gas within the insulation,  $k_s$  is the thermal conductivity of the powder, and  $V_r$  is the ratio of solid volume to the total volume. The amount of heat transport due to radiation through the powders can be reduced by the addition of metallic powders. A mixture containing approximately 40 to 50 wt % of a metallic powder gives the optimum performance.

**Foam Insulation** Since foams are not homogeneous materials, their apparent thermal conductivity is dependent upon the bulk density of the insulation, the gas used to foam the insulation, and the mean temperature of the insulation. Heat conduction through a foam is determined by convection and radiation within the cells and by conduction in the solid structure. Evacuation of a foam is effective in reducing its thermal conductivity, indicating a partially open cellular structure, but the resulting values are still considerably higher than either multilayer or evacuated powder insulations.

Data on the thermal conductivity for a variety of foams used at cryogenic temperatures have been presented by Kropschot (*Cryogenic Technology*, R. W. Vance, ed., Wiley, New York, 1963, p. 239). Of all the foams, polyurethane and polystyrene have received the widest use at

low temperatures. The major disadvantage of foams is that they tend to crack upon repeated thermal cycling and lose their insulation value.

**Storage and Transfer Systems** In general, heat leak into a storage or transfer system for a cryogen is by (1) radiation and conduction through the insulation, and (2) conduction through any inner shell or transfer-line supports, piping leads, and access ports. Conduction losses are reduced by introducing long heat-leak paths, by making the cross sections for heat flow small, and by using materials with low thermal conductivity. Radiation losses, a major factor in the heat leak through insulations, are reduced with the use of radiation shields, such as multilayer insulation, boil-off vapor-cooled shields, and opacifiers in powder insulation.

Several considerations must be met when designing the inner vessel. The material of construction selected must be compatible with the stored cryogen. Nine percent nickel steels are acceptable for the higher-boiling cryogens ( $T > 75$  K) while many aluminum alloys and austenitic steels are usually structurally acceptable throughout the entire temperature range. Because of its high thermal conductivity, aluminum is not a recommended material for piping and supports that must cross the insulation space. A change to a material of lower thermal conductivity for this purpose introduces a transition joint of a dissimilar material. Since such transition joints are generally mechanical in nature, leaks into the vacuum space develop upon repeated temperature cycling. In addition, the larger thermal coefficient of expansion of aluminum can pose still further support and cooldown problems.

Economic and cooldown considerations dictate that the shell of the storage container be as thin as possible. As a consequence, the inner container is designed to withstand only the internal pressure and bending forces while stiffening rings are used to support the weight of the fluid. The minimum thickness of the inner shell for a cylindrical vessel under such a design arrangement is given by Sec. VIII of the *ASME Boiler and Pressure Vessel Code*.

Since the outer shell of the storage container is subjected to atmospheric pressure on one side and evacuated conditions going down to  $1.3 \times 10^{-4}$  Pa on the other, consideration must be given to provide ample thickness of the material to withstand collapsing or buckling. Failure by elastic instability is covered by the ASME Code, in which design charts are available for the design of cylinders and spheres subjected to external pressure. Stiffening rings are also used on the outer shell to support the weight of the inner container and its contents as well as maintaining the sphericity of the shell.

The outer shell is normally constructed of carbon steel for economic reasons, unless aluminum is required to reduce the weight. Stainless-steel standoffs must be provided on the carbon steel outer shell for all piping penetrations to avoid direct contact with these penetrations when they are cold.

There are a variety of methods for supporting the inner shell within the outer shell and the cold transfer line within the outer line. Materials that have a high strength to thermal conductivity ratio are selected for these supports. Design of these supports for the inner shell must allow for shipping loads which may be several orders higher than in-service loads. Compression supports such as legs or pads may be used, but tension supports are more common. These may take the form of cables, welded straps, threaded bars, or a combination of these to provide restraint of the inner shell in several directions.

Most storage containers for cryogens are designed for a 10 percent ullage volume. The latter permits reasonable vaporization of the contents due to heat leak without incurring too rapid a buildup of the pressure in the container. This, in turn, permits closure of the container for short periods of time to either avoid partial loss of the contents or to transport flammable or hazardous cryogens safely from one location to another.

## CRYOGENIC INSTRUMENTATION

Even though the combined production of cryogenic nitrogen and oxygen exceeds the production of any other chemical in the United States, the cryogenic industry does not appear to warrant a separate product line of instruments for diagnostic and control purposes. Low-temperature thermometry is the one exception. The general approach

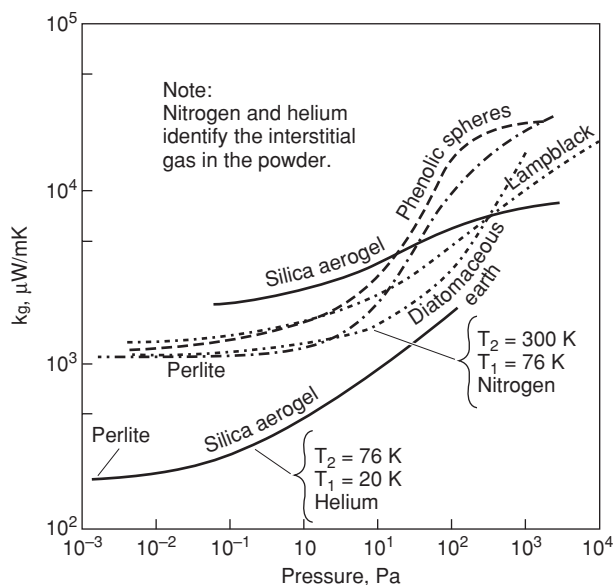


FIG. 11-121 Apparent mean thermal conductivities of several powder insulations as a function of interstitial gas pressure.

generally is that instruments developed for the usual CPI needs must be modified or accepted as is for cryogenic use.

Quite often problems arise when instruments for normal service are subjected to low temperature use. Since some metals become brittle at low temperatures, the instrument literally falls apart. Elastomeric gaskets and seals contract faster with decreasing temperatures than the surrounding metal parts, and the seal often is lost. Even hermetically sealed instruments can develop pin holes or small cracks to permit cryogenic liquids to enter these cases with time. Warming the instrument causes the trapped liquid to vaporize, sometimes generating excessive gas pressure and failure of the case.

Therefore, the first task in adapting normal instruments to cryogenic service is simply to give them a severe thermal shock by immersing them in liquid nitrogen repeatedly, and checking for mechanical integrity. This is the general issue; specific issues according to each type of measurement are discussed below.

**Pressure** This parameter is usually measured by the flush-mounted pressure transducer which consists of a force-summing device (bellow, diaphragm, bourdon tube, etc.) that translates the pressure into a displacement. The latter is then measured by an analog device (strain gage, piezoelectric crystal, variable distance between capacitor plates, and the like). Since these elements are likely to be made of different materials (bronze diaphragm, stainless-steel case, semiconductor strain gage), each will react to the temperature change in a different way. This is especially serious during cooldown, when the transient nature of material and construction prohibits all of the pressure-gage elements from being at the same temperature at the same time. Under steady-state conditions it is often possible to provide some temperature compensation through the well-known instrument technique of common-mode-rejection. Such compensation is generally not successful during transient temperature fluctuations. Only two courses of action are open: (1) hand-check each type of pressure transducer for thermal noise by thermally shocking it with immersion in liquid nitrogen; and (2) simplify the pressure-transducer construction to eliminate differences between materials. Some success has been observed in the latter area by manufacturers who make very small pressure sensing elements from a single semiconductor chip. The miniature size of these devices helps to reduce or eliminate temperature gradients across the device. The single-element nature of the pressure-gage assembly reduces differences in materials of construction.

**Liquid Level** The measurements for dense fluids such as liquid oxygen and liquid nitrogen are made in the conventional CPI approach using floats. Sight glasses cannot be used since radiation and thermal conduction would cause the cryogenic fluid within the sight glass to boil. The very light cryogens, liquid helium and liquid hydrogen, cannot sustain a float. Liquid hydrogen has the density of Styrofoam,<sup>TM</sup> about 70 g/l, making floating devices impractical. Some electrical analog is used for hydrogen and helium, most frequently a linear concentric-tube electrical capacitor. The dielectric constant of cryogens is related to their density by the Clausius-Mosotti relation. As the liquid level rises, the greater dielectric constant of the liquid between the tubes causes the overall capacitance to vary in a linear fashion. For best accuracy, these capacitance liquid-level measuring devices should be calibrated in place.

**Flow** The measurement of cryogenic fluids is most troublesome. Flow rate is not a natural physical parameter, like temperature, but is a derived quantity. A measurement of mass (or volume) must be made over a time interval to derive the flow rate. Because of this, any flow meter is only as good as its calibration. At this time, there is no national capability for calibrating cryogenic flowmeters. From data developed early in the nation's space program, considerable confidence has been developed in turbine-type flowmeters and in pressure-drop-type flowmeters. If all the usual ASTM guidelines are followed for meter installation, and if adequate temperature corrections are applied to changes in dimensions, then such meters can have an accuracy of  $\pm 1$  percent of their water calibrations. For very small flow applications, the Coriolis meters are promising. Vortex shedding flow meters appear useful for very large flow rates. Nonetheless, an actual calibration on the cryogen of interest is the only proof of accuracy.

**Temperature** The level of the temperature measurement (4 K, 20 K, 77 K, or higher) is the first issue to be considered. The second issue is the range needed (e.g., a few degrees around 90 K or 1 to 400 K). If the temperature level is that of air separation or liquefaction of natural gas (LNG), then the favorite choice is the platinum resistance thermometer (PRT). Platinum, as with all pure metals, has an electrical resistance that goes to zero as the absolute temperature decreases to zero. Accordingly, the lower useful limit of platinum is about 20 K, or liquid hydrogen temperatures. Below 20 K, semiconductor thermometers (germanium-, carbon-, or silicon-based) are preferred. Semiconductors have just the opposite resistance-temperature dependence of metals—their resistance increases as the temperature is lowered, as fewer valence electrons can be promoted into the conduction band at lower temperatures. Thus, semiconductors are usually chosen for temperatures from about 1 to 20 K.

If the temperature range of interest is large, say 1 to 400 K, then diode thermometers are recommended. Diodes have other advantages compared to resistance thermometers. By contrast, diode thermometers are very much smaller and faster. By selection of diodes all from the same melt, they may be made interchangeable. That is, one diode has the same calibration curve as another, which is not always the case with either semiconductor or metallic-resistance thermometers. It is well known, however, that diode thermometers may rectify an ac field, and thus may impose a dc noise on the diode output. Adequate shielding is required.

Special applications, such as in high-magnetic fields, require special thermometers. The carbon-glass and strontium-titanate resistance thermometers have the least magnetoresistance effects.

Thermocouples are unsurpassed for making temperature-difference measurements. The thermoelectric power of thermocouple materials makes them adequate for use at liquid-air temperatures and above. At 20 K and below, the thermoelectric power drops to a few  $\mu\text{V/K}$ , and their use in this range is as much art as science.

A descriptive flowchart has been prepared by Sparks (*Materials at Low Temperatures*, ASM, Metal Park, OH, 1983) to show the temperature range of cryogenic thermometers in general use today. Pavese and Molinar (*Modern Gas-Based Temperature and Pressure Measurements*, Plenum, New York, 1992) provide details on gas- and vapor-pressure thermometry at these temperatures.

## SAFETY

Past experience has shown that cryogenic fluids can be used safely in industrial environments as well as in typical laboratories provided all facilities are properly designed and maintained, and personnel handling these fluids are adequately trained and supervised. There are many hazards associated with cryogenic fluids. However, the principal ones are those associated with the response of the human body and the surroundings to the fluids and their vapors, and those associated with reactions between the fluids and their surroundings. Edeskuty and Stewart (*Safety in Handling Cryogenic Fluids*, Plenum Press, New York, 1996) provide a detailed examination of these various hazards.

**Physiological Hazards** Severe cold "burns" may be inflicted if the human body comes in contact with cryogenic fluids or with surfaces cooled by cryogenic fluids. Damage to the skin or tissue is similar to an ordinary burn. Because the body is composed mainly of water, the low temperature effectively freezes the tissue—damaging or destroying it. The severity of the burn depends upon the contact area and the contact time with prolonged contact resulting in deeper burns. Cold burns are accompanied by stinging sensations and pain similar to those of ordinary burns. The ordinary reaction is to withdraw that portion of the body that is in contact with the cold surface. Severe burns are seldom sustained if withdrawal is possible. Cold gases may not be damaging if the turbulence in the gas is low, particularly since the body can normally adjust for a heat loss of 95 J/m<sup>2</sup>s for an area of limited exposure. If the heat loss becomes much greater than this, the skin temperature drops and freezing of the affected area may ensue. Freezing of facial tissue will occur in about 100 s if the heat loss is 2,300 J/m<sup>2</sup>s.

**Materials and Construction Hazards** Construction materials for noncryogenic service usually are chosen on the basis of tensile strength, fatigue life, weight, cost, ease of fabrication, corrosion

resistance, and so on. When working with low temperatures the designer must consider the ductility of the material since low temperatures, as noted earlier, have the effect of making some construction materials brittle or less ductile. Some materials become brittle at low temperatures but still can absorb considerable impact, while others become brittle and lose their impact strength.

**Flammability and Explosion Hazards** In order to have a fire or an explosion requires the combination of an oxidant, a fuel, and an ignition source. Generally the oxidizer will be oxygen. The latter may be available from a variety of sources including leakage or spillage, condensation of air on cryogenically cooled surfaces below 90 K, and buildup, as a solid impurity in liquid hydrogen. The fuel may be almost any noncompatible material or flammable gas; compatible materials can also act as fuels in the presence of extreme heat (strong ignition sources). The ignition source may be a mechanical or electrostatic spark, flame, impact, heat by kinetic effects, friction, chemical reaction, and so on. Certain combinations of oxygen, fuel, and ignition sources will always result in fire or explosion. The order of magnitude of flammability and detonability limits for fuel-oxidant gaseous mixtures of two widely used cryogenics is shown in Table 11-27.

**High-Pressure Gas Hazards** Potential hazards also exist in highly compressed gases because of their stored energy. In cryogenic systems such high pressures are obtained by gas compression during liquefaction or refrigeration, by pumping of liquids to high pressure followed by evaporation, and by confinement of cryogenic liquids with

**TABLE 11-27 Flammability and Detonability Limits of Hydrogen and Methane Gas**

| Mixture                         | Flammability Limits (mol %) | Detonability Limits (mol %) |
|---------------------------------|-----------------------------|-----------------------------|
| H <sub>2</sub> -air             | 4-75                        | 20-65                       |
| H <sub>2</sub> -O <sub>2</sub>  | 4-95                        | 15-90                       |
| CH <sub>4</sub> -air            | 5-15                        | 6-14                        |
| CH <sub>4</sub> -O <sub>2</sub> | 5-61                        | 10-50                       |

subsequent evaporation. If this confined gas is suddenly released through a rupture or break in a line, a significant thrust may be experienced. For example, the force generated by the rupture of a 2.5-cm diameter valve located on a 13.9-MPa pressurized gas cylinder would be over 6670 N.

## SUMMARY

It is obvious that the best designed facility is no better than the attention that is paid to safety. The latter is not considered once and forgotten. Rather, it is an ongoing activity that requires constant attention to every conceivable hazard that might be encountered. Because of its importance, safety, particularly at low temperatures, has received a large focus in the literature with its own safety manual prepared by NIST as well as by the British Cryogenics Council.

## EVAPORATORS

**GENERAL REFERENCES:** Badger and Banchero, *Introduction to Chemical Engineering*, McGraw-Hill, New York, 1955. Standiford, *Chem. Eng.*, **70**, 158-176 (Dec. 9, 1963). *Testing Procedure for Evaporators*, American Institute of Chemical Engineers, 1979. *Upgrading Evaporators to Reduce Energy Consumption*, ERDA Technical Information Center, Oak Ridge, Tenn., 1977.

### PRIMARY DESIGN PROBLEMS

**Heat Transfer** This is the most important single factor in evaporator design, since the heating surface represents the largest part of evaporator cost. Other things being equal, the type of evaporator selected is the one having the highest heat-transfer cost coefficient under desired operating conditions in terms of J/s-K (British thermal units per hour per degree Fahrenheit) per dollar of installed cost. When power is required to induce circulation past the heating surface, the coefficient must be even higher to offset the cost of power for circulation.

**Vapor-Liquid Separation** This design problem may be important for a number of reasons. The most important is usually prevention of entrainment because of value of product lost, pollution, contamination of the condensed vapor, or fouling or corrosion of the surfaces on which the vapor is condensed. Vapor-liquid separation in the vapor head may also be important when spray forms deposits on the walls, when vortices increase head requirements of circulating pumps, and when short circuiting allows vapor or unflashed liquid to be carried back to the circulating pump and heating element.

Evaporator performance is rated on the basis of **steam economy**—kilograms of solvent evaporated per kilogram of steam used. Heat is required (1) to raise the feed from its initial temperature to the boiling temperature, (2) to provide the minimum thermodynamic energy to separate liquid solvent from the feed, and (3) to vaporize the solvent. The first of these can be changed appreciably by reducing the boiling temperature or by heat interchange between the feed and the residual product and/or condensate. The greatest increase in steam economy is achieved by reusing the vaporized solvent. This is done in a **multiple-effect evaporator** by using the vapor from one effect as the heating medium for another effect in which boiling takes place at

a lower temperature and pressure. Another method of increasing the utilization of energy is to employ a **thermoccompression** evaporator, in which the vapor is compressed so that it will condense at a temperature high enough to permit its use as the heating medium in the same evaporator.

**Selection Problems** Aside from heat-transfer considerations, the selection of type of evaporator best suited for a particular service is governed by the characteristics of the feed and product. Points that must be considered are crystallization, salting and scaling, product quality, corrosion, and foaming. In the case of a **crystallizing evaporator**, the desirability of producing crystals of a definite uniform size usually limits the choice to evaporators having a positive means of circulation. **Salting**, which is the growth on body and heating-surface walls of a material having a solubility that increases with increase in temperature, is frequently encountered in crystallizing evaporators. It can be reduced or eliminated by keeping the evaporating liquid in close or frequent contact with a large surface area of crystallized solid. **Scaling** is the deposition and growth on body walls, and especially on heating surfaces, of a material undergoing an irreversible chemical reaction in the evaporator or having a solubility that decreases with an increase in temperature. Scaling can be reduced or eliminated in the same general manner as salting. Both salting and scaling liquids are usually best handled in evaporators that do not depend on boiling to induce circulation. **Fouling** is the formation of deposits other than salt or scale and may be due to corrosion, solid matter entering with the feed, or deposits formed by the condensing vapor.

**Product Quality** Considerations of product quality may require low holdup time and low-temperature operation to avoid thermal degradation. The low holdup time eliminates some types of evaporators, and some types are also eliminated because of poor heat-transfer characteristics at low temperature. Product quality may also dictate special materials of construction to avoid metallic contamination or a catalytic effect on decomposition of the product. **Corrosion** may also influence evaporator selection, since the advantages of evaporators having high heat-transfer coefficients are more apparent when expensive materials of construction are indicated. Corrosion and erosion are frequently more severe in evaporators than in other types of



equipment because of the high liquid and vapor velocities used, the frequent presence of solids in suspension, and the necessary concentration differences.

## EVAPORATOR TYPES AND APPLICATIONS

Evaporators may be classified as follows:

1. Heating medium separated from evaporating liquid by tubular heating surfaces.
2. Heating medium confined by coils, jackets, double walls, flat plates, etc.
3. Heating medium brought into direct contact with evaporating liquid.
4. Heating by solar radiation.

By far the largest number of industrial evaporators employ tubular heating surfaces. Circulation of liquid past the heating surface may be induced by boiling or by mechanical means. In the latter case, boiling may or may not occur at the heating surface.

**Forced-Circulation Evaporators (Fig. 11-122 a, b, c)** Although it may not be the most economical for many uses, the forced-circulation (FC) evaporator is suitable for the widest variety of evaporator applications. The use of a pump to ensure circulation past the heating surface makes possible separating the functions of heat transfer, vapor-liquid separation, and crystallization. The pump withdraws liquor from the flash chamber and forces it through the heating element back to the flash chamber. Circulation is maintained regardless of the evaporation rate; so this type of evaporator is well suited to **crystallizing operation**, in which solids must be maintained in suspension at all times. The liquid velocity past the heating surface is limited only by the pumping power needed or available and by accelerated corrosion and erosion at the higher velocities. **Tube velocities** normally range from a minimum of about 1.2 m/s (4 ft/s) in salt evaporators with copper or brass tubes and liquid containing 5 percent or more solids up to about 3 m/s (10 ft/s) in caustic evaporators having nickel tubes and liquid containing only a small amount of solids. Even higher velocities can be used when corrosion is not accelerated by erosion.

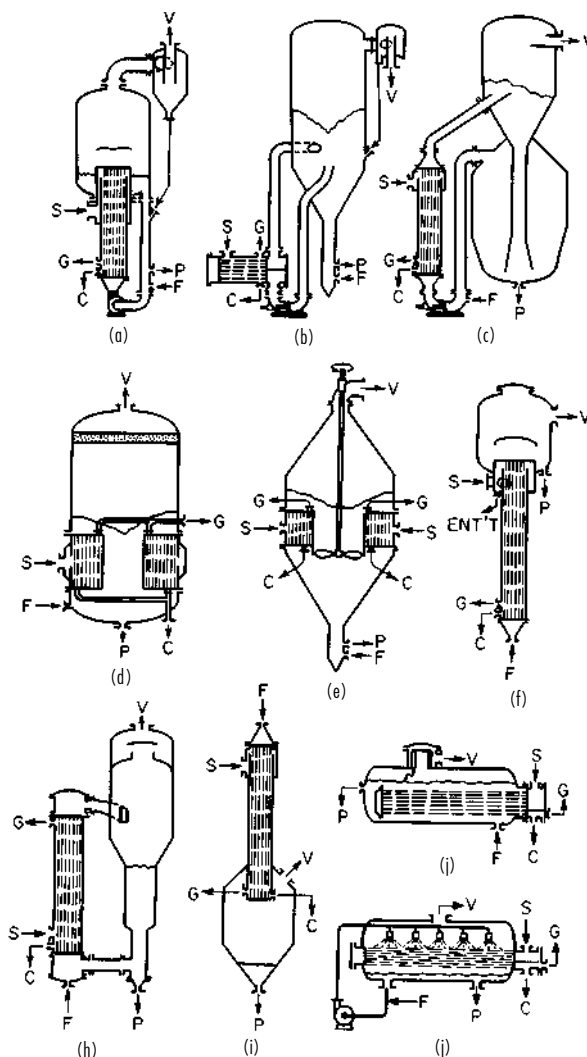
Highest heat-transfer coefficients are obtained in FC evaporators when the liquid is allowed to boil in the tubes, as in the type shown in Fig. 11-122a. The heating element projects into the vapor head, and the liquid level is maintained near and usually slightly below the top tube sheet. This type of FC evaporator is not well suited to salting solutions because boiling in the tubes increases the chances of salt deposit on the walls and the sudden flashing at the tube exits promotes excessive nucleation and production of fine crystals. Consequently, this type of evaporator is seldom used except when there are headroom limitations or when the liquid forms neither salt nor scale.

**Swirl Flow Evaporators** One of the most significant problems in the thermal design of once-through, tube-side evaporators is the poor predictability of the loss of  $\Delta T$  upon reaching the critical heat flux condition. This situation may occur through flashing due to a high wall temperature or due to process needs to evaporate most of, if not all, the liquid entering the evaporator. It is the result of sensible heating of the vapor phase which accumulates at the heat-transfer surface, dries out the tube wall, and blocks the transfer of heat to the remaining liquid.

In some cases, even with correctly predicted heat-transfer coefficients, the unexpected  $\Delta T$  loss can reduce the actual performance of the evaporator by as much as 200 percent below the predicted performance. The best approach is to maintain a high level of mixing of the phases through the heat exchanger near the heat-transfer surface.

The use of swirl flow, whereby a rotational vortex is imparted to the boiling fluid to centrifuge the liquid droplets out to the tube wall, has proved to be the most reliable means to correct for and eliminate this loss of  $\Delta T$ . The use of this technique almost always corrects the design to operate as well as or better than predicted. Also, the use of swirl flow eliminates the need to determine between horizontal or vertical orientation for most two-phase velocities. Both orientations work about the same in swirl flow.

Many commercially viable methods of inducing swirl flow inside of tubes are available in the form of either swirl flow tube inserts (twisted tapes, helical cores, spiral wire inserts) or special tube configurations



**FIG. 11-122** Evaporator types. (a) Forced circulation. (b) Submerged-tube forced circulation. (c) Oslo-type crystallizer. (d) Short-tube vertical. (e) Propeller calandria. (f) Long-tube vertical. (g) Recirculating long-tube vertical. (h) Falling film. (i, j) Horizontal-tube evaporators. C = condensate; F = feed; G = vent; P = product; S = steam; V = vapor; ENT = separated entrainment outlet.

(Twisted Tube®, internal spiral fins). All are designed to impart a natural swirl component to the flow inside the tubes. Each has been proved to solve the problem of tube-side vaporization at high vapor qualities up to and including complete tube-side vaporization.

By far the largest number of forced-circulation evaporators are of the submerged-tube type, as shown in Fig. 11-122b. The heating element is placed far enough below the liquid level or return line to the flash chamber to prevent boiling in the tubes. Preferably, the hydrostatic head should be sufficient to prevent boiling even in a tube that is plugged (and hence at steam temperature), since this prevents salting of the entire tube. Evaporators of this type sometimes have horizontal heating elements (usually two-pass), but the vertical single-pass heating element is used whenever sufficient headroom is available. The vertical element usually has a lower friction loss and is easier to clean or retube than a horizontal heater. The submerged-tube forced-circulation evaporator is relatively immune to salting in the tubes, since no supersaturation is generated by evaporation in the tubes. The

tendency toward scale formation is also reduced, since supersaturation in the heating element is generated only by a controlled amount of heating and not by both heating and evaporation.

The type of **vapor** head used with the FC evaporator is chosen to suit the product characteristics and may range from a simple centrifugal separator to the crystallizing chambers shown in Fig. 11-122*b* and *c*. Figure 11-122*b* shows a type frequently used for common salt. It is designed to circulate a slurry of crystals throughout the system. Figure 11-122*c* shows a submerged-tube FC evaporator in which heating, flashing, and crystallization are completely separated. The crystallizing solids are maintained as a fluidized bed in the chamber below the vapor head and little or no solids circulate through the heater and flash chamber. This type is well adapted to growing coarse crystals, but the crystals usually approach a spherical shape, and careful design is required to avoid production of fines in the flash chamber.

In a submerged-tube FC evaporator, all heat is imparted as sensible heat, resulting in a temperature rise of the circulating liquor that reduces the overall temperature difference available for heat transfer. Temperature rise, tube proportions, tube velocity, and head requirements on the circulating pump all influence the selection of circulation rate. Head requirements are frequently difficult to estimate since they consist not only of the usual friction, entrance and contraction, and elevation losses when the return to the flash chamber is above the liquid level but also of increased friction losses due to flashing in the return line and vortex losses in the flash chamber. Circulation is sometimes limited by vapor in the pump suction line. This may be drawn in as a result of inadequate vapor-liquid separation or may come from vortices near the pump suction connection to the body or may be formed in the line itself by short circuiting from heater outlet to pump inlet of liquor that has not flashed completely to equilibrium at the pressure in the vapor head.

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#### Advantages of forced-circulation evaporators:

1. High heat-transfer coefficients
2. Positive circulation
3. Relative freedom from salting, scaling, and fouling

#### Disadvantages of forced-circulation evaporators:

1. High cost
2. Power required for circulating pump
3. Relatively high holdup or residence time

#### Best applications of forced-circulation evaporators:

1. Crystalline product
2. Corrosive solutions
3. Viscous solutions

#### Frequent difficulties with forced-circulation evaporators:

1. Plugging of tube inlets by salt deposits detached from walls of equipment
  2. Poor circulation due to higher than expected head losses
  3. Salting due to boiling in tubes
  4. Corrosion-erosion
- 

**Short-Tube Vertical Evaporators (Fig. 11-122*d*)** This is one of the earliest types still in widespread commercial use. Its principal use at present is in the evaporation of cane-sugar juice. Circulation past the heating surface is induced by boiling in the tubes, which are usually 50.8 to 76.2 mm (2 to 3 in) in diameter by 1.2 to 1.8 m (4 to 6 ft) long. The body is a vertical cylinder, usually of cast iron, and the tubes are expanded into horizontal tube sheets that span the body diameter. The circulation rate through the tubes is many times the feed rate; so there must be a return passage from above the top tube sheet to below the bottom tube sheet. Most commonly used is a central well or **downtake** as shown in Fig. 11-122*d*. So that friction losses through the downtake do not appreciably impede circulation up through the tubes, the area of the downtake should be of the same order of magnitude as the combined cross-sectional area of the tubes. This results in a downtake almost half of the diameter of the tube sheet.

Circulation and heat transfer in this type of evaporator are strongly affected by the liquid "level." Highest heat-transfer coefficients are achieved when the level, as indicated by an external gauge glass, is only about halfway up the tubes. Slight reductions in level below the opti-

mum result in incomplete wetting of the tube walls with a consequent increased tendency to foul and a rapid reduction in capacity. When this type of evaporator is used with a liquid that can deposit salt or scale, it is customary to operate with the liquid level appreciably higher than the optimum and usually appreciably above the top tube sheet.

Circulation in the standard short-tube vertical evaporator is dependent entirely on boiling, and when boiling stops, any solids present settle out of suspension. Consequently, this type is seldom used as a crystallizing evaporator. By installing a propeller in the downtake, this objection can be overcome. Such an evaporator, usually called a **propeller calandria**, is illustrated in Fig. 11-122*e*. The propeller is usually placed as low as possible to reduce cavitation and is shrouded by an extension of the downtake well. The use of the propeller can sometimes double the capacity of a short-tube vertical evaporator. The evaporator shown in Fig. 11-122*e* includes an elutriation leg for salt manufacture similar to that used on the FC evaporator of Fig. 11-122*b*. The shape of the bottom will, of course, depend on the particular application and on whether the propeller is driven from above or below. To avoid salting when the evaporator is used for crystallizing solutions, the liquid level must be kept appreciably above the top tube sheet.

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#### Advantages of short-tube vertical evaporators:

1. High heat-transfer coefficients at high temperature differences
2. Low headroom
3. Easy mechanical descaling
4. Relatively inexpensive

#### Disadvantages of short-tube vertical evaporators:

1. Poor heat transfer at low temperature differences and low temperature
2. High floor space and weight
3. Relatively high holdup
4. Poor heat transfer with viscous liquids

#### Best applications of short-tube vertical evaporators:

1. Clear liquids
  2. Crystalline product if propeller is used
  3. Relatively noncorrosive liquids, since body is large and expensive if built of materials other than mild steel or cast iron
  4. Mild scaling solutions requiring mechanical cleaning, since tubes are short and large in diameter
- 

**Long-Tube Vertical Evaporators (Fig. 11-122*f, g, h*)** More total evaporation is accomplished in this type than in all others combined because it is normally the **cheapest per unit of capacity**. The long-tube vertical (LTV) evaporator consists of a simple one-pass vertical shell-and-tube heat exchanger discharging into a relatively small vapor head. Normally, no liquid level is maintained in the vapor head, and the residence time of liquor is only a few seconds. The tubes are usually about 50.8 mm (2 in) in diameter but may be smaller than 25.4 mm (1 in). Tube length may vary from less than 6 to 10.7 m (20 to 35 ft) in the rising film version and to as great as 20 m (65 ft) in the falling film version. The evaporator is usually operated single-pass, concentrating from the feed to discharge density in just the time that it takes the liquid and evolved vapor to pass through a tube. An extreme case is the caustic high concentrator, producing a substantially anhydrous product at 370°C (700°F) from an inlet feed of 50 percent NaOH at 149°C (300°F) in one pass up 22-mm- (8/8-in-) outside-diameter nickel tubes 6 m (20 ft) long. The largest use of LTV evaporators is for concentrating black liquor in the pulp and paper industry. Because of the long tubes and relatively high heat-transfer coefficients, it is possible to achieve higher single-unit capacities in this type of evaporator than in any other.

The LTV evaporator shown in Fig. 11-122*f* is typical of those commonly used, especially for black liquor. Feed enters at the bottom of the tube and starts to boil partway up the tube, and the mixture of liquid and vapor leaving at the top at high velocity impinges against a deflector placed above the tube sheet. This deflector is effective both as a primary separator and as a foam breaker.

In many cases, as when the ratio of feed to evaporation or the ratio of feed to heating surface is low, it is desirable to provide for **recirculation of product** through the evaporator. This can be done in the type shown in Fig. 11-122*f* by adding a pipe connection between

the product line and the feed line. Higher recirculation rates can be achieved in the type shown in Fig. 11-122g, which is used widely for condensed milk. By extending the enlarged portion of the vapor head still lower to provide storage space for liquor, this type can be used as a batch evaporator.

Liquid temperatures in the tubes of an LTV evaporator are far from uniform and are difficult to predict. At the lower end, the liquid is usually not boiling, and the liquor picks up heat as sensible heat. Since entering liquid velocities are usually very low, true heat-transfer coefficients are low in this nonboiling zone. At some point up the tube, the liquid starts to boil, and from that point on the liquid temperature decreases because of the reduction in static, friction, and acceleration heads until the vapor-liquid mixture reaches the top of the tubes at substantially vapor-head temperature. Thus the true temperature difference in the boiling zone is always less than the total temperature difference as measured from steam and vapor-head temperatures.

Although the true heat-transfer coefficients in the boiling zone are quite high, they are partially offset by the reduced temperature difference. The point in the tubes at which boiling starts and at which the maximum temperature is reached is sensitive to operating conditions, such as feed properties, feed temperature, feed rate, and heat flux. Figure 11-123 shows typical variations in liquid temperature in tubes of an LTV evaporator operating at a constant terminal temperature difference. Curve 1 shows the normal case in which the feed is not boiling at the tube inlet. Curve 2 gives an indication of the temperature difference lost when the feed enters at the boiling point. Curve 3 is for exactly the same conditions as curve 2 except that the feed contained 0.01 percent Teepol to reduce surface tension [Coulson and Mehta, *Trans. Inst. Chem. Eng.*, **31**, 208 (1953)]. The surface-active agent yields a more intimate mixture of vapor and liquid, with the result that liquid is accelerated to a velocity more nearly approaching the vapor velocity, thereby increasing the pressure drop in the tube. Although the surface-active agent caused an increase of more than 100 percent in the true heat-transfer coefficient, this was more than offset by the reduced temperature difference so that the net result was a reduction in evaporator capacity. This sensitivity of the LTV evaporator to changes in operating conditions is less pronounced at high than at low temperature differences and temperature levels.

The **falling-film** version of the LTV evaporator (Fig. 11-122h) eliminates these problems of hydrostatic head. Liquid is fed to the tops of the tubes and flows down the walls as a film. Vapor-liquid separation usually takes place at the bottom, although some evaporators of this type are arranged for vapor to rise through the tube counter-currently to the liquid. The pressure drop through the tubes is usually very small, and the boiling-liquid temperature is substantially the same as the vapor-head temperature. The falling-film evaporator is widely used for concentrating **heat-sensitive materials**, such as fruit juices, because the holdup time is very small, the liquid is not overheated during passage through the evaporator, and heat-transfer coefficients are high even at low boiling temperatures.

The principal problem with the falling-film LTV evaporator is that of **feed distribution** to the tubes. It is essential that all tube surfaces be wetted continually. This usually requires recirculation of the liquid unless the ratio of feed to evaporation is quite high. An alternative to the simple recirculation system of Fig. 11-122h is sometimes used when the feed undergoes an appreciable concentration change and the product is viscous and/or has a high boiling point rise. The feed chamber and vapor head are divided into a number of liquor com-

partments, and separate pumps are used to pass the liquor through the various banks of tubes in series, all in parallel as to steam and vapor pressures. The actual distribution of feed to the individual tubes of a falling-film evaporator may be accomplished by orifices at the inlet to each tube, by a perforated plate above the tube sheet, or by one or more spray nozzles.

Both rising- and falling-film LTV evaporators are generally unsuited to salting or severely scaling liquids. However, both are widely used for black liquor, which presents a mild scaling problem, and also are used to carry solutions beyond saturation with respect to a crystallizing salt. In the latter case, deposits can usually be removed quickly by increasing the feed rate or reducing the steam rate in order to make the product unsaturated for a short time. The falling-film evaporator is not generally suited to liquids containing solids because of difficulty in plugging the feed distributors. However, it has been applied to the evaporation of saline waters saturated with  $\text{CaSO}_4$  and containing 5 to 10 percent  $\text{CaSO}_4$  seeds in suspension for scale prevention (Anderson, ASME Pap. 76-WA/Pwr-5, 1976).

Because of their simplicity of construction, compactness, and generally high heat-transfer coefficients, LTV evaporators are well suited to service with corrosive liquids. An example is the reconcentration of rayon spin-bath liquor, which is highly acid. These evaporators employ impervious graphite tubes, lead, rubber-covered or impervious graphite tube sheets, and rubber-lined vapor heads. Polished stainless-steel LTV evaporators are widely used for food products. The latter evaporators are usually similar to that shown in Fig. 11-122g, in which the heating element is at one side of the vapor head to permit easy access to the tubes for cleaning.

#### Advantages of long-tube vertical evaporators:

1. Low cost
2. Large heating surface in one body
3. Low holdup
4. Small floor space
5. Good heat-transfer coefficients at reasonable temperature differences (rising film)
6. Good heat-transfer coefficients at all temperature differences (falling film)

#### Disadvantages of long-tube vertical evaporators:

1. High headroom
2. Generally unsuitable for salting and severely scaling liquids
3. Poor heat-transfer coefficients of rising-film version at low temperature differences
4. Recirculation usually required for falling-film version

#### Best applications of long-tube vertical evaporators:

1. Clear liquids
2. Foaming liquids
3. Corrosive solutions
4. Large evaporation loads
5. High temperature differences—rising film, low temperature differences—falling film
6. Low-temperature operation—falling film
7. Vapor compression operation—falling film

#### Frequent difficulties with long-tube vertical evaporators:

1. Sensitivity of rising-film units to changes in operating conditions
2. Poor feed distribution to falling-film units

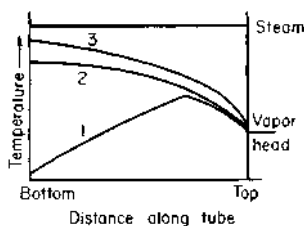


FIG. 11-123 Temperature variations in a long-tube vertical evaporator.

**Horizontal-Tube Evaporators (Fig. 11-122i)** In these types the steam is inside and the liquor outside the tubes. The submerged-tube version of Fig. 11-122i is seldom used except for the preparation of boiler feedwater. Low entrainment loss is the primary aim: the horizontal cylindrical shell yields a large disengagement area per unit of vessel volume. Special versions use deformed tubes between restrained tube sheets that crack off much of a scale deposit when sprayed with cold water. By showering liquor over the tubes in the version of Fig. 11-122f hydrostatic head losses are eliminated and heat-transfer performance is improved to that of the falling-film tubular type of Fig. 11-122h. Originally called the Lillie, this evaporator is now also called the spray-film or simply the horizontal-tube evaporator. Liquid distribution over the tubes is accomplished by sprays or perforated plates above the topmost tubes. Maintaining this distribution

through the bundle to avoid overconcentrating the liquor is a problem unique to this type of evaporator. It is now used primarily for seawater evaporation.

Advantages of horizontal-tube evaporators:

1. Very low headroom
2. Large vapor-liquid disengaging area—submerged-tube type
3. Relatively low cost in small-capacity straight-tube type
4. Good heat-transfer coefficients
5. Easy semiautomatic descaling—bent-tube type

Disadvantages of horizontal-tube evaporators:

1. Unsuitable for salting liquids
2. Unsuitable for scaling liquids—straight-tube type
3. High cost—bent-tube type
4. Maintaining liquid distribution—film type

Best applications of horizontal-tube evaporators:

1. Limited headroom
2. Small capacity
3. Nonscaling nonsalting liquids—straight-tube type
4. Severely scaling liquids—bent-tube type

**Miscellaneous Forms of Heating Surface** Special evaporator designs are sometimes indicated when heat loads are small, special product characteristics are desired, or the product is especially difficult to handle. **Jacketed kettles**, frequently with agitators, are used when the product is very viscous, batches are small, intimate mixing is required, and/or ease of cleaning is an important factor. Evaporators with steam in coiled **tubes** may be used for small capacities with scaling liquids in designs that permit “cold shocking,” or complete withdrawal of the coil from the shell for manual scale removal. Other designs for scaling liquids employ flat-plate heat exchangers, since in general a scale deposit can be removed more easily from a flat plate than from a curved surface. One such design, the **channel-switching evaporator**, alternates the duty of either side of the heating surface periodically from boiling liquid to condensing vapor so that scale formed when the surface is in contact with boiling liquid is dissolved when the surface is next in contact with condensing vapor.

**Agitated thin-film evaporators** employ a heating surface consisting of one large-diameter tube that may be either straight or tapered, horizontal or vertical. Liquid is spread on the tube wall by a rotating assembly of blades that either maintain a close clearance from the wall or actually ride on the film of liquid on the wall. The expensive construction limits application to the most difficult materials. High agitation [on the order of 12 m/s (40 ft/s) rotor-tip speed] and power intensities of 2 to 20 kW m<sup>2</sup> (0.25 to 2.5 hp/ft<sup>2</sup>) permit handling extremely viscous materials. Residence times of only a few seconds permit concentration of heat-sensitive materials at temperatures and temperature differences higher than in other types [Mutzenberg, Parker, and Fischer. *Chem. Eng.*, **72**, 175–190 (Sept. 13, 1965)]. High feed-to-product ratios can be handled without recirculation.

Economic and process considerations usually dictate that agitated thin-film evaporators be operated in single-effect mode. Very high temperature differences can then be used; many are heated with Dowtherm or other high-temperature media. This permits achieving reasonable capacities in spite of the relatively low heat-transfer coefficients and the small surface that can be provided in a single tube [to about 20 m<sup>2</sup> (200 ft<sup>2</sup>)]. The structural need for wall thicknesses of 6 to 13 mm (¼ to ½ in) is a major reason for the relatively low heat-transfer coefficients when evaporating water-like materials.

**Evaporators without Heating Surfaces** The **submerged-combustion** evaporator makes use of combustion gases bubbling through the liquid as the means of heat transfer. It consists simply of a tank to hold the liquid, a burner and gas distributor that can be lowered into the liquid, and a combustion-control system. Since there are no heating surfaces on which scale can deposit, this evaporator is well suited to use with severely scaling liquids. The ease of constructing the tank and burner of special alloys or nonmetallic materials makes practical the handling of highly corrosive solutions. However, since the vapor is mixed with large quantities of noncondensable gases, it is impossible to reuse the heat in this vapor, and installations are usually

limited to areas of low fuel cost. One difficulty frequently encountered in the use of submerged-combustion evaporators is a high entrainment loss. Also, these evaporators cannot be used when control of crystal size is important.

**Disk or cascade evaporators** are used in the pulp and paper industry to recover heat and entrained chemicals from boiler stack gases and to effect a final concentration of the black liquor before it is burned in the boiler. These evaporators consist of a horizontal shaft on which are mounted disks perpendicular to the shaft or bars parallel to the shaft. The assembly is partially immersed in the thick black liquor so that films of liquor are carried into the hot-gas stream as the shaft rotates.

Some forms of **flash evaporators** require no heating surface. An example is a recrystallizing process for separating salts having normal solubility curves from salts having inverse solubility curves, as in separating sodium chloride from calcium sulfate [Richards, *Chem. Eng.*, **59**(3), 140 (1952)]. A suspension of raw solid feed in a recirculating brine stream is heated by direct steam injection. The increased temperature and dilution by the steam dissolve the salt having the normal solubility curve. The other salt remains undissolved and is separated from the hot solution before it is flashed to a lower temperature. The cooling and loss of water on flashing cause recrystallization of the salt having the normal solubility curve, which is separated from the brine before the brine is mixed with more solid feed for recycling to the heater. This system can be operated as a multiple effect by flashing down to the lower temperature in stages and using flash vapor from all but the last stage to heat the recycle brine by direct injection. In this process no net evaporation occurs from the total system, and the process cannot be used to concentrate solutions unless heating surfaces are added.

## UTILIZATION OF TEMPERATURE DIFFERENCE

Temperature difference is the driving force for evaporator operation and usually is limited, as by compression ratio in vapor-compression evaporators and by available steam-pressure and heat-sink temperature in single- and multiple-effect evaporators. A fundamental objective of evaporator design is to make as much of this total temperature difference available for heat transfer as is economically justifiable. Some losses in temperature difference, such as those due to *boiling point rise* (BPR), are unavoidable. However, even these can be minimized, as by passing the liquor through effects or through different sections of a single effect in series so that only a portion of the heating surface is in contact with the strongest liquor.

Figure 11-124 shows approximate BPR losses for a number of process liquids. A correlation for concentrated solutions of many inorganic salts at the atmospheric pressure boiling point [Meranda and Furter, *J. Ch. and E. Data* **22**, 315-7 (1977)] is

$$\text{BPR} = 104.9N_2^{1.14} \quad (11-123)$$

where  $N_2$  is the mole fraction of salts in solution. Correction to other pressures, when heats of solution are small, can be based on a constant ratio of vapor pressure of the solution to that of water at the same temperature.

The principal reducible loss in  $\Delta T$  is that due to friction and to entrance and exit losses in vapor piping and entrainment separators. Pressure-drop losses here correspond to a reduction in condensing temperature of the vapor and hence a loss in available  $\Delta T$ . These losses become most critical at the low-temperature end of the evaporator, both because of the increasing specific volume of the vapor and because of the reduced slope of the vapor-pressure curve. Sizing of vapor lines is part of the economic optimization of the evaporator; extra costs of larger vapor lines being balanced against savings in  $\Delta T$ , which correspond to savings in heating-surface requirements. It should be noted that entrance and exit losses in vapor lines usually exceed by severalfold the straight-pipe friction losses, so they cannot be ignored.

## VAPOR-LIQUID SEPARATION

Product losses in evaporator vapor may result from foaming, splashing, or entrainment. Primary separation of liquid from vapor is accomplished in the vapor head by making the horizontal plan area large

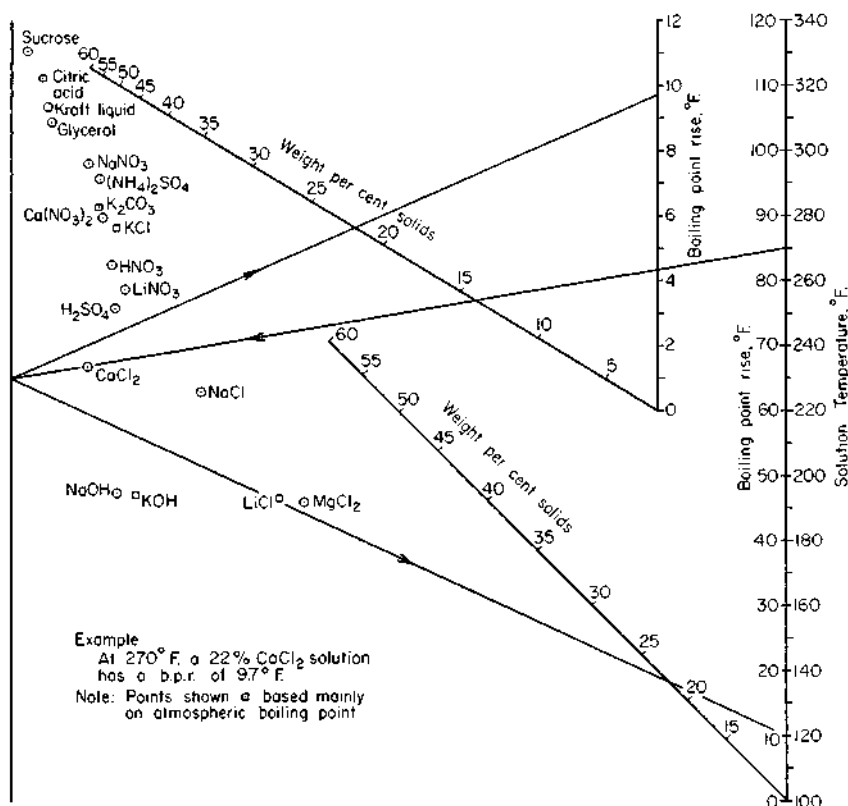


FIG. 11-124 Boiling-point rise of aqueous solutions.  $^{\circ}\text{C} = 5/9 (^{\circ}\text{F} - 32)$ .

enough so that most of the entrained droplets can settle out against the rising flow of vapor. Allowable velocities are governed by the Souders-Brown equation:  $V = k\sqrt{(\rho_1 - \rho_v)/\rho_v}$ , in which  $k$  depends on the size distribution of droplets and the decontamination factor  $F$  desired. For most evaporators and for  $F$  between 100 and 10,000,  $k \approx 0.245/(F - 50)^{0.4}$  (Standiford, *Chemical Engineers' Handbook*, 4th ed., McGraw-Hill, New York, 1963, p. 11-35). Higher values of  $k$  (to about 0.15) can be tolerated in the falling-film evaporator, where most of the entrainment separation occurs in the tubes, the vapor is scrubbed by liquor leaving the tubes, and the vapor must reverse direction to reach the outlet.

**Foaming losses** usually result from the presence in the evaporating liquid of colloids or of surface-tension depressants and finely divided solids. Antifoam agents are often effective. Other means of combating foam include the use of steam jets impinging on the foam surface, the removal of product at the surface layer, where the foaming agents seem to concentrate, and operation at a very low liquid level so that hot surfaces can break the foam. Impingement at high velocity against a baffle tends to break the foam mechanically, and this is the reason that the long-tube vertical, forced-circulation, and agitated-film evaporators are particularly effective with foaming liquids. Operating at lower temperatures and/or higher-dissolved solids concentrations may also reduce foaming tendencies.

**Splashing losses** are usually insignificant if a reasonable height has been provided between the liquid level and the top of the vapor head. The height required depends on the violence of boiling. Heights of 2.4 to 3.6 m (8 to 12 ft) or more are provided in short-tube vertical evaporators, in which the liquid and vapor leaving the tubes are projected upward. Less height is required in forced-circulation evaporators, in which the liquid is given a centrifugal motion or is projected downward as by a baffle. The same is true of long-tube vertical evaporators, in which the rising vapor-liquid mixture is projected against a baffle.

**Entrainment losses by flashing** are frequently encountered in an evaporator. If the feed is above the boiling point and is introduced above or only a short distance below the liquid level, entrainment losses may be excessive. This can occur in a short-tube-type evaporator if the feed is introduced at only one point below the lower tube sheet (Kerr, Louisiana Agric. Expt. Stn. Bull. 149, 1915). The same difficulty may be encountered in forced-circulation evaporators having too high a temperature rise through the heating element and thus too wide a flashing range as the circulating liquid enters the body. Poor vacuum control, especially during startup, can cause the generation of far more vapor than the evaporator was designed to handle, with a consequent increase in entrainment.

**Entrainment separators** are frequently used to reduce product losses. There are a number of specialized designs available, practically all of which rely on a change in direction of the vapor flow when the vapor is traveling at high velocity. Typical separators are shown in Fig. 11-122, although not necessarily with the type of evaporator with which they may be used. The most common separator is the cyclone, which may have either a top or a bottom outlet as shown in Fig. 11-122a and b or may even be wrapped around the heating element of the next effect as shown in Fig. 11-122f. The separation efficiency of a cyclone increases with an increase in inlet velocity, although at the cost of some pressure drop, which means a loss in available temperature difference. Pressure drop in a cyclone is from 10 to 16 velocity heads [Lawrence, *Chem. Eng. Prog.*, **48**, 241 (1952)], based on the velocity in the inlet pipe. Such cyclones can be sized in the same manner as a cyclone dust collector (using velocities of about 30 m/s (100 ft/s) at atmospheric pressure) although sizes may be increased somewhat in order to reduce losses in available temperature difference.

**Knitted wire mesh** serves as an effective entrainment separator when it cannot easily be fouled by solids in the liquor. The mesh is available in woven metal wire of most alloys and is installed as a blanket

across the top of the evaporator (Fig. 11-122d) or in a monitor of reduced diameter atop the vapor head. These separators have low-pressure drops, usually on the order of 13 mm ( $\frac{1}{2}$  in) of water, and collection efficiency is above 99.8 percent in the range of vapor velocities from 2.5 to 6 m/s (8 to 20 ft/s) [Carpenter and Othmer, *Am. Inst. Chem. Eng. J.*, **1**, 549 (1955)]. Chevron (hook-and-vane) type separators are also used because of their higher-allowable velocities or because of their reduced tendency to foul with solids suspended in the entrained liquid.

## EVAPORATOR ARRANGEMENT

**Single-Effect Evaporators** Single-effect evaporators are used when the required capacity is small, steam is cheap, the material is so corrosive that very expensive materials of construction are required, or the vapor is so contaminated that it cannot be reused. Single-effect evaporators may be operated in batch, semibatch, or continuous-batch modes or continuously. Strictly speaking, **batch evaporators** are ones in which filling, evaporating, and emptying are consecutive steps. This method of operation is rarely used since it requires that the body be large enough to hold the entire charge of feed and the heating element be placed low enough so as not to be uncovered when the volume is reduced to that of the product. The more usual method of operation is **semibatch**, in which feed is continually added to maintain a constant level until the entire charge reaches final density. **Continuous-batch** evaporators usually have a continuous feed and, over at least part of the cycle, a continuous discharge. One method of operation is to circulate from a storage tank to the evaporator and back until the entire tank is up to desired concentration and then finish in batches. **Continuous evaporators** have essentially continuous feed and discharge, and concentrations of both feed and product remain substantially constant.

**Thermocompression** The simplest means of reducing the energy requirements of evaporation is to compress the vapor from a single-effect evaporator so that the vapor can be used as the heating medium in the same evaporator. The compression may be accomplished by mechanical means or by a steam jet. In order to keep the compressor cost and power requirements within reason, the evaporator must work with a fairly narrow temperature difference, usually from about 5.5 to 11°C (10° to 20°F). This means that a large evaporator heating surface is needed, which usually makes the vapor-compression evaporator more expensive in first cost than a multiple-effect evaporator. However, total installation costs may be reduced when purchased power is the energy source, since the need for boiler and heat sink is eliminated. Substantial savings in operating cost are realized when electrical or mechanical power is available at a low cost relative to low-pressure steam, when only high-pressure steam is available to operate the evaporator, or when the cost of providing cooling water or other heat sink for a multiple-effect evaporator is high.

**Mechanical thermocompression** may employ reciprocating, rotary positive-displacement, centrifugal, or axial-flow compressors. Positive-displacement compressors are impractical for all but the smallest capacities, such as portable seawater evaporators. Axial-flow compressors can be built for capacities of more than 472 m<sup>3</sup>/s ( $1 \times 10^6$  ft<sup>3</sup>/min). Centrifugal compressors are usually cheapest for the intermediate-capacity ranges that are normally encountered. In all cases, great care must be taken to keep entrainment at a minimum, since the vapor becomes superheated on compression and any liquid present will evaporate, leaving the dissolved solids behind. In some cases a vapor-scrubbing tower may be installed to protect the compressor. A mechanical recompression evaporator usually requires more heat than is available from the compressed vapor. Some of this extra heat can be obtained by preheating the feed with the condensate and, if possible, with the product. Rather extensive heat-exchange systems with close approach temperatures are usually justified, especially if the evaporator is operated at high temperature to reduce the volume of vapor to be compressed. When the product is a solid, an elutriation leg such as that shown in Fig. 11-122b is advantageous, since it cools the product almost to feed temperature. The remaining heat needed to maintain the evaporator in operation must be obtained from outside sources.

While theoretical compressor power requirements are reduced slightly by going to lower evaporating temperatures, the volume of

vapor to be compressed and hence compressor size and cost increase so rapidly that low-temperature operation is more expensive than high-temperature operation. The requirement of low temperature for fruit-juice concentration has led to the development of an evaporator employing a **secondary fluid**, usually Freon or ammonia. In this evaporator, the vapor is condensed in an exchanger cooled by boiling Freon. The Freon, at a much higher vapor density than the water vapor, is then compressed to serve as the heating medium for the evaporator. This system requires that the latent heat be transferred through two surfaces instead of one, but the savings in compressor size and cost are enough to justify the extra cost of heating surface or the cost of compressing through a wider temperature range.

**Steam-jet thermocompression** is advantageous when steam is available at a pressure appreciably higher than can be used in the evaporator. The steam jet then serves as a reducing valve while doing some useful work. The efficiency of a steam jet is quite low and falls off rapidly when the jet is not used at the vapor-flow rate and terminal pressure conditions for which it was designed. Consequently multiple jets are used when wide variations in evaporation rate are expected. Because of the low first cost and the ability to handle large volumes of vapor, steam-jet thermocompressors are used to increase the economy of evaporators that must operate at low temperatures and hence cannot be operated in multiple effect. The steam-jet thermocompression evaporator has a heat input larger than that needed to balance the system, and some heat must be rejected. This is usually done by venting some of the vapor at the suction of the compressor.

**Multiple-Effect Evaporation** Multiple-effect evaporation is the principal means in use for economizing on energy consumption. Most such evaporators operate on a continuous basis, although for a few difficult materials a continuous-batch cycle may be employed. In a multiple-effect evaporator, steam from an outside source is condensed in the heating element of the first effect. If the feed to the effect is at a temperature near the boiling point in the first effect, 1 kg of steam will evaporate almost 1 kg of water. The first effect operates at (but is not controlled at) a boiling temperature high enough so that the evaporated water can serve as the heating medium of the second effect. Here almost another kilogram of water is evaporated, and this may go to a condenser if the evaporator is a double-effect or may be used as the heating medium of the third effect. This method may be repeated for any number of effects. Large evaporators having six and seven effects are common in the pulp and paper industry, and evaporators having as many as 17 effects have been built. As a first approximation, the **steam economy** of a multiple-effect evaporator will increase in proportion to the number of effects and usually will be somewhat less numerically than the number of effects.

The increased steam economy of a multiple-effect evaporator is gained at the expense of evaporator first cost. The total heat-transfer surface will increase substantially in proportion to the number of effects in the evaporator. This is only an approximation since going from one to two effects means that about half of the heat transfer is at a higher temperature level, where heat-transfer coefficients are generally higher. On the other hand, operating at lower temperature differences reduces the heat-transfer coefficient for many types of evaporator. If the material has an appreciable boiling-point elevation, this will also lower the available temperature difference. The only accurate means of predicting the changes in steam economy and surface requirements with changes in the number of effects is by detailed heat and material balances together with an analysis of the effect of changes in operating conditions on heat-transfer performance.

The approximate **temperature distribution** in a multiple-effect evaporator is under the control of the designer, but once built, the evaporator establishes its own equilibrium. Basically, the effects are a number of series resistances to heat transfer, each resistance being approximately proportional to  $1/U_n A_n$ . The total available temperature drop is divided between the effects in proportion to their resistances. If one effect starts to scale, its temperature drop will increase at the expense of the temperature drops across the other effects. This provides a convenient means of detecting a drop in heat-transfer coefficient in an effect of an operating evaporator. If the steam pressure and final vacuum do not change, the temperature in the effect that is scaling will decrease and the temperature in the preceding effect will increase.

The feed to a multiple-effect evaporator is usually transferred from one effect to another in series so that the ultimate product concentration is reached only in one effect of the evaporator. In **backward-feed** operation, the raw feed enters the last (coldest) effect, the discharge from this effect becomes the feed to the next-to-the-last effect, and so on until product is discharged from the first effect. This method of operation is advantageous when the feed is cold, since much less liquid must be heated to the higher temperature existing in the early effects. It is also used when the product is so viscous that high temperatures are needed to keep the viscosity low enough to give reasonable heat-transfer coefficients. When product viscosity is high but a hot product is not needed, the liquid from the first effect is sometimes flashed to a lower temperature in one or more stages and the flash vapor added to the vapor from one or more later effects of the evaporator.

In **forward-feed** operation, raw feed is introduced in the first effect and passed from effect to effect parallel to the steam flow. Product is withdrawn from the last effect. This method of operation is advantageous when the feed is hot or when the concentrated product would be damaged or would deposit scale at high temperature. Forward feed simplifies operation when liquor can be transferred by pressure difference alone, thus eliminating all intermediate liquor pumps. When the feed is cold, forward feed gives a low steam economy since an appreciable part of the prime steam is needed to heat the feed to the boiling point and thus accomplishes no evaporation. If forward feed is necessary and feed is cold, steam economy can be improved markedly by preheating the feed in stages with vapor bled from intermediate effects of the evaporator. This usually represents little increase in total heating surface or cost since the feed must be heated in any event and shell-and-tube heat exchangers are generally less expensive per unit of surface area than evaporator heating surface.

**Mixed-feed** operation is used only for special applications, as when liquor at an intermediate concentration and a certain temperature is desired for additional processing.

**Parallel feed** involves the introduction of raw feed and the withdrawal of product at each effect of the evaporator. It is used primarily when the feed is substantially saturated and the product is a solid. An example is the evaporation of brine to make common salt. Evaporators of the types shown in Fig. 11-122*b* or *e* are used, and the product is withdrawn as a slurry. In this case, parallel feed is desirable because the feed washes impurities from the salt leaving the body.

**Heat-recovery systems** are frequently incorporated in an evaporator to increase the steam economy. Ideally, product and evaporator condensate should leave the system at a temperature as low as possible. Also, heat should be recovered from these streams by exchange with feed or evaporating liquid at the highest possible temperature. This would normally require separate liquid-liquid heat exchangers, which add greatly to the complexity of the evaporator and are justifiable only in large plants. Normally, the loss in thermodynamic availability due to flashing is tolerated since the flash vapor can then be used directly in the evaporator effects. The most commonly used is a **condensate flash** system in which the condensate from each effect but the first (which normally must be returned to the boiler) is flashed in successive stages to the pressure in the heating element of each succeeding effect of the evaporator. Product flash tanks may also be used in a backward- or mixed-feed evaporator. In a forward-feed evaporator, the principal means of heat recovery may be by use of **feed preheaters** heated by vapor bled from each effect of the evaporator. In this case, condensate may be either flashed as before or used in a separate set of exchangers to accomplish some of the feed preheating. A feed preheated by last-effect vapor may also materially reduce condenser water requirements.

**Seawater Evaporators** The production of potable water from saline waters represents a large and growing field of application for evaporators. Extensive work done in this field to 1972 was summarized in the annual *Saline Water Conversion Reports* of the Office of Saline Water, U.S. Department of the Interior. **Steam economies** on the order of 10 kg evaporation/kg steam are usually justified because (1) unit production capacities are high, (2) fixed charges are low on capital used for public works (i.e., they use long amortization periods and have low interest rates, with no other return on investment considered), (3) heat-transfer performance is comparable with that of

pure water, and (4) properly treated seawater causes little deterioration due to scaling or fouling.

Figure 11-125*a* shows a **multiple-effect** (falling-film) flow sheet as used for seawater. Twelve effects are needed for a steam economy of 10. Seawater is used to condense last-effect vapor, and a portion is then treated to prevent scaling and corrosion. Treatment usually consists of acidification to break down bicarbonates, followed by deaeration, which also removes the carbon dioxide generated. The treated seawater is then heated to successively higher temperatures by a portion of the vapor from each effect and finally is fed to the evaporating surface of the first effect. The vapor generated therein and the partially concentrated liquid are passed to the second effect, and so on until the last effect. The feed rate is adjusted relative to the steam rate so that the residual liquid from the last effect can carry away all the salts in solution, in a volume about one-third of that of the feed. Condensate formed in each effect but the first is flashed down to the following effects in sequence and constitutes the product of the evaporator.

As the feed-to-steam ratio is increased in the flow sheet of Fig. 11-125*a*, a point is reached where all the vapor is needed to preheat the feed and none is available for the evaporator tubes. This limiting case is the **multistage flash evaporator**, shown in its simplest form in Fig. 11-125*b*. Seawater is treated as before and then pumped through a number of feed heaters in series. It is given a final boost in temperature with prime steam in a **brine heater** before it is flashed down in series to provide the vapor needed by the feed heaters. The amount of steam required depends on the approach-temperature difference in the feed heaters and the flash range per stage. Condensate from the feed heaters is flashed down in the same manner as the brine.

Since the flow being heated is identical to the total flow being flashed, the temperature rise in each heater is equal to the flash range in each flasher. This temperature difference represents a loss from the temperature difference available for heat transfer. There are thus two ways of increasing the steam economy of such plants: increasing the heating surface and increasing the number of stages. Whereas the number of effects in a multiple-effect plant will be about 20 percent greater than the steam economy, the number of stages in a flash plant will be 3 to 4 times the steam economy. However, a large number of stages can be provided in a single vessel by means of internal bulkheads. The heat-exchanger tubing is placed in the same vessel, and the tubes usually are continuous through a number of stages. This requires ferrules or special close tube-hole clearances where the tubes pass through the internal bulkheads. In a plant for a steam economy of 10, the ratio of flow rate to heating surface is usually such that the seawater must pass through about 152 m of 19-mm (500 ft of ¾-in) tubing before it reaches the brine heater. This places a limitation on the physical arrangement of the vessels.

Inasmuch as it requires a flash range of about 61°C (110°F) to produce 1 kg of flash vapor for every 10 kg of seawater, the multistage flash evaporator requires handling a large volume of seawater relative to the product. In the flow sheet of Fig. 11-125*b* all this seawater must be deaerated and treated for scale prevention. In addition, the last-stage vacuum varies with the ambient seawater temperature, and ejector equipment must be sized for the worst condition. These difficulties can be eliminated by using the **recirculating multistage flash** flow sheet of Fig. 11-125*c*. The last few stages, called the **reject stages**, are cooled by a flow of seawater that can be varied to maintain a reasonable last-stage vacuum. A small portion of the last-stage brine is blown down to carry away the dissolved salts, and the balance is recirculated to the **heat-recovery stages**. This arrangement requires a much smaller makeup of fresh seawater and hence a lower treatment cost.

The multistage flash evaporator is similar to a multiple-effect forced-circulation evaporator, but with all the forced-circulation heaters in series. This has the advantage of requiring only one large-volume forced-circulation pump, but the sensible heating and short-circuiting losses in available temperature differences remain. A disadvantage of the flash evaporator is that the liquid throughout the system is at almost the discharge concentration. This has limited its industrial use to solutions in which no great concentration differences are required between feed and product and to where the liquid can be heated through wide temperature ranges without scaling. A partial remedy is

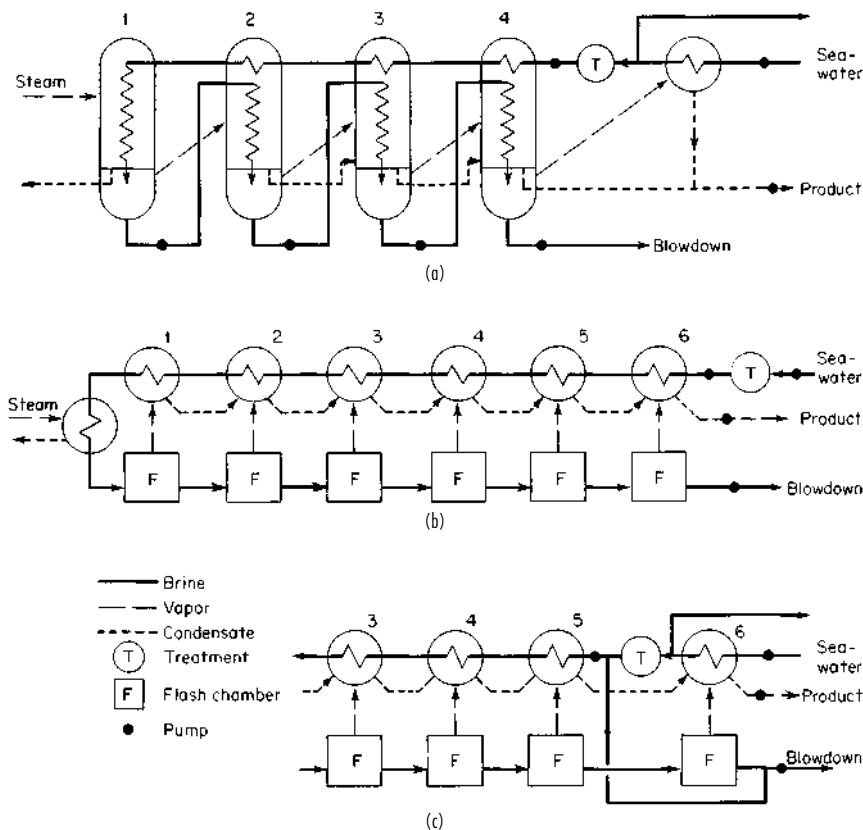


FIG. 11-125 Flow sheets for seawater evaporators. (a) Multiple effect (falling film). (b) Multistage flash (once-through). (c) Multistage flash (recirculating).

to arrange several multistage flash evaporators in series, the heat-rejection section of one being the brine heater of the next. This permits independent control of concentration but eliminates the principal advantage of the flash evaporator, which is the small number of pumps and vessels required. An unusual feature of the flash evaporator is that fouling of the heating surfaces reduces primarily the steam economy rather than the capacity of the evaporator. Capacity is not affected until the heat-rejection stages can no longer handle the increased flashing resulting from the increased heat input.

## EVAPORATOR CALCULATIONS

**Single-Effect Evaporators** The heat requirements of a single-effect continuous evaporator can be calculated by the usual methods of stoichiometry. If enthalpy data or specific heat and heat-of-solution data are not available, the heat requirement can be estimated as the sum of the heat needed to raise the feed from feed to product temperature and the heat required to evaporate the water. The latent heat of water is taken at the vapor-head pressure instead of at the product temperature in order to compensate partially for any heat of solution. If sufficient vapor-pressure data are available for the solution, methods are available to calculate the true latent heat from the slope of the Dühring line [Othmer, *Ind. Eng. Chem.*, **32**, 841 (1940)].

The heat requirements in batch evaporation are the same as those in continuous evaporation except that the temperature (and sometimes pressure) of the vapor changes during the course of the cycle. Since the enthalpy of water vapor changes but little relative to temperature, the difference between continuous and batch heat requirements is almost always negligible. More important usually is the effect of variation of

fluid properties, such as viscosity and boiling-point rise, on heat transfer. These can only be estimated by a step-by-step calculation.

In selecting the **boiling temperature**, consideration must be given to the effect of temperature on heat-transfer characteristics of the type of evaporator to be used. Some evaporators show a marked drop in coefficient at low temperature—more than enough to offset any gain in available temperature difference. The condenser cooling-water temperature and cost must also be considered.

**Thermocompression Evaporators** Thermocompression-evaporator calculations [Pridgeon, *Chem. Metall. Eng.*, **28**, 1109 (1923); Peter, *Chimia (Switzerland)*, **3**, 114 (1949); Petzold, *Chem. Ing. Tech.*, **22**, 147 (1950); and Weimer, Dolf, and Austin, *Chem. Eng. Prog.*, **76**(11), 78 (1980)] are much the same as single-effect calculations with the added complication that the heat supplied to the evaporator from compressed vapor and other sources must exactly balance the heat requirements. Some knowledge of compressor efficiency is also required. Large axial-flow machines on the order of 236-m<sup>3</sup>/s (500,000-ft<sup>3</sup>/min) capacity may have efficiencies of 80 to 85 percent. Efficiency drops to about 75 percent for a 14-m<sup>3</sup>/s (30,000-ft<sup>3</sup>/min) centrifugal compressor. Steam-jet compressors have thermodynamic efficiencies on the order of only 25 to 30 percent.

**Flash Evaporators** The calculation of a heat and material balance on a flash evaporator is relatively easy once it is understood that the temperature rise in each heater and temperature drop in each flasher must all be substantially equal. The steam economy  $E$ , kg evaporation/kg of 1055-kJ steam (lb/lb of 1000-Btu steam) may be approximated from

$$E = \left(1 - \frac{\Delta T}{1250}\right) \frac{\Delta T}{Y + R + \Delta T/N} \quad (11-124)$$



where  $\Delta T$  is the total temperature drop between feed to the first flasher and discharge from the last flasher, °C;  $N$  is the number of flash stages;  $Y$  is the approach between vapor temperature from the first flasher and liquid leaving the heater in which this vapor is condensed, °C (the approach is usually substantially constant for all stages); and  $R$ , °C (the sum of the boiling-point rise and the short-circuiting loss in the first flash stage). The expression for the mean effective temperature difference  $\Delta t$  available for heat transfer then becomes

$$\Delta t = \frac{\Delta T}{N \ln \frac{1 - \Delta T/1250 - RE/\Delta T}{1 - \Delta T/1250 - RE/\Delta T - E/N}} \quad (11-125)$$

**Multiple-Effect Evaporators** A number of approximate methods have been published for estimating performance and heating-surface requirements of a multiple-effect evaporator [Coates and Pressburg, *Chem. Eng.*, **67**(6), 157 (1960); Coates, *Chem. Eng. Prog.*, **45**, 25 (1949); and Ray and Carnahan, *Trans. Am. Inst. Chem. Eng.*, **41**, 253 (1945)]. However, because of the wide variety of methods of feeding and the added complication of feed heaters and condensate flash systems, the only certain way of determining performance is by detailed heat and material balances. Algebraic solutions may be used, but if more than a few effects are involved, trial-and-error methods are usually quicker. These frequently involve trial-and-error within trial-and-error solutions. Usually, if condensate flash systems or feed heaters are involved, it is best to start at the first effect. The basic steps in the calculation are then as follows:

1. Estimate temperature distribution in the evaporator, taking into account boiling-point elevations. If all heating surfaces are to be equal, the temperature drop across each effect will be approximately inversely proportional to the heat-transfer coefficient in that effect.
2. Determine total evaporation required, and estimate steam consumption for the number of effects chosen.
3. From assumed feed temperature (forward feed) or feed flow (backward feed) to the first effect and assumed steam flow, calculate evaporation in the first effect. Repeat for each succeeding effect, checking intermediate assumptions as the calculation proceeds. Heat input from condensate flash can be incorporated easily since the condensate flow from the preceding effects will have already been determined.
4. The result of the calculation will be a feed to or a product discharge from the last effect that may not agree with actual requirements. The calculation must then be repeated with a new assumption of steam flow to the first effect.
5. These calculations should yield liquor concentrations in each effect that make possible a revised estimate of boiling-point rises. They also give the quantity of heat that must be transferred in each effect. From the heat loads, assumed temperature differences, and heat-transfer coefficients, heating-surface requirements can be determined. If the distribution of heating surface is not as desired, the entire calculation may need to be repeated with revised estimates of the temperature in each effect.
6. If sufficient data are available, heat-transfer coefficients under the proposed operating conditions can be calculated in greater detail and surface requirements readjusted.

Such calculations require considerable judgment to avoid repetitive trials but are usually well worth the effort. Sample calculations are given in the American Institute of Chemical Engineers *Testing Procedure for Evaporators* and by Badger and Banchero, *Introduction to Chemical Engineering*, McGraw-Hill, New York, 1955. These balances may be done by computer but programming time frequently exceeds the time needed to do them manually, especially when variations in flow sheet are to be investigated. The MASSBAL program of SACDA, London, Ont., provides a considerable degree of flexibility in this regard. Another program, not specific to evaporators, is ASPEN PLUS by Aspen Tech., Cambridge, MA. Many such programs include simplifying assumptions and approximations that are not explicitly stated and can lead to erroneous results.

**Optimization** The primary purpose of evaporator design is to enable production of the necessary amount of satisfactory product at the lowest total cost. This requires economic-balance calculations that

may include a great number of variables. Among the possible variables are the following:

1. Initial steam pressure versus cost or availability.
2. Final vacuum versus water temperature, water cost, heat-transfer performance, and product quality.
3. Number of effects versus steam, water, and pump power cost.
4. Distribution of heating surface between effects versus evaporator cost.
5. Type of evaporator versus cost and continuity of operation.
6. Materials of construction versus product quality, tube life, evaporator life, and evaporator cost.
7. Corrosion, erosion, and power consumption versus tube velocity.
8. Downtime for retubing and repairs.
9. Operating-labor and maintenance requirements.
10. Method of feeding and use of heat-recovery systems.
11. Size of recovery heat exchangers.
12. Possible withdrawal of steam from an intermediate effect for use elsewhere.
13. Entrainment separation requirements.

The type of evaporator to be used and the materials of construction are generally selected on the basis of past experience with the material to be concentrated. The method of feeding can usually be decided on the basis of known feed temperature and the properties of feed and product. However, few of the listed variables are completely independent. For instance, if a large number of effects is to be used, with a consequent low temperature drop per effect, it is impractical to use a natural-circulation evaporator. If expensive materials of construction are desirable, it may be found that the forced-circulation evaporator is the cheapest and that only a few effects are justifiable.

The variable having the greatest influence on total cost is the number of effects in the evaporator. An economic balance can establish the optimum number where the number is not limited by such factors as viscosity, corrosiveness, freezing point, boiling-point rise, or thermal sensitivity. Under present United States conditions, savings in steam and water costs justify the extra capital, maintenance, and power costs of about seven effects in large commercial installations when the properties of the fluid are favorable, as in black-liquor evaporation. Under governmental financing conditions, as for plants to supply fresh water from seawater, evaporators containing from 12 to 30 or more effects can be justified.

As a general rule, the optimum number of effects increases with an increase in steam cost or plant size. Larger plants favor more effects, partly because they make it easier to install heat-recovery systems that increase the steam economy attainable with a given number of effects. Such recovery systems usually do not increase the total surface needed but do require that the heating surface be distributed between a greater number of pieces of equipment.

The most common evaporator design is based on the use of the same heating surface in each effect. This is by no means essential since few evaporators are "standard" or involve the use of the same patterns. In fact, there is no reason why all effects in an evaporator must be of the same type. For instance, the cheapest salt evaporator might use propeller calandrias for the early effects and forced-circulation effects at the low-temperature end, where their higher cost per unit area is more than offset by higher heat-transfer coefficients.

Bonilla [*Trans. Am. Inst. Chem. Eng.*, **41**, 529 (1945)] developed a simplified method for distributing the heating surface in a multiple-effect evaporator to achieve minimum cost. If the cost of the evaporator per unit area of heating surface is constant throughout, then minimum cost and area will be achieved if the ratio of area to temperature difference  $A/\Delta T$  is the same for all effects. If the cost per unit area  $z$  varies, as when different tube materials or evaporator types are used, then  $zA/\Delta T$  should be the same for all effects.

## EVAPORATOR ACCESSORIES

**Condensers** The vapor from the last effect of an evaporator is usually removed by a condenser. **Surface condensers** are employed when mixing of condensate with condenser cooling water is not desired. They are for the most part shell-and-tube condensers with

vapor on the shell side and a multipass flow of cooling water on the tube side. Heat loads, temperature differences, sizes, and costs are usually of the same order of magnitude as for another effect of the evaporator. Surface condensers use more cooling water and are so much more expensive that they are never used when a direct-contact condenser is suitable.

The most common type of direct-contact condenser is the counter-current **barometric condenser**, in which vapor is condensed by rising against a rain of cooling water. The condenser is set high enough so that water can discharge by gravity from the vacuum in the condenser. Such condensers are inexpensive and are economical on water consumption. They can usually be relied on to maintain a vacuum corresponding to a saturated-vapor temperature within 2.8°C (5°F) of the water temperature leaving the condenser [How, *Chem. Eng.*, **63**(2), 174 (1956)]. The ratio of water consumption to vapor condensed can be determined from the following equation:

$$\frac{\text{Water flow}}{\text{Vapor flow}} = \frac{H_v - h_2}{h_2 - h_1} \quad (11-126)$$

where  $H_v$  = vapor enthalpy and  $h_1$  and  $h_2$  = water enthalpies entering and leaving the condenser. Another type of direct-contact condenser is the **jet or wet condenser**, which makes use of high-velocity jets of water both to condense the vapor and to force noncondensable gases out the tailpipe. This type of condenser is frequently placed below barometric height and requires a pump to remove the mixture of water and gases. Jet condensers usually require more water than the more common barometric-type condensers and cannot be throttled easily to conserve water when operating at low evaporation rates.

**Vent Systems** Noncondensable gases may be present in the evaporator vapor as a result of leakage, air dissolved in the feed, or decomposition reactions in the feed. When the vapor is condensed in the succeeding effect, the noncondensables increase in concentration and impede heat transfer. This occurs partially because of the reduced partial pressure of vapor in the mixture but mainly because the vapor flow toward the heating surface creates a film of poorly conducting gas at the interface. (See page 11-14 for means of estimating the effect of noncondensable gases on the steam-film coefficient.) The most important means of reducing the influence of noncondensables on heat transfer is by properly channeling them past the heating surface. A positive vapor-flow path from inlet to vent outlet should be provided, and the path should preferably be tapered to avoid pockets of low velocity where noncondensables can be trapped. Excessive clearances and low-resistance channels that could bypass vapor directly from the inlet to the vent should be avoided [Standiford, *Chem. Eng. Prog.*, **75**, 59–62 (July 1979)].

In any event, noncondensable gases should be vented well before their concentration reaches 10 percent. Since gas concentrations are difficult to measure, the usual practice is to overvent. This means that an appreciable amount of vapor can be lost.

To help conserve steam economy, venting is usually done from the steam chest of one effect to the steam chest of the next. In this way, excess vapor in one vent does useful evaporation at a steam economy only about one less than the overall steam economy. Only when there are large amounts of noncondensable gases present, as in beet-sugar evaporation, is it desirable to pass the vents directly to the condenser to avoid serious losses in heat-transfer rates. In such cases, it can be worthwhile to recover heat from the vents in separate heat exchangers, which preheat the entering feed.

The noncondensable gases eventually reach the condenser (unless vented from an effect above atmospheric pressure to the atmosphere or to auxiliary vent condensers). These gases will be supplemented by air dissolved in the condenser water and by carbon dioxide given off on decomposition of bicarbonates in the water if a barometric condenser is used. These gases may be removed by the use of a water-jet-type condenser but are usually removed by a separate vacuum pump.

The vacuum pump is usually of the steam-jet type if high-pressure steam is available. If high-pressure steam is not available, more expensive mechanical pumps may be used. These may be either a water-ring (Hytor) type or a reciprocating pump.

The primary source of noncondensable gases usually is air dissolved in the condenser water. Figure 11-126 shows the dissolved-gas content

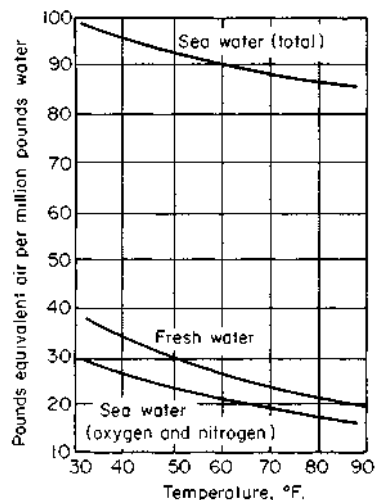


FIG. 11-126 Gas content of water saturated at atmospheric pressure. °C = 5/9 (°F – 32).

of fresh water and seawater, calculated as equivalent air. The lower curve for seawater includes only dissolved oxygen and nitrogen. The upper curve includes carbon dioxide that can be evolved by complete breakdown of bicarbonate in seawater. Breakdown of bicarbonates is usually not appreciable in a condenser but may go almost to completion in a seawater evaporator. The large increase in gas volume as a result of possible bicarbonate breakdown is illustrative of the uncertainties involved in sizing vacuum systems.

By far the largest load on the vacuum pump is water vapor carried with the noncondensable gases. Standard power-plant practice assumes that the mixture leaving a surface condenser will have been cooled 4.2°C (7.5°F) below the saturation temperature of the vapor. This usually corresponds to about 2.5 kg of water vapor/kg of air. One advantage of the countercurrent barometric condenser is that it can cool the gases almost to the temperature of the incoming water and thus reduce the amount of water vapor carried with the air.

In some cases, as with pulp-mill liquors, the evaporator vapors contain constituents more volatile than water, such as methanol and sulfur compounds. Special precautions may be necessary to minimize the effects of these compounds on heat transfer, corrosion, and condensate quality. They can include removing most of the condensate countercurrent to the vapor entering an evaporator-heating element, channeling vapor and condensate flow to concentrate most of the "foul" constituents into the last fraction of vapor condensed (and keeping this condensate separate from the rest of the condensate), and flashing the warm evaporator feed to a lower pressure to remove much of the foul constituents in only a small amount of flash vapor. In all such cases, special care is needed to properly channel vapor flow past the heating surfaces so there is a positive flow from steam inlet to vent outlet with no pockets, where foul constituents or noncondensibles can accumulate.

**Salt Removal** When an evaporator is used to make a crystalline product, a number of means are available for concentrating and removing the salt from the system. The simplest is to provide settling space in the evaporator itself. This is done in the types shown in Fig. 11-122*b*, *c*, and *e* by providing a relatively quiescent zone in which the salt can settle. Sufficiently high slurry densities can usually be achieved in this manner to reach the limit of pumpability. The evaporators are usually placed above barometric height so that the slurry can be discharged intermittently on a short time cycle. This permits the use of high velocities in large lines that have little tendency to plug.

If the amount of salts crystallized is on the order of a ton an hour or less, a salt trap may be used. This is simply a receiver that is connected to the bottom of the evaporator and is closed off from the evaporator periodically for emptying. Such traps are useful when insufficient headroom is available for gravity removal of the solids. However, traps require a

great deal of labor, give frequent trouble with the shutoff valves, and also can upset evaporator operation completely if a trap is reconnected to the evaporator without first displacing all air with feed liquor.

## EVAPORATOR OPERATION

The two principal elements of evaporator control are *evaporation rate* and *product concentration*. Evaporation rate in single- and multiple-effect evaporators is usually achieved by steam-flow control. Conventional-control instrumentation is used (see Sec. 22), with the added precaution that pressure drop across meter and control valve, which reduces temperature difference available for heat transfer, not be excessive when maximum capacity is desired. Capacity control of thermocompression evaporators depends on the type of compressor; positive-displacement compressors can utilize speed control or variations in operating pressure level. Centrifugal machines normally utilize adjustable inlet-guide vanes. Steam jets may have an adjustable spindle in the high-pressure orifice or be arranged as multiple jets that can individually be cut out of the system.

Product concentration can be controlled by any property of the solution that can be measured with the requisite accuracy and reliability. The preferred method is to impose control on rate of product withdrawal. Feed rates to the evaporator effects are then controlled by their

levels. When level control is impossible, as with the rising-film LTV, product concentration is used to control the feed rate—frequently by rationing of feed to steam with the ration reset by product concentration, sometimes also by feed concentration. Other controls that may be needed include vacuum control of the last effect (usually by air bleed to the condenser) and temperature-level control of thermocompression evaporators (usually by adding makeup heat or by venting excess vapor, or both as feed or weather conditions vary). For more control detail, see *Measurement and Control in Water Desalination*, N. Lior, ed., pp. 241–305, Elsevier Science Publ. Co., NY, 1986.

Control of an evaporator requires more than proper instrumentation. Operator logs should reflect changes in basic characteristics, as by use of **pseudo heat-transfer** coefficients, which can detect obstructions to heat flow, hence to capacity. These are merely the ratio of any convenient measure of heat flow to the temperature drop across each effect. **Dilution** by wash and seal water should be monitored since it absorbs evaporative capacity. Detailed tests, routine measurements, and operating problems are covered more fully in *Testing Procedure for Evaporators* (loc. cit.) and by Standiford [*Chem. Eng. Prog.*, **58**(11), 80 (1962)].

By far the best application of computers to evaporators is for working up operators' data into the basic performance parameters such as heat-transfer coefficients, steam economy, and dilution.

SECTION 12

Psychrometry,  
Evaporative  
Cooling, and  
Solids Drying

# PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



LARRY R. GENSKOW, WAYNE E. BEIMESCH  
JOHN P. HECHT, IAN KEMP, TIM LANGRISH  
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# Psychrometry, Evaporative Cooling, and Solids Drying\*

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### Nomenclature and Units

| Symbol           | Definition                                                                            | SI units           | U.S. Customary System units |
|------------------|---------------------------------------------------------------------------------------|--------------------|-----------------------------|
| $A$              | Area                                                                                  | $m^2$              | $ft^2$                      |
| $a_w$            | Water activity                                                                        | —                  | —                           |
| $a_w^{vapor}$    | Activity of water in the vapor phase                                                  | —                  | —                           |
| $a_w^{solid}$    | Activity of water in the solid                                                        | —                  | —                           |
| $c$              | Concentration                                                                         | $kg/m^3$           | $lb/ft^3$                   |
| $C_p$            | Specific heat capacity at constant pressure                                           | $J/(kg \cdot K)$   | $Btu/(lb \cdot ^\circ F)$   |
| $C_w$            | Concentration of water in the solid                                                   | $kg/m^3$           | $lbm/ft^3$                  |
| $\mathcal{D}(w)$ | Diffusion coefficient of water in a solid or liquid as a function of moisture content | $m^2/s$            | $ft^2/s$                    |
| $\mathcal{D}$    | Diffusion coefficient                                                                 | $m^2/s$            | $ft^2/s$                    |
| $d$              | Diameter (particle)                                                                   | $m$                | $in$                        |
| $E$              | Power                                                                                 | $W$                | $Btu/h$                     |
| $F$              | Solids or liquid mass flow rate                                                       | $kg/s$             | $lb/h$                      |
| $F$              | Mass flux of water at surface                                                         | $kg/(m^2 \cdot s)$ | $lbm/(ft^2 \cdot s)$        |

| Symbol           | Definition                                                      | SI units           | U.S. Customary System units         |
|------------------|-----------------------------------------------------------------|--------------------|-------------------------------------|
| $f$              | Relative drying rate                                            | —                  | —                                   |
| $G$              | Gas mass flow rate                                              | $kg/s$             | $lb/h$                              |
| $g$              | Acceleration due to gravity, $9.81 m/s^2$                       | $m/s^2$            | $ft/s^2$                            |
| $H$              | Enthalpy of a pure substance                                    | $J/kg$             | $Btu/lb$                            |
| $\Delta H_{vap}$ | Heat of vaporization                                            | $J/kg$             | $Btu/lb$                            |
| $h$              | Heat-transfer coefficient                                       | $W/(m^2 \cdot K)$  | $Btu/(ft^2 \cdot h \cdot ^\circ F)$ |
| $I$              | Humid enthalpy (dry substance and associated moisture or vapor) | $J/kg$             | $Btu/lb$                            |
| $J$              | Mass flux (of evaporating liquid)                               | $kg/(m^2 \cdot s)$ | $lb/(ft^2 \cdot h)$                 |
| $k$              | Mass-transfer coefficient                                       | $m/s$              | $lb/(ft^2 \cdot h \cdot atm)$       |
| $k_{air}$        | Thermal conductivity of air                                     | $W/(m \cdot K)$    | $Btu/(ft \cdot h \cdot ^\circ F)$   |
| $k_c$            | Mass-transfer coefficient for a concentration driving force     | $m/s$              | $ft^2/s$                            |
| $k_p$            | Mass transfer coefficient for a partial pressure driving force  | $kg/(m^2 \cdot s)$ | $lbm/(ft^2 \cdot s)$                |



## Nomenclature and Units (Concluded)

| Symbol                         | Definition                                                              | SI units               | U.S. Customary System units | Symbol               | Definition                                               | SI units  | U.S. Customary System units |
|--------------------------------|-------------------------------------------------------------------------|------------------------|-----------------------------|----------------------|----------------------------------------------------------|-----------|-----------------------------|
| $L$                            | Length; length of drying layer                                          | m                      | ft                          | $X$                  | Solids moisture content (dry basis)                      | —         | —                           |
| $M$                            | Molecular weight                                                        | kg/mol                 | lb/mol                      | $Y$                  | Mass ratio                                               | —         | —                           |
| $m$                            | Mass                                                                    | kg                     | lb                          | $z$                  | Distance coordinate                                      | m         | ft                          |
| $m_{\text{solids}}$            | Mass of dry solids                                                      | kg                     | lbm                         | Dimensionless groups |                                                          |           |                             |
| $N$                            | Specific drying rate ( $-dX/dt$ )                                       | 1/s                    | 1/s                         | Ar                   | Archimedes number, $(gd^3\rho_C/\mu^2)(\rho_P - \rho_C)$ | —         | —                           |
| $N$                            | Rotational speed (drum, impeller, etc.)                                 | 1/s                    | rpm                         | Bi                   | Biot number, $h \cdot L/\kappa$                          | —         | —                           |
| $P$                            | Total pressure                                                          | kg/(m $\cdot$ s $^2$ ) | lbf/in $^2$                 | Gr                   | Grashof number, $L^3\rho^2\beta g\Delta T/\mu^2$         | —         | —                           |
| $P_{\text{bulk}}$              | Partial pressure of water vapor in the air far from the drying material | kg/m $\cdot$ s $^2$    | lbf/in $^2$                 | Nu                   | Nusselt number, $hd_p/\kappa$                            | —         | —                           |
| $P_{\text{surface}}$           | Partial pressure of water vapor in the air at the solid interface       | kg/m $\cdot$ s $^2$    | lbf/in $^2$                 | Pr                   | Prandtl number, $\mu C_p/\kappa$                         | —         | —                           |
| $p$                            | Partial pressure/vapor pressure of component                            | kg/(m $\cdot$ s $^2$ ) | lbf/in $^2$                 | Re                   | Reynolds number, $\rho d_p U/\mu$                        | —         | —                           |
| $p_{\text{pure}}^{\text{sat}}$ | Pure component vapor pressure                                           | kg/(m $\cdot$ s $^2$ ) | lbf/in $^2$                 | Sc                   | Schmidt number, $\mu/\rho \mathcal{D}$                   | —         | —                           |
| $p_{\text{air}}$               | Partial pressure of water vapor in air                                  | kg/(m $\cdot$ s $^2$ ) | lbf/in $^2$                 | Sh                   | Sherwood number, $k_y d_p/\mathcal{D}$                   | —         | —                           |
| $Q$                            | Heat-transfer rate                                                      | W                      | Btu/h                       | Le                   | Lewis = Sc/Pr                                            | —         | —                           |
| $q$                            | Heat flux                                                               | W/m $^2$               | Btu/(ft $^2$ ·h)            | Greek letters        |                                                          |           |                             |
| $R$                            | Universal gas constant, 8314 J/(kmol $\cdot$ K)                         | J/(mol·K)              | Btu/(mol $\cdot$ °F)        | $\alpha$             | Slope                                                    | —         | —                           |
| $R$                            | Droplet radius                                                          | m                      | ft                          | $\beta$              | Psychrometric ratio                                      | —         | —                           |
| $r$                            | Radius; radial coordinate                                               | m                      | ft                          | $\varepsilon$        | Voidage (void fraction)                                  | —         | —                           |
| RH                             | Relative humidity                                                       | —                      | —                           | $\zeta$              | Dimensionless distance                                   | —         | —                           |
| $S$                            | Percentage saturation                                                   | —                      | —                           | $\eta$               | Efficiency                                               | —         | —                           |
| $s$                            | Solid-fixed coordinate                                                  | Depends on geometry    | —                           | $\theta$             | Dimensionless time                                       | —         | —                           |
| $T$                            | Absolute temperature                                                    | K                      | °R                          | $\kappa$             | Thermal conductivity                                     | W/(m·K)   | Btu/(ft·h·°F)               |
| $T, t$                         | Temperature                                                             | °C                     | °F                          | $\lambda$            | Latent heat of evaporation                               | J/kg      | Btu/lb                      |
| $t$                            | Time                                                                    | s                      | h                           | $\mu$                | Absolute viscosity                                       | kg/(m·s)  | lb/(ft·s)                   |
| $U$                            | Velocity                                                                | m/s                    | ft/s                        | $\mu_{\text{air}}$   | Viscosity of air                                         | kg/(m·s)  | lbm/(ft·s)                  |
| $u$                            | Mass of water/mass of dry solid                                         | —                      | —                           | $\rho$               | Density                                                  | kg/m $^3$ | lb/ft $^3$                  |
| $V$                            | Volume                                                                  | m $^3$                 | ft $^3$                     | $\rho_{\text{air}}$  | Air density                                              | kg/m $^3$ | lbm/ft $^3$                 |
| $V$                            | Air velocity                                                            | m/s                    | ft/s                        | $\rho_s$             | Mass concentration of solids                             | kg/m $^3$ | lbm/ft $^3$                 |
| $v$                            | Specific volume                                                         | m $^3$ /kg             | ft $^3$ /lb                 | $\rho_s^o$           | Density of dry solid                                     | kg/m $^3$ | lbm/ft $^3$                 |
| $v_{\text{droplet}}$           | Droplet volume                                                          | m $^3$                 | ft $^3$                     | $\rho_w^o$           | Density of pure water                                    | kg/m $^3$ | lbm/ft $^3$                 |
| $w$                            | Wet-basis moisture content                                              | —                      | —                           | $\tau$               | Residence time of solids                                 | s         | h                           |
| $w_{\text{avg dry-basis}}$     | Average wet-basis moisture content                                      | —                      | —                           | $\Phi$               | Characteristic (dimensionless) moisture content          | —         | —                           |
|                                |                                                                         |                        |                             | $\psi$               | Relative humidity                                        | %         | %                           |

## PSYCHROMETRY

**GENERAL REFERENCES** ASHRAE 2002 *Handbook: Fundamentals*, SI Edition, American Society of Heating, Refrigeration and Air-Conditioning Engineers, Atlanta, Ga., 2002, Chap. 6, "Psychrometrics," Chap. 19.2, "Sorbents and Desiccants." *Aspen Process Manual* (Internet knowledge base), Aspen Technology, 2000 onward. *Humidity and Dewpoint*, British Standard BS 1339 (rev.). Humidity and dewpoint, Pt. 1 (2002); Terms, definitions and formulae, Pt. 2 (2005); Psychrometric calculations and tables (including spreadsheet), Pt. 3 (2004); Guide to humidity measurement, British Standards Institution, Gunnersbury, United Kingdom. Cook and DuMont, *Process Drying Practice*, McGraw-Hill, New York, 1991, Chap. 6. Keey, *Drying of Loose and Particulate Materials*, Hemisphere, New York, 1992. Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2000. Earlier editions: 1st/2d editions, Reid and Sherwood (1958/1966); 3d ed., Reid, Prausnitz, and Sherwood (1977); 4th ed., Reid, Prausnitz, and Poling (1986). Soininen, "A Prospectively Transformed Psychrometric Chart and Its Application to Drying Calculations," *Drying Technol.* 4(2): 295–305 (1986). Sonntag, "Important New Values of the Physical Constants of 1986, Vapor Pressure Formulations Based on the ITS-90, and Psychrometer Formulae," *Zeitschrift für Meteorologie*, 40(5):340–344 (1990). Treybal, *Mass-Transfer Operations*, 3d ed., McGraw-Hill, New York, 1980. Wexler, *Humidity and Moisture*, vol. 1, Reinhold, New York, 1965.

Psychrometry is concerned with the determination of the properties of gas-vapor mixtures. These are important in calculations for

humidification and dehumidification, particularly in cooling towers, air-conditioning systems, and dryers. The first two cases involve the air-water vapor system at near-ambient conditions, but dryers normally operate at elevated temperatures and may also use elevated or subatmospheric pressures and other gas-solvent systems.

Principles involved in determining the properties of other systems are the same as with air-water vapor, with one major exception. Whereas the psychrometric ratio (ratio of heat-transfer coefficient to product of mass-transfer coefficient and humid heat, terms defined in the following subsection) for the air-water system can be taken as 1, the ratio for other systems in general does not equal 1. This has the effect of making the adiabatic saturation temperature different from the wet-bulb temperature. Thus, for systems other than air-water vapor, accurate calculation of psychrometric and drying problems is complicated by the necessity for point-to-point calculation of the temperature of the evaporating surface. For example, for the air-water system, the temperature of the evaporating surface will be constant during the constant-rate drying period even though the temperature and humidity of the gas stream change. For other systems, the temperature of the evaporating surface would change.



## 12-4 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

**TABLE 12-1 Interconversion Formulas for Air-Water System, to 3 Significant Figures**

| $T$ = temperature in <b>kelvins</b> (K); $P$ = total pressure in <b>pascals</b> (Pa or N/m <sup>2</sup> ) |                                         |                              |                             |                                           |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------|-----------------------------|-------------------------------------------|
| Convert from:                                                                                             | $Y$ (or ppm <sub>w</sub> ) <sup>a</sup> | $y$                          | $p$                         | $Y_e$                                     |
| Convert to:                                                                                               |                                         |                              |                             |                                           |
| Absolute humidity (mixing ratio) $Y$ (kg·kg <sup>-1</sup> )                                               | 1                                       | $Y = \frac{0.622Y}{1 - Y}$   | $Y = \frac{0.622p}{P - p}$  | $Y = \frac{0.622}{0.002167P/(Y_e T) - 1}$ |
| Mole fraction $y$ (mol·mol <sup>-1</sup> )                                                                | $y = \frac{Y}{0.622 + Y}$               | 1                            | $y = \frac{p}{P}$           | $y = \frac{461.5Y_e T}{P}$                |
| Vapor pressure $p$ (Pa)                                                                                   | $p = \frac{PY}{0.622 + Y}$              | $p = yP$                     | 1                           | $p = 461.5Y_e T$                          |
| Volumetric humidity $Y_e$ (kg·m <sup>-3</sup> )                                                           | $Y_e = \frac{0.002167PY}{(0.622 + Y)T}$ | $Y_e = \frac{0.002167yP}{T}$ | $Y_e = \frac{0.002167p}{T}$ | 1                                         |

### TERMINOLOGY

Terminology and nomenclature pertinent to psychrometry are given below. There is often considerable confusion between dry and wet basis, and between mass, molar, and volumetric quantities, in both definitions and calculations. Dry- and wet-basis humidity are similar at ambient conditions but can differ significantly at elevated humidities, e.g., in dryer exhaust streams. Complete interconversion formulas between four key humidity parameters are given in Table 12-1 for the air-water system and in Table 12-2 for a general gas-vapor system.

Definitions related to humidity, vapor pressure, saturation, and volume are as follows; the most useful are absolute humidity, vapor pressure, and relative humidity.

**Absolute humidity  $Y$**  Mass of water (or solvent) vapor carried by unit mass of dry air (or other carrier gas). It is also known as the **mixing ratio**, **mass ratio**, or **dry-basis humidity**. Preferred units are lb/lb or kg/kg, but g/kg and gr/lb are often used, as are ppm<sub>w</sub> and ppb<sub>w</sub> (parts per million/billion by weight); ppm<sub>w</sub> = 10<sup>6</sup> $Y$ , ppb<sub>w</sub> = 10<sup>9</sup> $Y$ .

**Specific humidity  $Y_w$**  Mass of vapor per unit mass of gas-vapor mixture. Also known as mass fraction or wet-basis humidity, and much more rarely used than dry-basis absolute humidity.  $Y_w = Y/(1 + Y)$ ;  $Y = Y_w/(1 - Y_w)$ .

**Mole ratio  $z$**  Number of moles of vapor per mole of gas (dry basis), mol/mol;  $z = (M_v/M_g)Y$ , where  $M_v$  = molecular weight of vapor and  $M_g$  = molecular weight of gas. It may also be expressed as ppm, and ppb<sub>v</sub> (parts per million/billion by volume); ppm<sub>v</sub> = 10<sup>6</sup> $z$ , ppb<sub>v</sub> = 10<sup>9</sup> $z$ .

**Mole fraction  $y$**  Number of moles of vapor per mole of gas-vapor mixture (wet basis);  $y = z/(1 + z)$ ;  $z = y/(1 - y)$ . If a mixture contains  $m_v$  kg and  $n_v$  mol of vapor (e.g., water) and  $m_g$  kg and  $n_g$  mol of non-condensable gas (e.g., air), with  $m_v = n_v M_v$  and  $m_g = n_g M_g$ , then the four quantities above are defined by

$$Y = \frac{m_v}{m_g} \quad Y_w = \frac{m_v}{m_g + m_v} \quad z = \frac{n_v}{n_g} \quad y = \frac{n_v}{n_g + n_v}$$

**Volumetric humidity  $Y_e$**  Mass of vapor per unit volume of gas-vapor mixture. It is sometimes, confusingly, called the absolute humidity, but it is really a vapor concentration; preferred units are kg/m<sup>3</sup> or lb/ft<sup>3</sup>, but g/m<sup>3</sup> and gr/ft<sup>3</sup> are also used. It is inconvenient for calculations because it depends on temperature and pressure and on the units system; absolute humidity  $Y$  is always preferable for heat and mass balances. It is proportional to the specific humidity (wet basis);  $Y_v = Y_w \rho_g$ , where  $\rho_g$  is the **humid gas density** (mass of gas-vapor mixture per unit volume, wet basis). Also

$$Y_e = \frac{M_v P n_v}{RT(n_g + n_v)}$$

**Vapor pressure  $p$**  Partial pressure of vapor in gas-vapor mixture, and is proportional to the mole fraction of vapor;  $p = yP$ , where  $P$  = total pressure, in the same units as  $p$  (Pa, N/m<sup>2</sup>, bar, atm, or psi). Hence

$$p = \frac{n_v}{n_g + n_v} P$$

**Saturation vapor pressure  $p_s$**  Pressure exerted by pure vapor at a given temperature. When the vapor partial pressure  $p$  in the gas-vapor mixture at a given temperature equals the saturation vapor pressure  $p_s$  at the same temperature, the air is **saturated** and the absolute humidity is designated the **saturation humidity**  $Y_s$ .

**Relative humidity RH or  $\Psi$**  The partial pressure of vapor divided by the saturation vapor pressure at the given temperature, usually expressed as a percentage. Thus  $RH = 100p/p_s$ .

**Percentage absolute humidity (percentage saturation)  $S$**  Ratio of absolute humidity to saturation humidity, given by  $S = 100Y/Y_s = 100p(P - p_s)/[p_s(P - p)]$ . It is much less commonly used than relative humidity.

**Dew point  $T_{dew}$  or saturation temperature** Temperature at which a given mixture of water vapor and air becomes saturated on cooling; i.e., the temperature at which water exerts a vapor pressure equal to the partial pressure of water vapor in the given mixture.

**TABLE 12-2 Interconversion Formulas for a General Gas-Vapor System**

| $M_g, M_v$ = molal mass of gas and vapor, respectively; $R$ = 8314 J/(kmol·K); $T$ = temperature in <b>kelvins</b> (K); $P$ = total pressure in <b>pascals</b> (Pa or N/m <sup>2</sup> ) |                                                |                                |                                 |                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|---------------------------------|--------------------------------------------|
| Convert from:                                                                                                                                                                            | $Y$ (or ppm <sub>w</sub> )                     | $y$                            | $p$                             | $Y_e$                                      |
| Convert to:                                                                                                                                                                              |                                                |                                |                                 |                                            |
| Absolute humidity (mixing ratio) $Y$ (kg·kg <sup>-1</sup> )                                                                                                                              | 1                                              | $Y = \frac{M_v y}{M_g(1 - Y)}$ | $Y = \frac{p M_v}{(P - p) M_g}$ | $Y = \frac{M_v}{M_g(P M_v / Y_e R T - 1)}$ |
| Mole fraction $y$ (mol·mol <sup>-1</sup> )                                                                                                                                               | $y = \frac{Y}{M_v/M_g + Y}$                    | 1                              | $y = \frac{p}{P}$               | $y = \frac{Y_e R T}{P M_v}$                |
| Vapor pressure $p$ (Pa)                                                                                                                                                                  | $p = \frac{PY}{M_v/M_g + Y}$                   | $p = yP$                       | 1                               | $p = \frac{Y_e R T}{M_v}$                  |
| Volumetric humidity $Y_e$ (kg·m <sup>-3</sup> )                                                                                                                                          | $Y_e = \frac{M_v}{R T} \frac{PY}{M_v/M_g + Y}$ | $Y_e = \frac{M_v y P}{R T}$    | $Y_e = \frac{M_v p}{R T}$       | 1                                          |

**Humid volume  $v$**  Volume in cubic meters (cubic feet) of 1 kg (1 lb) of dry air and the water vapor it contains.

**Saturated volume  $v_s$**  Humid volume when the air is saturated. Terms related to heat balances are as follows:

**Humid heat  $C_s$**  Heat capacity of unit mass of dry air and the moisture it contains.  $C_s = C_{pg} + C_{pv}Y$ , where  $C_{pg}$  and  $C_{pv}$  are the heat capacities of dry air and water vapor, respectively, and both are assumed constant. For approximate engineering calculations at near-ambient temperatures, in SI units,  $C_s = 1 + 1.9Y$  kJ/(kg·K) and in U.S. units,  $C_s = 0.24 + 0.45Y$  (Btu/(lb·°F)).

**Humid enthalpy  $H$**  Heat content at a given temperature  $T$  of unit mass of dry air and the moisture it contains, relative to a datum temperature  $T_0$ , usually 0°C. As water is liquid at 0°C, the humid enthalpy also contains a term for the latent heat of water. If heat capacity is invariant with temperature,  $H = (C_{pg} + C_{pv}Y)(T - T_0) + \lambda_0 Y$ , where  $\lambda_0$  is the latent heat of water at 0°C, 2501 kJ/kg (1075 Btu/lb). In practice, for accurate calculations, it is often easier to obtain the vapor enthalpy  $H_v$  from steam tables, when  $H = H_g + H_v = C_{pg}T + H_v$ .

**Adiabatic saturation temperature  $T_{as}$**  Final temperature reached by a small quantity of vapor-gas mixture into which water is evaporating. It is sometimes called the thermodynamic wet-bulb temperature.

**Wet-bulb temperature  $T_{wb}$**  Dynamic equilibrium temperature attained by a liquid surface from which water is evaporating into a flowing airstream when the rate of heat transfer to the surface by convection equals the rate of mass transfer away from the surface. It is very close to the adiabatic saturation temperature for the air-water system, but not for most other vapor-gas systems; see later.

## CALCULATION FORMULAS

Table 12-1 gives formulas for conversion between absolute humidity, mole fraction, vapor pressure, and volumetric humidity for the air-water system, and Table 12-2 does likewise for a general gas-vapor system. Where relationships are not included in the definitions, they are given below.

In U.S. units, the formulas are the same except for the volumetric humidity  $Y_v$ . Because of the danger of confusion with pressure units, it is recommended that in both Tables 12-1 and 12-2,  $Y_v$  be calculated in SI units and then converted.

Volumetric humidity is also related to absolute humidity and humid gas density by

$$Y_v = Y_w \rho_g = \frac{Y}{1 + Y} \rho_g \quad (12-1)$$

Two further useful formulas are as follows:

| Parameter                                                                              | General vapor-gas system                                                                       | Air-water system, SI units, to 3 significant figures | Eq. no. |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------|---------|
| <b>Density of humid gas (moist air)</b><br>$\rho_g$ (kg/m <sup>3</sup> )               | $\rho_g = \frac{M_g}{RT} \left( P - p + \frac{M_v}{M_g} p \right)$                             | $\rho_g = \frac{P - 0.378p}{287.1T}$                 | (12-2)  |
| <b>Humid volume <math>v</math></b><br>per unit mass of dry air<br>(m <sup>3</sup> /kg) | $v = \frac{RT}{M_g(P - p)} = \frac{RT}{P} \times \left( \frac{1}{M_g} + \frac{Y}{M_v} \right)$ | $v = \frac{461.5T}{P} (0.622 + Y)$                   | (12-3)  |

From Eq. (12-2), the density of dry air at 0°C (273.15 K) and 1 atm (101,325 Pa) is 1.292 kg/m<sup>3</sup> (0.08065 lb/ft<sup>3</sup>). Note that the density of moist air is always lower than that of dry air.

Equation (12-3) gives the humid volume of dry air at 0°C (273.15 K) and 1 atm as 0.774 m<sup>3</sup>/kg (12.4 ft<sup>3</sup>/lb). For moist air, humid volume is not the reciprocal of humid gas density;  $v = (1 + Y)/\rho_g$ .

The **saturation vapor pressure** of water is given by Sonntag (1990) in pascals (N/m<sup>2</sup>) at absolute temperature  $T$  (K).

Over water:

$$\ln p_s = -6096.9385T^{-1} + 21.2409642 - 2.711193 \times 10^{-3}T + 1.673952 \times 10^{-5}T^2 + 2.433502 \ln T \quad (12-4a)$$

Over ice:

$$\ln p_s = -6024.5282T^{-1} + 29.32707 + 1.0613868 \times 10^{-2}T - 1.3198825 \times 10^{-5}T^2 - 0.49382577 \ln T \quad (12-4b)$$

Simpler equations for saturation vapor pressure are the Antoine equation and Magnus formula. These are slightly less accurate, but easier to calculate and also easily reversible to give  $T$  in terms of  $p$ . For the Antoine equation, given below, coefficients for numerous other solvent-gas systems are given in Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, 2000.

$$\ln p_s = C_0 - \frac{C_1}{T - C_2} \quad T = \frac{C_1}{C_0 - \ln p_s} + C_2 \quad (12-5)$$

Values for Antoine coefficients for the air-water system are given in Table 12-3. The standard values give vapor pressure within 0.1 percent of steam tables over the range 50 to 100°C, but an error of nearly 3 percent at 0°C. The alternative coefficients give a close fit at 0 and 100°C and an error of less than 1.2 percent over the intervening range.

The Sonntag equation strictly only applies to water vapor with no other gases present (i.e., in a partial vacuum). The vapor pressure of a gas mixture, e.g., water vapor in air, is given by multiplying the pure liquid vapor pressure by an enhancement factor  $f$ , for which various equations are available (see British Standard BS 1339 Part 1, 2002). However, the correction is typically less than 0.5 percent, except at elevated pressures, and it is therefore usually neglected for engineering calculations.

## RELATIONSHIP BETWEEN WET-BULB AND ADIABATIC SATURATION TEMPERATURES

If a stream of air is intimately mixed with a quantity of water in an adiabatic system, the temperature of the air will drop and its humidity will increase. If the equilibration time or the number of transfer units approaches infinity, the air-water mixture will reach saturation. The **adiabatic saturation temperature**  $T_{as}$  is given by a heat balance between the initial unsaturated vapor-gas mixture and the final saturated mixture at thermal equilibrium:

$$C_s(T - T_{as}) = \lambda_{as}(Y_{as} - Y) \quad (12-6)$$

This equation has to be reversed and solved iteratively to obtain  $Y_{as}$  (absolute humidity at adiabatic saturation) and hence  $T_{as}$  (the calculation is divergent in the opposite direction). Approximate direct formulas are available from various sources, e.g., British Standard BS 1339 (2002) and Liley (*Int. J. Mech. Engg. Educ.* 21(2), 1993). The latent heat of evaporation evaluated at the adiabatic saturation temperature is  $\lambda_{as}$ ,

TABLE 12-3 Alternative Sets of Values for Antoine Coefficients for the Air-Water System

|                    |           | $C_0$   | $C_1$   | $C_2$   |             | $C_0$   | $C_1$   | $C_2$   |
|--------------------|-----------|---------|---------|---------|-------------|---------|---------|---------|
| Standard values    | $p$ in Pa | 23.1963 | 3816.44 | 46.13 K | $p$ in mmHg | 18.3036 | 3816.44 | 46.13 K |
| Alternative values | $p$ in Pa | 23.19   | 3830    | 44.83 K | $p$ in mmHg | 18.3    | 3830    | 44.87 K |

which may be obtained from steam tables; humid heat  $C_s$  is evaluated at initial humidity  $Y$ . On a psychrometric chart, the adiabatic saturation process almost exactly follows a **constant-enthalpy line**, as the sensible heat given up by the gas-vapor mixture exactly balances the latent heat of the liquid that evaporates back into the mixture. The only difference is due to the sensible heat added to the water to take it from the datum temperature to  $T_{as}$ . The adiabatic saturation line differs from the constant-enthalpy line as follows, where  $C_{PL}$  is the specific heat capacity of the liquid:

$$H_{as} - H = C_{PL} T_{as} (Y_{as} - Y) \quad (12-7)$$

Equation (12-7) is useful for calculating the adiabatic saturation line for a given  $T_{as}$  and gives an alternative iterative method for finding  $T_{as}$ , given  $T$  and  $Y$ ; compared with Eq. (12-6), it is slightly more accurate and converges faster, but the calculation is more cumbersome.

The **wet-bulb temperature** is the temperature attained by a fully wetted surface, such as the wick of a wet-bulb thermometer or a droplet or wet particle undergoing drying, in contact with a flowing unsaturated gas stream. It is regulated by the rates of vapor-phase heat and mass transfer to and from the wet bulb. Assuming mass transfer is controlled by diffusion effects and heat transfer is purely convective:

$$h(T - T_{wb}) = k_y \lambda_{wb} (Y_{wb} - Y) \quad (12-8)$$

where  $k_y$  is the corrected mass-transfer coefficient [ $\text{kg}/(\text{m}^2 \cdot \text{s})$ ],  $h$  is the heat-transfer coefficient [ $\text{kW}/(\text{m}^2 \cdot \text{K})$ ],  $Y_{wb}$  is the saturation mixing ratio at  $T_{wb}$ , and  $\lambda_{wb}$  is the latent heat ( $\text{kJ/kg}$ ) evaluated at  $T_{wb}$ . Again, this equation must be solved iteratively to obtain  $T_{wb}$  and  $Y_{wb}$ .

In practice, for any practical psychrometer or wetted droplet or particle, there is significant extra heat transfer from radiation. For an Assmann psychrometer at near-ambient conditions, this is approximately 10 percent. This means that any measured real value of  $T_{wb}$  is slightly higher than the "pure convective" value in the definition. It is often more convenient to obtain wet-bulb conditions from adiabatic saturation conditions (which are much easier to calculate) by the following formula:

$$\frac{T - T_{wb}}{Y_{wb} - Y} = \frac{T - T_{as}}{Y_{as} - Y} \beta \quad (12-9)$$

where the psychrometric ratio  $\beta = \overline{C_s} k_y / h$  and  $\overline{C_s}$  is the mean value of the humid heat over the range from  $T_{as}$  to  $T$ .

The advantage of using  $\beta$  is that it is approximately constant over normal ranges of temperature and pressure for any given pair of vapor and gas values. This avoids having to estimate values of heat- and mass-transfer coefficients  $\alpha$  and  $k_y$  from uncertain correlations. For the air-water system, considering convective heat transfer alone,  $\beta \sim 1.1$ . In practice, there is an additional contribution from radiation, and  $\beta$  is very close to 1. As a result, the wet-bulb and adiabatic saturation temperatures differ by less than  $1^\circ\text{C}$  for the air-water system at near-ambient conditions ( $0$  to  $100^\circ\text{C}$ ,  $Y < 0.1$   $\text{kg/kg}$ ) and can be taken as equal for normal calculation purposes. Indeed, typically the  $T_{wb}$  measured by a practical psychrometer or at a wetted solid surface is closer to  $T_{as}$  than to the "pure convective" value of  $T_{wb}$ .

However, for nearly all other vapor-gas systems, particularly for organic solvents,  $\beta < 1$ , and hence  $T_{wb} > T_{as}$ . This is illustrated in Fig. 12-5. For these systems the psychrometric ratio may be obtained by determining  $h/k_y$  from heat- and mass-transfer analogies such as the Chilton-Colburn analogy. The basic form of the equation is

$$\beta = \left( \frac{\text{Sc}}{\text{Pr}} \right)^n = \text{Le}^{-n} \quad (12-10)$$

$\text{Sc}$  is the Schmidt number for mass-transfer properties,  $\text{Pr}$  is the Prandtl number for heat-transfer properties, and  $\text{Le}$  is the Lewis number  $\kappa/(C_p \rho_g \mathcal{D})$ , where  $\kappa$  is the gas thermal conductivity and  $\mathcal{D}$  is the diffusion coefficient for the vapor through the gas. Experimental and theoretical values of the exponent  $n$  range from 0.56 [Bedingfield and Drew, *Ind. Eng. Chem.*, **42**:1164 (1950)] to  $\frac{2}{3} = 0.667$  [Chilton and Colburn, *Ind. Eng. Chem.*, **26**:1183 (1934)]. A detailed discussion is given by Keey (1992). Values of  $\beta$  for any system can be estimated from the specific heats, diffusion coefficients, and other data given in Sec. 2. See the example below.

For calculation of wet-bulb (and adiabatic saturation) conditions, the most commonly used formula in industry is the **psychrometer equation**. This is a simple, linear formula that gives vapor pressure directly if the wet-bulb temperature is known, and is therefore ideal for calculating humidity from a wet-bulb measurement using a psychrometer, although the calculation of wet-bulb temperature from humidity still requires an iteration.

$$p = p_{wb} - AP(T - T_{wb}) \quad (12-11)$$

where  $A$  is the psychrometer coefficient. For the air-water system, the following formulas based on equations given by Sonntag [*Zeitschrift für Meteorologie*, **40**(5): 340–344 (1990)] may be used to give  $A$  for  $T_{wb}$  up to  $30^\circ\text{C}$ ; they are based on extensive experimental data for Assmann psychrometers.

Over water (wet-bulb temperature):

$$A = 6.5 \times 10^{-4} (1 + 0.000944 T_{wb}) \quad (12-12a)$$

Over ice (ice-bulb temperature):

$$A_i = 5.72 \times 10^{-4} \quad (12-12b)$$

For other vapor-gas systems,  $A$  is given by

$$A = \frac{M_g C_s}{M_v \beta \lambda_{wb}} \quad (12-13)$$

Here  $\beta$  is the psychrometric coefficient for the system. As a cross-check, for the air-water system at  $20^\circ\text{C}$  wet-bulb temperature,  $50^\circ\text{C}$  dry-bulb temperature, and absolute humidity  $0.002$   $\text{kg/kg}$ ,  $C_s = (1.006 + 1.9 \times 0.002) = 1.01$   $\text{kJ}/(\text{kg} \cdot \text{K})$  and  $\lambda_{wb} = 2454$   $\text{kJ/kg}$ . Since  $M_g = 28.97$   $\text{kg/kmol}$  and  $M_v = 18.02$   $\text{kg/kmol}$ , Eq. (12-12) gives  $A$  as  $6.617 \times 10^{-4}/\beta$ , compared with Sonntag's value of  $6.653 \times 10^{-4}$  at this temperature, giving a value for the psychrometric coefficient  $\beta$  of  $0.995$ ; that is,  $\beta \approx 1$ , as expected for the air-water system.

## PSYCHROMETRIC CHARTS

Psychrometric charts are plots of humidity, temperature, enthalpy, and other useful parameters of a gas-vapor mixture. They are helpful for rapid estimates of conditions and for visualization of process operations such as humidification and drying. They apply to a given system at a given pressure, the most common of course being air-water at atmospheric pressure. There are four types, of which the Grosvenor and Mollier types are most widely used:

The **Grosvenor chart** plots temperature (abscissa) against humidity (ordinate). Standard charts produced by ASHRAE and other groups, or by computer programs, are usually of this type. The saturation line is a curve from bottom left to top right, and curves for constant relative humidity are approximately parallel to this. Lines from top left to bottom right may be of either constant wet-bulb temperature or constant enthalpy, depending on the chart. The two are not quite identical, so if only one is shown, correction factors are required for the other parameter. Examples are shown in Figs. 12-1 (SI units), 12-2a (U.S. Customary System units, medium temperature), and 12-2b (U.S. Customary System units, high temperature).

The **Bowen chart** is a plot of enthalpy (abscissa) against humidity (ordinate). It is convenient to be able to read enthalpy directly, especially for near-adiabatic convective drying where the operating line approximately follows a line of constant enthalpy. However, it is very difficult to read accurately because the key information is compressed in a narrow band near the saturation line. See Cook and DuMont, *Process Drying Practice*, McGraw-Hill, New York, 1991, chap. 6.

The **Mollier chart** plots humidity (abscissa) against enthalpy (lines sloping diagonally from top left to bottom right). Lines of constant temperature are shallow curves at a small slope to the horizontal. The chart is nonorthogonal (no horizontal lines) and hence a little difficult to plot and interpret initially. However, the area of greatest interest is expanded, and they are therefore easy to read accurately. They tend to cover a wider



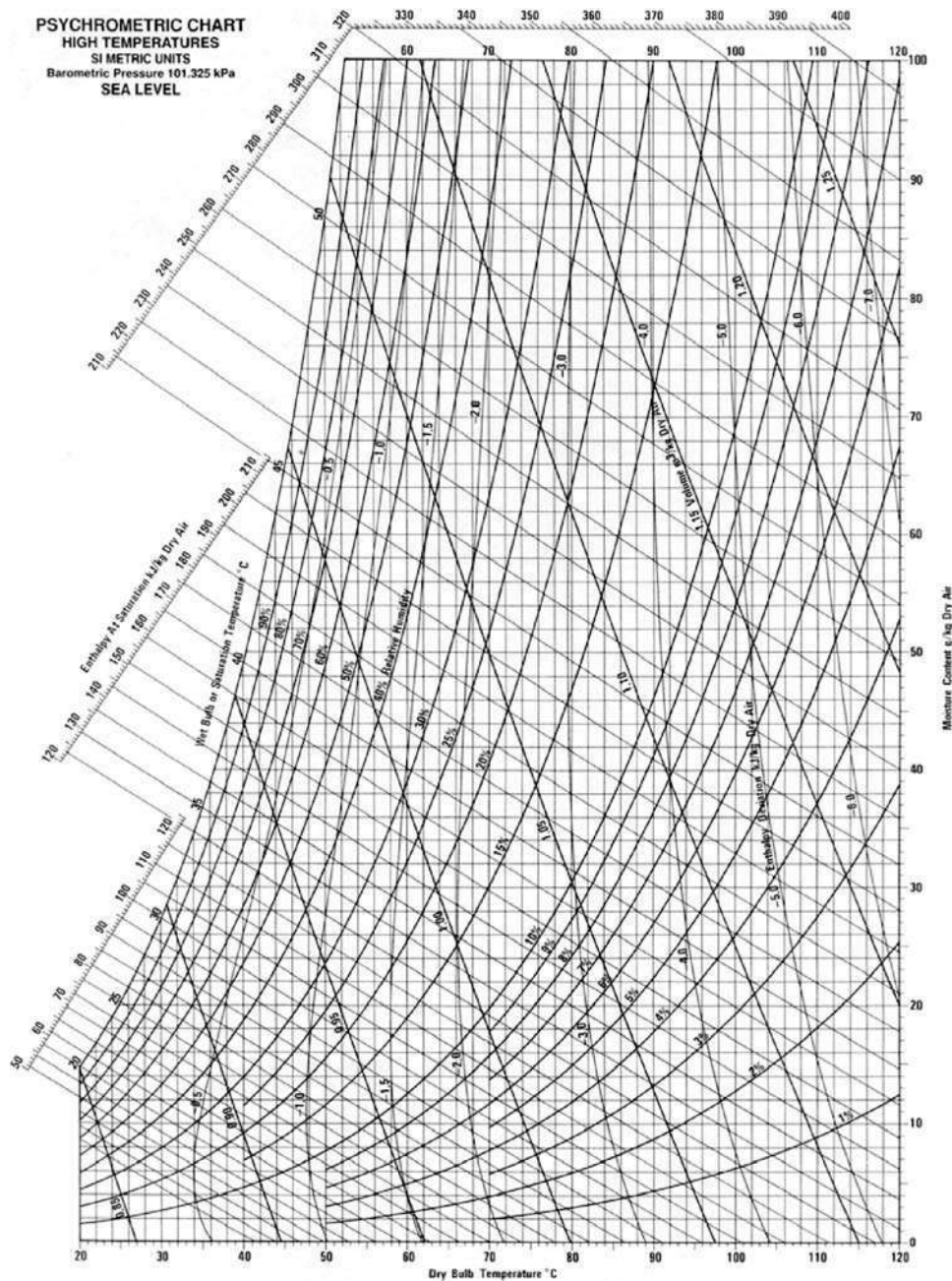


FIG. 12-1 Grosvenor psychrometric chart for the air-water system at standard atmospheric pressure, 101,325 Pa, SI units. (Courtesy Carrier Corporation.)

temperature range than Grosvenor charts, so are useful for dryer calculations. The slope of the enthalpy lines is normally  $-1/\lambda$ , where  $\lambda$  is the latent heat of evaporation. Adiabatic saturation lines are not quite parallel to constant-enthalpy lines and are slightly curved; the deviation increases as humidity increases. Figure 12-3 shows an example.

The **Salen-Soininen** perspective transformed **chart** is a triangular plot. It is tricky to plot and read, but covers a much wider range of humidity than do the other types of chart (up to 2 kg/kg) and is thus very effective for high-humidity mixtures and calculations near the

boiling point, e.g., in pulp and paper drying. See Soininen, *Drying Technol.* 4(2): 295-305 (1986).

Figure 12-4 shows a psychrometric chart for combustion products in air. The thermodynamic properties of moist air are given in Table 12-1. Figure 12-4 shows a number of useful additional relationships, e.g., specific volume and latent heat variation with temperature. Accurate figures should always be obtained from physical properties tables or by calculation using the formulas given earlier, and these charts should only be used as a quick check for verification.

## 12-8 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

In the past, psychrometric charts have been used to perform quite precise calculations. To do this, additive corrections are often required for enthalpy of added water or ice, and for variations in barometric pressure from the standard level (101,325 Pa, 14.696 lb/in<sup>2</sup>, 760 mmHg, 29.921 inHg). It is preferable to use formulas, which give an accurate figure at any set of conditions. Psychrometric charts and tables can be used as a rough cross-check that the result has been calculated correctly. Table 12-4 gives values of saturation humidity, specific volume, enthalpy, and entropy of saturated moist air at selected conditions. Below the freezing point, these become virtually identical to the values for dry air, as saturation humidity is very low. For pressure corrections, an altitude increase of approximately 900 ft gives a pressure decrease of 1 inHg (0.034 bar). For a recorded wet-bulb temperature of 50°F (10°C), this gives an increase in humidity of 1.9 gr/lb (0.00027 kg/kg) and the enthalpy increases by 0.29 Btu/lb (0.68 kJ/kg). This correction increases roughly proportionately for further changes in pressure, but climbs sharply as wet-bulb temperature is increased; when  $T_{wb}$  reaches 100°F (38°C),  $\Delta Y = 11.2$  gr/lb (0.0016 kg/kg) and  $\Delta H = 1.77$  Btu/lb (4.12 kJ/kg). Equivalent, more detailed tables in SI units can be found in the ASHRAE Handbook.

**Examples Illustrating Use of Psychrometric Charts** In these examples the following nomenclature is used:

- $t$  = dry-bulb temperatures, °F
- $t_w$  = wet-bulb temperature, °F
- $t_d$  = dewpoint temperature, °F
- $H$  = moisture content, lb water/lb dry air
- $\Delta H$  = moisture added to or rejected from the airstream, lb water/lb dry air
- $h'$  = enthalpy at saturation, Btu/lb dry air
- $D$  = enthalpy deviation, Btu/lb dry air
- $h = h' + D$  = true enthalpy, Btu/lb dry air
- $h_w$  = enthalpy of water added to or rejected from system, Btu/lb dry air

- $q_a$  = heat added to system, Btu/lb dry air
- $q_r$  = heat removed from system, Btu/lb dry air

Subscripts 1, 2, 3, etc., indicate entering and subsequent states.

**Example 1: Determination of Moist Air Properties** Find the properties of moist air when the dry-bulb temperature is 80°F and the wet-bulb temperature is 67°F.

**Solution:** Read directly from Fig. 12-2a (Fig. 12-6a shows the solution diagrammatically).

- Moisture content  $H = 78$  gr/lb dry air  
= 0.011 lb water/lb dry air
- Enthalpy at saturation  $h' = 31.6$  Btu/lb dry air
- Enthalpy deviation  $D = -0.1$  Btu/lb dry air
- True enthalpy  $h = 31.5$  Btu/lb dry air
- Specific volume  $v = 13.8$  ft<sup>3</sup>/lb dry air
- Relative humidity = 51 percent
- Dew point  $t_d = 60.3^\circ\text{F}$

**Example 2: Air Heating** Air is heated by a steam coil from 30°F dry-bulb temperature and 80 percent relative humidity to 75°F dry-bulb temperature. Find the relative humidity, wet-bulb temperature, and dew point of the heated air. Determine the quantity of heat added per pound of dry air.

**Solution:** Reading directly from the psychrometric chart (Fig. 12-2a),

- Relative humidity = 15 percent
- Wet-bulb temperature = 51.5°F
- Dew point = 25.2°F

The enthalpy of the inlet air is obtained from Fig. 12-2a as  $h_1 = h'_1 + D_1 = 10.1 + 0.06 = 10.16$  Btu/lb dry air; at the exit,  $h_2 = h'_2 + D_2 = 21.1 - 0.1 = 21$  Btu/lb dry air. The heat added equals the enthalpy difference, or

$$q_a = \Delta h = h_2 - h_1 = 21 - 10.16 = 10.84 \text{ Btu/lb dry air}$$

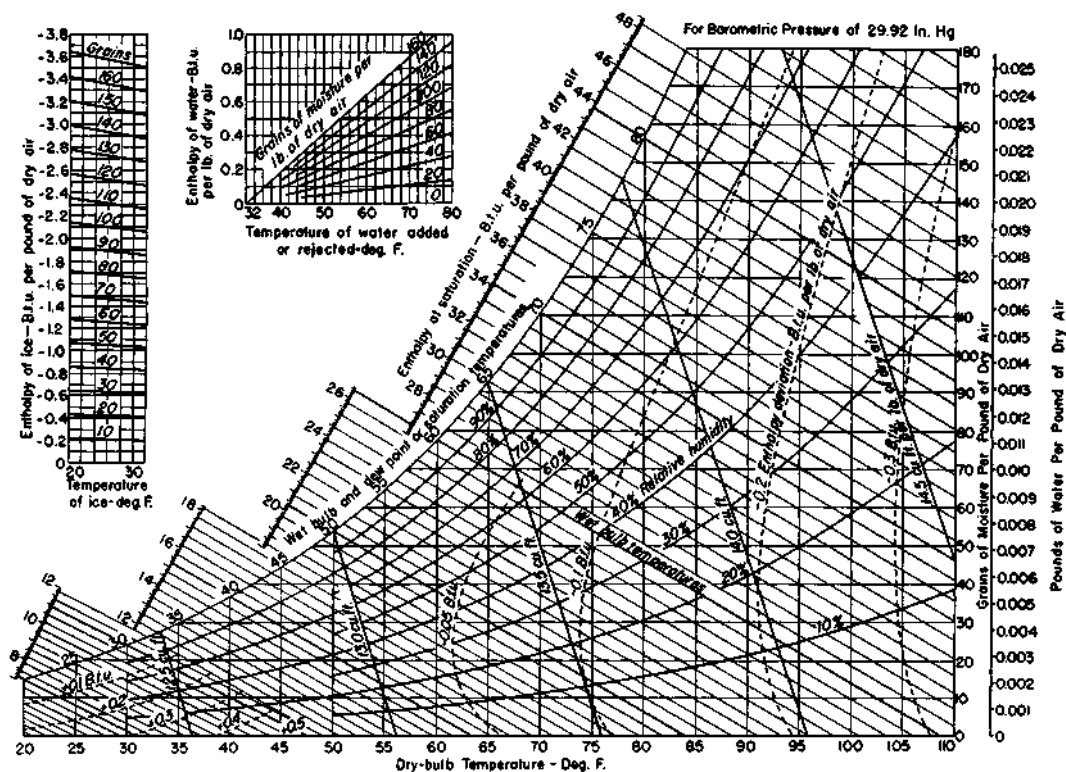


FIG. 12-2a Grosvenor psychrometric chart (medium temperature) for the air-water system at standard atmospheric pressure, 29.92 inHg, U.S. Customary units. (Courtesy Carrier Corporation.)





**FIG. 12-2b** Grosvenor psychrometric chart (high-temperature) for the air-water system at standard atmospheric pressure, 29.92 in. Hg, U.S. Customary units. (Source: Carrier Corporation.)

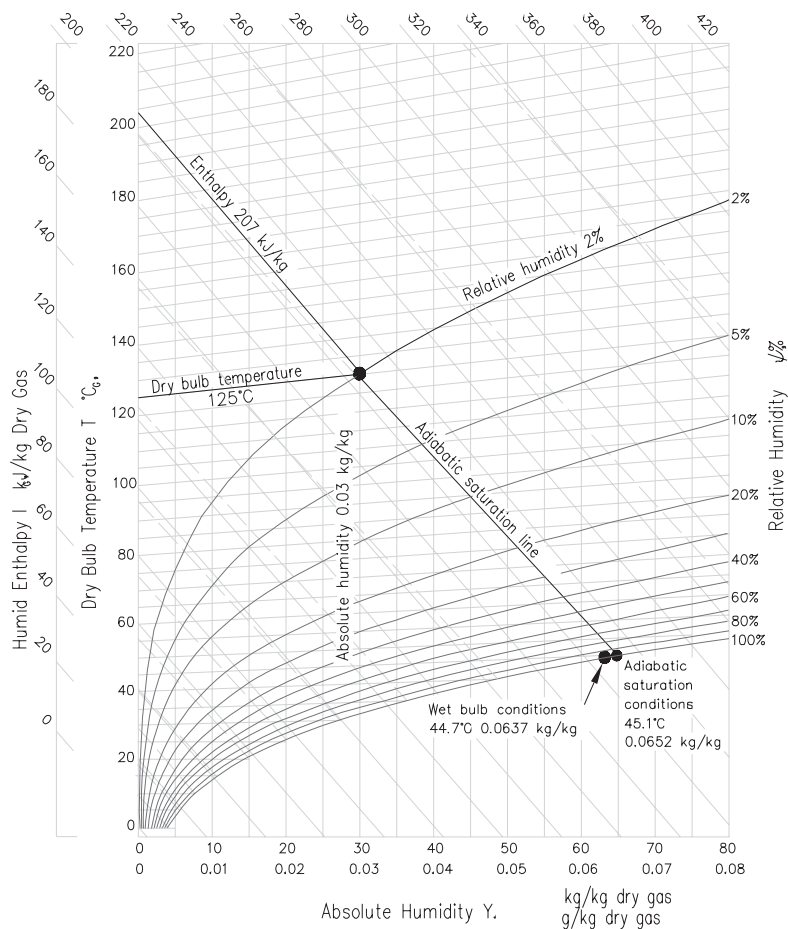
If the enthalpy deviation is ignored, the heat added  $q_a = \Delta h = 21.1 - 10.1 = 11$  Btu/lb dry air, or the result is 1.5 percent high. Figure 12-6b shows the heating path on the psychrometric chart.

**Example 3: Evaporative Cooling** Air at 95°F dry-bulb temperature and 70°F wet-bulb temperature contacts a water spray, where its relative humidity is increased to 90 percent. The spray water is recirculated; makeup water

enters at 70°F. Determine exit dry-bulb temperature, wet-bulb temperature, change in enthalpy of the air, and quantity of moisture added per pound of dry air.

**Solution:** Figure 12-6c shows the path on a psychrometric chart. The leaving dry-bulb temperature is obtained directly from Fig. 12-2a as 72.2°F. Since the spray water enters at the wet-bulb temperature of 70°F and there is no heat added to or removed from it, this is by definition an adiabatic process and there

## 12-10 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING



**FIG. 12-3** Mollier psychrometric chart for the air-water system at standard atmospheric pressure, 101,325 Pa SI units, plots humidity (abscissa) against enthalpy (lines sloping diagonally from top left to bottom right). (Source: Aspen Technology.)

will be no change in wet-bulb temperature. The only change in enthalpy is that from the heat content of the makeup water. This can be demonstrated as follows:

$$\begin{aligned}
 \text{Inlet moisture } H_1 &= 70 \text{ gr/lb dry air} \\
 \text{Exit moisture } H_2 &= 107 \text{ gr/lb dry air} \\
 \Delta H &= 37 \text{ gr/lb dry air} \\
 \text{Inlet enthalpy } h_1 &= h'_1 + D_1 = 34.1 - 0.22 \\
 &= 33.88 \text{ Btu/lb dry air} \\
 \text{Exit enthalpy } h_2 &= h'_2 + D_2 = 34.1 - 0.02 \\
 &= 34.08 \text{ Btu/lb dry air} \\
 \text{Enthalpy of added water } h_w &= 0.2 \text{ Btu/lb dry air (from small diagram,} \\
 &\quad 37 \text{ gr at } 70^\circ\text{F)} \\
 \text{Then} \quad q_a &= h_2 - h_1 + h_w \\
 &= 34.08 - 33.88 + 0.2 = 0
 \end{aligned}$$

**Example 4: Cooling and Dehumidification** Find the cooling load per pound of dry air resulting from infiltration of room air at 80°F dry-bulb temperature and 67°F wet-bulb temperature into a cooler maintained at 30°F dry-bulb and 28°F wet-bulb temperature, where moisture freezes on the coil, which is maintained at 20°F.

**Solution:** The path followed on a psychrometric chart is shown in Fig. 12-6d.

$$\begin{aligned}
 \text{Inlet enthalpy } h_1 &= h'_1 + D_1 = 31.62 - 0.1 \\
 &= 31.52 \text{ Btu/lb dry air}
 \end{aligned}$$

$$\begin{aligned}
 \text{Exit enthalpy } h_2 &= h'_2 + D_2 = 10.1 + 0.06 \\
 &= 10.16 \text{ Btu/lb dry air}
 \end{aligned}$$

$$\text{Inlet moisture } H_1 = 78 \text{ gr/lb dry air}$$

$$\text{Exit moisture } H_2 = 19 \text{ gr/lb dry air}$$

$$\text{Moisture rejected } \Delta H = 59 \text{ gr/lb dry air}$$

$$\text{Enthalpy of rejected moisture} = -1.26 \text{ Btu/lb dry air (from small diagram of Fig. 12-2a)}$$

$$\begin{aligned}
 \text{Cooling load } q_r &= 31.52 - 10.16 + 1.26 \\
 &= 22.62 \text{ Btu/lb dry air}
 \end{aligned}$$

Note that if the enthalpy deviations were ignored, the calculated cooling load would be about 5 percent low.

**Example 5: Cooling Tower** Determine water consumption and amount of heat dissipated per 1000 ft<sup>3</sup>/min of entering air at 90°F dry-bulb temperature and 70°F wet-bulb temperature when the air leaves saturated at 110°F and the makeup water is at 75°F.

**Solution:** The path followed is shown in Fig. 12-6e.

$$\text{Exit moisture } H_2 = 416 \text{ gr/lb dry air}$$

$$\text{Inlet moisture } H_1 = 78 \text{ gr/lb dry air}$$

$$\text{Moisture added } \Delta H = 338 \text{ gr/lb dry air}$$

$$\text{Enthalpy of added moisture } h_w = 2.1 \text{ Btu/lb dry air (from small diagram of Fig. 12-2b)}$$

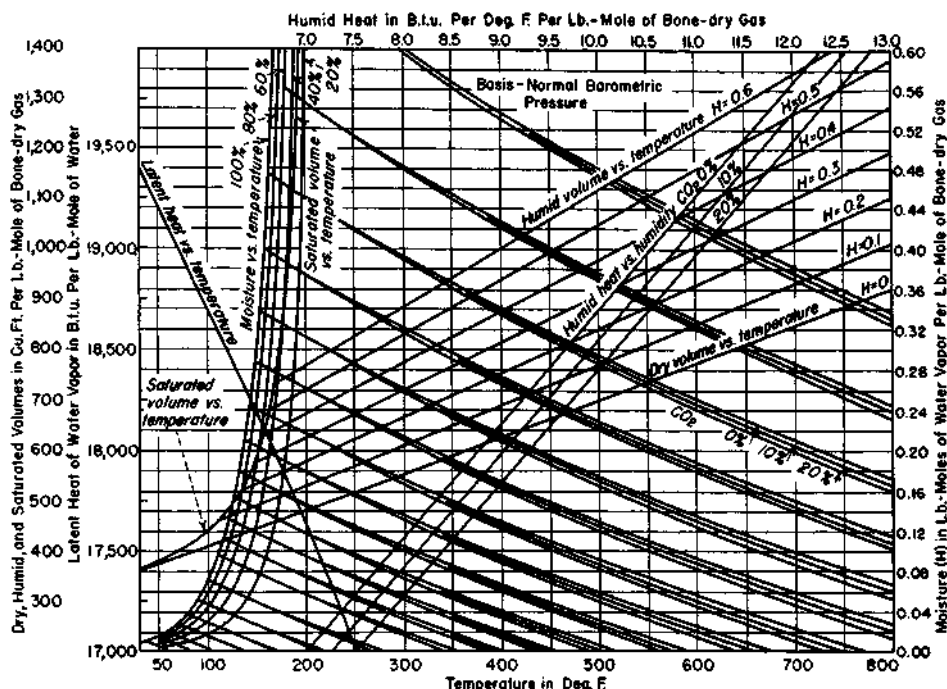


FIG. 12-4 Grosvenor psychrometric chart for air and flue gases at high temperatures, molar units [Hatta, Chem. Metall. Eng., 37:64 (1930)].

TABLE 12-4 Thermodynamic Properties of Saturated Air (U.S. Customary Units, at Standard Atmospheric Pressure, 29.921 inHg)

| Temp.<br>$T$ , °F | Saturation<br>humidity<br>$H_s$ | Volume,<br>ft <sup>3</sup> /lb dry air |          |        | Enthalpy,<br>Btu/lb dry air |          |        | Entropy,<br>Btu/(°F·lb dry air) |          |         | Condensed water              |                                      |                                  | Temp.<br>$T$ , °F |
|-------------------|---------------------------------|----------------------------------------|----------|--------|-----------------------------|----------|--------|---------------------------------|----------|---------|------------------------------|--------------------------------------|----------------------------------|-------------------|
|                   |                                 | $v_a$                                  | $v_{as}$ | $v_s$  | $h_a$                       | $h_{as}$ | $h_s$  | $s_a$                           | $s_{as}$ | $s_s$   | Enthalpy,<br>Btu/lb<br>$h_w$ | Entropy,<br>Btu/(<br>lb·°F)<br>$s_w$ | Vapor<br>pressure, inHg<br>$p_s$ |                   |
| -150              | $6.932 \times 10^{-9}$          | 7.775                                  | .000     | 7.775  | 36.088                      | .000     | 36.088 | 0.09508                         | .00000   | 0.09508 | 218.77                       | 0.4800                               | $3.301 \times 10^{-6}$           | -150              |
| -100              | $9.772 \times 10^{-7}$          | 9.046                                  | .000     | 9.046  | 24.037                      | .001     | 24.036 | 0.05897                         | .00000   | 0.05897 | 201.23                       | 0.4277                               | $4.666 \times 10^{-5}$           | -100              |
| -50               | $4.163 \times 10^{-5}$          | 10.313                                 | .001     | 10.314 | 12.012                      | .043     | 11.969 | 0.02766                         | .00012   | 0.02754 | 181.29                       | 0.3758                               | $1.991 \times 10^{-3}$           | -50               |
| 0                 | $7.872 \times 10^{-4}$          | 11.578                                 | .015     | 11.593 | 0.000                       | .835     | 0.835  | 0.00000                         | .00192   | 0.00192 | 158.93                       | 0.3244                               | $0.037645 \times 10^{-2}$        | 0                 |
| 10                | $1.315 \times 10^{-3}$          | 11.831                                 | .025     | 11.856 | 2.402                       | 1.401    | 3.803  | .00518                          | .00314   | .00832  | 154.17                       | 0.3141                               | 0.062858                         | 10                |
| 20                | $2.152 \times 10^{-3}$          | 12.084                                 | .042     | 12.126 | 4.804                       | 2.302    | 7.106  | .01023                          | .00504   | .01527  | 149.31                       | 0.3039                               | 0.10272                          | 20                |
| 30                | $3.454 \times 10^{-3}$          | 12.338                                 | .068     | 12.406 | 7.206                       | 3.709    | 10.915 | .01519                          | .00796   | .02315  | 144.36                       | 0.2936                               | 0.16452                          | 30                |
| 32                | $3.788 \times 10^{-3}$          | 12.388                                 | .075     | 12.463 | 7.686                       | 4.072    | 11.758 | .01617                          | .00870   | .02487  | 143.36                       | 0.2916                               | 0.18035                          | 32                |
| 32*               | $3.788 \times 10^{-3}$          | 12.388                                 | .075     | 12.463 | 7.686                       | 4.072    | 11.758 | .01617                          | .00870   | .02487  | 0.04                         | 0.0000                               | 0.18037                          | 32°               |
| 40                | $5.213 \times 10^{-3}$          | 12.590                                 | .105     | 12.695 | 9.608                       | 5.622    | 15.230 | .02005                          | .01183   | .03188  | 8.09                         | .0162                                | .24767                           | 40                |
| 50                | $7.658 \times 10^{-3}$          | 12.843                                 | .158     | 13.001 | 12.010                      | 8.291    | 20.301 | .02481                          | .01711   | .04192  | 18.11                        | .0361                                | .36240                           | 50                |
| 60                | $1.108 \times 10^{-2}$          | 13.096                                 | .233     | 13.329 | 14.413                      | 12.05    | 26.46  | .02948                          | .02441   | .05389  | 28.12                        | .0555                                | .52159                           | 60                |
| 70                | $1.582 \times 10^{-2}$          | 13.348                                 | .339     | 13.687 | 16.816                      | 17.27    | 34.09  | .03405                          | .03437   | .06842  | 38.11                        | .0746                                | .73915                           | 70                |
| 80                | $2.233 \times 10^{-2}$          | 13.601                                 | 0.486    | 14.087 | 19.221                      | 24.47    | 43.69  | 0.03854                         | 0.04784  | 0.08638 | 48.10                        | 0.0933                               | 1.0323                           | 80                |
| 90                | $3.118 \times 10^{-2}$          | 13.853                                 | .692     | 14.545 | 21.625                      | 34.31    | 55.93  | .04295                          | .06596   | .10890  | 58.08                        | .1116                                | 1.4219                           | 90                |
| 100               | $4.319 \times 10^{-2}$          | 14.106                                 | .975     | 15.081 | 24.029                      | 47.70    | 71.73  | .04729                          | .09016   | .13745  | 68.06                        | .1296                                | 1.9333                           | 100               |
| 110               | $5.944 \times 10^{-2}$          | 14.359                                 | 1.365    | 15.724 | 26.434                      | 65.91    | 92.34  | .05155                          | .1226    | .1742   | 78.03                        | .1472                                | 2.5966                           | 110               |
| 120               | $8.149 \times 10^{-2}$          | 14.611                                 | 1.905    | 16.516 | 28.841                      | 90.70    | 119.54 | .05573                          | .1659    | .2216   | 88.01                        | .1646                                | 3.4474                           | 120               |
| 130               | 0.1116                          | 14.864                                 | 2.652    | 17.516 | 31.248                      | 124.7    | 155.9  | .05985                          | .2245    | .2844   | 98.00                        | .1817                                | 4.5272                           | 130               |
| 140               | 0.1534                          | 15.117                                 | 3.702    | 18.819 | 33.655                      | 172.0    | 205.7  | .06390                          | .3047    | .3686   | 107.99                       | .1985                                | 5.8838                           | 140               |
| 150               | 0.2125                          | 15.369                                 | 5.211    | 20.580 | 36.063                      | 239.2    | 275.3  | .06787                          | .4169    | .4848   | 117.99                       | .2150                                | 7.5722                           | 150               |
| 160               | 0.2990                          | 15.622                                 | 7.446    | 23.068 | 38.472                      | 337.8    | 376.3  | .07179                          | .5793    | .6511   | 128.00                       | .2313                                | 9.6556                           | 160               |
| 170               | 0.4327                          | 15.874                                 | 10.938   | 26.812 | 40.882                      | 490.6    | 531.5  | .07565                          | .8273    | .9030   | 138.01                       | .2473                                | 12.203                           | 170               |
| 180               | 0.6578                          | 16.127                                 | 16.870   | 32.997 | 43.292                      | 748.5    | 791.8  | .07946                          | 1.240    | 1.319   | 148.03                       | .2631                                | 15.294                           | 180               |
| 190               | 1.099                           | 16.379                                 | 28.580   | 44.959 | 45.704                      | 1255     | 1301   | .08320                          | 2.039    | 2.122   | 158.07                       | .2786                                | 19.017                           | 190               |
| 200               | 2.295                           | 16.632                                 | 60.510   | 77.142 | 48.119                      | 2629     | 2677   | .08689                          | 4.179    | 4.266   | 168.11                       | .2940                                | 23.468                           | 200               |

NOTE: Compiled by John A. Goff and S. Gratch. See also Keenan and Kaye. *Thermodynamic Properties of Air*, Wiley, New York, 1945. Enthalpy of dry air taken as zero at 0°F. Enthalpy of liquid water taken as zero at 32°F.

To convert British thermal units per pound to joules per kilogram, multiply by 2326; to convert British thermal units per pound dry air-degree Fahrenheit to joules per kilogram-kelvin, multiply by 4186.8; and to convert cubic feet per pound to cubic meters per kilogram, multiply by 0.0624.

\* Extrapolated to represent metastable equilibrium with undercooled liquid.



## 12-12 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

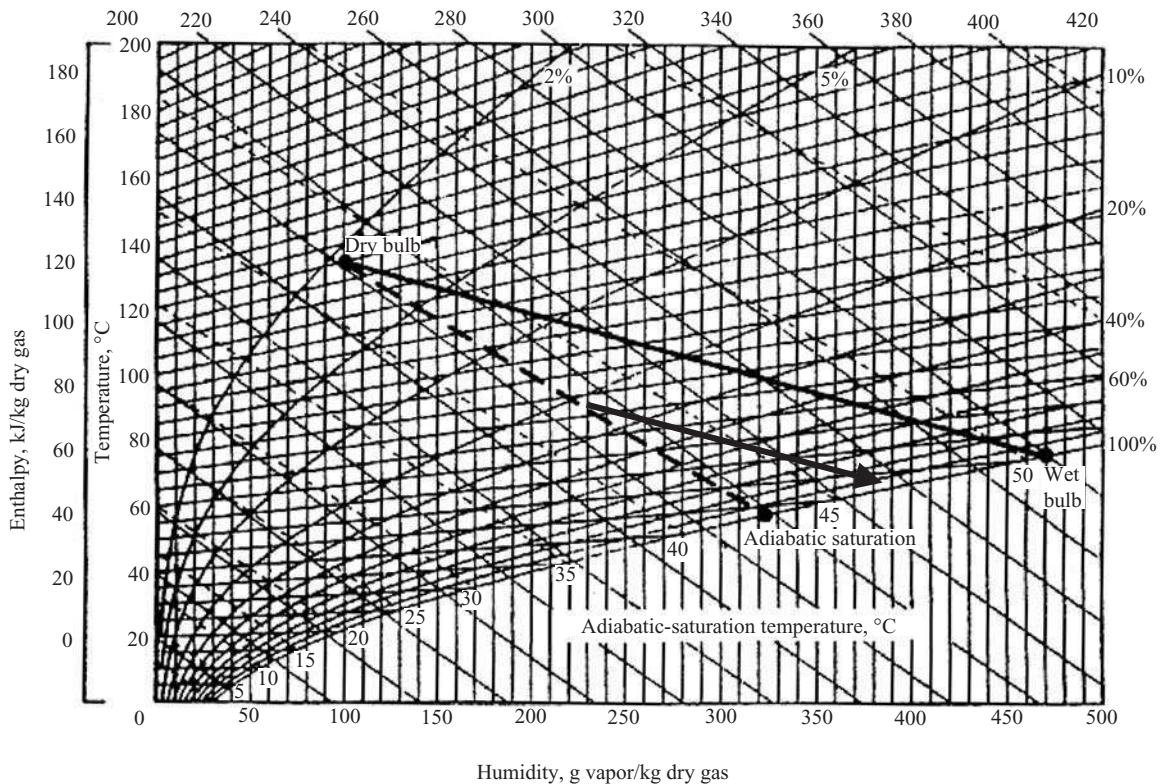


FIG. 12-5 Mollier chart showing changes in  $T_{wb}$  during an adiabatic saturation process for an organic system (nitrogen-toluene).

If greater precision is desired,  $h_w$  can be calculated as

$$h_w = (338/7000)(1)(75 - 32) = 2.08 \text{ Btu/lb dry air}$$

$$\text{Enthalpy of inlet air } h_1 = h'_1 + D_1 = 34.1 - 0.18 = 33.92 \text{ Btu/lb dry air}$$

$$\text{Enthalpy of exit air } h_2 = h'_2 + D_2 = 92.34 + 0 = 92.34 \text{ Btu/lb dry air}$$

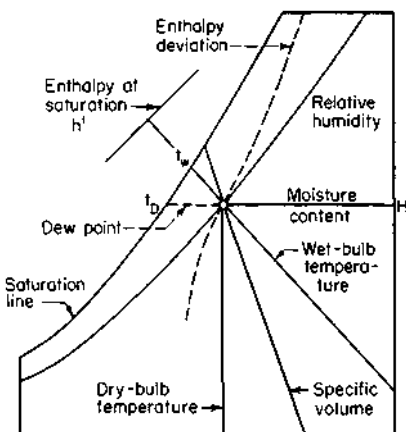


FIG. 12-6a Diagram of psychrometric chart showing the properties of moist air.

$$\begin{aligned} \text{Heat dissipated} &= h_2 - h_1 - h_w \\ &= 92.34 - 33.92 - 2.08 \\ &= 56.34 \text{ Btu/lb dry air} \end{aligned}$$

$$\text{Specific volume of inlet air} = 14.1 \text{ ft}^3/\text{lb dry air}$$

$$\text{Total heat dissipated} = \frac{(1000)(56.34)}{14.1} = 3990 \text{ Btu/min}$$

**Example 6: Recirculating Dryer** A dryer is removing 100 lb water/h from the material being dried. The air entering the dryer has a dry-bulb temperature of 180°F and a wet-bulb temperature of 110°F. The air leaves the dryer at 140°F. A portion of the air is recirculated after mixing with room air having a dry-bulb temperature of 75°F and a relative humidity of 60 percent. Determine the quantity of air required, recirculation rate, and load on the preheater if it is assumed that the system is adiabatic. Neglect heatup of the feed and of the conveying equipment.

**Solution:** The path followed is shown in Fig. 12-6f.

$$\text{Humidity of room air } H_1 = 0.0113 \text{ lb/lb dry air}$$

$$\text{Humidity of air entering dryer } H_3 = 0.0418 \text{ lb/lb dry air}$$

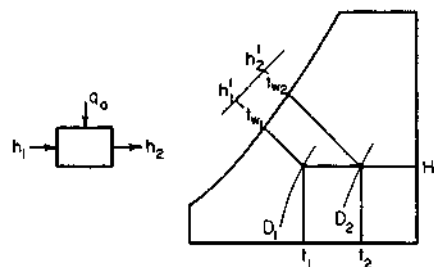


FIG. 12-6b Heating process

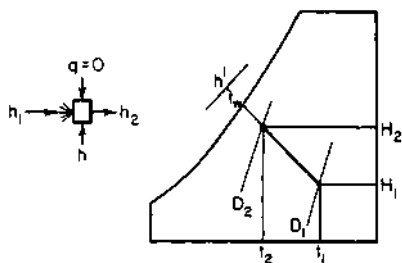


FIG. 12-6c Spray or evaporative cooling.

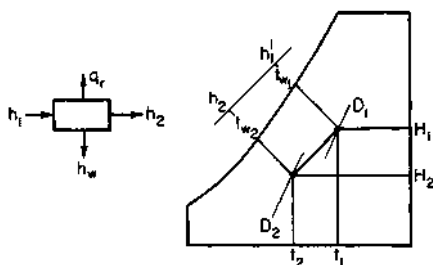


FIG. 12-6d Cooling and dehumidifying process.

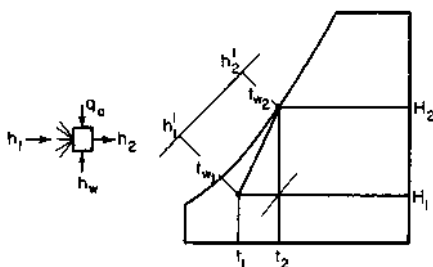


FIG. 12-6e Cooling tower.

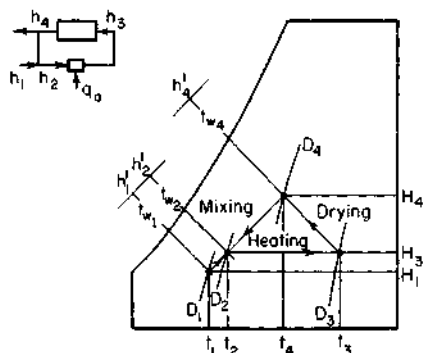


FIG. 12-6f Drying process with recirculation.

Humidity of air leaving dryer  $H_4 = 0.0518$  lb/lb dry air

Enthalpy of room air  $h_1 = 30.2 - 0.3$

$= 29.9$  Btu/lb dry air

Enthalpy of entering air  $h_3 = 92.5 - 1.3$

$= 91.2$  Btu/lb dry air

Enthalpy of leaving air  $h_4 = 92.5 - 0.55$

$= 91.95$  Btu/lb dry air

Quantity of air required is  $100/(0.0518 - 0.0418) = 10,000$  lb dry air/h. At the dryer inlet the specific volume is  $17.1$  ft<sup>3</sup>/lb dry air. Air volume is  $(10,000)(17.1)/60 = 2850$  ft<sup>3</sup>/min. Fraction exhausted is

$$\frac{X}{W_a} = \frac{0.0518 - 0.0418}{0.0518 - 0.0113} = 0.247$$

where  $X$  = quantity of fresh air and  $W_a$  = total airflow. Thus 75.3 percent of the air is recirculated. Load on the preheater is obtained from an enthalpy balance

$$q_a = 10,000(91.2) - 2470(29.9) - 7530(91.95) = 146,000 \text{ Btu/h}$$

## PSYCHROMETRIC CALCULATIONS

Table 12-5 gives the steps required to perform the most common humidity calculations, using the formulas given earlier.

Methods (i) to (iii) are used to find the humidity and dew point from temperature readings in a wet- and dry-bulb psychrometer. Method (iv) is used to find the humidity and dew point from a relative humidity measurement at a given temperature.

Methods (v) and (vi) give the adiabatic saturation and wet-bulb temperatures from absolute humidity (or relative humidity) at a given temperature.

Method (vii) gives the absolute and relative humidity from a dew point measurement.

Method (viii) allows the calculation of all the main parameters if the absolute humidity is known, e.g., from a mass balance on a process plant.

Method (ix) converts the volumetric form of absolute humidity to the mass form (mixing ratio).

Method (x) allows the dew point to be corrected for pressure. The basis is that the mole fraction  $y = p/P$  is the same for a given mixture composition at all values of total pressure  $P$ . In particular, the dew point measured in a compressed air duct can be converted to the dew point at atmospheric pressure, from which the humidity can be calculated. It is necessary to check that the temperature change associated with compression or expansion does not bring the dry-bulb temperature to a point where condensation can occur. Also, at these elevated pressures, it is strongly advisable to apply the enhancement factor (see BS 1339).

**Psychrometric Software and Tables** As an alternative to using charts or individual calculations, lookup tables have been published for many years for common psychrometric conversions, e.g., to find relative humidity given the dry-bulb and wet-bulb temperatures. These were often very extensive. To give precise coverage of  $T_{wb}$  in 1°C or 0.1°C steps, a complete table would be needed for each individual dry-bulb temperature.

Software is available that will perform calculations of humidity parameters for any point value, and for plotting psychrometric charts. Moreover, British Standard BS 1339 Part 2 (2006) provides functions as macros which can be embedded into any Excel-compatible spreadsheet. Users can therefore generate their own tables for any desired combination of parameters as well as perform point calculations. Hence, the need for published lookup tables has been eliminated. However, this software, like the previous lookup tables, is only valid for the air-water system. For other vapor-gas systems, the equations given in previous sections must be used.

Software may be effectively used to draw psychrometric charts or perform calculations. A wide variety of other psychrometric software may be found on the Internet, but quality varies considerably; the

## 12-14 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

**TABLE 12-5 Calculation Methods for Various Humidity Parameters**

|       | Known                        | Required                    | Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i.    | $T, T_{wb}$                  | $Y$                         | Find saturation vapor pressure $p_{ws}$ at wet-bulb temperature $T_{wb}$ from Eq. (12-4). Find actual vapor pressure $p$ at dry-bulb temperature $T$ from psychrometer equation (12-11). Find mixing ratio $Y$ by conversion from $p$ (Table 12-1).                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ii.   | $T, T_{wb}$                  | $T_{dp}, d_e$               | Find $p$ if necessary by method (i) above. Find dew point $T_{dp}$ from Eq. (12-4) by calculating the $T$ corresponding to $p$ [iteration required; Antoine equation (12-5) gives a first estimate]. Calculate volumetric humidity $Y_v$ using Eq. (12-1).                                                                                                                                                                                                                                                                                                                                                                                                                         |
| iii.  | $T, T_{wb}$                  | %RH ( $\psi$ )              | Use method (i) to find $p$ . Find saturation vapor pressure $p_s$ at $T$ from Eq. (12-4). Now relative humidity %RH = $100p/p_s$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| iv.   | $T, \%RH$                    | $Y, d_e$                    | Find saturation vapor pressure $p_s$ at $T$ from Eq. (12-4). Actual vapor pressure $p = p_s(\%RH/100)$ . Convert to $Y$ (Table 12-1). Find $Y_e$ from Eq. (12-1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v.    | $T, \%RH$ (or $T, Y$ )       | $T_{as}$                    | Use method (iv) to find $p$ and $Y$ . Make an initial estimate of $T_{as}$ , say, using a psychrometric chart. Calculate $Y_{as}$ from Eq. (12-6). Find $p$ from Table 12-1 and $T_{as}$ from Antoine equation (12-5). Repeat until iteration converges (e.g., using spreadsheet).<br>Alternative method: Evaluate enthalpy $H_{est}$ at these conditions and $H$ at initial conditions. Find $H_{as}$ from Eq. (12-7) and compare with $H_{est}$ . Make new estimate of $Y_{as}$ which would give $H_{est}$ equal to $H_{as}$ . Find $p$ from Table 12-1 and $T_{as}$ from Antoine equation (12-5). Reevaluate $H_{as}$ from Eq. (12-7) and iterate to refine value of $Y_{as}$ . |
| vi.   | $T, \%RH$ (or $T, Y$ )       | $T_{wb}$                    | Use method (iv) to find $p$ and $Y$ . Make an initial estimate of $T_{wb}$ , e.g., using a psychrometric chart, or (for air-water system) by estimating adiabatic saturation temperature $T_{as}$ . Find $p_{wb}$ from psychrometer equation (12-11). Calculate new value of $T_{wb}$ corresponding to $p_{wb}$ by reversing Eq. (12-4) or using the Antoine equation (12-5). Repeat last two steps to solve iteratively for $T_{wb}$ (computer program is preferable method).                                                                                                                                                                                                     |
| vii.  | $T, T_{dp}$                  | $Y, \%RH$                   | Find saturation vapor pressure at dew point $T_{dp}$ from Eq. (12-4); this is the actual vapor pressure $p$ . Find $Y$ from Table 12-1. Find saturation vapor pressure $p_s$ at dry-bulb temperature $T$ from Eq. (12-4). Now %RH = $100p/p_s$ .                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| viii. | $T, Y$                       | $T_{dp}, d_e, \%RH, T_{wb}$ | Find $p$ by conversion from $Y$ (Table 12-1). Then use method (ii), (iii), or (v) as appropriate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| ix.   | $T, Y_e$                     | $Y$                         | Find specific humidity $Y_w$ from Eqs. (12-2) and (12-1). Convert to absolute humidity $Y$ using $Y = Y_w/(1 - Y_w)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| x.    | $T_{dp}$ at $P_1$ (elevated) | $T_{dp}$ at $P_2$ (ambient) | Find vapor pressure $p_1$ at $T_{dp}$ and $P_1$ from Eq. (12-4). Convert to vapor pressure $p_2$ at new pressure $P_2$ by the formula $p_2 = p_1 P_2 / P_1$ . Find new dew point $T_{dp}$ from Eq. (12-4) by calculating the $T$ corresponding to $p_2$ [iteration required as in (ii)].                                                                                                                                                                                                                                                                                                                                                                                           |

source and basis of the calculation methods should be carefully checked before using the results. In particular, most methods only apply for the air-water system at moderate temperatures (below 100°C). For high-temperature dryer calculations, only software stated as suitable for this range should be used.

Reliable sources include the following:

1. The American Society of Agricultural Engineers (ASAE): <http://www.asae.org>. Psychrometric data in chart and equation form in both SI and English units. Charts for temperature ranges of -35 to 600°F in USCS units and -10 to 120°C in SI units. Equations and calculation procedures. Air-water system and Grosvenor (temperature-humidity) charts only.

2. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE): <http://www.ashrae.org>. Psychrometric Analysis CD with energy calculations and creation of custom charts at virtually any altitude or pressure. Detailed scientific basis given in ASHRAE Handbook. Air-water system and Grosvenor charts only.

3. Carrier Corporation, a United Technologies Company: <http://www.training.carrier.com>. PSYCH+, computerized psychrometric chart and instructional guide, including design of air conditioning processes and/or cycles. Printed psychrometric charts also supplied. Air-water system and Grosvenor charts only.

4. Linric Company: <http://www.linric.com>. PsycPro generates custom psychrometric charts in English (USCS) or metric (SI) units, based on ASHRAE formulas. Air-water system and Grosvenor charts only.

5. Aspen Technology: <http://www.aspentech.com>. PSYCHIC, one of the Process Tools, generates customized psychrometric charts. Mollier and Bowen enthalpy-humidity charts are produced in addition to Grosvenor. Any gas-vapor system can be handled as well as air-water; data supplied for common organic solvents. Can draw operating lines and spot points, as shown in Fig. 12-7.

6. British Standards Institution: <http://www.bsonline.bsi-global.com>. British Standard BS 1339 Part 2 is a spreadsheet-based software program providing functions based on latest internationally agreed

upon standards. It calculates all key psychrometric parameters and can produce a wide range of psychrometric tables. Users can embed the functions in their own spreadsheets to do psychrometric calculations. Air-water system only (although BS 1339 Part 1 text gives full calculation methods for other gas-vapor systems). SI (metric) units. It does not plot psychrometric charts.

7. Akton Associates provides digital versions of psychrometry charts.

### Psychrometric Calculations—Worked Examples

**Example 7: Determination of Moist Air Properties** An air-water mixture is found from the heat and mass balance to be at 60°C (333 K) and 0.025 kg/kg (25 g/kg) absolute humidity. Calculate the other main parameters for the mixture. Take atmospheric pressure as 101,325 Pa.

**Method:** Consult item (vi) in Table 12-5 for the calculation methodology.

From the initial terminology section, specific humidity  $Y_w = 0.02439$  kg/kg, mole ratio  $z = 0.0402$  kmol/kmol, mole fraction  $y = 0.03864$  kmol/kmol.

From Table 12-1, vapor pressure  $p = 3915$  Pa (0.03915 bar) and volumetric humidity  $Y_v = 0.02547$  kg/m<sup>3</sup>. Dew point is given by the temperature corresponding to  $p$  at saturation. From the reversed Antoine equation (12-5),  $T_{dp} = 3830/(23.19 - \ln 3915) + 44.83 = 301.58$  K = 28.43°C.

Relative humidity is the ratio of actual vapor pressure to saturation vapor pressure at dry-bulb temperature. From the Antoine equation (12-5),  $p_s = \exp [23.19 - 3830/(333.15 - 44.83)] = 20,053$  Pa (new coefficients), or  $p_s = \exp [23.1963 - 3816.44/(333.15 - 46.13)] = 19,921$  Pa (old coefficients).

From Sonntag equation (12-4),  $p_s = 19,948$  Pa; difference from Antoine is less than 0.5 percent. Relative humidity =  $100 \times 3915/19,948 = 19.6$  percent. From a psychrometric chart, e.g., Fig. 12-1, a humidity of 0.025 kg/kg at  $T = 60^\circ\text{C}$  lies very close to the adiabatic saturation line for 35°C. Hence a good first estimate for  $T_{as}$  and  $T_{wb}$  will be 35°C. Refining the estimate of  $T_{wb}$  by using the psychrometer equation and iterating gives

$$p_{wb} = 3915 + 6.46 \times 10^{-4} (1.033)(101,325)(60 - 35) = 5605$$

From the Antoine equation,

$$T_{wb} = 3830/(23.19 - \ln 5605) + 44.83 = 307.9$$
 K = 34.75°C

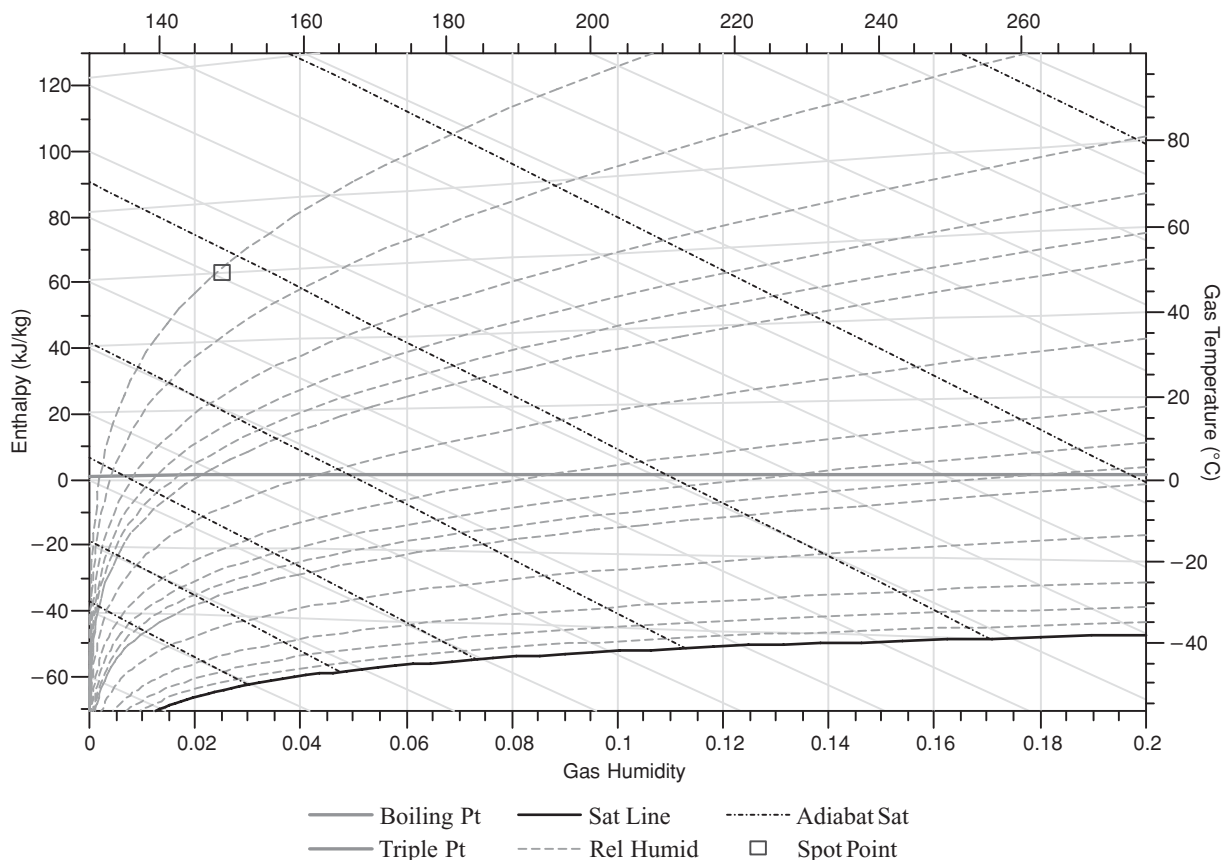
Second iteration:

$$p_{wb} = 3915 + 6.46 \times 10^{-4} (1.033)(101,325)(60 - 34.75) = 5622$$

$$T_{wb} = 307.96$$
 K = 34.81°C.

To a sensible level of precision,  $T_{wb} = 34.8^\circ\text{C}$ .

## Mollier Chart for Nitrogen/Acetone at 10 kPa



**FIG. 12-7** Mollier psychrometric chart (from PSYCHIC software program) showing determination of adiabatic saturation temperature plots humidity (abscissa) against enthalpy (lines sloping diagonally from top left to bottom right). (Courtesy AspenTech.)

From Table 12-1  $Y_{wb} = 5622 \times 0.622 / (101,325 - 5622) = 0.0365(4)$  kg/kg.

Enthalpy of original hot air is approximately given by  $H = (C_{p_g} + C_{p_v} Y)(T - T_0) + \lambda_0 Y = (1 + 1.9 \times 0.025) \times 60 + 2501 \times 0.025 = 62.85 + 62.5 = 125.35$  kJ/kg. A more accurate calculation can be obtained from steam tables;  $C_{p_g} = 1.005$  kJ/(kg·K) over this range,  $H_g$  at  $60^\circ\text{C} = 2608.8$  kJ/kg,  $H = 60.3 + 65.22 = 125.52$  kJ/kg.

Calculation (v), method 1: if  $T_{as} = 34.8$ , from Eq. (12-6), with  $C_s = 1 + 1.9 \times 0.025 = 1.048$  kJ/(kg·K),  $\lambda_{as} = 2419$  kJ/kg (steam tables),  $Y_{as} = 0.025 + 1.048/2419 (60 - 34.8) = 0.0359(2)$  kg/kg. From Table 12-1,  $p = 5530$  Pa. From the Antoine equation (12-5),  $T_{as} = 3830 / (23.19 - \ln 5530) + 44.83 = 307.65$  K =  $34.52^\circ\text{C}$ . Repeat until iteration converges (e.g., using spreadsheet). Final value  $T_{as} = 34.57^\circ\text{C}$ ,  $Y_{as} = 0.0360$  kg/kg.

Enthalpy check: From Eq. (12-7),  $H_{as} - H = 4.1868 \times 34.57 \times (0.036 - 0.025) = 1.59$  kJ/kg. So  $H_{as} = 127.11$  kJ/kg. Compare  $H_{as}$  calculated from enthalpies;  $H_g$  at  $34.57^\circ\text{C} = 2564$  kJ/kg,  $H_{est} = 34.90 + 92.29 = 127.19$  kJ/kg. The iteration has converged successfully.

Note that  $T_{as}$  is  $0.2^\circ\text{C}$  lower than  $T_{wb}$  and  $Y_{as}$  is  $0.0005$  kg/kg lower than  $Y_{wb}$ , both negligible differences.

**Example 8: Calculation of Humidity and Wet-Bulb Condition** A dryer exhaust which can be taken as an air-water mixture at  $70^\circ\text{C}$  (343.15 K) is measured to have a relative humidity of 25 percent. Calculate the humidity parameters and wet-bulb conditions for the mixture. Pressure is 1 bar (100,000 Pa).

**Method:** Consult item (v) in Table 12-5 for the calculation methodology.

From the Antoine equation (12-5), using standard coefficients (which give a better fit in this temperature range),  $p_s = \exp[23.1963 - 3816.44 / (343.15 - 46.13)] = 31,170$  Pa. Actual vapor pressure  $p = 25$  percent of 31,170 = 7792 Pa (0.078 bar).

From Table 12-1, absolute humidity  $Y = 0.05256$  kg/kg and volumetric humidity  $Y_v = 0.0492$  kg/m<sup>3</sup>. From the terminology section, mole fraction  $y = 0.0779$  kmol/kmol, mole ratio  $z = 0.0845$  kmol/kmol, specific humidity  $Y_w = 0.04994$  kg/kg.

Dew point  $T_{dp} = 3816.44 / (23.1963 - \ln 7792) + 46.13 = 314.22$  K =  $41.07^\circ\text{C}$ .

From the psychrometric chart, a humidity of  $0.0526$  kg/kg at  $T = 70^\circ\text{C}$  falls just below the adiabatic saturation line for  $45^\circ\text{C}$ . Estimate  $T_{as}$  and  $T_{wb}$  as  $45^\circ\text{C}$ . Refining the estimate of  $T_{wb}$  by using the psychrometer equation and iterating gives

$$p_{wb} = 7792 + 6.46 \times 10^{-4} (1.0425)(10^5)(70 - 45) = 9476$$

From the Antoine equation,

$$T_{wb} = 3816.44 / (23.1963 - \ln 9476) + 46.13 = 317.96$$
 K =  $44.81^\circ\text{C}$

Second iteration (taking  $T_{wb} = 44.8$ ):

$$p_{wb} = 9489 \quad T_{wb} = 317.99$$
 K =  $44.84^\circ\text{C}$

The iteration has converged.



**Example 9: Calculation of Psychrometric Properties of Acetone/Nitrogen Mixture** A mixture of nitrogen  $N_2$  and acetone  $CH_3COCH_3$  is found from the heat and mass balance to be at  $60^\circ\text{C}$  (333 K) and  $0.025\text{ kg/kg}$  (25 g/kg) absolute humidity (same conditions as in Example 7). Calculate the other main parameters for the mixture. The system is under vacuum at 100 mbar (0.1 bar, 10,000 Pa).

Additional data for acetone and nitrogen are obtained from *The Properties of Gases and Liquids* (Prausnitz et al.). Molecular weight (molal mass)  $M_g$  for nitrogen = 28.01 kg/kmol;  $M_g$  for acetone = 58.08 kg/kmol. Antoine coefficients for acetone are 16.6513, 2940.46, and 35.93, with  $p_s$  in **mmHg** and  $T$  in **K**. Specific heat capacity of nitrogen is approximately 1.014 kJ/(kg·K). Latent heat of acetone is 501.1 kJ/kg at the boiling point. The psychrometric ratio for the nitrogen-acetone system is not given, but the diffusion coefficient  $\mathcal{D}$  can be roughly evaluated as  $1.34 \times 10^{-5}$ , compared to  $2.20 \times 10^{-5}$  for water in air. As the psychrometric ratio is linked to  $\mathcal{D}^{2/3}$ , it can be estimated as 0.72, which is in line with tabulated values for similar organic solvents (e.g., propanol).

**Method:** Consult item (vi) in Table 12-5 for the calculation methodology.

From the terminology, specific humidity  $Y_w = 0.02439\text{ kg/kg}$ , the same as in Example 7. Mole ratio  $z = 0.0121\text{ kmol/kmol}$ , mole fraction  $y = 0.01191\text{ kmol/kmol}$ —lower than in Example 7 because molecular weights are different.

From the Antoine equation (12-5),

$$\ln p_s = C_0 - \frac{C_1}{T - C_2} = 16.6513 - \frac{2940.46}{T - 35.93}$$

Since  $T = 60^\circ\text{C}$ ,  $\ln p_s = 6.758$ ,  $p_s = 861.0\text{ mmHg}$ . Hence  $p_s = 1.148\text{ bar} = 1.148 \times 10^5\text{ Pa}$ . The saturation vapor pressure is higher than atmospheric pressure; this means that acetone at  $60^\circ\text{C}$  must be above its normal boiling point. Check;  $T_{bp}$  for acetone =  $56.5^\circ\text{C}$ .

Vapor pressure  $p = yP = 0.01191 \times 10,000 = 119.1\text{ Pa}$  (0.001191 bar)—much lower than before because of the reduced total pressure. This is 0.89 mmHg. Volumetric humidity  $Y_v = 0.0025\text{ kg/m}^3$ —again substantially lower than at 1 atm.

Dew point is the temperature where  $p_s$  equals  $p'$ . From the reversed Antoine equation (12-5),

$$T = \frac{C_1}{C_0 - \ln p_s} + C_2$$

so

$$T_{dp} = \frac{2940}{16.6513 - \ln 0.89} + 35.93 = 211.27\text{ K} = -61.88^\circ\text{C}$$

This very low dew point is due to the low boiling point of acetone and the low concentration.

Relative humidity is the ratio of actual vapor pressure to saturation vapor pressure at dry-bulb temperature. So  $p = 119.1\text{ Pa}$ ,  $p_s = 1.148 \times 10^5\text{ Pa}$ , RH = 0.104 percent—again very low.

A special psychrometric chart would need to be constructed for the acetone-nitrogen system to get first estimates (this can be done using PSYCHIC, as shown in Fig. 12-7). A humidity of  $0.025\text{ kg/kg}$  at  $T = 60^\circ\text{C}$  lies just below the adiabatic saturation line for  $-40^\circ\text{C}$ . The wet-bulb temperature will not be the same as  $T_{as}$  for this system; as the psychrometric ratio  $\beta$  is less than 1,  $T_{wb}$  should be significantly above  $T_{as}$ . However, let us assume no good first estimate is available and simply take  $T_{wb}$  to be  $0^\circ\text{C}$  initially.

When using the psychrometer equation, we will need to use Eq. (12-13) to obtain the value of the psychrometer coefficient. Using the tabulated values above, we obtain  $A = 0.00135$ , about double the value for air-water. We must remember that the estimate will be very rough because of the uncertainty in the value of  $\beta$ . Refining the estimate of  $T_{wb}$  by using the psychrometer equation and iterating gives

$$p_{wb} = 119.1 + 1.35 \times 10^{-3} (10^4) (60 - 0) = 932.3\text{ Pa} = 7.0\text{ mmHg}$$

From the Antoine equation,

$$T_{wb} = 2940 / (16.6513 - \ln 7) + 35.93 = 235.84\text{ K} = -37.3^\circ\text{C}$$

Second iteration:

$$p_{wb} = 119.1 + 1.35 \times 10^{-3} (10^4) (60 + 37.3) = 1433\text{ Pa} = 10.7\text{ mmHg}$$

$$T_{wb} = 241.85\text{ K} = -31.3^\circ\text{C}$$

Third iteration:

$$p_{wb} = 119.1 + 1.35 \times 10^{-3} (10^4) (60 + 31.3) = 1352\text{ Pa} = 10.1\text{ mmHg}$$

$$T_{wb} = 241.0\text{ K} = -32.1^\circ\text{C}$$

The iteration has converged successfully, despite the poor initial guess. The wet-bulb temperature is  $-32^\circ\text{C}$ ; given the levels of error in the calculation, it will be meaningless to express this to any greater level of precision.

In a similar way, adiabatic saturation temperature can be calculated from Eq. (12-6) by taking the first guess as  $-40^\circ\text{C}$  and assuming the humid heat to be  $1.05\text{ kJ/(kg} \cdot \text{K)}$  including the vapor:

$$Y_{as} = Y + \frac{C_s}{\lambda_{as}} (T - T_{as})$$

$$= 0.025 + \left( \frac{1.05}{501.1} \right) (60 + 40) = 0.235\text{ kg/kg}$$

From Table 12-2,

$$p_{as} = 1018\text{ Pa} = 7.63\text{ mmHg}$$

From Antoine,

$$T_{as} = 237.05\text{ K} = -36.1^\circ\text{C}$$

Second iteration:

$$Y_{as} = 0.025 + (1.05/501.1)(60 + 36.1) = 0.226\text{ kg/kg} \quad p_{as} = 984\text{ Pa} = 7.38\text{ mmHg}$$

From Antoine,

$$T_{as} = 236.6\text{ K} = -36.6^\circ\text{C}$$

This has converged. A more accurate figure could be obtained with more refined estimates for  $C_s$  and  $\lambda_{wb}$ .

## MEASUREMENT OF HUMIDITY

**Dew Point Method** The dew point of wet air is measured directly by observing the temperature at which moisture begins to form on an artificially cooled, polished surface.

Optical dew point hygrometers employing this method are the most commonly used fundamental technique for determining humidity. Uncertainties in temperature measurement of the polished surface, gradients across the surface, and the appearance or disappearance of fog have been much reduced in modern instruments. Automatic mirror cooling, e.g., thermoelectric, is more accurate and reliable than older methods using evaporation of a low-boiling solvent such as ether, or external coolants (e.g., vaporization of solid carbon dioxide or liquid air, or water cooling). Contamination effects have also been reduced or compensated for, but regular recalibration is still required, at least once a year.

**Wet-Bulb Method** In the past, probably the most commonly used method for determining the humidity of a gas stream was the measurement of wet- and dry-bulb temperatures. The wet-bulb temperature is measured by contacting the air with a thermometer whose bulb is covered by a wick saturated with water. If the process is adiabatic, the thermometer bulb attains the wet-bulb temperature. When the wet- and dry-bulb temperatures are known, the humidity is readily obtained from charts such as Figs. 12-1 through 12-4. To obtain reliable information, care must be exercised to ensure that the wet-bulb thermometer remains wet and that radiation to the bulb is minimized. The latter is accomplished by making the relative velocity between wick and gas stream high [a velocity of 4.6 m/s (15 ft/s) is usually adequate for commonly used thermometers] or by the use of radiation shielding. In the **Assmann psychrometer** the air is drawn past the bulbs by a motor-driven fan. Making sure that the wick remains wet is a mechanical problem, and the method used depends to a large extent on the particular arrangement. Again, as with the dew point method, errors associated with the measurement of temperature can cause difficulty.

For measurement of atmospheric humidities the **sling or whirling psychrometer** is widely used to give a quick and cheap, but inaccurate, estimate. A wet- and dry-bulb thermometer is mounted in a sling which is whirled manually to give the desired gas velocity across the bulb.

In addition to the mercury-in-glass thermometer, other temperature-sensing elements may be used for psychrometers. These include resistance thermometers, thermocouples, bimetal thermometers, and thermistors.

**Electric hygrometers** have been the fastest-growing form of humidity measurement in recent years. They measure the electrical resistance, capacitance, or impedance of a film of moisture-absorbing materials exposed to the gas. A wide variety of sensing

elements have been used. Often, it is relative humidity which is measured.

**Mechanical hygrometers** utilizing materials such as human hair, wood fiber, and plastics have been used to measure humidity. These methods rely on a change in dimension with humidity. They are not suitable for process use.

Other hygrometric techniques in process and laboratory use include electrolytic and piezoelectric hygrometers, infrared and mass

spectroscopy, and vapor pressure measurement, e.g., by a Pirani gauge.

The **gravimetric method** is accepted as the most accurate humidity-measuring technique. In this method a known quantity of gas is passed over a moisture-absorbing chemical such as phosphorus pentoxide, and the increase in weight is determined. It is mainly used for calibrating standards and measurements of gases with  $\text{SO}_x$  present.

## EVAPORATIVE COOLING

**GENERAL REFERENCES:** 2005 ASHRAE *Handbook of Fundamentals*, "Climatic Design Information," Chap. 28, ASHRAE, Atlanta, Ga.; ASHRAE *Handbook and Product Directory: Equipment*, ASHRAE, Atlanta, 2001.

### INTRODUCTION

Evaporative cooling, using recirculated cooling water systems, is the method most widely used throughout the process industries for employing water to remove process waste heat, rejecting that waste heat into the environment. Maintenance considerations (water-side fouling control), through control of makeup water quality and control of cooling water chemistry, form one reason for this preference. Environmental considerations, by minimizing consumption of potable water, minimizing the generation and release of contaminated cooling water, and controlling the release into the environment of chemicals from leaking heat exchangers (HX), form the second major reason.

Local ambient climatic conditions, particularly the maximum summer wet-bulb temperature, determine the design of the evaporative equipment. Typically, the wet-bulb temperature used for design is the 0.4 percent value, as listed in the ASHRAE *Handbook of Fundamentals*, equivalent to 35-h exceedance per year on average.

The first subsection below presents the classic cooling tower (CT), the evaporative cooling technology most widely used today. The second subsection presents the wet surface air cooler (WSAC), a more recently perfected technology, combining within one piece of equipment the functions of cooling tower, circulated cooling water system, and HX tube bundle. The most common application for WSACs is in the direct cooling of process streams. However, the closed-circuit cooling tower, employing WSACs for cooling the circulated cooling water (replacing the CT), is an important alternative WSAC application, presented at the end of this section.

**To minimize the total annualized costs for evaporative cooling** is a complex engineering task in itself, separate from classic process design (Sec. 24, "Minimizing the Annualized Costs for Process Energy"). The evaluation and the selection of the best option for process cooling impact many aspects of how the overall project will be optimally designed (utilities supply, reaction and separations design, pinch analyses, 3D process layout, plot plan, etc.). Therefore, evaluation and selection of the evaporative cooling technology system should be performed at the start of the project design cycle, during conceptual engineering (Sec. 9, "Process Economics," "Value Improving Practices"), when the potential to influence project costs is at a maximum value (Sec. 9, *VIP Figure 9-33*). The relative savings achievable for selection of the optimum heat rejection technology option can frequently exceed 25 percent, for the installed cost for the technology alone.

### PRINCIPLES

The processes of cooling water are among the oldest known. Usually water is cooled by exposing its surface to air. Some of the processes are slow, such as the cooling of water on the surface of a pond; others are comparatively fast, such as the spraying of water into air. These processes all involve the exposure of water surface to air in varying degrees.

The heat-transfer process involves (1) latent heat transfer owing to vaporization of a small portion of the water and (2) sensible heat transfer owing to the difference in temperatures of water and air. Approximately 80 percent of this heat transfer is due to latent heat and 20 percent to sensible heat.

Theoretical possible heat removal per pound of air circulated in a cooling tower depends on the temperature and moisture content of air. An indication of the moisture content of the air is its wet-bulb temperature. Ideally, then, the wet-bulb temperature is the lowest theoretical temperature to which the water can be cooled. Practically, the cold water temperature approaches but does not equal the air wet-bulb temperature in a cooling tower; this is so because it is impossible to contact all the water with fresh air as the water drops through the wetted fill surface to the basin. The magnitude of approach to the wet-bulb temperature is dependent on the tower design. Important factors are air-to-water contact time, amount of fill surface, and breakup of water into droplets. In actual practice, cooling towers are seldom designed for approaches closer than 2.8°C (5°F).

### COOLING TOWERS\*

**GENERAL REFERENCES:** *Counterflow Cooling Tower Performance*, Pritchard Corporation, Kansas City, Mo., 1957; Hensley, "Cooling Tower Energy," *Heat Piping Air Cond.* (October 1981); Kelley and Swenson, *Chem. Eng. Prog.* **52**: 263 (1956); McAdams, *Heat Transmission*, 3d ed., McGraw-Hill, New York, 1954, pp. 356-365; Merkel, *Z. Ver. Dtsch. Ing. Forsch.*, no. 275 (1925); *The Parallel Path Wet-Dry Cooling Tower*, Marley Co., Mission Woods, Kan., 1972; *Performance Curves*, Cooling Tower Institute, Houston, Tex., 1967; *Plume Abatement and Water Conservation with Wet-Dry Cooling Tower*, Marley Co., Mission Woods, Kan., 1973; Tech. Bull. R-54-P-5, R-58-P-5, Marley Co., Mission Woods, Kan., 1957; Wood and Betts, *Engineer*, **189**(4912), **377**(4913), **349** (1950); Zivi and Brand, *Refrig. Eng.*, **64**(8): 31-34, 90 (1956); Hensley, *Cooling Tower Fundamentals*, 2d ed., Marley Cooling Technologies, 1998; Mortensen and Gagliardo, *Impact of Recycled Water Use in Cooling Towers*, TP-04-12, Cooling Technology Institute, 2004; www.cti.org; www.ashrae.org; www.marleyct.com.

**Cooling Tower Theory** The most generally accepted theory of the cooling tower heat-transfer process is that developed by Merkel (op. cit.). This analysis is based upon **enthalpy potential difference** as the driving force.

Each particle of water is assumed to be surrounded by a film of air, and the enthalpy difference between the film and surrounding air provides the driving force for the cooling process. In the integrated form the Merkel equation is

$$\frac{KaV}{L} = \int_{T_2}^{T_1} \frac{C_L dT}{h' - h} \quad (12-14a)$$

where  $K$  = mass-transfer coefficient, lb water/(h·ft<sup>2</sup>);  $a$  = contact area, ft<sup>2</sup>/ft<sup>3</sup> tower volume;  $V$  = active cooling volume, ft<sup>3</sup>/ft<sup>2</sup> of plan area;  $L$  = water rate, lb/(h·ft<sup>2</sup>);  $C_L$  = heat capacity of water, Btu/(lb·°F);  $h'$  = enthalpy of saturated air at water temperature, Btu/lb;  $h$  = enthalpy of

\*The contributions of Ken Mortensen, and coworkers, of Marley Cooling Technologies, Overland Park, Kansas, toward the review and update of this subsection are acknowledged.

## 12-18 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

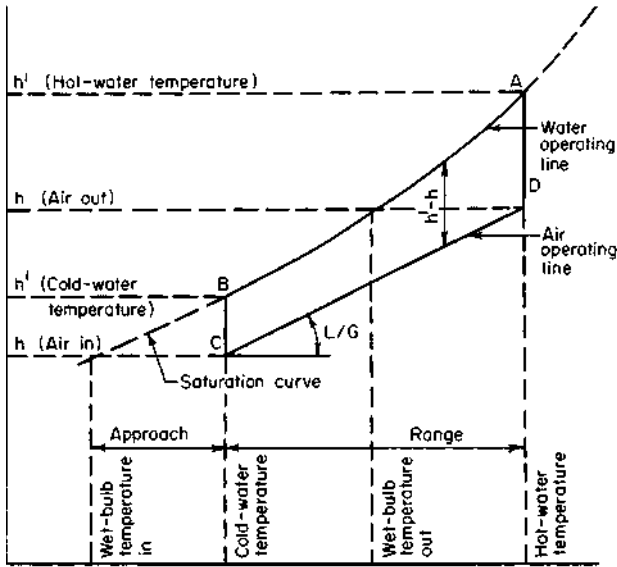


FIG. 12-8a Cooling-tower process heat balance. (Marley Co.)

airstream, Btu/lb; and  $T_1$  and  $T_2$  = entering and leaving water temperatures, °F. The right-hand side of Eq. (12-14a) is entirely in terms of air and water properties and is independent of tower dimensions.

Figure 12-8a illustrates water and air relationships and the driving potential which exist in a counterflow tower, where air flows parallel but opposite in direction to water flow. An understanding of this diagram is important in visualizing the cooling tower process.

The water operating line is shown by line AB and is fixed by the inlet and outlet tower water temperatures. The air operating line begins at C, vertically below B and at a point having an enthalpy corresponding to that of the entering wet-bulb temperature. Line BC represents the initial driving force  $h' - h$ . In cooling water at 1°F, the enthalpy per pound of air is increased 1 Btu multiplied by the ratio of pounds of water to pound of air. The liquid-gas ratio  $L/G$  is the slope of the operating line. The air leaving the tower is represented by point D. The cooling range is the projected length of line CD on the temperature scale. The cooling tower approach is shown on the diagram as the difference between the cold water temperature leaving the tower and the ambient wet-bulb temperature.

The coordinates refer directly to the temperature and enthalpy of any point on the water operating line but refer directly only to the enthalpy of a point on the air operating line. The corresponding wet-bulb temperature of any point on CD is found by projecting the point horizontally to the saturation curve, then vertically to the temperature coordinate. The integral [Eq. (12-14a)] is represented by the area ABCD in the diagram. This value is known as the **tower characteristic**, varying with the  $L/G$  ratio.

For example, an increase in entering wet-bulb temperature moves the origin C upward, and the line CD shifts to the right to maintain a constant  $KaV/L$ . If the cooling range increases, line CD lengthens. At a constant wet-bulb temperature, equilibrium is established by moving the line to the right to maintain a constant  $KaV/L$ . On the other hand, a change in  $L/G$  ratio changes the slope of CD, and the tower comes to equilibrium with a new  $KaV/L$ .

To predict tower performance, it is necessary to know the required tower characteristics for fixed ambient and water conditions. The tower characteristic  $KaV/L$  can be determined by integration. The Chebyshev method is normally used for numerically evaluating the integral, whereby

$$\frac{KaV}{L} = \int_{T_2}^{T_1} \frac{dT}{h_w - h_a} \cong \frac{T_1 - T_2}{4} \left( \frac{1}{\Delta h_1} + \frac{1}{\Delta h_2} + \frac{1}{\Delta h_3} + \frac{1}{\Delta h_4} \right)$$

where  $h_w$  = enthalpy of air-water vapor mixture at bulk water temperature, Btu/lb dry air

$h_a$  = enthalpy of air-water vapor mixture at wet-bulb temperature, Btu/lb dry air

$\Delta h_1$  = value of  $h_w - h_a$  at  $T_2 + 0.1(T_1 - T_2)$

$\Delta h_2$  = value of  $h_w - h_a$  at  $T_2 + 0.4(T_1 - T_2)$

$\Delta h_3$  = value of  $h_w - h_a$  at  $T_1 - 0.4(T_1 - T_2)$

$\Delta h_4$  = value of  $h_w - h_a$  at  $T_1 - 0.1(T_1 - T_2)$

### Example 10: Calculation of Mass-Transfer Coefficient Group

Determine the theoretically required  $KaV/L$  value for a cooling duty from 105°F inlet water, 85°F outlet water, 78°F ambient wet-bulb temperature, and an  $L/G$  ratio of 0.97.

From the air-water vapor-mixture tables, the enthalpy  $h_1$  of the ambient air at 78°F wet-bulb temperature is 41.58 Btu/lb.

$$h_2 (\text{leaving air}) = 41.58 + 0.97(105 - 85) = 60.98 \text{ Btu/lb}$$

| $T, ^\circ\text{F}$   | $h_{\text{water}}$ | $h_{\text{air}}$           | $h_w - h_a$          | $1/\Delta h$ |
|-----------------------|--------------------|----------------------------|----------------------|--------------|
| $T_2 = 85$            | 49.43              | $h_1 = 41.58$              |                      |              |
| $T_2 + 0.1(20) = 87$  | 51.93              | $h_1 + 0.1L/G(20) = 43.52$ | $\Delta h_1 = 8.41$  | 0.119        |
| $T_2 + 0.4(20) = 93$  | 60.25              | $h_1 + 0.4L/G(20) = 49.34$ | $\Delta h_2 = 10.91$ | 0.092        |
| $T_1 - 0.4(20) = 97$  | 66.55              | $h_2 - 0.4L/G(20) = 53.22$ | $\Delta h_3 = 13.33$ | 0.075        |
| $T_1 - 0.1(20) = 103$ | 77.34              | $h_2 - 0.1L/G(20) = 59.04$ | $\Delta h_4 = 18.30$ | 0.055        |
| $T_1 = 105$           | 81.34              | $h_2 = 60.98$              |                      | 0.341        |

$$\frac{KaV}{L} = \frac{105 - 85}{4} (0.341) = 1.71$$

A quicker but less accurate method is by the use of a nomograph (Fig. 12-8b) prepared by Wood and Betts (op. cit.).

Mechanical draft cooling towers normally are designed for  $L/G$  ratios ranging from 0.75 to 1.50; accordingly, the values of  $KaV/L$  vary from 0.50 to 2.50. With these ranges in mind, an example of the use of the nomograph will readily explain the effect of changing variables.

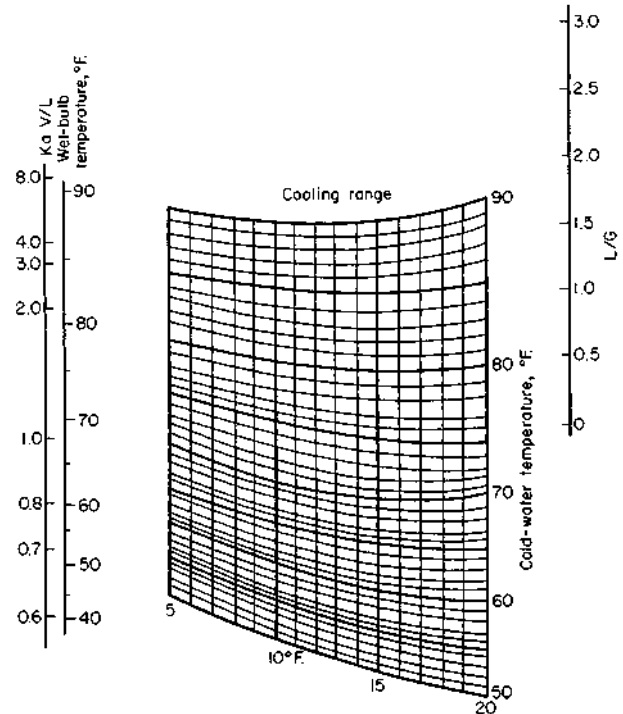


FIG. 12-8b Nomograph of cooling tower characteristics. [Wood and Betts, Engineer, 189(4912), 337 (1950).]

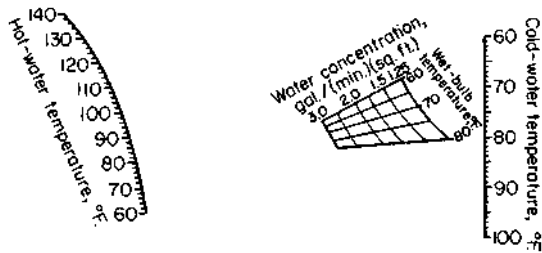


FIG. 12-8c Sizing chart for a counterflow induced-draft cooling tower. For induced-draft towers with (1) an upspray distributing system with 24 ft of fill or (2) a flume-type distributing system and 32 ft of fill. The chart will give approximations for towers of any height. (Ecodyne Corp.)

#### Example 11: Application of Nomograph for Cooling Tower Characteristics

If a given tower is operating with 20°F range, a cold water temperature of 80°F, and a wet-bulb temperature of 70°F, a straight line may be drawn on the nomograph. If the  $L/G$  ratio is calculated to be 1.0, then  $KaV/L$  may be established by a line drawn through  $L/G$  1.0 and parallel to the original line. The tower characteristic  $KaV/L$  is thus established at 1.42. If the wet-bulb temperature were to drop to 50°F, then  $KaV/L$  and  $L/G$  ratios may be assumed to remain constant. A new line parallel to the original will then show that for the same range the cold-water temperature will be 70°F.

The nomograph provides an approximate solution; degree of accuracy will vary with changes in cooling as well as from tower to tower. Once the theoretical cooling tower characteristic has been determined by numerical integration or from the nomograph for a given cooling duty, it is necessary to design the cooling tower fill and air distribution to meet the theoretical tower characteristic. The Pritchard Corporation (op. cit.) has developed performance data on various tower fill designs. These data are too extensive to include here, and those interested should consult this reference. See also Baker and Mart (Marley Co., Tech. Bull. R-52-P-10, Mission Woods, Kan.) and Zivi and Brand (loc. cit.).

**Mechanical Draft Towers** Two types of mechanical draft towers are in use today: the forced-draft and the induced-draft. In the **forced-draft tower** the fan is mounted at the base, and air is forced in at the bottom and discharged at low velocity through the top. This arrangement has the advantage of locating the fan and drive outside the tower, where it is convenient for inspection, maintenance, and repairs. Since the equipment is out of the hot, humid top area of the tower, the fan is not subjected to corrosive conditions. However, because of the low exit-air velocity, the forced-draft tower is subjected to excessive recirculation of the humid exhaust vapors back into the air intakes. Since the wet-bulb temperature of the exhaust air is considerably above the wet-bulb temperature of the ambient air, there is a decrease in performance evidenced by an increase in cold (leaving) water temperature.

The **induced-draft tower** is the most common type used in the United States. It is further classified into counterflow and cross-flow design, depending on the relative flow directions of water and air. Thermodynamically, the **counterflow arrangement** is more efficient, since the coldest water contacts the coldest air, thus obtaining maximum enthalpy potential. The greater the cooling ranges and the more difficult the approaches, the more distinct are the advantages of the counterflow type. For example, with an  $L/G$  ratio of 1, an ambient wet-bulb temperature of 25.5°C (78°F), and an inlet water temperature of 35°C (95°F), the counterflow tower requires a  $KaV/L$  characteristic of 1.75 for a 2.8°C (5°F) approach, while a cross-flow tower requires a characteristic of 2.25 for the same approach. However, if the approach is increased to 3.9°C (7°F), both types of tower have approximately the same required  $KaV/L$  (within 1 percent).

The **cross-flow tower** manufacturer may effectively reduce the tower characteristic at very low approaches by increasing the air quantity to give a lower  $L/G$  ratio. The increase in airflow is not necessarily achieved by increasing the air velocity but primarily by lengthening the tower to increase the airflow cross-sectional area. It appears then that the cross-flow fill can be made progressively longer in the direction perpendicular to the airflow and shorter in the direction of the airflow until it almost loses its inherent potential-difference disadvantage. However, as this is done, fan power consumption increases.

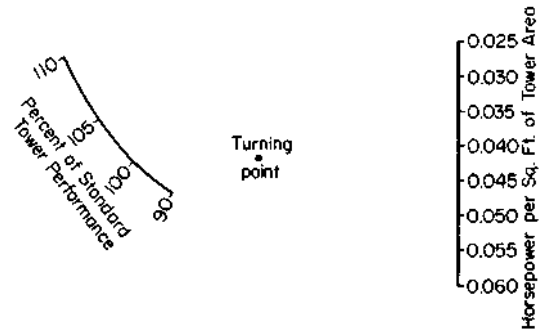


FIG. 12-8d Horsepower chart for a counterflow induced-draft cooling tower. [Fluor Corp. (now Ecodyne Corp.)]

Ultimately, the economic choice between counterflow and cross-flow is determined by the effectiveness of the fill, design conditions, water quality, and the costs of tower manufacture.

Performance of a given type of cooling tower is governed by the ratio of the weights of air to water and the time of contact between water and air. In commercial practice, the variation in the ratio of air to water is first obtained by keeping the air velocity constant at about 350 ft/(min·ft<sup>2</sup> of active tower area) and varying the water concentration, gal/(min·ft<sup>2</sup> of tower area). As a secondary operation, air velocity is varied to make the tower accommodate the cooling requirement.

**Time of contact** between water and air is governed largely by the time required for the water to discharge from the nozzles and fall through the tower to the basin. The time of contact is therefore obtained in a given type of unit by varying the height of the tower. Should the time of contact be insufficient, no amount of increase in the ratio of air to water will produce the desired cooling. It is therefore necessary to maintain a certain minimum height of cooling tower. When a wide approach of 8 to 11°C (15 to 20°F) to the wet-bulb temperature and a 13.9 to 19.4°C (25 to 35°F) cooling range are required, a relatively low cooling tower will suffice. A tower in which the water travels 4.6 to 6.1 m (15 to 20 ft) from the distributing system to the basin is sufficient. When a moderate approach and a cooling range of 13.9 to 19.4°C (25 to 35°F) are required, a tower in which the water travels 7.6 to 9.1 m (25 to 30 ft) is adequate. Where a close approach of 4.4°C (8°F) with a 13.9 to 19.4°C (25 to 35°F) cooling range is required, a tower in which the water travels from 10.7 to 12.2 m (35 to 40 ft) is required. It is usually not economical to design a cooling tower with an approach of less than 2.8°C (5°F).

Figure 12-8c shows the relationship of the hot water, cold water, and wet-bulb temperatures to the water concentration.<sup>6</sup> From this, the **minimum area** required for a given performance of a well-designed counterflow induced-draft cooling tower can be obtained. Figure 12-8d gives the horsepower per square foot of tower area required for a given performance. These curves do not apply to parallel or cross-flow cooling, since these processes are not so efficient as the counterflow process. Also, they do not apply when the approach to the cold water temperature is less than 2.8°C (5°F). These charts should be considered approximate and for preliminary estimates only. Since many factors not shown in the graphs must be included in the computation, the manufacturer should be consulted for final design recommendations.

The cooling performance of any tower containing a given depth of filling varies with the **water concentration**. It has been found that maximum contact and performance are obtained with a tower having a water concentration of 2 to 5 gal/(min·ft<sup>2</sup> of ground area). Thus the

<sup>6</sup>See also London, Mason, and Boelter, loc. cit.; Lichtenstein, loc. cit.; Simpson and Sherwood, *J. Am. Soc. Refrig. Eng.*, 52:535, 574 (1946); Simons, *Chem. Metall. Eng.*, 49(5):138; (6): 83 (1942); 46: 208 (1939); and Hutchinson and Spivey, *Trans. Inst. Chem. Eng.*, 20:14 (1942).



problem of calculating the size of a cooling tower becomes one of determining the proper concentration of water required to obtain the desired results. Once the necessary water concentration has been established, the tower area can be calculated by dividing the gallons per minute circulated by the water concentration in gallons per minute square foot. The required tower size then is a function of the following:

1. Cooling range (hot water temperature minus cold water temperature)
2. Approach to wet-bulb temperature (cold water temperature minus wet-bulb temperature)
3. Quantity of water to be cooled
4. Wet-bulb temperature
5. Air velocity through the cell
6. Tower height

### Example 12: Application of Sizing and Horsepower Charts

To illustrate the use of the charts, assume the following conditions:

Hot water temperature  $T_1$ , °F = 102  
Cold water temperature  $T_2$ , °F = 78  
Wet-bulb temperature  $t_w$ , °F = 70  
Water rate, gal/min = 2000

A straight line in Fig. 12-8c, connecting the points representing the design water and wet-bulb temperature, shows that a water concentration of 2 gal/(ft<sup>2</sup>·min) is required. The area of the tower is calculated as 1000 ft<sup>2</sup> (quantity of water circulated divided by water concentration).

Fan horsepower is obtained from Fig. 12-8d. Connecting the point representing 100 percent of standard tower performance with the turning point and extending this straight line to the horsepower scale show that it will require 0.041 hp/ft<sup>2</sup> of actual effective tower area. For a tower area of 1000 ft<sup>2</sup>, 41.0 fan hp is required to perform the necessary cooling.

Suppose that the actual commercial tower size has an area of only 910 ft<sup>2</sup>. Within reasonable limits, the shortage of actual area can be compensated for by an increase in air velocity through the tower. However, this requires boosting fan horsepower to achieve 110 percent of standard tower performance. From Fig. 12-8d, the fan horsepower is found to be 0.057 hp/ft<sup>2</sup> of actual tower area, or 0.057 × 910 = 51.9 hp.

On the other hand, if the actual commercial tower area is 1110 ft<sup>2</sup>, the cooling equivalent to 1000 ft<sup>2</sup> of standard tower area can be accomplished with less air and less fan horsepower. From Fig. 12-8d, the fan horsepower for a tower operating at 90 percent of standard performance is 0.031 hp/ft<sup>2</sup> of actual tower area, or 34.5 hp.

This example illustrates the sensitivity of fan horsepower to small changes in tower area. The importance of designing a tower that is slightly oversize in ground area and of providing plenty of fan capacity becomes immediately apparent.

### Example 13: Application of Sizing Chart

Assume the same cooling range and approach as used in Example 12 except that the wet-bulb temperature is lower. Design conditions would then be as follows:

Water rate, gal/min = 2000  
Temperature range  $T_1 - T_2$ , °F = 24  
Temperature approach  $T_2 - t_w$ , °F = 8  
Hot water temperature  $T_1$ , °F = 92  
Cold water temperature  $T_2$ , °F = 68  
Wet-bulb temperature  $t_w$ , °F = 60

From Fig. 12-8c, the water concentration required to perform the cooling is 1.75 gal/(ft<sup>2</sup>·min), giving a tower area of 1145 ft<sup>2</sup> versus 1000 ft<sup>2</sup> for a 70°F wet-bulb temperature. This shows that the lower the wet-bulb temperature for the same cooling range and approach, the larger the area of the tower required and therefore the more difficult the cooling job.

Figure 12-8e illustrates the type of performance curve furnished by the cooling tower manufacturer. This shows the variation in performance with changes in wet-bulb and hot water temperatures while the water quantity is maintained constant.

### Cooling Tower Operation

**Water Makeup** Makeup requirements for a cooling tower consist of the summation of evaporation loss, drift loss, and blowdown. Therefore,

$$W_m = W_e + W_d + W_b \quad (12-14b)$$

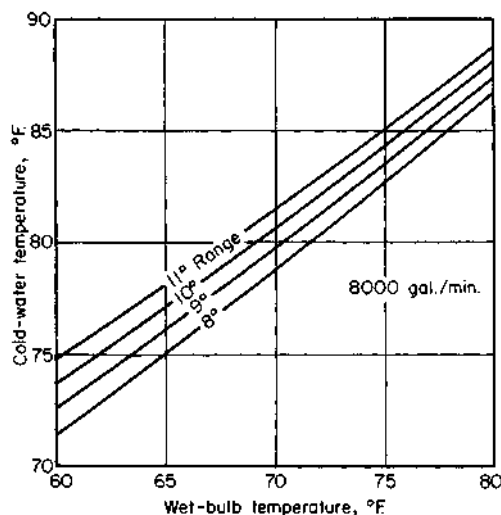


FIG. 12-8e Typical cooling-tower performance curve.

where  $W_m$  = makeup water,  $W_d$  = drift loss, and  $W_b$  = blowdown (consistent units: m<sup>3</sup>/h or gal/min).

Evaporation loss can be estimated by

$$W_e = 0.00085 W_c (T_1 - T_2) \quad (12-14c)$$

where  $W_c$  = circulating water flow, m<sup>3</sup>/h or gal/min at tower inlet, and  $T_1 - T_2$  = inlet water temperature minus outlet water temperature, °F. The 0.00085 evaporation constant is a good rule-of-thumb value. The actual evaporation rate will vary by season and climate.

Drift loss can be estimated by

$$W_d = 0.0002 W_c$$

Drift is entrained water in the tower discharge vapors. Drift loss is a function of the drift eliminator design and is typically less than 0.02 percent of the water supplied to the tower with the new developments in eliminator design.

Blowdown discards a portion of the concentrated circulating water due to the evaporation process in order to lower the system solids concentration. The amount of blowdown can be calculated according to the number of cycles of concentration required to limit scale formation. "Cycles of concentration" is the ratio of dissolved solids in the recirculating water to dissolved solids in the makeup water. Since chlorides remain soluble on concentration, cycles of concentration are best expressed as the ratio of the chloride contents of the circulating and makeup waters. Thus, the blowdown quantities required are determined from

$$\text{Cycles of concentration} = \frac{W_e + W_b + W_d}{W_b + W_d} \quad (12-14d)$$

or

$$W_b = \frac{W_e - (\text{cycles} - 1)W_d}{\text{cycles} - 1} \quad (12-14e)$$

Cycles of concentration involved with cooling tower operation normally range from three to five cycles. For water qualities where operating water concentrations must be below 3 to control scaling, blowdown quantities will be large. The addition of acid or scale-inhibiting chemicals can limit scale formation at higher cycle levels with such a water, and will allow substantially reduced water usage for blowdown.

The blowdown equation (12-14e) translates to calculated percentages of the cooling system circulating water flow exiting to drain, as listed in Table 12-6. The blowdown percentage is based on the cycles targeted and the cooling range. The range is the difference between the system hot water and cold water temperatures.

TABLE 12.6 Blowdown (%)

| Range, °F | 2X   | 3X   | 4X   | 5X   | 6X   |
|-----------|------|------|------|------|------|
| 10        | 0.83 | 0.41 | 0.26 | 0.19 | 0.15 |
| 15        | 1.26 | 0.62 | 0.41 | 0.30 | 0.24 |
| 20        | 1.68 | 0.83 | 0.55 | 0.41 | 0.32 |
| 25        | 2.11 | 1.04 | 0.69 | 0.51 | 0.41 |
| 30        | 2.53 | 1.26 | 0.83 | 0.62 | 0.49 |

It is the open nature of evaporative cooling systems, bringing in external air and water continuously, that determines the unique water problems these systems exhibit. Cooling towers (1) concentrate solids by the mechanisms described above and (2) wash air. The result is a buildup of dissolved solids, suspended contaminants, organics, bacteria, and their food sources in the circulating cooling water. These unique evaporative water system problems must be specifically addressed to maintain cooling equipment in good working order.

**Example 14: Calculation of Makeup Water** Determine the amount of makeup required for a cooling tower with the following conditions:

|                                               |               |
|-----------------------------------------------|---------------|
| Inlet water flow, m <sup>3</sup> /h (gal/min) | 2270 (10,000) |
| Inlet water temperature, °C (°F)              | 37.77 (100)   |
| Outlet water temperature, °C (°F)             | 29.44 (85)    |
| Drift loss, percent                           | 0.02          |
| Concentration cycles                          | 5             |

Evaporation loss:

$$W_e, \text{ m}^3/\text{h} = 0.00085 \times 2270 \times (37.77 - 29.44) \times (1.8^\circ\text{F}/^\circ\text{C}) = 28.9$$

$$W_e, \text{ gal/min} = 127.5$$

Drift loss

$$W_d, \text{ m}^3/\text{h} = 2270 \times 0.0002 = 0.45$$

$$W_d, \text{ gal/min} = 2$$

Blowdown

$$W_b, \text{ m}^3/\text{h} = 6.8$$

$$W_b, \text{ gal/min} = 29.9$$

Makeup

$$W_m, \text{ m}^3/\text{h} = 28.9 + 0.45 + 6.8 = 36.2$$

$$W_m, \text{ gal/min} = 159.4$$

**Fan Horsepower** In evaluating cooling tower ownership and operating costs, fan horsepower requirements can be a significant factor. Large air quantities are circulated through cooling towers at exit velocities of about 10.2 m/s (2000 ft/min) maximum for induced-draft towers. Fan airflow quantities depend upon tower design factors, including such items as type of fill, tower configuration, and thermal performance conditions.

The effective output of the fan is the static air horsepower (SAHP), which is obtained by the following equation:

$$\text{SAHP} = \frac{Q(h_s)(d)}{33,000(12)}$$

where  $Q$  = air volume, ft<sup>3</sup>/min;  $h_s$  = static head, in. of water; and  $d$  = density of water at ambient temperature, lb/ft<sup>3</sup>.

Cooling tower fan horsepower can be reduced substantially as the ambient wet-bulb temperature decreases if two-speed fan motors are used. Theoretically, operating at half speed will reduce airflow by 50 percent while decreasing horsepower to one-eighth of that of full-speed operation. However, actual half-speed operation will require about 17 percent of the horsepower at full speed as a result of the inherent motor losses at lighter loads.

Figure 12-8f shows a typical plot of outlet water temperatures when a cooling tower is operated (1) in the fan-off position, (2) with the fan

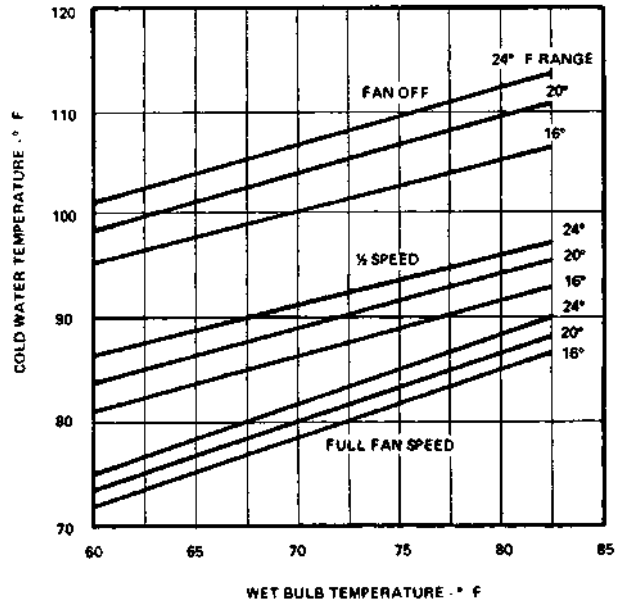


FIG. 12-8f Typical plot of cooling-tower performance at varying fan speeds.

at half speed, and (3) with the fan at full speed. Note that at decreasing wet-bulb temperatures the water leaving the tower during half-speed operation could meet design water temperature requirements of, say, 85°F. For example, for a 60°F wet-bulb, 20°F range, a leaving-water temperature slightly below 85°F is obtained with design water flow over the tower. If the fan had a 100-hp motor, 83 hp would be saved when operating it at half speed. In calculating savings, one should not overlook the advantage of having colder tower water available for the overall water circulating system.

Recent developments in cooling tower fan energy management also include automatic variable-pitch propeller-type fans and inverter-type devices to permit variable fan speeds. These schemes involve tracking the load at a constant outlet water temperature.

The variable-pitch arrangement at constant motor speed changes the pitch of the blades through a pneumatic signal from the leaving water temperature. As the thermal load and/or the ambient wet-bulb temperature decreases, the blade pitch reduces airflow and less fan energy is required.

Inverters make it possible to control a variable-speed fan by changing the frequency modulation. Standard alternating-current fan motors may be speed-regulated between 0 and 60 Hz. In using inverters for this application, it is important to avoid frequencies that would result in fan critical speeds.

Even though tower fan energy savings can result from these arrangements, they may not constitute the best system approach. Power plant steam condensers and refrigeration units, e.g., can take advantage of colder tower water to reduce power consumption. Invariably, these system savings are much larger than cooling tower fan savings with constant leaving water temperatures. A refrigeration unit condenser can utilize inlet water temperatures down to 12.8°C (55°F) to reduce compressor energy consumption by 25 to 30 percent.

**Pumping Horsepower** Another important factor in analyzing cooling tower selections, especially in medium to large sizes, is the portion of pump horsepower directly attributed to the cooling tower. A counterflow type of tower with spray nozzles will have a pumping head equal to static lift plus nozzle pressure loss. A cross-flow type of tower with gravity flow enables a pumping head to equal static lift. A reduction in tower height therefore reduces static lift, thus reducing pump horsepower:

$$\text{Pump bhp} = \frac{W_c h_t}{3960(\text{pump efficiency})} \quad (12-14f)$$

where  $W_c$  = water recirculation rate, gal/min, and  $h_t$  = total head, ft.

**Fogging and Plume Abatement** A phenomenon that occurs in cooling tower operation is fogging, which produces a highly visible plume and possible icing hazards. Fogging results from mixing warm, highly saturated tower discharge air with cooler ambient air that lacks the capacity to absorb all the moisture as vapor. While in the past visible plumes have not been considered undesirable, properly locating towers to minimize possible sources of complaints has now received the necessary attention. In some instances, guyed high fan stacks have been used to reduce ground fog. Although tall stacks minimize the ground effects of plumes, they can do nothing about water vapor saturation or visibility. The persistence of plumes is much greater in periods of low ambient temperatures.

More recently, environmental aspects have caused public awareness and concern over any visible plume, although many laypersons misconstrue cooling tower discharge as harmful. This has resulted in a new development for plume abatement known as a wet-dry cooling tower configuration. Reducing the relative humidity or moisture content of the tower discharge stream will reduce the frequency of plume formation. Figure 12-8g shows a "parallel path" arrangement that has been demonstrated to be technically sound but at substantially increased tower investment. Ambient air travels in parallel streams through the top dry-surface section and the evaporative section. Both sections benefit thermally by receiving cooler ambient air with the wet and dry airstreams mixing after leaving their respective sections. Water flow is arranged in series, first flowing to the dry coil section and then to the evaporation fill section. A "series path" airflow arrangement, in which dry coil sections can be located before or after the air traverses the evaporative section, also can be used. However, series-path airflow has the disadvantage of water impingement, which could result in coil scaling and restricted airflow.

Wet-dry cooling towers incorporating these designs are being used for large-tower industrial applications. At present they are not available for commercial applications.

**Thermal Performance** The thermal performance of the evaporative cooling tower is critical to the overall efficiency of cooling systems. Modern electronic measurement instrumentation allows accurate verification of cooling tower capability. Testing and tracking of the cooling tower capability are a substantial consideration in measuring cooling system performance. Cooling tower testing is a complex

activity that requires significant expertise in the art. Consult a competent testing company if such verification is desired.

**New Technologies** The cooling tower business is constantly changing in an attempt to improve efficiencies of evaporative cooling products. A significant thermal performance improvement over the splash-type fills, covered extensively in the writings above, can be achieved by using film-type fill. Film fills are formed plastic sheets separated by spacing knobs that allow water and air to flow easily between paired plastic surfaces. Fully wetted water flow over these panels creates an extensive "film" of evaporative surface on the plastic. Film fill is more sensitive to water quality than are splash-type fills.

These film fills are not sized via the graphical methods illustrated above for splash fills. They are selected by using manufacturers' proprietary sizing programs, which are based on extensive testing data. Such programs can be obtained by contacting manufacturers and/or industry trade organizations.

**Applications for Evaporative Cooling Towers** Cooling towers are commonly used in many commercial and industrial processes including

- Power generation (fossil fuel, nuclear)
- Industrial process (refinery, chemical production, plastic molding)
- Comfort cooling (HVAC)

**Natural Draft Towers, Cooling Ponds, Spray Ponds** Natural draft towers are primarily suited to very large cooling water quantities, and the reinforced concrete structures used are as large as 80 m in diameter and 105 m high.

When large ground areas are available, large cooling ponds offer a satisfactory method of removing heat from water. A pond may be constructed at a relatively small investment by pushing up earth in an earth dike 2 to 3 m high.

Spray ponds provide an arrangement for lowering the temperature of water by evaporative cooling and in so doing greatly reduce the cooling area required in comparison with a cooling pond.

Natural draft towers, cooling ponds, and spray ponds are infrequently used in new construction today in the chemical processing industry. Additional information may be found in previous Perry's editions.

## WET SURFACE AIR COOLER (WSAC)

**GENERAL REFERENCES:** Kals, "Wet Surface Aircoolers," *Chem. Engg.* July 1971; Kals, "Wet Surface Aircoolers: Characteristics and Usefulness," AIChE-ASME Heat Transfer Conference, Denver, Colo., August 6-9, 1972; Elliott and Kals, "Air Cooled Condensers," *Power*, January 1990; Kals, "Air Cooled Heat Exchangers: Conventional and Unconventional," *Hydrocarbon Processing*, August 1994; Hutton, "Properly Apply Closed Circuit Evaporative Cooling," *Chem. Engg. Progress*, October 1996; Hutton, "Improved Plant Performance through Evaporative Steam Condensing," ASME 1998 International Joint Power Conference, Baltimore, Md., August 23-26, 1998; <http://www.niagarablower.com/wsac.htm>; <http://www.baltimoreaircoil.com>.

**Principles** Rejection of waste process heat through a cooling tower (CT) requires transferring the heat in two devices in series, using two different methods of heat transfer. This requires two temperature driving forces in series: first, sensible heat transfer, from the process stream across the heat exchanger (HX) into the cooling water, and, second, sensible and latent heat transfer, from the cooling water to atmosphere across the CT. Rejecting process heat with a wet surface air cooler transfers the waste heat in a single device by using a single-unit operation. The single required temperature driving force is lower, because the WSAC does not require the use of cooling water sensible heat to transfer heat from the process stream to the atmosphere. A WSAC tube cross section (Fig. 12-8h) shows the characteristic external tube surface having a continuous flowing film of evaporating water, which cascades through the WSAC tube bundle. Consequently, process streams can be economically cooled to temperatures much closer to the ambient wet-bulb temperature (WBT), as low as to within 2.2°C (4°F), depending on the process requirements and economics for the specific application.

**Wet Surface Air Cooler Basics** The theory and principles for the design of WSACs are a combination of those known for evaporative cooling tower design and HX design. However, the design practices for engineering WSAC equipment remain a largely proprietary, technical

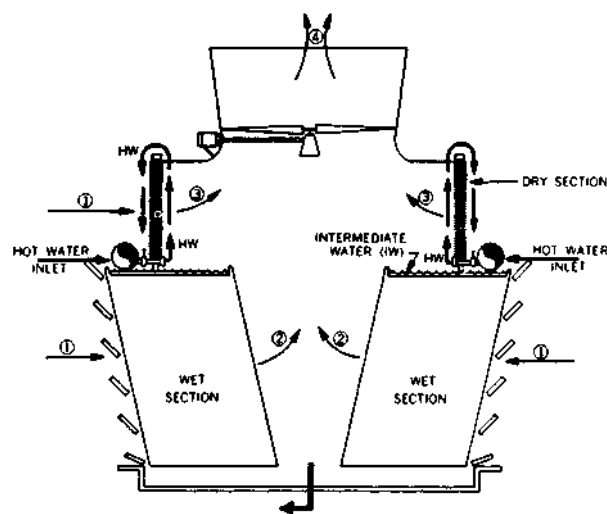
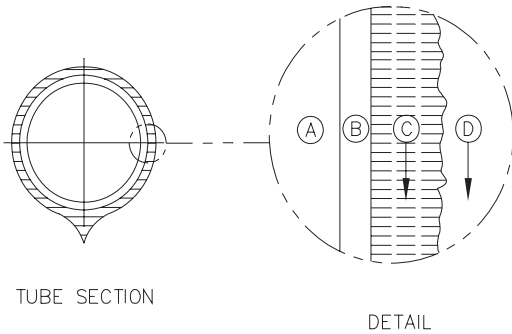


FIG. 12-8g Parallel-path cooling-tower arrangement for plume abatement. (Marley Co.)



**FIG. 12-8h** WSAC tube cross-section. Using a small T, heat flows from (A) the process stream, through (B) the tube, through (C) the flowing film of evaporating water, into (D) flowing ambient air.

art, and the details are not presented here. Any evaluation of the specifics and economics for any particular application requires direct consultation with a reputable vendor.

Because ambient air is contacted with evaporating water within a WSAC, from a distance a WSAC has a similar appearance to a CT (Fig. 12-8i). Economically optimal plot plan locations for WSACs can vary: integrated into, or with, the process structure, remote to it, in a pipe rack, etc.

In the WSAC the evaporative cooling occurs on the wetted surface of the tube bundle. The wetting of the tube bundle is performed by recirculating water the short vertical distance from the WSAC collection basin, through the spray nozzles, and onto the top of the bundle (Fig. 12-8j). The tube bundle is completely deluged with this cascading flow of water. Using water application rates between 12 and 24 ( $\text{m}^3/\text{h}/\text{m}^2$ ) (5 and 10  $\text{gpm}/\text{ft}^2$ ), the tubes have a continuous, flowing external water film, minimizing the potential for water-side biological fouling, sediment deposition, etc. Process inlet temperatures are limited to a maximum of about 85°C (185°F), to prevent external water-side mineral scaling. However, higher process inlet temperatures can be accepted, by incorporating bundles of dry, air-cooled finned tubing within the WSAC unit, to reduce the temperature of the process stream to an acceptable level before it enters the wetted evaporative tube bundles.

The WSAC combines within one piece of equipment the functions of cooling tower, circulated cooling water system, and water-cooled HX. In the basic WSAC configuration (Fig. 12-8k), ambient air is drawn in and

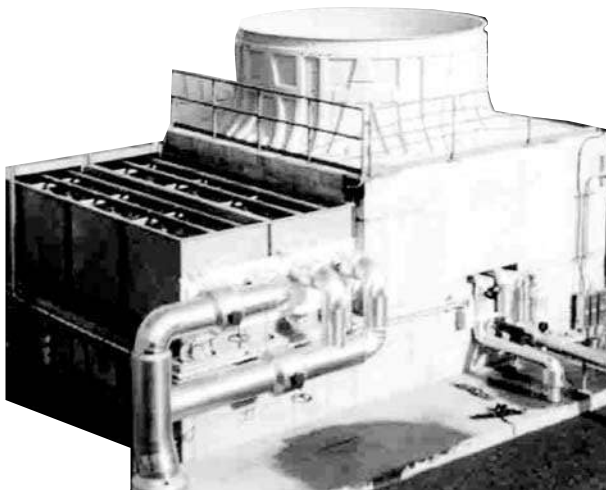


**FIG. 12-8j** Nozzles spraying onto wetted tube bundle in a WSAC unit.

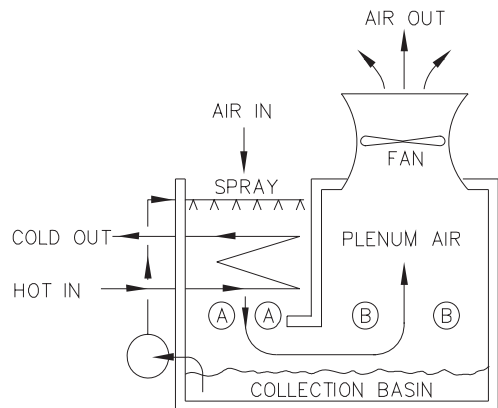
down through the tube bundle. This airflow is cocurrent with the evaporating water flow, recirculated from the WSAC collection basin sump to be sprayed over the tube bundles. This downward cocurrent flow pattern minimizes the generation of water mist (drift). At the bottom of the WSAC, the air changes direction through 180°, disengaging entrained fine water droplets. Drift eliminators can be added to meet very low drift requirements. Because heat is extracted from the tube surfaces by water latent heat (and not sensible heat), only about 75 percent as much circulating water is required in comparison to an equivalent CT-cooling water-HX application.

The differential head of the circulation water pump is relatively small, since dynamic losses are modest (short vertical pipe and a low  $\Delta P$  spray nozzle) and the hydraulic head is small, only about 6 m (20 ft) from the basin to the elevation of the spray header. Combined, the pumping energy demand is about 35 percent that for an equivalent CT application. The capital cost for this complete water system is also relatively small. The pumps and motors are smaller, the piping has a smaller diameter and is much shorter, and the required piping structural support is almost negligible, compared to an equivalent CT application. WSAC fan horsepower is typically about 25 percent less than that for an equivalent CT.

A WSAC is **inherently less sensitive to water-side fouling**. This is due to the fact that the deluge rate prevents the adhesion of waterborne material which can cause fouling within a HX. A WSAC



**FIG. 12-8i** Overhead view of a single-cell WSAC.



**FIG. 12-8k** Basic WSAC configuration.



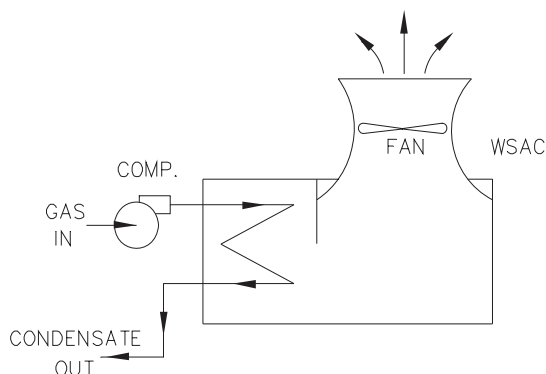


FIG. 12-8l WSAC configuration for condensing a compressed gas. A lower condensing pressure reduces compressor operating horsepower.

can accept relatively contaminated makeup water, such as CT blowdown, treated sewage plant effluent, etc. WSACs can endure **more cycles of concentration without fouling than can a CT** application. This higher practical operating concentration reduces the relative volume for the evaporative cooling blowdown, and therefore also reduces the relative volume of required makeup water. For facilities designed for **zero liquid discharge**, the higher practical WSAC blowdown concentration reduces the size and the operating costs for the downstream water treatment system. Since a hot process stream provides the unit with a heat source, a **WSAC has intrinsic freeze protection** while operating.

**Common WSAC Applications and Configurations** Employment of a WSAC can reduce process system operating costs that are not specific to the WSAC unit itself. A common WSAC application is **condensation of compressed gas** (Fig. 12-8l). A compressed gas can be condensed in a WSAC at a lower pressure, by condensing at a temperature closer to the ambient WBT, typically  $5.5^{\circ}\text{C}$  ( $10^{\circ}\text{F}$ ) above the WBT. This reduced condensation pressure reduces costs, by reducing gas compressor motor operating horsepower. Consequently, WSACs are widely applied for **condensing refrigerant gases**, for HVAC, process chillers, ice makers, gas-turbine inlet air cooling, chillers, etc. WSACs are also used directly to **condense lower-molecular-weight hydrocarbon streams**, such as ethane, ethylene, propylene, and LPG. A related WSAC application is the **cooling of compressed gases** ( $\text{CO}_2$ ,  $\text{N}_2$ , methane, LNG, etc.), which directly reduces gas compressor operating costs (inlet and interstage cooling) and indirectly reduces downstream condensing costs (aftercooling the compressed gas to reduce the downstream refrigeration load).

For combined cycle electric power generation, employment of a **WSAC increases steam turbine efficiency**. Steam turbine exhaust can be condensed at a lower pressure (higher vacuum) by condensing at a temperature closer to the ambient WBT, typically  $15^{\circ}\text{C}$  ( $27^{\circ}\text{F}$ ) above the WBT. This reduced condensation pressure results in a lower turbine discharge pressure, **increasing electricity generation** by increasing output shaft power (Fig. 12-8m). Due to standard WSAC configurations, a second cost advantage is gained at the turbine itself. The **steam turbine can be placed at grade**, rather than being mounted on an elevated platform, by venting horizontally into the WSAC, rather than venting downward to condensers located below the platform elevation, as is common for conventional water-cooled vacuum steam condensers.

A WSAC can **eliminate chilled water use**, for process cooling applications with required temperatures close to and just above the ambient WBT, typically about  $3.0$  to  $5.5^{\circ}\text{C}$  ( $5$  to  $10^{\circ}\text{F}$ ) above the WBT. This WSAC application can eliminate both chiller capital and operating costs. In such an application, either the necessary process temperature is below the practical CT water supply temperature, or they are so close to it that the use of CT water is uneconomical (a low-HX LMDT).

WSACs can be designed to **simultaneously cool several process streams in parallel** separate tube bundles within a single cell of a

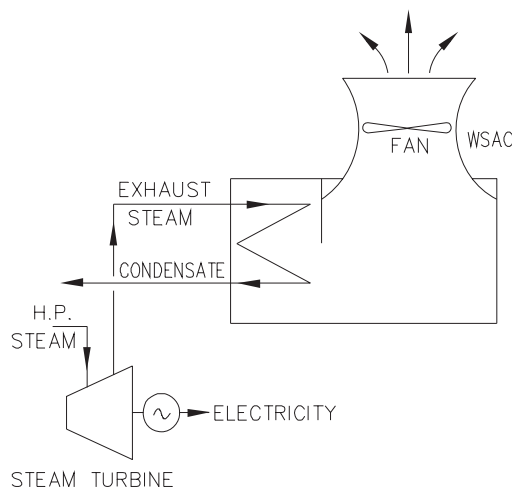


FIG. 12-8m WSAC configuration with electricity generation. A lower steam condensing pressure increases the turbine horsepower extracted.

WSAC (Fig. 12-8n). Often one of the streams is closed-circuit cooling water to be used for remote cooling applications. These might be applications not compatible with a WSAC (rotating seals, bearings, cooling jackets, internal reactor cooling coils, etc.) or merely numerous, small process streams in small HXs.

**WSAC for Closed-Circuit Cooling Systems** A closed-circuit cooling system as defined by the Cooling Technology Institute (CTI) employs a closed loop of circulated fluid (typically water) remotely as a cooling medium. By definition, this medium is cooled by **water evaporation involving no direct fluid contact** between the air and the enclosed circulated cooling medium. Applied in this manner, a WSAC can be used as the evaporative device to cool the circulated cooling medium, used remotely to cool process streams. This configuration completely isolates the cooling water (and the hot process streams) from the environment (Fig. 12-8o).

The closed circuit permits complete control of the cooling water chemistry, which permits **minimizing the cost for water-side materials of construction** and eliminating water-side fouling of, and fouling heat-transfer resistance in, the HXs (or jackets, reactor coils, etc.). **Elimination of water-side fouling** is particularly helpful for high-temperature cooling applications, especially where heat recovery may otherwise be impractical (quench oils, low-density polyethylene reactor cooling, etc.).

Closed-circuit cooling **minimizes circulation pumping horsepower**, which must overcome only dynamic pumping losses. This results through recovery of the returning circulated cooling water hydraulic head. A closed-circuit system can be designed for **operation at elevated pressures**, to guarantee that any process HX leak will be into the

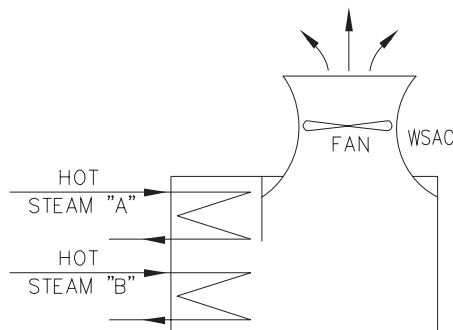


FIG. 12-8n WSAC configuration with parallel streams.

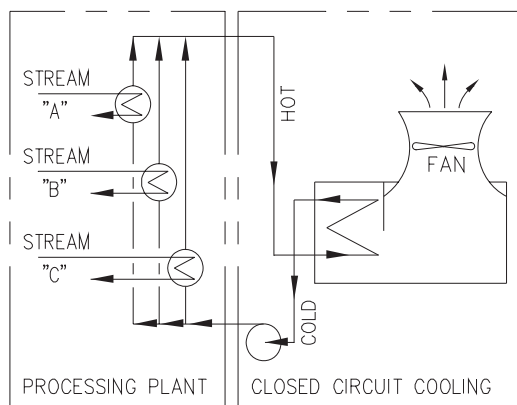


FIG. 12-8o WSAC configuration with no direct fluid contact.

process. Such high-pressure operation is economical, since the system overpressure is not lost during return flow to the circulation pump.

Closed-circuit cooling **splits the water chemistry needs** into two isolated systems: the **evaporating section**, exposed to the environment, and the **circulated cooling section**, isolated from the environment. Typically, this split reduces total water chemistry costs and water-related operations and maintenance problems. On the other hand, the split permits the effective **use of a low-quality or contaminated makeup water** for evaporative cooling, or a water source having severe seasonal quality problems, such as high sediment loadings.

If **highly saline water is used** for the evaporative cooling, a **reduced flow of makeup saline water** would need to be supplied to the WSAC. This reduction results from using latent cooling rather than sensible cooling to reject the waste heat. This consequence reduces the substantial capital investment required for the saline water supply and return systems (canal structures) and pump stations, and the saline supply pumping horsepower. (When saline water is used as the evaporative medium, special attention is paid to materials of construction and spray water chemical treatment due to the aggravated corrosion and scaling tendencies of this water.)

**Water Conservation Applications—"Wet-Dry" Cooling** A modified and hybridized form of a WSAC can be used to provide what is called "wet-dry" cooling for water conservation applications (Fig. 12-8p). A hybridized combination of air-cooled dry finned tubes, standard wetted bare tubes, and wet deck surface area permits the WSAC to **operate without water in cold weather**, reducing water consumption by about 75 percent of the total for an equivalent CT application.

Under design conditions of maximum summer WBT, the unit operates with spray water deluging the wetted tube bundle. The exiting water then flows down into and through the wet deck surface, where the water is cooled adiabatically to about the WBT, and then to the sump.

As the WBT drops, the process load is shifted from the wetted tubes to the dry finned tubes. By bypassing the process stream around the wetted tubes, cooling water evaporation (consumption) is proportionally reduced.

When the WBT drops to the "switch point," the process bypassing has reached 100 percent. This switch point WBT is at or above 5°C (41°F). As the ambient temperature drops further, adiabatic evaporative cooling continues to be used, to lower the dry-bulb temperature

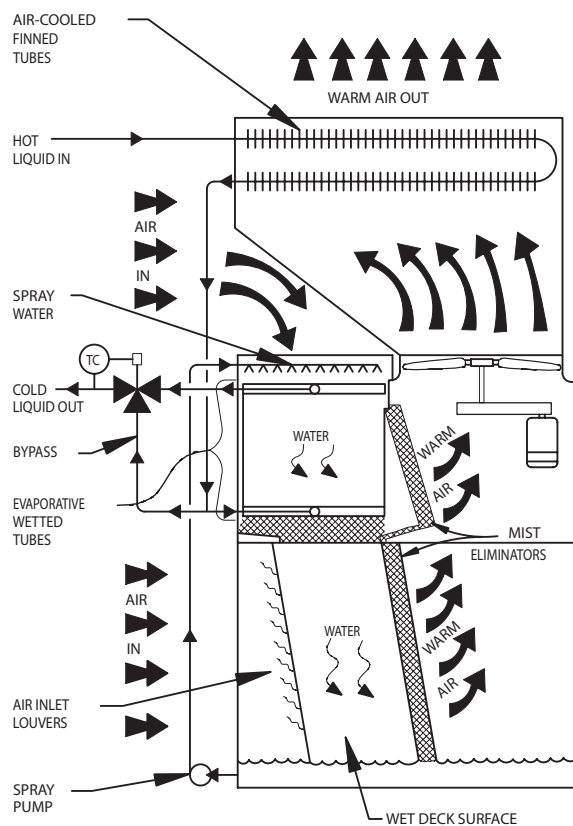


FIG. 12-8p As seasonal ambient temperatures drop, the "wet-dry" configuration for a WSAC progressively shifts the cooling load from evaporative to convective cooling.

to below the switch point temperature. This guarantees that the entire cooling load can be cooled in the dry finned tube bundle.

The use of water is discontinued after ambient dry-bulb temperatures fall below the switch point temperature, since the entire process load can be cooled using only cold fresh ambient air. By using this three-step load-shifting practice, total wet-dry cooling water consumption is about 25 percent of that consumption total experienced with an equivalent CT application.

Wet-dry cooling permits **significant reduction of water consumption**, which is useful where makeup water supplies are limited or where water treatment costs for blowdown are high. Because a WSAC (unlike a CT) has a heat source (the hot process stream), wet-dry cooling avoids various cold-weather-related CT problems. **Fogging and persistent plume formation can be minimized or eliminated** during colder weather. **Freezing and icing problems can be eliminated** by designing a wet-dry system for water-free operation during freezing weather, typically below 5°C (41°F). **In the arctic**, or regions of extreme cold, **elimination of freezing fog conditions** is realized by not evaporating any water during freezing weather.

## SOLIDS-DRYING FUNDAMENTALS

**GENERAL REFERENCES:** Cook and DuMont, *Process Drying Practice*, McGraw-Hill, New York, 1991. *Drying Technology—An International Journal*, Taylor and Francis, New York. Hall, *Dictionary of Drying*, Marcel Dekker, New York, 1979. Keey, *Introduction to Industrial Drying Operations*, Pergamon, New York, 1978. Keey, *Drying of Loose and Particulate Materials*, Hemisphere, New York, 1992. Masters, *Spray Drying Handbook*,

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## INTRODUCTION

Drying is the process by which volatile materials, usually water, are evaporated from a material to yield a solid product. Drying is a heat- and mass-transfer process. Heat is necessary to evaporate water. The latent heat of vaporization of water is about 2500 J/g, which means that the drying process requires a significant amount of energy. Simultaneously, the evaporating material must leave the drying material by diffusion and/or convection.

Heat transfer and mass transfer are not the only concerns when one is designing or operating a dryer. The product quality (color, particle density, hardness, texture, flavor, etc.) is also very strongly dependent on the drying conditions and the physical and chemical transformations occurring in the dryer.

Understanding and designing a drying process involves measurement and/or calculation of the following:

1. Mass and energy balances
2. Thermodynamics
3. Mass- and heat-transfer rates
4. Product quality considerations

The section below explains how these factors are measured and calculated and how the information is used in engineering practice.

## TERMINOLOGY

Generally accepted terminology and definitions are given alphabetically in the following paragraphs.

**Absolute humidity** is the mass ratio of water vapor (or other solvent mass) to dry air.

**Activity** is the ratio of the fugacity of a component in a system relative to the standard-state fugacity. In a drying system, it is the ratio of the vapor pressure of a solvent (e.g., water) in a mixture to the pure solvent vapor pressure at the same temperature. Boiling occurs when the vapor pressure of a component in a liquid exceeds the ambient total pressure.

**Bound moisture** in a solid is that liquid which exerts a vapor pressure less than that of the pure liquid at the given temperature. Liquid may become bound by retention in small capillaries, by solution in cell or fiber walls, by homogeneous solution throughout the solid, by chemical or physical adsorption on solid surfaces, and by hydration of solids.

**Capillary flow** is the flow of liquid through the interstices and over the surface of a solid, caused by liquid-solid molecular attraction.

**Constant-rate period (unhindered)** is that drying period during which the rate of water removal per unit of drying surface is constant, assuming the driving force is also constant.

**Convection** is heat or mass transport by bulk flow.

**Critical moisture content** is the average moisture content when the constant-rate period ends, assuming the driving force is also constant.

**Diffusion** is the molecular process by which molecules, moving randomly due to thermal energy, migrate from regions of high chemical potential (usually concentration) to regions of lower chemical potential.

**Dry basis** expresses the moisture content of wet solid as kilograms of water per kilogram of bone-dry solid.

**Equilibrium moisture content** is the limiting moisture to which a given material can be dried under specific conditions of air temperature and humidity.

**Evaporation** is the transformation of material from a liquid state to a vapor state.

**Falling-rate period (hindered drying)** is a drying period during which the instantaneous drying rate continually decreases.

**Fiber saturation point** is the moisture content of cellular materials (e.g., wood) at which the cell walls are completely saturated while the cavities are liquid-free. It may be defined as the equilibrium moisture content as the humidity of the surrounding atmosphere approaches saturation.

**Free moisture content** is that liquid which is removable at a given temperature and humidity. It may include bound and unbound moisture.

**Funicular state** is that condition in drying a porous body when capillary suction results in air being sucked into the pores.

**Hygroscopic material** is material that may contain bound moisture.

**Initial moisture distribution** refers to the moisture distribution throughout a solid at the start of drying.

**Internal diffusion** may be defined as the movement of liquid or vapor through a solid as the result of a concentration difference.

**Latent heat of vaporization** is the specific enthalpy change associated with evaporation.

**Moisture content** of a solid is usually expressed as moisture quantity per unit weight of the dry or wet solid.

**Moisture gradient** refers to the distribution of water in a solid at a given moment in the drying process.

**Nonhygroscopic material** is material that can contain no bound moisture.

**Pendular state** is that state of a liquid in a porous solid when a continuous film of liquid no longer exists around and between discrete particles so that flow by capillary cannot occur. This state succeeds the funicular state.

**Permeability** is the resistance of a material to bulk or convective, pressure-driven flow of a fluid through it.

**Relative humidity** is the partial pressure of water vapor divided by the vapor pressure of pure water at a given temperature. In other words, the relative humidity describes how close the air is to saturation.

**Sensible heat** is the energy required to increase the temperature of a material without changing the phase.

**Unaccomplished moisture change** is the ratio of the free moisture present at any time to that *initially* present.

**Unbound moisture** in a hygroscopic material is that moisture in excess of the equilibrium moisture content corresponding to saturation humidity. All water in a nonhygroscopic material is unbound water.

**Vapor pressure** is the partial pressure of a substance in the gas phase that is in equilibrium with a liquid or solid phase of the pure component.

**Wet basis** expresses the moisture in a material as a percentage of the weight of the wet solid. Use of a dry-weight basis is recommended since the percentage change of moisture is constant for all moisture levels. When the wet-weight basis is used to express moisture content, a 2 or 3 percent change at high moisture contents (above 70 percent) actually represents a 15 to 20 percent change in evaporative load. See Fig. 12-9 for the relationship between the dry- and wet-weight bases.

## MASS AND ENERGY BALANCES

The most basic type of calculation for a dryer is a mass and energy balance. This calculation only quantifies the conservation of mass and energy in the system; by itself it does not answer important questions of rate and quality.

Some examples here illustrate the calculations. Experimental determination of the values used in these calculations is discussed in a later section.

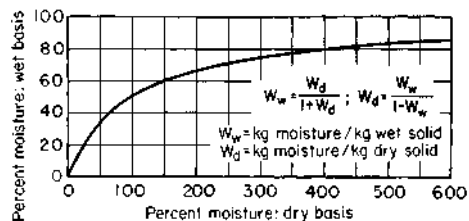


FIG. 12-9 Relationship between wet-weight and dry-weight bases.

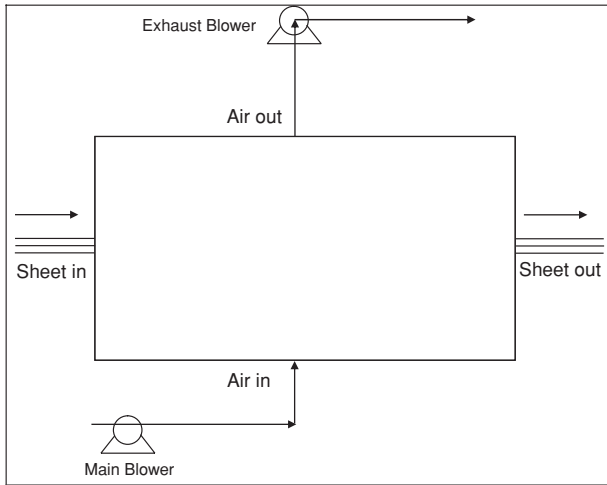


FIG. 12-10 Overall mass and energy balance diagram.

Example 15 illustrates a generic mass and energy balance. Other examples are given in the sections on fluidized bed dryers and rotary dryers.

**Example 15: Overall Mass and Energy Balance on a Sheet Dryer** Figure 12-10 shows a simple sheet drying system. Hot air enters the dryer and contacts a wet sheet. The sheet leaves a dryer with a lower moisture content, and the air leaves the dryer with a higher humidity.

**Given:** Incoming wet sheet mass flow rate is 100 kg/h. It enters with 20 percent water on a wet basis and leaves at 1 percent water on a wet basis. The air-flow rate is 1000 kg/h, with an absolute humidity of 0.01 g water/g dry air. The incoming air temperature is 170°C. The sheet enters at 20°C and leaves at 90°C.

**Relevant physical constants:**  $C_{p, \text{air}} = 1 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$ ,  $C_{p, \text{sheet}} = 2.5 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$ ,  $C_{p, \text{liquid water}} = 4.184 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$ ,  $C_{p, \text{water vapor}} = 2 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$  (for superheated steam at low partial pressures). Latent heat of vaporization of water at 20°C =  $\lambda_w = 2454 \text{ J/g}$

**Find the following:**

1. The absolute humidity of the exiting airstream
2. The exit air temperature

**Solution:** Answering the questions above involves an overall mass and energy balance. Only the mass and enthalpy of the streams need to be considered to answer the two questions above. Only the streams entering the overall process need to be considered. In this example, wet-basis moisture content (and therefore total mass flow rate including moisture) will be used. Since the same mass of air flows in and out of the dryer, there are no equations to solve for the dry air.

The mass balance is given by the following equations:

$$F_{\text{dry sheet in}} = F_{\text{dry sheet out}} \quad (12-15)$$

$$F_{\text{liquid water in}} = F_{\text{liquid water out}} + F_{\text{evaporated}} \quad (12-16)$$

$$G_{\text{dry air in}} = G_{\text{dry air out}} \quad (12-17)$$

$$G_{\text{water vapor in}} + F_{\text{evaporated}} = G_{\text{water vapor out}} \quad (12-18)$$

The wet-basis moisture contents of the incoming and outgoing sheet are given by

$$w_{\text{in}} = \frac{F_{\text{liquid water in}}}{F_{\text{liquid water in}} + F_{\text{dry sheet in}}} \quad (12-19)$$

$$w_{\text{out}} = \frac{F_{\text{liquid water out}}}{F_{\text{liquid water out}} + F_{\text{dry sheet out}}} \quad (12-20)$$

The relationship between the total airflow, the dry airflow, and the absolute humidity is given by

$$G_{\text{dry air}} = G_{\text{air}} \frac{1}{1 + Y} = 1000 \text{ kg/h} \frac{1}{1 + 0.01} = 990 \text{ kg/h}$$

The absolute humidity of each airstream is given by

$$Y_{\text{in}} = \frac{G_{\text{water vapor in}}}{G_{\text{dry air in}}} \quad (12-21)$$

$$Y_{\text{out}} = \frac{G_{\text{water vapor out}}}{G_{\text{dry air out}}} \quad (12-22)$$

The mass flow rates of the dry sheet and the liquid water in can be calculated from the overall sheet flow rate and the incoming moisture content:

$$G_{\text{liquid water in}} = G_{\text{sheet}} w_{\text{in}} = (100 \text{ kg/h})(0.2) = 20 \text{ kg/h} \quad (12-23)$$

$$F_{\text{dry sheet}} = F_{\text{sheet}}(1 - w_{\text{in}}) = (100 \text{ kg/h})(0.8) = 80 \text{ kg/h} \quad (12-24)$$

The mass flow rates of the dry air and incoming water vapor can be calculated from the overall air flow rate and the incoming absolute humidity:

$$G_{\text{water vapor in}} = G_{\text{dry air}} Y_{\text{in}} = (990 \text{ kg/h})(0.01) = 9.9 \text{ kg/h} \quad (12-25)$$

To calculate the exiting absolute humidity, Eq. (12-22) is used. But the evaporation rate  $G_{\text{evaporated}}$  is needed. This is calculated from Eqs. (12-16) and (12-20).

$$F_{\text{liquid water out}} = \frac{w_{\text{out}}}{1 - w_{\text{out}}} F_{\text{dry sheet out}} = \frac{0.01}{0.99} \cdot 80 \text{ kg/h} = 0.8 \text{ kg/h} \quad (12-20, \text{ rearranged})$$

$$G_{\text{evaporated}} = F_{\text{liquid water in}} - F_{\text{liquid water out}} = 20 - 0.8 \text{ kg/h} = 19.2 \text{ kg/h} \quad (12-26)$$

Equation (12-18) is now used to calculate the mass flow of water vapor out of the dryer:

$$G_{\text{water vapor out}} = 9.9 \text{ kg/h} + 19.2 \text{ kg/h} = 29.1 \text{ kg/h} \quad (12-27)$$

Now the absolute humidity of the exiting air is readily calculated from Eq. (12-22):

$$Y_{\text{out}} = \frac{G_{\text{water vapor out}}}{G_{\text{dry air}}} = \frac{29.1}{990} = 0.0294 \quad (12-28)$$

Next an energy balance must be used to estimate the outgoing air temperature. The following general equation is used:

$$H_{\text{dry air in}} + H_{\text{water vapor in}} + H_{\text{dry sheet in}} + H_{\text{liquid water in}} = H_{\text{dry air out}} + H_{\text{water vapor out}} + H_{\text{dry sheet out}} + H_{\text{liquid water out}} + \text{heat loss to surroundings} \quad (12-29)$$

Heat losses to the environment are often difficult to quantify, but they can be neglected for a first approximation. This assumption is more valid for large systems than small systems. It is neglected in this example.

Evaluation of the energy balance terms can be done in a couple of ways. Values of the enthalpies above can be calculated by using a consistent reference, or the equation can be rearranged in terms of enthalpy differences. The latter approach will be used here, as shown by Eq. (12-30).

$$\Delta H_{\text{dry air}} + \Delta H_{\text{water vapor}} + \Delta H_{\text{evaporation}} + \Delta H_{\text{liquid water}} + \Delta H_{\text{dry sheet}} = 0 \quad (12-30)$$

The enthalpy change due to evaporation  $\Delta H_{\text{evaporation}}$  is given by  $F_{\text{evaporated}} \lambda_w$ . To evaluate  $\lambda_w$  rigorously, a decision has to be made on the calculational path of the evaporating water since this water is both heating and evaporating. Typically, a two-step path is used—isoenthalpic evaporation and heating of either phase. The incoming liquid water can all be heated to the outlet temperature of the sheet, and then the heat of vaporization at the outlet temperature can be used; or the evaporation can be calculated as occurring at the inlet temperature, and the water vapor is heated from the inlet temperature to the outlet temperature. Alternatively a three-step path based on latent heat at the datum (0°C) may be used. All these methods of calculation are equivalent, since the enthalpy is a state function; but in this case, the second method is preferred since the outlet temperature is unknown. In the calculation, the water will be evaporated at 20°C, heated to the air inlet temperature 170°C, and then cooled to the outlet temperature. Alternatively, this enthalpy change can be calculated directly by using tabular enthalpy values available on the psychrometric chart or Mollier diagram.

The terms in these equations can be evaluated by using

$$\begin{aligned} \Delta H_{\text{dry air}} &= G_{\text{dry air in}} C_{p, \text{air}} (T_{\text{air in}} - T_{\text{air out}}) \\ &= (990.1 \text{ kg/h}) [1 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})] [(170 - T_{\text{air out}}) \text{ kJ/h}] \end{aligned} \quad (12-31)$$

$$\begin{aligned} \Delta H_{\text{water vapor}} &= G_{\text{water vapor out}} C_{p, \text{water vapor}} (T_{\text{air in}} - T_{\text{air out}}) \\ &= (29.1 \text{ kg/h}) [2 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})] [(170 - T_{\text{air out}}) \text{ kJ/h}] \end{aligned} \quad (12-32)$$



## 12-28 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

From steam tables,  $\Delta H_{\text{vap}}$  at  $20^\circ\text{C} = 2454 \text{ kJ/kg}$ ,  $h_l = 84 \text{ kJ/kg}$ , and  $h_g$  at  $170^\circ\text{C}$  (superheated, low pressure)  $= 2820 \text{ kJ/kg}$ .

$$\begin{aligned} -\Delta H_{\text{evaporation}} &= -C_{\text{evaporated}} \cdot \Delta H_{\text{vap}} \\ &= (-19.2 \text{ kg/h}) \\ [2736 \text{ kJ/kg (from steam table)}] \\ &= -52,530 \text{ kJ/h} \end{aligned} \quad (12-33)$$

$$\begin{aligned} \Delta H_{\text{liquid water}} &= F_{\text{liquid water}} C_{p, \text{liquid water}} (T_{\text{sheet, in}} - T_{\text{sheet, out}}) \\ &= (0.8 \text{ kg/h}) [4.18 \text{ kJ/(kg} \cdot ^\circ\text{C)}] [(20^\circ\text{C} - 90^\circ\text{C})] = -234 \text{ kJ/h} \end{aligned} \quad (12-34)$$

$$\begin{aligned} \Delta H_{\text{dry sheet}} &= F_{\text{dry sheet}} C_{p, \text{sheet}} (T_{\text{sheet, in}} - T_{\text{sheet, out}}) \\ &= (80 \text{ kg/h}) [2.5 \text{ kJ/(kg} \cdot ^\circ\text{C)}] [(20^\circ\text{C} - 90^\circ\text{C})] = -14,000 \text{ kJ/h} \end{aligned}$$

Putting this together gives

$$(990.1)(1)(170 - T_{\text{air, out}}) + (29.1)(2)(170 - T_{\text{air, out}}) - 52,530 - 293 - 14,000 = 0$$

$$T_{\text{air, out}} = 106^\circ\text{C}$$

### THERMODYNAMICS

The thermodynamic driving force for evaporation is the difference in chemical potential or water activity between the drying material and the gas phase. Although drying of water is discussed in this section, the same concepts apply analogously for solvent drying.

For a pure water drop, the driving force for drying is the difference between the vapor pressure of water and the partial pressure of water in the gas phase. The rate of drying is proportional to this driving force; please see the discussion on drying kinetics later in this chapter.

$$\text{Rate} \propto (p_{\text{pure}}^{\text{sat}} - p_{w, \text{air}})$$

The activity of water in the gas phase is defined as the ratio of the partial pressure of water to the vapor pressure of pure water, which is also related to the definition of relative humidity.

$$a_{w, \text{vapor}} = \frac{p_w}{p_{\text{pure}}^{\text{sat}}} = \frac{\% \text{RH}}{100}$$

The activity of water in a mixture or solid is defined as the ratio of the vapor pressure of water in the mixture to that of a reference, usually the vapor pressure of pure water. In solids drying or drying of solutions, the vapor pressure (or water activity) is lower than that for pure water. Therefore, the water activity value equals 1 for pure water and  $< 1$  when binding is occurring. This is caused by thermodynamic interactions between the water and the drying material. In many standard drying references, this is called *bound water*.

$$a_w^{\text{solid}} = \frac{p_{\text{mixture}}^{\text{sat}}}{p_{\text{pure}}^{\text{sat}}}$$

When a solid sample is placed into a humid environment, water will transfer from the solid to the air or vice versa until equilibrium is established. At thermodynamic equilibrium, the water activity is equal in both phases:

$$a_w^{\text{vapor}} = a_w^{\text{solid}} = a_w$$

Sorption isotherms quantify how tightly water is bound to a solid. The goal of obtaining a sorption isotherm for a given solid is to measure the equilibrium relationship between the percentage of water in the sample and the vapor pressure of the mixture. The sorption isotherm describes how dry a product can get if contacted with humid air for an infinite amount of time. An example of a sorption isotherm is shown in Fig. 12-11. In the sample isotherm, a feed material dried with 50 percent relative humidity air ( $a_w = 0.5$ ) will approach a moisture content of 10 percent on a dry basis. Likewise, a material kept in a sealed container will create a headspace humidity according to the isotherm; a 7 percent moisture sample in the example below will create a 20 percent relative humidity ( $a_w = 0.2$ ) headspace in a sample jar or package.

Strictly speaking, the equilibrium moisture content of the sample in a given environment should be independent of the initial condition of

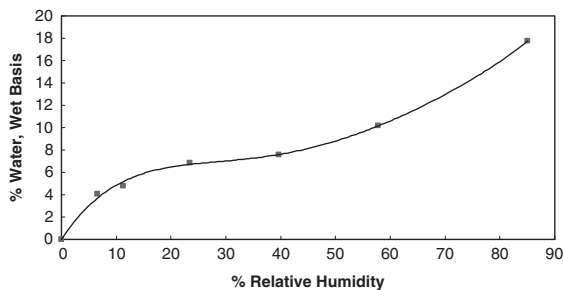


FIG. 12-11 Example of a sorption isotherm (coffee at  $22^\circ\text{C}$ ).

the sample. However, there are cases where the sorption isotherm of an initially wet sample (sometimes called a desorption isotherm) is different from that of an identical, but initially dry sample. This is called hysteresis and can be caused by irreversible changes in the sample during wetting or drying, micropore geometry in the sample, and other factors. Paper products are notorious for isotherm hysteresis. Most materials show little or no hysteresis.

Sorption isotherms cannot generally be predicted from theory. They need to be measured experimentally. The simplest method of measuring a sorption isotherm is to generate a series of controlled-humidity environments by using saturated salt solutions, allow a solid sample to equilibrate in each environment, and then analyze the solid for moisture content.

The basic apparatus is shown in Fig. 12-12, and a table of salts is shown in Table 12-7. It is important to keep each chamber sealed and to be sure that crystals are visible in the salt solution to ensure that the liquid is saturated. Additionally, the solid should be ground into a powder to facilitate mass transfer. Equilibration can take 2 to 3 weeks. Successive moisture measurements should be used to ensure that the sample has equilibrated, i.e., achieved a steady value. Care must be taken when measuring the moisture content of a sample; this is described later in the chapter.

Another common method of measuring a sorption isotherm is to use a dynamic vapor sorption device. This machine measures the weight change of a sample when exposed to humidity-controlled air. A series of humidity points are programmed into the unit, and it automatically delivers the proper humidity to the sample and monitors the weight. When the weight is stable, an equilibrium point is noted and the air humidity is changed to reflect the next setting in the series. When one is using this device, it is critical to measure and record the starting moisture of the sample, since the results are often reported as a percent of change rather than a percent of moisture.

There are several advantages to the dynamic vapor sorption device. First, any humidity value can be dialed in, whereas salt solutions are not available for every humidity value and some are quite toxic. Second, since the weight is monitored as a function of time, it is clear when equilibrium is reached. The dynamic devices also give the sorption/desorption rates, although these can easily be misused (see the drying kinetics section later). The salt solution method, on



FIG. 12-12 Sorption isotherm apparatus. A saturated salt solution is in the bottom of the sealed chamber; samples sit on a tray in the headspace.

TABLE 12-7 Maintenance of Constant Humidity

| Solid phase                                                                       | Max. temp., °C | % Humidity |
|-----------------------------------------------------------------------------------|----------------|------------|
| H <sub>3</sub> PO <sub>4</sub> ·aH <sub>2</sub> O                                 | 24.5           | 9          |
| ZnCl <sub>2</sub> ·aH <sub>2</sub> O                                              | 20             | 10         |
| KC <sub>2</sub> H <sub>3</sub> O <sub>2</sub>                                     | 168            | 13         |
| LiCl·H <sub>2</sub> O                                                             | 20             | 15         |
| KC <sub>2</sub> H <sub>3</sub> O <sub>2</sub>                                     | 20             | 20         |
| KF                                                                                | 100            | 22.9       |
| NaBr                                                                              | 100            | 22.9       |
| CaCl <sub>2</sub> ·6H <sub>2</sub> O                                              | 24.5           | 31         |
| CaCl <sub>2</sub> ·6H <sub>2</sub> O                                              | 20             | 32.3       |
| CaCl <sub>2</sub> ·6H <sub>2</sub> O                                              | 18.5           | 35         |
| CrO <sub>3</sub>                                                                  | 20             | 35         |
| CaCl <sub>2</sub> ·6H <sub>2</sub> O                                              | 10             | 38         |
| CaCl <sub>2</sub> ·6H <sub>2</sub> O                                              | 5              | 39.8       |
| K <sub>2</sub> CO <sub>3</sub> ·2H <sub>2</sub> O                                 | 24.5           | 43         |
| K <sub>2</sub> CO <sub>3</sub> ·2H <sub>2</sub> O                                 | 18.5           | 44         |
| Ca(NO <sub>3</sub> ) <sub>2</sub> ·4H <sub>2</sub> O                              | 24.5           | 51         |
| NaHSO <sub>4</sub> ·H <sub>2</sub> O                                              | 20             | 52         |
| Mg(NO <sub>3</sub> ) <sub>2</sub> ·6H <sub>2</sub> O                              | 24.5           | 52         |
| NaClO <sub>3</sub>                                                                | 100            | 54         |
| Ca(NO <sub>3</sub> ) <sub>2</sub> ·4H <sub>2</sub> O                              | 18.5           | 56         |
| Mg(NO <sub>3</sub> ) <sub>2</sub> ·6H <sub>2</sub> O                              | 18.5           | 56         |
| NaBr·2H <sub>2</sub> O                                                            | 20             | 58         |
| Mg(C <sub>2</sub> H <sub>3</sub> O <sub>2</sub> ) <sub>2</sub> ·4H <sub>2</sub> O | 20             | 65         |
| NaNO <sub>2</sub>                                                                 | 20             | 66         |
| (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub>                                   | 108.2          | 75         |
| (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub>                                   | 20             | 81         |
| NaC <sub>2</sub> H <sub>3</sub> O <sub>2</sub> ·3H <sub>2</sub> O                 | 20             | 76         |
| Na <sub>2</sub> S <sub>2</sub> O <sub>3</sub> ·5H <sub>2</sub> O                  | 20             | 78         |
| NH <sub>4</sub> Cl                                                                | 20             | 79.5       |
| NH <sub>4</sub> Cl                                                                | 25             | 79.3       |
| NH <sub>4</sub> Cl                                                                | 30             | 77.5       |
| KBr                                                                               | 20             | 84         |
| Ti <sub>2</sub> SO <sub>4</sub>                                                   | 104.7          | 84.8       |
| KHSO <sub>4</sub>                                                                 | 20             | 86         |
| Na <sub>2</sub> CO <sub>3</sub> ·10H <sub>2</sub> O                               | 24.5           | 87         |
| K <sub>2</sub> CrO <sub>4</sub>                                                   | 20             | 88         |
| NaBrO <sub>3</sub>                                                                | 20             | 92         |
| Na <sub>2</sub> CO <sub>3</sub> ·10H <sub>2</sub> O                               | 18.5           | 92         |
| Na <sub>2</sub> SO <sub>4</sub> ·10H <sub>2</sub> O                               | 20             | 93         |
| Na <sub>2</sub> HPO <sub>4</sub> ·12H <sub>2</sub> O                              | 20             | 95         |
| NaF                                                                               | 100            | 96.6       |
| Pb(NO <sub>3</sub> ) <sub>2</sub>                                                 | 20             | 98         |
| TiNO <sub>3</sub>                                                                 | 100.3          | 98.7       |
| TiCl <sub>3</sub>                                                                 | 100.1          | 99.7       |

For a more complete list of salts, and for references to the literature, see *International Critical Tables*, vol. 1, p. 68.

the other hand, is significantly less expensive to buy and maintain. Numerous samples can be placed in humidity chambers and run in parallel while a dynamic sorption device can process only one sample at a time.

An excellent reference on all aspects of sorption isotherms is by Bell and Labuza, *Moisture Sorption*, 2d ed., American Association of Cereal Chemists, 2000.

### MECHANISMS OF MOISTURE TRANSPORT WITHIN SOLIDS

Drying requires moisture to travel to the surface of a material. There are several mechanisms by which this can occur:

1. *Diffusion of moisture through solids.* Diffusion is a molecular process, brought about by random wanderings of individual molecules. If all the water molecules in a material are free to migrate, they tend to diffuse from a region of high moisture concentration to one of lower moisture concentration, thereby reducing the moisture gradient and equalizing the concentration of moisture.

2. *Convection of moisture within a liquid or slurry.* If a flowable solution is drying into a solid, then liquid motion within the material brings wetter material to the surface.

3. *Evaporation of moisture within a solid and gas transport out of the solid by diffusion and/or convection.* Evaporation can occur within a solid if it is boiling or porous. Subsequently vapor must move out of the sample.

4. *Capillary flow of moisture in porous media.* The reduction of liquid pressure within small pores due to surface tension forces causes liquid to flow in porous media by capillary action.

### DRYING KINETICS

This section discusses the rate of drying. The kinetics of drying dictates the size of industrial drying equipment, which directly affects the capital and operating costs of a process involving drying. The rate of drying can also influence the quality of a dried product since other simultaneous phenomena can be occurring, such as heat transfer and shrinkage due to moisture loss.

**Drying Curves and Periods of Drying** The most basic and essential kinetic information on drying is a *drying curve*. A drying curve describes the drying kinetics and how they change during drying. The drying curve is affected by the material properties, size or thickness of the drying material, and drying conditions. In this section, the general characteristics of drying curves and their uses are described. Experimental techniques to obtain drying curves are discussed in the "Experimental Methods" section and uses of drying curves for scale-up are discussed in "Dryer Modeling Design and Scale-up."

Several representations of a typical drying curve are shown in Fig. 12-13. The top plot, Fig. 12-13a, is the moisture content (dry basis) as a function of time. The middle plot, Fig. 12-13b, is the drying rate as a function of time, the derivative of the top plot. The bottom plot,

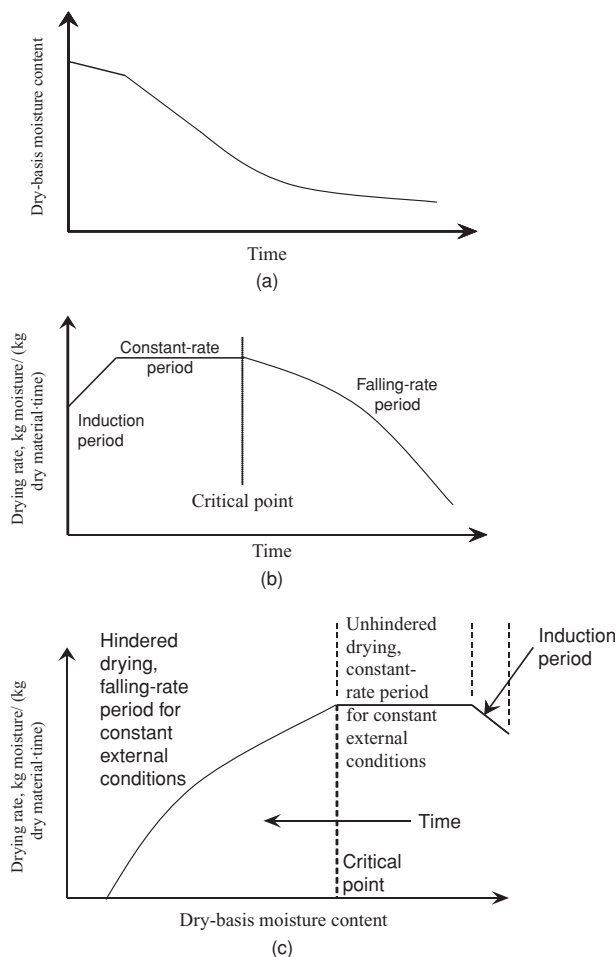


FIG. 12-13 Several common representations of a typical drying curve.

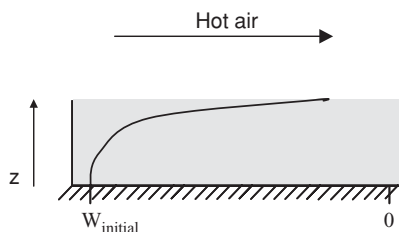


FIG. 12-14 Drying of a slab.

Fig. 12-13c, is the drying rate as affected by the average moisture content of the drying material. Since the material loses moisture as time passes, the progression of time in this bottom plot is from right to left.

Some salient features of the drying curve show the different periods of drying. These are common periods, but not all occur in every drying process. The first period of drying is called the induction period. This period occurs when material is being heated early in drying. The second period of drying is called the constant-rate period. During this period, the surface remains wet enough to maintain the vapor pressure of water on the surface. Once the surface dries sufficiently, the drying rate decreases and the falling-rate period occurs. This period can also be referred to as *hindered drying*.

Figure 12-13 shows the transition between constant- and falling-rate periods of drying occurring at the critical point. The *critical point* refers to the average moisture content of a material at this transition.

The sections below show examples of drying curves and the phenomena that give rise to common shapes.

**Introduction to Internal and External Mass-Transfer Control—Drying of a Slab** The concepts in drying kinetics are best illustrated with a simple example—air drying of a slab. Consider a thick slab of homogeneous wet material, as shown in Fig. 12-14. In this particular example, the slab is dried on an insulating surface under constant conditions. The heat for drying is carried to the surface with hot air, and air carries water vapor from the surface. At the same time, a moisture gradient forms within the slab, with a dry surface and a wet interior. The curved line is the representation of the gradient. At the bottom the slab ( $z = 0$ ), the material is wet and the moisture content is drier at the surface.

The following processes must occur to dry the slab:

1. Heat transfer from the air to the surface of the slab
2. Mass transfer of water vapor from the surface of the slab to the bulk air
3. Mass transfer of moisture from the interior of the slab to the surface of the slab

Depending on the drying conditions, thickness, and physical properties of the slab, any of the above steps can be rate-limiting. Figure 12-15 shows two examples of rate-limiting cases.

The top example shows the situation of *external rate control*. In this situation, the heat transfer to the surface and/or the mass transfer from the surface in the vapor phase is slower than mass transfer to the surface from the bulk of the drying material. In this limiting case, the moisture gradient in the material is minimal, and the rate of drying will be constant as long as the average moisture content remains high enough to maintain a high water activity (see the section on thermodynamics for a discussion of the relationship between moisture content and water vapor pressure). External rate control leads to the observation of a constant-rate period drying curve.

The bottom example shows the opposite situation: *internal rate control*. In the case of heating from the top, internal control refers to a slow rate of mass transfer from the bulk of the material to the surface of the material. Diffusion, convection, and capillary action (in the case of porous media) are possible mechanisms for mass transfer of moisture to the surface of the slab. In the internal rate control situation, moisture is removed from the surface by the air faster than moisture is transported to the surface. This regime is caused by relatively thick layers or high values of the mass- and heat-transfer coefficients in the air. Internal rate control leads to the observation of a falling-rate period drying curve.

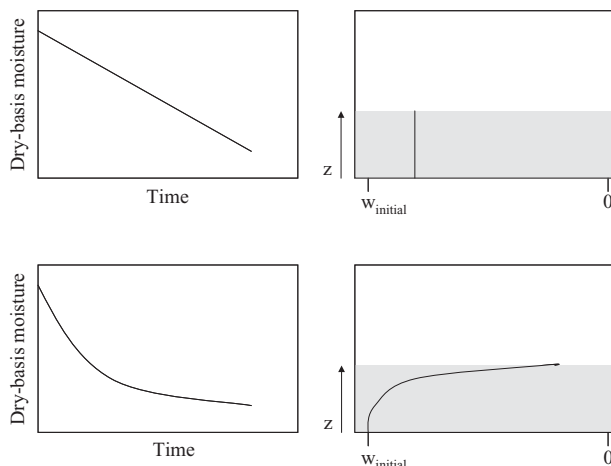


FIG. 12-15 Drying curves and corresponding moisture gradients for situations involving external heat and mass-transfer control and internal mass-transfer control.

Generally speaking, drying curves show both behaviors. When drying begins, the surface is often wet enough to maintain a constant-rate period and is therefore externally controlled. But as the material dries, the mass-transfer rate of moisture to the surface often slows, causing the rate to decrease since the lower moisture content on the surface causes a lower water vapor pressure. However, some materials begin dry enough that there is no observable constant-rate period.

## MATHEMATICAL MODELING OF DRYING

Mathematical models can be powerful tools to help engineers understand drying processes. Models can be either purchased or homemade. Several companies offer software packages to select dryers, perform scale-up calculations, and simulate dryers.

Homemade models are often mass and energy balance spreadsheets, simplified kinetic models, or the simultaneous solution of the convection diffusion and heat equations together with nonlinear isotherms. All levels of models have their place.

This section begins with the most rigorous and numerical models. These models are potentially the most accurate, but require physical property data and simultaneous solution of differential and algebraic equations. Generally speaking, simpler models are more accessible to engineers and easier to implement. They can be very useful as long as the inherent limitations are understood.

**Numerical Modeling of Drying Kinetics** This section summarizes a numerical approach toward modeling drying from a fundamental standpoint. In other words, predictions are made from the appropriate sets of differential and algebraic equations, together with physical properties of the drying medium and drying material. Statistical methods of data analysis, e.g., design of experiments, are not covered.

The approach in this section is lagrangian; i.e., the model is for a drying object (particle, drop, sheet, etc.) as it moves through the drying process in time. More complicated models can use an eulerian frame of reference by simulating the dryer with material moving into and out of the dryer.

The approach taken in this example also assumes that the mechanism of mass transport is by diffusion. This is not always the case and can be significantly incorrect, especially in the case of drying of porous materials.

Any fundamental mathematical model of drying contains mass and energy balances, constitutive equations for mass- and heat-transfer rates, and physical properties. Table 12-8 shows the differential mass balance equations that can be used for common geometries. Note there are two sets of differential mass balances—one including shrinkage and one not including shrinkage. When moisture leaves a drying material, the material can either shrink, or develop porosity, or both.

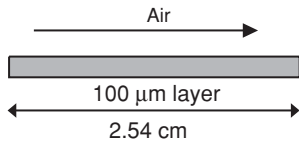
**TABLE 12-8 Mass-Balance Equations for Drying Modeling When Diffusion Is Mass-Transfer Mechanism of Moisture Transport**

| Case                 | Mass balance without shrinkage                                                                                                                                                                                                                                                                                                                                                                                            | Mass balance with shrinkage                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slab geometry        | $\frac{\partial C_w}{\partial t} = \frac{\partial}{\partial z} \left[ \mathcal{D}(w) \frac{\partial C_w}{\partial z} \right]$ <p>At top surface, <math>-\mathcal{D}(w) \frac{\partial C_w}{\partial z}_{\text{top surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At bottom surface, <math>\frac{\partial C_w}{\partial z}_{\text{bottom surface}} = 0</math></p> | $\frac{\partial u}{\partial t} = \frac{\partial}{\partial s} \left[ \mathcal{D}(w) \frac{\partial u}{\partial s} \right] \quad \frac{\partial s}{\partial z} = \rho_s$ <p>At top surface, <math>-\mathcal{D}(w) \frac{\partial u}{\partial s}_{\text{top surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At bottom surface, <math>\frac{\partial u}{\partial s}_{\text{bottom surface}} = 0</math></p>  |
| Cylindrical geometry | $\frac{\partial C_w}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[ r \mathcal{D}(w) \frac{\partial C_w}{\partial r} \right]$ <p>At surface, <math>-\mathcal{D}(w) \frac{\partial C_w}{\partial r}_{\text{surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At center, <math>\frac{\partial C_w}{\partial r}_{\text{center}} = 0</math></p>           | $\frac{\partial u}{\partial t} = \frac{\partial}{\partial s} \left[ \rho_s^2 \mathcal{D}(w) \frac{\partial u}{\partial s} \right] \quad \frac{\partial s}{\partial z} = r \rho_s$ <p>At surface, <math>-\mathcal{D}(w) r \frac{\partial u}{\partial s}_{\text{surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At center, <math>\frac{\partial u}{\partial s}_{\text{surface}} = 0</math></p>            |
| Spherical geometry   | $\frac{\partial C_w}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[ r^2 \mathcal{D}(w) \frac{\partial C_w}{\partial r} \right]$ <p>At surface, <math>-\mathcal{D}(w) \frac{\partial C_w}{\partial r}_{\text{surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At center, <math>\frac{\partial C_w}{\partial r}_{\text{center}} = 0</math></p>       | $\frac{\partial u}{\partial t} = \frac{\partial}{\partial s} \left[ \rho_s^4 \mathcal{D}(w) \frac{\partial u}{\partial s} \right] \quad \frac{\partial s}{\partial z} = r^2 \rho_s$ <p>At surface, <math>-\mathcal{D}(w) r^2 \frac{\partial u}{\partial s}_{\text{surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}</math></p> <p>At center, <math>\frac{\partial u}{\partial s}_{\text{bottom surface}} = 0</math></p> |

The variable  $u$  is the dry-basis moisture content. The equations that include shrinkage are taken from Van der Lijn, doctoral thesis, Wageningen (1976).

The equations in Table 12-8 are insufficient on their own. Some algebraic relationships are needed to formulate a complete problem, as illustrated in Example 16. Equations for the mass- and heat-transfer coefficients are also needed for the boundary conditions presented in Table 12-8. These require the physical properties of the air, the object geometry, and Reynolds number. Example 16 shows the solution for a problem using numerical modeling. This example shows some of the important qualitative characteristics of drying.

**Example 16: Air Drying of a Thin Layer of Paste** Simulate the drying kinetics of 100  $\mu\text{m}$  of paste initially containing 50 percent moisture (wet-basis) with dry air at 60°C, 0 percent relative humidity air at velocities of 1, 10, or 1000 m/s (limiting case) and at 60°C, 0 percent relative humidity air at 1 m/s. The diffusion coefficient of water in the material is constant at  $1 \times 10^{-10} \text{ m}^2/\text{s}$ . The length of the layer in the airflow direction is 2.54 cm.



**Physical property data:** Sorption isotherm data fit well to the following equation:

$$w = 3.10 \left( \frac{\%RH}{100} \right)^5 - 6.21 \left( \frac{\%RH}{100} \right)^4 + 4.74 \left( \frac{\%RH}{100} \right)^3 - 1.70 \left( \frac{\%RH}{100} \right)^2 + 0.378 \left( \frac{\%RH}{100} \right)$$

Solid density = 1150 kg/m<sup>3</sup>

Heat of vaporization = 2450 J/g

Solid heat capacity: 2.5 J/(g·K)

Water heat capacity: 4.184 J/(g·K)

**Solution:** The full numerical model needs to include shrinkage since the material is 50 percent water initially and the thickness will decrease from 100 to 46.5  $\mu\text{m}$  during drying. Assuming the layer is viscous enough to resist convection in the liquid, diffusion is the dominant liquid-phase transport mechanism.

Table 12-8 gives the mass balance equation:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial s} \left[ \mathcal{D}(w) \frac{\partial u}{\partial s} \right] \quad \frac{\partial s}{\partial z} = \rho_s$$

At top surface,

$$-\mathcal{D}(w) \frac{\partial u}{\partial s}_{\text{top surface}} = k_p \frac{P_w^{\text{bulk}} - P_w^{\text{surface}}}{P - P_w^{\text{surface}}}$$

At bottom surface,

$$\frac{\partial u}{\partial s}_{\text{bottom surface}} = 0$$

The temperature is assumed to be uniform through the thickness of the layer.

$$(1 + w_{\text{avg, dry-basis}}) \cdot m_{\text{solids}} C_p \cdot \frac{dT_{\text{layer}}}{dt} = [h(T_{\text{air}} - T_{\text{layer}}) - F \cdot \Delta H_{\text{vap}}] A$$

Mass- and heat-transfer coefficients are given by

$$\text{Nu} = \frac{hL}{k_{\text{air}}} = 0.664 \cdot \text{Re}^{0.5} \cdot \text{Pr}^{0.333}$$

$$\text{Sh} = \frac{k_c L}{\mathcal{D}_{\text{air/water}}} = 0.664 \cdot \text{Re}^{0.5} \cdot \text{Sc}^{0.333}$$

$$k_p = k_c \cdot \rho_{\text{air}}$$

The Reynolds number uses the length of the layer  $L$  in the airflow direction:

$$\text{Re} = \frac{VL\rho_{\text{air}}}{\mu_{\text{air}}}$$

where  $V$  = air velocity.

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The Prandtl and Schmidt numbers, Pr and Sc, for air are given by

$$Pr = \frac{C_{p,air} \mu_{air}}{k_{air}} = 0.70 \quad Sc = \frac{\mu_{air}}{\rho_{air} D_{air/water}} = 0.73$$

The following algebraic equations are also needed:

$$\frac{1}{\rho} = \frac{w}{\rho_w} + \frac{1-w}{\rho_s}$$
 density of wet material (assumes volume additivity)

$$C_w = w \cdot \rho$$
 concentration of water

$$\rho_s = (1-w)\rho$$
 concentration of solids

$$\frac{\%RH}{100} = \frac{P_{w,surface}}{P_{w,sat}}$$
 definition of relative humidity

Antoine equation for vapor pressure of water

$$P_{w,sat} = 0.01 \exp \left( 16.262 - \frac{3800}{226.3 + T_{liquid/solid}} \right)$$

Result: The results of the simulation are shown in Fig. 12-16. The top plot shows the average moisture content of the layer as a function of time, the middle plot shows the drying rate as a function of time, and the bottom plot shows the moisture gradient in each layer after 10 s of drying.

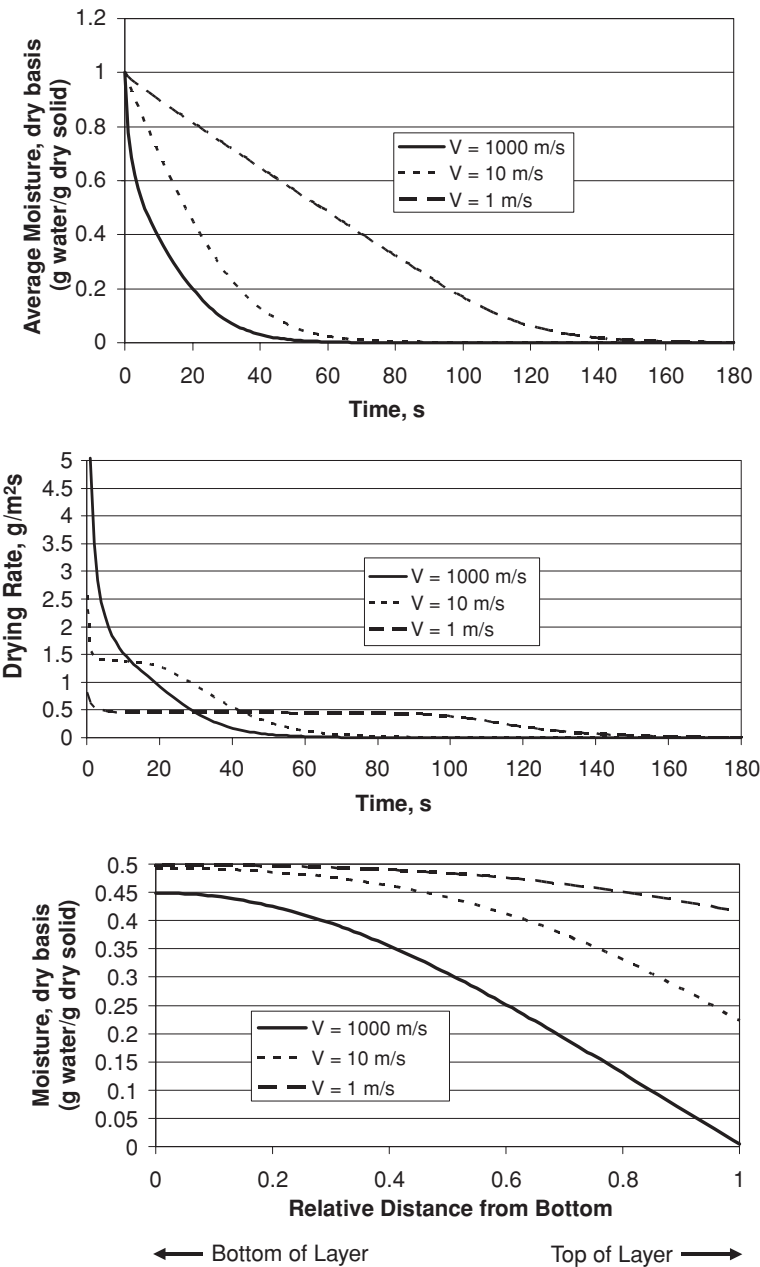


FIG. 12-16 Simulation results for thin layer drying example.

At a velocity of 1 m/s, drying occurs at a constant rate for nearly the entire process; at 10 m/s, drying begins at a high constant rate and then enters a falling-rate period; and at 1000 m/s (limiting case), there is no constant-rate period. These results illustrate the relationships between the external air conditions, drying rate, and moisture gradient. At high air velocity, the drying rate is faster, but becomes limited by internal diffusion and a steep moisture gradient forms. As the air velocity increases, the drying rate becomes less sensitive to air velocity.

The equation set in this example was solved by using a differential-algebraic equation solver called gPROMS from Process Systems Enterprises (www.pse.com). It can also be solved with other software and programming languages such as FORTRAN. Example 16 is too complicated to be done on a spreadsheet.

**Simplified Kinetic Models** This section presents several examples of simplified kinetic models. A model of the constant-rate period is shown in Example 17. During the constant-rate period, the drying rate is controlled by gas-phase mass and heat transfer. This is easier than modeling the falling-rate period, since the properties of air and water (or other gas-phase molecules) are well understood. Modeling the falling-rate period requires knowledge of and/or assumptions about the physical properties of the drying material.

**Example 17: Drying a Pure Water Drop** (Marshall, *Atomization & Spray Drying*, 1986.) Calculate the time to dry a drop of water, given the air temperature and relative humidity as a function of drop size.

**Solution:** Assume that the drop is drying at the wet-bulb temperature. Begin with an energy balance [Eq. (12-35)]

$$\text{Mass flux} = \frac{h(T_{\text{air}} - T_{\text{drop}})}{\Delta H_{\text{vap}}} \quad (12-35)$$

Next, a mass balance is performed on the drop. The change in mass equals the flux times the surface area.

$$\rho \frac{dV_{\text{droplet}}}{dt} = -A \cdot \text{mass flux} \quad (12-36)$$

Evaluating the area and volume for a sphere gives

$$\rho \cdot 4\pi R^2 \frac{dR}{dt} = -4\pi R^2 \cdot \text{mass flux} \quad (12-37)$$

Combining Eqs. (12-35) and (12-37) and simplifying gives

$$\rho \frac{dR}{dt} = \frac{-h(T_{\text{air}} - T_{\text{drop}})}{\Delta H_{\text{vap}}} \quad (12-38)$$

A standard correlation for heat transfer to a sphere is given by (Ranz and Marshall, 1952)

$$\text{Nu} = \frac{h(2R)}{k_{\text{air}}} = 2 + 0.6 \cdot \text{Re}^{0.5} \text{Pr}^{0.33} \quad (12-39)$$

For small drop sizes or for stagnant conditions, the Nusselt number has a limiting value of 2.

$$\text{Nu} = \frac{h(2R)}{k_{\text{air}}} = 2 \quad (12-40)$$

$$h = \frac{k_{\text{air}}}{R} \quad (12-41)$$

Inserting into Eq. (12-38) gives

$$R \frac{dR}{dt} = \frac{k_{\text{air}}(T_{\text{air}} - T_{\text{drop}})}{\rho \Delta H_{\text{vap}}} \quad (12-42)$$

Integration yields

$$\frac{R^2}{2} - \frac{R_0^2}{2} = \frac{k_{\text{air}}(T_{\text{air}} - T_{\text{drop}})t}{\rho \Delta H_{\text{vap}}} \quad (12-43)$$

where  $R_0$  = initial drop radius, m.

Now the total lifetime of a drop can be calculated from Eq. (12-43) by setting  $R = 0$ :

$$t = \frac{\rho \Delta H_{\text{vap}} R_0^2}{2k_{\text{air}}(T_{\text{air}} - T_{\text{drop}})} \quad (12-44)$$

The effects of drop size and air temperature are readily apparent from Eq. (12-44). The temperature of the drop is the wet-bulb temperature and can be obtained from a psychrometric chart, as described in the previous section. Sample results are plotted in Fig. 12-17.

The above solution for drying of a pure water drop cannot be used to predict the drying rates of drops containing solids. Drops containing solids will not shrink uniformly and will develop internal concentration gradients (falling-rate period) in most cases.

Modeling of the falling-rate period is usually done by treating the drying problem as a diffusion problem, where the rate-limiting step is the diffusion of moisture from deep within the solid to the surface.

One of the attractions of treating drying as a diffusion problem is its relative simplicity compared with more complex models for moisture movement. This renders the approach tractable for hand calculations, and these calculations are often appropriate given the wide variability in diffusion coefficients and permeabilities both within and between

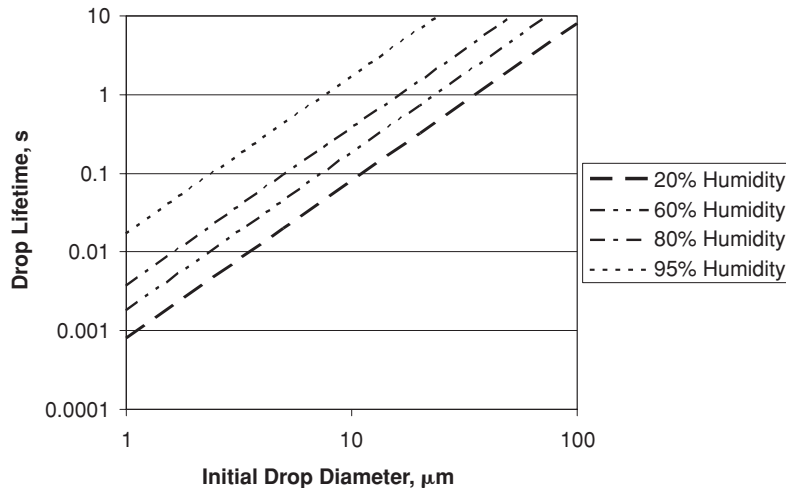


FIG. 12-17 Drying time of pure water drops as function of relative humidity at 25°C.



materials. The simplicity of this approach is also useful when one is optimizing processing conditions, where the number of calculations, even with modern workstations, is considerable. Moreover, this diffusion approach works well for predicting both average moisture contents and moisture-content profiles for some materials.

The three main driving forces which have been used within diffusion models (moisture content, partial pressure of water vapor, and chemical potential) will now be discussed. Attempts to predict diffusion coefficients theoretically will also be reviewed, together with experimental data for fitted diffusion coefficients and their dependence on temperature and moisture content.

Waananen et al. (1993), in their review of drying models, note that most models in their final form express the driving force for moisture movement in terms of a moisture concentration gradient. However, the true potential for transfer may be different, namely, differences in chemical potential, as explored in greater detail by Keey et al. (2000). In theory, the diffusion coefficient will be independent of moisture concentration only if the moisture is unbound, but concentration-independent diffusion coefficients have been successfully used in some cases over a wide range of moisture contents.

Since the true driving force is the chemical potential difference, transfer will occur between two moist bodies in the direction of falling chemical potential rather than decreasing moisture content. Moisture may flow from the drier body to the wetter one.

At low moisture contents, Perré and Turner (1996) suggest that there seems to be little difference between the predictions of drying models with driving forces based on gradients in chemical potential, moisture content, and partial pressure of water vapor, indicating that the simplest approach (a moisture content driving force) might be most practical. The majority of work involving the use of diffusion models has used moisture content driving forces. Hence, there is some empirical support for the use of moisture content driving forces.

In this model, described by Fick's second law, we have

$$\frac{\partial X}{\partial t} = \frac{\partial}{\partial z} \left( \mathcal{D} \frac{\partial X}{\partial z} \right) \quad (12-45)$$

where  $X$  is the free moisture content above the equilibrium moisture content,  $t$  is time,  $z$  is the distance coordinate perpendicular to the airstream, and  $\mathcal{D}$  is the diffusion coefficient. Sherwood (1929) was the first to use this approach, and he made the following additional assumptions:

- The diffusion coefficient  $\mathcal{D}$  is constant.
- The initial moisture content in the material is uniform.
- Surface material comes into equilibrium with the surrounding air instantaneously, so that the resistance of the boundary layer outside the material is negligible.

Translated to mathematical terms, the last two of these assumptions are

$$t = 0 \quad X = X_i \quad (12-46)$$

$$z = \delta \quad X = X_e \quad (\text{at the surface}) \quad (12-47)$$

where  $\delta$  is the half thickness of the board. This approach allowed Eq. (12-45) to be integrated to yield a predicted moisture content profile. This moisture content profile may be integrated to give average moisture contents, with the characteristic moisture content  $\Phi$  being defined as before,  $\Phi = (\bar{X} - X_e)/(X_i - X_e)$ , where  $\bar{X}$  is the volume-averaged moisture content and  $X_i$  and  $X_e$  are the initial and equilibrium moisture contents, respectively. The equation for the characteristic moisture content is

$$\Phi = \frac{8}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp \left[ - \left( \frac{2n+1}{2} \right)^2 \pi^2 \tau \right] \quad (12-48)$$

With this model, a characteristic parameter which governs the extent of drying is the mass-transfer Fourier number  $\tau$ , defined as follows:

$$\tau = \frac{\mathcal{D}t}{\delta^2} \quad (12-49)$$

If drying is controlled by diffusion, then for the same drying conditions, doubling the thickness of the material should increase the drying time to the same final moisture content fourfold.

If the diffusion coefficient is constant, the moisture content profile through a material for the steady-state movement of moisture through it would be linear. However, drying is not a steady-state process. When the moisture content change occurs over almost the entire half thickness of the material, in other words when the size of the fully wet region is very small, the moisture content profiles can be shown to be parabolic during drying if the diffusion coefficient is constant.

The surface of the material does not necessarily come instantly to equilibrium. The surface of the material is only at equilibrium with the drying air during the falling-rate period. Although dry patches have been seen and photographed on the surface of moist granular beds as they dry out (Oliver and Clarke, 1973), fine porous material can have a significant fraction of its exposed surface dry before the evaporation from the whole surface is affected (Suzuki et al., 1972; Schlinder, 1988) due to the buffering effect of the external boundary layer.

**Concept of a Characteristic Drying Rate Curve** In 1958, van Meel observed that the drying rate curves, during the falling-rate period, for a specific material often show the same shape (Figs. 12-18 and 12-19), so that a single characteristic drying curve can be drawn for the material being dried. Strictly speaking, the concept should only

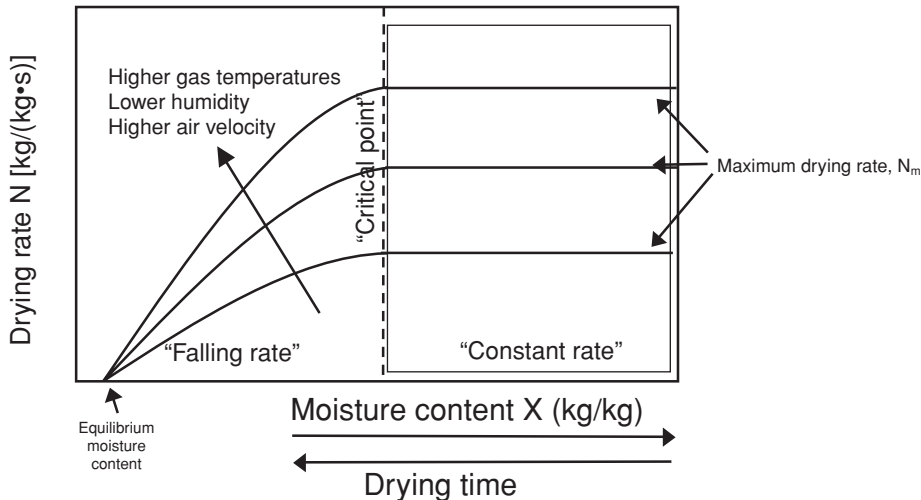


FIG. 12-18 Drying curves for a given material at different constant external conditions.

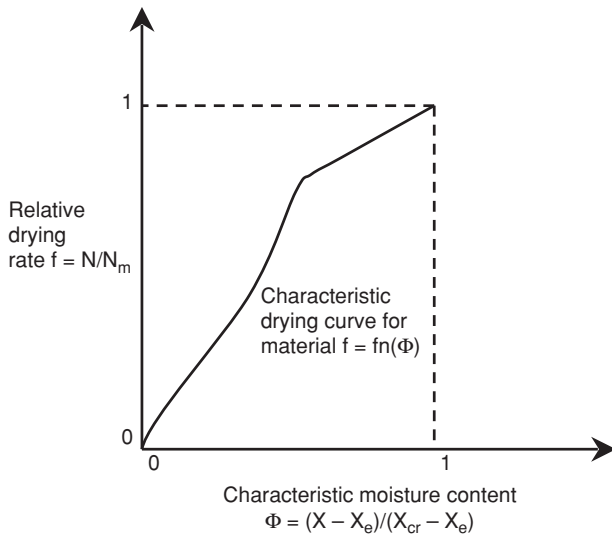


FIG. 12-19 Characteristic drying curve.

apply to materials of the same specific size (surface area to material ratio) and thickness, but Keey (1992) shows evidence that it applies over a somewhat wider range with reasonable accuracy. In the absence of experimental data, a linear falling-rate curve is often a reasonable first guess for the form of the characteristic function (good approximation for milk powder, fair for ion-exchange resin, silica gel). At each volume-averaged, free moisture content, it is assumed that there is a corresponding specific drying rate relative to the unhindered drying rate in the first drying period that is independent of the external drying conditions. *Volume-averaged* means averaging over the volume (distance cubed for a sphere) rather than just the distance. The relative drying rate is defined as

$$f = \frac{N}{N_m} \quad (12-50)$$

where  $N$  is the drying rate,  $N_m$  is the rate in the constant-rate period, and the characteristic moisture content becomes

$$\Phi = \frac{\bar{X} - X_e}{X_{cr} - X_e} \quad (12-51)$$

where  $\bar{X}$  is the volume-averaged moisture content,  $X_{cr}$  is the moisture content at the critical point, and  $X_e$  is that at equilibrium. Thus, the drying curve is normalized to pass through the point (1,1) at the critical point of transition in drying behavior and the point (0,0) at equilibrium.

This representation leads to a simple lumped-parameter expression for the drying rate in the falling-rate period, namely,

$$N = fN_m = f[k\phi_m(Y_w - Y_G)] \quad (12-52)$$

Here  $k$  is the external mass-transfer coefficient,  $\phi_m$  is the humidity-potential coefficient (corrects for the humidity not being a strictly true representation of the driving force; close to unity most of the time),  $Y_w$  is the humidity above a fully wetted surface, and  $Y_G$  is the bulk-gas humidity. Equation (12-52) has been used extensively as the basis for understanding the behavior of industrial drying plants owing to its simplicity and the separation of the parameters that influence the drying process: the material itself  $f$ , the design of the dryer  $k$ , and the process conditions  $\phi_m(Y_w - Y_G)f$ .

For example, suppose (with nonhygroscopic solids,  $X_e = 0$  kg/kg) that we have a linear falling-rate curve, with a maximum drying rate  $N_m$  of 0.5 kg moisture/(kg dry solids · s) from an initial moisture content of 1 kg moisture/kg dry solids. If the drying conditions around the sample are constant, what is the time required to dry the material to a moisture content of 0.2 kg moisture/kg dry solids?

$$N = \frac{X}{1 \text{ kg kg}^{-1}} 0.5 \text{ kg kg}^{-1} \text{ s}^{-1}$$

Given that the drying rate  $dX/dt$  is equal to  $N$ , we have

$$t = \int_{0.2}^1 \frac{dX}{N} = \int_{0.2}^1 \frac{dX}{X(0.5)} = 2 \int_{0.2}^1 \frac{dX}{X} = 2 \ln \frac{1}{0.2} = 3.21 \text{ s} \quad (12-53)$$

The characteristic drying curve, however, is clearly a gross approximation. A common drying curve will be found only if the volume-averaged moisture content reflects the moistness of the surface in some fixed way.

For example, in the drying of impermeable timbers, for which the surface moisture content reaches equilibrium quickly, there is unlikely to be any significant connection between the volume-averaged and the surface moisture contents, so the concept is unlikely to apply. While the concept might not be expected to apply to the same material with different thickness, e.g., Pang finds that it applies for different thicknesses in the drying of softwood timber (Keey, 1992), its applicability appears to be wider than the theory might suggest. A paper by Kemp and Oakley (2002) explains that many of the errors in the assumptions in this method often cancel out, meaning that the concept has wide applicability.

Keey and Suzuki (1974) have explored the conditions for which a characteristic curve might apply, using a simplified analysis based on an evaporative front receding through a porous mass. Their analysis shows that a unique curve pertains only when the material is thinly spread and the permeability to moisture is large. Internal diffusion often controls drying as the material becomes very dry, but the result of Keey and Suzuki suggests that the uniqueness of the curve, in theory, depends on drying not being significantly controlled by internal diffusion. One might expect, then, to find characteristic drying curves for small, microporous particles dried individually, and there is a sufficient body of data to suggest that a characteristic drying curve may be found to describe the drying of discrete particles below 20 mm in diameter over a range of conditions that normally exist within a commercial dryer. Nevertheless, Kemp and Oakley (1992) find that many of the deviations from the assumptions, in practice, cancel out, so that the limitation suggested by Keey and Suzuki (diffusion not controlling) is not as severe as might be expected.

An example of the application of a linear characteristic drying curve is given in the section on rotary dryers.

## EXPERIMENTAL METHODS

Lab-, pilot-, and plant-scale experiments all play important roles in drying research. Lab-scale experiments are often necessary to study product characteristics and physical properties; pilot-scale experiments are often used in proof-of-concept process tests and to generate larger quantities of sample material; and plant-scale experiments are often needed to diagnose processing problems and to start or change a full-scale process.

**Measurement of Drying Curves** Measuring and using experimental drying curves can be difficult. Typically, this is a three-step process. The first step is to collect samples at different times of drying, the second step is to analyze each sample for moisture, and the third step is to interpret the data to make process decisions.

Solid sample collection techniques depend on the type of dryer. Since a drying curve is the moisture content as a function of time, it must be possible to obtain material before the drying process is complete. There are several important considerations when sampling material for a drying curve:

1. The sampling process needs to be fast relative to the drying process. Drying occurring during or after sampling can produce misleading results. Samples must be sealed prior to analysis. Plastic bags do not provide a sufficient seal.

2. In heterogeneous samples, the sample must be large enough to accurately represent the composition of the mixture.

Table 12-9 outlines some sampling techniques for various dryer types. Moisture measurement techniques are critical to the successful collection and interpretation of drying data. The key message of this section is that the moisture value almost certainly depends on the measurement technique and that it is essential to have a *consistent*



**TABLE 12-9 Sample Techniques for Various Dryer Types**

| Dryer type                     | Sampling method                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fluid bed dryer<br>Sheet dryer | Sampling cup (see Fig. 12-20)<br>Collect at end of dryer. Increase speed to change the drying time.                                                                                       |
| Tray dryer<br>Indirect dryer   | Record initial moisture and mass of tray with time.<br>Decrease residence time with higher flow rate and sample at exit.                                                                  |
| Spray dryer                    | Residence time of product is difficult to determine and change. Special probes have been developed to sample partially dried powder in different places within the dryer (ref. Langrish). |

*technique* when measuring moisture. Table 12-10 compares and contrasts some different techniques for moisture measurement.

The most common method is gravimetric ("loss-on-drying"). A sample is weighed in a sample pan or tray and placed into an oven or heater at some high temperature for a given length of time. The sample is weighed again after drying. The difference in weight is then assumed to be due to the complete evaporation of water from the sample. The sample size, temperature, and drying time are all important factors. A very large or thick sample may not dry completely in the given time; a very small sample may not accurately represent the composition of a heterogeneous sample. A low temperature can fail to completely dry the sample, and a temperature that is too high can burn the sample, causing an artificially high loss of mass.

Usually, solid samples are collected as described, but in some experiments, it is more convenient to measure the change in humidity of the air due to drying. This technique requires a good mass balance of the system and is more common in lab-scale equipment than pilot- or plant-scale equipment.

**Performing a Mass and Energy Balance on a Large Industrial Dryer** Measuring a mass and energy balance on a large dryer is often necessary to understand how well the system is operating and how much additional capacity may be available. This exercise can also be used to detect and debug gross problems, such as leaks and product buildup.

There are several steps to this process.

1. Draw a sketch of the overall process including all the flows of mass into and out of the system. Look for places where air can leak into or out of the system. There is no substitute for physically walking around the equipment to get this information.

2. Decide on the envelope for the mass and energy balance. Some dryer systems have hot-air recycle loops and/or combustion or steam heating systems. It is not always necessary to include these to understand the dryer operation.

3. Decide on places to measure airflows and temperatures and to take feed and product samples. Drying systems and other process equipment are frequently not equipped for such measurements; the system may need minor modification, such as the installation of ports into pipes for pitot tubes or humidity probes. These ports must not leak when a probe is in place.

4. Take the appropriate measurements and calculate the mass and energy balances.

The measurements are inlet and outlet temperatures, humidities, and flow rates of the air inlets and outlets as well as the moisture and temperature of the feed and dry solids. The following are methods for each of the measurements:

**Airflow Rate** This is often the most difficult to measure. Fan curves are often available for blowers but are not always reliable. A small pitot tube can be used (see Sec. 22, "Waste Management," in this Handbook) to measure local velocity. The best location to use a pitot tube is in a straight section of pipe. Measurements at multiple positions in the cross section of the pipe or duct are advisable, particularly in laminar flow or near elbows and other flow disruptions.

**Air Temperature** A simple thermocouple can be used in most cases, but in some cases special care must be taken to ensure that wet or sticky material does not build up on the thermocouple. A wet thermocouple will yield a low temperature from evaporative cooling.

**Air Humidity** Humidity probes need to be calibrated before use, and the absolute humidity (or both the relative humidity and temperature) needs to be recorded. If the probe temperature is below the dew point of the air in the process, then condensation on the probe will occur until the probe heats.

**Feed and Exit Solids Rate** These are generally known, particularly for a unit in production. Liquids can be measured by using a bucket and stopwatch. Solids can be measured in a variety of ways.

**Feed and Exit Solids Moisture Content** These need to be measured using an appropriate technique, as described above. Use the same method for both the feed and exit solids. Don't rely on formula sheets for feed moisture information.

Figure 12-20 shows some common tools used in these measurements.

## DRYING OF NONAQUEOUS SOLVENTS

**Practical Considerations** Removal of nonaqueous solvents from a material presents several practical challenges. First, solvents are often flammable and require drying either in an inert environment, such as superheated steam or nitrogen, or in a gas phase comprised solely of solvent vapor. The latter will occur in indirect or

**TABLE 12-10 Moisture Determination Techniques**

| Method                              | Principle                                                                                                                                                                  | Advantages                                                                                                                                     | Disadvantages                                                                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gravimetric (loss on drying)        | Water evaporates when sample is held at a high temperature. Difference in mass is recorded.                                                                                | Simple technique. No extensive calibration methods are needed. Lab equipment is commonly available.                                            | Method is slow. Measurement time is several minutes to overnight (depending on material and accuracy). Generally not suitable for process control. Does not differentiate between water and other volatile substances. |
| IR/NIR                              | Absorption of infrared radiation by water is measured.                                                                                                                     | Fast method. Suitable for very thin layers or small particles.                                                                                 | Only surface moisture is detected. Extensive calibration is needed.                                                                                                                                                    |
| RF/microwave                        | Absorption of RF or microwave energy is measured.                                                                                                                          | Fast method. Suitable for large particles.                                                                                                     | Extensive calibration is needed.                                                                                                                                                                                       |
| Equilibrium relative humidity (ERH) | The equilibrium relative humidity headspace above sample in a closed chamber is measured. Sorption isotherm is used to determine moisture.                                 | Relatively quick method. Useful particularly if a final moisture specification is in terms of water activity (to retard microorganism growth). | May give misleading results since the surface of the material will equilibrate with the air. Large particles with moisture gradients can give falsely low readings. Measurement of relative humidity can be imprecise. |
| Karl Fischer titration              | Chemical titration that is water-specific. Material can be either added directly to a solvent or heated in an oven, with the headspace purged and bubbled through solvent. | Specific to water only and very precise. Units can be purchased with an autosampler. Measurement takes only a few minutes.                     | Equipment is expensive and requires solvents. Minimal calibration required. Sample size is small, which may pose a problem for heterogeneous mixtures.                                                                 |

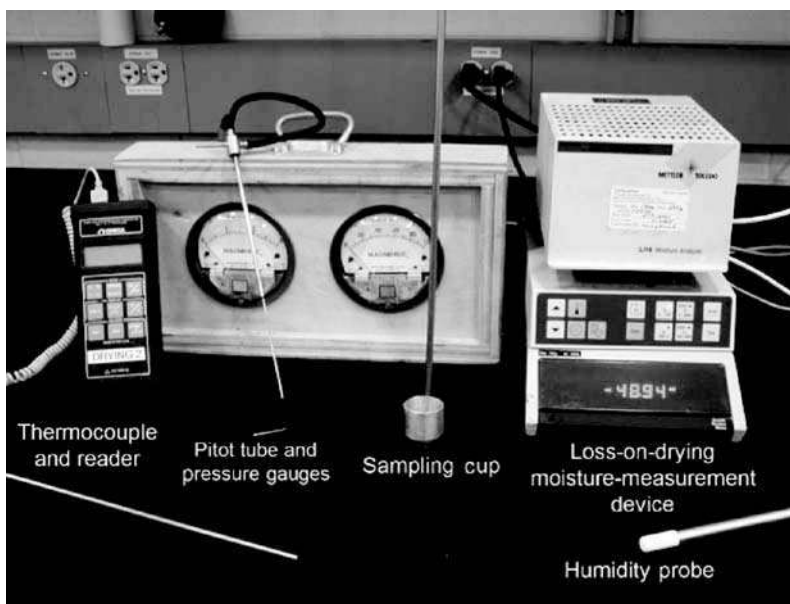


FIG. 12-20 Variety of tools used to measure mass and energy balances on dryers.

vacuum drying equipment. Second, the solvent vapor must be collected in an environmentally acceptable manner.

An additional practical consideration is the remaining solvent content that is acceptable in the final product. Failure to remove all the solvent can lead to problems such as toxicity of the final solid or can cause the headspace of packages, such as drums, to accumulate solvent vapor.

**Physical Properties** The physical properties that are important in solvent drying are the same as those for an aqueous system. The vapor pressure of a solvent is the most important property since it provides the thermodynamic driving force for drying. Acetone (BP 57°C), for example, can be removed from a solid at atmospheric pressure readily by boiling, but glycerol (BP 200°C) will dry only very slowly. Like water, a solvent may become bound to the solid and have a lower vapor pressure. This effect should be considered when one is designing a solvent-drying process.

**Example 18: Preparation of a Psychrometric Chart** Make a psychrometric chart for dipropylene glycol. It has a molecular weight of 134.2 g/mol and a normal boiling temperature of 228°C, and the latent heat of vaporization is 65.1 kJ/mol.

The Clausius-Clapeyron equation can be used to estimate the vapor pressure of dipropylene glycol as a function of temperature, with the boiling temperature as a reference.

$$\ln \frac{P_{T_1}^{\text{sat}}}{P_{T_2}^{\text{sat}}} = \frac{-\Delta H_{\text{vap}}}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$$

where  $P_{T_1}^{\text{sat}}$ ,  $P_{T_2}^{\text{sat}}$  = vapor pressures of the solvent at absolute temperatures  $T_1$  and  $T_2$

$$\Delta H_{\text{vap}} = \text{latent heat of vaporization, J/mol}$$

$$R = \text{gas constant, } 8.314 \text{ J/(mol} \cdot \text{K)}$$

Since the boiling temperature is 228°C, 501.15 K and 1 bar were used as  $T_2$  and  $P_2$ . The latent heat value is 65.1 kJ/mol.

Once the vapor pressure of dipropylene glycol is known at a given temperature, the mass of dipropylene glycol/mass of dry air can be calculated. Since dipropylene glycol is the only liquid, the partial pressure of dipropylene glycol equals the vapor pressure.

$$P_{\text{dipropylene glycol}} = P_{\text{dipropylene glycol}}^{\text{sat}}$$

The mole fraction of dipropylene glycol is the partial pressure divided by the total system pressure, taken to equal 1 bar.

$$y_{\text{dipropylene glycol}} = \frac{P_{\text{dipropylene glycol}}}{P}$$

The saturation mole ratio of dipropylene glycol to air is given by the following.

$$\text{Mole ratio} = \frac{y_{\text{dipropylene glycol}}}{1 - y_{\text{dipropylene glycol}}}$$

The saturation mass ratio of dipropylene glycol to air is calculated by multiplying by the molecular weights. The mass ratio as a function of temperature gives the saturation curve, as shown in Fig. 12-21.

$$\begin{aligned} \text{Saturation mass ratio} &= \frac{\text{g dipropylene glycol}}{\text{g dry air}} \\ &= \frac{\text{molecular weight of dipropylene glycol}}{\text{molecular weight of dry air}} \cdot \frac{y_{\text{dipropylene glycol}}}{1 - y_{\text{dipropylene glycol}}} \end{aligned}$$

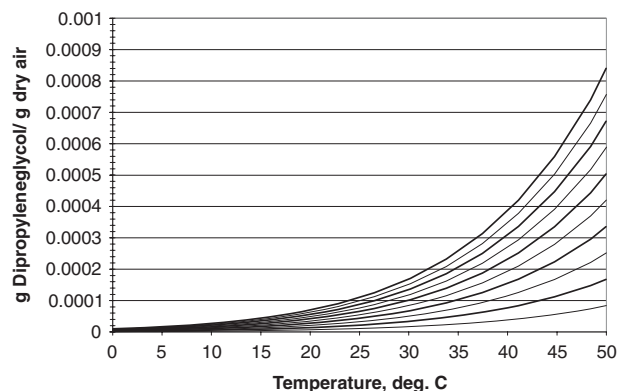
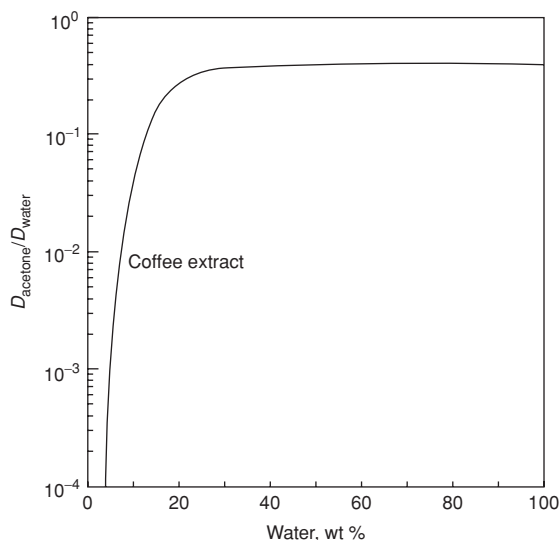


FIG. 12-21 An example of a solvent psychrometric chart.



**FIG. 12-22** The ratio of the diffusion coefficients of acetone to water in instant coffee as a function of moisture content (taken from Thijssen et al., *De Ingenieur*, JRG, 80, Nr. 47 (1968)]. Acetone has a much higher vapor pressure than water, but is selectively retained in coffee during drying.

Each relative humidity curve is proportional to the saturation value.

$$\text{Mass ratio} = \left( \frac{\%RH}{100} \right) \cdot \text{saturation mass ratio}$$

Diffusion of nonaqueous solvents through a material can be slow. The diffusion coefficient is directly related to the size of the diffusing molecule, so molecules larger than water typically have diffusion coefficients that have a much lower value. This phenomenon is known as *selective diffusion*. Large diffusing molecules can become kinetically trapped in the solid matrix. Solvents with a lower molecular weight will often evaporate from a material faster than a solvent with a higher molecular weight, even if the vapor pressure of the larger molecule is higher. Some encapsulation methods rely on selective diffusion; an example is instant coffee production using spray drying, where volatile flavor and aroma components are retained in particles more than water, even though they are more volatile than water, as shown in Fig. 12-22.

## PRODUCT QUALITY CONSIDERATIONS

**Overview** The drying operation usually has a very strong influence on final product quality and product performance measures. And the final product quality strongly influences the value of the product. Generally, a specific particle or unit size, a specific density, a specific color, and a specific target moisture are desired. Naturally every product is somewhat different, but these are usually the first things we need to get right.

**Target Moisture** This seems obvious, but it's very important to determine the right moisture target before we address other drying basics. Does biological activity determine the target, flowability of the powder, shelf life, etc.? Sometimes a very small (1 to 2 percent) change in the target moisture will have a very big impact on the size of the dryer required. This is especially true for difficult-to-dry products with flat falling-rate drying characteristics. Therefore, spend the time necessary to get clear on what really determines the moisture target. And as noted earlier in this subsection, care should be taken to define a moisture measurement method since results are often sensitive to the method.

**Particle Size** Generally a customer or consumer wants a very specific particle size—and the narrower the distribution, the better. No one wants lumps or dust. The problem is that some attrition and

sometimes agglomeration occur during the drying operation. We may start out with the right particle size, but we must be sure the dryer we've selected will not adversely affect particle size to the extent that it becomes a problem. And some dryers will treat particles more gently than others. Particle size is also important from a segregation standpoint. See Sec. 18, "Solid-Solid Operations and Equipment." And of course fine particles can also increase the risk of fire or explosion.

**Density** Customers and consumers are generally also very interested in getting the product density they have specified or expect. If the product is a consumer product and going into a box, then the density needs to be correct to fill the box to the appropriate level. If density is important, then product shrinkage during drying can be an important harmful transformation to consider. This is particularly important for biological products for which shrinkage can be very high. This is why freeze drying can be the preferred dryer for many of these materials.

**Solubility** Many dried products are rewet either during use by the consumer or by a customer during subsequent processing. Shrinkage can again be a very harmful transformation. Many times shrinkage is a virtually irreversible transformation which creates an unacceptable product morphology. Case hardening is a phenomenon that occurs when the outside of the particle or product initially shrinks to form a very hard and dense skin that does not easily rewet. A common cause is capillary collapse, discussed along with shrinkage below.

**Flowability** If we're considering particles, powders, and other products that are intended to flow, then this is a very important consideration. These materials need to easily flow from bins, hoppers, and out of boxes for consumer products. Powder flowability is a measurable characteristic using rotational shear cells (Peschl) or translational shear cells (Jenike) in which the powder is consolidated under various normal loads, and then the shear force is measured, enabling a complete yield locus curve to be constructed. This can be done at various powder moistures to create a curve of flowability versus moisture content. Some minimal value is necessary to ensure free flow. Additional information on these devices and this measure can be found in Sec. 21, "Solid-Solid Operations and Processing."

**Color** Product color is usually a very important product quality attribute, and a change in color can be caused by several different transformations.

**Transformations Affecting Product Quality** Drying, as with any other unit operation, has both productive and harmful transformations that occur. The primary productive transformation is water removal of course, but there are many harmful transformations that can occur and adversely affect product quality. The most common of these harmful transformations includes product shrinkage; attrition or agglomeration; loss of flavor, aroma, and nutritional value; browning reactions; discoloration; stickiness; and flowability problems. These were discussed briefly above, but are worth a more in-depth review.

**Shrinkage** Shrinkage is a particularly important transformation with several possible mechanisms to consider. It's usually especially problematic with food and other biological materials, but is a very broadly occurring phenomenon. Shrinkage generally affects solubility, wettability, texture and morphology, and absorptency. It can be observed when drying lumber when it induces stress cracking and during the drying of coffee beans prior to roasting. Tissue, towel, and other paper products undergo some shrinkage during drying. And many chemical products shrink as water evaporates, creating voids and capillaries prone to collapse as additional water evaporates. As we consider capillary collapse, there are several mechanisms worth mentioning.

**Surface tension**—the capillary suction created by a receding liquid meniscus can be extremely high.

**Plasticization**—an evaporating solvent which is also a plasticizer of polymer solute product will lead to greater levels of collapse and shrinkage.

**Electric charge effects**—the van der Waals and electrostatic forces can also be a strong driver of collapse and shrinkage.

**Surface Tension** These effects are very common and worth a few more comments. Capillary suction created by a receding liquid meniscus can create very high pressures for collapse. The quantitative

expression for the pressure differential across a liquid-fluid interface was first derived by Laplace in 1806. The meniscus, which reflects the differential, is affected by the surface tension of the fluid. Higher surface tensions create greater forces for collapse. These strong capillary suction pressures can easily collapse a pore. We can reduce these suction pressures by using low-surface-tension fluids or by adding surfactants, in the case of water, which will also significantly reduce surface tension (from 72 to 30 dyn/cm).

The collapse can also be reduced with some dryer types. Freeze drying and heat pump drying can substantially reduce collapse, but of course, the capital cost of these dryers sometimes makes them prohibitive. At the other extreme, dryers which rapidly flash off the moisture can reduce collapse. This mechanism can also be affected by particle size such that the drying is primarily boundary-layer-controlled. When the particle size becomes sufficiently small, moisture can diffuse to the surface at a rate sufficient to keep the surface wetted. This has been observed in a gel-forming food material when the particle size reached 150 to 200  $\mu\text{m}$  (Genskow, "Considerations in Drying Consumer Products," *Proceedings International Drying Symposium*, Versailles, France, 1988).

**Biochemical Degradation** Biochemical degradation is another harmful transformation that occurs with most biological products. There are four key reactions to consider: lipid oxidation, Maillard browning, protein denaturation, and various enzyme reactions. These reactions are both heat- and moisture-dependent such that control of temperature and moisture profiles can be very important during drying.

**Lipid oxidation.** Lipid oxidation is normally observed as a product discoloration and can be exacerbated with excess levels of bleach. It is catalyzed by metal ions, enzymes, and pigments. Acidic compounds can be used to complex the metal ions. Synthetic antioxidants, such as butylated hydroxytoluene (BHT) and butylated hydroxyanisole (BHA) can be added to the product, but are limited and coming under increased scrutiny due to toxicology concerns. It may be preferable to use natural antioxidants such as lecithin or vitamin E or to dry under vacuum or in an inert (nitrogen, steam) atmosphere.

**Protein denaturation.** Protein denaturation is normally observed as an increase in viscosity and a decrease in wettability. It is temperature-sensitive, generally occurring between 40 and 80°C. A common drying process scheme is to dry thermally and under wet-bulb drying conditions without overheating and then vacuum, heat-pump, or freeze-dry to the target moisture.

**Enzyme reactions.** Enzymatic browning is caused by the enzyme polyphenol oxidase which causes phenols to oxidize to ortho-quinones. The enzyme is active between pH 5 to 7. A viable process scheme again is to dry under vacuum or in an inert (nitrogen, steam) atmosphere.

**Maillard browning reaction.** This nonenzymatic reaction is observed as a product discoloration, which in some products creates an attractive coloration. The reaction is temperature-sensitive, and normally the rate passes through a maximum and then falls as the product becomes drier. The reaction can be minimized by minimizing the drying temperature, reducing the pH to acidic, or adding an inhibitor such as sulfur dioxide or metabisulfate. A viable process scheme again is to dry thermally and under wet-bulb drying conditions without overheating and then vacuum, heat-pump, or freeze-dry to the target moisture.

Some of the above reactions can be minimized by reducing the particle size and using a monodisperse particle size distribution. The small particle size will better enable wet-bulb drying, and the monodisperse size will reduce overheating of the smallest particles.

**Stickiness, Lumping, and Caking** These are not characteristics we generally want in our products. They generally connote poor product quality, but can be a desirable transformation if we are trying to enlarge particle size through agglomeration. Stickiness, lumping, and caking are phenomena which are dependent on product moisture and product temperature. The most general description of this phenomenon can be described by measuring the cohesion (particle to particle) of powders as described below. A related measure is adhesion—particle-to-wall interactions. Finally, sticky point is a special case for materials which undergo glass transitions.

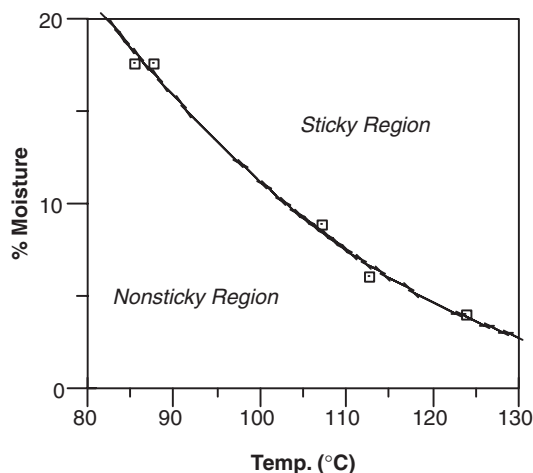


FIG. 12-23 Detergent stickiness curve.

The sticky point can be determined by using a method developed by Lazar and later by Downton [Downton, Flores-Luna, and King, "Mechanism of Stickiness in Hygroscopic, Amorphous Powders," *I&EC Fundamentals* 21: 447 (1982)]. In the simplest method, a sample of the product, at a specific moisture, is placed in a closed tube which is suspended in a water bath. A small stirrer is used to monitor the torque needed to "stir" the product. The water bath temperature is slowly increased until the torque increases. This torque increase indicates a sticky point temperature for that specific moisture. The test is repeated with other product moistures until the entire stickiness curve is determined. A typical curve is shown in Fig. 12-23.

As noted, a sticky point mechanism is a glass transition—the transition when a material changes from the glassy state to the rubbery liquid state. Glass transitions are well documented in food science (Levine and Slade). Roos and Karel [Roos and Karel, "Plasticizing Effect of Water on Thermal Behavior and Crystallization of Amorphous Food Models," *J. Food Sci.* 56(1): 38–43 (1991)] have demonstrated that for these types of products, the glass transition temperature follows the sticky point curve within about 2°C. This makes it straightforward to measure the stickiness curve by using a differential scanning calorimeter (DSC). Somewhat surprisingly, even materials which are not undergoing glass transitions exhibit this behavior, as demonstrated with the detergent stickiness curve above.

Lumping and caking can be measured by using the rotational shear cells (Peschl) or translational shear cells (Jenike) noted above for measuring flowability. The powder is consolidated under various normal loads, and then the shear force is measured, enabling a complete yield locus curve to be constructed. This can be done at various powder moistures to create a curve of "cake strength" versus moisture content. Slurries and dry solids are free-flowing, and there is a cohesion/adhesion peak at an intermediate moisture content, typically when voids between particles are largely full of liquid. A variety of other test methods for handling properties and flowability are available.

Product quality was addressed quite comprehensively by Evangelos Tsotsas at the 2d Nordic Drying Conference [Tsotsas, "Product Quality in Drying—Luck, Trial, Experience, or Science?" 2d Nordic Drying Conference, Copenhagen, Denmark, 2003]. Tsotsas notes that 31 percent of the papers at the 12th International Drying Symposium refer to product quality. The top 5 were color (12 percent), absence of chemical degradation (10 percent), absence of mechanical damage (9 percent), bulk density (8 percent), and mechanical properties (7 percent). These are all properties that are reasonably straightforward to measure. They are physical properties, and we are familiar with them for the most part. However, down the list at a rank



of 20 with only 2 percent of the papers dealing with it, we have sensory properties.

This is the dilemma—sensory properties should rank very high, but they don't because we lack the tools to measure them effectively. For the most part, these quality measures are subjective rather than objective, and frequently they require direct testing with consumers to determine efficacy of a particular product attribute. So the issue is really a lack of physical measurement tools that directly assess the performance measures important to the consumer of the product. The lack of objective performance measures and unknown mechanistic equations also makes mathematical modeling very difficult for addressing quality problems.

The good news is that there has been a shift from the macro to the meso and now to the microscale in drying science. We have some very powerful analytical tools to help us understand the transformations that are occurring at the meso and microscale.

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## SOLIDS-DRYING EQUIPMENT—GENERAL ASPECTS

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## CLASSIFICATION OF DRYERS

Drying equipment may be classified in several ways. Effective classification is vital in selection of the most appropriate dryer for the task and in understanding the key principles on which it operates. The main categories are as follows:

1. Form of feed and product—particulate (solid or liquid feed), sheet, slab
2. Mode of operation—batch or continuous
3. Mode of heat transfer—convective (direct), conductive (indirect), radiative, or dielectric
4. Condition of solids—static bed, moving bed, fluidized or dispersed
5. Gas-solids contacting—parallel flow, perpendicular flow, or through-circulation
6. Gas flow pattern—cross-flow, cocurrent, or countercurrent

Other important features of the drying system are the type of carrier gas (air, inert gas, or superheated steam/solvent), use of gas or solids recycle, type of heating (indirect or direct-fired), and operating pressure (atmospheric or vacuum). However, these are primarily related to the choice of the overall system and operating conditions, not to the individual dryer used, and are discussed briefly at the end of this section. The relative importance of the different categories depends on the purpose of the classification. For distinguishing differences in dryer design, construction, and operation, categories 2 and 3 are particularly useful. A classification chart of drying equipment on this basis is shown in Table 12-11, and the grouping in "Solids-Drying Equipment—Specific Types" follows this pattern. Simplified diagrams for batch and continuous dryers are shown in Figs. 12-24 and 12-25, respectively. However, in the selection of a group of dryers for preliminary consideration in a given drying problem, the most important factor is often category 1, the form, handling characteristics, and physical properties of the wet material. (See Table 12-12.)

In Table 12-11, dryers in round brackets are semicontinuous forms of batch dryers, not commonly used. Dryers in square brackets are semibatch forms of continuous dryers, also fairly rare.

The feed type is a very basic description; *particulate* can also include powders, granules, pastes, pellets, performs, etc.; *liquid/slurry* also includes solutions and sludges. Table 12-12 gives a more comprehensive classification based on particle size and handling properties.

## Description of Dryer Classification Criteria

**1. Form of Feed and Product** Dryers are specifically designed for particular feed and product forms; dryers handling films, sheets, slabs, and bulky artifacts form a clear subset. Most dryers are for particulate products, but the feed may range from a solution or slurry (free-flowing liquid) through a sticky paste to wet filter cakes, powders, or granules (again relatively free-flowing). The ability to successfully mechanically handle the feed and product is a key factor in dryer selection (see Table 12-12).

The drying kinetics (rate of drying, and hence required drying time) also depend strongly on solids properties, particularly particle size and porosity. The surface area/mass ratio and the internal pore structure control the extent to which an operation is diffusion-limited, i.e., diffusion into and out of the pores of a given solids particle, not through the voids among separate particles.

**2. Mode of Operation** Batch dryers are typically used for low throughputs (averaging under 50 kg/h), long drying times, or where the overall process is predominantly batch. Continuous dryers dominate for high throughputs (over 1 ton/h), high evaporation rates, and where the rest of the process is continuous. Often, there are batch and continuous dryers working on similar principles, but one batch dryer has two or more continuous equivalents, using different methods to move the solids through the dryer. For example, batch tray dryers (nonagitated solids) are equivalent to turbo-tray and plate dryers (vertical gravity transport) and to band dryers (horizontal mechanical transport). Also, dryers which are inherently continuous can be operated in semibatch mode (e.g., small-scale spray dryers) and vice versa.

### 3. Mode of Heat Transfer

**Direct (convective) dryers** The general operating characteristics of direct dryers are these:

**TABLE 12-11 Classification of Drying Equipment**

| Dryer group                                 | Feed type     | Dryer type                                                                                                      | Heating mode                                                                                                             | Synonyms and variants                                                                                                                            |
|---------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Batch tray<br>Nonagitated                   | Particulate   | Cross-circulated tray<br>Perforated tray<br>Contact/vacuum tray                                                 | Cross-circulation<br>Through-circulation<br>Conduction                                                                   | Atmospheric tray<br>Through-circulation, drying room<br>Vacuum oven, vacuum shelf                                                                |
| Batch agitated<br>Mechanical agitation      | Particulate   | Vertical pan<br>Conical<br>Spherical<br>Horizontal pan                                                          | Conduction<br>Conduction<br>Conduction<br>Conduction                                                                     | Vertical agitated<br>Sidescrew, Nauta<br>Turbosphere<br>Batch paddle, ploughshare                                                                |
| Continuous tray<br>Nonagitated              | Particulate   | Turbo-tray<br>Plate<br>Cascade<br>Moving bed                                                                    | Cross-circulation<br>Conduction<br>Through-circulation<br>Through-circulation                                            | Rotating tray/shelf, Wyssmont Turbo-dryer<br>Krauss-Maffei<br>Wenger<br>Tower, silo, gravity                                                     |
| Continuous band/tunnel<br>Nonagitated       | Particulate   | Tunnel<br>Perforated band<br>Contact/vacuum band                                                                | Cross-circulation<br>Through-circulation<br>Conduction                                                                   | Moving truck/trolleys<br>Atmospheric band/belt, vibrated bed<br>Vacuum belt, vibrated tray                                                       |
| Continuous agitated<br>Mechanical agitation | Particulate   | Paddle, low-speed<br>Paddle, high-speed<br>High-speed convective paddle                                         | Conduction<br>Conduction<br>Through-circulation                                                                          | Horizontal agitated, Disc, Porcupine, Nara<br>Solidaire<br>Rapid, Forberg                                                                        |
| Continuous rotary<br>Rotational agitation   | Particulate   | Indirect rotary<br>Rotary louvre<br>Cascading rotary                                                            | Conduction<br>Through-circulation<br>Dispersion                                                                          | Steam-tube, Louisville<br>Rotolouvre<br>Direct rotary, rotary drum                                                                               |
| Continuous dispersion<br>Airborne transport | Particulate   | Fluidized bed<br>Vibrofluidized bed<br>Pneumatic conveying<br>Spin-flash<br>Spouted bed<br>(Freeze)             | Dispersion<br>Dispersion<br>Dispersion<br>Dispersion<br>Dispersion<br>Conduction                                         | Well-mixed/plug-flow fluid bed<br>Vibrated fluid bed<br>Flash, ring, swept mill<br>Swirl fluidizer<br>Circulating fluid bed<br>Continuous freeze |
| Continuous special                          | Particulate   | Radiofrequency/microwave                                                                                        | Radiation                                                                                                                | Dielectric                                                                                                                                       |
| Continuous liquid feed                      | Liquid/slurry | Spray<br>Spray/fluidized bed<br>Fluid bed granulator<br>Thin-film<br>Drum<br>(Filter-dryer)<br>Centrifuge-dryer | Dispersion<br>Dispersion<br>Dispersion<br>Conduction<br>Conduction<br>Conduction<br>Through-circulation                  | Atomizing<br>Spray/belt<br>Recirculating inert balls<br>Evaporator-dryer, wiped-film, LUWA<br>Film-drum<br>Nutsche, Rosenmund<br>Henkel          |
| Continuous sheet/film                       | Film/sheet    | Cylinder<br>Yankee<br>Rotary through<br>Stenter<br>Flotation<br>Continuous oven<br>Infrared                     | Conduction<br>Conduction<br>Through-circulation<br>Through-circulation<br>Through-circulation<br>Conduction<br>Radiation | Paper machine, roller<br>Impingement<br><br>Tenter, range (textiles)<br>Coanda, floating web<br>Festoon, Spooner oven<br>Curing                  |

*a.* Direct contacting of hot gases with the solids is employed for solids heating and vapor removal.

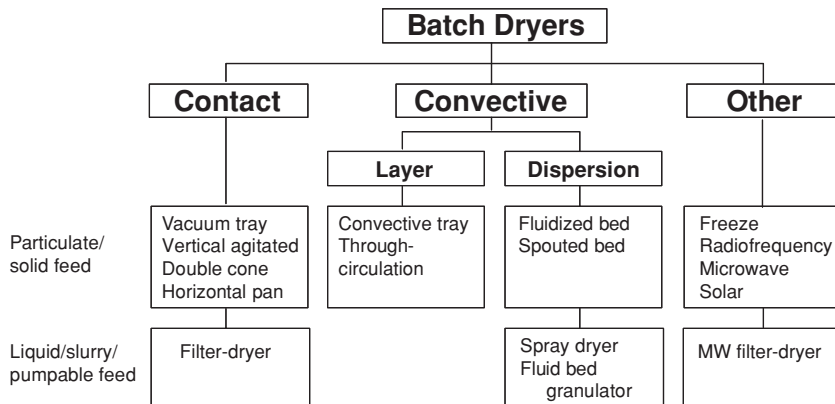
*b.* Drying temperatures may range up to 1000 K, the limiting temperature for most common structural metals. At higher temperatures, radiation becomes an important heat-transfer mechanism.

*c.* At gas temperatures below the boiling point, the vapor content of gas influences the rate of drying and the final moisture content of the solid. With gas temperatures above the boiling point throughout,

the vapor content of the gas has only a slight retarding effect on the drying rate and final moisture content. Thus, superheated vapors of the liquid being removed (e.g., steam) can be used for drying.

*d.* For low-temperature drying, dehumidification of the drying air may be required when atmospheric humidities are excessively high.

*e.* The lower the final moisture content, the more fuel per pound of water evaporated, that a direct dryer consumes.

**FIG. 12-24** Classification of batch dryers.

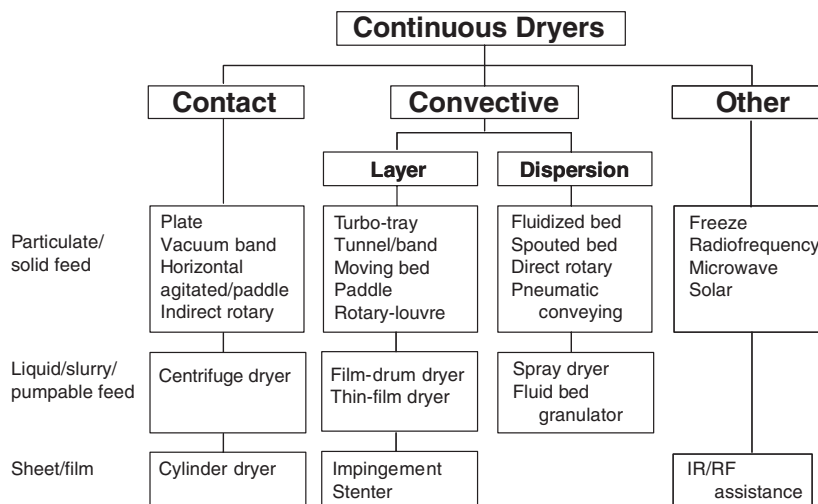


FIG. 12-25 Classification of continuous dryers.

f. Efficiency increases with an increase in the inlet gas temperature for a constant exhaust temperature.

g. Because large amounts of gas are required to supply all the heat for drying, dust recovery equipment may be very large and expensive, especially when drying very small particles.

**Indirect (contact or conductive) dryers** These differ from direct dryers with respect to heat transfer and vapor removal:

a. Heat is transferred to the wet material by conduction through a solid retaining wall, usually metallic.

b. Surface temperatures may range from below freezing in the case of freeze dryers to above 800 K in the case of indirect dryers heated by combustion products.

c. Indirect dryers are suited to drying under reduced pressures and inert atmospheres, to permit the recovery of solvents and to prevent the occurrence of explosive mixtures or the oxidation of easily decomposed materials.

d. Indirect dryers using condensing fluids as the heating medium are generally economical from the standpoint of heat consumption, since they furnish heat only in accordance with the demand made by the material being dried.

e. Dust recovery and dusty or hazardous materials can be handled more satisfactorily in indirect dryers than in direct dryers.

#### Miscellaneous dryers

a. **Infrared dryers** depend on the transfer of radiant energy to evaporate moisture. The radiant energy is supplied electrically by infrared lamps, by electric resistance elements, or by incandescent refractories heated by gas. The last method has the added advantage of convection heating. Infrared heating is not widely used in the chemical industries for the removal of moisture. Its principal use is in baking or drying paint films (curing) and in heating thin layers of materials. It is sometimes used to give supplementary heating on the initial rolls of paper machines (cylinder dryers).

b. **Dielectric dryers (radio-frequency or microwave)** have not as yet found a wide field of application, but are increasingly used. Their fundamental characteristic of generating heat within the solid indicates potentialities for drying massive geometric objects such as wood, sponge-rubber shapes, and ceramics, and for evening out moisture gradients in layers of solids. Power costs are generally much higher than the fuel costs of conventional methods; a small amount of dielectric heating (2 to 5 percent) may be combined with thermal heating to maximize the benefit at minimum operating cost. The high capital costs of these dryers must be balanced against product and process improvements.

**4. Condition of Solids** In solids-gas contacting equipment, the solids bed can exist in any of the following four conditions.

**Static** This is a dense bed of solids in which each particle rests upon another at essentially the settled bulk density of the solids phase. Specifically, *there is no relative motion among solids particles* (Fig. 12-26).

**Moving** This is a slightly expanded bed of solids in which the particles are separated only enough to flow one over another. Usually the flow is downward under the force of gravity (Fig. 12-27a), but upward motion by mechanical lifting or agitation may also occur within the process vessel (Fig. 12-27b). In some cases, lifting of the solids is accomplished in separate equipment, and solids flow in the presence of the gas phase is downward only. The latter is a moving bed as usually defined in the petroleum industry. In this definition, *solids motion is achieved by either mechanical agitation or gravity force*.

**Fluidized** This is an expanded condition in which the solids particles are supported by drag forces caused by the gas phase passing through the interstices among the particles at some critical velocity. The superficial gas velocity upward is less than the terminal settling velocity of the solids particles; the gas velocity is not sufficient to entrain and convey continuously all the solids. Specifically, the solids phase and the gas phase are intermixed and *together behave as a boiling fluid* (Fig. 12-28). The gas forms the continuous phase, but the bulk density is not much lower than a continuous packed bed of solids.

**Dispersed or dilute.** This is a fully expanded condition in which the solids particles are so widely separated that they exert essentially no influence upon one another. Specifically, the solids phase is so fully dispersed in the gas that the *density of the suspension is essentially that of the gas phase alone* (Fig. 12-29). Commonly, this situation exists when the gas velocity at all points in the system exceeds the terminal settling velocity of the solids and the particles can be lifted and continuously conveyed by the gas; however, this is not always true. Cascading rotary dryers, countercurrent-flow spray dryers, and gravity settling chambers such as prilling towers are three exceptions in which gas velocity is insufficient to entrain the solids completely.

Cascading (direct) rotary dryers with lifters illustrate all four types of flow in a single dryer. Particles sitting in the lifters (flights) are a static bed. When they are in the rolling bed at the bottom of the dryer, or rolling off the top of the lifters, they form a moving bed. They form a falling curtain which is initially dense (fluidized) but then spreads out and becomes dispersed.

Dryers where the solid forms the continuous phase (static and moving beds) are called **layer dryers**, while those where the gas forms the continuous phase (fluidized and dispersed solids) are classified as **dispersion dryers**. Gas-particle heat and mass transfer is much faster in dispersion dryers, and these are therefore often favored where high drying rates, short drying times, or high solids throughput is required.

**TABLE 12-12 Classification of Commercial Dryers Based on Feed Materials Handled**

|                                                                                                                                                           | Liquids                                                                                                                                | Slurries                                                                                                                          | Pastes and sludges                                                                                                                                          | Free-flowing powders                                                                                                                     | Granular, crystalline, or fibrous solids                                                                                                                                             | Large solids, special forms and shapes                                                             | Continuous sheets                                                       | Discontinuous sheets                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Type of dryer                                                                                                                                             | True and colloidal solutions; emulsions. Examples: inorganic salt solutions, extracts, milk, blood, waste liquors, rubber latex        | Pumpable suspensions. Examples: pigment slurries, soap and detergents, calcium carbonate, bentonite, clay slip, lead concentrates | Examples: filter-press cakes, sedimentation sludges, centrifuged solids, starch                                                                             | 100-mesh (150 $\mu\text{m}$ ) or less. Relatively free-flowing in wet state. Dusty when dry. Examples: centrifuged precipitates          | Larger than 100-mesh (150 $\mu\text{m}$ ). Examples: rayon staple, salt crystals, sand, ores, potato strips, synthetic rubber                                                        | Examples: pottery, brick, rayon cakes, shotgun shells, hats, painted objects, rayon skeins, lumber | Examples: paper, impregnated fabrics, cloth, cellophane, plastic sheets | Examples: veneer, wallboard, photograph prints, leather, foam rubber sheets |
| <b>Vacuum freeze.</b><br>Indirect type, batch or continuous operation                                                                                     | Expensive. Usually used only for high-value products such as pharmaceuticals; products, which are heat-sensitive and readily oxidized. | See comments under Liquids.                                                                                                       | See comments under Liquids.                                                                                                                                 | See comments under Liquids.                                                                                                              | Expensive. Usually used on pharmaceuticals and related products which cannot be dried successfully by other means. Applicable to fine chemicals                                      | See comments under Granular solids.                                                                | Applicable in special cases such as emulsion-coated films               | See comments under Granular solids.                                         |
| <b>Vacuum tray/shelf.</b><br>Indirect type, batch operation                                                                                               | Not applicable                                                                                                                         | Relatively expensive. Applicable for small-batch production                                                                       | Relatively expensive. Suitable for batch operation, small capacities. Useful for heat-sensitive or readily oxidizable materials. Solvents can be recovered. | See comments under Pastes and Sludges.                                                                                                   | Suitable for batch operation, small capacities. Useful for heat-sensitive or readily oxidizable materials. Solvents can be recovered.                                                | See comments under Granular solids.                                                                | Not applicable                                                          | See comments under Granular solids.                                         |
| <b>Pan.</b> Indirect type, batch operation including vertical agitated pan, spherical, conical, filter-dryer, double-cone tumbler                         | Atmospheric or vacuum. Suitable for small batches. Easily cleaned. Solvents can be recovered. Material agitated while dried.           | See comments under Liquids.                                                                                                       | See comments under Liquids.                                                                                                                                 | See comments under Liquids.                                                                                                              | Suitable for small batches. Easily cleaned. Material is agitated during drying, causing some degradation and/or balling up.                                                          | Not applicable                                                                                     | Not applicable                                                          | Not applicable                                                              |
| <b>Vacuum horizontal agitated and rotary.</b><br>Indirect type, batch operation. Includes indirect rotary, horizontal pan                                 | Not applicable, except when pumping slowly on dry "heel"                                                                               | May have application in special cases when pumping onto dry "heel"                                                                | Material usually cakes to dryer walls and agitator. Special precautions needed, e.g., cleaning hooks, twin screws. Solvents can be recovered.               | Suitable for nonsticking materials. Useful for large batches of heat-sensitive materials and for solvent recovery.                       | Useful for large batches of heat-sensitive materials or where solvent is to be recovered. Product will suffer some grinding action, or may ball up. Dust collectors may be required. | Not applicable                                                                                     | Not applicable                                                          | Not applicable                                                              |
| <b>Screw conveyor and indirect rotary.</b> Indirect type, continuous operation. Includes paddle, horizontal agitated and steam-tube dryers, rotary kilns. | Applicable with dry-product recirculation                                                                                              | Applicable with dry-product recirculation                                                                                         | Generally requires recirculation of dry product. Little dusting occurs.                                                                                     | Chief advantage is low dust loss. Well suited to most materials and capacities, particularly those requiring drying at steam temperature | Low dust loss. Material must not stick or be temperature-sensitive                                                                                                                   | Not applicable                                                                                     | Not applicable                                                          | Not applicable                                                              |



**TABLE 12-12 Classification of Commercial Dryers Based on Feed Materials Handled (Concluded)**

| Type of dryer                                                                                                                                        | Liquids                                                                                                                   | Slurries                                                                              | Pastes and sludges                                                                                                            | Free-flowing powders                                                  | Granular, crystalline, or fibrous solids                                                                                              | Large solids, special forms and shapes                                                                              | Continuous sheets                                                                                                                     | Discontinuous sheets                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Vibrating tray, vacuum band.</b><br>Indirect type, continuous operation                                                                           | Not applicable                                                                                                            | Not usually applicable. Belt with raised edges possible, but rare                     | Not usually applicable due to feed and discharge problems                                                                     | Suitable for free-flowing materials                                   | Suitable for free-flowing materials that can be conveyed on a vibrating tray or belt                                                  | Not applicable                                                                                                      | Not applicable                                                                                                                        | Not applicable                                                                                             |
| <b>Drum.</b> Indirect type, continuous operation                                                                                                     | Single, double, or twin. Atmospheric or vacuum operation. Product flaky and usually dusty. Maintenance costs may be high. | See comments under Liquids. Twin-drum dryers are widely used.                         | Can be used only when paste or sludge can be made to flow. See comments under Liquids.                                        | Not applicable                                                        | Not applicable                                                                                                                        | Not applicable                                                                                                      | Not applicable                                                                                                                        | Not applicable                                                                                             |
| <b>Cylinder.</b> Indirect type, continuous operation                                                                                                 | Not applicable                                                                                                            | Not applicable                                                                        | Not applicable                                                                                                                | Not applicable                                                        | Not applicable                                                                                                                        | Not applicable                                                                                                      | Suitable for thin or mechanically weak sheets which can be dried in contact with a heated surface. Special surface effects obtainable | Suitable for materials which need not be dried flat and which will not be injured by contact with hot drum |
| <b>Tray and compartment.</b><br>Direct type, batch operation. Includes cross-circulated tray                                                         | Not applicable                                                                                                            | For very small batch production. Laboratory drying                                    | Suited to batch operation. At large capacities, investment and operating costs are high. Long drying times                    | Dusting may be a problem. See comments under Pastes and Sludges.      | Suited to batch operation. At large capacities, investment and operating costs are high. Long drying times                            | See comments under Granular solids.                                                                                 | Not applicable                                                                                                                        | See comments under Granular solids.                                                                        |
| <b>Batch through-circulation.</b><br>Direct type, batch operation includes perforated tray, drying room                                              | Not applicable                                                                                                            | Not applicable                                                                        | Suitable only if material can be preformed. Suited to batch operation. Shorter drying time than tray dryers                   | Not applicable                                                        | Usually not suited for materials smaller than 30-mesh (0.5 mm). Suited to small capacities and batch operation                        | Primarily useful for small objects                                                                                  | Not applicable                                                                                                                        | Not applicable                                                                                             |
| <b>Tunnel/continuous tray.</b> Direct type, continuous operation. Includes tunnel, turbo-tray                                                        | Not applicable                                                                                                            | Not applicable                                                                        | Suitable for small- and large-scale production                                                                                | See comments under Pastes and Sludges. Vertical turbo-tray applicable | Essentially large-scale, semi-continuous tray drying                                                                                  | Suited to a wide variety of shapes and forms, especially tunnel type. Operation can be made continuous. Widely used | Not applicable                                                                                                                        | Suited for leather, wallboard, veneer                                                                      |
| <b>Continuous through-circulation (nonagitated).</b> Direct type, continuous operation. Includes perforated band, moving bed, centrifuge-dryer.      | Not applicable                                                                                                            | Only crystal filter dryer or centrifuge dryer may be suitable.                        | Suitable for materials that can be preformed. Will handle large capacities                                                    | Not applicable                                                        | Usually not suited for materials smaller than 30-mesh (0.5 mm). Material does not tumble or mix.                                      | Suited to smaller objects that can be loaded on each other. Can be used to convey materials through heated zones    | Not applicable                                                                                                                        | Special designs are required. Suited to veneers                                                            |
| <b>Continuous through-circulation (agitated/rotary).</b><br>Direct type, continuous operation. Includes high-speed convective paddle, rotary-louvre. | Not applicable                                                                                                            | Applicable with special high-speed fountain-type dryers, e.g., Hazemag Rapid, Forberg | Suitable for materials that can be preformed. Will handle large capacities. Rotary-louvre requires dry-product recirculation. | Not generally applicable, except rotary-louvre in certain cases       | Usually not suited for materials smaller than 30-mesh (0.5 mm). Material is tumbled and mixed, may suffer attrition in paddle dryers. | Not applicable                                                                                                      | Not applicable                                                                                                                        | Not applicable                                                                                             |

|                                                                                                                                                                 |                                                                                                                                                                                                        |                                                                                            |                                                                                                                                      |                                                                                                                               |                                                                                                                                                                    |                                                                        |                                                                                                                                        |                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Direct rotary.</b> Direct-type, continuous operation                                                                                                         | Applicable with dry-product recirculation                                                                                                                                                              | Applicable with dry-product recirculation                                                  | Suitable only if product does not stick to walls and does not dust. Recirculation of product may prevent sticking.                   | Suitable for most materials and especially for high capacities, provided dusting is not too severe                            | Suitable for most materials especially for high capacities. Dusting or crystal abrasion will limit its use.                                                        | Not applicable                                                         | Not applicable                                                                                                                         | Not applicable                                                    |
| <b>Fluid beds.</b> Direct type, batch or continuous                                                                                                             | Applicable only as fluid bed granulator with inert bed or dry-solids recirculator                                                                                                                      | See comments under Liquids.                                                                | See comments under Liquids.                                                                                                          | Suitable, if not too dusty. Internal coils can supplement heating, especially for fine powders. Suitable for high capacities. | Suitable for crystals, granules, and very short fibers. Suitable for high capacities                                                                               | Not applicable                                                         | Use hot inert particles for contacting; rare                                                                                           | Use hot inert particles for contacting; rare                      |
| <b>Spouted beds.</b> Direct type, batch or continuous                                                                                                           | Applicable only with inert bed or dry-solids recirculator. Usable to grow large particles by layering                                                                                                  | See comments under Liquids.                                                                | See comments under Liquids.                                                                                                          | Not applicable                                                                                                                | Suitable for large particles and granules over 20-mesh (800 $\mu\text{m}$ ) which are spoutable                                                                    | Not applicable unless objects are spoutable (conveyable in gas stream) | Use hot inert particles for contacting; rare                                                                                           | Use hot inert particles for contacting; rare                      |
| <b>Pneumatic conveying.</b> Direct-type, continuous operation includes flash, spin-flash, and ring dryers.                                                      | See comments under Slurries.                                                                                                                                                                           | Can be used only if product is recirculated (backmixed) to make feed suitable for handling | Usually requires recirculation of dry product to make suitable feed. Well suited to high capacities. Disintegration usually required | Suitable for materials that are easily suspended in a gas stream and lose moisture readily. Well suited to high capacities    | Suitable for materials conveyable in a gas stream. Well suited to high capacities. Only surface moisture usually removed. Product may suffer physical degradation. | Not applicable                                                         | Not applicable                                                                                                                         | Not applicable                                                    |
| <b>Spray.</b> Direct type, continuous operation. Rotary atomizer, pressure nozzle, or two-fluid nozzle. Includes combined spray-fluid bed and spray-belt dryers | Suited for large capacities. Product is usually powdery, spherical, and free-flowing. High temperatures can sometimes be used with heat-sensitive materials. Products generally have low bulk density. | See comments under Liquids. Pressure-nozzle atomizers subject to erosion                   | Requires special pumping equipment to feed the atomizer. See comments under Liquids.                                                 | Not applicable unless feed is pumpable                                                                                        | Not applicable                                                                                                                                                     | Not applicable                                                         | Not applicable                                                                                                                         | Not applicable                                                    |
| <b>Continuous sheeting.</b> Direct-type, continuous operation includes stenter, Yankee, impingement.                                                            | Not applicable                                                                                                                                                                                         | Not applicable                                                                             | Not applicable                                                                                                                       | Not applicable                                                                                                                | Not applicable                                                                                                                                                     | Not applicable                                                         | Generally high capacity. Different types are available for different requirements. Suitable for drying without contacting hot surfaces | Not applicable                                                    |
| <b>Infrared.</b> Batch or continuous operation. Electric heating or gas-fired                                                                                   | Only for thin films. Can be used in combination with other dryers such as drum.                                                                                                                        | See comments under Liquids.                                                                | See comments under Liquids (only for thin layers).                                                                                   | Only for thin layers                                                                                                          | Primarily suited to drying surface moisture. Not suited for thick layers                                                                                           | Specially suited for drying and baking paint and enamels               | Useful when space is limited. Usually used in conjunction with other methods, e.g., in drying paper coatings                           | Useful for laboratory work or in conjunction with other methods   |
| <b>Dielectric.</b> Batch or continuous operation includes microwave, radio-frequency (RF)                                                                       | Expensive, may be used in small batch filter-dryers, often as supplement to thermal heating. Sometimes useful in combination with other dryers.                                                        | See comments under Liquids.                                                                | See comments under Liquids.                                                                                                          | Expensive, may be used on small batch dryers, often as supplement to thermal heating                                          | Expensive, can assist thermal drying especially to dry center of large granules/pellets                                                                            | Rapid drying of large objects suited to this method                    | Applications for final stages of paper and textile dryers                                                                              | Successful on foam rubber. Not fully developed on other materials |



FIG. 12-26 Solids bed in static condition (tray dryer).

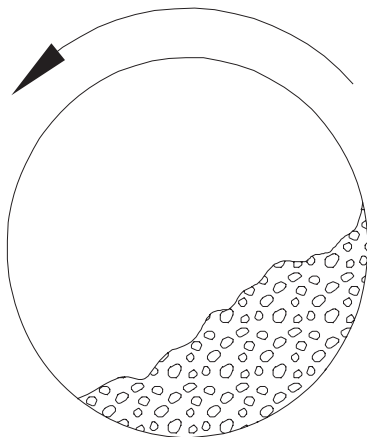


FIG. 12-27a Horizontal moving bed.

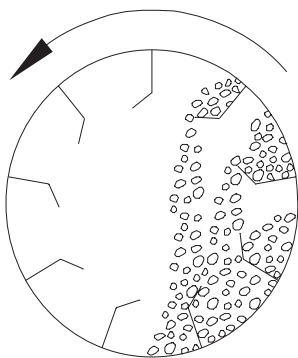


FIG. 12-27b Moving solids bed in a rotary dryer with lifters.

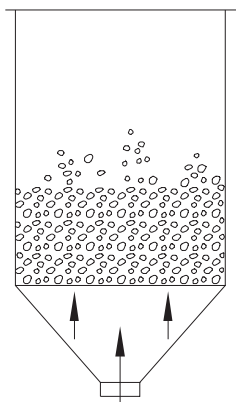


FIG. 12-28 Fluidized solids bed.

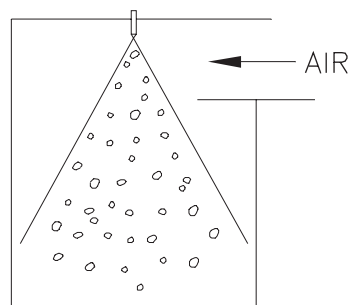


FIG. 12-29 Solids in a dilute condition near the top of a spray dryer.

Layer dryers are very suitable for slow-drying materials requiring a long residence time.

Because in a gas-solids-contacting operation heat transfer and mass transfer take place at the solids' surfaces, maximum process efficiency can be expected with a maximum exposure of solids surface to the gas phase, together with thorough mixing of gas and solids. Both are important. Within any arrangement of particulate solids, gas is present in the voids among the particles and contacts all surfaces except at the points of particle contact. When the solids are fluidized or dispersed, the gas moves past them rapidly, and external heat- and mass-transfer rates are high. When the solids bed is in a static or slightly moving condition, however, gas within the voids is cut off from the main body of the gas phase and can easily become saturated, so that local drying rates fall to low or zero values. Some transfer of energy and mass may occur by diffusion, but it is usually insignificant. The problem can be much reduced by using through-circulation of gas instead of cross-circulation, or by agitating and mixing the solids.

*Solids Agitation and Mixing* There are four alternatives:

1. No agitation, e.g., tray and band dryers. This is desirable for friable materials. However, drying rates can be extremely low, particularly for cross-circulation and vacuum drying.
2. Mechanical agitation, e.g., vertical pan and paddle dryers. This improves mixing and drying rates, but may give attrition depending on agitator speed; and solids may stick to the agitator, as shown in Fig. 12-30.
3. Vessel rotation, e.g., double-cone and rotary dryers. Mixing and heat transfer are better than for static dryers but may be less than for mechanical agitation. Formation of balls and lumps may be a problem.
4. Airborne mixing, e.g., fluidized beds and flash and spray dryers. Generally there is excellent mixing and mass transfer, but feed must be dispersible and entrainment and gas cleaning are higher. Mechanical vibration may also be used to assist solids movement in some dryers.

*Solids transport* In continuous dryers, the solids must be moved through the dryer. The main methods of doing this are

1. Gravity flow (usually vertical), e.g., turbo-tray, plate and moving-bed dryers, and rotary dryers (due to the slope)

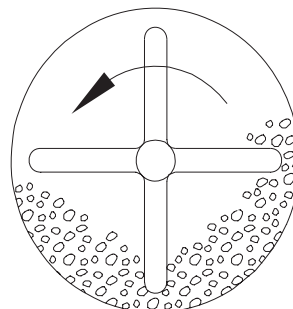


FIG. 12-30 Paddle dryer.

2. Mechanical conveying (usually horizontal), e.g., band, tunnel, and paddle dryers

3. Airborne transport, e.g., fluidized beds and flash and spray dryers

**Solids flow pattern** For most continuous dryers, the solids are basically in plug-flow; backmixing is low for nonagitated dryers but can be extensive for mechanical, rotary, or airborne agitation. Exceptions are well-mixed fluidized beds, fluid-bed granulators, and spouted beds (well-mixed) and spray and spray/fluidized-bed units (complex flow patterns).

**5. Gas-Solids Contacting** Where there is a significant gas flow, it may contact a bed of solids in the following ways:

*a. Parallel flow or cross-circulation.* The direction of gas flow is parallel to the surface of the solids phase. Contacting is primarily at the interface between phases, with possibly some penetration of gas into the voids among the solids near the surface. The solids bed is usually in a static condition (Fig. 12-31).

*b. Perpendicular flow or impingement.* The direction of gas flow is normal to the phase interface. The gas impinges on the solids bed. Again the solids bed is usually in a static condition (Fig. 12-32). This most commonly occurs when the solids are a continuous sheet, film, or slab.

*c. Through circulation.* The gas penetrates and flows through interstices among the solids, circulating more or less freely around the individual particles (Fig. 12-33). This may occur when solids are in static, moving, fluidized, or dilute conditions.

**6. Gas Flow Pattern in Dryer** Where there is a significant gas flow, it may be in cross-flow, cocurrent, or countercurrent flow compared with the direction of solids movement.

*a. Cocurrent gas flow.* The gas phase and solids particles both flow in the same direction (Fig. 12-34).

*b. Countercurrent gas flow.* The direction of gas flow is exactly opposite to the direction of solids movement.

*c. Cross-flow of gas.* The direction of gas flow is at a right angle to that of solids movement, across the solids bed (Fig. 12-35). The difference between these is shown most clearly in the gas and solids temperature profiles along the dryer. For cross-flow dryers, all solids particles are exposed to the same gas temperature, and the solids temperature approaches the gas temperature near the end of

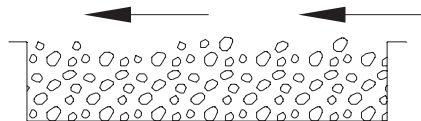


FIG. 12-31 Parallel gas flow over a static bed of solids.

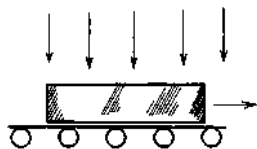


FIG. 12-32 Circulating gas impinging on a large solid object in perpendicular flow, in a roller-conveyor dryer.

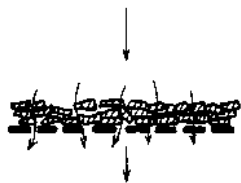


FIG. 12-33 Gas passing through a bed of preformed solids, in through-circulation on a perforated-band dryer.

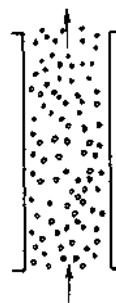


FIG. 12-34 Cocurrent gas-solids flow in a vertical-lift dilute-phase pneumatic conveyor dryer.

drying (Fig. 12-36). In cocurrent dryers, the gas temperature falls throughout the dryer, and the final solids temperature is much lower than that for the cross-flow dryer (Fig. 12-37). Hence cocurrent dryers are very suitable for drying heat-sensitive materials, although it is possible to get a solids temperature peak inside the dryer. Conversely, countercurrent dryers give the most even temperature gradient throughout the dryer, but the exiting solids come into contact with the hottest, driest gas (Fig. 12-38). These can be used to heat-treat the solids or to give low final moisture content (minimizing the local equilibrium moisture content) but are obviously unsuitable for thermally sensitive solids.

### Subclassifications

**Heater** This may be an indirect heat exchanger or a direct-fired burner, or heating may be electrical (including RF/microwave absorption).

**Gas Circuit** This may be open-cycle (once-through) or closed-cycle (gas recycle, often using inert gas). A closed-cycle system with a direct-fired burner can be operated as a self-inerting system with reduced oxygen concentration.

**Solids Feeders** These convey the solids into the dryer and may also perform as a metering or sealing dryer. Dry solids may be back-mixed into the wet feed if the latter is sticky and difficult to handle. See Sec. 21.

**Gas-Solids Separations** After the solids and gas have been brought together and mixed in a gas-solids contactor, it becomes necessary to separate the two phases, particularly for dispersion dryers where the solids loading in the exhaust gas can be very high. If the solids are sufficiently coarse and the gas velocity sufficiently low, it is possible to effect a complete gravitational separation in the primary contactor. Applications of this type are rare, however, and supplementary dust collection equipment is commonly required. The recovery step may even dictate the type of primary contacting device selected. For example, in treating an extremely friable solid material, a deep fluidized-solids contactor might overload the collection system with fines, whereas the more gentle contacting of a traveling-screen contactor

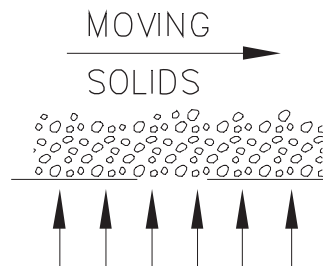


FIG. 12-35 Cross-flow of gas and solids in a fluid bed or band dryer.

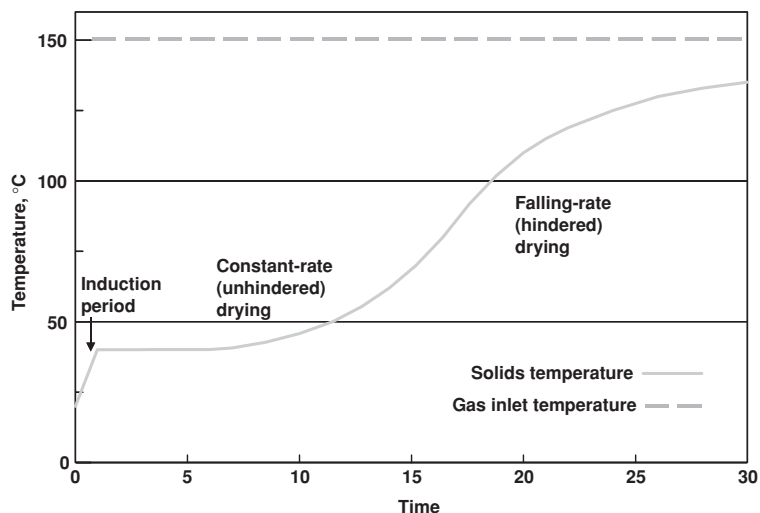


FIG. 12-36 Temperature profiles along a continuous plug-flow dryer for cross-flow of gas and solids. (Aspen Technology Inc.)

would be expected to produce a minimum of fines by attrition. Therefore, although gas solids separation is usually considered as separate and distinct from the primary contacting operation, it is usually desirable to evaluate the separation problem at the same time as contacting methods are evaluated. Methods are noted later in "Environmental Considerations." The subject is covered in depth in Sec. 17, "Gas-Solids Operations."

### SELECTION OF DRYING EQUIPMENT

**Dryer Selection Considerations** Dryer selection is a challenging task and rarely clear-cut. For 500- $\mu\text{m}$  particles, there may be several different dryer types which are likely to handle the task well, at similar cost. For 5- $\mu\text{m}$  particles, there may be no dryers that are fully suitable, and the task is to find the "least bad"!

Dryer classification often helps to reveal the broad choices for which equipment is suitable. For instance:

- Batch dryers are almost invariably used for mean throughputs below 50 kg/h and continuous dryers above 1 ton/h; in the intervening range, either may be suitable.
- Liquid or slurry feeds, large artifacts, or continuous sheets and films require completely different equipment to particulate feeds.
- Particles and powders below 1 mm are effectively dried in dispersion or contact dryers, but most through-circulation units are unsuitable. Conversely, for particles of several millimeters or above, through-circulation dryers, rotary dryers, and spouted beds are very suitable.
- Through-circulation and dispersion convective dryers (including fluidized-bed, rotary, and pneumatic types), and agitated or rotary contact dryers, generally give better drying rates than nonagitated cross-circulated or contact tray dryers.
- Nonagitated dryers (including through-circulation) may be preferable for fragile particles where it is desired to avoid attrition.
- For organic solvents, or solids which are highly flammable, are toxic, or decompose easily, contact dryers are often preferable to

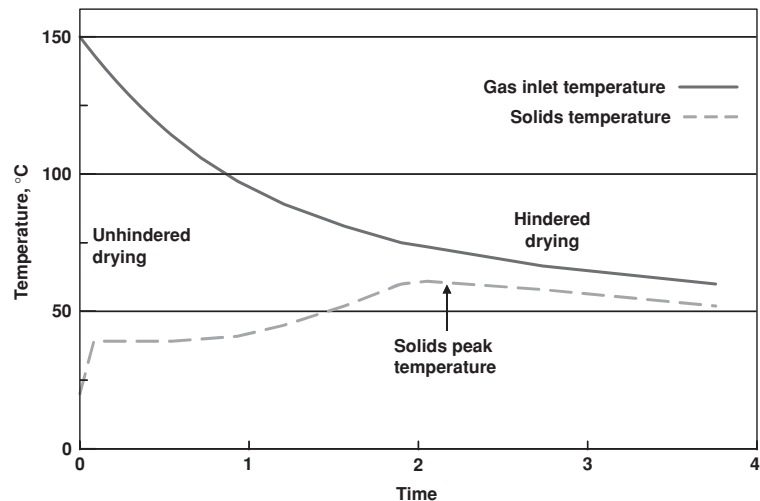


FIG. 12-37 Temperature profiles along a continuous plug-flow dryer for cocurrent flow of gas and solids. (Aspen Technology Inc.)

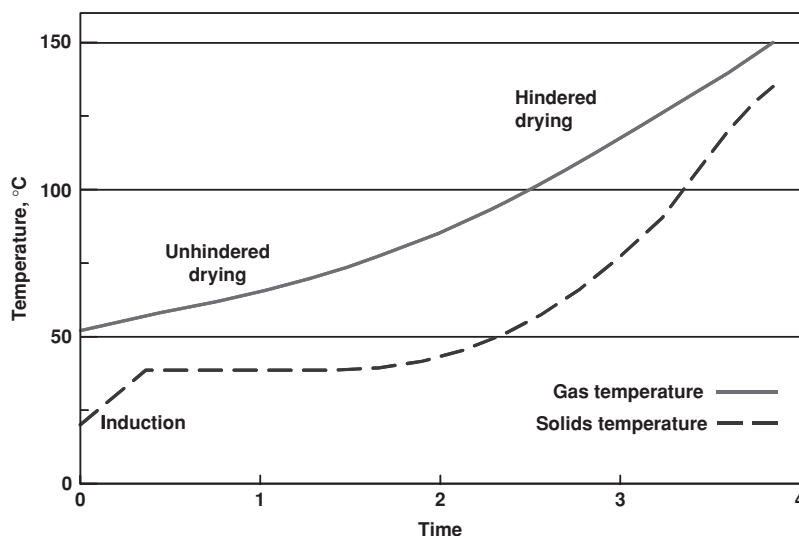


FIG. 12-38 Temperature profiles along a continuous plug-flow dryer for countercurrent flow of gas and solids. (Aspen Technology Inc.)

convective, as containment is better and environmental emissions are easier to control. If a convective dryer is used, a closed-cycle system using an inert carrier gas (e.g., nitrogen) is often required.

- Cocurrent, vacuum, and freeze dryers can be particularly suitable for heat-sensitive materials.

A detailed methodology for dryer selection, including the use of a rule-based expert system, has been described by Kemp [*Drying Technol.* **13**(5-7): 1563-1578 (1995) and **17**(7 and 8): 1667-1680 (1999)].

A simpler step-by-step procedure is given here.

1. *Initial selection of dryers.* Select those dryers which appear best suited to handling the wet material and the dry product, which fit into the continuity of the process as a whole, and which will produce a product of the desired physical properties. This preliminary selection can be made with the aid of Table 12-12, which classifies the various types of dryers on the basis of the materials handled.

2. *Initial comparison of dryers.* The dryers so selected should be evaluated approximately from available cost and performance data. From this evaluation, those dryers which appear to be uneconomical or unsuitable from the standpoint of performance should be eliminated from further consideration.

3. *Drying tests.* Drying tests should be conducted in those dryers still under consideration. These tests will determine the optimum operating conditions and the product characteristics and will form the basis for firm quotations from equipment vendors.

4. *Final selection of dryer.* From the results of the drying tests and quotations, the final selection of the most suitable dryer can be made.

The important factors to consider in the preliminary selection of a dryer are the following:

- Properties of the material being handled
  - Physical characteristics when wet (stickiness, cohesiveness, adhesiveness, flowability)
  - Physical characteristics when dry
  - Corrosiveness
  - Toxicity
  - Flammability
  - Particle size
  - Abrasiveness
- Drying characteristics of the material
  - Type of moisture (bound, unbound, or both)
  - Initial moisture content (maximum and range)
  - Final moisture content (maximum and range)
  - Permissible drying temperature

- Probable drying time for different dryers
  - Level of nonwater volatiles
- Flow of material to and from the dryer
    - Quantity to be handled per hour (or batch size and frequency)
    - Continuous or batch operation
    - Process prior to drying
    - Process subsequent to drying
  - Product quality
    - Shrinkage
    - Contamination
    - Uniformity of final moisture content
    - Decomposition of product
    - Overdrying
    - State of subdivision
    - Product temperature
    - Bulk density
  - Recovery problems
    - Dust recovery
    - Solvent recovery
  - Facilities available at site of proposed installation
    - Space
    - Temperature, humidity, and cleanliness of air
    - Available fuels
    - Available electric power
    - Permissible noise, vibration, dust, or heat losses
    - Source of wet feed
    - Exhaust-gas outlets

The physical nature of the material to be handled is the primary item for consideration. A slurry will demand a different type of dryer from that required by a coarse crystalline solid, which, in turn, will be different from that required by a sheet material (Table 12-12).

Following preliminary selection of suitable types of dryers, a rough evaluation of the size and cost should be made to eliminate those which are obviously uneconomical. Information for this evaluation can be obtained from material presented under discussion of the various dryer types. When data are inadequate, preliminary cost and performance data can usually be obtained from the equipment manufacturer. In comparing dryer performance, the factors in the preceding list which affect dryer performance should be properly weighed. The possibility of eliminating or simplifying processing steps which precede or follow drying, such as filtration, grinding, or conveying, should be carefully considered.



**Drying Tests** These tests should establish the optimum operating conditions, the ability of the dryer to handle the material physically, product quality and characteristics, and dryer size. The principal manufacturers of drying equipment are usually prepared to perform the required tests on dryers simulating their equipment. Occasionally, simple laboratory experiments can serve to reduce further the number of dryers under consideration.

Once a given type and size of dryer have been installed, the product characteristics and drying capacity can be changed only within relatively narrow limits. Thus it is more economical and far more satisfactory to experiment in small-scale units than on the dryer that is finally installed.

On the basis of the results of the drying tests that establish size and operating characteristics, formal quotations and guarantees should be obtained from dryer manufacturers. Initial costs, installation costs, operating costs, product quality, dryer operability, and dryer flexibility can then be given proper weight in final evaluation and selection.

Effective scale-up from tests to industrial equipment is obviously very important, and it was covered in a special issue of *Drying Technol.* 12 (1 and 2): 1–452 (1994).

## DRYER MODELING, DESIGN, AND SCALE-UP

**General Principles** Models and calculations on dryers can be categorized in terms of (1) the level of complexity used and (2) the purpose or type of calculation (design, performance rating, or scale-up). A fully structured approach to dryer modeling can be developed from these principles, as described below and in greater detail by Kemp and Oakley (2002).

**Levels of Dryer Modeling** Modeling can be carried out at four different levels, depending on the amount of data available and the level of detail and precision required in the answer.

**Level 1. Heat and mass balances.** These balances give information on the material and energy flows to and from the dryer, but say nothing about the required equipment size or the performance which a given dryer is capable of.

**Level 2. Scoping** Approximate or scoping calculations give rough sizes and throughputs (mass flow rates) for dryers, using simple data and making some simplifying assumptions. Either heat-transfer control or first-order drying kinetics is assumed.

**Level 3. Scaling** Scaling calculations give overall dimensions and performance figures for dryers by scaling up drying curves from small-scale or pilot-plant experiments.

**Level 4. Detailed** Rigorous or detailed methods aim to track the temperature and drying history of the solids and find local conditions inside the dryer. Naturally, these methods use more complex modeling techniques with many more parameters and require many more input data.

**Types of Dryer Calculations** The user may wish to either design a new dryer or improve the performance of an existing one. Three types of calculations are possible:

- Design of a new dryer to perform a given duty, using information from the process flowsheet and physical properties databanks
- Performance calculations for an existing dryer at a new set of operating conditions
- Scale-up from laboratory-scale or pilot-plant experiments to a full-scale dryer

Solids drying is very difficult to model reliably, particularly in the falling-rate period which usually has the main effect on determining the overall drying time. Falling-rate drying kinetics depend strongly on the internal moisture transport within a solid. This is highly dependent on the internal structure, which in turn varies with the upstream process, the solids formation step, and often between individual batches. Hence, many key drying parameters within solids (e.g., diffusion coefficients) cannot be predicted from theory alone, or obtained from physical property databanks; practical measurements are required. Because of this, experimental work is almost always necessary to design a dryer accurately, and scale-up calculations are more reliable than design based only on thermodynamic data. The experiments are used to verify the theoretical model and find the difficult-to-measure parameters; the full-scale dryer can then be modeled more realistically.

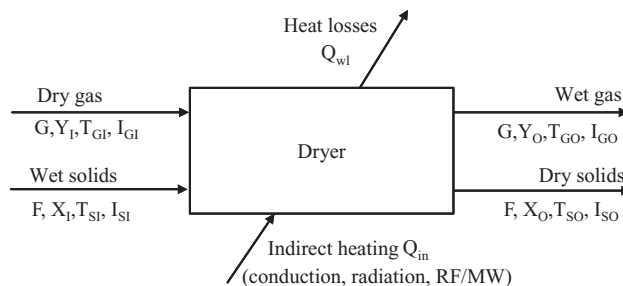


FIG. 12-39 Heat and material flows around a continuous dryer.

**Heat and Mass Balance** The heat and mass balance on a generic continuous dryer is shown schematically in Fig. 12-39. In this case, mass flows and moisture contents are given on a dry basis.

The **mass balance** is usually performed on the principal solvent and gives the evaporation rate  $\dot{E}$  (kg/s). In a contact or vacuum dryer, this is approximately equal to the exhaust vapor flow, apart from any noncondensibles. In a convective dryer, this gives the increased outlet humidity of the exhaust. For a continuous dryer at steady-state operating conditions,

$$\dot{E} = F(X_1 - X_0) = G(Y_0 - Y_1) \quad (12-54)$$

This assumes that the dry gas flow  $G$  and dry solids flow  $F$  do not change between dryer inlet and outlet. Mass balances can also be performed on the overall gas and solids flows to allow for features such as air leaks and solids entrainment in the exhaust gas stream.

In a design mode calculation (including scale-up), the required solids flow rate, inlet moisture content  $X_1$  and outlet moisture  $X_0$  are normally specified, and the evaporation rate and outlet gas flow are calculated. In performance mode, the calculation is normally reversed; the evaporation rate under new operating conditions is found, and the new solids throughput or outlet moisture content is back-calculated.

For a batch dryer with a dry mass  $m$  of solids, a mass balance only gives a snapshot at one point during the drying cycle and an instantaneous drying rate, given by

$$\dot{E} = m \left( \frac{-dX}{dt} \right) = G(Y_0 - Y_1) \quad (12-55)$$

The **heat balance** on a continuous dryer takes the generic form

$$GI_{G1} + FI_{S1} + Q_{in} = GI_{G0} + FI_{S0} + Q_{wl} \quad (12-56)$$

Here  $I$  is the enthalpy (kJ/kg dry material) of the solids or gas plus their associated moisture. Enthalpy of the gas includes the latent heat term for the vapor. Expanding the enthalpy terms gives

$$G(C_{s1}T_{G1} + \lambda Y_1) + F(C_{PS} + X_1 C_{PL})T_{S1} + Q_{in} = G(C_{s0}T_{G0} + \lambda Y_0) + F(C_{PS} + X_0 C_{PL})T_{S0} + Q_{wl} \quad (12-57)$$

Here  $C_s$  is the humid heat  $C_{PG} + YC_{PY}$ . In convective dryers, the left-hand side is dominated by the sensible heat of the hot inlet gas  $GC_{s1}T_{G1}$ ; in contact dryers, the heat input from the jacket  $Q_{in}$  is dominant. In both cases, the largest single term on the right-hand side is the latent heat of the vapor  $G\lambda Y_0$ . Other terms are normally below 10 percent. This shows why the operating line of a convective dryer on a psychrometric chart is roughly parallel to a constant-enthalpy line.

The corresponding equation for a batch dryer is

$$GI_{G1} + Q_{in} = GI_{G0} + m \frac{dI_s}{dt} + Q_{wl} \quad (12-58)$$

Further information on heat and mass balances, including practical challenges on industrial dryers and a worked example, is given in the "Drying Fundamentals" section.

**Scoping Design Calculations** In scoping calculations, some approximate dryer dimensions and drying times are obtained based mainly on a heat and mass balance, without measuring a drying curve or other experimental drying data. They allow the cross-sectional area of convective dryers and the volume of batch dryers to be estimated quite accurately, but are less effective for other calculations and can give overoptimistic results.

**Continuous Convective Dryers** In design mode, the required solids throughput  $F$  and the inlet and outlet moisture content  $X_I$  and  $X_O$  are known, as is the ambient humidity  $Y_I$ . If the inlet gas temperature  $T_{CI}$  is chosen, the outlet gas conditions (temperature  $T_{CO}$  and humidity  $Y_O$ ) can be found, either by calculation or (more simply and quickly) by using the constant-enthalpy lines on a psychrometric chart. However, it may be necessary to allow for heat losses and sensible heating of solids, which typically reduce the useful enthalpy of the inlet gas by 10 to 20 percent. Also, if tightly bound moisture is being removed, the heat of wetting to break the bonds should be allowed for. The gas mass flow rate  $G$  can now be calculated, as it is the only unknown in the mass balance on the solvent [Eq. (12-56)]. A typical gas velocity  $U_G$  along the dryer is now chosen, for example, 20 m/s for a flash dryer, 0.5 m/s for a fluidized bed, and 3 m/s for a cocurrent rotary dryer. For through-circulation and dispersion dryers, the cross-sectional area  $A$  is given by

$$A = \frac{G}{\rho_{GI}U_G} = \frac{F(X_I - X_O)}{\rho_{GI}U_G(Y_O - Y_I)} \quad (12-59)$$

The dryer diameter, or linear dimensions of a rectangular bed, can then be calculated. The result is usually accurate within 10 percent, and can be further improved by better estimates of velocity and heat losses. In performance mode, the equation is reversed to find the gas flow rate from  $G = \rho_{GI}U_GA$ .

The method gives no information about solids residence time or dryer length. A minimum drying time  $t_{\min}$  can be calculated by evaluating the maximum (unhindered) drying rate  $N_{cr}$ , assuming gas-phase heat-transfer control and estimating a gas-to-solids heat-transfer coefficient. The simple equation (12-60) then applies:

$$t_{\min} = \frac{X_I - X_O}{N_{cr}} \quad (12-60)$$

Alternatively, it may be assumed that first-order falling-rate kinetics apply throughout the drying process, and scale the estimated drying time by using Eq. (12-63). However, these crude methods can give serious underestimates of the required drying time, and it is much better to measure the drying time experimentally and apply scaling (level 3) methods.

**Continuous Contact Dryers** The key parameter is the area of the heat-transfer surface  $A_s$ . In design mode, this can be found from the equation:

$$A_s = \frac{Q}{h_{ws} \Delta T_{ws}} = \frac{E \lambda_{ev}}{h_{ws} \Delta T_{ws}} = \frac{F(X_I - X_O) \lambda_{ev}}{h_{ws} \Delta T_{ws}} \quad (12-61)$$

Here  $Q$  is the rate of heat transfer from the heated wall to the solids, and  $\Delta T_{ws}$  is the temperature driving force. The latent heat of evaporation  $\lambda_{ev}$  should allow for bound moisture and heating of solids and vapor to the final temperature. A typical wall-to-solids heat-transfer coefficient  $h_{ws}$  for the given dryer type should be used. The calculation is less accurate than the one for convective dryers. Again, the heat-transfer rate is assumed to be the overall limiting factor.

If the drying process is strongly limited by falling-rate drying kinetics, the calculated size of dryer corresponding to the given heating surface  $A_s$  may not give sufficient solids residence time to reach the desired final moisture content. Again, experimental measurement of a drying curve is strongly recommended.

**Batch Dryers** If the batch size is stipulated, the requirement is simply that the dryer be able to physically contain the volume of the solids, and the dryer volume and dimensions can thus be calculated directly. Solids residence time must then be calculated. Equation (12-61) can be reversed and modified to give

$$t_{CR} = \frac{m_s(X_I - X_O) \lambda_{ev}}{h_{ws} \Delta T_{ws} A_s} \quad (12-62)$$

Heat-transfer control and constant-rate drying are assumed. Again, the calculation will be inaccurate and overoptimistic for falling-rate drying, and it is preferable to measure a drying curve and use a scaling calculation, as outlined in the next section. It is possible to compare the surface area/volume ratios of various types of dryers and deduce how their drying times will compare with each other (see "Drying Equipment—Batch Agitated and Rotary Dryers").

**Falling-Rate Kinetics** To correct from a calculated constant-rate (unhindered) drying time  $t_{CR}$  to first-order falling-rate kinetics, the following equation is used, where  $X_1$  is the initial,  $X_2$  the final, and  $X_E$  the equilibrium moisture content (all must be dry-basis):

$$\frac{t_{FR}}{t_{CR}} = \frac{X_1 - X_E}{X_1 - X_2} \ln \left( \frac{X_1 - X_E}{X_2 - X_E} \right) \quad (12-63)$$

Note that  $t_{FR} \geq t_{CR}$ . Likewise, to convert to a two-stage drying process with constant-rate drying down to  $X_{cr}$  and first-order falling-rate drying beyond, the equation is

$$\frac{t_{2S}}{t_{CR}} = \frac{X_1 - X_{cr}}{X_1 - X_2} + \frac{X_{cr} - X_E}{X_1 - X_2} \ln \left( \frac{X_{cr} - X_E}{X_2 - X_E} \right) \quad (12-64)$$

**Scale-up Effects** As dryers get larger, if the drying rate is either controlled by heat transfer (unhindered or constant-rate drying) or proportional to it (first-order drying kinetics following the characteristic drying curve), then the drying rate  $N$  (kg/kg/s) and drying time  $t$  will be proportional to the ratio between the area over which heat enters and the mass or volume of solids.

For most types of dryer, it is found that the specific drying rate (SDR), which is a mass flux (evaporation rate per unit area), is constant for a given set of operating conditions. The concept is described by Moyers [Drying Technol. 12(1 and 2):393 (1994)]. For convective layer dryers, both through-circulation and cross-circulation, mass increases proportionately to bed area if layer depth remains constant; hence the drying time should remain the same. This is also true of fluidized beds and of contact dryers where the solids rest as a layer on a heated plate (tray, vacuum band, plate, film-drum, and thin-film dryers). However, for mechanically agitated and rotating contact dryers (vertical pan, conical, double-cone tumbler, and paddle), the heat-transfer surface area increases as the square of dryer diameter and volume as the cube, and hence drying time increases with the cube root of batch size:

$$\frac{t_2}{t_1} = \left( \frac{m_2}{m_1} \right)^{1/3} \quad (12-65)$$

Providing additional internal heating surfaces, such as heated agitators or steam tubes in paddle or rotary dryers, gives a higher area/volume ratio and faster drying, so these will be the preferred contact dryer types for large batches or high throughput. This applies if the drying rate is proportional to the rate of heat supply. For a continuous dryer, heating the agitator allows a smaller dryer for a given solids throughput; for a batch dryer with fixed batch size, a heated agitator shortens the required drying time. However, if a minimum residence time is required to allow removal of tightly bound moisture, there will be little or no gain from providing very large amounts of heat-transfer area.

Again, these methods take no account of the actual drying kinetics of the particle, which are included in the next section.

**Example 19: Drying of Particles** A convective dryer is to be used to dry 720 kg/h (0.2 kg/s) of particulate material from 0.2 to 0.02 kg/kg moisture content (all flows and moistures on dry basis), using air at 180°C and 0.005 kg/kg humidity. Estimate the required air flow rate and dryer size for a fluidized-bed dryer (0.5 m/s inlet velocity) and a pneumatic conveying dryer (20 m/s inlet velocity). Assume outlet RH is approximately 20 percent. What is the effect of 10 percent heat losses?

**Solution:** Using a psychrometric chart with  $T_{CI} = 180^\circ\text{C}$ , the outlet gas temperature is approximately 70°C,  $Y_O = 0.048$  kg/kg with no losses, or 0.040 kg/kg with 10 percent losses.

From the mass balance, Eq. (12-54):  $0.2(0.2 - 0.02) = W_G(Y_O - 0.005)$ . Hence  $W_G = 0.837$  kg/s (no losses) or 1.03 kg/s (10 percent losses).



The cross-sectional area of the dryer is obtained from Eq. (12-59), by taking  $\rho_G$  at 180°C as 0.78 kg/m<sup>3</sup>. For the fluidized-bed dryer, assuming 10 percent losses;  $A_B = 1.03/(0.5 \times 0.78) = 2.64$  m<sup>2</sup>. For a circular bed,  $D = 1.83$  m. For the pneumatic conveying dryer, assuming 10 percent losses,  $A_{ss} = 1.03/(20 \times 0.78) = 0.066$  m<sup>2</sup>. For a circular duct,  $D = 0.29$  m; for a square duct,  $D = 0.257$  m.

A similar example for batch dryers may be found in the section "Batch Agitated and Rotating Dryers," including constant-rate and falling-rate kinetics and scale-up from an experimental test result.

## SCALING MODELS

These models use experimental data from drying kinetics tests in a laboratory, pilot-plant or full-scale dryer, and are thus more accurate and reliable than methods based only on estimated drying kinetics. They treat the dryer as a complete unit, with drying rates and air velocities averaged over the dryer volume, except that, if desired, the dryer can be subdivided into a small number of sections. These methods are used for layer dryers (tray, oven, horizontal-flow band, and vertical-flow plate types) and for a simple estimate of fluidized-bed dryer performance. For batch dryers, they can be used for scale-up by refining the scoping design calculation.

The basic principle is to take an experimental drying curve and perform two transformations: (1) from test operating conditions to full-scale operating conditions and (2) for test dimensions to full-scale dryer dimensions. If the operating conditions of the test (e.g., temperature, gas velocity, agitation rate) are the same as those for the full-scale plant, the first correction is not required.

Scaling models are the main design method traditionally used by dryer manufacturers. Pilot-plant test results are scaled to a new set of conditions on a dryer with greater airflow or surface area by empirical rules, generally based on the external driving forces (temperature, vapor pressure, or humidity driving forces). By implication, therefore, a characteristic drying curve concept is again being used, scaling the external heat and mass transfer and assuming that the internal mass transfer changes in proportion. A good example is the set of rules described under "Fluidized-Bed Dryers," which include the effects of temperature, gas velocity, and bed depth on drying time in the initial test and the full-scale dryer.

The integral model is a development of a simple scale-up model which allows for mixing and residence time effects, first suggested for fluidized beds by Vanecek et al. (1964, 1966). The mean outlet moisture content is given by summing the product of the particle moisture content and the probability that it emerges at time  $t$ :

$$\bar{X} = \int_0^{\infty} E(t)X(t) dt \quad (12-66)$$

Here  $X(t)$  is the drying curve, corrected as before to the new scale and new operating conditions, and  $E(t)$  is the residence time function, which must be known. This approach has been used successfully for well-mixed fluidized beds. For pure plug flow,  $E(t)$  is a spike (Green's function) and  $\bar{X} = X(t)$ .

**Scale-up of Batch Dryers** We can use the same equations as before but base drying time on an experimental value rather than one obtained from an unhindered drying calculation.

**Example 20: Scaling of Data** An experimental batch drying curve has been measured at 100°C, and drying time was 2 h. Estimate the drying time at (a) 100°C and (b) 150°C for (i) a fluidized-bed dryer and (ii) a conical vacuum dryer at 100-mbar absolute pressure, for a batch size 10 times greater than that of the test. Assume for the fluidized bed that temperature driving forces are proportional to  $T - T_{wb}$  and batch drying time is proportional to bed depth, and for the conical dryer that the solids temperature is equal to the saturation temperature at 100-mbar pressure (46°C for water vapor).

**Solution:**

(i) For the fluid bed,  $T = 100^\circ\text{C}$  and  $150^\circ\text{C}$ ,  $T_{wb} = 30^\circ\text{C}$  and  $38^\circ\text{C}$ , respectively. The increase in heat transfer and drying rate for case (b) is a factor of  $112/70 = 1.6$ .

The bed could be scaled up by increasing the bed area by a factor of 10 and keeping depth  $z$  constant, in which case drying time will remain at 2 h for case (a) and become  $2/1.6 = 1.25$  h for case (b).

Alternatively, all dimensions could be scaled up proportionately; as  $V = \rho D^2 z$ ,  $D$  and  $z$  will increase by  $\sqrt[3]{10} = 2.16$ . Drying time then becomes 4.32 h for (a) and 2.70 h for (b).

(ii) For the conical dryer, in case (b), temperature driving force increases by a factor of  $104/54 = 1.93$ . From Eq. (12-65), linear dimensions and drying time all scale up by a factor of  $\sqrt{10}$ . Drying time becomes 4.32 h for (a) and  $2\sqrt{10}/1.93 = 2.24$  h for (b).

## DETAILED OR RIGOROUS MODELS

These models aim to predict local conditions within the dryer and the transient condition of the particles and gas in terms of temperature, moisture content, velocity, etc. Naturally, they require much more input data. There are many published models of this type in the academic literature. They give the possibility of more detailed results, but the potential cumulative errors are also greater.

- Incremental models track the local conditions of the gas and particles through the dryer, mainly in one dimension. They are especially suitable for cocurrent and countercurrent dryers, e.g., flash (pneumatic conveying) and rotary dryers. The air conditions are usually treated as uniform across the cross-section and dependent only on axial position. This method can also be used to determine local conditions (e.g., temperature) where a simpler model has been used to find the overall drying rate. A two- or three-dimensional grid can also be used, e.g., modeling vertical and horizontal variations in a band dryer or plug-flow fluidized bed.
- Complex three-dimensional models, e.g., CFD (computational fluid dynamics), aim to solve the gas conditions and particle motion throughout the dryer. They are the only effective models for spray dryers because of the complex swirling flow pattern; they can also be used to find localized conditions in other dryers.

**Incremental Model** The one-dimensional incremental model is a key analysis tool for several types of dryers. A set of simultaneous equations is solved at a given location (Fig. 12-40), and the simulation moves along the dryer axis in a series of steps or increments—hence the name. The procedure may be attempted by hand if a few large steps (say, 5 to 10) are used; but for an accurate simulation, a computer program is needed and thousands of increments may be used.

Increments may be stated in terms of time ( $dt$ ), length ( $dz$ ), or moisture content ( $dX$ ). A set of six simultaneous equations is then solved, and ancillary calculations are also required, e.g., to give local values of gas and solids properties. The generic set of equations (for a time increment  $\Delta t$ ) is as follows:

$$\text{Heat transfer to particle: } Q_P = h_{PG}A_P(T_G - T_s) \quad (12-67)$$

Mass transfer from particle:

$$\frac{-dX}{dt} = \text{function}(X, Y, T_P, T_G, h_{PG}, A_P) \quad (12-68)$$

$$\text{Mass balance on moisture: } G\Delta Y = -F\Delta X = F \frac{-dX}{dt} \Delta t \quad (12-69)$$

$$\text{Heat balance on particle: } \Delta T_s = \frac{Q_P \Delta t - \lambda_{ws} m_P \Delta X}{m_P(C_{PS} + C_{PL}X)} \quad (12-70)$$

Heat balance for increment:

$$-\Delta T_G = \frac{F(C_{PS} + C_{PL}X)\Delta T_s + G(\lambda_{G0} + C_{PY}T_G)\Delta Y + \Delta Q_{wl}}{G(C_{PG} + C_{PY}Y)} \quad (12-71)$$

$$\text{Particle transport: } \Delta z = U_s \Delta t \quad (12-72)$$

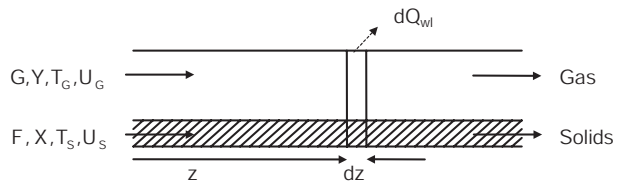


FIG. 12-40 Principle of the incremental model.

The mass and heat balance equations are the same for any type of dryer, but the particle transport equation is completely different, and the heat- and mass-transfer correlations are also somewhat different as they depend on the environment of the particle in the gas (i.e., single isolated particles, agglomerates, clusters, layers, fluidized beds, or packed beds). The mass-transfer rate from the particle is regulated by the drying kinetics and is thus obviously material-dependent (at least in falling-rate drying).

The model is effective and appropriate for dryers where both solids and gas are approximately in axial plug flow, such as pneumatic conveying and cascading rotary dryers. However, it runs into difficulties where there is recirculation or radial flow.

The incremental model is also useful for measuring variations in local conditions such as temperature, solids moisture content, and humidity along the axis of a dryer (e.g., plug-flow fluidized bed), through a vertical layer (e.g., tray or band dryers), or during a batch drying cycle (using time increments, not length). It can be applied in these situations even though the integral model has been used to determine the overall kinetics and drying time.

**Example 21: Sizing of a Cascading Rotary Dryer** The average gas velocity passing through a cocurrent, adiabatic, cascading rotary dryer is 4 m/s. The particles moving through the dryer have an average diameter of 5 mm, a solids density of 600 kg/m<sup>3</sup>, and a shape factor of 0.75. The particles enter with a moisture content of 0.50 kg/kg (dry basis) and leave with a moisture content of 0.15 kg/kg (dry basis). The drying rate may be assumed to decrease linearly with average moisture content, with no unhindered ("constant-rate") drying period. In addition, let us assume that the solids are nonhygroscopic (so that the equilibrium moisture content is zero; hygroscopic means that the equilibrium moisture content is nonzero).

The inlet humidity is 0.10 kg/kg (dry basis) due to the use of a direct-fired burner, and the ratio of the flow rates of dry solids to dry gas is unity ( $F/G = 1$ ). The gas temperature at the inlet to the dryer is 800°C, and the gas may be assumed to behave as a pure water vapor/air mixture.

What is the gas-phase residence time that is required?

Data:

|                               |                               |                              |
|-------------------------------|-------------------------------|------------------------------|
| $U = 4 \text{ m/s}$           | $X_i = 0.50 \text{ kg/kg}$    | $F/G = 1$                    |
| $d_{PSM} = 0.005 \text{ m}$   | $X_o = 0.15 \text{ kg/kg}$    | $T_{GI} = 800^\circ\text{C}$ |
| $\rho_p = 600 \text{ kg/m}^3$ | $X_{cr} = 0.50 \text{ kg/kg}$ | $Y_i = 0.10 \text{ kg/kg}$   |
| $\alpha_p = 0.75$             | $X_{eq} = 0.0 \text{ kg/kg}$  |                              |

Application of concept of characteristic drying curve: A linear-falling rate curve implies the following equation for the drying kinetics:

$$f = \Phi \quad \text{assumption of linear drying kinetics} \quad (12-73)$$

where  $f$  is the drying rate relative to the initial drying rate.

$$f = \frac{N}{N_{\text{initial}}}$$

Since the material begins drying in the falling-rate period, the critical moisture content can be taken as the initial moisture content. The equilibrium moisture content is zero since the material is not hygroscopic.

$$\Phi = \frac{\bar{X} - X_{eq}}{X_{cr} - X_{eq}} = \frac{\bar{X}}{0.5}$$

**Application of mass balances (theory):** A mass balance around the inlet and any section of the dryer is shown in Fig. 12-41.

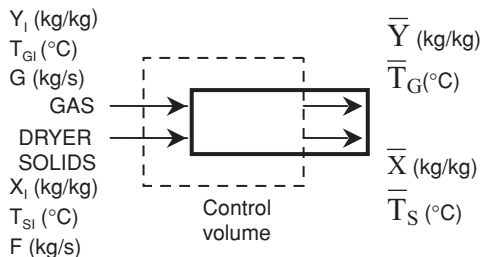


FIG. 12-41 Mass balance around a typical section of a cocurrent dryer.

The essential idea is to calculate the average gas humidity  $\bar{Y}$  at each average moisture content  $\bar{X}$ .

A differential mass balance on the air at any position in the bed is given below.

$$F \cdot dX = -G \cdot dY$$

$$-\frac{dY}{dX} = \frac{F}{G} = \frac{Y - Y_i}{X_i - \bar{X}} \quad (12-74)$$

where  $Y$  = gas humidity, kg moisture/kg dry gas

$X$  = solids moisture content, kg moisture/kg dry solids

$W_G$  = flow rate of dry gas, kg dry gas/s

$W_S$  = flow rate of dry solids, kg dry solids/s

**Application of mass balances:** Plugging in the numbers gives the relationship between absolute humidity and moisture in the solids at any position.

$$\frac{\bar{Y} - 0.1}{0.5 - \bar{X}} = 1 = \frac{F}{G}$$

$$\bar{Y} = 0.6 - \bar{X}$$

$$Y_o = 0.6 - 0.15 = 0.45 \text{ kg/kg}$$

From Mollier chart:  $T_{ub} = 79^\circ\text{C}$   $Y_s^* = 0.48 \text{ kg/kg}$

For the whole dryer,  $\bar{Y} = 0.275 \text{ kg/kg}$

The mass balance information is important, but not the entire answer to the question. Now the residence time can be calculated from the kinetics.

**Application of concept of characteristic drying curve to estimating drying rates in practice (theory):** The overall (required) change in moisture content is divided into a number of intervals of size  $\Delta X$ . The sizes of the intervals need not be the same and should be finer where the fastest moisture content change occurs. For the sake of simplicity, this example will use intervals of uniform size. Then the application of the concept of a characteristic drying curve gives the following outcomes.

$$-\frac{d\bar{X}}{dt} = \text{drying rate in interval}$$

$$= \frac{f \cdot k \cdot \phi \cdot \frac{A_p}{\rho_p} \cdot \frac{A_p}{V_p} (Y_s^* - \bar{Y})}{V_p} \quad (12-75)$$

where  $f$  = relative drying rate in interval (dimensionless)

$\bar{Y}$  = average humidity in interval, kg/kg

$\phi$  = humidity potential coefficient, close to unity

$\rho_p$  = density of dry solids, kg/m<sup>3</sup>

$Y_s^*$  = humidity at saturation, from the adiabatic saturation contour on Mollier chart

$$\frac{A_p}{V_p} = \frac{\text{particle surface area}}{\text{particle volume}} = \frac{6}{\phi_p \cdot d_{PSM}} \quad (12-76)$$

$d_{PSM}$  = Sauter-mean particle diameter for mixture (volume-surface diameter), m

$\phi_p$  = particle shape factor, unity for spheres (dimensionless)

$k$  = **mass-transfer coefficient**, kg/(m<sup>2</sup>·s), obtained from the heat-transfer coefficient (often easier to obtain) using the Chilton-Colburn analogy

$$k \cdot \phi = \frac{\delta}{C_{PY}} h \quad (12-77)$$

$\delta$  = psychrometric ratio, close to unity for air/water vapor system

$C_{PY}$  = humid heat capacity =  $C_{PG} + YC_{PV}$

$C_{PG}$  = specific heat capacity of dry gas (air), J/(kg·K)

$C_{PV}$  = specific heat capacity of water vapor, J/(kg·K)

$h$  = **heat-transfer coefficient**, W/(m<sup>2</sup>·K)

Define

$Re_p$  = particle Reynolds number

$$= \frac{U \cdot d_{PSM}}{\nu} \quad (12-78)$$

$U$  = relative velocity between gas and particles; in cascading rotary dryers, this is almost constant throughout the dryer and close to the superficial gas velocity  $U_{Gsuper}$

$\nu$  = kinematic viscosity of gas at average  $T_G$  in dryer, m<sup>2</sup>/s



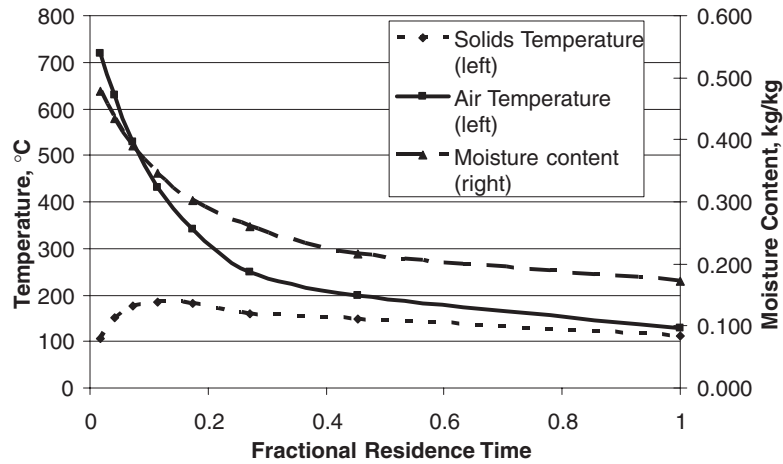


FIG. 12-42 Typical variation of process conditions through a cocurrent cascading rotary dryer.

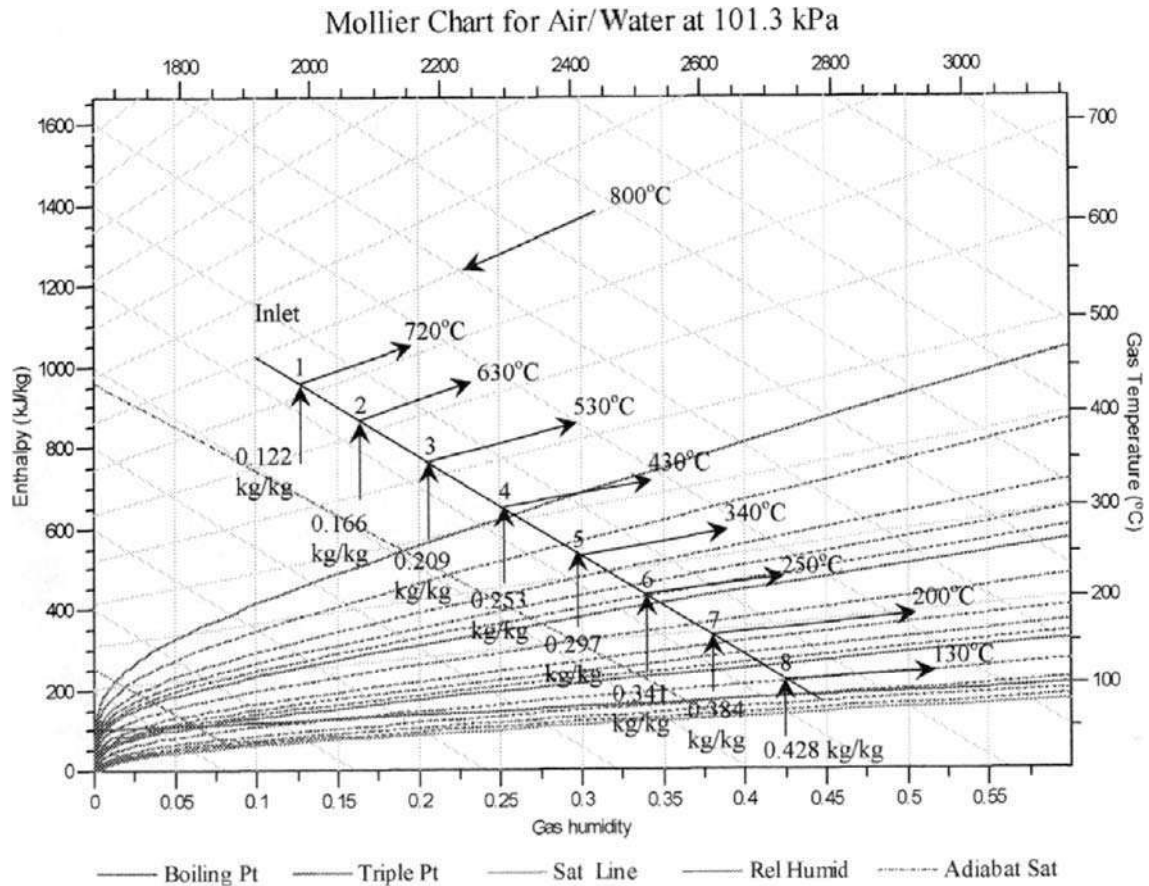


FIG. 12-43 Enthalpy humidity chart used to generate the results in Table 12-13 plots humidity (abscissa) against enthalpy (lines sloping diagonally from top left to bottom right).



**Agitated and Rotating Batch Dryers** Scale up from pilot-plant tests in a small-scale dryer at the same temperature and pressure and similar agitation conditions. As noted under scoping design, scale-up depends on the surface area/volume ratio, and hence normally to the one-third power of mass. Results from one dryer type may be extrapolated to a different type if assumptions are made on the heat transfer coefficients in both dryers; obviously this is less reliable than measurements on the same dryer type.

**Fluidized-Bed Dryers** In design mode, the required gas flow rate can be obtained from a heat and mass balance. Bed cross-sectional area is found from the scoping design calculation; the required gas velocity should be found from fluidization tests, but for initial design purposes, a typical value is 0.5 m/s.

For scale-up based on an experimentally recorded batch drying curve, including performance mode calculations and altering operating conditions, Kemp and Oakley (2002) showed that the drying time for a given range of moisture content  $\Delta X$  scales according to the relationship

$$\frac{\Delta\tau_2}{\Delta\tau_1} = Z = \frac{(m_B/A)_2 G_1 (T_{Cl} - T_{ub})_1 (1 - e^{-fNTU_1})}{(m_B/A)_1 G_2 (T_{Cl} - T_{ub})_2 (1 - e^{-fNTU_2})} \quad (12-83)$$

where 1 denotes experimental or original conditions and 2 denotes full-scale or new conditions;  $Z$  is the normalization factor;  $G$  is gas mass flux;  $m_B/A$  is bed mass per unit area, proportional to bed depth  $z$ ;  $NTU$  is number of transfer units through the bed; and  $f$  is falling-rate kinetics factor. This method can be used to scale a batch drying curve section by section. Almost always, one of two simplified limiting cases applies, known as type A and type B normalization. In type A,  $f(NTU)$  is high, the exponential term is negligible, and the drying time is proportional to  $G(T_{Cl} - T_{ub})/(m_B/A)$ . This applies to all fast-drying materials and the vast majority of other materials, even well into the falling-rate period. In type B,  $f(NTU)$  is low, and expanding the exponential term shows that drying time is simply proportional to  $T_{Cl} - T_{ub}$ . This applies to a few very slow-drying materials, at very low moisture contents or where drying kinetics is completely controlled by internal moisture movement (e.g., wheat and grain, which have thick cell walls).

For a typical pilot-plant experiment, the fluidization velocity and temperature driving forces are similar to those of the full-size bed, but the bed diameter and depth are much less. Hence, for type A normalization, the  $m_B/A$  term dominates,  $Z$  is much greater than 1, and the drying time in the full-scale bed is typically 5 to 10 times that in the pilot-plant.

The drying time, bed area, solids throughput, and bed depth expressed as  $m_B/A$  are linked by

$$W_s = \frac{A_B}{\tau_s} \left( \frac{m_B}{A_B} \right) \quad (12-84)$$

*The consequence is that increasing gas velocity is beneficial for type A normalization (giving reduced drying time and either a higher throughput or a smaller bed area) but gives no real benefit for type B; likewise, increasing bed depth is beneficial for type B (giving either a higher throughput or a smaller bed area with the same drying time) but not type A. However, using unnecessarily high gas velocity or an unnecessarily deep bed increases pressure drop and operating costs.*

**Cascading Rotary Dryers** In design mode, the required gas flow rate can be obtained from a heat and mass balance. Bed cross-sectional area is found from the scoping design calculation (a typical gas velocity is 3 m/s for cocurrent and 2 m/s for countercurrent units). Length is normally between 5 and 10 times drum diameter (an  $L/D$  value of 8 can be used for initial estimation) or can be calculated by using an incremental model (see worked example).

**Entrainment Dryers** In design mode, the required gas flow rate can be obtained from a heat and mass balance. For pneumatic conveying dryers, duct cross-sectional area and diameter are found from the scoping design calculation (if required gas velocity is unknown, a typical value is 20 m/s). Duct length can be estimated by an incremental model, but some parameters are hard to obtain and conditions change rapidly near the feed point, so the model is most effective for scaling up from pilot-plant data; see Kemp and Oakley (2002). Spray

dryer chamber design is complex, and sizing should normally be done by manufacturers.

## ADDITIONAL READING

- Kemp, Bahu, and Oakley, "Modeling Vertical Pneumatic Conveying Dryers," *Drying '91* (7th Int. Drying Symp., Prague, Czechoslovakia, Aug. 1990), Mujumdar et al. (eds.), Elsevier, 1991, pp. 217–227.
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- Vanecek, Picka, and Najmr, *Fluidized Bed Drying*, Leonard Hill, London, 1966.

## DRYER DESCRIPTIONS

**GENERAL REFERENCES:** *Aspen Process Manual* (Internet knowledge base), Aspen Technology, 2000 onward. Baker (ed.), *Industrial Drying of Foods*, Blaikie, London, 1997; Cook and DuMont, *Process Drying Practice*, McGraw-Hill, 1991. *Drying Technology—An International Journal*, Marcel Dekker, New York, 1982 onward. Keey, *Drying of Loose and Particulate Materials*, Hemisphere, New York, 1992. Masters, *Spray Drying Handbook*, Wiley, New York, 1990. Mujumdar (ed.), *Handbook of Industrial Drying*, Marcel Dekker, New York, 1995. Nonhebel and Moss, *Drying of Solids in the Chemical Industry*, CRC Press, Cleveland Ohio, 1971. van't Land, *Industrial Drying Equipment*, Marcel Dekker, New York, 1991.

### Batch Tray Dryers

**Description** A tray or compartment dryer is an enclosed, insulated housing in which solids are placed upon tiers of trays in the case of particulate solids or stacked in piles or upon shelves in the case of large objects. Heat transfer may be *direct* from gas to solids by circulation of large volumes of hot gas or *indirect* by use of heated shelves, radiator coils, or refractory walls inside the housing. In indirect-heat units, excepting vacuum-shelf equipment, circulation of a small quantity of gas is usually necessary to sweep moisture vapor from the compartment and prevent gas saturation and condensation. Compartment units are employed for the heating and drying of lumber, ceramics, sheet materials (supported on poles), painted and metal objects, and all forms of particulate solids.

**Classification** Batch; nonagitated; layer; convective (cross-circulation or through-circulation) or contact/conduction.

**Field of Application** Because of the high labor requirements usually associated with loading or unloading the compartments, batch compartment equipment is rarely economical except in the following situations:

1. A long heating cycle is necessary because the size of the solid objects or permissible heating temperature requires a long holdup for internal diffusion of heat or moisture. This case may apply when the cycle will exceed 12 to 24 h.
2. The production of several different products requires strict batch identity and thorough cleaning of equipment between batches. This is a situation existing in many small multiproduct plants, e.g., for pharmaceuticals or specialty chemicals.
3. The quantity of material to be processed does not justify investment in more expensive, continuous equipment. This case would apply in many pharmaceutical drying operations.

Further, because of the nature of solids-gas contacting, which is usually by parallel flow and rarely by through-circulation, heat transfer and mass transfer are comparatively inefficient. For this reason, use of tray and compartment equipment is restricted primarily to ordinary drying and heat-treating operations. Despite these harsh limitations, when the listed situations do exist, economical alternatives are difficult to develop.

**Auxiliary Equipment** If noxious gases, fumes, or dust is given off during the operation, dust or fume recovery equipment will be

necessary in the exhaust gas system. Wet scrubbers are employed for the recovery of valuable solvents from dryers. To minimize heat losses, thorough insulation of the compartment with brick, asbestos, or other insulating compounds is necessary. Modern fabricated dryer compartment panels usually have 7.5 to 15 cm of blanket insulation placed between the internal and external sheet-metal walls. Doors and other access openings should be gasketed and tight. In the case of tray and truck equipment, it is usually desirable to have available extra trays and trucks so that they can be preloaded for rapid emptying and loading of the compartment between cycles. Air filters and gas dryers are occasionally employed on the inlet air system for direct-heat units.

Vacuum-shelf dryers require auxiliary stream jets or other vacuum-producing devices, intercondensers for vapor removal, and occasionally wet scrubbers or (heated) bag-type dust collectors.

Uniform depth of loading in dryers and furnaces handling particulate solids is essential to consistent operation, minimum heating cycles, or control of final moisture. After a tray has been loaded, the bed should be leveled to a uniform depth. Special preform devices, noodle extruders, pelletizers, etc., are employed occasionally for preparing pastes and filter cakes so that screen bottom trays can be used and the advantages of through-circulation approached.

Control of tray and compartment equipment is usually maintained by control of the circulating air temperature (and humidity) and rarely by the solids temperature. On vacuum units, control of the absolute pressure and heating-medium temperature is utilized. In direct dryers, cycle controllers are frequently employed to vary the air temperature or velocity across the solids during the cycle; e.g., high air temperatures may be employed during a constant-rate drying period while the solids surface remains close to the air wet-bulb temperature. During the falling-rate periods, this temperature may be reduced to prevent case hardening or other degrading effects caused by overheating the solids surfaces. In addition, higher air velocities may be employed during early drying stages to improve heat transfer; however, after surface drying has been completed, this velocity may need to be reduced to prevent dusting. Two-speed circulating fans are employed commonly for this purpose.

**Direct-Heat Tray Dryers** Satisfactory operation of tray-type dryers depends on maintaining a constant temperature and a uniform air velocity over all the material being dried.

Circulation of air at velocities of 1 to 10 m/s is desirable to improve the surface heat-transfer coefficient and to eliminate stagnant air pockets. Proper airflow in tray dryers depends on sufficient fan capacity, on the design of ductwork to modify sudden changes in direction, and on properly placed baffles. *Nonuniform airflow is one of the most serious problems in the operation of tray dryers.*

Tray dryers may be of the tray-truck or the stationary-tray type. In the former, the trays are loaded on trucks which are pushed into the dryer; in the latter, the trays are loaded directly into stationary racks within the dryer. Trucks may be fitted with flanged wheels to run on tracks or with flat swivel wheels. They may also be suspended from and moved on monorails. Trucks usually contain two tiers of trays, with 18 to 48 trays per tier, depending upon the tray dimensions.

Trays may be square or rectangular, with 0.5 to 1 m<sup>2</sup> per tray, and may be fabricated from any material compatible with corrosion and temperature conditions. When the trays are stacked in the truck, there should be a clearance of not less than 4 cm between the material in one tray and the bottom of the tray immediately above. When material characteristics and handling permit, the trays should have screen bottoms for additional drying area. Metal trays are preferable to non-metallic trays, since they conduct heat more readily. Tray loadings range usually from 1 to 10 cm deep.

Steam is the usual heating medium, and a standard heater arrangement consists of a main heater before the circulating fan. When steam is not available or the drying load is small, electric heat can be used. For temperatures above 450 K, products of combustion can be used, or indirect-fired air heaters.

Air is circulated by propeller or centrifugal fans; the fan is usually mounted within or directly above the dryer. Above 450 K, external or water-cooled bearings become necessary. Total pressure drop through

the trays, heaters, and ductwork is usually in the range of 2.5 to 5 cm of water. Air recirculation is generally in the order of 80 to 95 percent except during the initial drying stage of rapid evaporation. Fresh air is drawn in by the circulating fan, frequently through dust filters. In most installations, air is exhausted by a separate small exhaust fan with a damper to control air recirculation rates.

**Prediction of heat- and mass-transfer coefficients in direct heat tray dryers** In convection phenomena, heat-transfer coefficients depend on the geometry of the system, the gas velocity past the evaporating surface, and the physical properties of the drying gas. In estimating drying rates, the use of heat-transfer coefficients is preferred because they are usually more reliable than mass-transfer coefficients. In calculating mass-transfer coefficients from drying experiments, the partial pressure at the surface is usually inferred from the measured or calculated temperature of the evaporating surface. Small errors in temperature have negligible effect on the heat-transfer coefficient but introduce relatively large errors in the partial pressure and hence in the mass-transfer coefficient.

For many cases in drying, the heat-transfer coefficient is proportional to  $U_g^n$ , where  $U_g$  is an appropriate local gas velocity. For flow parallel to plane plates, the exponent  $n$  has been reported to range from 0.35 to 0.8. The differences in exponent have been attributed to differences in flow pattern in the space above the evaporating surface, particularly whether it is laminar or turbulent, and whether the length is sufficient to allow fully developed flow. In the absence of applicable specific data, the heat-transfer coefficient for the parallel-flow case can be taken, for estimating purposes, as

$$h = \frac{8.8J^{0.8}}{D_c^{0.2}} \quad (12-85)$$

where  $h$  is the heat-transfer coefficient, W/(m<sup>2</sup>·K) [or J/s·m<sup>2</sup>·K];  $J$  is the gas mass flux, kg/(m<sup>2</sup>·S); and  $D_c$  is a characteristic dimension of the system. The experimental data have been weighted in favor of an exponent of 0.8 in conformity with the usual Colburn  $j$  factor, and average values of the properties of air at 370 K have been incorporated. Typical values are in the range 10 to 50 W/(m<sup>2</sup>·K).

Experimental data for drying from flat surfaces have been correlated by using the equivalent diameter of the flow channel or the length of the evaporating surface as the characteristic length dimension in the Reynolds number. However, the validity of one versus the other has not been established. The proper equivalent diameter probably depends at least on the geometry of the system, the roughness of the surface, and the flow conditions upstream of the evaporating surface. For most tray drying calculations, the equivalent diameter (4 times the cross-sectional area divided by the perimeter of the flow channel) should be used.

For airflow impinging normally to the surface from slots, nozzles, or perforated plates, the heat-transfer coefficient can be obtained from the data of Friedman and Mueller (*Proceedings of the General Discussion on Heat Transfer*, Institution of Mechanical Engineers, London, and American Society of Mechanical Engineers, New York, 1951, pp. 138–142). These investigators give

$$h = \alpha J^{0.78} \quad (12-86)$$

where the gas mass flux  $J$  is based on the total heat-transfer area and is dependent on the plate open area, hole or slot size, and spacing between the plate, nozzle, or slot and the heat-transfer surface.

Most efficient performance is obtained with plates having open areas equal to 2 to 3 percent of the total heat-transfer area. The plate should be located at a distance equal to four to six hole (or equivalent) diameters from the heat-transfer surface.

Data from tests employing multiple slots, with a correction calculated for slot width, were reported by Korger and Kizek [*Int. J. Heat Mass Transfer*, London, 9:337 (1966)].

Another well-known correlation has been used to predict heat- and mass-transfer coefficients for air impinging on a surface from arrays of holes (jets). This correlation uses relevant geometric properties such as the diameter of the holes, the distance between the holes, and the distance between the holes and the sheet [Martin, "Heat and Mass Transfer Between Impinging Gas Jets and Solid

## 12-58 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

Surfaces," *Advances in Heat Transfer*, vol. 13, Academic Press, 1977, pp. 1-66].

$$\frac{Sh}{Sc^{0.42}} = \frac{Nu}{Pr^{0.42}} = \left[ 1 + \left( \frac{H/D}{0.6/\sqrt{f}} \right)^6 \right]^{-0.05} \\ \times \sqrt{f} \frac{1 - 2.2\sqrt{f}}{1 + 0.2(H/D - 6)\sqrt{f}} \times Re^{2/3}$$

where  $D$  = diameter of nozzle, m

$$f = \frac{\pi}{2\sqrt{3}} \left( \frac{D}{L_D} \right)^2$$

$H$  = distance from nozzle to sheet, m

$L_D$  = average distance between nozzles, m

$Nu$  = Nusselt number

$Pr$  = Prandtl number

$$Re = \frac{wD}{\nu}, \text{ Reynolds number}$$

$Sc$  = Schmidt number

$Sh$  = Sherwood number

$w$  = velocity of air at nozzle exit, m/s

$\nu$  = kinematic viscosity of air, m<sup>2</sup>/s

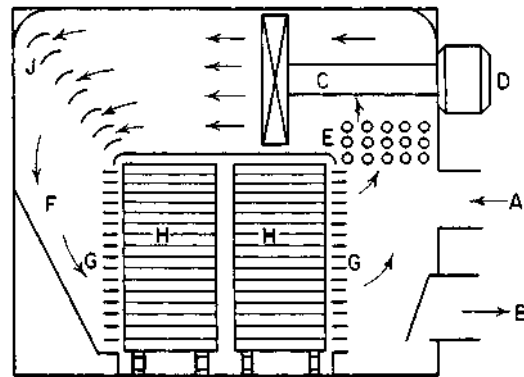
The heat- and mass-transfer coefficients were then calculated from the definitions of the Nusselt and Sherwood numbers.

$$Nu = \frac{hD}{k_{th}} \quad \text{where } k_{th} = \text{thermal conductivity of air, W/(m}\cdot\text{K)}$$

$D$  = diameter of holes in air bars, m

$$Sh = \frac{k_m D}{diff} \quad \text{where } diff = \text{diffusion coefficient of water vapor in air, m}^2/\text{s}$$

Air impingement is commonly employed for drying sheets, film, thin slabs, and coatings. The temperature driving force must also be found. When radiation and conduction are negligible, the temperature of the evaporating surface approaches the wet-bulb temperature and is readily obtained from the humidity and dry-bulb temperatures. Frequently, however, radiation and conduction cause the temperature of the evaporating surface to exceed the wet-bulb temperature. When this occurs, the true surface temperature must be estimated. The easiest way is to use a psychrometric chart and to change the slope of the adiabatic saturation line; a typical figure for the additional radiation is about 10 percent. In many cases this is canceled out by heat losses and



**FIG. 12-44** Double-truck tray dryer. (A) Air inlet duct. (B) Air exhaust duct with damper. (C) Adjustable-pitch fan 1 to 15 hp. (D) Fan motor. (E) Fin heaters. (F) Plenum chamber. (G) Adjustable air blast nozzles. (H) Trucks and trays. (J) Turning vanes.

the heat required to warm the solid, leaving the dryer approximately adiabatic.

As with many drying calculations, the most reliable design method is to perform experimental tests and to scale up. By measuring performance on a single tray with similar layer depth, air velocity, and temperature, the SDR (specific drying rate) concept can be applied to give the total area and number of trays required for the full-scale dryer.

**Performance data for direct heat tray dryers** A standard two-truck dryer is illustrated in Fig. 12-44. Adjustable baffles or a perforated distribution plate is normally employed to develop 0.3 to 1.3 cm of water pressure drop at the wall through which air enters the truck enclosure. This will enhance the uniformity of air distribution, from top to bottom, among the trays. In three (or more) truck ovens, air reheat coils may be placed between trucks if the evaporative load is high. Means for reversing airflow direction may also be provided in multiple-truck units.

Performance data on some typical tray and compartment dryers are tabulated in Table 12-14. These indicate that an overall rate of evaporation of 0.0025 to 0.025 kg water/(s·m<sup>2</sup>) of tray area may be expected from tray and tray-truck dryers. The thermal efficiency of this type of dryer will vary from 20 to 50 percent, depending on the drying temperature used and the humidity of the exhaust air. In drying to very low moisture contents under temperature restrictions, the thermal efficiency may be on the order of 10 percent. The major operating cost for a tray dryer is the labor involved in loading and unloading the trays. About 2 labor-hours is required to load and unload a standard

**TABLE 12-14 Manufacturer's Performance Data for Tray and Tray-Truck Dryers\***

| Material                                        | Color       | Chrome yellow  | Toluidine red | Half-finished Titone | Color         |
|-------------------------------------------------|-------------|----------------|---------------|----------------------|---------------|
| Type of dryer                                   | 2-truck     | 16-tray dryer  | 16-tray       | 3-truck              | 2-truck       |
| Capacity, kg product/h                          | 11.2        | 16.1           | 1.9           | 56.7                 | 4.8           |
| Number of trays                                 | 80          | 16             | 16            | 180                  | 120           |
| Tray spacing, cm                                | 10          | 10             | 10            | 7.5                  | 9             |
| Tray size, cm                                   | 60 × 75 × 4 | 65 × 100 × 2.2 | 65 × 100 × 2  | 60 × 70 × 3.8        | 60 × 70 × 2.5 |
| Depth of loading, cm                            | 2.5 to 5    | 3              | 3.5           | 3                    |               |
| Initial moisture, % bone-dry basis              | 207         | 46             | 220           | 223                  | 116           |
| Final moisture, % bone-dry basis                | 4.5         | 0.25           | 0.1           | 25                   | 0.5           |
| Air temperature, °C                             | 85-74       | 100            | 50            | 95                   | 99            |
| Loading, kg product/m <sup>2</sup>              | 10.0        | 33.7           | 7.8           | 14.9                 | 9.28          |
| Drying time, h                                  | 33          | 21             | 41            | 20                   | 96            |
| Air velocity, m/s                               | 1.0         | 2.3            | 2.3           | 3.0                  | 2.5           |
| Drying, kg water evaporated/(h·m <sup>2</sup> ) | 0.59        | 65             | 0.41          | 1.17                 | 0.11          |
| Steam consumption, kg/kg water evaporated       | 2.5         | 3.0            | —             | 2.75                 |               |
| Total installed power, kW                       | 1.5         | 0.75           | 0.75          | 2.25                 | 1.5           |

\*Courtesy of Wolverine Proctor & Schwartz, Inc.

two-truck tray dryer. In addition, about one-third to one-fifth of a worker's time is required to supervise the dryer during the drying period. Power for tray and compartment dryers will be approximately 1.1 kW per truck in the dryer. Maintenance will run from 3 to 5 percent of the installed cost per year.

**Batch Through-Circulation Dryers** These may be either of shallow bed or deep bed type. In the first type of batch through-circulation dryer, heated air passes through a stationary permeable bed of the wet material placed on removable screen-bottom trays suitably supported in the dryer. This type is similar to a standard tray dryer except that hot air passes through the wet solid instead of across it. The pressure drop through the bed of material does not usually exceed about 2 cm of water. In the second type, deep perforated-bottom trays are placed on top of plenum chambers in a closed-circuit hot air circulating system. In some food-drying plants, the material is placed in finishing bins with perforated bottoms; heated air passes up through the material and is removed from the top of the bin, reheated, and recirculated. The latter types involve a pressure drop through the bed of material of 1 to 8 cm of water at relatively low air rates. Table 12-15 gives performance data on three applications of batch through-circulation dryers. Batch through-circulation dryers are restricted in application to granular materials (particle size typically 1 mm or greater) that permit free flow-through circulation of air. Drying times are usually much shorter than in parallel-flow tray dryers. Design methods are included in the subsection "Continuous Through-Circulation Dryers."

**Contact Tray and Vacuum-Shelf Dryers** Vacuum-shelf dryers are indirectly heated batch dryers consisting of a vacuum-tight chamber usually constructed of cast iron or steel plate, heated, supporting shelves within the chamber, a vacuum source, and usually a condenser. One or two doors are provided, depending on the size of the chamber. The doors are sealed with resilient gaskets of rubber or similar material. It is also possible, but much less common, to operate at atmospheric pressure without vacuum.

Hollow shelves of flat steel plate are fastened permanently inside the vacuum chamber and are connected in parallel to inlet and outlet headers. The heating medium, entering through one header and passing through the hollow shelves to the exit header, is generally steam, ranging in pressure from 700 kPa gauge to subatmospheric pressure for low-temperature operations. Low temperatures can be provided by circulating hot water, and high temperatures can be obtained by circulating hot oil or Dowtherm. Some small dryers employ electrically heated shelves. The material to be dried is placed in pans or trays on the heated shelves. The trays are generally of metal to ensure good heat transfer between the shelf and the tray.

**TABLE 12-15 Performance Data for Batch Through-Circulation Dryers\***

| Kind of material                                             | Granular polymer | Vegetable          | Vegetable seeds |
|--------------------------------------------------------------|------------------|--------------------|-----------------|
| Capacity, kg product/h                                       | 122              | 42.5               | 27.7            |
| Number of trays                                              | 16               | 24                 | 24              |
| Tray spacing, cm                                             | 43               | 43                 | 43              |
| Tray size, cm                                                | 91.4 × 104       | 91.4 × 104         | 85 × 98         |
| Depth of loading, cm                                         | 7.0              | 6                  | 4               |
| Physical form of product                                     | Crumbs           | 0.6-cm diced cubes | Washed seeds    |
| Initial moisture content, % dry basis                        | 11.1             | 669.0              | 100.0           |
| Final moisture content, % dry basis                          | 0.1              | 5.0                | 9.9             |
| Air temperature, °C                                          | 88               | 77 dry-bulb        | 36              |
| Air velocity, superficial, m/s                               | 1.0              | 0.6 to 1.0         | 1.0             |
| Tray loading, kg product/m <sup>2</sup>                      | 16.1             | 5.2                | 6.7             |
| Drying time, h                                               | 2.0              | 8.5                | 5.5             |
| Overall drying rate, kg water evaporated/(h·m <sup>2</sup> ) | 0.89             | 11.86              | 1.14            |
| Steam consumption, kg/kg water evaporated                    | 4.0              | 2.42               | 6.8             |
| Installed power, kW                                          | 7.5              | 19                 | 19              |

\*Courtesy of Wolverine Proctor & Schwartz, Inc.

Vacuum-shelf dryers may vary in size from 1 to 24 shelves, the largest chambers having overall dimensions of 6 m wide, 3 m long, and 2.5 m high.

Vacuum is applied to the chamber, and vapor is removed through a large pipe which is connected to the chamber in such a manner that if the vacuum is broken suddenly, the in-rushing air will not greatly disturb the bed of material being dried. This line leads to a condenser where moisture or solvent that has been vaporized is condensed. The noncondensable exhaust gas goes to the vacuum source, which may be a wet or dry vacuum pump or a steam-jet ejector.

Vacuum-shelf dryers are used extensively for drying pharmaceuticals, temperature-sensitive or easily oxidizable materials, and materials so valuable that labor cost is insignificant. They are particularly useful for handling small batches of materials wet with toxic or valuable solvents. Recovery of the solvent is easily accomplished without danger of passing through an explosive range. Dusty materials may be dried with negligible dust loss. Hygroscopic materials may be completely dried at temperatures below that required in atmospheric dryers. The equipment is employed also for freeze-drying processes, for metallizing-furnace operations, and for the manufacture of semiconductor parts in controlled atmospheres. All these latter processes demand much lower operating pressures than do ordinary drying operations.

**Design methods for vacuum-shelf dryers** Heat is transferred to the wet material by conduction through the shelf and bottom of the tray and by radiation from the shelf above. The critical moisture content will not be necessarily the same as for atmospheric tray drying, as the heat-transfer mechanisms are different.

During the constant-rate period, moisture is rapidly removed. Often 50 percent of the moisture will evaporate in the first hour of a 6- to 8-h cycle. The drying time has been found to be proportional to between the first and second power of the depth of loading. Shelf vacuum dryers operate in the range of 1 to 25 mmHg pressure. For size-estimating purposes, a heat-transfer coefficient of 20 J/(m<sup>2</sup>·s·K) may be used. The area employed in this case should be the shelf area in direct contact with the trays. Trays should be maintained as flatly as possible to obtain maximum area of contact with the heated shelves. For the same reason, the shelves should be kept free from scale and rust. Air vents should be installed on steam-heated shelves to vent noncondensable gases. The heating medium should not be applied to the shelves until after the air has been evacuated from the chamber, to reduce the possibility of the material's overheating or boiling at the start of drying. Case hardening can sometimes be avoided by retarding the rate of drying in the early part of the cycle.

**Performance data for vacuum-shelf dryers** The purchase price of a vacuum-shelf dryer depends upon the cabinet size and number of shelves per cabinet. For estimating purposes, typical prices (1985) and auxiliary equipment requirements are given in Table 12-16. Installed cost of the equipment will be roughly 100 percent of the carbon-steel purchase cost.

The thermal efficiency of a vacuum-shelf dryer is usually on the order of 60 to 80 percent. Table 12-17 gives operating data for one organic color and two inorganic compounds. Labor may constitute 50 percent of the operating cost; maintenance, 20 percent. Annual maintenance costs amount to 5 to 10 percent of the total installed cost. Actual labor costs will depend on drying time, facilities for loading and unloading trays, etc. The power required for these dryers is only that for the vacuum system; for vacuums of 680 to 735 mmHg, the power requirements are on the order of 0.06 to 0.12 kW/m<sup>2</sup> tray surface.

**Continuous Tray and Gravity Dryers** Continuous tray dryers are equivalent to batch tray dryers, but with the solids moving between trays by a combination of mechanical movement and gravity. Gravity (moving-bed) dryers are normally through-circulation convective dryers with no internal trays where the solids gradually descend by gravity. In all these types, the net movement of solids is vertically downward.

**Classification** Continuous; nonagitated (except for turnover when falling between trays); layer; convective (cross-circulation or through-circulation) or contact/conduction; vertical solids movement by gravity and mechanical agitation.

**Turbo-Tray Dryers** The turbo-tray dryer (also known as rotating tray, rotating shelf, or Wyssmont TURBO-DRYER®) is a continuous



TABLE 12-16 Standard Vacuum-Shelf Dryers\*

| Shelf area,<br>m <sup>2</sup> | Floor space,<br>m <sup>2</sup> | Weight average,<br>kg | Pump capacity,<br>m <sup>3</sup> /s | Pump motor,<br>kW | Condenser<br>area, m <sup>2</sup> | Price/m <sup>2</sup> (1995) |                        |
|-------------------------------|--------------------------------|-----------------------|-------------------------------------|-------------------|-----------------------------------|-----------------------------|------------------------|
|                               |                                |                       |                                     |                   |                                   | Carbon<br>steel             | 304 stainless<br>steel |
| 0.4–1.1                       | 4.5                            | 540                   | 0.024                               | 1.12              | 1                                 | \$110                       | \$170                  |
| 1.1–2.2                       | 4.5                            | 680                   | 0.024                               | 1.12              | 1                                 | 75                          | 110                    |
| 2.2–5.0                       | 4.6                            | 1130                  | 0.038                               | 1.49              | 4                                 | 45                          | 65                     |
| 5.0–6.7                       | 5.0                            | 1630                  | 0.038                               | 1.49              | 4                                 | 36                          | 65                     |
| 6.7–14.9                      | 6.4                            | 3900                  | 0.071                               | 2.24              | 9                                 | 27                          | 45                     |
| 16.7–21.1                     | 6.9                            | 5220                  | 0.071                               | 2.24              | 9                                 | 22                          | 36                     |

\*Stokes Vacuum, Inc.

dryer consisting of a stack of rotating annular shelves in the center of which turbo-type fans revolve to circulate the air over the shelves. Wet material enters through the roof, falling onto the top shelf as it rotates beneath the feed opening. After completing 1 r, the material is wiped by a stationary wiper through radial slots onto the shelf below, where it is spread into a uniform pile by a stationary leveler. The action is repeated on each shelf, with transfers occurring once in each revolution. From the last shelf, material is discharged through the bottom of the dryer (Fig. 12-45). The steel-frame housing consists of removable insulated panels for access to the interior. All bearings and lubricated parts are exterior to the unit with the drives located under the housing. Parts in contact with the product may be of steel or special alloy. The trays can be of any sheet material.

The rate at which each fan circulates air can be varied by changing the pitch of the fan blades. In final drying stages, in which diffusion controls or the product is light and powdery, the circulation rate is considerably lower than in the initial stage, in which high evaporation rates prevail. In the majority of applications, air flows through the dryer upward in counterflow to the material. In special cases, required drying conditions dictate that airflow be cocurrent or both counter-current and cocurrent with the exhaust leaving at some level between solids inlet and discharge. A separate cold-air-supply fan is provided if the product is to be cooled before being discharged.

By virtue of its vertical construction, the turbo-type tray dryer has a stack effect, the resulting draft being frequently sufficient to operate the dryer with natural draft. Pressure at all points within the dryer is maintained close to atmospheric. Most of the roof area is used as a breeching, lowering the exhaust velocity to settle dust back into the dryer.

Heaters can be located in the space between the trays and the dryer housing, where they are not in direct contact with the product, and thermal efficiencies up to 3500 kJ/kg (1500 Btu/lb) of water evaporated can be obtained by reheating the air within the dryer. For materials which have a tendency to foul internal heating surfaces, an external heating system is employed.

The turbo-tray dryer can handle materials from thick slurries [1 million N·s/m<sup>2</sup> (100,000 cP) and over] to fine powders. Filterpress cakes are granulated before feeding. Thixotropic materials are fed directly from a rotary filter by scoring the cake as it leaves the drum. Pastes can be extruded onto the top shelf and subjected to a hot blast of air to make them firm and free-flowing after 1 r.

TABLE 12-17 Performance Data of Vacuum-Shelf Dryers

| Material                                 | Sulfur black         | Calcium carbonate    | Calcium phosphate    |
|------------------------------------------|----------------------|----------------------|----------------------|
| Loading, kg dry material/m <sup>2</sup>  | 25                   | 17                   | 33                   |
| Steam pressure, kPa gauge                | 410                  | 410                  | 205                  |
| Vacuum, mmHg                             | 685–710              | 685–710              | 685–710              |
| Initial moisture content, % (wet basis)  | 50                   | 50.3                 | 30.6                 |
| Final moisture content, % (wet basis)    | 1                    | 1.15                 | 4.3                  |
| Drying time, h                           | 8                    | 7                    | 6                    |
| Evaporation rate, kg/(s·m <sup>2</sup> ) | $8.9 \times 10^{-4}$ | $7.9 \times 10^{-4}$ | $6.6 \times 10^{-4}$ |

The turbo-tray dryer is manufactured in sizes from package units 2 m in height and 1.5 m in diameter to large outdoor installations 20 m in height and 11 m in diameter. Tray areas range from 1 m<sup>2</sup> up to about 2000 m<sup>2</sup>. The number of shelves in a tray rotor varies according to space available and the minimum rate of transfer required, from as few as 12 shelves to as many as 58 in the largest units. Standard construction permits operating temperatures up to 615 K, and high-temperature heaters permit operation at temperatures up to 925 K.

A recent innovation has enabled TURBO-DRYER® to operate with very low inert gas makeup. Wyssmont has designed a tank housing that is welded up around the internal structure rather than the column-and-gasket panel design that has been the Wyssmont standard for many years. In field-erected units, the customer does the welding in the field; in packaged units, the tank-type welding is done in the shop. The tank-type housing finds particular application for operation under positive pressure. On the standard design, doors with explosion latches and gang latch operators are used. In the tank-type design,

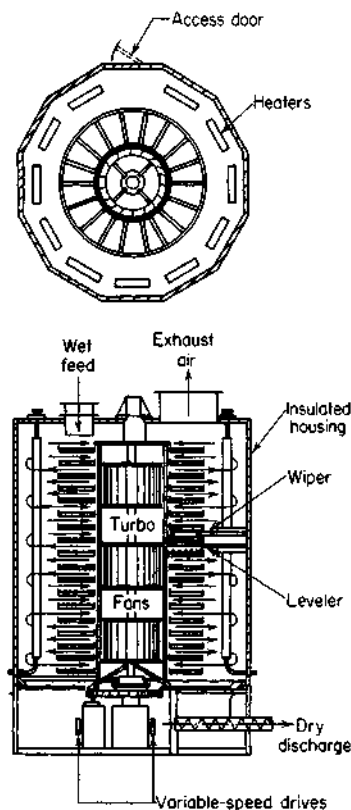


FIG. 12-45 TURBO-DRYER®. (Wyssmont Company, Inc.)

tight-sealing manway-type openings permit access to the interior. Tank-type housing designs have been requested when drying solvent wet materials and for applications where the material being dried is highly toxic and certainty is required that no toxic dust get out.

**Design methods for turbo-tray dryers** The heat- and mass-transfer mechanisms are similar to those in batch tray dryers, except that constant turning over and mixing of the solids significantly improve drying rates. Design must usually be based on previous installations or pilot tests by the manufacturer; apparent heat-transfer coefficients are typically 30 to 60 J/(m<sup>2</sup>·s·K) for dry solids and 60 to 120 J/(m<sup>2</sup>·s·K) for wet solids. Turbo-tray dryers have been employed successfully for the drying and cooling of calcium hypochlorite, urea crystals, calcium chloride flakes, and sodium chloride crystals. The Wyssmont "closed-circuit" system, as shown in Fig. 12-46, consists of the turbo-tray dryer with or without internal heaters, recirculation fan, condenser with receiver and mist eliminators, and reheater. Feed and discharge are through a sealed wet feeder and lock, respectively. This method is used for continuous drying without leakage of fumes, vapors, or dust to the atmosphere. A unified approach for scaling up dryers such as turbo-tray, plate, conveyor, or any other dryer type that forms a defined layer of solids next to a heating source is the SDR (specific drying rate) method described by Moyers [*Drying Technol.* 12(1 & 2): 393-417 (1994)].

**Performance and cost data for turbo-tray dryers** Performance data for four applications of closed-circuit drying are included in Table 12-18. Operating, labor, and maintenance costs compare favorably with those of direct heat rotating equipment.

**Plate Dryers** The plate dryer is an indirectly heated, fully continuous dryer available for three modes of operation: atmospheric, gastight, or full vacuum. The dryer is of vertical design, with horizontal, heated plates mounted inside the housing. The plates are heated by hot water, steam, or thermal oil, with operating temperatures up to 320°C possible. The product enters at the top and is conveyed through the dryer by a product transport system consisting of a central-rotating shaft with arms and plows. (See dryer schematic, Fig. 12-47.) The thin product layer [approximately ½-in (12-mm) depth] on the surface of the plates, coupled with frequent product turnover by the conveying system, results in short retention times (approximately 5 to 40 min), true plug flow of the material, and uniform drying. The vapors are removed from the dryer by a small amount of heated purge gas or by vacuum. The material of construction of the plates and housing is normally stainless steel, with special metallurgies also available. The drive unit is located at the bottom of the dryer and supports the central-rotating shaft. Typical speed of the dryer is 1 to 7 rpm. Full-opening doors are located on two adjacent sides of the dryer for easy access to dryer internals.

The plate dryer may vary in size from 5 to 35 vertically stacked plates with a heat-exchange area between 3.8 and 175 m<sup>2</sup>. The largest unit available has overall dimensions of 3 m (w) by 4 m (l) by 10 m (h). Depending

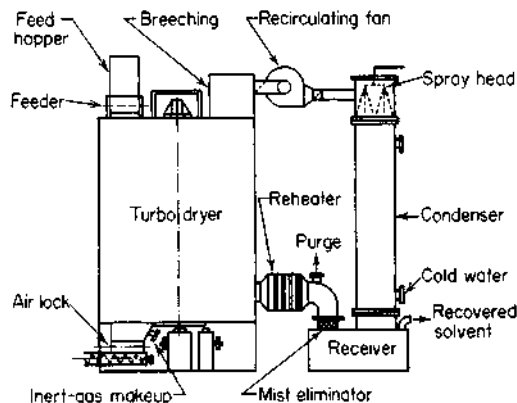


FIG. 12-46 TURBO-DRYER® in closed circuit for continuous drying with solvent recovery. (Wyssmont Company, Inc.)

upon the loose-bulk density of the material and the overall retention time, the plate dryer can process up to 5000 kg/h of wet product.

The plate dryer is limited in its scope of applications only in the consistency of the feed material (the products must be friable, free-flowing, and not undergo phase changes) and drying temperatures up to 320°C. Applications include specialty chemicals, pharmaceuticals, foods, polymers, pigments, etc. Initial moisture or volatile level can be as high as 65 percent, and the unit is often used as a final dryer to take materials to a bone-dry state, if necessary. The plate dryer can also be used for heat treatment, removal of waters of hydration (bound moisture), solvent removal, and as a product cooler.

The atmospheric plate dryer is a dust-tight system. The dryer housing is an octagonal, panel construction, with operating pressure in the range of  $\pm 0.5$  kPa gauge. An exhaust air fan draws the purge air through the housing for removal of the vapors from the drying process. The purge air velocity through the dryer is in the range of 0.1 to 0.15 m/s, resulting in minimal dusting and small dust filters for the exhaust air. The air temperature is normally equal to the plate temperature. The vapor-laden exhaust air is passed through a dust filter or a scrubber (if necessary) and is discharged to the atmosphere. Normally, water is the volatile to be removed in this type of system.

The gastight plate dryer, together with the components of the gas recirculation system, forms a closed system. The dryer housing is semi-cylindrical and is rated for a nominal pressure of 5 kPa gauge. The flow rate of the recirculating purge gas must be sufficient to absorb the vapors generated from the drying process. The gas temperature must be adjusted according to the specific product characteristics and the

TABLE 12-18 Turbo-Dryer® Performance Data in Wyssmont Closed-Circuit Operations\*

|                                  |                          |                          |                          |                          |
|----------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Material dried                   | Antioxidant              | Water-soluble polymer    | Antibiotic filter cake   | Petroleum coke           |
| Dried product, kg/h              | 500                      | 85                       | 2400                     | 227                      |
| Volatiles composition            | Methanol and water       | Xylene and water         | Alcohol and water        | Methanol                 |
| Feed volatiles, % wet basis      | 10                       | 20                       | 30                       | 30                       |
| Product volatiles, % wet basis   | 0.5                      | 4.8                      | 3.5                      | 0.2                      |
| Evaporation rate, kg/h           | 53                       | 16                       | 910                      | 302                      |
| Type of heating system           | External                 | External                 | External                 | External                 |
| Heating medium                   | Steam                    | Steam                    | Steam                    | Steam                    |
| Drying medium                    | Inert gas                | Inert gas                | Inert gas                | Inert gas                |
| Heat consumption, J/kg           | $0.56 \times 10^6$       | $2.2 \times 10^6$        | $1.42 \times 10^6$       | $1.74 \times 10^6$       |
| Power, dryer, kW                 | 1.8                      | 0.75                     | 12.4                     | 6.4                      |
| Power, recirculation fan, kW     | 5.6                      | 5.6                      | 37.5                     | 15                       |
| Materials of construction        | Stainless-steel interior | Stainless-steel interior | Stainless-steel interior | Carbon steel             |
| Dryer height, m                  | 4.4                      | 3.2                      | 7.6                      | 6.5                      |
| Dryer diameter, m                | 2.9                      | 1.8                      | 6.0                      | 4.5                      |
| Recovery system                  | Shell-and-tube condenser | Shell-and-tube condenser | Direct-contact condenser | Shell-and-tube condenser |
| Condenser cooling medium         | Brine                    | Chilled water            | Tower water              | Chilled water            |
| Location                         | Outdoor                  | Indoor                   | Indoor                   | Indoor                   |
| Approximate cost of dryer (2004) | \$300,000                | \$175,000                | \$600,000                | \$300,000                |
| Dryer assembly                   | Packaged unit            | Packaged unit            | Field-erected unit       | Field-erected unit       |

\*Courtesy of Wyssmont Company, Inc.

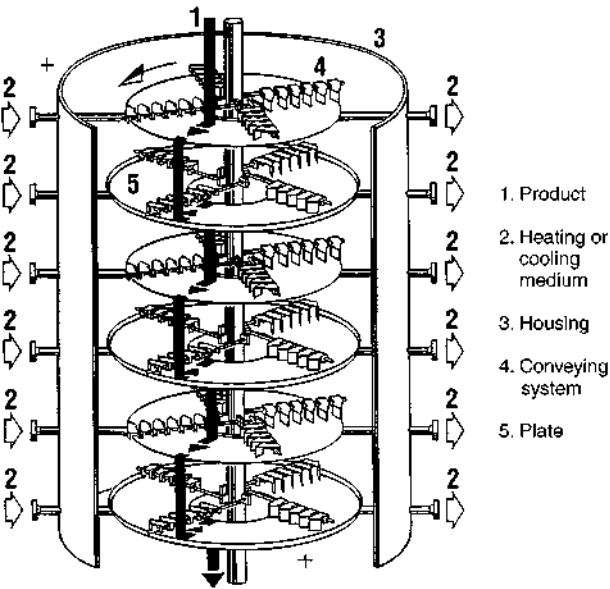


FIG. 12-47 Indirect heat continuous plate dryer for atmospheric, gastight, or full-vacuum operation. (Krauss Maffei.)

type of volatile. After condensation of the volatiles, the purge gas (typically nitrogen) is recirculated back to the dryer via a blower and heat exchanger. Solvents such as methanol, toluene, and acetone are normally evaporated and recovered in the gastight system.

The vacuum plate dryer is provided as part of a closed system. The vacuum dryer has a cylindrical housing and is rated for full-vacuum operation (typical pressure range of 3 to 27 kPa absolute). The exhaust vapor is evacuated by a vacuum pump and is passed through a condenser for solvent recovery. There is no purge gas system required for operation under vacuum. Of special note in the vacuum-drying system are the vacuum feed and discharge locks, which allow for continuous operation of the plate dryer under full vacuum.

*Comparison data—plate dryers* Comparative studies have been done on products under both atmospheric and vacuum drying conditions. See Fig. 12-48. These curves demonstrate (1) the improvement in drying achieved with elevated temperature and (2) the impact to

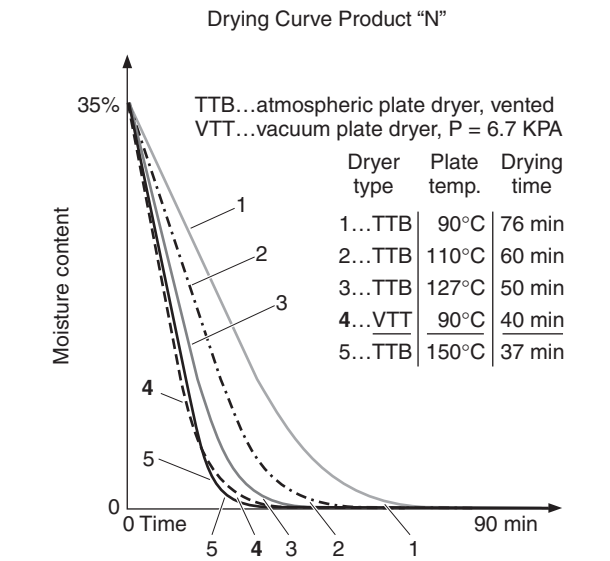


FIG. 12-48 Plate dryer drying curves demonstrating impact of elevated temperature and/or operation under vacuum. (Krauss Maffei.)

the drying process obtained with vacuum operation. Note that curve 4 at 90°C, pressure at 6.7 kPa absolute, is comparable to the atmospheric curve at 150°C. Also, the comparative atmospheric curve at 90°C requires 90 percent more drying time than the vacuum condition. The dramatic improvement with the use of vacuum is important to note for heat-sensitive materials.

The above drying curves have been generated via testing on a plate dryer simulator. The test unit duplicates the physical setup of the production dryer; therefore linear scale-up from the test data can be made to the full-scale dryer. Because of the thin product layer on each plate, drying in the unit closely follows the normal type of drying curve in which the constant-rate period (steady evolution of moisture or volatiles) is followed by the falling-rate period of the drying process. This results in higher heat-transfer coefficients and specific drying capacities on the upper plates of the dryer as compared to the lower plates. The average specific drying capacity for the plate dryer is in the range of 2 to 20 kg/(m<sup>2</sup>·h) (based on final dry product). Performance data for typical applications are shown on Table 12-19.

TABLE 12-19 Plate Dryer Performance Data for Three Applications\*

|                                       |                  |                 |                 |
|---------------------------------------|------------------|-----------------|-----------------|
| Product                               | Plastic additive | Pigment         | Foodstuff       |
| Volatiles                             | Methanol         | Water           | Water           |
| Production rate, dry                  | 362 kg/hr        | 133 kg/hr       | 2030 kg/hr      |
| Inlet volatiles content               | 30%              | 25%             | 4%              |
| Final volatiles content               | 0.1%             | 0.5%            | 0.7%            |
| Evaporative rate                      | 155 kg/hr        | 44 kg/hr        | 70 kg/hr        |
| Heating medium                        | Hot water        | Steam           | Hot water       |
| Drying temperature                    | 70°C             | 150°C           | 90°C            |
| Dryer pressure                        | 11 kPa abs       | Atmospheric     | Atmospheric     |
| Air velocity                          | NA               | 0.1 m/sec       | 0.2 m/sec       |
| Drying time, min                      | 24               | 23              | 48              |
| Heat consumption, kcal/kg dry product | 350              | 480             | 100             |
| Power, dryer drive                    | 3 kW             | 1.5 kW          | 7.5 kW          |
| Material of construction              | SS 316L/316Ti    | SS 316L/316Ti   | SS 316L/316Ti   |
| Dryer height                          | 5 m              | 2.6 m           | 8.2 m           |
| Dryer footprint                       | 2.6 m diameter   | 2.2 m by 3.0 m  | 3.5 m by 4.5 m  |
| Location                              | Outdoors         | Indoors         | Indoors         |
| Dryer assembly                        | Fully assembled  | Fully assembled | Fully assembled |
| Power, exhaust fan                    | NA               | 2.5 kW          | 15 kW           |
| Power, vacuum pump                    | 20 kW            | NA              | NA              |

\*Krauss Maffei

**Gravity or Moving-Bed Dryers** A body of solids in which the particles, consisting of granules, pellets, beads, or briquettes, flow downward by gravity at substantially their normal settled bulk density through a vessel in contact with gases is defined frequently as a **moving-bed** or **tower dryer**. Moving-bed equipment is frequently used for grain drying and plastic pellet drying, and it also finds application in blast furnaces, shaft furnaces, and petroleum refining. Gravity beds are also employed for the cooling and drying of extruded pellets and briquettes from size enlargement processes.

A gravity dryer consists of a stationary vertical, usually cylindrical housing with openings for the introduction of solids (at the top) and removal of solids (at the bottom), as shown schematically in Fig. 12-49. Gas flow is through the solids bed and may be cocurrent or countercurrent and, in some instances, cross-flow. By definition, the rate of gas flow upward must be less than that required for fluidization.

**Fields of application** One of the major advantages of the gravity-bed technique is that it lends itself well to true intimate countercurrent contacting of solids and gases. This provides for efficient heat transfer and mass transfer. Gravity-bed contacting also permits the use of the solid as a heat-transfer medium, as in pebble heaters.

Gravity vessels are applicable to coarse granular free-flowing solids which are comparatively dust-free. The solids must possess physical properties in size and surface characteristics so that they will not stick together, bridge, or segregate during passage through the vessel. The presence of significant quantities of fines or dust will close the passages among the larger particles through which the gas must penetrate, increasing pressure drop. Fines may also segregate near the sides of the bed or in other areas where gas velocities are low, ultimately completely sealing off these portions of the vessel. The high efficiency of gas-solids contacting in gravity beds is due to the uniform distribution of gas throughout the solids bed; hence choice of feed and its preparation are important factors to successful operation. Preforming techniques such as pelleting and briquetting are employed frequently for the preparation of suitable feed materials.

Gravity vessels are suitable for low-, medium-, and high-temperature operation; in the last case, the housing will be lined completely with refractory brick. Dust recovery equipment is minimized in this type of operation since the bed actually performs as a dust collector itself, and dust in the bed will not, in a successful application, exist in large quantities.

Other advantages of gravity beds include flexibility in gas and solids flow rates and capacities, variable retention times from minutes to several hours, space economy, ease of start-up and shutdown, the

potentially large number of contacting stages, and ease of control by using the inlet and exit gas temperatures.

Maintenance of a uniform rate of solids movement downward over the entire cross-section of the bed is one of the most critical operating problems encountered. For this reason gravity beds are designed to be as high and narrow as practical. In a vessel of large cross section, discharge through a conical bottom and center outlet will usually result in some degree of "ratholing" through the center of the bed. Flow through the center will be rapid while essentially stagnant pockets are left around the sides. To overcome this problem, multiple outlets may be provided in the center and around the periphery; table unloaders, rotating plows, wide moving grates, and multiple-screw unloaders are employed; insertion of inverted cone baffles in the lower section of the bed, spaced so that flushing at the center is retarded, is also a successful method for improving uniformity of solids movement. Fortunately, the problems are less critical in gravity dryers, which are usually for slow drying of large particles, than in applications such as catalytic reactors, where disengagement of gas from solids at the top of the tower can also present serious difficulties.

**Continuous Band and Tunnel Dryers** This group of dryers is variously known as band, belt, conveyor, or tunnel dryers.

**Classification** Continuous; nonagitated; layer; convective (cross-circulation or through-circulation) or contact/conduction; horizontal movement by mechanical means.

**Continuous tunnels are batch truck or tray compartments, operated in series.** The solids to be processed are placed in trays or on trucks which move progressively through the tunnel in contact with hot gases. Operation is **semicontinuous**; when the tunnel is filled, one truck is removed from the discharge end as each new truck is fed into the inlet end. In some cases, the trucks move on tracks or monorails, and they are usually conveyed mechanically, employing chain drives connecting to the bottom of each truck.

**Belt-conveyor and screen-conveyor (band) dryers are truly continuous** in operation, carrying a layer of solids on an endless conveyor.

Continuous tunnel and conveyor dryers are more suitable than (multiple) batch compartments for large-quantity production, usually giving investment and installation savings. In the case of truck and tray tunnels, labor savings for loading and unloading are not significant compared with those for batch equipment. Belt and screen conveyors which are truly continuous represent major labor savings over batch operations but require additional investment for automatic feeding and unloading devices.

Airflow can be totally **cocurrent**, **countercurrent**, or a combination of both. In addition, **cross-flow** designs are employed frequently, with the heating air flowing back and forth across the trucks or belt in series. Reheat coils may be installed after each cross-flow pass to maintain constant-temperature operation; large propeller-type circulating fans are installed at each stage, and air may be introduced or exhausted at any desirable points. Tunnel equipment possesses maximum flexibility for any combination of airflow and temperature staging. When handling granular, particulate solids which do not offer high resistance to airflow, perforated or screen-type belt conveyors are employed with **through-circulation** of gas to improve heat- and mass-transfer rates, almost invariably in cross-flow. Contact drying is also possible, usually under vacuum, with the bands resting on heating plates (vacuum band dryer).

**Tunnel Dryers** In tunnel equipment, the solids are usually heated by direct contact with hot gases. In high-temperature operations, radiation from walls and refractory lining may be significant also. The air in a direct heat unit may be heated directly or indirectly by combustion or, at temperature below 475 K, by finned steam coils.

Applications of tunnel equipment are essentially the same as those for batch tray and compartment units previously described, namely, practically all forms of particulate solids and large solid objects. Continuous tunnel or conveyor ovens are employed also for drying refractory shapes and for drying and baking enameled pieces. In many of these latter, the parts are suspended from overhead chain conveyors.

Auxiliary equipment and the special design considerations discussed for batch trays and compartments apply also to tunnel equipment. For size-estimating purposes, tray and truck tunnels and

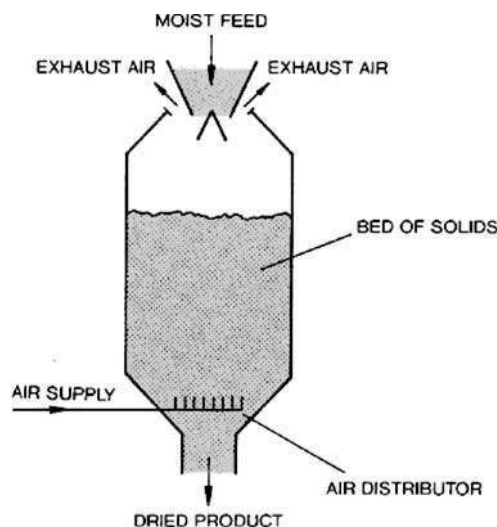


FIG. 12-49 Moving-bed gravity dryer.



furnaces can be treated in the same manner as discussed for batch equipment.

**Ceramic tunnel kilns** handling large irregular-shaped objects must be equipped for precise control of temperature and humidity conditions to prevent cracking and condensation on the product. The internal mechanism causing cracking when drying clay and ceramics have been studied extensively. Information on ceramic tunnel kiln operation and design is reported fully in publications such as *The American Ceramic Society Bulletin*, *Ceramic Industry*, and *Transactions of the British Ceramic Society*.

Another use of tunnel dryers is for drying leather. Moisture content is initially around 50 percent but must not be reduced below about 15 percent, or else the leather will crack and be useless. To avoid this, a high-humidity atmosphere is maintained by gas recycle, giving a high equilibrium moisture content.

**Continuous Through-Circulation Band Dryers** Continuous through-circulation dryers operate on the principle of blowing hot air through a permeable bed of wet material passing continuously through the dryer. Drying rates are high because of the large area of contact and short distance of travel for the internal moisture.

The most widely used type is the **horizontal conveyor dryer** (also called perforated band or conveying-screen dryer), in which wet material is conveyed as a layer, 2 to 15 cm deep (sometimes up to 1 m), on a horizontal mesh screen, belt, or perforated apron, while heated air is blown either upward or downward through the bed of material. This dryer consists usually of a number of individual sections, complete with fan and heating coils, arranged in series to form a housing or tunnel through which the conveying screen travels. As shown in the sectional view in Fig. 12-50, the air circulates through the wet material and is reheated before reentering the bed. It is not uncommon to circulate the hot gas upward in the wet end and downward in the dry end, as shown in Fig. 12-51. A portion of the air is exhausted continuously by one or more exhaust fans, not shown in the sketch, which handle air from several sections. Since each section can be operated independently, extremely flexible operation is possible, with high temperatures usually at the wet end, followed by lower temperatures; in some cases a unit with cooled or specially humidified air is employed for final conditioning. The **maximum pressure drop** that can be taken through the bed of solids without developing leaks or air bypassing is roughly 50 mm of water.

Through-circulation drying requires that the wet material be in a state of granular or pelleted subdivision so that hot air may be readily blown through it. Many materials meet this requirement without special preparation. Others require special and often elaborate pretreatment to render them suitable for through-circulation drying. The process of converting a wet solid to a form suitable for through-circulation of air is called **preforming**, and often the success or failure of this contacting method depends on the preforming step. Fibrous, flaky, and coarse granular materials are usually amenable to drying without preforming. They can be loaded directly onto the conveying screen by suitable spreading feeders of the oscillating-belt or vibrating type or by spiked drums or belts feeding from bins.

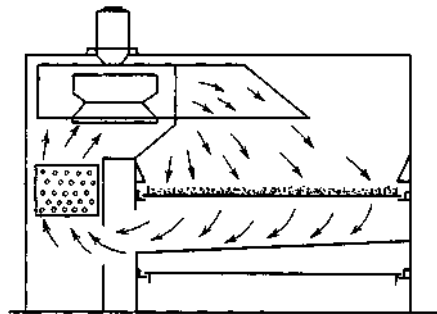


FIG. 12-50 Section view of a continuous through-circulation conveyor dryer. (Proctor & Schwartz, Inc.)

When materials must be preformed, several methods are available, depending on the physical state of the wet solid.

1. Relatively dry materials such as centrifuge cakes can sometimes be granulated to give a suitably porous bed on the conveying screen.

2. Pasty materials can often be preformed by extrusion to form spaghetti-like pieces, about 6 mm in diameter and several centimeters long.

3. Wet pastes that cannot be granulated or extruded may be predried and preformed on a steam-heated finned drum. Preforming on a finned drum may be desirable also in that some predrying is accomplished.

4. Thixotropic filter cakes from rotary vacuum filters that cannot be preformed by any of the above methods can often be scored by knives on the filter, the scored cake discharging in pieces suitable for through-circulation drying.

5. Material that shrinks markedly during drying is often reloaded during the drying cycle to 2 to 6 times the original loading depth. This is usually done after a degree of shrinkage which, by opening the bed, has destroyed the effectiveness of contact between the air and solids.

6. In a few cases, powders have been pelleted or formed in briquettes to eliminate dustiness and permit drying by through-circulation. Table 12-20 gives a list of materials classified by preforming methods suitable for through-circulation drying.

Steam-heated air is the usual heat-transfer medium employed in these dryers, although combustion gases may be used also. Temperatures above 600 K are not usually feasible because of the problems of lubricating the conveyor, chain, and roller drives. Recirculation of air is in the range of 60 to 90 percent of the flow through the bed. Conveyors may be made of wire-mesh screen or perforated-steel plate. The minimum practical screen opening size is about 30-mesh (0.5 mm). Multiple bands in series may be used.

Vacuum band dryers utilize heating by conduction and are a continuous equivalent of vacuum tray (shelf) dryers, with the moving

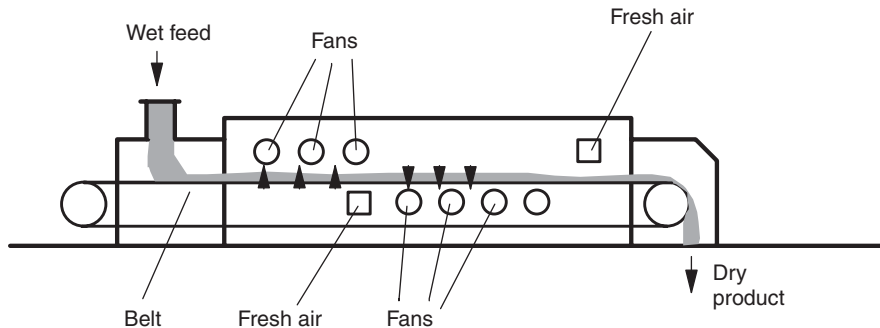


FIG. 12-51 Longitudinal view of a continuous through-circulation conveyor dryer with intermediate airflow reversal.

TABLE 12-20 Methods of Preforming Some Materials for Through-Circulation Drying

| No preforming required                                                                                                                                                  | Scored on filter           | Granulation                                                                                 | Extrusion                                                                                                                     | Finned drum                                                          | Flaking on chilled drum | Briquetting and squeezing                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------|--------------------------------------------|
| Cellulose acetate<br>Silica gel<br>Scoured wool<br>Sawdust<br>Rayon waste<br>Fluorspar<br>Tapioca<br>Breakfast food<br>Asbestos fiber<br>Cotton linters<br>Rayon staple | Starch<br>Aluminum hydrate | Kaolin<br>Cryolite<br>Lead arsenate<br>Cornstarch<br>Cellulose acetate<br>Dye intermediates | Calcium carbonate<br>White lead<br>Lithopone<br>Titanium dioxide<br>Magnesium carbonate<br>Aluminum stearate<br>Zinc stearate | Lithopone<br>Zinc yellow<br>Calcium carbonate<br>Magnesium carbonate | Soap flakes             | Soda ash<br>Cornstarch<br>Synthetic rubber |

bands resting on heating plates. Drying is usually relatively slow, and it is common to find several bands stacked above one another, with material falling to the next band and flowing in opposite directions on each pass, to reduce dryer length and give some product turnover.

**Design Methods for Continuous Band Dryers** In actual practice, design of a continuous through-circulation dryer is best based upon data taken in pilot-plant tests. Loading and distribution of solids on the screen are rarely as nearly uniform in commercial installations as in test dryers; 50 to 100 percent may be added to the test drying time for commercial design.

A mathematical method of a through-circulation dryer has been developed by Thygeson [*Am. Inst. Chem. Eng. J.* **16**(5):749 (1970)]. Rigorous modeling is possible with a two-dimensional incremental model, with steps both horizontally along the belt and vertically through the layer; nonuniformity of the layer across the belt could also be allowed for if desired. Heat-transfer coefficients are typically in the range of 100 to 200 W/(m<sup>2</sup>·K) and the relationship  $h_c = 12(\rho_g U_g d_p)^{0.5}$  may be used for a first estimate, where  $\rho_g$  is gas density (kg/m<sup>3</sup>);  $U_g$ , local gas velocity (m/s); and  $d_p$ , particle diameter (m). For 5-mm particles and air at 1 m/s, 80°C and 1 kg/m<sup>3</sup> [mass flux 1 kg/(m<sup>2</sup>·s)] this gives  $h_c = 170$  W/(m<sup>2</sup>·K).

**Performance and Cost Data for Continuous Band and Tunnel Dryers** Experimental performance data are given in Table 12-21 for numerous common materials. Performance data from several commercial through-circulation conveyor dryers are given in Table 12-22. Labor requirements vary depending on the time required for feed adjustments, inspection, etc. These dryers may consume as little as 1.1 kg of steam/kg of water evaporated, but 1.4 to 2 is a more common range. Thermal efficiency is a function of final moisture required and percent air recirculation.

Conveying-screen dryers are fabricated with conveyor widths from 0.3- to 4.4-m sections 1.6 to 2.5 m long. Each section consists of a sheet-metal enclosure, insulated sidewalls and roof, heating coils, a circulating fan, inlet air distributor baffles, a fines catch pan under the conveyor, and a conveyor screen (Fig. 12-51). Table 12-23 gives approximate purchase costs for equipment with type 304 stainless-steel hinged conveyor screens and includes steam-coil heaters, fans, motors, and a variable-speed conveyor drive. Cabinet and auxiliary equipment fabrication is of aluminized steel or stainless-steel materials. Prices do not include temperature controllers, motor starters, preform equipment, or auxiliary feed and discharge conveyors. These may add \$75,000 to \$160,000 to the dryer purchase cost (2005 costs).

### Batch Agitated and Rotating Dryers

**Description** An *agitated dryer* is defined as one on which the housing enclosing the process is stationary while solids movement is accomplished by an internal mechanical agitator. A rotary dryer is one in which the outer housing rotates. Many forms are in use, including batch and continuous versions. The batch forms are almost invariably heated by conduction with operation under vacuum. Vacuum is used in conjunction with drying or other chemical operations when low solids temperatures must be maintained because heat will cause damage to the product or change its nature; when air combines with the

product as it is heated, causing oxidation or an explosive condition; when solvent recovery is required; and when materials must be dried to extremely low moisture levels.

Vertical agitated pan, spherical and conical dryers are mechanically agitated; tumbler or double-cone dryers have a rotating shell. All these types are typically used for the drying of solvent or water-wet, free-flowing powders in small batch sizes of 1000 L or less, as frequently found in the pharmaceutical, specialty chemical, and fine chemicals industries. Corrosion resistance and cleanability are often important, and common materials of construction include SS 304 and 316, and Hastelloy. The batch nature of operation is of value in the pharmaceutical industry to maintain batch identification. In addition to pharmaceutical materials, the conical mixer dryer is used to dry polymers, additives, inorganic salts, and many other specialty chemicals. As the size increases, the ratio of jacket heat-transfer surface area to volume falls, extending drying times. For larger batches, horizontal agitated pan dryers are more common, but there is substantial overlap of operating ranges. Drying times may be reduced for all types by heating the internal agitator, but this increases complexity and cost.

**Classification** Batch; mechanical or rotary agitation; layer; contact/conduction.

**Mechanical versus rotary agitation** Agitated dryers are applicable to processing solids which are relatively free-flowing and granular when discharged as product. Materials which are not free-flowing in their feed condition can be treated by recycle methods as described in the subsection "Continuous Rotary Dryers." In general, agitated dryers have applications similar to those of rotating vessels. Their chief advantages compared with the latter are twofold. (1) Large-diameter rotary seals are not required at the solids and gas feed and exit points because the housing is stationary, and for this reason gas leakage problems are minimized. Rotary seals are required only at the points of entrance of the mechanical agitator shaft. (2) Use of a mechanical agitator for solids mixing introduces shear forces which are helpful for breaking up lumps and agglomerates. Balling and pelleting of sticky solids, an occasional occurrence in rotating vessels, can be prevented by special agitator design. The problems concerning dusting of fine particles in direct-heat units are identical to those discussed under "Continuous Rotary Dryers."

**Vacuum processing** All these types of dryer usually operate under vacuum, especially when drying heat-sensitive materials or when removing flammable organic solvents rather than water. The heating medium is hot water, steam, or thermal oil, with most applications in the temperature range of 50 to 150°C and pressures in the range of 3 to 30 kPa absolute. The vapors generated during the drying process are evacuated by a vacuum pump and passed through a condenser for recovery of the solvent. A dust filter is normally mounted over the vapor discharge line as it leaves the dryer, thus allowing any entrapped dust to be pulsed back into the process area. Standard cloth-type dust filters are available, along with sintered metal filters.

In vacuum processing and drying, a major objective is to create a large temperature-driving force between the jacket and the product. To accomplish this purpose at fairly low jacket temperatures, it is necessary to reduce the internal process pressure so that the liquid being removed will boil at a lower vapor pressure. It is not always economical, however,

TABLE 12-21 Experimental Through-Circulation Drying Data for Miscellaneous Materials

| Material             | Physical form             | Moisture contents, kg/kg dry solid |          |        | Inlet-air temperature, K | Depth of bed, cm | Loading, kg product/m <sup>2</sup> | Air velocity, m/s × 10 <sup>1</sup> | Experimental drying time, s × 10 <sup>-2</sup> |
|----------------------|---------------------------|------------------------------------|----------|--------|--------------------------|------------------|------------------------------------|-------------------------------------|------------------------------------------------|
|                      |                           | Initial                            | Critical | Final  |                          |                  |                                    |                                     |                                                |
| Alumina hydrate      | Briquettes                | 0.105                              | 0.06     | 0.00   | 453                      | 6.4              | 60.0                               | 6.0                                 | 18.0                                           |
| Alumina hydrate      | Scored filter cake        | 9.60                               | 4.50     | 1.15   | 333                      | 3.8              | 1.6                                | 11.0                                | 90.0                                           |
| Alumina hydrate      | Scored filter cake        | 5.56                               | 2.25     | 0.42   | 333                      | 7.0              | 4.6                                | 11.0                                | 108.0                                          |
| Aluminum stearate    | 0.7-cm extrusions         | 4.20                               | 2.60     | 0.003  | 350                      | 7.6              | 6.5                                | 13.0                                | 36.0                                           |
| Asbestos fiber       | Flakes from squeeze rolls | 0.47                               | 0.11     | 0.008  | 410                      | 7.6              | 13.6                               | 9.0                                 | 5.6                                            |
| Asbestos fiber       | Flakes from squeeze rolls | 0.46                               | 0.10     | 0.0    | 410                      | 5.1              | 6.3                                | 9.0                                 | 3.6                                            |
| Asbestos fiber       | Flakes from squeeze rolls | 0.46                               | 0.075    | 0.0    | 410                      | 3.8              | 4.5                                | 11.0                                | 2.7                                            |
| Calcium carbonate    | Preformed on finned drum  | 0.85                               | 0.30     | 0.003  | 410                      | 3.8              | 16.0                               | 11.5                                | 12.0                                           |
| Calcium carbonate    | Preformed on finned drum  | 0.84                               | 0.35     | 0.0    | 410                      | 8.9              | 25.7                               | 11.7                                | 18.0                                           |
| Calcium carbonate    | Extruded                  | 1.69                               | 0.98     | 0.255  | 410                      | 1.3              | 4.9                                | 14.3                                | 9.0                                            |
| Calcium carbonate    | Extruded                  | 1.41                               | 0.45     | 0.05   | 410                      | 1.9              | 5.8                                | 10.2                                | 12.0                                           |
| Calcium stearate     | Extruded                  | 2.74                               | 0.90     | 0.0026 | 350                      | 7.6              | 8.8                                | 5.6                                 | 57.0                                           |
| Calcium stearate     | Extruded                  | 2.76                               | 0.90     | 0.007  | 350                      | 5.1              | 5.9                                | 6.0                                 | 42.0                                           |
| Calcium stearate     | Extruded                  | 2.52                               | 1.00     | 0.0    | 350                      | 3.8              | 4.4                                | 10.2                                | 24.0                                           |
| Cellulose acetate    | Granulated                | 1.14                               | 0.40     | 0.09   | 400                      | 1.3              | 1.4                                | 12.7                                | 1.8                                            |
| Cellulose acetate    | Granulated                | 1.09                               | 0.35     | 0.0027 | 400                      | 1.9              | 2.7                                | 8.6                                 | 7.2                                            |
| Cellulose acetate    | Granulated                | 1.09                               | 0.30     | 0.0041 | 400                      | 2.5              | 4.1                                | 5.6                                 | 10.8                                           |
| Cellulose acetate    | Granulated                | 1.10                               | 0.45     | 0.004  | 400                      | 3.8              | 6.1                                | 5.1                                 | 18.0                                           |
| Clay                 | Granulated                | 0.277                              | 0.175    | 0.0    | 375                      | 7.0              | 46.2                               | 10.2                                | 19.2                                           |
| Clay                 | 1.5-cm extrusions         | 0.28                               | 0.18     | 0.0    | 375                      | 12.7             | 100.0                              | 10.7                                | 43.8                                           |
| Cryolite             | Granulated                | 0.456                              | 0.25     | 0.0026 | 380                      | 5.1              | 34.2                               | 9.1                                 | 24.0                                           |
| Fluorspar            | Pellets                   | 0.13                               | 0.066    | 0.0    | 425                      | 5.1              | 51.4                               | 11.6                                | 7.8                                            |
| Lead arsenate        | Granulated                | 1.23                               | 0.45     | 0.043  | 405                      | 5.1              | 18.1                               | 11.6                                | 18.0                                           |
| Lead arsenate        | Granulated                | 1.25                               | 0.55     | 0.054  | 405                      | 6.4              | 22.0                               | 10.2                                | 24.0                                           |
| Lead arsenate        | Extruded                  | 1.34                               | 0.64     | 0.024  | 405                      | 5.1              | 18.1                               | 9.4                                 | 36.0                                           |
| Lead arsenate        | Extruded                  | 1.31                               | 0.60     | 0.0006 | 405                      | 8.4              | 26.9                               | 9.2                                 | 42.0                                           |
| Kaolin               | Formed on finned drum     | 0.28                               | 0.17     | 0.0009 | 375                      | 7.6              | 44.0                               | 9.2                                 | 21.0                                           |
| Kaolin               | Formed on finned drum     | 0.297                              | 0.20     | 0.005  | 375                      | 11.4             | 56.3                               | 12.2                                | 15.0                                           |
| Kaolin               | Extruded                  | 0.443                              | 0.20     | 0.008  | 375                      | 7.0              | 45.0                               | 10.16                               | 18.0                                           |
| Kaolin               | Extruded                  | 0.36                               | 0.14     | 0.0033 | 400                      | 9.6              | 40.6                               | 15.2                                | 12.0                                           |
| Kaolin               | Extruded                  | 0.36                               | 0.21     | 0.0037 | 400                      | 19.0             | 80.7                               | 10.6                                | 30.0                                           |
| Lithopone (finished) | Extruded                  | 0.35                               | 0.065    | 0.0004 | 408                      | 8.2              | 63.6                               | 10.2                                | 18.0                                           |
| Lithopone (crude)    | Extruded                  | 0.67                               | 0.26     | 0.0007 | 400                      | 7.6              | 41.1                               | 9.1                                 | 51.0                                           |
| Lithopone            | Extruded                  | 0.72                               | 0.28     | 0.0013 | 400                      | 5.7              | 28.9                               | 11.7                                | 18.0                                           |
| Magnesium carbonate  | Extruded                  | 2.57                               | 0.87     | 0.001  | 415                      | 7.6              | 11.0                               | 11.4                                | 17.4                                           |
| Magnesium carbonate  | Formed on finned drum     | 2.23                               | 1.44     | 0.0019 | 418                      | 7.6              | 13.2                               | 8.6                                 | 24.0                                           |
| Mercuric oxide       | Extruded                  | 0.163                              | 0.07     | 0.004  | 365                      | 3.8              | 66.5                               | 11.2                                | 24.0                                           |
| Silica gel           | Granular                  | 4.51                               | 1.85     | 0.15   | 400                      | 3.8-0.6          | 3.2                                | 8.6                                 | 15.0                                           |
| Silica gel           | Granular                  | 4.49                               | 1.50     | 0.215  | 340                      | 3.8-0.6          | 3.4                                | 9.1                                 | 63.0                                           |
| Silica gel           | Granular                  | 4.50                               | 1.60     | 0.218  | 325                      | 3.8-0.6          | 3.5                                | 9.1                                 | 66.0                                           |
| Soda salt            | Extruded                  | 0.36                               | 0.24     | 0.008  | 410                      | 3.8              | 22.8                               | 5.1                                 | 51.0                                           |
| Starch (potato)      | Scored filter cake        | 0.866                              | 0.55     | 0.069  | 400                      | 7.0              | 26.3                               | 10.2                                | 27.0                                           |
| Starch (potato)      | Scored filter cake        | 0.857                              | 0.42     | 0.082  | 400                      | 5.1              | 17.7                               | 9.4                                 | 15.0                                           |
| Starch (corn)        | Scored filter cake        | 0.776                              | 0.48     | 0.084  | 345                      | 7.0              | 26.4                               | 7.4                                 | 54.0                                           |
| Starch (corn)        | Scored filter cake        | 0.78                               | 0.56     | 0.098  | 380                      | 7.0              | 27.4                               | 7.6                                 | 24.0                                           |
| Starch (corn)        | Scored filter cake        | 0.76                               | 0.30     | 0.10   | 345                      | 1.9              | 7.7                                | 6.7                                 | 15.0                                           |
| Titanium dioxide     | Extruded                  | 1.2                                | 0.60     | 0.10   | 425                      | 3.0              | 6.8                                | 13.7                                | 6.3                                            |
| Titanium dioxide     | Extruded                  | 1.07                               | 0.65     | 0.29   | 425                      | 8.2              | 16.0                               | 8.6                                 | 6.0                                            |
| White lead           | Formed on finned drum     | 0.238                              | 0.07     | 0.001  | 355                      | 6.4              | 76.8                               | 11.2                                | 30.0                                           |
| White lead           | Extruded                  | 0.49                               | 0.17     | 0.0    | 365                      | 3.8              | 33.8                               | 10.2                                | 27.0                                           |
| Zinc stearate        | Extruded                  | 4.63                               | 1.50     | 0.005  | 360                      | 4.4              | 4.2                                | 8.6                                 | 36.0                                           |

to reduce the internal pressure to extremely low levels because of the large vapor volumes thereby created. It is necessary to compromise on operating pressure, considering leakage, condensation problems, and the size of the vapor lines and pumping system. Very few vacuum dryers operate below 5 mmHg pressure on a commercial scale. Air in-leakage through gasket surfaces will be in the range of 0.2 kg/(h-linear m of gasketed surface) under these conditions. To keep vapor partial pressure and solids temperature low without pulling excessively high vacuum, a nitrogen bleed may be introduced, particularly in the later stages of drying. The vapor and solids surface temperatures then fall below the vapor boiling point, toward the wet-bulb temperature.

**Vertical Agitated Dryers** This classification includes vertical pan dryers, filter dryers, and spherical and conical dryers.

**Vertical pan dryer** The basic vertical pan dryer consists of a short, squat vertical cylinder (Fig. 12-52 and Table 12-24) with an outer heating jacket and an internal rotating agitator, again with the axis

vertical, which mixes the solid and sweeps the base of the pan. Heat is supplied by circulation of hot water, steam, or thermal fluid through the jacket; it may also be used for cooling at the end of the batch cycle, using cooling water or refrigerant. The agitator is usually a plain set of solid blades, but may be a ribbon-type screw or internally heated blades. Product is discharged from a door at the lower side of the wall. Sticky materials may adhere to the agitator or be difficult to discharge.

**Filter dryer** The basic Nutsche filter dryer is like a vertical pan dryer, but with the bottom heated plate replaced by a filter plate. Hence, a slurry can be fed in and filtered, and the wet cake dried in situ. These units are especially popular in the pharmaceutical industry, as containment is good and a difficult wet solids transfer operation is eliminated by carrying out both filtration and drying in the same vessel. Drying times tend to be longer than for vertical pan dryers as the bottom plate is no longer heated. Some types (e.g., Mitchell Thermovac,

TABLE 12-22 Performance Data for Continuous Through-Circulation Dryers\*

|                                                                           | Kind of material         |                     |                                       |                |                       |                         |
|---------------------------------------------------------------------------|--------------------------|---------------------|---------------------------------------|----------------|-----------------------|-------------------------|
|                                                                           | Inorganic pigment        | Cornstarch          | Fiber staple                          |                | Charcoal briquettes   | Inorganic chemical      |
| Capacity, kg dry product/h                                                | 712                      | 4536                | 1724                                  |                | 5443                  | 862                     |
| Approximate dryer area, m <sup>2</sup>                                    | 22.11                    | 66.42               | Stage A, 57.04                        | Stage B, 35.12 | 52.02                 | 30.19                   |
| Depth of loading, cm                                                      | 3                        | 4                   |                                       |                | 16                    | 4                       |
| Air temperature, °C                                                       | 120                      | 115 to 140          | 130 to 100                            | 100            | 135 to 120            | 32 to 52                |
| Loading, kg product/m <sup>2</sup>                                        | 18.8                     | 27.3                | 3.5                                   | 3.3            | 182.0                 | 9.1                     |
| Type of conveyor, mm                                                      | 1.59 by 6.35 slots       | 1.19 by 4.76 slots  | 2.57-diameter holes, perforated plate |                | 8.5 × 8.5 mesh screen | 4.23 × 4.23 mesh screen |
| Preforming method or feed                                                 | Rolling extruder         | Filtered and scored | Fiber feed                            |                | Pressed               | Extrusion               |
| Type and size of preformed particle, mm                                   | 6.35-diameter extrusions | Scored filter cake  | Cut fiber                             |                | 64 × 51 × 25          | 2-diameter extrusions   |
| Initial moisture content, % bone-dry basis                                | 120                      | 85.2                | 110                                   |                | 37.3                  | 300                     |
| Final moisture content, % bone-dry basis                                  | 0.5                      | 13.6                | 9                                     |                | 5.3                   | 11.1                    |
| Drying time, min                                                          | 35                       | 24                  | 11                                    |                | 105                   | 192                     |
| Drying rate, kg water evaporated/(h·m <sup>2</sup> )                      | 38.39                    | 42.97               | 17.09                                 |                | 22.95                 | 9.91                    |
| Air velocity (superficial), m/s                                           | 1.27                     | 1.12                | 0.66                                  |                | 1.12                  | 1.27                    |
| Heat source per kg water evaporated, steam kg/kg gas (m <sup>3</sup> /kg) | Gas                      | Steam               | Steam                                 |                | Waste heat            | Steam                   |
| Installed power, kW                                                       | 29.8                     | 119.3               | 194.0                                 |                | 82.06                 | 179.0                   |

\*Courtesy of Wolverine Proctor & Schwartz, Inc.

Krauss-Maffei TNT) invert the unit between the filtration and drying stages to avoid this problem.

**Spherical dryer** Sometimes called the *turbosphere*, this is another agitated dryer with a vertical axis mixing shaft, but rotation is typically faster than in the vertical pan unit, giving improved mixing and heat transfer. The dryer chamber is spherical, with solids discharge through a door or valve near the bottom.

**Conical mixer dryer** This is a vertically oriented conical vessel with an internally mounted rotating screw. Figure 12-53 shows a schematic of a typical conical mixer dryer. The screw rotates about its own axis (speeds up to 100 rpm) and around the interior of the vessel (speeds up to 0.4 rpm). Because it rotates around the full circumference of the vessel, the screw provides a self-cleaning effect for the heated vessel walls, as well as effective agitation; it may also be internally heated. Either top-drive (via an internal rotating arm) or bottom-drive (via a universal joint) may be used; the former is more common. The screw is cantilevered in the vessel and requires no additional support (even in vessel sizes up to 20-m<sup>3</sup> operating volume). Cleaning of the dryer is facilitated with CIP systems that can be used for cleaning, and/or the vessel can be completely flooded with water or solvents. The dryer makes maximum use of the product-heated areas—the filling volume of the vessel (up to the knuckle of the dished head) is the usable product loading. In some recent applications, microwaves have been used to provide additional energy input and shorten drying times.

In the bottom-drive system, the vessel cover is free of drive components, allowing space for additional process nozzles, manholes, explosion venting, etc., as well as a temperature lance for direct, continuous product temperature measurement in the vessel. The top cover of the vessel is easily heated by either a half-pipe coil or heat tracing, which ensures that no vapor condensation will occur in the process area.

TABLE 12-23 Conveyor-Screen-Dryer Costs\*

| Length | 2.4-m-wide conveyor   | 3.0-m-wide conveyor   |
|--------|-----------------------|-----------------------|
| 7.5 m  | \$8600/m <sup>2</sup> | \$7110/m <sup>2</sup> |
| 15 m   | \$6700/m <sup>2</sup> | \$5600/m <sup>2</sup> |
| 22.5 m | \$6200/m <sup>2</sup> | \$5150/m <sup>2</sup> |
| 30 m   | \$5900/m <sup>2</sup> | \$4950/m <sup>2</sup> |

\*National Drying Machinery Company, 1996.

Because there are no drive components in the process area, the risk of batch failures due to contamination from gear lubricants is eliminated. However, the bottom joint requires especially careful design, maintenance, and sealing. The disassembly of the unit is simplified, as all work on removing the screw can be done without vessel entry. For disassembly, the screw is simply secured from the top, and the drive components are removed from the bottom of the dryer.

**Horizontal Pan Dryer** This consists of a stationary cylindrical shell, mounted horizontally, in which a set of agitator blades mounted on a revolving central shaft stirs the solids being treated. They tend to be used for larger batches than vertical agitated or batch rotating dryers. Heat is supplied by circulation of hot water, steam, or Dowtherm through the jacket surrounding the shell and, in larger units, through the hollow central shaft. The agitator can be of many different forms, including simple paddles, ploughshare-type blades, a single discontinuous spiral, or a double continuous spiral. The outer blades are set as closely as possible to the wall without touching, usually leaving a gap of 0.3 to 0.6 cm. Modern units occasionally employ spring-loaded shell scrapers mounted on the blades. The dryer is charged through a port at the top and emptied through one or more discharge nozzles at the bottom. Vacuum is applied and maintained by any of the conventional methods, i.e., steam jets, vacuum pumps, etc.

A similar type, the batch indirect rotary dryer, consists of a rotating horizontal cylindrical shell, suitably jacketed. Vacuum is applied to this unit through hollow trunnions with suitable packing glands. Rotary glands must be used also for admitting and removing the heating medium from the jacket. The inside of the shell may have lifting bars, welded longitudinally, to assist agitation of the solids. Continuous rotation is needed while emptying the solids, and a circular dust hood is frequently necessary to enclose the discharge-nozzle turning circle and prevent serious dust losses to the atmosphere during unloading. A typical vacuum rotary dryer is illustrated in Fig. 12-54. Sealing tends to be more difficult where the entire shell rotates compared to the horizontal pan, where only the central agitator shaft rotates, since the seal diameter is smaller in the latter case. Conversely, a problem with a stationary shell is that it can be difficult to empty the final “heel” of material out of the bottom of the cylinder. If batch integrity is important, this is an advantage for the rotary variant over the horizontal pan.

**Heated Agitators** For all agitated dryers, in addition to the jacket heated area, heating the agitator with the same medium as the





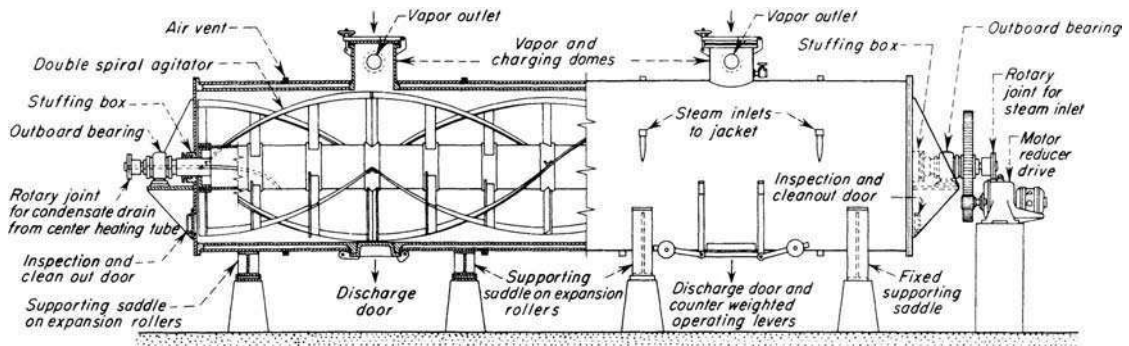


FIG. 12-54 A typical horizontal pan vacuum dryer. (Blaw-Knox Food & Chemical Equipment, Inc.)

jacket (hot water, steam, or thermal oil) will increase the heat-exchange area. This is usually accomplished via rotary joints. Obviously, heating the screw or agitator will mean shorter batch drying times, which yields higher productivity and better product quality due to shorter exposure to the drying temperature, but capital and maintenance costs will be increased. In pan and conical dryers the area is increased only modestly, by 15 to 30 percent; but in horizontal pan and paddle dryers, the opportunity is much greater and indeed the majority of the heat may be supplied through the agitator.

Also, the mechanical power input of the agitator can be a significant additional heat source, and microwave assistance has also been used in filter dryers and conical dryers to shorten drying times (and is feasible in other types).

**Tumbler or Double-Cone Dryers** These are rotating batch vacuum dryers, as shown in Fig. 12-55. Some types are an offset cylinder, but a double-cone shape is more common. They are very common in the pharmaceutical and fine chemicals industries. The gentle rotation can give less attrition than in some mechanically agitated dryers; on the other hand, formation of lumps and balls is more likely. The sloping walls of the cones permit more rapid emptying of solids when the dryer is in a stationary position, compared to a horizontal cylinder, which requires continuous rotation during emptying to convey product to the discharge nozzles. Several new designs of the double-cone type employ internal tubes or plate coils to provide additional heating surface.

On all rotating dryers, the vapor outlet tube is stationary; it enters the shell through a rotating gland and is fitted with an elbow and an upward extension so that the vapor inlet, usually protected by a felt dust filter, will be at all times near the top of the shell.

**Design, Scale-up, and Performance** Like all batch dryers, agitated and rotating dryers are primarily sized to physically contain the required batch volume. Note that the nominal capacity of most dryers is significantly lower than their total internal volume, because of the headspace needed for mechanical drives, inlet ports, suction lines,

dust filters, etc. Care must be taken to determine whether a stated "percentage fill" is based on nominal capacity or geometric volume. Vacuum dryers are usually filled to 50 to 65 percent of their total shell volume.

The standard scoping calculation methods for batch conduction drying apply. The rate of heat transfer from the heating medium through the dryer wall to the solids can be expressed by the usual formula

$$Q = hA \Delta T_m \quad (12-87)$$

where  $Q$  = heat flux, J/s [Btu/h];  $h$  = overall heat-transfer coefficient, J/(m<sup>2</sup>·s·K) [Btu/(h·ft<sup>2</sup>·°F)];  $A$  = total jacket area, m<sup>2</sup> (ft<sup>2</sup>); and  $\Delta T_m$  = log-mean-temperature driving force from heating medium to the solids, K (°F).

The overall heat-transfer rate is almost entirely dependent upon the film coefficient between the inner jacket wall and the solids, which depends on the dryer type and agitation rate, and to a large extent on the solids characteristics. Overall coefficients may range from 30 to 200 J/(m<sup>2</sup>·s·K), based upon total area if the dryer walls are kept reasonably clean. Coefficients as low as 5 or 10 may be encountered if caking on the walls occurs.

For estimating purposes without tests, a reasonable coefficient for ordinary drying, and without taking the product to absolute dryness, may be assumed at  $h = 50$  J/(m<sup>2</sup>·s·K) for mechanically agitated dryers (although higher figures have been quoted for conical and spherical dryers) and 35 J/(m<sup>2</sup>·s·K) for rotating units. The true heat-transfer coefficient is usually higher, but this conservative assumption makes some allowance for the slowing down of drying during the falling-rate period. However, if at all possible, it is always preferable to do pilot-plant tests to establish the drying time of the actual material. Drying trials are conducted in small pilot dryers (50- to 100-L batch units) to determine material handling and drying retention times. Variables such as drying temperature, vacuum level, and screw speed are analyzed during the

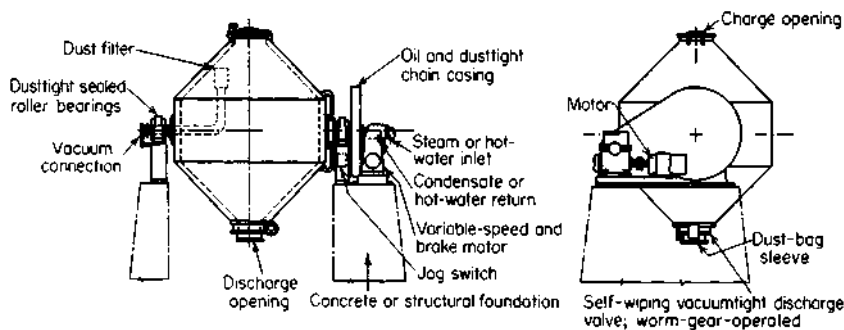


FIG. 12-55 Rotating (double-cone) vacuum dryer. (Stokes Vacuum, Inc.)

## 12-70 PSYCHROMETRY, EVAPORATIVE COOLING, AND SOLIDS DRYING

**TABLE 12-24 Dimensions of Vertical Pan Dryers (Buflovak Inc.)**

| I.D., ft | Product depth, ft | Working volume, ft <sup>3</sup> | USG | Jacketed height, ft | Jacketed area, ft <sup>2</sup> |        |       |                    |    |
|----------|-------------------|---------------------------------|-----|---------------------|--------------------------------|--------|-------|--------------------|----|
|          |                   |                                 |     |                     | Cylinder wall                  | Bottom | Total | Discharge door, in |    |
| 3        | 0.75              | 5.3                             | 40  | 1.0                 | 9                              | 7      | 16    | 5                  | 8  |
| 4        | 1                 | 12.6                            | 94  | 2.0                 | 25                             | 13     | 38    | 6                  | 8  |
| 5        | 1                 | 19.6                            | 147 | 2.0                 | 31                             | 20     | 51    | 8                  | 9  |
| 6        | 1                 | 28.3                            | 212 | 2.0                 | 38                             | 28     | 66    | 8                  | 9  |
| 8        | 1                 | 50.3                            | 377 | 2.0                 | 50                             | 50     | 101   | 8                  | 9  |
| 10       | 1.5               | 117.8                           | 884 | 3.0                 | 94                             | 79     | 173   | 12                 | 12 |

test trials. Scale-up to larger units is done based upon the area/volume ratio of the pilot unit versus the production dryer. In most applications, the overall drying time in the production models is in the range of 2 to 24 h.

Agitator or rotation speeds range from 3 to 8 rpm. Faster speeds yield a slight improvement in heat transfer but consume more power and in some cases, particularly in rotating units, can cause more “balling up” and other stickiness-related problems.

In all these dryers, the surface area tends to be proportional to the square of the diameter  $D^2$ , and the volume to diameter cubed  $D^3$ . Hence the area/volume ratio falls as diameter increases, and drying times increase. It can be shown that the ratio of drying times in the production and pilot-plant dryers is proportional to the cube root of the ratio of batch volumes. However, if the agitator of the production unit is heated, the drying time increase can be reduced or reversed. Table 12-25 gives basic geometric relationships for agitated and rotating batch dryers, which can be used for approximate size estimation or (with great caution) for extrapolating drying times obtained from one dryer type to another. Note that these do not allow for nominal capacity or partial solids fill. For the paddle (horizontal pan) dryer with heated agitator,  $R$  is the ratio of the heat transferred through the agitator to that through the walls, which is proportional to the factor  $hA$  for each case.

**Example 22: Calculations for Batch Dryer** For a 10-m<sup>3</sup> batch of material containing 5000 kg dry solids and 30 percent moisture (dry basis), estimate the size of vacuum dryers required to contain the batch at 50 percent volumetric fill. Jacket temperature is 200°C, applied pressure is 100 mbar (0.1 bar), and the solvent is water (take latent heat as 2400 kJ/kg). Assuming the heat-transfer coefficient based on the total surface area to be 50 W/(m<sup>2</sup>·K) for all

types, calculate the time to dry to 5 percent for (a) unhindered (constant-rate) drying throughout, (b) first-order falling-rate (hindered) drying throughout, (c) if experiment shows the actual drying time for a conical dryer to be 12.5 h and other cases are scaled accordingly. Take  $R = 5$  with the heated agitator. Assume material is nonhygroscopic (equilibrium moisture content  $X_E = 0$ ).

**Solution:** The dryer volume  $V$  must be 20 m<sup>3</sup>, and the diameter is calculated from column 4 of Table 12-25, assuming the default  $L/D$  ratios. Table 12-26 gives the results. Water at 100 mbar boils at 46°C so take  $\Delta T$  as 200 – 46 = 154°C. Then  $Q$  is found from Eq. (12-87). The methods used are given in the section “Equipment—General, Scoping Design.” For constant-rate drying throughout, drying time  $t_{CR}$  = evaporation rate/heat input rate and was given by Eq. (12-62):

$$t_{CR} = \frac{m_S(X_O - X_E)\lambda_{ev}}{h_{WS}\Delta T_{WS}A_S} = \frac{5000(0.3 - 0.05)2400}{0.05(154A_S)} \quad (12-62)$$

This gives  $t_{CR}$  as 389,610/ $A_S$  s or 108.23/ $A_S$  h. Values for  $A_S$  and calculated times for the various dryer types are given in Table 12-26.

For falling-rate drying throughout, time  $t_{FR}$  is given by Eq. (12-63); the multiplying factor for drying time is  $1.2 \ln 6 = 2.15$  for all dryer types.

$$\frac{t_{FR}}{t_{CR}} = \frac{X_1 - X_E}{X_1 - X_2} \ln \left( \frac{X_1 - X_E}{X_2 - X_E} \right) = \frac{0.3}{0.25} \ln \frac{0.3}{0.05} \quad (12-63)$$

If the material showed a critical moisture content, the calculation could be split into two sections for constant-rate and falling-rate drying. Likewise, the experimental drying time  $t_{exp}$  for the conical dryer is 12.5 h which is a factor of 3.94 greater than the constant-rate drying time. A very rough estimate of drying times for the other dryer types has then been made by applying the same scaling factor (3.94) to their constant-rate drying times. Two major sources of error are possible: (1) The drying kinetics could differ between dryers; and (2) if the

**TABLE 12-25 Calculation of Key Dimensions for Various Batch Contact Dryers (Fig. 12-55a Shows the Geometries)**

| Dryer type                   | Volume as $f(D)$                                    | Typical $L/D$ | Diameter as $f(V)$                              | Surface area as $f(D)$                                                                  | Ratio $A/V$                                                                                   |
|------------------------------|-----------------------------------------------------|---------------|-------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Tumbler/double-cone          | $V = \frac{\pi D^3}{12} \left( \frac{L}{D} \right)$ | 1.5           | $D = \left[ \frac{12V}{\pi(L/D)} \right]^{1/3}$ | $A = \frac{\pi D^2}{2} \left[ \left( \frac{L}{D} \right)^2 + 1 \right]^{1/2}$           | $\frac{A}{V} = \frac{6}{D} \left[ 1 + \left( \frac{D}{L} \right)^2 \right]^{1/2}$             |
| Vertical pan                 | $V = \frac{\pi D^3}{4} \left( \frac{L}{D} \right)$  | 0.5           | $D = \left[ \frac{4V}{\pi(L/D)} \right]^{1/3}$  | $A = \pi D^2 \left( \frac{L}{D} + \frac{1}{4} \right)$                                  | $\frac{A}{V} = \frac{4}{D} \left( 1 + \frac{D}{4L} \right)$                                   |
| Spherical                    | $V = \frac{\pi D^3}{6} \left( \frac{L}{D} \right)$  | 1             | $D = \left[ \frac{6V}{\pi(L/D)} \right]^{1/3}$  | $A = \pi D^2 \left( \frac{L}{D} \right)$                                                | $\frac{A}{V} = \frac{6}{D}$                                                                   |
| Filter dryer                 | $V = \frac{\pi D^3}{4} \left( \frac{L}{D} \right)$  | 0.5           | $D = \left[ \frac{4V}{\pi(L/D)} \right]^{1/3}$  | $A = \pi D^2 \left( \frac{L}{D} \right)$                                                | $\frac{A}{V} = \frac{4}{D}$                                                                   |
| Conical agitated             | $V = \frac{\pi D^3}{12} \left( \frac{L}{D} \right)$ | 1.5           | $D = \left[ \frac{12V}{\pi(L/D)} \right]^{1/3}$ | $A = \frac{\pi D^2}{2} \left[ \left( \frac{L}{D} \right)^2 + \frac{1}{4} \right]^{1/2}$ | $\frac{A}{V} = \frac{6}{D} \left[ 1 + \frac{1}{4} \left( \frac{D}{L} \right)^2 \right]^{1/2}$ |
| Paddle (horizontal agitated) | $V = \frac{\pi D^3}{4} \left( \frac{L}{D} \right)$  | 5             | $D = \left[ \frac{4V}{\pi(L/D)} \right]^{1/3}$  | $A = \pi D^2 \left( \frac{L}{D} \right)$                                                | $\frac{A}{V} = \frac{4}{D}$                                                                   |
| Paddle, heated agitator      | $V = \frac{\pi D^3}{4} \left( \frac{L}{D} \right)$  | 5             | $D = \left[ \frac{4V}{\pi(L/D)} \right]^{1/3}$  | $A = \pi D^2 \left( \frac{L}{D} \right) (1 + R)$                                        | $\frac{A}{V} = \frac{4}{D} (1 + R)$                                                           |

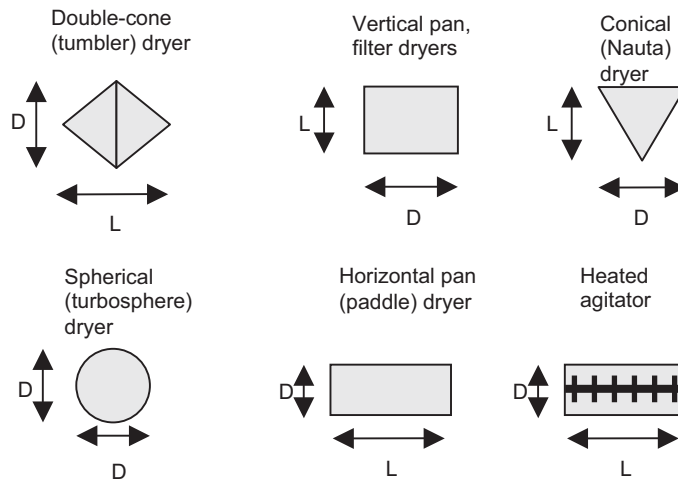


FIG. 12-55a Basic geometries for batch dryer calculations.

estimated heat-transfer coefficient for either the base case or the new dryer type is in error, the scaling factor will be wrong. All drying times have been shown in hours, as this is more convenient than seconds.

The paddle with heated agitator has the shortest drying time, and the filter dryer the longest (because the bottom plate is unheated). Other types are fairly comparable. The spherical dryer would usually have a higher heat-transfer coefficient and shorter drying time than shown.

#### Performance and Cost Data for Batch Vacuum Rotary Dryers

Typical performance data for horizontal pan vacuum dryers are given in Table 12-27. Size and cost data for rotary agitator units are given in Table 12-28. Data for double-cone rotating units are in Table 12-29.

#### Continuous Agitated Dryers

**Description** These dryers, often known as paddle or horizontal agitated dryers, consist of one or more horizontally mounted shells with internal mechanical agitators, which may take many different forms. They are a continuous equivalent of the horizontal pan dryer and are similar in construction, but usually of larger dimensions. They have many similarities to continuous indirect rotary dryers and are sometimes classed as rotary dryers, but this is a misnomer because the outer shell does not rotate, although in some types there is an inner shell which does. Frequently, the internal agitator is heated, and a wide variety of designs exist. Often, two intermeshing agitators are used. There are important variants with high-speed agitator rotation and supplementary convective heating by hot air.

**Classification** Continuous; mechanical agitation and transport; layer; contact/conduction or convective (through-circulation).

The basic differences are in type of agitator, the two key factors being heat-transfer area and solids handling/stickiness characteristics. Unfortunately, the types giving the highest specific surface area (multiple tubes and coils) are often also the ones most liable to fouling and blockage and most difficult to clean. Figure 12-56 illustrates a number of different agitator types.

The most common problem with paddle dryers (and with their closely related cousins, steam-tube and indirect rotary dryers) is the

buildup of sticky deposits on the surface of the agitator or outer jacket. This leads, first, to reduced heat-transfer coefficients and slower drying and, second, to blockages and stalling of the rotor. Also, thermal decomposition and loss of product quality can result. The problem is usually most acute at the feed end of the dryer, where the material is wettest and stickiest. A wide variety of different agitator designs have been devised to try to reduce stickiness problems and enhance cleanability while providing a high heat-transfer area. Many designs incorporate a high torque drive combined with rugged shaft construction to prevent rotor stall during processing, and stationary mixing elements are installed in the process housing which continually clean the heat-exchange surfaces of the rotor to minimize any crust buildup and ensure an optimum heat-transfer coefficient at all times. Another alternative is to use two parallel intermeshing shafts, as in the Nara paddle dryer (Fig. 12-57). Suitably designed continuous paddle and batch horizontal pan dryers can handle a wide range of product consistencies (dilute slurries, pastes, friable powders) and can be used for processes such as reactions, mixing, drying, cooling, melting, sublimation, distilling, and vaporizing. Bearing supports are usually provided at both ends of the unit for shaft support.

**Design Methods for Paddle Dryers** Product trials are conducted in small pilot dryers (8- to 60-L batch or continuous units) to determine material handling and process retention times. Variables such as drying temperature, pressure level, and shaft speed are analyzed during the test trials. For initial design purposes, the heat-transfer coefficient for paddle dryers is typically in the range of 10 W/(m<sup>2</sup>·K) (light, free-flowing powders) up to 150 W/(m<sup>2</sup>·K) (dilute slurries). However, it is preferable to scale up from the test results, finding the heat-transfer coefficient by backcalculation and scaling up on the basis of total area of heat-transfer surfaces, including heated agitators. Typical length/diameter ratios are between 5 and 8, similar to rotary dryers and greater than some batch horizontal pan dryers.

**Continuous Rotary Dryers** A rotary dryer consists of a cylinder that rotates on suitable bearings and that is usually slightly inclined to the horizontal. The cylinder length may range from 4 to more than 10 times

TABLE 12-26 Comparative Dimensions and Drying Times for Various Batch Contact Dryers

| Dryer type                   | $h$ , kW/(m <sup>2</sup> ·K) | $L/D$ | $D$ , m | $L$ , m | $A$ , m <sup>2</sup> | $t_{CR}$ , h | $t_{FR}$ , h | $t_{\text{expt}}$ , h |
|------------------------------|------------------------------|-------|---------|---------|----------------------|--------------|--------------|-----------------------|
| Tumbler/double-cone          | 0.05                         | 1.5   | 3.71    | 5.56    | 38.91                | 2.78         | 5.98         | 11.0                  |
| Vertical pan                 | 0.05                         | 0.5   | 3.71    | 1.85    | 32.37                | 3.34         | 7.19         | 13.2                  |
| Spherical                    | 0.05                         | 1     | 3.37    | 3.37    | 35.63                | 3.04         | 6.54         | 12.0                  |
| Filter dryer                 | 0.05                         | 0.5   | 3.71    | 1.85    | 21.58                | 5.01         | 10.77        | 19.8                  |
| Conical agitated             | 0.05                         | 1.5   | 3.71    | 5.56    | 34.12                | 3.17         | 6.82         | 12.5                  |
| Paddle (horizontal agitated) | 0.05                         | 5     | 1.72    | 8.60    | 46.50                | 2.33         | 5.01         | 9.2                   |
| Paddle, heated agitator      | 0.05                         | 5     | 1.72    | 8.60    | 278.99               | 0.39         | 0.83         | 1.52                  |



TABLE 12-27 Performance Data of Vacuum Rotary Dryers\*

| Material                      | Diameter × length, m | Initial moisture, % dry basis | Steam pressure, Pa × 10 <sup>3</sup> | Agitator speed, r/min | Batch dry weight, kg | Final moisture, % dry basis | Pa × 10 <sup>3</sup> | Time, h | Evaporation, kg/(h·m <sup>2</sup> ) |
|-------------------------------|----------------------|-------------------------------|--------------------------------------|-----------------------|----------------------|-----------------------------|----------------------|---------|-------------------------------------|
| Cellulose acetate             | 1.5 × 9.1            | 87.5                          | 97                                   | 5.25                  | 610                  | 6                           | 90–91                | 7       | 1.5                                 |
| Starch                        | 1.5 × 9.1            | 45–48                         | 103                                  | 4                     | 3630                 | 12                          | 88–91                | 4.75    | 7.3                                 |
| Sulfur black                  | 1.5 × 9.1            | 50                            | 207                                  | 4                     | 3180                 | 1                           | 91                   | 6       | 4.4                                 |
| Fuller's earth/mineral spirit | 0.9 × 3.0            | 50                            | 345                                  | 6                     | 450                  | 2                           | 95                   | 8       | 5.4                                 |

\*Stokes Vacuum, Inc.

the diameter, which may vary from less than 0.3 to more than 3 m. Solids fed into one end of the drum are carried through it by gravity, with rolling, bouncing and sliding, and drag caused by the airflow either retarding or enhancing the movement, depending on whether the dryer is cocurrent or countercurrent. It is possible to classify rotary dryers into direct-fired, where heat is transferred to the solids by direct exchange between the gas and the solids, and indirect, where the heating medium is separated from physical contact with the solids by a metal wall or tube. Many rotary dryers contain flights or lifters, which are attached to the inside of the drum and which cascade the solids through the gas as the drum rotates.

For handling large quantities of granular solids, a cascading rotary dryer is often the equipment of choice. If the material is not naturally free-flowing, recycling of a portion of the final dry product may be used to precondition the feed, either in an external mixer or directly inside the drum. Hanging link chains and/or scrapper chains are also used for sticky feed materials.

Their operating characteristics when performing heat- and mass-transfer operations make them suitable for the accomplishment of drying, chemical reactions, solvent recovery, thermal decompositions, mixing, sintering, and agglomeration of solids. The specific types included are the following:

**Direct cascading rotary dryer (cooler).** This is usually a bare metal cylinder but with internal flights (shelves) which lift the material and drop it through the airflow. It is suitable for low- and medium-temperature operations, the operating temperature being limited primarily by the strength characteristics of the metal employed in fabrication.

**Direct rotary dryer (cooler).** As above but without internal flights.

**Direct rotary kiln.** This is a metal cylinder lined on the interior with insulating block and/or refractory brick. It is suitable for high-temperature operations.

**Indirect steam-tube dryer.** This is a bare metal cylinder provided with one or more rows of metal tubes installed longitudinally in the shell. It is suitable for operation up to available steam temperatures or in processes requiring water cooling of the tubes.

**Indirect rotary calciner.** This is a bare metal cylinder surrounded on the outside by a fired or electrically heated furnace. It is suitable for operation at medium temperatures up to the maximum that can be tolerated by the metal wall of the cylinder, usually 650 to 700 K for carbon steel and 800 to 1025 K for stainless steel.

**Direct Roto-Louvre dryer.** This is one of the more important special types, differing from the direct rotary unit in that true *through-circulation* of gas through the solids bed is provided. Like the direct rotary, it is suitable for low- and medium-temperature operation.

**Direct heat rotary dryer.** The direct heat units are generally the simplest and most economical in operation and construction, when the solids and gas can be permitted to be in contact. In design mode, the required gas flow rate can be obtained from a heat and mass balance. Bed cross-sectional area is found from a scoping design calculation (a typical gas velocity is 3 m/s for cocurrent and 2 m/s for countercurrent units). Length is normally between 5 and 10 times drum diameter (an  $L/D$  value of 8 can be used for initial estimation) or can be calculated by using an incremental model (see Examples 21 and 23).

A typical schematic diagram of a rotary dryer is shown in Fig. 12-58, while Fig. 12-59 shows typical lifting flight designs.

**Classification** Continuous; agitation and transport by rotation/gravity; layer (dispersion for cascading rotary dryers); convective (through-circulation) or contact/conduction.

**Residence Time, Standard Configuration** The residence time in a rotary dryer  $\tau$  represents the average time that particles are present in the equipment, so it must match the required drying time.

**Traditional approaches** For rotary kilns, without lifting flights, Sullivan et al. (U.S. Bureau of Mines Tech. Paper 384) gave an early formula:

$$\tau = \frac{106.2L\sqrt{\gamma}}{N_m D \tan \alpha} \quad (12-88)$$

Here, the natural angle of repose of the material is  $\gamma$ , which increases as the material becomes more cohesive and less free-flowing, and the residence time  $\tau$  is in seconds, but the rotation rate  $N_m$  is in revolutions per minute (rpm), not per second. The Friedman and Marshall equation [Chem. Eng. Progr. 45(8): 482 (1949)] is derived from this, with an additional term to account for air drag on the solids:

$$\tau = \frac{13.8L}{N_m^{0.9} D \alpha} \pm \frac{590.6LG}{d_p^{0.5} G} \quad (12-89)$$

Here  $d_p$  is the particle size, in micrometers, while  $F$  and  $G$  are the mass flow rates of solids and gas, respectively. This formula has been frequently reported and includes a correction factor to the initial constant term to reflect actual experimental results. Friedman and Marshall took the angle of repose for the solids to be 40° and introduced a 0.9 power for the rotational speed, which had questionable justification within the accuracy of the data. The second term represents the airflow drag term and is negative for cocurrent flow and positive for countercurrent flow.

TABLE 12-28 Standard Rotary Vacuum Dryers\*

| Diameter, m | Length, m | Heating surface, m <sup>2</sup> | Working capacity, m <sup>3</sup> † | Agitator speed, r/min | Drive, kW | Weight, kg | Purchase price (1995) |                       |
|-------------|-----------|---------------------------------|------------------------------------|-----------------------|-----------|------------|-----------------------|-----------------------|
|             |           |                                 |                                    |                       |           |            | Carbon steel          | Stainless steel (304) |
| 0.46        | 0.49      | 0.836                           | 0.028                              | 7½                    | 1.12      | 540        | \$ 43,000             | \$ 53,000             |
| 0.61        | 1.8       | 3.72                            | 0.283                              | 7½                    | 1.12      | 1,680      | 105,000               | 130,000               |
| 0.91        | 3.0       | 10.2                            | 0.991                              | 6                     | 3.73      | 3,860      | 145,000               | 180,000               |
| 0.91        | 4.6       | 15.3                            | 1.42                               | 6                     | 3.73      | 5,530      | 180,000               | 205,000               |
| 1.2         | 6.1       | 29.2                            | 3.57                               | 6                     | 7.46      | 11,340     | 270,000               | 380,000               |
| 1.5         | 7.6       | 48.1                            | 6.94                               | 6                     | 18.7      | 15,880     | 305,000               | 440,000               |
| 1.5         | 9.1       | 57.7                            | 8.33                               | 6                     | 22.4      | 19,050     | 330,000               | 465,000               |

\*Stokes Vacuum, Inc. Prices include shell, 50-lb/in<sup>2</sup>-gauge jacket, agitator, drive, and motor; auxiliary dust collectors, condensers.

†Loading with product level on or around the agitator shaft.

TABLE 12-29 Standard (Double-Cone) Rotating Vacuum Dryers\*

| Working capacity, m <sup>3</sup> | Total volume, m <sup>3</sup> | Heating surface, m <sup>2</sup> | Drive, kW | Floor space, m <sup>2</sup> | Weight, kg | Purchase cost (1995) |                 |
|----------------------------------|------------------------------|---------------------------------|-----------|-----------------------------|------------|----------------------|-----------------|
|                                  |                              |                                 |           |                             |            | Carbon steel         | Stainless steel |
| 0.085                            | 0.130                        | 1.11                            | .373      | 2.60                        | 730        | \$ 32,400            | \$ 38,000       |
| 0.283                            | 0.436                        | 2.79                            | .560      | 2.97                        | 910        | 37,800               | 43,000          |
| 0.708                            | 1.09                         | 5.30                            | 1.49      | 5.57                        | 1810       | 50,400               | 57,000          |
| 1.42                             | 2.18                         | 8.45                            | 3.73      | 7.15                        | 2040       | 97,200               | 106,000         |
| 2.83                             | 4.36                         | 13.9                            | 7.46      | 13.9                        | 3860       | 198,000              | 216,000         |
| 4.25                             | 6.51                         | 17.5                            | 11.2      | 14.9                        | 5440       | 225,000              | 243,000         |
| 7.08                             | 10.5                         | *38.7                           | 11.2      | 15.8                        | 9070       | 324,000              | 351,000         |
| 9.20                             | 13.9                         | *46.7                           | 11.2      | 20.4                        | 9980       | 358,000              | 387,000         |
| 11.3                             | 16.0                         | *56.0                           | 11.2      | 26.0                        | 10,890     | 378,000              | 441,000         |

\*Stokes Vacuum, Inc. Price includes dryer, 15-lb/in<sup>2</sup> jacket, drive with motor, internal filter, and trunnion supports for concrete or steel foundations. Horsepower is established on 65 percent volume loading of material with a bulk density of 50 lb/ft<sup>3</sup>. Models of 250 ft<sup>3</sup>, 325 ft<sup>3</sup>, and 400 ft<sup>3</sup> have extended surface area.

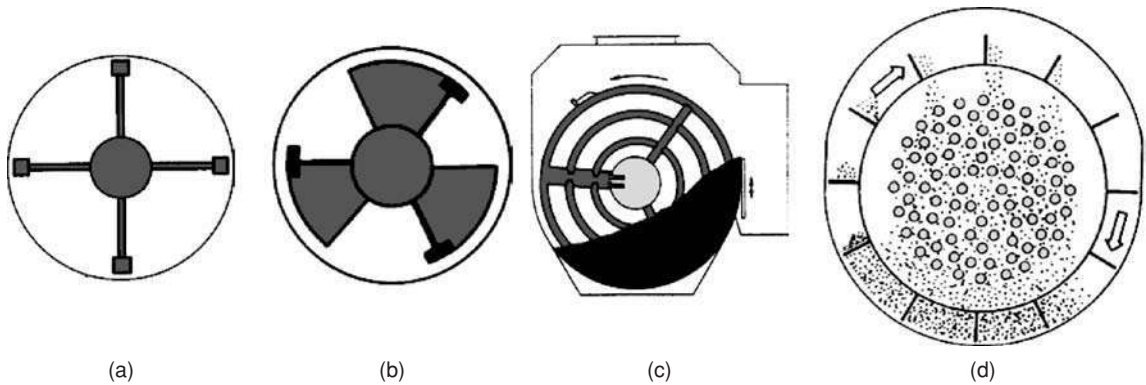


FIG. 12-56 Typical agitator designs for paddle (horizontal agitated) dryers. (a) Simple unheated agitator. (b) Heated cut-flight agitator. (c) Multicoil unit. (d) Tube bundle.

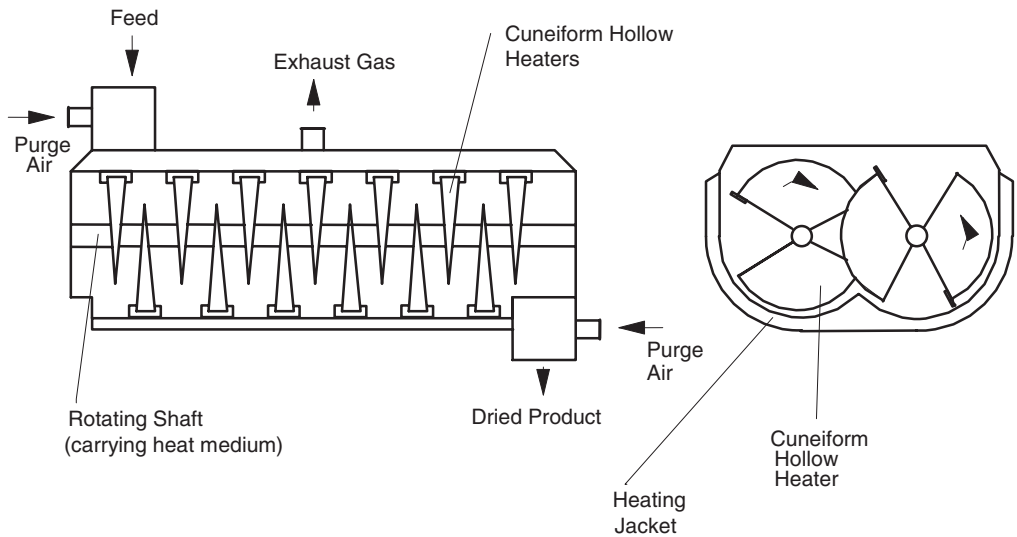


FIG. 12-57 Nara twin-shaft paddle dryer.

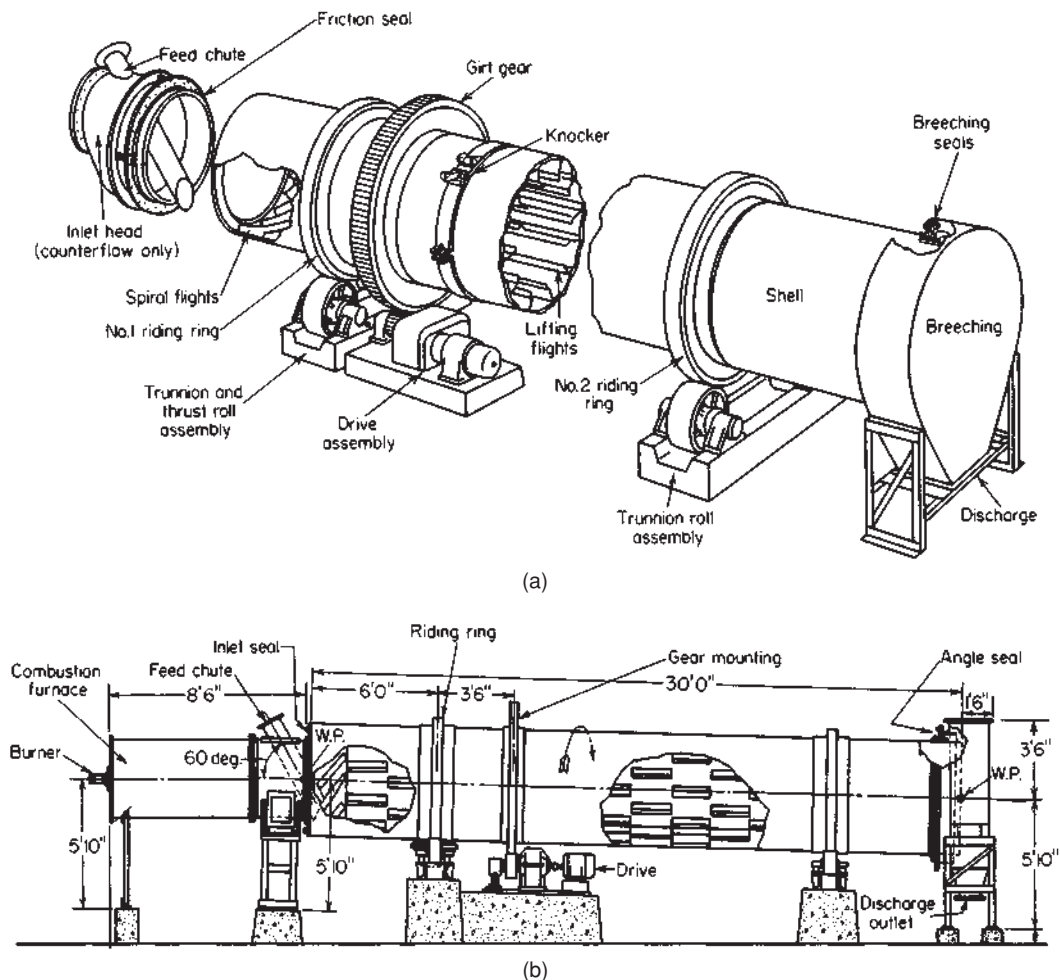


FIG. 12-58 Component arrangement (a) and elevation (b) of countercurrent direct-heat rotary dryer. (Air Preheater Company, Raymond® & Bartlett Snow™ Products.)

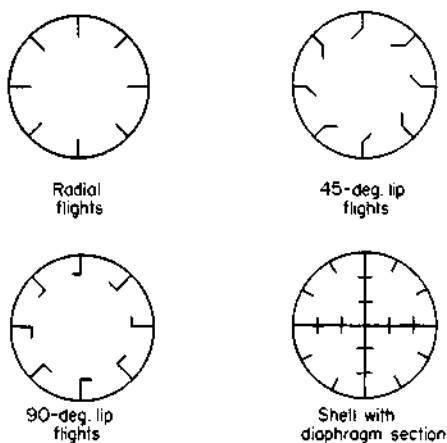


FIG. 12-59 Typical lifting flight designs.

Saeman and Mitchell [*Chem. Eng. Progr.* **50**(9):467 (1954)] proposed the following expression:

$$\tau = \frac{L}{f_H N D (\tan \alpha \pm k_m U_G)} \quad (12-90)$$

Here  $f_H$  is a cascade factor, with values typically between 2 and  $\pi$ , increasing as solids holdup increases, and  $k_m$  is an empirical constant (dimensional) for a given material. The superficial gas velocity through the empty drum is  $U_G$ . It was assumed that the airborne particle velocity was proportional to the air velocity. Two empirical constants  $f_H$  and  $k_m$  are also required to use the equation, and these are not generally available.

Schofield and Glikin [*Trans. IChemE* **40**:183 (1962)] analyzed particle motion from flights and airborne drag, obtaining

$$\tau = \frac{L}{\bar{y} \bar{\theta} N [\sin \alpha - (K U_G^2 / g)]} \quad (12-91)$$

Here  $\bar{y}$  is the mean distance of fall of the particles,  $\bar{\theta}$  is the mean angle moved by particles in flights, and  $K$  is a dimensional drag constant. At

small angles  $\alpha$ ,  $\sin \alpha \approx \tan \alpha$ , and they noted that  $\bar{y} \bar{\theta} \approx 2D$ , so their final Eq. (12-91) is similar to that of Saeman and Mitchell (1954).

The above equations mainly differ in whether the drag term is additive or subtractive (as with Friedman and Marshall) or in the denominator (as with Saeman and Mitchell, and Schofield and Glikin). Some workers, including Sullivan et al. (1927), have neglected the effect of air drag completely. However, the general experience with rotary dryers is that the effect of air velocity and hence of air drag is very substantial, suggesting that neglecting air drag in any equation or analysis is unlikely to be sufficient unless the air velocity is very low. The formulas link  $L$  and  $\tau$ , which is reasonably convenient for dryer performance assessment, but inconvenient for dryer design, where neither  $L$  nor  $\tau$  is initially known.

**Modern analysis** Matchett and Baker [J. Sep. Proc. Technol. 8:11 (1987)] provided a complete analysis of particle motion in rotary dryers. They considered both the airborne phase (particles falling through air) and the dense phase (particles in the flights or the rolling bed at the bottom). Typically, particles spend 90 to 95 percent of the time in the dense phase, but the majority of the drying takes place in the airborne phase. In the direction parallel to the dryer axis, most particle movement occurs through four mechanisms: by gravity and air drag in the airborne phase, and by bouncing, and sliding and rolling, in the dense phase. The combined particle velocity in the airborne phase is  $U_{p1}$ , which is the sum of the gravitational and air drag components for cocurrent dryers and the difference between them for countercurrent dryers. The dense-phase velocity, arising from bouncing, sliding, and rolling, is denoted  $U_{p2}$ .

Papadakis et al. [Dry. Tech. 12(1&2):259 (1994)] rearranged the Matchett and Baker model from its original "parallel" form into a more computationally convenient "series" form. The sum of the calculated residence times in the airborne and dense phases,  $\tau_G$  and  $\tau_s$ , respectively, is the total solids residence time. The dryer length is simply the sum of the distances travelled in the two phases.

$$\tau = \tau_G + \tau_s \quad (12-92)$$

$$L = \tau_G U_{p1} + \tau_s U_{p2} \quad (12-93)$$

For airborne phase motion, the velocity  $U_{p1}^o$  due to the gravitational component is given approximately by

$$U_{p1}^o = \sqrt{\frac{gD_e}{2 \cos \alpha}} K_{fall} \tan \alpha \quad (12-94)$$

where  $D_e$  is the effective diameter, which is the distance actually fallen by the particles. When one is designing a dryer, this parameter will not be known until the flight width is decided. And  $K_{fall}$  is a parameter that allows for particles falling from a number of positions, with different times of flight and lifting times, and is generally between 0.7 and 1. The velocity  $U_{p1}^o$  due to the gravitational component is most conveniently expressed as

$$U_{p1}^o = \sqrt{\frac{gD}{2}} \tan \alpha K_{fall} \sqrt{\frac{D_e}{D \cos \alpha}} = \sqrt{\frac{gD}{2}} \tan \alpha K_K \quad (12-95)$$

The drag force gives a velocity component  $U_{p1}^d$  that must be obtained from experimental correlations, and combining these components gives  $U_{p1}$ .

Bouncing, rolling, and sliding are not so easily analyzed theoretically. Matchett and Baker suggested that the dense-phase velocity could be characterized in terms of a dimensionless dense-phase velocity number  $a$ , through the equation

$$a = \frac{U_{p2}}{N \cdot D \cdot \tan \alpha} \quad (12-96)$$

Other workers suggested that, in underloaded and design-loaded dryers, bouncing was a significant transport mechanism, whereas for overloaded dryers, rolling (kilning) was important. Bouncing mechanisms can depend on the airborne phase velocity  $U_{p1}$ , since this affects the angle at which the particles hit the bottom of the kiln and the dis-

tance they move forward. Rolling mechanisms would be expected to depend on the depth of the bottom bed, and hence on the difference between the actual holdup  $H$  and the design-loaded holdup  $H^*$ .

As an example of the typical numbers involved, Matchett and Baker [J. Sep. Proc. Technol., 9:5 (1988)] used their correlations to assess the data of Saeman and Mitchell for an industrial rotary dryer with  $D = 1.83$  m and  $L = 10.67$  m, with a slope of  $4^\circ$ ,  $0.067$  m/m. For a typical run with  $U_G = 0.98$  m/s and  $N = 0.08$  r/s, they calculated that  $U_{p1}^o = 0.140$  m/s,  $U_{p1}^d = -0.023$  m/s,  $U_{p1} = 0.117$  m/s, and  $U_{p2} = -0.02$  m/s. The dryer modeled was countercurrent and therefore had a greater slope and lower gas velocity than those of a cocurrent unit; for the latter,  $U_{p1}^o$  would be lower and  $U_{p1}^d$  positive and larger. The ratio  $\tau_s/\tau_G$  is approximately 12 in this case, so that the distance traveled in dense-phase motion would be about twice that in the airborne phase.

Kemp and Oakley [Dry. Tech., 20(9):1699 (2002)] showed that the ratio  $\tau_G/\tau_s$  can be found by comparing the average time of flight from the top of the dryer to the bottom  $t_f$  to the average time required for the particles to be lifted by the flights  $t_d$ . They derived the following equation:

$$\frac{\tau_s}{\tau_G} = \frac{t_d}{t_f} = \frac{K_{fl}}{N} \sqrt{\frac{g}{D}} \quad (12-97)$$

Here all the unknowns have been rolled into a single dimensionless parameter  $K_{fl}$ , given by

$$K_{fl} = \frac{\theta}{\pi \sqrt{2} \sin \theta} \sqrt{\frac{D}{D_e}} \quad (12-98)$$

Here  $D_e$  is the effective diameter (internal diameter between lips of flights), and the solids are carried in the flights for an angle  $2\theta$ , on average, before falling. Kemp and Oakley concluded that  $K_{fl}$  can be taken to be 0.4 to a first (and good) approximation. For overloaded dryers with a large rolling bed,  $K_{fl}$  will increase. The form of Eq. (12-97) is very convenient for design purposes since it does not require  $D_e$ , which is unknown until a decision has been made on the type and geometry of the flights.

The model of Matchett and Baker has been shown by Kemp (Proc. IDS 2004, B, 790) to be similar in form to that proposed by Saeman and Mitchell:

$$\tau = \frac{1.1L}{ND \left[ \tan \alpha \cdot (K_K/K_{fl} \sqrt{2} + a) + (1/K_{fl}) \sqrt{\frac{1}{gD}} \cdot U_{p1}^d \right]} \quad (12-99)$$

In Eq. (12-99),  $K_K/(K_{fl} \sqrt{2})$  will typically be on the order of unity, and reported values of  $a$  are in the range of 1 to 4. The airborne gravity component is usually smaller than the dense-phase motion but is not negligible. The sum of these two terms is essentially equivalent to the factor  $f_{fl}$  in Saeman and Mitchell's equation.

**Heat- and Mass-Transfer Estimates** Many rotary dryer studies have correlated heat- and mass-transfer data in terms of an overall volumetric heat-transfer coefficient  $U_a$  [W/(m<sup>3</sup>·K)], defined by

$$Q = U_a a \cdot V_{dryer} \cdot \Delta T_m \quad (12-100)$$

Here  $Q$  is the overall rate of heat transfer between the gas and the solids (W),  $V_{dryer}$  is the dryer volume (m<sup>3</sup>), and  $\Delta T_m$  is an average temperature driving force (K). When one is calculating the average temperature driving force, it is important to distinguish between the case of heat-transfer with dry particles, where the change in the particle temperature is proportional to the change in the gas temperature, and the case of drying particles, where the particle temperature does not change so significantly. Where the particles are dry, the average temperature difference is the logarithmic mean of the temperature differences between the gas and the solids at the inlet and outlet of the dryer, although Miller et al. (1942) took the logarithmic temperature difference as the average temperature difference even when the particles were drying. The volumetric heat-transfer coefficient itself consists of a heat-transfer coefficient  $U_e$  based on the effective area of contact between the gas and the solids, and the ratio  $a$  of this area to



**TABLE 12-30 Values of the Index  $n$  in Correlations for the Volumetric Heat-Transfer Coefficient (after Baker, 1983)**

| Author(s)                    | Exponent $n$ |
|------------------------------|--------------|
| Saeman and Mitchell (1954)   | 0            |
| Friedman and Marshall (1949) | 0.16         |
| Aiken and Polsak (1982)      | 0.37         |
| Miller et al. (1942)         | 0.46–0.60    |
| McCormick (1962)             | 0.67         |
| Myklestad (1963)             | 0.80         |

the dryer volume. Thus, this procedure eliminates the need to specify where most of the heat-transfer occurs (e.g., to material in the air, on the flights, or in the rolling bed). Empirical correlations are of the form

$$U_s a = \frac{K' U_{\text{Gsuper}}^n}{D} \quad (12-101)$$

where  $K'$  depends on the solids properties, the flight geometry, the rotational speed, and the dryer holdup. Table 12-30 gives the values of  $n$  chosen by various authors.

McCormick (1962) reworked the data of Miller et al. (1942), Friedman and Marshall (1949), and Saeman and Mitchell (1954) with a view to obtaining a single correlation of the form of Eq. (12-101) for the volumetric heat-transfer coefficient. He demonstrated that all the data could be correlated with values of the exponent  $n$  from 0.46 to 0.67. Although the evidence was far from conclusive, he believed that a value of 0.67 for the exponent  $n$  was most reliable. Individual values of the constant  $K'$  were obtained from the results of each of the workers cited above. He found that it was a function of the solids properties, the flight geometry, the rotational speed, and the dryer holdup, but that there was insufficient evidence available to relate  $K'$  to these parameters.

A comparison between the correlations of various workers was made by Baker (1983), and this is given in Table 12-31. A 2-m-diameter dryer containing 16 flights was chosen as the basis for the comparisons. With the exception of the results of Myklestad (1963), the values of  $U_s a$  were calculated by using the values of  $K'$  and a value of  $n$  of 0.67, as obtained by McCormick (1963). A 17-fold variation in the predicted values of  $U_s a$  can be observed at both 1 and 3 m/s. The reason for this is not readily apparent. With the exception of the commercial data correlation of Miller et al. (1942), the results were all obtained in pilot-scale rigs having diameters ranging from 0.2 to 0.3 m. Differences in equipment size are therefore not likely to be the cause of the variation. Hence the variation must be attributed to a combination of experimental errors and differences in the experimental conditions which are unaccounted for in the correlations.

An alternative procedure is the use of a conventional film heat-transfer coefficient  $h_f$  [W/(m<sup>2</sup>·K)]

$$Q = h_f \cdot A_s \cdot \Delta T \quad (12-102)$$

Here  $Q$  is the local heat-transfer rate (W),  $A_s$  is the total surface area of all the particles (m<sup>2</sup>), and  $\Delta T$  is the temperature difference between the gas and the solids (K). The method has the advantages that  $h_f$  can

**TABLE 12-31 Summary of the Predictions Using the Correlations for the Volumetric Heat-Transfer Coefficients of Various Authors (after Baker, 1983)**

| Author(s)                    | $U_s a$ , W/(m <sup>2</sup> ·K) |                             |
|------------------------------|---------------------------------|-----------------------------|
|                              | $U_{\text{Gsuper}} = 1$ m/s     | $U_{\text{Gsuper}} = 3$ m/s |
| Miller et al. (1942)         |                                 |                             |
| Commercial data              | 248                             | 516                         |
| Pilot-scale data             | 82                              | 184                         |
| Friedman and Marshall (1949) | 67                              | 138                         |
| Saeman and Mitchell (1954)   | 495–1155                        | 1032–2410                   |
| Myklestad (1963)             | 423                             | 1019                        |

be determined by relatively simple tests (or calculated from appropriate correlations in the literature), variations in operating conditions can be allowed for, and analogies between heat and mass transfer allow the film coefficients for these processes to be related. However, the area for heat transfer must be estimated under the complex conditions of gas-solids interaction present in particle cascades. Schofield and Glikin (1962) estimated this area to be the surface area of particles per unit mass  $6/(\rho_p d_p)$ , multiplied by the fraction of solids in the drum that are cascading through the gas at any moment, which was estimated as the fraction of time spent by particles cascading through the gas:

$$A_s = \frac{6}{\rho_p d_p} \frac{t_f}{t_f + t_d} \quad (12-103)$$

Schofield and Glikin estimated the heat-transfer coefficient by using the correlation given by McAdams (1954), which correlates data for gas-to-particle heat transfer in air to about 20 percent over a range of Reynolds numbers ( $Re_p$ , defined in the previous section) between 17 and 70,000:

$$Nu_p = 0.33 \cdot Re_p^{1/2} \quad (12-104)$$

Here the particle Nusselt number is  $Nu_p$ , where  $Nu_p = h_f d_p / k_G$ , and  $k_G$  is the thermal conductivity of the gas [W/(m·K)]. They stated that the heat-transfer rates predicted by this procedure were much larger than those measured on an industrial cooler, which is probably due to the particles on the inside of the cascades not experiencing the full gas velocity. Kamke and Wilson (1986) used a similar approach to model the drying of wood chips, but used the Ranz-Marshall (1952) equation to predict the heat-transfer coefficient:

$$Nu_p = 2 + 0.6 \cdot Re_p^{1/2} \cdot Pr_G \quad (12-105)$$

where  $Pr_G$  is the Prandtl number of the gas.

**Drying Time Estimates** Sometimes, virtually all the drying takes place in the airborne phase. Under such circumstances, the airborne-phase residence time  $\tau_c$  and the drying time are virtually the same, and the required drying time can be estimated from equivalent times in drying kinetics experiments, e.g., using a thin-layer test (Langrish, D.Phil. thesis, 1988).

An example of how to incorporate the concept of the characteristic drying curve into a design calculation is given in Example 23.

**Example 23: Sizing of a Cascading Rotary Dryer** The average gas velocity passing through a cocurrent, adiabatic, cascading rotary dryer is 4 m/s. The particles moving through the dryer have an average diameter of 5 mm, a solids density of 600 kg/m<sup>3</sup>, and a shape factor of 0.75. The particles enter with a moisture content of 0.50 kg/kg (dry basis) and leave with a moisture content of 0.15 kg/kg (dry basis). The drying kinetics may be assumed to be linear, with no unhindered (constant-rate) drying period. In addition, let us assume that the solids are nonhygroscopic (so that the equilibrium moisture content is zero; hygroscopic means that the equilibrium moisture content is nonzero).

The inlet humidity is 0.10 kg/kg (dry basis) due to the use of a direct-fired burner, and the ratio of the flow rates of dry solids to dry gas is unity. The gas temperature at the inlet to the dryer is 800°C, and the gas may be assumed to behave as a pure water vapor/air mixture.

The gas-phase residence time that is required was calculated in the fundamentals section to be 38.0 s.

How does this gas-phase residence time relate to the total residence time that is required and to the dryer dimensions?

**Application of residence time calculations (practice):** Suppose that this dryer has a slope  $\alpha$  of 4° and a diameter  $D$  of 1.5 m, operating at a rotational speed  $N$  of 0.04 r/s. We already know that the gas velocity through the drum  $U_{\text{Gsuper}}$  is 4 m/s, and that the particles have a mean diameter  $d_p$  of 5 mm and a particle density  $\rho_p$  of 600 kg/m<sup>3</sup>. As a first estimate, suppose that the gas density  $\rho_G$  is 1 kg/m<sup>3</sup> and the gas viscosity  $\mu_G$  is  $1.8 \times 10^{-5}$  kg/(m·s).

Now  $K_R/(K_R \sqrt{2}) \approx 1$ ,  $K_R \approx 0.4$ ,  $K_{\text{fall}} \approx 1$ , and  $a$  is within the range of 1 to 4, say, 2.5, and  $U_{p1}^*$  is estimated by the following calculation, for Reynolds numbers up to 220.

$$U_{p1}^* = 7.45 \times 10^{-4} Re^{0.22} \frac{\mu U_{\text{Gsuper}} t^*}{\rho_p d_p^2} \quad (12-106)$$

Above this Reynolds number, the following equation was recommended by Matchett and Baker (1987):

$$U_{p1}^d = 125 \frac{\mu U_{Gsuper}^*}{\rho_p d_p^2} \quad (12-107)$$

Here Re is the Reynolds number ( $U_{Gsuper} d_p \rho / \mu$ ) and  $t_f$  is the average time of flight of a particle in the airborne phase.

$$t_f = \left( \frac{2D}{g \cos \alpha} \right)^{1/2} K_{fall} \quad (12-108)$$

Substituting in the numbers gives

$$\begin{aligned} t_f &= \left( \frac{2 \cdot 1.5 \text{ m}}{9.81 \text{ m/s}^2 \cdot \cos 4^\circ} \right)^{1/2} 1.0 = 0.554 \text{ s} \\ \text{Re} &= \frac{4 \text{ m/s} \cdot 0.005 \text{ m} \cdot 1 \text{ kg/m}^3}{1.8 \times 10^5 (\text{kg/m} \cdot \text{s})} = 1100 \\ U_{p1}^d &= 125 \frac{1.8 \times 10^5 \text{ kg/(m} \cdot \text{s)} \cdot 4 \text{ m/s} \cdot 0.554 \text{ s}}{(600 \text{ kg/m}^3)(0.005 \text{ m})^2} \\ &= 0.332 \text{ m/s} \\ \frac{\tau}{L} &= \frac{1.1}{0.04 \text{ s}^{-1} \cdot 1.5 \text{ m}} \\ &\times \left[ \frac{\tan 4^\circ \cdot (1 + 2.5) + \frac{1}{0.4 \sqrt{9.81 \text{ m/s}^2 \cdot 1.5 \text{ m}}}}{0.4 \sqrt{9.81 \text{ m/s}^2 \cdot 1.5 \text{ m}}} \cdot 0.332 \text{ m/s} \right] \\ &= 30 \text{ s/m} \\ \frac{\tau_s}{\tau_G} &= \frac{t_d}{t_f} = \frac{K_d}{N} \sqrt{\frac{g}{D}} = \frac{0.4}{0.04 \text{ s}^{-1}} \sqrt{\frac{9.81 \text{ m/s}^2}{1.5 \text{ m}}} = 25.6 \end{aligned}$$

Now, the required gas-phase residence time  $\tau_G$  is 38.0 s. The ratio of solids to gas-phase residence times now gives us the required solids-phase residence time  $\tau_s$  of  $25.6 \times 38.0 \text{ s} = 972 \text{ s}$ , and a total residence time of  $972 + 38 = 1010 \text{ s}$ . If the total residence time per unit length is 30 s/m, then the required dryer length is  $1010 \text{ s}/(30 \text{ s/m}) = 34.2 \text{ m}$ . The dryer length/diameter ratio is therefore  $34.2 \text{ m}/1.5 \text{ m} = 22.8$ , which is significantly larger than the recommended ratio of between 5:1 and 10:1. The remedy would then be to use a larger dryer diameter and repeat these calculations. The larger dryer diameter would decrease the gas velocity, slowing the particle velocity along the drum, increasing the residence time per unit length, and hence decreasing the required drum length, to give a more normal length/diameter ratio.

**Performance and Cost Data for Direct Heat Rotary Dryers** Table 12-32 gives estimating-price data for direct rotary dryers employing steam-heated air. Higher-temperature operations requiring combustion chambers and fuel burners will cost more. The total installed cost of rotary dryers including instrumentation, auxiliaries, allocated building space, etc., will run from 150 to 300 percent of the purchase cost. Simple erection costs average 10 to 20 percent of the purchase cost.

Operating costs will include 5 to 10 percent of one worker's time, plus power and fuel required. Yearly maintenance costs will range from 5 to 10 percent of total installed costs. Total power for fans, dryer drive, and feed and product conveyors will be in the range of  $0.5D^2$  to  $1.0D^2$ . Thermal efficiency of a high-temperature direct heat rotary dryer will range from 55 to 75 percent and, with steam-heated air, from 30 to 55 percent.

A representative list of materials dried in direct heat rotary dryers is given in Table 12-33.

**Indirect Heat Rotary Steam-Tube Dryers** Probably the most common type of indirect heat rotary dryer is the steam-tube dryer (Fig. 12-60). Steam-heated tubes running the full length of the cylinder are fastened symmetrically in one, two, or three concentric rows inside the cylinder and rotate with it. Tubes may be simple pipe with condensate draining by gravity into the discharge manifold or bayonet type. Bayonet-type tubes are also employed when units are used as water-tube coolers. When handling sticky materials, one row of tubes is preferred. These are occasionally shielded at the feed end of the dryer to prevent buildup of solids behind them. Lifting flights are usually inserted behind the tubes to promote solids agitation.

Wet feed enters the dryer through a chute or screw feeder. The product discharges through peripheral openings in the shell in ordinary dryers. These openings also serve to admit purge air to sweep moisture or other evolved gases from the shell. In practically all cases, gas flow is countercurrent to solids flow. To retain a deep bed of material within the dryer, normally 10 to 20 percent fillage, the discharge openings are supplied with removable chutes extending radially into the dryer. These, on removal, permit complete emptying of the dryer.

Steam is admitted to the tubes through a revolving steam joint into the steam side of the manifold. Condensate is removed continuously, by gravity through the steam joint to a condensate receiver and by means of lifters in the condensate side of the manifold. By employing simple tubes, noncondensables are continuously vented at the other ends of the tubes through Sarco-type vent valves mounted on an auxiliary manifold ring, also revolving with the cylinder.

Vapors (from drying) are removed at the feed end of the dryer to the atmosphere through a natural-draft stack and settling chamber or wet scrubber. When employed in simple drying operations with  $3.5 \times 10^5$  to

**TABLE 12-32 Warm-Air Direct-Heat Cocurrent Rotary Dryers: Typical Performance Data\***

| Dryer size, m × m                      | 1.219 × 7.62 | 1.372 × 7.621 | 1.524 × 9.144 | 1.839 × 10.668 | 2.134 × 12.192 | 2.438 × 13.716 | 3.048 × 16.767 |
|----------------------------------------|--------------|---------------|---------------|----------------|----------------|----------------|----------------|
| Evaporation, kg/h                      | 136.1        | 181.4         | 226.8         | 317.5          | 408.2          | 544.3          | 861.8          |
| Work, 10 <sup>6</sup> J/h              | 3.61         | 4.60          | 5.70          | 8.23           | 1.12           | 1.46           | 2.28           |
| Steam, kg/h at kg/m <sup>2</sup> gauge | 317.5        | 408.2         | 521.6         | 725.7          | 997.9          | 131.5          | 2041           |
| Discharge, kg/h                        | 408          | 522           | 685           | 953            | 1270           | 1633           | 2586           |
| Exhaust velocity, m/min                | 70           | 70            | 70            | 70             | 70             | 70             | 70             |
| Exhaust volume, m <sup>3</sup> /min    | 63.7         | 80.7          | 100.5         | 144.4          | 196.8          | 257.7          | 399.3          |
| Exhaust fan, kW                        | 3.7          | 3.7           | 5.6           | 7.5            | 11.2           | 18.6           | 22.4           |
| Dryer drive, kW                        | 2.2          | 5.6           | 5.6           | 7.5            | 14.9           | 18.6           | 37.3           |
| Shipping weight, kg                    | 7700         | 10,900        | 14,500        | 19,100         | 35,800         | 39,900         | 59,900         |
| Price, FOB Chicago                     | \$158,000    | \$168,466     | \$173,066     | \$204,400      | \$241,066      | \$298,933      | \$393,333      |

\*Courtesy of Swenson Process Equipment Inc.

NOTE:

Material: heat-sensitive solid

Maximum solids temperature: 65°C

Feed conditions: 25 percent moisture, 27°C

Product conditions: 0.5 percent moisture, 65°C

Inlet-air temperature: 165°C

Exit-air temperature: 71°C

Assumed pressure drop in system: 200 mm

System includes finned air heaters, transition piece, dryer, drive, product collector, duct, and fan.

Prices are for carbon steel construction and include entire dryer system (November, 1994).

For 304 stainless-steel fabrication, multiply the prices given by 1.5.

**TABLE 12-33 Representative Materials Dried in Direct-Heat Rotary Dryers\***

| Material dried          | Moisture content, %<br>(wet basis) |           | Heat efficiency, % |
|-------------------------|------------------------------------|-----------|--------------------|
|                         | Initial                            | Final     |                    |
| High-temperature:       |                                    |           |                    |
| Sand                    | 10                                 | 0.5       | 61                 |
| Stone                   | 6                                  | 0.5       | 65                 |
| Fluorspar               | 6                                  | 0.5       | 59                 |
| Sodium chloride         | 3                                  | 0.04      | 70-80              |
| (vacuum salt)           |                                    |           |                    |
| Sodium sulfate          | 6                                  | 0.1       | 60                 |
| Ilmenite ore            | 6                                  | 0.2       | 60-65              |
| Medium-temperature:     |                                    |           |                    |
| Copperas                | 7                                  | 1 (moles) | 55                 |
| Ammonium sulfate        | 3                                  | 0.10      | 50-60              |
| Cellulose acetate       | 60                                 | 0.5       | 51                 |
| Sodium chloride         | 25                                 | 0.06      | 35                 |
| (grainer salt)          |                                    |           |                    |
| Cast-iron borings       | 6                                  | 0.5       | 50-60              |
| Styrene                 | 5                                  | 0.1       | 45                 |
| Low-temperature:        |                                    |           |                    |
| Oxalic acid             | 5                                  | 0.2       | 29                 |
| Vinyl resins            | 30                                 | 1         | 50-55              |
| Ammonium nitrate prills | 4                                  | 0.25      | 30-35              |
| Urea prills             | 2                                  | 0.2       | 20-30              |
| Urea crystals           | 3                                  | 0.1       | 50-55              |

\*Taken from *Chem. Eng.*, June 19, 1967, p. 190, Table III.

$10 \times 10^5$  Pa steam, draft is controlled by a damper to admit only sufficient outside air to sweep moisture from the cylinder, discharging the air at 340 to 365 K and 80 to 90 percent saturation. In this way, shell gas velocities and dusting are minimized. When used for solvent recovery or other processes requiring a sealed system, sweep gas is recirculated through a scrubber-gas cooler and blower.

Steam manifolds for pressures up to  $10^6$  Pa are of cast iron. For higher pressures, the manifold is fabricated from plate steel, stay-bolted, and welded. The tubes are fastened rigidly to the manifold faceplate and are supported in a close-fitting annular plate at the other end to permit expansion. Packing on the steam neck is normally graphite asbestos. Ordinary rotating seals are similar in design with

allowance for the admission of small quantities of outside air when the dryer is operated under a slight negative internal pressure.

Steam-tube dryers are used for the continuous drying, heating, or cooling of granular or powdery solids which cannot be exposed to ordinary atmospheric or combustion gases. They are especially suitable for fine dusty particles because of the low gas velocities required for purging of the cylinder. Tube sticking is avoided or reduced by employing recycle, shell knockers, etc., as previously described; tube scaling by sticky solids is one of the major hazards to efficient operation. The dryers are suitable for drying, solvent recovery, and chemical reactions. Steam-tube units have found effective employment in soda ash production, replacing more expensive indirect-heat rotary calciners.

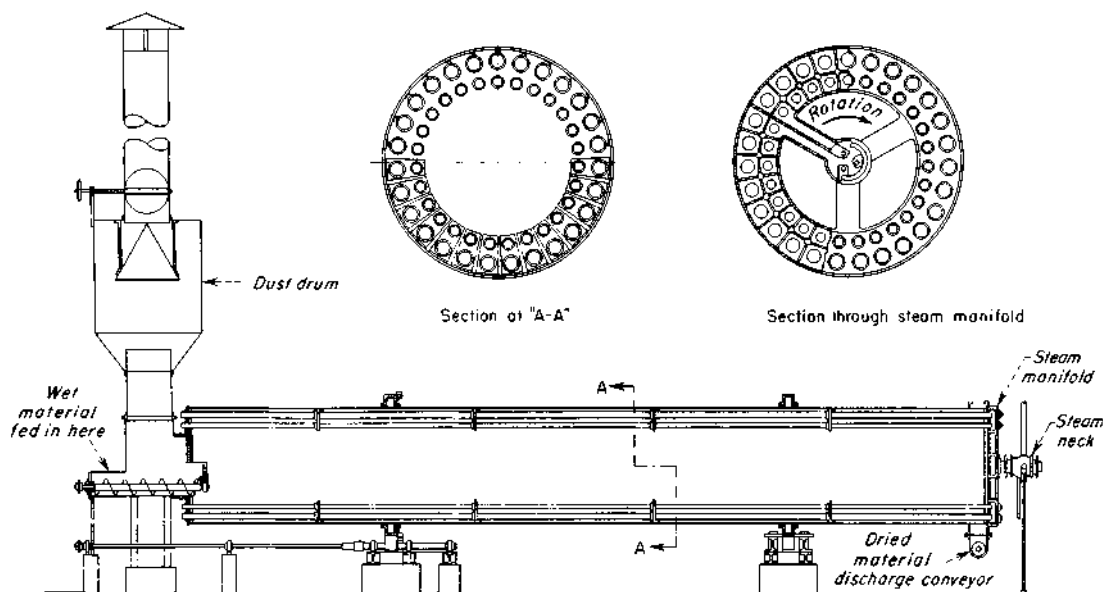
Special types of steam-tube dryers employ packed and purged seals on all rotating joints, with a central solids-discharge manifold through the steam neck to reduce the seal diameter. This manifold contains the product discharge conveyor and a passage for the admission of sweep gas. Solids are removed from the shell by special volute lifters and dropped into the discharge conveyor. Units have been fabricated for operation at 76 mm of water, internal shell pressure, with no detectable air leakage.

*Design methods for indirect heat rotary steam-tube dryers* Heat-transfer coefficients in steam-tube dryers range from 30 to 85  $W/(m^2 \cdot K)$ . Coefficients will increase with increasing steam temperature because of increased heat transfer by radiation. In units carrying saturated steam at 420 to 450 K, the heat flux  $UT$  will range from 6300  $W/m^2$  for difficult-to-dry and organic solids to 1890 to 3790  $W/m^2$  for finely divided inorganic materials. The effect of steam pressure on heat-transfer rates up to  $8.6 \times 10^5$  Pa is illustrated in Fig. 12-61.

*Performance and cost data for indirect heat rotary steam-tube dryers* Table 12-34 contains data for a number of standard sizes of steam-tube dryers. Prices tabulated are for ordinary carbon steel construction. Installed costs will run from 150 to 300 percent of purchase cost.

The thermal efficiency of steam-tube units will range from 70 to 90 percent, if a well-insulated cylinder is assumed. This does not allow for boiler efficiency, however, and is therefore not directly comparable with direct heat units such as the direct heat rotary dryer or indirect heat calciner.

Operating costs for these dryers include 5 to 10 percent of one person's time. Maintenance will average 5 to 10 percent of total installed cost per year.

**FIG. 12-60** Steam-tube rotary dryer.

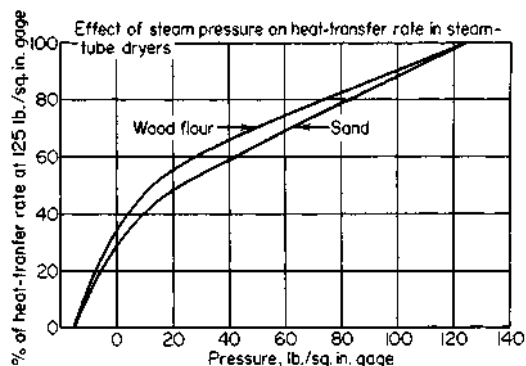


FIG. 12-61 Effect of steam pressure on the heat-transfer rate in steam-tube dryers.

Table 12-35 outlines typical performance data from three drying applications in steam-tube dryers.

**Indirect Rotary Calciners and Kilns** These large-scale rotary processors are used for very high temperature operations. Operation is similar to that of rotary dryers. For additional information, refer to Perry's 7th Edition, pages 12-56 to 12-58.

**Indirect Heat Calciners** Indirect heat rotary calciners, either batch or continuous, are employed for heat treating and drying at higher temperatures than can be obtained in steam-heated rotating equipment. They generally require a minimum flow of gas to purge the cylinder, to reduce dusting, and are suitable for gas-sealed operation with oxidizing, inert, or reducing atmospheres. Indirect calciners are widely utilized, and some examples of specific applications are as follows:

1. Activating charcoal
2. Reducing mineral high oxides to low oxides
3. Drying and devolatilizing contaminated soils and sludges
4. Calcination of alumina oxide-based catalysts
5. Drying and removal of sulfur from cobalt, copper, and nickel
6. Reduction of metal oxides in a hydrogen atmosphere
7. Oxidizing and "burning off" of organic impurities
8. Calcination of ferrites

This unit consists essentially of a cylindrical retort, rotating within a stationary insulation-lined furnace. The latter is arranged so that fuel combustion occurs within the annular ring between the retort and the furnace. The retort cylinder extends beyond both ends of the furnace. These end extensions carry the riding rings and drive gear. Material may be fed continuously at one end and discharged continuously at the other. Feeding and solids discharging are usually accomplished with screw feeders or other positive feeders to prevent leakage of gases into or out of the calciner.

In some cases in which it is desirable to cool the product before removal to the outside atmosphere, the discharge end of the cylinder is provided with an additional extension, the exterior of which is water-spray-cooled. In cocurrent flow calciners, hot gases from the interior of the heated portion of the cylinder are withdrawn through a special extraction tube. This tube extends centrally through the cooled section to prevent flow of gas near the cooled-shell surfaces and possible condensation. Frequently a separate cooler is used, isolated from the calciner by an air lock.

To prevent sliding of solids over the smooth interior of the shell, agitating flights running longitudinally along the inside wall are frequently provided. These normally do not shower the solids as in a direct heat vessel but merely prevent sliding so that the bed will turn over and constantly expose new surface for heat and mass transfer. To prevent scaling of the shell interior by sticky solids, cylinder scraper and knocker arrangements are occasionally employed. For example, a scraper chain is fairly common practice in soda ash calciners, while knockers are frequently utilized on metallic-oxide calciners.

Because indirect heat calciners frequently require close-fitting gas seals, it is customary to support all parts on a self-contained frame, for sizes up to approximately 2 m in diameter. The furnace can employ electric heating elements or oil and/or gas burners as the heat source for the process. The hardware would be zoned down the length of the furnace to match the heat requirements of the process. Process control is normally by shell temperature, measured by thermocouples or radiation pyrometers. When a special gas atmosphere must be maintained inside the cylinder, positive rotary gas seals, with one or more pressurized and purged annular chambers, are employed. The diaphragm-type seal is suitable for pressures up to 5 cm of water, with no detectable leakage.

In general, the temperature range of operation for indirect heat calciners can vary over a wide range, from 475 K at the low end to approximately 1475 K at the high end. All types of carbon steel, stainless, and

TABLE 12-34 Standard Steam-Tube Dryers\*

| Size, diameter × length, m | Tubes       |             | m <sup>2</sup> of free area | Dryer speed, r/min | Motor size, hp | Shipping weight, kg | Estimated price |
|----------------------------|-------------|-------------|-----------------------------|--------------------|----------------|---------------------|-----------------|
|                            | No. OD (mm) | No. OD (mm) |                             |                    |                |                     |                 |
| 0.965 × 4.572              | 14 (114)    |             | 21.4                        | 6                  | 2.2            | 5,500               | \$152,400       |
| 0.965 × 6.096              | 14 (114)    |             | 29.3                        | 6                  | 2.2            | 5,900               | 165,100         |
| 0.965 × 7.620              | 14 (114)    |             | 36.7                        | 6                  | 3.7            | 6,500               | 175,260         |
| 0.965 × 9.144              | 14 (114)    |             | 44.6                        | 6                  | 3.7            | 6,900               | 184,150         |
| 0.965 × 10.668             | 14 (114)    |             | 52.0                        | 6                  | 3.7            | 7,500               | 196,850         |
| 1.372 × 6.096              | 18 (114)    | 18 (63.5)   | 58.1                        | 4.4                | 3.7            | 10,200              | 203,200         |
| 1.372 × 7.620              | 18 (114)    | 18 (63.5)   | 73.4                        | 4.4                | 3.7            | 11,100              | 215,900         |
| 1.372 × 9.144              | 18 (114)    | 18 (63.5)   | 88.7                        | 5                  | 5.6            | 12,100              | 228,600         |
| 1.372 × 10.668             | 18 (114)    | 18 (63.5)   | 104                         | 5                  | 5.6            | 13,100              | 243,840         |
| 1.372 × 12.192             | 18 (114)    | 18 (63.5)   | 119                         | 5                  | 5.6            | 14,200              | 260,350         |
| 1.372 × 13.716             | 18 (114)    | 18 (63.5)   | 135                         | 5.5                | 7.5            | 15,000              | 273,050         |
| 1.829 × 7.62               | 27 (114)    | 27 (76.2)   | 118                         | 4                  | 5.6            | 19,300              | 241,300         |
| 1.829 × 9.144              | 27 (114)    | 27 (76.2)   | 143                         | 4                  | 5.6            | 20,600              | 254,000         |
| 1.829 × 10.668             | 27 (114)    | 27 (76.2)   | 167                         | 4                  | 7.5            | 22,100              | 266,700         |
| 1.829 × 12.192             | 27 (114)    | 27 (76.2)   | 192                         | 4                  | 7.5            | 23,800              | 278,400         |
| 1.829 × 13.716             | 27 (114)    | 27 (76.2)   | 217                         | 4                  | 11.2           | 25,700              | 292,100         |
| 1.829 × 15.240             | 27 (114)    | 27 (76.2)   | 242                         | 4                  | 11.2           | 27,500              | 304,800         |
| 1.829 × 16.764             | 27 (114)    | 27 (76.2)   | 266                         | 4                  | 14.9           | 29,300              | 317,500         |
| 1.829 × 18.288             | 27 (114)    | 27 (76.2)   | 291                         | 4                  | 14.9           | 30,700              | 330,200         |
| 2.438 × 12.192             | 90 (114)    |             | 394                         | 3                  | 11.2           | 49,900              | 546,100         |
| 2.438 × 15.240             | 90 (114)    |             | 492                         | 3                  | 14.9           | 56,300              | 647,700         |
| 2.438 × 18.288             | 90 (114)    |             | 590                         | 3                  | 14.9           | 63,500              | 736,600         |
| 2.438 × 21.336             | 90 (114)    |             | 689                         | 3                  | 22.4           | 69,900              | 838,200         |
| 2.438 × 24.387             | 90 (114)    |             | 786                         | 3                  | 29.8           | 75,300              | 927,100         |

\*Courtesy of Swenson Process Equipment Inc. (prices from November, 1994). Carbon steel fabrication; multiply by 1.75 for 304 stainless steel.



TABLE 12-35 Steam-Tube Dryer Performance Data

|                                                         | Class 1                                                                          | Class 2                                                                | Class 3                                                                                         |
|---------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Class of materials handled                              | High-moisture organic, distillers' grains, brewers' grains, citrus pulp          | Pigment filter cakes, blanc fixe, barium carbonate, precipitated chalk | Finely divided inorganic solids, water-ground mica, water-ground silica, flotation concentrates |
| Description of class                                    | Wet feed is granular and damp but not sticky or muddy and dries to granular meal | Wet feed is pasty, muddy, or sloppy; product is mostly hard pellets    | Wet feed is crumbly and friable; product is powder with very few lumps                          |
| Normal moisture content of wet feed, % dry basis        | 233                                                                              | 100                                                                    | 54                                                                                              |
| Normal moisture content of product, % dry basis         | 11                                                                               | 0.15                                                                   | 0.5                                                                                             |
| Normal temperature of wet feed, K                       | 310–320                                                                          | 280–290                                                                | 280–290                                                                                         |
| Normal temperature of product, K                        | 350–355                                                                          | 380–410                                                                | 365–375                                                                                         |
| Evaporation per product, kg                             | 2                                                                                | 1                                                                      | 0.53                                                                                            |
| Heat load per lb product, kJ                            | 2250                                                                             | 1190                                                                   | 625                                                                                             |
| Steam pressure normally used, kPa gauge                 | 860                                                                              | 860                                                                    | 860                                                                                             |
| Heating surface required per kg product, m <sup>2</sup> | 0.34                                                                             | 0.4                                                                    | 0.072                                                                                           |
| Steam consumption per kg product, kg                    | 3.33                                                                             | 1.72                                                                   | 0.85                                                                                            |

alloy construction are used, depending upon temperature, process, and corrosion requirements. Fabricated-alloy cylinders can be used over the greater part of the temperature range; however, the greater creep-stress abilities of cast alloys makes their use desirable for the highest calciner cylinder temperature applications.

**Design methods for calciners** In indirect heat calciners, heat transfer is primarily by radiation from the cylinder wall to the solids bed. The thermal efficiency ranges from 30 to 65 percent. By utilization of the furnace exhaust gases for preheated combustion air, steam production, or heat for other process steps, the thermal efficiency can be increased considerably. The limiting factors in heat transmission lie in the conductivity and radiation constants of the shell metal and solids bed. If the characteristics of these are known, equipment may be accurately sized by employing the Stefan-Boltzmann radiation equation. Apparent heat-transfer coefficients will range from 17 W/(m<sup>2</sup>·K) in low-temperature operations to 85 W/(m<sup>2</sup>·K) in high-temperature processes.

**Cost data for calciners** Power, operating, and maintenance costs are similar to those previously outlined for direct and indirect heat rotary dryers. Estimating purchase costs for preassembled and frame-mounted rotary calciners with carbon steel and type 316 stainless-steel cylinders are given in Table 12-36 together with size, weight, and motor requirements. Sale price includes the cylinder, ordinary angle seals, furnace, drive, feed conveyor, burners, and controls. Installed cost may be estimated, not including building or foundation costs, at up to 50 percent of the purchase cost. A layout of a typical continuous calciner with an extended cooler section is illustrated in Fig. 12-62.

Small batch retorts, heated electrically or by combustion, are widely used as carburizing furnaces and are applicable also to chemical processes involving the heat treating of particulate solids. These are mounted on a structural-steel base, complete with cylinder, furnace, drive motor, burner, etc. Units are commercially available in diame-

ters from 0.24 to 1.25 m and lengths of 1 to 2 m. Continuous retorts with helical internal spirals are employed for metal heat-treating purposes. Precise retention control is maintained in these operations. Standard diameters are 0.33, 0.5, and 0.67 m with effective lengths up to 3 m. These vessels are employed in many small-scale chemical process operations which require accurate control of retention. Their operating characteristics and applications are identical to those of the larger indirect heat calciners.

**Direct Heat Roto-Louvre Dryer** One of the more important special types of rotating equipment is the Roto-Louvre dryer. As illustrated in Fig. 12-63, hot air (or cooling air) is blown through louvres in a double-wall rotating cylinder and up through the bed of solids. The latter moves continuously through the cylinder as it rotates. Constant turnover of the bed ensures uniform gas contacting for heat and mass transfer. The annular gas passage behind the louvres is partitioned so that contacting air enters the cylinder only beneath the solids bed. The number of louvres covered at any one time is roughly 30 percent. Because air circulates through the bed, fillages of 13 to 15 percent or greater are employed.

Roto-Louvre dryers range in size from 0.8 to 3.6 m in diameter and from 2.5 to 11 m long. The largest unit is reported capable of evaporating 5500 kg/h of water. Hot gases from 400 to 865 K may be employed. Because gas flow is through the bed of solids, high pressure drop, from 7 to 50 cm of water, may be encountered within the shell. For this reason, both a pressure inlet fan and an exhaust fan are provided in most applications to maintain the static pressure within the equipment as closely as possible to atmospheric. This prevents excessive in-leakage or blowing of hot gas and dust to the outside. For pressure control, one fan is usually operated under fixed conditions, with an automatic damper control on the other, regulated by a pressure detector-controller.

In heating or drying applications, when cooling of the product is desired before discharge to the atmosphere, cool air is blown through

TABLE 12-36 Indirect-Heat Rotary Calciners: Sizes and Purchase Costs\*

| Diameter, ft | Overall cylinder length | Heated cylinder length | Cylinder drive motor hp | Approximate Shipping weight, lb | Approximate sale price in carbon steel construction† | Approximate sale price in No. 316 stainless construction |
|--------------|-------------------------|------------------------|-------------------------|---------------------------------|------------------------------------------------------|----------------------------------------------------------|
| 4            | 40 ft                   | 30 ft                  | 7.5                     | 50,000                          | \$275,000                                            | \$325,000                                                |
| 5            | 45 ft                   | 35 ft                  | 10                      | 60,000                          | 375,000                                              | 425,000                                                  |
| 6            | 50 ft                   | 40 ft                  | 20                      | 75,000                          | 475,000                                              | 550,000                                                  |
| 7            | 60 ft                   | 50 ft                  | 30                      | 90,000                          | 550,000                                              | 675,000                                                  |

\* ABB Raymond (Bartlett-Snow™).

† Prices for November, 1994.

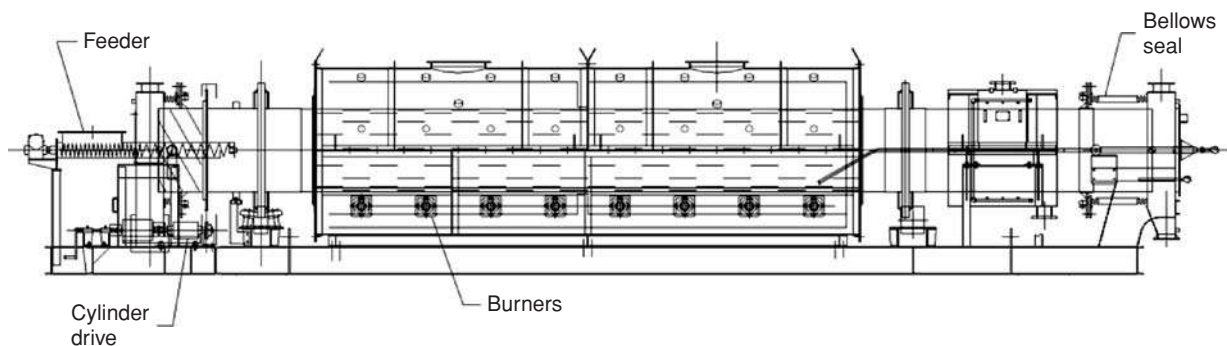


FIG. 12-62 Gas-fired rotary calciner with integral cooler. (Air Preheater Company, Raymond & Bartlett Snow™ Products.)

a second annular space, outside the inlet hot-air annulus, and released through the louvers at the solids discharge end of the shell.

Roto-Louvre dryers are suitable for processing coarse granular solids which do not offer high resistance to airflow, do not require intimate gas contacting, and do not contain significant quantities of dust.

Heat transfer and mass transfer from the gas to the surface of the solids are extremely efficient; hence the equipment size required for a given duty is frequently less than that required when an ordinary direct heat rotary vessel with lifting flights is used. Purchase price savings are partially balanced, however, by the more complex construction of the Roto-Louvre unit. A Roto-Louvre dryer will have a capacity roughly 1.5 times that of a single-shell rotary dryer of the same size under equivalent operating conditions. Because of the cross-flow method of heat exchange, the average  $t$  is not a simple function of inlet and outlet  $t$ 's. There are currently no published data which permit the sizing of equipment without pilot tests as recommended by the manufacturer. Three applications of Roto-Louvre dryers are outlined in Table 12-37. Installation, operating, power, and maintenance costs will be similar to those experienced with ordinary direct heat rotary dryers. Thermal efficiency will range from 30 to 70 percent.

#### Additional Reading

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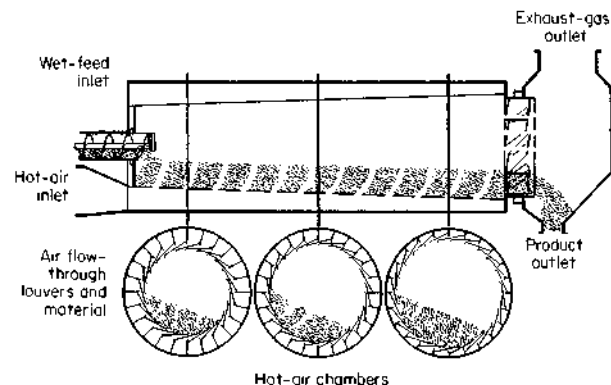


FIG. 12-63 FMC Link-Belt Roto-Louvre Dryer.

TABLE 12-37 Manufacturer's Performance Data for FMC Link-Belt Roto-Louvre Dryers\*

| Material dried                               | Ammonium sulfate | Foundry sand             | Metallurgical coke |
|----------------------------------------------|------------------|--------------------------|--------------------|
| Dryer diameter                               | 2 ft 7 in        | 6 ft 4 in                | 10 ft 3 in         |
| Dryer length                                 | 10 ft            | 24 ft                    | 30 ft              |
| Moisture in feed, % wet basis                | 2.0              | 6.0                      | 18.0               |
| Moisture in product, % wet basis             | 0.1              | 0.5                      | 0.5                |
| Production rate, lb/h                        | 2500             | 32,000                   | 38,000             |
| Evaporation rate, lb/h                       | 50               | 2130                     | 8110               |
| Type of fuel                                 | Steam            | Gas                      | Oil                |
| Fuel consumption                             | 255 lb/h         | 4630 ft <sup>3</sup> /h  | 115 gal/h          |
| Calorific value of fuel                      | 837 Btu/lb       | 1000 Btu/ft <sup>3</sup> | 150,000 Btu/gal    |
| Efficiency, Btu, supplied per lb evaporation | 4370             | 2170                     | 2135               |
| Total power required, hp                     | 4                | 41                       | 78                 |

\*Material Handling Systems Division, FMC Corp. To convert British thermal units to kilojoules, multiply by 1.06; to convert horsepower to kilowatts, multiply by 0.746.

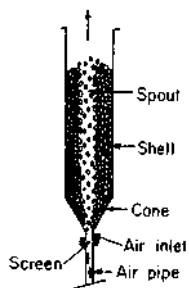


FIG. 12-64 Schematic diagram of spouted bed. [Mathur and Gishler, *Am. Inst. Chem. Eng. J.*, **1**, 2, 15 (1955).]

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### Fluidized and Spouted Bed Dryers

**Spouted Beds** The spouted bed technique was developed primarily for solids which are too coarse to be handled in fluidized beds.

Although their applications overlap, the methods of gas-solids mixing are completely different. A schematic view of a spouted bed is given in Fig. 12-64. Mixing and gas-solids contacting are achieved first in a fluid "spout," flowing upward through the center of a loosely packed bed of solids. Particles are entrained by the fluid and conveyed to the top of the bed. They then flow downward in the surrounding annulus as in an ordinary gravity bed, countercurrently to gas flow. The mechanisms of gas flow and solids flow in spouted beds were first described by Mathur and Gishler [*Am. Inst. Chem. Eng. J.* **1**(2):157-164 (1955)]. Drying studies have been carried out by Cowan [*Eng. J.* **41**:5, 60-64 (1958)], and a theoretical equation for predicting the minimum fluid velocity necessary to initiate spouting was developed by Madonna and Lama [*Am. Inst. Chem. Eng. J.* **4**(4):497 (1958)]. Investigations to determine maximum spoutable depths and to develop theoretical relationships based on vessel geometry and operating variables have been carried out by Lefroy [*Trans. Inst. Chem. Eng.* **47**(5):T120-128 (1969)] and Reddy [*Can. J. Chem. Eng.* **46**(5):329-334 (1968)].

Gas flow in a spouted bed is partially through the spout and partially through the annulus. About 30 percent of the gas entering the system immediately diffuses into the downward-flowing annulus. Near the top of the bed, the quantity in the annulus approaches 66 percent of the total gas flow; the gas flow through the annulus at any point in the bed equals that which would flow through a loosely packed solids bed under the same conditions of pressure drop. Solids flow in the annulus is both downward and slightly inward. As the fluid spout rises in the bed, it entrains more and more particles, losing velocity and gas into the annulus. The volume of solids displaced by the spout is roughly 6 percent of the total bed.

On the basis of experimental studies, Mathur and Gishler derived an empirical correlation to describe the minimum fluid flow necessary for spouting, in 3- to 12-in.-diameter columns:

$$u = \frac{D_p}{D_c} \left( \frac{D_o}{D_c} \right)^{0.33} \left[ \frac{2gL(\rho_s - \rho_f)}{\rho_f} \right]^{0.5} \quad (12-109)$$

where  $u$  = superficial fluid velocity through the bed, ft/s;  $D_p$  = particle diameter, ft;  $D_c$  = column (or bed) diameter, ft;  $D_o$  = fluid inlet orifice diameter, ft;  $L$  = bed height, ft;  $\rho_s$  = absolute solids density, lb/ft<sup>3</sup>;  $\rho_f$  = fluid density, lb/ft<sup>3</sup>; and  $g = 32.2$  ft/s<sup>2</sup>, gravity acceleration.

To convert feet per second to meters per second, multiply by 0.305; to convert pounds per cubic foot to kilograms per cubic meter, multiply by 16. In SI units,  $g = 9.8$  m/s<sup>2</sup>. The inlet orifice diameter, air rate, bed diameter, and bed depth were all found to be critical and interdependent:

1. In a given-diameter bed, deeper beds can be spouted as the gas inlet orifice size is decreased. Using air, a 12-in.-diameter bed containing 0.125- by 0.250-in. wheat can be spouted at a depth of over 100 in with a 0.8-in. orifice, but at only 20 in with a 2.4-in. orifice.

2. Increasing bed diameter increases spoutable depth. By employing a bed/orifice diameter ratio of 12 for air spouting, a 9-in.-diameter bed was spouted at a depth of 65 in while a 12-in.-diameter bed was spouted at 95 in.

3. As indicated by Eq. (12-109), the superficial fluid velocity required for spouting increases with bed depth and orifice diameter and decreases as the bed diameter is increased.

Employing wood chips, Cowan's drying studies indicated that the volumetric heat-transfer coefficient obtainable in a spouted bed is at least twice that in a direct heat rotary dryer. By using 20- to 30-mesh Ottawa sand, fluidized and spouted beds were compared. The volumetric coefficients in the fluid bed were 4 times those obtained in a spouted bed. Mathur dried wheat continuously in a 12-in.-diameter spouted bed, followed by a 9-in.-diameter spouted bed cooler. A drying rate of roughly 100 lb/h of water was obtained by using 450 K inlet air. Six hundred pounds per hour of wheat was reduced from 16 to 26 percent to 4 percent moisture. Evaporation occurred also in the cooler by using sensible heat present in the wheat. The maximum drying bed temperature was 118°F, and the overall thermal efficiency of the system was roughly 65 percent. Some aspects of the spouted bed technique are covered by patent (U.S. Patent 2,786,280).

Cowan reported that significant size reduction of solids occurred when cellulose acetate was dried in a spouted bed, indicating its possible limitations for handling other friable particles.

**Direct Heat Vibrating Conveyor Dryers** Information on vibrating conveyors and their mechanical construction is given in Sec. 19, "Solid-Solid Operations and Equipment." The vibrating conveyor dryer is a modified form of fluidized-bed equipment, in which fluidization is maintained by a combination of pneumatic and mechanical forces. The heating gas is introduced into a plenum beneath the conveying deck through ducts and flexible hose connections and passes up through a screen, perforated, or slotted conveying deck, through the fluidized bed of solids, and into an exhaust hood (Fig. 12-65). If ambient air is employed for cooling, the sides of the plenum may be open and a simple exhaust system used; however, because the gas distribution plate may be designed for several inches of water pressure drop to ensure a uniform velocity distribution through the bed of solids, a combination pressure-blower exhaust-fan system is desirable to balance the pressure above the

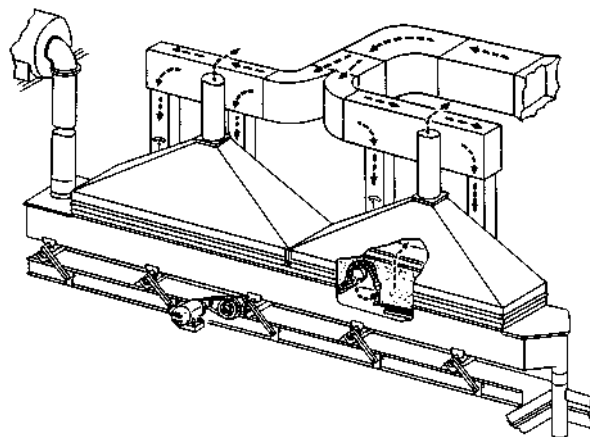


FIG. 12-65 Vibrating conveyor dryer. (Carrier Vibrating Equipment, Inc.)



**TABLE 12-38 Table for Estimating Maximum Superficial Air Velocities through Vibrating-Conveyor Screens\***

| Mesh size | Velocity, m/s        |                      |
|-----------|----------------------|----------------------|
|           | 2.0 specific gravity | 1.0 specific gravity |
| 200       | 0.22                 | 0.13                 |
| 100       | 0.69                 | 0.38                 |
| 50        | 1.4                  | 0.89                 |
| 30        | 2.6                  | 1.8                  |
| 20        | 3.2                  | 2.5                  |
| 10        | 6.9                  | 4.6                  |
| 5         | 11.4                 | 7.9                  |

\*Carrier Vibrating Equipment, Inc.

deck with the outside atmosphere and prevent gas in-leakage or blowing at the solids feed and exit points.

Units are fabricated in widths from 0.3 to 1.5 m. Lengths are variable from 3 to 50 m; however, most commercial units will not exceed a length of 10 to 16 m per section. Power required for the vibrating drive will be approximately 0.4 kW/m<sup>2</sup> of deck.

In general, this equipment offers an economical heat-transfer area for first cost as well as operating cost. Capacity is limited primarily by the air velocity which can be used without excessive dust entrainment. Table 12-38 shows limiting air velocities suitable for various solids particles. Usually, the equipment is satisfactory for particles larger than 100 mesh in size. [The use of indirect heat conveyors eliminates the problem of dust entrainment, but capacity is limited by the heat-transfer coefficients obtainable on the deck (see Sec. 11)].

When a stationary vessel is employed for fluidization, all solids being treated must be fluidized; nonfluidizable fractions fall to the bottom of the bed and may eventually block the gas distributor. The addition of mechanical vibration to a fluidized system offers the following advantages:

1. Equipment can handle nonfluidizable solids fractions. Although these fractions may drop through the bed to the screen, directional-throw vibration will cause them to be conveyed to the discharge end of the conveyor. Prescreening or sizing of the feed is less critical than in a stationary fluidized bed.

2. Because of mechanical vibration, incipient channeling is reduced.

3. Fluidization may be accomplished with lower pressures and gas velocities. This has been evidenced on vibratory units by the fact that fluidization stops when the vibrating drive is stopped. Vibrating conveyor dryers are suitable for free-flowing solids containing mainly surface moisture. Retention is limited by conveying speeds which range from 0.02 to 0.12 m/s. Bed depth rarely exceeds 7 cm, although units are fabricated to carry 30- to 46-cm-deep beds; these also employ plate and pipe coils suspended in the bed to provide additional heat-transfer area. Vibrating dryers are not suitable for fibrous materials which mat or for sticky solids which may ball or adhere to the deck.

For estimating purposes for direct heat drying applications, it can be assumed that the average exit gas temperature leaving the solids bed will approach the final solids discharge temperature on an ordinary unit carrying a 5- to 15-cm-deep bed. Calculation of the heat load and selection of an inlet air temperature and superficial velocity (Table 12-38) will then permit approximate sizing, provided an approximation of the minimum required retention time can be made.

Vibrating conveyors employing direct contacting of solids with hot, humid air have also been used for the agglomeration of fine powders, chiefly for the preparation of agglomerated water-dispersible food products. Control of inlet air temperature and dew point permits the uniform addition of small quantities of liquids to solids by condensation on the cool incoming-particle surfaces. The wetting section of the conveyor is followed immediately by a warm-air-drying section and particle screening.

**Fluidized-Bed Dryers** The basic principles of fluid-bed technology are thoroughly described in Sec. 17, "Gas-Solid Operations and Equipment." Originally conceived as a heterogeneous chemical reactor, the use of this technology in connection with drying processes has increased considerably during the last several decades. The technol-

ogy offers the following advantages when compared with other drying methods:

- It has no moving parts.
- It provides rapid heat and mass exchange between gas and particles.
- It provides high heat-transfer rates between the gas/particle bed and immersed objects such as heating panels.
- It provides intensive mixing of solids, leading to homogeneous conditions and reliable control of the drying process.

The fluid-bed technology can be applied to continuous as well as batch processes.

As described in Sec. 17, the process parameter of the highest importance is the gas velocity in the fluidized bed, referred to as the fluidizing velocity or the superficial gas velocity. This velocity is of nominal character since the flow field will be disturbed and distorted by the presence of the solid phase and the turbulent fluctuations created by the gas/solid interaction.

The fluid bed consists of a layer of particles suspended partly by a bed plate with perforations or a grid and partly by the fluidizing gas flowing through the bed plate. If the gas velocity is low, the gas will merely percolate through a bed of particles that appear to be fixed. At a higher velocity the particles will start to move under influence of the aerodynamic forces, and at the point where the pressure drop reaches the equivalent of the weight per unit of area, all the particles will tend to be moving in suspension, also called the incipiently fluidized state. The particle layer behaves as a liquid, and the bed volume expands considerably. At even higher velocities the motion will be stronger, and the excess gas flow will tend to appear as bubbles. In this state the particle layer will undergo vigorous mixing, while still appearing as a dense layer of fluidlike material or a boiling liquid. If the gas velocity is further increased, the solid phase will change into a slugging mode where gas bubbles throw lumps of solid material away from the bed surface. Even further increase will result in the solid phase being entrained by the gas flow and will appear as a lean phase undergoing transport or movement by the gas phase. Most fluid-bed drying processes are adjusted to operate safely below slugging conditions. Figure 12-66 shows a view into an operating drying fluid bed.



**FIG. 12-66** Fluid bed for drying in operation. (Niro A/S.)

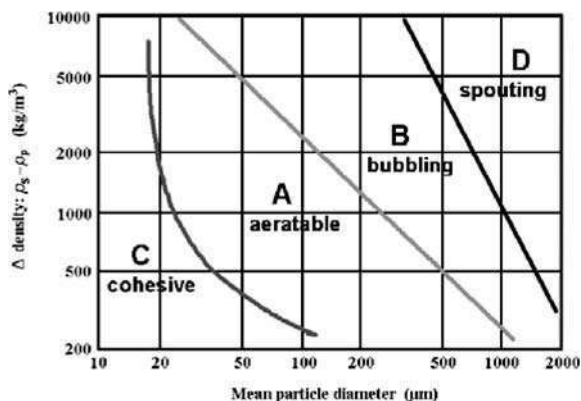


FIG. 12-67 Geldart diagram.

Proper design and operation of a fluid bed installation for drying requires consideration of several important topics. Among these are

- Ability of the material to be fluidized
- Drying characteristics of the material
- The fluidization velocity
- The design of the fluid-bed plate
- The operating conditions
- The mode of operation

**Ability of the material to be fluidized** has been investigated by Geldart, resulting in the well-known Geldart diagram, a version of which is shown in Fig. 12-67. The general knowledge to be derived from the Geldart diagram is that particulate material can be handled successfully in a fluid bed only if it is not too fine or too coarse. It must also have flowability. Fluid beds are best suited for particles that are regular in shape, not too sticky, and with a mean particle size between 20 μm and 10 mm. Particles of needle- or leaflike shape should be considered as nonfluidizable.

**Drying characteristics of the material** can be difficult to determine, but a test in a small batch fluid bed can reveal the drying curve of the material, as shown in Fig. 12-68. The drying curve clearly shows that the surface moisture is rapidly evaporated while the material is maintained at a low temperature close to the wet-bulb temperature of the drying gas. At a certain time the surface water has disappeared, and the so-called transition point has been reached. From here on the drying rate is controlled by internal diffusion inside the material, and the drying curve becomes characteristic for the individual material. While the moisture content of the material decreases, the bed temperature increases while approaching the inlet temperature of the drying air. The total drying time to reach the final moisture and the

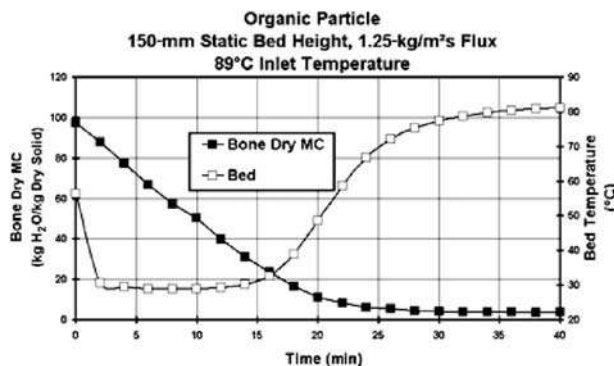


FIG. 12-68 Drying curve of organic material.

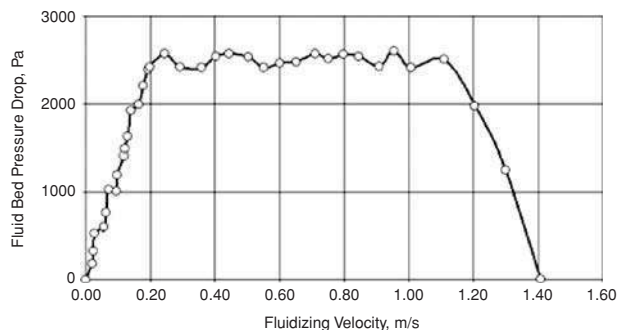


FIG. 12-69 Fluid-bed pressure drop versus fluidizing velocity. (Niro A/S.)

heat sensitivity of the material are important parameters for design of an industrial plant.

**The fluidization velocity** is of major importance, as indicated in the introduction. Each material will have individual requirements for the gas velocity and pressure drop to provide good fluidization. An investigation of the relationship between fluidization velocity and bed pressure drop for a given material may result in a diagram such as shown in Fig. 12-69. The results are illustrative and intended to give a clear picture of the relationship. The minimum fluidization velocity may be calculated from the Wen and Yu correlation given in Sec. 17.

At a fluidizing velocity below the value required for minimum or incipient fluidization, the pressure drop over the bed will increase proportionally with the velocity. Above a certain critical velocity, the pressure drop corresponds to the weight of the fluidized mass of material and remains roughly at this value even at higher velocities. The critical velocity for a given material may be estimated by methods mentioned in Sec. 17. At a much higher value of the fluidizing velocity, the material in the bed ceases to appear as a moving layer, and it is gradually carried away. Accordingly the pressure drop falls to zero. The fluidizing velocity value that will serve a drying task best cannot be derived exactly from the diagram. However, as a general recommendation, a value between the critical value and the value where the pressure drop falls off will be right. A first choice could be a factor of 2 to 5 times the minimum fluidization velocity. Further clarification must be derived from test work with the actual material.

**The design of the fluid-bed plate** is important for several reasons. First, the plate is responsible for the distribution of the drying or fluidization gas. This requires an even pattern of orifices in the plate and a sufficient pressure drop over the plate. As a general rule, the following guideline may be recommended:

$$\Delta P_{\text{plate}} = \frac{1}{2} \Delta P_{\text{powder}}$$

with the following limits:  $\Delta P_{\text{plate}}$  minimum 500 Pa,  $\Delta P_{\text{plate}}$  maximum 2500 Pa.

The estimation of the pressure drop in design situations may be difficult except for the case of the traditional perforated sheet with cylindrical holes perpendicular to the plane of the plate, as shown in Fig. 12-70. For this type of plate the formula of McAllister et al. may be useful. A calculation using this formula will show that a plate giving a required pressure drop of 1500 Pa and a typical fluidizing velocity of 0.35 m/s will need an open area of roughly 1 percent. Provided by a plate of 1-mm thickness and 1-mm-diameter holes, this requires approximately 12,500 holes per square meter.

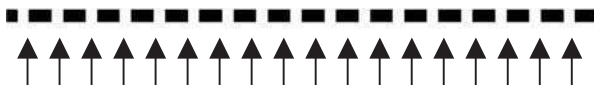


FIG. 12-70 Traditional perforated plate for fluid-bed application.

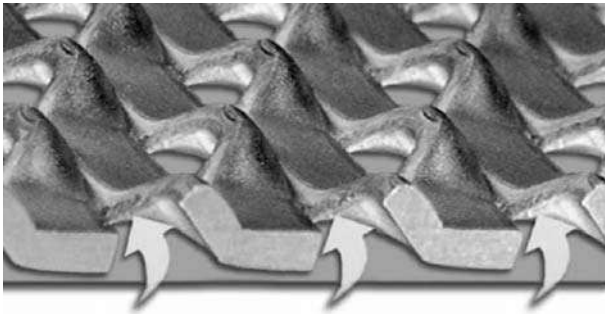


FIG. 12-71 Conidur® plate for fluid-bed application. (Hein, Lehmann Trenn- und Fördertechnik GmbH.)

However, this type of plate is being replaced in most fluid-bed applications due to its inherent disadvantages, which are caused by the difficulties of punching holes of smaller diameter than the thickness of the plate itself. The result is that the plates are weak and are prone to sifting back of the finer particles. The perpendicular flow pattern also means that the plate does not provide a transport capacity for lumps of powder along the plane of the plate.

This transport capacity is provided by plate of so-called gill types of which there are two distinct categories. One category is the type where plates are punched in a very fine regular pattern, not only to provide holes or orifices but also to deform the plate so that each orifice acquires a shape suited for acceleration of the gas flow in magnitude and direction. An example of this type is shown in Fig. 12-71, representing the Conidur® trademark.

The particular feature of Conidur® sheets is the specific hole shape which creates a directional airflow to help in discharging the product and to influence the retention time in the fluid bed. The special method of manufacturing Conidur® sheets enables finishing of fine perforations in sheets with an initial thickness many times over the hole width. Perforations of only 100  $\mu\text{m}$  in an initial sheet thickness of 0.7 mm are possible. With holes this small 1  $\text{m}^2$  of plate may comprise several hundred thousand individual orifices.

The capacity of contributing to the transport of powder in the plane of the plate due to the horizontal component of the gas velocity is also the present for the second category of plates of the gill-type. Figure 12-72 shows an example.

In this type of plate, the holes or orifices are large and the number of gills per square meter is just a few thousand. The gas flow through each of the gills has a strong component parallel to the plate, providing powder transport capacity as well as a cleaning effect. The gills are punched individually or in groups and can be oriented individually to provide a possibility of articulating the horizontal transport effect.

In certain applications in the food and pharmaceutical industries, the nonsifting property of a fluid-bed plate is particularly appreciated. This property of a gill-type plate can be enhanced as illustrated in Fig. 12-73, where the hole after punching is additionally deformed so that the gill overlaps the orifice.

The fifth and final type of fluid-bed plate to be mentioned here is the so-called bubble plate type. Illustrated in Fig. 12-74, it is in principle a gill-type plate. The orifice is cut out of the plate, and the bub-



FIG. 12-72 GILL PLATE™ for fluid-bed application. (Niro A/S.)



FIG. 12-73 NON-SIFTING GILL PLATE™. (Patented by Niro A/S.)



FIG. 12-74 BUBBLE PLATE™. (Patented by Niro A/S.)

ble is subsequently pressed so that the orifice is oriented in a predominantly horizontal direction. A fluid-bed plate will typically have only 1600 holes per square meter. By this technology a combination of three key features is established. The plate is nonsifting, it has transport capacity that can be articulated through individual orientation of bubbles, and it is totally free of cracks that may compromise sanitary aspects of the installation.

The **operating conditions** of a fluid bed are to a high degree dictated by the properties of the material to be dried, as already indicated. One parameter can be chosen regardless of the fluidization process, namely, the fluidization air temperature. For most products, however, the temperature is of primary importance, since the fluidized state results in very high heat-transfer rates so that heat sensitivity may restrict temperature and thereby prolong process time.

To achieve the most favorable combination of conditions to carry out a fluid-bed drying process, it is necessary to consider the different modes of fluid-bed drying available.

**Industrial fluid-bed drying** The first major distinction between fluid-bed types is the choice of mode: batch or continuous.

**Batch fluid beds** may appear in several forms. The process chamber has a perforated plate or screen in the bottom and a drying gas outlet at the top, usually fitted with an internal filter. The drying gas enters the fluid bed through a plenum chamber below the perforated plate and leaves through the filter arrangement. The batch of material is enclosed in the process chamber for the duration of the process.

Figure 12-75 shows a sketch of a typical batch fluid-bed dryer as used in the food and pharmaceutical industries. The process chamber is conic in order to create a freeboard velocity in the upper part of the chamber that is lower than the fluidizing velocity just above the plate.

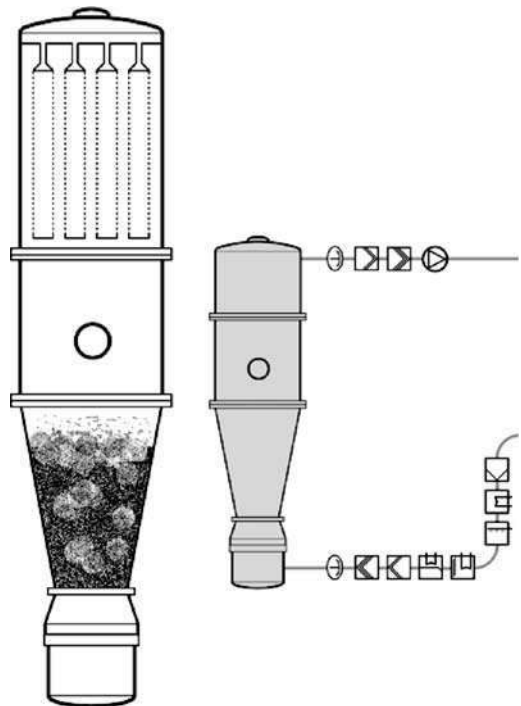


FIG. 12-75 Batch-type fluid-bed. (Aeromatic-Fieldler.)



The enclosed product batch is prevented from escaping the process chamber and will therefore allow a freer choice of fluidizing velocity than is the case in a continuous fluid bed, as described later.

The right-hand side of Fig. 12-75 illustrates in symbol form the drying gas supply system comprising fan, filters of various grade, pre-heater, moisturizer, dehumidifier, final heater, and fast-closure valves. This arrangement is necessary for products with extreme quality requirements such as found in pharmaceutical production.

The drying can be carried out very like the process indicated in Fig. 12-68. The versatile drying gas supply system will allow the drying gas temperature and humidity to be controlled throughout the drying process to optimize process time and to minimize overheating of the product.

**Continuous fluid beds** may be even more varied than batch fluid beds. The main distinction between continuous fluid beds will be according to the solids flow pattern in the dryer. The continuous fluid bed will have an inlet point for moist granular material to be dried and an outlet for the dried material. If the moist material is immediately fluidizable, it can be introduced directly onto the plate and led through the bed in a plug-flow pattern that will enhance control of product residence time and temperature control. If the moist granular material is sticky or cohesive due to surface moisture and therefore needs a certain degree of drying before fluidization, it can be handled by a backmix fluid bed, to be described later.

**Continuous plug-flow beds** are designed to lead the solids flow along a distinct path through the bed. Baffles will be arranged to prevent or limit solids mixing in the horizontal direction. Thereby the residence time distribution of the solids becomes narrow. The bed may be of cylindrical or rectangular shape.

The temperature and moisture contents of the solids will vary along the path of solids through the bed and thereby enable the solids to come close to equilibrium with the drying gas. A typical plug-flow fluid bed is shown in Fig. 12-76.

Continuous plug-flow beds of stationary as well as vibrating type may benefit strongly from use of the gill-type fluid-bed plates with the capacity for controlling the movement of powder along the plate and around bends and corners created by baffles. Proper use of these means may make it possible to optimize the combination of fluidization velocity, bed layer height, and powder residence time.

**Continuous backmix beds** are used in particular when the moist granular material needs a certain degree of drying before it can fluidize. By distributing the material over the surface of an operating fluid bed arranged for total solids mixing, also called backmix flow, it will be absorbed by the dryer material in the bed, and lumping as well as sticking to the chamber surfaces will be avoided. The distribution of the feed can be arranged in different ways, among which a rotary thrower

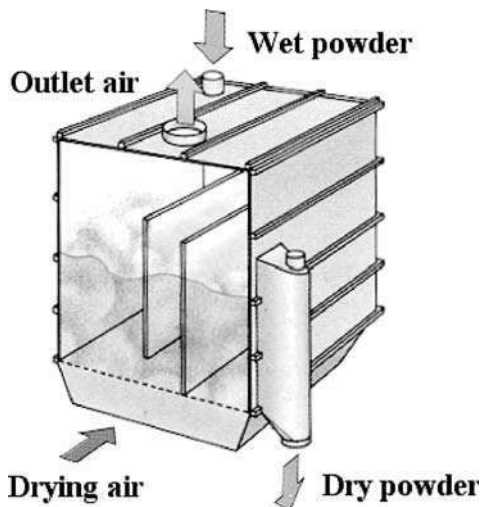


FIG. 12-76 Continuous plug-flow fluid bed. (Niro A/S.)

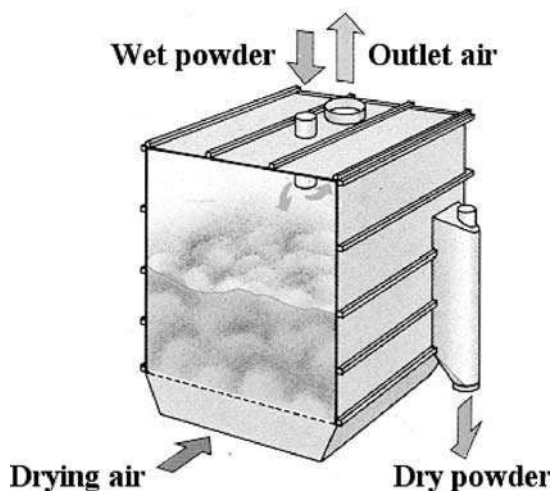


Figure 12-77 Continuous back-mix fluid bed. (Niro A/S.)

is an obvious choice. A typical backmix fluid bed is shown in Fig. 12-77. Backmix fluid beds can be of box-shaped design or cylindrical.

The whole mass of material in the backmix fluid bed will be totally mixed, and all powder particles in the bed will experience the same air temperature regardless of their position on the drying curve illustrated in Fig. 12-68. The residence time distribution becomes very wide, and part of the material may get a very long residence time while another part may get a very short time.

**Continuous contact fluid beds** are common in the chemical industry as the solution to the problem arising from materials requiring low fluidizing air temperature due to heat sensitivity and high energy input to complete the drying operation. An illustration of a Niro CONTACT FLUIDIZER™ is shown in Fig. 12-78.

The main feature of the contact fluid bed is the presence of heating panels, which are plate or tube structures submerged in the fluidized-bed layer and heated internally by an energy source such as steam, water, or oil. The fluidized state of the bed provides very high heat-transfer rates between the fluidizing gas, the fluidized material, and any objects submerged in the bed. The result is that a very significant portion of the required energy input can be provided by the heating panels without

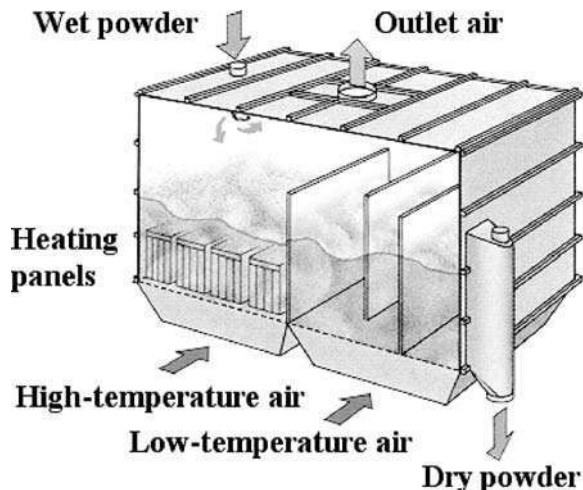


FIG. 12-78 Continuous CONTACT FLUIDIZER™. (Niro A/S.)

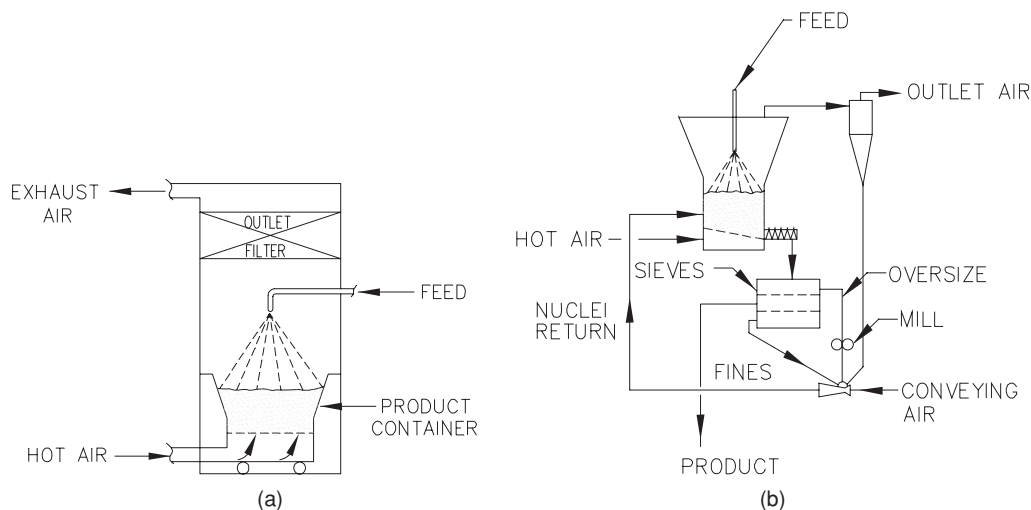


FIG. 12-79 Fluid-bed granulators. (a) Batch; (b) continuous.

risk of overheating the material. The fluidized state of the bed ensures that the material in the bed will flow with little restriction around the heating panels.

The CONTACT FLUIDIZER™ shown in Fig. 12-78 has a number of other features which in combination lead to compact design, high thermal efficiency, and low gas throughput: The first section of the bed is a backmix bed complete with rotary powder distributor and high-temperature fluidizing air supply. It takes care of the drying of the surface moisture, which is controlled mainly by heat supply. The heating panels are distributed over the whole bed volume of this section. The second section of the bed is a plug-flow bed with a fluidizing gas supply adjusted in both temperature and velocity to fit the requirements for the time-controlled diffusion drying of the powder present in this section.

The CONTACT FLUIDIZER™ is primarily used in the polymer industry for drying of polymer powders in high tonnages. Sizewise the individual units become very large, and units with a total fluid-bed area in excess of 60 m<sup>2</sup> are in operation.

**Design methods for fluid beds** When fluid-bed technology can be applied to drying of granular products, significant advantages compared to other drying processes can be observed. Design variables such as fluidizing velocity, critical moisture content for fluidization, and residence time required for drying to the specified residual moisture must, however, be established by experimental or pilot test before design steps can be taken. Reliable and highly integrated fluid-bed systems of either batch or continuous type can be designed, but only by using a combination of such pilot test and industrial experience. Scale-up rules are given by Kemp and Oakley (2002).

#### Additional Reading

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**Dryers with Liquid Feeds** If the feed is a liquid, paste, slurry, or solution, special equipment is required. The available choices are as follows:

**Spray Dryers** A pumpable feed is atomized into droplets by a rotary or nozzle atomizer, as described under “Entrainment Dryers.” An integral fluid bed or belt may be added below the dryer to give longer residence time and some agglomeration. Semibatch and continuous operation is possible.

**Fluidized-Bed Granulator** A slurry or solution is sprayed onto a fluidized bed of particles, as shown in Fig. 12-79. The difference from the spray fluid bed is that the spray is still liquid when it contacts the particles, so that layered growth or surface agglomeration occurs, producing stronger large particles or agglomerates. Both batch and continuous forms exist (the latter involving continuous solids recycle with classification). In an older variant, the bed may be of inert balls, and the solid forming on the outside is periodically knocked off. Dryer construction and operation are largely as described under “Fluid-Bed Dryers.”

**Drum (Film-Drum) Dryers** A film of liquid or paste is spread onto the outer surface of a rotating, internally heated drum. Drying occurs by conduction, and at the end of the revolution the dry product, which can be in the form of powder, flakes, or chips and typically is 100 to 300 μm thick, is removed by a doctor's knife. Drum dryers cannot handle feedstocks which do not adhere to metal, products which dry to a glazed film, or thermoplastics. The drum is heated normally by condensing steam or in vacuum drum dryers by hot water. Figure 12-80 shows three of the many possible forms. The dip feed system is the simplest and most common arrangement but is not suitable for viscous or pasty materials. The nip feed system is usually employed on double-drum dryers, especially for viscous materials, but it cannot handle lumpy or abrasive solids. The latter are usually applied by roller, and this is also effective for sticky and pasty materials. Spray and splash devices are used for feeding heat-sensitive, low-viscosity materials. Vacuum drum dryers are simply conventional units encased in a vacuum chamber with a suitable air lock for product discharge. Air impingement is also used as a secondary heat source on drum and can dryers, as shown in Fig. 12-81.

**Contact Drying** (Special thanks to R. B. Keey for the following example of contact drying.) In contact drying, the moist material covers a hot surface which supplies the heat required for the drying process.

Let us consider a moist material lying on a hot flat plate of infinite extent. Figure 12-82 illustrates the temperature profile for the fall in temperature from  $T_H$  in the heating fluid to  $T_C$  in the surrounding air. It is assumed that the temperatures remain steady, unhindered drying takes place, and there is no air-gap between the material being dried and the heating surface.

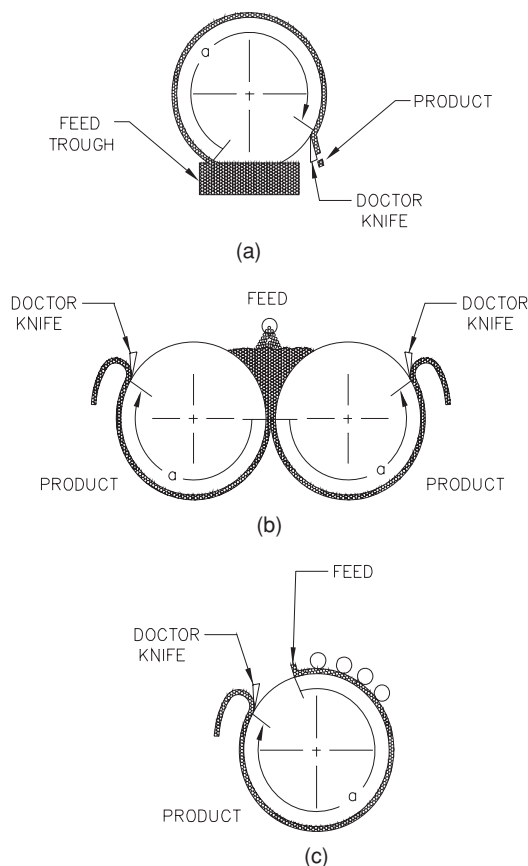


FIG. 12-80 Main types of drum dryers. (a) Dip; (b) nip; (c) roller.

The heat conducted through the wall and material is dissipated by evaporation of moisture and convection from the moist surface to the surrounding air. A heat balance yields

$$U(T_H - T_S) = N_W \Delta H_{VS} + h_C(T_S - T_G) \quad (12-110)$$

where  $U$  is the overall heat-transfer coefficient. This coefficient is found from the reciprocal law of summing resistances in series:

$$\frac{1}{U} = \frac{1}{h_H} + \frac{b_B}{\lambda_B} + \frac{b_s}{\lambda_s} \quad (12-111)$$

in which  $h_H$  is the heat-transfer coefficient for convection inside the heating fluid. If condensing steam is used, this coefficient is very large normally and the corresponding resistance  $1/h_H$  is negligible. Rearrangement of Eq. (12-110) yields an expression for the maximum drying rate

$$N_W = \frac{U(T_H - T_S) - h_C(T_S - T_G)}{\Delta H_{VS}} \quad (12-112)$$

Equation (12-112), as it stands, would give an overestimate of the maximum drying rate for the case of contact drying over heated rolls, when there are significant heat losses from the ends of the drum and only part of the drum's surface can be used for drying. In the roller drying arrangements shown in Fig. 12-80, only a fraction  $a$  of the drum's periphery is available from the point of pickup to the point where the solids are peeled off.

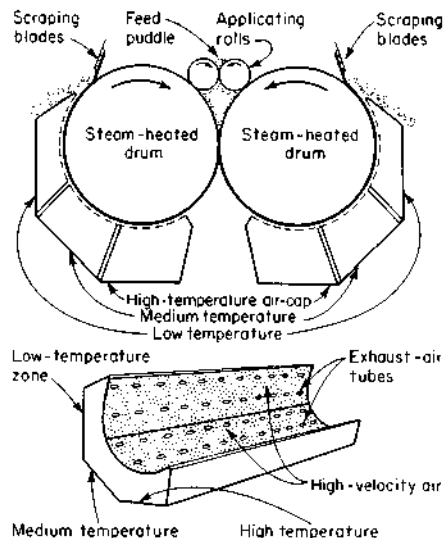


FIG. 12-81 Example of the use of air impingement in drying as a secondary heat source on a double-drum dryer. (Chem. Eng., 197, June 19, 1967.)

Let  $q_E$  be the heat loss per unit area from the ends. The ratio of the end areas to cylindrical surface, from a drum of diameter  $D$  and length  $L$ , is  $2(\frac{1}{4}\pi D^2)/\pi DL$  or  $D/2L$ . Equation (12-112) for the maximum drying rate under roller drying conditions thus becomes

$$N_W = \frac{aU(T_H - T_S) - h_C(T_S - T_G) - Dq_E/2L}{\Delta H_{VS}} \quad (12-113)$$

The total evaporation from the drum is  $N_W a(\pi DL)$ . Equation (12-113) could be refined further, as it neglects the effect caused by the small portion of the drum's surface being covered by the slurry in the feed trough, as well as thermal conduction through the axial shaft to the bearing mounts. The use of Eq. (12-113) to estimate the maximum drying rate is illustrated in Example 24.

**Example 24: Heat-Transfer Calculations** A single rotating drum of 1.250-m diameter and 3 m wide is internally heated by saturated steam at 0.27 MPa. As the drum rotates, a film of slurry 0.1 mm thick is picked up and dried. The dry product is removed by a knife, as shown in Fig. 12-80a. About three-quarters of the drum's surface is available for evaporating moisture. Estimate the maximum drying rate when the outside air temperature  $T_G$  is 15°C and the surface temperature 50°C, and compare the effectiveness of the unit with a dryer without end effects and in which all the surface could be used for drying.

Data:

|                                                      |                            |
|------------------------------------------------------|----------------------------|
| Heat-transfer coefficient $h_H$                      | 50 W/(m <sup>2</sup> ·K)   |
| Thickness of cylinder wall $b_B$                     | 10 mm                      |
| Thermal conductivity of wall $\lambda_B$             | 40 W/(m·K)                 |
| Thermal conductivity of slurry film $\lambda_s$      | 0.10 W/(m·K)               |
| Film transfer coefficient for condensing steam $h_H$ | 2.5 kW/(m <sup>2</sup> ·K) |

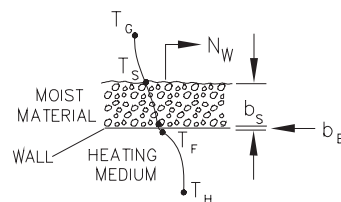


FIG. 12-82 Temperature profile in conductive drying.

Overall heat-transfer coefficient  $U$ : The thermal resistances are as follows:

$$\begin{aligned}\text{Steamside} & 1/2.5 = 0.40 \text{ m}^2\text{K/kW} \\ \text{Wall} & 0.01/0.04 = 0.25 \text{ m}^2\text{K/kW} \\ \text{Filmside} & 0.0001/0.1 \times 10^{-3} = 1.0 \text{ m}^2\text{K/kW} \\ \therefore \text{Overall resistance} & = 0.40 + 0.25 + 1.0 = 1.65 \text{ m}^2\text{K/kW} \\ U & = 1/1.65 = 0.606 \text{ kW}/(\text{m}^2 \cdot \text{K})\end{aligned}$$

Wall temperature  $T_B$ : At 0.27 MPa, the steam temperature is 130°C. If it is assumed that the temperature drops between the steam and the film surface are directionally proportional to the respective thermal resistances, it follows that

$$\begin{aligned}\frac{T_H - T_B}{T_H - T_S} & = \frac{0.40 + 0.25}{1.65} = 0.3939 \\ \therefore T_B & = T_H - 0.3939(T_H - T_S) \\ & = 130 - 0.3939(130 - 50) \\ & = 98.5^\circ\text{C}\end{aligned}$$

Heat losses from ends  $q_E$ : For an emissivity  $\sim 1$  and an air temperature of 15°C with a drum temperature of 98.5°C, one finds [see Eq. (12-119)],

$$q_E = 1184 \text{ W/m}^2$$

Maximum drying rate  $N_W$ : From Eq. (12-113),

$$\begin{aligned}N_W & = \frac{aU(T_H - T_S) - h_c(T_S - T_G) - Dq_E/2L}{\Delta H_{VS}} \\ & = \frac{0.75 \times 0.606(130 - 50) - 0.05(50 - 15) - (1.25 \times 1.184)/6}{2382} \\ & = 0.0144 \text{ kg}/(\text{m}^2 \cdot \text{s})\end{aligned} \quad (12-114)$$

The ideal maximum rate is given by Eq. (12-112) for an endless surface:

$$\begin{aligned}N_N & = \frac{U(T_H - T_S) - h_c(T_S - T_G)}{\Delta H_{VS}} \\ & = \frac{0.606(130 - 50) - 0.05(50 - 15)}{2382} \\ & = 0.0196 \text{ kg}/(\text{m}^2 \cdot \text{s})\end{aligned} \quad (12-115)$$

Therefore the effectiveness of the dryer is  $0.0144/0.0196 = 0.735$ .

The predicted thermal efficiency  $\eta$  is

$$\begin{aligned}\eta & = 1 - \frac{h_c(T_S - T_G + Dq_E/2L)}{aU(T_H - T_S)} \\ & = 1 - \frac{0.05(50 - 15) + (1.25 \times 1.184)/6}{0.75 \times 0.606(130 - 50)} \\ & = 0.945\end{aligned} \quad (12-116)$$

These estimates may be compared with the range of values found in practice, as shown in Table 12-39 (Nonhebel and Moss, *Drying of Solids in the Chemical Industry*, Butterworths, London, 1971, p. 168).

The typical performance is somewhat less than the estimated maximum evaporation capacity, although values as high as 25 g/(m<sup>2</sup>·s) have been reported. As the solids dry out, so the thermal resistance of the film increases and the evaporation falls off accordingly. Heat losses through the bearing of the drum shaft have been neglected, but the effect of radiation is accounted for in the value of  $h_c$  taken. In the case of drying organic pastes, the heat losses have been determined to be 2.5 kW/m<sup>2</sup> over the whole surface, compared with 1.75 kW/m<sup>2</sup> estimated here for the cylindrical surface. The inside surface of the drum has been assumed to be clean, and scale would reduce the heat transfer markedly.

For constant hygrothermal conditions, the base temperature  $T_B$  is directly proportional to the thickness of the material over the hot surface. When the wet-

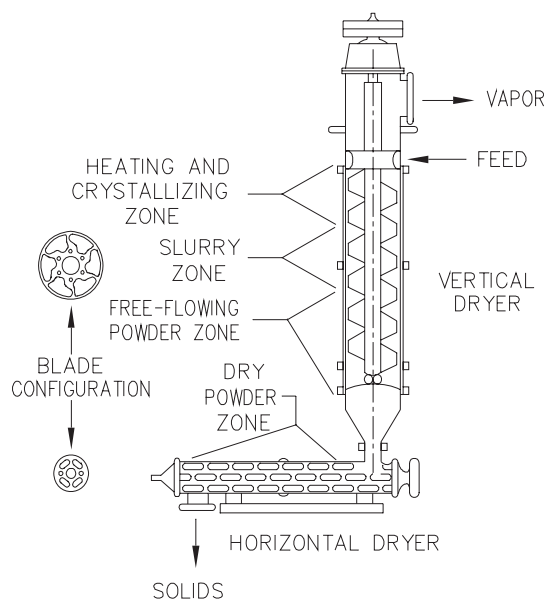


FIG. 12-83 Continuous thin-film dryer.

bulb temperature is high and the layer of material is thick enough, the temperature  $T_B$  will reach the boiling point of the moisture. Under these conditions, a mixed vapor-air layer interposes between the material and the heating surface. This is known as the Leidenfrost effect, and the phenomenon causes a greatly increased thermal resistance to heat transfer to hinder drying.

**Thin-Film Dryers** Evaporation and drying take place in a single unit, normally a vertical chamber with a vertical rotating agitator which almost touches the internal surface. The feed is distributed in a thin layer over the heated inner wall and may go through liquid, slurry, paste, and wet solid forms before emerging at the bottom as a dry solid. These dryers are based on wiped-film or scraped-surface (Luwa-type) evaporators and can handle viscous materials and deal with the "cohesion peak" experienced by many materials at intermediate moisture contents. They also offer good containment. Disadvantages are complexity, limited throughput, and the need for careful maintenance. Continuous or semibatch operation is possible. A typical unit is illustrated in Fig. 12-83.

**Filter Dryers** Basically this is a Nutsche filter (Sec. 18, "Liquid-Solid Operations and Equipment") followed by a batch dryer, usually of vertical pan type (see "Batch Agitated and Rotating Dryers" section). They are popular in the pharmaceutical and specialty chemicals industries as two unit operations are performed in the same piece of equipment without intermediate solids transfer, and containment is good.

**Centrifuge Dryers** Usually they are batch or continuous filtering centrifuges (Sec. 18, "Liquid-Solid Operations and Equipment") with hot air being blown over the solids in the discharge section. Manufacturers include Heinkel and Bird-Humboldt.

Pastelike feeds can be handled by some dryers for particulate materials, if either they do not require free-flowing feeds or some dry product can be backmixed with the wet feed to improve its handling.

**Dryers for Films and Sheets** The construction of dryers where both the feed and the product are in the form of a sheet, web, or film is markedly different from that for dryers used in handling particulate materials. The main users are the paper and textile industries. Almost invariably the material is formed into a very long sheet (often hundreds or thousands of meters long) which is dried in a continuous process. The sheet is wound onto a bobbin at the exit from the dryer; again, this may be 1 or 2 m in diameter. Alternatively, the sheet may be chopped into shorter sections.

**Cylinder Dryers and Paper Machines** The most common type of dryer in papermaking is the cylinder dryer (Fig. 12-84), which is a

TABLE 12-39 Operating Information

|                                             | This estimate | Typical range |
|---------------------------------------------|---------------|---------------|
| Specific evaporation, g/(m <sup>2</sup> ·s) | 14.4          | 7–11          |
| Thermal efficiency                          | 0.945         | 0.4–0.7       |



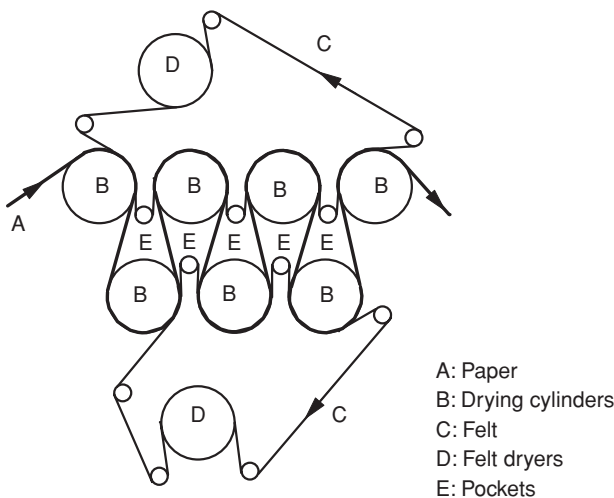


FIG. 12-84 Cylinder dryer (paper machine).

contact dryer. The paper web is taken on a convoluted path during which it wraps around the surface of cylinders which are internally heated by steam or hot water. In papermaking, the sheet must be kept taut, and a large number of cylinders are used, with only short distances between them and additional small unheated rollers to maintain the tension. Normally, a continuous sheet of felt is also used to hold the paper onto the cylinders, and this also becomes damp and is dried on a separate cylinder.

Most of the heating is conductive, through contact with the drums. However, infrared assistance is frequently used in the early stages of modern paper machines. This gets the paper sheet up to the wet-bulb temperature more rapidly, evaporates more surface moisture, and allows the number of cylinders to be reduced for a given throughput. Hot air jets (jet foil dryer) may also be used to supplement heating at the start of the machine. Infrared and dielectric heating may also be used in the later stages to assist the drying of the interior of the sheet.

Although paper is the most common application, multicylinder dryers can also be used for polymer films and other sheet-type feeds.

Convective dryers may be used as well in papermaking. In the Yankee dryer (Fig. 12-85), high-velocity hot airstreams impinging on the

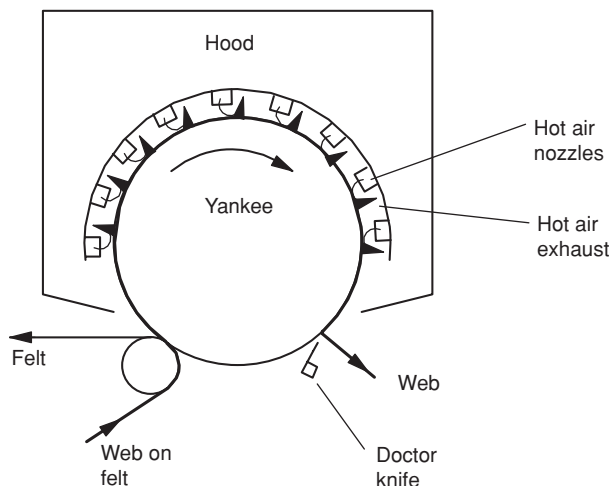


Fig. 12-85 Yankee dryer.

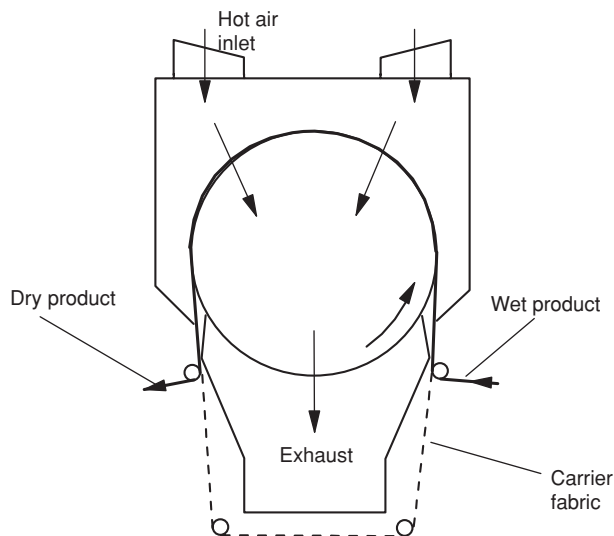


FIG. 12-86 Rotary through-dryer.

web surface give heating by cross-convection. The "Yankees" are bars holding the web in place. Normally the cylinder is also internally heated, giving additional conduction heating of the lower bed surface. In the rotary through-dryer (Fig. 12-86), the drum surface is perforated and hot air passes from the outside to the center of the drum, so that it is a through-circulation convective dryer.

Another approach to drying of sheets has been to suspend or "float" the web in a stream of hot gas, using the Coanda effect, as illustrated in Fig. 12-87. Air is blown from both sides, and the web passes through as an almost flat sheet (with a slight "ripple"). The drying time is reduced because the heat transfer from the impinging hot air jets is faster than that from stagnant hot air in a conventional oven. It is essential to control the tension of the web very accurately. The technique is particularly useful for drying coated paper, as the expensive surface coating can stick to cylinder drums.

**Stenters (Tenters) and Textile Dryers** These are the basic type of dryer used for sheets or webs in the textile industry. The sheet is held by its edges by clips (clip stenter) or pins (pin stenter), which not only suspend the sheet but also keep it taut and regulate its width—a vital consideration in textile drying. Drying is by convection; hot air is introduced from one or both sides, passes over the surface of the sheet, and permeates through it. Infrared panels may also be used to supply additional heat. A schematic diagram of the unit is shown in Fig. 12-88. A typical unit is 1.4 m wide and handles 2 to 4 t/h of material.

Heavy-duty textiles with thick webs may need a long residence time, and the web can be led up and down in "festoons" to reduce dryer length. Substantial improvements in drying rates have been obtained with radio-frequency heating assistance.

Air impingement dryers as in Fig. 12-87 may also be used for textiles.

**Spray Dryers** Spray drying is a drying process for transformation of a pumpable liquid feed in the form of a solution, dispersion, slurry, or paste into a particulate dried product in one single operation. The process comprises atomization of the feed followed by intense contact with hot air.

Due to the very large surface area created by the atomization of the feed, rapid evaporation occurs from the surface of each particle or droplet in the spray. The magnitude of the surface area can be illustrated by a simple calculation. Atomization of 1 L of water into a uniform spray of 100- $\mu\text{m}$  droplets results in approximately  $1.9 \times 10^9$  individual particles with a combined surface area of 60  $\text{m}^2$ . A realistic spray with variation of the droplet size may have a substantially higher number of droplets and a somewhat higher surface area. The dry

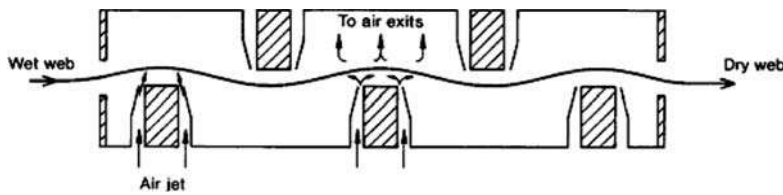


FIG. 12-87 Air flotation (impingement) dryer.

particulate product is formed while the spray droplets are still suspended in the hot drying air. The spray drying process is concluded by product recovery and separation from the drying air.

Spray drying belongs to the family of suspended particle processing (SPP) systems. Other members of this family are fluid-bed drying, flash drying, spray granulation, spray reaction, spray cooling, and spray absorption.

**Drying Principles** In the spray drying process or operation, the liquid to be removed by drying is predominantly water. Certain special products are produced with use of organic solvents, which are removed in a spray drying process. The drying principles involved for aqueous as well as nonaqueous systems are the same.

The liquid or moisture in a spray droplet is present in two basic forms: bound and unbound moisture. The nature of the solid and the liquid matter determines the drying characteristics of the product.

The category of bound moisture comprises water retained in small capillaries in the solid, water absorbed on solid surfaces, water bound as solutions in cells or fiber walls, and water bound as crystal water in chemical combination with the solid. Bound water exerts an equilibrium vapor pressure lower than that of pure water at the same temperature.

The category of unbound moisture can be described as the moisture in excess of the bound moisture. A hygroscopic material may contain bound as well as unbound moisture. A nonhygroscopic material contains unbound moisture only. The equilibrium vapor pressure of unbound water is equal to that of pure water at the same temperature.

The free moisture in a particle is the moisture in excess of the equilibrium moisture and may consist of unbound and some bound moisture. Only free moisture can be removed by evaporation during spray drying.

The mechanism of moisture flow in a droplet during spray drying is mainly diffusion supplemented by capillary flow. The drying characteristics of the droplet depend on the balance of bound and unbound as each category has distinct features.

The presence of unbound moisture in the droplet means that the drying proceeds at a constant high rate as long as the moisture diffusion within the droplet is able to maintain saturated surface conditions. When the diffusional and capillary flows can no longer maintain these conditions, a critical point is reached and the drying rate will decline until equilibrium moisture content is reached. The evaporation of bound moisture is strongly dependent on the nature of the solid matter in the spray droplet.

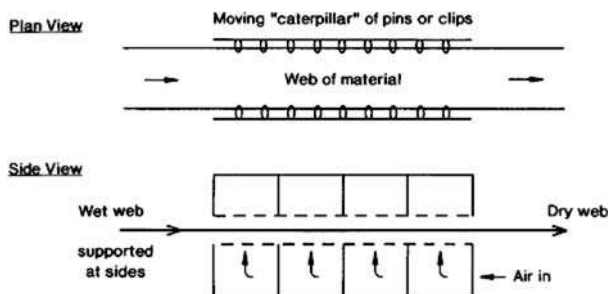


FIG. 12-88 Stenter or tenter for textile drying.

A spray drying plant comprises four process stages, as shown in Table 12-40.

**Atomization Stage** Spray drying is often used in industrial processes characterized by high production rates. Although the three different methods of atomization indicated in Table 12-40 are the same as those for many other atomization or spray forming processes, the relative weight of the methods is special for spray drying with rotary atomizers and hydraulic pressure nozzles having a very broad application, while two-fluid nozzles are only used to a smaller extent in specialized applications.

**Rotary Atomizer** Figure 12-89 shows a rotary atomizer in operation. The liquid feed is supplied to the atomizer by gravity or hydraulic pressure. A liquid distributor system leads the feed to the inner part of a rotating wheel. Since the wheel is mounted on a spindle supported by bearings in the atomizer structure, the liquid distributor is usually formed as an annular gap or a ring of holes or orifices concentric with the spindle and wheel. The liquid is forced to follow the wheel either by friction or by contact with internal vanes in the wheel. Due to the high centrifugal forces acting on the liquid, it moves rapidly toward the rim of the wheel, where it is ejected as a film or a series of jets or ligaments. By interaction with the surrounding air the liquid breaks up to form a spray of droplets of varying size. The spray pattern is virtually horizontal with a spray angle said to be  $180^\circ$ . The mean droplet size of the spray depends strongly on the atomizer wheel speed and to a much lesser degree on the feed rate and the feed physical properties such as viscosity. More details about spray characteristics such as droplet size distribution will be given below.

As indicated above, the atomizer wheel speed is the important parameter influencing the spray droplet size and thus the particle size of the final product. The atomizer machine will normally have the capability to operate the wheel at the required speed. More important for the atomization process is the selection of a wheel capable of handling a specific liquid feed with characteristic properties such as abrasiveness, high viscosity, nonnewtonian behavior, or tendency to coagulate.

The most common design of atomizer wheel has radial vanes, as shown in Fig. 12-90. This wheel type is widely used in the chemical industry and is virtually blockage-free and simple to operate, even at very high speed. For high-capacity applications, the number and height of the vanes may be increased to maintain limited liquid film thickness conditions on each vane.

Wheels with radial vanes have one important drawback, i.e., their capacity for pumping large amounts of air through the wheel. This so-called air pumping effect causes unwanted product aeration, resulting in powders of low bulk density for some sensitive spray dried products.

TABLE 12-40 Stages of Spray Drying

| Process stages of spray drying | Methods                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------|
| 1. Atomization                 | Rotary atomization<br>Pressure nozzle atomization<br>Two-fluid nozzle atomization |
| 2. Spray/hot air contact       | Cocurrent flow<br>Countercurrent flow<br>Mixed flow                               |
| 3. Evaporation                 | Drying<br>Particle shape formation                                                |
| 4. Product recovery            | Drying chamber<br>Dry collector<br>Wet collectors                                 |



FIG. 12-89 Rotary atomizer operation. (Niro.)

Unwanted air pumping effect and product aeration can be reduced through careful wheel design involving change of the shape of the vanes that may appear as forward-curved. This wheel type is used widely in the dairy industry to produce powders of high bulk density. The powder bulk density may increase as much as 15 percent when a curved vane wheel is replacing a radial vane wheel of standard design.

Another way of reducing the air pumping effect is to reduce the space between the vanes so that the liquid feed takes up a larger fraction of the available cross-sectional area. This feature is used with consequence in the so-called bushing wheels such as shown in Fig.12-91. This wheel combines two important design aspects. The air pumping effect is reduced by reducing the flow area to a number of circular orifices, each 5 to 10 mm in diameter. By placing these orifices or nozzles in replaceable bushings or inserts made of very hard materials such as technical ceramics, i.e., alumina or silicon carbide, a substantially abrasion-resistant atomizer wheel design is achieved. This feature is very important in a number of spray drying applications with abrasive feeds, which would wear down a standard vaned wheel in a matter of hours. With an abrasion-resistant wheel, almost unlimited lifetime can be expected for the atomizer wheel structure and several thousand hours for replaceable bushings.

The rotary atomizer machines are high-speed machines traditionally built with a step-up gear to increase the speed from the 3000 or 3600 rpm of the standard two-pole electric motors to 10,000 to 20,000 rpm normally required to achieve sufficiently fine atomization. Newer designs feature high-speed electric motors with frequency control of the atomizer speed. Table 12-41 gives the main operational param-

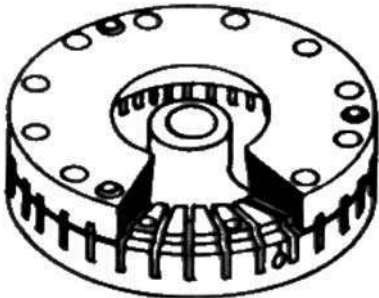


FIG. 12-90 Rotary atomizer wheel with radial vanes. (Niro.)

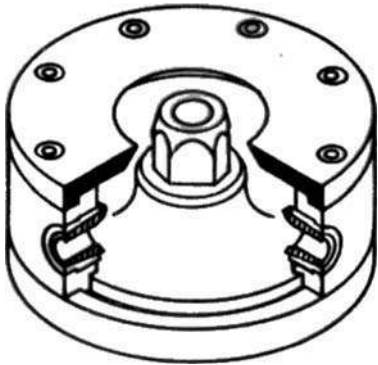


FIG. 12-91 Abrasion-resistant bushing atomizer wheel. (Niro.)

ters for three typical atomizers covering the wide range of capacity and size.

The F800 atomizer is the largest rotary atomizer offered to industry today. It has the capability of handling up to 200 t/h in one single atomizer. The capacity limit of an atomizer is normally its maximum power rating. As indicated above, the atomizer wheel speed is the important parameter influencing the spray droplet size. The wheel speed also determines the power consumption of the atomizer. It can be shown that the atomizer power consumption exclusive mechanical losses amount to

$$P_s = \frac{U^2}{3600}$$

where  $P_s$  = specific power consumption, kWh/t and  $U$  = peripheral velocity, m/s.

Since the atomizer wheel peripheral speed is proportional to the rotational speed, the maximum feed rate that can be handled by a rotary atomizer declines with the square of the rotational speed. The maximum feed rates indicated in Table 12-41 are therefore not available in the higher end of the speed ranges.

The rotary atomizer has one distinct advantage over other means of atomization. The degree or fineness of atomization achieved at a given speed is only slightly affected by changes in the feed rate. In other words, the rotary atomizer has a large turndown capability.

The larger atomizer machines cited in Table 12-41 represent a range of very large rotary atomizers available to industry. They are equipped with epicyclic-type gearboxes complete with a lubrication system. An extensive monitoring system is integrated in each machine.

Many atomization duties involve much lower capacities than foreseen for this range of atomizers. A full range of smaller rotary atomizers are available with nominal capacities down to less than 100 kg/h. Various designs may be seen with either belt drive or worm gears. Designs without gears are available with high-speed electric motor drive. Table 12-41 gives data for a smaller atomizer machine (FS1.5). It belongs to a family of high-speed machines without gears and lubrication systems capable of operating under the strictest requirements

| TABLE 12-41 Operational Parameters for Atomizers (Niro) |       |        |        |        |  |
|---------------------------------------------------------|-------|--------|--------|--------|--|
| Rotary atomizer designs                                 |       |        |        |        |  |
| Atomizer type                                           |       | FS1.5  | F160   | F800   |  |
| Nominal power rating                                    | kW    | 1.5    | 160    | 1000   |  |
| Maximal feed rate                                       | t/h   | 0.3    | 50     | 200    |  |
| Atomizer wheel diameter                                 | mm    | 90     | 240    | 350    |  |
| Typical gear ratio                                      | #     | 1:1    | 4.4:1  | 2.9:1  |  |
| Minimum speed                                           | rpm   | 10,000 | 6,000  | 8,800  |  |
| Maximum speed                                           | rpm   | 30,000 | 18,200 | 11,500 |  |
| Typical peripheral velocity                             | m/s   | 141    | 165    | 161    |  |
| Typical specific power                                  | kWh/t | 5.5    | 7.6    | 7.2    |  |

for noncontamination of the product and in explosion-prone environments.

**Hydraulic pressure nozzle** In hydraulic pressure nozzle atomizers, the liquid feed is fed to the nozzle under pressure. In the nozzle orifice the pressure energy is converted to kinetic energy. The internal parts of the nozzle are normally designed to apply a certain amount of swirl to the feed flow so that it issues from the orifice as a high-speed film in the form of a cone with a desired vertex angle. This film disintegrates readily into droplets due to instability. The vertex or spray angle is normally on the order of 50° to 80°, a much narrower spray pattern than is seen with rotary atomizers. This means that spray drying chamber designs for pressure nozzle atomization differ substantially from designs used with rotary atomizers. The droplet size distribution produced by a pressure nozzle atomizer varies inversely with the pressure and to some degree with feed rate and viscosity. The capacity of a pressure nozzle varies with the square root of the pressure. To obtain a certain droplet size, the pressure nozzle must operate very close to the design pressure and feed rate. This implies that the pressure nozzle has very little turndown capability.

Hydraulic pressure nozzles cannot combine the capability for fine atomization with high feed capacity in one single unit. Many spray dryer applications, where pressure nozzles are applied, therefore require multinozzle systems with the consequence that start-up, operational control, and shutdown procedures become more complicated.

**Two-fluid nozzle atomization** In two-fluid nozzle atomizers, the liquid feed is fed to the nozzle under marginal or no pressure conditions. An additional flow of gas, normally air, is fed to the nozzle under pressure. Near the nozzle orifice, internally or externally, the two fluids (feed and pressurized gas) are mixed and the pressure energy is converted to kinetic energy. The flow of feed disintegrates into droplets during the interaction with the high-speed gas flow which may have sonic velocity.

The spray angle obtained with two-fluid nozzles is normally on the order of 10° to 20°, a very narrow spray pattern that is related to the spread of a free jet of gas. Spray drying chamber designs for two-fluid nozzle atomization are very specialized according to the application.

The droplet size produced by a two-fluid nozzle atomizer varies inversely with the ratio of gas to liquid and with the pressure of the atomization gas. The capacity of a two-fluid nozzle is not linked to its atomization performance. Therefore two-fluid nozzles can be attributed with some turndown capability.

Two-fluid nozzles share with pressure nozzles the lack of high feed capacity combined with fine atomization in one single unit. Many spray dryer applications with two-fluid nozzle atomization have a very high number of individual nozzles. The main advantage of two-fluid nozzles is the capability to achieve very fine atomization.

**Choice of atomizer system** The choice of atomizer system for a specific spray drying operation depends upon the particle size distribution required in the final dried product. It also depends upon the physical and chemical properties of the feed liquid.

In cases where the different types of atomizer means produce similar particle size distributions, the rotary atomizer may be preferred due to its greater flexibility and ease of operation. When one is comparing the atomizer types, the rotary atomizer has distinct advantages. (1) It can handle high feed rates in one single unit, (2) it can handle abrasive feeds with minimal wear, and (3) it has negligible blockage tendencies due to the large flow ports in the atomizer wheel. (4) It is a low-pressure system that can be served by a simple feed supply system, and (5) droplet size control is simple through wheel speed adjustment.

Although it lacks the flexibility of the rotary atomizer, the pressure nozzle is nevertheless widely used in spray drying applications. For many products the requirement for nondusty appearance calls for large mean particle size and lack of a fines fraction that cannot be met with a rotary atomizer. In the other end of the particle size range, some products require finer particles than are practically achievable with a rotary atomizer. This is the range where two-fluid nozzles are applied. The following guidelines may be used as an indication of the particle sizes obtainable in spray dryers:

- For spray dryers with rotary atomizer, the mean size of the dried product varies from 40 to 110  $\mu\text{m}$ , although larger product mean sizes can be produced in large-diameter chambers.

- For spray dryers with pressure nozzle atomization, the mean particle size of the dried product varies in the range from 50 to 250  $\mu\text{m}$ .
- For spray dryers with two-fluid nozzle atomization, the mean particle size of the dried product varies in the range from 15 to 50  $\mu\text{m}$ .

The different means of atomization can also be compared in terms of energy power consumption. As indicated in Table 12-41, typical specific power figures for rotary atomizers are in the range of 5 to 11 kWh/t. Similar figures can be calculated for pressure and two-fluid nozzle systems, i.e., the pumping energy of the feed and the compression energy of the atomization gas. Any such calculation will show that similar median particle sizes are obtained for a given atomization energy independent of the means of atomization. None of the three types stand out as being energy-efficient. The hydraulic pressure nozzle is best suited for relatively coarse atomization, because pressures higher than 300 bar are impractical. Rotary atomizers are limited, because the wheel peripheral speeds required for very fine atomization put the wheel material under extreme tensile stress.

Droplet size distributions obtained with any means mentioned here are relatively well represented by a Rosin-Rammler distribution with an exponent of approximately 2. This means that approximately 80 percent of the droplet population mass is in the range of 0.39 to 1.82 times the median droplet size.

Theoretical prediction of mean particle sizes is difficult and of little practical importance, since the selection of spray drying operational parameters is based on experience and pilot-scale test work. The scientific literature, however, contains numerous estimation formulas to help predict the droplet sizes in sprays. Table 12-42 provides nomenclature for these estimation formulas.

For rotary atomizers median droplet sizes can be estimated from the following empirical equation of obscure origin:

$$d_{50} = K_r \times \dot{m}_L^{0.15} \times D^{-0.8} \times N^{-0.05} \times \omega^{-0.75} \times \mu_L^{0.07}$$

For hydraulic pressure nozzles the following formula proposed by Lefebvre may be used:

$$d_{50} = K_p \times \dot{m}_L^{0.25} \times \Delta P_L^{-0.5} \times (\sigma_L \times \mu_L)^{0.25} \times \rho_A^{-0.25}$$

Similarly, a range of equations or formulas are available for prediction of droplet size for sprays from two-fluid nozzles. The most widely cited in the literature is the Nukiyama-Tanasawa equation, which, however, is complicated and of doubtful validity at high flow rates. A much simpler equation has been proposed by Geng Wang et al.:

$$d_{50} = K_t \times \rho_A^{-0.325} \times \left( \frac{\dot{m}_L}{\dot{m}_L + U_L + \dot{m}_A \times U_A} \right)^{0.55}$$

If any difference between the atomization means mentioned here were to be pointed out, it would be the tendency for two-fluid nozzles to have the wider particle size distribution and narrower pressure nozzles with rotary atomizers in between.

**TABLE 12-42 Nomenclature for Atomization Equations**

|                                     |                   |
|-------------------------------------|-------------------|
| $d_{50}$ = mass median droplet size | m                 |
| $K_r$ = empirical factor            | 0.008             |
| $K_p$ = empirical factor            | 4.0               |
| $K_t$ = empirical factor            | 0.1               |
| $\dot{m}_L$ = liquid feed rate      | kg/s              |
| $D$ = wheel diameter                | m                 |
| $N$ = number of vanes               | #                 |
| $\omega$ = atomizer wheel speed     | rad/s             |
| $\mu_L$ = liquid viscosity          | Pa·s              |
| $\Delta P_L$ = atomization pressure | Pa                |
| $\rho_A$ = air density              | kg/m <sup>3</sup> |
| $\sigma_L$ = liquid surface tension | N/m               |
| $\dot{m}_A$ = atomization gas rate  | kg/s              |
| $U_L$ = liquid velocity             | m/s               |
| $U_A$ = atomization gas velocity    | m/s               |



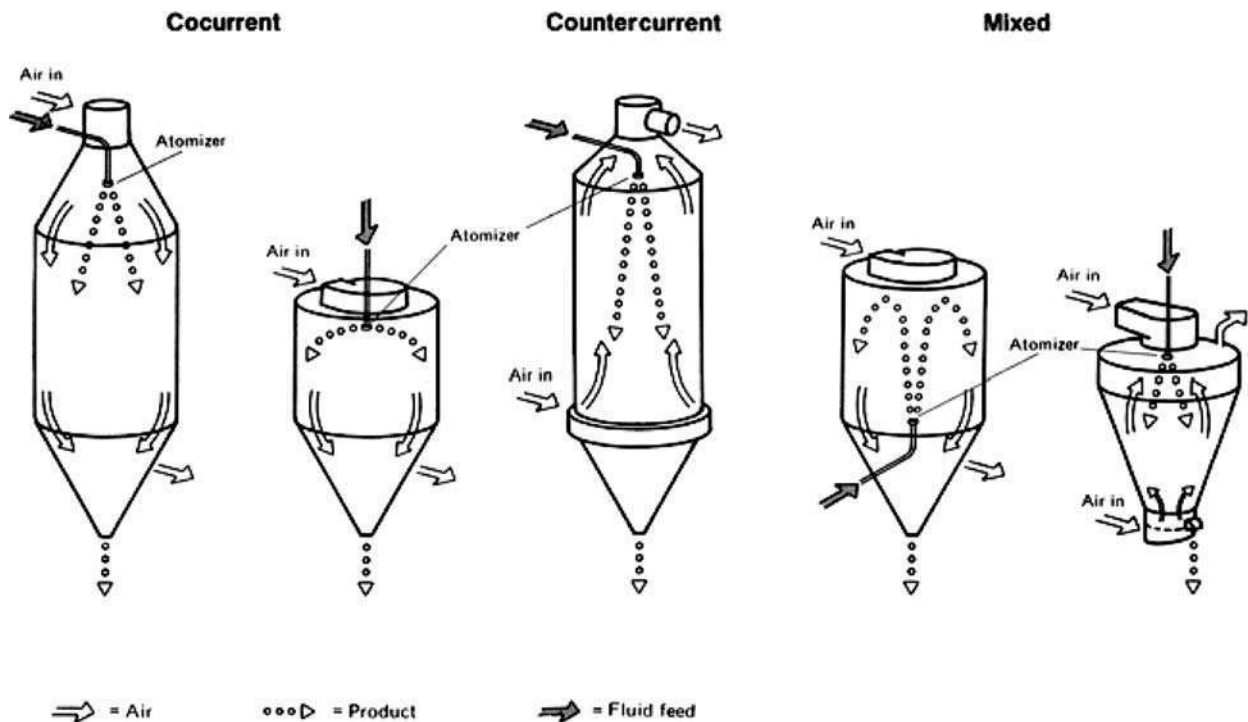


FIG. 12-92 Different forms of spray/hot air contact. (Niro.)

**Spray/Hot Air Contact** Atomization is first and most important process stage in spray drying. The final result of the process does, however, to a very large degree depend on the second stage, the spray/hot air contact. The way the spray of droplets is contacted by the hot air or gas carrying the thermal energy required to evaporate the moisture in the droplets is important for the quality of the product. In general terms three possible forms can be defined. These are as depicted in Fig. 12-92:

- Cocurrent flow
- Countercurrent flow
- Mixed flow

Different drying chamber forms and different methods of hot air introduction accompany the different flow pattern forms and are selected according to

- Required particle size in product specification
- Required particle form
- Temperature or heat sensitivity of the dried particle

In general terms, selection of chamber design and flow pattern form follows these guidelines:

- Use cocurrent spray drying for heat-sensitive products of fine as well as coarse particle size, where the final product temperature must be kept lower than the dryer outlet temperature.
- Use countercurrent spray drying for products which are not heat-sensitive, but may require some degree of heat treatment to obtain a special characteristic, i.e., porosity or bulk density. In this case the final powder temperature may be higher than the dryer outlet temperature.
- Use mixed-flow spray drying when a coarse product is required and the product can withstand short time exposure to heat without adverse effects on dried product quality.

**Evaporation Stage** Evaporation takes place from a moisture film which establishes on the droplet surface. The droplet surface temperature is kept low and close to the adiabatic saturation temper-

ature of the drying air. As the temperature of the drying air drops off and the solids content of the droplet/particle increases, the evaporation rate is reduced. The drying chamber design must provide a sufficient residence time in suspended condition for the particle to enable completion of the moisture removal.

During the evaporation stage the atomized spray droplet size distribution may undergo changes as the droplets shrink, expand, collapse, fracture, or agglomerate.

**Dry Product Recovery** Product recovery is the last stage of the spray drying process. Two distinct systems are used:

- In two-point discharge, primary discharge of a coarse powder fraction is achieved by gravity from the base of the drying chamber. The fine fraction is recovered by secondary equipment downstream of the chamber air exit.
- In single-point discharge, total recovery of dry product is accomplished in the dryer separation equipment.

Collection of powder from an airstream is a large subject of its own. In spray drying, dry collection of powder in a nondestructive way is achieved by use of cyclones, filters with textile bags or metallic cartridges, and electrostatic precipitators.

With the current emphasis on environmental protection, many spray dryers are equipped with additional means to collect even the finest fraction. This collection is often destructive to the powder. Equipment in use are wet scrubbers, bag or other kinds of filters, and in a few cases incinerators.

**Industrial Designs and Systems** Thousands of different products are processed in spray dryers representing a wide range of feed and product properties as well as drying conditions. The flexibility of the spray drying concept, which is the main reason for this wide application, is described by the following systems.

**Plant Layouts** Figure 12-93a shows a standard cocurrent cone-based chamber with roof gas disperser. The chamber can have either single- or two-point discharge and can be equipped with rotary or

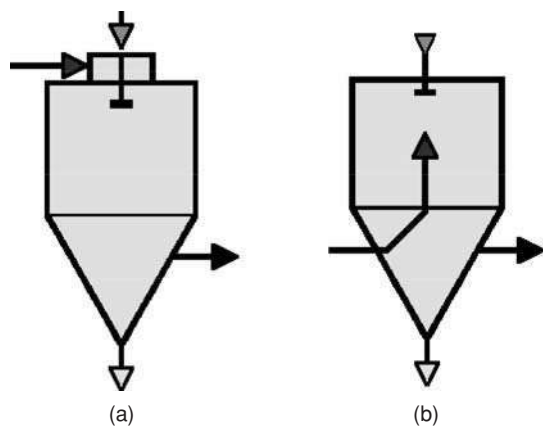


FIG. 12-93 (a) Standard cocurrent and (b) high-temperature chambers.

nozzle atomization. Fine or moderately coarse powders can be produced. This type of dryer finds application in dairy, food, chemical, pharmaceutical, agrochemical, and polymer industries.

Figure 12-93b shows a high-temperature chamber with the hot gas distributor arranged internally on the centerline of the chamber. The atomizer is rotary. Inlet temperature in the range of 600 to 1000°C can be utilized in the drying of non-heat-sensitive products in the chemical and mining industries. Kaolin and mineral flotation concentrates are typical examples.

Figure 12-94a shows a cocurrent cone-based tall form chamber with roof gas disperser. This chamber design is used primarily with pressure nozzle atomization to produce powders of large particle sizes with a minimum of agglomeration. The chamber can be equipped with an oversize cone section to maximize powder discharge from the chamber bottom. This type of dryer is used for dyestuffs, baby foods, detergents, and instant coffee powder.

Figure 12-94b shows a countercurrent flow chamber with pressure nozzle atomization. This design is in limited use because it cannot produce heat-sensitive products. Detergent powder is the main application.

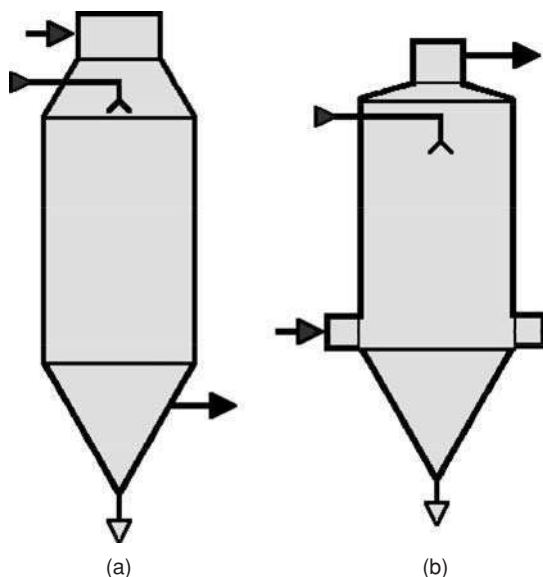


FIG. 12-94 Tall form: (a) cocurrent and (b) countercurrent chambers.

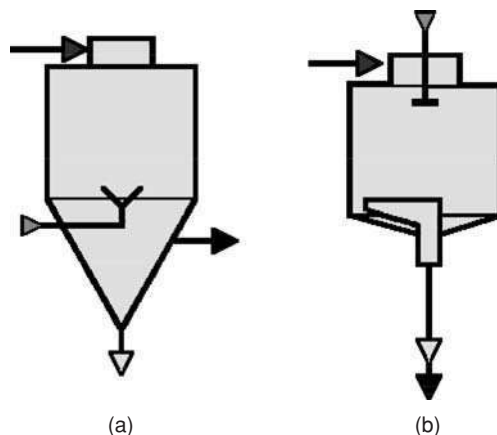


FIG. 12-95 (a) Mixed-flow and (b) flat chambers.

Figure 12-95a shows a mixed-flow chamber with pressure nozzle atomization arranged in so-called fountain mode. This design is ideal for producing a coarse product in a limited-size low-cost drying chamber. This type of dryer is used extensively for ceramic products. Figure 12-95b shows a flat-based cocurrent chamber as used with limited building height. Powder removal requires a sweeping suction device. One of few advantages is ease of access for manual cleaning. These are widely used in production of flavoring materials.

Figure 12-96a shows an integrated fluid-bed chamber which represents the latest development in spray dryer design. The final stage of the drying process is accomplished in a fluid bed located in the lower cone of the chamber. This type of operation allows lower outlet temperatures to be used, leading to fewer temperature effects on the powder and higher energy efficiency. Figure 12-96b shows an integrated belt chamber where product is sprayed onto a moving belt, which also acts as the air exhaust filter. It is highly suitable for slowly crystallizing and high-fat products. Previous operational difficulties derived from hygienic problems on the belt have been overcome, and the integrated belt dryer is now moving the limits of products that can be dried by spray drying.

**Atomization/Gas Disperser Arrangement** Some of the above-mentioned layouts allow a choice of atomization means while others are restricted to a particular choice. The arrangement of the gas distributor means will be closely related to the choice of atomizer. A

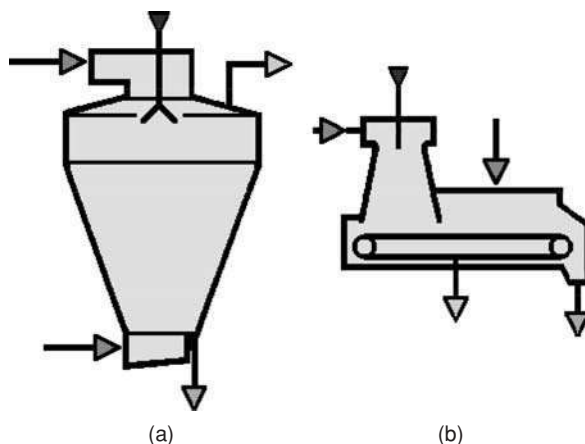


FIG. 12-96 (a) Integrated fluid-bed and (b) belt designs.

rotary atomizer will generally be arranged in a roof gas disperser as suited for the chambers in Figs. 12-93 and 12-95. The hot gas or air enters through a scroll-shaped housing which distributes the air evenly into an annular gap entry with adjustable guide vanes. The geometry and adjustment of the entry gap may determine the success of the drying process. Figure 12-95b shows an alternative arrangement of a rotary atomizer with a central gas disperser such as suited for the high-temperature spray dryer layout.

**Hot Air Supply System** All the above-mentioned chamber layouts can be used in open-cycle, partial recycle, or closed-cycle layouts. The selection is based on the needs of operation, feed, and powder specification and on environmental considerations.

An open-cycle layout is by far the most common in industrial spray drying. The open layout involves intake of drying air from the atmosphere and discharge of exhaust air to the atmosphere. Drying air can be supplemented by a waste heat source to reduce overall fuel consumption. The heater may be direct, i.e., natural gas burner, or indirect by steam-heated heat exchanger.

A closed-cycle layout is used for drying inflammable or toxic solvent feedstocks. The closed-cycle layout ensures complete solvent recovery and prevents explosion and fire risks. The reason for the use of a solvent system is often to avoid oxidation/degradation of the dried product. Consequently closed-cycle plants are gastight installations operating with an inert drying medium, usually nitrogen. These plants operate at a slight gauge pressure to prevent inward leakage of air.

Partial recycle is used in a plant type applied for products of moderate sensitivity toward oxygen. The atmospheric drying air is heated in a direct fuel-burning heater. Part of the exhaust air, depleted of its oxygen content by the combustion, is condensed in a condenser and recycled to the heater. This type of plant is also designated self-inertizing.

**Industrial Applications** As mentioned above, thousands of products are spray dried. The most common products may be classified as follows:

- Agrochemicals
- Catalysts
- Ceramics
- Chemicals
- Dyestuffs
- Foodstuffs
- Pharmaceuticals

Table 12-43 shows some of the operational parameters associated with specific and typical products. For each of these product groups and any other product, successful drying depends on the proper selection of a plant concept and proper selection of operational parameters, in particular inlet and outlet temperatures and the atomization method. These parameters are traditionally established through pilot-scale test work, and leading suppliers on the spray drying market often have extensive test stations to support their sales efforts.

Table 12-43 shows the variety of process parameters used in practical applications of spray drying. The air temperatures are traditionally established through experiments and test work. The inlet temperatures reflect the heat sensitivity of the different products, and the outlet temperatures the willingness of the products to release moisture. The percent water in feed parameter is an indication of feed viscosity

and other properties that influence the pumpability and behavior under atomization of the individual feeds.

As a consequence, the amount of drying air or gas required for drying one unit of feed or product varies considerably. Table 12-43 shows for the individual products the ratio of drying gas to evaporation as well as the ratio of drying gas to product on a mass basis. The calculation behind the table neglects the variation of thermodynamic properties with temperature and the variation of residual moisture in each product.

A quick scoping estimate of the size of an industrial spray dryer can be made on this basis. The required evaporation rate or product rate can be multiplied by the relevant ratio from the table to give the mass flow rate of the drying gas. The next step would be to calculate the size of a spray drying chamber to allow the drying gas at outlet conditions approximately 25 s of residence time. A cylindrical chamber with diameter  $D$  and height  $H$  equal to  $D$  and a 60° conical bottom has a nominal volume of

$$V_{\text{chamber}} = \frac{\pi}{4} D^3 \times \left( H + \frac{\sqrt{3}}{2} D \right) = 1.47 \times D^3$$

Accordingly a zinc sulfate spray dryer with a drying capacity of 2 t/h would require a drying gas flow rate of approximately 8.45 kg/s. With an outlet gas density of 0.89 kg/m<sup>3</sup> and the above-mentioned gas residence time, this results in a required chamber volume of

$$V_{\text{chamber}} = 8.44 \text{ kg/s} / 0.89 \text{ kg/m}^3 \times 25 \text{ s} = 237 \text{ m}^3$$

The chamber size now becomes

$$D = \sqrt[3]{\frac{237}{1.47}} = 5.5 \text{ m}$$

A similar calculation for the other products based on a powder capacity of 2 t/h would reveal a variation of gas flow rates from 8.4 to 114 kg/s and chamber diameters from 5.5 to 12.7 m.

The selection of the plant concept involves the drying modes illustrated in Figs. 12-93 through 12-96. For different products a range of plant concepts are available to secure successful drying at the lowest cost. Three different concepts are illustrated in Figs. 12-97, 12-98, and 12-99.

Figure 12-97 shows a traditional spray dryer layout with a cone-based chamber and roof gas disperser. The chamber has two-point discharge and rotary atomization. The powder leaving the chamber bottom as well as the fines collected by the cyclone is conveyed pneumatically to a conveying cyclone from where the product discharges. A bag filter serves as the common air pollution control system.

Figure 12-98 shows closed-cycle spray dryer layout used to dry certain products with a nonaqueous solvent in an inert gas flow. The background for this may be product sensitivity to water and oxygen or severe explosion risk. Typical products can be tungsten carbide or pharmaceuticals.

Figure 12-99 shows an integrated fluid-bed chamber layout of the type used to produce agglomerated product. The drying process is accomplished in several stages, the first being a spray dryer with atomization. The second stage is an integrated static fluid bed located in the lower cone of the chamber. The final stages are completed in external

**TABLE 12-43 Some Products That Have Been Successfully Spray Dried**

| Product             | Air temperature, K |     | Water in feed, % | Air/evap. ratio, kg/kg | Air/prod. ratio, kg/kg | Product           | Air temperature, K |     | Water in feed, % | Air/evap. ratio, kg/kg | Air/prod. ratio, kg/kg |
|---------------------|--------------------|-----|------------------|------------------------|------------------------|-------------------|--------------------|-----|------------------|------------------------|------------------------|
|                     | In                 | Out |                  |                        |                        |                   | In                 | Out |                  |                        |                        |
| Animal blood        | 440                | 345 | 65               | 27.6                   | 51.3                   | Detergent A       | 505                | 395 | 50               | 25.4                   | 25.4                   |
| Yeast               | 500                | 335 | 86               | 15.7                   | 96.2                   | Detergent B       | 510                | 390 | 63               | 22.8                   | 38.8                   |
| Zinc sulfate        | 600                | 380 | 55               | 12.4                   | 15.2                   | Detergent C       | 505                | 395 | 40               | 25.8                   | 17.2                   |
| Lignin              | 475                | 365 | 63               | 24.3                   | 41.4                   | Manganese sulfate | 590                | 415 | 50               | 16.3                   | 16.3                   |
| Aluminum hydroxide  | 590                | 325 | 93               | 9.7                    | 128.4                  | Aluminum sulfate  | 415                | 350 | 70               | 40.5                   | 94.4                   |
| Silica gel          | 590                | 350 | 95               | 10.9                   | 206.5                  | Urea resin A      | 535                | 355 | 60               | 14.8                   | 22.1                   |
| Magnesium carbonate | 590                | 320 | 92               | 9.5                    | 108.7                  | Urea resin B      | 505                | 360 | 70               | 18.3                   | 42.7                   |
| Tanning extract     | 440                | 340 | 46               | 26.4                   | 22.5                   | Sodium sulfide    | 500                | 340 | 50               | 16.5                   | 16.5                   |
| Coffee extract      | 420                | 355 | 70               | 40.6                   | 94.8                   | Pigment           | 515                | 335 | 73               | 14.4                   | 39.0                   |



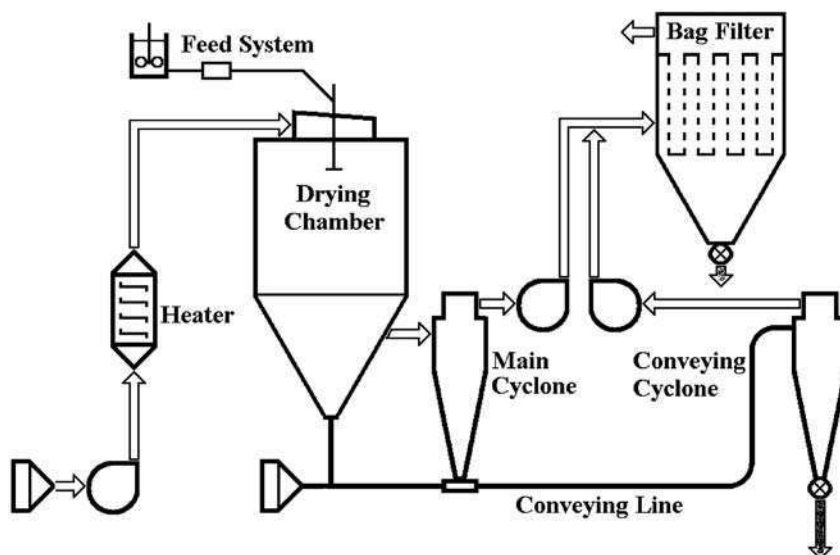


FIG. 12-97 Spray dryer with rotary atomizer and pneumatic powder conveying. (Niro.)

fluid beds of the vibrating type. This type of operation allows lower outlet temperatures to be used, leading to fewer temperature effects on the powder and higher energy efficiency. The chamber has a mixed-flow concept with air entering and exiting at the top of the chamber. This chamber is ideal for heat-sensitive, sticky products. It can be used with pressure nozzle as well as rotary atomization. An important feature is the return of fine particles to the chamber to enhance the agglomeration effect. Many products have been made feasible for spray drying by the development of this concept, which was initially aimed at the food and dairy industry. Recent applications have, however, included dyestuffs, agrochemicals, polymers, and detergents.

#### Additional Reading

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**Pneumatic Conveying Dryers** A gas-solids contacting operation in which the solids phase exists in a dilute condition is termed a *dispersion system*. It is often called a pneumatic system because, in most cases, the quantity and velocity of the gas are sufficient to lift and convey the solids against the forces of gravity and friction. (These systems are sometimes incorrectly called flash dryers when in fact the moisture is not actually "flashed" off. True flash dryers are sometimes used for soap drying to describe moisture removal when pressure is

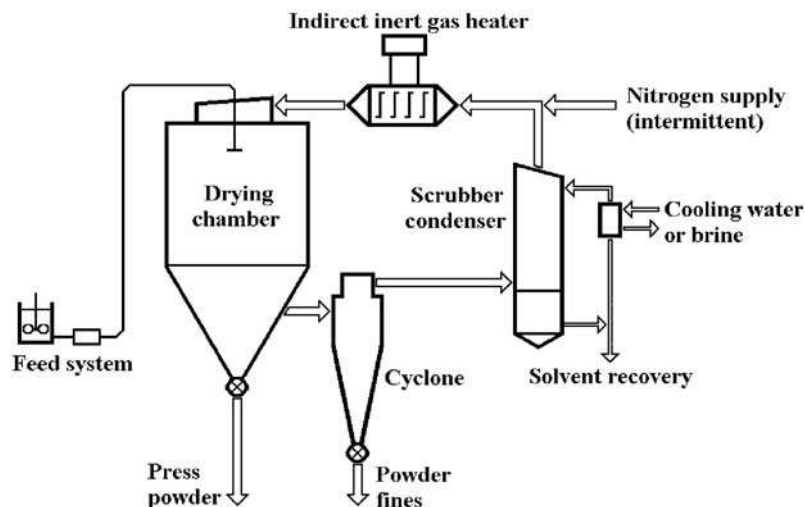


FIG. 12-98 Spray dryer with rotary atomizer and closed-cycle layout. (Niro.)

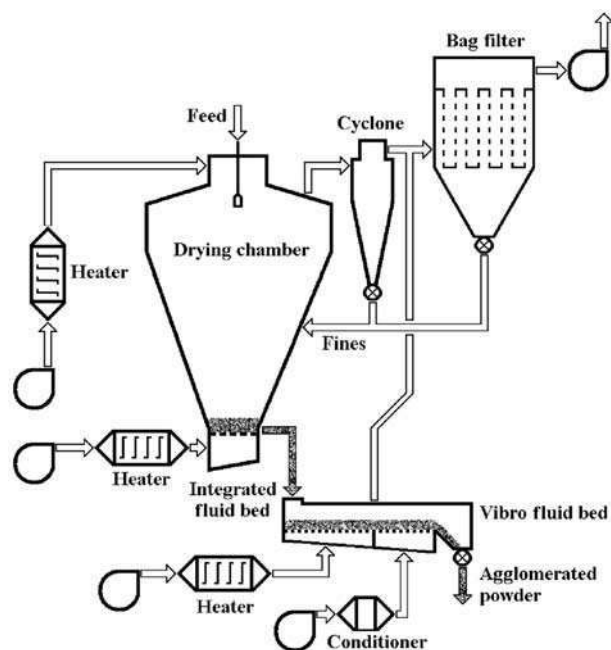


FIG. 12-99 Spray dryer with nozzle atomizer and integrated fluid bed. (Niro.)

quickly reduced.) Pneumatic systems may be distinguished by two characteristics:

1. Retention of a given solids particle in the system is on the average very short, usually no more than a few seconds. This means that any process conducted in a pneumatic system cannot be diffusion-controlled. The reaction must be mainly a surface phenomenon, or the solids particles must be so small that heat transfer and mass transfer from the interiors are essentially instantaneous.

2. On an energy-content basis, the system is balanced at all times; i.e., there is sufficient energy in the gas (or solids) present in the system at any time to complete the work on all the solids (or gas) present at the same time. This is significant in that there is no lag in response to control changes or in starting up and shutting down the system; no partially processed residual solids or gas need be retained between runs.

It is for these reasons that pneumatic equipment is especially suitable for processing heat-sensitive, easily oxidized, explosive, or flammable materials which cannot be exposed to process conditions for extended periods.

Gas flow and solids flow are usually cocurrent, one exception being a countercurrent flow spray dryer. The method of gas-solids contacting is best described as through-circulation; however, in the dilute condition, solids particles are so widely dispersed in the gas that they exhibit apparently no effect upon one another, and they offer essentially no resistance to the passage of gas among them.

**Pneumatic Conveyor Dryers** Pneumatic conveyor dryers, often also referred to as flash dryers, comprise a long tube or duct carrying a gas at high velocity, a fan to propel the gas, a suitable feeder for addition and dispersion of particulate solids in the gas stream, and a cyclone collector or other separation equipment for final recovery of solids from the gas.

The solids feeder may be of any type: Screw feeders, venturi sections, high-speed grinders, and dispersion mills are employed. For pneumatic conveyors, selection of the correct feeder to obtain thorough initial dispersion of solids in the gas is of major importance. For example, by employing an air-swept hammer mill in a drying operation, 65 to 95 percent of the total heat may be transferred within the mill itself if all the drying gas is passed through it. Fans may be of the

induced-draft or the forced-draft type. The former is usually preferred because the system can then be operated under a slight negative pressure. Dust and hot gas will not be blown out through leaks in the equipment. Cyclone separators are preferred for low investment. If maximum recovery of dust or noxious fumes is required, the cyclone may be followed by a wet scrubber or bag collector.

In ordinary heating and cooling operations, during which there is no moisture pickup, continuous recirculation of the conveying gas is frequently employed. Also, solvent recovery operations employing continuously recirculated inert gas with intercondensers and gas reheaters are carried out in pneumatic conveyors.

Pneumatic conveyors are suitable for materials which are granular and free-flowing when dispersed in the gas stream, so they do not stick on the conveyor walls or agglomerate. Sticky materials such as filter cakes may be dispersed and partially dried by an air-swept disintegrator in many cases. Otherwise, dry product may be recycled and mixed with fresh feed, and then the two dispersed together in a disintegrator. Coarse material containing internal moisture may be subjected to fine grinding in a hammer mill. The main requirement in all applications is that the operation be instantaneously completed; internal diffusion of moisture must not be limiting in drying operations, and particle sizes must be small enough that the thermal conductivity of the solids does not control during heating and cooling operations. Pneumatic conveyors are rarely suitable for abrasive solids. Pneumatic conveying can result in significant particle size reduction, particularly when crystalline or other friable materials are being handled. This may or may not be desirable but must be recognized if the system is selected. The action is similar to that of a fluid-energy grinder.

Pneumatic conveyors may be single-stage or multistage. The former is employed for evaporation of small quantities of surface moisture. Multistage installations are used for difficult drying processes, e.g., drying heat-sensitive products containing large quantities of moisture and drying materials initially containing internal as well as surface moisture.

Typical single- and two-stage drying systems are illustrated in Figs. 12-100, 12-101, and 12-102. Figure 12-100 illustrates the flow diagram of a single-stage dryer with a paddle mixer, a screw conveyor followed by a rotary disperser for introduction of the feed into the airstream at the throat of a venturi section. The drying takes place in the drying column after which the dry product is collected in a cyclone. A diverter introduces the option of recycling part of the product into the mixer in order to handle somewhat sticky products. The environmental requirements are met with a wet scrubber in the exhaust stream.

Figure 12-101 illustrates a two-stage dryer where the initial feed material is dried in a flash dryer by using the spent drying air from the second stage. This semidried product is then introduced into the second-stage flash dryer for contact with the hottest air. This concept is in use in the pulp and paper industry. Its use is limited to materials that are dry enough on the surface after the first-stage to avoid plugging of the first-stage cyclone. The main advantage of the two-stage concept is the heat economy which is improved considerably over that of the single-stage concept.

Figure 12-102 is an elevation view of an actual single-stage dryer, employing an integral coarse-fraction classifier, used to separate undried particles for recycle.

Several typical products dried in pneumatic conveyors are described in Table 12-44.

**Design methods for pneumatic conveyor dryers** Depending upon the temperature sensitivity of the product, inlet air temperatures between 125 and 750°C are employed. With a heat-sensitive solid, a high initial moisture content should permit use of a high inlet air temperature. Evaporation of surface moisture takes place at essentially the wet-bulb air temperature. Until this has been completed, by which time the air will have cooled significantly, the surface-moisture film prevents the solids temperature from exceeding the wet-bulb temperature of the air. Pneumatic conveyors are used for solids having initial moisture contents ranging from 3 to 90 percent, wet basis. The air quantity required and solids-to-gas loading are fixed by the moisture load, the inlet air temperature, and, frequently, the exit air humidity. If the last is too great to permit complete drying, i.e., if the

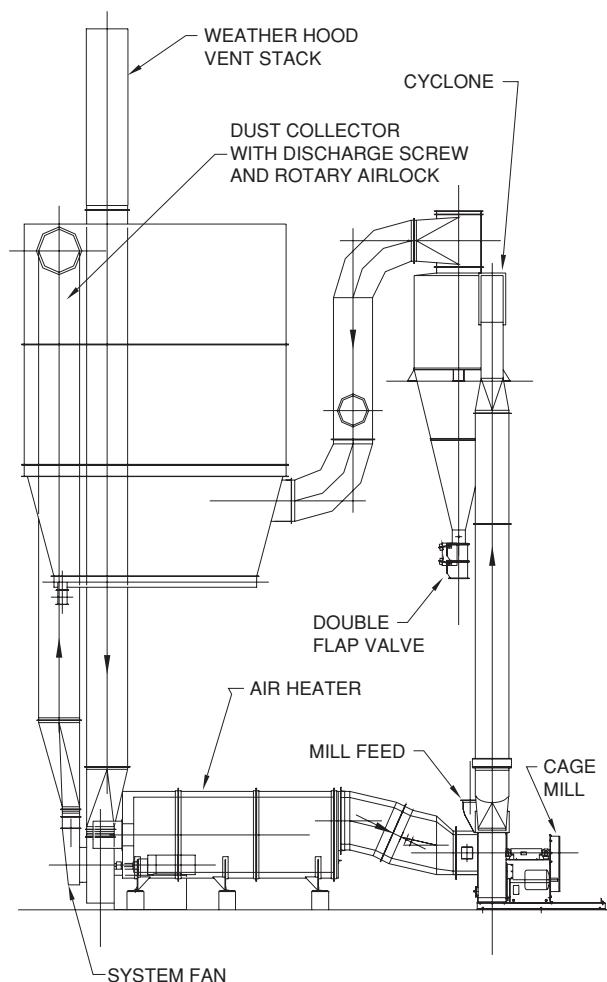


FIG. 12-100 Flow diagram of single-stage flash dryer. (Air Preheater Company, Raymond® & Bartlett Snow™ Products.)

exit air humidity is above that in equilibrium with the product at required dryness, then the solids/gas ratio must be reduced together with the inlet air temperature.

The gas velocity in the conveying duct must be sufficient to convey the largest particle. This may be calculated accurately by methods given in Sec. 17, "Gas-Solids Operations and Equipment." For estimating purposes, a velocity of 25 m/s, calculated at the exit air temperature, is frequently employed. If mainly surface moisture is present, the temperature driving force for drying will approach the log mean of the inlet and exit gas wet-bulb depressions. (The exit solids temperature will approach the exit gas dry-bulb temperature.)

Observation of operating conveyors indicates that the solids are rarely uniformly dispersed in the gas phase. With infrequent exceptions, the particles move in a streaklike pattern, following a streamline along the duct wall where the flow velocity is at a minimum. Complete or even partial diffusion in the gas phase is rarely experienced even with low-specific-gravity particles. Air velocities may approach 20 to 30 m/s. It is doubtful, however, that even finer and lighter materials reach more than 80 percent of this speed, while heavier and larger fractions may travel at much slower rates [Fischer, *Mech. Eng.*, 81(11): 67-69 (1959)]. Very little information and few operating data

on pneumatic conveyor dryers which would permit a true theoretical basis for design have been published.

Therefore, firm design always requires pilot tests. It is believed, however, that the significant velocity effect in a pneumatic conveyor is the difference in velocities between gas and solids, which is strongly linked to heat- and mass-transfer coefficients and is the reason why a major part of the total drying actually occurs in the feed input section.

For estimating purposes, the conveyor cross-section is fixed by the assumed air velocity and quantity. The standard scoping design method is used, obtaining the required gas flow rate from a heat and mass balance, and the duct cross-sectional area and diameter from the gas velocity (if unknown, a typical value is 20 m/s). An incremental mode may be used to predict drying conditions along the duct. However, several parameters are hard to obtain, and conditions change rapidly near the feed point. Hence, for reliable estimates of drying time and duct length, pilot-plant tests should always be used. A conveyor length larger than 50 diameters is rarely required. The length of the full-scale dryer should always be somewhat larger than required in pilot-plant tests, because wall effects are higher in small-diameter ducts. This gives greater relative velocity (and thus higher heat transfer) and lower particle velocity in the pilot-plant dryer, both effects giving a shorter length than the full-scale dryer for a given amount of drying. If desired, the length difference on scale-up can be predicted by using the incremental model and using the pilot-plant data to back-calculate the uncertain parameters; see Kemp, *Drying Technol.* 12(1&2):279 (1994) and Kemp and Oakley (2002).

An alternative method of estimating dryer size very roughly is to estimate a volumetric heat-transfer coefficient [typical values are around 2000 J/(m<sup>3</sup>·s·K)] and thus calculate dryer volume.

Pressure drop in the system may be computed by methods described in Sec. 6, "Fluid and Particle Dynamics." To prevent excessive leakage into or out of the system, which may have a total pressure drop of 2000 to 4000 Pa, rotary air locks or screw feeders are employed at the solids inlet and discharge.

The conveyor and collector parts are thoroughly insulated to reduce heat losses in drying and other heating operations. Operating control is maintained usually by control of the exit gas temperature, with the inlet gas temperature varied to compensate for changing feed conditions. A constant solids feed rate must be maintained.

**Ring Dryers** The ring dryer is a development of flash, or pneumatic conveyor, drying technology, designed to increase the versatility of application of this technology and overcome many of its limitations.

One of the great advantages of flash drying is the very short retention time, typically no more than a few seconds. However, in a conventional flash dryer, residence time is fixed, and this limits its application to materials in which the drying mechanism is not diffusion-controlled and where a range of moisture within the final product is acceptable. The ring dryer offers two advantages over the flash dryer. First, residence time is controlled by the use of an adjustable internal classifier that allows fine particles, which dry quickly, to leave while larger particles, which dry slowly, have an extended residence time within the system. Second, the combination of the classifier with an internal mill can allow simultaneous grinding and drying with control of product particle size and moisture. Available with a range of different feed systems to handle a variety of applications, the ring dryer provides wide versatility.

The essential difference between a conventional flash dryer and the ring dryer is the manifold centrifugal classifier. The manifold provides classification of the product about to leave the dryer by using differential centrifugal force. The manifold, as shown in Fig. 12-103, uses the centrifugal effect of an airstream passing around the curve to concentrate the product into a moving layer, with the dense material on the outside and the light material on the inside.

This enables the adjustable splitter blades within the manifold classifier to segregate the denser, wetter material and return it for a further circuit of drying. Fine, dried material is allowed to leave the dryer with the exhaust air and to pass to the product collection system. This selective extension of residence time ensures a more evenly dried material than is possible from a conventional flash

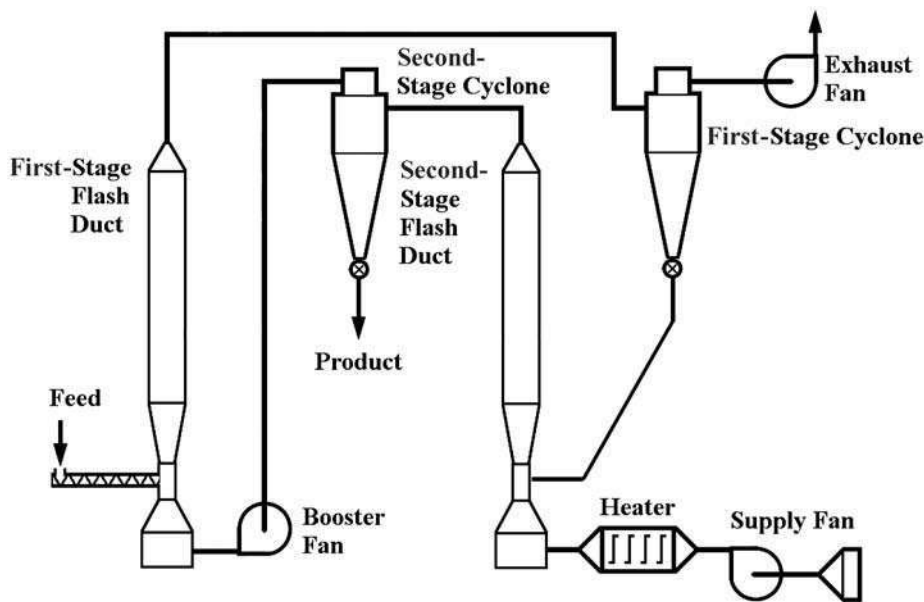


FIG. 12-101 Flow diagram of countercurrent two-stage flash dryer. (Niro.)

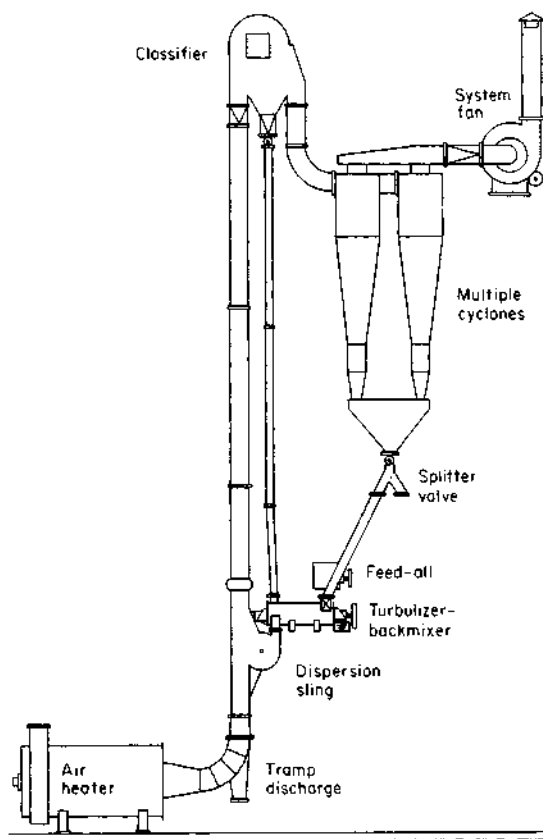


FIG. 12-102 Flow diagram of Strong Scott flash dryer with integral coarse-fraction classifier. (Bepex Corp.)

dryer. Many materials that have traditionally been regarded as difficult to dry can be processed to the required moisture content in a ring dryer. The recycle requirements of products in different applications can vary substantially depending upon the scale of operation, ease of drying, and finished-product specification. The location of reintroduction of undried material back into the drying medium has a significant impact upon the dryer performance and final-product characteristics.

Three configurations of the ring dryer have been developed to offer flexibility in design and optimal performance:

1. *Single-stage manifold-vertical configuration* The feed ring dryer (see Fig. 12-104) is similar to a flash dryer but incorporates a single-stage classifier, which diverts 40 to 60 percent of the product back to the feed point. The feed ring dryer is ideally suited for materials which neither are heat-sensitive nor require a high degree of classification. An advantage of this configuration is that it can be manufactured to very large sizes to achieve high evaporative capacities.

2. *Full manifold-horizontal configuration* The full ring dryer (see Fig. 12-105) incorporates a multistage classifier which allows much higher recycle rates than the single-stage manifold. This configuration usually incorporates a disintegrator which provides adjustable amounts of product grinding depending upon the speed and manifold setting. For sensitive or fine materials, the disintegrator can be omitted. Alternative feed locations are available to suit the material sensitivity and the final-product requirements. The full ring configuration gives a very high degree of control of both residence time and particle size, and is used for a wide variety of applications from small production rates of pharmaceutical and fine chemicals to large production rates of food products, bulk chemicals, and minerals. This is the most versatile configuration of the ring dryer.

3. *P-type manifold-vertical configuration* The P-type ring dryer (see Fig. 12-106) incorporates a single-stage classifier and was developed specifically for use with heat-sensitive materials. The undried material is reintroduced into a cool part of the dryer in which it recirculates until it is dry enough to leave the circuit.

An important element in optimizing the performance of a flash or ring dryer is the degree of dispersion at the feed point. Maximizing the product surface area in this region of highest evaporative

TABLE 12-44 Typical Products Dried in Pneumatic Conveyor Dryers (Barr-Rosin)

| Material                       | Initial moisture, wet basis, % | Final moisture, wet basis, % | Plant configuration  |
|--------------------------------|--------------------------------|------------------------------|----------------------|
| Expandable polystyrene beads   | 3                              | 0.1                          | Single-stage flash   |
| Coal fines                     | 23                             | 1.0                          | Single-stage flash   |
| Polycarbonate resin            | 25                             | 10                           | Single-stage flash   |
| Potato starch                  | 42                             | 20                           | Single-stage flash   |
| Aspirin                        | 22                             | 0.1                          | Single-stage flash   |
| Melamine                       | 20                             | 0.05                         | Single-stage flash   |
| Com gluten meal                | 60                             | 10                           | Feed-type ring dryer |
| Maize fiber                    | 60                             | 18                           | Feed-type ring dryer |
| Distillers dried grains (DDGs) | 65                             | 10                           | Feed type ring dryer |
| Vital wheat gluten             | 70                             | 7                            | Full-ring dryer      |
| Casein                         | 50                             | 10                           | Full-ring dryer      |
| Tricalcium phosphate           | 30                             | 0.5                          | Full-ring dryer      |
| Zeolite                        | 45                             | 20                           | Full-ring dryer      |
| Orange peels                   | 82                             | 10                           | Full-ring dryer      |
| Modified com starch            | 40                             | 10                           | P-type ring dryer    |
| Methylcellulose                | 45                             | 25                           | P-type ring dryer    |

driving force is a key objective in the design of this type of dryer. Ring dryers are fed using similar equipment to conventional flash dryers. Ring dryers with vertical configuration are normally fed by a flooded screw and a disperser which propels the wet feed into a high-velocity venturi, in which the bulk of the evaporation takes place. The full ring dryer normally employs an air-swept disperser or mill within the drying circuit to provide screenless grinding when required. Together with the manifold classifier this ensures a product with a uniform particle size. For liquid, slurry, or pasty feed materials, backmixing of the feed with a portion of the dry product will be carried out to produce a conditioned friable material. This further increases the versatility of the ring dryer, allowing it to handle sludge and slurry feeds with ease.

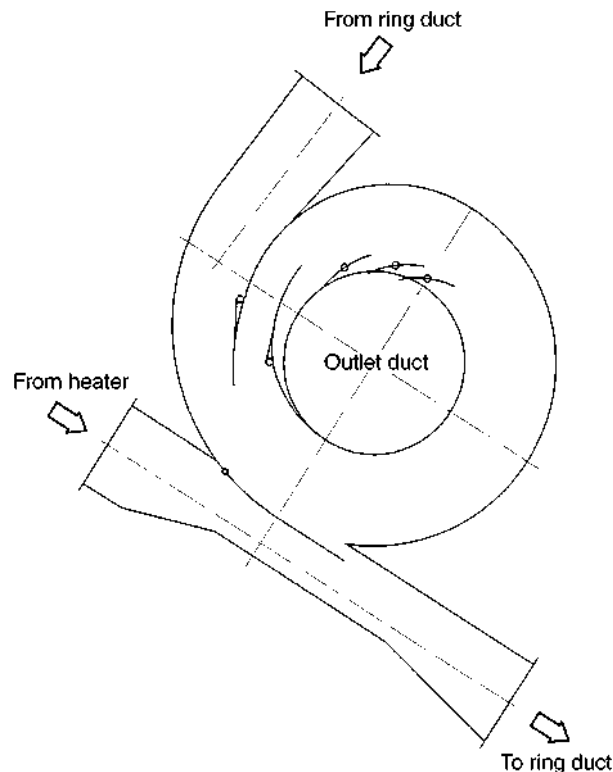


FIG. 12-103 Full manifold classifier for ring dryer. (Barr-Rosin.)

Dried product is collected in either cyclones or bag filters depending upon the product-particle properties. When primary collection is carried out in cyclones, secondary collection in a bag filter or scrubber is usually necessary to comply with environmental regulations. A rotary valve is used to provide an air lock at the discharge point. Screws are utilized to combine product from multiple cyclones or large bag filters. If required, a portion of the dried product is separated from the main stream and returned to the feed system for use as backmix.

**Design methods for ring dryers** Depending on the temperature sensitivity of the material to be processed, air inlet temperatures as high as 750°C can be utilized. Even with heat-sensitive solids, high feed moisture content may permit the use of high air inlet temperature since evaporation of surface moisture takes place at the wet-bulb air temperature. Until the surface moisture has been removed, it will prevent the solids temperature from exceeding the air wet-bulb temperature, by which time the air will generally have cooled significantly. Ring dryers have been used to process materials with feed moisture contents between 2 and 95 percent, weight fraction. The product moisture content has been controlled to values from 20 percent down to less than 1 percent.

The air velocity required and air/solids ratio are determined by the evaporative load, the air inlet temperature, and the exhaust air humidity. Too high an exhaust air humidity would prevent complete drying, so then a lower air inlet temperature and air/solids ratio would be required. The air velocity within the dryer must be sufficient to convey the largest particle, or agglomerate. The air/solids ratio must be high enough to convey both the product and backmix, together with internal recycle from the manifold. For estimating purposes a velocity of 25 m/s, calculated at dryer exhaust conditions, is appropriate both for pneumatic conveyor and ring dryers.

**Agitated Flash Dryers** Agitated flash dryers produce fine powders from feeds with high solids contents, in the form of filter cakes, pastes, or thick, viscous liquids. Many continuous dryers are unable to dry highly viscous feeds. Spray dryers require a pumpable feed. Conventional flash dryers often require backmixing of dry product to the feed in order to fluidize. Other drying methods for viscous pastes and filter cakes are well known, such as contact, drum, band, and tray dryers. They all require long processing time, large floor space, high maintenance, and aftertreatment such as milling.

The agitated flash dryer offers a number of process advantages, such as ability to dry pastes, sludges, and filter cakes to a homogeneous, fine powder in a single-unit operation; continuous operation; compact layout; effective heat- and mass-transfer short drying times; negligible heat loss and high thermal efficiency; and easy access and cleanliness.

The agitated flash dryer (Fig. 12-107) consists of four major components: feed system, drying chamber, heater, and exhaust air system. Wet feed enters the feed tank, which has a slow-rotating impeller to break up large particles. The level in the feed tank is maintained by a



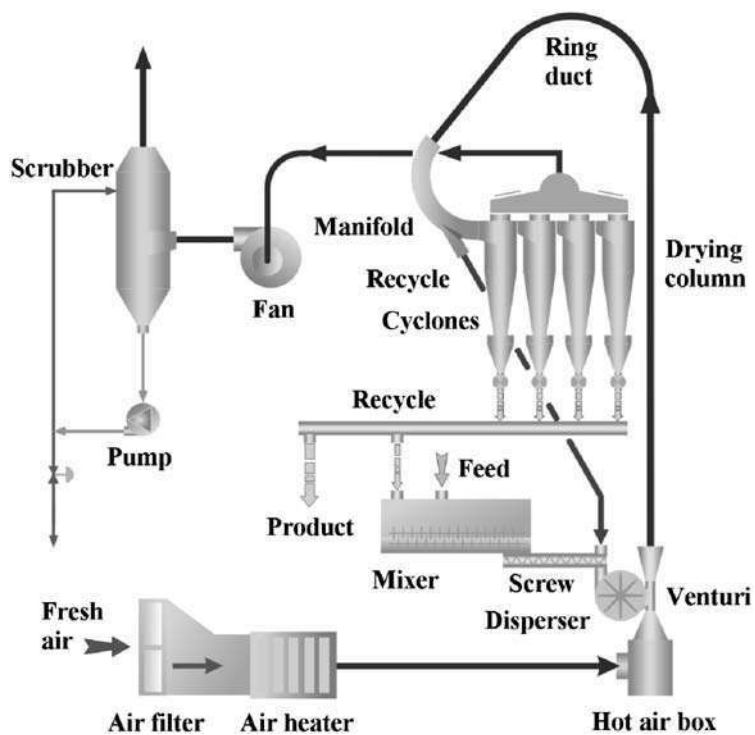


FIG. 12-104 Flow diagram of feed-type ring dryer. (Barr-Rosin.)

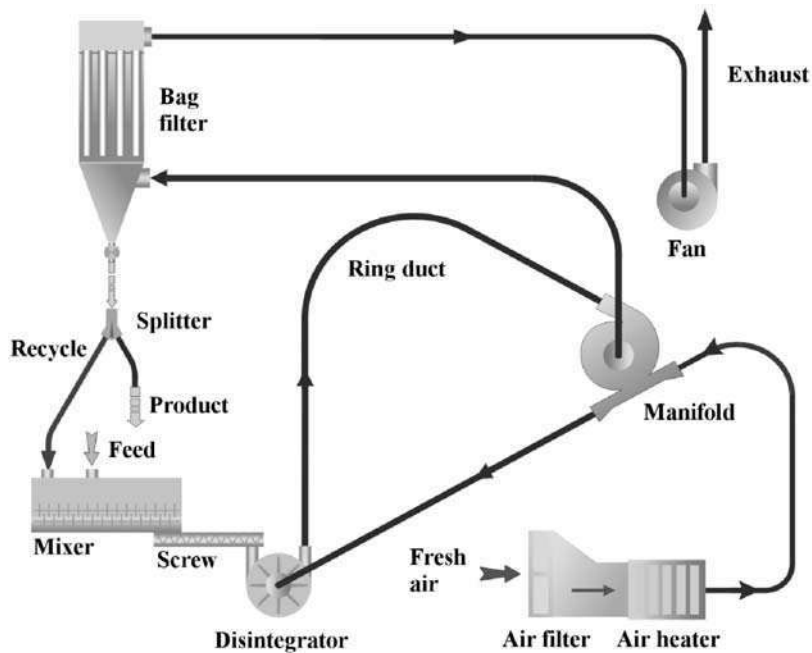


FIG. 12-105 Flow diagram of full manifold-type ring dryer. (Barr-Rosin.)

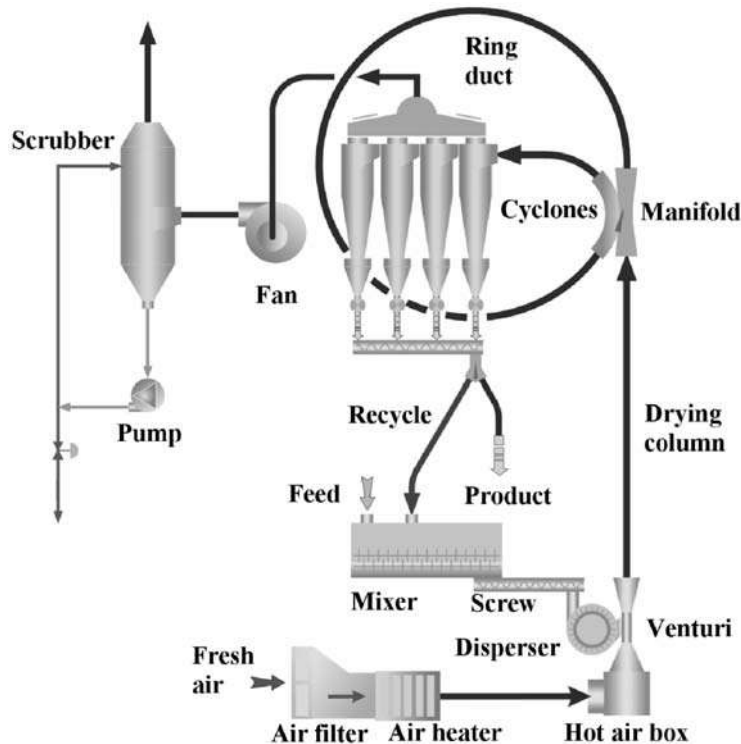


FIG. 12-106 Flow diagram of P-type ring dryer. (Barr-Rosin.)

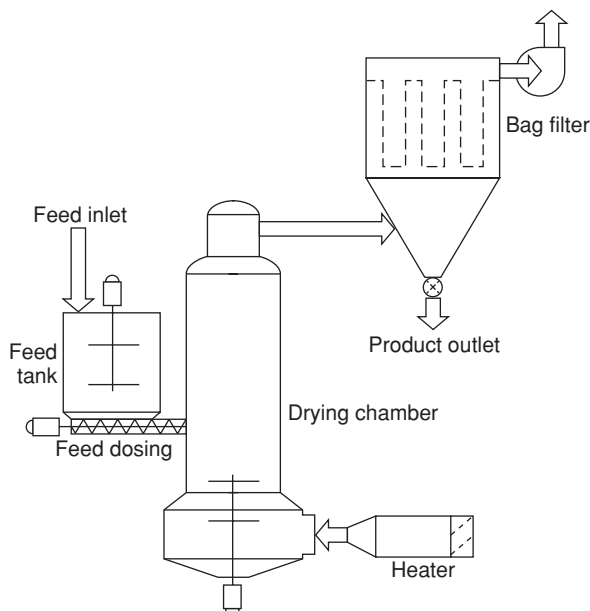


FIG. 12-107 Agitated flash dryer with open cycle. (Niro, Inc.)

level controller. The feed is metered at a constant rate into the drying chamber via a screw conveyor mounted under the feed tank. If the feed is shear thinning and can be pumped, the screw feeder can be replaced by a positive displacement pump.

The drying chamber is the heart of the system consisting of three important components: air disperser, rotating disintegrator, and drying section. Hot, drying air enters the air disperser tangentially and is introduced into the drying chamber as a swirling airflow. The swirling airflow is established by a guide-vane arrangement. The rotating disintegrator is mounted at the base of the drying chamber. The feed, exposed to the hot, swirling airflow and the agitation of the rotating disintegrator, is broken up and dried. The fine dry particles exit with the exhaust air and are collected in the bag filter. The speed of the rotating disintegrator controls the particle size. The outlet air temperature controls the product moisture content.

The drying air is heated either directly or indirectly, depending upon the feed material, powder properties, and available fuel source. The heat sensitivity of the product determines the drying air temperature. The highest possible value is used to optimize thermal efficiency. A bag filter is usually recommended for collecting the fine particles produced. The exhaust fan maintains a slight vacuum in the dryer, to prevent powder leakage into the surroundings. The appropriate process system is selected according to the feed and powder characteristics, available heating source, energy utilization, and operational health and safety requirements.

Open systems use atmospheric air for drying. In cases where products pose a potential for dust explosion, plants are provided with pressure relief or suppression systems. For recycle systems, the drying system medium is recycled, and the evaporated solvent recovered as condensate. There are two alternative designs. In the self-inertizing mode, oxygen content is held below 5 percent by combustion control at the heater. This is recommended for products with serious dust



explosion hazards. In the inert mode, nitrogen is the drying gas. This is used when an organic solvent is evaporated or product oxidation during drying must be prevented.

**Design methods** The size of the agitated flash dryer is based on the evaporation rate required. The operating temperatures are product-specific. Once established, they determine the airflow requirements. The drying chamber is designed based on air velocity (approximately 3 to 4 m/s) and residence time (product-specific).

### Other Dryer Types

**Freeze Dryer** Industrial freeze drying is carried out in two steps:

1. Freezing of the food or beverage product
2. Freeze drying, i.e., sublimation drying of the ice content and desorption drying of the bound or crystal water content

Freeze drying differs from conventional drying in that when ice is sublimated, only water vapor is transported within the product, causing no displacement of soluble substances such as sugars, salts, and acids. In all conventional drying systems in which water is dried, the water containing the soluble substances is transported to the product surface by capillary action. The water will evaporate from the surface, leaving the soluble substances displaced on the product surface. The major advantages of freeze drying are therefore

- Preservation of original flavor, aroma, color, shape, and texture
- Very little shrinkage, resulting in excellent and instant rehydration characteristics
- Negligible product loss
- Minimal risk of cross-contamination

The freeze drying process is today used widely for a number of products including vegetables, fruits, meat, fish, and beverage products, such as

- Instant coffee for which excellent flavor and aroma retention are of special importance
- Strawberries for which excellent color preservation is of special importance
- Chives for which shape preservation is of special importance

**Freezing** The freezing methods applied for solid products are all conventional freezing methods such as blast freezing, individual quick freezing (IQF), or similar.

The products maintain their natural cell structure, and the aim is to freeze the free water to pure ice crystals, leaving the soluble substances as high concentrates or even crystallized. To ensure good stability of the product during storage, a product temperature of  $-20$  to  $-30^{\circ}\text{C}$  should be achieved to ensure that more than 95 percent of the free water is frozen.

Liquid products have no cell structure, thus the structure of the freeze dried products is formed by the freezing process. The intercrystalline matrix of the concentrated product giving the structure of the freeze dried product is formed around the ice crystals. The size of the ice crystals is a function of the freezing time. Quick freezing results in small ice crystals, slow freezing in large ice crystals. The structure of the matrix determines the freeze drying performance as well as the appearance, mechanical strength, and solubility rate. Small ice crystals lead to light color (high surface reflection of light), diffusion restrictions for vapor transport inside the product, and a good mechanical strength of the freeze dried product. Large ice crystals lead to the opposite results.

Thus the freezing method must be carefully adapted to the quality criteria of the finished product. The preferred methods are

- Drum freezing, by which a thin slab of 1.5 to 3 mm is frozen within 1.5 to 3 min
- Belt freezing, by which a slab of 6 to 10 mm passing through different freezing zones is frozen during 10 to 20 min
- Foaming, used to influence the structure and mainly to control the density of the freeze dried product

**Freeze drying** Freeze drying of foods takes place in a freeze dryer at vacuum levels of 0.4 to 1.3 mbar absolute, corresponding to sublimation temperatures from  $-30$  to  $-17^{\circ}\text{C}$  depending on the product requirements. The main components of the freeze dryer are

- The vacuum chamber, heating plates, and vapor traps, all built into the freeze dryer

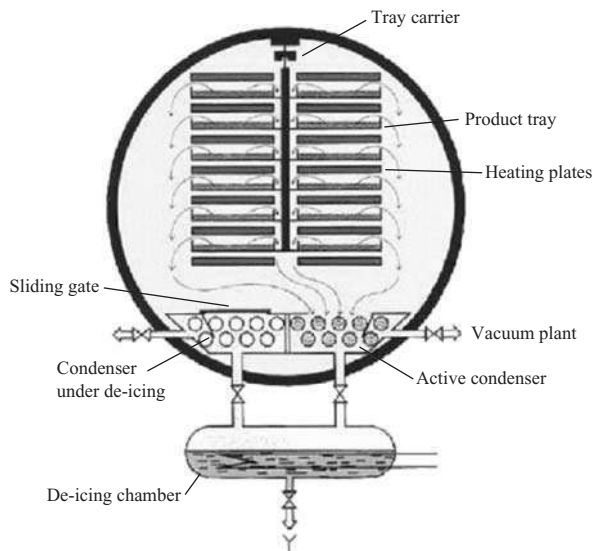


FIG. 12-108 Cross-section of RAY™ batch freeze dryer. (Niro A/S.)

- The external systems, such as the transport system for the product trays, the deicing system, and the support systems for supply of heat, vacuum, and refrigeration

**Batch freeze drying** The frozen product is carried in trays, and the trays are carried in tray trollies suspended in an overhead rail system for easy transport and quick loading and unloading. The freeze dryer as illustrated in Fig. 12-108 is charged with 3 to 6 trolley loads depending on the size of the freeze dryer. The trollies place the trays between the heating plates for radiation heat transfer. Radiation is preferred to ensure an even heat transfer over the large heating surface, typically  $2 \times (70 \text{ to } 140 \text{ m}^2)$ . The distribution of the heating medium (water or thermal oil) to the heating plates and the flow rate inside the plates are very important factors. To avoid uneven drying, the surface temperature difference of the heating plates should not exceed  $2$  to  $3^{\circ}\text{C}$  at maximum load.

When the loading is completed, the freeze dryer is closed and vacuum applied.

The operation vacuum should be reached quickly (within 10 min) to avoid the risk of product melting. For the same reason, the heating plates are cooled to approximately  $25^{\circ}\text{C}$ . When the operation vacuum is achieved, the heating plate temperature is raised quickly to the maximum drying temperature restricted by the capacity of the vapor traps, to perform the sublimation drying as quickly as possible for capacity reasons. During this period, the product is kept cool by the sublimation, and approximately 75 to 80 percent of the free water is sublimated.

The capability of the freeze drying plant to perform during this period is vital for efficient operation. To maintain the required sublimation temperature, the surface temperature of the ice layer on the vapor trap condenser must compensate for the pressure loss of the vapor flow from the sublimation front to the condenser.

The evaporation temperature of the refrigerant must further compensate for the temperature difference through the ice layer to the evaporating refrigerant.

With the flow rate at 1 mbar of approximately  $1 \text{ m}^3/(\text{s} \cdot \text{m}^2 \text{ of tray area})$ , the thermodynamic design of the vapor trap is the main issue for a well-designed freeze dryer.

A built-in vapor trap allowing a large opening for the vapor flow to the condenser and a continuous deicing (CDI) system, reducing the ice layer on the condenser to a maximum of 6 to 8 mm, are important features of a modern freeze drying plant. Approximately 75 percent of the energy costs relate to the refrigeration plant, and if the requirement

TABLE 12-45 Freeze Dryer, Performance Data, Niro RAY™ and CONRAD™ Types

|                                | Tray area, m <sup>2</sup> | Typical sublimation capacity |                   | Electricity consumption,<br>kWh/kg. sublimated | Steam consumption,<br>kg/kg sublimated |
|--------------------------------|---------------------------|------------------------------|-------------------|------------------------------------------------|----------------------------------------|
|                                |                           | Flat tray, kg/h              | Ribbed tray, kg/h |                                                |                                        |
| RAY Batch Plant—1 mbar         |                           |                              |                   |                                                |                                        |
| RAY 75                         | 68                        | 68                           | 100               | 1.1                                            | 2.2                                    |
| RAY 100                        | 91                        | 91                           | 136               | 1.1                                            | 2.2                                    |
| RAY 125*                       | 114                       | 114                          | 170               | 1.1                                            | 2.2                                    |
| CONRAD Continuous Plant—1 mbar |                           |                              |                   |                                                |                                        |
| CONRAD 300                     | 240                       | 240                          | 360               | 1.0                                            | 2.0                                    |
| CONRAD 400                     | 320                       | 320                          | 480               | 1.0                                            | 2.0                                    |
| CONRAD 500*                    | 400                       | 400                          | 600               | 1.0                                            | 2.0                                    |

\*Other sizes available.

of the evaporation temperature is 10°C lower than optimum, the energy consumption of the refrigeration plant will increase by approximately 50 percent.

At the end of the sublimation drying, the product surface temperature reaches the maximum allowable product temperature, requiring that the temperature of the heating plates be lowered gradually, and the drying will change to desorption drying. The temperature will finally be kept constant at the level of the maximum allowable product temperature until the residual moisture has been reduced to 2 to 3 percent, which is a typical level for a freeze dried product.

**Continuous freeze drying** From the description of batch freeze drying, it can be seen that the utility requirements vary considerably. During sublimation drying the requirements are 2 to 2.5 times the average requirement. To overcome this peak load and to meet the market request for high unit capacities, continuous freeze dryer designs have been developed. The special features are twofold:

- The tray transport system is a closed-loop system in which the trays pass one by one under the tray filler, where frozen product is automatically filled into the trays at a preset weight. The full tray is charged to the vacuum lock which is then evacuated to the drier vacuum level. Then the tray is pushed into the dryer and grabbed by an elevator which is filled stepwise with a stack of trays. Next a full stack of trays is pushed into the drying area whereby each of the stacks inside the drying area will move one step forward. Thus the last stack containing the finished, freeze dried product will be pushed out of the drying area to an outlet elevator which will be emptied stepwise by discharge of the trays through the outlet vacuum lock. From the outlet vacuum lock the trays are pushed to the emptying station for emptying and then returned to the tray filler.
- As the tray stacks are pushed forward through the freeze dryer, they pass through various temperature zones. The temperature zones form the heating profile, high temperatures during the sublimation drying, medium temperatures during the transition period toward desorption drying, and low temperatures during the final desorption drying. The temperature profile is selected so that overheating of the dry surface is avoided.

**Design methods** The size of the freeze drying plant is based on the average sublimation capacity required as well as on the product type and form. The external systems for batch plants must be designed for a peak load of 2 to 2.5 times the average capacity in the case of a single plant. Further, a batch plant is not available for drying all the time. A modern batch freeze dryer with the CDI system loses approximately 30 min per batch. Typically, 2 to 3 batches will be freeze dried per day. The evaporation temperature of the refrigeration plant depends on the required vacuum. At 1 mbar it will be -35 to -40°C depending on the vapor trap performance. Sample data are shown in Table 12-45.

#### Field Effects Drying—Drying with Infrared, Radio-Frequency, and Microwave Methods

**Dielectric Methods (Radio-Frequency and Microwave)** Schiffmann (1995) defines dielectric (radio-frequency) frequencies as covering the range of 1 to 100 MHz, while microwave frequencies range from 300 MHz to 300 GHz. The devices used for generating microwaves are called magnetrons and klystrons.

Water molecules are dipolar (i.e., they have an asymmetric charge center), and they are normally randomly oriented. The rapidly changing polarity of a microwave or radio-frequency field attempts to pull these dipoles into alignment with the field. As the field changes polarity, the dipoles return to a random orientation before being pulled the other way. This buildup and decay of the field, and the resulting stress on the molecules, causes a conversion of electric field energy to stored potential energy, then to random kinetic or thermal energy. Hence dipolar molecules such as water absorb energy in these frequency ranges. The power developed per unit volume  $P_e$  by this mechanism is

$$P_e = kE^2 f \epsilon' \tan \delta = kE^2 f \epsilon'' \quad (12-117)$$

where  $k$  is a dielectric constant, depending on the units of measurement,  $E$  is the electric field strength (V/m<sup>3</sup>),  $f$  is the frequency,  $\epsilon'$  is the relative dielectric constant or relative permeability,  $\tan \delta$  is the loss tangent or dissipation factor, and  $\epsilon''$  is the loss factor.

The field strength and the frequency are dependent on the equipment, while the dielectric constant, dissipation factor, and loss factor are material-dependent. The electric field strength is also dependent on the location of the material within the microwave/radio-frequency cavity (Turner and Ferguson, 1995), which is one reason why domestic microwave ovens have rotating turntables (so that the food is exposed to a range of microwave intensities). This mechanism is the major one for the generation of heat within materials by these electromagnetic fields.

There is also a heating effect due to ionic conduction, since the ions (sodium, chloride, and hydroxyl) in the water inside materials are accelerated and decelerated by the changing electric field. The collisions which occur as a result of the rapid accelerations and decelerations lead to an increase in the random kinetic (thermal) energy of the material. This type of heating is not significantly dependent on either temperature or frequency, and the power developed per unit volume  $P_e$  from this mechanism is

$$P_e = E^2 q n \mu \quad (12-118)$$

where  $q$  is the amount of electric charge on each of the ions,  $n$  is the charge density (ions/m<sup>3</sup>), and  $\mu$  is the level of mobility of the ions.

Schiffmann (1995) indicates that the dielectric constant of water is over an order of magnitude higher than that of most underlying materials, and the overall dielectric constant of most materials is usually nearly proportional to moisture content up to a critical moisture content, often around 20 to 30 percent. Hence microwave and radio-frequency methods preferentially heat and dry wetter areas in most materials, a process which tends to give more uniform final moisture contents. The dielectric constant of air is very low compared with that of water, so lower density usually means lower heating rates. For water and other small molecules, the effect of increasing temperature is to decrease the heating rate slightly, hence leading to a self-limiting effect.

Other effects (frequency, conductivity, specific heat capacity, etc.) are discussed by Schiffmann (1995), but are less relevant because the range of available frequencies (which do not interfere with radio transmissions) is small (2.45 GHz, 910 MHz). Lower frequencies lead

to greater penetration depths into material than higher frequencies, with 2.45-GHz frequencies sometimes having penetration depths as low as 1 in. For in-depth heating ("volumetric heating"), radio frequencies, with lower frequencies and longer wavelengths, are often used.

**Infrared Methods** Infrared radiation is commonly used in the dehydration of coated films and to even out the moisture content profiles in the drying of paper and boards. The mode of heating is essentially on the material surface, and IR sources are relatively inexpensive compared with dielectric sources.

The heat flux obtainable from an IR source is given by

$$q = F\alpha\epsilon(T_{\text{source}}^4 - T_{\text{drying material}}^4) \quad (12-119)$$

where  $q$  = heat flux,  $\text{W}/\text{m}^2$ ;  $\alpha$  = Stefan-Boltzmann constant =  $5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$ ;  $\epsilon$  = emissivity;  $F$  = view factor; and  $T$  = absolute temperature of the source or drying material.

The emissivity is a property of the material. The limiting value is 1 (blackbody); shiny surfaces have a low value of emissivity. The view factor is a fractional value that depends on the geometric orientation of the source with respect to the heating object.

It is very important to recognize the  $T^4$  dependence on the heat flux. IR sources need to be very hot to give appreciable heat fluxes. Therefore, IR sources should not be used with flammable materials. Improperly designed IR systems can also overheat materials and equipment.

## OPERATION AND TROUBLESHOOTING

**Troubleshooting** Dryer troubleshooting is not extensively covered in the literature, but a systematic approach has been proposed by Kemp and Gardiner (2001). The main steps of the algorithm are as follows:

- **Problem definition**—definition of the dryer problem to be solved.
- **Data gathering**—collection of relevant information, e.g., plant operating data
- **Data analysis**—e.g., heat and mass balance—and *identification of the cause of the problem*
- **Conclusions and actions**—selection and implementation of a solution in terms of changes to process conditions, equipment, or operating procedures
- **Performance auditing**—monitoring to ensure that the problem was permanently solved

There is often a danger in practice that the pressure to get the plant back into production as soon as possible may lead to some of these stages being omitted. Even if a short-term fix has been found, it is highly desirable to make sure what the problem really was, to see whether there are better ways of solving it in the long term, and to check that the problem really has been solved (sometimes it reappears later, e.g., when a temporarily cleaned heat exchanger becomes fouled again, or climatic conditions return to previous values).

The algorithm might also be considered as a "plant doctor." The doctor collects data, or symptoms, and makes a diagnosis of the cause or causes of the problem. Then alternative solutions, or treatments, are considered and a suitable choice is made. The results of the treatment are reviewed (i.e., the process is monitored) to ensure that the "patient" has returned to full health. See Fig. 12-109.

The algorithm is an excellent example of the "divergent-convergent" (brainstorming) method of problem solving. It is important to list all possible causes and solutions, no matter how ridiculous they may initially seem; there may actually be some truth in them, or they may lead to a new and better idea.

**Problem Categorization** In the problem definition stage, it is extremely useful to categorize the problem, as the different broad groups require different types of solution. Five main categories of dryer problems can be identified:

1. Drying performance (outlet moisture content too high, throughput too low)
2. Materials handling (dried material too sticky to get out of dryer, causing blockage)
3. Product quality (too many fines in product or bulk density too low)

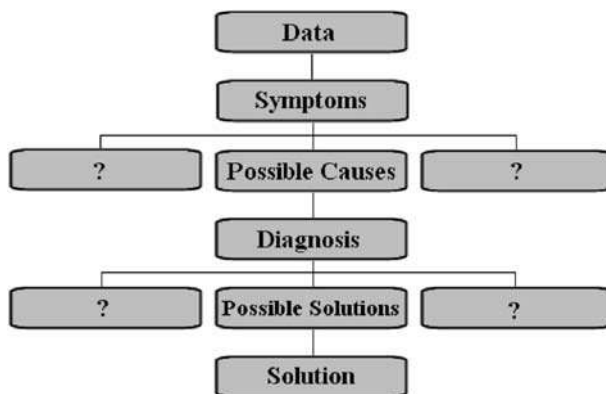


FIG. 12-109 Schematic diagram of algorithm for dryer troubleshooting.

4. Mechanical breakdown (catastrophic sudden failure)
5. Safety, health, and environmental (SHE) issues

Experience suggests that the majority of problems are of the first three types, and these are about equally split over a range of industries and dryer types. Ideally, unforeseen SHE problems will be rare, as these will have been identified in the safety case before the dryer is installed or during commissioning. Likewise, major breakdowns should be largely avoided by a planned maintenance program.

**Drying Performance Problems** Performance problems can be further categorized as

1. Heat and mass balance deficiencies (not enough heat input to do the evaporation)
2. Drying kinetics (drying too slowly, or solids residence time in dryer too short)
3. Equilibrium moisture limitations (reaching a limiting value, or regaining moisture in storage)

For the heat and mass balance, the main factors are

- Solids throughput
- Inlet and outlet moisture content
- Temperatures and heat supply rate
- Leaks and heat losses

As well as problem-solving, these techniques can be used for performance improvement and debottlenecking.

Drying kinetics, which are affected by temperature, particle size, and structure, are limited by external heat and mass transfer to and from the particle surface in the early stages, but internal moisture transport is the main parameter at lower moisture.

Equilibrium moisture content increases with higher relative humidity, or with lower temperature. Problems that depend on the season of the year, or vary between day and night (both suggesting a dependence on ambient temperature and humidity), are often related to equilibrium moisture content.

**Materials Handling Problems** The vast majority of handling problems in a dryer concern sticky feedstocks. Blockages can be worse than performance problems as they can close down a plant completely, without warning. Most stickiness, adhesion, caking, and agglomeration problems are due to *mobile liquid bridges* (surface moisture holding particles together). These are extensively described in particle technology textbooks. Unfortunately, these forces tend to be at a maximum when the solid forms the continuous phases and surface moisture is present, which is the situation for most filter and centrifuge cakes at discharge. By comparison, slurries (where the liquid forms the continuous phase) and dry solids (where all surface moisture has been eliminated) are relatively free-flowing and give fewer problems.

Other sources of problems include electrostatics (most marked with fine and dry powders) and *immobile liquid bridges*, the so-called sticky-point phenomenon. This latter is sharply temperature-dependent, with only a weak dependence on moisture content, in contrast to mobile

liquid bridges. It occurs for only a small proportion of materials, but is particularly noticeable in amorphous powders and foods and is often linked to the glass transition temperature.

**Product Quality Problems** (These do not include moisture level of the main solvent.) Many dryer problems either concern product quality or cannot be solved without considering the effect of any changes on product quality. Thus it is a primary consideration in most troubleshooting, although product quality measurements are specific to the particular product, and it is difficult to generalize. However, typical properties may include color, taste (not easily quantifiable), bulk density, viscosity of a paste or dispersion, dispersibility, or rate of solution. Others are more concerned with particle size, size distribution (e.g., coarse or fine fraction), or powder handling properties such as rate of flow through a standard orifice. These property measurements are nearly always made off-line, either by the operator or by the laboratory, and many are very difficult to characterize in a rigorous quantitative manner. (See also "Fundamentals" Section.)

Storage problems, very common in industry, result if the product from a dryer is free-flowing when packaged, but has caked and formed solid lumps when received by the customer. Sometimes, the entire internal contents of a bag or drum have welded together into a huge lump, making it impossible to discharge.

Depending on the situation, there are at least three different possible causes:

1. *Equilibrium moisture content*—hygroscopic material is absorbing moisture from the air on cooling.
2. *Incomplete drying*—product is continuing to lose moisture in storage.
3. *Psychrometry*—humid air is cooling and reaching its dew point.

The three types of problem have some similarities and common features, but the solution to each one is different. Therefore, it is essential to understand which mechanism is actually occurring.

**Option 1:** The material is hygroscopic and is absorbing moisture back from the air in storage, where the cool air has a higher relative humidity than the hot dryer exhaust. **Solution:** Pack and seal the solids immediately on discharge in tough impermeable bags (usually double- or triple-lined to reduce the possibility of tear and pinholes), and minimize the ullage (airspace above the solids in the bags) so that the amount of moisture that can be absorbed is too low to cause any significant problem. Dehumidifying the air to the storage area is also possible, but often very expensive.

**Option 2:** The particles are emerging with some residual moisture, and continue to dry after being stored or bagged. As the air and solids cool down, the moisture in the air comes out as dew and condenses on the surface of the solids, causing caking by mobile liquid bridges. **Solution:** If the material is meeting its moisture content specification, cool the product more effectively before storage, to stop the drying process. If the outlet material is wetter than specification, alter dryer operating conditions or install a postdryer.

**Option 3:** Warm, wet air is getting into the storage area or the bags, either because the atmosphere is warm with a high relative humidity (especially in the tropics) or because dryer exhaust air has been allowed to enter. As in option 2, when the temperature falls, the air goes below its dew point and condensation occurs on the walls of the storage area or inside the bags, or on the surface of the solids, leading to caking. **Solution:** Avoid high-humidity air in the storage area. Ensure the dryer exhaust is discharged a long way away. If the ambient air humidity is high, consider cooling the air supply to storage to bring it below its dew point and reduce its absolute humidity.

See Kemp and Gardiner, "An Outline Method for Troubleshooting and Problem-Solving in Dryers," *Drying Technol.* 19(8):1875–1890 (2001).

### Dryer Operation

**Start-up Considerations** It is important to start up the heating system before introducing product into the dryer. This will minimize condensation and subsequent product buildup on dryer walls. It is also important to minimize off-quality production by not overdrying or underdrying during the start-up period. Proper control system design can aid in this regard. The dryer turndown ratio is also an

important consideration during start-up. Normally the dryer is started up at the lowest end of the turndown ratio, and it is necessary to match heat input with capacity load.

**Shutdown Considerations** The sequence for dryer shutdown is also very important and depends on the type of dryer. The sequence must be thoroughly thought through to prevent significant off-quality product or a safety hazard. The outlet temperature during shutdown is a key operating variable to follow.

**Energy Considerations** The first consideration is to minimize moisture content of the dryer feed, e.g., with dewatering equipment, and to establish as high an outlet product moisture target as possible. Other energy considerations vary widely by dryer type. In general, heating with gas, fuel oil, and steam is significantly more economical than heating with electricity. Hence RF, microwave, and infrared drying is energy-intensive. Direct heating is more efficient than indirect in most situations. Sometimes air recycle (direct or indirect) can be effective to reduce energy consumption. And generally operating at high inlet temperatures is more economical.

**Recycle** In almost all situations, the process system must be able to accommodate product recycle. The question is, How to handle it most effectively, considering product quality, equipment size, and energy?

**Improvement Considerations** The first consideration is to evaluate mass and energy balances to identify problem areas. This will identify air leaks and excessive equipment heat losses and will enable determination of overall energy efficiency.

A simplified heat balance will show what might need to be done to debottleneck a convective (hot gas) dryer, i.e., increase its production rate  $F$ .

$$F(X_I - X_O)\lambda_{ev} \approx GC_{PG}(T_{GI} - T_{GO}) - Q_{ul}$$

Before proceeding along this line, however, it is necessary to establish that the dryer is genuinely heat and mass balance limited. If the system is controlled by kinetics or equilibria, changing the parameters may have undesirable side effects, e.g., increasing the product moisture content.

The major alternatives are then as follows (assuming gas specific heat capacity  $C_{PG}$  and latent heat of evaporation  $\lambda_{ev}$  are fixed):

1. Increase gas flow rate  $G$ —usually increases pressure drop, so new fans and gas cleaning equipment may be required.
2. Increase inlet gas temperature  $T_{GI}$ —usually limited by risk of thermal damage to product.
3. Decrease outlet gas temperature  $T_{GO}$ —but note that this increases NTUs, outlet humidity, and relative humidity, and reduces both temperature and humidity driving forces. Hence it may require a longer drying time and a larger dryer, and may also increase equilibrium and outlet moistures  $X_E$  and  $X_O$ .
4. Reduce inlet moisture content  $X_I$ , say, by dewatering by gas blowing, centrifuging, vacuum or pressure filtration, or a predryer.
5. Reduce heat losses  $Q_{ul}$  by insulation, removing leaks, etc.

**Dryer Safety** This section discusses some of the key considerations in dryer safety. General safety considerations are discussed in Sec. 23, "Safety and Handling of Hazardous Materials," and should be referred to for additional guidance.

Fires, explosions, and, to a lesser extent, runaway decompositions are the primary hazards associated with drying operations. The outbreak of fire is a result of ignition which may or may not be followed by an explosion. A hazardous situation is possible if

1. The product is combustible
2. The product is wetted by a flammable solvent
3. The dryer is direct-fired

An explosion can be caused by dust or flammable vapors, both of which are fires that rapidly propagate, causing a pressure rise in a confined space.

**Dust Explosions** Dispersion dryers can be more hazardous than layer-type dryers if we are drying a solid combustible material which is then dispersed in air, particularly if the product is a fine particle size. If this finely dispersed product is then exposed to an ignition source, an explosion can result. The following conditions (van't Land, *Industrial Drying Equipment*, Marcel Dekker, New York, 1991) will be conducive to fire and explosion hazard:



1. Small particle sizes, generally less than 75  $\mu\text{m}$ , which are capable of propagating a flame
2. Dust concentrations within explosive limits, generally 10 to 60  $\text{g}/\text{m}^3$
3. Ignition source energy of 10 to 1000 mJ or as low as 5 mJ for highly explosive dust sources
4. Atmosphere supporting combustion

Since most product and hence dust compositions vary widely, it is generally necessary to do quantitative testing in approved test equipment.

**Flammable Vapor Explosions** This can be a problem for products wetted by flammable solvents if the solvent concentration exceeds 0.2% v/v in the vapor phase. The ignition energy of vapor-air mixtures is lower ( $< 1$  mJ) than that of dust-air suspensions. Many of these values are available in the literature, but testing may sometimes be required.

**Ignition Sources** There are many possible sources of an ignition, and they need to be identified and addressed by both designers and operators. A few of the most common ignition sources are

1. Spontaneous combustion
2. Electrostatic discharge
3. Electric or frictional sparks
4. Incandescent solid particles from heating system

Safety hazards must be addressed with proper dryer design specifications. The following are a few key considerations in dryer design.

**Inert system design** The dryer atmosphere is commonly inerted with nitrogen, but superheated steam or self-inertized systems are also possible. Self-inertized systems are not feasible for flammable solvent systems. These systems must be operated with a small overpressure to ensure no oxygen ingress. And continuous on-line oxygen concentration monitoring is required to ensure that oxygen levels remain well below the explosion hazard limit.

**Relief venting** Relief vents that are properly sized relieve and direct dryer explosions to protect the dryer and personnel if an explosion does occur. Normally they are simple pop-out panels with a minimum length of ducting to direct the explosion away from personnel or other equipment.

**Suppression systems** Suppression systems typically use an inert gas such as carbon dioxide to minimize the explosive peak pressure rise and fire damage. Dryer operating pressure must be properly monitored to detect the initial pressure rise followed by shutdown of the dryer operating systems and activation of the suppression system.

**Clean design** Care should be taken in the design of both the dryer and dryer ancillary (cyclones, filters, etc.) equipment to eliminate ledges, crevices, and other obstructions which can lead to dust and product buildup. Smooth drying equipment walls will minimize deposits. This can go a long way in prevention. No system is perfect, of course, and a routine cleaning schedule is also recommended.

**Start-up and shutdown** Start-up and shutdown situations must be carefully considered when designing a dryer system. These situations can create higher than normal dust and solvent concentrations. This coupled with elevated temperatures can create a hazard well beyond normal continuous operation.

**Environmental Considerations** Environmental considerations are continuing to be an increasingly important aspect of dryer design and operation as environmental regulations are tightened. The primary environmental problems associated with drying are particulate and volatile organic compound (VOC) emissions. Noise can be an issue with certain dryer types.

**Environmental Regulations** These vary by country, and it is necessary to know the specific regulations in the country in which the dryer will be installed. It is also useful to have some knowledge of the direction of regulations so that the environmental control system is not obsolete by the time it becomes operational.

Particulate emission problems can span a wide range of hazards. Generally there are limits on both toxic and nontoxic particles in terms of annual and peak emissions limits. Particles can present toxic, bacterial, viral, and other hazards to human, animal, and plant life.

Likewise, VOC emissions can span a wide range of hazards and issues from toxic gases to smelly gases.

**Environmental Control Systems** We should consider environmental hazards before the drying operation is even considered. The focus should be on minimizing the hazards created in the upstream

processing operations. After potential emissions are minimized, these hazards must be dealt with during dryer system design and then subsequently with proper operational and maintenance procedures.

**Particle Emission Control Equipment** The four most common methods of particulate emissions control are as follows:

1. **Cyclone separators** The advantage of cyclones is they have relatively low capital and operating costs. The primary disadvantage is that they become increasingly ineffective as the particle size decreases. As a general rule of thumb, we can say that they are 100 percent efficient with particles larger than 20  $\mu\text{m}$  and 0 percent efficient with particles smaller than 1  $\mu\text{m}$ . Cyclones can also be effective precleaning devices to reduce the load on downstream bag filters.

2. **Scrubbers** The more general classification is wet dedusters, the most common of which is the wet scrubber. The advantage of wet scrubbers is that they can remove fine particles that the cyclone does not collect. The disadvantages are they are more costly than cyclones and they can turn air contamination into water contamination, which may then require additional cleanup before the cleaning water is put to the sewer.

3. **Bag filters** The advantages of filters are that they can remove very fine particles and bag technologies continue to improve and enable ever-smaller particles to be removed without excessive pressure drops or buildup. The primary disadvantages are higher cost relative to cyclones and greater maintenance costs, especially if frequent bag replacement is necessary.

4. **Electrostatic precipitators** The capital cost of these systems is relatively high, and maintenance is critical to effective operation.

**VOC Control Equipment** The four most prevalent equipment controls are

1. **Scrubbers** Similar considerations as above apply.
2. **Absorbers** These systems use a high-surface-area absorbent, such as activated carbon, to remove the VOC absorbate.
3. **Condensers** These systems are generally only feasible for recovering solvents from nonaqueous wetted products.
4. **Thermal and catalytic incinerators** These can be quite effective and are generally a low capital and operating cost solution, except in countries with high energy costs.

**Noise** Noise analysis and abatement is a very specialized area. Generally, the issue with dryers is associated with the fans, particularly for systems requiring fans that develop very high pressures. Noise is a very big issue that needs to be addressed with pulse combustion dryers and can be an issue with very large dryers such as rotary dryers and kilns.

Additional considerations regarding environmental control and waste management can be found in Secs. 22, "Waste Management," and 23, "Process Safety."

**Control and Instrumentation** The purpose of the control and instrumentation system is to provide a system that enables the process to produce the product at the desired moisture target and that meets other quality control targets discussed earlier (density, particle size, color, solubility, etc.). This segment discusses key considerations for dryer control and instrumentation. Additional more detailed information can be found in Sec. 8, "Process Control."

Proper control of product quality starts with the dryer selection and design. Sometimes two-stage or multistage systems are required to meet product quality targets. Multistage systems enable us to better control temperature and moisture profiles during drying. Assuming the proper dryer design has been selected, we must then design the control and instrumentation system to ensure we meet all product quality targets.

**Manual versus Automatic Control** Dryers can be controlled either manually or automatically. Generally lab-, pilot-, and small-scale production units are controlled manually. These operations are usually batch systems, and manual operation provides lower cost and greater flexibility. The preferred mode for large-scale, continuous dryers is automatic.

**Key Control Variables** Product moisture and product temperature are key control variables. Ideally both moisture and temperature measurement are done on-line, but frequently moisture measurement is done off-line and temperature (or exhaust air temperature) becomes the primary control variable. And generally, inlet temperature

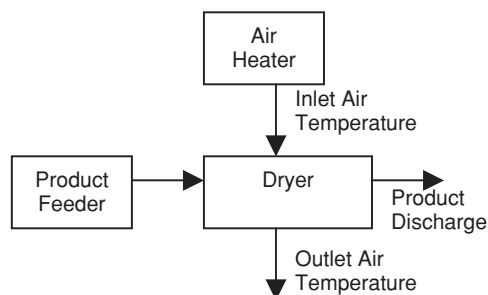


FIG. 12-110 Typical dryer system.

will control the rate of production and outlet temperature will control the product moisture and other product quality targets.

**Common Control Schemes** Two relatively simple, but common control schemes in many dryer systems (Fig. 12-110) are as follows:

1. Outlet air temperature is controlled by feed rate regulation with inlet temperature controlled by gas heater regulation.
2. Outlet air temperature is controlled by heater regulation with feed rate held constant.

Alternatively, product temperatures can replace air temperatures with the advantage of better control and the disadvantage of greater maintenance of the product temperature sensors.

#### Other Instrumentation and Control

**Pressure** Pressure and equipment pressure drops are important to proper dryer operation. Most dryers are operated under vacuum. This prevents dusting to the environment, but excess leakage in decreases dryer efficiency. Pressure drops are especially important for stable fluid-bed operation.

**Air (gas) flow rate** Obviously gas flows are another important parameter for proper dryer operation. Pitot tubes are useful when a system has no permanent gas flow sensors. Averaging pitot tubes work well in permanent installations. The devices work best in straight sections of ductwork which are sometimes difficult to find and make accurate measurement a challenge.

**Product feed rate** It's important to know that product feed rates and feed rate changes are sometimes used to control finished product moistures. Weigh belts are common for powdered products, and there is a wide variety of equipment available for liquid feeds. Momentum devices are inexpensive but less accurate.

**Humidity** The simplest method is sometimes the best. Wet- and dry-bulb temperature measurement to get air humidity is simple and works well for the occasional gas humidity measurement. The problem

with permanent humidity measurement equipment is the difficulty of getting sensors robust enough to cope with a hot, humid, and sometimes dusty environment.

**Interlocks** Interlocks are another important feature of a well-designed control and instrumentation system. Interlocks are intended to prevent damage to the dryer system or to personnel, especially during the critical periods of start-up and shutdown. The following are a few key interlocks to consider in a typical dryer system.

**Drying chamber damage** This type of damage can occur when the chamber is subjected to significant vacuum when the exhaust fans are started up before the supply fans.

**Personnel injury** This interlock is to prevent injury due to entering the dryer during operation, but more typically to prevent dryer start-up with personnel in the main chamber or inlet or exhaust air ductwork on large dryers. This typically involves microswitches on access doors coupled with proper door lock devices and tags.

**Assurance of proper startup and shutdown** These interlocks ensure, e.g., that the hot air system is started up before the product feed system and that the feed system is shut down before the hot air system.

**Heater system** There are a host of important heater system interlocks to prevent major damage to the entire drying system. Additional details can be found in Sec. 23, "Process Safety."

**Drying Software** Several software programs for psychrometric charts and calculations are available and are described in the "Psychrometry" section.

Dryers are included as modules in standard process simulators such as Aspen Plus and HYSYS (Aspen Technology), Pro/II (Simsci/Invensys) and Unisim (Honeywell), and the prototype Solidsim solids process simulator. These are confined (as of 2006) to heat and mass balances or, at most, simple scoping design.

Many higher-level dryer models have been produced by researchers and universities, but they are not commercially available. Windows-based drying programs are available in the Process Tools (Aspen Technology), including a psychrometric chart, dryer selection expert system, dryer scoping design, and fluid-bed dryer simulation. Some CFD programs (e.g., Fluent, CFX) include a module for spray dryers.

In addition to textbooks, a detailed online knowledge base, the Process Manual, is available from Aspen Technology (see [www.processmanual.com](http://www.processmanual.com)). This covers equipment, scientific background, design, and operation for drying and 10 other technical areas in solids and separation processes. Company and university licenses are available.

Detailed reviews of drying software packages given by Menshutina and Kudra, *Drying Technol.* **19**(8):1825–1850 (2001); and by Kemp, Chap. 7 in *Modern Drying Technology*, vol. 1, Wiley-VCH (2007), and *Drying Technol.* **25** (2007).

SECTION **13**

Distillation

# PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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M. F. MALONE, R. TAYLOR



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# Distillation\*

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\* Certain portions of this section draw heavily on the work of J. D. Seader, Jeffrey J. Sirola, and Scott D. Barnicki, authors of this section in the 7th edition.

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## Nomenclature and Units

| Symbol           | Definition                                     | SI units                   | U.S. Customary System units | Symbol                      | Definition                                                      | SI units          | U.S. Customary System units |
|------------------|------------------------------------------------|----------------------------|-----------------------------|-----------------------------|-----------------------------------------------------------------|-------------------|-----------------------------|
| $A$              | Absorption factor                              |                            |                             | $h$                         | Height                                                          | m                 | ft                          |
| $A$              | Area                                           | m <sup>2</sup>             | ft <sup>2</sup>             | $h$                         | Heat-transfer coefficient                                       | kW/m <sup>2</sup> | Btu/(ft <sup>2</sup> ·h)    |
| $C$              | Number of chemical species                     |                            |                             | $k$                         | Mass-transfer coefficient                                       | m/s               | ft/h                        |
| $D$              | Distillate flow rate                           | kg·mol/s                   | lb·mol/h                    | $l$                         | Component flow rate in liquid                                   | kg·mol/s          | lb·mol/h                    |
| $D$              | Diffusion coefficient                          | m <sup>2</sup> /s          | ft <sup>2</sup> /h          | $p$                         | Pressure                                                        | kPa               | psia                        |
| $E$              | Efficiency                                     |                            |                             | $q$                         | Measure of thermal condition of feed                            |                   |                             |
| $E$              | Energy flux                                    | kW/m <sup>2</sup>          | Btu/(ft <sup>2</sup> ·h)    | $q$                         | Heat flux                                                       | kW/m <sup>2</sup> | Btu/(ft <sup>2</sup> ·h)    |
| $\mathcal{E}$    | Energy transfer rate                           | kW                         | Btu/h                       | $q_c$                       | Condenser duty                                                  | kW                | Btu/h                       |
| $F$              | Feed flow rate                                 | kg·mol/s                   | lb·mol/h                    | $q_r$                       | Reboiler duty                                                   | kW                | Btu/h                       |
| $H$              | Column height                                  | m                          | ft                          | $r$                         | Sidestream ratio                                                |                   |                             |
| $H$              | Enthalpy                                       | J/(kg·mol)                 | Btu/(lb·mol)                | $s$                         | Liquid-sidestream ratio                                         |                   |                             |
| $H$              | Liquid holdup                                  | kg·mol                     | lb·mol                      | $t$                         | Time                                                            | s                 | h                           |
| $H$              | Height of a transfer unit                      | m                          | ft                          | $u$                         | Velocity                                                        | m/s               | ft/h                        |
| $K$              | Vapor-liquid equilibrium ratio ( $K$ value)    |                            |                             | $v$                         | Component flow rate in vapor                                    | kg·mol/s          | lb·mol/h                    |
| $K_D$            | Chemical equilibrium constant for dimerization |                            |                             | $w$                         | Weight fraction                                                 |                   |                             |
| $K_d$            | Liquid-liquid distribution ratio               |                            |                             | $x$                         | Mole fraction in liquid                                         |                   |                             |
| $L$              | Liquid flow rate                               | kg·mol/s                   | lb·mol/h                    | $y$                         | Mole fraction in vapor                                          |                   |                             |
| $N$              | Number of equilibrium stages                   |                            |                             | $z$                         | Mole fraction in feed                                           |                   |                             |
| $N_c$            | Number of relationships                        |                            |                             | Greek Symbols               |                                                                 |                   |                             |
| $N_i$            | Number of design variables                     |                            |                             | $\alpha$                    | Relative volatility                                             |                   |                             |
| $N_{\min}$       | Minimum number of equilibrium stages           |                            |                             | $\gamma$                    | Activity coefficient                                            |                   |                             |
| $N_p$            | Number of phases                               |                            |                             | $\epsilon$                  | TBK efficiency                                                  |                   |                             |
| $N_r$            | Number of repetition variables                 |                            |                             | $\xi$                       | Dimensionless time                                              |                   |                             |
| $N_o$            | Number of variables                            |                            |                             | $\rho$                      | Density                                                         | kg/m <sup>3</sup> | lb/ft <sup>3</sup>          |
| $\mathcal{N}$    | Rate of mass transfer                          | kg·mol/s                   | lb·mol/h                    | $\mu$                       | Viscosity                                                       | N/m <sup>2</sup>  |                             |
| $N$              | Molar flux                                     | kg·mol/(m <sup>2</sup> ·s) | lb·mol/(ft <sup>2</sup> ·h) | $\sigma$                    | Surface tension                                                 | N/m               |                             |
| $\mathcal{N}$    | Number of transfer units                       |                            |                             | $\theta$                    | Time for batch distillation                                     | s                 | h                           |
| $P$              | Pressure                                       | Pa                         | psia                        | $\Theta$                    | Parameter in Underwood equations                                |                   |                             |
| $P^{\text{sat}}$ | Vapor pressure                                 | Pa                         | psia                        | $\Phi$                      | Fugacity coefficient of pure component                          |                   |                             |
| $Q$              | Heat-transfer rate                             | kW                         | Btu/h                       | $\hat{\Phi}$                | Fugacity coefficient in mixture                                 |                   |                             |
| $Q_c$            | Condenser duty                                 | kW                         | Btu/h                       | $\Phi_A$                    | Fraction of a component in feed vapor that is not absorbed      |                   |                             |
| $Q_r$            | Reboiler duty                                  | kW                         | Btu/h                       | $\Phi_s$                    | Fraction of a component in entering liquid that is not stripped |                   |                             |
| $R$              | External-reflux ratio                          |                            |                             | $\Psi$                      | Factor in Gilliland correlation                                 |                   |                             |
| $R_{\min}$       | Minimum-reflux ratio                           |                            |                             | Subscripts and Superscripts |                                                                 |                   |                             |
| $S$              | Sidestream flow rate                           | kg·mol/s                   | lb·mol/h                    | EQ                          | Equilibrium                                                     |                   |                             |
| $S$              | Stripping factor                               |                            |                             | $f$                         | Froth                                                           |                   |                             |
| $S$              | Vapor-sidestream ratio                         |                            |                             | $hk$                        | Heavy key                                                       |                   |                             |
| $Sc$             | Schmidt number                                 |                            |                             | $i$                         | Component index                                                 |                   |                             |
| $T$              | Temperature                                    | K                          | °R                          | $j$                         | Stage index                                                     |                   |                             |
| $U$              | Liquid-sidestream rate                         | kg·mol/s                   | lb·mol/h                    | $L$                         | Liquid                                                          |                   |                             |
| $V$              | Vapor flow rate                                | kg·mol/s                   | lb·mol/h                    | $lk$                        | Light key                                                       |                   |                             |
| $W$              | Vapor-sidestream rate                          | kg·mol/s                   | lb·mol/h                    | $MV$                        | Murphree vapor                                                  |                   |                             |
| $X$              | Relative mole fraction in liquid phase         |                            |                             | $o$                         | Overall                                                         |                   |                             |
| $Y$              | Relative mole fraction in vapor phase          |                            |                             | $s$                         | Superficial                                                     |                   |                             |
| $a$              | Activity                                       |                            |                             | $t$                         | Mixture or total                                                |                   |                             |
| $a$              | Area                                           | m <sup>2</sup>             | ft <sup>2</sup>             | $V$                         | Vapor                                                           |                   |                             |
| $b$              | Component flow rate in bottoms                 | kg·mol/s                   | lb·mol/h                    | $^{\circ}$                  | Equilibrium composition                                         |                   |                             |
| $c$              | Number of chemical species                     |                            |                             | Acronyms                    |                                                                 |                   |                             |
| $c$              | Molar density                                  | kg·mol/m <sup>3</sup>      | lb·mol/ft <sup>3</sup>      | HETP                        | Height equivalent to a theoretical plate                        |                   |                             |
| $d$              | Component flow rate in distillate              | kg·mol/s                   | lb·mol/h                    | n.b.p.                      | Normal boiling point (1-atm pressure)                           |                   |                             |
| $d$              | Mass-transfer driving force                    |                            |                             | NTU                         | Number of transfer units                                        |                   |                             |
| $e$              | Rate of heat transfer                          | kW                         | Btu/h                       |                             |                                                                 |                   |                             |
| $f$              | Component flow rate in feed                    | kg·mol/s                   | lb·mol/h                    |                             |                                                                 |                   |                             |
| $f$              | Fugacity                                       | Pa                         | psia                        |                             |                                                                 |                   |                             |

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## INTRODUCTION TO DISTILLATION OPERATIONS

### GENERAL PRINCIPLES

Separation operations achieve their objective by the creation of two or more coexisting zones which differ in temperature, pressure, composition, and/or phase state. Each molecular species in the mixture to be separated responds in a unique way to differing environments offered by these zones. Consequently, as the system moves toward equilibrium, each species establishes a different concentration in each zone, and this results in a separation between the species.

The separation operation called *distillation* utilizes vapor and liquid phases at essentially the same temperature and pressure for the coexisting zones. Various kinds of devices such as *random* or *structured packings* and *plates* or *trays* are used to bring the two phases into intimate contact. Trays are stacked one above the other and enclosed in a cylindrical shell to form a *column*. Packings are also generally contained in a cylindrical shell between hold-down and support plates. The column may be operated continuously or in batch mode depending on a number of factors such as scale and flexibility of operations and solids content of feed. A typical tray-type continuous distillation column plus major external accessories is shown schematically in Fig. 13-1.

The *feed* material, which is to be separated into fractions, is introduced at one or more points along the column shell. Because of the difference in density between vapor and liquid phases, liquid runs down the column, cascading from tray to tray, while vapor flows up the column, contacting liquid at each tray.

Liquid reaching the bottom of the column is partially vaporized in a heated *reboiler* to provide *boil-up*, which is sent back up the column. The remainder of the bottom liquid is withdrawn as *bottoms*, or bottom product. Vapor reaching the top of the column is cooled and condensed to liquid in the *overhead condenser*. Part of this liquid is returned to the column as *reflux* to provide liquid overflow. The remainder of the overhead stream is withdrawn as *distillate*, or overhead product. In some cases only part of the vapor is condensed so that a vapor distillate can be withdrawn.

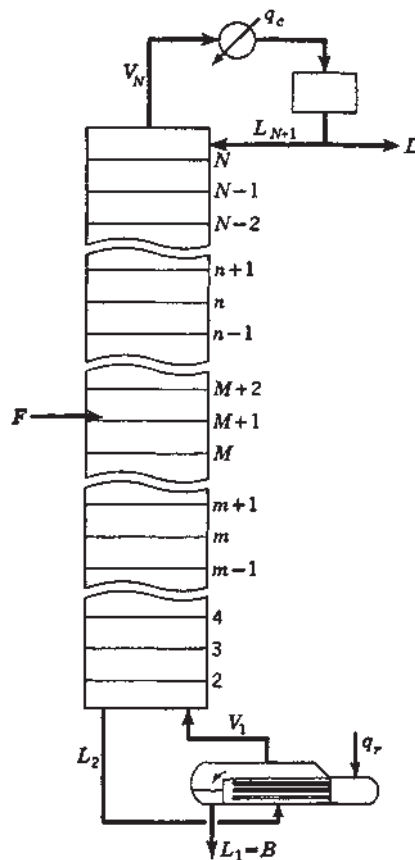
This overall flow pattern in a distillation column provides counter-current contacting of vapor and liquid streams on all the trays through the column. Vapor and liquid phases on a given tray approach thermal, pressure, and composition equilibria to an extent dependent upon the efficiency of the contacting tray.

The *lighter* (lower-boiling temperature) components tend to concentrate in the vapor phase, while the *heavier* (higher-boiling temperature) components concentrate in the liquid phase. The result is a vapor phase that becomes richer in light components as it passes up the column and a liquid phase that becomes richer in heavy components as it cascades downward. The overall separation achieved between the distillate and the bottoms depends primarily on the *relative volatilities* of the components, the number of contacting trays in each column section, and the ratio of the liquid-phase flow rate to the vapor-phase flow rate in each section.

If the feed is introduced at one point along the column shell, the column is divided into an upper section, which is often called the *rectifying* section, and a lower section, which is often referred to as the *stripping* section. In *multiple-feed* columns and in columns from

which a liquid or vapor *sidestream* is withdrawn, there are more than two column sections between the two end-product streams. The notion of a column section is a useful concept for finding alternative *systems* (or *sequences*) of columns for separating multicomponent mixtures, as described below in the subsection Distillation Systems.

All separation operations require energy input in the form of heat or work. In the conventional distillation operation, as typified in Fig. 13-1, energy required to separate the species is added in the form of heat to the reboiler at the bottom of the column, where the temperature is highest. Also heat is removed from a condenser at the top of the column, where the temperature is lowest. This frequently results in a



**FIG. 13-1** Schematic diagram and nomenclature for a simple continuous distillation column with one feed, a total overhead condenser, and a partial reboiler.

large energy-input requirement and low overall thermodynamic efficiency, especially if the heat removed in the condenser is wasted. Complex distillation operations that offer higher thermodynamic efficiency and lower energy-input requirements have been developed and are also discussed below in the subsection Distillation Systems.

Batch distillation is preferred for small feed flows or seasonal production which is carried out intermittently in "batch campaigns." In this mode the feed is charged to a still which provides vapor to a column where the separation occurs. Vapor leaving the top of the column is condensed to provide liquid reflux back to the column as well as a distillate stream containing the product. Under normal operation, this is the only stream leaving the device. In addition to the batch rectifier just described, other batch configurations are possible as discussed in the subsection Batch Distillation. Many of the concepts and methods discussed for continuous distillation are useful for developing models and design methods for batch distillation.

## EQUILIBRIUM AND NONEQUILIBRIUM-STAGE CONCEPTS

The transfer processes taking place in an actual distillation column are a complicated interplay between the thermodynamic phase equilibrium properties of the mixture, rates of intra- and interphase mass and energy transport, and multiphase flows. Simplifications are necessary to develop tractable models. The landmark concept of the *equilibrium-stage model* was developed by Sorel in 1893, in which the liquid in each stage is considered to be well mixed and such that the vapor and liquid streams leaving the stage are in thermodynamic equilibrium with each other. This is needed so that thermodynamic phase equilibrium relations can be used to determine the temperature and composition of the equilibrium streams at a given pressure. A hypothetical column composed of equilibrium stages (instead of actual contact trays) is

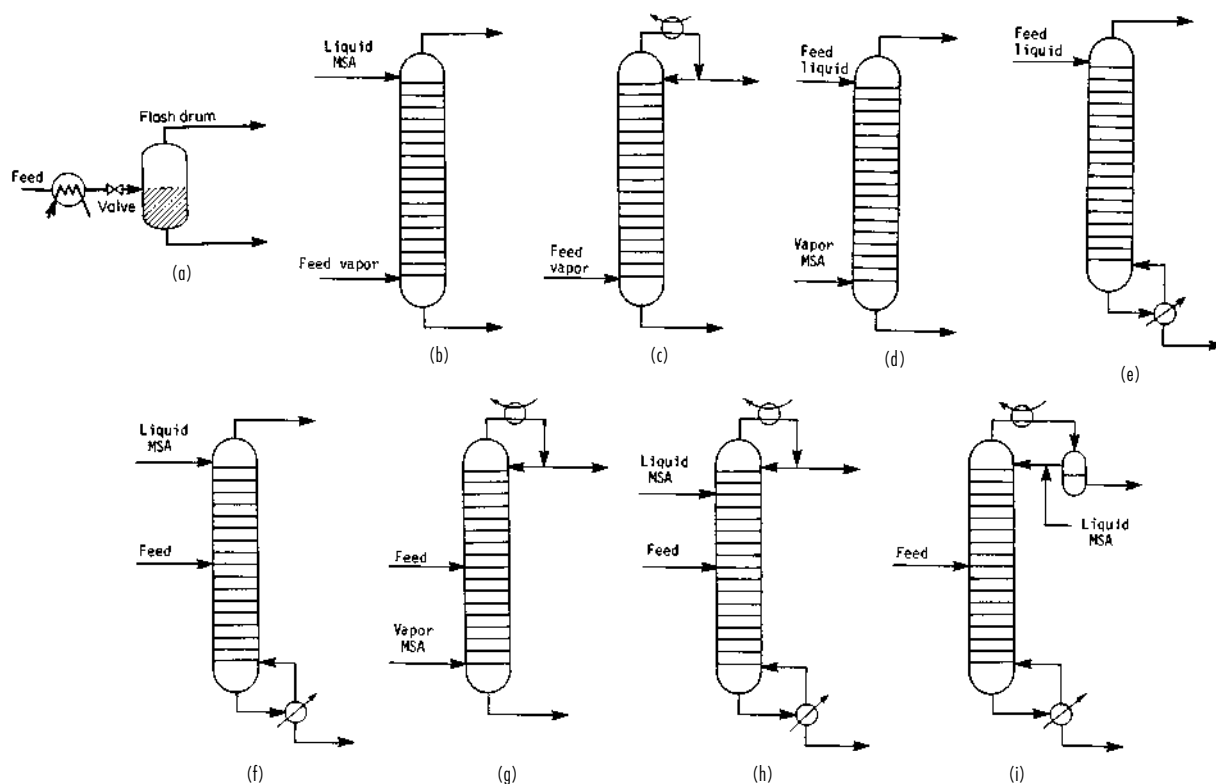
designed to accomplish the separation specified for the actual column. The number of hypothetical equilibrium stages required is then converted to a number of actual trays by means of *tray efficiencies*, which describe the extent to which the performance of an actual contact tray duplicates the performance of an equilibrium stage. Alternatively and preferably, tray inefficiencies can be accounted for by using rate-based models that are described below.

Use of the equilibrium-stage concept separates the design of a distillation column into three major steps: (1) Thermodynamic data and methods needed to predict equilibrium-phase compositions are assembled. (2) The number of equilibrium stages and the energy input required to accomplish a specified separation, or the separation that will be accomplished in a given number of equilibrium stages for a given energy input, are calculated. (3) The number of equilibrium stages is converted to an equivalent number of actual contact trays or height of packing, and the column diameter is determined. Much of the third step is eliminated if a rate-based model is used. This section deals primarily with equilibrium and rate-based models of distillation. Section 4 covers the first step, but a summary of methods and some useful data are included in this section. Section 14 covers equipment design.

## RELATED SEPARATION OPERATIONS

The simple and complex distillation operations just described all have two things in common: (1) Both rectifying and stripping sections are provided so that a separation can be achieved between two components that are adjacent in volatility; and (2) the separation is effected only by the addition and removal of energy and not by the addition of any mass separating agent (MSA) such as in liquid-liquid extraction.

Sometimes, alternative single- or multiple-stage vapor-liquid separation operations, of the types shown in Fig. 13-2, may be more suitable than distillation for the specified task.



**FIG. 13-2** Separation operations related to distillation. (a) Flash vaporization or partial condensation. (b) Absorption. (c) Rectifier. (d) Stripping. (e) Reboiled stripping. (f) Reboiled absorption. (g) Refluxed stripping. (h) Extractive distillation. (i) Azeotropic distillation.

A single-stage flash, as shown in Fig. 13-2a, may be appropriate if (1) the relative volatility between the two components to be separated is very large; (2) the recovery of only one component in one of the two product streams is to be achieved, without regard to the separation of the other components; or (3) only a partial separation is to be made. A common example is the separation of light gases such as hydrogen and methane from aromatics. The desired temperature and pressure of a flash may be established by the use of heat exchangers, a valve, a compressor, and/or a pump upstream of the vessel, used to separate the product vapor and liquid phases. Depending on the original condition of the feed, it may be partially condensed or partially vaporized in a so-called flash operation.

If the recovery of only one component is required rather than a sharp separation between two components of adjacent volatility, their absorption or stripping in a single section of stages may be sufficient. If the feed is vapor at separation conditions, absorption is used either with a liquid MSA absorbent of relatively low volatility, as in Fig. 13-2b, or with reflux produced by an overhead partial condenser, as in Fig. 13-2c. The choice usually depends on the ease of partially condensing the overhead vapor or of recovering and recycling the absorbent. If the feed is liquid at separation conditions, stripping is used, either with an externally supplied vapor stripping agent of relatively high volatility, as shown in Fig. 13-2d, or with boil-up produced by a partial reboiler, as in Fig. 13-2e. The choice depends on the ease of partially reboiling the bottoms or of recovering and recycling the stripping agent.

If a relatively sharp separation is required between two components of adjacent volatility, but either an undesirably low temperature is required to produce reflux at the column operating pressure or an undesirably high temperature is required to produce boil-up, then refluxed stripping, as shown in Fig. 13-2g, or reboiled absorption, as shown in Fig. 13-2f, may be used. In either case, the choice of MSA follows the same consideration given for simple absorption and stripping.

When the volatility difference between the two components to be separated is so small that a very large number of stages would be required, then extractive distillation, as shown in Fig. 13-2h, should be considered. Here, an MSA is selected that increases the volatility difference sufficiently to reduce the stage requirement to a reasonable number. Usually, the MSA is a polar compound of low volatility that leaves in the bottoms, from which it is recovered and recycled. It is introduced in an appreciable amount near the top stage of the column so as to affect the volatility difference over most of the stages. Some reflux to the top stage is used to minimize the MSA content in the distillate. An alternative to extractive distillation is azeotropic distillation, which is shown in Fig. 13-2i in just one of its many modes. In a common mode, an MSA that forms a heterogeneous minimum-boiling azeotrope with one or more components of the feed is used. The azeotrope is taken overhead, and the MSA-rich phase is decanted and returned to the top of the column as reflux.

Numerous other multistaged configurations are possible. One important variation of a stripper, shown in Fig. 13-2d, is a refluxed stripper, in which an overhead condenser is added. Such a configuration is sometimes used to steam-strip sour water containing  $\text{NH}_3$ ,  $\text{H}_2\text{O}$ , phenol, and HCN.

All the separation operations shown in Fig. 13-2, as well as the simple and complex distillation operations described earlier, are referred to here as distillation-type separations because they have much in common with respect to calculations of (1) thermodynamic properties, (2) vapor-liquid equilibrium stages, and (3) column sizing. In fact, as will be evident from the remaining treatment of this section, the trend is toward single generalized digital computer program packages that compute many or all distillation-type separation operations.

This section also includes a treatment of distillation-type separations from a rate-based point of view that uses principles of mass- and heat-transfer rates. Section 14 also presents details of that subject as applied to absorption and stripping.

## THERMODYNAMIC DATA AND MODELS

Reliable thermodynamic data are essential for the accurate design or analysis of distillation columns. Failure of equipment to perform at specified levels is often attributable, at least in part, to the lack of such data.

This subsection summarizes and presents examples of phase equilibrium data currently available to the designer. The thermodynamic concepts used are presented in the subsection Thermodynamics of Sec. 4.

### PHASE EQUILIBRIUM DATA

For a binary mixture, pressure and temperature fix the equilibrium vapor and liquid compositions. Thus, experimental data are frequently presented in the form of tables of vapor mole fraction  $y$  and liquid mole fraction  $x$  for one constituent over a range of temperature  $T$  for a fixed pressure  $P$  or over a range of pressure for a fixed temperature. A small selection of such data, at a pressure of 101.3 kPa (1 atm, 1.013 bar), for four nonideal binary systems is given in Table 13-1. More extensive presentations and bibliographies of such data may be found in Hala, Wichterle, Polak, and Boublik (*Vapor-Liquid Equilibrium Data at Normal Pressures*, Pergamon, Oxford, 1968); Hirata, Ohe, and Nagahama (*Computer Aided Data Book of Vapor-Liquid Equilibria*, Elsevier, Amsterdam, 1975); Wichterle, Linek, and Hala (*Vapor-Liquid Equilibrium Data Bibliography*, Elsevier, Amsterdam, 1973, Supplement I, 1976, Supplement II, 1979); Ohe (*Vapor-Liquid Equilibrium Data*, Elsevier, Amsterdam, 1989); Ohe (*Vapor-Liquid Equilibrium Data at High Pressure*, Elsevier, Amsterdam, 1990); Walas (*Phase Equilibria in Chemical Engineering*, Butterworth, Boston, 1985); and, particularly, Gmehling and Onken [*Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chemistry Data ser., vol. 1 (parts 1–10), Frankfurt, 1977]. Extensive databases of phase equilibrium measurements are readily available in most process simulators together with models for correlating, interpolating, and extrapolating

(care is needed here) the data. Many of these simulators also provide graphical display of the data for easy visualization and interpretation.

For application to distillation (a nearly isobaric process) binary-mixture data are frequently plotted, for a fixed pressure, as  $y$  versus  $x$ , with a line of 45° slope included for reference, and as  $T$  versus  $y$  and  $x$ , as shown in Figs. 13-3 to 13-8. In some binary systems, one of the components is more volatile than the other over the entire composition range. This is the case in Figs. 13-3 and 13-4 for the benzene-toluene system at pressures of both 101.3 and 202.6 kPa (1 and 2 atm), where benzene is more volatile than toluene.

For other binary systems, one of the components is more volatile over only a part of the composition range. Two systems of this type, ethyl acetate-ethanol and chloroform-acetone, are shown in Figs. 13-5 to 13-7. Figure 13-5 shows that chloroform is less volatile than acetone below a concentration of 66 mol % chloroform and that ethyl acetate is more volatile than ethanol below a concentration of 53 mol % ethyl acetate. Above these concentrations, volatility is reversed. Such mixtures are known as azeotropic mixtures, and the composition in which the reversal occurs, which is the composition in which vapor and liquid compositions are equal, is the azeotropic composition, or azeotrope. The azeotropic liquid may be homogeneous or heterogeneous (two immiscible liquid phases). Two of the binary mixtures of Table 13-1 form homogeneous azeotropes. Non-azeotrope-forming mixtures such as benzene and toluene in Figs. 13-3 and 13-4 can be separated by simple distillation into two essentially pure products. By contrast, simple distillation of azeotropic mixtures will at best yield the azeotrope and one essentially pure species. The distillate and bottoms products obtained depend upon the feed composition and whether a minimum-boiling azeotrope is formed as with the ethyl acetate-ethanol mixture in Fig. 13-6 or a maximum-boiling azeotrope is formed as with the chloroform-acetone mixture in Fig. 13-7. For example, if a mixture of 30 mol % chloroform and 70 mol % acetone is fed to a simple distillation column, such as that



TABLE 13-1 Constant-Pressure Liquid-Vapor Equilibrium Data for Selected Binary Systems

| Component       |            | Temperature,<br>°C | Mole fraction A in |        | Total pressure,<br>kPa | Reference |
|-----------------|------------|--------------------|--------------------|--------|------------------------|-----------|
| A               | B          |                    | Liquid             | Vapor  |                        |           |
| Acetone         | Chloroform | 62.50              | 0.0817             | 0.0500 | 101.3                  | 1         |
|                 |            | 62.82              | 0.1390             | 0.1000 |                        |           |
|                 |            | 63.83              | 0.2338             | 0.2000 |                        |           |
|                 |            | 64.30              | 0.3162             | 0.3000 |                        |           |
|                 |            | 64.37              | 0.3535             | 0.3500 |                        |           |
|                 |            | 64.35              | 0.3888             | 0.4000 |                        |           |
|                 |            | 64.02              | 0.4582             | 0.5000 |                        |           |
|                 |            | 63.33              | 0.5299             | 0.6000 |                        |           |
|                 |            | 62.23              | 0.6106             | 0.7000 |                        |           |
|                 |            | 60.72              | 0.7078             | 0.8000 |                        |           |
|                 |            | 58.71              | 0.8302             | 0.9000 |                        |           |
|                 |            | 57.48              | 0.9075             | 0.9500 |                        |           |
| Acetone         | Water      | 74.80              | 0.0500             | 0.6381 | 101.3                  | 2         |
|                 |            | 68.53              | 0.1000             | 0.7301 |                        |           |
|                 |            | 65.26              | 0.1500             | 0.7716 |                        |           |
|                 |            | 63.59              | 0.2000             | 0.7916 |                        |           |
|                 |            | 61.87              | 0.3000             | 0.8124 |                        |           |
|                 |            | 60.75              | 0.4000             | 0.8269 |                        |           |
|                 |            | 59.95              | 0.5000             | 0.8387 |                        |           |
|                 |            | 59.12              | 0.6000             | 0.8532 |                        |           |
|                 |            | 58.29              | 0.7000             | 0.8712 |                        |           |
|                 |            | 57.49              | 0.8000             | 0.8950 |                        |           |
|                 |            | 56.68              | 0.9000             | 0.9335 |                        |           |
|                 |            | 56.30              | 0.9500             | 0.9627 |                        |           |
| Ethyl acetate   | Ethanol    | 78.3               | 0.0                | 0.0    | 101.3                  | 3         |
|                 |            | 76.6               | 0.050              | 0.102  |                        |           |
|                 |            | 75.5               | 0.100              | 0.187  |                        |           |
|                 |            | 73.9               | 0.200              | 0.305  |                        |           |
|                 |            | 72.8               | 0.300              | 0.389  |                        |           |
|                 |            | 72.1               | 0.400              | 0.457  |                        |           |
|                 |            | 71.8               | 0.500              | 0.516  |                        |           |
|                 |            | 71.8               | 0.540              | 0.540  |                        |           |
|                 |            | 71.9               | 0.600              | 0.576  |                        |           |
|                 |            | 72.2               | 0.700              | 0.644  |                        |           |
|                 |            | 73.0               | 0.800              | 0.726  |                        |           |
|                 |            | 74.7               | 0.900              | 0.837  |                        |           |
|                 |            | 76.0               | 0.950              | 0.914  |                        |           |
| Ethylene glycol | Water      | 77.1               | 1.000              | 1.000  | 30.4                   | 4         |
|                 |            | 69.5               | 0.0                | 0.0    |                        |           |
|                 |            | 76.1               | 0.23               | 0.002  |                        |           |
|                 |            | 78.9               | 0.31               | 0.003  |                        |           |
|                 |            | 83.1               | 0.40               | 0.010  |                        |           |
|                 |            | 89.6               | 0.54               | 0.020  |                        |           |
|                 |            | 103.1              | 0.73               | 0.06   |                        |           |
|                 |            | 118.4              | 0.85               | 0.13   |                        |           |
|                 |            | 128.0              | 0.90               | 0.22   |                        |           |
|                 |            | 134.7              | 0.93               | 0.30   |                        |           |
|                 |            | 145.0              | 0.97               | 0.47   |                        |           |
|                 |            | 160.7              | 1.00               | 1.00   |                        |           |

NOTE: To convert degrees Celsius to degrees Fahrenheit,  $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$ . To convert kilopascals to pounds-force per square inch, multiply by 0.145.

<sup>1</sup>Kojima, Kato, Sunaga, and Hashimoto, *Kagaku Kogaku*, **32**, 337 (1968).

<sup>2</sup>Kojima, Tochigi, Seki, and Watase, *Kagaku Kogaku*, **32**, 149 (1968).

<sup>3</sup>Chu, Getty, Brennecke, and Paul, *Distillation Equilibrium Data*, New York, 1950.

<sup>4</sup>Trimble and Potts, *Ind. Eng. Chem.*, **27**, 66 (1935).

shown in Fig. 13-1, operating at 101.3 kPa (1 atm), the distillate could approach pure acetone and the bottoms could approach the maximum-boiling azeotrope.

An example of heterogeneous-azeotrope formation is shown in Fig. 13-8 for the water–normal butanol system at 101.3 kPa. At liquid compositions between 0 and 3 mol % butanol and between 40 and 100 mol % butanol, the liquid phase is homogeneous. Phase splitting into two separate liquid phases (one with 3 mol % butanol and the other with 40 mol % butanol) occurs for any overall liquid composition between 3 and 40 mol % butanol. A minimum-boiling heterogeneous azeotrope occurs at 92°C (198°F) when the vapor composition and the overall composition of the two liquid phases are 25 mol % butanol.

For mixtures containing more than two species, an additional degree of freedom is available for each additional component. Thus, for a four-component system, the equilibrium vapor and liquid compositions are fixed only if the pressure, temperature, and mole fractions of two com-

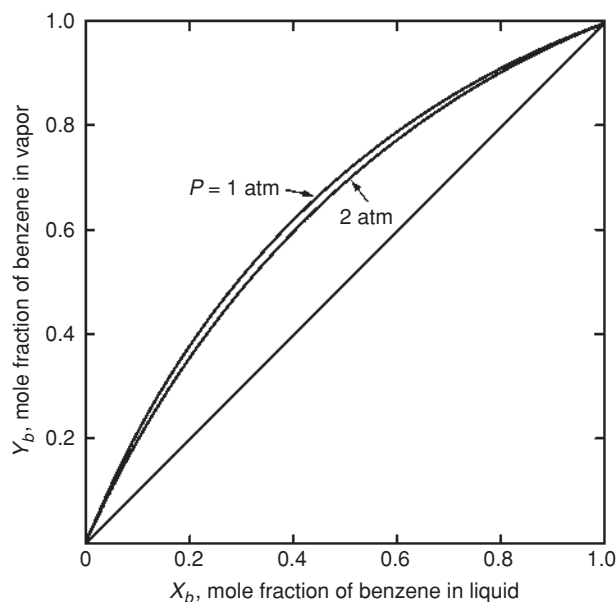
ponents are set. Representation of multicomponent vapor-liquid equilibrium data in tabular or graphical form of the type shown earlier for binary systems is either difficult or impossible. Instead, such data, as well as binary-system data, are commonly represented in terms of  $K$  values (vapor-liquid equilibrium ratios), which are defined by

$$K_i = \frac{y_i}{x_i} \quad (13-1)$$

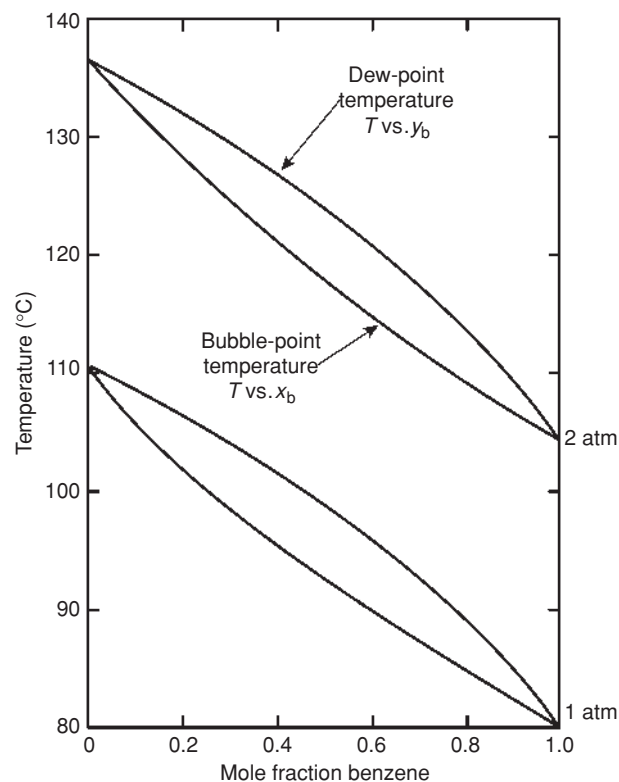
and are correlated empirically or theoretically in terms of temperature, pressure, and phase compositions in the form of tables, graphs, and equations. The  $K$  values are widely used in multicomponent distillation calculations, and the ratio of the  $K$  values of two species, called the relative volatility,

$$\alpha_{ij} = \frac{K_i}{K_j} \quad (13-2)$$

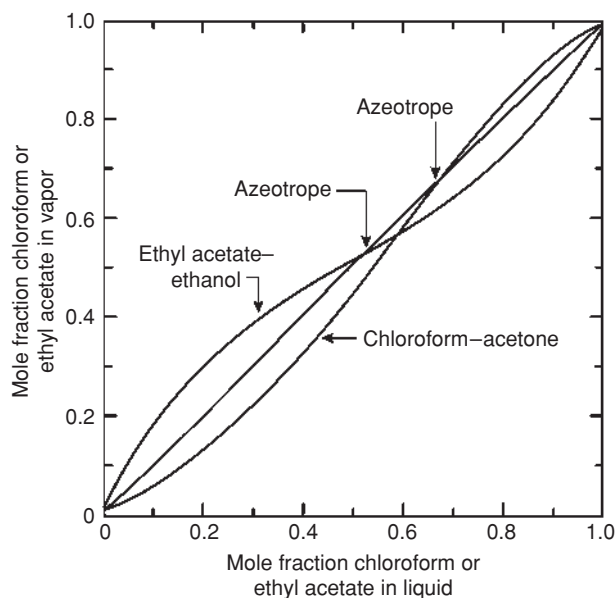
## 13-8 DISTILLATION



**FIG. 13-3** Isobaric  $y$ - $x$  curves for benzene-toluene. (Brian, Staged Cascades in Chemical Processing, Prentice-Hall, Englewood Cliffs, N.J., 1972.)



**FIG. 13-4** Isobaric vapor-liquid equilibrium curves for benzene-toluene. (Brian, Staged Cascades in Chemical Processing, Prentice-Hall, Englewood Cliffs, N.J., 1972.)

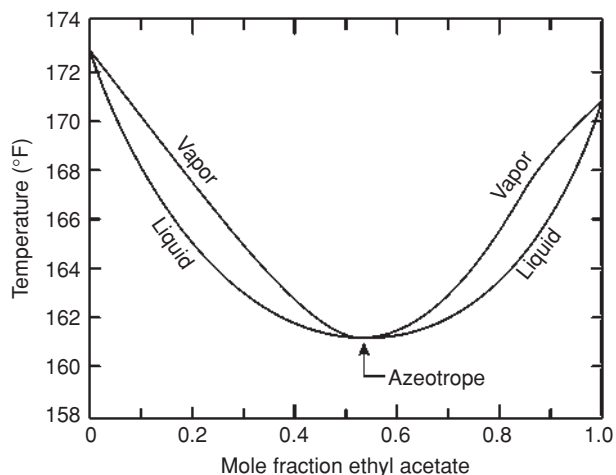


**FIG. 13-5** Vapor-liquid equilibria for the ethyl acetate-ethanol and chloroform-acetone systems at 101.3 kPa (1 atm).

is a convenient index of the relative ease or difficulty of separating components  $i$  and  $j$  by distillation. Rarely is distillation used on a large scale if the relative volatility is less than 1.05, with  $i$  more volatile than  $j$ .

### GRAPHICAL $K$ VALUE CORRELATIONS

As discussed in Sec. 4, the  $K$  value of a species is a complex function of temperature, pressure, and equilibrium vapor- and liquid-phase compositions. However, for mixtures of compounds of similar molecular structure and size, the  $K$  value depends mainly on temperature and pressure. For example, several major graphical  $K$  value correlations are available for light-hydrocarbon systems. The easiest to use are the DePriester charts [Chem. Eng. Prog. Symp. Ser. 7, **49**, 1 (1953)],



**FIG. 13-6** Liquid boiling points and vapor condensation temperatures for minimum-boiling azeotrope mixtures of ethyl acetate and ethanol at 101.3-kPa (1-atm) total pressure.

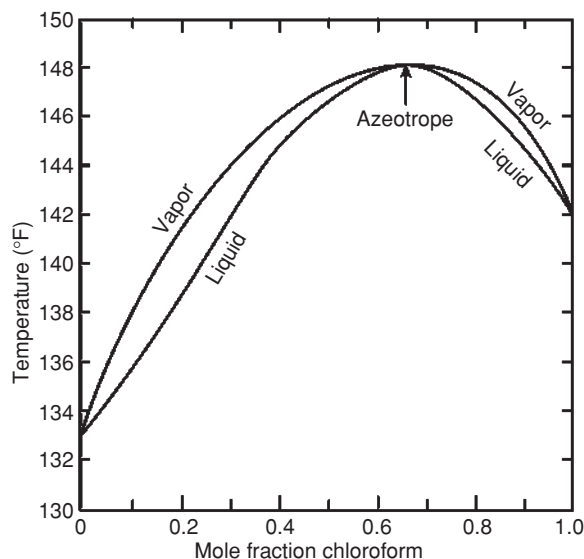


FIG. 13-7 Liquid boiling points and vapor condensation temperatures for maximum-boiling azeotrope mixtures of chloroform and acetone at 101.3-kPa (1-atm) total pressure.

which cover 12 hydrocarbons (methane, ethylene, ethane, propylene, propane, isobutane, isobutylene, *n*-butane, isopentane, *n*-pentane, *n*-hexane, and *n*-heptane). These charts are a simplification of the Kellogg charts (*Liquid-Vapor Equilibria in Mixtures of Light Hydrocarbons*, MWK Equilibrium Constants, *Polyco Data*, 1950) and include additional experimental data. The Kellogg charts, and hence the DePriester charts, are based primarily on the Benedict-Webb-Rubin equation of state [Chem. Eng. Prog., **47**, 419 (1951); **47**, 449 (1951)], which can represent both the liquid and the vapor phases and can predict *K* values quite accurately when the equation constants are available for the components in question.

A trial-and-error procedure is required with any *K* value correlation that takes into account the effect of composition. One cannot calculate *K* values until phase compositions are known, and those cannot be known until the *K* values are available to calculate them. For *K* as a

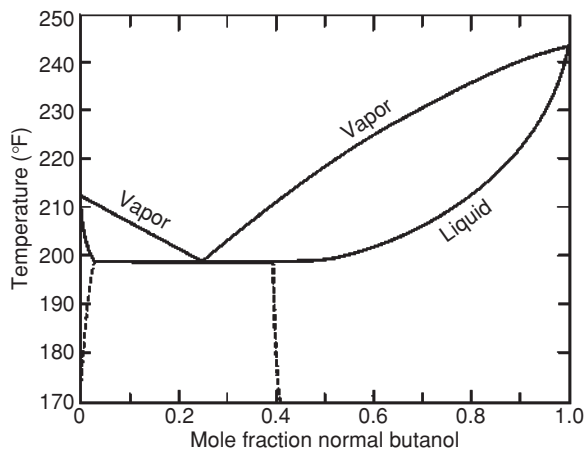


FIG. 13-8 Vapor-liquid equilibrium data for an *n*-butanol-water system at 101.3 kPa (1 atm); phase splitting and heterogeneous azeotrope formation.

function of *T* and *P* only, the DePriester charts provide good starting values for the iteration. These nomographs are shown in Fig. 13-9a and b. SI versions of these charts have been developed by Dadyburtov [Chem. Eng. Prog., **74**(4), 85 (1978)].

The Kellogg and DePriester charts and their subsequent extensions and generalizations use the molar average boiling points of the liquid and vapor phases to represent the composition effect. An alternative measure of composition is the convergence pressure of the system, which is defined as that pressure at which the *K* values for all the components in an isothermal mixture converge to unity. It is analogous to the critical point for a pure component in the sense that the two phases become indistinguishable. The behavior of a complex mixture of hydrocarbons for a convergence pressure of 34.5 MPa (5000 psia) is illustrated in Fig. 13-10.

Two major graphical correlations based on convergence pressure as the third parameter (besides temperature and pressure) are the charts published by the Gas Processors Association (GPA, *Engineering Data Book*, 9th ed., Tulsa, 1981) and the charts of the American Petroleum Institute (API, *Technical Data Book—Petroleum Refining*, New York, 1966) based on the procedures from Hadden and Grayson [Hydrocarbon Process., *Pet. Refiner*, **40**(9), 207 (1961)]. The former uses the method proposed by Hadden [Chem. Eng. Prog. Symp. Ser. 7, **49**, 53 (1953)] for the prediction of convergence pressure as a function of composition. The basis for Hadden's method is illustrated in Fig. 13-11, where it is shown that the critical loci for various mixtures of methane-propane-pentane fall within the area circumscribed by the three binary loci. (This behavior is not always typical of more nonideal systems.) The critical loci for the ternary mixtures vary linearly, at constant temperature, with weight percent propane on a methane-free basis. The essential point is that critical loci for mixtures are independent of the concentration of the lightest component in a mixture. This permits representation of a multicomponent mixture as a pseudobinary. The light component in this pseudobinary is the lightest species present (to a reasonable extent) in the multicomponent mixture. The heavy component is a pseudosubstance whose critical temperature is an average of all other components in the multicomponent mixture. This pseudocritical point can then be located on a *P-T* diagram containing the critical points for all compounds covered by the charts, and a critical locus can be drawn for the pseudobinary by interpolation between various real binary curves. Convergence pressure for the mixture at the desired temperature is read from the assumed loci at the desired system temperature. This method is illustrated in the left half of Fig. 13-12 for the methane-propane-pentane ternary. Associated *K* values for pentane at 104°C (220°F) are shown to the right as a function of mixture composition (or convergence pressure).

The GPA convergence pressure charts are primarily for alkane and alkene systems but do include charts for nitrogen, carbon dioxide, and hydrogen sulfide. The charts may not be valid when appreciable amounts of naphthenes or aromatics are present; the API charts use special procedures for such cases. Useful extensions of the convergence pressure concept to more varied mixtures include the nomographs of Winn [Chem. Eng. Prog. Symp. Ser. 2, **48**, 121 (1952)], Hadden and Grayson (op. cit.), and Cajander, Hipkin, and Lenoir [J. Chem. Eng. Data, **5**, 251 (1960)].

## ANALYTICAL *K* VALUE CORRELATIONS

The widespread availability and use of digital computers for distillation calculations have given impetus to the development of analytical expressions for *K* values. McWilliams [Chem. Eng., **80**(25), 138 (1973)] presents a regression equation and accompanying regression coefficients that represent the DePriester charts of Fig. 13-9. Regression equations and coefficients for various versions of the GPA convergence pressure charts are available from the GPA.

Preferred analytical correlations are less empirical and most often are theoretically based on one of two exact thermodynamic formulations, as derived in Sec. 4. When a single pressure-volume-temperature (*P-V-T*) equation of state is applicable to both vapor and liquid phases, the formulation used is

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V} \quad (13-3)$$

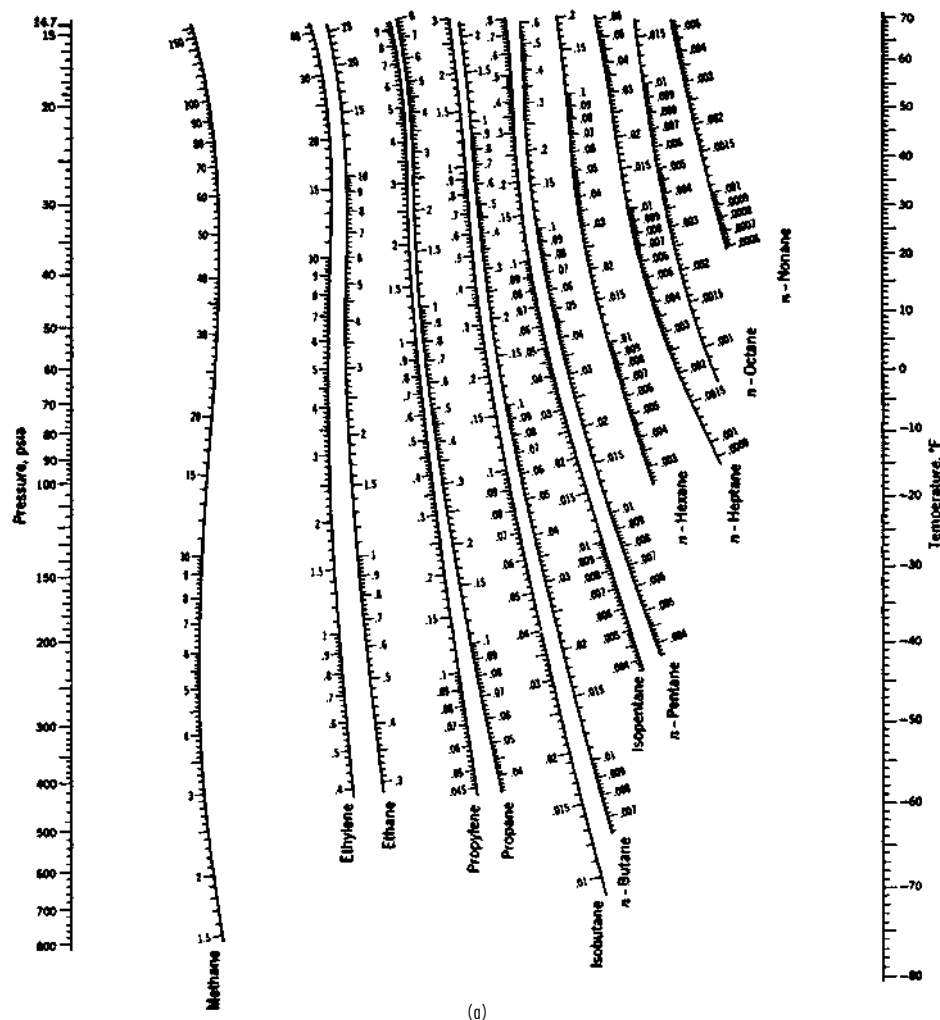


FIG. 13-9 The  $K$  values ( $K = y/x$ ) in light-hydrocarbon systems. (a) Low-temperature range. (b) High-temperature range. [DePriester, Chem. Eng. Prog. Symp., Ser. 7, 49, 1 (1953).]

where the mixture fugacity coefficients  $\hat{\Phi}_i^L$  for the liquid and  $\hat{\Phi}_i^V$  for the vapor are derived by classical thermodynamics from the  $P$ - $V$ - $T$  expression. Consistent equations for enthalpy can be similarly derived.

Until recently, equations of state that have been successfully applied to Eq. (13-3) have been restricted to mixtures of nonpolar compounds, namely, hydrocarbons and light gases. These equations include those of Benedict-Webb-Rubin (BWR), Soave (SRK) [Chem. Eng. Sci., 27, 1197 (1972)], who extended the remarkable Redlich-Kwong equation, and Peng-Robinson (PR) [Ind. Eng. Chem. Fundam., 15, 59 (1976)]. The SRK and PR equations belong to a family of so-called cubic equations of state. The Starling extension of the BWR equation (Fluid Thermodynamic Properties for Light Petroleum Systems, Gulf, Houston, 1973) predicts  $K$  values and enthalpies of the normal paraffins up through  $n$ -octane, as well as isobutane, isopentane, ethylene, propylene, nitrogen, carbon dioxide, and hydrogen sulfide, including the cryogenic region. Computer programs for  $K$  values derived from the SRK, PR, and other equations of state are widely available in all computer-aided process design and simulation programs. The ability of the SRK correlation to predict  $K$  values even when the pressure approaches the convergence pressure is shown for a multicomponent system in Fig. 13-13. Similar results are achieved with the PR correlation. The Wong-Sandler

mixing rules for cubic equations of state now permit such equations to be extended to mixtures of organic chemicals, as shown in a reformulated version by Orbey and Sandler [AIChE J., 41, 683 (1995)].

An alternative  $K$  value formulation that has received wide application to mixtures containing polar and/or nonpolar compounds is

$$K_i = \frac{\gamma_i^L \Phi_i^L}{\Phi_i^V} \quad (13-4)$$

where different equations of state may be used to predict the pure-component liquid fugacity coefficient  $\Phi_i^L$  and the vapor-mixture fugacity coefficient, and any one of a number of mixture free-energy models may be used to obtain the liquid activity coefficient  $\gamma_i^L$ . At low to moderate pressures, accurate prediction of the latter is crucial to the application of Eq. (13-4).

When either Eq. (13-3) or Eq. (13-4) can be applied, the former is generally preferred because it involves only a single equation of state applicable to both phases and thus would seem to offer greater consistency. In addition, the quantity  $\Phi_i^L$  in Eq. (13-4) is hypothetical for any components that are supercritical. In that case, a modification of Eq. (13-4) that uses Henry's law is sometimes applied.

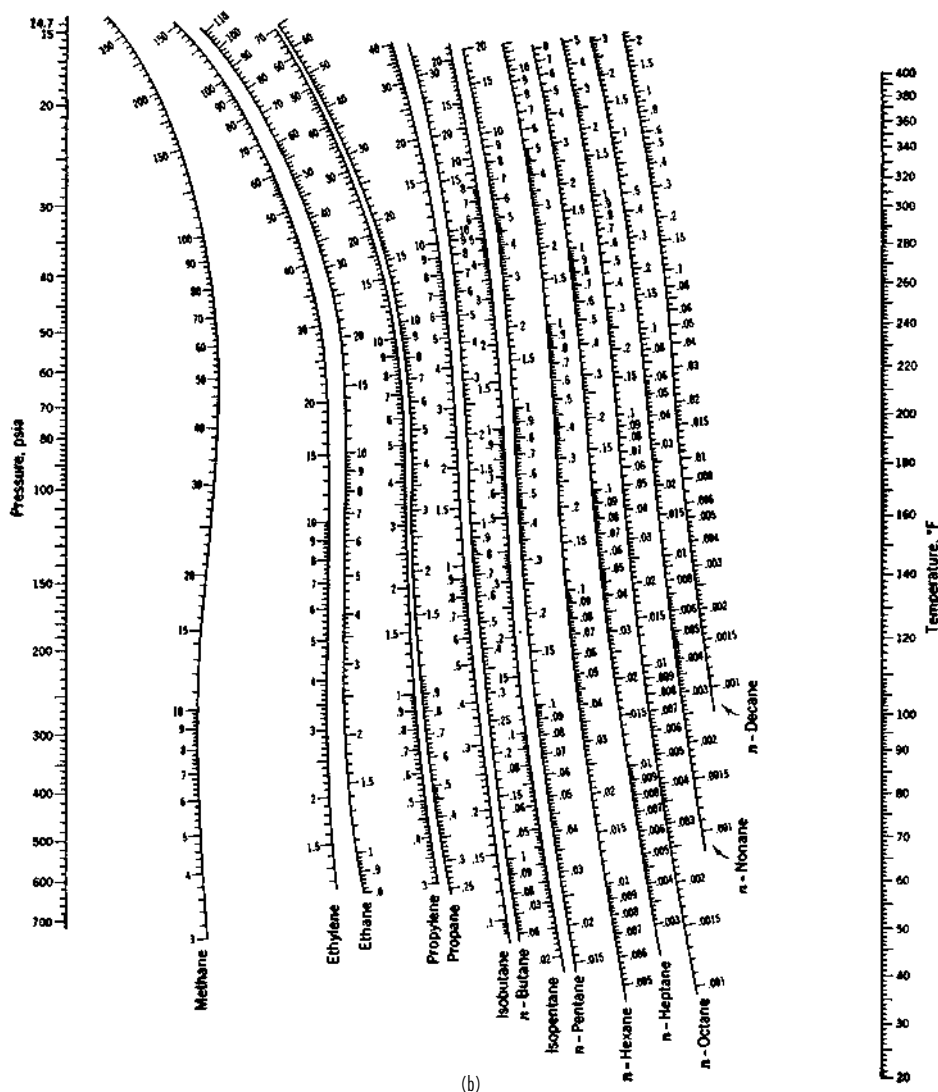


FIG. 13-9 (Continued)

For mixtures of hydrocarbons and light gases, Chao and Seader (CS) [AICHE J., 7, 598 (1961)] applied Eq. (13-4) by using an empirical expression for  $\Phi_i^L$  based on the generalized corresponding-states  $P$ - $V$ - $T$  correlation of Pitzer et al., the Redlich-Kwong equation of state for  $\Phi_i^V$ , and the regular solution theory of Scatchard and Hildebrand for  $\gamma_i^L$ . The predictive ability of the last-named theory is exhibited in Fig. 13-14 for the heptane-toluene system at 101.3 kPa (1 atm). Five pure-component constants for each species ( $T_v$ ,  $P_v$ ,  $\omega$ ,  $\delta$ , and  $v^L$ ) are required to use the CS method which, when applied within the restrictions discussed by Lenoir and Koppány [Hydrocarbon Process., 46(11), 249 (1967)], gives good results. Revised coefficients of Grayson and Streed (GS) (Paper 20-P07, Sixth World Pet. Conf. Frankfurt, June, 1963) for the  $\Phi_i^L$  expression permit application of the CS correlation to higher temperatures and pressures and give improved predictions for hydrogen. Jin, Greenkorn, and Chao [AICHE J., 41, 1602 (1995)] present a revised correlation for the standard-state liquid fugacity of hydrogen, applicable from 200 to 730 K.

For mixtures containing polar substances, more complex predictive equations for  $\gamma_i^L$  that involve binary-interaction parameters for each

pair of components in the mixture are required for use in Eq. (13-4), as discussed in Sec. 4. Six popular expressions are the Margules, van Laar, Wilson, NRTL, UNIFAC, and UNIQUAC equations. The preferred expressions for representing activity coefficients are the NRTL and UNIQUAC equations. Extensive listings of binary-interaction parameters for use in all but the UNIFAC equation are given by Gmehling and Onken (op. cit.). They obtained the parameters for binary systems at 101.3 kPa (1 atm) from best fits of the experimental  $T$ - $y$ - $x$  equilibrium data by setting  $\Phi_i^V$  and  $\Phi_i^L$  to their ideal-gas, ideal-solution limits of 1.0 and  $P^{\text{sat}}/P$ , respectively, with the vapor pressure  $P^{\text{sat}}$  given by a three-constant Antoine equation, whose values they tabulate. Table 13-2 lists their parameters for selected binary systems based on the binary system activity coefficient equation forms given in Table 13-3.

Consistent Antoine vapor pressure constants and liquid molar volumes are listed in Table 13-4. The Wilson equation is particularly useful for systems that are highly nonideal but do not undergo phase splitting, as exemplified by the ethanol-hexane system, whose activity coefficients are shown in Fig. 13-15. For systems such as this, in which activity coefficients in dilute regions may exceed values of

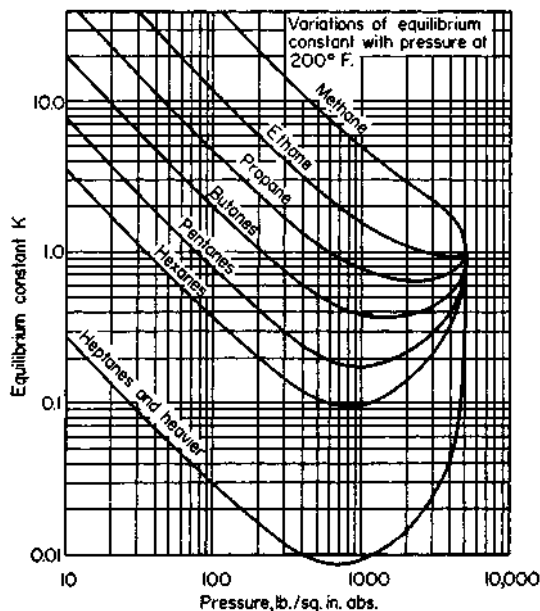


FIG. 13-10 Typical variation of  $K$  values with total pressure at constant temperature for a complex mixture. Light hydrocarbons in admixture with crude oil. [Katz and Hachmuth, *Ind. Eng. Chem.*, **29**, 1072 (1937).]

approximately 7.5, the van Laar equation erroneously predicts phase splitting.

Tables 13-2 and 13-4 include data on formic acid and acetic acid, two substances that tend to dimerize in the vapor phase according to the chemical equilibrium expression

$$K_D = \frac{P_D}{P_M^2} = 10^{A+B/T} \quad (13-5)$$

where  $K_D$  is the chemical equilibrium constant for dimerization,  $P_D$  and  $P_M$  are partial pressures of dimer and monomer, respectively, in torr,

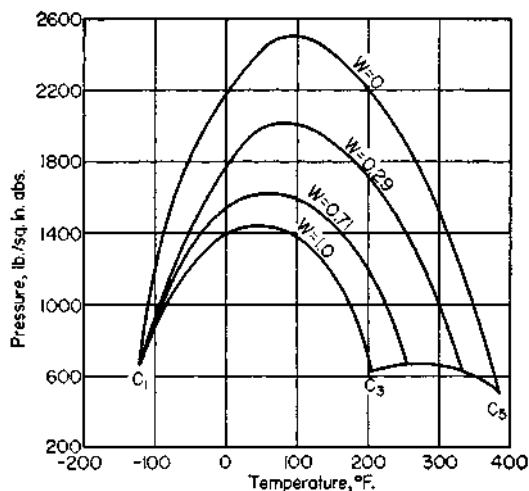


FIG. 13-11 Critical loci for a methane-propane-pentane system according to Hadden [Chem. Eng. Prog. Symp. Sec. 7, **49**, 53 (1953)]. Parameter  $W$  is weight fraction propane on a methane-free basis.

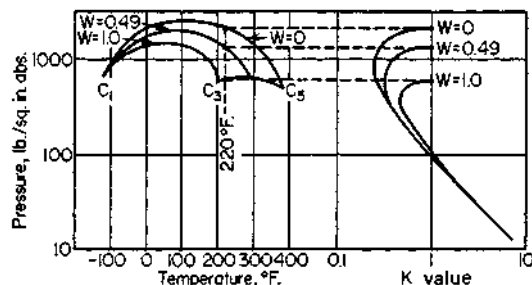


FIG. 13-12 Effect of mixture composition on  $K$  value for  $n$ -pentane at  $104^\circ\text{C}$  ( $220^\circ\text{F}$ ). The  $K$  values are shown for various values of  $W$ , weight fraction propane on a methane-free basis for the methane-propane-pentane system. [Hadden, Chem. Eng. Prog. Symp. Sec. 7, **49**, 58 (1953).]

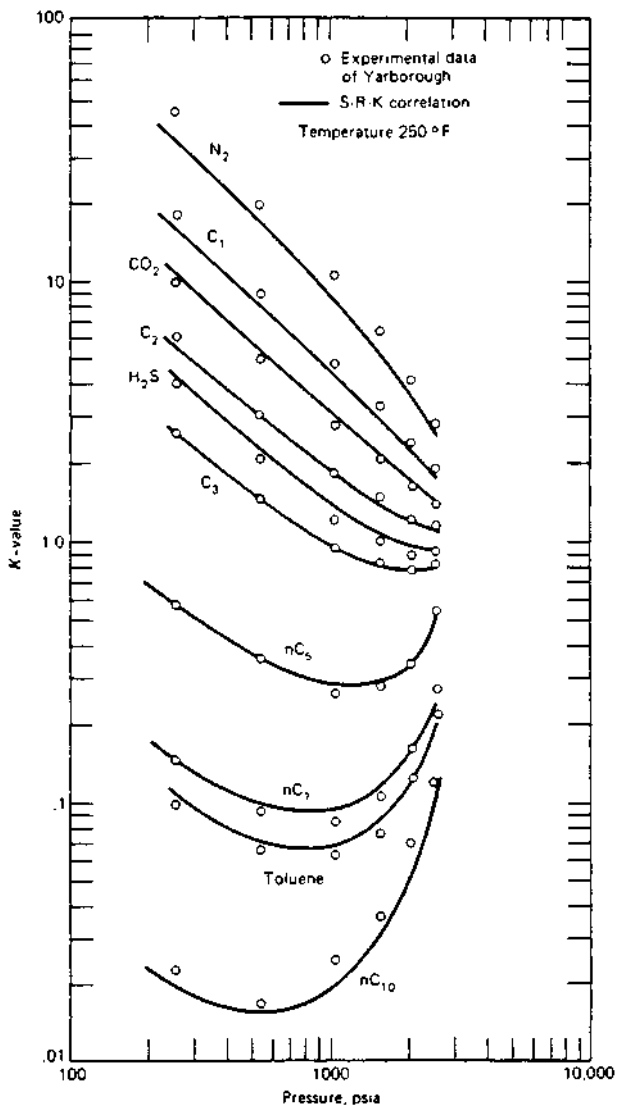


FIG. 13-13 Comparison of experimental  $K$  value data and SRK correlation. [Hendley and Seader, *Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981; data of Yarborough, *J. Chem. Eng. Data*, **17**, 129 (1972).]



TABLE 13-2 Binary-Interaction Parameters\*

| System                                | Margules       |                | van Laar |          | Wilson (cal/mol)                |                                 |
|---------------------------------------|----------------|----------------|----------|----------|---------------------------------|---------------------------------|
|                                       | $\bar{A}_{12}$ | $\bar{A}_{21}$ | $A_{12}$ | $A_{21}$ | $(\lambda_{12} - \lambda_{11})$ | $(\lambda_{21} - \lambda_{22})$ |
| Acetone (1), chloroform (2)           | -0.8404        | -0.5610        | -0.8643  | -0.5899  | 116.1171                        | -506.8519                       |
| Acetone (1), methanol (2)             | 0.6184         | 0.5788         | 0.6184   | 0.5797   | -114.4047                       | 545.2942                        |
| Acetone (1), water (2)                | 2.0400         | 1.5461         | 2.1041   | 1.5555   | 344.3346                        | 1482.2133                       |
| Carbon tetrachloride (1), benzene (2) | 0.0948         | 0.0922         | 0.0951   | 0.0911   | 7.0459                          | 59.6233                         |
| Chloroform (1), methanol (2)          | 0.8320         | 1.7365         | 0.9356   | 1.8860   | -361.7944                       | 1694.0241                       |
| Ethanol (1), benzene (2)              | 1.8362         | 1.4717         | 1.8570   | 1.4785   | 1264.4318                       | 266.6118                        |
| Ethanol (1), water (2)                | 1.6022         | 0.7947         | 1.6798   | 0.9227   | 325.0757                        | 953.2792                        |
| Ethyl acetate (1) ethanol (2)         | 0.8557         | 0.7476         | 0.8552   | 0.7526   | 58.8869                         | 570.0439                        |
| n-Hexane (1), ethanol (2)             | 1.9398         | 2.7054         | 1.9195   | 2.8463   | 320.3611                        | 2189.2896                       |
| Methanol (1), benzene (2)             | 2.1411         | 1.7905         | 2.1623   | 1.7925   | 1666.4410                       | 227.2126                        |
| Methanol (1), ethyl acetate (2)       | 1.0016         | 1.0517         | 1.0017   | 1.0524   | 982.2689                        | -172.9317                       |
| Methanol (1), water (2)               | 0.7923         | 0.5434         | 0.8041   | 0.5619   | 82.9876                         | 520.6458                        |
| Methyl acetate (1), methanol (2)      | 0.9605         | 1.0120         | 0.9614   | 1.0126   | -93.8900                        | 847.4348                        |
| 1-Propanol (1), water (2)             | 2.7070         | 0.7172         | 2.9095   | 1.1572   | 906.5256                        | 1396.6398                       |
| 2-Propanol (1), water (2)             | 2.3319         | 0.8976         | 2.4702   | 1.0938   | 659.5473                        | 1230.2080                       |
| Tetrahydrofuran (1), water (2)        | 2.8258         | 1.9450         | 3.0216   | 1.9436   | 1475.2583                       | 1844.7926                       |
| Water (1), acetic acid (2)            | 0.4178         | 0.9533         | 0.4973   | 1.0623   | 705.5876                        | 111.6579                        |
| Water (1), 1-butanol (2)              | 0.8608         | 3.2051         | 1.0996   | 4.1760   | 1549.6600                       | 2050.2569                       |
| Water (1), formic acid (2)            | -0.2966        | -0.2715        | -0.2935  | -0.2757  | -310.1060                       | 1180.8040                       |

\* Abstracted from Gmehling and Onken, *Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chemistry Data ser., vol. 1 (parts 1-10). Frankfurt, 1977.

and  $T$  is in kelvins. Values of  $A$  and  $B$  for the first four normal aliphatic acids are

|                  | $A$     | $B$  |
|------------------|---------|------|
| Formic acid      | -10.743 | 3083 |
| Acetic acid      | -10.421 | 3166 |
| n-Propionic acid | -10.843 | 3316 |
| n-Butyric acid   | -10.100 | 3040 |

As shown by Marek and Standart [*Collect. Czech. Chem. Commun.*, **19**, 1074 (1954)], it is preferable to correlate and use liquid-phase activity coefficients for the dimerizing component by considering separately the partial pressures of the monomer and dimer. For example, for a binary system of components 1 and 2, when only compound 1 dimerizes in the vapor phase, the following equations apply if an ideal gas is assumed:

$$P_1 = P_D + P_M \quad (13-6)$$

$$y_1 = \frac{P_M + 2P_D}{P} \quad (13-7)$$

These equations when combined with Eq. (13-5) lead to the following equations for liquid-phase activity coefficients in terms of measurable quantities:

$$\gamma_1 = \frac{Py_1}{P_1^{\text{sat}}x_1} \left\{ \frac{1 + (1 + 4K_D P_1^{\text{sat}})^{0.5}}{1 + [1 + 4K_D Py_1(2 - y_1)]^{0.5}} \right\} \quad (13-8)$$

$$\gamma_2 = \frac{Py_2}{P_2^{\text{sat}}x_2} \left( \frac{2(1 - y_1 + [1 + 4K_D Py_1(2 - y_1)]^{0.5})}{(2 - y_1)[1 + [1 + 4K_D Py_1(2 - y_1)]^{0.5}]} \right) \quad (13-9)$$

Detailed procedures, including computer programs for evaluating binary-interaction parameters from experimental data and then using these parameters to predict  $K$  values and phase equilibria, are given in terms of the UNIQUAC equation by Prausnitz et al. (*Computer Calculations for Multicomponent Vapor-Liquid and Liquid-Liquid Equilibria*, Prentice-Hall, Englewood Cliffs, N.J., 1980) and in terms of the UNIFAC group contribution method by Fredenslund, Gmehling, and Rasmussen (*Vapor-Liquid Equilibria Using UNIFAC*, Elsevier, Amsterdam, 1980). Both use the method of Hayden and O'Connell [*Ind. Eng. Chem. Process Des. Dev.*, **14**, 209 (1975)] to compute  $\Phi_1^V$  in Eq. (13-4). When the system temperature is greater than the critical

TABLE 13-3 Activity-Coefficient Equations in Binary Form for Use with Parameters and Constants in Tables 13-2 and 13-4

| Type of equation | Adjustable parameters         | Equations in binary form                                                                                                                                  |
|------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Margules         | $\bar{A}_{12}$                | $\ln \gamma_1 = [\bar{A}_{12} + 2(\bar{A}_{21} - \bar{A}_{12})x_1]x_2^2$                                                                                  |
|                  | $\bar{A}_{21}$                | $\ln \gamma_2 = [\bar{A}_{21} + 2(\bar{A}_{12} - \bar{A}_{21})x_2]x_1^2$                                                                                  |
| van Laar         | $A_{12}$                      | $\ln \gamma_1 = A_{12} \left( \frac{A_{21}x_2}{A_{12}x_1 + A_{21}x_2} \right)^2$                                                                          |
|                  | $A_{21}$                      | $\ln \gamma_2 = A_{21} \left( \frac{A_{12}x_1}{A_{12}x_1 + A_{21}x_2} \right)^2$                                                                          |
| Wilson           | $\lambda_{12} - \lambda_{11}$ | $\ln \gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2 \left( \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{\Lambda_{21}x_1 + x_2} \right)$ |
|                  | $\lambda_{21} - \lambda_{22}$ | $\ln \gamma_2 = -\ln(x_2 + \Lambda_{21}x_1) - x_1 \left( \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{\Lambda_{21}x_1 + x_2} \right)$ |

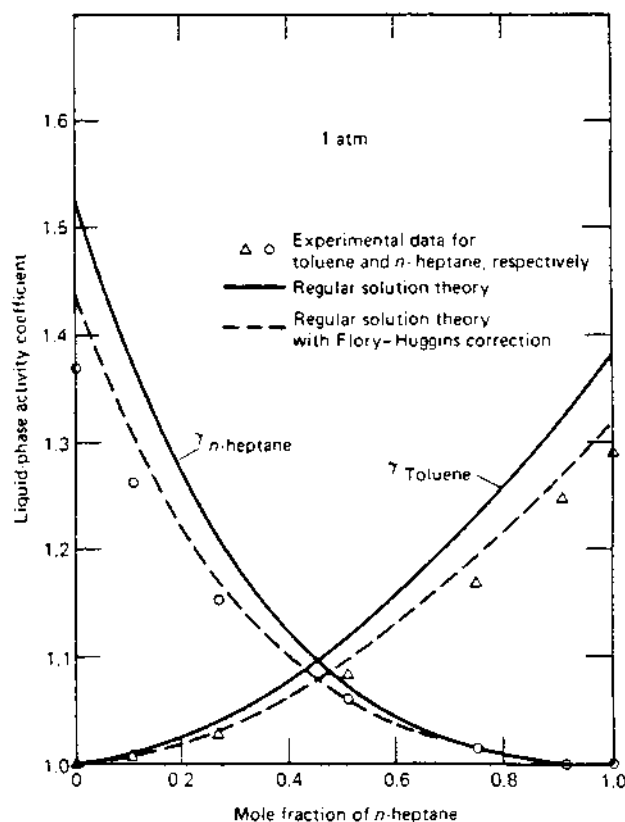
$$\text{where } \Lambda_{12} = \frac{v_2^L}{v_1^L} \exp\left(-\frac{\lambda_{12} - \lambda_{11}}{RT}\right) \quad \Lambda_{21} = \frac{v_1^L}{v_2^L} \exp\left(-\frac{\lambda_{21} - \lambda_{22}}{RT}\right)$$

$v_i^L$  = molar volume of pure-liquid component  $i$

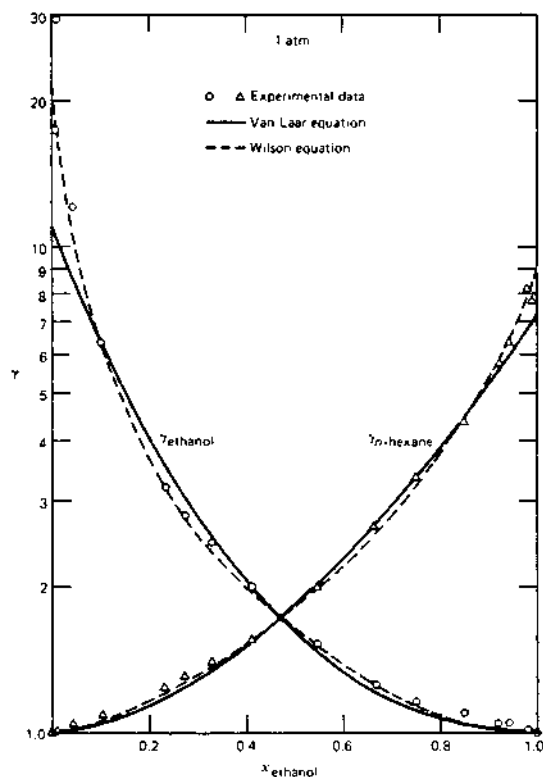
$\lambda_{ij}$  = interaction energy between components  $i$  and  $j$ ,  $\lambda_{ij} = \lambda_{ji}$



## 13-14 DISTILLATION



**FIG. 13-14** Liquid-phase activity coefficients for an *n*-heptane-toluene system at 101.3 kPa (1 atm). [Henley and Seader, *Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981; data of Yerazunis *et al.*, *AIChE J.*, **10**, 660 (1964).]



**FIG. 13-15** Liquid-phase activity coefficients for an ethanol-*n*-hexane system. [Henley and Seader, *Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981; data of Sinor and Weber, *J. Chem. Eng. Data*, **5**, 243-247 (1960).]

**TABLE 13-4 Antoine Vapor-Pressure Constants and Liquid Molar Volume\***

| Species              | Antoine constants† |          |         | Applicable temperature region, °C | $v^L$ , liquid molar volume, cm <sup>3</sup> /g-mol |
|----------------------|--------------------|----------|---------|-----------------------------------|-----------------------------------------------------|
|                      | A                  | B        | C       |                                   |                                                     |
| Acetic acid          | 8.02100            | 1936.010 | 258.451 | 18-118                            | 57.54                                               |
| Acetone              | 7.11714            | 1210.595 | 229.664 | (-13)-55                          | 74.05                                               |
| Benzene              | 6.87987            | 1196.760 | 219.161 | 8-80                              | 89.41                                               |
| 1-Butanol            | 7.36366            | 1305.198 | 173.427 | 89-126                            | 91.97                                               |
| Carbon tetrachloride | 6.84083            | 1177.910 | 220.576 | (-20)-77                          | 97.09                                               |
| Chloroform           | 6.95465            | 1170.966 | 226.232 | (-10)-60                          | 80.67                                               |
| Ethanol              | 7.58670            | 1281.590 | 193.768 | 78-203                            | 58.68                                               |
| Ethanol              | 8.11220            | 1592.864 | 226.184 | 20-93                             | 58.68                                               |
| Ethyl acetate        | 7.10179            | 1244.951 | 217.881 | 16-76                             | 98.49                                               |
| Formic acid          | 6.94459            | 1295.260 | 218.000 | 36-108                            | 37.91                                               |
| <i>n</i> -Hexane     | 6.91058            | 1189.640 | 226.280 | (-30)-170                         | 131.61                                              |
| Methanol             | 8.08097            | 1582.271 | 239.726 | 15-84                             | 40.73                                               |
| Methyl acetate       | 7.06524            | 1157.630 | 219.726 | 2-56                              | 79.84                                               |
| 1-Propanol           | 8.37895            | 1788.020 | 227.438 | (-15)-98                          | 75.14                                               |
| 2-Propanol           | 8.87829            | 2010.320 | 252.636 | (-26)-83                          | 76.92                                               |
| Tetrahydrofuran      | 6.99515            | 1202.290 | 226.254 | 23-100                            | 81.55                                               |
| Water                | 8.07131            | 1730.630 | 233.426 | 1-100                             | 18.07                                               |

\* Abstracted from Gmehling and Onken, *Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chemistry Data ser., vol. 1 (parts 1-10), Frankfurt, 1977.

† Antoine equation is  $\log P^{\text{sat}} = A - B/(T + C)$  with  $P^{\text{sat}}$  in torr and  $T$  in °C.

NOTE: To convert degrees Celsius to degrees Fahrenheit, °F = 1.8°C + 32. To convert cubic centimeters per gram-mole to cubic feet per pound-mole, multiply by 0.016.

temperature of one or more components in the mixture, Prausnitz et al. use a Henry's law constant  $H_{i,M}$  in place of the product  $\gamma_i^L \Phi_i^L$  in Eq. (13-4). Otherwise  $\Phi_i^L$  is evaluated from vapor pressure data with a Poynting saturated-vapor fugacity correction. When the total pressure is less than about 202.6 kPa (2 atm) and all components in the mixture have a critical temperature that is greater than the system temperature, then  $\Phi_i^L = P_i^{\text{sat}}/P$  and  $\Phi_i^V = 1.0$ . Equation (13-4) then reduces to

$$K_i = \frac{\gamma_i^L P_i^{\text{sat}}}{P} \quad (13-10)$$

which is referred to as a modified Raoult's law  $K$  value. If, furthermore, the liquid phase is ideal, then  $\gamma_i^L = 1.0$  and

$$K_i = \frac{P_i^{\text{sat}}}{P} \quad (13-11)$$

which is referred to as a Raoult's law  $K$  value that is dependent solely on the vapor pressure  $P_i^{\text{sat}}$  of the component in the mixture. The UNIFAC method is being periodically updated with new group contributions; e.g., see Hansen et al. [*Ind. Eng. Chem. Res.*, **30**, 2352 (1991)].

## SINGLE-STAGE EQUILIBRIUM FLASH CALCULATIONS

The simplest continuous distillation process is the adiabatic single-stage equilibrium flash process pictured in Fig. 13-16. Feed temperature and the pressure drop across the valve are adjusted to vaporize the feed to the desired extent, while the drum provides disengaging space to allow the vapor to separate from the liquid. The expansion across the valve is at constant enthalpy, and this fact can be used to calculate  $T_2$  (or  $T_1$  to give a desired  $T_2$ ).

A degrees-of-freedom analysis indicates that the variables subject to the designer's control are  $\bar{C} + 3$  in number. The most common way to use these is to specify the feed rate, composition, and pressure ( $C + 1$  variables) plus the drum temperature  $T_2$  and pressure  $P_2$ . This operation will give one point on the *equilibrium flash curve* shown in Fig. 13-17. This curve shows the relation at constant pressure between the fraction  $V/F$  of the feed flashed and the drum temperature. The temperature at  $V/F = 0.0$  when the first bubble of vapor is about to form (saturated liquid) is the *bubble point* temperature of the feed mixture, and the value at  $V/F = 1.0$  when the first droplet of liquid is about to form (saturated vapor) is the *dew point* temperature.

### BUBBLE POINT AND DEW POINT

For a given drum pressure and feed composition, the bubble and dew point temperatures bracket the temperature range of the equilibrium flash. At the bubble point temperature, the total vapor pressure exerted by the mixture becomes equal to the confining drum pressure, and it follows that  $\sum y_i = 1.0$  in the bubble formed. Since  $y_i = K_i x_i$  and since the  $x_i$ 's still equal the feed compositions (denoted by  $z_i$ ), calculation of the bubble point temperature involves a trial-and-error search for the temperature which, at the specified pressure, makes  $\sum K_i z_i = 1.0$ . If instead the temperature is specified, one can find the bubble point pressure that satisfies this relationship.

At the dew point temperature  $y_i$  still equals  $z_i$  and the relationship  $\sum x_i = \sum z_i/K_i = 1.0$  must be satisfied. As in the case of the bubble point, a trial-and-error search for the dew point temperature at a specified pressure is involved. Or, if the temperature is specified, the dew point pressure can be calculated.

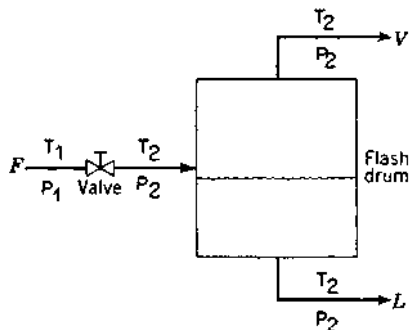


FIG. 13-16 Equilibrium flash separator.

### ISOTHERMAL FLASH

The calculation for a point on the flash curve that is intermediate between the bubble point and the dew point is referred to as an isothermal flash calculation because  $T_2$  is specified. Except for an ideal binary mixture, procedures for calculating an isothermal flash are iterative. A popular and recommended method is the following, due to Rachford and Rice [*J. Pet. Technol.*, **4**(10), sec. 1, p. 19, and sec. 2, p. 3 (October 1952)]. The component mole balance ( $Fz_i = Vy_i + Lx_i$ ), phase distribution relation ( $K_i = y_i/x_i$ ), and total mole balance ( $F = V + L$ ) can be combined to give

$$x_i = \frac{z_i}{1 + (V/F)(K_i - 1)} \quad (13-12)$$

$$y_i = \frac{K_i z_i}{1 + (V/F)(K_i - 1)} \quad (13-13)$$

Since  $\sum x_i - \sum y_i = 0$ ,

$$f\left(\frac{V}{F}\right) = \sum_i \frac{z_i(1 - K_i)}{1 + (V/F)(K_i - 1)} = 0 \quad (13-14)$$

Equation (13-14) is solved iteratively for  $V/F$ , followed by the calculation of values of  $x_i$  and  $y_i$  from Eqs. (13-12) and (13-13) and  $L$  from the total mole balance. Any one of a number of numerical root-finding procedures such as the Newton-Raphson, secant, false-position, or bisection method can be used to solve Eq. (13-14). Values of  $K_i$  are constants if they are independent of liquid and vapor compositions. Then the resulting calculations are straightforward. Otherwise, the  $K_i$  values must be periodically updated for composition effects, perhaps

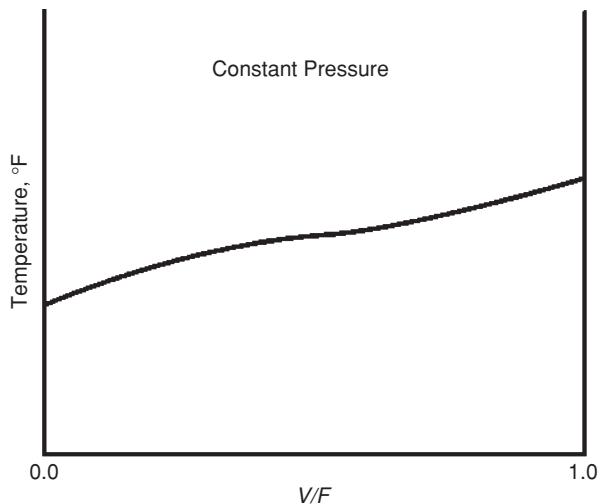


FIG. 13-17 Equilibrium flash curve.

## 13-16 DISTILLATION

after each iteration, using prorated values of  $x_i$  and  $y_i$  from Eqs. (13-12) and (13-13). Generally the iterations are continued until the change in the absolute value of  $V/F$  is sufficiently small and until the absolute value of the residual  $f(V/F)$  is close to zero. When converged,  $\sum x_i$  and  $\sum y_i$  will each be very close to a value of 1, and, if desired,  $T_1$  can be computed from an energy balance around the valve if no heat exchanger is used. Alternatively, if  $T_1$  is fixed, as mentioned earlier, a heat exchanger must be added before, after, or in place of the valve with the required heat duty being calculated from an energy balance. The limits of applicability of Eqs. (13-12) to (13-14) are the bubble point, at which  $V = 0$  and  $x_i = z_i$ , and the dew point, at which  $L = 0$  and  $y_i = z_i$ . At these limits Eq. (13-14) reduces to the bubble point equation

$$\sum_i K_i x_i = 1 \quad (13-15)$$

and the dew point equation, respectively,

$$\sum_i \frac{y_i}{K_i} = 1 \quad (13-16)$$

For a *binary feed*, specification of the flash drum temperature and pressure fixes the equilibrium-phase compositions, which are related to the  $K$  values by

$$x_1 = \frac{1 - K_2}{K_1 - K_2} \quad \text{and} \quad y_1 = \frac{K_1 K_2 - K_1}{K_2 - K_1}$$

The mole balance can be rearranged to

$$\frac{V}{F} = \frac{z_1(K_1 - K_2)/(1 - K_2) - 1}{K_1 - 1}$$

If  $K_1$  and  $K_2$  are functions of temperature and pressure only (ideal solutions), the flash curve can be calculated directly without iteration.

### ADIABATIC FLASH

In Fig. 13-16, if  $P_2$  and the feed-stream conditions (that is,  $F$ ,  $z_i$ ,  $T_1$ ,  $P_1$ ) are known, then the calculation of  $T_2$ ,  $V$ ,  $L$ ,  $y_i$ , and  $x_i$  is referred to as an adiabatic flash. In addition to Eqs. (13-12) to (13-14) and the total mole balance, the following energy balance around both the valve and the flash drum combined must be included:

$$H^F F = H^V V + H^L L \quad (13-17)$$

By taking a basis of  $F = 1.0$  mol and eliminating  $L$  with the total mole balance, Eq. (13-17) becomes

$$f_2[V, T_2] = H^F - V(H^V - H^L) - H^L = 0 \quad (13-18)$$

With  $T_2$  now unknown, Eq. (13-14) becomes

$$f_1[V, T_2] = \sum_i \frac{z_i(1 - K_i)}{1 + V(K_i - 1)} = 0 \quad (13-19)$$

A number of iterative procedures have been developed for solving Eqs. (13-18) and (13-19) simultaneously for  $V$  and  $T_2$ . Frequently, and especially if the feed contains components of a narrow range of volatility, convergence is rapid for a tearing method in which a value of  $T_2$  is assumed, Eq. (13-19) is solved iteratively by the isothermal flash procedure, and, using that value of  $V$ , Eq. (13-18) is solved iteratively for a new approximation of  $T_2$ , which is then used to initiate the next cycle

until  $T_2$  and  $V$  converge. However, if the feed contains components of a wide range of volatility, it may be best to invert the sequence and assume a value for  $V$ , solve Eq. (13-19) for  $T_2$ , solve Eq. (13-18) for  $V$ , and then repeat the cycle. If the  $K$  values and/or enthalpies are sensitive to the unknown phase compositions, it may be necessary to solve Eqs. (13-18) and (13-19) simultaneously by a Newton or other suitable iterative technique. Alternatively, the two-tier method of Boston and Britt [*Comput. Chem. Eng.*, **2**, 109 (1978)], which is also suitable for difficult isothermal flash calculations, may be applied.

### OTHER FLASH SPECIFICATIONS

Flash drum specifications in addition to ( $P_2$ ,  $T_2$ ) and ( $P_2$ , adiabatic) are possible but must be applied with care, as discussed by Michelsen [*Comp. Chem. Engng.*, **17**, 431 (1993)]. Most computer-aided process design and simulation programs permit a wide variety of flash specifications.

### THREE-PHASE FLASH

Single-stage equilibrium flash calculations become considerably more complex when an additional liquid phase can form, as in mixtures of water with hydrocarbons, water with ethers, and water with higher alcohols (containing four or more carbon atoms). Procedures for computing such situations are referred to as three-phase flash methods, which are given for the general case by Henley and Rosen (*Material and Energy Balance Computations*, Wiley, New York, 1968, chap. 8). When the two liquid phases are almost mutually insoluble, they can be considered separately and relatively simple procedures apply, as discussed by Smith (*Design of Equilibrium Stage Processes*, McGraw-Hill, New York, 1963). Condensation of such mixtures may result in one liquid phase being formed before the other. Computer-aided process design and simulation programs all contain a Gibbs free-energy routine that can compute a three-phase flash by minimization of the Gibbs free energy. Many important and subtle aspects of three-phase flash calculations are discussed by Michelsen [*Fluid Phase Equil.*, **9**, 1, 21 (1982)], McDonald and Floudas [*AIChE J.*, **41**, 1798 (1995)], and Wasylkiewicz et al. [*Ind. Eng. Chem. Research*, **35**, 1395 (1996)].

### COMPLEX MIXTURES

Feed analyses in terms of component compositions are usually not available for complex hydrocarbon mixtures with a final normal boiling point above about 38°C (100°F) (*n*-pentane). One method of handling such a feed is to break it down into pseudocomponents (narrow-boiling fractions) and then estimate the mole fraction and  $K$  value for each such component. Edmister [*Ind. Eng. Chem.*, **47**, 1685 (1955)] and Maxwell (*Data Book on Hydrocarbons*, Van Nostrand, Princeton, N.J., 1958) give charts that are useful for this estimation. Once  $K$  values are available, the calculation proceeds as described above for multicomponent mixtures. Another approach to complex mixtures is to obtain an American Society for Testing and Materials (ASTM) or true-boiling point (TBP) curve for the mixture and then use empirical correlations to construct the atmospheric-pressure equilibrium flash vaporization (EFV) curve, which can then be corrected to the desired operating pressure. A discussion of this method and the necessary charts is presented in a later subsection Petroleum and Complex-Mixture Distillation.

## GRAPHICAL METHODS FOR BINARY DISTILLATION

Multistage distillation under continuous, steady-state operating conditions is widely used in practice to separate a variety of mixtures. Table 13-5, taken from the study of Mix, Dweck, Weinberg, and Armstrong [*AIChE Symp. Ser.* **76**, 192, 10 (1980)] lists key components along with typical stage requirements to perform the separation for 27 industrial distillation processes. The design of multistage columns can be accomplished by graphical techniques when the feed mixture contains only two components. The  $x$ - $y$  diagram method developed by McCabe and Thiele [*Ind. Eng. Chem.*, **17**, 605 (1925)]

uses only phase equilibrium and mole balance relationships. The method assumes an adiabatic column (no heat losses through the column walls) and constant latent heat for the binary mixture at all compositions (which requires, among other things, equal latent heat for both components). The method is exact only for those systems in which energy effects on vapor and liquid rates leaving the stages are negligible. However, the approach is simple and gives a useful first estimate of the column design which can be refined by using the enthalpy composition diagram method of Ponchon [*Tech. Mod.*, **13**,

**TABLE 13-5 Key Components and Typical Number of (Real) Stages Required to Perform the Separation for Distillation Processes of Industrial Importance**

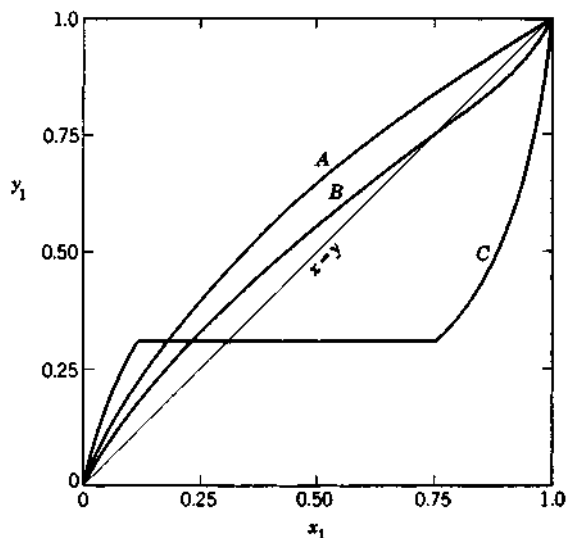
| Key components                      | Typical number of trays |
|-------------------------------------|-------------------------|
| Hydrocarbon systems                 |                         |
| Ethylene-ethane                     | 73                      |
| Propylene-propane                   | 138                     |
| Propyne-1-3-butadiene               | 40                      |
| 1-3 Butadiene-vinyl acetylene       | 130                     |
| Benzene-toluene                     | 34, 53                  |
| Benzene-ethyl benzene               | 20                      |
| Benzene-diethyl benzene             | 50                      |
| Toluene-ethyl benzene               | 28                      |
| Toluene-xylenes                     | 45                      |
| Ethyl benzene-styrene               | 34                      |
| <i>o</i> -Xylene- <i>m</i> -xylene  | 130                     |
| Organic systems                     |                         |
| Methanol-formaldehyde               | 23                      |
| Dichloroethane-trichloroethane      | 30                      |
| Acetic acid-acetic anhydride        | 50                      |
| Acetic anhydride-ethylene diacetate | 32                      |
| Vinyl acetate-ethyl acetate         | 90                      |
| Ethylene glycol-diethylene glycol   | 16                      |
| Cumene-phenol                       | 38                      |
| Phenol-acetophenone                 | 39, 54                  |
| Aqueous systems                     |                         |
| HCN-water                           | 15                      |
| Acetic acid-water                   | 40                      |
| Methanol-water                      | 60                      |
| Ethanol-water                       | 60                      |
| Isopropanol-water                   | 12                      |
| Vinyl acetate-water                 | 35                      |
| Ethylene oxide-water                | 50                      |
| Ethylene glycol-water               | 16                      |

20, 55 (1921)] and Savarit [*Arts Metiers*, **65**, 142, 178, 241, 266, 307 (1922)]. This approach uses the energy balance in addition to mole balance and phase equilibrium relationships and is rigorous when enough calorimetric data are available to construct the diagram without assumptions.

With the widespread availability of computers, the preferred approach to design is equation-based since it provides answers rapidly and repeatedly without the tedium of redrawing graphs. Such an approach is especially useful for sensitivity analysis, which gives insight into how a design changes under variations or uncertainty in design parameters such as thermodynamic properties; feed flow rate, composition, temperature, and pressure; and desired product compositions. Nevertheless, diagrams are useful for quick approximations, for interpreting the results of equation-based methods, and for demonstrating the effect of various design variables. The  $x$ - $y$  diagram is the most convenient for these purposes, and its use is developed in detail here. The use of the enthalpy composition diagram is given by Smith (*Design of Equilibrium Stage Processes*, McGraw-Hill, New York, 1963) and Henley and Seader (*Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981). An approximate equation-based approach based on the enthalpy composition diagram was proposed by Peters [*Ind. Eng. Chem.*, **14**, 476 (1922)] with additional aspects developed later by others. Doherty and Malone (*Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, app. A) describe this method for binary mixtures and extend it to multicomponent systems. The approach is exact when the enthalpy composition surfaces are linear.

## PHASE EQUILIBRIUM DIAGRAMS

Three types of binary phase equilibrium curves are shown in Fig. 13-18. The  $y$ - $x$  diagram is almost always plotted for the component that is the more volatile (denoted by the subscript 1) in the region where distillation is to take place. Curve A shows the common case in which component 1 remains more volatile over the entire composition range. Curve B is typical of many systems (e.g., ethanol-water) in which the



**FIG. 13-18** Typical binary equilibrium curves. Curve A, system with normal volatility. Curve B, system with homogeneous azeotrope (one liquid phase). Curve C, system with heterogeneous azeotrope (two liquid phases in equilibrium with one vapor phase).

component that is more volatile at low values of  $x_1$  becomes less volatile than the other component at high values of  $x_1$ . The vapor and liquid compositions are identical for the homogeneous azeotrope where curve B crosses the 45° diagonal (that is,  $x_1 = y_1$ ). A heterogeneous azeotrope is formed by curve C, in which there are two equilibrium liquid phases and one equilibrium vapor phase.

An azeotrope limits the separation that can be obtained between components by simple distillation. For the system described by curve B, the maximum overhead-product concentration that could be obtained from a feed with  $z_1 = 0.25$  is the azeotropic composition. Similarly, a feed with  $x_1 = 0.9$  could produce a bottom-product composition no lower than the azeotrope.

The phase rule permits only two variables to be specified arbitrarily in a binary two-phase mixture at equilibrium. Consequently, the curves in Fig. 13-18 can be plotted at either constant temperature or constant pressure but not both. The latter is more common, and data in Table 13-1 correspond to that case. The  $y$ - $x$  diagram can be plotted in mole, weight, or volume fractions. The units used later for the phase flow rates must, of course, agree with those used for the equilibrium data. Mole fractions, which are almost always used, are applied here.

It is sometimes permissible to assume constant *relative volatility* to approximate the equilibrium curve quickly. Then by applying Eq. (13-2) to components 1 and 2,

$$\alpha = \frac{K_1}{K_2} = \frac{y_1 x_2}{x_1 y_2}$$

which can be rewritten as (using  $x_2 = 1 - x_1$  and  $y_2 = 1 - y_1$ )

$$y_1 = \frac{x_1 \alpha}{1 + (\alpha - 1)x_1} \quad (13-20)$$

With a constant value for  $\alpha$  this equation provides a simple, approximate expression for representing the equilibrium  $y = x$  diagram. Doherty and Malone (*Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, sec. 2.3) discuss this approximation in greater detail and give a selection of binary mixtures for which the approximation is reasonable. At a constant pressure of 1 atm these include benzene + toluene,  $\alpha = 2.34$ ; benzene + *p*-xylene,  $\alpha = 4.82$ ; and hexane + *p*-xylene,  $\alpha = 7.00$ .

## McCABE-THIELE METHOD

**Operating Lines** The McCabe-Thiele method is based upon representation of the material balance equations as operating lines on the  $y$ - $x$  diagram. The lines are made straight by the assumption of *constant molar overflow*, which eliminates the need for an energy balance. The liquid-phase flow rate is assumed to be constant from tray to tray in each section of the column between addition (feed) and withdrawal (product) points. If the liquid rate is constant, the vapor rate must also be constant.

The constant-molar-overflow assumption rests on several underlying thermodynamic assumptions. The most important one is equal molar heats of vaporization for the two components. The other assumptions are adiabatic operation (no heat leaks) and no heat of mixing or sensible heat effects. These assumptions are most closely approximated for close-boiling isomers. The result of these assumptions on the calculation method can be illustrated with Fig. 13-19, which shows two material balance envelopes cutting through the top section (above the top feed stream or sidestream) of the column. If the liquid flow rate  $L_{n+1}$  is assumed to be identical to  $L_{n-1}$ , then  $V_n = V_{n-2}$  and the component material balance for both envelopes 1 and 2 can be represented by

$$y_n = \left(\frac{L}{V}\right)x_{n+1} + \frac{Dx_D}{V} \quad (13-21)$$

where  $y$  and  $x$  have a stage subscript  $n$  or  $n+1$ , but  $L$  and  $V$  need be identified only with the section of the column to which they apply. Equation (13-21) has the analytical form of a straight line where  $L/V$  is the slope and  $Dx_D/V$  is the  $y$  intercept at  $x = 0$ .

The effect of a sidestream withdrawal point is illustrated by Fig. 13-20. The material balance equation for the column section below the sidestream is

$$y_n = \frac{L'}{V'}x_{n+1} + \frac{Dx_D + Sx_s}{V'} \quad (13-22)$$

where the primes designate the  $L$  and  $V$  below the sidestream. Since the sidestream must be a saturated phase,  $V = V'$  if a liquid sidestream is withdrawn and  $L = L'$  if it is a vapor.

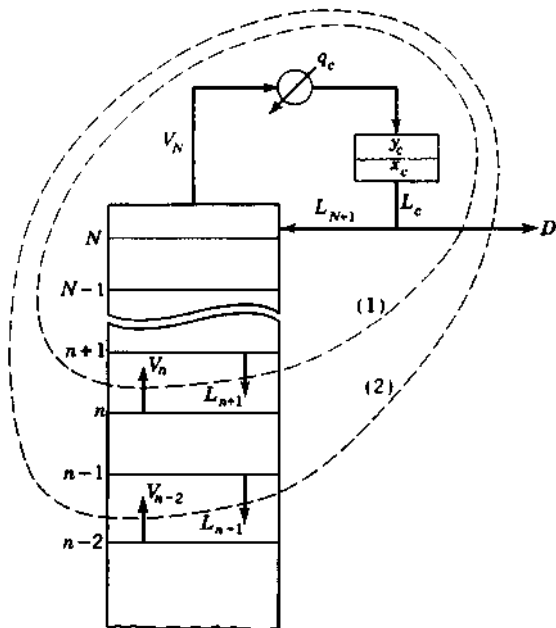


FIG. 13-19 Two material balance envelopes in the top section of a distillation column.

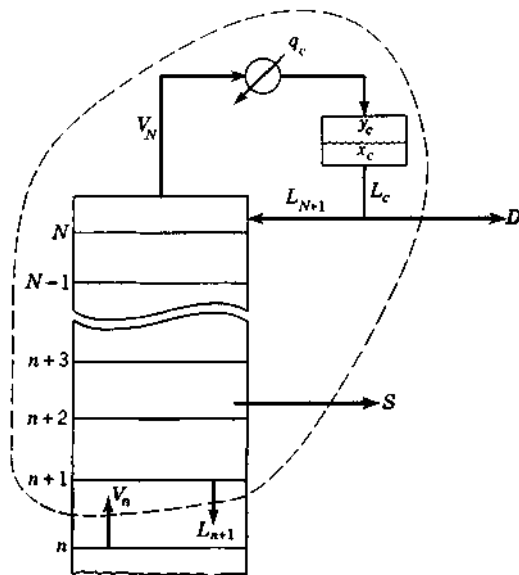


FIG. 13-20 Material balance envelope which contains two external streams  $D$  and  $S$ , where  $S$  represents a sidestream product withdrawn above the feed plate.

If the sidestream in Fig. 13-20 is a feed (not necessarily a saturated liquid or vapor), the balance for the section below the feed becomes

$$y_n = \frac{L'}{V'}x_{n+1} + \frac{Dx_D - Fz_F}{V'} \quad (13-23)$$

Similar equations can be written for the bottom section of the column. For the envelope shown in Fig. 13-21,

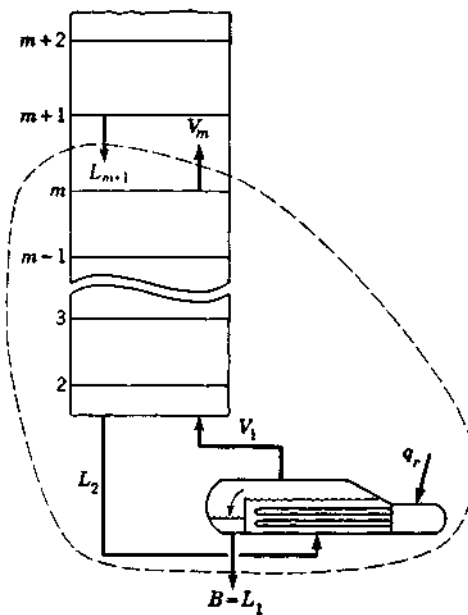


FIG. 13-21 Material balance envelope around the bottom end of the column. The partial reboiler is equilibrium stage 1.



$$y_m = \frac{L'}{V'} x_{m+1} - \frac{Bx_B}{V'} \quad (13-24)$$

where the subscript  $m$  is used to identify the stage number in the bottom section.

Equations such as (13-21) through (13-24), when plotted on the  $y$ - $x$  diagram, furnish a set of *operating lines*. A point on an operating line represents two *passing streams*, and the operating line itself is the locus of all possible pairs of passing streams within the column section to which the line applies.

An operating line can be located on the  $y$ - $x$  diagram if (1) two points on the line are known or (2) one point and the slope are known. The known points on an operating line are usually its intersection with the  $y$ - $x$  diagonal and/or its intersection with another operating line.

The slope  $L/V$  of the operating line is termed the *internal reflux ratio*. This ratio in the operating line equation for the top section of the column [see Eq. (13-21)] is related to the *external reflux ratio*  $R = L_{N+1}/D$  by

$$\frac{L}{V} = \frac{L_{N+1}}{V_N} = \frac{RD}{(1+R)D} = \frac{R}{1+R} \quad (13-25)$$

when the reflux stream  $L_{N+1}$  is a saturated liquid.

**Thermal Condition of the Feed** The slope of the operating line changes whenever a feed stream or a sidestream is passed. To calculate this change, it is convenient to introduce a quantity  $q$  which is defined by the following equations for a feed stream  $F$ :

$$L' = L + qF \quad (13-26)$$

$$V = V' + (1 - q)F \quad (13-27)$$

The primes denote the streams below the stage to which the feed is introduced. The value of  $q$  is a measure of the thermal condition of the feed and represents the moles of saturated liquid formed in the feed stage per mole of feed. The value of  $q$  for a particular feed can be estimated from

$$q = \frac{\text{energy to convert 1 mol of feed to saturated vapor}}{\text{molar heat of vaporization}}$$

It takes on the following values for various thermal conditions of the feed:

|                         |             |
|-------------------------|-------------|
| Subcooled liquid feed:  | $q > 1$     |
| Saturated liquid feed:  | $q = 1$     |
| Partially flashed feed: | $0 < q < 1$ |
| Saturated vapor feed:   | $q = 0$     |
| Superheated vapor feed: | $q < 0$     |

Equations analogous to (13-26) and (13-27) can be written for a sidestream, but the value of  $q$  will be either 1 or 0 depending upon whether the sidestream is taken from the liquid or the vapor stream.

The quantity  $q$  can be used to derive the " $q$  line equation" for a feed stream or a sidestream. The  $q$  line is the locus of all points of intersection of the two operating lines, which meet at the feed stream or sidestream stage. This intersection must occur along that section of the  $q$  line between the equilibrium curve and the  $y = x$  diagonal. At the point of intersection, the same  $y, x$  point must satisfy both the operating line equation above the feed stream (or sidestream) stage and the one below the feed stream (or sidestream) stage. Subtracting one equation from the other gives for a feed stage

$$(V - V')y = (L - L')x + Fz_F$$

which, when combined with Eqs. (13-26) and (13-27), gives the  $q$  line equation

$$y = \frac{q}{q-1}x - \frac{z_F}{q-1} \quad (13-28)$$

A  $q$  line construction for a partially flashed feed is given in Fig. 13-22. It is easily shown that the  $q$  line must intersect the diagonal at  $z_F$ .

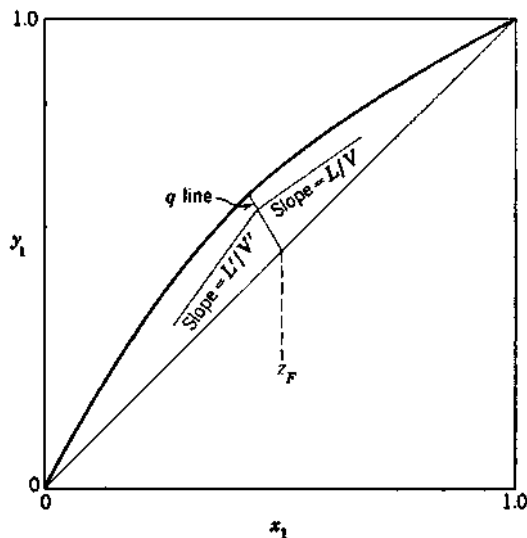


FIG. 13-22 Typical intersection of the two operating lines at the  $q$  line for a feed stage. The  $q$  line shown is for a partially flashed feed.

The slope of the  $q$  line is  $q/(q-1)$ . All five  $q$  line cases are shown in Fig. 13-23. Note that when  $q = 1$ , the  $q$  line has infinite slope and is vertical.

The derivation of Eq. (13-28) assumes a single-feed column and no sidestream. However, the same result is obtained for other column configurations. Typical  $q$  line constructions for sidestream stages are shown in Fig. 13-24. Note that the  $q$  line for a sidestream must always intersect the diagonal at the composition ( $y_s$  or  $x_s$ ) of the sidestream. Figure 13-24 also shows the intersections of the operating lines with the diagonal construction line. The top operating line must always intersect the diagonal at the overhead-product composition  $x_D$ . This can be shown by substituting  $y = x$  in Eq. (13-21) and using  $V - L = D$  to reduce the resulting equation to  $x = x_D$ . Similarly (except for columns in which open steam is introduced at the bottom), the bottom operating line must always intersect the diagonal at the bottom-product composition  $x_B$ .

**Equilibrium-Stage Construction** Use of the equilibrium curve and the operating lines to "step off" equilibrium stages is illustrated in Fig. 13-25. The plotted portions of the equilibrium curve (curved) and the operating line (straight) cover the composition range existing in the column section shown in the lower right-hand corner of the figure. If  $y_n$  and  $x_n$  represent the compositions (in terms of the more volatile

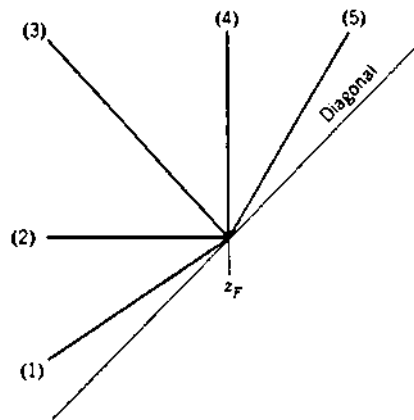


FIG. 13-23 All five cases of  $q$  lines: (1) superheated vapor feed, (2) saturated vapor feed, (3) partially vaporized feed, (4) saturated liquid feed, and (5) subcooled liquid feed. Slope of  $q$  line is  $q/(q-1)$ .

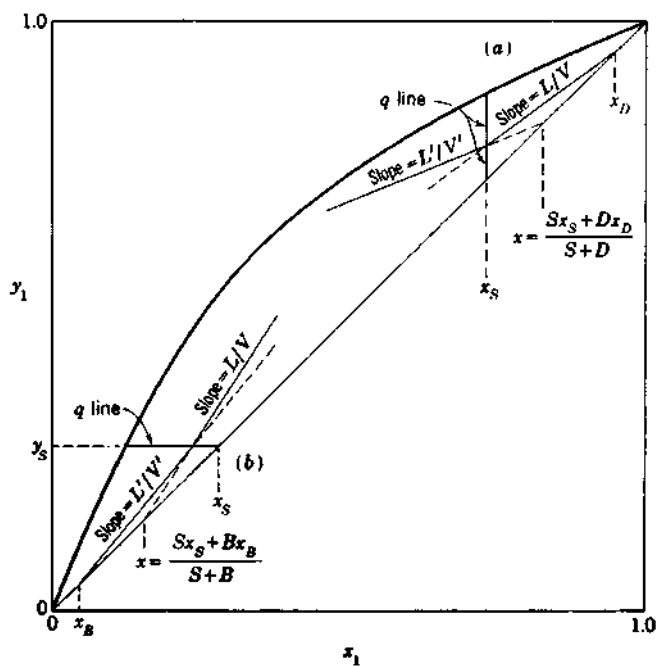


FIG. 13-24 Typical construction for a sidestream showing the intersection of the two operating lines with the  $q$  line and with the  $x$ - $y$  diagonal. (a) Liquid sidestream near the top of the column. (b) Vapor sidestream near the bottom of the column.

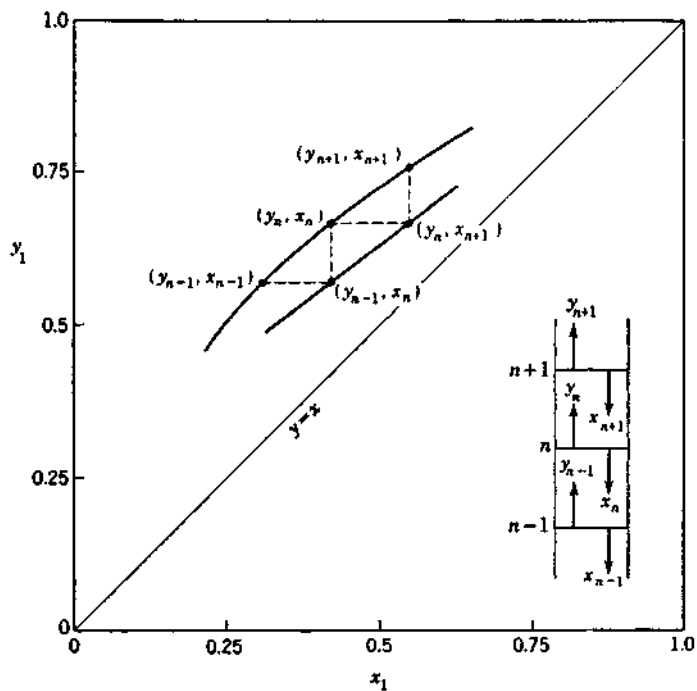


FIG. 13-25 Illustration of how equilibrium stages can be located on the  $x$ - $y$  diagram through the alternating use of the equilibrium curve and the operating line.



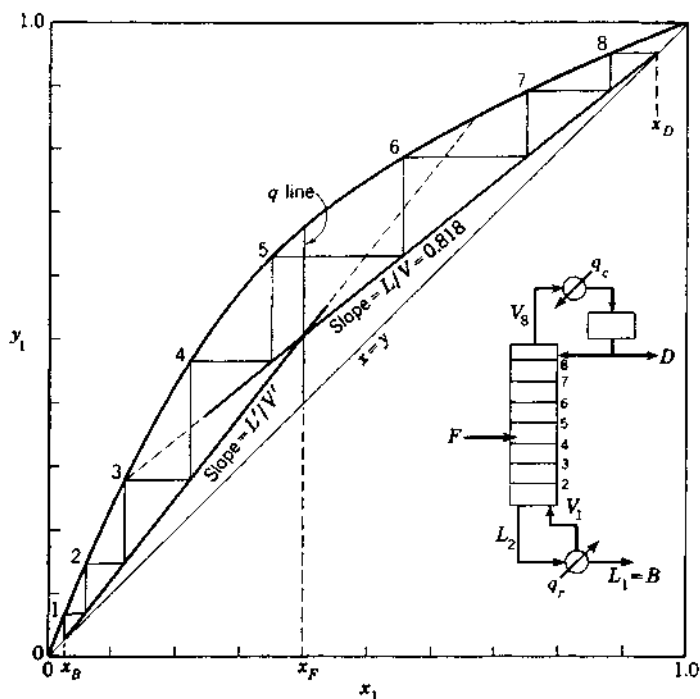


FIG. 13-26 Construction for a column with a bubble point feed, a total condenser, and a partial reboiler.

component) of the equilibrium vapor and liquid leaving stage  $n$ , then point  $(y_n, x_n)$  on the equilibrium curve must represent the equilibrium stage  $n$ . The operating line is the locus for compositions of all possible pairs of passing streams within the section, and therefore a horizontal line (dashed) at  $y_n$  must pass through the point  $(y_n, x_{n+1})$  on the operating line since  $y_n$  and  $x_{n+1}$  represent passing streams. Likewise, a vertical line (dashed) at  $x_n$  must intersect the operating line at point  $(y_{n-1}, x_n)$ . The equilibrium stages above and below stage  $n$  can be located by a vertical line through  $(y_n, x_{n+1})$  to find  $(y_{n+1}, x_{n+1})$  and a horizontal line through  $(y_{n-1}, x_n)$  to find  $(y_{n-1}, x_{n-1})$ . This procedure can be repeated by alternating the use of equilibrium and operating lines upward or downward through the column to find the total number of equilibrium stages.

**Total Column Construction** The graphical construction for an entire column is shown in Fig. 13-26. The process, pictured in the lower right-hand corner of the diagram, is an existing column with a number of actual trays equivalent to eight equilibrium stages. A partial reboiler (equivalent to an equilibrium stage) and a total condenser are used. This column configuration has  $C + 2N + 9$  design variables (degrees of freedom) which must be specified to define one unique operation [see subsection Degrees of Freedom and Design Variables, especially Fig. 13-62 and Eq. (13-111)]. These may be used as follows as the basis for a graphical solution:

| Specifications                           | Degrees of freedom |
|------------------------------------------|--------------------|
| Stage pressures (including reboiler)     | $N$                |
| Condenser pressure                       | 1                  |
| Stage heat leaks (except reboiler)       | $N - 1$            |
| Pressure and heat leak in reflux divider | 2                  |
| Feed stream                              | $C + 2$            |
| Feed-stage location                      | 1                  |
| Total number of stages $N$               | 1                  |
| One overhead purity                      | 1                  |
| Reflux temperature                       | 1                  |
| External reflux ratio                    | 1                  |
|                                          | $C + 2N + 9$       |

Pressures can be specified at any level below the safe working pressure of the column. The condenser pressure will be set at 275.8 kPa (40 psia), and all pressure drops within the column will be neglected. The equilibrium curve in Fig. 13-26 represents data at that pressure. All heat leaks will be assumed to be zero. The feed composition is 40 mol % of the more volatile component 1, and the feed rate is 0.126 kg-mol/s (1000 lb-mol/h) of saturated liquid ( $q = 1$ ). The feed-stage location is fixed at stage 4 and the total number of stages at eight.

The overhead purity is specified as  $x_D = 0.95$ . The reflux temperature is the bubble point temperature (saturated reflux), and the external reflux ratio is set at  $R = 4.5$ .

Answers are desired to the following two questions. First, what bottom-product composition  $x_B$  will the column produce under these specifications? Second, what is the value of the top vapor rate  $V_N$  in this operation, and will it exceed the maximum vapor rate capacity for this column, which is assumed to be 0.252 kg-mol/s (2000 lb-mol/h) at the top-tray conditions?

The solution is started by using Eq. (13-25) to convert the external reflux ratio of 4.5 to an internal reflux ratio of  $L/V = 0.818$ . The distillate composition  $x_D = 0.95$  is then located on the diagonal, and the upper operating line is drawn as shown in Fig. 13-26.

If the  $x_B$  value were known, the bottom operating line could be immediately drawn from the  $x_B$  value on the diagonal up to its required intersection point with the upper operating line on the feed  $q$  line. In this problem, since the number of stages is fixed, the value of  $x_B$  which gives a lower operating line that will require exactly eight stages must be found by trial and error. An  $x_B$  value is assumed, and the resulting lower operating line is drawn. The stages can be stepped off by starting from either  $x_B$  or  $x_D$ ;  $x_B$  was used in this case.

Note that the lower operating line is used until the fourth stage is passed, at which time the construction switches to the upper operating line. This is necessary because the vapor and liquid streams passing each other between the fourth and fifth stages must fall on the upper line.

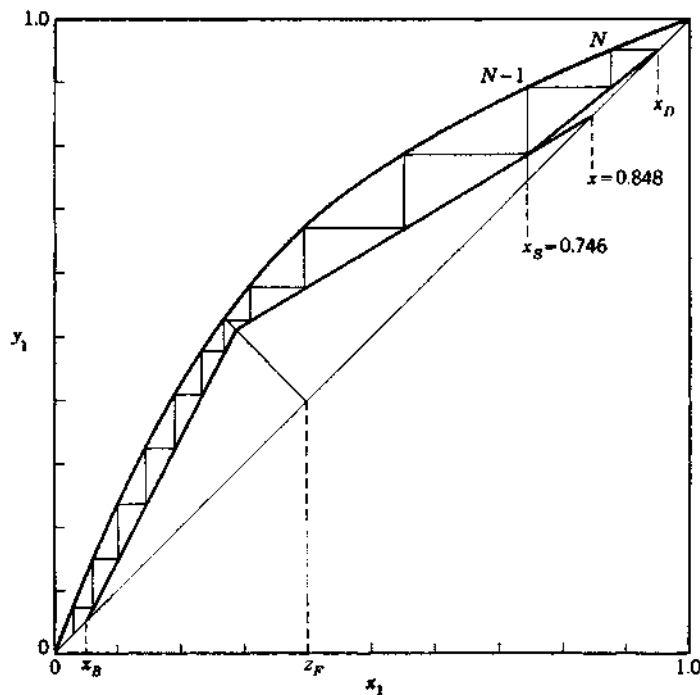


FIG. 13-27 Graphical solution for a column with a partially flashed feed, a liquid sidestream, and a total condenser.

The  $x_B$  that requires exactly eight equilibrium stages is  $x_B = 0.026$ . An overall component balance gives  $D = 0.051$  kg-mol/s (405 lb-mol/h). Then

$$\begin{aligned} V_N = V_B = L_{N+1} + D = D(R + 1) &= 0.051(4.5 + 1.0) \\ &= 0.280 \text{ kg-mol/s (2230 lb-mol/h)} \end{aligned}$$

which exceeds the column capacity of 0.252 kg-mol/s (2007 lb-mol/h). This means that the column cannot provide an overhead-product yield of 40.5 percent at 95 percent purity. Either the purity specification must be reduced, or we must be satisfied with a lower yield. If the distillate specification ( $x_D = 0.95$ ) is retained, the reflux rate must be reduced. This will cause the upper operating line to pivot upward around its fixed point of  $x_D = 0.95$  on the diagonal. The new intersection of the upper line with the  $q$  line will lie closer to the equilibrium curve. The  $x_B$  value must then move upward along the diagonal because the eight stages will not "reach" as far as before. The higher  $x_B$  composition will reduce the recovery of component 1 in the 95 percent overhead product.

Another entire column with a partially vaporized feed, a liquid sidestream rate equal to  $D$  withdrawn from the second stage from the top, and a total condenser is shown in Fig. 13-27. The specified compositions are  $z_F = 0.40$ ,  $x_B = 0.05$ , and  $x_D = 0.95$ . The specified  $L/V$  ratio in the top section is 0.818. These specifications permit the top operating line to be located and the two top stages stepped off to determine the liquid sidestream composition  $x_s = 0.746$ . The operating line below the sidestream must intersect the diagonal at the "blend" of the sidestream and the overhead stream. Since  $S$  was specified to be equal to  $D$  in rate, the intersection point is

$$x = \frac{(1.0)(0.746) + (1.0)(0.95)}{1.0 + 1.0} = 0.848$$

This point plus the point of intersection of the two operating lines on the sidestream  $q$  line (vertical at  $x_s = 0.746$ ) permits the location of the

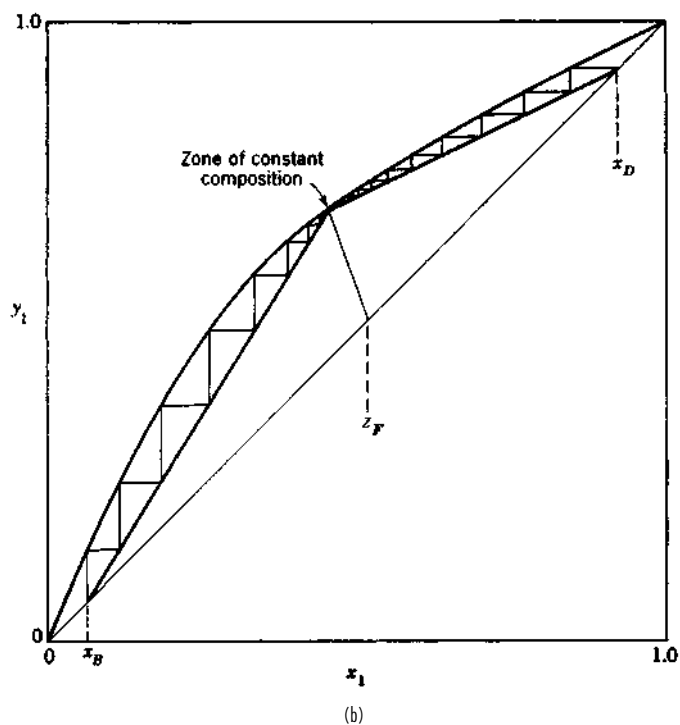
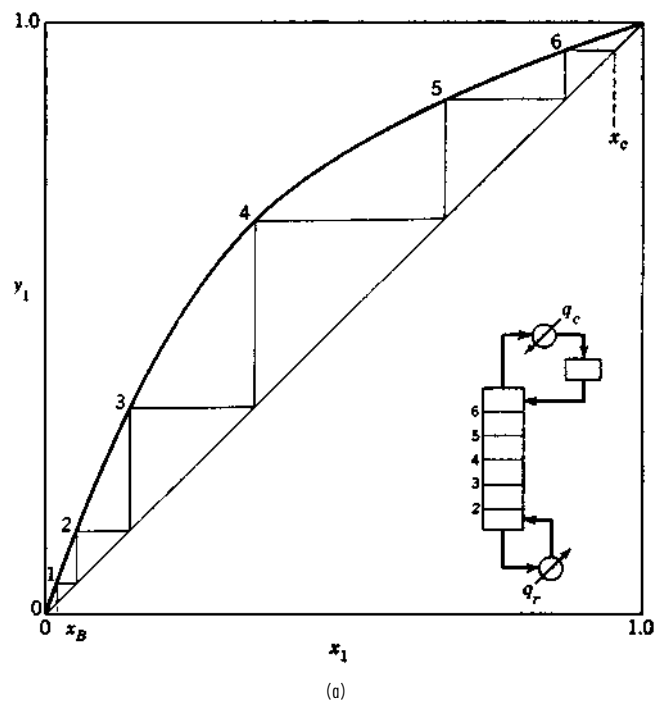
middle operating line. (The slope of the middle operating line could also have been used.) The lower operating line must run from the specified  $x_B$  value on the diagonal to the required point of intersection on the feed  $q$  line. The stages are stepped off from the top down in this case. The sixth stage from the top is the feed stage, and a total of about 11.4 stages are required to reach the specified  $x_B = 0.05$ .

Fractional equilibrium stages have meaning. The 11.4 will be divided by a tray efficiency, and the rounding up to an integral number of actual trays should be done after that division. For example, if the average tray efficiency for the process modeled in Fig. 13-27 is 80 percent, then the number of actual trays required is  $11.4/0.8 = 14.3$ , which is rounded up to 15.

**Feed-Stage Location** The optimum feed-stage location is that location which, with a given set of other operating specifications, will result in the widest separation between  $x_D$  and  $x_B$  for a given number of stages. Or, if the number of stages is not specified, the optimum feed location is the one that requires the lowest number of stages to accomplish a specified separation between  $x_D$  and  $x_B$ . Either of these criteria will always be satisfied if the operating line farthest from the equilibrium curve is used in each step, as in Fig. 13-26.

It can be seen from Fig. 13-26 that the optimum feed location would have been the fifth tray for that operation. If a new column were being designed, that should be the designer's choice. However, when an existing column is being modeled, the feed stage on the diagram should correspond as closely as possible to the actual feed tray in the column. It can be seen that a badly mislocated feed (a feed that requires one to remain with an operating line until it closely approaches the equilibrium curve) can be very wasteful insofar as the effectiveness of the stages is concerned.

**Minimum Stages** A column operating at total reflux is represented in Fig. 13-28a. Enough material has been charged to the column to fill the reboiler, the trays, and the overhead condensate drum to their working levels. The column is then operated with no feed and with all the condensed overhead stream returned as reflux ( $L_{N+1} = V_N$  and  $D = 0$ ). Also all the liquid reaching the reboiler is



**FIG. 13-28** McCabe-Thiele diagrams for limiting cases. (a) Minimum stages for a column operating at total reflux with no feeds or products. (b) Minimum reflux for a binary system of normal volatility.

vaporized and returned to the column as vapor. Since  $F$ ,  $D$ , and  $B$  are all zero,  $L_{n+1} = V_n$  at all points in the column. With a slope of unity ( $L/V = 1.0$ ), the operating line must coincide with the diagonal throughout the column. Total reflux operation gives the minimum number of stages required to effect a specified separation between  $x_B$  and  $x_D$ .

**Minimum Reflux** The minimum reflux ratio is defined as that ratio which if decreased by an infinitesimal amount would require an infinite number of stages to accomplish a specified separation between two components. The concept has meaning only if a separation between two components is specified and the number of stages is not specified. Figure 13-28b illustrates the minimum reflux condition. As the reflux ratio is reduced, the two operating lines swing upward, pivoting around the specified  $x_B$  and  $x_D$  values, until one or both touch the equilibrium curve. For equilibrium curves shaped like the one shown, the contact occurs at the feed  $q$  line, resulting in a *feed pinch point*. Often an equilibrium curve will dip down closer to the diagonal at higher compositions. In such cases, the upper operating line may make contact before its intersection point on the  $q$  line reaches the equilibrium curve, resulting in a *tangent pinch point*. Wherever the contact appears, the intersection of the operating line with the equilibrium curve produces a pinch point which contains a very large number of stages, and a zone of constant composition is formed (see Doherty and Malone, 2001, chap. 3 and sec. 4.6 for additional information).

**Intermediate Reboilers and Condensers** When a large temperature difference exists between the ends of the column due to a wide boiling point difference between the components, intermediate reboilers and/or condensers may be used to add heat at a lower temperature, or remove heat at a higher temperature, respectively. [A distillation column of this type is shown in *Perry's Chemical Engineers' Handbook*, 7th ed. (1986), Fig. 13-2a.] A column operating with an intermediate reboiler and an intermediate condenser in addition to a regular reboiler and a condenser is illustrated with the solid lines in Fig. 13-29. The dashed lines correspond to simple distillation with only a bottoms reboiler and an overhead condenser. Total boiling and condensing heat loads are the same for both columns. As shown by Kayihan [AIChE Symp. Ser. 76, 192, 1 (1980)], the addition of intermediate reboilers and intermediate condensers increases thermodynamic efficiency but requires additional stages, as is clear from the positions of the operating lines in Fig. 13-29.

**Optimum Reflux Ratio** The general effect of the operating reflux ratio on fixed costs, operating costs, and the sum of these is shown in Fig. 13-30. In ordinary situations, the minimum on the total cost curve will generally occur at an operating reflux ratio in the inter-

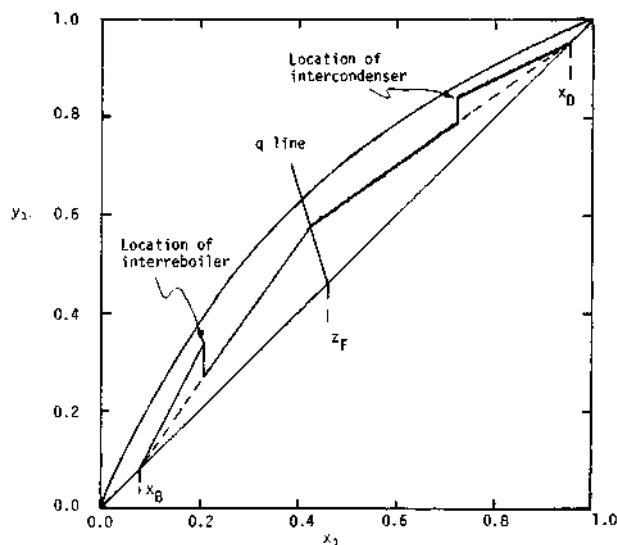


FIG. 13-29 McCabe-Thiele diagram for columns with and without an intermediate reboiler and an intermediate condenser.

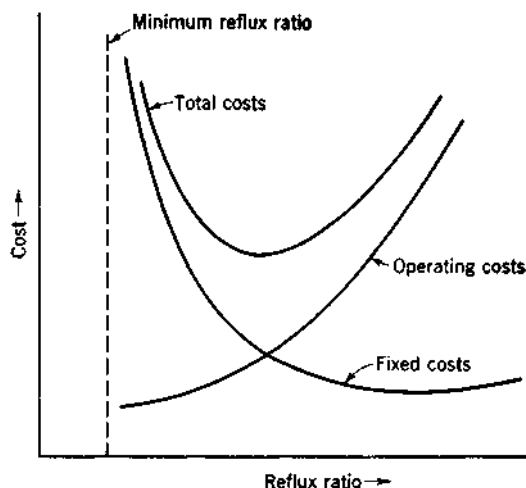


FIG. 13-30 Location of the optimum reflux for a given feed and specified separation.

val 1.1 to 2 times the minimum value. Generally, the total cost curve rises slowly from its minimum value as the operating reflux ratio increases, and very steeply as the operating reflux ratio decreases. In the absence of a detailed cost analysis for the specific separation of interest, it is recommended to select operating reflux ratios closer to 1.5 to 2.0 times the minimum value (see Doherty and Malone, 2001, chap. 6 for additional discussion).

**Difficult Separations** Some binary separations may pose special problems because of extreme purity requirements for one or both

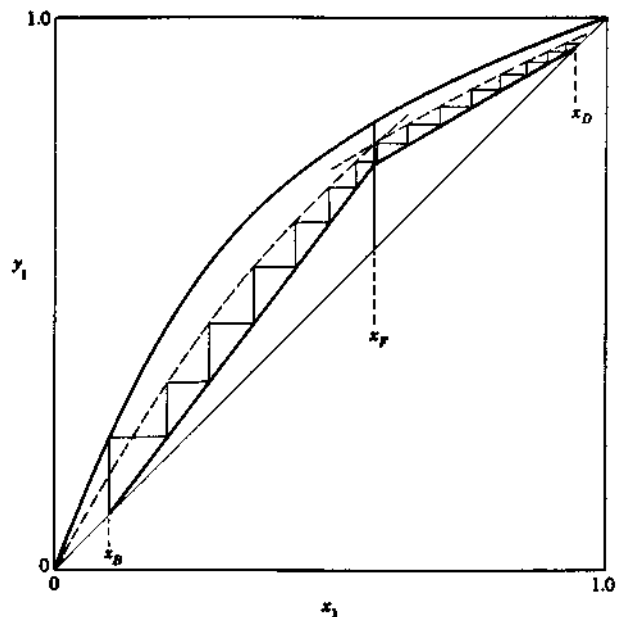


FIG. 13-31 Application of a 50 percent Murphree vapor-phase efficiency to each stage (excluding the reboiler) in the column. Each step in the diagram corresponds to an actual stage.

products or because of a relative volatility close to 1. The  $y$ - $x$  diagram is convenient for stepping off stages at extreme purities if it is plotted on log-log paper. However, such cases are best treated by equation-based design methods.

**Equation-Based Design Methods** Exact design equations have been developed for mixtures with constant relative volatility. Minimum stages can be computed with the Fenske equation, minimum reflux from the Underwood equation, and the total number of stages in each section of the column from either the Smoker equation (*Trans. Am. Inst. Chem. Eng.*, **34**, 165 (1938); the derivation of the equation is shown, and its use is illustrated by Smith, op. cit.), or Underwood's method. A detailed treatment of these approaches is given in Doherty and Malone (op. cit., chap. 3). Equation-based methods have also been developed for nonconstant relative volatility mixtures (including nonideal and azeotropic mixtures) by Julka and Doherty [*Chem. Eng. Sci.*, **45**, 1801 (1990); *Chem. Eng. Sci.*, **48**, 1367 (1993)], and Fidkowski et al. [*AIChE J.*, **37**, 1761 (1991)]. Also see Doherty and Malone (op. cit., chap. 4).

**Stage Efficiency** The use of the *Murphree plate efficiency* is particularly convenient on  $y$ - $x$  diagrams. The *Murphree efficiency* is defined for the vapor phase as

$$E^{MV} = \frac{y_n - y_{n-1}}{y_n^* - y_{n-1}} \quad (13-29)$$

where  $y_n^*$  is the composition of the vapor that would be in equilibrium with the liquid leaving stage  $n$  and is the value read from the equilibrium curve. The  $y_{n-1}$  and  $y_n$  are the actual (nonequilibrium) values for vapor streams leaving the  $n-1$  and  $n$  stages, respectively. Note that

## APPROXIMATE MULTICOMPONENT DISTILLATION METHODS

Some approximate calculation methods for the solution of multicomponent, multistage separation problems continue to serve useful purposes even though computers are available to provide more rigorous solutions. The available phase equilibrium and enthalpy data may not be accurate enough to justify the longer rigorous methods. Or in extensive design and optimization studies, a large number of cases can be worked quickly and cheaply by an approximate method to define roughly the optimum specifications, which can then be investigated more exactly with a rigorous method.

Two approximate multicomponent shortcut methods for simple distillation are the Smith-Brinkley (SB) method, which is based on an analytical solution of the finite-difference equations that can be written for staged separation processes when stages and interstage flow rates are known or assumed, and the Fenske-Underwood-Gilliland (FUG) method, which combines Fenske's total reflux equation and Underwood's minimum reflux equation with a graphical correlation by Gilliland that relates actual column performance to total and minimum reflux conditions for a specified separation between two key components. Thus, the SB and FUG methods are rating and design methods, respectively. Both methods work best when mixtures are nearly ideal.

The SB method is not presented here, but is presented in detail in the 6th edition of *Perry's Chemical Engineers' Handbook*. Extensions of the SB method to nonideal mixtures and complex configurations are developed by Eckert and Hlavacek [*Chem. Eng. Sci.*, **33**, 77 (1978)] and Eckert [*Chem. Eng. Sci.*, **37**, 425 (1982)], respectively, but are not discussed here. However, the approximate and very useful method of Kremser [*Nat. Pet. News*, **22**(21), 43 (May 21, 1930)] for application to absorbers and strippers is discussed at the end of this subsection.

### FENSKE-UNDERWOOD-GILLILAND (FUG) SHORTCUT METHOD

In this approach, Fenske's equation [*Ind. Eng. Chem.*, **24**, 482 (1932)] is used to calculate  $N_{\min}$ , which is the number of plates required to make a specified separation at total reflux, i.e., the minimum value of  $N$ . Underwood's equations [*J. Inst. Pet.*, **31**, 111 (1945); **32**, 598 (1946);

for the  $y_{n-1}$  and  $y_n$  values we assume that the vapor streams are completely mixed and uniform in composition. An analogous efficiency can be defined for the liquid phase.

The application of a 50 percent *Murphree vapor-phase efficiency* on a  $y$ - $x$  diagram is illustrated in Fig. 13-31. A pseudoequilibrium curve is drawn halfway (on a vertical line) between the operating lines and the true equilibrium curve. The true equilibrium curve is used for the first stage (the partial reboiler is assumed to be an equilibrium stage), but for all other stages the vapor leaving each stage is assumed to approach the equilibrium value  $y_n^*$  only 50 percent of the way. Consequently, the steps in Fig. 13-31 represent actual trays.

In general, application of a constant efficiency to each stage as in Fig. 13-31 will not give the same answer as obtained when the number of equilibrium stages (obtained by using the true equilibrium curve) is divided by the same efficiency factor.

The prediction and use of stage efficiencies are described in detail in Sec. 14. Alternative approaches based on mass-transfer rates are preferred, as described in the subsection below, *Nonequilibrium Modeling*.

**Miscellaneous Operations** The  $y$ - $x$  diagrams for several other column configurations have not been presented here. The omitted items are *partial condensers*, *rectifying columns* (feed introduced to the bottom stage), *stripping columns* (feed introduced to the top stage), total reflux in the top section but not in the bottom section, multiple feeds, and introduction of *open steam* to the bottom stage to eliminate the reboiler. These configurations are discussed in Smith (op. cit.) and Henley and Seader (op. cit.), who also describe the more rigorous Ponchon-Savarit method, which is not covered here.

**32**, 614 (1946); and *Chem. Eng. Prog.*, **44**, 603 (1948)] are used to estimate the minimum reflux ratio  $R_{\min}$ . The empirical correlation of Gilliland [*Ind. Eng. Chem.*, **32**, 1220 (1940)] shown in Fig. 13-32 then uses these values to give  $N$  for any specified  $R$ , or  $R$  for any specified  $N$ . Limitations of the Gilliland correlation are discussed by Henley and Seader (*Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981). The following equation, developed by Molokanov et al. [*Int. Chem. Eng.*, **12**(2), 209 (1972)], satisfies the endpoints and fits the Gilliland curve reasonably well:

$$\frac{N - N_{\min}}{N + 1} = 1 - \exp \left( \frac{1 + 54.4\Psi}{11 + 117.2\Psi} \times \frac{\Psi - 1}{\Psi^{0.5}} \right) \quad (13-30)$$

where  $\Psi = (R - R_{\min})/(R + 1)$ .

The Fenske total reflux equation can be written as

$$\left( \frac{x_i}{x_r} \right)_D = (\alpha_i)^{N_{\min}} \left( \frac{x_i}{x_r} \right)_B \quad (13-31)$$

$$\text{or as} \quad N_{\min} = \frac{\log[(Dx_D/Bx_B)/(Bx_B/Dx_D)]}{\log \alpha_i} \quad (13-32)$$

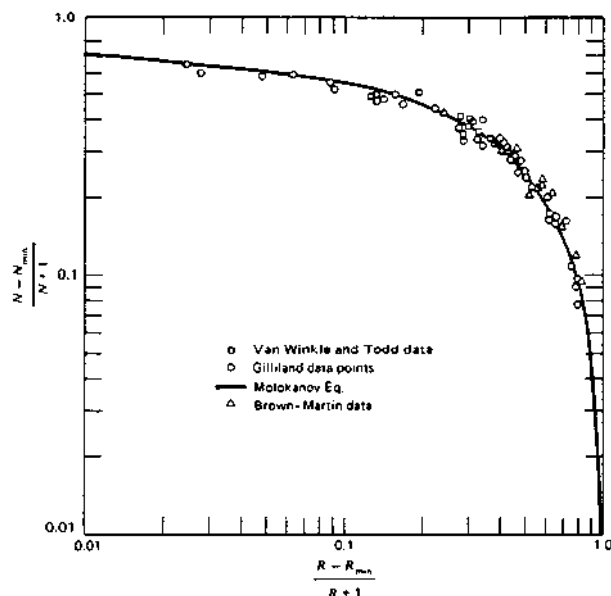
where  $i$  is any component and  $r$  is an arbitrarily selected reference component in the definition of relative volatilities

$$\alpha_i = \frac{K_i}{K_r} = \frac{y_i x_r}{y_r x_i} \quad (13-33)$$

The particular value of  $\alpha_i$  used in Eqs. (13-31) and (13-32) is the effective value calculated from Eq. (13-34) defined in terms of values for each stage in the column by

$$\alpha_i^N = \alpha_{iN} \alpha_{iN-1} \cdots \alpha_{i2} \alpha_{i1} \quad (13-34)$$

Equations (13-31) and (13-32) are exact relationships between the splits obtained for components  $i$  and  $r$  in a column at total reflux.



**FIG. 13-32** Comparison of rigorous calculations with Gilliland correlation. [Henley and Seader, *Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981; data of Van Winkle and Todd, *Chem. Eng.*, **78**(21), 136 (Sept. 20, 1971); data of Gilliland, *Elements of Fractional Distillation*, 4th ed., McGraw-Hill, New York, 1950; data of Brown and Martin, *Trans. Am. Inst. Chem. Eng.*, **35**, 679 (1939).]

However, the value of  $\alpha_i$  must always be estimated, and this is where the approximation enters. It is usually estimated from

$$\alpha_i = (\alpha_{\text{top}} \alpha_{\text{bottom}})^{1/2} \quad (13-35)$$

or

$$\alpha_i = (\alpha_{\text{top}} \alpha_{\text{middle}} \alpha_{\text{bottom}})^{1/3} \quad (13-36)$$

As a side note, the separation that will be accomplished in a column with a known number of equilibrium stages can often be reasonably well estimated by specifying the split of one component (designated as the reference component  $r$ ), setting  $N_{\text{min}}$  equal to 40 to 60 percent of the number of equilibrium stages (not actual trays), and then using Eq. (13-32) to estimate the splits of all the other components. This is an iterative calculation because the component splits must first be arbitrarily assumed to give end compositions that can be used to give initial end-temperature estimates. The  $\alpha_{\text{top}}$  and  $\alpha_{\text{bottom}}$  values corresponding to these end temperatures are used in Eq. (13-5) to give  $\alpha_i$  values for each component. The iteration is continued until the  $\alpha_i$  values do not change from trial to trial.

The Underwood minimum reflux equations of main interest are those that apply when some of the components do not appear in either the distillate or the bottom products at minimum reflux. These equations are

$$\sum_i \frac{\alpha_i (x_{iD})_{\text{min}}}{\alpha_i - \Theta} = R_{\text{min}} + 1 \quad (13-37)$$

and

$$\sum_i \frac{\alpha_i z_{iF}}{\alpha_i - \Theta} = 1 - q \quad (13-38)$$

The relative volatilities  $\alpha_i$  are defined by Eq. (13-33),  $R_{\text{min}}$  is the minimum reflux ratio, and  $q$  describes the thermal condition of the feed (1 for a saturated liquid feed and 0 for a saturated vapor feed). The  $z_{iF}$  values are available from the given feed composition. The  $\Theta$  is the common root for the top section equations and the bottom section

equations developed by Underwood for a column at minimum reflux with separate zones of constant composition in each section. The common root value must fall between  $\alpha_{hk}$  and  $\alpha_{lk}$ , where  $hk$  and  $lk$  stand for *heavy key* and *light key*, respectively. The *key components* are the ones the designer wants to separate. In the butane-pentane splitter problem in Example 1, the light key is  $n\text{-C}_4$  and the heavy key is  $i\text{-C}_5$ .

The  $\alpha_i$  values in Eqs. (13-37) and (13-38) are effective values obtained from Eq. (13-35) or Eq. (13-36). Once these values are available,  $\Theta$  can be calculated in a straightforward iteration from Eq. (13-38). Since the  $\alpha\text{-}\Theta$  difference can be small,  $\Theta$  should be determined to four decimal places to avoid numerical difficulties.

The  $(x_{iD})_{\text{min}}$  values in Eq. (13-37) are minimum reflux values, i.e., the overhead composition that would be produced by the column operating at the minimum reflux with an infinite number of stages. When the light key and the heavy key are adjacent in relative volatility and the specified split between them is sharp or the relative volatilities of the other components are not close to those of the two keys, only the two keys will distribute at minimum reflux and the  $(x_{iD})_{\text{min}}$  values are easily determined. This is often the case and is the only one considered here. Other cases in which some of or all the nonkey components distribute between distillate and bottom products are discussed in detail by Henley and Seader (op. cit.).

The FUG method is convenient for new column design with the following specifications:

1. A value for  $R/R_{\text{min}}$
2. Desired split on the reference component (usually chosen as the heavy key)
3. Desired split on one other component (usually the light key)

However, the total number of equilibrium stages  $N$ ,  $N/N_{\text{min}}$ , or the external reflux ratio can be substituted for one of these three specifications. Note that the feed location is automatically specified as the optimum one; this is assumed in the Underwood equations. The assumption of saturated liquid reflux is also inherent in the Fenske and Underwood equations (i.e., the reflux is *not* subcooled). An important limitation on the Underwood equations is the assumption of constant molar overflow. As discussed by Henley and Seader (op. cit.), this assumption can lead to a prediction of the minimum reflux that is considerably lower than the actual value. No such assumption is inherent in the Fenske equation. An exact calculation technique for minimum reflux is given by Tavana and Hansen [*Ind. Eng. Chem. Process Des. Dev.*, **18**, 154 (1979)]. Approximate explicit expressions for minimum reflux for various types of splits in three- and four-component mixtures were developed by Glinos and Malone [*Ind. Eng. Chem. Process Des. Dev.*, **23**, 764 (1984)] as well as lumping rules for applying their expressions to mixtures containing more than four components. These expressions are fairly accurate (usually within 5 percent of the exact value for  $R_{\text{min}}$ ) and are extremely convenient for using in the FUG method since they remove the tedious calculation of  $R_{\text{min}}$  via the Underwood equations. A computer program for the FUG method is given by Chang [*Hydrocarbon Process.*, **60**(8), 79 (1980)]. The method is best applied to mixtures that form ideal or nearly ideal solutions, and should not be used for strongly nonideal or azeotropic mixtures.

**Example 1: Application of FUG Method** A large butane-pentane splitter is to be shut down for repairs. Some of its feed will be diverted temporarily to an available smaller column, which has only 11 trays plus a partial reboiler. The feed enters on the middle tray. Past experience with similar feeds indicates that the 11 trays plus the reboiler are roughly equivalent to 10 equilibrium stages and that the column has a maximum top vapor capacity of 1.75 times the feed rate on a mole basis. The column will operate at a condenser pressure of 827.4 kPa (120 psia). The feed will be at its bubble point ( $q = 1.0$ ) at the feed tray conditions and has the following composition on the basis of 0.0126 kg-mol/s (100 lb-mol/h):

| Component      | $Fx_{iF}$ |
|----------------|-----------|
| $C_3$          | 5         |
| $i\text{-C}_4$ | 15        |
| $n\text{-C}_4$ | 25        |
| $i\text{-C}_5$ | 20        |
| $n\text{-C}_5$ | 35        |
|                | 100       |



The original column normally has less than 7 mol %  $i$ -C<sub>5</sub> in the overhead and less than 3 mol %  $n$ -C<sub>4</sub> in the bottom product when operating at a distillate rate of  $D/F = 0.489$ . Can these product purities be produced on the smaller column at  $D/F = 0.489$ ?

Pressure drops in the column will be neglected, and the  $K$  values will be read at 827 kPa (120 psia) in both column sections from the DePriester nomograph in Fig. 13-9b. When constant molar overflow is assumed in each section, the rates in pound moles per hour in the upper and lower sections are as follows:

| Top section                                   | Bottom section           |
|-----------------------------------------------|--------------------------|
| $D = (0.489)(100) = 48.9$                     | $B = 100 - 48.9 = 51.1$  |
| $V = (1.75)(100) = 175$                       | $V' = V = 175$           |
| $L = 175 - 48.9 = 126.1$                      | $L' = L + F = 226.1$     |
| $\frac{V}{L} = 1.388$                         | $\frac{V'}{L'} = 0.7739$ |
| $\frac{L}{L'} = \frac{126.1}{226.1} = 0.5577$ |                          |
| $R = \frac{126.1}{48.9} = 2.579$              |                          |

NOTE: To convert pound-moles per hour to kilogram-moles per second, multiply by  $1.26 \times 10^{-3}$ .

Application of the FUG method is demonstrated on the splitter. Specifications necessary to model the existing column include these:

1.  $N = 10$ , total number of equilibrium stages
2. Optimum feed location (which may or may not reflect the actual location)
3. Maximum  $V/F$  at the top tray of 1.75
4. Split on one component given in the following paragraphs

The solution starts with an assumed arbitrary split of all the components to give estimates of top and bottom compositions that can be used to get initial end temperatures. The  $\alpha_i$ 's evaluated at these temperatures are averaged with the  $\alpha$  at the feed-stage temperature (assumed to be the bubble point of the feed) by using Eq. (13-36). The initial assumption for the split on  $i$ -C<sub>5</sub> is  $Dx_D/Bx_B = 3.15/16.85$ . As mentioned earlier,  $N_{\min}$  usually ranges from 0.4N to 0.6N, and the initial  $N_{\min}$  value assumed here will be  $(0.6)(10) = 6.0$ . Equation (13-32) can be rewritten as

$$\left( \frac{Dx_D}{F x_F - Dx_D} \right)_i = \alpha_i^{6.0} \left( \frac{3.15}{16.85} \right) = \alpha_i^{6.0}(0.1869)$$

or 
$$Dx_{iD} = \frac{0.1869\alpha_i^{6.0}}{1 + 0.1869\alpha_i^{6.0}} F x_{iF}$$

The evaluation of this equation for each component is as follows:

| Component           | $\alpha_i$ | $\alpha_i^{6.0}$ | $0.1869\alpha_i^{6.0}$ | $F x_{iF}$ | $Dx_{iD}$ | $Bx_{iB}$ |
|---------------------|------------|------------------|------------------------|------------|-----------|-----------|
| C <sub>3</sub>      | 5.00       |                  |                        | 5          | 5.0       | 0.0       |
| $i$ -C <sub>4</sub> | 2.63       | 330              | 61.7                   | 15         | 14.8      | 0.2       |
| $n$ -C <sub>4</sub> | 2.01       | 66               | 12.3                   | 25         | 25.1      | 1.9       |
| $i$ -C <sub>5</sub> | 1.00       | 1.00             | 0.187                  | 20         | 3.15      | 16.85     |
| $n$ -C <sub>5</sub> | 0.843      | 0.36             | 0.0672                 | 35         | 2.20      | 32.80     |
|                     |            |                  |                        | 100        | 48.25     | 51.75     |

The end temperatures corresponding to these product compositions are 344 K (159°F) and 386 K (236°F). These temperatures plus the feed bubble point temperature of 358 K (185°F) provide a new set of  $\alpha_i$ 's which vary only slightly from those used earlier. Consequently, the  $D = 48.25$  value is not expected to vary greatly and will be used to estimate a new  $i$ -C<sub>5</sub> split. The desired distillate

composition for  $i$ -C<sub>5</sub> is 7 percent; so it will be assumed that  $Dx_D = (0.07)(48.25) = 3.4$  for  $i$ -C<sub>5</sub> and that the split on that component will be 3.4/16.6. The results obtained with the new  $\alpha_i$ 's and the new  $i$ -C<sub>5</sub> split are as follows:

| Component           | $\alpha_i^{6.0}$ | $0.2048\alpha_i^{6.0}$ | $F x_{iF}$ | $Dx_{iD}$ | $Bx_{iB}$ | $x_{iD}$ | $x_{iB}$ |
|---------------------|------------------|------------------------|------------|-----------|-----------|----------|----------|
| C <sub>3</sub>      |                  |                        | 5          | 5.0       | 0.0       | 0.102    | 0.000    |
| $i$ -C <sub>4</sub> | 322              | 65.9                   | 15         | 14.8      | 0.2       | 0.301    | 0.004    |
| $n$ -C <sub>4</sub> | 68               | 13.9                   | 25         | 23.3      | 1.7       | 0.473    | 0.033    |
| $i$ -C <sub>5</sub> | 1.00             | 0.205                  | 20         | 3.4       | 16.6      | 0.069    | 0.327    |
| $n$ -C <sub>5</sub> | 0.415            | 0.085                  | 35         | 2.7       | 32.3      | 0.055    | 0.636    |
|                     |                  |                        | 100        | 49.2      | 50.8      | 1.000    | 1.000    |

The calculated  $i$ -C<sub>5</sub> composition in the overhead stream is 6.9 percent, which is close enough to the target value of 7.0 for now.

Table 13-6 shows subsequent calculations using the Underwood minimum reflux equations. The  $\alpha$  and  $x_D$  values in Table 13-6 are those from the Fenske total reflux calculation. As noted earlier, the  $x_D$  values should be those at minimum reflux. This inconsistency may reduce the accuracy of the Underwood method; but to be useful, a shortcut method must be fast, and it has not been shown that a more rigorous estimation of  $x_D$  values results in an overall improvement in accuracy. The calculated  $R_{\min}$  is 0.9426. The actual reflux assumed is obtained from the specified maximum top vapor rate of 0.022 kg·mol/s [175 lb·(mol/h)] and the calculated  $D$  of 49.2 (from the Fenske equation).

$$L_{N+1} = V_N - D$$

$$R = \frac{V_N}{D} - 1 = \frac{175}{49.2} - 1 = 2.557$$

The values of  $R_{\min} = 0.9426$ ,  $R = 2.557$ , and  $N = 10$  are now used with the Gilliland correlation in Fig. 13-32 or Eq. (13-30) to check the initially assumed value of 6.0 for  $N_{\min}$ . Equation (13-30) gives  $N_{\min} = 6.95$ , which differs from the assumed value.

Repetition of the calculations with  $N_{\min} = 7$  gives  $R = 2.519$ ,  $R_{\min} = 0.9782$ , and a calculated check value of  $N_{\min} = 6.85$ , which is close enough. The final product compositions and the  $\alpha$  values used are as follows:

| Component           | $\alpha_i$ | $Dx_{iD}$ | $Bx_{iB}$ | $x_{iD}$ | $x_{iB}$ |
|---------------------|------------|-----------|-----------|----------|----------|
| C <sub>3</sub>      | 4.98       | 5.00      | 0         | 0.1004   | 0.0      |
| $i$ -C <sub>4</sub> | 2.61       | 14.91     | 0.09      | 0.2996   | 0.0017   |
| $n$ -C <sub>4</sub> | 2.02       | 24.16     | 0.84      | 0.4852   | 0.0168   |
| $i$ -C <sub>5</sub> | 1.00       | 3.48      | 16.52     | 0.0700   | 0.3283   |
| $n$ -C <sub>5</sub> | 0.851      | 2.23      | 32.87     | 0.0448   | 0.6532   |
|                     |            | 49.78     | 50.32     | 1.0000   | 1.0000   |

These results indicate that the 7 percent composition of  $i$ -C<sub>5</sub> in  $D$  and the 3 percent composition of  $i$ -C<sub>4</sub> in  $B$  obtained in the original column can also be obtained with the smaller column. These results disagree somewhat with the answers obtained from a rigorous computer solution, as shown in the following comparison. However, given the approximations that went into the FUG method, the agreement is good.

| Component           | $x_{iD}$ |       | $x_{iB}$ |       |
|---------------------|----------|-------|----------|-------|
|                     | Rigorous | FUG   | Rigorous | FUG   |
| C <sub>3</sub>      | 0.102    | 0.100 | 0.0      | 0.0   |
| $i$ -C <sub>4</sub> | 0.299    | 0.300 | 0.006    | 0.002 |
| $n$ -C <sub>4</sub> | 0.473    | 0.485 | 0.037    | 0.017 |
| $i$ -C <sub>5</sub> | 0.073    | 0.070 | 0.322    | 0.328 |
| $n$ -C <sub>5</sub> | 0.053    | 0.045 | 0.635    | 0.653 |
|                     | 1.000    | 1.000 | 1.000    | 1.000 |

TABLE 13-6 Application of Underwood Equations

| Component                | $x_F$       | $\alpha$ | $\alpha x_F$ | $\theta = 1.36$   |                                      | $\theta = 1.365$  |                                      | $x_D$        | $\alpha x_D$ | $\alpha - \theta$ | $\frac{\alpha x_D}{\alpha - \theta}$ |
|--------------------------|-------------|----------|--------------|-------------------|--------------------------------------|-------------------|--------------------------------------|--------------|--------------|-------------------|--------------------------------------|
|                          |             |          |              | $\alpha - \theta$ | $\frac{\alpha x_F}{\alpha - \theta}$ | $\alpha - \theta$ | $\frac{\alpha x_F}{\alpha - \theta}$ |              |              |                   |                                      |
|                          |             |          |              |                   |                                      |                   |                                      |              |              |                   |                                      |
| C <sub>3</sub>           | 0.05        | 4.99     | 0.2495       | 3.63              | 0.0687                               | 3.625             | 0.0688                               | 0.102        | 0.5090       | 3.6253            | 0.1404                               |
| <i>i</i> -C <sub>4</sub> | 0.15        | 2.62     | 0.3930       | 1.26              | 0.3119                               | 1.255             | 0.3131                               | 0.301        | 0.7886       | 1.2553            | 0.6282                               |
| <i>n</i> -C <sub>4</sub> | 0.25        | 2.02     | 0.5050       | 0.66              | 0.7651                               | 0.655             | 0.7710                               | 0.473        | 0.9555       | 0.6553            | 1.4581                               |
| <i>i</i> -C <sub>5</sub> | 0.20        | 1.00     | 0.2000       | -0.36             | -0.5556                              | -0.365            | -0.5479                              | 0.069        | 0.0690       | -0.3647           | -0.1892                              |
| <i>n</i> -C <sub>5</sub> | <u>0.35</u> | 0.864    | 0.3024       | -0.496            | <u>-0.6097</u>                       | -0.501            | <u>-0.6036</u>                       | <u>0.055</u> | 0.0475       | -0.5007           | <u>-0.0949</u>                       |
|                          | 1.00        |          |              |                   | -0.0196                              |                   | +0.0014                              | 1.000        |              |                   | 1.9426 = $R_w + 1$                   |

Interpolation gives  $\theta = 1.3647$ .



## KREMSEER EQUATION

Starting with the classical method of Kremser (op. cit.), approximate methods of increasing complexity have been developed to calculate the behavior of groups of equilibrium stages for a countercurrent cascade, such as is used in simple absorbers and strippers of the type shown in Fig. 13-2*b* and *d*. However, none of these methods can adequately account for stage temperatures that are considerably higher or lower than the two entering stream temperatures for absorption and stripping, respectively, when appreciable composition changes occur. Therefore, only the simplest form of the Kremser method is presented here. Fortunately, rigorous computer methods described later can be applied when accurate results are required. The Kremser method is most useful for making preliminary estimates of absorbent and stripping agent flow rates or equilibrium-stage requirements. The method can also be used to extrapolate quickly results of a rigorous solution to a different number of equilibrium stages.

Consider the general adiabatic countercurrent cascade of Fig. 13-33 where  $v$  and  $\ell$  are molar component flow rates. Regardless of whether the cascade is an absorber or a stripper, components in the entering vapor will tend to be absorbed and components in the entering liquid will tend to be stripped. If more moles are stripped than absorbed, the cascade is a stripper; otherwise, the cascade is an absorber. The Kremser method is general and applies to either case. Application of component material balance and phase equilibrium equations successively to stages  $N-1$ , 1 through  $N-2$ , etc., as shown by Henley and Seader (op. cit.), leads to the following equations originally derived by Kremser. For each component  $i$ ,

$$(v_i)_N = (v_i)_0(\Phi_i)_A = (\ell_i)_{N+1}[1 - (\Phi_i)_s] \quad (13-39)$$

where

$$(\Phi_i)_A = \frac{(A_i)_e - 1}{(A_i)_{e,N+1} - 1} \quad (13-40)$$

is the fraction of component  $i$  in the entering vapor that is not absorbed,

$$(\Phi_i)_s = \frac{(S_i)_e - 1}{(S_i)_{e,N+1} - 1} \quad (13-41)$$

is the fraction of component  $i$  in the entering liquid that is not stripped,

$$(A_i)_e = \left( \frac{L}{KV} \right)_e \quad (13-42)$$

is the effective or average absorption factor for component  $i$ , and

$$(S_i)_e = \frac{1}{(A_i)_e} \quad (13-43)$$

is the effective or average stripping factor for component  $i$ . When the entering streams are at the same temperature and pressure and negligible absorption and stripping occur, effective component absorption and stripping factors are determined simply by entering stream conditions. Thus, if  $K$  values are composition-independent, then

$$(A_i)_e = \frac{1}{(S_i)_e} = \frac{L_{N+1}}{K_i(T_{N+1}, P_{N+1})V_0} \quad (13-44)$$

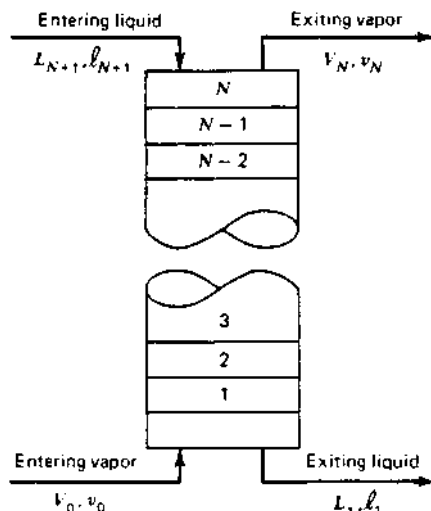


FIG. 13-33 General adiabatic countercurrent cascade for simple absorption or stripping.

When entering stream temperatures differ and/or moderate to appreciable absorption and/or stripping occurs, values of  $A_i$  and  $S_i$  should be based on effective average values of  $L$ ,  $V$ , and  $K_i$  in the cascade. However, even then Eq. (13-44) with  $T_{N+1}$  replaced by  $(T_{N+1} + T_0)/2$  may be able to give a first-order approximation of  $(A_i)_e$ . In the case of an absorber,  $L_{N+1} < L_e$  and  $V_0 > V_e$  will be compensated to some extent by  $K_i[(T_{N+1} + T_0)/2, P] < K_i(T_e, P)$ . A similar compensation, but in opposite directions, will occur in the case of a stripper. Equations (13-40) and (13-41) are plotted in Fig. 13-34. Components having large values of  $A_e$  or  $S_e$  absorb or strip, respectively, to a large extent. Corresponding values of  $\Phi_A$  and  $\Phi_s$  approach a value of 1 and are almost independent of the number of equilibrium stages.

An estimate of the minimum absorbent flow rate for a specified amount of absorption from the entering gas of some key component  $K$  for a cascade with an infinite number of equilibrium stages is obtained from Eq. (13-40) as

$$(L_{N+1})_{\min} = K_K V_0 [1 - (\Phi_K)_A] \quad (13-45)$$

The corresponding estimate of minimum stripping agent flow rate for a stripper is obtained as

$$(V_0)_{\min} = \frac{L_{N+1} [1 - (\Phi_K)_s]}{K_K} \quad (13-46)$$

**Example 2: Calculation of Kremser Method** For the simple absorber specified in Fig. 13-35, a rigorous calculation procedure as described below gives the results in Table 13-7. Values of  $\Phi$  were computed from component

TABLE 13-7 Results of Calculations for Simple Absorber of Fig. 13-35

| Component    | N = 6 (rigorous method) |              |              |              | N = 12 (Kremser method) |           |              |              |              |              |
|--------------|-------------------------|--------------|--------------|--------------|-------------------------|-----------|--------------|--------------|--------------|--------------|
|              | (lb-mol)/h              |              | $(\Phi_i)_A$ | $(\Phi_i)_s$ | $(A_i)_e$               | $(S_i)_e$ | (lb-mol)/h   |              | $(\Phi_i)_A$ | $(\Phi_i)_s$ |
|              | $(v_i)_6$               | $(\ell_i)_1$ |              |              |                         |           | $(v_i)_{12}$ | $(\ell_i)_1$ |              |              |
| $C_1$        | 147.64                  | 12.36        | 0.9228       |              | 0.0772                  |           | 147.64       | 12.36        | 0.9228       |              |
| $C_2$        | 276.03                  | 94.97        | 0.7460       |              | 0.2541                  |           | 275.98       | 94.02        | 0.7459       |              |
| $C_3$        | 105.42                  | 134.58       | 0.4393       |              | 0.5692                  |           | 103.46       | 136.54       | 0.4311       |              |
| $nC_4$       | 1.15                    | 23.85        | 0.0460       |              | 1.3693                  |           | 0.16         | 24.84        | 0.0063       |              |
| $nC_5$       | 0.0015                  | 4.9985       | 0.0003       |              | 3.6                     |           | 0            | 5.0          | 0.0          |              |
| Absorber oil | 0.05                    | 164.95       |              | 0.9997       |                         | 0.0003    | 0.05         | 164.95       |              | 0.9997       |
| Totals       | 530.29                  | 435.71       |              |              |                         |           | 527.29       | 437.71       |              |              |

NOTE: To convert pound-moles per hour to kilogram-moles per hour, multiply by 0.454.

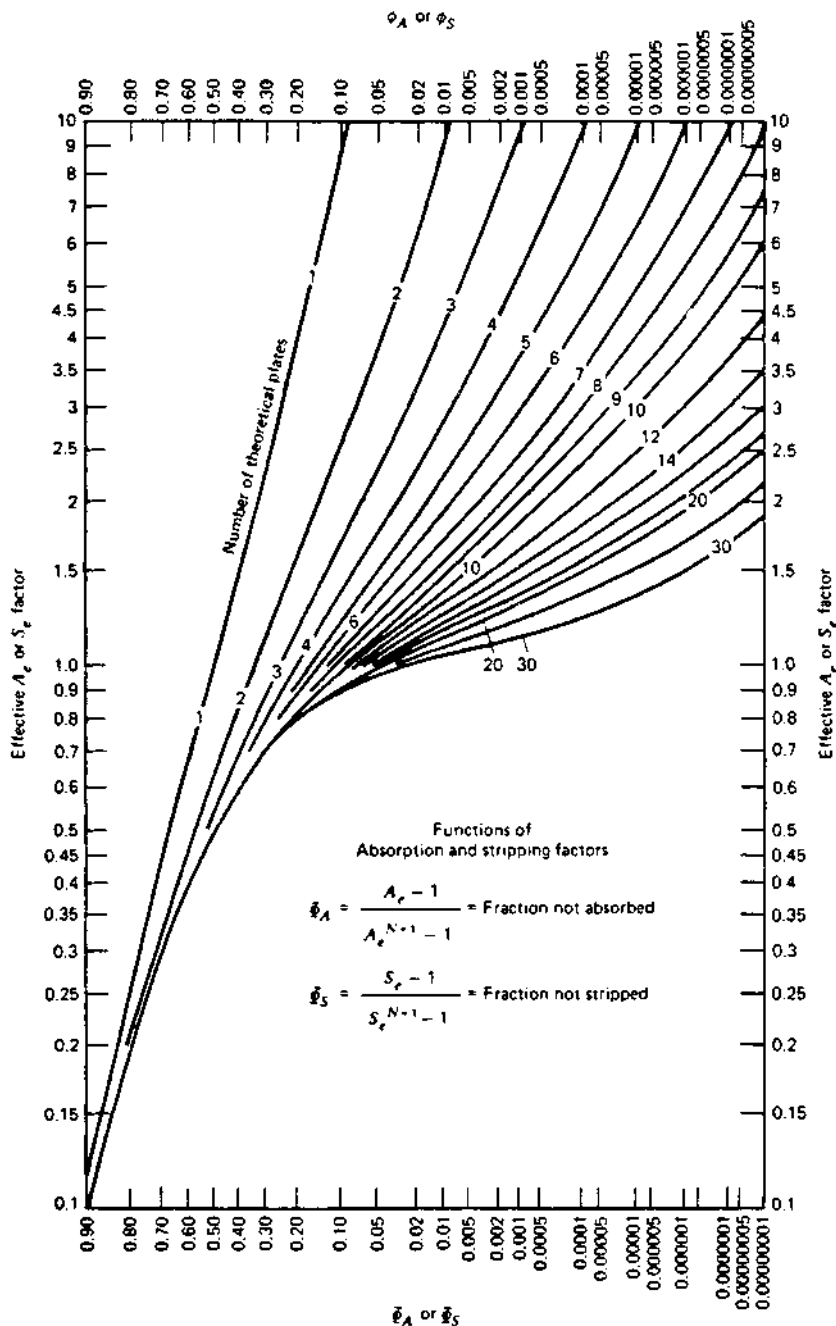


FIG. 13-34 Absorption and stripping factors. [W. C. Edmister, *AIChE J.*, 3, 165-171 (1957).]

product flow rates, and corresponding effective absorption and stripping factors were obtained by iterative calculations in using Eqs. (13-40) and (13-41) with  $N=6$ . Use the Kremser method to estimate component product rates if  $N$  is doubled to a value of 12.

Assume that values of  $A_e$  and  $S_e$  will not change with a change in  $N$ . Application of Eqs. (13-40), (13-41), and (13-39) gives the results in the last four

columns of Table 13-7. Because of its small value of  $A_e$ , the extent of absorption of  $C_1$  is unchanged. For the other components, somewhat increased amounts of absorption occur. The degree of stripping of the absorber oil is essentially unchanged. Overall, only an additional 0.5 percent of absorption occurs. The greatest increase in absorption occurs for  $n\text{-C}_6$  to the extent of about 4 percent.

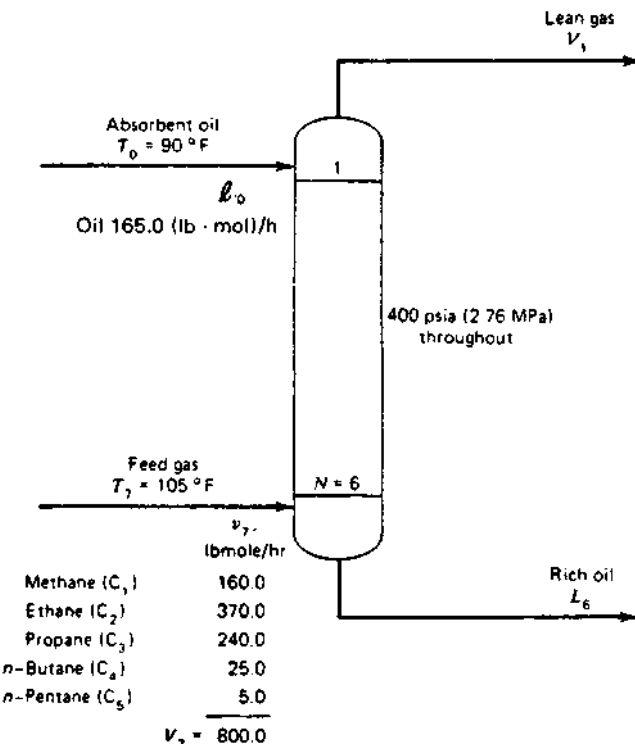


FIG. 13-35 Specifications for the absorber example.

## SIMULATION OF DISTILLATION PROCESSES

Chemical engineers have been solving distillation problems by using the *equilibrium-stage model* since 1893 when Sorel outlined the concept to describe the distillation of alcohol. Since that time, it has been used to model a wide variety of distillation-like processes, including simple distillation (single-feed, two-product columns), complex distillation (multiple-feed, multiple-product columns), extractive and azeotropic distillation, petroleum distillation, absorption, liquid-liquid extraction, stripping, and supercritical extraction.

Real distillation processes, however, nearly always operate away from equilibrium. In recent years it has become possible to simulate distillation and absorption as the mass-transfer rate-based operations that they really are, using what have become known as *nonequilibrium* (NEQ) or *rate-based* models [Taylor et al., *CEP* (July 28, 2003)].

The computer simulation of distillation processes, whether done using equilibrium or nonequilibrium models, requires us to address the following topics:

- Formulation of the model equations
  - Physical property data and calculations (see Sec. 4 of this handbook)
  - Degrees-of-freedom analysis
  - Solution of large linear and strongly nonlinear systems of equations
- These are the topics we consider in what follows. We also discuss the use of simulation tools in the modeling of real distillation columns.

### EQUILIBRIUM-STAGE MODELING

A schematic diagram of an equilibrium stage is shown in Fig. 13-36a. Vapor from the stage below and liquid from a stage above are brought into contact on the stage together with any fresh or recycle feeds. The

vapor and liquid streams leaving the stage are assumed to be in equilibrium with each other. A complete separation process is modeled as a sequence of these *equilibrium stages*, as shown in Fig. 13-36b.

**The MESH Equations (the 2c + 3 Formulation)** The equations that model equilibrium stages often are referred to as the MESH equations. The M equations are the material balance equations, E stands for equilibrium equations, S stands for mole fraction summation equations, and H refers to the heat or enthalpy balance equations.

There are two types of material balance: the total material balance

$$V_{j+1} + L_{j-1} + F_j - (1 + r_j^V)V_j - (1 + r_j^L)L_j = 0 \quad (13-47)$$

and the component material balance

$$V_{j+1}y_{i,j+1} + L_{j-1}x_{i,j-1} + F_jz_{i,j} - (1 + r_j^V)V_jy_{i,j} - (1 + r_j^L)L_jx_{i,j} = 0 \quad (13-48)$$

In the material balance equations given above,  $r_j$  is the ratio of side-stream flow to interstage flow:

$$r_j^V = \frac{U_j}{V_j} \quad r_j^L = \frac{W_j}{L_j} \quad (13-49)$$

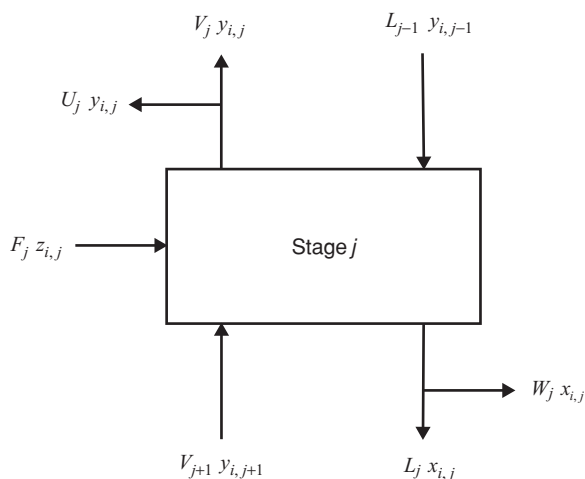
Mole fractions must be forced to sum to unity, thus

$$\sum_{i=1}^c x_{i,j} = 1 \quad \sum_{i=1}^c y_{i,j} = 1 \quad \sum_{i=1}^c z_{i,j} = 1 \quad (13-50)$$

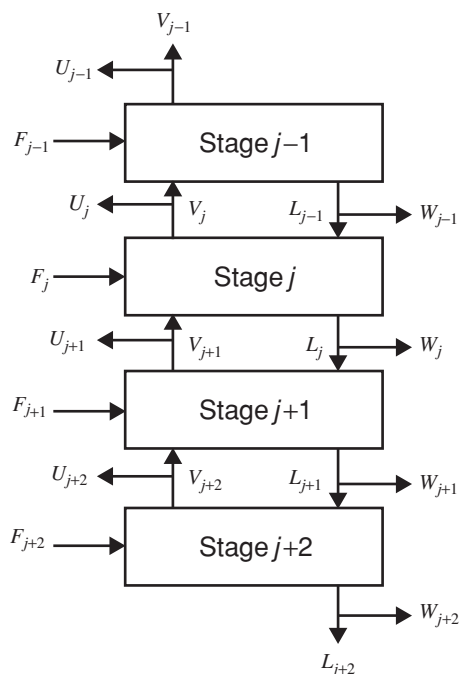
The enthalpy balance is given by

$$V_{j+1}H_{j+1}^V + L_{j-1}H_{j-1}^L + F_jH_j^F - (1 + r_j^V)V_jH_j^V - (1 + r_j^L)L_jH_j^L - Q_j = 0 \quad (13-51)$$

The superscripted  $H$ 's are the enthalpies of the appropriate phase.



(a)



(b)

FIG. 13-36 (a) The equilibrium stage. (b) Multistage column.

To complete the model, it is usual to add equations that relate the compositions of the two streams leaving the stage. In the standard model of a distillation column, we assume that these two streams are in equilibrium with each other. Thus, the mole fractions in the exiting streams are related by the familiar equations of phase equilibrium:

$$y_{i,j} = K_{i,j} x_{i,j} \quad (13-52)$$

The  $K_{i,j}$  are the equilibrium ratios or  $K$  values for species  $i$  on stage  $j$ .

**Degrees-of-Freedom Analysis and Problem Formulation**  
Table 13-8 summarizes the equations for a single equilibrium stage (in a sequence of stages). There are  $2c + 4$  equations per stage, of which

TABLE 13-8 Equations for an Equilibrium Stage

| Equation                          | Equation no. | Number   |
|-----------------------------------|--------------|----------|
| Total mass balance                | (13-47)      | 1        |
| Component mass balances           | (13-48)      | $c$      |
| Total energy balance              | (13-51)      | 1        |
| Mole fraction summation equations | (13-50)      | 2        |
| Equilibrium equations             | (13-52)      | $c$      |
|                                   |              | $2c + 4$ |

only  $2c + 3$  are independent. Thus, one of these equations must be ignored in a (computer-based) method to solve the equations. In some methods we disregard the total material balance; an alternative is to combine the vapor and liquid mole fraction summation equations (as was done for flash calculations by Rachford and Rice, op. cit.). Note that the mole fraction summation equations for the feed are omitted here, as are the mole fraction summation equations for interstage vapor and liquid traffic; the latter “belong” to the equation set for adjacent stages (see, however, the subsection Degrees of Freedom below in which this topic is revisited in greater detail).

The quantities for stage  $j$  that appear in these equations are summarized in Table 13-9. The total number of variables appearing in these equations is  $3c + 10$ . Note that the  $K$  values and enthalpies that also appear in the MESH equations are not included in the table of variables, nor are equations for their estimation included in the list of equations. Thermodynamic properties are functions of temperature, pressure, and composition, quantities that do appear in the table of variables.

The  $2c + 3$  unknown variables normally determined by solving the MESH equations are the  $c$  vapor mole fractions  $y_{i,j}$ ; the  $c$  liquid mole fractions  $x_{i,j}$ ; the stage temperature  $T_j$ ; and the vapor and liquid flow  $V_j$  and  $L_j$ . The remaining variables,  $c + 7$  in number (the difference between the number of variables and the number of independent equations), that need to be specified are the stage pressure  $P_j$ , the feed flow rate,  $c - 1$  mole fractions in the feed (the last is then computed directly from the feed mole fraction summation equation, which was not included in the equations for stage  $j$  in Table 13-8), temperature and pressure of the feed, the stage heat duty  $Q_j$ , and the sidestream flows  $U_j$  and  $W_j$ . It is important to recognize that the other flows and composition variables appearing in the MESH equations are associated with the equivalent equations for adjacent stages.

For a column of  $N$  stages, we must solve  $N(2c + 3)$  equations. The table below shows how we may need to solve hundreds or even thousands of equations.

| $c$ | $N$ | $N(2c + 3)$ |
|-----|-----|-------------|
| 2   | 10  | 70          |
| 3   | 20  | 180         |
| 5   | 50  | 650         |
| 10  | 30  | 690         |
| 40  | 100 | 8300        |

TABLE 13-9 Variables for an Equilibrium Stage

| Variable                    | Symbol    | Number    |
|-----------------------------|-----------|-----------|
| Vapor flow rate             | $V_j$     | 1         |
| Liquid flow rate            | $L_j$     | 1         |
| Feed flow rate              | $F_j$     | 1         |
| Vapor sidestream flow       | $W_j$     | 1         |
| Liquid sidestream flow      | $U_j$     | 1         |
| Vapor-phase mole fractions  | $y_{i,j}$ | $c$       |
| Liquid-phase mole fractions | $x_{i,j}$ | $c$       |
| Feed composition            | $z_{i,j}$ | $c$       |
| Stage temperature           | $T_j$     | 1         |
| Stage pressure              | $P_j$     | 1         |
| Feed temperature            |           | 1         |
| Feed pressure               |           | 1         |
| Stage heat duty             | $Q_j$     | 1         |
|                             |           | $3c + 10$ |

**The 2c + 1 Formulation** An alternative form of the MESH equations is used in many algorithms. In this variation, we make use of the component flow defined by

$$v_{ij} = V_j y_{ij} \quad l_{ij} = L_j x_{ij} \quad f_{ij} = F_j z_{ij} \quad (13-53)$$

In terms of the component flow, the component material balance becomes

$$v_{i,j+1} + l_{i,j-1} + f_{i,j} - (1 + r_j^v) v_{i,j} - (1 + r_j^l) l_{i,j} = 0 \quad (13-54)$$

Since the total vapor and liquid flow rates are, by definition, the sum of the component flow rates of the respective phases, the summation equations and total mass balance equation are satisfied automatically. Thus, the number of equations and variables per stage has been reduced by 2 to  $2c + 1$ .

**The c + 3 Formulation** The number of unknown variables per stage can be reduced to only  $c + 3$  if we use the equilibrium relations to eliminate the vapor-phase mole fractions from the component mass balances

$$V_{j+1} K_{i,j+1} x_{i,j+1} + L_{j-1} x_{i,j-1} + F_j z_{i,j} - (1 + r_j^v) V_j K_{i,j} x_{i,j} - (1 + r_j^l) L_j x_{i,j} = 0 \quad (13-55)$$

and from the summation equation

$$\sum_{i=1}^c K_{i,j} x_{i,j} = 1 \quad (13-56)$$

This equation is familiar to us from bubble point calculations. In this formulation of the MESH equations, the vapor-phase mole fractions no longer are independent variables but are defined by Eq. (13-52). This formulation of the MESH equations has been used in quite a number of algorithms. It is less useful if vapor-phase nonideality is important (and, therefore, the  $K$  values depend on the vapor-phase composition).

**Condenser and Reboiler** The MESH equations given above apply to all the interior stages of a column. In addition, any reboiler and condenser must be considered. On an actual plant, a total condenser may be followed by a reflux accumulator and a stream divider. The accumulator does not add any equations to a steady-state model (but it is important to consider in an unsteady-state model), but the stream splitter is a separate unit with its own temperature, pressure, and heat loss, and modeled by the appropriate balance equations (see the section below on Degrees of Freedom where we address this topic in greater detail). In practice the condenser and reflux splitter often are modeled as a single combined unit, and the MESH equations described above may be used to model these stages with only some minor modifications. For example, for a total condenser *and* the reflux stream splitter at the top of a distillation column, the liquid distillate is  $U_1$  and the reflux ratio  $R = 1/r_1^l$ . For a partial condenser (with no stream splitter needed) the vapor product is  $V_1$  and the reflux ratio  $R = L_1/V_1$ . A condenser/splitter device unit that provides both vapor and liquid products is given by a combination of these two units. Finally, for a partial reboiler at the base of a column, the bottoms flow rate is  $L_N$ . Note that an equilibrium stage with a sidestream is considered here to be a single unit in essentially the same way.

In computer-based methods for solving the MESH equations, it is common to replace the energy balance of the condenser (with or without the associated stream splitter) and reboiler with a specification equation. Possible specifications include

- The flow rate of the distillate/bottoms product stream
- The mole fraction of a component in either the distillate or bottoms product stream
- Component flow rate in either the distillate or bottoms product stream
- A reflux/reboil ratio or rate
- The temperature of the condenser or reboiler
- The heat duty to the condenser or reboiler

If the condenser and/or reboiler heat duties are not specified, it is possible to calculate them from the energy balances after all the other model equations have been solved.

For a total condenser, the vapor composition used in the equilibrium relations is that determined during a bubble point calculation based on the actual pressure and liquid compositions found in the condenser. These vapor mole fractions are not used in the component mass balances since there is no vapor stream from a total condenser. It often happens that the temperature of the reflux stream is below the bubble point temperature of the condensed liquid (subcooled condenser). In such cases it is necessary to specify either the actual temperature of the reflux stream or the difference in temperature between the reflux stream and the bubble point of the condensate.

**Solution of the MESH Equations** We may identify several classes of methods of solving the *equilibrium*-stage model equations:

- **Graphical methods.** These methods were developed before modern computer methods became widely adopted. Some graphical methods retain their value for a variety of reasons and were discussed at length in a prior subsection.
- **Approximate methods.** In these a great many simplifying assumptions are made to obtain solutions of the model equations. These methods were the subject of the immediately preceding subsection.
- **Computer-based methods.** Indeed, from the late 1950s to the early 1990s hardly a year passed without the publication of at least one (and usually many more than one) new algorithm for solving these equations (Seader, op. cit.). It is possible to make the case that it was the equilibrium model that brought computing into chemical engineering in the first place! One of the incentives for this activity has always been a desire to solve problems with which existing methods have trouble. The evolution of algorithms for solving the stage model equations has been influenced by, among other things, the availability (or lack) of sufficient computer storage and speed, the development of mathematical techniques that can be exploited, the complexity of physical property ( $K$  value and enthalpy) correlations, and the form of the model equations being solved. We continue with a brief discussion of computer-based methods. Readers interested in complete details of the methods discussed should consult the references cited below. In addition, many computer-based methods are discussed at length in a number of textbooks [see, e.g., Holland, op. cit.; King, op. cit.; Seader and Henley, op. cit.; Haas (in Kister op. cit. 1992)]. Seader (op. cit.) has written an interesting history of equilibrium-stage simulation.

The MESH equations form a set of nonlinear, algebraic equations (in the sense that no derivatives or integrals are involved) and must, therefore, be solved by some iterative process. There are two key steps in the development of an algorithm for solving systems of nonlinear equations:

- The selection of particular numerical methods
- The selection of the order in which the equations are to be solved

We may identify several classes of methods for numerically solving the MESH equations:

1. Tearing methods in which the equations are divided into groups and each group is solved separately
2. Inside-out (IO) methods
3. Simultaneous convergence (SC) methods in which all the equations are solved at the same time simultaneously by using Newton's method (or a variant thereof)
4. Relaxation methods in which the MESH equations are cast in unsteady-state form and integrated numerically until the steady-state solution has been found
5. Continuation methods
6. Collocation methods
7. Optimization methods

The methods for solving the MESH equations that are used most widely fall into categories 1 through 3 and 5—tearing methods, inside-out methods, simultaneous convergence methods, and continuation method. Following the sections on each of these three classes of method, we discuss homotopy continuation methods. For many years the older tearing methods formed the backbone of sequential modular flow sheet simulation programs; such methods have largely been replaced by the inside-out methods that now are standard in all major (and many less widely used) flow sheet simulation programs. Simultaneous convergence methods also now are available in many of these systems. Even homotopy methods (sometimes considered to belong



to the domain of specialists) are finding increasing use in commercial software.

**Tearing Methods** Equation tearing involves breaking a system of nonlinear equations into small groups and solving each group of equations in turn. In solving any subset of the complete set of equations, only a corresponding number of variables, the "tear" variables, can be determined. To start the calculations, therefore, it is necessary to assume values for the remaining variables. The "torn" set of equations is then solved for the "tear" variables, assuming that the values assigned to all other variables are correct. Successive groups of equations and variables are *torn* or *decoupled* from the full set of equations and variables until all the variables have been updated. At this point, the process starts over and is repeated until all the equations are satisfied simultaneously.

A great many tearing methods for solving the MESH equations have been proposed. According to Friday and Smith (1964), tearing methods may be analyzed in the following terms:

- The order in which the equations are grouped
- The order in which each group of equations is solved
- The selection of which variables are to be computed from which equation

Subsequent issues are

- The method(s) used to solve the M and E equations
- The method of calculating the new stage temperatures
- The method of calculating the new flow rates  $V_j$  and  $L_j$

Most tearing methods are based on keeping all the equations of a given type together, e.g., the M equations or the H equations, for all the stages at once.

The classic papers by Lewis and Matheson [*Ind. Eng. Chem.*, **24**, 496 (1932)] and Thiele and Geddes [*Ind. Eng. Chem.*, **25**, 290 (1933)] represent the first attempts at solving the MESH equations for multicomponent systems numerically (the graphical methods for binary systems discussed earlier had already been developed by Ponchon, by Savarit, and by McCabe and Thiele). At that time the computer had yet to be invented, and since modeling a column could require hundreds, possibly thousands, of equations, it was necessary to divide the MESH equations into smaller subsets if hand calculations were to be feasible. Despite their essential simplicity and appeal, stage-to-stage calculation procedures are not used now as often as they used to be.

Matrix techniques were first used in separation process calculations by Amundson and Pontinen [*Ind. Eng. Chem.*, **50**, 730 (1958)] who demonstrated that the combined M + E equations (the  $c + 3$  formulation) could be conveniently written in tridiagonal matrix form.

Three quite different approaches to the problem of computing the flow rates and temperatures have evolved:

1. In the bubble point (BP) method, the S + E equations are used to determine the stage temperatures from a bubble point calculation. The vapor and liquid flow rates are computed from the energy balances and total material balances. The BP method was introduced by Amundson and Pontinen (op. cit.) who used matrix inversion to solve equations. A significant improvement in the BP method was introduced by Wang and Henke [*Hydrocarbon Process.*, **45**(8), 155 (1966)] who used a form of Gaussian elimination known as the Thomas algorithm that can be used to solve a tridiagonal system very efficiently. Holland and coworkers (Holland, op. cit.) have combined some of these ideas with the "theta method" of convergence acceleration.

2. In the sum rates (SR) method, the vapor and liquid flow rates are computed directly from the summation equations and the stage temperatures determined from the energy balances. The SR algorithm was due originally to Sujata [*Hydrocarbon Process.*, **40**(12), 137 (1961)] and Friday and Smith (op. cit.). Sujata used the  $2c + 1$  version of the MESH equations in his work and the tridiagonal system of combined M + E equations was written in terms of the component flow rates in the liquid phase. Burningham and Otto [*Hydrocarbon Process.*, **46**(10), 163 (1967)] incorporated the Thomas algorithm in their implementation of the SR method. Sridhar and Lucia [*Comput. Chem. Eng.*, **14**, 901 (1990)] developed a modified SR method that is capable of solving problems involving narrow, intermediate, and wide-boiling mixtures alike.

3. In the Newton-Raphson methods, the bubble point equations and energy balances are solved simultaneously for the stage temperatures and vapor flow rates; the liquid flow rates follow from the total material balances [Tierney and coworkers, *AIChE J.*, **13**, 556 (1967); **15**, 897 (1969); Billingsley and Boynton, *AIChE J.*, **17**, 65 (1971)]. Similar methods are described by Holland (op. cit.) and by Tomich [*AIChE J.*, **16**, 229 (1970)].

**Inside-Out Methods** First proposed by Boston and Sullivan [*Can. J. Chem. Engr.*, **52**, 52 (1974)] and further developed by Boston and coworkers [*Comput. Chem. Engrg.*, **2**, 109 (1978); *ACS Symp. Ser. No. 124*, 135 (1980); *Comput. Chem. Engrg.*, **8**, 105 (1984); and *Chem. Eng. Prog.*, **86** (8), 45-54 (1990)], Jelinek [*Comput. Chem. Engrg.*, **12**, 195 (1988)], and Simandl and Svrcek [*Comput. Chem. Engrg.*, **15**, 337 (1991)], inside-out methods really belong in the group of tearing methods; however, their very widespread use in many commercial simulation programs demands that they be granted a category all to themselves. In these methods, complicated equilibrium and enthalpy expressions are replaced by simple models, and the iteration variables  $T$ ,  $V$ ,  $x$ ,  $y$  are replaced by variables within the simple models that are relatively free of interactions with each other. Seader and Henley (op. cit.) provide complete details of the somewhat lengthy algorithm. Inside-out methods have replaced many of the older algorithms in commercial simulation programs, the modified version of Russell [*Chem. Eng.*, **90**, (20), 53 (1983)] being the basis for some of them.

**Simultaneous Convergence Methods** One drawback of some tearing methods is their relatively limited range of application. For example, the BP methods are more successful for distillation, and the SR-type methods are considered better for mixtures that exhibit a wide range of (pure-component) boiling points (see, however, our remarks above on modified BP and SR methods). Other possible drawbacks (at least in some cases) include the number of times physical properties must be evaluated (several times per outer loop iteration) if temperature- and composition-dependent physical properties are used. It is the physical properties calculations that generally dominate the computational cost of chemical process simulation problems. Other problems can arise if any of the iteration loops are hard to converge.

The development of methods for solving all the equations at the same time was tackled independently by a number of investigators. Simultaneous solution of all the MESH equations was suggested as a method of last resort by Friday and Smith (op. cit.) in a classic paper analyzing the reasons why other algorithms fail. They did not, however, implement such a technique. The two best-known papers are those of Goldstein and Stanfield [*Ind. Eng. Chem. Process Des. Dev.*, **9**, 78 (1970)] and Naphtali and Sandholm [*AIChE J.*, **17**, 148 (1971)], the latter providing more details of an application of Newton's method described by Naphtali at an AIChE meeting in May 1965. A method to solve all the MESH equations for all stages at once by using Newton's method was implemented by Whitehouse [Ph.D. Thesis, University of Manchester Institute of Science and Technology (1964)] (see Stainthorpe and Whitehouse [*I. Chem. E. Symp. Series*, **23**, 181 (1967)]). Among other things, Whitehouse's code allowed for specifications of purity,  $T$ ,  $V$ ,  $L$ , or  $Q$  on any stage. Interlinked systems of columns and nonideal solutions could be dealt with even though no examples of the latter type were solved by Whitehouse.

Many others have since employed Newton's method or a related method to solve the MESH equations. Such methods have become standard in commercial process simulation programs, most often for simulating systems for which tearing and inside-out methods are less successful (typically those that involve strongly nonideal mixtures). Simultaneous convergence procedures have shown themselves to be generally fast and reliable, having a locally quadratic convergence rate in the case of Newton's method, and these methods are much less sensitive to difficulties associated with nonideal solutions than are tearing methods. Seader (Sec. 13, *Perry's Chemical Engineers' Handbook*, 7th ed., 1986) discusses the design of simultaneous convergence methods.

**Continuation Methods (for Really Difficult Problems)** For those problems that other methods fail to solve, we may use continuation

methods. These methods begin with a known solution of a companion set of equations and follow a path to the desired solution of the set of equations to be solved. In most cases, the path exists and can be followed. We can identify the following categories of continuation method that have been used for solving the equilibrium-stage MESH equations:

1. Mathematical methods which place the MESH equations into a homotopy equation of purely mathematical origin

2. Physical continuation methods in which the nature of the equations being solved is exploited in some way  
The first to use the Newton homotopy for separation process problems were Hlavacek and coworkers [*Chem. Eng. Sci.*, **36**, 1599 (1981); *Chem. Eng. Commun.*, **28**, 165 (1984)]. In papers of considerable significance Wayburn and Seader [*Comput. Chem. Engng.*, **11**, 7–25 (1987); *Proc. Second Intern. Conf. Foundations of Computer-Aided Process Design*, CACHE, Austin, Tex., 765–862 (1984); *AIChE Monograph Series*, AIChE, New York, **81**, No. 15 (1985)] used the Newton homotopy in their solution of the MESH equations for interlinked distillation columns. Kovach and Seider [*Comput. Chem. Engng.*, **11**, 593 (1987)] used the Newton homotopy for solving distillation problems involving highly nonideal mixtures.

In the category of physical continuation methods are the thermodynamic homotopies of Vickery and Taylor [*AIChE J.*, **32**, 547 (1986)] and a related method due to Frantz and Van Brunt (AIChE National Meeting, Miami Beach, 1986). Thermodynamic continuation has also been used to find azeotropes in multicomponent systems by Fidkowski et al. [*Comput. Chem. Engng.*, **17**, 1141 (1993)]. Parametric continuation methods may be considered to be physical continuation methods. The reflux ratio or bottoms flow rate has been used in parametric solutions of the MESH equations [Jelinek et al., *Chem. Eng. Sci.*, **28**, 1555 (1973)].

Bryne and Baird [*Comput. Chem. Engng.*, **9**, 593 (1985)] describe the use of proprietary continuation methods. Woodman (Ph.D. thesis, University of Cambridge, 1989) has combined several continuation methods to solve problems that involved nonstandard specifications and multiple liquid phases.

For detailed descriptions of homotopy methods in chemical engineering, see Seader [*Computer Modeling of Chemical Processes*, AIChE Monograph Series 15, **81**, (1985)].

**Other Methods** Relaxation techniques differ from both tearing algorithms and simultaneous convergence methods in that the relaxation techniques do not solve steady-state problems. Rather, at least one set of the MESH equations is cast in an unsteady-state form. The equations are then integrated from some initial state (guess) until successive values of the variables do not change. The appeal of the relaxation methods lies in their extreme stability. A drawback is that convergence is generally a very slow process, slowing even more as the solution is approached. For this reason, the method is used (only) for problems which are very difficult to converge or for situations in which a knowledge of how the steady state is achieved is important. Key papers are by Rose, Sweeney, and Schrodt [*Ind. Eng. Chem.*, **50**, 737 (1958)]; Ball (AIChE National Meeting, New Orleans, 1961); Verneuil and Oleson (ACS National Mtg., Los Angeles, 1971); and Ketchum [*Chem. Eng. Sci.*, **34**, 387 (1979)].

Collocation methods are widely used for solving systems of partial differential equations. Despite their considerable potential for certain types of equilibrium-stage simulation problems, collocation methods have not become part of the mainstream of equilibrium-stage simulation. Key papers are by Stewart et al. [*Chem. Eng. Sci.*, **40**, 409 (1985)], Swartz and Stewart [*AIChE J.*, **32**, 1832 (1986)], Cho and Joseph [*AIChE J.*, **29**, 261, 270 (1983); *Comput. Chem. Engng.*, **8**, 81 (1984)], and Seferlis and Hrymak [*Chem. Eng. Sci.*, **49**, 1369 (1994); *AIChE J.*, **40**, 813 (1994)].

Methods that have been developed for finding the optimal solution(s) to engineering models have also been used to solve difficult equilibrium-stage separation process problems. An example of a method in this category is that of Lucia and Wang [*Comput. Chem. Engng.*, **28**, 2541 (2004)].

**Examples** In what follows we illustrate possible column specifications by considering four examples from Seader (*Perry's Chemical Engineers' Handbook*, 7th ed., 1997). They are a simple distillation

column, a more complicated distillation column, an absorber, and a reboiled stripper. A simultaneous convergence method was used for the calculations reported below. The computer program that was used for these exercises automatically generated initial estimates of all the unknown variables (flows, temperatures, and mole fractions). In most cases the results here differ only very slightly from those obtained by Seader (almost certainly due to small differences in physical property constants).

**Example 3: Simple Distillation Column** Compute stage temperatures, interstage vapor and liquid flow rates and compositions, and reboiler and condenser duties for the butane-pentane splitter studied in Example 1. The specifications for this problem are summarized below and in Fig. 13-37.

The specifications made in this case are summarized below:

| Variable                                                 | Number      | Value                      |
|----------------------------------------------------------|-------------|----------------------------|
| Number of stages                                         | 1           | 11                         |
| Feed stage location                                      | 1           | 6                          |
| Component flows in feed                                  | $c = 5$     | 5, 15, 25, 20, 35 lb-mol/h |
| Feed pressure                                            | 1           | 120 psia                   |
| Feed vapor fraction                                      | 1           | 0                          |
| Pressure on each stage                                   |             |                            |
| including condenser and reboiler                         | $N = 11$    | $P_j = 120$ psia           |
| Heat duty on each stage except reboilers and condensers  | $N - 2 = 9$ | $Q_j = 0$                  |
| Vapor flow to condenser (replaces heat duty of reboiler) | 1           | $V_2 = 175$ lb-mol/h       |
| Distillate flow rate (replaces heat duty of condenser)   | 1           | $D = 48.9$ lb-mol/h        |
| Total                                                    | 31          |                            |

In addition, we have assumed that the pressure of the reflux divider is the same as the pressure of the condenser, the heat loss from the reflux divider is zero, and the reflux temperature is the boiling point of the condensed overhead vapor.

The Peng-Robinson equation of state was used to estimate  $K$  values and enthalpy departures [as opposed to the De Priester charts used in Example 1 and by Seader (ibid.) who solved this problem by using the Thiele-Geddes (op. cit.) method].

With 11 stages and 5 components the equilibrium-stage model has 143 equations to be solved for 143 variables (the unknown flow rates, temperatures, and mole fractions). Convergence of the computer algorithm was obtained in just four iterations. Computed product flows are shown in Fig. 13-37.

A pseudobinary McCabe-Thiele diagram for this multicomponent system is shown in Fig. 13-38. For systems with more than two components, these diagrams can only be computed from the results of a computer simulation. The axes are defined by the relative mole fractions:

$$X = \frac{x_{lk}}{x_{lk} + x_{hk}} \quad Y = \frac{y_{lk}}{y_{lk} + y_{hk}}$$

where the subscripts  $lk$  and  $hk$  refer to light and heavy key, respectively. The lines in the diagram have similar significance as would be expected from our knowledge of McCabe-Thiele diagrams for binary systems discussed earlier in this section; the triangles correspond to equilibrium stages. The operating lines are not straight because of heat effects and because the feed is not in the best location.

The fact that the staircase of triangles visible in Fig. 13-38 fails to come close to the corners of the diagram where  $X = Y = 1$  and  $X = Y = 0$  shows that the separation is not especially sharp. It is worth asking what can be done to improve the separation obtained with this column. The parameters that have a significant effect on the separation are the numbers of stages in the sections above and below the feed, the reflux ratio, and a product flow rate (or reflux flow). Figure 13-39 shows how the mole fraction of *i*-pentane in the overhead and of *n*-butane in the bottom product changes with the reflux ratio. For the base case considered above, the reflux ratio is 2.58 (calculated from the results of the simulation). It is clear that increasing the reflux ratio has the desired effect of improving product purity. This improvement in purity is, however, accompanied by an increase in both the operating cost, indicated in Fig. 13-39 by the increase in reboiler duty, and capital cost, because a larger column would be needed to accommodate the increased internal flow. Note, however, that the curves that represent the mole fractions of the keys in the overhead and bottoms appear to flatten, showing that product purity will not increase indefinitely as the reflux ratio increases. Further improvement in product purity can best be made by changing a different specification.

Figure 13-40 shows the tradeoff in product purities when we change the specified distillate flow rate, maintaining all other specifications at the values



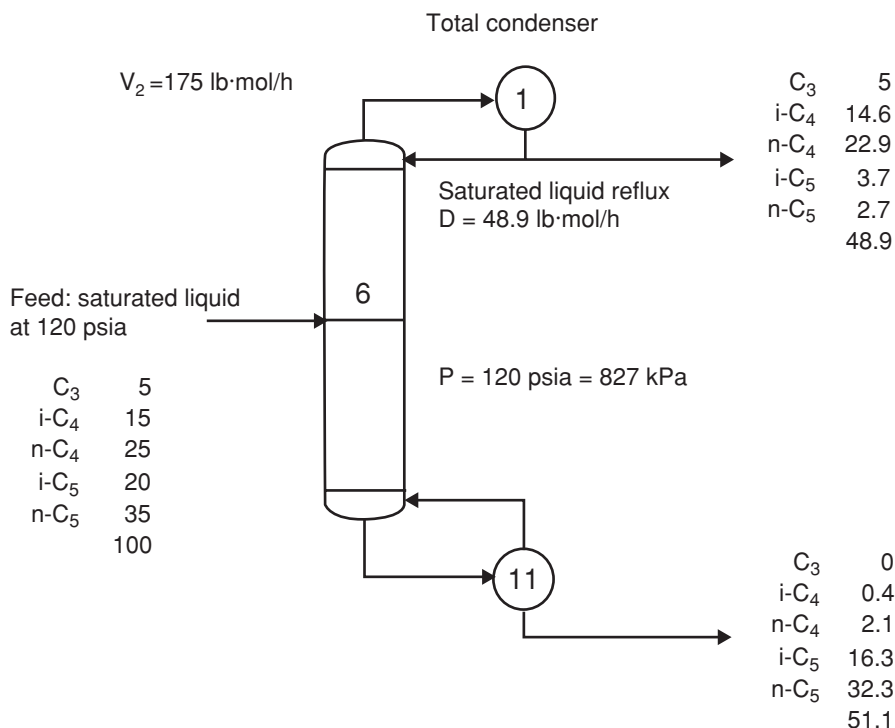


FIG. 13-37 Specifications and calculated product stream flows for butane-pentane splitter. Flows are in pound-moles per hour.

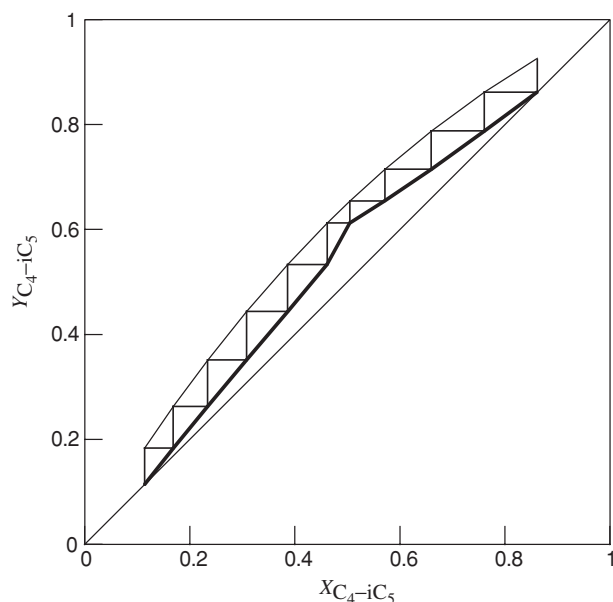


FIG. 13-38 Multicomponent McCabe-Thiele diagram for butane-pentane splitter in Fig. 13-37.

specified in the base case. At lower distillate flow rates the mole fraction of heavy key (*i*-C<sub>5</sub>) in the distillate is small, and at higher distillate flow rates the mole fraction of light key (*n*-C<sub>4</sub>) in the bottoms is small. Both cannot be small simultaneously. From this result we see that the “best” overall product purities are obtained when the distillate rate is in the vicinity of 45 lb-mol/h. On reflection this should not come as a surprise; the flow rate of the light key (*n*-butane) and all components with a higher volatility is 45 lb-mol/h. However, even with the distillate flow rate set to 45 lb-mol/h there remains room for improvement in the separation.

The other key design specifications here are the total number of stages and the location of the feed stage. In most cases, increasing the number of stages will improve the separation. On increasing the number of stages to 26, with the feed to stage 12, increasing the overhead vapor flow to 195 lb-mol/h, and decreasing the distillate rate to 45 lb-mol/h, we obtain the following products:

| Mole flows, lb-mol/h | Feed   | Top   | Bottom |
|----------------------|--------|-------|--------|
| Propane              | 5.00   | 5.00  | 0.00   |
| Isobutane            | 15.00  | 15.00 | 0.00   |
| N-Butane             | 25.00  | 24.81 | 0.19   |
| Isopentane           | 20.00  | 0.16  | 19.84  |
| N-Pentane            | 35.00  | 0.03  | 34.97  |
| Total                | 100.00 | 45.00 | 55.00  |

The McCabe-Thiele diagram for this configuration, shown in Fig. 13-41, shows that the product purities have improved significantly.

The temperature and liquid phase composition profiles for this final case are shown in Fig. 13-42. The temperature increases from top to bottom of the column. This is normally the case in distillation columns (exceptions may occur with cold feeds or feeds with boiling points significantly lower than that of the mixture on stages above the feed stage). The composition profiles also are as expected. The components more volatile than the light key (*n*-butane) are

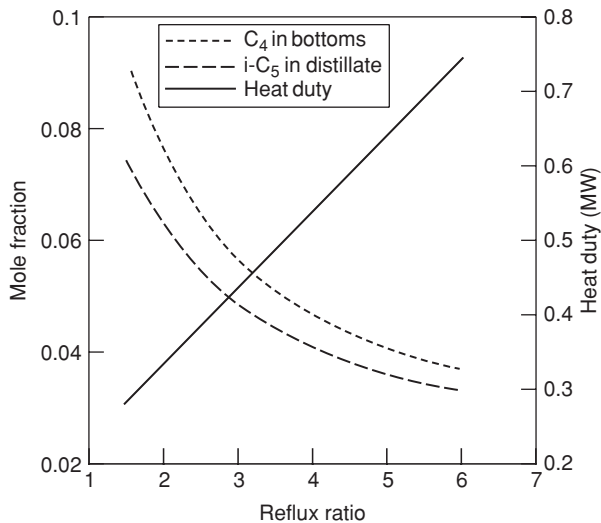


FIG. 13-39 Product mole fractions and reboiler heat duty as a function of the reflux ratio for butane-pentane splitter in Fig. 13-37.

concentrated above the feed; those compounds less volatile than the heavy key (*i*-pentane) are concentrated below the feed. The mole fractions of the two keys exhibit maxima, the light key above the feed stage and the heavy key below the feed stage. The decrease in the mole fraction of light key over the top few stages is necessary to accommodate the increase in the composition of the lighter compounds. Similar arguments pertain to the decrease in the mole fraction of the heavy key over the stages toward the bottom of the column.

Flow profiles are shown in Fig. 13-42c. Note the step change in the liquid flow rate around the feed stage. Had the feed been partially vaporized, we would have observed changes in both vapor and liquid flows around the feed stage, and a saturated vapor feed would significantly change only the vapor flow

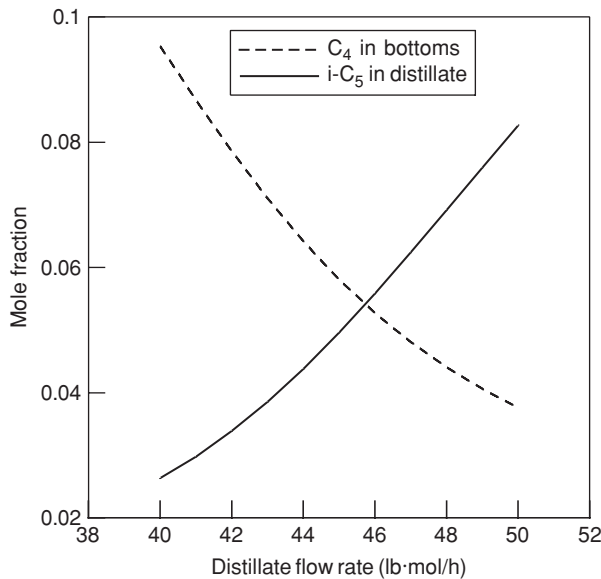


FIG. 13-40 Product mole fraction duty as a function of the distillate rate for butane-pentane splitter in Fig. 13-37.

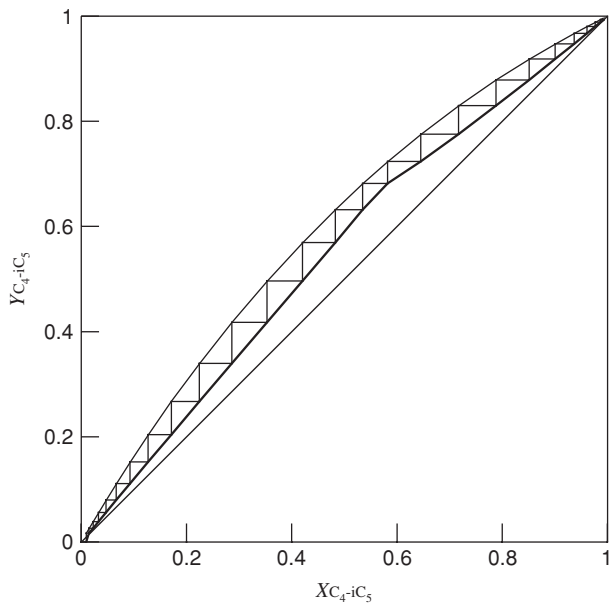


FIG. 13-41 Multicomponent McCabe-Thiele diagram for butane-pentane splitter after optimization to improve product purities.

profile. The slight (in this case) curvature in the flow profiles is due to enthalpy changes.

**Example 4: Light Hydrocarbon Distillation** Compute stage temperatures, interstage vapor and liquid flow rates and compositions, and reboiler and condenser duties for the light hydrocarbon distillation column shown in Fig. 13-43. How might the separation be improved?

This more complicated example features a partial condenser (with vapor product) and a vapor sidestream withdrawn from the 13th stage. The SRK equation of state may be used for estimating the *K* values and enthalpy departures for thermodynamic properties.

The specifications made in this case are summarized below:

| Variable                                                | Number       | Value                     |
|---------------------------------------------------------|--------------|---------------------------|
| Number of stages                                        | 1            | 17                        |
| Feed stage location                                     | 1            | 9                         |
| Component flows in feed                                 | $c = 5$      | 3, 20, 37, 35, 5 lb-mol/h |
| Feed pressure                                           | 1            | 260 psia                  |
| Feed vapor fraction                                     | 1            | 0                         |
| Pressure on each stage including condenser and reboiler | $N = 17$     | $P_j = 250$ psia          |
| Heat duty on each stage except reboilers and condensers | $N - 2 = 15$ | $Q_j = 0$                 |
| Reflux rate (replaces heat duty of reboiler)            | 1            | $L_1 = 150$ lb-mol/h      |
| Distillate flow rate (replaces heat duty of condenser)  | 1            | $D = 23$ lb-mol/h         |
| Sidestream stage                                        | 1            | 13                        |
| Sidestream flow rate and phase                          | 2            | 37 lb-mol/h vapor         |
| Total                                                   | 46           |                           |

As in Example 3, we have assumed that the pressure of the reflux divider is the same as the pressure of the condenser, the heat loss from the reflux divider is zero, and the reflux temperature is the boiling point of the condensed overhead vapor.

The specifications were selected to obtain three products: a vapor distillate rich in  $C_2$  and  $C_3$ , a vapor sidestream rich in  $n$ - $C_4$ , and a bottoms rich in  $n$ - $C_5$  and  $n$ - $C_6$ , as summarized in the table below.

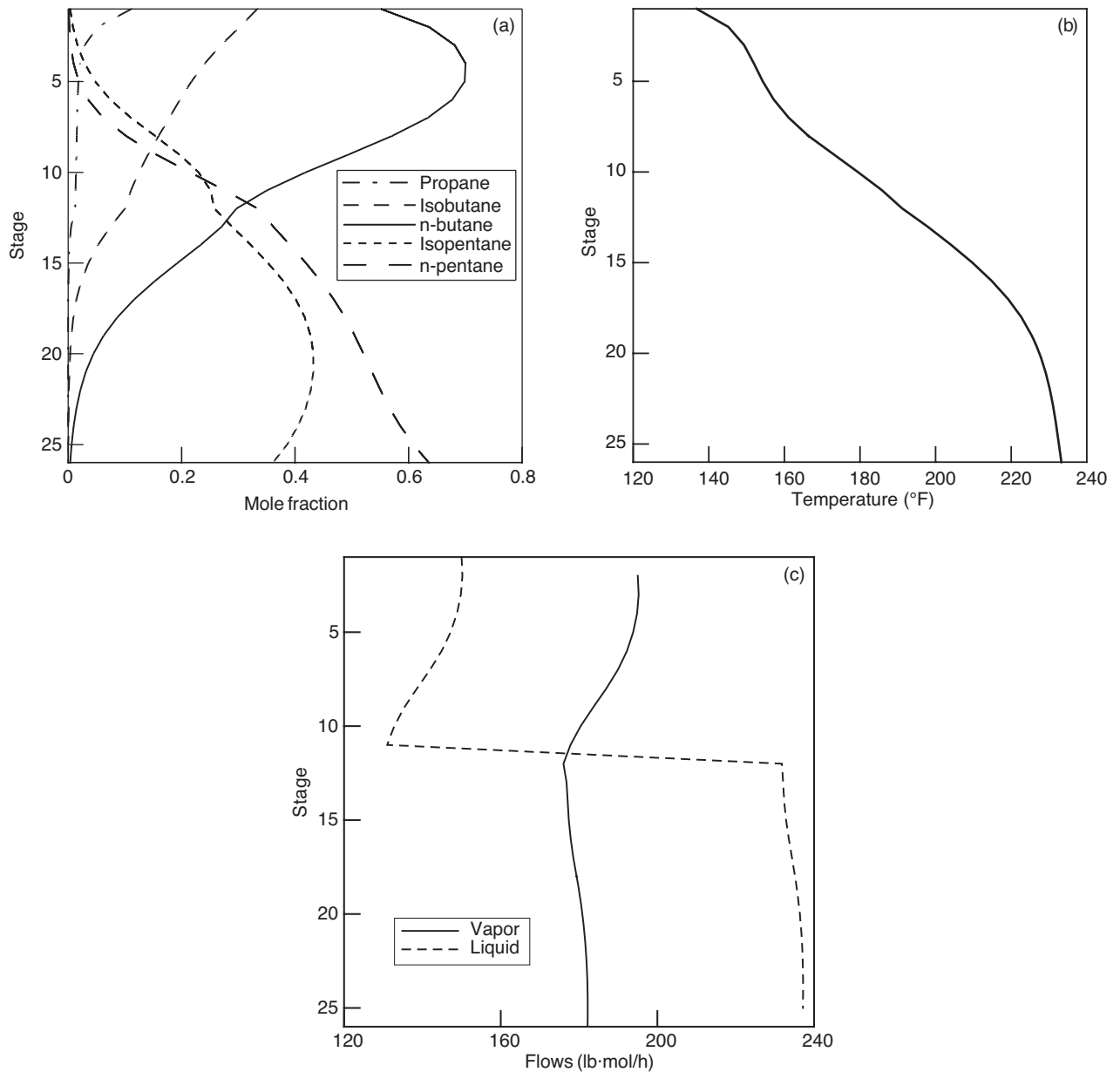


FIG. 13-42 (a) Composition, (b) temperature, and (c) flow profiles in butane-pentane splitter.

| Mole flows, lb-mol/h | Feed   | Top  | Bottom | Sidestream |
|----------------------|--------|------|--------|------------|
| Ethane               | 3.00   | 3.00 | 0.0    | 0.0        |
| Propane              | 20.00  | 18.3 | 0.0    | 1.6        |
| N-Butane             | 37.00  | 1.7  | 9.6    | 25.7       |
| N-Pentane            | 35.00  | 0.0  | 25.8   | 9.2        |
| N-Hexane             | 5.00   | 0.0  | 4.5    | 0.5        |
| Total molar flow     | 100.00 | 23.0 | 40.0   | 37.0       |

Convergence of the simultaneous convergence method was obtained in five iterations.

Further improvement in the purity of the sidestream as well as of the other two products could be obtained by increasing the reflux flow rate (or reflux ratio) and the number of stages in each section of the column. If, e.g., we increase the

number of stages to 25 (including condenser and reboiler in this total), with the feed to stage 7 and the sidestream removed from stage 17, we obtain the following:

| Mole flows, lb-mol/h | Feed  | Top   | Bottom | Sidestream |
|----------------------|-------|-------|--------|------------|
| Ethane               | 3.00  | 3.00  | 0.00   | 0.00       |
| Propane              | 20.00 | 19.46 | 0.00   | 0.54       |
| N-Butane             | 37.00 | 0.54  | 7.46   | 29.00      |
| N-Pentane            | 35.00 | 0.00  | 27.93  | 7.07       |
| N-Hexane             | 5.00  | 0.00  | 4.61   | 0.39       |
| Total                | 100.0 | 23.0  | 40.0   | 37.00      |

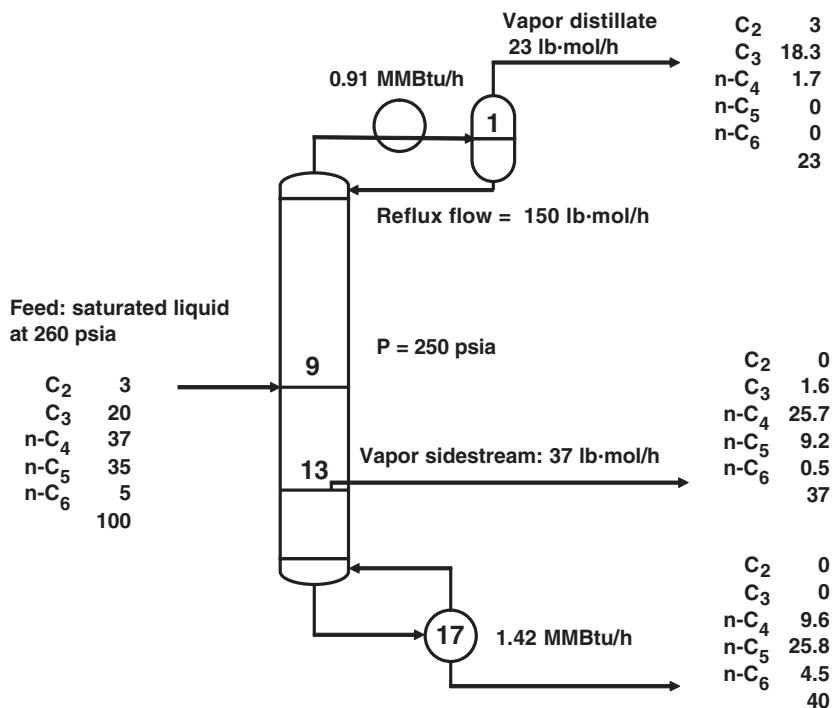


FIG. 13-43 Specifications and calculated product stream flows and heat duties for light hydrocarbon still. Flows are in pound-moles per hour.

The McCabe-Thiele diagram for this design, showing that the feed is to the optimum stage, is shown in Fig. 13-44. The flow profiles are shown in Fig. 13-45; note the step changes due to both the feed and the sidestream. As was the case in Example 3, the curvature in the flow profiles is due to enthalpy changes.

**Example 5: Absorber** Compute stage temperatures and interstage vapor and liquid flow rates and compositions for the absorber specifications shown in Fig. 13-46. Note that a second absorber oil feed is used in addition to the main absorber oil and that heat is withdrawn from the seventh theoretical stage. The oil may be taken to be *n*-dodecane.

The specifications made in this case are summarized below:

| Variable                  | Number  | Value(s)                                                 |
|---------------------------|---------|----------------------------------------------------------|
| Number of stages          | 1       | 8                                                        |
| Location of feed 1        | 1       | 1                                                        |
| Component flows in feed 1 | $c = 6$ | 0, 0, 0, 0, 250 lb-mol/h                                 |
| Pressure of feed 1        | 1       | 400 psia                                                 |
| Temperature of feed 1     | 1       | 80°F                                                     |
| Location of feed 2        | 1       | 4                                                        |
| Component flows in feed 2 | $c = 6$ | 13, 3, 4, 5, 5, 135 lb-mol/h                             |
| Pressure of feed 2        | 1       | 400 psia                                                 |
| Temperature of feed 2     | 1       | 80°F                                                     |
| Location of feed 3        | 1       | 8                                                        |
| Component flows in feed 3 | $c = 6$ | 360, 40, 25, 15, 15, 10 lb-mol/h                         |
| Pressure of feed 3        | 1       | 400 psia                                                 |
| Temperature of feed 3     | 1       | 80°F                                                     |
| Pressure on each stage    | $N = 8$ | $P_j = 400$ psia; $j = 1, 2, \dots, 8$                   |
| Heat duty on each stage   | $N = 8$ | $Q_j = 0, j = 1, 2, \dots, 6, 8$ ; $Q_7 = 150,000$ Btu/h |
| Total                     | 44      |                                                          |

The simultaneous solution method solved this example in four iterations. The Peng-Robinson equation of state was used to estimate *K* values and enthalpy

departures and by Seader (*Perry's Chemical Engineers' Handbook*, 7th ed.) who solved this problem by using the SR method.

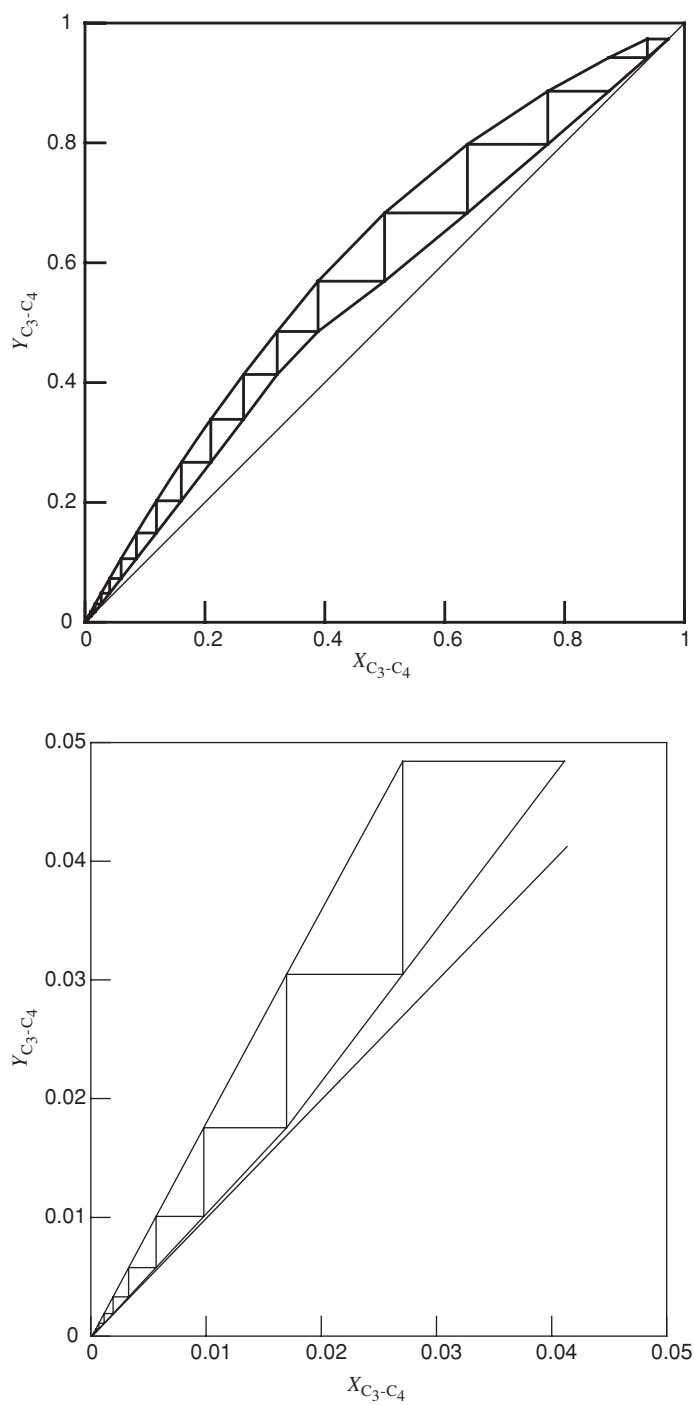
The computed product flows are summarized below:

| Mole flows, lb-mol/h | Lean oil | Secondary oil | Rich gas | Lean gas | Rich oil |
|----------------------|----------|---------------|----------|----------|----------|
| Methane              | 0.0      | 13.0          | 360.0    | 303.7    | 69.3     |
| Ethane               | 0.0      | 3.0           | 40.0     | 10.7     | 32.3     |
| Propane              | 0.0      | 4.0           | 25.0     | 0.2      | 28.8     |
| N-Butane             | 0.0      | 4.0           | 15.0     | 0.0      | 19.0     |
| N-Pentane            | 0.0      | 5.0           | 10.0     | 0.0      | 15.0     |
| N-Dodecane           | 250.0    | 135.0         | 0.0      | 0.0      | 385.0    |
| Total molar flow     | 250.0    | 164.0         | 450.0    | 314.6    | 549.4    |

The energy withdrawn from stage 7 has the effect of slightly increasing the absorption of the more volatile species in the rich oil leaving the bottom of the column. Temperature and flow profiles are shown in Fig. 13-47. The temperature profile shows the rise in temperature toward the bottom of the column that is typical of gas absorption processes. The bottom of the column is where the bulk of the absorption takes place, and the temperature rise is a measure of the heat of absorption. The liquid flow profile exhibits a step change due to the secondary oil feed at the midpoint of the column.

**Example 6: Reboiled Stripper** Compute stage temperatures and interstage vapor and liquid flow rates and compositions and reboiler heat duty for the reboiled stripper shown in Fig. 13-48. Thermodynamic properties may be estimated by using the Grayson-Streed modification of the Chao-Seader method.

The specifications made in this case are summarized in Fig. 13-48 and in the table below. The specified bottoms rate is equivalent to removing most of the *n*-C<sub>5</sub> and *n*-C<sub>6</sub> in the bottoms.



**FIG. 13-44** Multicomponent McCabe-Thiele diagram for the hydrocarbon distillation in Fig. 13-43.

13-40 DISTILLATION

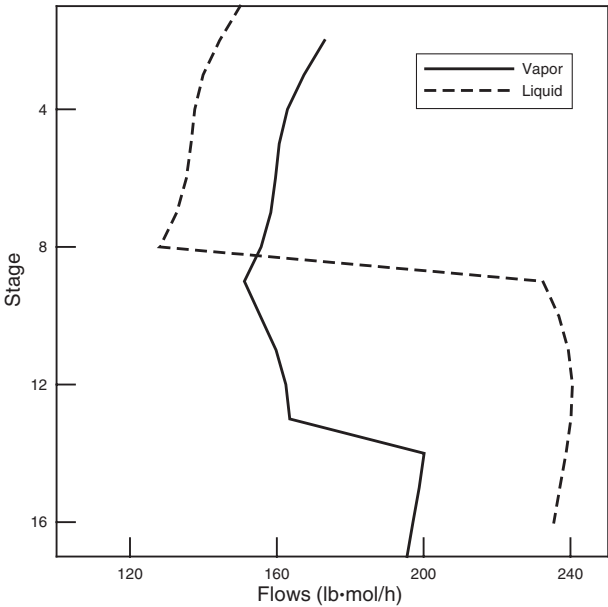


FIG. 13-45 Flow profiles in hydrocarbon distillation in Fig. 13-43.

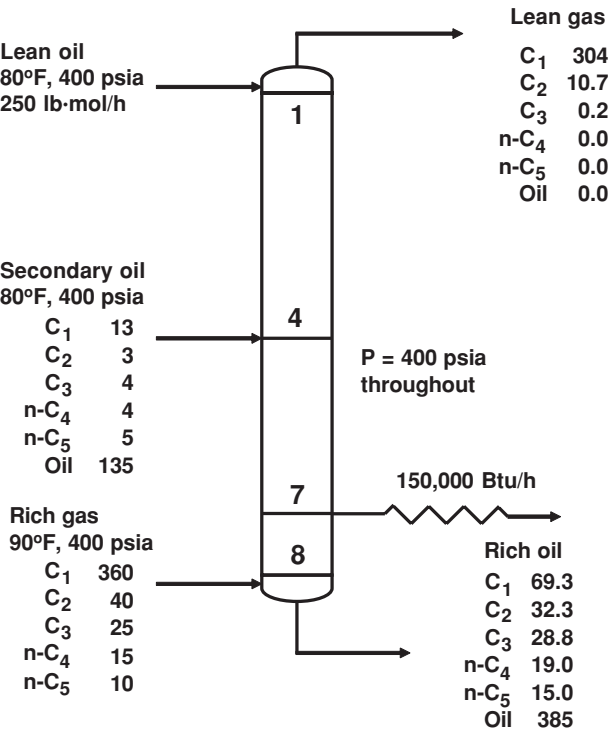
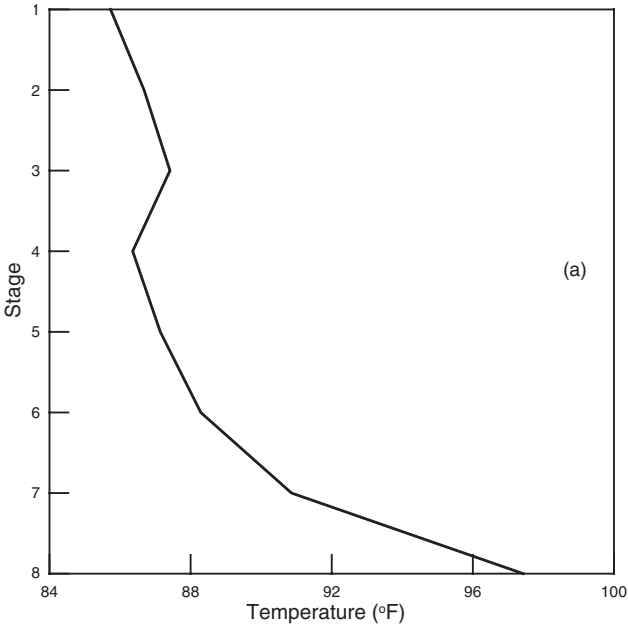


FIG. 13-46 Specifications and calculated product stream flows and heat duties for absorber. Flows are in pound-moles per hour.

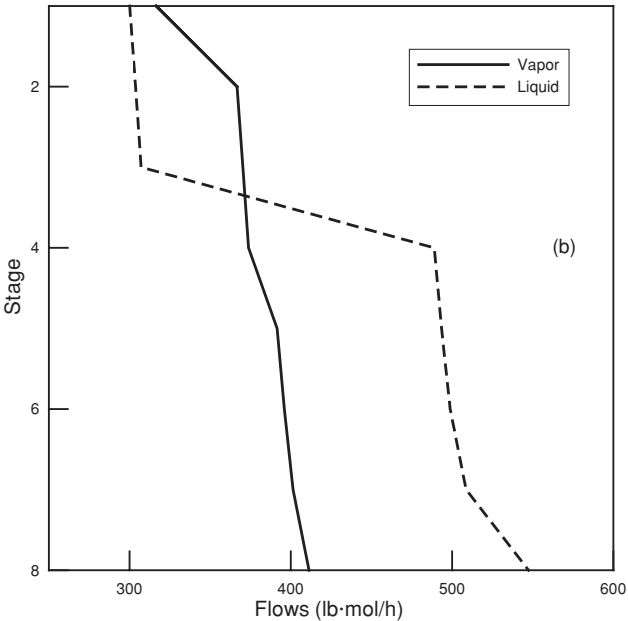


FIG. 13-47 (a) Temperature and (b) flow profiles in absorber in Fig. 13-46.

| Variable                                  | Number      | Value(s)                                                |
|-------------------------------------------|-------------|---------------------------------------------------------|
| Number of stages                          | 1           | 8                                                       |
| Component flows in feed                   | $c = 7$     | 0.22, 59.51, 73.57, 153.2, 173.2, 58.22, 33.63 lb-mol/h |
| Pressure of feed                          | 1           | 150 psia                                                |
| Temperature of feed                       | 1           | 40.8°F                                                  |
| Pressure on each stage                    | $N = 8$     | $P_j = 150 \text{ psia}; j = 1, \dots, 8$               |
| Heat duty on each stage (except reboiler) | $N - 1 = 7$ | $Q_j = 0, j = 1, \dots, 7$                              |
| Bottom product flow rate                  | 1           | 99.33 lb-mol/h                                          |
| Total                                     | 26          |                                                         |

Convergence in this case was obtained after four iterations. The computed product flows and reboiler duty are shown alongside the specifications in Fig. 13-48 and in the table below.

| Mole flows, lb-mol/h | Feed   | Overhead | Bottoms |
|----------------------|--------|----------|---------|
| Nitrogen             | 0.22   | 0.02     | 0.20    |
| Methane              | 59.51  | 59.51    | 0.00    |
| Ethane               | 73.57  | 73.57    | 0.00    |
| Propane              | 153.22 | 153.16   | 0.06    |
| <i>N</i> -Butane     | 173.22 | 150.09   | 23.13   |
| <i>N</i> -Pentane    | 58.22  | 13.23    | 44.99   |
| <i>N</i> -Hexane     | 33.63  | 2.69     | 30.94   |
| Total                | 551.59 | 452.26   | 99.33   |

Computed temperature, flow rates, and vapor-phase mole fraction profiles, shown in Fig. 13-49, are not of the shapes that might have been expected. Vapor and liquid flow rates for *n*-C<sub>4</sub> change dramatically from stage to stage.

**Example 7: An Industrial *i*-Butane/*n*-Butane Fractionator**  
Klemola and Ilme [*Ind. Eng. Chem.*, **35**, 4579 (1996)] and Ilme (Ph.D. thesis, University of Lappeenranta, Finland, 1997) report data from an industrial *i*-butane/*n*-butane fractionator that is used here as the basis for this example.

The column has 74 valve trays, the design details of which can be found in Example 11. The feed was introduced onto tray 37.

To properly model an existing column, it is necessary to know all feed and product conditions (flow rate, temperature, pressure, and composition). Flows, temperatures, and pressures often are available from standard instrumentation. It may be necessary to obtain additional samples of these streams to determine

their composition. Such sampling should be scheduled as part of a plant trial to ensure that measured data are consistent. Ideally, multiple sets of plant measurements should be obtained at different operating conditions, and care should be taken to ensure that operating data are obtained at steady state since a steady-state model can only be used to describe a column at steady state. Measurements should be taken over a time interval longer than the residence time in the column and time-averaged to avoid a possible mismatch between feed and product data. Condenser and reboiler heat duties should be known (or available from the appropriate energy balance) whenever possible.

The measured compositions and flow rates of the feed and products for the C<sub>4</sub> splitter are summarized in the table below.

Measured Feed and Product Flows and Compositions  
(Mass %) for *i*-Butane/*n*-Butane Fractionator (Ilme op. cit.)

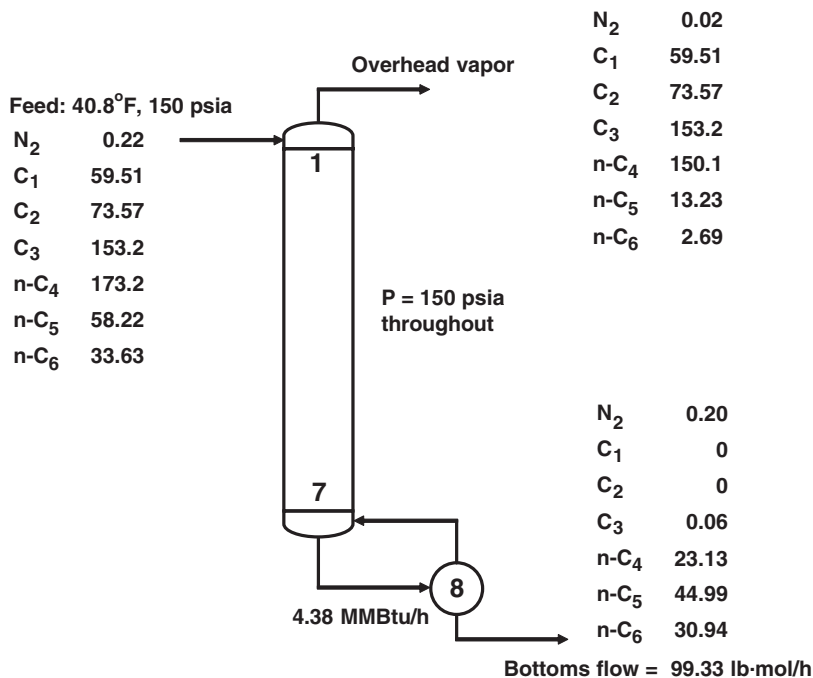
| Species                | Feed   | Top  | Bottom |
|------------------------|--------|------|--------|
| Propane                | 1.50   | 5.30 | 0.00   |
| Isobutane              | 29.4   | 93.5 | 0.30   |
| <i>n</i> -Butane       | 67.7   | 0.20 | 98.1   |
| C <sub>4</sub> olefins | 0.50   | 1.00 | 0.20   |
| Neopentane             | 0.10   | 0.00 | 0.20   |
| Isopentane             | 0.80   | 0.00 | 1.10   |
| <i>n</i> -Pentane      | 0.10   | 0.00 | 0.10   |
| Total flow, kg/h       | 26,234 | 8011 | 17,887 |

Other measured parameters are as follows:

Other Details of the *i*-Butane/*n*-Butane Fractionator

|                             |        |
|-----------------------------|--------|
| Reflux flow rate, kg/h      | 92,838 |
| Reflux temperature, °C      | 18.5   |
| Column top pressure, kPa    | 658.6  |
| Pressure drop per tray, kPa | 0.47   |
| Feed pressure, kPa          | 892.67 |
| Boiler duty, MW             | 10.24  |

Rarely, and this is a case in point, are plant data in exact material balance, and it will be necessary to reconcile errors in such measurements before continuing. The feed and product compositions, as adjusted by Ilme so that they satisfy material balance constraints, are provided below. Note how the C<sub>4</sub> olefins are assigned to isobutene and 1-butene.



**FIG. 13-48** Specifications and calculated product stream flows and reboiler heat duty for a reboiled stripper. Flows are in pound-moles per hour.



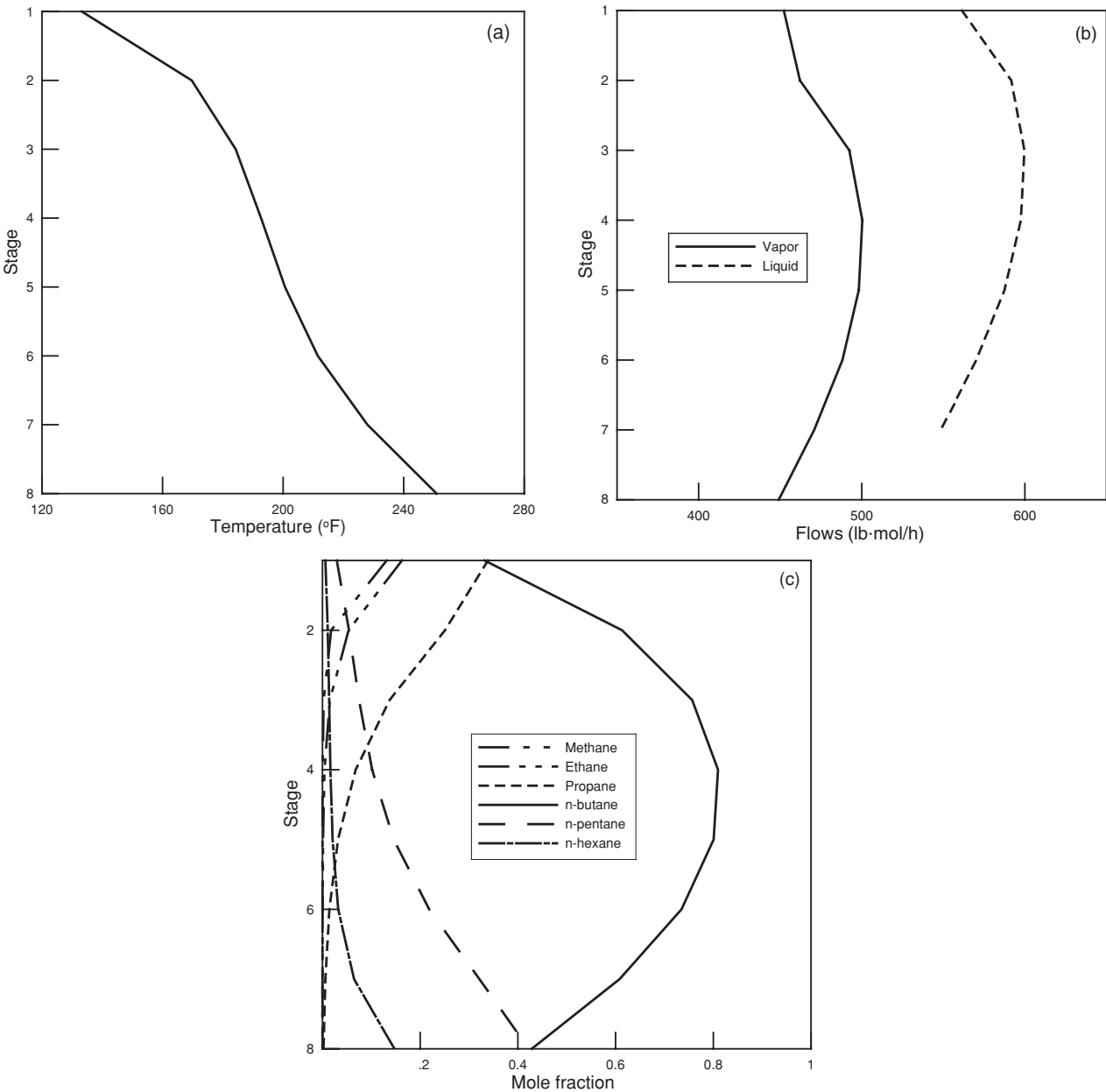


FIG. 13-49 (a) Temperature, (b) flow, and (c) vapor mole fraction profiles in reboiled stripper in Fig. 13-48.

Adjusted Feed and Product Compositions (Mass %) and Flows for *i*-Butane/  
*n*-Butane Fractionator (Ilme op. cit.)

| Species           | Feed   | Top  | Bottom |
|-------------------|--------|------|--------|
| Propane           | 1.54   | 4.94 | 0.00   |
| Isobutane         | 29.5   | 94.2 | 0.3    |
| <i>n</i> -Butane  | 67.7   | 0.20 | 98.1   |
| Isobutene         | 0.13   | 0.23 | 0.08   |
| 1-Butene          | 0.20   | 0.41 | 0.10   |
| Neopentane        | 0.11   | 0.00 | 0.17   |
| Isopentane        | 0.77   | 0.00 | 1.12   |
| <i>n</i> -Pentane | 0.08   | 0.00 | 0.11   |
| Total flow, kg/h  | 26,122 | 8123 | 17,999 |

To proceed with building a model of this column, we specify the number of stages equal to the number of trays plus condenser and reboiler ( $N = 76$ ). The common arrangement of locating the actual feed between stages may require modeling as two separate feeds, the liquid portion to the stage below and the vapor portion to the stage above. In this particular illustration the feed is (assumed to be) saturated liquid, and we provide just a single feed to stage 38.

Upon computing the bubble point of the overhead product, we find that the measured reflux temperature is well below the estimated boiling point. Thus, we choose the subcooled condenser model. The steady-state concept of the “subcooled” condenser often does not exist in practice. Instead, the condenser is in vapor-liquid equilibrium with the vapor augmented by a blanket of noncondensable gas (that has the effect of lowering the dew point of the overhead vapor). The subcooled condenser is a convenient work-around for steady-state models (as is needed here), but not for dynamic models. We assume a partial reboiler.

The pressure of the top stage is specified (at 658.6 kPa). The pressures of all trays below the top tray can then be fixed from the knowledge of the per-tray pressure drop (0.47 kPa). The pressure of the condenser is not known here. Thus, in the absence of further information, we make the condenser pressure equal to the top tray pressure (knowing that in practice it will be lower).

It is advisable to use the plant set points in building a model. For example, it is possible that a column simulation might involve the specification of the reflux ratio and bottoms flow rate because such specifications are (relatively) easy to converge. It is quite likely that the column may be controlled by using the temperature at some specific location (e.g., the temperature of tray 48). This specification should be used in building the model. In this case we do not know the set points used for controlling the column, but we do have sufficient information to allow us to compute the reflux ratio from plant flow data ( $R = 11.6$ ). Finally, the bottom product flow is specified as equal to the adjusted value reported above (17,999 kg/h).

The specifications made to model this column are summarized below:

| Variable                                                | Number       | Value                     |
|---------------------------------------------------------|--------------|---------------------------|
| Number of stages                                        | 1            | $N = 76$                  |
| Feed stage location                                     | 1            | 39                        |
| Component flows in feed                                 | $c = 8$      | See previous table        |
| Feed pressure                                           | 1            | 120 psia                  |
| Feed vapor fraction                                     | 1            | 0                         |
| Pressure at the top of the column                       | 1            | 658.6 kPa                 |
| Pressure drop per stage                                 | $N - 1 = 75$ | 0.47 kPa                  |
| Heat duty on each stage except reboilers and condensers | $N - 2 = 74$ | $Q_j = 0$                 |
| Reflux ratio (replaces heat duty of condenser)          | 1            | $R = 11.588$              |
| Bottoms flow rate (replaces heat duty of reboiler)      | 1            | $B = 17,999 \text{ kg/h}$ |
| Temperature of reflux                                   | 1            | 291.65 K                  |
| Total                                                   | 165          |                           |

Finally, we must select appropriate methods of estimating thermodynamic properties. Ilme (op. cit.) used the SRK equation of state to model this column, whereas Klemola and Ilme (op. cit.) had earlier used the UNIFAC model for liquid-phase activity coefficients, the Antoine equation for vapor pressures, and the SRK equation for vapor-phase fugacities only. For this exercise we used the Peng-Robinson equation of state. Computed product compositions and flow rates are shown in the table below.

Specified Feed and Computed Product Compositions (Mass %) and Flows for *i*-Butane/*n*-Butane Fractionator (Ilme op. cit.)

| Compound          | Feed   | Top     | Bottom |
|-------------------|--------|---------|--------|
| Propane           | 1.54   | 4.95    | 0.00   |
| Isobutane         | 29.49  | 93.67   | 0.53   |
| <i>n</i> -Butane  | 67.68  | 0.73    | 97.89  |
| Isobutene         | 0.13   | 0.29    | 0.06   |
| 1-Butene          | 0.20   | 0.36    | 0.13   |
| Neopentane        | 0.11   | 0.00    | 0.16   |
| Isopentane        | 0.77   | 0.00    | 1.12   |
| <i>n</i> -Pentane | 0.08   | 0.00    | 0.12   |
| Total flow, kg/h  | 26,122 | 8123.01 | 17,999 |

The agreement with the adjusted material balance (tabulated previously) appears to be quite good, and to a first approximation it seems that we have a good model of the column.

Note that although this column is distilling a mixture containing at least eight identifiable compounds, only two are present in significant amounts, and therefore this is essentially a binary separation. It is usually relatively straightforward to match product compositions in processes involving only two different species simply by adjusting the number of equilibrium stages. We return to this point later.

**Efficiencies** In actual operation the trays of a distillation column rarely, if ever, operate at equilibrium despite attempts to approach this condition by proper design and choice of operating conditions. The usual way of dealing with departures from equilibrium in multistage towers is through the use of stage and/or overall efficiencies.

The overall column efficiency is defined by

$$E_o = \frac{N_{EQ}}{N_{\text{actual}}} \quad (13-57)$$

where  $N_{EQ}$  is the number of equilibrium stages.

There are many different definitions of stage (or tray) efficiency, with that of Murphree [*Ind. Eng. Chem.*, **17**, 747–750, 960–964 (1925)] being by far the most widely used in separation process calculations:

$$E_i^{MV} = \frac{y_{i,j} - y_{i,j+1}}{y_{i,j}^* - y_{i,j+1}} \quad (13-58)$$

Here  $E_i^{MV}$  is the Murphree vapor efficiency for component  $i$  on stage  $j$ , and  $y_{i,j}^*$  is the composition of the vapor in equilibrium with the liquid. Other types of efficiency include that of Hausen [*Chemie Ingr. Tech.*, **25**, 595 (1953)], vaporization (see Holland, op. cit.), and generalized Hausen [Standart, *Chem. Eng. Sci.*, **20**, 611 (1965)]. There is by no means a consensus on which is best. Arguments for and against various types are presented by, among others, Standart [op. cit.; *Chem. Eng. Sci.*, **26**, 985 (1971)], Holland and McMahon [*Chem. Eng. Sci.*, **25**, 431 (1972)], and Medina et al. [*Chem. Eng. Sci.*, **33**, 331 (1978), **34**, 1105 (1979)]. Possibly the most soundly based definition, the generalized Hausen efficiency of Standart (op. cit.), is never used in industrial practice. Seader [*Chem. Eng. Progress*, **85**(10), 41 (1989)] summarizes the shortcomings of efficiencies.

The Murphree (and Hausen) efficiencies of both components in a binary mixture are equal; although they cannot be less than 0, they may be greater than 1. A table of typical values of Murphree tray efficiency can be found in Sec. 14. Also described in Sec. 14 are methods for estimating Murphree efficiencies when they are not known.

For multicomponent systems (i.e., those with more than two components) there are  $c - 1$  independent component efficiencies, and there are sound theoretical reasons as well as experimental evidence for not assuming the individual component efficiencies to be alike; indeed, they may take values between plus and minus infinity. Component efficiencies are more likely to differ for strongly nonideal mixtures. While models exist for estimating efficiencies in multicomponent systems [see chapter 13 in Taylor and Krishna, (op. cit.) for a review of the literature], they are not widely used and have not (yet) been included in any of the more widely used commercial simulation programs.

The fact that component efficiencies in multicomponent systems are unbounded means that the arithmetic average of the component Murphree efficiencies is useless as a measure of the performance of a multicomponent distillation process. Taylor, Baur, and Krishna [*AIChE J.*, **50**, 3134 (2004)] proposed the following efficiency for multicomponent systems:

$$\epsilon_j = \frac{\sqrt{\sum_{i=1}^c (\Delta y_{i,j})^2}}{\sqrt{\sum_{i=1}^c (\Delta y_{i,j}^*)^2}} \quad (13-59)$$

This efficiency has a simple and appealing physical significance; it is the ratio of the length of the actual composition profile (in mole fraction space) to the length of the composition profile given by the equilibrium-stage model. The Taylor-Baur-Krishna (TBK) efficiency has just one value per stage regardless of the number of components in the mixture; it can never be negative. For binary systems in tray columns the TBK average efficiency simplifies to the Murphree efficiency [Taylor, Baur, and Krishna (op. cit.)].

Murphree efficiencies are easily incorporated within simultaneous convergence algorithms (something that is not always easy, or even possible, with some tearing methods). (As an aside, note that vaporization

efficiencies are very easily incorporated in all computer algorithms, a fact that has helped to prolong the use of these quantities in industrial practice despite the lack of any convenient way to relate them to the fundamental processes of heat and mass transfer. Unfortunately, it is not at all easy to include the more fundamentally sound TBK efficiencies in a computer method for equilibrium-stage simulations.)

The Murphree-stage efficiency also makes a good continuation parameter for cases that are hard to converge. For vanishingly small stage efficiencies the column performs no separation, and the streams leaving the stage have essentially the same flow rates, composition, and temperature as the combined feeds to the stage. This fact can be exploited in a simple continuation method for solving difficult equilibrium-stage separation process problems [Muller, Ph.D. Thesis in Chem. Engng., ETH Zurich, 1979; Sereno, Ph.D. Thesis in Chem. Engng., University of Porto, 1985; Vickery, Ferrari, and Taylor, *Comput. Chem. Engng.*, **12**, 99 (1988)]. These methods are very effective at solving difficult problems involving *standard specifications* (in which the reflux ratio and bottoms flow rate are specified); however, they cannot easily handle problems involving *nonstandard specifications* (e.g., when a product stream purity is specified).

Efficiencies are often used to fit actual operating data, along with the number of equilibrium stages in each section of the column (between feed and product takeoff points). The maximum number of these efficiencies is the number of independent efficiencies per stage ( $c - 1$ ) times the number of stages—potentially a very large number, indeed. This many adjustable parameters may lead to a model that fits the data very well, but has no predictive ability (i.e., cannot describe how the column will behave when something changes). At the other extreme, the overall efficiency defined by Eq. (13-57) is just a single parameter that can improve the robustness of the model and speed of convergence, but it may be difficult to match actual temperature and/or composition profiles since there is unlikely to be a one-to-one correspondence between the model stages and actual trays. A compromise often used in practice is to use just one value for all components and all stages in a single section of a column. Efficiencies should not be used to model condensers and reboilers; it is usually safe to assume that they are equilibrium devices. It is also unwise to employ Murphree efficiencies for trays with a vapor product since any Murphree efficiency less than 1 will necessarily lead to the prediction of a subcooled vapor.

**Example 8: The Industrial *i*-Butane/*n*-Butane Fractionator (Again)** With the material on efficiencies in mind, we return to the model of the  $C_4$  splitter that we developed in Example 7.

It is possible to estimate the overall efficiency for a column such as this one simply by adjusting the number of equilibrium stages in each section of the column that is needed to match the mass fractions of *i*-butane in the distillate and *n*-butane in the bottoms. Using the SRK equation of state for estimating thermodynamic properties, Ilme (op. cit.) found that 82 equilibrium stages (plus condenser and reboiler) and the feed to stage 38 were required. This corresponds to an overall column efficiency of  $82/74 = 111$  percent. Klemola and Ilme (op. cit.) used the UNIFAC model for liquid-phase activity coefficients, the Antoine equation for vapor pressures, and the SRK equation for vapor-phase fugacities only and found that 88 ideal stages were needed; this corresponded to an overall efficiency of 119 percent. With the Peng-Robinson equation of state for the estimation of thermodynamic properties, we find that 84 stages are needed (while maintaining the feed to the center stage as is the case here); the overall column efficiency for this model is 114 percent. The differences between these efficiencies are not large in this case, but the important point here is that efficiencies—all types—depend on the choice of model used to estimate the thermodynamic properties. Caution must therefore be exercised when one is using efficiencies determined in this way to predict column performance.

As an alternative to varying the number of stages, we may prefer to maintain a one-to-one correspondence between the number of stages and the number of actual trays, 74 in this case (plus condenser and reboiler), with the feed to tray 38. Using the Peng-Robinson equation of state and a Murphree stage efficiency of 116 percent, we find the product mass fractions that are in excellent agreement with the plant data. The McCabe-Thiele (Hengstebeck) diagram for this case, assembled from the results of the simulation, is shown in Fig. 13-50. Composition profiles computed from this model are shown in Fig. 13-51. Note that the mole fractions are shown on a logarithmic axis so that all the composition profiles can easily be seen.

It must be remembered that this is essentially a binary separation and that it is usually relatively straightforward to match product compositions in processes involving only two different species. In other cases involving a greater number

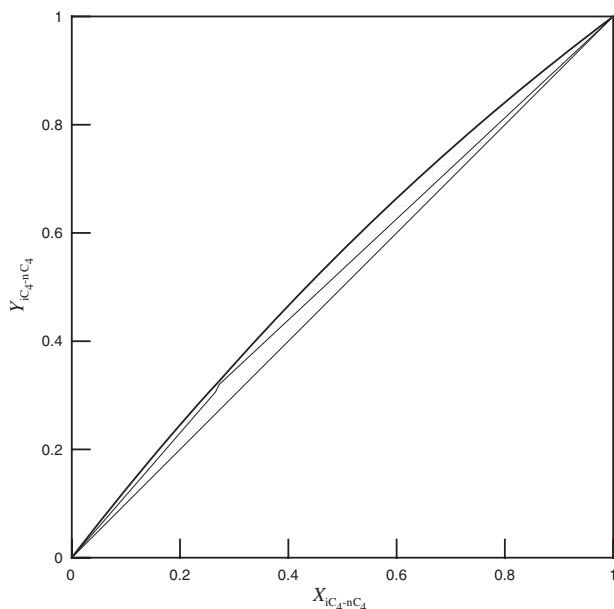


FIG. 13-50 McCabe-Thiele diagram for  $C_4$  splitter.

of species with significant concentrations, it will likely be necessary to vary both the number of stages and the component efficiencies to match plant data. We do not recommend adjusting thermodynamic model parameters to fit plant data since this can have unfortunate consequences on the prediction of product distributions, process temperatures, and/or pressures.

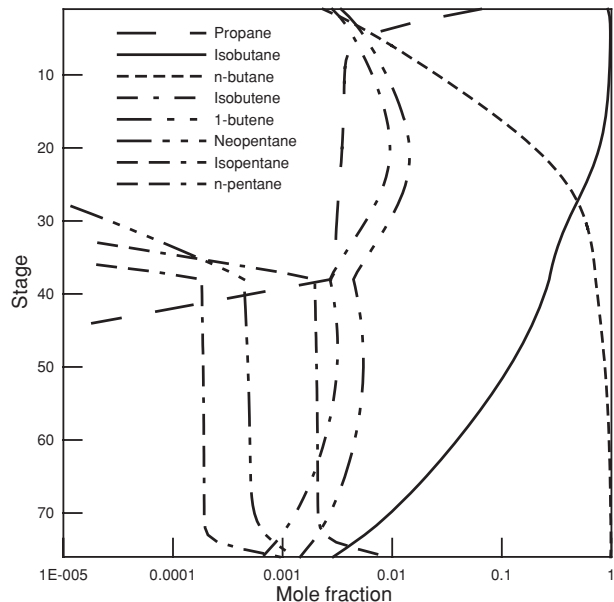


FIG. 13-51 Liquid-phase mole fraction profiles for *i*-butane/*n*-butane fractionator.

The performance of a packed column often is expressed in terms of the height equivalent to a theoretical plate (HETP) for packed columns. The HETP is related to the height of packing  $H$  by

$$\text{HETP} = \frac{H}{N_{\text{EQ}}} \quad (13-60)$$

In this case  $N_{EQ}$  is the number of equilibrium stages (theoretical plates) needed to accomplish the separation that is possible in a real packed column of height  $H$ .

**Example 9: HETP of a Packed Absorber** McDaniel, Bassiyoni, and Holland [*Chem. Engng. Sci.*, **25**, 633 (1970)] presented the results of field tests on a packed absorber in a gas plant. The packed section was 23 ft in height, and the column was 3 ft in diameter and filled with 2-in metallic Pall rings. The measured feeds are summarized in the table below.

| Stream               | Lean oil | Rich gas |
|----------------------|----------|----------|
| Pressure, psia       | 807      | 807      |
| Temperature, °F      | 2.9      | 0        |
| Mole flows, lb-mol/h |          |          |
| Carbon dioxide       | 0.0      | 14.1     |
| Nitrogen             | 0.0      | 5.5      |
| Methane              | 0.0      | 2655.8   |
| Ethane               | 0.0      | 199.9    |
| Propane              | 0.0      | 83.2     |
| Isobutane            | 0.0      | 19.1     |
| <i>n</i> -Butane     | 0.0      | 10.9     |
| Isopentane           | 0.2      | 3.5      |
| <i>n</i> -Pentane    | 0.2      | 1.5      |
| <i>n</i> -Hexane     | 4.5      | 0.4      |
| <i>n</i> -Heptane    | 17.2     | 0.2      |
| <i>n</i> -Octane     | 54.5     | 0.1      |
| <i>n</i> -Nonane     | 50.5     | 0.0      |
| <i>n</i> -Decane     | 61.7     | 0.0      |
| Total molar flow     | 188.9    | 2994.2   |

Determine the number of equilibrium stages needed to match the lean gas product flow for this column (2721.1 lb-mol/h, 93.6% methane).

As a first step, we choose an appropriate thermodynamic model that will be used to estimate  $K$  values and enthalpies. Either the Chao-Seader method or the Peng-Robinson equation of state could be considered for this system. It turns out not to be possible to match the plant data with the Chao-Seader method since even one equilibrium stage overpredicts the separation by a very significant amount. It is not even possible to match the exit flows by using the Chao-Seader model combined with a stage efficiency as low as 0.00001. With the Peng-Robinson equation of state, however, it is possible to obtain reasonable agreement with the measured overall product flows by using precisely one equilibrium stage! This suggests that the HETP for this column is 23 ft, a value much higher than any HETP ever published by a packing vendor. In this case the reason for the unrealistic estimate of the HETP has nothing to do with the packing; it is the extreme sensitivity of the simulation to the thermodynamic model, emphasizing the need for caution when one is using efficiencies and HETPs to model some absorption (and distillation) processes.

**Using a Simulator to Solve Distillation Problems** Computer-based methods for solving distillation (and related) column simulation problems now are reasonably reliable. Nevertheless, at times such methods fail to converge. The principal cause of convergence failures is generally a poor initial estimate of the variables being computed. Below we discuss some of the reasons that a simulation might be difficult to converge, along with suggestions on what might make the problem more amenable to solution. The key idea is to modify the problem that is difficult to solve as posed into one that is easy to solve—essentially to provide an improved initial estimate that is more likely to lead to convergence. Note that most simulators allow a calculation to be restarted from an older converged solution. The solution to the “easy” problem may then be used as a starting point for the more difficult problem whose solution is desired. By doing this we are employing a form of continuation (albeit executed manually, at least in part) to solve those problems with which the algorithm at hand may have trouble.

As a rule, the degree of difficulty increases with increasing nonideality. Simultaneous convergence methods are often recommended for simulating strongly nonideal systems (as opposed to tearing or inside-out methods), but even SC methods can experience difficulties with strongly nonideal systems. A possible remedy is to make the system “less” nonideal. It is likely that an activity coefficient model is part of the model used to describe the thermodynamics of these systems, and the source of the convergence difficulties often encountered with such systems. First solving an equivalent

ideal system that omits entirely the activity coefficient model (i.e., using Raoult's law) may provide a converged solution that may be an adequate starting point for the nonideal system of interest. However, since many simulators use ideal solution thermodynamic models in any self-initialization method, this technique may not be of sufficient help.

A measure of the nonideality of the system is given by the magnitude of the interaction parameters for the activity coefficient model. It is possible, therefore, to lessen the degree of nonideality by reducing the interaction parameters sufficiently to make the problem easy to solve. The parameters may then be increased in size in a series of steps until the desired values are reached, each time using the solution converged by using the previous set of parameter values as the starting point. It is essential that the parameters return to their correct values in the final step because intermediate solutions have no meaning, serving merely as an aid to convergence. Using the stage efficiency as a continuation parameter also is useful for such cases, provided that the simulation employs standard specifications (more on this topic below).

The most strongly nonideal systems are those that may exhibit two liquid phases. We have avoided detailed discussion of such systems in this section because special algorithms are needed for these cases; see, however, the section on azeotropic distillation and, for example, chapter 8 of Doherty and Malone (op. cit.) for entry points to the literature.

Large heat effects can lead to convergence difficulties. For such systems it is the enthalpies that are the source of the nonlinearity that leads to convergence failures. It is generally not straightforward to modify enthalpies in a simulator because no adjustable parameters exert their influence over the enthalpy in a way comparable to that of the interaction parameters in the activity coefficient model. Use of a constant-enthalpy model in distillation calculations, if available, will lead to constant molar flows from stage to stage (within each separate section of the column), a condition often approached in many real distillation (but not absorption) columns. Thus, if the simulator includes a constant-enthalpy model, then this can be used to obtain a converged solution that may provide a good starting point for the problem with a more realistic enthalpy model.

High pressure adds to the difficulties of converging simulation models. It is likely that an equation of state will be used to estimate fugacity coefficients and enthalpy departures in such systems. Mixtures become increasingly nonideal as the pressure is raised. In some cases the column may operate close to the critical point at which the densities of both phases approach each other. In other cases the iterations may take the estimates of temperature and composition into regions where the equation of state can provide only one mathematically real root for density or compressibility. Occurrences of this behavior often are a source of convergence difficulties. For such systems we suggest reducing the pressure until the problem becomes easy to solve. A converged solution obtained in this way may be used as the starting point for subsequent calculations at increasingly higher pressures (up to that desired).

Very large numbers of stages can pose their own kind of convergence difficulty. A possible remedy is to reduce the number of stages until a converged solution can be obtained. This solution can then be used as the starting point for a problem with more stages. Interpolation will have to be used to estimate values of the flows, temperatures, and mole fractions for any added stages, something not available in all programs.

Nonstandard specifications are very likely to be the source of convergence difficulties. It is all too easy to specify a desired product purity or component flow rate that simply cannot be attained with the specified column configuration. There is always (at least) one solution if the reflux ratio and bottoms flow rate are specified (the so-called standard specifications), which is likely to converge easily. Other specifications that can cause difficulties for similar reasons include specifying temperatures and compositions anywhere in the column and specifying condenser and/or reboiler heat duties. A way to circumvent this kind of difficulty is first to obtain a converged solution for a case involving standard specifications. Once the behavior of the column is

understood, it will be possible to make sensible nonstandard specifications, again using an old converged result as a starting point.

Columns in which temperature and/or compositions change over a wide range in a limited number of stages pose their own particular difficulties. Some highly nonideal systems exhibit this kind of behavior (see Example 11 below). For cases such as this, it is wise to limit per-iteration changes to temperature and composition. Most modern computer methods will do this as a matter of course, and problems with this cause are not the source of convergence difficulties that they once were.

This is by no means an exhaustive list of the reasons that computer-based simulations fail. Indeed, in many cases it is a combination of more than one of the above factors that leads to difficulty. In those cases it may be necessary to combine several of the strategies outlined above to solve the simulation problem. Often, however, there is no substitute for trial and error. Haas [chap. 4 in Kister (op. cit.), 1992] offers some additional insight on using simulators to solve distillation column models.

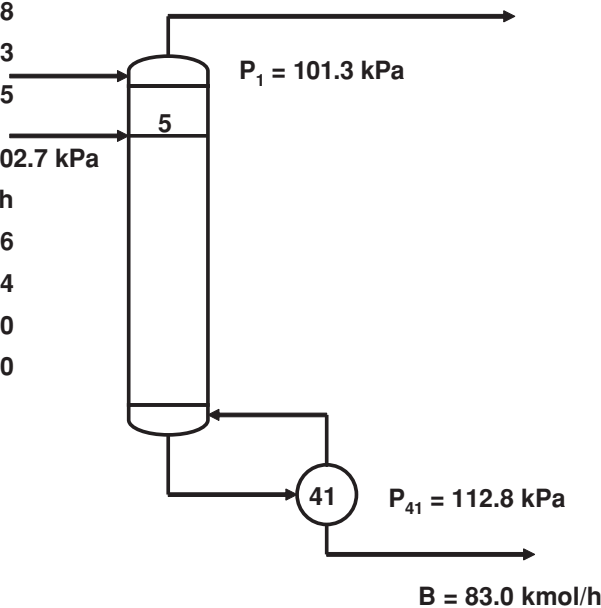
**Example 10: Multiple Steady States in Distillation** This example is one of the most famous in the entire literature on distillation column modeling, having been studied, in one form or another, by many investigators including Magnussen et al. [*I. Chem. E. Symp. Series*, **56** (1979)], Prokopakis and Seider [*AIChE J.*, **29**, 49 (1983)], and Venkataraman and Lucia [*Comput. Chem. Engng.*, **12**, 55 (1988)]. The column simulated here is adapted from the work of Prokopakis and Seider and shown in Fig. 13-52. The ethanol-benzene-water ternary system actually splits into two liquid phases when the overhead vapor is condensed and cooled below its bubble point. One liquid phase is sent to a second column, and the other is returned to the main column as (cold) reflux. Here, in common with others, this column is modeled by ignoring the condenser and decanter. Reflux is simulated by a feed of appropriate composition, temperature, and pressure to the top of the column. The UNIQUAC method was used for estimating the activity coefficients, with parameters given by Prokopakis and Seider. The numerical results are very sensitive to the choice of activity coefficient model and associated parameters; qualitatively, however, the behavior illustrated below is typical of many systems.

**Reflux: 298 K and 101.3 kPa**

| Reflux  | kmol/h |
|---------|--------|
| Ethanol | 103.2  |
| Water   | 215.8  |
| Benzene | 56.3   |
| Total   | 375.5  |

**Feed: 311 K and 102.7 kPa**

| Feed    | kmol/h |
|---------|--------|
| Ethanol | 85.6   |
| Water   | 14.4   |
| Benzene | 0.0    |
| Total   | 100.0  |



**FIG. 13-52** Azeotropic distillation tower for distillation of an ethanol-water mixture using benzene as a mass separating agent. [After Prokopakis and Seider (op. cit.).]

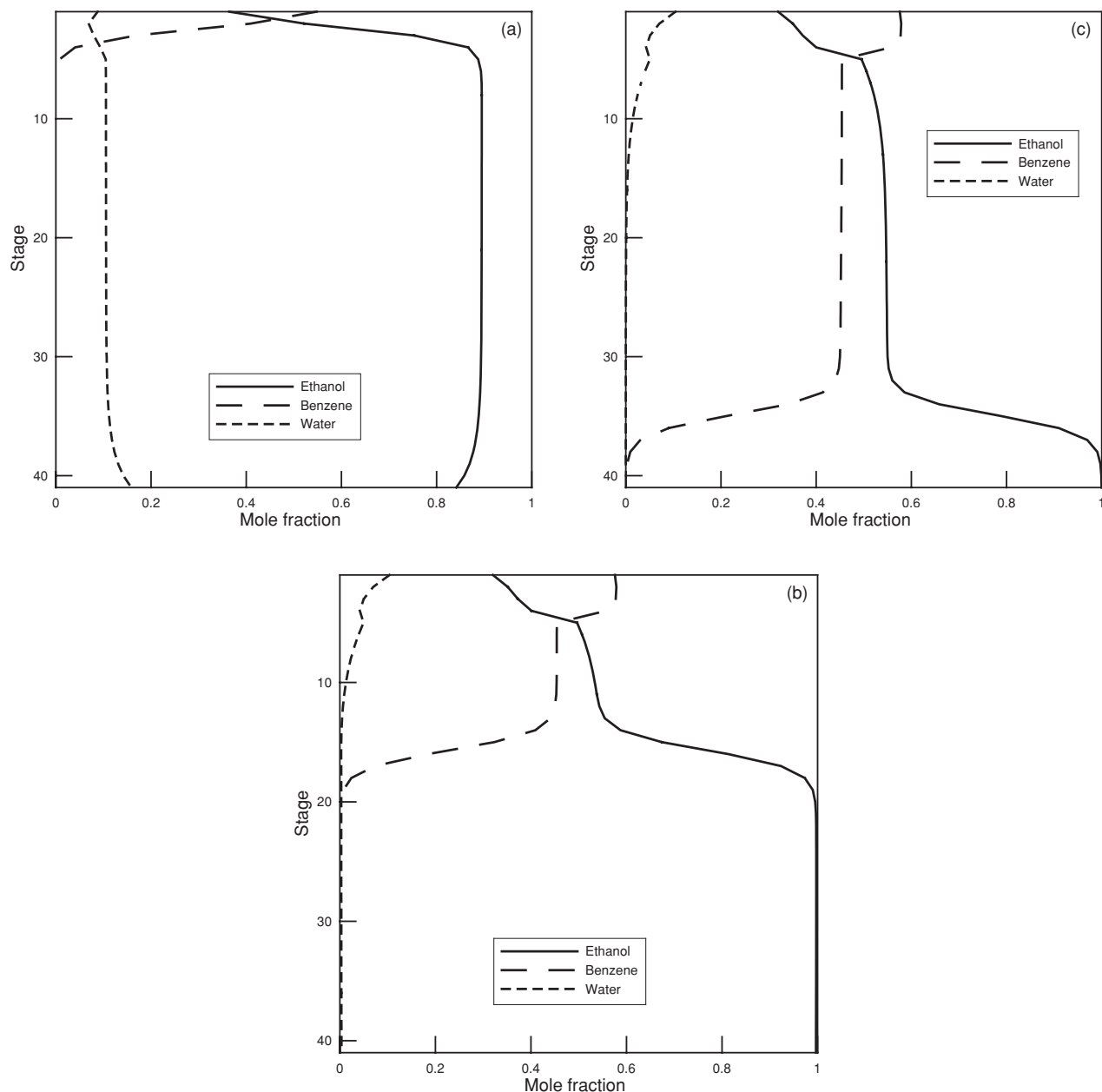
This system is considered difficult because convergence of the MESH equations can be difficult to obtain with any algorithm. In fact, for the specifications considered here, there are no less than three solutions; the composition profiles are shown in Fig. 13-53. The goal of the distillation is to recover high-purity ethanol in the bottom stream from the column by using benzene as a mass separating agent. The low-purity profile in Fig. 13-53a, containing a large amount of water in the bottom product, is easily obtained from an ideal solution starting point (but with severe restrictions on the maximum allowed temperature change per iteration). The intermediate profile in Fig. 13-53b is rather more difficult to obtain. We were able to find it by using, as a starting point, a profile that had been converged at a stage efficiency of 0.7. The high-purity solution in Fig. 13-53c, containing very little water in the bottom product, is also easily obtained from an initial profile calculated by assuming that the stage efficiency is quite low (0.3). Multiple solutions for this column have been reported by many authors (including the three cited above). In fact, with the parameters used here, the three solutions exist over a narrow range of ethanol feed flows. Taylor, Achuthan, and Lucia [*Comput. Chem. Engng.*, **20**, 93 (1996)] found complex-valued solutions to the MESH equations for values outside this range.

Multiple steady-state solutions of the MESH equations have been found for many systems, and the literature on this topic is quite extensive. An introduction to the literature is provided by Bekiaris, Guttinger, and Morari [*AIChE J.*, **46**, 955 (2000)]. Chavez, Seader, and Wayburn [*Ind. Eng. Chem. Fundam.*, **25**, 566 (1986)] used homotopy methods to find multiple solutions for some systems of interlinked columns. Parametric continuation has been used to detect multiple solutions of the MESH equations [Ellis et al., *Comput. Chem. Engng.*, **10**, 433 (1986); Kovach and Seider (op. cit.); Burton, Ph.D. Thesis in Chem. Engng., Cambridge University, 1986]. That real distillation columns can possess multiple steady states has been confirmed by the experimental work of Kienle et al. [*Chem. Engng. Sci.*, **50**, 2691 (1995)], Koggersbol et al. [*Comput. Chem. Engng.*, **20**, S835 (1996)], Gaubert et al. [*Ind. Eng. Chem. Res.*, **40**, 2914 (2001)], and others.

## NONEQUILIBRIUM MODELING

Although the widely used equilibrium-stage models for distillation, described above, have proved to be quite adequate for binary and close-boiling, ideal and near-ideal multicomponent vapor-liquid mixtures,





**FIG. 13-53** Liquid-phase mole fraction profiles for ethanol-benzene-water distillation. (a) Low-purity profile. (b) Intermediate-purity profile. (c) High-purity profile.

their deficiencies for general multicomponent mixtures have long been recognized. Even Murphree (op. cit.), who formulated the widely used plate efficiencies that carry his name, pointed out clearly their deficiencies for multicomponent mixtures and when efficiencies are small. Walter and Sherwood [*Ind. Eng. Chem.*, **33**, 493 (1941)] showed that experimentally measured efficiencies could cover an enormous range, with some values less than 10 percent. Toor [*AIChE J.*, **3**, 198 (1957)] predicted that the Murphree vapor efficiencies in some multicomponent systems could cover the entire range of values from minus infinity to plus infinity, a result that was verified experimentally by others.

In recent years a new approach to the modeling of distillation and absorption processes has become available: the *nonequilibrium* or *rate-based* models. These models treat these classical separation processes as the mass-transfer rate governed processes that they really are, and avoid entirely the (a priori) use of concepts such as efficiency and HETP [Krishnamurthy and Taylor, *AIChE J.*, **31**, 449–465 (1985); Taylor, Kooijman, and Hung, *Comput. Chem. Engng.*, **18**, 205–217 (1994)].

A schematic diagram of a nonequilibrium (NEQ) stage is given in Fig. 13-54. This NEQ stage may represent (part of) the two phases on

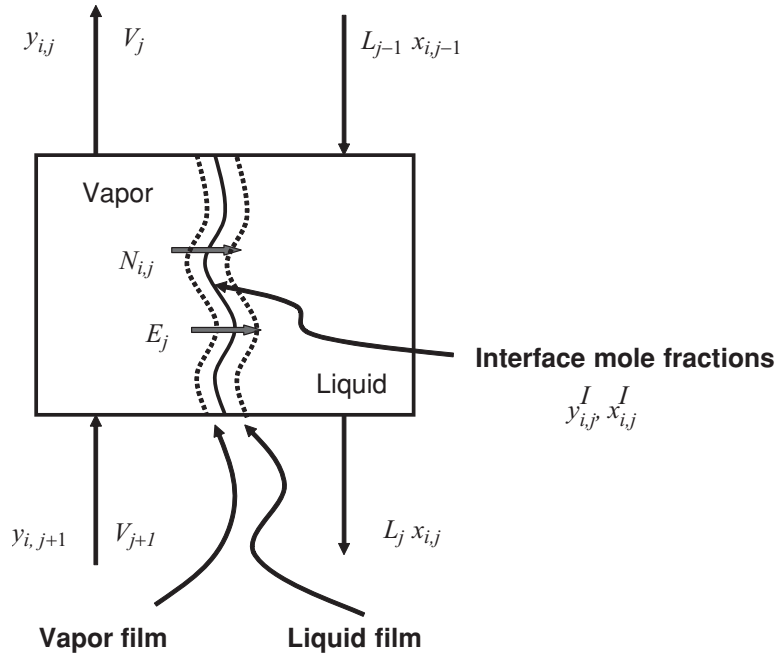


FIG. 13-54 Schematic diagram of a nonequilibrium stage.

a tray or within a section of a packed column. The wavy line in the middle of the box represents the phase interface. This illustration is intended only to aid in understanding the basic principles of nonequilibrium modeling; the actual flow patterns and the shape of the phase boundary are very complicated and depend upon, among other things, the equipment design, the column operation, and the physical properties of the system.

In a nonequilibrium model, separate balance equations are written for each distinct phase. The material balances for each species in the vapor and liquid phases on an arbitrary stage  $j$  are

$$(1 + r_j^V)V_j y_{i,j} - V_{j+1} y_{i,j+1} - f_{i,j}^V + \mathcal{N}_{i,j}^V = 0 \quad (13-61)$$

$$(1 + r_j^L)L_j x_{i,j} - L_{j-1} x_{i,j-1} - f_{i,j}^L - \mathcal{N}_{i,j}^L = 0 \quad (13-62)$$

where  $r_j^V$  and  $r_j^L$  are the ratios of sidestream to interstage flows and are defined by Eqs. (13-49);  $f_{i,j}^p$  is the external feed flow rate of species  $i$  in phase  $p$  to stage  $j$ ; and  $\mathcal{N}_{i,j}$  is the rate of mass transfer across the phase interface (with units of moles per second or equivalent). Formally, we may write

$$\mathcal{N}_{i,j}^p = \int N_{i,j}^p da_j \quad (13-63)$$

where  $N_{i,j}^p$  is the molar flux in phase  $p$  [with units of  $\text{mol}/(\text{m}^2 \cdot \text{s})$  or equivalent] at a particular point in the two-phase dispersion and  $da_j$  is the portion of interfacial area through which that flux passes. A material balance around the interface yields

$$\mathcal{N}_{i,j}^V = \mathcal{N}_{i,j}^L \quad (13-64)$$

The sum of the phase and interface balances yields the component material balance for the stage as a whole, the equation used in the equilibrium-stage model.

The energy balance is treated in a similar way, split into two parts—one for each phase, each part containing a term for the rate of energy transfer across the phase interface.

$$(1 + r_j^V)V_j H_j^V - V_{j+1} H_{j+1}^V - F_j H_j^{VF} + \mathcal{E}_j^V + Q_j^V = 0 \quad (13-65)$$

$$(1 + r_j^L)L_j H_j^L - L_{j-1} H_{j-1}^L - F_j H_j^{LF} - \mathcal{E}_j^L + Q_j^L = 0 \quad (13-66)$$

where  $\mathcal{E}_j^p$  is the rate of energy transfer across the phase interface in phase  $p$  and is defined by

$$\mathcal{E}_j^p = \int_a E_j^p da \quad (13-67)$$

where  $a$  is the interfacial area and  $E_j^p$  is the energy flux across the interface from/to phase  $p$ . An energy balance at the phase interface yields

$$\mathcal{E}_j^V = \mathcal{E}_j^L \quad (13-68)$$

In addition, we need summation equations for the mole fractions in the vapor and liquid phases.

A review of early applications of NEQ models is available in chapter 14 of Taylor and Krishna (op. cit.).

It is worth emphasizing that Eqs. (13-61) to (13-68) hold regardless of the models used to calculate the interphase transport rates  $\mathcal{N}_{i,j}$  and  $\mathcal{E}_j^p$ . With a mechanistic model of sufficient complexity it is possible, at least in principle, to account for mass transfer from bubbles in the froth on a tray as well as to entrained droplets in a spray, as well as transport between the phases flowing over and through the elements of packing in a packed column. However, a completely comprehensive model for estimating mass-transfer rates in all the possible flow regimes does not exist at present, and simpler approaches are used.

The simplest approach is to say that the molar fluxes at a vapor-liquid interface may be expressed as

$$N_i^V = c_i^V k_i^V (y_i^V - y_i^I) + y_i^V N_i^V \quad (13-69)$$

with a similar expression for the liquid phase.

$$N_i^L = c_i^L k_i^L (x_i^L - x_i^I) + x_i^L N_i^L \quad (13-70)$$

In these equations  $c_i^V$  and  $c_i^L$  are the molar densities of the superscripted phases,  $y_i^V$  is the mole fraction in the bulk vapor phase,  $x_i^L$  is the mole fraction in the bulk liquid phase, and  $x_i^I$  and  $y_i^I$  are the mole fractions of species  $i$  at the phase interface. Also  $N_i^p$  is the total molar flux in phase  $p$ , and  $k_i^V$  and  $k_i^L$  are the mass-transfer coefficients for



the vapor and liquid phases (with units of velocity), respectively. Methods for estimating mass-transfer coefficients in distillation processes are discussed briefly below and at greater length in Section 5 of this handbook.

The second term on the right-hand sides of Eqs. (13-69) and (13-70) is not often important in distillation (its neglect is equivalent to the assumption of equimolar counterflows in the column), but can be quite significant in gas absorption.

The energy fluxes are related by

$$E^V = q^V + \sum_{i=1}^c N_i^V \bar{H}_i^V = E = q^L + \sum_{i=1}^c N_i^L \bar{H}_i^L = E^L \quad (13-71)$$

$$\text{with} \quad q^V = h^V(T^V - T^I) \quad q^L = h^L(T^I - T^L) \quad (13-72)$$

where  $h^V$  and  $h^L$  are the heat-transfer coefficients in the vapor and liquid phases, respectively.

The inclusion in the model of the mass and energy transport equations introduces the mole fractions and temperature at the interface. It is common in almost all treatments of mass transfer across a phase boundary to assume that the mole fractions in the vapor and liquid phases at the interface are in equilibrium with each other. We may, therefore, use the very familiar equations from phase equilibrium thermodynamics to relate the interface mole fractions

$$y_i^I = K_i x_i^I \quad (13-73)$$

where the superscript  $I$  denotes the interface compositions and  $K_i$  is the vapor-liquid equilibrium ratio (or  $K$  value) for component  $i$ . In equilibrium-stage calculations, the equilibrium equations are used to relate the composition of the streams leaving the stage, and the  $K$  values are evaluated at the composition of the two exiting streams and the stage temperature (usually assumed to be the same for both phases). In nonequilibrium models the  $K$  values are evaluated at the interface composition and temperature by using exactly the same thermodynamic property models as that used in equilibrium-stage simulations. The interface composition and temperature must, therefore, be computed during a nonequilibrium column simulation. Strictly speaking, the composition and temperature at the interface vary with position in two-phase dispersion that exists on a tray or within the confines of a packed bed. In most NEQ models, the interface state is assumed uniform on the stage; thus, the model of a single stage includes one set of mass-transfer, heat-transfer, and phase equilibrium equations.

In equilibrium-stage models, the compositions of the leaving streams are related through the assumption that they are in equilibrium (or by use of an efficiency equation). It is important to recognize that efficiencies are not used in a nonequilibrium model; they may, however, be calculated from the results obtained by solving the model equations.

**Degrees of Freedom** Table 13-10 summarizes the equations for a single nonequilibrium stage. There are  $6c + 5$  independent equations

**TABLE 13-10 Equations for a Nonequilibrium Stage**

| Equation                                                        | Equation no. | Number                     |
|-----------------------------------------------------------------|--------------|----------------------------|
| Vapor-phase component balance                                   | (13-61)      | $c$                        |
| Liquid-phase component balance                                  | (13-62)      | $c$                        |
| Interface material balance                                      | (13-64)      | $c$                        |
| Vapor bulk mole fraction summation equation                     |              | 1                          |
| Liquid bulk mole fraction summation equation                    |              | 1                          |
| Vapor-phase energy balance                                      | (13-65)      | 1                          |
| Liquid-phase energy balance                                     | (13-66)      | 1                          |
| Equilibrium at the interface                                    | (13-73)      | $c$                        |
| Mass transfer in the vapor phase                                | (13-69)      | $c - 1$                    |
| Mass transfer in the liquid phase                               | (13-70)      | $c - 1$                    |
| Energy balance at phase interface                               | (13-71)      | 1                          |
| Summation equation for vapor mole fractions at phase interface  |              | 1                          |
| Summation equation for liquid mole fractions at phase interface |              | 1                          |
| <b>Total</b>                                                    |              | <b><math>6c + 5</math></b> |

**TABLE 13-11 Variables for a Nonequilibrium Stage**

| Variable                                | Symbol               | Number                      | Specified                  |
|-----------------------------------------|----------------------|-----------------------------|----------------------------|
| Vapor and liquid flow rates             | $V_j, L_j$           | 2                           |                            |
| Sidestream flow rates                   | $U_j, W_j$           | 2                           | Yes                        |
| Vapor-phase bulk composition            | $y_{ij}$             | $c$                         |                            |
| Liquid-phase bulk composition           | $x_{ij}$             | $c$                         |                            |
| Vapor interface composition             | $y_{ij}^I$           | $c$                         |                            |
| Liquid interface composition            | $x_{ij}^I$           | $c$                         |                            |
| Component feed flow rates               | $f_{ij}^V, f_{ij}^L$ | $2c$                        | Yes                        |
| Feed pressure and temperature           |                      | 2                           | Yes                        |
| Vapor-phase temperature                 | $T_j^V$              | 1                           |                            |
| Liquid-phase temperature                | $T_j^L$              | 1                           |                            |
| Interface temperature                   | $T_j^I$              | 1                           |                            |
| Stage pressure                          | $P_j$                | 1                           | Yes                        |
| Heat loss from vapor phase              | $Q_j^V$              | 1                           | Yes                        |
| Heat loss from liquid phase             | $Q_j^L$              | 1                           | Yes                        |
| Mass-transfer rates in the vapor phase  | $\mathcal{N}_{ij}^V$ | $c$                         |                            |
| Mass-transfer rates in the liquid phase | $\mathcal{N}_{ij}^L$ | $c$                         |                            |
| <b>Total</b>                            |                      | <b><math>8c + 12</math></b> | <b><math>2c + 7</math></b> |

per stage. As with the equilibrium-stage model discussed above, we have not included the feed mole fraction summation equation, or those for the vapor and liquid streams coming from adjacent stages.

The variables appearing in these equations are summarized in Table 13-11.

It is important to recognize that we have not included mass- and heat-transfer coefficients in the table of variables. These quantities are considered analogous to the thermodynamic properties of the equilibrium-stage model and are functions of other variables (as discussed in greater detail below).

The  $6c + 5$  variables for each stage determined during the solution of the nonequilibrium model equations are the vapor and liquid flow  $V_j$  and  $L_j$ , respectively; the bulk vapor mole fractions,  $y_{ij}$  ( $c$  in number); the bulk liquid mole fractions  $x_{ij}$  ( $c$ ); the vapor and liquid temperatures  $T_j^V$  and  $T_j^L$ , respectively; the interface mole fractions and temperature  $y_{ij}^I$  ( $c$ ),  $x_{ij}^I$  ( $c$ ), and  $T_j^I$ ; and the mass-transfer rates  $\mathcal{N}_{ij}^V$  ( $c$ ) and  $\mathcal{N}_{ij}^L$  ( $c$ ). Note that the interface material balance, Eq. (13-64), means that only one set of mass-transfer rates really needs to be counted in the set of variables for this stage, say,  $\mathcal{N}_{ij}$  (this reduces the number of variables being computed). The remaining variables,  $2c + 7$  in number, that need to be specified, are identified in Table 13-11. It is important to recognize that the other flows and composition variables appearing in the nonequilibrium-stage model equations are associated with the equivalent equations for adjacent stages. Although it appears that the number of degrees of freedom is higher for this more complicated model, this is misleading. The additional variables that are specified here take into account that there is one additional heat duty (one per phase) and that the feed is split into vapor and liquid fractions. In practice, the overall feed flow, pressure, temperature (or vapor fraction), and composition would be specified, and the vapor and liquid component flows in the feed determined from an adiabatic flash.

**Physical Properties** The only physical properties needed for an equilibrium-stage simulation are those needed to estimate  $K$  values and enthalpies; these same properties are needed for nonequilibrium models as well. Enthalpies are required for the energy balance equations; vapor-liquid equilibrium ratios are needed for the calculation of driving forces for mass and heat transfer. The need for mass- (and heat-) transfer coefficients means that nonequilibrium models are rather more demanding of physical property data than are equilibrium-stage models. These coefficients may depend on a number of other physical properties, as summarized in Table 13-12.

Methods for estimating physical and transport properties are described in Sections 4 and 5 of this handbook [see also Poling et al., *The Properties of Gases and Liquids*, McGraw-Hill, 5th ed. (2001)].

**Flow Models** In a real column the composition of the vapor and liquid phases changes due to flow across a tray or over and around packing. Thus, the bulk phase mole fractions that appear in the rate equations (13-69) and (13-70) vary with position and should not automatically be assumed to be equal to the average exit mole

**TABLE 13-12 Physical Property Needs of Equilibrium and Nonequilibrium Models**

| Property                   | EQ model | NEQ model | Used for                              |
|----------------------------|----------|-----------|---------------------------------------|
| K values                   | Yes      | Yes       | Driving forces                        |
| Enthalpy                   | Yes      | Yes       | Energy balances                       |
| Activity coefficient       | Yes      | Yes       | K values, enthalpies                  |
| Fugacity coefficients      | Yes      | Yes       | K values, enthalpies                  |
| Vapor pressure             | Yes      | Yes       | K values                              |
| Heat capacity              | Yes      | Yes       | Enthalpies, heat-transfer coefficient |
| Mass-transfer coefficients |          | Yes       | Mass-transfer rate equations          |
| Heat-transfer coefficients |          | Yes       | Energy-transfer rate equation         |
| Density                    |          | Yes       | Mass-transfer coefficients            |
| Diffusion coefficients     |          | Yes       | Mass-transfer coefficients            |
| Viscosity                  |          | Yes       | Mass-transfer coefficients            |
| Surface tension            |          | Yes       | Mass-transfer coefficients            |
| Thermal conductivity       |          | Yes       | Heat-transfer coefficients            |

fractions that appear in the material balance equations. In practice, we assume a flow pattern for the vapor and liquid phases, and this allows us to determine appropriate average mole fractions for use in the rate equations. There are three flow models in general use: mixed flow, plug flow, and dispersion flow (Lockett, op. cit.). A flow model needs to be identified for each phase. If both phases are assumed well mixed, then the average mole fractions are indeed equal to the mole fractions in the exit streams. This is the simplest (and an often used) approach that leads to the most conservative simulation (lowest mass-transfer rates, tallest column); at the opposite extreme is plug flow. The most realistic model is dispersion flow (see Lockett, op. cit.), but this model is not included in most computer implementations of NEQ models as it is quite complicated. For further discussion of the importance of flow models and the equations used to estimate average compositions, see Taylor and Krishna (op. cit.) and Kooijman and Taylor [*Chem. Engng. J.*, **57**, 177 (1995)].

**Mass-Transfer Coefficients** Mass-transfer coefficients (and the equally important interfacial area, a parameter with which they frequently are combined) may be computed from empirical correlations or theoretical models.

The mass-transfer performance of trays often is expressed by way of a dimensionless group called the number of transfer units [see Lockett (op. cit.), Kister (op. cit.), and Sec. 14 for additional background]. These dimensionless numbers are defined by

$$N^V = k^V a' t_V = \frac{k^V a h_f}{u_s} \quad (13-74)$$

$$N^L = k^L \bar{a} t_L = \frac{k^L a h_f Z}{Q_L / W} \quad (13-75)$$

where  $h_f$  = froth height, m

$Z$  = liquid flow path length, m

$W$  = weir length, m

$Q_L = \frac{L}{c_L'} =$  volumetric liquid flow rate, m<sup>3</sup>/s

$u_s = \frac{V}{c_L' A_{\text{pub}}} =$  superficial vapor velocity, m/s

$A_{\text{pub}}$  = bubbling area of tray, m<sup>2</sup>

$h_L$  = clear liquid height on tray, m

$a'$  = interfacial area per unit volume of vapor, m<sup>2</sup>/m<sup>3</sup>

$\bar{a}$  = interfacial area per unit volume of liquid, m<sup>2</sup>/m<sup>3</sup>

These areas are related to the interfacial area per unit volume of froth  $a$  by

$$a' = \frac{1}{1 - \alpha} \quad \bar{a} = \frac{\alpha}{\alpha} \quad (13-76)$$

where  $\alpha = h_L/h_f$  is the relative froth density. Also  $t_V$  and  $t_L$  are the vapor- and liquid-phase residence times, defined by

$$t_V = (1 - \alpha) h_f / u_s \quad (13-77)$$

$$t_L = \frac{Z}{u_L} = \frac{h_L Z W}{Q_L} \quad (13-78)$$

The *AICHE Bubble Tray Design Manual*, published in 1958 [see also Gerster et al., *Tray Efficiencies in Distillation Columns*, AIChE

(1958)], presented the first comprehensive procedure for estimating the numbers of transfer units in distillation. For many years this work represented the only such procedure available in the open literature; the work of organizations such as Fractionation Research Incorporated (FRI) was available only to member companies. Other comprehensive procedures for trays appeared in the 1980s [Zuiderweg, *Chem. Engng. Sci.*, **37**, 1441 (1982); Chan and Fair, *Ind. Engng. Chem. Proc. Des. Dev.*, **23**, 814, 820 (1984)]. Readers are referred to Kister (op. cit.), Lockett (op. cit.), Kle-mola and Ilme (op. cit.), and Sec. 14 of this handbook for summaries and references to what is available in the open literature.

#### Example 11: Mass-Transfer Coefficient in a Tray Column

Consider again the C<sub>4</sub> splitter that formed the basis of Examples 7 and 8. The key design parameters for the valve trays are given in the table below (from Kle-mola and Ilme, op. cit.).

|                         |                     |                           |                      |
|-------------------------|---------------------|---------------------------|----------------------|
| Column height           | 51.8 m              | Downcomer area center     | 0.86 m <sup>2</sup>  |
| Column diameter         | 2.9 m               | Tray spacing              | 0.6 m                |
| Number of trays         | 74                  | Hole diameter             | 39 mm                |
| Weir length, side       | 1.859 m             | Total hole area           | 0.922 m <sup>2</sup> |
| Weir length, center     | 2.885 m             | Outlet weir height        | 51 mm                |
| Liquid flow path length | 0.967 m             | Tray thickness            | 2 mm                 |
|                         | per pass            |                           |                      |
| Active area             | 4.9 m <sup>2</sup>  | Number of valves per tray | 772                  |
| Downcomer area, side    | 0.86 m <sup>2</sup> | Free fractional hole area | 18.82%               |

Estimate the mass-transfer coefficients for tray 7 where the flow and physical properties are estimated to be as summarized below:

|                                | Gas/vapor            | Liquid                |
|--------------------------------|----------------------|-----------------------|
| Flow, mol/s                    | 590                  | 550                   |
| Density, kg/m <sup>3</sup>     | 16.8                 | 520                   |
| Viscosity, N/m <sup>2</sup>    | $8.6 \times 10^{-6}$ | $1.35 \times 10^{-4}$ |
| Molecular weight, kmol/kg      | 58.0                 | 58.0                  |
| Diffusivity, m <sup>2</sup> /s | $800 \times 10^{-9}$ | $1.0 \times 10^{-9}$  |
| Surface tension, N/m           |                      | 0.014                 |

We use the AIChE correlation to illustrate the general approach, noting that the correlation was not developed specifically for valve trays (few methods were). In this model the number of transfer units is given by

$$N^V = \frac{0.776 + 4.57 h_w - 0.238 F_s + 104.8 Q_L / W}{\sqrt{Sc^V}} \quad (13-79)$$

$$N^L = 19,700 \sqrt{D^L} (0.4 F_s + 0.17) t_L \quad (13-80)$$

In the expressions above  $h_w$  is the weir height (m). The vapor-phase Schmidt number  $Sc^V$  is defined by  $Sc^V = \mu^V / (\rho^V D^V)$ , which here takes the value  $Sc^V = 0.640$ .

The superficial velocity is computed next from  $u_s^V = V / (c_L' A_{\text{pub}}) = 0.42$  m/s. Here  $F_s$  is the so-called  $F$  factor and is  $F_s = u_s^V \sqrt{\rho^V} = 1.7$  (kg/ms)<sup>1/2</sup>. The volumetric liquid flow is  $Q_L = L / c_L' = 0.061$  m<sup>3</sup>/s.

The froth height on the tray is estimated (by using the methods in Sec. 14 of this handbook—see also Lockett, 1986; Kister, 1992) to be  $h_f = 0.143$  m. The liquid-phase residence time is  $t_L = h_L Z W / Q_L = 4.67$  s.

The number of transfer units follows for the vapor phase from Eq. (13-79) as  $N^V = 2.45$  and for the liquid phase from Eq. (13-80) as  $N^L = 2.33$ . The products of the vapor- and liquid-phase mass-transfer coefficients and the interfacial area follow directly from the second parts of the same equations. Note that it is not possible with these correlations to separate the mass-transfer coefficient from the interfacial area. In practice this is not a concern since it is the mass-transfer rates that are needed rather than the fluxes and the product suffices for NEQ model computations.

The diffusivities used in this example were for the light key–heavy key pair of components. For systems similar to this one, the diffusion coefficients of all binary pairs in the mixture would be expected to have similar values. This will not be the case for mixtures of components that differ sharply in their fundamental properties (e.g., size, polarity). For these more highly nonideal mixtures, it is necessary to estimate the mass-transfer coefficients for each of the binary pairs.

The number of transfer units for packed columns is defined by

$$\mathbb{N}^V = \frac{k^V a' H}{u_V} \quad (13-81)$$

$$\mathbb{N}^L = \frac{k^L a' H}{u_L} \quad (13-82)$$

where  $u_V = V/(c_V^V A_c)$  and  $u_L = L/(c_V^L A_c)$  are the superficial vapor and liquid velocities, with  $A_c$  the cross-sectional area of the column;  $a'$  is the interfacial area per unit volume. The height of a transfer unit (HTU) is defined as

$$\mathbb{H}^V = \frac{H}{\mathbb{N}^V} = \frac{u_V}{k^V a'} \quad (13-83)$$

$$\mathbb{H}^L = \frac{H}{\mathbb{N}^L} = \frac{u_L}{k^L a'} \quad (13-84)$$

Methods of estimating numbers and/or heights of transfer units and mass-transfer coefficients and interfacial areas in packed columns are reviewed by Ponter and Au Yeung (in *Handbook of Heat and Mass Transfer*, Gulf Pub., 1986), by Wang et al. [*Ind. Eng. Chem. Res.*, **44**, 8715 (2005)], and in Sec. 5 of this handbook; one such method is illustrated below.

#### Example 12: Mass-Transfer Coefficients in a Packed Column

Estimate the mass-transfer coefficients at the top of the packed gas absorber in Example 9. The column has 23 ft of 2-in metallic Pall rings and is 3 ft in diameter. The specific surface area of this packing is 112 m<sup>2</sup>/m<sup>3</sup>. The flows and physical properties are estimated to be as summarized below.

|                                | Gas/vapor             | Liquid                |
|--------------------------------|-----------------------|-----------------------|
| Flow, mol/s                    | 332                   | 37                    |
| Temperature, K                 | 267                   | 266                   |
| Density, kg/m <sup>3</sup>     | 51.2                  | 705                   |
| Viscosity, N/m <sup>2</sup>    | $1.17 \times 10^{-5}$ | $2.43 \times 10^{-4}$ |
| Molecular weight, kmol/kg      | 16.8                  | 87.9                  |
| Diffusivity, m <sup>2</sup> /s | $100 \times 10^{-9}$  | $4.6 \times 10^{-9}$  |
| Surface tension, N/m           | 0.0085                |                       |

We will use the well-known correlation of Onda et al. (see Sec. 14) to estimate the mass-transfer coefficients. The vapor-phase coefficient is given by

$$\frac{k^V}{a_p D^V} = A \left( \frac{\rho^V u^V}{\mu^V a_p} \right)^{0.7} \left( \frac{\mu^V}{\rho^V D^V} \right)^{0.33} (a_p d_p)^{-2} \quad (13-85)$$

where  $d_p$  is the nominal packing size (2 in = 0.0508 m), and  $a_p$  is the specific surface area of the packing. Also  $A$  is a constant that has the value 2 if nominal packing size is less than 0.012 m, otherwise,  $A = 5.23$ .

The liquid-phase mass-transfer coefficient is given by

$$k^L = 0.0051 \left( \frac{\rho^L}{\mu^L g} \right)^{-1/3} \left( \frac{u_L}{a' \mu^L} \right)^{2/3} \left( \frac{\mu^L}{\rho^L D^L} \right)^{1/2} (a_p d_p)^{0.4} \quad (13-86)$$

Finally, the interfacial area per unit volume is given by

$$a' = a_p \left\{ 1 - \exp \left[ -1.45 \left( \frac{\sigma_c}{\sigma^L} \right)^{0.75} \left( \frac{\rho^L u_L}{a_p \mu^L} \right)^{0.1} \left( \frac{\mu^L u_L^2}{g} \right)^{0.05} \left( \frac{u^2 \rho^L}{a_p \sigma^L} \right)^{0.2} \right] \right\} \quad (13-87)$$

The vapor and liquid velocities are calculated to be  $u_V = V/(c_V^V A_c) = 0.42$  m/s and  $u_L = L/(c_V^L A_c) = 0.0017$  m/s. Substituting the values provided above into Eqs. (13-85) to (13-87) gives  $k^V = 0.0021$  m/s,  $a' = 96.2$  m<sup>2</sup>/m<sup>3</sup>, and  $k^L = 3.17 \times 10^{-4}$  m/s.

**Solving the NEQ Model Equations** In general, a nonequilibrium model of a column has many more equations than does an equivalent equilibrium-stage model. Nevertheless, we use may essentially the same computational approaches to solve the nonequilibrium model equations: simultaneous convergence (Krishnamurthy and Taylor, op. cit.) and continuation methods [Powers et al., *Comput. Chem. Engng.*, **12**, 1229 (1988)]. Convergence of a nonequilibrium model is likely to be slower than that of the equilibrium model because of the greater

number of model equations and the associated overhead in evaluating a greater number of physical properties. Finally, we note that the strategies outlined above for helping to converge equilibrium-stage simulation may prove equally useful when simulating distillation operations using the nonequilibrium models described here.

**Equipment Design** As we have already seen, the estimation of mass-transfer coefficients and interfacial areas from empirical correlations nearly always requires us to know something about the column design. At the very least we need to know the diameter and type of internal column (although usually we need to know more than that since most empirical correlations for mass-transfer coefficients have some dependency on equipment design parameters, e.g., weir height of trays). This need for more or less complete equipment design details suggests that nonequilibrium models cannot be used in preliminary process design (before any actual equipment design has been carried out). This is not true, however. Column design methods are available in the literature as well in most process simulation programs, and it is straightforward to carry out equipment sizing calculations at the same time as the stage equations are being solved (Taylor et al., op. cit.). This does not add significantly to the difficulty of the calculation, while providing the very significant advantage of allowing nonequilibrium or rate-based models to be used at all stages of process simulation.

**Example 13: A Nonequilibrium Model of a C<sub>4</sub> Splitter** Consider, again the C<sub>4</sub> splitter that formed the basis of Examples 7, 8, and 11.

When we create a nonequilibrium model of this—or any—column, we do not need to guess how many stages to use in each section of the column. The real column had 74 valve trays; the model column includes 74 model trays with the feed to tray 38 [plus a (subcooled) condenser and a reboiler, both of which are modeled as equilibrium stages, as described above]. All operating specifications are the same as for the corresponding equilibrium-stage model and are given in Examples 7 and 8. It is necessary to choose models that allow for the estimation of the rates of interphase mass transfer; that means selecting vapor and liquid flow models and correlations to estimate the mass-transfer coefficients in each phase, as discussed above. In this case the AIChE correlations were used. It is known that this method is more conservative than others (i.e., the predicted efficiencies are lower). The importance of the flow model is clear from the simulation results tabulated below. The predicted component Murphree efficiencies computed with Eq. (13-58) vary more widely from stage to stage and from component to component than might be expected for a system such as this. The TBK efficiency, on the other hand, does not change by more than a few percentage points over the height of the column; the value in the table below is an average of that computed for each tray from the simulation results using Eq. (13-59).

| Vapor flow model | Liquid flow model | <i>i</i> -C <sub>4</sub> in distillate, % | <i>n</i> -C <sub>4</sub> in bottoms, % | TBK efficiency, % |
|------------------|-------------------|-------------------------------------------|----------------------------------------|-------------------|
| Mixed            | Mixed             | 90.2                                      | 96.3                                   | 63                |
| Plug             | Mixed             | 92.2                                      | 97.2                                   | 78                |
| Plug             | Dispersion        | 93.9                                      | 98.0                                   | 106               |

Internal vapor and/or liquid composition data rarely are available, but such data are the best possible for model discrimination and validation. It is often relatively easy to match even a simple model only to product compositions. In the absence of composition profiles, the internal temperature profile can often be as useful provided that it is known to which phase a measured temperature pertains. The table below compares the few available measured tray temperatures with those computed during the simulation. The agreement is quite good.

| Tray | Temperature, °C |           |
|------|-----------------|-----------|
|      | Measured        | Predicted |
| 9    | 47.5            | 48.6      |
| 65   | 62.2            | 62.5      |
| 74   | 63.2            | 63.1      |

A portion of the McCabe-Thiele diagram for the simulation involving plug flow of vapor and dispersion flow of the liquid is shown in Fig. 13-55. For a nonequilibrium column these diagrams can only be constructed from the results of a computer simulation. Note that the triangles that represent the stages extend beyond the curve that represents the equilibrium line; this is so because the efficiencies are greater than 100 percent.

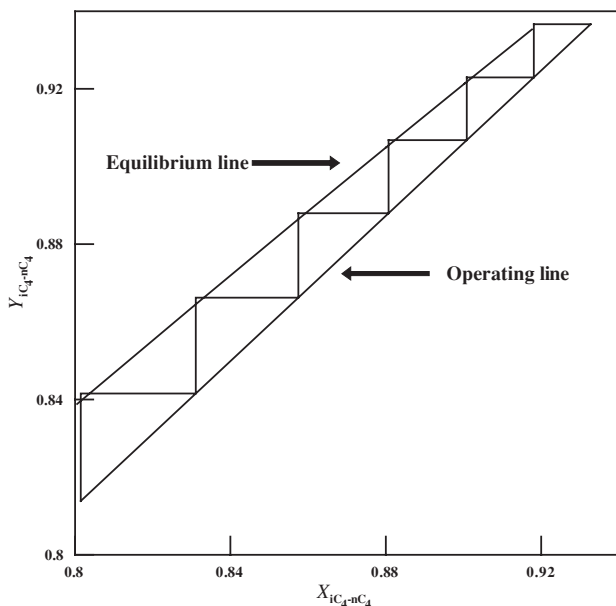


FIG. 13-55 Expanded view near upper right corner of McCabe-Thiele diagram for  $C_4$  splitter.

In this particular case the converged composition and temperature profiles have the same shape as those obtained with the equilibrium-stage model (with specified efficiency) and, therefore, are not shown. The reason for the similarity is that, as noted above, this is basically a binary separation of very similar compounds. The important point here is that, unlike the equilibrium-stage model simulations, the nonequilibrium model predicted how the column would perform; no parameters were adjusted to provide a better fit to the plant data. That is not to say, of course, that NEQ models cannot be used to fit plant data. In principle, the mass-transfer coefficients and interfacial area (or parameters in the equations used to estimate them) can be tuned to help the model better fit plant data.

### MAXWELL-STEFAN APPROACH

Strictly speaking, Eqs. (13-69) and (13-70) are valid only for describing mass transfer in binary systems under conditions where the rates of mass transfer are low. Most industrial distillation and absorption processes, however, involve more than two different chemical species. The most fundamentally sound way to model mass transfer in multi-component systems is to use the Maxwell-Stefan (MS) approach (Taylor and Krishna, op. cit.).

The MS equation for diffusion in a mixture with any number of different species can be written as

$$d_i = - \sum_{k=1}^c \frac{x_i x_k (u_i - u_k)}{\mathfrak{D}_{ik}} \quad (13-88)$$

where  $\mathfrak{D}_{ik}$  is the Maxwell-Stefan diffusion coefficient for the binary  $i$ - $k$  pair of components. Methods for estimating these coefficients are discussed by Taylor and Krishna (op. cit.) (see also Section 5 of this handbook and Poling et al., op. cit.).

In Eq. (13-88),  $d_i$  is termed the generalized driving force. For an ideal gas mixture the driving force is related to the partial pressure gradient and the mole fraction gradient as follows:

$$d_i = \frac{1}{P} \frac{dp_i}{dz} = \frac{dx_i}{dz} \quad (13-89)$$

For a nonideal fluid, the driving force is related to the chemical potential gradient

$$d_i = \frac{x_i}{RT} \frac{d\mu_i}{dz} \quad (13-90)$$

Equations (13-88) for ideal gas mixtures may be derived by using nothing more complicated than Newton's second law: *The sum of the forces acting on the molecules of a particular species is directly proportional to the rate of change of momentum.* The rate of change of momentum between different species is proportional to the concentrations (mole fractions) of the different species and to their relative velocity [see also Taylor and Krishna (op. cit.) for a more complete derivation]. Equation (13-88) is more familiar in the form

$$d_i = - \sum_{k=1}^c \frac{x_i N_k - x_k N_i}{c_t \mathfrak{D}_{ik}} \quad (13-91)$$

where we have replaced the velocities by the molar fluxes  $N_i = c u_i$  (see Section 5). Only  $c - 1$  of Eqs. (13-91) are independent; the mole fraction of the last component is obtained by the mole fraction summation equations for both phases.

Solving the MS equations can be quite involved (see Taylor and Krishna, op. cit.). Most often a simple film model solution of Eqs. (13-91) is used, leading to a simple difference approximation to the MS equations

$$\Delta x_i = - \sum_{k=1}^c \frac{\bar{x}_i N_k - \bar{x}_k N_i}{c_t \kappa_{i,k}} \quad (13-92)$$

where  $\kappa_{i,k}$  is the mass-transfer coefficient for the binary  $i$ - $k$  pair of components. The Maxwell-Stefan mass-transfer coefficients can be estimated from existing correlations for mass-transfer coefficients using the binary MS diffusion coefficients.

For a nonideal fluid the driving force is related to the chemical potential gradient

$$d_i = \frac{x_i}{RT} \frac{d\mu_i}{dz} \quad (13-93)$$

The difference approximation of this expression is somewhat more involved since we have to include the derivative of the activity (or fugacity) coefficient in the approximation. If, as often is assumed to be the case (not always with justification), the resistance to mass transfer in the liquid phase is negligible, then the MS equations for the liquid phase can safely be replaced by

$$\Delta x_i = x_i^I - x_i^L = 0 \quad (13-94)$$

The use of the MS equations in place of the simpler Eqs. (13-69) and (13-70) does not change the number of independent model equations or the number of degrees of freedom.

**Example 14: The Need for Rigorous Maxwell-Stefan-Based NEQ Models** Design a distillation column to separate a feed of 20 mol/s methanol, 10 mol/s isopropanol, and 20 mol/s water. The bottom product is to contain no more than 0.5 mol % methanol, and the distillate is to contain at least 99 mol % methanol but no more than 50 ppm water.

As a first step we attempt to design the column by using the equilibrium-stage model. Following Doherty and Malone (op. cit.), the NRTL model was used for the activity coefficients and the Antoine equation for the vapor pressures. Doherty and Malone estimate the minimum reflux as 5; we used a value 50 percent higher in this example and specified the bottoms product rate at 30 mol/s; this choice provides a consistent basis for the comparison of different models. The number of stages and the location of the feed were varied until a column configuration was obtained that met the desired product purity: 80 total stages (including total condenser and partial reboiler) with the feed to stage 16.

Efficiencies of alcohol-water and alcohol-alcohol systems obtained experimentally in sieve tray columns vary from 60 to 100 percent (Sec. 14 in the seventh

edition of this handbook). After specifying an average efficiency of 80 percent, we find that 99 total stages with the feed to stage 21 were needed to get the distillate product below 50 ppm water.

If we use the nonequilibrium model to design a sieve tray column, it is found that a column with 84 trays (plus condenser and reboiler) and with the feed to tray 21 (stage 22) will produce an overhead product of the desired purity. The reflux ratio and bottoms flows were maintained at the values employed for the equilibrium-stage design. The AIChE method was used for estimating the mass-transfer coefficient–interfacial area products, and the vapor and liquid phases were assumed to be in plug flow. The pressure was assumed constant in the column (an assumption that would need to be relaxed at a later stage of the design exercise). The computer simulation also provided a preliminary tray design; that for the trays above the feed is summarized in the table below.

|                                 |           |
|---------------------------------|-----------|
| Column diameter, m              | 1.76      |
| Total tray area, m <sup>2</sup> | 2.43      |
| Number of flow passes           | 2         |
| Tray spacing, m                 | 0.6       |
| Liquid flow path length, m      | 0.75      |
| Active area, % total            | 91.4      |
| Total hole area, % active       | 14        |
| Downcomer area, % total         | 4.3       |
| Hole diameter, mm               | 5         |
| Hole pitch, mm                  | 12        |
| Weir type                       | Segmental |
| Combined weir length, m         | 1.55      |
| Weir height, mm                 | 50        |

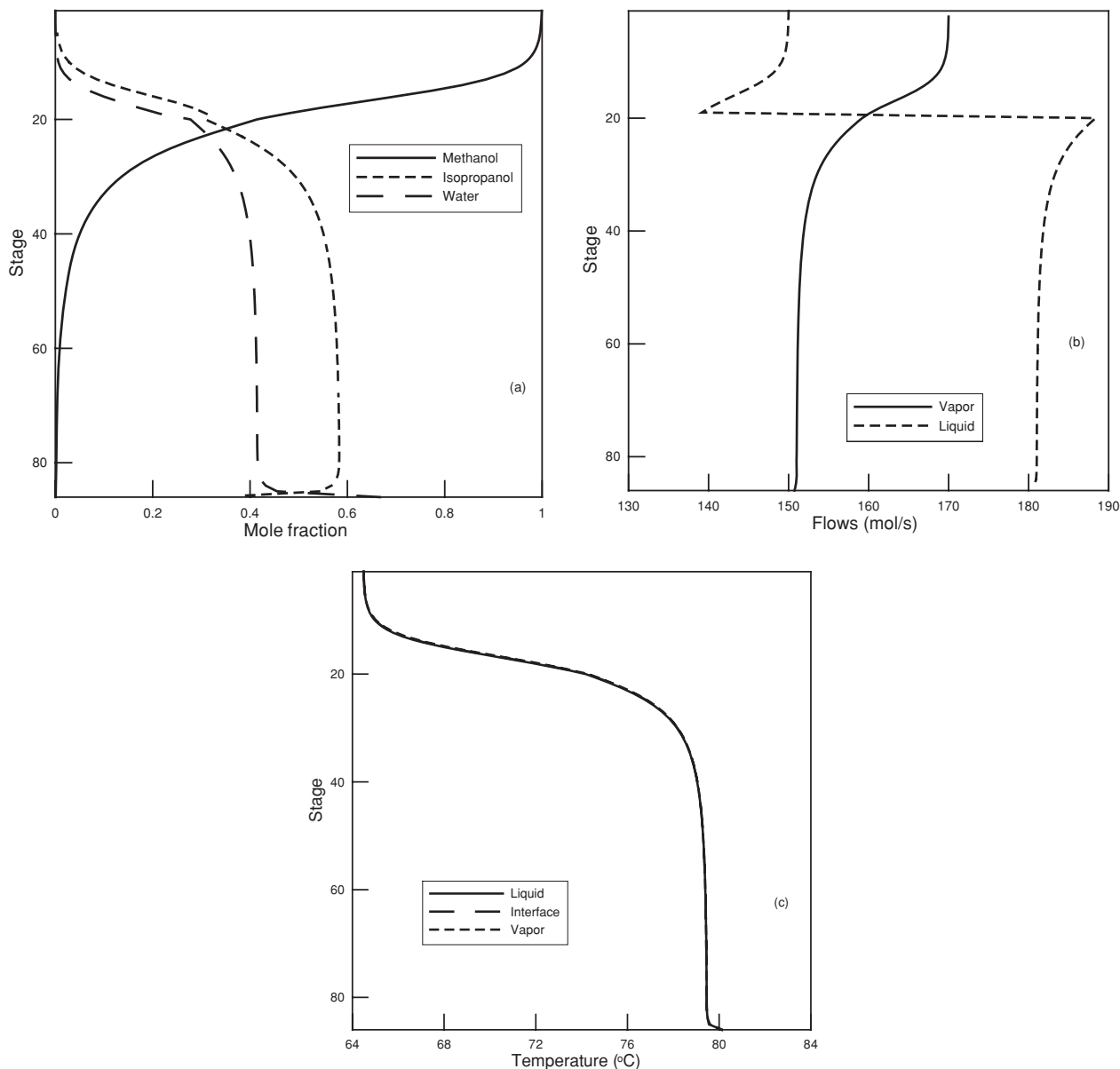


FIG. 13-56 Composition, flow, and temperature profiles in nonideal distillation process.



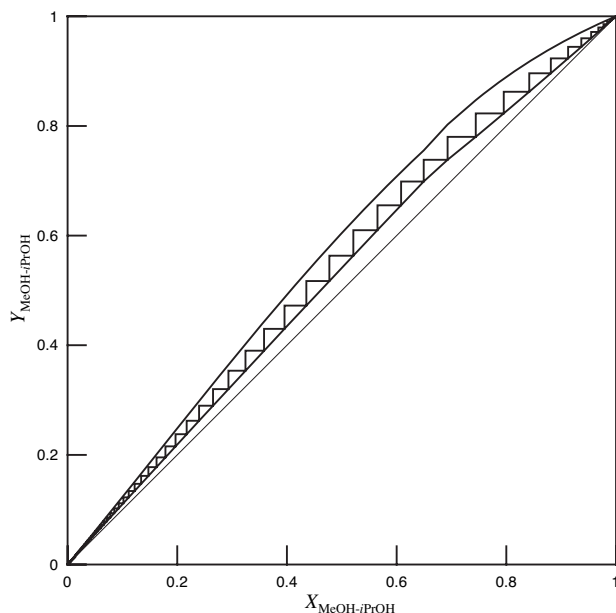


FIG. 13-57 McCabe-Thiele diagram for nonideal distillation column.

To converge the nonequilibrium model at the specified reflux ratio, it was necessary first to solve the problem at a much lower reflux ratio  $R = 2$  and then increase  $R$  in steps until the desired value of 7.5 was reached.

The liquid composition, flow, and temperature profiles are shown in Fig. 13-56. In this particular system the vapor and liquid temperatures estimated by the rate-based model are quite close (as often is the case in distillation operations).

The McCabe-Thiele diagram for this column is shown in Fig. 13-57. Note that in this case the triangles that represent the stages do not touch the equilibrium line. The length of the vertical section of each step in Fig. 13-57 is a measure of the efficiency of that tray. The component Murphree efficiencies calculated from the simulation results and Eq. (13-58), as well as the TBK average efficiency defined by Eq. (13-59), are shown in Fig. 13-58. The efficiency of methanol in the stripping section is seen to be around 80 percent, that of isopropanol to be approximately 75 percent, while that of water is close to 90 percent in the bulk of the column before falling off on the bottom few trays. All component efficiencies are found to be lower in the rectifying section. The TBK average efficiency, also shown in Fig. 13-58, is close to the Murphree efficiency of methanol and varies from 60 percent in the top of the column to 78 percent. Thus, the constant value of 80 percent used above appears to be appropriate, and yet, the column designed with the constant-efficiency model required no less than 99 stages (97 trays)!

With 84 trays as opposed to 78 equilibrium stages (not counting condenser and reboiler in either case), we find an overall efficiency of 93 percent, a figure that is quite at odds with the values of the individual component efficiencies seen in Fig. 13-58. How, then, is it possible that the nonequilibrium model suggests that the column needs only 6 trays more than the number of equilibrium stages? It is, in fact, because the efficiency of water is so much higher than that of the alcohols that leads to a column design that can produce high-purity methanol while producing the 50-ppm methanol bottom product in so few extra stages. Note that nonequilibrium models will not always lead to a design with fewer trays than might be suggested by a constant-efficiency model; it is just as likely for the mass-transfer rate-based model to predict that more stages will be needed—it all depends on the differences between the component efficiencies.

Individual component efficiencies can vary as much as they do in this example only when the diffusion coefficients of the three binary pairs that exist in this system differ significantly. For ideal or nearly ideal systems, all models lead to essentially the same results. This example demonstrates the importance of mass-transfer models for nonideal systems, especially when trace components are a concern. For further discussion of this example, see Doherty and Malone (op. cit.) and Baur et al. [*AIChE J.* **51**, 854 (2005)]. It is worth noting that there exists extensive experimental evidence for mass-transfer effects for this system, and it is known that nonequilibrium models accurately describe the behavior of this system, whereas equilibrium models (and equal-efficiency models) sometime

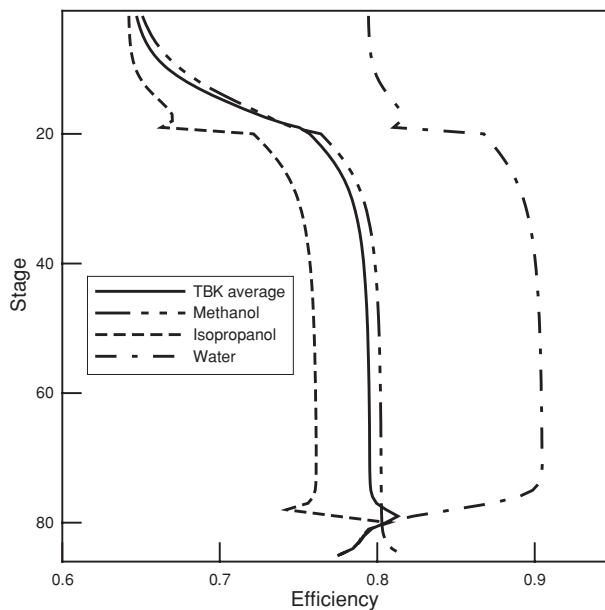


FIG. 13-58 Component Murphree efficiencies and TBK average efficiency [defined by Eq. (13-59)] for nonideal distillation.

predict completely erroneous product compositions [Pelkonen et al., *Ind. Engng. Chem. Res.*, **36**, 5392 (1997) and *Chem. Eng. Process*, **40**, 235 (2001); Baur et al., *Trans. I. Chem. E.*, **77**, 561 (1999)].

Nonequilibrium models should be preferred to equilibrium models when efficiencies are unknown, cannot be reliably predicted, and are low and/or highly variable; in nonideal systems and in processes where trace components are a concern.

There is a rapidly growing body of literature on nonequilibrium modeling of distillation and absorption processes. An extended bibliography is available at [www.chemsep.org/publications](http://www.chemsep.org/publications). A brief review of other applications follows.

Simulation methods currently in use for three-phase systems and systems involving chemical reaction employ the equilibrium-stage model [Doherty and Malone (op. cit.)]. Three-phase distillation remains relatively poorly understood compared to conventional distillation operations involving just a single liquid phase. It is important to be able to correctly predict the location of the stages where a second liquid phase can form (e.g., to determine the appropriate location for a sidestream decanter). The limited experimental data available suggest that efficiencies are low and highly variable with between 25 percent and 50 percent being not uncommon. Clearly, a model based on the assumption of equilibrium on every stage cannot hope to be able to predict column performance. Cairns and Furzer [*Ind. Engng. Chem. Res.*, **29**, 5392 (1997)] explicitly warn against incorporating Murphree efficiencies into the equilibrium-stage model for three-phase systems, although Müller and Marquardt [*Ind. Engng. Chem. Res.*, **36**, 5410 (1997)] find that an efficiency modified EQ stage model to be perfectly adequate for their column for the dehydration of ethanol using cyclohexane.

It is possible to develop nonequilibrium models for systems with more than two phases, as shown by Lao and Taylor [*Ind. Engng. Chem. Res.*, **33**, 2367 (1994)], Eckert and Vaněk [*Comput. Chem. Eng.*, **25**, 603 (2001)], and Higler et al. [*Comput. Chem. Eng.*, **28**, 2021 (2004)]. Experimental work that can be used to evaluate these models is scarce; see, however, Cairns and Furzer (op. cit.) and Springer et al. [*Chem. Eng. Res. Design*, **81**, 413 (2003)].

There is now an extensive literature on using nonequilibrium models for reactive distillation; see, e.g., Taylor and Krishna [*Chem. Eng.*

**TABLE 13-13 Selected List of Suppliers of Column Simulation Software**

| Supplier                     | Website               | EQ model | NEQ model |
|------------------------------|-----------------------|----------|-----------|
| Aspen Tech                   | www.aspentech.com     | Yes      | Yes       |
| Bryan Research & Engineering | www.bre.com           | Yes      |           |
| ChemSep                      | www.chemsep.com       | Yes      | Yes       |
| Chemstations                 | www.chemstations.net  | Yes      | Yes       |
| Deerhaven Technical Software | www.deerhaventech.com | Yes      |           |
| Honeywell                    | www.honeywell.com     | Yes      | Yes       |
| Process Systems Engineering  | www.psenterprise.com  | Yes      | Yes       |
| ProSim                       | www.prosim.net        | Yes      |           |
| SimSci-ESSCOR                | www.simsi-esscor.com  | Yes      | Yes       |
| VMG                          | virtualmaterials.com  | Yes      |           |

*Sci.*, **55**, 6139 (2000)], Sundmacher and Kienle (*Reactive Distillation: Status and Future Directions*, Wiley-VCH, 2003), Noeres et al. [*Chem. Engng. Processing*, **42**, 157 (2003)], and Klöcker et al. [*Chem. Engng. Processing*, **44**, 617 (2005)]. Gas absorption accompanied by chemical reaction for a long time has been modeled by using mass-transfer rate-based concepts; see, e.g., Cornelisse et al. [*Chem. Eng. Sci.*, **35**, 1245 (1980)], Pacheco and Rochelle [*Ind. Eng. Chem. Res.*, **37**, 4107 (1998)], and a review by Kenig et al. [*Chem. Eng. Technol.*, **26**, 631 (2003)]. For such systems the chemical reaction influences the efficiencies to such an extent that the concept loses its meaning.

Even at steady state, efficiencies vary from component to component and with position in a column. Thus, if the column is not at steady state, then efficiencies also must vary with time as a result of changes to flow rates and composition inside the column. Thus, equilibrium-stage models with efficiencies should not be used to model the dynamic behavior of distillation and absorption columns. Nonequilibrium models for studying column dynamics are described by, e.g., Kooijman and Taylor [*AIChE J.*, **41**, 1852 (1995)], Baur et al. [*Chem.*

*Eng. Sci.*, **56**, 2085 (2001)], Gunaseelan and Wankat [*Ind. Eng. Chem. Res.*, **41**, 5775 (2002)], Peng et al. [*Chem. Eng. Sci.*, **58**, 2671 (2003)], and Kenig et al. [*Chem. Eng. Sci.*, **54**, 5195 (1999)].

### SOFTWARE FOR DISTILLATION COLUMN SIMULATIONS

Computer software for equilibrium-stage and nonequilibrium column models is available from a number of suppliers. Many other models have been implemented primarily for research purposes and are not available commercially.

In Table 13-13 we list several suppliers of column simulation models that are available commercially, without in any way claiming that this list is exhaustive or providing an endorsement of any particular package. We recommend that users interested in any of these (or other) packages carry out an independent evaluation that focuses on the ability of the package to tackle the simulation problems of direct interest. The simulations described in this subsection were carried out with *ChemSep*.

## DEGREES OF FREEDOM AND DESIGN VARIABLES

### DEFINITIONS

In the models described in previous subsections, we have determined the degrees of freedom on a case-by-case basis. We now develop a general approach to the topic.

For separation processes, a design solution is possible if the number of independent equations equals the number of unknowns

$$N_i = N_v - N_c \quad (13-95)$$

where  $N_v$  is the total number of variables (unknowns) involved in the process under consideration,  $N_c$  is the number of restricting relationships among the unknowns (independent equations), and  $N_i$  is the degrees of freedom that must be specified for there to be exactly the same number of unknowns as there are independent equations in the model. The concept of degrees of freedom in this context is similar to the same concept that appears in the Gibbs phase rule. The degrees of freedom is the number of design variables that must be specified to define one unique operation (solution) of the process.

The variables  $N_i$  with which the designer of a separation process must be concerned are

1. Stream concentrations (e.g., mole fractions)
2. Temperatures
3. Pressures
4. Stream flow rates
5. Repetition variables  $N_r$

The first three are intensive variables. The fourth is an extensive variable that is not considered in the usual phase rule analysis. The fifth is neither an intensive nor an extensive variable but is a single degree of freedom that the designer uses in specifying how often a particular element is repeated in a unit. For example, a distillation column section is composed of a series of equilibrium stages, and when the designer specifies the number of stages that the section contains,

he or she uses the single degree of freedom represented by the repetition variable ( $N_r = 1.0$ ). If the distillation column contains more than one section (such as above and below a feed stage), the number of stages in each section must be specified, and as many repetition variables exist as there are sections, that is,  $N_r = 2$ .

The various restricting relationships  $N_c$  can be classified as

1. Inherent
2. Mass balance
3. Energy balance
4. Phase distribution
5. Chemical equilibrium

The inherent restrictions are usually the result of definitions and take the form of identities. For example, the concept of the equilibrium stage involves the inherent restrictions that  $T^v = T^l$  and  $P^v = P^l$  where the superscripts  $V$  and  $L$  refer to the equilibrium exit streams.

The mass balance restrictions are the  $C$  balances written for the  $C$  components present in the system. (Since we will deal with only non-reactive mixtures, each chemical compound present is a phase rule component.) An alternative is to write  $C - 1$  component balances and one overall mass balance.

The phase distribution restrictions reflect the requirement that  $f_i^v = f_i^l$  at equilibrium where  $f$  is the fugacity. This may be expressed by Eq. (13-1). In vapor-liquid systems, it should always be recognized that all components appear in both phases to some extent and there will be such a restriction for each component in the system. In vapor-liquid-liquid systems, each component will have three such restrictions, but only two are independent. In general, when all components exist in all phases, the number of restricting relationships due to the distribution phenomenon will be  $C(N_p - 1)$ , where  $N_p$  is the number of phases present.

For the analysis here, the forms in which the restricting relationships are expressed are unimportant. Only the number of such restrictions is important.



## ANALYSIS OF ELEMENTS

An *element* is defined as part of a more complex *unit*. The unit may be all or only part of an operation or the entire *process*. Our strategy will be to analyze all elements that appear in a separation process and to determine the number of design variables associated with each. The appropriate elements can then be quickly combined to form the entire process. Of course allowance must be made for the connecting streams (*interstreams*) whose variables are counted twice when elements or units are joined.

The simplest element is a *single homogeneous stream*. The variables necessary to define it are

|              | $N_v^e$           |
|--------------|-------------------|
| Compositions | $C - 1$           |
| Temperature  | 1                 |
| Pressure     | 1                 |
| Flow rate    | $\frac{1}{C + 2}$ |

There are no restricting relationships when the stream is considered only at a point. Henley and Seader (*Equilibrium-Stage Separation Operations in Chemical Engineering*, Wiley, New York, 1981) count all  $C$  compositions as variables, but then have to include as a restriction.

$$\sum_i x_i = 1 \quad \text{or} \quad \sum_i y_i = 1 \quad (13-96)$$

A stream divider simply splits a stream into two or more streams of the same composition. Consider Fig. 13-59, which shows the division of the condensed overhead liquid  $L_c$  into distillate  $D$  and reflux  $L_{N+1}$ . The divider is permitted to operate nonadiabatically if desired.

Three mass streams and one possible "energy stream" are involved, so

$$N_v^e = 3(C + 2) + 1 = 3C + 7 \quad (13-97)$$

Each mass stream contributes  $C + 2$  variables, but an energy stream has only its rate  $q$  as a variable. The independent restrictions are as follows:

|                                                  | $N_r^e$  |
|--------------------------------------------------|----------|
| $T$ and $P$ identities between $L_{N+1}$ and $D$ | 2        |
| Composition identities between $L_{N+1}$ and $D$ | $C - 1$  |
| Mass balances                                    | $C$      |
| Energy balance                                   | 1        |
|                                                  | $2C + 2$ |

The number of design variables for the element is given by

$$N_f^e = N_v^e - N_r^e = (3C + 7) - (2C + 2) = C + 5 \quad (13-98)$$

Specification of the feed stream  $L_c$  ( $C + 2$  variables), the ratio  $L_{N+1}/D$ , the "heat leak"  $q$ , and the pressure of either stream leaving the divider uses these design variables and defines one unique operation of the divider.

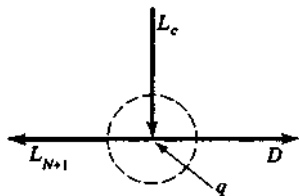


FIG. 13-59 Stream divider.

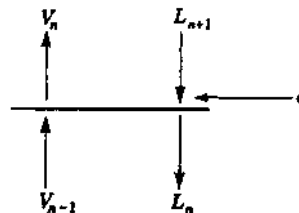


FIG. 13-60 Simple equilibrium stage.

A simple equilibrium stage (no feed or sidestreams) is depicted in Fig. 13-60. Four mass streams and a heat leak (or heat addition) stream provide the following number of variables:

$$N_v^e = 4(C + 2) + 1 = 4C + 9 \quad (13-99)$$

Vapor and liquid streams  $V_n$  and  $L_n$ , respectively, are in equilibrium with each other by definition and therefore are at the same  $T$  and  $P$ . These two inherent identities when added to  $C$ -component balances, one energy balance, and the  $C$  phase distribution relationships give

$$N_r^e = 2C + 3 \quad (13-100)$$

Then

$$N_f^e = N_v^e - N_r^e \quad (13-101)$$

$$= (4C + 9) - (2C + 3) = 2C + 6 \quad (13-102)$$

These design variables can be used as follows:

| Specifications                    | $N_f^e$            |
|-----------------------------------|--------------------|
| Specification of $L_{n+1}$ stream | $C + 2$            |
| Specification of $V_{n-1}$ stream | $C + 2$            |
| Pressure of either leaving stream | 1                  |
| Heat leak $q$                     | $\frac{1}{2C + 6}$ |

The results of the analyses for all the various elements commonly encountered in distillation processes are summarized in Table 13-14. Details of the analyses are given by Smith (*Design of Equilibrium Stage Processes*, McGraw-Hill, New York, 1967) and in a somewhat different form by Henley and Seader (op. cit.).

## ANALYSIS OF UNITS

A *unit* is defined as a combination of elements and may or may not constitute the entire process. By definition

$$N_v^u = N_r + \sum_i N_v^e \quad (13-103)$$

and

$$N_f^u = N_v^u - N_r^u \quad (13-104)$$

TABLE 13-14 Design Variables  $N_f^e$  for Various Elements

| Element                        | $N_v^e$   | $N_r^e$  | $N_f^e$  |
|--------------------------------|-----------|----------|----------|
| Homogeneous stream             | $C + 2$   | 0        | $C + 2$  |
| Stream divider                 | $3C + 7$  | $2C + 2$ | $C + 5$  |
| Stream mixer                   | $3C + 7$  | $C + 1$  | $2C + 6$ |
| Pump                           | $2C + 5$  | $C + 1$  | $C + 4$  |
| Heater                         | $2C + 5$  | $C + 1$  | $C + 4$  |
| Cooler                         | $2C + 5$  | $C + 1$  | $C + 4$  |
| Total condenser                | $2C + 5$  | $C + 1$  | $C + 4$  |
| Total reboiler                 | $2C + 5$  | $C + 1$  | $C + 4$  |
| Partial condenser              | $3C + 7$  | $2C + 3$ | $C + 4$  |
| Partial reboiler               | $3C + 7$  | $2C + 3$ | $C + 4$  |
| Simple equilibrium stage       | $4C + 9$  | $2C + 3$ | $2C + 6$ |
| Feed stage                     | $5C + 11$ | $2C + 3$ | $3C + 8$ |
| Sidestream stage               | $5C + 11$ | $3C + 4$ | $2C + 7$ |
| Adiabatic equilibrium flash    | $3C + 6$  | $2C + 3$ | $C + 3$  |
| Nonadiabatic equilibrium flash | $3C + 7$  | $2C + 3$ | $C + 4$  |

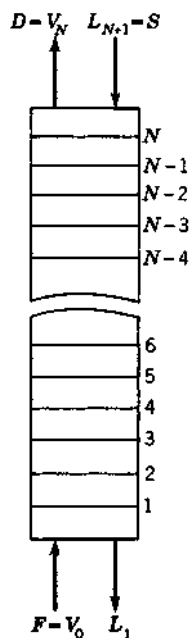


FIG. 13-61 Simple absorption column.

where  $N_c^u$  refers to *new* restricting relationships (identities) that may arise when elements are combined. Here  $N_c^u$  does not include any of the restrictions considered in calculating the  $N_i^e$ 's for the various elements. It includes only the stream identities that exist in each interstream between two elements. The interstream variables ( $C + 2$ ) were counted in each of the two elements when their respective  $N_i^e$ 's were calculated. Therefore,  $C + 2$  new restricting relationships must be counted for each interstream in the combination of elements to prevent redundancy.

The simple absorber column shown in Fig. 13-61 is analyzed here to illustrate the procedure. This unit consists of a series of simple equilibrium stages of the type in Fig. 13-60. Specification of the number of stages  $N$  uses the single repetition variable and

$$N_c^u = N_r + \sum_i N_i^e = 1 + N(2C + 6) \quad (13-105)$$

since  $N_i^e = 2C + 6$  for a simple equilibrium stage in Table 13-14. There are  $2(N - 1)$  interstreams, and therefore  $2(N - 1)(C + 2)$  new identities (not previously counted) come into existence when elements are combined. Subtraction of these restrictions from  $N_c^u$  gives  $N_i^u$ , the design variables that must be specified.

$$N_i^u = N_c^u - N_c^e = N_r + \sum_i N_i^e - N_c^e \quad (13-106)$$

$$= [1 + N(2C + 6)] - 2[(N - 1)(C + 2)] \quad (13-107)$$

$$= 2C + 2N + 5 \quad (13-108)$$

These might be used as follows:

| Specifications                               | $N_i^u$       |
|----------------------------------------------|---------------|
| Two feed streams                             | $2C + 4$      |
| Number of stages $N$                         | 1             |
| Pressure of either stream leaving each stage | $N$           |
| Heat leak for each stage                     | $N$           |
|                                              | $2C + 2N + 5$ |

A more complex unit is shown in Fig. 13-62, which is a schematic diagram of a distillation column with one feed, a total condenser, and

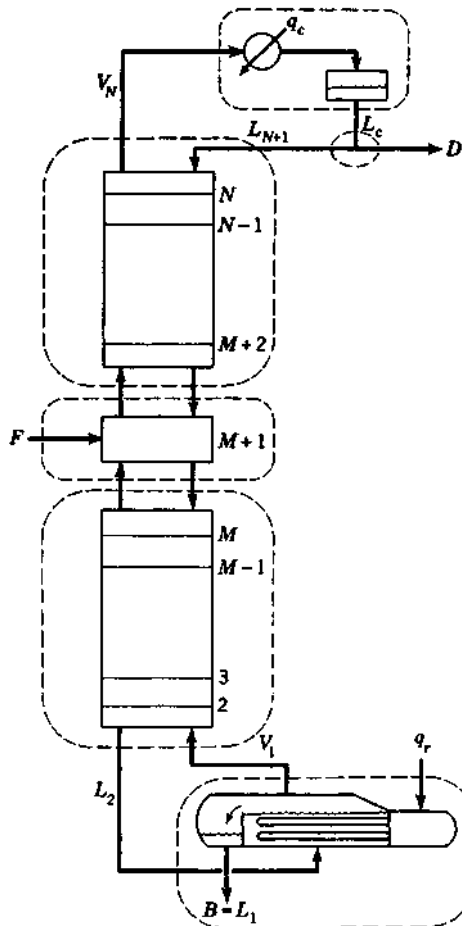


FIG. 13-62 Distillation column with one feed, a total condenser, and a partial reboiler.

a partial reboiler. Dotted lines encircle the six connected elements (or units) that constitute the distillation operation. The variables  $N_c^u$  that must be considered in the analysis of the entire process are just the sum of the  $N_i^e$ 's for these six elements since here  $N_r = 0$ . Using Table 13-14, we get the following:

| Element (or unit)                | $N_c^u = \sum_i N_i^e$  |
|----------------------------------|-------------------------|
| Total condenser                  | $C + 4$                 |
| Reflux divider                   | $C + 5$                 |
| $N - (M + 1)$ equilibrium stages | $2C + 2(N - M - 1) + 5$ |
| Feed stage                       | $3C + 8$                |
| $M - 1$ equilibrium stages       | $2C + 2(M - 1) + 5$     |
| Partial reboiler                 | $C + 4$                 |
|                                  | $10C + 2N + 27$         |

Here, the two units of  $N - (M + 1)$  and  $M - 1$  stages are treated just as elements. Nine interstreams are created by the combination of elements, so

$$N_c^e = 9(C + 2) = 9C + 18 \quad (13-109)$$

The number of design variables is

$$N_i^u = C + 2N + 9N_c^u = (10C + 2N + 27) - (9C + 18) \quad (13-110)$$

$$= C + 2N + 9 \quad (13-111)$$

One set of specifications that is particularly convenient for computer solutions is the following:

| Specifications                                                    | $N_i^u$      |
|-------------------------------------------------------------------|--------------|
| Pressure of either stream leaving each stage (including reboiler) | $N$          |
| Pressure of stream leaving condenser                              | 1            |
| Pressure of either stream leaving reflux divider                  | 1            |
| Heat leak for each stage (excluding reboiler)                     | $N - 1$      |
| Heat leak for reflux divider                                      | 1            |
| Feed stream                                                       | $C + 2$      |
| Reflux temperature                                                | 1            |
| Total number of stages $N$                                        | 1            |
| Number of stages below feed $M$                                   | 1            |
| Distillate rate $D/F$                                             | 1            |
| Reflux ratio $L_{N+1}/D$                                          | 1            |
|                                                                   | $C + 2N + 9$ |

Other specifications often used in place of one or more of the last four listed are the fractional recovery of one component in either  $D$  or  $B$  and/or the composition of one component in either  $D$  or  $B$ .

### OTHER UNITS AND COMPLEX PROCESSES

In Table 13-15, the number of design variables is summarized for several distillation-type separation operations, most of which are shown

**TABLE 13-15 Design Variables  $N_i^u$  for Separation Units**

| Unit                                                       | $N_i^u$ *      |
|------------------------------------------------------------|----------------|
| Distillation (partial reboiler-total condenser)            | $C + 2N + 9$   |
| Distillation (partial reboiler-partial condenser)          | $C + 2N + 6$   |
| Absorption                                                 | $2C + 2N + 5$  |
| Rectification (partial condenser)                          | $C + 2N + 3$   |
| Stripping                                                  | $2C + 2N + 5$  |
| Reboiled stripping (partial reboiler)                      | $C + 2N + 3$   |
| Reboiled absorption (partial reboiler)                     | $2C + 2N + 6$  |
| Refluxed stripping (total condenser)                       | $2C + 2N + 9$  |
| Extractive distillation (partial reboiler-total condenser) | $2C + 2N + 12$ |

\* $N$  includes reboiler, but not condenser.

in Fig. 13-2. For columns not shown in Figs. 13-1 or 13-2 that involve additional feeds and/or sidestreams, add  $C + 3$  degrees of freedom for each additional feed ( $C + 2$  to define the feed and 1 to designate the feed stage) and 2 degrees of freedom for each sidestream (1 for the sidestream flow rate and 1 to designate the sidestream-stage location). Any number of elements or units can be combined to form complex processes. No new rules beyond those developed earlier are necessary for the analysis. Further examples are given in Henley and Seader (op. cit.). An alternative method for determining the degrees of freedom for equipment and processes is given by Pham [*Chem. Eng. Sci.*, **49**, 2507 (1994)].

## DISTILLATION SYSTEMS

Distillation systems for the separation of nonazeotropic mixtures are discussed in this subsection. Many of the results extend also to azeotropic mixtures when the desired splits do not attempt to break azeotropes or cross a distillation boundary.

Whenever we desire to separate a mixture into multiple products, various combinatorial possibilities of column arrangements are feasible and the optimal (usually the least expensive) column configurations are sought. For example, there are at least two possible ways of separating a ternary mixture of components A, B, and C into pure-product streams (where A is the most volatile and C the least volatile component):

1. *Direct split*, where component A is separated from BC first (A/BC split) and then mixture BC is distilled to separate B from C
2. *Indirect split*, where C is removed first (AB/C) and then mixture AB is distilled

The difference in energy required for these splits can be assessed by simply comparing the total minimum vapor flows (summed over all the columns) for each column sequence. Example calculations were performed for a mixture with relative volatilities  $\alpha_A = 4$ ,  $\alpha_B = 2$ , and  $\alpha_C = 1$  containing 90 percent of A, 5 percent of B, and 5 percent of C in the feed stream. The indirect split requires 58 percent more energy than the direct split, assuming that columns are connected by a liquid stream in both cases. Moreover, the system using more energy requires bigger heat exchangers and larger column diameters, which increase the capital investment costs. Therefore, the direct split configuration would clearly be a better choice in this case.

One of the most important factors that determines the column configuration is the formulation (or goals) of the separation task with respect to the total flow sheet. Although a mixture may consist of  $C$  components, it does not mean that all  $C$  products are necessary. The components contained in streams recycled into the process (e.g., unreacted reactants recycled to the reactor) usually do not have to be separated from each other. Also separation of streams that are later mixed (blended) should be avoided, if possible. The separation system needs to be optimized together with the entire plant, either simultaneously or in a hierarchical approach, as described by Douglas (*The Conceptual Design of Chemical Processes*, McGraw-Hill, New York, 1988).

Heuristic methods have been widely used for synthesis of distillation sequences to avoid lengthy calculations. Many heuristics are intuitive, e.g., "Remove corrosive components first," "Remove most plentiful

components first," or "Remove the lightest component first." Since heuristics are just rules of thumb, they sometimes conflict with one another and may provide wrong answers even if they are not in conflict among themselves [Malone, Glins, Marques, and Douglas, *AIChE J.*, **31**, 683 (1985)]. More exact, algorithmic synthesis methods and cost optimization should be used in practice. The total energy of separation has been identified as the major component of the total separation system cost and energy-saving distillation schemes for light hydrocarbons with high levels of flexibility and operability, were discussed by Petterson and Wells [*Chem. Engng.*, **84**, (20), 78, 1977]. Tedder and Rudd presented a three-part paper [*AIChE J.*, **24**, 303 (1978)], where they compared eight distillation systems for the separation of ternary mixtures and determined the regions of economic optimality (with respect to feed compositions and relative volatilities). They proposed several heuristics to compare various configurations. They evaluated several rank order functions that allow for comparison of various configurations without detailed cost calculations. Interestingly, one of the rank order functions used successfully to compare various systems was the total minimum vapor flow. Minimum vapor flow will also be used to compare various distillation systems presented here. Finally, in the third part of their paper, Tedder and Rudd proposed a design method for various column networks.

In other approaches, Gomez-Munoz and Seader [*Comp. & Chem. Eng.*, **9**, 311 (1985)] proposed an optimization algorithm based on maximum thermodynamic efficiency. However, Smith and Linhoff [*Chem. Eng. Res. Des.*, **66**, 195 (1988)] pointed out the importance of a simultaneous design of the separation network and the rest of the process. They developed a pinch method based on the temperature-heat duty diagram to accomplish this task. Nishida, Stephanopoulos, and Westerberg [*AIChE J.*, **27**, 321 (1981)] presented a comprehensive review of process synthesis, including distillation. Westerberg [*Comp. & Chem. Eng.*, **9**, 421 (1985)] discussed methods for synthesis of distillation systems that include sharp splits, nonsharp splits, thermal linking, and heat integration. More recently, Agrawal [*AIChE J.*, **49**, 379 (2003)] presented a method for systematic generation of distillation configurations, including conventional and complex columns—a detailed discussion of these systems is given below. Sorting through the alternatives and selecting the low-cost systems can be quite tedious and are best done with computer-aided optimization strategies such as those proposed by Caballero and Grossmann [*Ind. Eng. Chem. Res.*, **40**, 2260 (2001)].

## POSSIBLE CONFIGURATIONS OF DISTILLATION COLUMNS

This subsection describes how to generate the feasible combinatorial possibilities of distillation column configurations for separation of mixtures that do not form azeotropes. Components are named A, B, C, D, . . . and they are listed in the order of decreasing volatility (or increasing boiling temperature). We limit our considerations to splits where the most volatile (lightest) component and the least volatile (heaviest) component do not distribute between the top and bottom product. For simplicity, we consider only separations where final products are relatively pure components. Systems containing simultaneously simple and complex distillation columns are considered. Simple columns are the conventional columns with one feed stream and two product streams; complex columns have multiple feeds and/or multiple product streams.

The combinatorial possibilities for separating a three-component mixture ABC into two product streams in which the most volatile component and the least volatile component do not distribute between the top and bottom product are

1. A/BC—Top product A is separated from bottom product BC.
2. AB/BC—Component B distributes between both product streams.
3. AB/C—Top product AB is separated from bottoms C.

These separations are also frequently referred to as splits; sharp splits (none of the components distribute) in cases (1) and (3), and a non-sharp split in case (2). When we add binary separations, the column configuration corresponding to split (1) is known as the direct split; the column configuration corresponding to split (2) is called the prefractionator system, and configuration (3) is called the indirect split (see Fig. 13-63). In the prefractionator system the binary columns separating components AB and BC may be stacked together, forming one column with three products A, B, and C as indicated by the dashed envelope in Fig. 13-63b. These systems are called the *basic column configurations*. Basic column configurations are configurations in which the types of interconnecting streams are not defined. The arrows on the flow sheets symbolize the net material flow, but the types of streams connecting the columns (liquid, vapor, two-phase, multiple streams) are not specified. Reboilers and condensers are deliberately not shown in Fig. 13-63, because for some types of interconnecting streams they are not necessary.

Symbolic-network representations of these separation systems, called *state-task networks* (STNs), are shown in Fig. 13-64. In this representation, the states (feeds, intermediate mixtures, and products) are represented by the nodes (ABC, AB, BC, A, B, C) in the network, and the tasks (separations) are depicted as lines (1, 2, . . . , 6) connecting the nodes, where arrows denote the net flow of material. This STN representation was used by Sargent to represent distillation systems [Comp. & Chem. Eng., **22**, 31 (1998)] and has been widely used ever since. Originally STNs were introduced by Kondili, Pan-

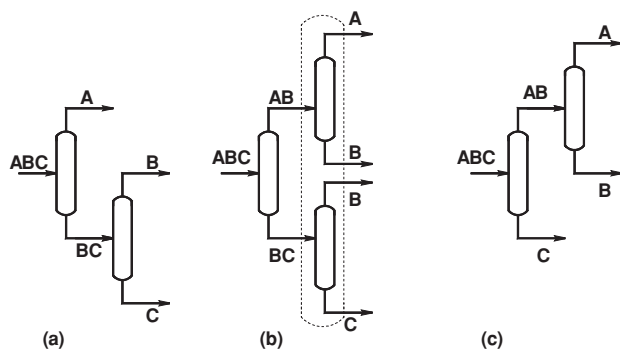


FIG. 13-63 Basic column configurations. (a) Direct split. (b) Prefractionator system. (c) Indirect split.

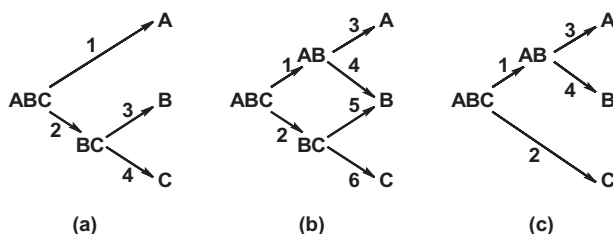


FIG. 13-64 State-task networks. (a) Direct split. (b) Prefractionator system. (c) Indirect split.

telides, and Sargent [Comp. & Chem. Eng., **17**, 211 (1993)] for representing batch processes.

Interconnecting streams may be liquids, vapors, or two-phase mixtures. The total energy of separation can be minimized for the direct split if the two columns have a liquid connection; for the indirect split it is minimized if the columns have a vapor connection, and for the prefractionator system if the top connection is vapor and the bottom connection is liquid. See Fidkowski and Krolikowski [AIChE J., **33**, 643 (1987); **36**, 1275 (1990)].

A complete direct split configuration, including reboilers and condensers, is shown in Fig. 13-65a. The numbers used to represent the column sections in this figure correspond to the numbers used to represent the tasks in the STN (Fig. 13-64a). By eliminating the reboiler from the first column in Fig. 13-65a and supplying the boil-up from the second column, we obtain the side rectifier arrangement, shown in Fig. 13-65b. An alternative side rectifier arrangement is seen more clearly in Fig. 13-65c, where the stripping section of the binary BC column (section 4) has been moved and lumped with the first column. Note that all these three configurations can be represented by one state-task network, shown in Fig. 13-64a. In each configuration in Fig. 13-65, feed ABC is separated in section 1 to get component A and in section 2 to get mixture BC. Then mixture BC is further separated in section 3 to give component B and in section 4 to produce C. Therefore, all these three systems are topologically equivalent to the same basic column configuration represented by the STN in Fig. 13-64a and the column arrangement in Fig. 13-63a. However, if we take into account reboilers and condensers, we see that only the two side rectifier configurations (Fig. 13-65b and c) are topologically equivalent. Side stripper configurations can be obtained from the indirect split in an analogous way (Fig. 13-66a, b, and c).

By eliminating the reboiler and condenser in the prefractionator column in Fig. 13-67a (the column containing sections 1 and 2) we obtain a *thermally coupled system*, also known as a Petlyuk system, shown in Fig. 13-67b [Petlyuk, Platonov, and Slavinskii, Int. Chem. Eng., **5**, 555 (1965)]. Side stripper, side rectifier, and Petlyuk systems can also be built as divided wall columns, as explained in detail in the subsection below on thermally coupled systems.

There are many other possible ternary column systems that have different interconnecting streams between the columns. For example, if we eliminate only the reboiler (or only the condenser) from the prefractionator column in Fig. 13-67a, we obtain a partially thermally coupled system [Agrawal and Fidkowski, AIChE J., **45**, 485 (1999), U.S. Patent 5,970,742]. In other instances, one may significantly increase the thermodynamic efficiency of the direct split and the indirect split if a portion of the interconnecting stream is vaporized and fed to the second column below the liquid connection [Agrawal and Fidkowski, Ind. Eng. Chem. Res., **38**, 2065 (1999)].

There are six possible types of splits for a quaternary mixture. The ternary mixtures resulting from these splits may be separated in one of the three possible ternary splits, as described above. Table 13-16 summarizes the resulting number of possible basic column configurations. This number does not account for the various possible types of interconnecting streams.

This method of generating various column configurations [Fidkowski, AIChE J., **52**, 2098 (2006)] is very similar to the methods

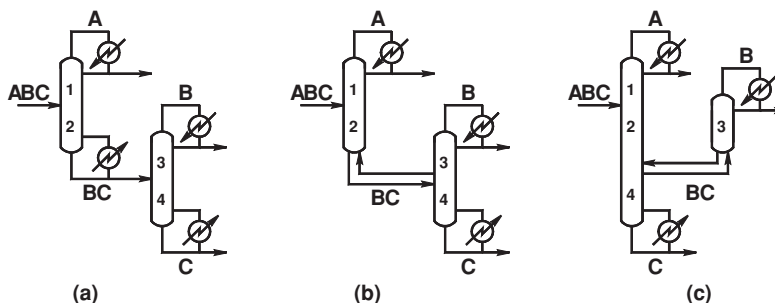


FIG. 13-65 (a) Direct split configuration. (b), (c) Side rectifier configurations.

used previously for conventional systems; see, e.g., Rathore, Van Wormer, and Powers [*AIChE J.*, **20**, 491 (1974); **20**, 940 (1974)]. The only difference is that in addition to sharp splits, the nonsharp splits are included here, which leads to unconventional systems. Similar methods were proposed by Rong et al. [Rong, Kraslawski, and Turunen, *Ind. Eng. Chem. Res.*, **42**, 1204 (2003)] to generate all the possible quaternary thermally coupled configurations.

The basic column configurations for all 22 quaternary distillation systems are shown in Fig. 13-68. Five of these configurations consist of only sharp splits; hence each species appears in only one product stream. Seventeen of the configurations have at least one nonsharp split, which results in each distributing component appearing in product streams from two different locations. Again, the interconnecting streams could be liquid, vapor, two-phase, or two-way liquid and vapor, as in thermally linked columns. There might also be several alternatives for column stacking, which increases the number of possible configurations even further. Subsequently, one can analyze possible splits for a five-component mixture, etc. These splits create quaternary and ternary products that can be further separated by one of the column configurations discussed above.

### THERMALLY COUPLED SYSTEMS AND DIVIDING WALL COLUMNS

In recent years there has been significant interest in thermally coupled systems and dividing wall columns for ternary mixtures. In this subsection we discuss such column arrangements, their energy requirements, design and optimization methods, controllability and operability, experimental and industrial experience, and extension to more than three components.

Two columns are thermally coupled if a vapor (liquid) stream is sent from the first column to the second column and then a return liquid (vapor) stream is implemented between the same locations. These

streams, when introduced at the top or bottom of a column, provide (at least partial) reflux or boil-up to this column.

The development of thermally coupled systems started with attempts to find energy-saving schemes for the separation of ternary mixtures into three products. One of the first industrial applications was the side rectifier configuration for air separation. The side stripper configuration followed naturally. By combining the two we obtain the fully thermally coupled system of Petlyuk, Platonov, and Slavinskii [*Int. Chem. Eng.*, **5**, 555 (1965)]; see Fig. 13-67b. It consists of the prefractionator which accepts the ternary feed stream followed by the main column that produces the products (product column).

The dividing wall column was invented as a way of producing three pure products from a single column [Monro, U.S. Patent 2,134,882 (1933); Wright, U.S. Patent 2,471,134 (1949)]. In some cases it is possible to achieve high purity of the intermediate component in the side-stream of a distillation column. This is possible when the sidestream is withdrawn above the feed as a liquid or below the feed as a vapor and when relative volatilities of components differ significantly. In many applications, however, a sidestream is contaminated to an appreciable extent by either the light or the heavy component. For example, if a sidestream is withdrawn from the rectifying section, it must contain not only the intermediate component but also some of the most volatile component. This contamination problem can be eliminated by adding a dividing wall that prevents the most volatile component (and the heaviest component) from entering the zone where the intermediate component is withdrawn; see Fig. 13-69.

The dividing wall column is topologically equivalent to the fully thermally coupled system (Fig. 13-67b). The prefractionator and the main product column are built in one shell, separated by a dividing wall (Fig. 13-69c). Similarly, the side rectifier (Fig. 13-65c) and side stripper (Fig. 13-66c) configurations can be built in one shell as dividing wall columns [Agrawal, *Ind. Eng. Chem. Res.*, **40**, 4258 (2001)]. The corresponding dividing wall columns are shown in Fig. 13-69a

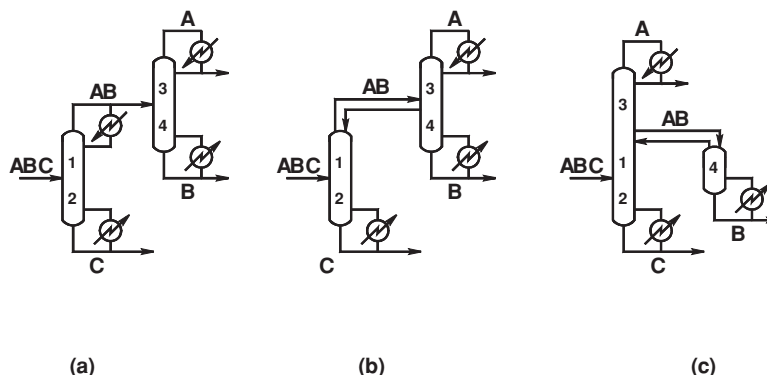


FIG. 13-66 (a) Indirect split configuration. (b), (c) Side stripper configurations.



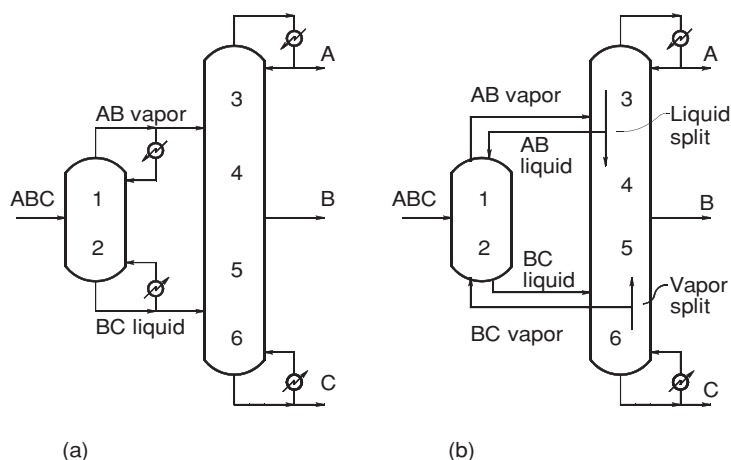


FIG. 13-67 (a) Prefractionator system. (b) Thermally coupled system.

and *b*, respectively. The corresponding sections in the thermally coupled systems and the dividing wall columns have exactly the same numbers and perform exactly the same separation tasks, indicating that the column arrangements are topologically equivalent.

Various design, simulation, and evaluation methods have been developed for the distillation systems shown in Figs. 13-65 to 13-67. These include those by Stupin and Lockhart [*Chem. Eng. Prog.*, **68**

(10), 71 (1972)]; Fidkowski and Krolikowski [*AIChE J.*, **32**, 537 (1986)]; Nikolaidis and Malone [*Ind. Eng. Chem. Res.*, **27**, 811, (1988)]; Rudd [Distillation Supplement to the Chemical Engineer, S14, (1992)]; Triantafyllou and Smith [*Trans. Inst. Chem. Eng.*, **70**, Part A, 118 (1992)]; Finn [*Gas Sep. Purif.*, **10**, 169 (1996)]; Annakou and Mizsey [*Ind. Eng. Chem. Eng.*, **35**, 1877 (1996)]; Hernandez and Jimenez [*Comp. & Chem. Eng.*, **23**, 1005 (1999)]; Dunnebie and Pantelides [*Ind. Eng. Chem. Res.*, **38**, 162 (1999)]; Watzdorf, Bausa, and Marquardt [*AIChE J.*, **45**, 1615 (1999)]; Kim [*J. Chem. Eng. Japan*, **34**, 236 (2001)]. The fully thermally coupled system uses less energy than any other ternary column configuration [Fidkowski and Krolikowski, *AIChE J.*, **33**, 643–653 (1987)]. The energy savings may be on the order of 30 to 50 percent, depending on the feed composition and volatilities of the components. Similar energy savings are possible in partially thermally coupled columns, where only one connection (top or bottom) between the columns is thermally coupled and the other is just a single liquid or vapor stream together with an associated condenser or reboiler, respectively [Agrawal and Fidkowski, *AIChE J.*, **45**, 485 (1999)]. Fidkowski and Krolikowski [*AIChE J.*, **32**, 537 (1986)]

TABLE 13-16 Number of Basic Column Configurations for Separation of a Four-Component Mixture

| Split | Distillate/bottoms | Number of configurations |
|-------|--------------------|--------------------------|
| 1     | A/BCD              | 3                        |
| 2     | AB/BCD             | 3                        |
| 3     | AB/CD              | 1                        |
| 4     | ABC/BCD            | $3 \times 3 = 9$         |
| 5     | ABC/CD             | 3                        |
| 6     | ABC/D              | 3                        |
| Total |                    | 22                       |

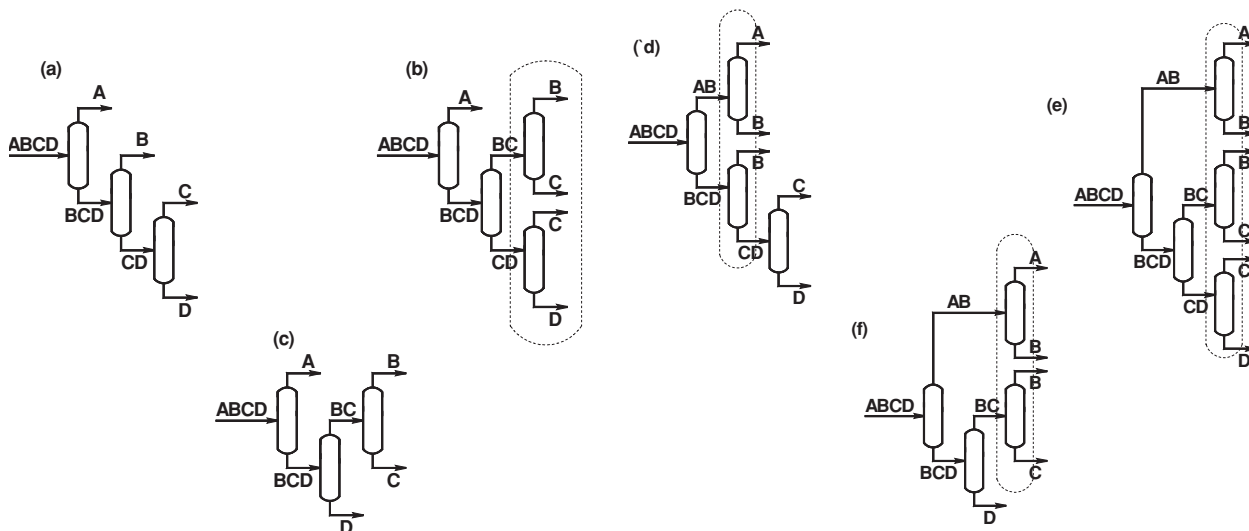


FIG. 13-68 Basic column configurations for separation of a four-component mixture.

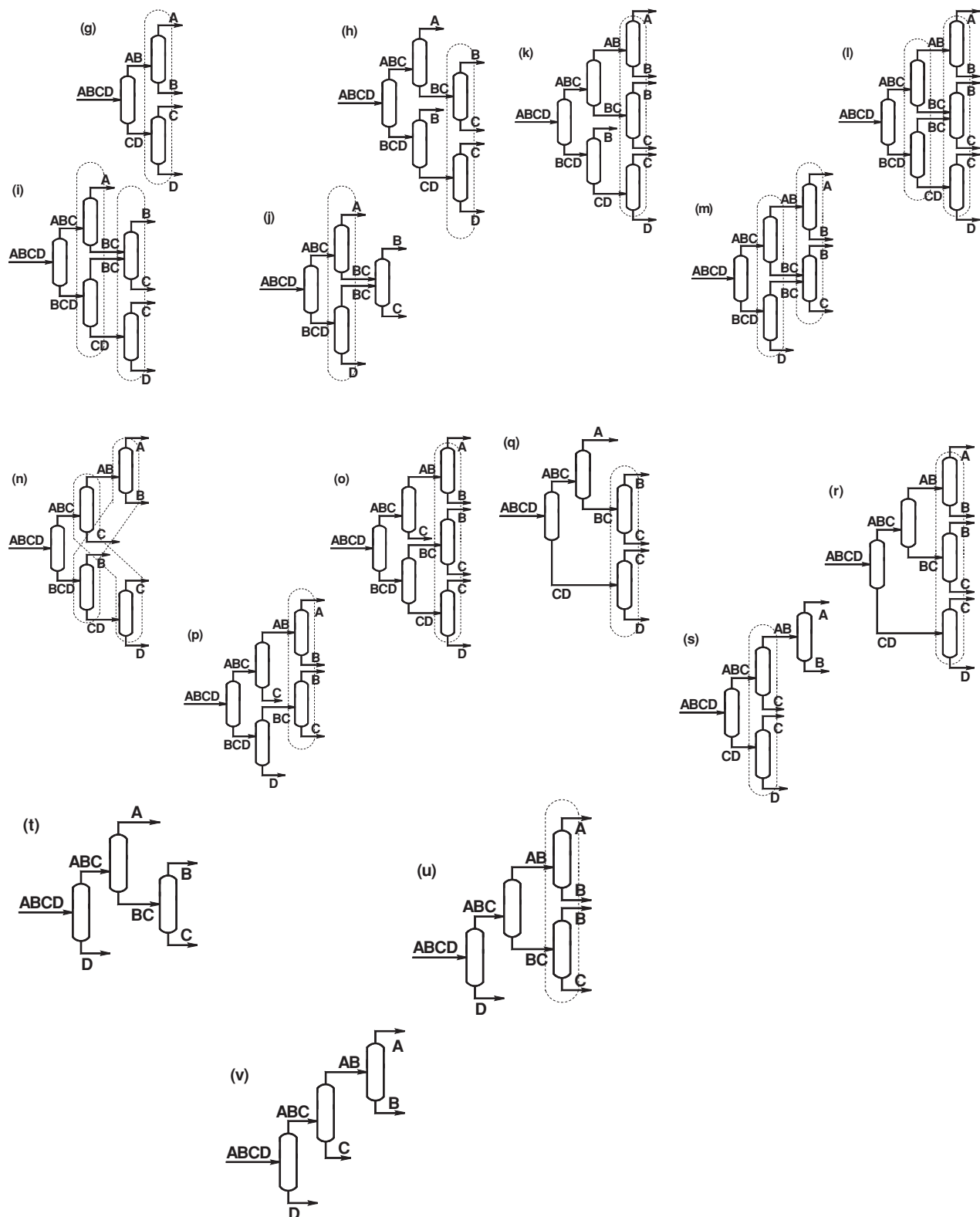


FIG. 13-68 (Continued)



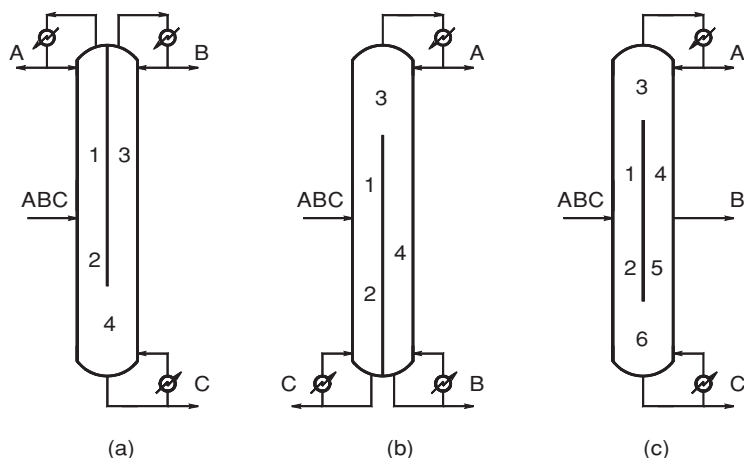


FIG. 13-69 Dividing wall columns equivalent to (a) side rectifier configuration, (b) side stripper configuration, (c) thermally coupled system.

solved analytically the optimization problem for the minimum vapor boil-up rate from the reboiler in the main column for the fully thermally coupled system (shown in Fig. 13-67b), assuming constant molar overflow and constant relative volatilities among the components. The solution depends on the splits of vapor and liquid between the main product column and the prefractionator [or between both sides of the dividing wall (Fig. 13-69c)]. The minimum vapor flow for each column in the system is shown in Fig. 13-70 as a function of  $\beta$ , where the parameter  $\beta$  is defined as the fractional recovery of the distributing component B in the top product of the prefractionator

$$\beta = \frac{V_1 y_B - L_1 x_B}{f_B} \quad (13-112)$$

where  $V_1$  and  $L_1$  are the vapor and liquid flows at the top of the prefractionator (section 1),  $y_B$  and  $x_B$  are corresponding mole fractions of component B, and  $f_B$  denotes the molar flow of component B in the feed stream. The lower line in Fig. 13-70 is the minimum vapor flow in the first column, the prefractionator. At  $\beta = 0$  it corresponds to the A/BC split and at  $\beta = 1$  to the AB/C split. There is a minimum in the minimum vapor flow in the prefractionator for  $\beta = \beta_p$ , the so-called transition split;  $\beta_p$  can be calculated as

$$\beta_p = \frac{\alpha_B - \alpha_C}{\alpha_A - \alpha_C} \quad (13-113)$$

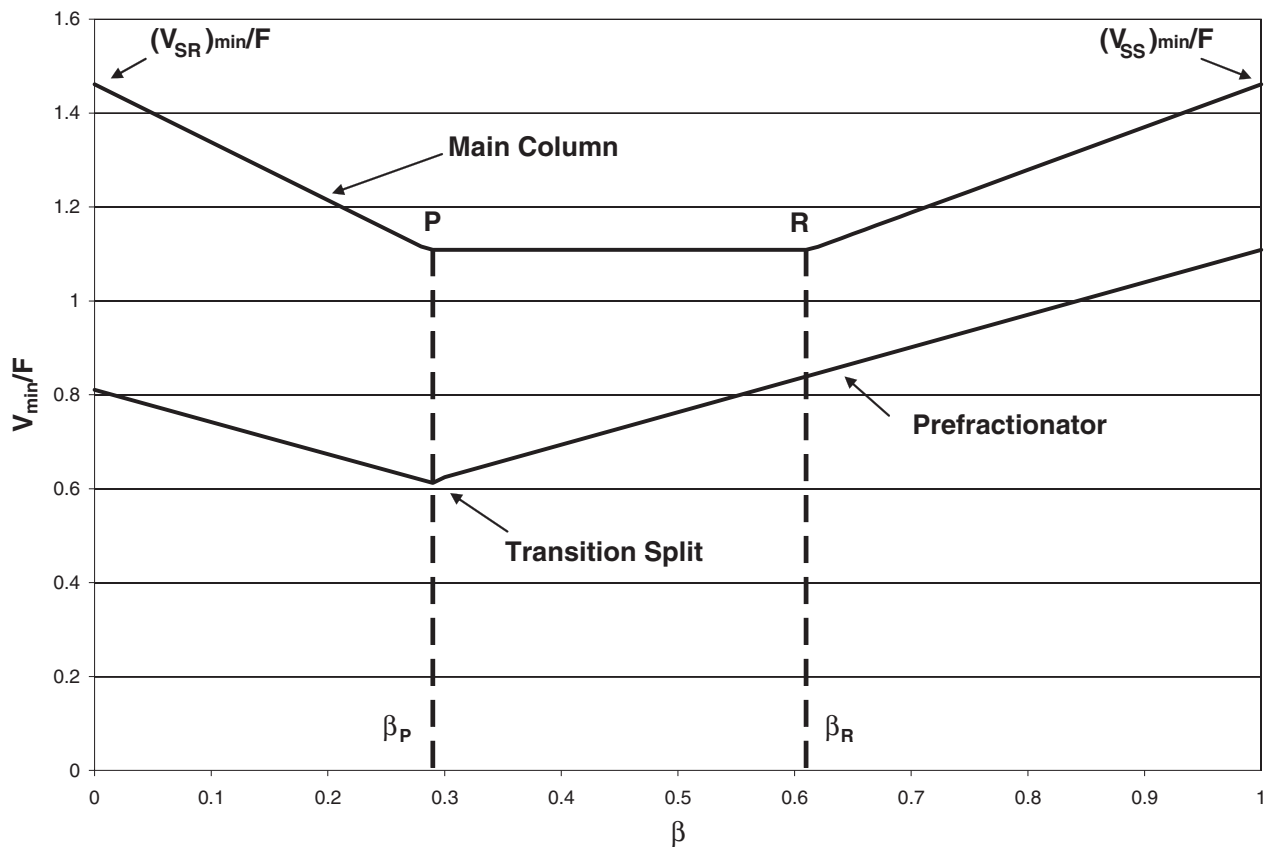
where the  $\alpha$ 's are relative to any reference component.

The transition split divides direct-type splits from indirect-type splits as discussed by Doherty and Malone (*Conceptual Design of Distillation Systems*, 2001, chaps. 4 and 5); also see Fidkowski, Doherty, and Malone [*AIChE J.*, **39**, 1301(1993)]. The upper line in Fig. 13-70 is the minimum vapor flow leaving the reboiler of the main column, which also corresponds to the minimum vapor flow for the entire system since all the vapor for the total system is generated by this reboiler. For  $\beta = 0$  the minimum vapor flow for the entire thermally coupled system (i.e., main column) becomes equal to the minimum vapor flow for the side rectifier system (i.e., main column of the side-rectifier system; see Fig. 13-65b or c) ( $V_{SR})_{\min}$ ; for  $\beta = 1$  it is equal to the minimum vapor flow of the entire side stripper system ( $V_{SS})_{\min}$  (which is the sum of the vapor flows from both the reboilers in this system; see Fig. 13-66b or c). Coincidentally, the values of these two minimum vapor flows are always the same:  $(V_{SR})_{\min} = (V_{SS})_{\min}$ . For  $\beta = \beta_R$  the main column is pinched at both feed locations; i.e., the minimum vapor flows for separations A/B and B/C are equal.

The minimum vapor flow for the entire thermally coupled system is flat over a wide range of  $\beta$ :  $\beta_p \leq \beta \leq \beta_R$ . This is the reason why dividing wall columns usually work well without tight control of the vapor or liquid split between both sides of the partition. The optimally designed fully thermally coupled system should operate with a fractional recovery of B in the top product of the prefractionator placed somewhere between points  $P$  and  $R$ . The transition split  $P$  is located at one end of the optimal section  $PR$ , and it is not a recommended design point for normal operation because process disturbances may move the operating point outside the optimal section  $PR$  shown in Fig. 13-70.

Although invented long ago, dividing wall columns and fully thermally coupled distillation systems were not implemented in practice until the late 1980s. The major objections concerned controllability and operability. Recently, however, several papers have shown that control of these systems is possible; see, e.g., Hernandez and Jimenez [*Ind. Eng. Chem. Res.*, **38**, 3957 (1999)]; Mutalib, Zeglam, and Smith [*Trans. Inst. Chem. Eng.*, **76**, 319 (1998)]; Halvorsen and Skogestad [*Comp. Chem. Eng.*, **21**, Suppl., S249 (1997)]. A major control problem was attributed to inability to set the vapor split between the main column and prefractionator due to conflicting pressure drop requirements between sections 6 and 2, as well as sections 1 and 3 in Fig. 13-67b. For the BC vapor to flow from section 6 to section 2, the pressure in the main column at the top of section 6 must be *higher* than the pressure in the prefractionator at the bottom of section 2. But the pressure at the bottom of section 3 in the main column must be *lower* than the pressure at the top of section 1 in the prefractionator, or else the AB vapor stream will not flow from the prefractionator to the main column. Liquid split control is easier to realize in practice, by using liquid collectors, overflows, or pumps. How much vapor flows straight up the main column and how much vapor splits off to the prefractionator depends on the pressure drops in the middle sections of the main column (sections 4 and 5 in Fig. 13-67b, between the interconnecting streams) and in the prefractionator. These pressure drops depend on the height of these sections, type of packing or stages, and liquid flows. Therefore, these pressure drops cannot be easily controlled in the configuration shown in Fig. 13-67b; moreover, they may even be such that vapor flows in the wrong direction.

Agrawal and Fidkowski [*AIChE J.*, **44**, 2565 (1998)]; U.S. Patent 6,106,674 proposed robustly operable two-column configurations that cleverly overcome this design and control problem. One is shown in Fig. 13-71. This new configuration is topologically equivalent to the original thermally coupled configuration in Fig. 13-67b and retains its energy advantage. In the new Agrawal and Fidkowski configuration, column section 6 is just shifted from the product column to the



**FIG. 13-70** Minimum vapor flows in the thermally coupled system; bottom curve, minimum vapor flow in prefractionator; top curve, minimum vapor flow from the reboiler of the main column (total minimum vapor flow for entire system) for  $\alpha_A = 6.25$ ,  $\alpha_B = 2.5$ ,  $\alpha_C = 1.0$  and feed mole fractions  $z_A = 0.33$ ,  $z_B = 0.33$ , and  $z_C = 0.34$ .

bottom of prefractionator; thus the two interconnecting vapor streams flow in the *same direction*. The first column (with sections 1, 2, and 6) in Fig. 13-71 operates at a slightly higher pressure than the second column, and the relative flows of the vapor streams can be changed by using a valve on one of them, as shown in the figure. The Agrawal and Fidkowski thermally coupled systems are expected to have higher investment cost than dividing wall columns, and the same investment cost as a Petlyuk thermally coupled system. However, for certain feed compositions and volatilities, the energy optimum in Fig. 13-70 may be narrow (i.e., the interval  $PR$  may be short) and dividing wall columns may not be able to operate at the optimum. On the other hand, the Agrawal and Fidkowski configurations are able to operate at the optimum because of better control. Also the Agrawal and Fidkowski configurations are useful in cases where high-purity products (especially B) are required. This is so because the two columns are built in separate shells which may contain more stages in one of the shells than could be accommodated in the corresponding side of the dividing wall column.

Experimental tests of dividing wall columns were carried out by Mutalib, Zeglam, and Smith [*Trans. Inst. Chem. Eng.*, **76**, 319 (1998)]. Today there are about 60 dividing wall columns in operation; 42 are owned by BASF (Parkinson, CEP, p. 10, July 2005).

Thermally coupled systems can also be devised for multicomponent mixtures. Sargent and Gaminibandara (*Optimization in Action*, L. W. C. Dixon, ed., Academic Press, London, 1976, p. 267) presented a natural extension of the Petlyuk column sequence to multicomponent systems. Agrawal [*Ind. Eng. Chem. Res.*, **35**, 1059 (1996); *Trans. Inst. Chem. Eng.*, **78**, 454 (2000)] presented a method for generating an even more complete superstructure from which all the

known column configurations (including thermally coupled systems) can be derived. Fidkowski and Agrawal [*AIChE J.*, **47**, 2713 (2001)] presented a method for calculating the minimum vapor flows in multicomponent thermally coupled systems. They analyzed the quaternary fully thermally coupled system in detail (there are many equivalent column configurations, all with the minimum number of column sections, which is 10, as well as one reboiler and one condenser; one of the configurations is shown in Fig. 13-72). They showed that one of the optimum solutions (with the minimum value of the total minimum vapor flow rate from the single reboiler in the system) occurs when the quaternary feed column (far left column in Fig. 13-72) and both ternary columns (the two middle columns in Fig. 13-72) perform transition splits. They also concluded that the optimized quaternary fully coupled system always requires less energy than the five sharp-split conventional systems (where each column performs a sharp split and has one feed, two products, one reboiler, and one condenser). The basic configurations for these five sharp-split systems are shown in Fig. 13-68a, c, g, t, and v. Selecting the best thermally coupled column configuration can be tedious without computer-aided tools such as the disjunctive programming approach developed by Caballero and Grossmann [*Ind. Eng. Chem. Res.*, **40**, 2260 (2001)].

All the multicomponent thermally coupled configurations have a corresponding dividing wall column equivalent. Keibel [*Chem. Eng. Technol.*, **10**, 92 (1987)] has shown examples of columns with multiple dividing walls, separating three, four, and six components. Agrawal [*Ind. Eng. Chem. Res.*, **40**, 4258 (2001)] presented several examples of quaternary columns with partitions and multiple reboilers and condensers. One of these examples is shown in Fig. 13-73.

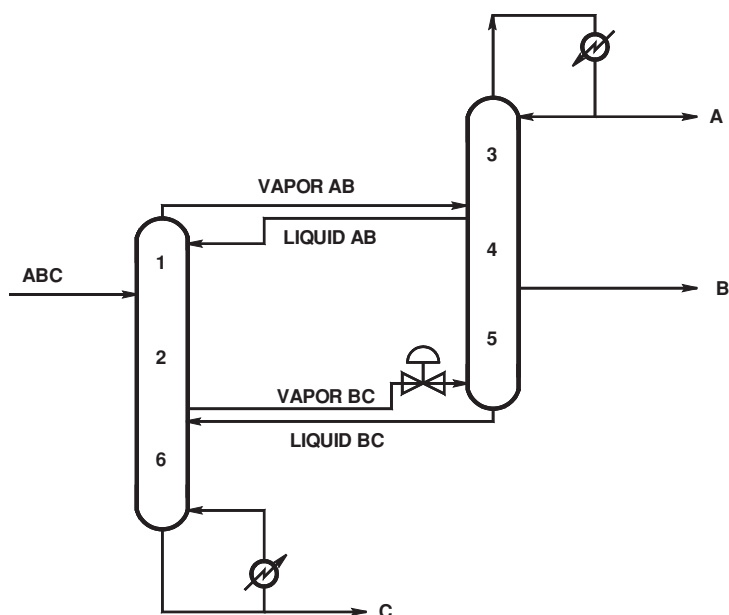


FIG. 13-71 Agrawal and Fidkowski thermally coupled system (topologically equivalent to the Petlyuk system shown in Fig. 13-67b).

### THERMODYNAMIC EFFICIENCY

Thermodynamic efficiency can be a useful figure of merit (in place of total cost, or total vapor rate) for comparing alternative column configurations. This is especially true for cryogenic distillations where very low temperatures are needed and highly efficient “cold box” designs are needed for achieving them. Thermodynamic efficiency of thermally coupled and other distillation systems for the separation of ternary mixtures was analyzed by Agrawal and Fidkowski [*Ind. Eng. Chem. Res.*, **37**, 3444 (1998)]. Feed composition regions for column configurations with the highest thermodynamic efficiency are shown in Fig. 13-74. Often, the efficiency of the direct split or the indirect

split is better than the efficiency of thermally coupled systems. This is primarily due to the ability of these configurations to accept or reject heat at the intermediate boiling or condensing temperatures of the binary submixtures. The fully thermally coupled system can only accept heat at the temperature of the highest-boiling component (boiling point of C) and reject heat at the lowest temperature (condensation temperature of A). This conclusion gave rise to new, more thermodynamically efficient thermally coupled configurations, as discussed by Agrawal and Fidkowski [*Ind. Eng. Chem. Res.*, **38**, 2065 (1999); U.S. Patent 6,116,051].

### HEAT INTEGRATION

In this subsection we describe heat pumps, multieffect distillation of binary mixtures, synthesis of multicomponent distillation systems with heat integration, and multieffect distillation for thermally coupled configurations.

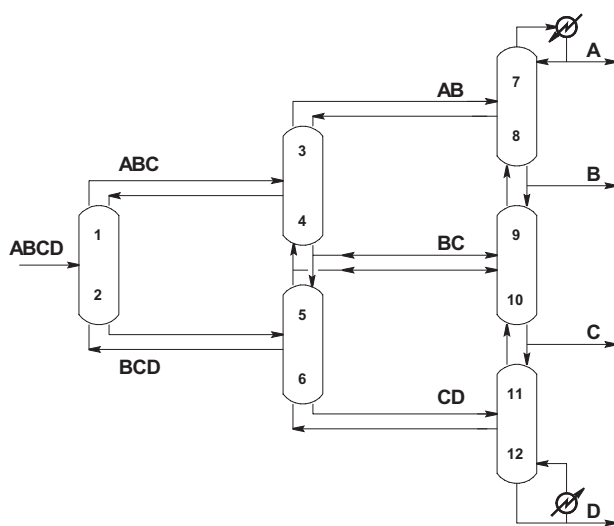


FIG. 13-72 One configuration of a fully thermally coupled system for separation of a quaternary mixture.

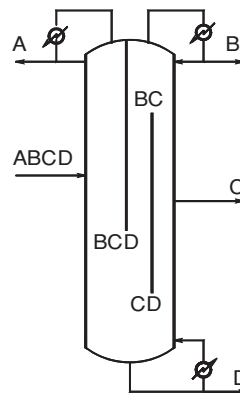
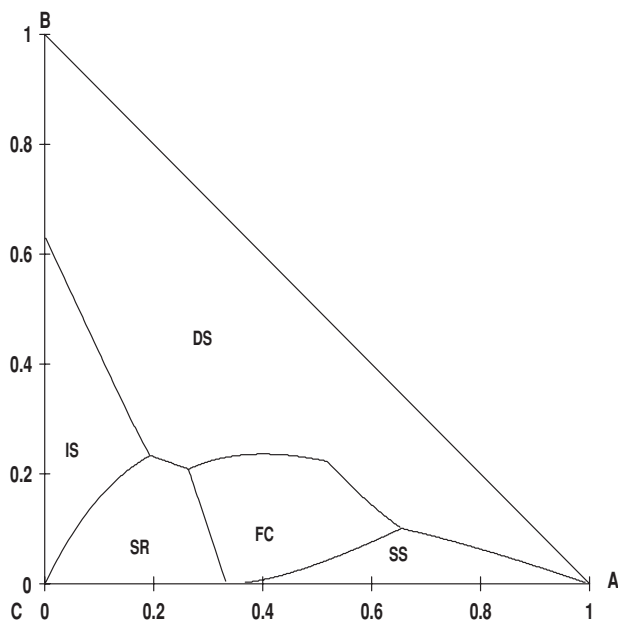


FIG. 13-73 One possible dividing wall column for separation of a quaternary mixture.



**FIG. 13-74** Feed composition regions of column configurations with highest thermodynamic efficiency; DS—direct split, IS—indirect split, SR—side rectifier, SS—side stripper, FC—fully thermally coupled. Example for  $\alpha_A = 4.0$ ,  $\alpha_B = 2.0$ , and  $\alpha_C = 1.0$ .

Two columns are heat-integrated when they exchange heat indirectly through a heat exchanger. This is different than in thermally linked configurations, where the heat exchange between columns is direct, through the material stream connecting the columns. The objective of heat integration in distillation systems is to save energy. Heat integration is realized by matching heat sources (usually condensers) with heat sinks (usually reboilers). The other heat exchangers considered for heat integration might be feed preheaters and product coolers. Typical examples of heat integration schemes are heat transfer from a condenser to a reboiler, or heat exchange between a (hot) column feed and a reboiler. If the temperature of the heat source

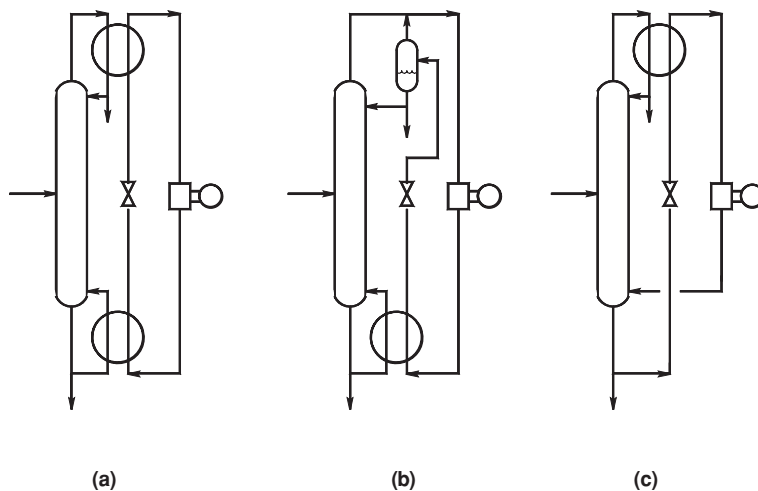
(condenser) is sufficiently higher than the temperature of the heat sink (reboiler), the opportunity for the match is straightforward. If the condenser temperature is too low, one may increase the condensing pressure or use a heat pump.

For example, the reboiler and condenser from the same column may be heat-integrated by using a heat pump, as discussed by Null [*Chem. Eng. Prog.*, **72**(7), 58 (1976)]. A heat pump may use an external fluid which is vaporized in the condenser, then compressed and condensed in the column reboiler (Fig. 13-75a). Another heat pump (Fig. 13-75b) uses column overhead vapor, which is compressed and condensed in the reboiler and then returned to the top of the column as reflux. A third possibility is to use the column bottoms, which is let down in pressure and vaporized in the condenser, then compressed and fed back to the bottom of the column as boil-up (Fig. 13-75c).

The entire rectifying section can be pressurized, and the heat can be transferred between any desired stages of the rectifying and stripping sections. This is called *secondary reflux and vaporization* (SRV) distillation. It reduces the consumption of both hot and cold utilities, and the sizes of the reboiler and condenser. However, capital cost is increased by additional intermediate heat exchangers; moreover, since the process is more thermodynamically reversible, it requires more stages to achieve the desired separation.

In multiple-column systems, possibilities for heat integration may be created by increasing the pressure in one of the columns, to increase the temperature of the condenser. When the temperature of the condenser becomes higher than the temperature of some other column reboiler, it is possible to heat-integrate these streams via a heat exchanger and reuse the heat rejected from the condenser. However, there are several drawbacks to this procedure. The required heat-transfer area in the integrated exchangers needs to be increased due to smaller temperature differences between the process streams than would normally exist when hot and cold external utilities are used to provide boil-up and condensation. Higher-pressure columns need hotter external heating utilities that are more expensive. Separation in higher-pressure columns is more difficult (because relative volatilities tend to decrease with increasing pressure), and more stages and energy may be required. Finally, higher-pressure column shells and piping may be more expensive, although this is not always certain, since the overall dimensions decrease with pressure.

One of the first industrial applications of heat-integrated distillation was a double column for air separation, developed by Linde in the beginning of the 20th century (Fig. 13-76). Air is compressed (typically to about 6 bar) and fed in the bottom of the high-pressure column. Nitrogen is condensed at the top of the high-pressure column.



**FIG. 13-75** Heat pumps transfer heat from the top to the bottom of the column. (a) External fluid heat pump. (b) Heat pump using column overhead. (c) Heat pump using column bottoms.

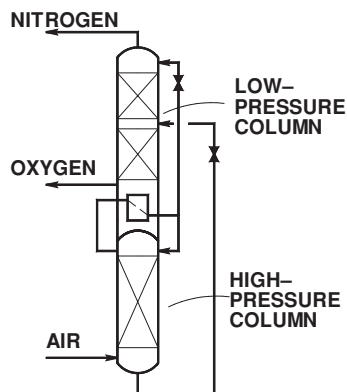


FIG. 13-76 Double-column arrangement for air separation.

The heat of condensation provides boil-up to the low-pressure column, by vaporizing oxygen in the sump located above the nitrogen condenser. Another industrial example of heat integration occurs in the large-scale methanol-water separation in the production of methanol from synthesis gas; see Siirola [Adv. Chem. Engng., **23**, 1 (1996)]. Several alternative designs were evaluated, and the regions of superior cost were developed on volatility-feed composition diagrams. Three heat-integrated designs were better than a single column:

1. *Split feed configuration*, in which the feed is split between the high-pressure and the low-pressure columns

2. *Sloppy first bottom split, second column pressurized*, in which the bottoms from the first column is further separated in the high-pressure column

3. *Sloppy first bottom split, first column pressurized*, in which the bottoms from the first, high-pressure column is further separated in the low-pressure column

These heat-integrated systems were economically advantageous because of the large feed flow rate. The last configuration was built.

Design heuristics and computer-aided design methods for heat-integrated distillation systems have been developed in numerous publications. Wankat [Ind. Eng. Chem. Res., **32**, 894 (1993)] analyzed 23 different multieffect distillation systems obtained by dividing a single column into two or three columns and operating them at various pressures or compositions, to obtain the temperatures necessary for heat integration. He developed heuristics to devise a feasible system. Rathore, Van Wormer, and Powers [AIChE J., **20**, 491 (1974) and **20**, 940 (1974)] presented an algorithmic synthesis method for multicomponent separation systems with energy integration. The five-component separation problem was solved by decomposing it into all the possible sharp splits and then examining all the possible integrations of heat exchangers. Thermally coupled systems, however, were not considered. Umeda, Niida, and Shiriko [AIChE J., **25**, 423 (1979)] published a thermodynamic approach to heat integration in distillation systems based on pinching heat source and heat sink curves on the temperature-heat duty diagram. Linhoff and coworkers (A User Guide on Process Integration for the Efficient Use of Energy, Institution of Chemical Engineers, Rugby, U.K., 1982) developed the very successful *pinch technology* approach, commonly used for heat exchanger synthesis and heat integration of entire plants. But, most successfully, Andrichowich and Westerberg [AIChE J., **31**, 1461 (1985)] presented a simple conceptual design approach wherein temperature-enthalpy diagrams are used to select the best column stacking. This design method is the recommended starting point for heat integration studies.

Even thermally coupled systems can be heat-integrated, as discovered by Agrawal [AIChE J., **46**, 2211 (2000)]. To achieve heat integration without compressors, the vapor connections between the columns must be eliminated. This is shown in Fig. 13-77a. The configuration is constructed by extending the prefractionator to a full

column with a reboiler and condenser, and adding sections 3x and 6x (analogous to sections 3 and 6 in the product column). By balancing heat duties, only liquid interconnecting streams remain, and the configuration is still thermodynamically equivalent to the original thermally coupled system (Fig. 13-67b). It is now possible to heat-integrate the columns by pressurizing one of them and allowing heat transfer from the high-pressure condenser to the low-pressure reboiler (Fig. 13-77b). The thermally coupled system and dividing wall column are equally energy-efficient. Therefore, the thermally coupled system with heat integration is even more energy-efficient than the dividing wall column.

There is a significant cost associated with heat integration (e.g., for heat pump compressors, taller columns with more stages, thicker walls of high-pressure equipment, higher cost of high-temperature utilities) which is why such configurations are not widely used. The operational flexibility of a heat-integrated system also becomes more constrained. However, heat-integrated systems can be quite economical for some applications. Typical applications where heat-integrated systems are preferred include

- Cryogenic separation, where very low-temperature cooling utilities do not exist and need to be created, which is very expensive. Therefore, it makes sense to reuse the expensive refrigeration.
- Very large-scale processes, with large feed and product flow rates. Because of the large feed rate, multiple columns may be required to process a given feed anyway. These cases have the potential to save a lot of energy due to heat integration. Also, pressurizing some equipment in these large-scale processes allows for decreased equipment sizes or increased production rate at a fixed equipment size.

## IMBALANCED FEEDS

In many practical cases, feed compositions are far from equimolar, with some components present in very small amount (e.g., less than 2 percent). In these cases the top or bottom product flow rate is less than 2 percent of the feed flow rate. The design for imbalanced feeds may include

- Various column diameters
- Intermittent pumping of heavies from the reboiler
- Intermittent firing up a specially devoted column to purify the dirty product made at a small rate
- In a continuous operation an overrefluxed section (to keep the column diameter the same over the height of the column)
- Producing an impure product to be purified later or discarded

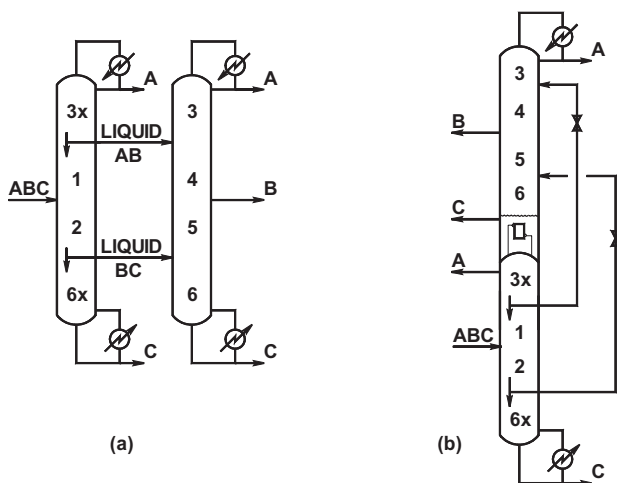


FIG. 13-77 (a) Configuration equivalent to thermally coupled system, but without vapor interconnecting streams. (b) Heat-integrated thermally coupled system.



## ENHANCED DISTILLATION

In distillation operations, separation results from differences in vapor- and liquid-phase compositions arising from the partial vaporization of a liquid mixture or the partial condensation of a vapor mixture. The vapor phase becomes enriched in the more volatile components while the liquid phase is depleted of those same components. In many situations, however, the change in composition between the vapor and liquid phases in equilibrium becomes small (so-called pinched condition), and a large number of successive partial vaporizations and partial condensations are required to achieve the desired separation. Alternatively, the vapor and liquid phases may have identical compositions, because of the formation of an azeotrope, and no separation by simple distillation is possible.

Several enhanced distillation-based separation techniques have been developed for close-boiling or low-relative-volatility systems, and for systems exhibiting azeotropic behavior. All these special techniques are ultimately based on the same differences in the vapor and liquid compositions as ordinary distillation; but, in addition, they rely on some additional mechanism to further modify the vapor-liquid behavior of the key components. These enhanced techniques can be classified according to their effect on the relationship between the vapor and liquid compositions:

1. *Azeotropic distillation and pressure-swing distillation.* Methods that cause or exploit azeotrope formation or behavior to alter the boiling characteristics and separability of the mixture.
2. *Extractive distillation and salt distillation.* Methods that primarily modify liquid-phase behavior to alter the relative volatility of the components of the mixture.
3. *Reactive distillation.* Methods that use chemical reaction to modify the composition of the mixture or, alternatively, use existing vapor-liquid differences between reaction products and reactants to enhance the performance of a reaction.

## AZEOTROPY

At low to moderate pressure ranges typical of most industrial applications, the fundamental composition relationship between the vapor and liquid phases in equilibrium can be expressed as a function of the total system pressure, the vapor pressure of each pure component, and the liquid-phase activity coefficient of each component  $i$  in the mixture:

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \quad (13-114)$$

In systems that exhibit ideal liquid-phase behavior, the activity coefficients  $\gamma_i$  are equal to unity and Eq. (13-114) simplifies to Raoult's law. For nonideal liquid-phase behavior, a system is said to show negative deviations from Raoult's law if  $\gamma_i < 1$  and, conversely, positive deviations from Raoult's law if  $\gamma_i > 1$ . In sufficiently nonideal systems, the deviations may be so large that the temperature-composition phase diagrams exhibit extrema, as shown in Figs. 13-6, 13-7, and 13-8. At such maxima or minima, the equilibrium vapor and liquid compositions are identical. Thus,

$$y_i = x_i \quad \text{for all } i = 1, \dots, c \quad (13-115)$$

and the system is said to form an azeotrope (from the Greek word meaning "to boil unchanged"). Azeotropic systems show a minimum in the  $T$  vs.  $x$ ,  $y$  diagram when the deviations from Raoult's law are positive (Fig. 13-6) and a maximum in the  $T$  vs.  $x$ ,  $y$  diagram when the deviations from Raoult's law are negative (Fig. 13-7). If, at these two conditions, a single liquid phase is in equilibrium with the vapor phase, the azeotrope is homogeneous. If multiple-liquid-phase behavior is exhibited at the azeotropic condition, the azeotrope is heterogeneous. For heterogeneous azeotropes, the vapor-phase composition is equal to the overall composition of the two (or more) liquid phases (Fig. 13-8). These conditions are consequences of the general definition of an azeotrope in any kind of mixture (i.e., homogeneous, heterogeneous, reactive, or in any combination), which is as follows: An *azeotropic state* is one in which mass transfer occurs between phases in a closed system while the composition of each phase remains constant, but not necessarily equal (see Prigogine and Defay, *Chemical Thermodynamics*, 4th ed., Longmans Green and Co., London, 1967;

Rowlinson, *Liquids and Liquid Mixtures*, 2d ed., Butterworths, London, 1969; Doherty and Malone, *Conceptual Design of Distillation Systems*, McGraw-Hill, 2001, chaps. 5, 8, App. C).

Mixtures with only small deviations from Raoult's law (i.e., ideal or nearly ideal mixtures) may form an azeotrope but only if the saturated vapor pressure curves of the two pure components cross each other (such a point is called a Bancroft point). In such a situation, the azeotrope occurs at the temperature and pressure where the curves cross, and perhaps also in the vicinity close to the Bancroft point [e.g., cyclohexane (n.b.p. 80.7°C) and benzene (n.b.p. 80.1°C) form an almost ideal mixture yet exhibit a minimum-boiling azeotrope with roughly equal proportions of each component]. As the boiling point difference between the components increases, the composition of the azeotrope shifts closer to one of the pure components (toward the lower-boiling pure component for minimum-boiling azeotropes, and toward the higher-boiling pure component for maximum-boiling azeotropes). For example, the minimum-boiling azeotrope between methanol (n.b.p. 64.5°C) and toluene (n.b.p. 110.6°C) occurs at  $\approx 90$  mol % methanol, and the minimum-boiling azeotrope between methyl acetate (n.b.p. 56.9°C) and water (n.b.p. 100°C) occurs at  $\approx 90$  mol % methyl acetate. Mixtures of components whose boiling points differ by more than about 50°C generally do not exhibit azeotropes distinguishable from the pure components even if large deviations from Raoult's law are present. As a qualitative guide to liquid-phase activity coefficient behavior, Robbins [*Chem. Eng. Prog.*, **76**(10), 58 (1980)] developed a matrix of chemical families, shown in Table 13-17, which indicates expected deviations from Raoult's law.

The formation of two liquid phases within some boiling temperature range is generally an indication that the system will also exhibit a minimum-boiling azeotrope, since two liquid phases may form when deviations from Raoult's law are large and positive. The fact that immiscibility does occur, however, does not guarantee that the azeotrope will be heterogeneous since the azeotropic composition may not necessarily fall within the composition range of the two-liquid phase region, as is the case for the methyl acetate–water and tetrahydrofuran–water systems. Since strong positive deviations from Raoult's law are required for liquid-liquid phase splitting, maximum-boiling azeotropes ( $\gamma_i < 1$ ) are never heterogeneous.

Additional general information on the thermodynamics of phase equilibria and azeotropy is available in Malesinski (*Azeotropy and Other Theoretical Problems of Vapour-Liquid Equilibrium*, Interscience, London, 1965), Swietoslawski (*Azeotropy and Polyazeotropy*, Pergamon, London, 1963), Van Winkle (*Distillation*, McGraw-Hill, New York, 1967), Smith and Van Ness (*Introduction to Chemical Engineering Thermodynamics*, McGraw-Hill, New York, 1975), Wiziak [*Chem. Eng. Sci.*, **38**, 969 (1983)], and Walas (*Phase Equilibria in Chemical Engineering*, Butterworths, Boston, 1985). Horsley (*Azeotropic Data-III*, American Chemical Society, Washington, 1983) compiled an extensive list of experimental azeotropic boiling point and composition data for binary and some multicomponent mixtures. Another source for azeotropic data and activity coefficient model parameters is the multivolume *Vapor-Liquid Equilibrium Data Collection* (DECHEMA, Frankfurt, 1977), a compendium of published experimental VLE data. Most of the data have been tested for thermodynamic consistency and have been fitted to the Wilson, UNIQUAC, Van Laar, Margules, and NRTL equations. An extensive two-volume compilation of azeotropic data for 18,800 systems involving 1700 compounds, entitled *Azeotropic Data* by Gmehling et al., was published in 1994 by VCH Publishers, Deerfield Beach, Fla. A computational method for determining the temperatures and compositions of all homogeneous azeotropes of a multicomponent mixture, from liquid-phase activity coefficient correlations, by a differential arclength homotopy continuation method is given by Fidkowski, Malone, and Doherty [*Computers and Chem. Eng.*, **17**, 1141 (1993)]. The method was generalized to determine all homogeneous and heterogeneous azeotropes by Wasylkiewicz, Doherty, and Malone [*Ind. Eng. Chem. Res.*, **38**, 4901 (1999)].